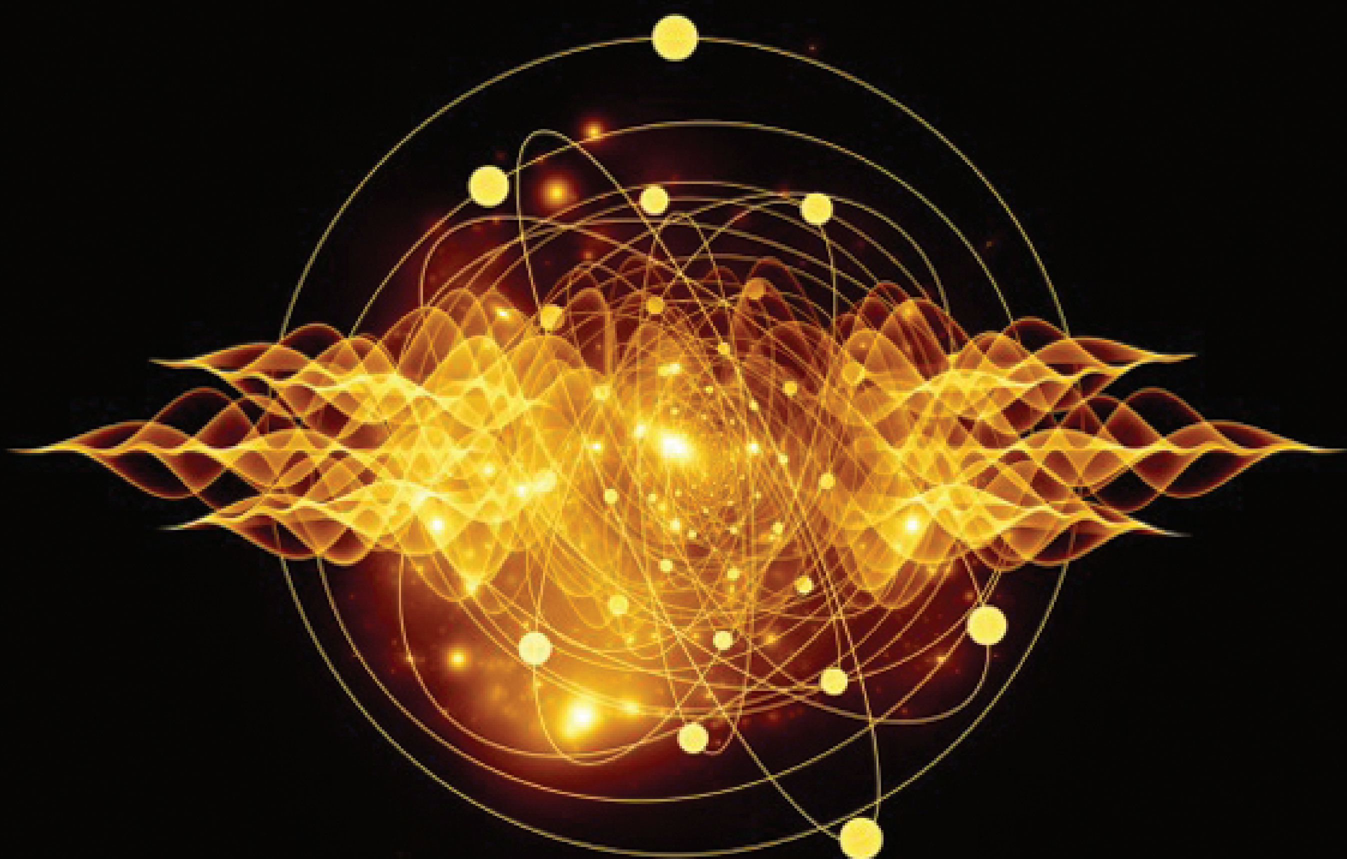
Problems in PHYSICS for

IIT-JEE ADVANCED

Fully Solved

Volume II

Heat | E&M | Optics | Modern Physics



Shashi Bhushan Tiwari

McGraw Hill Education

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PHYSICS *for*

IIT-JEE ADVANCED

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About the Author

Shashi Bhushan Tiwari is a distinguished academician and Physics guru. He graduated from IIT Kharagpur in year 1995 and has been mentoring students for IIT JEE for more than two decades.

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VOLUME II

Shashi Bhushan Tiwari

McGraw Hill Education (India) Private Limited
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McGraw Hill Education (India) Private Limited

Published by McGraw Hill Education (India) Private Limited,
444/1, Sri Ekambara Naicker Industrial Estate,
Alapakkam, Porur, Chennai - 600 116, Tamil Nadu, India

Problems in Physics for JEE Advanced, Volume II

ISBN (13): 978-93-5260-440-1
ISBN (10): 93-5260-440-7

Typeset at Bharati Composers, D-6/159, Sector-VI, Rohini, Delhi 110 085, and text and cover printed at

Cover Designer: India Binding House

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Dedicated to my beloved wife

Mrs. Kanti Tiwari

Preface

In continuation of Volume 1, the problems in this book too will challenge you on conceptual clarity and analytical skills besides judging your comprehension abilities. Most of the problems are interesting and intriguing, and will take your preparation to next level by upgrading your ability to apply fundamental laws of Physics in most diverse conditions.

Nobel laureate, Richard Feynman once said, "*You do not know anything until you have practiced.*" This book has a large number of challenging problems to give you a thorough practice.

This book has a very simple objective — it will test and mature you on all the required parameters to excel in JEE exam. The problems are strictly based on JEE Advanced syllabus.

Similar to Volume 1, every chapter in the book has been divided into three sections – Level 1, 2 and 3-in the increasing order of difficulty.

You will find this collection of problems as fresh and challenging. Attempt and enjoy learning physics!

Any suggestion towards the improvement of the book is welcome.

S.B. Tiwari

Problem Solving in Physics

Problem solving is moving towards a target when the path is not known. If you know the path for sure then it is not a 'problem'. Therefore, there is no single strategy or path for solving all problems.

Despite this fact, to develop a clear path, we need proper visualisation of the given situation in every problem. Therefore, it is recommended that students should always make a neat diagram wherever possible and jot down all necessary information. Then think about the applicable principles and relevant links between them. This will help you to get organised. Remember, practice will not make you perfect; only organised practice will make you perfect. You must know why you are doing a set of calculations or why you are writing an equation.

Please don't panic in any situation. Read the problem again – understand it word by word. Every word is important. You need to focus on key words such as massless, uniform, steady, constant, horizontal, vertical, etc. This will help you in constructing the problem quickly and finalising the relevant principles.

If a solution is getting lengthy but you are confident of your approach, just keep going Many problems are tough only because of their mathematical rigour.

Always make a habit of checking the units and dimensions of your answers or expressions that look odd. Checking the answers for extreme cases is must. It gives valuable insights and confidence about correctness of the solution.

Contents

CHAPTER 1

Temperature and Thermal Expansion

LEVEL 1

Q. 1: In a temperature scale X ice point of water is assigned a value of 20° X and the boiling point of water is assigned a value of 220° X. In another scale Y the ice point of water is assigned a value of –20° Y and the boiling point is given a value of 380° Y. At what temperature the numerical value of temperature on both the scales will be same?

Q. 2: The length of the mercury column in a mercury-in-glass thermometer is 5.0 cm at triple point of water. The length is 6.84 cm at the steam point. If length of the mercury column can be read with a precision of 0.01 cm, can this thermometer be used to distinguish between the ice point and the triple point of water?

Q. 3: What effect the following changes will makes to the range, sensitivity and responsiveness of a mercury in glass thermometer–

(a) Increase in size of the bulb.

(b) Increase in diameter of the capillary bore.

(c) Increase in length of the stem.

(d) Use of thicker glass for the bulb.

Q. 4: The focal length of a spherical mirror is given by $f = \frac{R}{2}$, where R is radius of curvature of the mirror. For a given spherical mirror made of steel the focal length is f = 24.0 cm. Find its new focal length if temperature increases by 50°C. Given $\alpha_{\text{steel}} = 1.2 \times 10^{-5}\,°\text{C}^{-1}$

Q. 5: A glass rod when measured using a metal scale at 30°C appears to be of length 100 cm. It is known that the scale was calibrated at 0°C. Find true length of the glass rod at–

(a) 30°C (b) 0°C

$\alpha_{\text{glass}} = 8 \times 10^{-6}\,°\text{C}^{-1}$ and $\alpha_{\text{metal}} = 26 \times 10^{-6}\,°\text{C}^{-1}$

Q. 6: A pendulum based clock keeps correct time in an aeroplane flying uniformly at a height h above the surface of the earth. The cabin temperature inside the plane is 10°C. The same pendulum keeps correct time on the surface of the earth when temperature is 30°C. Find the coefficient of linear expansions of the material of the pendulum. You can assume that $h << R$ (radius of the earth)

Q. 7: A liquid having coefficient of volume expansion γ_0 is filled in a cylindrical glass vessel. Glass has a coefficient of linear expansion of α_g. The liquid along with the container is heated to raise their temperature by ΔT. Mass of the container is negligible.

(a) Find relationship between α_g and γ_0 if it was found that the centre of mass of the system did not move due to heating.

(b) Find relationship between α_g and γ_0 if the fraction of volume of the container occupied by the liquid does not change due to heating.

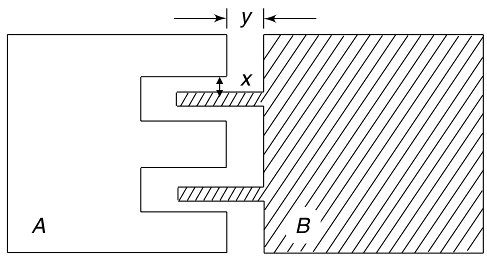
LEVEL 2

Q. 8: A water in glass thermometer has density of water marked on its stem [Density of water is the thermometric property in this case]. When this thermometer is dipped in liquid A the density of water read is 0.99995 g cm^{-3}. Thereafter it is dipped in liquid B and the reading remains unchanged. Maximum density of water is 1.00000 g cm^{-3}.

(a) Can we say that liquid A and B are necessarily in thermal equilibrium?

(b) If two liquids are mixed and the thermometer is inserted in the mixture, the height of water column in stem is found to change (i.e. reading is different from 0.99995 g cm^{-3}). Has the height increased or decreased?

Q. 9: Two metal plates A and B made of same material are placed on a table as shown in the figure. If the plates are heated uniformly, will the gap indicated by x and y in the figure increase or decrease?

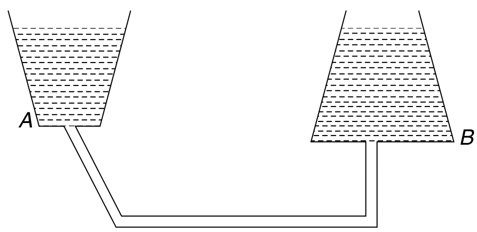


Q. 10: Containers A and B contain a liquid up to same height. They are connected by a tube (see figure).

(a) If the liquid in container A is heated, in which direction will the liquid flow through the tube.

(b) If the liquid in the container B is heated in which direction will the liquid flow through the tube?

Assume that the containers do not expand on heating.

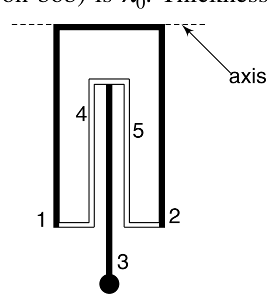


Q. 11: Height of mercury in a barometer is h_0 = 76. 0 cm at a temperature of θ_1 = 20°C. If the actual atmospheric pressure does not change, but the temperature of the air, and hence the temperature of the mercury and the tube rises to θ_2 = 35°C; what will be the height of mercury column in the barometer now? Coefficient of volume expansion of mercury and coefficient of linear expansion of glass are

$$\gamma_{Hg} = 1.8 \times 10^{-4}{}^\circ C^{-1};\ \alpha_g = 0.09 \times 10^{-4}{}^\circ C^{-1}$$

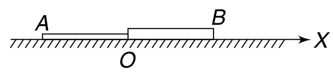
Q. 12: In the last problem if the scale for reading the height of mercury column is marked on the glass tube of the barometer, what reading will it show when temperature rises to θ_2 = 35°C?

Q. 13: Pendulum of a clock consists of very thin sticks of iron and an alloy. At room temperature the iron sticks 1 and 2 have length L_0 each. Length of each of the two alloy sticks 4 and 5 is ℓ_0 and the length of iron stick 3 (measured up to the centre of the iron bob) is λ_0. Thickness of connecting strips are negligible and mass of everything except the bob is negligible. The pendulum oscillates about the horizontal axis shown in the figure. It is desired that the time period of the pendulum should not change even if temperature of the room changes. Find the coefficient of linear expansion (α) of the alloy if the coefficient of linear expansion for iron is α_0.



Q. 14: Two samples of a liquid have volumes 400 cc and 220 cc and their temperature are 10°C and 110°C respectively. Find the final temperature and volume of the mixture if the two samples are mixed. Assume no heat exchange with the surroundings. Coefficient of volume expansion of the liquid is $\gamma = 10^{-3}{}^\circ C^{-1}$ and its specific heat capacity is a constant for the entire range of temperature.

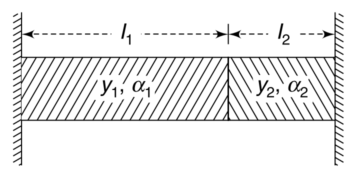
Q. 15: A composite bar has two segments of equal length L each. Both segments are made of same material but cross sectional area of segment OB is twice that of OA. The bar is kept on a smooth table with the joint at the origin of the co - ordinate system attached to the table. Temperature of the composite bar is uniformly raised by $\Delta\theta$. Calculate the x co-ordinate of the joint if coefficient of linear thermal expansion for the material is $\alpha{}^\circ C^{-1}$.



Q. 16: Two rods of different metals having the same area of cross section A, are placed between the two massive walls as shown in figure. The first rod has a length l_1, coefficient of linear expansion α_1 and Young's modulus Y_1. The corresponding quantities for second rod are are l_2, α_2 and Y_2. The temperature of both the rods is now raised uniformly by T degrees.

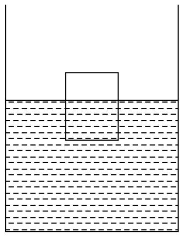
(a) Find the force with which the rods act on each other (at higher temperature) in terms of given quantities.

(b) Also find the length of the rods at higher temperature.



Q. 17: Two rods of equal cross-sections, one of copper and the other of steel, are joined to form a composite rod of length 2.0 m. At 30°C the length of the copper rod is 0.5 m. When the temperature is raised to 130°C the length of the composite rod increases to 2.002 m. If the composite rod is fixed between two rigid walls and is thus not allowed to expand, it is found that the lengths of the two component rods also do not change with the increase in temperature. Calculate the Young's modulus and the coefficient of linear expansion of steel. Given: Young's modulus of copper 1.3×10^{11} N/m^2, coefficient of linear expansion of copper = 1.6×10^{-5} per°C.

Q. 18: A beaker contains a liquid of volume V_0. A solid block of volume V floats in the liquid with 90% of its volume submerged in the liquid. The whole system is heated to raise its temperature by $\Delta\theta$. It is observed that the height of liquid in the beaker does not change and the solid in now floating with its entire volume submerged. Calculate $\Delta\theta$. It is given that coefficient of volume expansion of the solid and the glass (beaker) are γ_s and γ_g respectively.

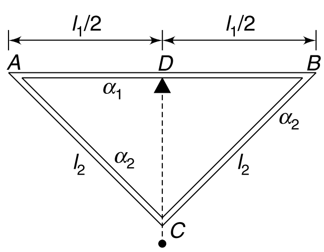


Q. 19: Assume that the coefficient of linear expansion of the material of a rod remains constant, equal to $\alpha°C^{-1}$ for a fairly large range of temperature. Length of the rod is L_0 at temperature θ_0.

(a) Find the length of the rod at a high temperature θ.

(b) Approximate the answer obtained in (a) to show that the length of the rod for small changes in temperature is given by $L = L_0[1 + \alpha(\theta - \theta_0)]$

Q. 20: In a compensated pendulum a triangular frame ABC is made using two different metals. AB of length ℓ_1 is made using a metal having coefficient of linear expansion α_1. BC and AC of length ℓ_2 each have coefficient of linear expansion α_2. A heavy bob is attached at C. Pendulum can oscillate about the pivot D. Find $\frac{\ell_2}{\ell_1}$ so that distance of bob from the pivot point D does not change with change in temperature.

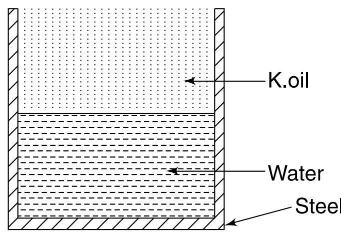


Q. 21: A thin uniform rod of mass M and length l is rotating about a frictionless axis passing through one of its ends and perpendicular to the rod. The rod is heated uniformly to increases its temperature by $\Delta\theta$. Calculate the percentage change in rotational kinetic energy of the rod. Explain why the answer is not zero. Take coefficient of linear expansion of the material of the rod to be α.

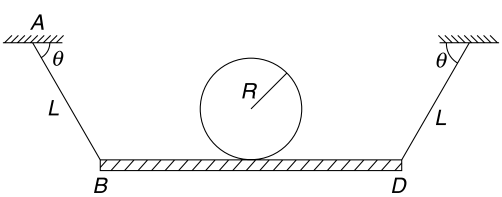
Q. 22: (a) A steel tank has internal volume V_0 (= 100 litre). It contains half water $\left(\text{volume} = \frac{V_0}{2}\right)$ and half kerosene oil at temperature $\theta_1 = 10°C$ Calculate the mass of kerosene that flows out of the tank at temperature of $\theta_2 = 40°C$. Coefficient of cubical expansion for different substances are:

$\gamma_k = 10^{-3}\,°C^{-1}$; $\gamma_w = 2 \times 10^{-4}\,°C^{-1}$; $\gamma_{steel} = 1.2 \times 10^{-5}\,°C^{-1}$. Density of kerosene at 10°C is $\rho_1 = 0.8$ kg/litre

(b) In the last problem the height of water in the container at $\theta_1 = 10°C$ is $H_1 = 1.0$ m. Find the height of water at $\theta_2 = 40°C$.



Q. 23: A metal cylinder of radius R is placed on a wooden plank BD. The plank is kept horizontal suspended with the help of two identical string AB and CD each of length L. The temperature coefficient of linear expansion of the cylinder and the strings are α_1 and α_2 respectively. Angle θ shown in the figure is 30°. It was found that with change in temperature the centre of the cylinder did not move. Find the ratio $\frac{\alpha_1}{\alpha_2}$, if it is know that $L = 4R$. Assume that change in value of θ is negligible for small temperature changes.

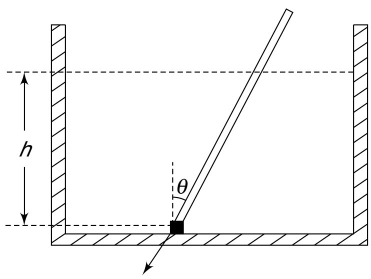


Q. 24: A vernier calliper has 10 divisions on vernier scale coinciding with 9 main scale divisions. It is made of a material whose coefficient of linear expansion is $\alpha = 10^{-3}\,°C^{-1}$. At 0°C each main scale division = 1mm. An object has a length of 10 cm at a temperature of 0°C and its material has coefficient of linear expansion equal to $\alpha_1 = 1 \times 10^{-4}\,°C^{-1}$. The length of this object is measured using the said vernier calliper when room temperature is 50°C.

(a) Find the reading on the main scale and the vernier scale

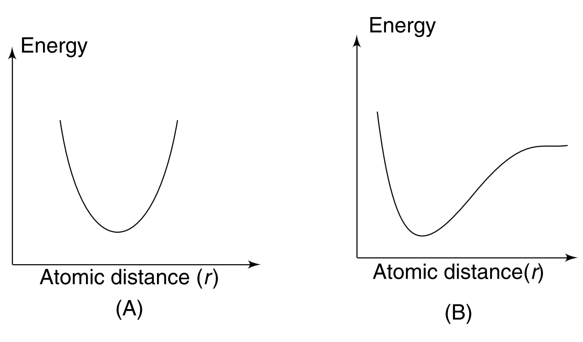
(b) The same object is measured (at 50°C) using a wooden scale whose least count is 1mm. Write the measured reading using this scale assuming it to be correct at all temperature.

Q. 25: A rectangular tank contains water to a height h. A metal rod is hinged to the bottom of the tank so that it can rotate freely in the vertical plane. The length of the rod is L and it remains at rest with a part of it lying above the water surface. In this position the rod makes an angle θ with the vertical. Assume that $y = \cos\theta$ and find fractional change in value of y when temperature of the system increases by a small value ΔT. Coefficient of linear expansion of material of rod and the tank are α_1 and α_2 respectively. Coefficient of volume expansion of water is γ. What is necessary condition for θ to increase?

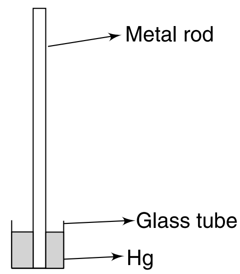


LEVEL 3

Q. 26: In the given figure graph B shows the variation of potential energy versus atomic separation (r) in a material. Argue qualitatively to show that if the potential energy graph was a symmetrical one as depicted in graph A, there would have been no thermal expansion on heating.



Q. 27: In design of a compensated pendulum, a light metal rod of length L_0 = 1.0 m is attached to a glass tube filled with mercury. Neglecting the mass of the glass tube as well, calculate the height of mercury column in the glass tube so that centre of mass of this system does not rise or fall with temperature Given : $\gamma_{Hg} = 1.82 \times 10^{-4}\,K^{-1}$; $\alpha_{glass} = 9 \times 10^{-6}\,K^{-1}$; $\alpha_{metal} = 1.2 \times 10^{-5}\,K^{-1}$

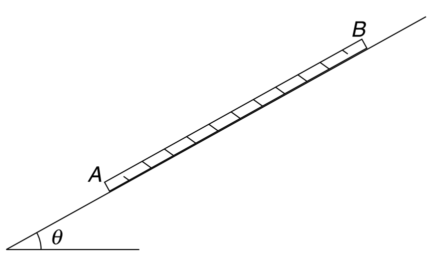


Q. 28: A uniform metal rod (AB) of mass m and length L is lying on a rough incline. The inclination of the incline and coefficient of friction between the rod and the incline is $\theta = 37°$ and $\mu = 1.0$ respectively. $\tan 37° = \frac{3}{4}$

(a) If temperature increases the rod expands. However, there is a point P on the rod which does not move. Find the distance of this point from the lower end of the rod.

(b) If the temperature falls the rod contracts. Once again there is a point Q which does not move. Find distance of Q from the lower end the rod.

(c) Will the repeated expansion and contraction cause the rod to slide down?



ANSWERS

1. 40° X = 40° Y

2. No

3.

Change made to thermometer	change in range	change in sensitivity	Change in responsiveness
Increase in size of bulb	shorter range (reach a lower maximum temperature)	More sensitive	Less responsiveness (longer response time)
Increase in diameter of capillary bore	Longer range	Less sensitive	No change
Increase in length of stem	Longer range	No change	No change
Use of thick glass in bulb	No change	No change	Less responsive

4. 24.0144 cm

5. (a) 100.078 cm (b) 100.054 cm

6. $\alpha = \dfrac{h}{10R}$

7. (a) $\gamma_0 = 2\alpha_g$ (b) $\gamma_0 = 3\alpha_g$

8. (a) No (b) decreased

9. x increases, y decreases

10. In both cases the liquid flows from B to A.

11. $h = h_0[1 + \gamma_{Hg}\Delta\theta] = 76.205$ cm

12. $\dfrac{h_0[1 + \gamma_{Hg}\Delta\theta]}{[1 + \alpha_g\Delta\theta]} \simeq 76.195$ cm

13. $\alpha = \dfrac{(L_0 + \lambda_0)\ \alpha_0}{\ell_0}$

14. 43.33°C; 620 cc

15. $-\dfrac{L\alpha\Delta\theta}{6}$

16. (a) $F = \dfrac{AT(l_1\alpha_1 + l_2\alpha_2)}{\left(\dfrac{l_1}{Y_1} + \dfrac{l_2}{Y_2}\right)}$

(b) $\left(l_1 + l_1\alpha_1T - \dfrac{F}{A}\dfrac{l_1}{Y_1}\right), \left(l_2 + l_2\alpha_2T - \dfrac{F}{A}\dfrac{l_2}{Y_2}\right)$

17. $Y_s = 2.6 \times 10^{11}$ N/m^2 $\alpha_s = 0.8 \times 10^{-5}$/°C

18. $\Delta\theta = \dfrac{0.1(V_0 - V)}{(0.9\ V_0 + V)\gamma_s - (V_0 + 0.9\ V)\ \gamma_g}$

19. $L = L_0 e^{\alpha(\theta - \theta_0)}$

20. $\dfrac{1}{2}\sqrt{\dfrac{\alpha_1}{\alpha_2}}$

21. $-200\,\alpha\Delta\theta\%$

22. (a) $\Delta m = \dfrac{V_0\rho_1}{2}\dfrac{(\gamma_k + \gamma_w - 2\gamma_s)}{(1 + \gamma_k\Delta\theta)}\ \Delta\theta$;

$= 1.37$ kg [where $\Delta\theta = \theta_2 - \theta_1$]

(b) $\dfrac{H_1[1 + \gamma_w\Delta\theta]}{\left[1 + \frac{2}{3}\gamma_s\Delta\theta\right]} = 1.0057$ m

23. $\dfrac{\alpha_1}{\alpha_2} = \dfrac{8}{1}$

24. (a) $MSR = 95$ mm; $VSR = 7$

(b) 100 ± 1 mm

25. $\dfrac{1}{2}(\alpha_1 + \gamma - 4\alpha_2)\Delta T$; $4\alpha_2 > \alpha_1 + \gamma$

27. $\dfrac{2L_0\,\alpha_{metal}}{\gamma_m - 2\alpha_g} = 0.146$ m

28. (a) $0.875\ L$ (b) $0.125\ L$

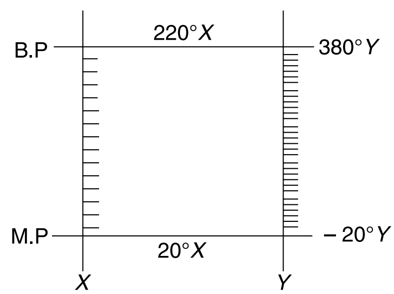
(c) Yes

SOLUTIONS

1. Change of 1° X = change of 2° Y

At a particular temperature, if we are x divisions away from 20° X and y divisions away from $-20°$ Y then–

$$20 + x = -20 + 2x \Rightarrow x = 40$$



2. The length Hg column increases linearly with temperature.

Steam point = 100°C

Triple point = 0.01°C

∴ Change in temperature of 99.99°C causes a change in length equal to 1.84 cm

∴ Change in temperature of 0.01°C will cause a change in length equal to

$$\frac{1.84}{99.99} \times 0.01 = 0.00018 \text{ cm}$$

Hence, the thermometer will fail to distinguish between the ice point and the triple point .

4.
$$R_0 = 2f = 48.0 \text{ cm}$$
$$R = R_0(1 + \alpha\Delta\theta) = 48.0\ [1 + 1.2 \times 10^{-5} \times 50]$$
$$= 48.0[1.0006] = 48.0288 \text{ cm}$$

∴ New focal length is $f = \dfrac{48.0288}{2} = 24.0144$ cm

5. (i) Two markings on the metal scale at a separation of 100 cm at a temperature of 30°C, correspond to true length given by $\ell = 100(1 + \alpha_{\text{metal}}\Delta\theta)$

$$= 100\,(1 + 26 \times 10^{-6} \times 30) = 100.078 \text{ cm}$$

Hence, true length of glass at 30°C is 100.78 cm

(ii) If ℓ_0 = length of glass at 0°C, then

$$100.78 = \ell_0[1 + \alpha_g \times 30]$$

$$\ell_0 = \frac{100.78}{1 + 8 \times 10^{-6} \times 30} = 100.054 \text{ cm}$$

6. Acceleration due to gravity at height h is $g' \simeq g\left(1 - \frac{2h}{R}\right)$

$\therefore$ Time period at height h is

$$T' = 2\pi\sqrt{\frac{\ell}{g\left(1 - \frac{2h}{R}\right)}} \quad \text{where} \quad \ell = \text{length at } 10°\text{C}$$

Time period on the surface of the earth is $T = 2\pi\sqrt{\frac{\ell(1 + \alpha\Delta\theta)}{g}}$

Since $$T' = T$$

$$\left(1 - \frac{2h}{R}\right)^{-\frac{1}{2}} = (1 + \alpha\Delta\theta)^{\frac{1}{2}} \Rightarrow 1 + \frac{h}{R} = 1 + \frac{1}{2}\alpha\Delta\theta$$

$$\alpha = \frac{2h}{R\Delta\theta} = \frac{2h}{R\cdot(20)} = \frac{h}{10\cdot R}$$

7. (a) The COM will not move if the height of liquid column in the container does not change.

Let original volume of liquid, area of cross section of the container and height of liquid be V_0, A_0 and H_0 respectively.

$$H_0 = \frac{V_0}{A_0}$$

If temperature increases by $\Delta\theta$

$$V = V_0[1 + \gamma_0\Delta\theta] \quad \text{and} \quad A = A_0[1 + 2\alpha_g\Delta\theta]$$

$\therefore$ Height of liquid

$$H = \frac{V_0[1 + \gamma_0\Delta\theta]}{A_0[1 + 2\alpha_g\Delta\theta]} = \frac{H_0[1 + \gamma_0\Delta\theta]}{[1 + 2\alpha_g\Delta\theta]}$$

If $$H = H_0 \quad \text{then} \quad 1 + \gamma_0\Delta\theta = 1 + 2\alpha_g\Delta\theta$$

$\Rightarrow$ $$\gamma_0 = 2\alpha_g$$

(b) Let original value of the liquid and the container be $V_{\ell 0}$ and V_{C0}.

At increased temperature

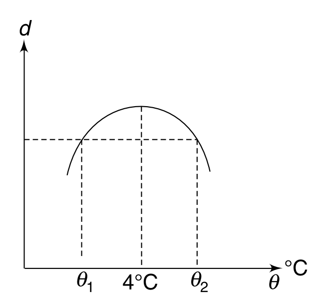
$$V_\ell = V_{\ell 0}[1 + \gamma_0\Delta\theta] \quad \text{and} \quad V_C = V_{C0}[1 + 3\alpha_g\Delta\theta]$$

$$\frac{V_\ell}{V_C} = \frac{V_{\ell 0}[1 + \gamma_0\Delta\theta]}{V_{C0}[1 + 3\alpha_g\Delta\theta]}$$

For $\frac{V_\ell}{V_C} = \frac{V_{\ell 0}}{V_{C0}}$ we must have

$$1 + \gamma_0\Delta\theta = 1 + 3\alpha_g\Delta\theta$$

$\Rightarrow$ $$\gamma_0 = 3\alpha_g$$



8. (a) The density of water changes with temperature as shown in the figure.

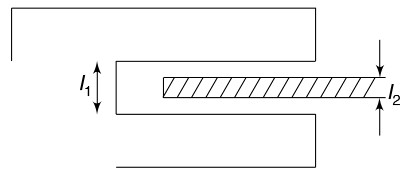
If is possible that the two liquids are at temperature θ_1 (< 4°C) and θ_2 (> 4°C) and therefore, the density of water is same.

(b) When liquids at θ_1 and θ_2 are mixed, the mixture will have a temperature between θ_1 and θ_2. It means density of water will increases and height of water column in the stem will decreases.

9. Change in length ℓ_1 due to increases in temperature $\Delta\ell_1 = \ell_1 \alpha \Delta\theta$

Change in length ℓ_2 is $\Delta\ell_2 = \ell_2 \alpha \Delta\theta$

Since $\ell_1 > \ell_2$ $\quad\therefore\quad \Delta\ell_1 > \Delta\ell_2$



$\therefore$ Gap indicated by x will increase.

It is trivial to understand that y will decrease.

10. Consider a cylindrical container containing a liquid. It is easy to see that weight of liquid divided by the area of the base of the container is pressure at the bottom. Since neither the area nor the weight of the liquid changes on heating, the pressure remains constant.

If the liquid is heated, its density decreases and volume increases but the pressure ($P = \rho gh$) at the bottom does not change.

When liquid in A is heated, the height change is less compared to the change in a cylindrical container but density change is identical in two case. Hence pressure at the bottom of A decreases and liquid flows from B to A.

When B is heated, height change of liquid is larger than the case of a cylindrical container and pressure at bottom of B increases. Hence the liquid flows from B to A.

11. $\theta_2 - \theta_1 = \Delta\theta = 15°C$

Let density of Hg at θ_1 be ρ_0

Then $\quad P_0 = \rho_0 g h_0 \quad$ [P_0 = atmospheric pressure] ...(i)

Now density of Hg at temperature θ_2 is–

$$\rho = \frac{\rho_0}{1 + \gamma_{Hg}\Delta\theta} \qquad \therefore \quad P_0 = \frac{\rho_0 gh}{1 + \gamma_{Hg}\Delta\theta} \qquad \text{...(ii)}$$

From (i) and (ii)

$$\frac{h}{1 + \gamma_{Hg}\Delta\theta} = h_0$$

$$\therefore \qquad h = h_0[1 + \gamma_{Hg}\Delta\theta]$$

$$= 76.0[1 + 1.8 \times 10^{-4} \times 15] = 76.0 \times 1.0027 = 76.205 \text{ cm}$$

12. Actual height of Hg column at changed temperature is

$$h = h_0[1 + \gamma_{Hg}\Delta\theta] \qquad \text{...(i)}$$

But the glass scale also expands and the reading shown by it will be less than h given in (i). A reading of 1 cm on glass scale at θ_2 actually represents a length of

$$= 1 \text{ cm}[1 + \alpha_g \Delta\theta]$$

$\therefore$ A length h will be read (by the scale) as

$$\frac{1}{1[1 + \alpha_g\Delta\theta]} h = \frac{h_0[1 + \gamma_{Hg}\Delta\theta]}{1 + \alpha_g\Delta\theta}$$

$$= \frac{76.205}{1 + 0.09 \times 10^{-4} \times 15} = \frac{76.205}{1.000135} = 76.195 \text{ cm}$$

13. Length of the pendulum is $L = L_0 + \lambda_0 - \ell_0$

$$\Delta L = \Delta L_0 + \Delta\lambda_0 - \Delta\ell_0$$

$$\therefore \qquad 0 = L_0\alpha_0\Delta\theta + \lambda_0\alpha_0\Delta\theta - \ell_0\alpha\Delta\theta [\Delta\theta = \text{change in temperature}]$$

$$\Rightarrow \qquad \alpha = \frac{(L_0 + \lambda_0)\,\alpha_0}{\ell_0}$$

14. Mass of sample at 10°C is $m_1 = \rho_{10}V_{10} = \rho_{10}(400)$

Mass of sample at 110°C is $m_2 = \rho_{110}V_{110} = \rho_{110}(220)$

$$= \frac{\rho_{10}}{1 + \gamma \times 100} \cdot 220 = \frac{\rho_{10}}{1.1} \times 220 = \rho_{10}(200) = \frac{m_1}{2}$$

When mixed, let the final temperature be θ.

$$m_1 \cdot s \cdot (\theta - 10) = m_2 \cdot s \cdot (110 - \theta)$$

$$\Rightarrow \quad \theta - 10 = \frac{110 - \theta}{2} \Rightarrow \theta = \frac{130}{3} = 43.33°C$$

$$\text{Mass of mixture} = m_1 + \frac{m_1}{2} = \frac{3m_1}{2}$$

$\therefore$ Volume of $\frac{3m_1}{2}$ mass of liquid at 10°C will be $= \frac{3}{2} \times 400 = 600$ cc

At $\frac{130}{3}$ °C this volume will become

$$V = 600\left[1 + \left(\frac{130}{3} - 10\right)\gamma\right] = 600\left[1 + \frac{100}{3} \times 10^{-3}\right] = 600\left[\frac{3.1}{3}\right] = 620 \text{ cc}$$

15. Mass of $OA = m$ and Mass of $OB = 2$ m

Let the joint shift by x to left.

The COM of the composite rod will not move.

The COM of segment OA will move to left by $x + \frac{L\alpha\Delta\theta}{2}$

The COM of segment BO will move to right by $\frac{L\alpha\Delta\theta}{2} - x$

For COM of the composite rod to remain unmoved, we must have

$$m\left(x + \frac{L\alpha\Delta\theta}{2}\right) = 2m\left(\frac{L\alpha\Delta\theta}{2} - x\right)$$

$$\Rightarrow \quad 3x = \frac{L\alpha\Delta\theta}{2} \Rightarrow x = \frac{L\alpha\Delta\theta}{6}$$

$\therefore$ x co-ordinate of joint is $-\frac{L\alpha\Delta\theta}{6}$

16. (a) When the temperature is raised by T, then

Increase in length of first rod $= l_1\alpha_1 T$

and increase in length of second rod $= l_2\alpha_2 T$

$\therefore$ Total increase in length $l_1\alpha_1 T + l_2\alpha_2 T = T(l_1\alpha_1 + l_2\alpha_2)$ (i)

As the walls are rigid, the above increase will not be possible. This will be compensated by the force F producing decrease in the length of the rods.

$$\text{Decrease in length of first rod} = \frac{F \times l_1}{Y_1 \times A}$$

$$\text{Decrease in length of second rod} = \frac{F \times l_2}{Y_2 \times A}$$

$\therefore$ Total decrease in length due to $F = \frac{F}{A}\left(\frac{l_1}{Y_1} + \frac{l_2}{Y_2}\right)$ (ii)

From eq. (i) and (ii), we have

$$\frac{F}{A}\left(\frac{l_1}{Y_1}+\frac{l_1}{Y_1}\right) = T(l_1\alpha_1 + l_2\alpha_2)$$

Or, $$F = \frac{AT(l_1\alpha_1 + l_2\alpha_2)}{\left(\frac{l_1}{Y_1}+\frac{l_2}{Y_2}\right)} \quad \text{...(iii)}$$

(b) Length of the first rod = (original length) + (increase in length due to temp) – (decrease in length due to force F)

$$= \left(l_1 + l_1\alpha_1 T - \frac{F}{A}\frac{l_1}{Y_1}\right) \quad \text{...(iv)}$$

$$\text{Length of the second rod} = \left(l_2 + l_2\alpha_2 T - \frac{F}{A}\frac{l_2}{Y_2}\right) \quad \text{...(v)}$$

The total length will remain unaltered.

17. For copper rod

$$(l_T)_{cu} - (l_0)_{cu} = \alpha_{cu} \times (l_0)_{cu} \times (T_2 - T_1) = \alpha_{cu} \times 0.5 \times (130 - 30) = 50\alpha_{cu}$$

Similarly, for steel rod

$$(l_T)_s - (l_0)_s = \alpha_s \times 1.5 \times 100 = 150\alpha_s$$

$$\text{Total change in length} = 50\alpha_{cu} + 150\alpha_s = 0.002$$

Here $$\alpha_{cu} = 1.6 \times 10^{-5}/°C$$

$$\text{Solving for } \alpha_s \text{ gives } \alpha_s = 0.8 \times 10^{-5}/°C$$

According to the given question, there is no change in length of individual rod. So the length change due to stress is balanced by the length change due to thermal expansion.

Stress in steel rod = $Y_s \times$ strain = $Y_s(\Delta l/l_0)_s = Y_s \times \alpha_s \times \Delta T$

Similarly stress in copper rod = $Y_{cu} \times \alpha_{cu} \times \Delta T$

But, stress in steel rod = Stress in copper rod

$\therefore$ $$\frac{Y_s}{Y_{cu}} = \frac{\alpha_{cu}}{\alpha_s} \quad \text{Or,} \quad Y_s = Y_{cu}\left(\frac{\alpha_{cu}}{\alpha_s}\right)$$

Putting the values we get $Y_s = 2.6 \times 10^{11}$ N/m^2.

18. Let ρ_ℓ and ρ_s be initial densities of the liquid and the solid respectively.

According to the problem $$\frac{\rho_s}{\rho_\ell} = 0.9 \quad \text{...(i)}$$

On increasing the temperature the two densities become equal.

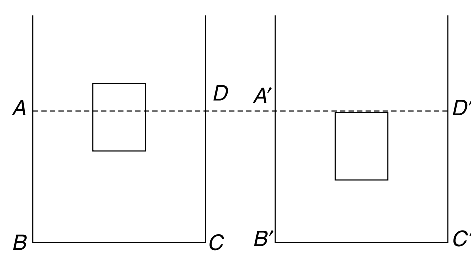
$$\rho_s[1 + \gamma_s\Delta\theta] = \rho_\ell[1 + \gamma_\ell\Delta\theta] \Rightarrow 0.9(1 + \gamma_s\Delta\theta) = (1 + \gamma_l\Delta\theta)$$

V_0 = original volume of liquid

V = original volume of solid

Volume $ABCD$ = V_0 + 0.9 V

Volume $A'B'C'D'$ = $V_0(1 + \gamma_\ell\Delta\theta) + V(1 + \gamma_s\Delta\theta)$



The liquid level will not change if volume $ABCD$ of the container expands to be equal to volume $A'B'C'D'$

$$\Rightarrow \quad (V_0 + 0.9V)(1 + \gamma_g\,\Delta\theta) = V_0(1 + \gamma_\ell\Delta\theta) + V(1 + \gamma_s\Delta\theta)$$

$$\Rightarrow \quad (V_0 + 0.9V)(1 + \gamma_g\,\Delta\theta) = 0.9V_0(1 + \gamma_s\Delta\theta) + V(1 + \gamma_s\Delta\theta) \quad \text{[using (ii)]}$$

$$\Rightarrow \quad (V_0 + 0.9V)(1 + \gamma_g\,\Delta\theta) = (0.9V_0 + V)(1 + \gamma_s\Delta\theta)$$

$$\Rightarrow \quad (V_0 + 0.9V) - (0.9V_0 + V) = [(0.9\,V_0 + V)\gamma_s - (V_0 + 0.9V)\gamma_g]\,\Delta\theta$$

$$\Rightarrow \quad \frac{0.1(V_0 - V)}{(0.9V_0 + V)\gamma_s - (V_0 + 0.9V)\gamma_s} = \Delta\theta$$

19. (a) Let the length of the rod be L at temperature θ. A small change in temperature by $d\theta$ will cause the length to change by $dL = \alpha L d\theta$

$$\Rightarrow \quad \int_{L_0}^{L} \frac{dL}{L} = \alpha \int_{\theta_0}^{\theta} d\theta$$

$$\Rightarrow \quad \ell n\left(\frac{L}{L_0}\right) = \alpha(\theta - \theta_0)$$

$$\Rightarrow \quad L = L_0 e^{\alpha(\theta - \theta_0)}$$

(b) It is known that $e^x = 1 + x + \frac{x^2}{2!} + \ldots$

When x is small: $e^x \simeq 1 + x$

$$L = L_0 e^{\alpha(\theta - \theta_0)} \qquad \therefore \quad L \simeq L_0[1 + \alpha(\theta - \theta_0)]$$

20.

$$y^2 = \ell_2^2 - \frac{\ell_1^2}{4}$$

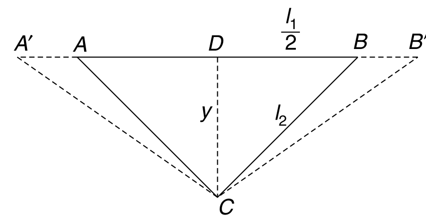
$$2y\Delta y = 2\ell_2 \Delta\ell_2 - \frac{2}{4}\ell_1 \Delta\ell_1$$

For

$$\Delta y = 0$$

$$4\ell_2\Delta\ell_2 = \ell_1\Delta\ell_1 \quad \Rightarrow \quad 4\ell_2^2\alpha_2\Delta T = \ell_1^2\alpha_1\Delta T$$

$$\frac{\ell_2}{\ell_1} = \frac{1}{2}\sqrt{\frac{\alpha_1}{\alpha_2}}$$



21. Moment of inertia of the rod about the rotation axis is $I_0 = \frac{1}{3}Ml_0^2$

When temperature rises, l changes and hence I change

$$I = \frac{1}{3}Ml^2$$

But

$$l = l_0(1 + \alpha\Delta\theta)$$

$$l^2 = l_0^2(1 + \alpha\Delta\theta)^2 \simeq l_0^2(1 + 2\alpha\Delta\theta)$$

$$\Rightarrow \quad I = I_0(1 + 2\alpha\Delta\theta)$$

Conservation of angular momentum gives $L = L_0$

Rotational KE before and after heating are

$$k_0 = \frac{1}{2}I_0\omega_0^2 = \frac{L_0^2}{2I_0}$$

And

$$k = \frac{1}{2}I\omega^2 = \frac{L^2}{2I}$$

$$\therefore \quad \frac{k}{k_0} = \frac{I_0}{I} \qquad \therefore \quad \frac{k}{k_0} = \frac{1}{1 + 2\alpha\Delta\theta}$$

$$\frac{k}{k_0} = [1 + 2\alpha\Delta\theta]^{-1} \simeq [1 - 2\alpha\Delta\theta]$$

$$\Rightarrow \quad \frac{k}{k_0} - 1 = -2\alpha\Delta\theta \qquad \Rightarrow \quad \frac{k - k_0}{k_0} = -2\alpha\Delta\theta$$

$$\Rightarrow \quad \frac{\Delta k}{k_0} \times 100 = -200\,\alpha\Delta\theta\%$$

22. (a) Volume of kerosene that spills out

$$\Delta V = \Delta V_k + \Delta V_w - \Delta V_s$$

$$= \frac{V_0}{2}\gamma_k\Delta\theta + \frac{V_0}{2}\gamma_w\Delta\theta - V_0\gamma_s\Delta\theta = \frac{V_0\Delta\theta}{2}(\gamma_k + \gamma_w - 2\gamma_s)$$

$$= \frac{100 \times 30}{2}[10^{-3} + 0.2 \times 10^{-3} - 2 \times 0.012 \times 10^{-3}]$$

$$= 1.5\,[1.2 - 0.024] = 1.76 \text{ litre}$$

Density of kerosene at θ_2 is

$$\rho_2 = \frac{\rho_1}{1 + \gamma_k(\theta_2 - \theta_1)} = \frac{0.8}{1 + 10^{-3} \times 30} = \frac{0.8}{1.03} = 0.78 \text{ kg/litre}$$

$\therefore$ Mass of K.oil that flows out is

$$\Delta m = \Delta V \rho_2 = \frac{V_0 \rho_1}{2} \frac{(\gamma_k + \gamma_w - 2\gamma_s)}{1 + \gamma_k(\theta_2 - \theta_1)}(\theta_2 - \theta_1)$$

$$= 0.78 \times 1.76 = 1.37 \text{ kg}$$

(b) Volume of water at θ_2 is $V = \frac{V_0}{2}[1 + \gamma_w \Delta\theta]$ [where $\Delta\theta = \theta_2 - \theta_1$]

The area of cross section of the tank at θ_1 is $A_1 = \frac{V_0}{2H_1}$

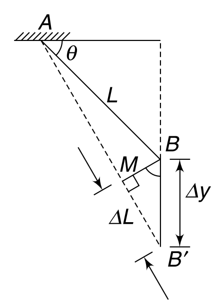
At θ_2 cross section will be $A_2 = A_1[1 + \beta\Delta\theta] = A_1\left[1 + \frac{2}{3}\gamma_s\Delta\theta\right] = \frac{V_0}{2H_1}\left[1 + \frac{2}{3}\gamma_s\Delta\theta\right]$

Height of water column is

$$H_2 = \frac{V}{A_2} = \frac{\frac{V_0}{2}[1 + \gamma_w\Delta\theta]}{\frac{V_0}{2H_1}\left[1 + \frac{2}{3}\gamma_s\Delta\theta\right]}$$

$$= H_1\left[\frac{1 + \gamma_w\Delta\theta}{1 + \frac{2}{3}\gamma_s\Delta\theta}\right] = \frac{1 + 2 \times 10^{-4} \times 30}{1 + \frac{2}{3} \times 1.2 \times 10^{-5} \times 30} = 1.0057 \text{ m}$$

23. Let change in temperature be ΔT



Length of a string changes by $\Delta L = L\alpha_2 \Delta T$

The wooden plank descends by $\Delta y = \frac{\Delta L}{\sin\theta}$

$$\Delta y = 2\Delta L \quad \left[\because \sin\theta = \frac{1}{2}\right]$$

Change in radius of the ball: $\Delta R = R\alpha_1 \Delta T$

The centre of the ball will not move if $\Delta y = \Delta R$

$$\Rightarrow \quad 2L\alpha_2\Delta T = R\alpha_1\Delta T \Rightarrow 8R\alpha_2 = R\alpha_1 \Rightarrow \frac{\alpha_1}{\alpha_2} = \frac{8}{1}$$

24. Length of object at 50°C is

$$L = 100[1 + 10^{-4} \times 50] \text{ mm} = 100.50 \text{ mm}$$

$$1\ MSD \text{ at } 50°C = 1[1 + 10^{-3} \times 50] = 1.050 \text{ mm}$$

$$1\ VSD \text{ at } 50°C = 0.9[1 + 10^{-3} \times 50] = 0.945 \text{ mm}$$

Least count $LC = 1\ MSD - 1\ VSD = 0.105$ mm

$$L = 95 \times MSD + 0.75 \text{ mm}$$

$$\therefore \quad MS \text{ reading} = 95$$

$$VS \text{ reading} = \left[\frac{0.75}{0.105}\right] = 7$$

(b) wooden scale reading = 100 ± 1 mm

25. In equilibrium the torque on the rod about the hinge is zero.

$$\therefore \quad Mg\,\frac{L}{2}\sin\theta = F_B\left(\frac{h\sec\theta}{2}\right)\sin\theta$$

Where buoyancy is $\quad F_B = A\,h\sec\theta\cdot\rho\cdot g$

A = area of cross section of rod, ρ = density of water, M = mass of the rod

$$\therefore \quad MgL = A\,(h\sec\theta)^2\rho g \;\Rightarrow\; \cos^2\theta = \frac{A\rho h_2}{ML}$$

$$\Rightarrow \quad \cos\theta = \left(\frac{A\rho}{ML}\right)^{1/2} h \;\Rightarrow\; y = \left(\frac{A\rho}{ML}\right)^{1/2}\frac{V}{A_0}$$

V = volume of water, A_0 = cross section of tank

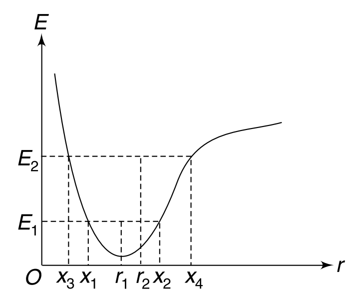
$$\therefore \quad \frac{\Delta y}{y} = \frac{1}{2}\frac{\Delta A}{A} + \frac{1}{2}\frac{\Delta\rho}{\rho} - \frac{1}{2}\frac{\Delta L}{L} + \frac{\Delta V}{V} - \frac{\Delta A_0}{A_0}$$

$$= \frac{1}{2}(2\alpha_1\Delta T) + \frac{1}{2}(-\gamma\Delta T) - \frac{1}{2}(\alpha_1\Delta T) + (\gamma\Delta T) - (2\alpha_2\Delta T)$$

$$= \frac{1}{2}(\alpha_1 + \gamma - 4\alpha_2)\,\Delta T$$

If θ increases y $(= \cos\theta)$ will decreases. This is possible if $\Delta y < 0 \Rightarrow 4\alpha_2 > \alpha_1 + \gamma$

26. With rise in temperature the energy rises and atoms oscillate with higher amplitude.



If energy is E_1 at a temperature T_1, the inter atomic separation oscillates between x_1 to x_2 and the mean separation is r_1.

If temperature rises to T_2 the energy becomes higher at E_2. The atomic separation now oscillates between x_3 and x_4 with atoms spending more time at greater distances (due to reduced force as can be seen from the graph). Thus the average separation r_2 becomes higher than r_1 and the material expands.

27. Let m = mass of Hg; L_0 = 1.0 m = Length of metal rod

h_0 = height of Hg

A_0 = area of cross section of the tube.

Position of COM from top end of metal rod is $y_{cm}^0 = L_0 - \dfrac{h_0}{2}$

When temperature increases by ΔT

$$y_{cm}^T = L^T - \frac{h^T}{2}$$

$$= L_0[1 + \alpha_{metal}\Delta T] - \frac{1}{2}\,\frac{[A_0 h_0]\,[1 + \gamma_m\Delta T]}{A_0[1 + 2\alpha_g\Delta T]}$$

$$\simeq L_0[1 + \alpha_{metal}\Delta T] - \frac{h_0}{2}[1 + (\gamma_m - 2\alpha_g)\,\Delta T]$$

$$[\because\;(1 + \gamma_m\Delta T)(1 + 2\alpha_g\Delta T)^{-1} \simeq (1 + \gamma_m\Delta T)(1 - 2\alpha_g\Delta T)$$

$$= 1 + \gamma_m\Delta T - 2\alpha_g\Delta T - 2\alpha_g\gamma_m\Delta T^2 \simeq 1 + (\gamma_m - 2\alpha_g)\,\Delta T]$$

$$y_{cm}^T = y_{cm}^0 + L_0\,\alpha_{metal}\,\Delta T - \frac{h_0}{2}(\gamma_m - 2\alpha_g)\,\Delta T$$

For $\quad y_{cm}^T = y_{cm}^0$

$$L_0\alpha_{metal} = \frac{h_0}{2}(\gamma_m - 2\alpha_g)$$

$$\therefore \quad h_0 = \frac{2L_0\,\alpha_{metal}}{\gamma_m - 2\alpha_g} = 0.146 \text{ m}$$

28. (a) Let the required distance be ℓ_1. The lower part AP of the rod has mass $\frac{m}{L}\ \ell_1$ and the upper part BP has mass $\frac{m}{L}\ (L - \ell_1)$. When heated, the part AP will be moving down and friction on it (f_1) will be up the incline. The part BP will move up and friction on it will be down (say f_2).

For equilibrium of the entire rod

$$f_1 - f_2 = mg\sin\theta$$

$$\mu\frac{m}{L}\ell_1 g\cos\theta - \mu\frac{m}{L}(L - \ell_1)\, g\cos\theta = mg\sin\theta$$

$$\Rightarrow \quad \frac{\mu}{L}(\ell_1 - L + \ell_1) = \tan\theta \quad \Rightarrow \quad 2\ell_1 = L + \frac{L}{\mu}\tan\theta$$

$$\Rightarrow \quad \ell_1 = \frac{L}{2}\left[1 + \frac{\tan\theta}{\mu}\right] = 0.875\text{ L}$$

(b) In this case the friction on upper part will be up and that on the lower part will be down. Proceeding in similar way as above we can get

$$\ell_2 = \frac{L}{2}\left[1 - \frac{\tan\theta}{\mu}\right] = 0.125\text{ L}$$

(c) During expansion P is fixed but Q moves down. Then during contraction Q remains fixed. Therefore, in one cycle the point Q has moved down.

CHAPTER 2

Calorimetry

LEVEL 1

Q. 1: You are on a picnic and you make tea for yourself and your friend. However, your friend has gone out to bring something for you. You observed that the fire (that you ignited for making tea) has heated two nearby blocks of stones – one of sand stone and other of granite – to 90°C. Both blocks have nearly same mass but granite has higher specific heat than marble. To keep the tea hot for your friend you decided to place the tea pot on one of the stones. Which stone will you choose – granite or marble?

Q. 2: An electric kettle is filled with 1.3 kg of water at 20°C. The power of the heating coil of the kettle is 2.0 KW. After switching it on the water begins to boil in 220s. If the kettle was kept on for a further interval of Δt it was observed that only 200g water remained in the kettle and remaining water vaporized (the vapor is allowed to escape through a small vent). The specific latent heat of vaporization of water at 100°C (boiling point) is L = 2.26 KJ/g. Calculate the specific heat capacity of water and the interval Δt. Assume that heat supplied by the heater is completely absorbed by the water.

Q. 3: A heavy machine rejects a liquid at 60°C which is to be cooled to 30°C before it is fed back to the machine. The liquid rejected by the machine is kept flowing through a long tube while it is cooled by 60 liter water surrounding the tube. The initial temperature of the cooling water is 10°C and it is 20°C when it is changed after 1 hour. Calculate the amount of liquid that passes through the tube in one hour. Specific heat capacity of the liquid and water are 0.5 $\text{cal}\,\text{g}^{-1}\,{}^\circ\text{C}^{-1}$ and 1.0 $\text{cal}\,\text{g}^{-1}\,{}^\circ\text{C}^{-1}$ respectively.

Q. 4: A solid metal cube has side length L and density d. Its specific heat capacity and coefficient of linear expansion are s and α respectively. How much heat must be added to the cube to increase its volume by 2%?

Q. 5: A certain mass of a solid exists at its melting temperature of 20°C. When a heat Q is added $\frac{4}{5}$ of the material melts. When an additional Q amount of heat is added the material transforms to its liquid state at 50°C. Find the ratio of specific latent heat of fusion (in J/g) to the specific heat capacity of the liquid (in $\text{J}\,\text{g}^{-1}\,{}^\circ\text{C}^{-1}$) for the material.

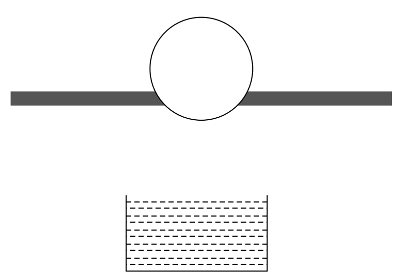
Q. 6: The temperature of samples of three liquids A, B and C are 12°C, 19°C and 28°C respectively. The temperature when A and B are mixed is 16°C and when B and C are mixed it is 23°C.

(i) What should be the temperature when A and C are mixed?

(ii) What is final temperature if all the three liquids are mixed?

Assume no heat loss to the surrounding.

Q. 7: A table top is made of aluminium and has a hole of diameter 2 cm. An iron sphere of diameter 2.004 m is resting on this hole. Below the hole, an insulated container has 2 kg of water in it. Everything is at ambient temperature of 25°C. The table top along with the iron sphere is heated till the ball falls through the hole into the water. Find the equilibrium temperature of the ball and water system.



Neglect any heat loss from ball–water system to the surrounding and assume the heat capacity of the container to be negligible.

Relevant data: Coefficient of linear expansion for aluminium and iron are 2.4×10^{-5} °C^{-1} and 1.2×10^{-5} °C^{-1} respectively. Specific heat capacity of water and iron are 4200 J °C^{-1} g^{-1} and 450 J °C^{-1} g^{-1} respectively. Density of iron at 25°C is 8000 kg/m^3.

Q. 8: A 50 *g* ice at 0°C is added to 200 g water at 70°C taken in a flask. When the ice has melted completely, the temperature of the flask and the contents is reduced to 40°C. Now to bring down the temperature of the contents to 20°C, find a further amount of ice that is to be added.

Q. 9: The latent heat of vaporization of water at its boiling point is L_V. But water can evaporate at temperatures below the boiling point – for example it evaporates at body temperature when you perspire. Will the energy needed to evaporate unit mass of water at body temperature be more than or less than L_V?

Q. 10: A vessel contains a small amount of water at 0°C. If the air in the vessel is rapidly pumped out, it causes freezing of the water. Why? What percentage of the water in the container can be frozen by this method? Latent heat of vaporization and fusion are L_V = 540 cal g^{-1} and L_f = 80 cal g^{-1} respectively.

Q. 11: A vessel containing 100 g ice at 0°C is suspended in a room where temperature is 35°C. It was found that the entire ice melted in 10 hour. Now the same vessel containing 100 g of water at 0°C is suspended in the same room. How much time will it take for the temperature of water to rise to 0.5°C. Neglect the heat capacity of the vessel. Specific heat of water and specific latent heat of fusion of ice are 1 cal g^{-1} °C^{-1} and 80 cal g^{-1} respectively.

Q. 12: A calorimeter of negligible heat capacity contains ice at 0°C. 50 g metal at 100°C is dropped in the calorimeter. When thermal equilibrium is attained the volume of the content of the calorimeter was found to reduce by 0.5×10^{-6} m^3. Calculate the specific heat capacity of the metal. Neglect the change in volume of the metal. Specific latent heat of fusion of ice is $L = 300 \times 10^3$ J kg^{-1} and its relative density is 0.9.

Q. 13: A refrigerator converts 1.3 kg of water at 20°C into ice at –15°C in 1 hour. Calculate the effective power of the refrigerator.

Specific latent heat of fusion of ice = 3.4×10^5 J kg^{-1}

Specific heat capacity of water = 4.2×10^3 J kg^{-1} K^{-1}

Specific heat capacity of ice = 2.1×10^3 J kg^{-1} K^{-1}

Q. 14: A calorimeter of water equivalent 10 g contains a liquid of mass 50 g at 40 °C. When *m* gram of ice at –10°C is put into the calorimeter and the mixture is allowed to attain equilibrium, the final temperature was found to be 20°C. It is known that specific heat capacity of the liquid changes with temperature as $S = \left(1 + \frac{\theta}{500}\right)$ cal g^{-1} °C^{-1} where θ is temperature in °C. The specific heat capacity of ice, water and the calorimeter remains constant and values are S_{ice} = 0.5 cal g^{-1} °C^{-1}; S_{water} = 1.0 cal g^{-1} °C^{-1} and latent heat of fusion of ice is L_f = 80 cal g^{-1}. Assume no heat loss to the surrounding and calculate the value of *m*.

Q. 15: A well insulated container has a mixture of ice and water, at 0 °C. The mixture is supplied heat at a constant rate of 420 watt by switching on an electric heater at time *t* = 0. The temperature of the mixture was recorded at time *t* = 150*s*, 273*s* and 378*s* and the readings were 0 °C, 10 °C and 20 °C respectively. Calculate the mass of water and ice in the mixture.

Specific heat of water = 4.2 J g^{-1} °C^{-1}, Specific latent heat of fusion of ice = 336 J g^{-1}.

Assume that the mixture is stirred slowly to maintain a uniform temperature of its content.

Q. 16: An insulated container has 60 g of ice at –10 °C. 10 g steam at 100 °C, sourced from a boiler, is mixed to the ice inside the container. When thermal equilibrium was attained, the entire content of the container was liquid water at 0 °C. Calculate the percentage of steam (in terms of mass) that was condensed before it was fed to the container of ice. Specific heat and latent heat values are

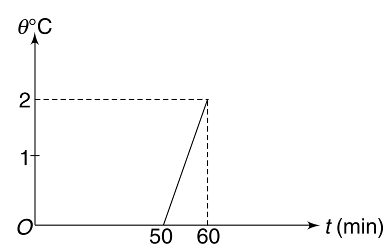
S_{ice} = 0.5 cal g^{-1} °C^{-1}; S_{water} = 1.0 cal g^{-1} °C^{-1}

L_{fusion} = 80 cal g^{-1}; $L_{vaporization}$ = 540 cal g^{-1}

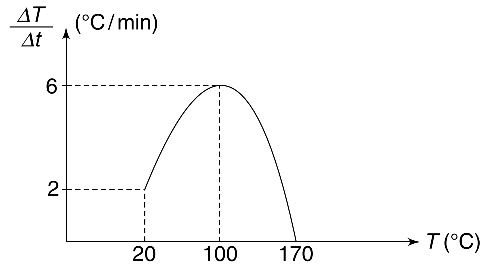
Q. 17: A container contains 5 kg of water at 0°C mixed to an unknown mass of ice in thermal equilibrium. The water equivalent of the container is 100 g. At time *t* = 0, a heater is switched on which supplies heat at a constant rate to the container. The temperature of the mixture is measured at various times and the result has been plotted in the given figure. Neglect any heat loss from the mixture – container system to the surrounding and calculate the initial mass of the ice.

Given: Sp. latent heat of fusion of ice is L_f = 80 cal g^{-1}

Sp. heat capacity of water = 1 cal g^{-1} °C^{-1}



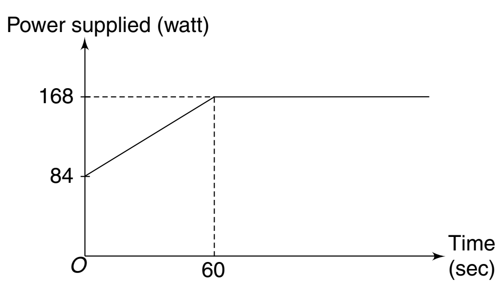
Q. 18: A liquid kept in a beaker is supplied heat. The rate of change of temperature of the liquid is plotted versus its temperature. Which intrinsic property of the liquid can be inferred from the graph? What is its value?



Q. 19: A meteorite has mass of 500 kg and is composed of a metal. The temperature of the meteor is −20°C and its speed is 10 km/hr when it is at large distance from a planet. The meteorite crashes into the planet and its entire kinetic energy gets converted into heat. This heat is equally shared between the planet and the meteorite. Assume that the heating of meteorite is uniform and the average specific heat capacity of the metal; for its solid, liquid and vapour phase; is 1200 J kg^{-1} °C^{-1}. The latent heat of fusion and vaporization of the metal are $L_f = 4 \times 10^5$ J kg^{-1} and $L_v = 1.1 \times 10^7$ J kg^{-1} respectively. The melting point and boiling points are 380 °C and 2380 °C respectively. Find the temperature of the meteorite material immediately after the impact.

Take: $G = 6.6 \times 10^{11}$ N m^2 kg^{-2}; mass of planet $M = 6 \times 10^{24}$ kg; radius of planet $R = 6600$ km

Q. 20: 100 g of ice at −40 °C is supplied heat using a heater. The heater is switched on at time $t = 0$ and its power increases linearly for first 60 second and thereafter it becomes constant as shown in the graph. Heater is kept on for 5 minutes. The specific heat capacity for ice and water are known to be 2.1 $\frac{\text{J}}{\text{g°C}}$ and 4.2 $\frac{\text{J}}{\text{g°C}}$ respectively. The specific latent heat for fusion of ice is 336 J/g.



The temperature of the ice sample kept on increasing till time t_1 and then remained constant in the interval $t_1 \le t \le t_2$.

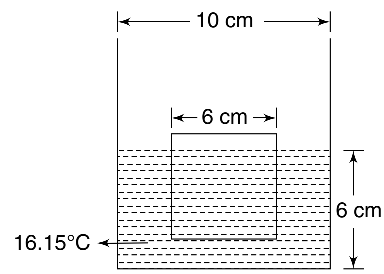
(i) Find t_1 and t_2

(ii) Find final temperature of the sample when the heater is switched off.

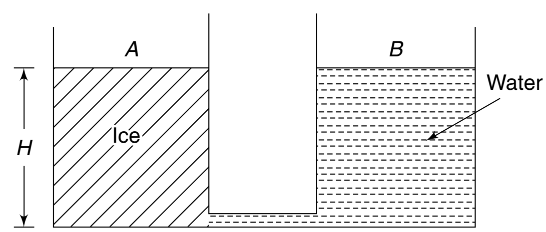
LEVEL 2

Q. 21: A container has a square cross-section of 10 cm × 10 cm. A cubical ice block of side length 6 cm is floating in water in the container. Water level in the container is 6 cm high. The ice block is at a temperature of 0°C and the water is at 16.15°C. Assume that heat exchange take place between the ice block and water only. What length of ice block will remain submerged in water when the system reaches thermal equilibrium? Assume that the ice block maintains its cubical shape as it melts. Take - density of ice = 0.9 g/cc, density of water = 1.0 g/cc

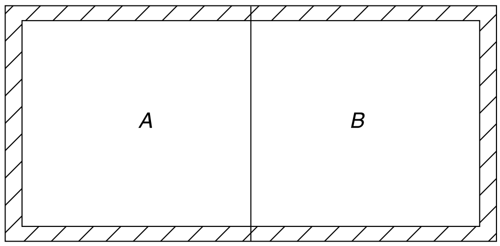
Specific heat capacity of water = 1 cal g^{-1} °C^{-1}, Specific latent heat of fusion of ice = 80 cal g^{-1}



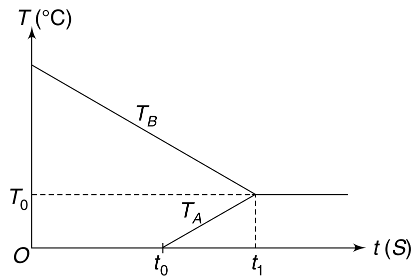
Q. 22: Two identical cylindrical containers A and B are interconnected by a tube of negligible dimensions. Container A is filled with an ice block up to height $H = 1.8$ m and container B is filled up to same height with water. Ice is at 0°C and water is at 40°C. Due to heat exchange between water and ice, the ice block begins to melt. Assume that the ice block melt in horizontal layers starting from the bottom. The thickness of ice block reduces uniformly over the entire cross section of the container. The ice block moves without friction inside the container and no water enters between the vertical wall of the container and the ice block. Heat is exchanged only between the ice block and the water and there is no heat exchange with containers or atmosphere. Calculate the height of water in container B when thermal equilibrium is attained. Relative density and specific latent heat of fusion of ice are 0.9 and 80 cal g^{-1} respectively. Specific heat capacity of water is 1 cal g^{-1} °C^{-1}.



Q. 23: A well insulated box has two compartments A and B with a conducting wall between them. 100 g of ice at 0°C is kept in compartment A and 100 g of water at 100°C is kept in B at time $t = 0$. The temperature of the two parts A and B is monitored and a graph is plotted for temperatures T_A and T_B versus time (t) [Fig. (b)]. Assume that temperature inside each compartment remains uniform.



(a)



(b)

(a) Is it correct to assert that the conducting wall conducts heat at a uniform rate, irrespective of the temperature difference between A and B?

(b) Find the value of time t_1 and temperature T_0 shown in the graph, if it is known that $t_0 = 200\ s$.

Specific heat of ice = 0.5 cal g^{-1} $°C^{-1}$

Specific heat of water = 1.0 cal g^{-1} $°C^{-1}$

Latent heat of fusion of ice = 80 cal g^{-1}

Q. 24: An ice ball has a metal piece embedded into it. The temperature of the ball is $-\theta$ °C and it contains mass M of ice. When placed in a large tub containing water at 0°C, it sinks. Assume that the water in immediate contact with the ice ball freezes and thereby size of the ball grows. What is the maximum possible mass of the metal piece so that the ball can eventually begin to float. Densities of ice, water and metal are σ, ρ and d respectively. Specific heat capacity of ice is s and its specific latent heat is L. Neglect the heat capacity of the metal piece.

Q. 25: Water from a reservoir maintained at a constant temperature of 80°C is added at a slow and steady rate of $\mu = 3\ gs^{-1}$ to a calorimeter initially containing 1000 g of water at 20°C. The water in the calorimeter is stirred slowly to make the temperature uniform. Assume heat loss to the surrounding and work done in stirring is negligible and heat capacity of the calorimeter is negligible. Write the temperature of water in the calorimeter as a function of time.

Q. 26: A cylindrical container has a cross sectional area of $A_0 = 1\ cm^2$ at 0°C. A scale has been marked on vertical surface of the container which shows correct reading at 0°C. A liquid is poured in the container. When the liquid and container is heated to 100°C, the scale shows the height of the liquid as 83.33 cm. The coefficient of volume expansion for the liquid is $\gamma = 0.001°C^{-1}$ and the coefficient of linear expansion of the material of cylindrical container is $\alpha = 0.0005°C^{-1}$. A beaker has 300 cm^3 of same liquid at 0°C. The two liquids are mixed. Find the final temperature of the mixture assuming that heat exchange takes place between the liquids only, and its specific heat capacity is independent of temperature.

LEVEL 3

Q. 27: A copper calorimeter has mass of 180 g and contains 450 g of water and 50 g of ice; all at 0°C. Dry steam is passed into the calorimeter until a certain temperature (θ) is reached. The mass of the calorimeter and its contents at the end of the experiment increased by 25 g. Assume no heat loss to the surrounding and take specific heat capacities of water and copper to be 4200 J kg^{-1} K^{-1} and 390 J kg^{-1} K^{-1}, respectively. Take specific latent heat of vaporization of water to be 3.36×10^5 J kg^{-1} and 2.26×10^6 J kg^{-1} respectively.

(a) Find the final temperature θ

(b) If steam enters into the system at a steady rate of 5 g min^{-1}, plot the variation of temperature of the system till final temperature θ is attained.

ANSWERS

1. Granite

2. $s = 4.23$ kJ Kg^{-1} $°C^{-1}$; $\Delta t = 1250$ s

3. 40 kg

4. $\frac{1}{150}\frac{L^3 \cdot d \cdot s}{\alpha}$

5. 60

6. (i) 20.26°C (ii) 19.76°C

7. 27.4°C

8. 50 g

9. More than L_V.

10. 87%

11. 3.75 min.

12. 270 J kg^{-1} k^{-1}

13. 164.5 W

14. 12 g

15. $m_\omega = 840$ g; $m_{ice} = 210$ g

16. 24%

17. 729 g

18. Boiling point, 170°C

19. Nearly 36400°C

20. (i) $t_1 = 65\,\text{s}$; $t_2 = 265\,\text{s}$ (ii) 14°C

21. 4.5 cm

22. 1.71 m

23. (a) yes (b) $T_0 = 13.3°\text{C}$, $t_1 = 217$ s

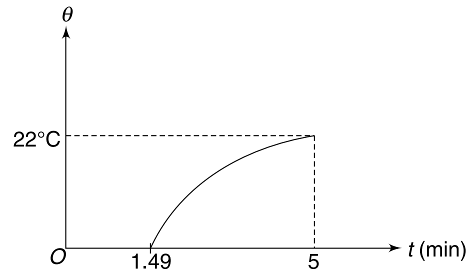
24. $\dfrac{d}{\sigma}\dfrac{(\rho - \sigma)}{(d - \rho)}\left[1 + \dfrac{s\theta}{L}\right]$

25. $\theta = 80 - 60\left(1 + \dfrac{3t}{1000}\right)^{-1}$

26. 25°C

27. (a) 22°C

(b)



SOLUTIONS

2. Heat supplied in 220 s

$$\Delta Q = 2 \times 10^3 \text{ W} \times 220 \text{ s} = 4.4 \times 10^5 \text{ J}$$

$$\therefore \quad \text{ms}\,\Delta\theta = 4.4 \times 10^5 \text{ J}$$

$$(1.3 \text{ kg})(\text{s})(80°\text{C}) = 4.4 \times 10^5 \text{ J}$$

$$\text{s} = 4.23 \times 10^3 \text{ JKg}^{-1}°\text{C}^{-1} = 4.23 \text{ K J Kg}^{-1}°\text{C}^{-1}$$

Heat required for vaporization of 1.1 kg water

$$\Delta Q' = mL = 1.1 \times 2.26 \times 10^3 \left(\frac{\text{KJ}}{\text{Kg}}\right) = 2.49 \times 10^3 \text{ KJ}$$

$$\therefore \quad \Delta t\,(2 \times 10^3 \text{ Js}^{-1}) = 2.49 \times 10^3 \times 10^3 \text{ J}$$

$$\Delta t = 1.25 \times 10^3 \text{ s} = 1250 \text{ s}$$

3. Heat absorbed by the water $= ms\Delta\theta = 60 \times 10^3 \times 1.0 \times (20 - 10) = 6 \times 10^5$ cal.

$[m = 60 \times 10^3 \text{ g}]$

If M is mass of liquid that is passed through the tube in 1 hour, heat rejected by it is –

$$Ms_l(60 - 30) = M \times 0.5 \times 30$$

$$\therefore \quad M \times 0.5 \times 30 = 6 \times 10^5$$

$$M = 4 \times 10^4 \text{ g} = 40 \text{ kg}$$

4.

$$M = V \cdot d = L^3 d$$

$$\frac{\Delta V}{V} = (3\alpha)\Delta\theta \quad \Rightarrow \quad \Delta\theta = \frac{1}{3\alpha}\frac{\Delta V}{V}$$

$$\therefore \quad \Delta Q = Ms\Delta\theta = L^3\,ds\,\frac{1}{3\alpha}\frac{\Delta V}{V}$$

$$= \frac{L^3 ds}{3\alpha}(0.02) = \frac{1}{150}\frac{L^3 d \cdot s}{\alpha} \qquad \left[\because \frac{\Delta V}{V} = 0.02\right]$$

5.

$$Q = \frac{3}{4}m \cdot L \qquad \ldots\text{(i)}$$

$$2Q = mL + ms(50 - 20)$$

$$\Rightarrow \quad \frac{3}{2}mL = mL + 30\,ms$$

$$\Rightarrow \quad \frac{L}{2} = 30s \quad \Rightarrow \quad \frac{L}{s} = 60$$

6. When A and B are mixed

$$\text{Heat gained by } A = \text{Heat lost by } B$$

$$m_A s_A(16 - 12) = m_B s_B(19 - 16)$$

$$\therefore \quad m_B s_B = \frac{4}{3} m_A s_A \qquad \text{...(i)}$$

Similarly, when B and C are mixed –

$$m_B s_B(23 - 19) = m_C s_C(28 - 13)$$

$$\Rightarrow \quad m_C s_C = \frac{4}{5} m_B s_B \qquad \text{...(ii)}$$

Using (i) and (ii) $\quad m_C s_C = \frac{4}{5} \times \frac{4}{3} m_A s_A = \frac{16}{15} m_A s_A$

(i) When A and C are mixed, let the final temperature be θ.

$$m_A s_A(\theta - 12) = m_C s_C(28 - \theta)$$

$$\Rightarrow \quad \theta - 12 = \frac{16}{15}(28 - \theta) \quad \Rightarrow \quad \theta = \frac{628}{31} = 20.26°\text{C}$$

(ii) When all three are mixed, let the final temperature be θ
Heat gained by A + Heat gained by B = Heat lost by C

$$m_A s_A(\theta - 12) + m_B s_B(\theta - 19) = m_C s_C(28 - \theta)$$

$$\Rightarrow \quad (\theta - 12) + \frac{m_B s_B}{m_A s_A}(\theta - 19) = \frac{m_C s_C}{m_A s_A}(28 - \theta)$$

$$\Rightarrow \quad \theta - 12 + \frac{4}{3}(\theta - 19) = \frac{16}{15}(28 - \theta)$$

$$\Rightarrow \quad 15\theta - 180 + 20\theta - 380 = 448 - 16\theta$$

$$\Rightarrow \quad 51\theta = 1008 \quad \Rightarrow \quad \theta = 19.76°\text{C}$$

7. The sphere falls when temperature rises to T, so that diameter of the sphere and hole become equal.

$$(\text{diameter of hole})_{\text{Al}} = (\text{diameter of sphere})_{\text{Fe}}$$

$$2[1 + \alpha_{\text{Al}}(T - 25)] = 2.004[1 + \alpha_{\text{Fe}}(T - 25)]$$

Solving, $\quad T = 191.7°\text{C}$

When hot ball falls into water, the ball loses heat and water gains heat. Let the equilibrium temperature be T_0.

$$\text{Heat lost by ball} = \text{Heat gained by water}$$

$$m_{\text{ball}} \cdot s_{\text{Fe}}(191.7 - T_0) = 2 \times s_\omega \times (T_0 - 25)$$

$$\left[m_{\text{ball}} = \frac{4}{3}\pi(2.004)^3 \times 8 \simeq 268 \text{ g} = 0.268\right]$$

$$\therefore \quad 0.268 \times 450 \times (191.7 - T_0) = 2 \times 4200 \times (T_0 - 25)$$

$$T_0 = 27.4°\text{C}$$

8. Let W = water equivalent of flask

$$50 \times 80 + 50 \times 1 \times (40 - 0) = (200 + W) \times 1 \times (70 - 40)$$

$$W = 0 \text{ (flask will not absorb heat)}$$

Let m gram ice be required to reduce the temperature to 20°C

$$m \times 80 + m \times 1 \times (20 - 0) = 250 \times 1 \times (40 - 20) \quad \Rightarrow \quad m = 50 \text{ gram}$$

10. At any temperature, there is some vapour above the surface of water. The vapour pressure depends on temperature. When the air is pumped out, the vapour gets removed and the vapour pressure falls. There is further evaporation of the liquid to saturate the air. The latent heat for evaporation is supplied by the water itself and therefore a part of it freezes.

Let

$$m = \text{initial mass of water}$$

$$m_1 = \text{mass that evaporates}$$

$$m_2 = \text{mass that freezes}$$

$$m_1 L_V = m_2 L_f \quad \therefore \quad m_2 = m_1 \frac{L_V}{L_f}$$

$$\therefore \quad \frac{m_2}{m} = \frac{m_2}{m_1 + m_2} = \frac{m_1 \dfrac{L_V}{L_f}}{m_1 + m_1 \dfrac{L_V}{L_f}} = \frac{L_V}{L_f + L_V}$$

$$\therefore \quad \frac{m_2}{m} \times 100 = \frac{L_V}{L_f + L_V} \times 100 = \frac{540}{80 + 540} \times 100 = 87\%$$

11. Rate of heat gain for the vessel and its content $= \dfrac{\text{mL}}{10\,\text{hr}}$.

Since conditions are unchanged, rate of heat gain will remain same for water filled container.

$$\therefore \quad ms_\omega \frac{\Delta\theta}{\Delta t} = \frac{\text{mL}}{10\,\text{hr}}$$

$$\therefore \quad \Delta t = \frac{10 \times s_\omega \times \Delta\theta}{L} = \frac{10 \times 1 \times 0.5}{80} = \frac{1}{16}\text{hr} = 3.75 \text{ min}$$

12. Volume of 1 kg of water $= \dfrac{1}{1000}\text{m}^3$

Volume of 1 kg of ice $= \dfrac{1}{900}\text{m}^3$

Contraction in volume when 1 kg ice melts

$$= \frac{1}{900} - \frac{1}{1000} = \frac{1}{9000}\text{m}^3$$

$\Rightarrow$ $1\,\text{m}^3$ Contraction in volume $\equiv$ melting of 9000 kg ice

$\therefore$ $0.5 \times 10^{-6}\text{m}^3$ Contraction $\equiv 9000 \times 0.5 \times 10^{-6}$ kg melting of ice

$$= 4500 \times 10^{-6} \text{ kg}$$

$$= 4.5 \times 10^{-3} \text{ kg ice melts.}$$

Heat gained by ice $= mL = 4.5 \times 10^{-3} \times 3 \times 10^{5} = 1350$ J.

Heat lost by metal in cooling from 100°C to 0°C

$$= ms\Delta\theta = 50 \times 10^{-3} \times s \times 100 = 5.s$$

$\therefore$ $$5.s = 1350 \quad \Rightarrow \quad s = 270 \text{ J kg}^{-1} \text{ K}^{-1}$$

13. Heat removed to cool 1.3 kg water from 20°C to 0°C is

$$E_1 = ms\Delta\theta = 1.3 \times 4.2 \times 10^{3} \times 20 = 1.092 \times 10^{5} \text{ J}$$

Heat removed in converting 1.3 kg water at 0°C into ice at 0°C

$$E_2 = mL_f = 1.3 \times 3.4 \times 10^{5} = 4.42 \times 10^{5} \text{ J}$$

Heat removed in cooling 1.3 kg ice from 0°C to −15°C

$$E_3 = ms\Delta\theta = 1.3 \times 2.1 \times 10^{3} \times 15 = 4.095 \times 10^{4} \text{ J}$$

Total heat removed in 1 hour $E = E_1 + E_2 + E_3 = 5.922 \times 10^{5}$ J

$$\text{Power} = \frac{5.922 \times 10^{5}\,\text{J}}{60 \times 60\,\text{s}} = 164.5\,\text{W}$$

14. Heat gained by ice $= mS_{\text{ice}} \times 10 + mS_{\omega} \times 20 + mL_f = 105$ m. cal.

Heat lost by calorimeter $= 10 \times 1 \times 20 = 200$ cal.

$$\text{Heat lost by liquid} = -50 \int_{40}^{20} \left(1 + \frac{\theta}{500}\right) d\theta$$

$$= 50\left[\theta + \frac{\theta^2}{1000}\right]_{20}^{40}$$

$$= 50\left[\left(40 + \frac{1600}{1000}\right) - \left(20 + \frac{400}{1000}\right)\right] = 50 \times 21.2 = 1060 \text{ cal.}$$

$\therefore$ $$105\,\text{m} = 1060 + 200 \quad \Rightarrow \quad m = 12 \text{ g}$$

15. Let mass of mixture be M (in g) and mass of ice be m.

In the interval of 105 s between 273 s to 378 s, the content was only water and its temperature increased by 10°C.

$\therefore$ $$Ms\Delta\theta = P\Delta t$$

$$M \times 4.2 \times 10 = P \times 105 \quad \Rightarrow \quad M = 1050 \qquad \ldots\text{(i)}$$

In the interval of 273 s [Between $t = 0$ to $t = 273\,s$] the mass m of ice melted and mass M of water was heated from 0°C to 10°C

$\therefore$ $$mL + MS\Delta\theta = P\Delta t$$

$$336m + 1050 \times 4.2 \times 10 = 420 \times 273$$

$$8m = 2730 - 1050$$

$$m = 210 \text{ g}$$

$\therefore$ $$\text{Mass of water} = M - m = 1050 - 210 = 840 \text{ g}$$

16. Heat required by ice to convert itself to water at 0°C is

$$Q_1 = ms\Delta\theta + mL_f = 60 \times 0.5 \times 10 + 60 \times 80 = 5100 \text{ cal.}$$

If the steam has 'x' gram vapour and $(10 - x)$ gram condensed water, heat released by it when it converts to water at 0°C is

$$Q_2 = x \times 540 + 10 \times 1.0 \times 100 = 540x + 1000$$

$$\text{Because } Q_1 = Q_2$$

$$\therefore \quad 540x + 1000 = 5100$$

$$x = 7.6 \text{ g}$$

Percentage of condensed component in steam $= \dfrac{2.4}{10} \times 100 = 24\%$

17. Let m = mass of ice

Rate of heat addition in first 50 min is

$$\frac{dQ}{dt} = \frac{mL_f}{50\,\text{min}} = \frac{80m}{50}\,\frac{\text{cal}}{\text{min}} \qquad \text{...(1)}$$

In first 50 min, the heat is used to melt the ice.

From 50 to 60 min

$$\frac{dQ}{dt} = \frac{(m + 5000 + 100) \times (1 \text{ cal}^{-1} \text{ g}\,°\text{C}^{-1}) \times (2°\text{C})}{10\,\text{min}} \qquad \text{...(2)}$$

From (1) and (2)

$$\frac{80m}{50} = \frac{(5100 + m) \times 2}{10}$$

$$16m = 10200 + 2m \quad \Rightarrow \quad 14m = 10200$$

$$m = 729 \text{ g}$$

18. Once the liquid reaches its boiling point, its temperature stops increasing and $\dfrac{\Delta T}{\Delta t} = 0$. Hence boiling point = 170°C.

19. From energy conservation, we get the KE of the meteorite just before collision as

$$KE = \frac{1}{2}mu^2 + \frac{GMm}{R} = m\left[\frac{u^2}{2} + \frac{GM}{R}\right]$$

$$= 500\left[\frac{(10 \times 10^3)^2}{2} + \frac{6.6 \times 10^{-11} \times 6 \times 10^{24}}{6.6 \times 10^6}\right]$$

$$= 500 \times 10^8\,[0.5 + 0.6] = 1.1 \times 5 \times 10^{10} \text{ J}$$

Half the KE is absorbed by the meteorite.

$$\therefore \quad Q = \frac{1}{2} \times 1.1 \times 5 \times 10^{10} = 275 \times 10^8 \text{ J}$$

Heat needed to raise the temperature from –20°C to 380°C is

$$Q_1 = ms\Delta\theta = 500 \times 1200 \times 400 = 2.4 \times 10^8 \text{ J}$$

Latent heat for melting $\qquad Q_2 = mL_f = 500 \times 4 \times 10^5 = 2 \times 10^8 \text{ J}$

Heat needed to raise the temperature from 380°C to 2380°C is

$$Q_3 = ms\Delta\theta = 500 \times 1200 \times 2000 = 12 \times 10^8 \text{ J}$$

Heat needed for varporization $\quad Q_4 = mL_v = 500 \times 1.1 \times 10^7 = 55 \times 10^8$ J

Heat that is left to raise the temperature further is

$$Q_5 = [275 - 2.4 - 2 - 12 - 55] \times 10^8 = 203.6 \times 10^8 \text{ J}$$

$\therefore$ Further rise in temperature (above 2380°C) is

$$\Delta\theta = \frac{Q_5}{ms} = \frac{203.6 \times 10^8}{500 \times 1200} = \frac{203.6}{6} \times 10^3 \text{°C}$$

$$\simeq 34 \times 10^3 = 34000\text{°C}$$

$\therefore$ Final temperature $= 2380 + 34000 = 36400$°C

20. Heat supplied in 60 second is $\Delta Q_1 = \frac{1}{2} \times 60 \times (84 + 168) = 7560$ J

Heat required to heat 100 g ice at –40°C to 0°C is

$$\Delta Q = ms\Delta\theta = 100 \times 2.1 \times 40 = 8400 \text{ J}$$

$\therefore$ $8400 - 7560 = 840$ J of heat is supplied at a constant rate of 168 J/s.

It will take $\frac{840}{168} = 5$ sec (beyond 60 sec) for temperature to reach 0°C

$\therefore$ $t_1 = 65$ sec

To melt the ice, heat required is $\Delta Q_L = mL_f = 100 \times 336 = 33600$ J

At a rate of 168 J/s it will need $\frac{33600}{168} = 200$ sec.

$\therefore$ $t_2 = 265$ sec

Heat supplied in interval $265 \le t \le 300$ s is $\Delta Q = 168 \times 35 = 5880$ J

This will raise the temperature of 100 g water by

$$\Delta\theta = \frac{\Delta Q}{ms} = \frac{5880}{100 \times 4.2} = 14\text{°C}$$

$\therefore$ Final temperature is 14°C.

21. Volume of ice block $= 6 \times 6 \times 6 = 216 \text{cm}^3$

Volume of water $= 10 \times 10 \times 6 - 6 \times 6 \times 5.4 = 405.6 \text{cm}^3$

$\left[\because \text{ Length of ice block submerged} = \frac{\rho_{\text{ice}}}{\rho_\omega} \times 6 = \frac{0.9}{1} \times 6 = 5.4 \text{ cm}\right]$

Heat rejected by water when it cools from 16.15°C to 0°C

$$Q = ms\Delta\theta = 405.6 \times 1 \times 16.15 = 6550.4 \text{ cal}$$

$$\text{Mass of ice that melts} = \frac{6550.4}{80} = 81.88 \text{ g}$$

$$\text{Volume of ice that melts} = \frac{81.88}{0.9} = 91 \text{ cm}^3$$

$\therefore$ Volume of ice block in thermal equilibrium at 0°C will be $= 6^3 - 91 = 125 \text{ cm}^3$

$\therefore$ Side length = 5 cm

$\therefore$ length of the submerged part $= 0.9 \times 5 = 4.5$ cm.

22. It can be easily verified that heat rejected by water till its temperature reaches 0°C is insufficient to melt the ice block completely. Let x be the length of ice block that melts by the time the entire water goes to 0°C.

Heat gained by ice = Heat lost by water

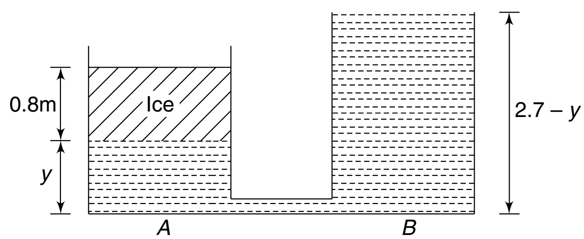
$$Ax\rho_{ice}L_f = AH\rho_\omega S_\omega \theta \quad [A = \text{Area of cross section of container }]$$

$$\Rightarrow \quad x \times 0.9 \times 80 = 1.8 \times 1 \times 40$$

$$\Rightarrow \quad x = 1.0 \text{ m.}$$

When 10 m length of ice melts completely it will produce 0.9 m length of water in the containers.

In final position, let the height of water column in container A be y.



Considering pressure at the bottom of the ice block, we get

$$\rho_{ice}g(0.8) = \rho_\omega g(2.7 - 2y)$$

$$\Rightarrow \quad 0.9 \times 0.8 = 2.7 - 2y$$

$$\Rightarrow \quad y = 1.35 - 0.36 = 0.99 \text{ m}$$

$\therefore$ Required answer is $2.70 - 0.99 = 1.71$ m

23. (a) Since temperature of the water is decreasing at a uniform rate, it is correct to assert that the wall conducts equal amount of heat in equal intervals of time.

(b) Initially, temperature of water falls and the heat gained by ice is used to melt it. Once the ice is completely melted, (at time t_0), its temperature begins to rise till both compartments acquire same temperature (T_0) at time t_1.

$$\text{Heat lost by water} = \text{Heat gained by ice}$$

$$100 \times 1 \times (100 - T_0) = 100 \times 80 + 100 \times 0.5 \times T_0$$

$$100 - T_0 = 80 + 0.5\, T_0$$

$$T_0 = \frac{20}{1.5} = \frac{40}{3} = 13.3\text{°C}$$

Amount of heat transfer by the time ice melts is

$$Q = m_{ice}L = 100 \times 80 = 8000 \text{ cal.}$$

$\therefore$ Constant rate of heat transfer through the wall is

$$\frac{dQ}{dt} = \frac{8000\,\text{cal}}{200\,\text{s}} = 40 \text{ cal s}^{-1}$$

$$\text{Total heat transfer} = m_\omega s_\omega(100 - T_0) = 100 \times 1 \times 86.7 = 8670 \text{ cal.}$$

$$\Rightarrow \qquad t_1 = \frac{8670}{40} \simeq 217\,\text{s}$$

24. Let maximum mass of water that can get frozen around the ball be m_0.

$$\text{Heat lost by freezing water} = \text{Heat gained by the ball}$$

$$m_0 L = Ms\theta$$

$$m_0 = \frac{Ms\theta}{L} \qquad \text{...(i)}$$

Volume of the ball now becomes

$$V = \frac{(M + m_0)}{\sigma} + \frac{m}{d} \quad [m = \text{mass of metal}]$$

$$\text{Buoyancy } F_B = V\rho g = \left[\frac{M + m_0}{\sigma} + \frac{m}{d}\right] \rho g$$

The ball will just get buoyed if $F_B >$ Weight

$$\left[\frac{M + m_0}{\sigma} + \frac{m}{d}\right] \rho g > (M + m_0 + m) g$$

Using (i) we get $\left[\frac{M}{\sigma} + \frac{Ms\theta}{L\sigma} + \frac{m}{d}\right] \rho > M + \frac{Ms\theta}{L} + m$

$$M\left[\frac{\rho}{\sigma} + \frac{s\theta\rho}{L\sigma} - \frac{s\theta}{L} - 1\right] > m \left[1 - \frac{\rho}{d}\right]$$

$$\Rightarrow \qquad \left[\frac{\rho - \sigma}{\sigma} + \frac{s\theta}{L\sigma}(\rho - \sigma)\right] M > m \left(\frac{d - \rho}{d}\right)$$

$$\Rightarrow \qquad \frac{(\rho - \sigma)}{\sigma}\left[1 + \frac{s\theta}{L}\right] \frac{d}{(d - \rho)} > m.$$

25. Mass of water in the calorimeter at time t is $1000 + \mu t = (1000 + 3t)\,g$

Let temperature at time t be θ.

Mass of hot water (at 80°C) added in interval dt is $\mu dt = 3\,dt$.

Heat lost = Heat gained

$$3dt(80 - \theta)\cdot s = (1000 + 3t)\cdot s \cdot d\theta$$

$$\Rightarrow \qquad 3\int_0^t \frac{dt}{1000 + 3t} = \int_{20}^{\theta} \frac{d\theta}{80 - \theta}$$

$$\Rightarrow \qquad [\ln(1000 + 3t)]_0^t = -[\ln(80 - \theta)]_{20}^{\theta}$$

$$\ln\left(\frac{1000 + 3t}{1000}\right) = -\ln\left(\frac{80 - \theta}{60}\right)$$

$$\ln\left(1+\frac{3t}{1000}\right)^{-1} = \ln\left(\frac{4}{3}-\frac{\theta}{60}\right)$$

$$\Rightarrow \quad \left(1+\frac{3t}{1000}\right)^{-1} = \frac{4}{3}-\frac{\theta}{60}$$

$$\Rightarrow \quad \frac{\theta}{60} = \frac{4}{3}-\left(1+\frac{3t}{1000}\right)^{-1}$$

$$\Rightarrow \quad \theta = 80-60\left(1+\frac{3t}{1000}\right)^{-1}$$

26. Correct height of liquid in the cylindrical container is $h = 95.24[1+\alpha\times 100]$

$$= 95.24[1+0.0005\times 100] = 100 \text{ cm}$$

Cross sectional area of the container at 100°C is $A = A_0[1+2\alpha\times 100]$

$$= 1[1+2\times 0.0005\times 100] = 1.1 \text{ cm}^2$$

Volume of liquid in the container at 110°C = Ah = 110 cm^3

Volumer of this liquid at 0°C will be

$$\therefore \quad \frac{110}{1+\gamma\times 100} = \frac{110}{1+0.001\times 100} = 100 \text{ cm}^3$$

If mass of liquid in container = m

Then mass of liquid in beaker = 3m

If θ = equilibrium temperature; $3ms(\theta-0°) = ms(100-\theta)$

$$\Rightarrow \quad 3\theta = 100-\theta \quad\Rightarrow\quad \theta = 25°\text{C}$$

27. (a) Energy released when steam condenses at 100°C is

$$E_1 = mL_v = 0.025\times 2.26\times 10^6 = 5.65\times 10^4 \text{ J}$$

Energy released when condensed steam (actually you should read "water") cools from 100°C to θ is

$$E_2 = 0.025\times 4200\times(100-\theta) = 1.05\times 10^4 - 105\,\theta$$

Energy absorbed by 50 g ice on melting at 0°C is

$$E_3 = mL_f = 0.05\times 3.36\times 10^5 = 1.68\times 10^4 \text{ J}$$

Energy absorbed by 500 g water (original 450 g + 50 g of melted ice) in moving from 0°C to θ°C

$$E_4 = 0.5\times 4200\times\theta = 2100\,\theta$$

Energy absorbed by Cu is $E_5 = ms\Delta\theta = 0.18\times 390\times\theta = 70.2\,\theta$

Heat lost = Heat gained

$$5.65\times 10^4 + 1.05\times 10^4 - 105\,\theta = 1.68\times 10^4 + 2100\,\theta + 70.2\,\theta$$

$$\Rightarrow \quad \theta = 22°\text{C}$$

(b) Initially, steam condenses and ice melts. Energy released by condensed steam per minute is -

$$E_1 = mL_v = 0.005\times 2.26\times 10^6 = 1.13\times 10^4 \text{ J min}^{-1}$$

Rate of melting of ice (kg min^{-1}) is given by

$$\frac{1.13\times 10^4\,\text{J min}^{-1}}{3.36\times 10^5\,\text{J kg}^{-1}} = 33.6\times 10^{-3} \text{ kg min}^{-1} = 33.6 \text{ g min}^{-1}$$

Time needed to melt 50 g ice is $t_1 = \dfrac{50}{33.6}$ min = 1.49 min

Till this time the temperature remains constant at 0°C. After this it increases to become θ at time 't' [For simplicity, let's begin counting time from the instant the complete ice melts.]

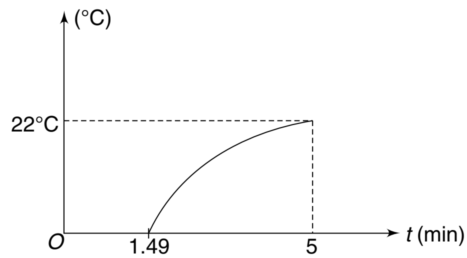
Proceeding as in part (a)

$$0.005t \times 2.26 \times 10^6 + 0.005t \times 4200\,(100 - \theta) = 0.5 \times 4200 \times \theta + 0.18 \times 390 \times \theta$$

$$\Rightarrow \quad 11.3 \times 10^3 t + 2.1 \times 10^3 t = (2100 + 70.2 + 21t)\,\theta$$

$$\theta = \frac{13.4 \times 10^3 t}{(2170.2 + 21t)}$$

One can find $\dfrac{d\theta}{dt}$ to check that slope of θ versus t graph will decrease with increasing 't'. Graph is as shown.

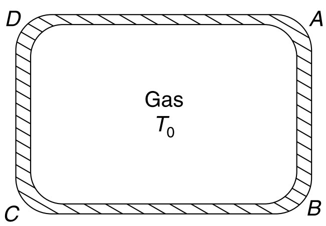


CHAPTER 3

Kinetic Theory of Gases and Gas Laws

LEVEL 1

Q. 1: An ideal gas at temperature T_0 is contained in a container. By some mechanism, the temperature of the wall AB is suddenly increased to $T(> T_0)$. Will the pressure exerted by the gas on wall AB change suddenly? Will it be higher or lower than pressure on wall CD?



Q. 2: Find the number of atoms in molecule of a gas for which the ratio of specific heats at constant pressure and constant volume (γ) becomes $\frac{25}{21}$ times if the rotational degree of freedom of its molecules are frozen. Assume that the gas molecules originally had translational and rotational degree of freedom.

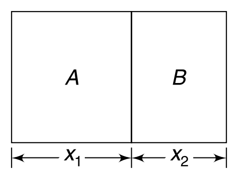
Q. 3: An ideal gas expands following a relation $\frac{P^2}{\rho}$ = constant, where P = pressure and ρ = density of the gas. The gas is initially at temperature T and density ρ and finally its density becomes $\frac{\rho}{3}$.

(a) Find the final temperature of the gas.

(b) Draw the $P - T$ graph for the process.

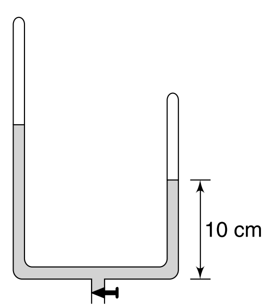
Q. 4: A container having volume V contains N atoms of a gas X. $2N$ atoms of another gas Y is injected into the container and temperature is raised to T and maintained constant. At this temperature the atoms of X and Y combine to form molecule X_2Y. Find the pressure inside the container after the reaction is completed.

Q. 5: A conducting piston separates a cylindrical tube into two compartments A and B. The two compartments contain equal mass of two different gases. The molar mass of two gases are M_A = 32 g and M_B = 28 g. Find the ratio of lengths $(X_1 : X_2)$ of the two compartments in equilibrium.



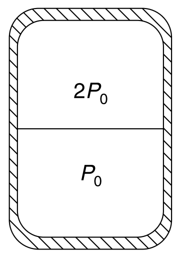
Q. 6: A container of volume V_1 has an ideal gas of molar mass M_1 at pressure P_1 and temperature T_1. Another container of volume V_2 has another ideal gas of molar mass M_2 at pressure P_2 and temperature T_2. Two gases are mixed in a vessel and acquire an equilibrium temperature and pressure of T_0 and P_0 respectively. Find the density of the mixture.

Q. 7: A U shaped tube has two arms of equal cross section and lengths ℓ_1 = 80 cm and ℓ_2 = 40 cm. The open ends are sealed with air in the tube at a pressure of 80 cm of mercury. Some mercury is now introduced in the tube through a stopcock connected at the bottom (the air is not allowed to leak out). In steady condition the length of mercury column in the shorter arm was found to be 10 cm. Find the length of the mercury column in the longer arm. Neglect the volume of the part of the tube connecting two arms and assume that the temperature is constant.

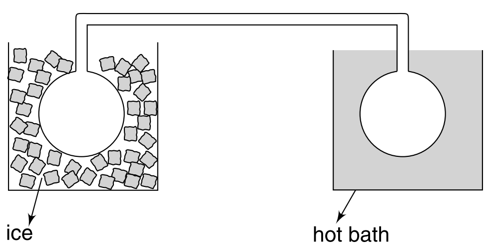


Q. 8: Tidal volume is that volume of air which is inhaled (and exhaled) in one breath by a human during quiet breathing. For a normal man this volume is close to 0.6 litre. 35% of oxygen (in terms of molecular count) gets converted into carbon – di – oxide in the exhaled air. Nearly 21% of the air that we breath in is oxygen. Assume that a man is breathing when the air is at STP and the moisture content of inhaled and exhaled air is same. Estimate the difference in mass of an inhaled breath and exhaled breath.

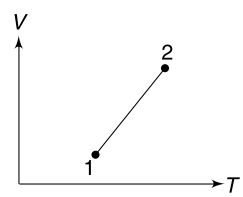
Q. 9: A cylindrical container of volume V_0 is divided into two parts by a thin conducting separator of negligible mass. The walls of the container are adiabatic. Ideal gases are filled in the two parts such that the pressures are P_0 and $2P_0$ when the separator is held in the middle of the container (see figure). Now the separator is slowly slid and released in a position where it stays in equilibrium. Find the volume of the two parts.



Q. 10: Two identical glass bulbs are interconnected by a thin tube of negligible volume. An ideal gas is filled in the bulbs at STP. One bulb is placed in a tub of melting ice and the other bulb is placed in a hot bath. The gas pressure in the bulbs becomes 1.5 times. Find the tem perature of the hot bath. Which bulb has more gas?



Q. 11: Two states –1 and 2 – of an ideal gas has been shown in *VT* graph. In which state is the pressure of the gas higher?



Q. 12: A cylindrical tube of cross sectional area A and length L is filled with an ideal gas. Temperature of the gas varies linearly from T_0 at one end to $2T_0$ at the other end. Calculate the number of mole of the gas in the tube assuming pressure to be uniform throughout equal to P.

Q. 13: Calculate mean free path length (λ) for molecules of an ideal gas at STP. It is known that molecular diameter is 2×10^{-10} m.

Q. 14: A container contains a gas and a few drops of water at TK. The pressure in the container is 830 mm of Hg. The temperature of the container is reduced by 1%. Calculate the new pressure in the container. It is known that the gas is saturated with water vapour and the saturated vapour pressure at two temperatures are 30 mm and 25 mm of Hg.

Q. 15: A close container of volume 0.02 m^3 contains a mixture of neon and another gas A of unknown molecular mass. The container contains 4 g of Neon and 24 g of gas A. At a temperature of 27°C the pressure of the mixture is 10^5 N/m^2. Is there any possibility of finding gas A in the atmosphere of a planet of radius 600 km and mean density $\rho = 5 \times 10^3$ kg/m^3?

Temperature of the planet is 2200°C.

Molar molecular mass of neon = 20 g;

Gas constant $R = 8.314$ J mol^{-1} K^{-1}

Gravitational constant $G = 6.67 \times 10^{-11}$ N·m^2 kg^{-2}

Q. 16: We know that volume (V) occupied by a substance scales as x^3, where x is the average distance between its molecules. Assume that water vapour at 100°C and atmospheric pressure has average intermolecular separation equal to x_v and for liquid water at 100°C the intermolecular separation is x_w. Find the ratio $\frac{x_v}{x_w}$ considering density of water at 100°C to be 1.0×10^3 kg/m^3 and taking water vapour as an ideal gas.

Atmospheric pressure $P_0 = 1.0 \times 10^5$ N/m^2; Gas constant $R = 8.3$ J mol^{-1} K^{-1}.

Q. 17: The diameter of molecules of a gas, considered as sphere, is about 3×10^{-10} m. Assume the entire volume occupied by the gas to be divided into cubic cells with one molecule per cell. Estimate the distance between molecules in terms of molecular diameter under standard conditions.

Q. 18: The core of the sun is a plasma. It is at a temperature of the order of 10^7 K and contains equal number of protons and electrons. The density of the core of the Sun is 10^5 kg m^{-3}. Assume that molar mass of proton is 1 g mol^{-1} and the mass of an electron to be negligible compared to the mass of a proton. Estimate the pressure at the core of the sun. Assume that plasma behaves like an ideal gas.

Q. 19: Calculate the *rms* speed of He and N_2 molecules in the atmosphere at 300 K. Explain why our atmosphere has only a small amount of helium though it has large nitrogen content. The sun has temperature close to 6000 K. How can you explain the existence of He in the Sun.

Q. 20: A cylinder of oxygen, used as medical aid, has a volume of 16 L. It is filled to a pressure of 1.37×10^7 Nm^{-2} above the atmospheric pressure at room temperature of 300 K. When in use, the flow rate at atmospheric pressure is 2.4 L/min. How long will the cylinder last? Atmospheric pressure is 10^5 Nm^{-2}.

Q. 21. It is known that pressure of a gas increases if–

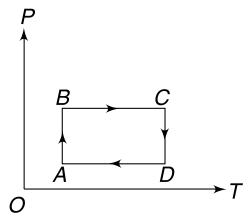
(A) You increase the temperature of the gas while holding its volume constant.

(B) You compress the gas holding its temperature constant.

(i) In which of the two cases [A or B] the average impulse imparted by a gas molecule to the container wall during a collision increases?

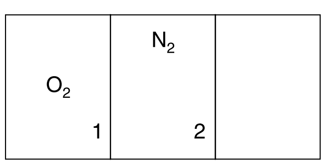
(ii) In which of the two cases the frequency of collisions of gas molecules with the container wall increases.

Q. 22: An ideal gas undergoes a cyclic process $A \rightarrow B \rightarrow C \rightarrow D$ for which $P - T$ graph is as shown. Draw $P - V$ and $V - T$ of graph to represent the same process.

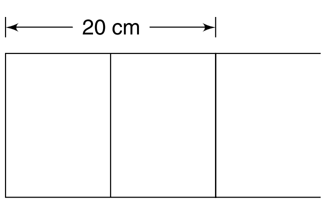


Q. 23: An insulated cylindrical vessel is divided into three identical parts by two partitions 1 and 2. The left part contains O_2 gas, the middle part has N_2 and the third chamber has vacuum. The average molecular speed in oxygen chamber is V_0 and that in nitrogen chamber is $\sqrt{\frac{8}{7}}V_0$. Pressure of the gases in two chambers is same. Partition 1 is removed and the gases are allowed to mix. Now the stopper holding the partition 2 is removed and it slides to the right wall of the container, so that the mixture of gases occupy the entire volume of the container.

Find the average speed of O_2 molecules now.



Q. 24: In a cylindrical container two pistons enclose gas in two compartments as shown in the figure. The pistons have negligible thickness and can move without friction. The system is originally in equilibrium. The outer piston is slowly moved out by 10 cm and the inner piston is found to move by 4 cm. Find the distance of the inner piston in equilibrium from the closed end of the cylinder if the outer piston is slowly moved out of the cylinder. Assume temperature of the gas to remain constant.

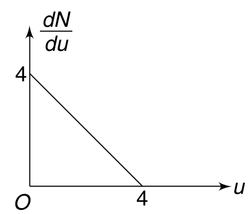


LEVEL 2

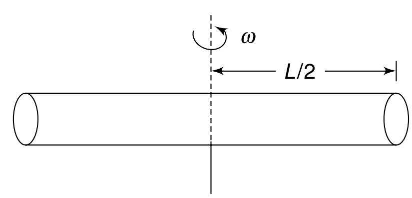
Q. 25: The *rms* speed of molecules of an ideal gas is v_{rms}. Obtain the expression for *rms* value of relative speed for all pair of molecules in a gas sample.

Q. 26: On a cold winter night you switch on a room heater to keep your room warm. Does it mean that the total energy of the air inside the room increases after you switch on the heater?

Q. 27: A hypothetical gas sample has its molecular speed distribution graph as shown in the figure. The speed (u) and $\frac{dN}{du}$ have appropriate units. Find the root mean square speed of the molecules. Do not worry about units.



Q. 28: A cylindrical container of length $2L$ is rotating with an angular speed ω about an axis passing through its centre and perpendicular to its length. Its contains an ideal gas of molar mass M. Calculate the ratio of gas pressure at the end of the container to the pressure at its centre. Neglect gravity and assume that temperature of the gas throughout the container is T.

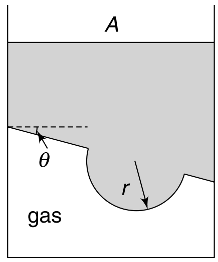


Q. 29: A container of volume $V = 16.6$ litre contains a mixture of hydrogen and helium at a temperature of 300 K. The pressure in the container is 6×10^5 Pa and mass of the mixture is 10 g. Let f_1 and f_2 be the number of collisions made by the hydrogen and helium molecules with unit area of the container wall in unit time. Calculate the ratio $\frac{f_1}{f_2}$. [$R = 8.3$ J mol^{-1} K^{-1}].

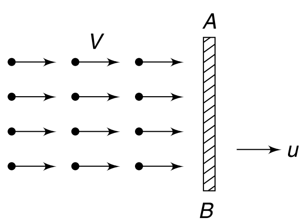
Q. 30: An ideal gas is enclosed in a cylinder having cross sectional area A. The piston of mass M has its lower face inclined at θ to the horizontal. The lower face of the piston also has a hemispherical bulge of radius r. The atmospheric pressure is P_0.

(a) Find pressure of the gas.

(b) The piston is slowly pulled up by a distance x. During the process the piston is always maintained in equilibrium by adding heat to the gas. [It means if the piston is left at any stage it will stay there in equilibrium]. Find change in temperature of the gas. Number of mole of the gas in the cylinder is one.



Q. 31: A beam of gas molecules is incident normally on a plate AB. Each gas molecule has mass m and velocity V. The incident beam falls on an area A on the plate and all the molecule strike the plate elastically. Number of molecules in unit volume of the beam is n. When the plate is moved to right (see fig.) a force F_1 is needed to keep it moving with constant velocity $u (< V)$. When the plate is moved to left with constant velocity u, an external force F_2 is needed. Find $F_2 - F_1$.



Q. 32: Atmosphere of a planet contains only an ideal gas of molar mass M_0. The temperature of the atmosphere varies with height such that the density of atmosphere remains same throughout. The planet is a uniform sphere of mass M and radius 'a'. Thickness of atmosphere is small compared to 'a' so that acceleration due to gravity can be assumed to be uniform throughout the atmosphere.

(a) Find the temperature difference between the surface of the planet and a point at height H in its atmosphere

(b) If the *rms* speed of gas molecules near the surface of the planet is half the escape speed, calculate the temperature of the atmosphere at a height H above the surface. Assume $\dfrac{C_P}{C_V} = \gamma$ for the atmospheric gas.

Q. 33: The Maxwell-Boltzmann distribution of molecular speeds in a sample of an ideal gas can be expressed as

$$f = \frac{4}{\sqrt{\pi}}\left(\frac{m}{2kT}\right)^{3/2} v^2 e^{-\frac{mv^2}{2kT}} \cdot dv$$

Where f represent the fraction of total molecules that have speeds between v and $v + dv \cdot m$, k and T are mass of each molecule, Boltzmann constant and temperature of the gas.

(a) What will be value of $\int\limits_{v=0}^{v=\infty} f dv$?

(b) It is given that $\int\limits_0^\infty v^3 e^{-av^2} dv = \dfrac{1}{2a^2}$

Find the average speed of gas molecules at temperature T.

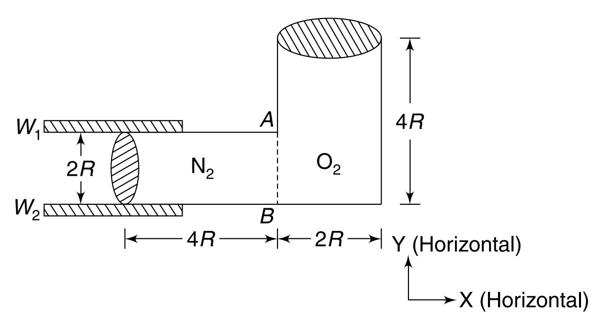
Q. 34. A large cylindrical tower is kept vertical with its ends closed. An ideal gas having molar mass M fills the tower. Assume that temperature is constant throughout, acceleration due to gravity (g) is constant throughout and the pressure at the top of the tower is zero.

(a) Calculate the fraction of total weight of the gas inside the tower that lies above certain height h.

(b) At what height h_0 the quantity of gas above and below is same. How does the value of h_0 change with temperature of the gas in the tower?

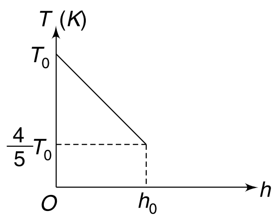
Q. 35: A physics book was found on a newly discovered island in 18th century. A problem in the book was as follows. "1 *pinch* of an ideal gas is kept in a container of volume 1.5 *volka*. When the temperature is 40 *tapu*, the gas pressure is 25 *phatka*. When the temperatue is reduced to –20 *tapu* the gas pressure becomes 10 *phatka*. Find the temperature of absolute zero in *tapu*.

Q. 36: A L shaped container has dimensions shown in the fig. It has circular cross section of radius R. It is placed on a smooth horizontal table. The container is divided into two equal sections by a membrane AB. One section contains nitrogen and the other one contains oxygen. Temperature of both sides is same but pressure in the compartment having N_2 is 4 times that in the other compartment. Due to some reason the membrane gets punctured. Find the distance moved by the cylinder if the cylinder is constrained to move in x direction only [with the help of guiding walls W_1 and W_2]. There is no friction anywhere and mass of the cylinder and membrane is negligible.



Q. 37: (a) Assume the atmosphere as an ideal gas in static equilibrium at constant temperature T_0. The pressure on the ground surface is P_0. The molar mass of the atmosphere is M. Calculate the atmospheric pressure at height h above the ground.

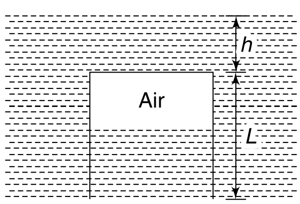
(b) To be more realistic, let as assume that the temperature in the troposphere (lower part of the atmosphere) decreases with height as shown in the figure. Now calculate the atmospheric pressure at a height h_0 above the ground.



Q. 38: A cylindrical container has cross sectional area of $A = 0.05\ m^2$ and length $L = 0.775$ m. Thickness of the wall of the container as well as mass of the container is negligible. The container is pushed into a water tank with its open end down. It is held in a position where its closed end is h = 5.0 m below the water surface. What force is required to hold the container in this position? Assume temperature of air to remain constant.

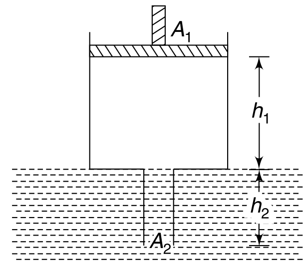
Atmospheric pressure $P_0 = 1 \times 10^5$ Pa; Acceleration due to gravity $g = 10\ m/s^2$

Density of water = $10^3\ kg/m^3$



Q. 39: A syringe has two cylindrical parts of cross sectional area $A_1 = 4\ cm^2$ and $A_2 = 1\ cm^2$. A mass less piston can slide on the inner wall of the part having cross section A_1. The end of the part having cross section A_2 is open. Syringe is dipped in a large water tank such that the narrower part remains completely submerged and the wider part is filled with air. The length shown in figure are $h_1 = 55$ cm and $h_2 = 100$ cm. The piston is pushed down by a distance x such that 25% of the air inside the syringe is expelled out through the open end of the narrower part. Assume that the temperature of air remains constant and calculate x.

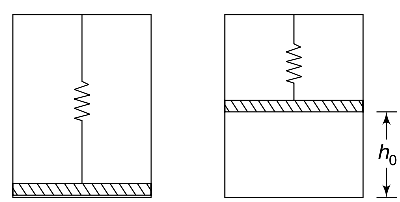
Density of water = $10^3\ k\ gm^{-3}$; g = 10 ms^{-2}; Atmospheric pressure = $10^5\ Nm^{-2}$.



Q. 40: If an ideal gas is allowed to undergo free expansion it does not cool. But a real gas cools during free expansion. Why?

Q. 41: A container has a gas that consists of positively charged ions. The gas undergoes a free expansion in which there is no heat exchange with surrounding. Does the temperature of the gas increases or decreases? Why?

Q. 42: A freely sliding massive piston is supported by a spring inside a vertical cylinder as shown. When all air is pumped out of the container, the piston remains in equilibrium with only a tiny gap between the piston and the bottom surface of the cylinder. An ideal gas at temperature T_0 is slowly injected under the piston so that it rises to height h_0 (see figure). Calculate the height of the piston from the bottom of the container if the temperature of the gas is slowly raised to $4T_0$.



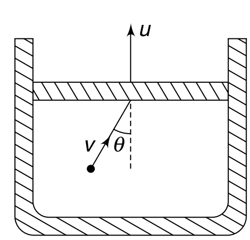
LEVEL 3

Q. 43: An ideal gas is inside a cylinder with a piston that can move freely. The walls of the cylinder and piston are non- conducting. The piston is being moved out of the cylinder at a constant speed u.

(a) Consider a gas molecule of mass m moving with speed $v (>> u)$. It hits the piston elastically at an angle of incidence θ. Calculate the loss in kinetic energy of the molecule.

(b) If the area of piston is A and pressure of the gas is P, calculate the rate of decreases of molecular kinetic energy of the gas sample.

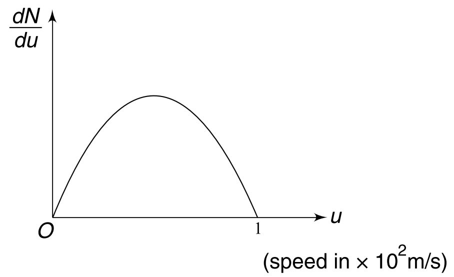
(c) If u >> molecular velocities, at what rate will the gas lose its molecular kinetic energy?



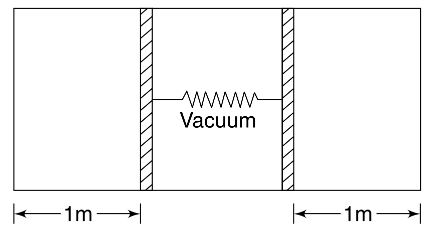
Q. 44: The speed distribution of molecules in a sample of a gas is shown in the figure. The graph between $\frac{dN}{du}$ and u is a parabola and total number of molecules in the sample is N_0.

(a) Calculate the *rms* speed of the molecules.

(b) Calculate the total translational kinetic energy of molecules if mass of the sample is 10 g.



Q. 45: A cylindrical container is divided into three parts by two tight fitting pistons. The pistons are connected by a spring. The region between the pistons is vacuum and the other two parts have same number of moles of an ideal gas. Initially, both the gas chambers are at temperature T_0 and the spring is compressed by 1 m. Length of both gas chambers is 1 m in this position. Now the temperature of the left and right chambers are raised to $\frac{4T_0}{3}$ and $\frac{5T_0}{3}$ respectively. Find the final compression in the spring in equilibrium. Assume that the pistons slide without friction.



Q. 46: A helium balloon has its rubber envelope of weight w_{Rubber} and its is used to lift a weight of w_{Load}. The volume of fully inflated balloon is V_0. The atmospheric temperature is T_0 throughout and the atmospheric pressure is known to change with height y according to equation $P = P_0 e^{-ky}$ where k is a positive constant and P_0 is atmospheric pressure at ground level. The balloon is inflated with sufficient amount of helium so that the net upward force on is F_0 at the ground level. Assume that pressure inside the balloon is always equal to the outside atmospheric pressure. Density of air and helium at ground level is ρ_a and ρ_{He} respectively.

(a) Find the number of moles (n) of the helium in the balloon.

For following two questions assume n to be a known quantity.

(b) Find the height (y_0) at which the balloon is fully inflated.

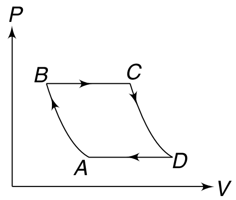
(c) Prove that the balloon will be able to rise to height y_0 if

$$\frac{nRT_0 g}{P_0}(\rho_a - \rho_{\text{Hg}}) > w_{\text{Rubber}} + w_{\text{Load}}.$$

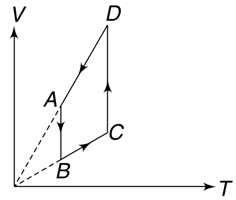
ANSWERS

1. Pressure on AB will be higher than pressure on CD.
2. Two
3. (a) $\sqrt{3}\,T$
4. $P = \left(\frac{2N}{N_A}\right)\frac{RT}{V}$
5. $\frac{7}{8}$
6. $\frac{(P_1V_1M_1T_2 + P_2V_2M_2T_1)P_0}{(P_1V_1T_2 + P_2V_2T_1)RT_0}$
7. 16.27 cm
8. 0.24 mg
9. $\frac{2V_0}{3}$ and $\frac{V_0}{3}$
10. 819 K
11. State 1
12. $\frac{PAL}{RT_0}\ln 2$
13. 2.0×10^{-7} m
14. 817 mm of Hg
15. No
16. 12
17. Ten times the molecular diameter
18. $2 \times 10^{16}\ P_a$
20. 920 min
21. (a) A (ii) A and B

22.



(a)



(b)

23. V_0

24. 8 cm.

25. $\sqrt{2}\, v_{rms}$

26. No

27. $u_{rms} = \sqrt{\dfrac{8}{3}}$

28. $e^{\frac{M\omega^2 L^2}{2RT}}$

29. $3\sqrt{2}$

30. (a) $P_0 + \dfrac{\text{Mg}}{A}$ (b) $\dfrac{(P_0 A + \text{Mg})x}{R}$

31. 8 m *An Vu*

32. (a) $\Delta T = \dfrac{GMM_0 H}{Ra^2}$ (b) $T_H = \dfrac{GMM_0}{R_a}\left[\dfrac{1}{2\gamma} - \dfrac{H}{a}\right]$

33. (a) 1 (b) $\sqrt{\dfrac{8\text{kT}}{\pi m}}$

34. (a) $e^{\frac{-Mgh}{RT}}$ (b) $\left(\dfrac{RT}{\text{Mg}}\right)\ln 2$

(c) h_0 increase with T.

35. -60 *tapu*

36. 0.83 R towards left

37. (a) $P = P_0 e^{\frac{-Mgh}{RT}}$ (b) $P = P_0\left(\dfrac{5}{4}\right)^{-\frac{5Mgh_0}{RT_0}}$

38. 250 N

39. 42.5 cm

41. Temperature increases

42. $2h_0$

43. (a) $2mvu\cos\theta$ (b) PAu (c) zero

44. (a) $v_{rms} = \sqrt{\dfrac{3}{10}} \times 100$ m/s

(b) $K_T = 15$ J

45. $\dfrac{\sqrt{13} - 1}{2}$ m

46. (a) $n = \dfrac{P_0(F_0 + w_{\text{Rubber}} + w_{\text{Load}})}{RT_0(\rho_a - \rho_{\text{He}})g}$

(b) $y_0 = \dfrac{1}{k}\ln\left(\dfrac{P_0 V_0}{nRT_0}\right)$

SOLUTIONS

1. **Hint:** When a molecule strikes AB, it will gain energy & rebound with higher speed. Change in momentum will be higher.

2. Let degree of freedom for gas molecules be f

$$\gamma = \frac{f+2}{f}$$

After freezing of rotational degree of freedoms the molecules can have translational kinetic energy only and degree of freedom will become = 3

$$\gamma' = \frac{3+2}{3} = \frac{5}{3}$$

Given $$\frac{25}{21}\,\frac{f+2}{f} = \frac{5}{3} \Rightarrow f = 5$$

In means the gas is diatomic

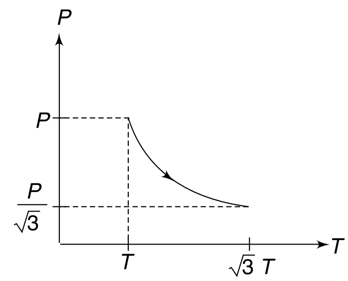
3. (a) $$\frac{P^2}{\rho} = \text{a const} \qquad \text{...(i)}$$

From $$PV = nRT$$

$$\frac{P}{\rho} = \frac{RT}{M} \qquad \text{...(ii)}$$

Using (i) and (ii) $\qquad PT = \text{a constant} \qquad$ ...(iii)

Using (i) we can say that when density changes from ρ to $\frac{\rho}{3}$, the pressure will change from P to $\frac{P}{\sqrt{3}}$ and using (iii) it can be concluded that temperature will change from T be $\sqrt{3}\,T$



(b) PT = constant

4. 2 atoms of X combine with 1 atoms of Y

$\therefore \quad \frac{N}{2}$ molecule of X_2Y is formed and $\frac{3N}{2}$ atoms of Y is left as it is.

$\therefore \quad$ Total number of particles $= \frac{N}{2} + \frac{3N}{2} = 2N$

Pressure is given by $PV = nRT$

$$\Rightarrow \qquad P = \left(\frac{2N}{N_A}\right)\frac{RT}{V}$$

5. For an ideal gas $PV = nRT$

$$\Rightarrow \qquad PV = \frac{m}{M}RT \quad \therefore \quad MV = \frac{mRT}{P} = \text{a constant}$$

This is because the two gases have equal mass, same pressure and temperature in equilibrium.

$$\therefore \qquad M_A V_A = M_B V_B$$

$$32\,(X_1 A) = 28\,(X_2 A) \quad \therefore \quad \frac{X_1}{X_2} = \frac{7}{8}$$

6. $$n_1 = \frac{P_1 V_1}{RT_1} \text{ and } n_2 = \frac{P_2 V_2}{RT_2}$$

For the mixture $\qquad P_0 V_0 = (n_1 + n_2)RT_0$

$$V_0 = (n_1 + n_2)\frac{RT_0}{P_0}$$

$\therefore \quad$ Density of mixture $\qquad \rho = \frac{n_1 M_1 + n_2 M_2}{V_0} = \frac{(n_1 M_1 + n_2 M_2)P_0}{(n_1 + n_2)RT_0}$

$$= \frac{\left(\frac{P_1 V_1 \mathrm{M}_1}{RT_1} + \frac{P_2 V_2 \mathrm{M}_2}{RT_2}\right)P_0}{\left(\frac{P_1 V_1}{RT_1} + \frac{P_2 V_2}{RT_2}\right)RT_0} = \frac{(P_1 V_1 T_2 M_1 + P_2 V_2 T_1 M_2)P_0}{(P_1 V_1 T_2 + P_2 V_2 T_1)RT_0}$$

7. Let P_1 and P_2 be pressure (in cm of Hg) of the air trapped in the shorter and the longer arms respectively. Let the required height of Hg column in the longer tube be h.

For air in shorter arm (PV = const)

$$(80\text{ cm})(40\text{ cm. }A) = P_1(40 - 10)A \quad \Rightarrow \quad P_1 = \frac{320}{3}\text{ cm}$$

Similarly, for the other arm

$$(80\text{ cm})(80\text{ cm. A}) = P_2(80 - h)A$$

$$\therefore \qquad P_2 = \frac{80 \times 80}{80 - h}$$

Equating the pressure in Hg just below the surface in shorter arm to the pressure at same horizontal level in the other arm–

$$P_2 + (h - 10) = P_1$$

$$\frac{80 \times 80}{80 - h} + h - 10 = \frac{320}{3}$$

$$3[6400 + 90h - 800 - h^2] = 25600 - 320h$$

$$3h^2 - 590h + 8800 = 0$$

$$h = \frac{590 \pm \sqrt{348100 - 105600}}{6}$$

$$= \frac{590 \pm 492.4}{6} = \frac{97.6}{6} = 16.27 \text{ cm}$$

[+ve sign is unacceptable]

8. Number of moles of air in one breath $= \dfrac{0.6 \text{ litre}}{22.4 \text{ litre}} = 0.027$ mole

Number of moles of O_2 in inhaled air $= 0.027 \times 0.21$

No. of moles of O_2 converted into CO_2 is

$$= 0.027 \times 0.21 \times 0.35 = 1.98 \times 10^{-3}$$

$\therefore$ Difference in mass of air

$$= 1.98 \times 10^{-3}\left[M_{CO_2} - M_{O_2}\right]$$

$$= 1.98 \times 10^{-3}\left[44\frac{\text{g}}{\text{mol}} - 32\frac{\text{g}}{\text{mol}}\right]$$

$$= 2.4 \times 10^{-2} \text{ g} = 0.24 \text{ mg}$$

9. Since the separator is conducting, the temperature in both parts is same (say, T_0). Number of moles in two chambers are

$$n_1 = \frac{2P_0V_0}{2RT} = \frac{P_0V_0}{RT} \text{ and } n_2 = \frac{P_0V_0}{2RT}$$

$$\therefore \quad \frac{n_1}{n_2} = \frac{2}{1}$$

When the separator is in equilibrium, the pressure on two sides is same.

Let volume of upper and lower parts be V_1 and V_2 respectively and the common pressure and temperature be P and T

$$PV_1 = n_1RT \quad \text{and} \quad PV_2 = n_2RT$$

$$\therefore \quad \frac{V_1}{V_2} = \frac{n_1}{n_2} = \frac{2}{1}$$

Also $\quad V_1 + V_2 = V_0$

$$\therefore \quad V_1 = \frac{2V_0}{3} \quad \text{and} \quad V_2 = \frac{V_0}{3}$$

10. Let volume of each bulb be V_0

Number of moles in each bulb $n_0 = \dfrac{P_0V_0}{RT_0}$

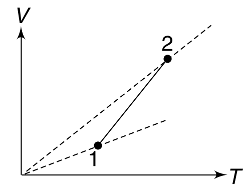
Number of moles in the bulb kept in ice $= \dfrac{1.5P_0V_0}{RT_0}$

Number of moles in the other bulb = $\dfrac{1.5 P_0 V_0}{RT}$

$$\therefore \quad \frac{1.5 P_0 V_0}{RT} + \frac{1.5 P_0 V_0}{RT_0} = \frac{2 P_0 V_0}{RT_0}$$

$$\frac{1.5}{T} + \frac{1.5}{T_0} = \frac{2}{T_0}$$

$$\Rightarrow \quad \frac{1.5}{T} = \frac{0.5}{T_0} \Rightarrow T = 3T_0 = 819 \text{ K}$$

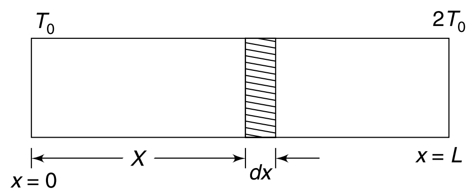


11.
$$PV = nRT$$

$$\Rightarrow \quad V = \frac{nR}{P} T$$

If we draw constant pressure line on *VT* graph, the slope will be higher for lower pressure. Hence $P_2 < P_1$.

12. Temperature at distance x from the colder end is–



$$T = T_0 + \frac{T_0}{L} x$$

Consider a volume $A\,dx$ as shown in the figure.

Number of moles in this volume is

$$dn = \frac{P(A\,dx)}{RT} = \frac{PA}{RT_0} \frac{dx}{\left(1 + \frac{x}{L}\right)}$$

$$\therefore \quad n = \frac{PA}{RT_0} \int_0^L \frac{dx}{1 + \frac{x}{L}} = \frac{PAL}{RT_0} \left[\ln\left(1 + \frac{x}{L}\right)\right]_0^L$$

$$= \frac{PAL}{RT_0} \ln 2$$

13. 22.4 litre volume of the gas has 6.02×10^{23} molecules.

Number of molecule per unit volume is $n = \dfrac{6.02 \times 10^{23}}{22.4 \times 10^{-3}} = 2.7 \times 10^{25}$ molecules m^{-3}

Mean free path length is given by $\lambda = \dfrac{1}{\sqrt{2}\,\pi n d^2}$

$$= \frac{1}{1.414 \times 3.14 \times 2.7 \times 10^{25} (2 \times 10^{-10})^2} = 2.0 \times 10^{-7} \text{ m}$$

14. By Dalton's law of partial pressure, the partial pressure of gas at *TK* is = (830 – 30) mm of Hg

Let the pressure in container be P at new temperature of $\left(T - \dfrac{T}{100}\right)K$

The partial pressure of gas is (P – 25) mm of Hg

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

$$\therefore \quad \frac{800}{T} = \frac{(P-25)}{T\cdot\frac{99}{100}}$$

$$\Rightarrow \quad P = 25 + \frac{800 \times 99}{100} = 817\,\text{mm of Hg}$$

15. Number of moles of gas mixture in the container

$$n = \frac{PV}{RT} = \frac{10^5 \times 0.02}{8.314 \times 300}$$

$$n_{\text{Ne}} + n_A = 0.8$$

$$\frac{4}{20} + n_A = 0.8 \quad \Rightarrow \quad n_A = 0.6$$

$\therefore$ Molar mass of gas A is $\quad m = \frac{24}{0.6} = 40$ g

Let's find the *rms* speed of gas molecules of A at temperature of 2200°C = 2473 K

$$V_{rms} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3 \times 8.314 \times 2473}{40 \times 10^{-3}}} = 1242 \text{ m/s}$$

The escape speed on the surface of a planet is

$$V_e = \sqrt{\frac{2GM}{R}} = \sqrt{\frac{2G\frac{4}{3}\pi R^3 \cdot \rho}{R}} = \sqrt{\frac{8\pi}{3}G\rho}\,.R$$

$$= \sqrt{\frac{8 \times 3.14 \times 6.67 \times 10^{-11} \times 5 \times 10^3}{3}} \times 600 \times 10^3$$

$$= 16.7 \times 10^{-4} \times 6 \times 10^5 \simeq 1000\,\text{m/s}$$

Because $V_{rms} > V_e$, it is unlikely that gas A will be found in atmosphere of the planet.

16. Consider m mass of water as well as vapour. Volumes are

$$V_w = \frac{m}{\rho_w} \text{ and}$$

$$V_v = \frac{nRT}{P_0} \qquad [\because \quad PV = nRT]$$

$$= \frac{mRT}{MP_0} \quad [M = \text{molar mass of water} = 1.8 \times 10^{-2}\ \text{kg mol}^{-1}]$$

$$\therefore \quad \frac{V_v}{V_w} = \frac{RT\rho_w}{MP_0}$$

$$\therefore \quad \left(\frac{x_v}{x_w}\right)^3 = \frac{RT\rho_w}{MP_0} = \frac{8.3 \times 373 \times 1.0 \times 10^3}{1.8 \times 10^{-2} \times 1.0 \times 10^5} = 1720$$

$$\therefore \quad \frac{x_v}{x_w} \simeq 12$$

17. Volume of 1 mole at STP is 22.4 litre = $22.4 \times 10^{-3}\,m^3$

Number of molecules in one mole = 6.02×10^{23}

$\therefore$ Volume of each cubic cell = $\dfrac{22.4 \times 10^{-3}}{6.02 \times 10^{23}} = 3.7 \times 10^{-26}\,m^3$

Side length of each cell = $(3.7 \times 10^{-26})^{1/3} = 3.3 \times 10^{-9}\,m$

Molecular diameter = $3 \times 10^{-10}\,m$

$\therefore$ Distance between molecules is about 10 times the molecular diameter.

18. $1\,m^3$ volume of matter has a mass of 10^5 kg and is mostly due to protons.

n_p = number of moles of proton in $1\,m^3$ volume.

$$= \frac{\text{mass of protons}}{\text{Molar mass of proton}} = \frac{10^5}{10^{-3}} = 10^8 \text{ mol.}$$

The matter has equal number of electrons. Hence total count of particles is

$$2n_p = 2 \times 10^8 \text{ mol.}$$

$$\therefore \quad P = \frac{nRT}{V} = \frac{2 \times 10^8 \times 8.314 \times 10^7}{1} = 2 \times 10^{16}\,P_a$$

19.

$$V_{rms} = \sqrt{\frac{3RT}{M}}$$

$$V_{\text{He}} = \sqrt{\frac{3 \times 8.214 \times 300}{4 \times 10^{-3}}} = 1.37 \times 10^3 \text{ ms}^{-1} = 1.37 \text{ kms}^{-1}$$

$$V_{N_2} = \sqrt{\frac{3 \times 8.314 \times 300}{28 \times 10^{-3}}} = 0.5 \times 10^3 \text{ ms}^{-1} = 0.5 \text{ kms}^{-1}$$

The escape speed from the surface of the earth is 11.2 kms^{-1}. A significant fraction of He molecules will have speed in excess of escape speed. Thus, they leak away into space. Over time the He content has decreased to almost nothing.

Escape speed on the sun is too high due to its large mass.

20. Let the volume of entire gas at atmospheric pressure be V.

$$P_1 V_1 = P_2 V_2 \qquad [P_2 = 1.37 \times 10^7 + 10^5 \simeq 1.38 \times 10^7]$$

$$10^5 V = 1.38 \times 10^7 \times 16$$

$$\therefore \quad V = 1.38 \times 16 \times 10^2 \text{ litre}$$

$\therefore$ Time for which the cylinder can last is $t = \dfrac{1.38 \times 16 \times 10^2}{2.4} = 920$ min.

21. (i) In case (A) the speed of molecules increase. Therefore, they impart more impulse during each collision. In case (B) the speed of molecules do not change since temperature is held constant.

(ii) Frequency of collisions will increase in both cases. In case (A) it happens because of more speed and in case (B) it happens because of lesser distance to be travelled by molecules between two collisions.

23.
$$V_0 \propto \sqrt{\frac{T_0}{M_0}} \quad \text{and} \quad V_N \propto \sqrt{\frac{T_N}{M_N}}$$

$$\therefore \quad \frac{V_N}{V_0} = \sqrt{\frac{M_0 T_N}{M_N T_0}}$$

$$\frac{8}{7} = \frac{32 T_N}{28 T_0} \quad \therefore \quad T_N = T_0$$

Since pressure and volume of two chambers are also same we conclude that number of moles of both gases must be same.

$\therefore$ After mixing the temperature remains unchanged. In pushing partition 2 the gas does no work since there is vacuum to the right of the partition. It means internal energy of gas mixture does not change. Thus average speed of molecules do not change.

24. Initial pressure in both chambers = atmospheric pressure (P_0)

Let initial volume of the left and right chambers be V_1 and V_2 respectively, and the area of cross section of the cylinder be A.

For left chamber using PV = constant we get–

$$P(V_1 + 4A) = P_0 V_1 \qquad \text{...(i)}$$

$4A$ is change in volume of the compartment.

Similarly for right chamber $P(V_2 + 6A) = P_0 V_2$ ...(ii)

Where P = final common pressure of the two chambers after the outer piston has been moved by 10 cm

(i) ÷ (ii) gives
$$\frac{V_1 + 4A}{V_2 + 6A} = \frac{V_1}{V_2}$$

$$\Rightarrow \quad 4V_2 = 6V_1 = \frac{V_1}{V_2} = \frac{2}{3}$$

$\therefore$ If original distance of inner piston from the closed end is x_1 then $x_1 = \frac{20}{5} \times 2 = 8$ cm

After the outer piston is removed, the inner piston will come to equilibrium at its original location only; because pressure will become equal to atmospheric pressure.

Hence answer is 8 cm.

25. Consider two molecules having velocities $\vec{v_1}$ and $\vec{v_2}$. Relative velocity of these two molecules

$$\overrightarrow{v_{12}} = \vec{v_1} - \vec{v_2}$$

$$\therefore \quad \overrightarrow{v_{12}} \cdot \overrightarrow{v_{12}} = (\vec{v_1} - \vec{v_2}) \cdot (\vec{v_1} - \vec{v_2})$$

$$v_{12}^2 = v_1^2 + v_2^2 - 2\vec{v_1} \cdot \vec{v_2}$$

Taking average for all pair of molecules

$$\langle v_{12}^2 \rangle = \langle v_1^2 \rangle + \langle v_2^2 \rangle - 2\langle \vec{v}_1 \cdot \vec{v}_2 \rangle$$

But
$$\langle \vec{v}_1 \cdot \vec{v}_2 \rangle = \langle v_1 \cdot v_2 \cos\theta \rangle = 0$$

This is because $\cos\theta$ will be negative for obtuse angle between $\vec{v}_1$ and $\vec{v}_2$ and positive for acute angles. Number of molecules is large.

$$\therefore \quad \langle v_{12}^2 \rangle = \langle v_1^2 \rangle + \langle v_2^2 \rangle$$

$$\langle v_1^2 \rangle = \langle v_2^2 \rangle = \langle v^2 \rangle \text{ [say]} = \langle v_{rms}^2 \rangle$$

$$\therefore \qquad \langle v_{12}^2 \rangle = v_{rms}^2$$

$$\sqrt{\langle v_{12}^2 \rangle} = \sqrt{2}\, v_{rms}$$

26. The pressure inside the room = outside pressure (P_0)
Volume of the room V_0 is a constant.

$$\therefore \qquad P_0 V_0 = nRT$$

$$\Rightarrow \qquad nT = \frac{P_0 V_0}{R} = \text{a constant}$$

$$\therefore \qquad U = nC_v T = \text{a constant}$$

Total energy of air in the room does not change (!)
However, with rise in temperature the average energy of molecules does increase. Actually, some molecules are expelled out of the room and total number of molecules in the room decreases.

27. The equation of the straight line is

$$\frac{dN}{du} = -u + 4$$

$$u_{rms}^2 = \frac{\int u^2\, dN}{\int dN} = \frac{\int_0^4 u^2 \frac{dN}{du} du}{\int_0^4 \frac{dN}{du} du} = \frac{\int_0^4 u^2(-u+4)\, du}{\text{area under the graph}}$$

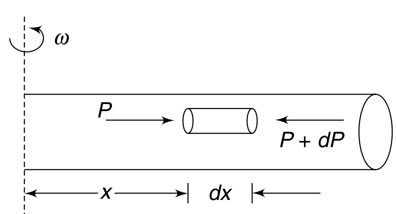
$$= \frac{\int_0^4 (-u^3)\, du + 4\int_0^4 u^2\, du}{\frac{1}{2} \times 4 \times 4} = \frac{\left[-\frac{u^4}{4} + 4 \cdot \frac{u^3}{3}\right]_0^4}{8}$$

$$u_{rms}^2 = \frac{8}{3} \quad \Rightarrow u_{rms} = \sqrt{\frac{8}{3}}$$

28. For an ideal gas

$$\frac{P}{\rho} = \frac{RT}{M}$$

$$\therefore \qquad \rho = \frac{PM}{RT} \qquad \text{...(i)}$$



Consider a cylindrical element of the gas at a distance x from the axis of rotation.
Mass of element = $\rho\, A\, dx$
[A = cross section of the element]
Net force on the element towards centre is = $A\, dP$

This must be equal to the centripetal force

$$\therefore \quad AdP = (\rho A\,dx)\,\omega^2 x$$

$$dP = \rho\omega^2 x dx$$

$$dP = \frac{PM}{RT}\omega^2 x\,dx \quad \text{[using (i)]}$$

$$\therefore \quad \int_{P_0}^{P} \frac{dP}{P} = \frac{M\omega^2}{RT}\int_0^L x\,dx$$

[P_0 = pressure at centre, P = pressure at the end]

$$\therefore \quad \ln\frac{P}{P_0} = \frac{M\omega^2 L^2}{2RT}$$

$$\therefore \quad \frac{P}{P_0} = e^{\frac{M\omega^2 L^2}{2RT}}$$

29. For the gas mixture $nRT = pV$

$$\Rightarrow \quad \left(\frac{m_{\text{H}}}{m_{\text{H}}} + \frac{m_{\text{He}}}{M_{\text{He}}}\right)RT = PV$$

$$\Rightarrow \quad \left(\frac{m_{\text{H}}}{2} + \frac{m_{\text{He}}}{4}\right) \times 8.3 \times 300 = 6 \times 10^5 \times 16.6 \times 10^{-3}$$

$$\Rightarrow \quad \frac{m_{\text{H}}}{2} + \frac{m_{\text{He}}}{4} = \frac{6 \times 16.6 \times 10^2}{8.3 \times 300} = 4 \text{ gram}$$

$$\Rightarrow \quad 2m_{\text{H}} + m_{\text{He}} = 16 \qquad \text{...(i)}$$

Given $\quad m_{\text{H}} + m_{\text{He}} = 10$

Solving $\quad m_{\text{H}} = 6\text{g};\ m_{\text{He}} = 4\text{g}$

$$n_{\text{H}} = 3\,\text{mol.};\ n_{\text{He}} = 1 \text{ mol.}$$

Partial pressure of the two gases

$$\frac{P_{\text{H}}}{P_{\text{He}}} = \frac{n_{\text{H}}}{n_{\text{He}}} = \frac{3}{1}$$

Now pressure = (change in momentum of a molecule during one collision) × (frequency of collisions per unit area)

$$\therefore \quad P = 2mV_{rms}\cdot f$$

$$\therefore \quad \frac{P_{\text{H}}}{P_{\text{He}}} = \frac{(2mV_{rms}f)_{\text{H}}}{(2mV_{rms}f)_{\text{He}}}$$

$$\Rightarrow \quad \frac{3}{1} = \frac{M_{\text{H}}}{M_{\text{He}}}\sqrt{\frac{M_{\text{He}}}{M_{\text{H}}}}\cdot\frac{f_1}{f_{\text{He}}}$$

$$\Rightarrow \quad \frac{f_1}{f_2} = \frac{3}{1}\sqrt{\frac{M_{\text{He}}}{M_{\text{H}}}} = 3\sqrt{2}$$

30. (a) The gas applies force on the piston, normal to its surface. When we take projection of this force in vertical direction it will be simply PA.

$$\therefore \quad PA = P_0A + Mg$$

$$\therefore \quad P = P_0 + \frac{Mg}{A} \quad \text{...(i)}$$

(b) Pressure of the gas is always P as given by (i)

Change in volume $\Delta V = Ax$

$$PV = nRT$$

$$\therefore \quad P\Delta V = nR\Delta T \quad [\because P = \text{a constant}]$$

$$\therefore \quad \Delta T = \frac{P\Delta V}{nR} = \frac{\left(P_0 + \frac{Mg}{A}\right)A \cdot x}{nR}$$

$$= \frac{(P_0A + Mg)x}{nR} = \frac{(P_0A + Mg)x}{R}$$

31. When plate is moving to right:

Molecules bounce back with speed $V - 2u$

$\therefore$ Change in momentum of one molecule

$$\Delta P_1 = m(V - 2u) - (-mV) = 2mV - 2mu$$

Number of molecule hitting per unit time $n_1 = A(V - u)n$

$$\therefore \quad F_1 = n_1\Delta P_1 = An(V - u)2m(V - u) = 2mnA(V - u)^2$$

When plate is moving to left:

Molecule bounce back with speed $V + 2u$

$\therefore$ Change in momentum of one molecule

$$\Delta P_2 = m(V + 2u) - (-mV) = 2mV + 2mu$$

And number of molecule hitting per unit time

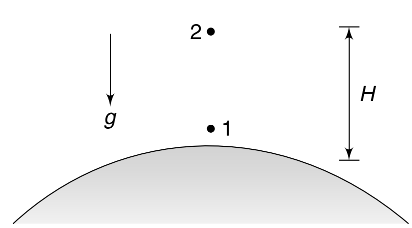
$$n_2 = A(V + u)\cdot n$$

$$\therefore \quad F_2 = n_2\Delta P_2 = An(V + u)2m(V + u) = 2mAn(V + u)^2$$

$$\therefore \quad F_2 - F_1 = 2mnA[(V + u)^2 - (V - u)^2]$$

$$= 8mnAV\cdot u$$

32. (a) For an ideal gas $PV = nRT$



$$PV = \frac{m}{M_0}RT$$

$$\frac{P}{\rho} = \frac{RT}{M_0} \quad \left[\rho = \frac{m}{V} = \text{density}\right]$$

Pressure difference between a point on surface and a point at a height H is given by

$$P_1 - P_2 = \rho g H \quad \Rightarrow \quad \frac{P_1}{\rho} - \frac{P_2}{\rho} = gH$$

$$\frac{RT_1}{M_0} - \frac{RT_2}{M_0} = gH \quad \therefore \quad T_1 - T_2 = gH\frac{M_0}{R}$$

But $$g = \frac{GM}{a^2}$$

$\therefore$ $$T_1 - T_2 = \frac{GMM_0H}{Ra^2} \qquad ...(i)$$

(b) Escape speed $$V_e = \sqrt{\frac{2GM}{a}}$$

$\therefore$ $$\sqrt{\frac{\gamma RT_1}{M_0}} = \sqrt{\frac{GM}{2a}}$$

$\therefore$ $$T_1 = \frac{GMM_0}{2\gamma Ra}$$

From (i) $$T_2 = \frac{GMM_0}{Ra}\left[\frac{1}{2\gamma} - \frac{H}{a}\right]$$

33. (a) The given integral must be equal to 1. f is fraction of molecules having speed between v and $v + dv$. If all such fractions are summed up it must be 1.

(b) $$v_{av} = \int_0^\infty vf\,dv$$

$$= \frac{4}{\sqrt{\pi}}\left(\frac{m}{2kT}\right)^{3/2}\int_0^\infty v^3\, e^{-\frac{mv^2}{2kT}}\,dv$$

Let $$\frac{m}{2kT} = a$$

$$v_{av} = \frac{4}{\sqrt{\pi}}a^{3/2}\int_0^\infty v^3 e^{-av^2}\,dv$$

$$= \frac{4}{\sqrt{\pi}}a^{3/2}\frac{a^{-2}}{2} = \frac{2}{\sqrt{\pi a}}$$

$$v_{av} = \sqrt{\frac{8kT}{\pi m}}$$

34. P = pressure at height y

$P + dP$ = pressure at height $y + dy$

$$dP = -\rho g\,dy \qquad ...(i) \quad [\rho = \text{density of gas}]$$

From ideal gas equation $$PV = nRT$$

$$\Rightarrow \qquad \frac{P}{\rho} = \frac{RT}{M}$$

$$\therefore \qquad \text{from (i) } dP = -\frac{Mg}{RT} P\,dy$$

$$\int_{P_0}^{P} \frac{dP}{P} = -\frac{Mg}{RT}\int_0^h dy \quad \Rightarrow \quad \ln\frac{P}{P_0} = -\frac{Mgh}{RT}$$

$$\therefore \qquad P = P_0 e^{-\frac{Mgh}{RT}} \qquad \text{...(ii)}$$

(a) Pressure at height $h = \dfrac{\text{wt. of gas above}}{\text{area}}$

Pressure at bottom $= \dfrac{\text{wt. of complete gas}}{\text{area}}$

$$\therefore \qquad \frac{P}{P_0} = \frac{\text{wt. of gas above '}h\text{'}}{\text{Total weight of gas}}$$

$$\therefore \qquad \text{Required fraction} = \frac{P}{P_0} = e^{-\frac{Mgh}{RT}}$$

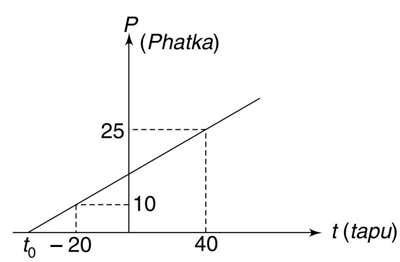
(b) $$\frac{P}{P_0} = \frac{1}{2}$$

$$\Rightarrow \qquad e^{-\frac{Mgh_0}{RT}} = \frac{1}{2}$$

$$\frac{Mgh_0}{RT} = \ln 2 \quad \therefore \quad h_0 = \frac{RT}{Mg}\ln 2$$

35. *Pinch*, *volka*, *tapu* and *phatka* are unit of quantity of gas, volume, temperature and pressure.

Volume is constant. Hence P change linearly with temperature 't'. Graph of P (in *phatka*) vs temperature t (in *tapu*) is as shown.



The equation of the line is

$$P = \frac{1}{4}t + 15$$

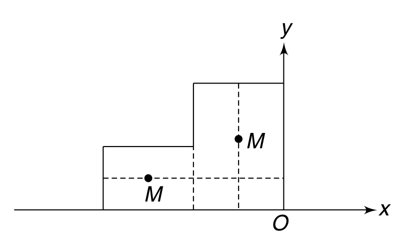
At absolute zero temperature, $P = 0$

$$\therefore \qquad t_0 = -60 \textit{ tapu}$$

36. Pressure in N_2 chamber = 4 times the pressure in O_2 chamber

$\therefore$ Number of moles in N_2 chamber = 4 times the number of moles in O_2 chamber

$$n_{N_2} = 4n_{O_2} = 4n \text{ (say)}$$



Mass of gas in N_2 chamber

$$M_{N_2} = 28 \times 4n = 112\,n$$

Mass of gas in O_2 chamber, $M_{O_2} = 32\,n$

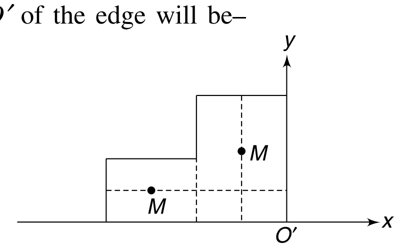
Let's find the x co-ordinate of the COM of the gaseous system with edge O as origin

$$x_{cm} = \frac{M_{N_2}(-4R) + M_{O_2}(-R)}{M_{N_2} + M_{O_2}} = -\frac{112n \times 4R + 32n \times R}{112n + 32n}$$

$$= -\frac{480R}{144} = -3.33R$$

After mixing of gases, the x co-ordinate of COM with respect to new position O' of the edge will be–

$$x'_{cm} = \frac{M(-4R) + M(-R)}{2M} = -2.5\ R$$



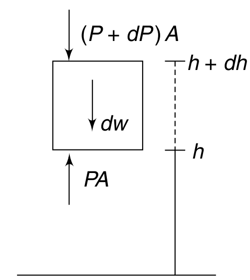
But x co-ordinate of COM will not get displaced in absence of any external force in x direction.

For this, distance travelled by edge O must be

$$(3.33 - 2.5)R = 0.83\ R \text{ towards left.}$$

37. (a) For an ideal gas density $\rho = \dfrac{MP}{RT}$

Consider a cylindrical element of atmospheric air at height h. For its equilibrium



$$PA - (P + dP)A = dw$$

$$-dP \cdot A = \rho \cdot A\,dhg$$

$$-dP = -\rho g dh$$

$$\frac{dP}{P} = -\frac{Mg}{RT}dh \qquad \text{...(A)}$$

$$\int_{P_0}^{P}\frac{dP}{P} = -\frac{Mg}{RT}\int_0^h dh$$

$$\Rightarrow \qquad \ln\frac{P}{P_0} = -\frac{Mgh}{RT} \Rightarrow P = P_0 e^{-\frac{Mgh}{RT}}$$

(b) The temperature changes with height as

$$T = -\left(\frac{T_0}{5h_0}\right)h + T_0$$

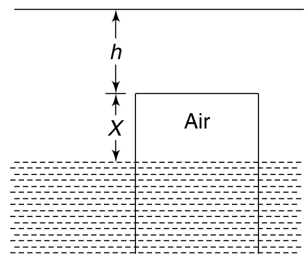
$$\therefore \text{ from (A)} \qquad \frac{dP}{P} = -\frac{Mg}{R}\frac{dh}{T_0 - \left(\frac{T_0}{5h_0}\right)h}$$

$$\int_{P_0}^{P}\frac{dP}{P} = -\frac{Mg}{RT_0}\int_0^{h_0}\frac{dh}{\left(1 - \frac{h}{5h_0}\right)}$$

$$\ell n \frac{P}{P_0} = \frac{-\dfrac{Mg}{RT_0}}{-\dfrac{1}{5h_0}} \left[\ell n \left[1 - \frac{h}{5h_0} \right] \right]_0^{h_0}$$

$$\ln \frac{P}{P_0} = -\frac{5Mgh_0}{RT_0} \ln \frac{5}{4} \quad \Rightarrow \quad P = P_0 \left(\frac{5}{4}\right)^{-\frac{5Mgh_0}{RT_0}}$$

38. Volume of air in the container before it is placed inside water $V_1 = A \cdot L$



When inside water, let the length of air column be x.

Volume of air $V_2 = A \cdot x$

Pressure of air $= P_2 = P_0 + \rho g(x + h)$

For isothermal condition $P_2 V_2 = P_1 V_1$

$$P_2 \cdot A \cdot x = P_0 AL$$

$$[P_0 + \rho g(x + h)]x = P_0 L \quad \Rightarrow \quad [10^5 + 10^4(x + 5)]x = 10^5 \times 0.775$$

$$10x + x^2 + 5x = 7.75 \quad \Rightarrow \quad x^2 + 15x - 7.75 = 0$$

$$\Rightarrow \qquad x = 0.5 \text{ m}$$

Force required = Buoyancy on 0.5 m length of air column $= Ax\rho \cdot g$

$$= 0.05 \times 0.5 \times 10^3 \times 10 = 250\ N$$

39. Initial volume of air $= A_1 h_1 = 4 \times 55$ cm^3

Final volume of air $= V_2$

Initial air pressure $P_1 = P_0 = 10^5$ Nm^{-2}

Initial number of moles of air $= n$

Final no. of moles $= 0.75\ n$

Final air pressure

$$P_2 = 10^5 + \rho g h_2 = 10^5 + 10^3 \times 10 \times 1 = 1.1 \times 10^5 \text{ Nm}^{-2}$$

Now $$\frac{P_2 V_2}{n_2} = \frac{P_1 V_1}{n_1}$$

$$\frac{1.1 \times 10^5 \times V_2}{0.75 n} = \frac{10^5 \times 4 \times 55}{n}$$

$$\therefore \qquad V_2 = 150 \text{ cm}^3$$

Volume of air in narrow part = 100 cm^3,

$\therefore$ air in wider part = 50 cm^3

$\therefore$ length of air column in wider part $= \dfrac{50 \text{ cm}^3}{4 \text{ cm}^2} = 12.5$ cm

$$\therefore \qquad x = 55 - 12.5 = 42.5 \text{ cm}$$

40. In case of an ideal gas we assume no inter-molecular force. When gas undergoes free expansion, the kinetic energy of molecules do not change, and hence the temperature does not change.

In real gases molecules do exert force on each other. These forces increases the potential energy during expansion [in fact, two attracting particles will have more potential energy the farther apart they are]. Increase in potential energy results in decrease in kinetic energy. This results in fall in temperature.

41. Repelling particles will have lesser potential energy the farther they are. A decrease in potential energy means an increase in kinetic energy. This results in rise in temperature.

42. Let initial extension in the spring be x_0

$$kx_0 = mg \qquad \text{...(i)}$$

Let pressure of the gas at temperature T_0 be P_0

For equilibrium of the piston $P_0A + k(x_0 - h_0) = mg$

Using (i) we get $$P_0A = kh_0 \qquad \text{...(ii)}$$

After the temperature is raised to $4T_0$ let the pressure be P and height be h.

A similar working, as above, can show that

$$PA = kh \qquad \text{...(iii)}$$

(iii) ÷ (ii) $$\frac{P}{P_0} = \frac{h}{h_0} \qquad \text{...(iv)}$$

But $$\frac{PV}{T} = \text{constant}$$

$\therefore$ $$\frac{P(Ah_0)}{T_0} = \frac{PAh}{4T_0} \Rightarrow \frac{P}{P_0} = \frac{4h_0}{h} \qquad \text{...(iv)}$$

From (iv) and (v) $$\frac{h}{h_0} = \frac{4h_0}{h}$$

$\Rightarrow$ $$h = 2h_0$$

43. (a) The velocity component of the molecule parallel to the wall do not change. Velocity component perpendicular to the wall is $v\cos\theta$ and after the collision it becomes $v\cos\theta - 2u$.

[since, speed of approach = speed of separation in an elastic collision]

Loss in kinetic energy of the molecule is

$$\frac{1}{2}m(v\cos\theta)^2 - \frac{1}{2}m(v\cos\theta - 2u)^2$$

$$= \frac{1}{2}m(v\cos\theta)^2 - \frac{1}{2}m(v\cos\theta)^2 - \frac{1}{2}m(2u)^2 + 2mvu\cos\theta \simeq 2mvu\cos\theta.$$

(b) The product $Fu = PAu$ is the rate at which work is done on the piston or the power developed by the expanding gas. This is the rate at which the molecules will lose kinetic energy as they are not receiving energy from any other source.

(c) Molecules will not collide with the piston in this case. There is no loss in kinetic energy and no change in temperature.

44. (a) Let the equation of parabola be

$$\frac{dN}{du} = a(u - u^2)$$

$$\therefore \quad \int_0^1 \left(\frac{dN}{du}\right) du = N_0 \Rightarrow a\left[\frac{u^2}{2} - \frac{u^3}{3}\right]_0^1 = N_0$$

$$\Rightarrow \quad \frac{a}{6} = N_0 \Rightarrow a = 6N_0$$

$$\frac{dN}{du} = 6N_0(u - u^2)$$

$$v_{rms} = \sqrt{\frac{\int u^2 dN}{N_0}}$$

But
$$\int u^2 dN = \int u^2\left(\frac{dN}{du}\right) du = a\int_0^1 (u^3 - u^4)\, du$$

$$= a\left[\frac{1}{4} - \frac{1}{5}\right] = \frac{a}{20} = \frac{6N_0}{20} = \frac{3N_0}{10}$$

$$\therefore \quad v_{rms} = \sqrt{\frac{3}{10}} \text{ unit}$$

Since the speed axis shows speed in 10^2 m/s

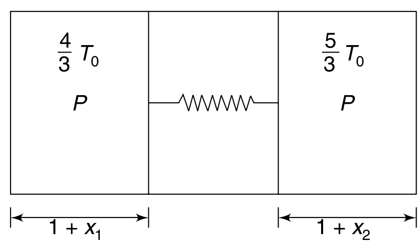
$$\therefore \quad v_{rms} = \sqrt{\frac{3}{10}} \times 100 \text{ m/s}$$

(b)
$$K_T = \frac{1}{2} mv_{rms}^2 = \frac{1}{2} \times (10 \times 10^{-3}) \times 3000 = 15 \text{ J}$$

45. Let the left piston move by $x_1(\rightarrow)$ ad the right piston move by $x_2(\leftarrow)$

Further compression in the spring $x = x_1 + x_2$

Let initial and final pressure of both gases be P_0 and P respectively.



$$\frac{P_0A(1)}{T_0} = \frac{PA(1 + x_1)}{\frac{4}{3}T_0} = \frac{PA(1 + x_2)}{\frac{5}{3}T_0}$$

$$\Rightarrow \quad \frac{P_0}{P} = \frac{3}{4}(1 + x_1) \qquad \text{...(i)}$$

And
$$5(1 + x_1) = 4(1 + x_2) \Rightarrow 5x_1 = 4x_2 - 1 \qquad \text{...(ii)}$$

But
$$x_2 = x - x_1$$

$$5x_1 = 4(x - x_1) - 1 \Rightarrow x_1 = \frac{4x - 1}{9} \qquad \text{...(iii)}$$

Put this in (i) to get
$$\frac{P_0}{P} = \frac{2 + x}{3} \qquad \text{...(iv)}$$

For initial and final equilibrium

$$P_0A = kx_0 \Rightarrow P_0 = \frac{k(1)}{A} = \frac{k}{A} \text{ and } PA = k(1 + x) \Rightarrow P = \frac{k(1 + x)}{A}$$

$$\therefore \quad \frac{P_0}{P} = \frac{1}{1 + x} \qquad \text{...(v)}$$

From (iv) and (v) $\dfrac{2+x}{3} = \dfrac{1}{1+x}$

$$\Rightarrow \quad x^2 + 3x - 1 = 0$$

$$\Rightarrow \quad x = \frac{\sqrt{13} - 3}{2}$$

$\therefore$ Final compression $= x + 1 = \dfrac{\sqrt{13} - 1}{2}$ m

46. (a) Net upward force on the balloon is

$$F_0 = F_B - w_{\text{Rubber}} - w_{\text{Load}} - w_{\text{He}}$$

$$F_0 = V\rho_a g - w_{\text{Rubber}} - w_{\text{Load}} - V\rho_{\text{He}} g$$

$$V = \frac{F_0 + w_{\text{Rubber}} + w_{\text{Load}}}{(\rho_a - \rho_{\text{He}})g} \quad \text{...(i)}$$

$$n = \frac{PV}{RT} = \frac{P_0(F_0 + w_{\text{Rubber}} + w_{\text{Load}})}{RT_0(\rho_a - \rho_{\text{He}})g}$$

(b) Atmospheric pressure at height y

$$P = P_0 e^{-ky}$$

$$\Rightarrow \quad \ln\left(\frac{P}{P_0}\right) = -ky$$

$$\Rightarrow \quad y = \frac{1}{k}\ln\left(\frac{P_0}{P}\right) \quad \text{...(ii)}$$

But $\quad P = \dfrac{nRT}{V}$

For fully inflated balloon $\quad P = \dfrac{nRT_0}{V_0}$

$$\therefore \quad y_0 = \frac{1}{k}\ln\left(\frac{P_0 V_0}{nRT_0}\right) \quad \text{...(iii)}$$

(c) Density of air decreases with height. If temperature remains constant then

$$\frac{P}{\rho} = \text{a constant}$$

$$\Rightarrow \quad \frac{P}{P_0} = \frac{\rho_y}{\rho_a} \Rightarrow \rho_y = \frac{P}{P_0}\rho_a \quad \text{...(iv)}$$

$\therefore$ Buoyancy at height y is

$$F_{By} = \frac{P}{P_0}\rho_a V g = \rho_a V g e^{-ky}$$

Net upward force on the balloon at height y is

$$F_{\text{net}} = F_{\text{By}} - w_{\text{Rubber}} - w_{\text{Load}} - w_{\text{He}}$$

$$= \rho_a V g e^{-ky} - w_{\text{Rubber}} - w_{\text{Load}} - \rho_{\text{He}} V g e^{-ky}$$

[since density of He will also change in a way similar to the air]

$$\therefore \quad F_{\text{net}} = V g e^{-ky}(\rho_a - \rho_{\text{He}}) - w_{\text{Rubber}} - w_{\text{Load}}$$

Balloon will rise to height $y = y_0 = \frac{1}{k}\ln\left(\frac{P_0 V_0}{nRT_0}\right)$

If $F_{\text{net}} > 0$

$$\Rightarrow \quad V_0 g\, e^{-\ln\left(\frac{P_0 V_0}{nRT_0}\right)}(\rho_a - \rho_{\text{He}}) > w_{\text{Rubber}} + w_{\text{Load}}$$

$$\Rightarrow \quad V_0 \text{g}\frac{nRT_0}{P_0 V_0}(\rho_a - \rho_{\text{He}}) > w_{\text{Rubber}} + w_{\text{Load}}$$

$$\Rightarrow \quad \frac{nRT_0 g}{P_0}(\rho_a - \rho_{\text{Hg}}) > w_{\text{Rubber}} + w_{\text{Load}}$$

CHAPTER 4

First Law of Thermodynamics

LEVEL 1

Q. 1: In different experiments an ideal gas is expanded through

(i) Isothermal

(ii) Adiabatic

(iii) Isobaric process

In which of the processes mentioned the internal energy of the gas may decrease?

Q. 2: Which of the words-out of reversible, irreversible, adiabatic, isothermal, isochoric and isobaric-will you choose to describe the following processes–

(i) A bullet stops in a target [system is bullet plus target].

(ii) A gas, enclosed in a metallic cylinder provided with a piston, is slowly expanded [System is gas]. There is friction between piston and cylinder.

(iii) A piece of hot stone (which has coefficient of thermal expansion equal to zero) is dipped into cold water [System is stone].

Q. 3: Calculate the amount of heat required in calorie to change 1 g of ice at $-10°C$ to steam at $120°C$. The entire process is carried out at atmospheric pressure. Specific heat of ice and water are 0.5 cal $g^{-1}°C^{-1}$ and 1.0 cal $g^{-1}°C^{-1}$ respectively. Latent heat of fusion of ice and vaporization of water are 80 cal g^{-1} and 540 cal g^{-1} respectively. Assume steam to be an ideal gas with its molecules having 6 degrees of freedom. Gas constant $R = 2$ cal $mol^{-1} K^{-1}$.

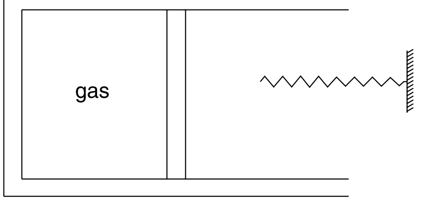
Q. 4: A mixture of 1 mole helium and 1 mole nitrogen is enclosed in a vessel of constant volume at 300 K. Find the quantity of heat absorbed by the mixture if the root mean square speed of its molecules get doubled. Give your answer in terms of universal gas constant R.

Q. 5: The average number of degree of freedom per molecule for a gas is 7. A sample of the gas perform 30 J of work when it expands at constant pressure. Find the heat absorbed by the gas in the process.

Q. 6: An ideal gas is made to undergo a process $T = T_0 e^{\alpha V}$ where T_0 and α are constants. Find the molar specific heat capacity of the gas in the process if its molar specific heat capacity at constant volume is C_v. Express your answer as a function of volume (V).

Q. 7: An ideal diatomic gas undergoes a process in which the pressure is proportional to the volume. Calculate the molar specific heat capacity of the gas for the process.

Q. 8: (i) A horizontal cylinder is fitted with a smooth movable piston. The cylinder contains an ideal gas. The gas is heated slowly so that the piston gradually moves out. After moving out for some distance the piston encounters an ideal spring and compresses it while moving out. Draw $P - V$ diagram for the entire process.



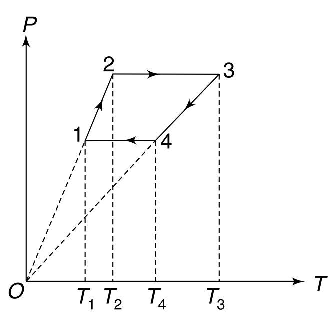
(ii) One mole of an ideal gas is expanded isothermally at temperature T_0 to double its volume from V_0 to $2V_0$. Draw a graph showing the variation of volume (V) of the gas versus the amount of heat (Q) added to it.

Q. 9: A spherical balloon contains air at pressure P_1 and is placed in vacuum. It has an initial diameter of D_1. The balloon is heated until its diameter becomes $D_2 = 2D_1$. It is known that pressure in the balloon is proportional to its diameter. Calculate the work done by the gas in expansion.

Q. 10: (i) An adiabatic cylinder contains an ideal gas. It is fitted with a freely movable insulating piston. In one experiment the piston is pulled out very fast to double the volume of the gas. In another experiment starting from same initial state, the piston is pulled out very slowly to double the volume of the gas. At the end of which experiment the final pressure of the gas will be higher?

(ii) An ideal gas is contained in a cylinder fitted with a movable piston. In an experiment 'A' the gas is allowed to perform a work $W(>0)$ on the surrounding during an isobaric process and thereafter the pressure of the gas is reduced isochorically to half the initial value. At the end of the experiment the temperature of the gas is T_A. In a different experiment 'B' the pressure of the gas is reduced to half in an isochoric process and then the gas performs a work W on the surrounding during an isobaric process. At the end of the experiment the gas temperature is T_B. Which is higher, T_A or T_B?

Q. 11: n moles of an ideal gas is taken through a four step cyclic process as shown in the diagram. Calculate work done by the gas in a cycle in terms of temperatures T_1, T_2, T_3 and T_4.



Q. 12: Two thermally insulated vessels are filled with an ideal gas. The pressure, volume and temperature in the two vessels are P_1, V_1, T_1 and P_2, V_2, T_2 respectively. Now, the two vessels are connected using a short insulating tube.

(a) Find final temperature of the gas.

(b) Express the final pressure of the gas in terms of P_1, V_1, P_2 and V_2 only.

Q. 13: Two tanks are connected by a valve. One tank contains 2 kg of an ideal gas at 77°C and 0.7 atm pressure. The other tank has 8kg of same gas at 27°C and 1.2 atm pressure. The valve is opened and the gases are allowed to mix. The final equilibrium temperature was found to be 42°C.

(a) Find the equilibrium pressure in both tanks.

(b) How much heat was transferred from surrounding to the tanks during the mixing process. Given: C_v for the gas is 0.745 $\text{KJ kg}^{-1}\text{K}^{-1}$.

Q. 14: The ratio of specific heats (C_p and C_v) for an ideal gas is γ. Volume of one mole sample of the gas is varied according to the law $V = \dfrac{a}{T^2}$ where T is temperature and a is a constant. Find the heat absorbed by the gas if its temperature changes by ΔT.

Q. 15: A sample of an ideal gas has diatomic molecules at a temperature at which effective degree of freedom is 5. Under the action of a suitable radiation the molecules split into two atoms. The ratio of the number of dissociated molecules to the total number of molecules is α. Plot the ratio of molar specific heats $\gamma\left(=\dfrac{C_p}{C_v}\right)$ as a function of α.

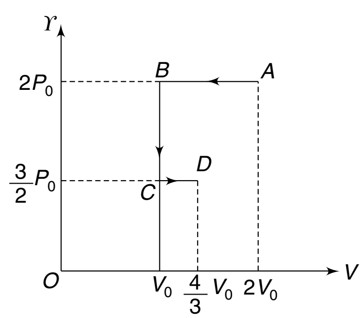
Q. 16: An ideal gas undergoes a series of processes represented by $A \to B \to C \to D$ on the P-V diagram. Answer the following questions.

(a) Is the internal energy of the gas at B and D equal?

(b) Find work done by the gas in the process $A \to B \to C \to D$.

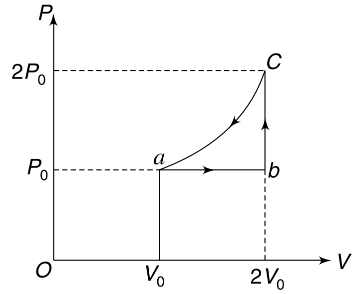
(c) Is it right to say that point B and D lie on an isotherm?

(d) Find the ratio of internal energy of the gas in state A to that in state D.



Q. 17: 30 people gather in a 10 m × 5 m × 3 m room for a confidential meeting. The room is completely sealed off and insulated. Calculate the rise in temperature of the room in half an hour. Assume that average energy thrown off by the body of a person is 2500 kcal/day, density of air is 1.2 kg/m^3 and specific heat specify capacity of air at constant volume is 0.24 $\text{kcal kg}^{-1}\,°\text{C}^{-1}$. Neglect volume occupied by human bodies.

Q. 18: One mole of an ideal monoatomic gas is taken through a cycle a-b-c-a as shown in figure. Find the difference in maximum and minimum value of internal energy of the gas during the cycle.

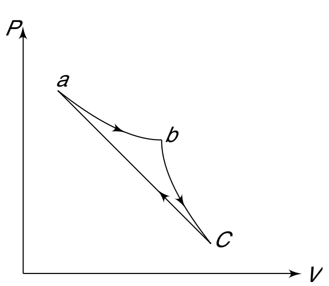


Q. 19: An ideal mono atomic gas A is supplied heat so as to expand without changing its temperature. In another process, starting with the same state, it is supplied heat at constant pressure. In both the cases a graph of work done by the gas (W) is plotted versus heat added (Q) to the gas. The ratio of slope of the graphs obtained in first and second process is η_1. The same ratio obtained for an ideal diatomic gas in η_2. Find the ratio $\dfrac{\eta_1}{\eta_2}$.

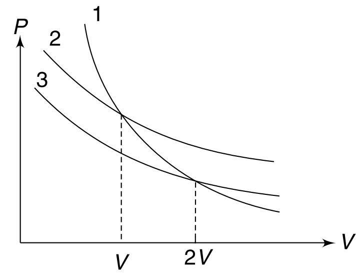
Q. 20: For an ideal gas the ratio of specific heats is $\dfrac{C_p}{C_v} = \gamma$. The gas undergoes a polytropic process PV^n = a constant. Find the values of n for which the temperature of the gas increases when it rejects heat to the surrounding.

Q. 21: For an ideal gas the slope of V-T graph during an adiabatic process is $\dfrac{dV}{dT} = -m$ at a point where volume and temperature are V_0 and T_0. Find the value of C_p of the gas. It is given that m is a positive number.

Q. 22: A gas undergoes a cyclic process a-b-c-a which is as shown in the PV diagram. The process a-b is isothermal, b-c is adiabatic and c-a is a straight line on P-V diagram. Work done in process ab and bc is 5 J and 4 J respectively. Calculate the efficiency of the cycle, if the area enclosed by the diagram abca in the figure is 3 J.

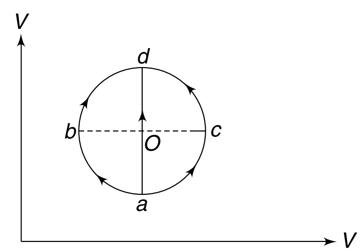


Q. 23: In the shown figure curve 1 represents an adiabat for n moles of an ideal mono atomic gas. Curve 2 and 3 are two isotherms for the same sample of the gas. Calculate the ratio of work done by the gas in doubling its volume from V to 2V along the isotherms 2 and 3.

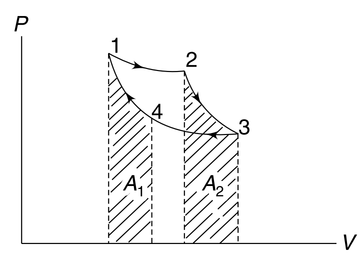


Q. 24: A mass less piston divides a thermally insulated cylinder into two parts having volumes V = 2.5 litre and 3 V = 7.5 litre. 0.1 mole of an ideal gas is confined into the part with volume V at a pressure of $P = 10^5$ N/m^2. The other part of the cylinder is empty. The piston is now released and the gas expands to occupy the entire volume of the cylinder. Now the piston is pressed back to its initial position. Find final temperature of the gas. $\left[\text{Take } R = \dfrac{25}{3}\ \text{J mol}^{-1}\ \text{K}^{-1} \text{ and } \gamma = 1.5 \text{ for the gas}\right]$

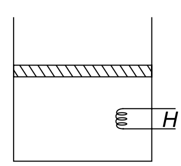
Q. 25: An ideal gas is taken from its initial state a to its find state d in three different quasi static processes marked as $a - b - d$, $a - o - d$ and $a - c - d$. Rank the net heat absorbed by the gas in the three processes. The diagram shown is a circle with centre at o.



Q. 26: The figure shows a Carnot cycle for an ideal gas on a P-V diagram. Which of the areas A_1 or A_2 is larger?



Q. 27: Air is contained in a vertical piston – cylinder assembly fitted with an electric heater. The piston has a mass of 50 kg and cross sectional area of 0.1 m^2. Mass of the air inside the cylinder is 0.3 kg. The heater is switched on and the volume of the air slowly increases by 0.045 m^3. It was found that the internal energy of the air increased by 32.2 kJ/kg

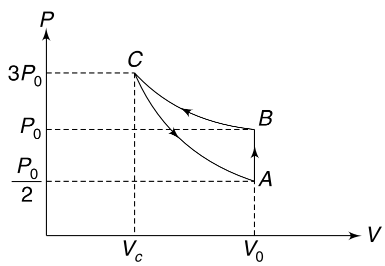


and that of the piston increased by 0.06 kJ/kg. Assume that the container walls and outer surface of the piston are well insulated and there is no friction. The atmospheric pressure is 100 kPa. Determine the heat transferred by the heater to the system consisting of air and the piston.

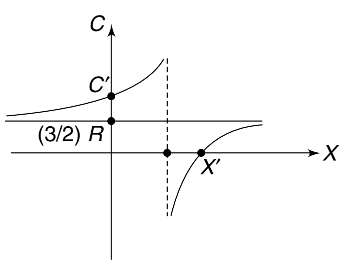
Q. 28: One mole of an ideal mono atomic gas is at a temperature T_0 = 1000 K and its pressure is P_0. The gas is adiabatically cooled so that its pressure becomes $\frac{2}{3}P_0$. Thereafter, the gas is cooled at constant volume to reduce its pressure to $\frac{P_0}{3}$. Calculate the total heat absorbed by the gas during the process.

Take $R = \frac{25}{3}\ \text{J mol}^{-1}\text{K}^{-1}$ and $\left(\frac{2}{3}\right)^{2/5} = 0.85$

Q. 29: One mole of an ideal gas is carried through a thermodynamics cycle as shown in the figure. The cycle consists of an isochoric, an isothermal and an adiabatic process. Find the adiabatic exponent of the gas.

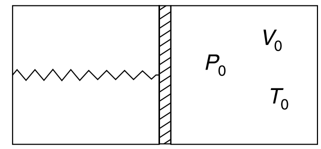


Q. 30: One mole of an ideal monatomic gas is taken along the process in which PV^x = constant. The graph shown represents the variation of molar heat capacity of such a gas with respect to 'x'. Find the values of C' and x' indicated in the diagram.



LEVEL 2

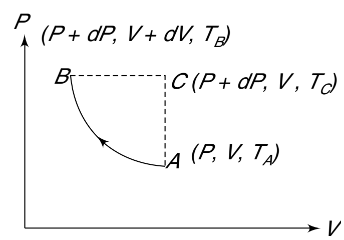
Q. 31: A container has a tight fitting movable piston which can slide without friction. The compartment containing spring has vacuum and to the left of the piston there is diatomic gas. If vacuum is created in the right compartment also the piston touches the right wall and the spring is relaxed. Find the heat capacity of the system neglecting the heat capacities of the material of spring, container and the piston. Express your answer in terms of P_0, V_0 and T_0.



Q. 32: One mole of a gas is in state $A[P, V, T_A]$. A small adiabatic process causes the state of the gas to change to $B[P + dP, V + dV, T_B]$. The changes dV & dP are infinitesimally small and dV is negative. An alternative process takes the gas from state A to B via $A \to C \to B$. $A \to C$ is isochoric and $C \to B$ is isobaric path. State at C is $[P + dP, V, T_C]$.

(a) Rank the temperatures T_A, T_B and T_C from highest to lowest.

(b) Find γ of the gas in terms of T_A, T_B and T_C.



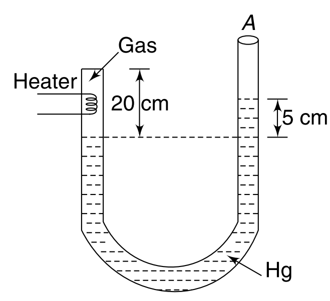
Q. 33: A metal cylinder of density d, cross sectional area A and height h is standing on a horizontal surface. Coefficient of linear expansion of the metal is α°C^{-1} and its specific heat capacity is S. Calculate the rise in temperature of the cylinder if a heat ΔQ is supplied to it. Assume no atmosphere.

Q. 34: n moles of an ideal mono atomic gas is initially at pressure $32P_0$ and volume V_0. Its volume is doubled by an isobaric process. After this the gas is adiabatically expanded so as to make its volume $16V_0$. Now the gas is isobarically expanded. Finally, the gas is made to return to its initial state by an isothermal process.

(a) Represent the process on a P–V diagram.

(b) Calculate work done by the gas in one cycle.

Q. 35: One end of an insulating U tube is sealed using insulating material. A mono atomic gas at temperature 300 K occupies 20 cm length of the tube as shown. The level of mercury on two sides of the tube differ by 5 cm. The other end of the tube is open to atmosphere.

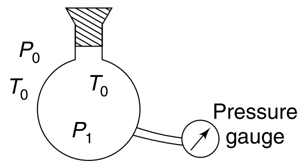


Area of cross section of the tube is uniform and is equal to 0.01 m^2. The gas in the tube is heated by an electric heater so as to raise its temperature to 562.5 K. Assume that no heat is conducted to mercury by the gas.

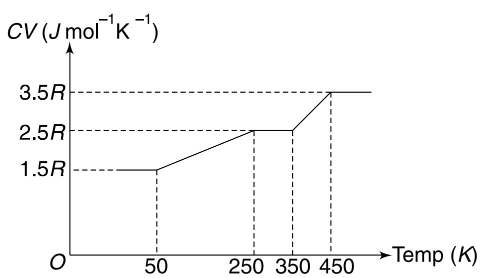
(a) Find the final length of the gas column.

(b) Find the amount of heat supplied by the heater to the gas.

Q. 36: Air is filled inside a jar which has a pressure gauge connected to it. The temperature of the air inside the jar is same as outside temperature ($=T_0$) but pressure (P_1) is slightly larger than the atmospheric pressure (P_0). The stopcock is quickly opened and quickly closed, so that the pressure inside the jar becomes equal to the atmospheric pressure P_0. The jar is now allowed to slowly warm up to its original temperature T_0. At this time the pressure of the air inside is P_2 ($P_0 < P_2 < P_1$). Assume air to be an ideal gas. Calculate the ratio of specific heats ($=\gamma$) for the air, in terms of P_0, P_1 and P_2.



Q. 37: The molar specific heat capacity at constant volume (C_V) for an ideal gas changes with temperature as shown in the graph. Find the amount of heat supplied at constant pressure in raising the temperature of one mole of the gas from 200 K to 400 K.

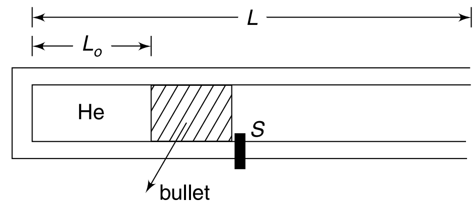


Q. 38: A sample of oxygen is heated in a process for which the molar specific heat capacity is $2R$. During the process the temperature becomes $(32)^{1/3}$ times of the original temperature. How does the volume of the gas change?

Q. 39: A gas-gun has a cylindrical bore made in an insulating material. Length of the bore is L. A small bullet having mass m just fits inside the bore and can move frictionlessly inside it. Initially n moles of helium gas is filled in the bore to a length L_0. The bullet does not allow the gas to leak and the bullet itself is kept at rest by a stopper S. The gas is at temperature T_0. The gun fires if the stopper S is removed suddenly. Neglect atmospheric pressure in your calculations [Think that the gun is in space].

(a) Calculate the speed with which the bullet is ejected from the gun.

(b) Find the maximum possible speed that can be imparted to the bullet by using n moles of helium at temperature T_0.



Q. 40: During a process carried on an ideal gas it was found that $\eta(<1)$ times the heat supplied to the gas is equal to increase in internal energy of the gas. Write the process equation in terms of pressure (P) and volume (V) of the gas. It is given that ratio of specific heats for the gas is $\frac{C_P}{C_V} = \gamma$.

Q. 41: An ideal gas is taken through a thermodynamic cycle $ABCDA$. In state A pressure and volume are P_0 and V_0 respectively. During the process $A \rightarrow B$, work done by the gas is zero and its temperature increases two fold. During the process $B \rightarrow C$, internal energy of the gas remains constant but work done by it is $W_{BC} = -P_0V_0 \ln 2$. In the process $C \rightarrow D$, the temperature decreases by 50% while the volume does not change. In the process $D \rightarrow A$ the temperature of the gas does not change.

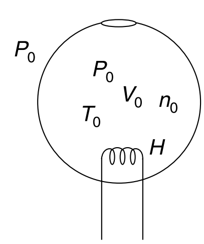
(a) Draw pressure – volume (P–V) and pressure – density ($P-\rho$) graph for the cyclic process.

(b) Calculate work done on the gas during the cycle.

Q. 42 (i) A conducting piston divides a closed thermally insulated cylinder into two equal parts. The piston can slide inside the cylinder without friction. The two parts of the cylinder contain equal number of moles of an ideal gas at temperature T_0. An external agent slowly moves the piston so as to increase the volume of one part and decrease that of the other. Write the gas temperature as a function of ratio (β) of the volumes of the larger and the smaller parts. The adiabatic exponent of the gas is γ.

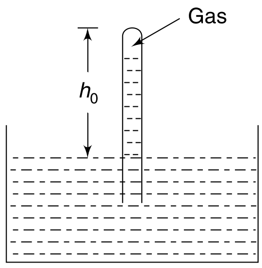
(ii) In a closed container of volume V there is a mixture of oxygen and helium gases. The total mass of gas in the container is m gram. When Q amount of heat is added to the gas mixture its temperature rises by ΔT. Calculate change in pressure of the gas.

Q. 43: A spherical container made of non conducting wall has a small orifice in it. Initially air is filled in it at atmospheric pressure (P_0) and atmospheric temperature (T_0). Using a small heater, heat is slowly supplied to the air inside the container at a constant rate of H J/s. Assuming air to be an



ideal diatomic gas find its temperature as a function of time inside the container.

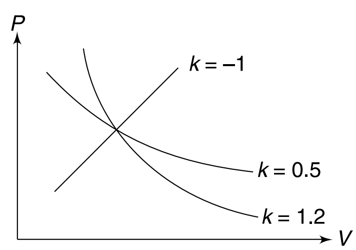
Q. 44: A glass tube is inverted and dipped in mercury as shown. One mole of an ideal monoatomic gas is trapped in the tube and the tube is held so that length of the tube above the mercury level is always h_0 meter. The atmospheric pressure is equal to h_0 meter of mercury. The mercury vapour pressure, heat capacity of mercury, tube and the container are negligible. How much heat must be supplied to the gas inside the tube so as to increase its temperature by ΔT?



Q. 45: Figure shows P versus V graph for various processes performed by an ideal gas. All the processes are polytropic following the process equation PV^k = constant.

(i) Find the value of k for which the molar specific heat of the gas for the process is $\dfrac{C_P + C_V}{2}$. Does any of the graph given in figure represent this process?

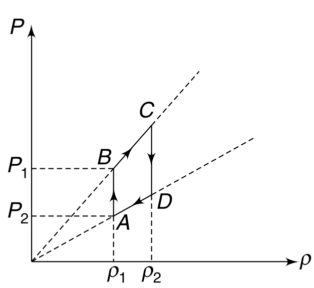
(ii) Find the value of k for which the molar specific heat of the gas is $C_V + C_P$. Assume that gas is mono atomic. Draw approximately the P versus V graph for this process in the graph given above.



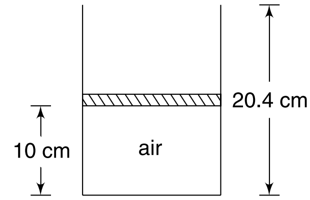
Q. 46: 'n' moles of an ideal gas having molar mass M is made to undergo a cyclic process $ABCDA$. The cycle has been represented on a pressure (P) density (ρ) diagram.

(a) Draw the corresponding $P - V$ diagram

(b) Calculate the work done by the gas in the cycle.



Q. 47: In the arrangement shown in figure, the piston can smoothly move inside the cylinder. The mass of the piston is m = 100 g and its cross sectional area is A = 10 cm^2. The length of air column at temperature of T = 27°C is 10 cm. Overall length of the cylinder is 20.4 cm. The container is turned upside down and the length of the air column in equilibrium was found to be l at 27°C.

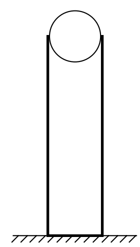


Take $R = \dfrac{25}{3}\ \text{J mol}^{-1}\text{K}^{-1}$ and assume air to be diatomic gas. g = 10 m/s^2, atmospheric pressure is 1.01×10^5 Nm^{-2}

(a) Find l

(b) If the air in the container is supplied heat in upside down position, the piston slowly begins to move down, and ultimately it gets ejected out of the cylinder. Calculate the amount of heat that the air must absorb for the piston to come out.

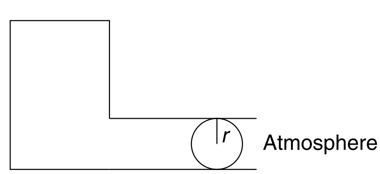
Q. 48: There is a long vertical tube of radius r containing air at atmospheric pressure. A steel ball is held at the mouth of the tube and dropped. The ball has radius r and it just fits inside the tube. The tube wall is perfectly smooth and no air can leak from the tube as the ball falls inside it. The ball falls through half the length of the tube before coming to rest. Assume that wall of the tube is perfectly conducting and temperature of the air inside the tube remains constant. Density of steel is d and atmospheric pressure is P_0 and take $L >> r$. Take air to be an ideal gas.



(a) Find the radius (r) of the tube.

(b) At what depth from the top of the tube the ball will be in equilibrium?

Q. 49: A ball of radius r fits tightly inside a tube attached to a container. There is no friction between the tube wall and the ball. Volume of air inside the container is V_0 when the ball is in equilibrium. Density of the material of the ball is d and atmospheric pressure is P_0. If the ball is displaced a little from its equilibrium position and released, find time period of its oscillation. Assume that temperature in the container remains constant and that air is an ideal gas.

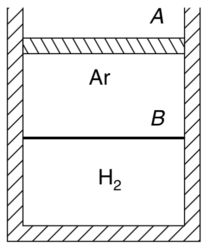


Q. 50: An ideal mono atomic gas is at temperature T_0. The pressure and volume are quasi-statically doubled such that the process traces a straight line on the PV diagram.

(a) Calculate the heat absorbed by the gas in the process if number of moles of the gas in the sample is n.

(b) Calculate the average molar specific heat capacity of the gas for the process.

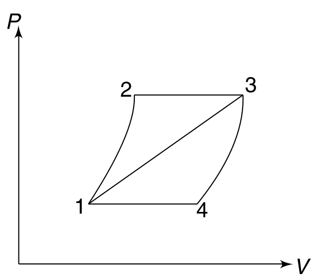
Q. 51: A cylinder contains equal volumes of Ar and H_2, separated by a freely movable piston B. Piston A can also move without friction. Volume of each gas in equilibrium is V_0. All walls of the container including piston A are non conducting.



(i) Piston A is pushed down slowly till the volume occupied by argon becomes $\frac{V_0}{4}$. Find the volume of H_2. Assume that piston B is also non conducting.

(ii) Now assume that piston B is conducting and assume that each gas has n moles. The external agent performs work W_0 in slowly pushing down the piston A. Find rise in temperature of each gas.

Q. 52: An ideal gas is taken through cycle 1231 (see figure) and the efficiency of the cycle was found to be 25%. When the same gas goes through the cycle 1341 the efficiency is 10%. Find the efficiency of the cycle 12341.



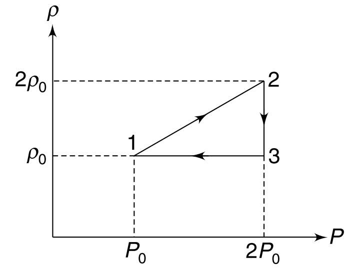
Q. 53: A heat engine is based on a gaseous cycle comprising of four processes viz. isothermal expansion at temperature (T_1), isochoric cooling to temperature T_2, isothermal compression (at T_2) and isochoric heating back to temperature T_1. The engine has been designed so as to completely use the heat rejected during isochoric cooling, in the isochoric heating process. Calculate the efficiency of this reversible cycle. Show the process on a PV graph.

Q.54: An ideal gas with a known γ, completes a cycle consisting of two isotherms and two isobars. The isothermal processes are executed at temperatures T and T' ($< T$) and isobaric processes are completed at pressures P_0 and eP_0 [e = base of natural logarithm]. Find the efficiency of the cycle.

Q. 55: One mole of a mono atomic gas of molar mass M undergoes a cyclic process as shown in the figure. Here ρ is density and P is pressure of the gas.

(a) Calculate the heat rejected by the gas in one complete cycle.

(b) Find the efficiency of the cycle.



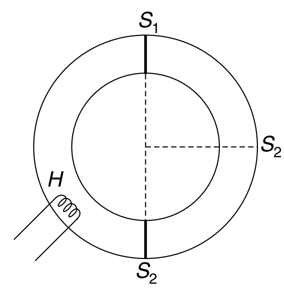
Q. 56: Helum gas is used as working substance in an engine working on a thermodynamic cycle $A - B - C - D - A$. Process AB is isobaric, BC is adiabatic compression. During process CD, pressure is increased keeping the volume constant and DA is an isothermal process. The gas has maximum volume at A and the ratio of maximum to minimum volume during the entire cycle is $8\sqrt{2}$. Also, the ratio of maximum to minimum absolute temperature is 4.

(a) Represent the cycle on a $P - V$ diagram.

(b) Calculate efficiency of the cycle in percentage if it is used as an engine.

[Take $ln\ 2 = 0.693$]

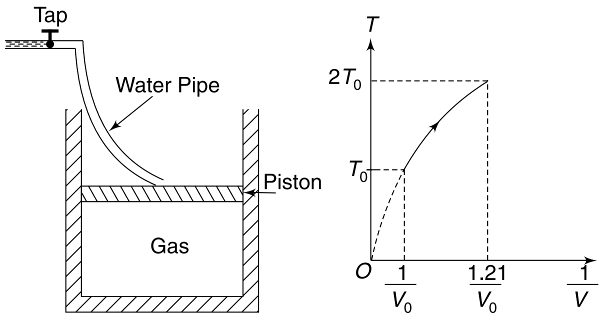
Q. 57: A ring shaped tube has uniform cross sectional area and its entire volume is $2V_0$. The tube is well insulated from the surrounding. Inside the tube there is an adiabatic fixed wall S_1 and another movable adiabatic partition S_2. Initially, the movable partition is diametrically opposite to S_1 and the two halves of the tube have equal amount of an ideal gas ($\gamma = 1.5$) at same pressure P_0. Now, a heater H is switched on which supplies heat slowly to one of the chambers. Heater is kept on till the partition S_2 moves through the quarter of the circle. At this position the heater is switched off and the partition S_2 remains in equilibrium. Neglect any friction as well as heat loss to the surrounding through the walls of the tube. Find the heat supplied by the heater to the gas.



Q. 58: A cylindrical container has insulating wall and an insulating piston which can freely move up and down without any friction. It contains a mixture of ideal gases. Originally the gas is at atmospheric pressure P_0 and temperature (T_0). A tap positioned above the container is opened and it supplies water at a constant rate of $\frac{dm}{dt} = 0.25$ kg/s. The water collects above the piston in the container and the gas compresses. The tap is kept open till the temperature of the gas is doubled. During the process the T vs $\frac{1}{V}$ graph for the gas was recorded and found to be a parabola with its vertex at origin as shown in the graph. Area of piston $A = 1.515 \times 10^{-3}$ m^2 and atmospheric pressure = 10^5 N/m^2

(a) Find the ratio of V_{rms} and speed of sound in the gaseous mixture.

(b) For how much time the tap was kept open?



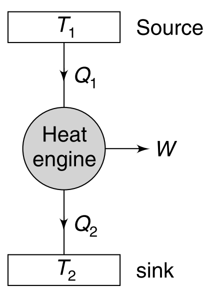
Q. 59: On a hot summer day the temperature inside is house is T_0 and the outside temperature is $T_0 + \Delta T$. How does the energy consumed by the air conditioner depend on T_0 and ΔT? Assume that the air conditioner operates ideally at its maximum coefficient of performance.

Q. 60: A room air conditioner is a Carnot cycle based heat engine run is reverse. An amount of heat Q_2 is absorbed from the room at a temperature T_2 into coils having a working gas (these gases are not good for environment!). The gas is compressed adiabatically to the outside temperature T_1. Then the gas is compressed isothermally in the unit outside the room, giving off an amount of heat Q_1. The gas expands adiabatically back to the temperature T_2 and the cycle is repeated. The electric motor electric consumes power P.

(i) Find the maximum rate at which heat can be removed from the room.

(i) Heat flows into the room at a constant rate of $k\Delta T$ where k is a constant and ΔT is temperature difference between the outside and inside of the room. Find the smallest possible room temperature in terms of T_1, k and P.

Q. 61: A Carnot cycle based ideal heat engine operates between two tanks each having same mass m of water. The source tank has an initial temperature of T_1 = 361 K and the sink tank has an initial temperature of T_2 = 289 K. Assume that the two tanks are isolated from the surrounding and exchange heat with the engine only. Specific heat of water is s.



(a) Find the final common temperature of the two tanks.

(b) Find the total work that the engine will be able to deliver by the time the two tanks reach common temperature.

Q. 62: A cylindrical container of height $3L$ and cross sectional area A is fitted with a smooth movable piston of negligible weight. It contains an ideal diatomic gas. Under normal atmospheric pressure P_0 the piston stays in equilibrium at a height L above the base of the container. The gas chamber is provided with a heater and a copper coil through which a cold liquid can be circulated to extract heat from the gas. Volume occupied by the heater and the liquid coil is negligible. Following set of operations are performed to take the gas through a cyclic process.

(1) Heater is switched on. At the same time a tap above the cylinder is opened. Water fills slowly in the container above the piston and it is observed that the piston does not move. Water is allowed to fill the container so that the height of water column becomes L. Now the tap is closed.

(2) The heater is kept on and the piston slowly moves up. Heater is switched off at the time water is at brink of overflowing.

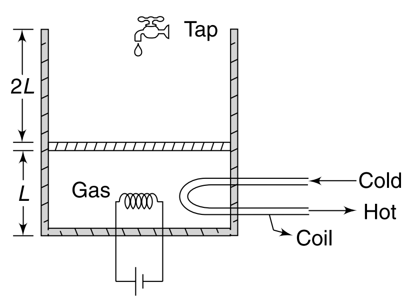
(3) Now the cold liquid is allowed to pass through the coil. The liquid extracts heat from the gas. Water is removed from the container so as to keep the position of piston fixed. Entire water is removed and the gas is brought back to atmospheric pressure.

(4) The circulation of cold liquid is continued and the piston slowly falls down to the original height L above the base of the container. Circulation of liquid is stopped.

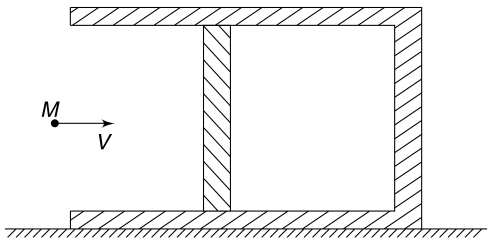
Assume that the container is made of adiabatic wall and density of water is ρ. Force on piston due to impact of falling water may be neglected.

(a) Draw the entire cycle on a P–V graph.

(b) Find the amount of heat supplied by the heater and the amount of heat extracted by the cold liquid from the gas during the complete cycle.



Q. 63: An adiabatic cylindrical chamber with a frictionless movable piston has been placed on a smooth horizontal surface as shown. One mole an ideal monotonic gas is enclosed inside the chamber. Mass of the piston is M and mass of the remaining chamber including the gas is 4 M. The gas is at atmospheric pressure and temperature. A particle of mass M moving horizontally with speed v, strikes the piston elastically. Find the change in temperature of the gas when the compression is maximum.



Q. 64: (a) A polytropic process for an ideal gas is represented by PV^x = constant, where $x \neq 1$. Show that molar specific heat capacity for such a process is given by $C = C_v + \dfrac{R}{1-x}$.

(b) An amount Q of heat is added to a mono atomic ideal gas in a process in which the gas performs a work $\dfrac{Q}{2}$ on its surrounding. Show that the process is polytropic and find the molar heat capacity of the gas in the process.

LEVEL 3

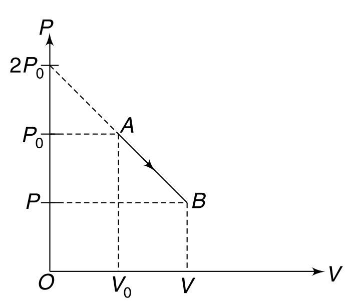
Q. 65: An adiabatic cylinder of cross section A is fitted with a mass less conducting piston of thickness d and thermal conductivity K. Initially, a monatomic gas at temperature T_0 and pressure P_0 occupies a volume V_0 in the cylinder. The atmospheric pressure is P_0 and the atmospheric temperature is $T_1 (> T_0)$. Find

(a) the temperature of the gas as a function of time

(b) the height raised by the piston as a function of time

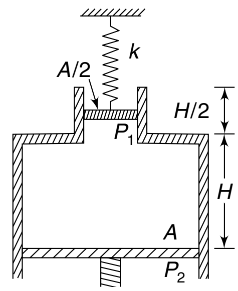
Neglect friction and heat capacities of the container and the piston.

Q. 66: One mole of an ideal gas is expanded from the state $A(P_0, V_0)$ to final state B having volume V. The process follows a path represented by a straight line on the P–V diagram (see figure). Up to what volume (V) the gas shall be expanded so that final temperature is half the maximum temperature during the process.



Q. 67: An ideal gas, in initial state $1(P_1, V_1, T_1)$ is cooled to a state $3(P_3, V_3, T_3)$ by a process which can be represented by a straight one on the P–V graph. The same gas in a different initial state $2(P_2, V_2, T_1)$ is cooled to same final state $3(P_3, V_3, T_3)$ by a process which can also be represented by a straight line on the same P–V graph. Q_1 and Q_2 are heat rejected by the gas in the two processes. Which is larger Q_1 or Q_2. It is given that $P_1 > P_2$.

Q. 68: In ideal gas is enclosed in an adiabatic container having cross section $A = 27\,\text{cm}^2$ for a part of it and $\dfrac{A}{2}$ for remaining part. Pistons P_1 and P_2 can move freely without friction along the inner wall. In the position shown in the figure the spring attached to the piston P_1 is relaxed. Sprig constant the spring is $K = 3700$ N/m. Piston P_2 is pushed up gradually through a distance $\dfrac{H}{2}$ and its was observed that the piston P_1 goes up by $\dfrac{3H}{32}$. Take $\gamma = 1.5$, mass of piston $P_1 = 13.5$ kg and atmospheric pressure $P_0 = 1 \times 10^5$ N/m^2



(a) Find H.

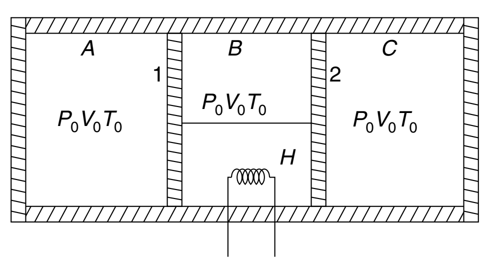
(b) Find final temperature of the gas if its initial temperature is 300 K.

Q. 69: An insulated cylinder is divided into three parts A, B and C. Pistons 1 and 2 are connected by a rigid rod and can slide without friction inside the cylinder. Piston 1 is perfectly conducting while piston 2 is perfectly insulating. Equal quantity of an ideal gas is filled in three compartments and the state of gas in every part is same $(P_0 V_0 T_0)$. Adiabatic exponent of the gas is $\gamma = 1.5$. The compartment B is slowly given heat through a heater H such that the final volume of gas in part C becomes $\dfrac{4V_0}{9}$

(a) Calculate the heat supplied by the heater.

(b) Calculate the amount of heat flow through piston 1.

(c) If heater were in compartment A, instead of B how would your answers to (a) and (b) change?

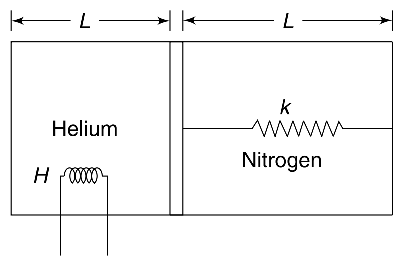


Q. 70: An adiabatic cylinder has length $2L$ and cross sectional area A. A freely moving non conducting piston of negligible thickness divides the cylinder into two equal parts. The piston is connected to the right face of the cylinder with an ideal spring of force constant k. The right chamber contains 28 g nitrogen in which one third of the molecules are dissociated into atoms. The left chamber container 4 g helium. With piston in equilibrium and spring relaxed the pressure in both chamber is P_0. The helium chamber is slowly given heat using an electric heater (H), till the piston moves to right by a distance $\frac{3L}{4}$. Neglect the volume occupied by the spring and the heating coil. Also neglect heat capacity of the spring.

(a) Find the ratio of C_P and C_V for nitrogen gas in right chamber.

(b) Calculate change in temperature of helium.

(c) Calculate heat supplied by the heater.



ANSWERS

1. Adiabatic

2. (i) irreversible adiabatic
(ii) isothermal
(iii) irreversible isochoric

3. 733.8 cal

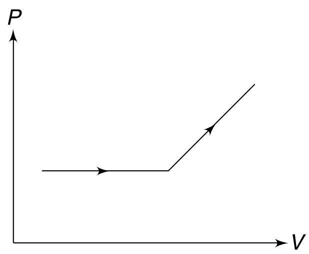
4. 3600R

5. 135 J

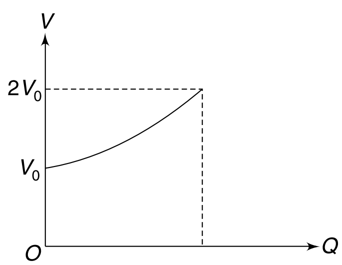
6. $C = C_v + \frac{R}{\alpha V}$

7. $3R$

8. (i)



(ii)



9. $W = \frac{15\pi}{4} P_1 D_1^3$

10. (i) Experiment 1 (ii) T_B

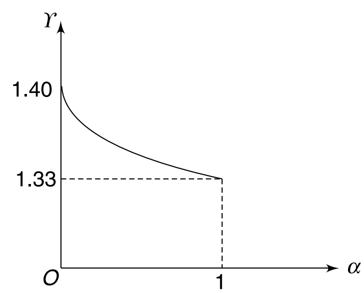
11. $W = nR[T_1 + T_3 - T_2 - T_4]$

12. (a) $\frac{T_1 T_2(P_1V_1 + P_2V_2)}{(P_1V_1T_2 + P_2V_2T_1)}$ (b) $\frac{P_1V_1 + P_1V_2}{V_1 + V_2}$

13. (a) 1.05 atm (b) 37.25 KJ

14. $\Delta Q = R\left(\frac{3 - 2\gamma}{\gamma - 1}\right)\Delta T$

15.



16. (a) Yes (b) $-\frac{3}{2} P_0V_0$

(c) Yes (d) 2 : 1

17. 36°C

18. $\frac{9}{2} P_0V_0$

19. $\frac{5}{7}$

20. $1 < n < \gamma$

21. $\left(\frac{mT_0}{V_0} + 1\right) R$

22. 0.6

23. $\frac{W_1}{W_2} = 2^{2/3}$

24. 600 K

25. $\Delta Q_{abd} > \Delta Q_{aod} > \Delta Q_{acd}$

26. $A_1 = A_2$

27. 17.38 kJ

28. 5312.5 J

29. $\gamma = \dfrac{\ell n6}{\ell n3}$

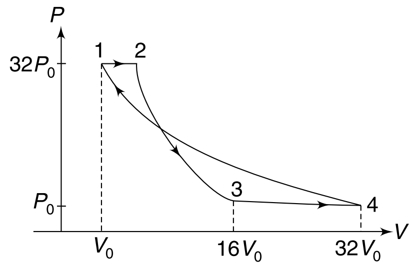
30. $C' = \dfrac{5}{2}R \quad x' = \dfrac{5}{3}$

31. $C = \dfrac{3P_0V_0}{T_0}$

32. (a) $T_C > T_B > T_A$ (b) $\gamma = \dfrac{T_C - T_A}{T_C - T_B}$

33. $\dfrac{2\Delta Q}{A\cdot d\cdot h(2s + \alpha\cdot g\cdot h)}$

34. (a)



(b) $W = 8\ P_0V_0$

35. (a) 30 cm (b) 406 J

36. $\gamma = \dfrac{\ln(P_0/P_1)}{\ln(P_2/P_1)}$

37. 706.25 R

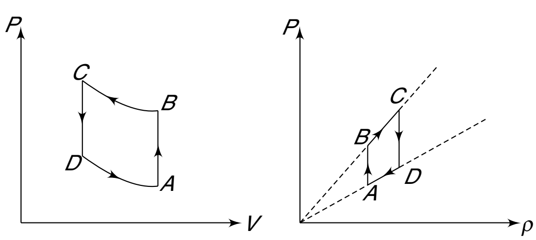
38. Volume doubles

39. (a) $u = \sqrt{\dfrac{3nRT_0}{m}\left[1 - \left(\dfrac{L_0}{L}\right)^{2/3}\right]}$

(b) $u_{\max} = \sqrt{\dfrac{3nRT_0}{m}}$

40. $PV^{\left[1 - \frac{\eta(\gamma-1)}{(1-\eta)}\right]}$

41. (a)



(b) $W = \dfrac{1}{2}P_0V_0\ln 2$

42. (i) $T = T_0\left[\dfrac{4\beta}{(1+\beta)^2}\right]^{\frac{1-\gamma}{2}}$

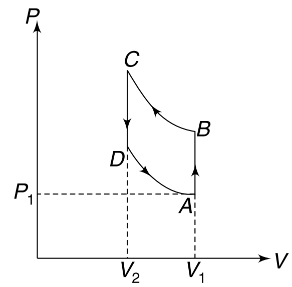
(ii) $\dfrac{28Q - mR\Delta T}{38\text{ V}}$

43. $T = T_0e^{\frac{2Ht}{7n_0T_0R}}$

44. $2R\Delta T$

45. (i) $k = -1$ (ii) $k = 0.6$

46. (a)



(b) $W = -\dfrac{nM}{\rho_1}(P_2 - P_1)\ln\left(\dfrac{\rho_2}{\rho_1}\right)$

47. (a) 10.2 cm (b) 35.7 J

48. (a) $r = \dfrac{3}{2}\ln(2)\dfrac{P_0}{dg}$ (b) $L\left[1 - \dfrac{1}{2\ln(2)}\right]$

49. $T = 4\sqrt{\dfrac{\pi V_0 d}{3P_0 r}}$

50. (a) $\Delta Q = 6nRT_0$ (b) $2R$

51. (i) $\dfrac{V_0}{(2)^{\frac{25}{21}}}$ (ii) $\dfrac{W_0}{4nR}$

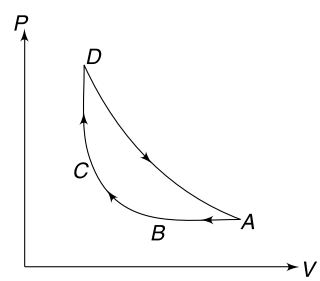
52. 32.5%

53. $\eta = \dfrac{T_1 - T_2}{T_1}$

54. $\dfrac{(T - T')(\gamma - 1)}{(2\gamma - 1)T - \gamma T'}$

55. (a) $\dfrac{P_0M}{\rho_0}\left(\dfrac{3}{2} + \ln 2\right)$ (b) $\dfrac{2}{5}(1 - \ln 2)$

56. (a)



(b) 41%

57. $4(2\sqrt{2} - 1)\ P_0V_0$

58. (a) $\sqrt{2}$ (b) 20 second

59. Energy consumed $\propto \dfrac{(\Delta T)^2}{T_0}$

60. (i) $P\left(\dfrac{T_2}{T_1 - T_2}\right)$

(ii) $T_1 - \dfrac{P}{k}\left[\sqrt{1 + \dfrac{4kT_1}{P}} - 1\right]$

61. (a) $T_0 = \sqrt{T_1 T_2} = 323$ K

(b) $W = ms(T_1 + T_2 - 2\sqrt{T_1 T_2}) = 4$ ms

62 (b) $\Delta Q_H = \dfrac{1}{2}(7P_0 + 12\rho gh)\ AL;$

$\Delta Q_L = \dfrac{1}{2}(7P_0 + 10\rho gL)\ AL$

63. $\dfrac{4Mv^2}{15R}$

64. $PV^{1/3} =$ constant; $C = 3R$

65. (a) $T = T_1 - (T_1 - T_0)\ e^{-\beta t}$

(b) $\Delta h = \dfrac{V_0(T_1 - T_0)}{AT_0}[1 - e^{-\beta t}]$ where $\beta = \dfrac{2T_0KA}{5P_0V_0d}$

66. $V = \dfrac{\sqrt{2}+1}{\sqrt{2}} V_0$

67. $Q_2 > Q_1$

68. (a) $\dfrac{16}{15}$ m (b) 400 K

69. (a) $18\ P_0V_0$ (b) $\dfrac{19}{2}\ P_0V_0$

(c) Answer to (a) does not change. Answer to (b) is $\dfrac{17}{2}\ P_0V_0$

70. (a) $\gamma = \dfrac{3}{2}$ (b) $\dfrac{1}{R}\left[\dfrac{15kL^2}{16} + 9P_0AL\right]$

(c) $\dfrac{27}{16}kL^2 + \dfrac{35}{2}P_0AL$

SOLUTIONS

1. Work done by gas is positive during all processes as the gas expands.

In an isothermal process there in no change in internal energy of the gas. In an isobaric process the temperature will increase as PV increases with expansion.

In adiabatic expansion the work done by the gas is at the expense of its internal energy.

3. Molar specific heat of steam at constant pressure will be

$$C_p = 4R = 8 \text{ cal mol}^{-1}\text{K}^{-1}$$

$$= \frac{8}{18} \text{ cal g}^{-1}\text{K}^{-1} = 0.44 \text{ cal g}^{-1}\text{K}^{-1}$$

$\therefore$ Q = Heat needed to heat the ice from –10°C to 0°C + Heat needed to melt the ice + Heat needed to heat water from 0°C to 100°C + Heat needed to convert water into steam + Heat needed to heat the steam from 100°C to 120°C.

$$= 1 \times 0.5 \times 10 + 1 \times 80 + 1 \times 1 \times 100 + 1 \times 540 + 1 \times 0.44 \times 20$$

$$= 5 + 80 + 100 + 540 + 8.8 = 733.8 \text{ cal.}$$

4. To double the rms speed, the temperature must be made 4 times.

$\therefore$ $$T_1 = 300 \text{ K}, \quad T_2 = 4 \times 300 = 1200 \text{ K}$$

$\therefore$ $$\Delta Q = n_1 C_{v_1}\ \Delta T + n_2 C_{v_2}\ \Delta T$$

$$= 1 \times \frac{3}{2} R \times 900 + 1 \times \frac{5}{2} R \times 900 = 3600\ R$$

5. $$\Delta U = nC_v\ \Delta T = n\frac{f}{2}R\Delta T = n\frac{7}{2}R\Delta T$$

$$W = P\Delta V = nR\Delta T = \frac{2\Delta U}{7}$$

$\therefore$ $$\Delta U = \frac{30 \times 7}{2} = 105 \text{ J}$$

$\therefore$ $$\Delta Q = \Delta U + W = 105 + 30 = 135 \text{ J}$$

6. $$T = T_0\ e^{\alpha V}$$

$\Rightarrow \quad \ln T = \ln T_0 + \alpha V$

$\therefore \quad \dfrac{dT}{T} = \alpha dV \qquad ...(1)$

First law of thermodynamics–

$$dQ = dU + dW$$

$$nCdT = nC_v dT + PdV = nC_v dT + \left(\frac{nRT}{V}\right)\left(\frac{dT}{\alpha T}\right)$$

$$\therefore \quad C = C_v + \frac{R}{\alpha V}$$

7. $$\Delta W = \int_{V_1}^{V_2} PdV$$

$$\because \quad P = KV$$

$$\therefore \quad \Delta W = K\int_{V_1}^{V_2} VdV = \frac{K}{2}(V_2^2 - V_1^2)$$

$$= \frac{(KV_2)(V_2) - (KV_1)(V_1)}{2} = \frac{P_2V_2 - P_1V_1}{2}$$

$$= \frac{nRT_2 - nRT_1}{2} = nR\frac{\Delta T}{2}$$

$$\Delta U = nC_v\Delta T = \frac{5}{2}nR\Delta T$$

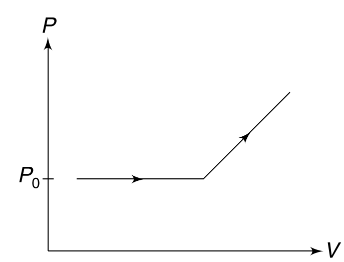
First law of thermodynamics

$$\Delta Q = \Delta U + W$$

$$nC\Delta T = \frac{5}{2}nR\Delta T + nR\frac{\Delta T}{2}$$

$$C = 3\ R$$

8. (i) Till the piston hits the spring, the gas pressure remains constant at atmospheric pressure (P_0). After that the gas pressure increases linearly with change in volume since spring force can be written as $kx = k\dfrac{\Delta V}{A}$ where ΔV = change in volume and A is area of cross sectional area of the piston.

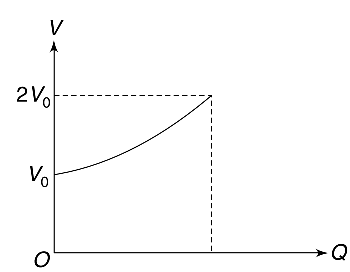


(ii) For isothermal process

$$Q = W = nRT_0 \ln\left(\frac{V}{V_0}\right)$$

$$\Rightarrow \quad \frac{Q}{nRT_0} + \ln V_0 = \ln V$$

$$\Rightarrow \quad V = e^{\left(\frac{Q}{nRT_0} + \ln V_0\right)}$$



9. $$V = \frac{4}{3}\pi\left(\frac{D}{2}\right)^3 \Rightarrow D = \left(\frac{6V}{\pi}\right)^{1/3} \qquad ...(1)$$

Given $\quad P = KD$

$$\Rightarrow \quad P = K\left(\frac{6V}{\pi}\right)^{1/3} \qquad [K = a \text{ Constant}]$$

This is the process equation.

$$\therefore \quad W = \int_{V_1}^{V_2} PdV$$

$$= K\left(\frac{6}{\pi}\right)^{1/3}\int_{V_1}^{V_2} V^{1/3}\,dV$$

$$= K\left(\frac{6}{\pi}\right)^{1/3}\left(\frac{3}{4}\right)\left[V_2^{4/3} - V_1^{4/3}\right]$$

$$= K\left(\frac{6}{\pi}\right)^{1/3}\left(\frac{3}{4}\right)V_1^{4/3}\left[\left(\frac{V_2}{V_1}\right)^{4/3} - 1\right]$$

From initial state $\quad P_1 = K\left(\frac{6}{\pi}\right)^{1/3}\cdot V_1^{1/3}$

$\Rightarrow \quad K = P_1\left(\frac{6V_1}{\pi}\right)^{-1/3}$

$\therefore \quad W = P_1\left(\frac{6V_1}{\pi}\right)^{-1/3}\left(\frac{3}{4}\right)V_1^{4/3}\left[\left(\frac{V_2}{V_1}\right)^{4/3} - 1\right]$

$$= \frac{3}{4}P_1V_1\left[\left(\frac{V_2}{V_1}\right)^{4/3} - 1\right]$$

Put $\quad V_1 = \frac{4}{3}\pi\left(\frac{D_1}{2}\right)^3$ and $V_2 = \frac{4}{3}\pi\left(\frac{D_2}{2}\right)^3$

$$W = \frac{\pi}{4}P_1D_1^3\left[\left(\frac{D_2}{D_1}\right)^4 - 1\right]$$

$$= \frac{\pi}{4}P_1D_1^3\,[2^4 - 1] = \frac{15\pi}{4}P_1D_1^3$$

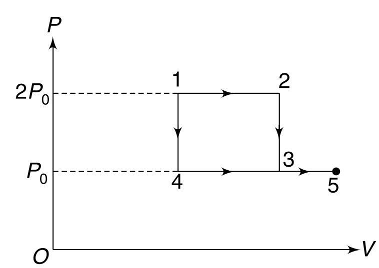
10. (i) When the piston is pulled out very fast, it is like free expansion of the gas. The gas temperature does not change. Therefore, PV = a constant and at the end the pressure becomes half the original value.

In second experiment the gas performs work. Therefore, its internal energy falls (notice that $\Delta Q = 0$). It means final temperature of the gas is less than its initial temperature.

$\therefore \quad P_1V_1 > P_2V_2$

$\therefore$ If $V_2 = 2V_1$, it means final pressure will be less then the original value.

(ii) The $P - V$ diagram for two processes is as shown in the figure.



Experiment A is represented by 1-2-3

Experiment B is represented by 1-4-5

Since $\quad P_5V_5 > P_3V_3$

$\therefore \quad T_5 > T_3$

$\Rightarrow \quad T_B > T_A$

Notice that $\quad W_{12} = W_{45}$

$\Rightarrow \quad 2P_0(\Delta V_{12}) = P_0(\Delta V_{45})$

$\Rightarrow \quad 2\cdot\Delta V_{12} = \Delta V_{45}$

11. Process 1 – 2 and 3 – 4 are isochoric process.

$\therefore \quad W_{12} = W_{34} = 0$

2 – 3 and 4 – 1 are isobaric processes.

$\therefore \quad W_{23} = P_2\Delta V = P_2V_3 - P_2V_2$

$$= nRT_3 - nRT_2 = nR(T_3 - T_2)$$

$$W_{41} = P_1\Delta V = P_1V_1 - P_1V_4$$

$$= nR(T_1 - T_4) = -nR(T_4 - T_1)$$

$\therefore \quad W = nR[T_3 - T_2 - T_4 + T_1]$

12. (a)
$$\Delta Q = 0,\ \Delta W = 0$$
$$\therefore \quad \Delta U = 0$$
$$U_{\text{initial}} = n_1 C_V T_1 + n_2 C_V T_2$$
$$= \frac{C_v}{R}[n_1 R T_1 + n_2 R T_2]$$
$$U_{\text{final}} = (n_1 + n_2)\ C_v T = \frac{C_v}{R}(n_1 + n_2)\ RT$$
$$\because \quad U_{\text{final}} = U_{\text{initial}}$$
$$\therefore \quad (n_1 + n_2)\ T = n_1 T_1 + n_2 T_2$$
$$T = \frac{n_1 T_1 + n_2 T_2}{n_1 + n_2} \quad ...(1)$$
$$\frac{\frac{P_1 V_1}{R} + \frac{P_2 V_2}{R}}{\frac{P_1 V_1}{R T_1} + \frac{P_2 V_2}{R T_2}} = \frac{T_1 T_2 (P_1 V_1 + P_2 V_2)}{(P_1 V_1 T_2 + P_2 V_2 T_1)}$$

(b)
$$P = \frac{(n_1 + n_2)\ RT}{V_1 + V_2}$$

Using (1)
$$P = \frac{R(n_1 T_1 + n_2 T_2)}{V_1 + V_2} = \frac{P_1 V_1 + P_2 V_2}{V_1 + V_2}$$

13. (a)
$$V_1 = \frac{n_1 R T_1}{P_1};\ V_2 = \frac{n_2 R T_2}{P_2}$$

After mixing
$$P = \frac{(n_1 + n_2) R T_0}{V_1 + V_2}$$
$$[T_0 = \text{final temperature} = 42 + 273 = 315\ \text{K}]$$
$$= \frac{(n_1 + n_2) R T_0}{\frac{n_1 R T_1}{P_1} + \frac{n_2 R T_2}{P_2}}$$
$$= \frac{\left(\frac{m_1}{M} + \frac{m_2}{M}\right) R T_0}{\frac{m_1 R T_1}{M P_1} + \frac{m_2 R T_2}{M P_2}} = \frac{(m_1 + m_2) T_0}{\frac{m_1 T_1}{P_1} + \frac{m_2 T_2}{P_2}}$$
$$= \frac{(2\ \text{kg} + 8\ \text{kg})\ (315\ \text{K})}{\frac{(2\ \text{kg})(350\ \text{K})}{0.7\ \text{atm}} + \frac{(8\ \text{kg})(300\ \text{K})}{1.2\ \text{atm}}} = 1.05\ \text{atm}$$

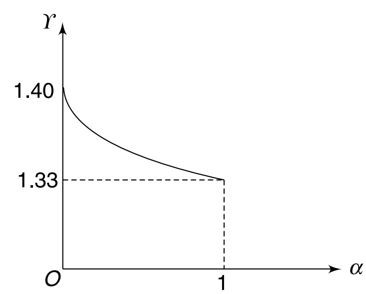
(b)
$$W = 0$$
$$\therefore \quad \Delta Q = \Delta U = U_f - U_i$$
$$= (m_1 + m_2) C_V T_0 - m_1 C_V T_1 - m_2 C_V T_2$$
$$= (10\ \text{kg})\left(0.745\ \frac{\text{kJ}}{\text{kgK}}\right)(315\ \text{K}) - (2\ \text{kg})\left(0.745\ \frac{\text{kJ}}{\text{kgK}}\right)(350\ \text{K})$$
$$- (8\ \text{kg})\left(0.745 \frac{\text{kJ}}{\text{kgK}}\right)(300\ \text{K})$$
$$= 37.25\ \text{kJ}$$

14.
$$VT^2 = a$$
$$\therefore \quad T^2 dV + V(2TdT) = 0$$
$$\therefore \quad TdV = -2VdT$$
$$\therefore \quad dV = -2V\frac{dT}{T}$$
$$PdV = -2PV\frac{dT}{T} = -nRT\frac{dT}{T} = -2nRdT$$

First law of thermodynamics for an infinitesimal change will be

$$dQ = dU + dW$$
$$= nC_V dT + PdV$$
$$= n\frac{R}{\gamma - 1}dT + (-2nR\,dT)$$
$$= R\left(\frac{3-2\gamma}{\gamma-1}\right)dT \qquad [\because \; n = 1]$$
$$\therefore \quad \Delta Q = R\left(\frac{3-2\gamma}{\gamma-1}\right)\Delta T$$

15. Consider 1 mole sample of the diatomic gas. Due to radiation α mole molecules split into 2α mole of atoms and $1 - \alpha$ mole remain as diatomic gas.



C_v for the mixture will be

$$C_v = \frac{2\alpha \cdot \frac{3}{2}R + (1-\alpha)\frac{5}{2}R}{1} = \frac{(\alpha+5)}{2}R$$
$$C_p = C_v + R = \left(\frac{\alpha+7}{2}\right)R$$
$$\therefore \quad \gamma = \frac{C_p}{C_V} = \frac{\alpha+7}{\alpha+5} = 1 + \frac{2}{\alpha+5}$$

Graph is as shown

16. (a) Product PV is same at B and D. It means that temperature is same at these two points.

(b) $W = (2P_0)V_0 - \frac{3}{2}P_0\left(\frac{V_0}{3}\right) = -\frac{3}{2}P_0V_0$

(c) Yes, reason being what has been said in part (a).

(d) Internal energy is proportional to temperature, which is proportional to product of pressure and volume. Hence the answer is 2.

17. Work done by air in the room $W = 0$

$$\therefore \quad \Delta U = Q = 30 \times 2500 = 75000 \text{ kcal/day}$$
$$= \frac{75000}{24 \times 60}\text{ kcal/min} = 52 \text{ kcal/min}$$

Volume of air in the room $\quad V = 10 \times 5 \times 3 = 150 \text{ m}^3$

$\therefore$ Mass of air in the room $\quad m = 1.20 \times 150 = 180$ kg

$$\therefore \quad mc\Delta T = 52 \times 30 \text{ Kcal}$$
$$\Delta T = \frac{52 \times 30}{180 \times 0.24} = 36°\text{C (!)}$$

18. Let temperature in state 'a' be T_0. Temperature at 'b' will be $2T_0$ [$\because P_BV_B = 2P_AV_A$] and temperature at C is $4T_0$ [$\because P_CV_C = 4P_AV_A$]

The product PV is least at 'a' and is maximum at 'c'.

$$\therefore \quad U_{min} = U_a \quad \text{and} \quad U_{max} = U_c$$

$$U_{max} - U_{min} = U_c - U_a = nC_V(T_c - T_a)$$

$$= 1 \times \frac{3R}{2} \times 3T_0 = \frac{9}{2} RT_0$$

$$= \frac{9}{2} P_0V_0 \qquad [\because \quad P_0V_0 = 1 \cdot R \cdot T_0]$$

19. First law $\quad Q = W + \Delta U$

For isothermal process $\quad \Delta U = 0$

$\therefore \quad Q = W$

Graph of W vs Q is a straight line with slope $m_1 = 1$

For isobaric process:

$$\frac{Q}{\Delta U} = \frac{nC_p\Delta T}{nC_v\Delta T} = \gamma = \frac{5}{3}$$

$$\therefore \quad \frac{Q}{Q - W} = \frac{5}{3}$$

$$\Rightarrow \quad W = \frac{2}{5} Q$$

Slope of W vs Q graph is $m_2 = \frac{2}{5}$

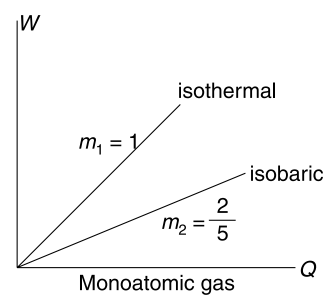
$$\therefore \quad \eta_1 = \frac{m_1}{m_2} = \frac{5}{2}$$

Similarly, for Diatomic gas

$$m_1 = 1 \quad \text{and} \quad m_2 = \frac{2}{7}$$

$$\therefore \quad \eta_2 = \frac{m_1}{m_2} = \frac{7}{2}$$

$$\therefore \quad \frac{\eta_1}{\eta_2} = \frac{5/2}{7/2} = \frac{5}{7}$$



20. Temperature of the gas increases when it rejects heat to the surrounding. This means that specific heat capacity of the gas for the given polytropic process must be negative.

For one mole gas–

$$W = \int PdV = \frac{R(T_1 - T_2)}{n - 1} \quad \text{[prove yourself by integration]}$$

$$\Delta U = C_v\Delta T = \frac{R(T_2 - T_1)}{\gamma - 1}$$

$$\therefore \quad \Delta Q = \Delta U + W$$

$$C\Delta T = \frac{R}{\gamma - 1}(\Delta T) - \frac{R}{n - 1}(\Delta T)$$

$$\therefore \quad C = \frac{R}{\gamma - 1} - \frac{R}{n - 1}$$

$$\therefore \quad C = \frac{R(n - \gamma)}{(n - 1)(\gamma - 1)}$$

γ is always > 1

$\therefore \quad C$ is negative if $(n - \gamma)$ is negative and $(n - 1)$ is positive

$$\Rightarrow \quad 1 < n < \gamma$$

21. $TV^{\gamma-1}$ = Constant

$$\ell n T + (\gamma - 1)\ \ell n V = \text{Const}$$

$$\frac{1}{T} + \frac{\gamma - 1}{V}\frac{dV}{dT} = 0$$

$$\Rightarrow \quad \frac{dV}{dT} = -\frac{V}{T(\gamma - 1)}$$

Given $\left(\frac{dV}{dT}\right)_{V_0, T_0} = -m$

$$\therefore \quad \frac{V_0}{T_0(\gamma - 1)} = m$$

$$\Rightarrow \quad \frac{1}{\gamma - 1} = \frac{mT_0}{V_0}$$

$$C_v = \frac{R}{\gamma - 1} = \frac{mRT_0}{V_0}$$

$$C_p = C_v + R = \left(\frac{mT_0}{V_0} + 1\right) R$$

22.

$$W_{ab} = 5 \text{ J}$$

$$\Delta U_{ab} = 0 \quad \text{(isothermal)}$$

$$Q_{ab} = 5 \text{ J} \quad \text{(First law)}$$

$$W_{bc} = 4 \text{ J}$$

$$Q_{bc} = 0 \quad \text{(Adiabatic)}$$

$$\Delta U_{bc} = -4 \text{ J} \quad \text{(Adiabatic)}$$

$$\Delta U_{bc} = -4 \text{ J} \quad \text{(First law)}$$

$$\because \quad \Delta U_{ab} + \Delta U_{bc} + \Delta U_{ca} = 0$$

$$\therefore \quad \Delta U_{ca} = 4 \text{ J}$$

Also, $W_{ab} + W_{bc} + W_{ca} = 3$ J (Area inside cycle)

$$5 + 4 + W_{ca} = 3$$

$$W_{ca} = -6 \text{ J}$$

First law for process *ca* $\quad Q_{ca} = 4 - 6 = -2$ J

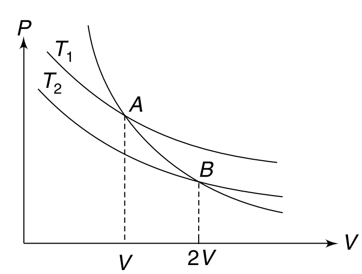
Efficiency of cycle $\quad \eta = \frac{W_{\text{cycle}}}{Q_{ab}} = \frac{3}{5} = 0.6$

23. If temperature along two isotherms are T_1 & T_2 then,

$$T_1 V^{\gamma-1} = T_2 (2V)^{\gamma-1} \quad [\because \ A \text{ and } B \text{ lie on an adiabat}]$$

Put $\gamma = \frac{5}{3}$ to get $\frac{T_1}{T_2} = 2^{2/3}$

Work done in doubling the volume:



Along 1st isotherm $\quad W_1 = nRT_1 \ell n\left(\frac{2V}{V}\right)$

Along 2nd isotherm $\quad W_2 = nRT_2 \ell n\left(\frac{2V}{V}\right)$

$$\therefore \quad \frac{W_1}{W_2} = \frac{T_1}{T_2} = 2^{2/3}$$

24. Expansion:

$$P_1V_1 = PV$$

$$P_1(4V) = PV \Rightarrow P_1 = \frac{P}{4} \qquad ...(1)$$

During expansion, no work is done and

$$\Delta Q = 0 \qquad \therefore \quad \Delta U = 0$$

$\therefore$ Temperature does not change.

Initial temperature $\qquad T = \dfrac{PV}{nR} = \dfrac{10^5 \times 2.5 \times 10^{-3}}{0.1 \times \frac{25}{3}} = 300 \text{ K}$

Compression:

$$T_2V_2^{\gamma-1} = T_1V_1^{\gamma-1}$$

$$T_2V^{0.5} = 300 \times (4V)^{0.5}$$

$$T_2 = 600 \text{ K}$$

25. In all three processes ΔU is same.

$$W_{abd} = \text{area } (abdoa)$$

$$W_{aod} = 0$$

$$W_{acd} = -\text{area}(acdoa)$$

$\therefore \qquad W_{abd} > W_{aod} > W_{acd}$

$\therefore \qquad \Delta Q_{abd} > \Delta Q_{aod} > \Delta Q_{acd}$

26. 2 – 3 and 4 – 1 are adiabatic processes and 1 – 2 and 3 – 4 are isotherms (Why?)

$\therefore \qquad T_1 = T_2; \quad T_3 = T_4$

Work done in an adiabatic process depends on change in temperature. Change in temperature in both 2 – 3 and 4 – 1 is same.

$\therefore \qquad A_1 = A_2$

27. $\qquad \Delta U = 32.2 \times 0.3 + 0.06 \times 50 = 12.66 \text{ kJ}$

Rise in gravitational potential energy of the piston

$$\Delta U_g = \text{mgh} = 50 \times 9.8 \times \frac{0.045}{0.1} = 0.221 \text{ kJ}$$

Work done by (air + piston) system against the atmospheric pressure is

$$W = P_0\,\Delta V = (100 \text{ kPa}) \times (0.045 \text{ m}^3) = 4.5 \text{ kJ}$$

$\therefore \qquad \Delta Q = W + \Delta U + \Delta U_g$

$$= 4.5 + 12.66 + 0.221 = 17.38 \text{ kJ}$$

28. For the adiabatic process we have

$$P_2^{1-\gamma}\,T_2^{\gamma} = P_1^{1-\gamma}T_1^{\gamma}$$

$\Rightarrow \qquad \dfrac{T_2}{T_1} = \left(\dfrac{P_1}{P_2}\right)^{\frac{1-\gamma}{\gamma}}$

$\Rightarrow \qquad T_2 = 1000\left(\dfrac{3}{2}\right)^{\frac{3}{5}-1} = 1000\left(\dfrac{2}{3}\right)^{2/5} = 1000 \times 0.85 = 850 \text{ K}$

For the isochoric process

$$\frac{P_3}{T_3} = \frac{P_2}{T_2} \Rightarrow T_3 = \left(\frac{P_3}{P_2}\right)T_2$$

$\Rightarrow \qquad T_3 = \left(\dfrac{P_0/3}{2P_0/3}\right) \times 850$

$\Rightarrow \qquad T_3 = \frac{850}{2} = 425 \text{ K}$

Heat is lost in 2nd process only.

$$\text{Heat lost} = nCv\,\Delta T = 1 \times \frac{3}{2} R \times (850 - 425)$$

$$= \frac{3}{2} \times \frac{25}{3} \times 425 = 5312.5 \text{ J}$$

29. Since BC is isothermal process $(3P_0)V_C = P_0V_0 \quad \Rightarrow \quad V_C = \frac{V_0}{3}$

For adiabatic process CA $\qquad (3P_0)\left(\frac{V_0}{3}\right)^{\gamma} = \left(\frac{P_0}{2}\right)(V_0^{\gamma})$

$\Rightarrow \qquad \gamma = \frac{\ell n6}{\ell n3}$

30. For $\qquad PV^x = \text{constant}$

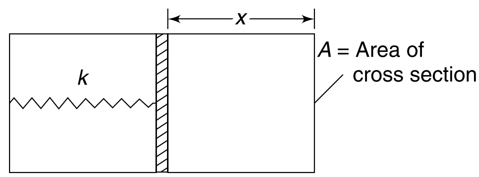
$$C = \frac{R}{\gamma - 1} - \frac{R}{x - 1} \quad \left(\gamma = \frac{5}{3} \text{ for monatomic gas}\right)$$

C' corresponds to $x = 0 \quad \Rightarrow \quad C' = \frac{R}{\frac{5}{3} - 1} + R = \frac{5}{2} R$

x' corresponds to $C = 0 \quad \Rightarrow \quad x' = \gamma = \frac{5}{3}$

31. Assume that the system is given dQ amount of heat. This is used in increasing the internal energy of the gas and spring potential energy.

If the spring compresses further by dx when heat dQ is supplied, change in PE can be calculated as



$$U = \frac{1}{2} kx^2$$

$\Rightarrow \qquad \frac{dU}{dx} = kx \quad \Rightarrow \quad dU = kxdx$

$\therefore \qquad dQ = nC_v dT + kxdx$

$$CdT = \frac{5}{2} nR\,dT + kxdx \qquad \text{...(1)}$$

But $\qquad kx = PA \qquad$ ...(2)

and $\qquad PV = nRT$

$\therefore \qquad PAx = nRT$

$\Rightarrow \qquad PA = \frac{nRT}{x}$

Put in (2) $\qquad kx = \frac{nRT}{x}$

$\therefore \qquad kx^2 = nRT$

$\therefore \qquad 2kxdx = nRdT$

Putting in (1) $\qquad CdT = \frac{5}{2} nR\,dT + \frac{1}{2} nRdT$

$\Rightarrow \qquad C = 3nR = 3 \cdot \frac{P_0V_0}{T_0}$

32. $A \rightarrow B$ (Adiabatic)

$$dW = PdV$$

$\because \qquad dQ = 0$

$\therefore \qquad dU = -dW = -PdV$

$A \to C$ (Isochoric)

$$dW = 0$$

$$dU_1 = dQ = C_V(T_C - T_A)$$

$C \to B$ (isobaric)

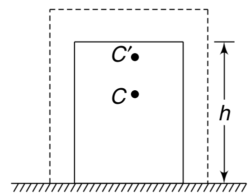
$$dU_2 = dQ - dW = -C_p(T_C - T_B) - PdV$$

$\because \qquad dU = dU_1 + dU_2$

$\therefore \qquad -PdV = C_V(T_C - T_A) - C_p(T_C - T_B) - PdV$

$\therefore \qquad \dfrac{C_p}{C_V} = \dfrac{T_C - T_A}{T_C - T_B}$

33. Let the temperature rise by $\Delta\theta$



Increase in height of the centre of the cylinder is

$$\Delta h_{cm} = \alpha \frac{h}{2} \Delta\theta$$

$\therefore$ Rise in gravitational potential energy of the cylinder is

$$\Delta U = mg\,\Delta h_{\text{cm}}$$

$$= Ahdg\,\alpha \frac{h}{2}\Delta\theta = \frac{1}{2}\alpha Ad\cdot g\cdot h^2\Delta\theta$$

This is actually work done (W) by the expanding cylinder against gravity.
From first law of thermodynamics

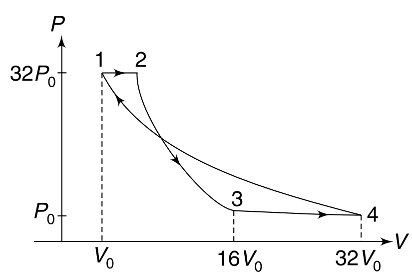
$$\Delta Q = W + \text{change in internal energy}$$

$$\Delta Q = \frac{1}{2}\alpha Adgh^2\,\Delta\theta + mS\Delta\theta$$

$$= \frac{1}{2}\alpha Ad\cdot g\cdot h^2\Delta\theta + A\cdot d\cdot h\cdot S\cdot\Delta\theta$$

$\therefore \qquad \Delta\theta = \dfrac{2\Delta Q}{A\cdot d\cdot h\cdot(2S + \alpha\cdot g\cdot h)}$

34. (a)



(b) $\qquad W_{12} = 32P_0V_0$

2 – 3 is an adiabatic process

$$P_3V_3^{\gamma} = P_2V_2^{\gamma}$$

$$P_3(16V_0)^{5/3} = 32P_0(2V_0)^{5/3}$$

$\Rightarrow \qquad P_3 = P_0$

$$W_{23} = \frac{P_2V_2 - P_3V_3}{\gamma - 1} = \frac{(32P_0)\,(2V_0) - P_0(16V_0)}{\frac{5}{3} - 1}$$

$$= 72\,P_0V_0$$

Since final process is isothermal

$\therefore \quad P_4V_4 = P_1V_1$

$\Rightarrow \quad P_0 \cdot V_4 = 32\,P_0 \cdot V_0 \quad \Rightarrow \quad V_4 = 32V_0$

$\therefore \quad W_{34} = P_0(32V_0 - 16V_0) = 16P_0V_0$

And $\quad W_{41} = P_4V_4 \ell n \dfrac{V_1}{V_4} = 32P_0V_0 \ell n\left(\dfrac{1}{32}\right) = -160P_0V_0 \ell n 2$

$\therefore$ Total work done is

$$W = 32P_0V_0 + 72P_0V_0 + 16P_0V_0 - 160\,\ell n 2 P_0V_0$$

$$= 120P_0V_0 - 160 \times 0.7P_0V_0$$

$$= 8P_0V_0$$

35. Initial pressure of gas $P_1 = 75 + 5 = 80$ cm of Hg.

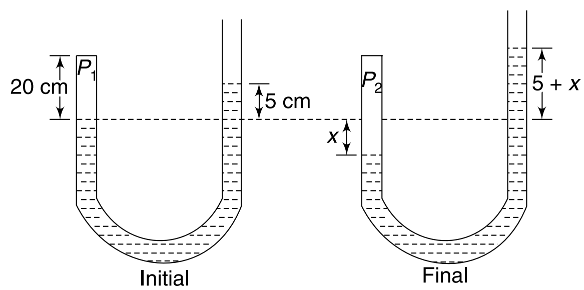
Let its final pressure be P_2

$$\frac{P_2V_2}{T_2} = \frac{P_1V_1}{T_1}$$

$$\frac{(75 + 5 + 2x)[A(20 + x)]}{562.5} = \frac{(80)\cdot(A \cdot 20)}{300}$$

$\Rightarrow \quad 3(80 + 2x)(20 + x) = 9000 \quad \Rightarrow \quad 2x^2 + 120x + 1600 = 3000$

$\Rightarrow \quad x^2 + 60x - 700 = 0$



Solving this quadratic equation gives $x = 10$ cm

Work done by gas = Work done against atmospheric pressure in pushing the Hg column by x + Work done against gravity (i.e., rise in gravitational P.E. of Hg)

$\therefore \quad W = P_{atm}Ax + mg(5 + x)$

Where m = mass of Hg in column of length x.

$\therefore \quad W = 10^5 \times 0.01 \times 0.1 + 13.6 \times 10^3 \times 0.01 \times 0.1 \times 10 \times 0.15$

$$= 100 + 20.4 = 120.4 \text{ J}$$

Change in internal energy of the gas

$$\Delta U = nC_V\,\Delta T = \left(\frac{P_1V_1}{RT_1}\right)\left(\frac{3}{2}R\right)(T_2 - T_1)$$

$$= \frac{0.8 \times 13.6 \times 10^3 \times 10 \times 0.01 \times 0.2}{300} \times \frac{3}{2} \times 262.5 = 285.6 \text{ J}$$

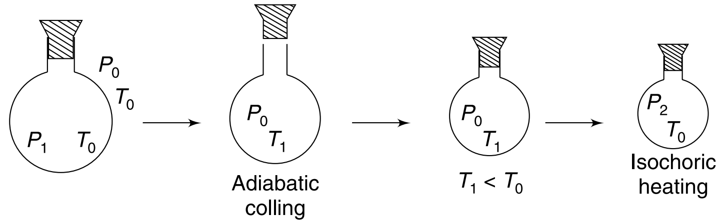
$\therefore$ Heat supplied by heater

$$Q = W + \Delta U = 120.4 + 285.6 = 406 \text{ J}$$

36. The expansion of air on opening the stopcock is sudden. The process is close to adiabatic.

$$P_1^{-(\gamma-1)}T_0^{\gamma} = P_0^{-(\gamma-1)}T_1^{\gamma}$$

$$\Rightarrow \quad \left(\frac{P_1}{P_0}\right)^{-(\gamma-1)} = \left(\frac{T_1}{T_0}\right)^{\gamma} \quad ...(1)$$



The slightly colder air inside the jar picks up heat from the surrounding and warms up to temperature T_0. The process is isochoric.

$$\therefore \quad \frac{P_2}{T_0} = \frac{P_0}{T_1}$$

$$\therefore \quad \left(\frac{T_1}{T_0}\right) = \frac{P_0}{P_2} \quad ...(2)$$

From (1) and (2) $$\left(\frac{P_1}{P_0}\right)^{-(\gamma-1)} = \left(\frac{P_0}{P_2}\right)^{\gamma}$$

$$\left(\frac{P_0}{P_1}\right)^{\frac{\gamma-1}{\gamma}} = \frac{P_0}{P_2}$$

$$\therefore \quad \left(1-\frac{1}{\gamma}\right)\ln\left(\frac{P_0}{P_1}\right) = \ln\left(\frac{P_0}{P_2}\right)$$

$$\Rightarrow \quad \ln\left(\frac{P_0}{P_1}\right) - \ln\left(\frac{P_0}{P_2}\right) = \frac{1}{\gamma}\ln\frac{P_0}{P_1}$$ simplifying gives-

$$\therefore \quad \gamma = \frac{\ln\left(\dfrac{P_0}{P_1}\right)}{\ln\left(\dfrac{P_2}{P_1}\right)}$$

37. $$C_p = C_V + R$$

$$\Delta Q = \int_{T_1}^{T_2} nC_p\,dT = 1\cdot\int_{200}^{400}(C_V + R)\,dT$$

$$\Delta Q = \int_{200}^{400} C_V dT + R \times 200 \quad ...(1)$$

The integral can be obtained by area under the graph.

At $T = 200$ K, $C_V = 2.25\ R$

and $T = 400$ K, $C_V = 3.0\ R$

$$\therefore \quad \int_{200}^{400} C_V\,dT = \frac{1}{2}\times 50\times(2.25\ R + 2.5\ R) + 100\times 2.5\ R + \frac{1}{2}\times 50\times(2.5\ R + 3\ R)$$

$$= (118.75 + 250 + 137.5)\text{R} = 506.25\ \text{R}$$

$\therefore$ From (1) $\Delta Q = 706.25\ R$

38. Consider an infinitesimal change in the state of the gas.

Let dQ = heat added dT = change in temperature C = molar heat capacity

$$dQ = nCdT \Rightarrow dQ = 2nRdT \qquad ...(i)$$

Work done by the gas $dW = PdV$

Since $PV = nRT$

$$\Rightarrow \quad dW = \frac{nRT}{V}dV \qquad ...(ii)$$

Change in internal energy

$$dU = nC_V dT = \frac{7}{5}nRdT \quad \left[\because \; C_V = \frac{7}{5}R\right] \qquad ...(iii)$$

First law of thermodynamics says

$$dQ = dU + dW$$

$$2nRdT = \frac{7}{5}nRdT + nRT\frac{dV}{V}$$

$$\Rightarrow \quad \frac{3}{5}dT = T\frac{dV}{V} \Rightarrow \frac{3}{5}\frac{dT}{T} = \frac{dV}{V}$$

$$\Rightarrow \quad \frac{3}{5}\int\frac{dT}{T} = \int\frac{dV}{V}$$

$$\Rightarrow \quad \frac{3}{5}\ln T = \ln V + \ln k \qquad [k = a \quad \text{constant}]$$

$$\Rightarrow \quad \ln T^{3/5} = \ln(kV)$$

$$\Rightarrow \quad T^{3/5} = kV$$

$$\Rightarrow \quad V \propto T^{3/5}$$

$\therefore$ If temperature is made $(32)^{1/3}$ times,

The volume becomes $(32)^{1/5} = 2$ times.

39. (a) The gas expands adiabatically.

Its volume changes from $V_0 = AL_0$ to $V = AL$ as the bullet moves out of the gun.

$$\therefore \quad T_0 V_0^{\gamma-1} = TV^{\gamma-1}$$

Where T = temperature of gas as the bullet moves out

$$\therefore \quad T_0(AL_0)^{\gamma-1} = T(AL)^{\gamma-1}$$

$$\Rightarrow \quad T = T_0\left(\frac{L_0}{L}\right)^{\gamma-1} \qquad ...(i)$$

Kinetic energy gained by the bullet is equal to work done by the gas in adiabatic expansion.

$$\therefore \quad \frac{1}{2}mu^2 = \frac{nR(T_0 - T)}{\gamma - 1}$$

$$\therefore \quad u^2 = \frac{2nRT_0}{m(\gamma-1)}\left[1 - \left(\frac{L_0}{L}\right)^{\gamma-1}\right]$$

For helium $\gamma = \frac{5}{3}$

$$\therefore \quad u^2 = \frac{3nRT_0}{m}\left[1 - \left(\frac{L_0}{L}\right)^{2/3}\right]$$

$$u = \sqrt{\frac{3nRT_0}{m}\left[1 - \left(\frac{L_0}{L}\right)^{2/3}\right]}$$

(b) Speed will be maximum when $\frac{L_0}{L} << 1$

$$u_{max} = \sqrt{\frac{3nRT_0}{m}}$$

In this case ($L >> L_0$) the entire internal energy of the gas gets converted into kinetic energy of the bullet. Therefore, we may also write

$$\frac{1}{2} mu_{max}^2 = nC_V T_0$$

$$\Rightarrow \quad \frac{1}{2} mu_{max}^2 = n\frac{3R}{2} T_0$$

$$\Rightarrow \quad u_{max} = \sqrt{\frac{3nRT_0}{m}}$$

40.

$$dU = \eta \cdot dQ$$

$$\therefore \quad dW = dQ - dU = \left(\frac{1}{\eta} - 1\right) dU$$

$$PdV = \left(\frac{1-\eta}{\eta}\right) nC_V\, dT \qquad \text{...(i)}$$

Ideal gas equation $\quad PV = nRT$

$$\Rightarrow \quad PdV + VdP = nRdT$$

$$\Rightarrow \quad PdV + VdP = \frac{\eta R}{(1-\eta)\, C_V} PdV \qquad \text{[using (i)]}$$

$$\Rightarrow \quad PdV \left[1 - \frac{\eta R}{(1-\eta)\, C_V}\right] + VdP = 0$$

$$\Rightarrow \quad \left[1 - \frac{\eta R}{(1-\eta)\, C_V}\right] \frac{dV}{V} + \frac{dP}{P} = 0$$

$$\Rightarrow \quad \left[1 - \frac{\eta R}{(1-\eta)\, C_V}\right] \int \frac{dV}{V} + \int \frac{dP}{P} = \text{a constant}$$

$$\Rightarrow \quad \left[1 - \frac{\eta R}{(1-\eta)\, C_V}\right] \ln V + \ln P = \text{constant}$$

$$\Rightarrow \quad PV^{\left[1 - \frac{\eta R}{(1-\eta)\, C_V}\right]} = \text{constant} \quad \text{Put } C_V = \frac{R}{\gamma - 1}$$

$$\Rightarrow \quad PV^{\left(1 - \frac{(\gamma-1)\eta}{(1-\eta)}\right)} = \text{a constant}$$

41. $A \rightarrow B$

$$W = 0$$

$$\Rightarrow \quad \Delta V = 0 \quad \Rightarrow \quad \Delta\rho = 0 \text{ (density and volume constant)}$$

$$\therefore \quad P \propto T$$

$\therefore$ Temperature changes from T_0 to $2T_0$ and pressure changes from P_0 to $2P_0$

$B \rightarrow C$

$$\therefore \quad \Delta U = 0 \quad \Rightarrow \quad T = \text{constant at } 2T_0$$

Process is isothermal

$$\therefore \qquad W = 2P_0 \cdot V_0 \ln \frac{V_C}{V_B}$$

$$\Rightarrow \qquad -P_0V_0 \ln 2 = 2P_0V_0 \ln \frac{V_C}{V_0}$$

$$\Rightarrow \qquad \left(\frac{V_C}{V_0}\right)^2 = \frac{1}{2} \quad \Rightarrow \quad V_C = \frac{V_0}{\sqrt{2}}$$

$$\therefore \qquad P_C = \frac{2P_0V_0}{V_C} = 2\sqrt{2}\,P_0$$

$$\Rightarrow \qquad 2\sqrt{2}P_0V_C = 2P_0V_0 \quad \Rightarrow \quad V_C = \frac{V_0}{\sqrt{2}}$$

$C \to D$

Constant volume process

$$\therefore \qquad P \propto T$$

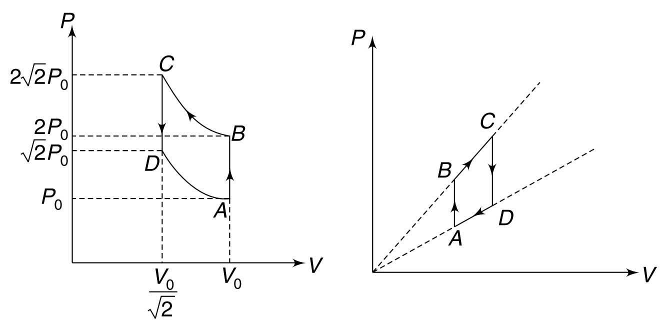
Temperature becomes half ($2T_0 \to T_0$) and pressure also becomes half $= \dfrac{P_C}{2} = \sqrt{2}P_0$

$D \to A$

Isothermal process $\left(\text{Volume changes from } \dfrac{V_0}{\sqrt{2}} \text{ to } V_0\right)$

$$W = (\sqrt{2}\ P_0)\left(\frac{V_0}{\sqrt{2}}\right) \ln\sqrt{2} = \frac{1}{2} P_0V_0 \ln 2$$

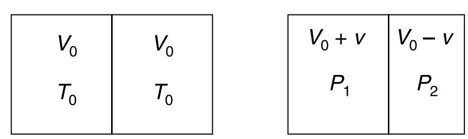
(a)



(c) Work done by the gas

$$W = -P_0V_0 \ln 2 + \frac{1}{2} P_0V_0 \ln 2 = -\frac{1}{2} P_0V_0 \ln 2$$

$\therefore$ Work done on the gas $\qquad W = \dfrac{1}{2} P_0V_0 \ln 2$



42. (i) Piston is conducting. It means temperature in both compartments is always same.

Let volume of two parts be $(V_0 + v)$ and $(V_0 - v)$ at an instant.

$$\beta = \frac{V_0 + v}{V_0 - v} \quad \Rightarrow \quad V = \left(\frac{\beta - 1}{\beta + 1}\right)V_0 \qquad \ldots\text{(i)}$$

If T is temperature of the two parts then

$$P_1 = \frac{nRT}{V_0 + v};\ P_2 = \frac{nRT}{V_0 - v}$$

Work done by external agent in further changing the volume of two parts by dv is

$$dW_{\text{ext}} = F_{\text{ext}} \cdot dv = (P_2 - P_1)\ dv$$

$$= nRT\left(\frac{1}{V_0 - v} - \frac{1}{V_0 + v}\right) dv \qquad \ldots\text{(ii)}$$

For the entire gas as system, $\quad dQ = 0$

$$\therefore \quad dW_{\text{ext}} = dU$$

$$\therefore \quad nRT\left(\frac{2v}{V_0^2 - v^2}\right) dv = 2 \cdot nC_V dT$$

$$\Rightarrow \quad R\int_0^V \frac{vdv}{V_0^2 - v^2} = \frac{R}{\gamma - 1}\int_{T_0}^{T} \frac{dT}{T}$$

$$\Rightarrow \quad -\frac{(\gamma - 1)}{2}\left[\ln\left[V_0^2 - v^2\right]\right]_0^v = [\ln T]_{T_0}^{T}$$

$$\Rightarrow \quad -\left(\frac{\gamma - 1}{2}\right)\ln\left[\frac{V_0^2 - v^2}{V_0^2}\right] = \ln\left(\frac{T}{T_0}\right)$$

$$\Rightarrow \quad \left[1 - \frac{v^2}{V_0^2}\right]^{-\frac{\gamma - 1}{2}} = \frac{T}{T_0}$$

$$\Rightarrow \quad T = T_0\left[1 - \left(\frac{\beta - 1}{\beta + 1}\right)^2\right]^{\frac{1-\gamma}{2}} = T_0\left[\frac{4\beta}{(\beta + 1)^2}\right]^{\frac{1-\gamma}{2}}$$

(ii) Let n_1 and n_2 be number of moles of oxygen and helium respectively

$$32n_1 + 4n_2 = m \qquad \text{...(i)}$$

Since volume of gas does not change, therefore $W = 0$

$$Q = \Delta U = (n_1 C_{VO_2} + C_{VHe})\ \Delta T$$

$$\Rightarrow \quad Q = \left(\frac{5}{2}\ n_1 R + \frac{3}{2}\ n_2 R\right)\ \Delta T$$

$$\Rightarrow \quad 5n_1 + 3n_2 = \frac{2Q}{R\Delta T} \qquad \text{...(ii)}$$

Solving (i) and (ii) gives

$$n_1 = \frac{1}{76}\left[3m - \frac{8Q}{R\Delta T}\right] = \frac{3m}{76} - \frac{2Q}{19R\Delta T}$$

And

$$n_2 = -\frac{5m}{76} + \frac{16Q}{19R\Delta T}$$

$\therefore$ Total number of moles of gas

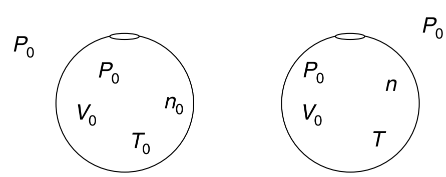
$$n = n_1 + n_2 = \frac{14Q}{19R\Delta T} - \frac{m}{38}$$

$\therefore$ change in pressure is given by

$$\Delta P = \frac{nR\Delta T}{V} = \frac{14Q}{19V} - \frac{mR\Delta T}{38V}$$

$$= \frac{28Q - mR\Delta T}{38V}$$

43. Let number of moles of air inside the container at time t be n, and temperature be T. It is pressure and volume that is not changing. In time dt, heat supplied is



$$dQ = Hdt$$

$$\therefore \quad Hdt = nC_P dT \quad \Rightarrow \quad Hdt = n \cdot \frac{7}{2}\ RdT$$

But

$$nT = n_0 T_0 = \frac{P_0 V_0}{R}$$

$$\therefore \quad n = \frac{n_0 T_0}{T}$$

$$\therefore \quad \frac{7}{2}\frac{n_0 T_0}{T} R\,dT = H\,dt$$

$$\therefore \quad \frac{7 n_0 T_0 R}{2H}\int_{T_0}^{T}\frac{dT}{T} = \int_0^t dt$$

$$\therefore \quad \ln\frac{T}{T_0} = \frac{2Ht}{7 n_0 T_0 R}$$

$$\therefore \quad T = T_0 e^{\frac{2Ht}{7 n_0 T_0 R}}$$

44. Let P_1 = initial pressure of the gas inside the tube

$$P_0 = P_1 + \rho g h_1 \qquad [h_1 \text{ height of Hg column}]$$

$$\rho g h_0 = P_1 + \rho g h_1$$

$$\therefore \quad P_1 = \rho g (h_0 - h_1)$$

Temperature of the gas is given by

$$T_1 = \frac{P_1 V_1}{nR} = \frac{\rho g (h_0 - h_1) A (h_0 - h_1)}{R} \qquad [A = \text{area of cross section}]$$

$$\therefore \quad T_1 = \frac{(h_0 - h_1)^2 \rho g A}{R}$$

After increasing the temperature, let the height of Hg column be h_2.

$$T_2 = \frac{(h_0 - h_2)^2 \rho g A}{R}$$

$$\therefore \quad \Delta T = T_2 - T_1 = \frac{\rho g A}{R}\left[(h_0 - h_2)^2 - (h_0 - h_1)^2\right] \qquad \text{...(i)}$$

Let's calculate the work done by the gas in pushing the Hg column.

$$W = \int P\,dV = -\int_{h_1}^{h_2} \rho g (h_0 - h)\, A\,dh \qquad [-ve \text{ sign as } dh \text{ is negative}]$$

$$= \rho g A \left[\frac{(h_0 - h)^2}{2}\right]_{h_1}^{h_2}$$

$$= \frac{\rho g A}{2}\left[(h_0 - h_2)^2 - (h_0 - h_1)^2\right] = \frac{R\Delta T}{2} \qquad [\text{using (i)}]$$

First law of thermodynamics

$$\Delta Q = n C_V \Delta T + W$$

$$= \frac{3}{2} R\Delta T + \frac{R\Delta T}{2} = 2R\Delta T$$

45. PV^k = constant

For small changes in P, V and T, we can write–

$$PkV^{k-1}\Delta V + \Delta P V^k = 0 \qquad \text{...(i)}$$

For an ideal gas

$$PV = nRT \qquad \text{...(ii)}$$

$$\therefore \quad P\Delta V + V\Delta P = nR\Delta T \qquad \text{...(iii)}$$

From (i) $\quad (V\Delta P)V^{k-1} = -kPV^{k-1}\Delta V$

$\therefore \quad V\Delta P = -kP\Delta V$

Putting in (iii) $\quad P\Delta V - kP\Delta V = nR\Delta T$

$\Rightarrow \quad P\Delta V(1-k) = nR\Delta T \quad$ ... (iv)

First law of thermodynamics–

$$\Delta Q = \Delta U + W$$

$$nC\Delta T = nC_V\Delta T + P\Delta V$$

$$nC\Delta T = nC_V\Delta T + \frac{nR\Delta T}{1-k} \quad \text{[using (iv)]}$$

$$\therefore \quad C = C_V + \frac{R}{1-k} \quad \text{... (v)}$$

(i) $\quad C = \dfrac{C_P + C_V}{2} = \dfrac{C_V + R + C_V}{2} = C_V + \dfrac{R}{2}$

$\therefore \quad k = -1$

(ii) $\quad C = C_P + C_V$

For mono atomic gas $\quad C_P = \dfrac{5R}{2}$

$\therefore$ from (v) $\quad \dfrac{R}{1-k} = \dfrac{5R}{2}$

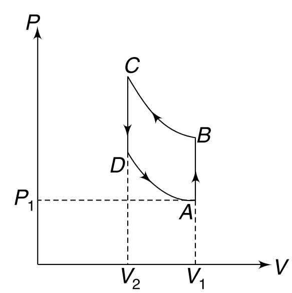
$\Rightarrow \quad 2 = 5 - 5k \Rightarrow k = 0.6$

46. (a) AB and CD are isochoric.

BC and DA are isothermal processes because $P \propto \rho$ means T is constant as shown below-

$$PV = nRT \Rightarrow PV = \frac{m}{M}RT$$

$$\Rightarrow \quad P = \frac{m}{V}\frac{RT}{M} \Rightarrow P = \rho\frac{RT}{M}$$



$P \propto \rho$ means T is constant.

(b) $\quad V_1 = \dfrac{nM}{\rho_1}$ and $V_2 = \dfrac{nM}{\rho_2}$

$$W_{AB} = W_{CD} = 0$$

$$W_{BC} = P_BV_B\ln\left(\frac{V_C}{V_B}\right) = P_2V_1\ln\left(\frac{V_2}{V_1}\right) = P_2\frac{nM}{\rho_1}\ln\left(\frac{\rho_1}{\rho_2}\right)$$

$$W_{DA} = P_AV_A\ln\left(\frac{V_A}{V_D}\right) = P_1V_1\ln\left(\frac{V_1}{V_2}\right) = P_1\frac{nM}{\rho_1}\ln\left(\frac{\rho_2}{\rho_1}\right)$$

$$\therefore \quad W = \frac{nM}{\rho_1}\left[-P_2\ln\left(\frac{\rho_2}{\rho_1}\right) + P_1\ln\left(\frac{\rho_2}{\rho_1}\right)\right]$$

$$= -\frac{nM}{\rho_1}\ln\left(\frac{\rho_2}{\rho_1}\right)(P_2 - P_1)$$

47. (a) In upside down position

$$P_2A = P_0A - mg$$

$$\therefore \quad P_2 = P_0 - \frac{mg}{A}$$

$$= 1.01 \times 10^5 - \frac{0.1 \times 10}{10 \times 10^{-4}} = 1.0 \times 10^5 \text{ N/m}^2$$

In original piston

$$P_1A = P_0A + mg$$

$$\Rightarrow \quad P_1 = 1.02 \times 10^5 \text{ N/m}^2$$

Since, temperature remained constant

$$P_2V_2 = P_1V_1$$

$$\therefore \quad V_2 = 1.02 \times 100 = 102\,cc$$

$$\therefore \text{ length of air column} \quad l_2 = \frac{V_2}{A} = \frac{102}{10} = 10.2 \text{ cm}$$

(b) As the piston reaches the mouth of the cylinder, length of air column becomes 20.4 cm, i.e., volume of air doubles. The piston moves slowly, it means pressure remains constant.

$$\therefore \quad W = P_2\Delta V = 1.0 \times 10^5 \times (10.2 \times 10 \times 10^{-6})$$

$$= 10.2 \text{ J}$$

Keeping pressure constant, the volume has been doubled. It means temperature has doubled.

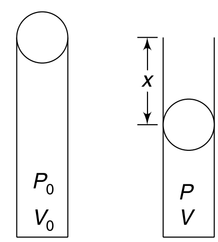
$$\therefore \quad \Delta U = nC_V\Delta T = n\cdot\frac{5}{2}R \times 300$$

$$= (nR \times 300) \times \frac{5}{2} = P_1V_1 \times \frac{5}{2}$$

$$= 1.02 \times 10^5 \times 100 \times 10^{-6} \times \frac{5}{2} = 25.5 \text{ J}$$

$$\therefore \quad \Delta Q = 25.5 + 10.2 = 35.7 \text{ J}$$

48. $V_0 = AL;\ V = A(L - x) \qquad [A = \pi r^2]$



For isothermal compression of air

$$PV = P_0V_0$$

$$PA[L - x] = P_0AL \quad \Rightarrow \quad P = P_0\left(\frac{L}{L - x}\right)$$

(a) Work done by the gas on the ball $= -P_0V_0\ln\left(\frac{L}{L - x}\right)$

Work done by the gravity on the ball $= mgx$

If the ball comes to rest

$$mgx - P_0V_0\ln\left(\frac{L}{L - x}\right) = 0$$

Given $\quad x = \frac{L}{2}$

$$mg\,\frac{L}{2} = P_0AL\ln\left(\frac{L}{L - \frac{L}{2}}\right)$$

$$\therefore \quad \frac{4}{3}\pi r^3\cdot d\cdot g\cdot\frac{1}{2} = P_0\pi r^2\ln(2)$$

$$\therefore \quad r = \frac{3}{2}\,\frac{P_0\ln(2)}{d\cdot g\cdot}$$

(b) In equilibrium, $\quad P\pi r^2 = mg$

$$P_0\left(\frac{L}{L - x}\right)\pi r^2 = \frac{4}{3}\pi r^3\cdot d\cdot g$$

$$L - x = \frac{3P_0L}{4rdg} = \frac{3}{4}L\cdot\frac{2}{3\ln(2)} = \frac{L}{2\ln(2)}$$

$$x = L\left[1 - \frac{1}{2\ln(2)}\right]$$

49. Ball is in equilibrium means pressure inside the container is equal to the atmospheric pressure (P_0).

If the ball is displaced a little; volume of air inside the container changes and hence, the pressure changes. This creates a restoring force which causes oscillations.

Assume that the ball is displaced by a small distance x towards right. Change in pressure can be calculated as follows.

$$PV = \text{constant}$$

$$\ln P + \ln V = \text{constant}$$

$$\frac{1}{P}\frac{\Delta P}{\Delta V} + \frac{1}{V} = 0$$

$$\therefore \quad \Delta P = -\frac{P\Delta V}{V}$$

[Negative sign means pressure drops when volume increases and vice-versa]

$$\Delta P = -\frac{P_0}{V_0}\pi r^2\cdot x$$

$\therefore$ Restoring force towards left = (outside atmospheric pressure – Inside gas pressure) × A

$$= \Delta P\cdot \pi r^2$$

$$\therefore \quad \frac{4}{3}\pi r^3\cdot d\left(\frac{d^2x}{dt^2}\right) = -\frac{P_0}{V_0}\pi r^2\cdot x\cdot \pi r^2$$

$$\frac{d^2x}{dt^2} = -\frac{3P_0\pi r}{4V_0 d}\cdot x$$

$\therefore$ Motion is SHM.

$$\omega = \sqrt{\frac{3\pi P_0 r}{4V_0 d}}$$

$$T = 4\pi\sqrt{\frac{V_0 d}{3\pi P_0\cdot r}}$$

$$T = 4\sqrt{\frac{\pi V_0 d}{3P_0 r}}$$

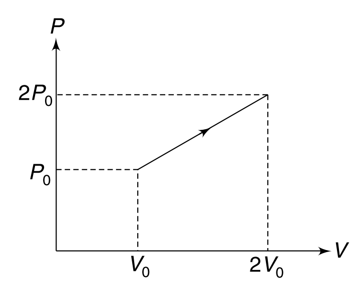
50. (a)

$$P_0V_0 = nRT_0$$

$$W = \text{area under the graph} = \frac{1}{2}V_0[P_0 + 2P_0]$$

$$= \frac{3}{2}P_0V_0 = \frac{3}{2}nRT_0$$

And $\quad \Delta U = nC_V\Delta T = n\cdot\frac{3}{2}R(T_f - T_i)$



Final temperature will be 4 times the initial temperature since $P_fV_f = 4P_iV_i$

$$\therefore \quad \Delta U = \frac{3}{2}nR(4T_0 - T_0) = \frac{9}{2}nRT_0$$

$$\therefore \quad \Delta Q = W + \Delta U = \frac{3}{2}nRT_0 + \frac{9}{2}nRT_0 = 6nRT_0$$

(b) $nC\Delta T = \Delta Q$

$\therefore \quad nC(3T_0) = 6nRT_0$

$\therefore \quad C = 2R$

51. (i) The process is adiabatic for each gas.

For argon: $\gamma = \dfrac{5}{3}$

$$\therefore \quad P_{Ar}\left(\frac{V_0}{2}\right)^{\frac{5}{3}} = P_0 V_0^{5/3}$$

$$\Rightarrow \quad P_{\text{Ar}} = 2^{\frac{5}{3}} P_0 \quad \text{...(i)}$$

For H_2: $\gamma = \dfrac{7}{5}$

$$P_{H_2} V_{H_2}^{7/5} = P_0 V_0^{7/5} \quad \text{...(ii)}$$

But $P_{Ar} = P_{H_2}$ (process is slow and pressure is same in both chambers at all instants)

$\therefore$ from (1) and (2) $\quad V_{H_2} = \dfrac{V_0}{(2)^{\frac{25}{21}}}$

(ii) There is exchange of heat between the gases but no exchange of heat from surrounding.

For both gases as system

$$\Delta Q = 0$$

$$\therefore \quad W + \Delta U = 0$$

$$\therefore \quad \Delta U = -W$$

$$nC_{V_{Ar}}\Delta T + nC_{V_{H_2}}\Delta T = W_0 \qquad [\text{Work done by gas } W = -W_0]$$

$$n \cdot \frac{3}{2} R\Delta T + n \cdot \frac{5}{2} R\Delta T = W_0$$

$$4nR\Delta T = W_0$$

$$\therefore \quad \Delta T = \frac{W_0}{4nR}$$

52. Cycle 1–2–3–1

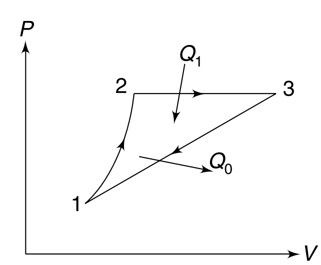
From $1 \to 2$, work done is $+ve$ ($\because$ volume increases) and ΔU is also positive ($\because$ temperature increased).

Similarly, for $2 \to 3$, W as well as ΔU are positive.

In both the processes $1 \to 2$ and $2 \to 3$ the gas absorbs heat.

Let the total heat absorbed in the process $1 \to 2 \to 3$ be Q_1.

Similarly, one can argue that the gas rejects heat (say Q_0) to the surrounding in the process $3 \to 1$. Let work done in the cycle be W_1.



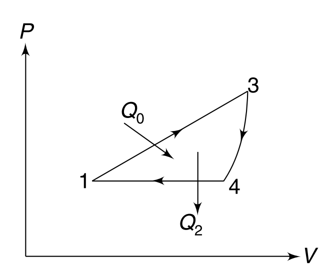
Using first law of thermodynamics for complete cycle–

$$\Delta Q = \Delta U + W \qquad [\Delta U_{\text{cycle}} = 0]$$

$$Q_1 - Q_0 = W_1 \quad \text{...(i)}$$

And efficiency $\quad \eta = \dfrac{W_1}{Q_1}$

$$\frac{1}{4} = \frac{W_1}{Q_1} \quad \text{...(ii)}$$



Cycle 1 – 3 – 4 – 1

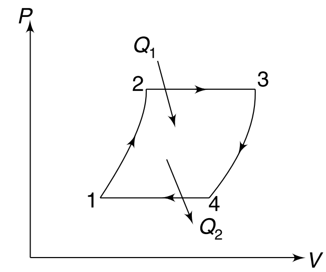
Q_0 = heat gained by the gas in process $1 \rightarrow 3$.

Q_2 = heat rejected by the gas in process $3 \rightarrow 4 \rightarrow 1$

W_2 = work done in cycle.

$$\eta = \frac{W_2}{Q_0}$$

$$\frac{1}{10} = \frac{W_2}{Q_1 - W_1} \qquad \text{[using (i)]}$$



$$\Rightarrow \qquad Q_1 - W_1 = 10W_2$$

$$\Rightarrow \qquad Q_1 = W_1 + 10W_2 \qquad \text{...(iii)}$$

Using (i) and (iii)

$$4W_1 = W_1 + 10W_2$$

$$3W_1 = 10W_2 \qquad \text{...(iv)}$$

Efficiency of cycle 1 – 2 – 3 – 4 – 1

$$\eta = \frac{W_1 + W_2}{Q_1} = \frac{W_1 + W_2}{W_1 + 10W_2}$$

$$= \frac{1 + \frac{W_2}{W_1}}{1 + 10\,\frac{W_2}{W_1}} = \frac{1 + \frac{3}{10}}{1 + 3} \qquad \text{[using 4]}$$

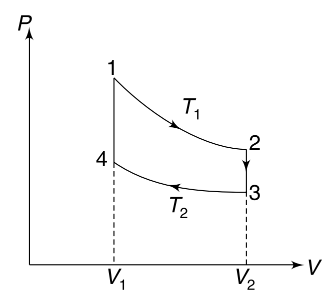
$$= \frac{13}{40}$$

In percentage: $$\eta = \frac{13}{40} \times 100 = 32.5\%$$

53. $$W_{\text{cycle}} = W_{12} + W_{34} = nRT_1 \ln \frac{V_2}{V_1} + nRT_2 \ln \frac{V_1}{V_2}$$

$$= \left(nR \ln \frac{V_2}{V_1}\right)(T_1 - T_2)$$

$$Q_{12} = W_{12}$$

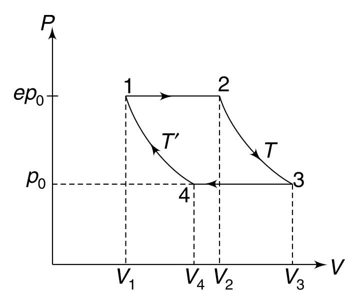


Heat rejected during 3-4 is a wastage.

$\therefore$ Efficiency $$\eta = \frac{\left(nR \ln \frac{V_2}{V_1}\right)(T_1 - T_2)}{\left(nR \ln \frac{V_2}{V_1}\right)(T_1)}$$

$$\therefore \qquad \eta = \frac{T_1 - T_2}{T_1}$$

54. Isobaric process 1-2



$$W_{12} = eP_0(V_2 - V_1) = nR(T - T')$$

$$\Delta U_{12} = nC_V \Delta T = \frac{nR}{\gamma - 1}(T - T')$$

$$\therefore \qquad \Delta Q_{12} = nR(T - T')\left[1 + \frac{1}{\gamma - 1}\right] = \frac{nR\gamma}{\gamma - 1}(T - T')$$

Isothermal process 2-3:

$$W_{23} = nRT\ln\left(\frac{P_2}{P_3}\right) = nRT\ln e = nRT$$

$$\Delta Q_{23} = W_{23} = nRT$$

Isobaric process 3-4:

$$W_{34} = P_0(V_4 - V_3) = P_1V_1 - P_2V_2 = -W_{12}$$

$$\Delta U < 0$$

Heat is rejected in the process.

Isothermal process 4-1:

$$W_{41} = nRT'\ln\left(\frac{P}{eP_0}\right) = -nRT'$$

Total work done in cycle = $nRT - nRT' = nR(T - T')$

Total heat absorbed in cycle $= \dfrac{nR\gamma}{\gamma - 1}(T - T') + nRT$

$$= \frac{nR}{\gamma - 1}[\gamma T - \gamma T' + \gamma T - T]$$

$$= \frac{nR[(2\gamma - 1)T - \gamma T']}{\gamma - 1}$$

$\therefore$ Efficiency $\quad \eta = \dfrac{\text{Total work done}}{\text{total heat absorbed}} = \dfrac{(T - T')(\gamma - 1)}{(2\gamma - 1)T - \gamma T'}$

55. The line 1-2 passes through origin if extended. It means $\rho \propto P$. The process must be isothermal.

$$\Delta W_{12} = nRT\ln\frac{P_1}{P_2} = P_1V_1\ln\frac{1}{2} = P_0\frac{M}{\rho_0}\ln\left(\frac{1}{2}\right) = -\frac{P_0M}{\rho_0}\ln 2$$

$$\Delta U_{12} = 0$$

$\therefore$

$$\Delta Q_{12} = -\frac{P_0M}{\rho_0}\ln 2$$

$$\text{Heat rejected} = \frac{P_0M}{\rho_0}\ln 2$$

Process 2-3 is isobaric

$$\Delta W_{23} = 2P_0(V_3 - V_2) = 2P_0\left(\frac{M}{\rho_3} - \frac{M}{\rho_2}\right)$$

$$= 2P_0M\left(\frac{1}{\rho_0} - \frac{1}{2\rho_0}\right) = \frac{P_0M}{\rho_0}$$

$$\Delta U_{23} = nC_V\Delta T = 1 \times \frac{3}{2}R[T_3 - T_2]$$

$$= \frac{3}{2}[P_3V_3 - P_2V_2] = \frac{3}{2}\cdot 2P_0\left(\frac{M}{\rho_0} - \frac{M}{2\rho_0}\right) = \frac{3}{2}\frac{P_0M}{\rho_0}$$

$$\Delta Q_{23} = \Delta W_{23} + \Delta U_{23} = \frac{5}{2}\frac{P_0M}{\rho_0}$$

Process 3-1 is isochoric.

$$\Delta W_{31} = 0$$

$$\Delta U_{31} = nC_V\Delta T = n\cdot\frac{3}{2}R[T_1 - T_3]$$

$$= \frac{3}{2}[P_1V_1 - P_3V_3] = \frac{3}{2}\left[\frac{P_0M}{\rho_0} - \frac{2P_0M}{\rho_0}\right]$$

$$= -\frac{3}{2}\frac{P_0M}{\rho_0}$$

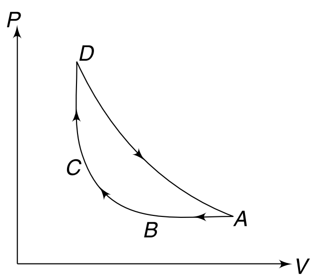
$$\Delta Q_1 = -\frac{3}{2}\frac{P_0M}{\rho_0}$$

(a) $$\Delta Q_{\text{rejected}} = |\Delta Q_{12}| + |\Delta Q_{31}| = \frac{P_0M}{\rho_0}\left(\frac{3}{2} + \ln 2\right)$$

(b) $$\eta = \frac{\text{Work done}}{\text{Heat supplied}} = \frac{\frac{P_0M}{\rho_0} - \frac{P_0M}{\rho_0}\ln 2}{\frac{5}{2}\frac{P_0M}{\rho_0}}$$

$$= \frac{1 - \ln 2}{\frac{5}{2}} = \frac{2}{5}(1 - \ln 2)$$

56. (a)



(b) Let $V_C = V_D = V_0$ (minimum volume)

$V_A = 8\sqrt{2}V_0$ (maximum volume)

Obviously T is maximum at A and D ($\because T \propto PV$) and minimum at B. In adiabatic compression temperature is increased. Hence $T_C > T_B$.

Let $T_B = T_0$; then $T_A = T_D = 4T_0$

For process AB $V \propto T$

$$\therefore \quad V_B = T_B\left(\frac{V_A}{T_A}\right) = T_0\left(\frac{8\sqrt{2}V_0}{4T_0}\right) = 2\sqrt{2}V_0$$

Process BC is adiabatic

$$\therefore \quad T_C = T_B\left(\frac{V_B}{V_C}\right)^{\gamma - 1} = T_0\left(\frac{2\sqrt{2}\,V_0}{V_0}\right)^{\frac{5}{3} - 1} = 2T_0 \qquad \left[\gamma = \frac{5}{3} \text{ for He}\right]$$

$$\therefore \quad T_A = 4T_0,\ T_B = T_0,\ T_C = 2T_0 \text{ and } T_D = 4T_0$$

Now in process AB (isobaric)-

$$\Delta W_{AB} = \Delta Q_{AB} - \Delta U_{AB} = nC_P\Delta T - nC_V\Delta T = nR\Delta T$$

$$= nR(T_B - T_A) = -3nRT_0$$

Also ΔQ_{AB} is $-ve$

In process BC (adiabatic) $\Delta Q_{BC} = 0$

$$\Delta W_{BC} = -\Delta U_{BC} = -nC_V\Delta T = n\left(\frac{3}{2}R\right)(T_B - T_C) = -\frac{3}{2}nRT_0$$

For CD (isochoric); $\Delta W_{CD} = 0$

$$\Delta Q_{CD} = \Delta U_{CD} = nCV(T_D - T_C) = n\left(\frac{3}{2}R\right)(2T_0) = 3nRT_0$$

For DA (isothermal): $\Delta U_{DA} = 0$

$$\Delta Q_{DA} = \Delta W_{DA} = nRT_D \ln \frac{V_A}{V_B} = nR(4T_0)\ln(8\sqrt{2})$$

$$= 14nRT_0 \ln 2$$

$\therefore$ Total work done; $\quad W = -3nRT_0 - \frac{3}{2}nRT_0 + 14nRT_0 \ln 2$

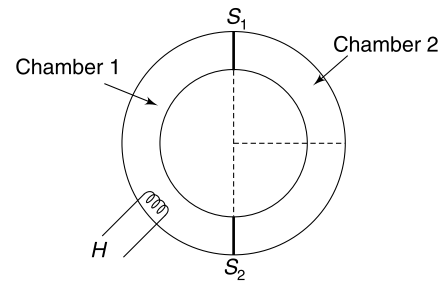
$$= nRT_0\left[14\ln 2 - \frac{9}{2}\right] = 5.202\ nRT_0$$

Heat absorbed $\quad Q = 3nRT_0 + 14nRT_0 \ln(2)$

$$= 12.702\ nRT_0$$

$\therefore$
$$\eta = \frac{W}{Q} \times 100 = \frac{5.202}{12.702} \times 100 = 41\%$$

57. The process in chamber 2 is adiabatic.



Let final pressure in chamber 2 be P_2.

$$P_2\left(\frac{V_0}{2}\right)^{\gamma} = P_0V_0^{\gamma}$$

$\Rightarrow$
$$P_2 = (2)^{3/2}P_0 \qquad \left[\because\ \gamma = \frac{3}{2}\right]$$

$$= 2\sqrt{2}\,P_0$$

In equilibrium, pressure in both chambers must be same, hence final pressure in chamber 1 is

$$P_1 = P_2 = 2\sqrt{2}\,P_0$$

Work done on the gas in chamber 2 by gas in chamber 1 is

$$W = \frac{P_2V_2 - P_0V_0}{\gamma - 1} = \frac{2\sqrt{2}\cdot P_0\frac{V_0}{2} - P_0V_0}{\frac{3}{2} - 1}$$

$$W = 2(\sqrt{2} - 1)P_0V_0$$

Change in internal energy of the gas in chamber 1

$$\Delta U = nC_V\Delta T = \frac{nR\Delta T}{\gamma - 1} = 2nR\Delta T$$

$$= 2[P_1V_1 - P_0V_0] = 2\left[2\sqrt{2}P_0\cdot\frac{3}{2}V_0 - P_0V_0\right]$$

$$= 2(3\sqrt{2} - 1)\ P_0V_0$$

$\therefore$
$$\Delta Q = \Delta U + W = 2(3\sqrt{2} - 1)\ P_0V_0 + 2(2\sqrt{2} - 1)\ P_0V_0$$

$$= 2P_0V_0[3\sqrt{2} - 1 + \sqrt{2} - 1] = 4P_0V_0[2\sqrt{2} - 1]$$

58. (a) Equation of a parabola passing through origin will be of the form $y^2 = kx$

$\Rightarrow$
$$T^2 = \frac{k}{V} \quad \Rightarrow \quad T^2V = k$$

$\Rightarrow$
$$TV^{1/2} = k$$

Because the process is adiabatic, $TV^{\gamma - 1}$ = Const.

$\therefore$
$$\gamma - 1 = \frac{1}{2} \quad \Rightarrow \quad \gamma = \frac{3}{2}$$

$$\frac{V_{\text{rms}}}{V_{\text{sound}}} = \frac{\sqrt{\frac{3RT}{M}}}{\sqrt{\frac{\gamma RT}{M}}} = \sqrt{\frac{3}{\gamma}} = \sqrt{2}$$

(b) During the process volume changed from V_0 to $\frac{V_0}{1.21}$

$\therefore \quad P_0 V_0^{\gamma} = P\left(\frac{V_0}{1.21}\right)^{\gamma}$

$\Rightarrow \quad P = P_0(1.21)^{3/2} = P_0(1.1)^3 = 1.33 P_0$

Increase in pressure

$$\Delta P = 0.03 P_0 = 0.33 \times 10^5 \text{ N/m}^2$$

$\therefore$ Weight of water over piston

$$mg = \Delta P A = 0.33 \times 10^5 \times 1.515 \times 10^{-3}$$
$$= 0.5 \times 10^2 = 50 \text{ N}$$

$\therefore \quad m = 5 \text{ kg}$

$\therefore$ Time required $\quad t = \frac{5}{0.25} = 20 \text{ sec}$

59. The rate at which energy flows into the house

$$\frac{dQ}{dt} \propto \Delta T \quad \Rightarrow \quad \frac{dQ}{dt} = k\Delta T$$

k is a constant that depends on geometry and material of the walls of the room. To maintain the temperature the air conditioner must remove heat at the same rate.

If β is COP, then $\frac{dQ/dt}{dW/dt} = \beta$

$\Rightarrow \quad \frac{dW}{dt} = \frac{1}{\beta}\frac{dQ}{dt} = \frac{k}{\beta}\Delta T$

Now $\quad \beta = \frac{T_0}{\Delta T}$

$\therefore \quad \frac{dW}{dt} = \frac{k(\Delta T)^2}{T_0}$

60. (i) $\quad \frac{Q_1}{Q_2} = \frac{T_1}{T_2}$

And $\quad Q_1 = Q_2 + W$

$\Rightarrow \quad Q_2 = Q_1 - W = Q_2\left(\frac{T_1}{T_2}\right) - W$

$\Rightarrow \quad W = Q_2\left(\frac{T_1}{T_2} - 1\right) \quad \Rightarrow \quad Q_2 = W\left(\frac{T_2}{T_1 - T_2}\right)$

$$\frac{dQ_2}{dt} = \frac{dW}{dt}\left(\frac{T_2}{T_1 - T_2}\right)$$

$\therefore \quad \frac{dQ_2}{dt} = P\left(\frac{T_2}{T_1 - T_2}\right) \quad$ ...(i)

(ii) From $\quad k\Delta T = P\,\frac{T_2}{\Delta T}$

$$\Rightarrow \qquad k\Delta T^2 = P(T_1 - \Delta T) \;\Rightarrow\; k\Delta T^2 + P\Delta T - PT_1 = 0$$

$$\therefore \qquad \Delta T = \frac{-P \pm \sqrt{P^2 + 4PkT_1}}{2k}$$

Only +ve sign is meaningful

$$\therefore \qquad \Delta T = \frac{P}{k}\left[\sqrt{1 + \frac{4kT_1}{P}} - 1\right]$$

$$\therefore \qquad T_2 = T_1 - \Delta T = T_1 - \frac{P}{k}\left[\sqrt{1 + \frac{4kT_1}{P}} - 1\right]$$

61. Let instantaneous temperature of the two tanks be θ_1 and θ_2 respectively. A small amount of heat (dQ_1) is extracted from the source and its temperature falls by ($-d\theta_1$). In the same time a heat (dQ_2) is rejected into the sink and its temperature increases by $d\theta_2$.

(a) We know that–

$$\frac{dQ_1}{\theta_1} = \frac{dQ_2}{\theta_2}$$

$$\Rightarrow \qquad -\frac{msd\theta_1}{\theta_1} = \frac{msd\theta_2}{\theta_2} \;\Rightarrow\; -\int_{T_1}^{T_0}\frac{d\theta_1}{\theta_1} = \int_{T_2}^{T_0}\frac{d\theta_2}{\theta_2}$$

$$\Rightarrow \qquad \ln\frac{T_1}{T_0} = \ln\frac{T_0}{T_2} \;\Rightarrow\; T_0 = \sqrt{T_1T_2} = \sqrt{361 \times 289}$$

$$= 323K$$

(b)

$$W = \Delta Q_1 - \Delta Q_2$$

$$= ms(T_1 - T_0) - ms(T_0 - T_2)$$

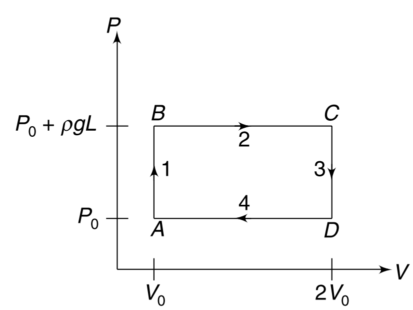
$$= ms(T_1 + T_2 - 2T_0) = 4ms$$

62. (a) Process 1 is a constant volume process. Pressure is increased from P_0 to $P_0 + \rho gL$

Process 2 is constant pressure process. Volume of gas in increased from $V_0(=AL)$ to $2V_0(=2AL)$

Process 3 is again a constant volume process in which pressure reduced from $P_0 + \rho gL$ to P_0.

Process 4 is again an isobaric process in which the gas is brought back to original volume. The $P - V$ graph is as shown.



(b) Heater supplies heat in the process 1 and 2

$$\Delta Q_H = \Delta Q_1 + \Delta Q_2 = nC_v\Delta T_1 + nC_p\Delta T_2$$

$$= \frac{5}{2}nR(T_B - T_A) + \frac{7}{2}nR(T_C - T_B)$$

$$= \frac{5}{2}[(P_0 + \rho gL)V_0 - P_0V_0] + \frac{7}{2}[(P_0 + \rho gL)\,2V_0 - (P_0 - \rho gL)\,V_0]$$

$$= \frac{5}{2}\rho gLV_0 + \frac{7}{2}(P_0 + \rho gL)\,V_0$$

$$= \frac{1}{2}(7P_0 + 12\rho gL)V_0$$

In the entire cycle $\qquad dU = 0$

First law of thermodynamics for entire cycle

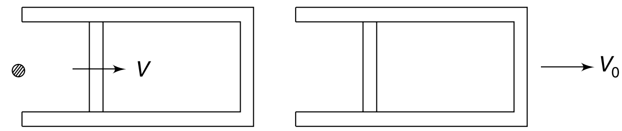
$$\Delta Q = \Delta U + W$$

$$\Delta Q = 0 + \rho g L V_0 \quad [W = \text{area under } P - V \text{ graph}]$$

$\therefore$ Heat extracted by cold liquid

$$\Delta Q_L = \Delta Q_H - \Delta Q = \frac{1}{2}(7P_0 + 10\rho g L)V_0 = \frac{1}{2}(7P_0 + 10\rho g L)AL$$

63. Immediately after collision, the particle comes to rest and the piston starts moving with velocity v, because the collision is elastic and two colliding objects have equal mass.



The compression is maximum when the piston stops moving relative to the cylinder.

Let v_0 = velocity of entire system at the point of maximum compression

The first figure shows situation immediately after collision and the second figure shows situation at maximum compression.

Momentum conservation gives–

$$5Mv_0 = Mv \quad \Rightarrow \quad v_0 = \frac{v}{5}$$

Energy conservation gives–

$$nC_v \Delta T = \frac{1}{2}Mv^2 - \frac{1}{2}(5M)\left(\frac{V}{5}\right)^2$$

$$1 \times \frac{3}{2}R\Delta T = \frac{2}{5}Mv^2 \quad \left[\text{for monatomic gas } C_v = \frac{3}{2}R\right]$$

$$\Delta T = \frac{4Mv^2}{15R}$$

64. (a)

$$C = \frac{\Delta Q}{\Delta T} \quad (\text{for } n = 1)$$

$$\therefore \quad C = \frac{\Delta U}{\Delta T} + \frac{\Delta W}{\Delta T} = C_v + \frac{\Delta W}{\Delta T}$$

Let's find ΔW in the process $PV^x = k$ as the state of gas changes from (P_1, V_1, T_1) to (P_2, V_2, T_2)

$$\Delta W = \int_{V_1}^{V_2} Pdv = k\int_{V_1}^{V_2} V^{-x}dv = \left[\frac{kV^{-x+1}}{-x+1}\right]_{V_1}^{V_2}$$

$$= \frac{kV_2^{-x+1} - kV_1^{-x+1}}{-x+1}$$

$$= \frac{P_2V_2^xV_2^{-x+1} - P_1V_1^xV_1^{-x+1}}{1-x} = \frac{P_2V_2 - P_1V_1}{1-x} \quad [\because \; P_1V_1^x = P_2V_2^x = k]$$

$$\therefore \quad \Delta W = \frac{RT_2 - RT_1}{1-x} = \frac{R\Delta T}{1-x}$$

$$\frac{\Delta W}{\Delta T} = \frac{R}{1-x}$$

$$\therefore \quad C = C_v + \frac{R}{1-x}$$

(b)

$$Q = \Delta U + \Delta W$$

$$\Rightarrow \quad Q = \Delta U + \frac{Q}{2} \quad \Rightarrow \quad \Delta U = \frac{Q}{2}$$

$$\therefore \qquad \Delta U = \Delta W \quad \Rightarrow \quad nC_v dT = PdV$$

$$n\frac{3R}{2}\,dT = PdV \qquad [\because \quad PV = nRT \;\therefore\; Pdv + VdP = nRdT]$$

$$\Rightarrow \qquad \frac{3}{2}[PdV + VdP] = PdV$$

$$\Rightarrow \qquad PdV = -3VdP \quad \Rightarrow \quad \int \frac{dV}{V} = -3\int \frac{dP}{P}$$

$$\Rightarrow \qquad \ln V = -3\ln P + \ln K$$

$$\Rightarrow \qquad \ln P^3 V = \ln K$$

$$\Rightarrow \qquad P^3 V = \text{constant}$$

As shown in part (a) $\qquad C = C_V + \dfrac{R}{1-x}$

$$\therefore \qquad C = \frac{3}{2}R + \frac{R}{1-\frac{1}{3}} = 3R$$

Alternatively, we can simply say

$$dQ = dU + dW$$

$$\Rightarrow \qquad dQ = 2dU \qquad [\because \quad dU = dW]$$

$$\Rightarrow \qquad nCdT = 2nC_V dT \quad \Rightarrow \quad C = 2C_V$$

$$= 2\cdot\frac{3}{2}R = 3R$$

65. (a) Rate of teat flow through the piston is $\dfrac{dQ}{dt} = \dfrac{KA(T_1 - T)}{d}$

Where T is the instantaneous temperature of the gas.

The gas receives this heat at a constant pressure. Hence,

$$nC_p\frac{dT}{dt} = \frac{dQ}{dt}$$

$$\Rightarrow \qquad nC_p\frac{dT}{dt} = \frac{KA(T_1 - T)}{d}$$

$$\Rightarrow \qquad \frac{P_0V_0}{RT_0}\cdot\frac{5}{2}R\frac{dT}{dt} = \frac{KA}{d}(T_1 - T)$$

$$\Rightarrow \qquad \frac{5P_0V_0d}{2T_0KA}\int_{T_0}^{T}\frac{dT}{T_1 - T} = \int_0^t dt$$

$$\Rightarrow \qquad \ln\left(\frac{T_1 - T}{T_1 - T_0}\right) = -\frac{2T_0KA}{5P_0V_0d}\,t$$

$$\Rightarrow \qquad T_1 - T = (T_1 - T_0)\,e^{\left(\frac{-2T_0KAt}{5P_0V_0d}\right)}$$

$$\therefore \qquad T = T_1 - (T_1 - T_0)\,e^{\left(\frac{-2T_0KAt}{5P_0V_0d}\right)} \qquad \text{...(i)}$$

(b) Let the height at time t be h and initial height be $h_0\left(=\dfrac{V_0}{A}\right)$

$$\frac{PV}{RT} = \frac{P_0V_0}{RT_0}$$

$$\therefore \qquad P_0Ah = P_0V_0\left(\frac{T}{T_0}\right)$$

$$\Rightarrow \qquad h = \frac{V_0}{A}\left(\frac{T}{T_0}\right)$$

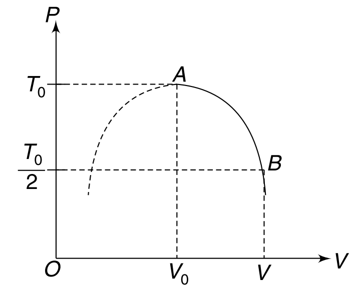
$\therefore$ Height raised $\quad \Delta h = h - h_0 = h - \dfrac{V_0}{A}$

$$= \frac{V_0}{A}\left[\frac{T}{T_0} - 1\right] = \frac{V_0}{AT_0}(T - T_0)$$

$$= \frac{V_0}{AT_0}\left[(T_1 - T_0) - (T_1 - T_0)e^{\frac{-2T_0KAt}{5P_0V_0d}}\right]$$

$$= \frac{V_0(T_1 - T_0)}{AT_0}\left[1 - e^{\frac{-2T_0KAt}{5P_0V_0d}}\right]$$

66. Equation of the given straight line is the process equation.



$$P = -\left(\frac{P_0}{V_0}\right)V + 2P_0 \quad \text{...(i)}$$

But for an ideal gas $\quad PV = nRT$

$$\therefore \quad \left[-\frac{P_0}{V_0}V + 2P_0\right]V = nRT$$

$$\Rightarrow \quad T = -\frac{P_0}{V_0R}V^2 + \frac{2P_0}{R}V \quad [\text{As } n = 1] \quad \text{...(ii)}$$

This is equation of a parabola.

The vertex of parabola corresponds to $\dfrac{dT}{dV} = 0$

$$-\frac{2P_0}{V_0R}V + \frac{2P_0}{R} = 0 \quad \Rightarrow \quad V = V_0$$

Temperature in this state is

$$T_0 = \frac{P_0V_0}{nR}$$

$$\because \quad n = 1 \quad \therefore \quad T_0 = \frac{P_0V_0}{R}$$

This is the maximum temperature during the process. If final temperature at B is $\dfrac{T_0}{2} = \dfrac{P_0V_0}{2R}$ then volume can be obtained using (ii)

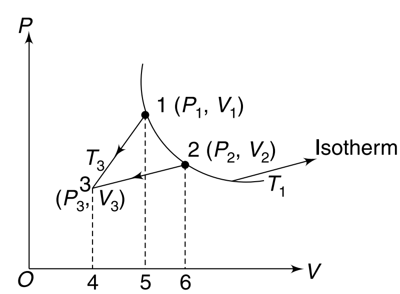
$$\frac{P_0V_0}{2R} = -\frac{P_0}{RV_0}V^2 + \frac{2P_0}{R}V$$

$$\Rightarrow \quad \frac{V_0}{2} = -\frac{V^2}{V_0} + 2V$$

$$\Rightarrow \quad 2V^2 - 4V_0V + V_0^2 = 0$$

$$\therefore \quad V = \left(\frac{\sqrt{2} + 1}{\sqrt{2}}\right)V_0$$

67. State 1 and 2 lie on an isotherm



$$\therefore \quad \Delta U_{1-3} = \Delta U_{2-3}$$

Hence, more heat will be rejected in the process in which work done on the gas is higher.

$$W_{1-3}^{\text{on gas}} = \text{area}(1345)$$

$$= \frac{1}{2} \times (V_1 - V_3)(P_1 + P_3)$$

$$= \frac{1}{2}[P_1V_1 + P_3V_1 - P_1V_3 - P_3V_3]$$

$$W_{2-3}^{\text{on gas}} = \text{area } (2346)$$

$$= \frac{1}{2} \times (V_2 - V_3)\ (P_2 + P_3)$$

$$= \frac{1}{2}[V_2P_2 + V_2P_3 - V_3P_2 - V_3P_3]$$

$$W_{1-3}^{\text{on gas}} - W_{2-3}^{\text{on gas}} = \frac{1}{2}[P_3(V_1 - V_2) + V_3(P_2 - P_1)]$$

But $V_1 - V_2 < 0$ and $P_2 - P_1 < 0$

$\therefore$ $W_{1-3}^{\text{on gas}} - W_{2-3}^{\text{on gas}} < 0$

$\therefore$ $W_{1-3}^{\text{on gas}} < W_{2-3}^{\text{on gas}}$

$\therefore$ $Q_1 < Q_2$

68. Initial volume

$$V_1 = AH + \frac{A}{2}\frac{H}{2} = \frac{5AH}{4}$$

Find volume $V_2 = A\frac{H}{2} + \frac{A}{2}\left[\frac{H}{2} - \frac{3H}{32}\right] = \frac{45AH}{64}$

Process is adiabatic

$\therefore$ $P_2V_2^{\gamma} = P_1V_1^{\gamma}$

$$P_2\left(\frac{45AH}{64}\right)^{3/2} = P_1\left(\frac{5AH}{4}\right)^{3/2}$$

$$P_2\left(\frac{9}{16}\right)^{3/2} = P_1 \quad \Rightarrow \quad P_2 = \frac{64}{27}P_1 \qquad \text{...(i)}$$

But $P_1\frac{A}{2} = Mg + P_0\frac{A}{2}$ ...(ii)

[P_0 = atmospheric pressure, M = mass of piston P_1]

And $P_2\frac{A}{2} = Mg + \frac{P_0A}{2} + k\frac{3H}{32}$ ...(ii)

$\Rightarrow$ $\frac{64}{27}P_1\frac{A}{2} = Mg + \frac{P_0A}{2} + k\frac{3H}{32}$

$\Rightarrow$ $\frac{64}{27}\left[Mg + \frac{P_0A}{2}\right] = \left(Mg + P_0\frac{A}{2}\right) + \frac{3kH}{32}$

$\Rightarrow$ $\frac{37}{27}\left[Mg + \frac{P_0A}{2}\right] = \frac{3kH}{32}$

$\therefore$ $H = \frac{37}{27} \times \frac{32}{3} \times \frac{1}{3700}\left[\frac{27}{2} \times 10 + \frac{10^5 \times 27 \times 10^{-4}}{2}\right] = \frac{16}{15}\ m$

$$T_2V_2^{\gamma-1} = T_1V_1^{\gamma-1}$$

$$T_2\left[\frac{45}{64}AH\right]^{1/2} = T_1\left[\frac{5}{4}AH\right]^{1/2}$$

$\Rightarrow$ $T_2 = \frac{4}{3} \times T_1$

$\Rightarrow$ $T_2 = 400\ \text{K}$

69. Gas in A and B are always at equal temperature. Process for gas in C is adiabatic.

$$\therefore \quad P_0V_0^{3/2} = P_C\left(\frac{4V_0}{9}\right)^{3/2} \quad \therefore \quad P_C = \frac{27}{8}P_0$$

For mechanical equilibrium $\quad P_A = P_C$

$$\therefore \quad P_A = P_C = \frac{27}{8}P_0$$

Again for C: $\quad T_C\left(\frac{4V_0}{9}\right)^{3/2-1} = T_0V_0^{\frac{3}{2}-1} \Rightarrow T_C = \frac{3}{2}T_0$

For A: $\quad P_0V_0 = nRT_0$

And $\quad P_A\left[V_0 + \frac{5V_0}{9}\right] = nRT_A$

$$\left[\because \text{ volume change} = V_0 - \frac{4V_0}{9} = \frac{5V_0}{9}\right]$$

$$\therefore \quad \frac{27}{8}P_0\frac{14}{9}V_0 = nRT_A$$

$$\therefore \quad T_A = \frac{21}{4}T_0$$

Also $\quad T_A = T_B = \frac{21}{4}T_0$

(a) Work done on gas in C = work done by gas in A.

$$W_A = -\left(\frac{P_0V_0 - P_CV_C}{\gamma - 1}\right) = -\frac{P_0V_0 - \frac{27}{8}P_0\frac{4V_0}{9}}{0.5} = P_0V_0$$

Work done by gas in B is zero.

Change in internal energy of $A + B$ is

$$\Delta U_{A+B} = 2nC_V\Delta T = 2n\frac{R}{\gamma - 1}\left(\frac{21}{4} - 1\right)T_0$$

$$= \frac{2nR}{0.5} \times \frac{17}{4}T_0 = 17nRT_0 = 17P_0V_0$$

$$Q_{\text{Heater}} = \Delta U_{A+B} + W_{\text{by}A} = 18P_0V_0$$

(b) Heat flow through piston 1

$$H = \Delta U_A + W_{\text{by}A}$$

$$= \frac{17}{2}P_0V_0 + P_0V_0 = \frac{19}{2}P_0V_0$$

(c) Answer to (a) does not change answer to (b) becomes

$$H = \Delta U_B = \frac{17}{2}P_0V_0$$

70. The right chamber has 1 mole N_2. Out of this $\frac{1}{3}$ mole N_2 dissociates into atoms. Therefore, the chamber has a mixture of $\frac{2}{3}$ mole of diatomic gas and $\frac{2}{3}$ mole of mono atomic gas

$$\therefore \quad C_V = \frac{n_1C_{V_1} + n_2C_{V_2}}{n_1 + n_2} \quad \left[n_1 = n_2 = \frac{2}{3}\right]$$

$$= \frac{C_{V_1} + C_{V_2}}{2} = \frac{\frac{3}{2}R + \frac{5}{2}R}{2} = 2R$$

$$C_P = C_V + R = 3R$$

$$\therefore \quad \gamma = \frac{C_P}{C_V} = \frac{3}{2}$$

For the adiabatic compression in this part $PV^\gamma = P_0V_0^\gamma$

$$P\left(\frac{AL}{4}\right)^\gamma = P_0(AL)^\gamma \quad \Rightarrow \quad P = (4)^{\frac{3}{2}}P_0 = 8P_0$$

Work done by nitrogen during the process

$$W_{Ad} = \frac{P_0V_0 - PV}{\gamma - 1} = \frac{P_0V_0 - 8P_0\dfrac{V_0}{4}}{\dfrac{3}{2} - 1}$$

$$= -4P_0V_0 = -4P_0AL$$

Work done on nitrogen by the gas in the other chamber is

$$W = 4P_0AL$$

Let the final pressure of He chamber be P_1. For equilibrium

$$P_1A = k\left(\frac{3L}{4}\right) + PA$$

$$\therefore \quad P_1 = \frac{3}{4}\frac{kL}{A} + 8P_0$$

Change in temperature of He is

$$\Delta T = \frac{P_1V_1 - P_0V_0}{n \cdot R} \quad [n = 1]$$

$$= \frac{\dfrac{3}{4}\dfrac{kL}{A}\dfrac{5}{4}AL + 8P_0\dfrac{5}{4}AL - P_0AL}{R}$$

$$= \frac{1}{R}\left[\frac{15}{16}kL^2 + 9P_0AL\right]$$

$\therefore$ Change in internal energy of He

$$\Delta U = nC_V\Delta T = \frac{3}{2}\left[\frac{15}{16}kL^2 + 9P_0AL\right]$$

$$= \frac{45}{32}kL^2 + \frac{27}{2}P_0AL$$

$$\therefore \quad \Delta Q = \Delta U + \text{work to compress the spring} + \text{work on nitrogen}$$

$$\therefore \quad \Delta Q = \frac{45}{32}kL^2 + \frac{27}{2}P_0AL + \frac{1}{2}k\left(\frac{3L}{4}\right)^2 + 4P_0AL$$

$$= \frac{27}{16}kL^2 + \frac{35}{2}P_0AL$$

Chapter 5

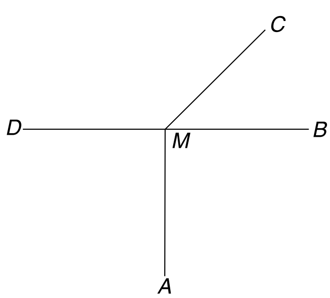
Heat Transfer

LEVEL 1

Q. 1: Four metal rods each of length L and cross sectional area A are joined at point M. Thermal conductivities of MA, MB and MD are equal and that of MC is thrice that of MA. The end points A, B, C and D are kept in large reservoirs. Heat flows into the junction from B at a rate of $P(\text{Js}^{-1})$ and from C at a rate of $3P$. Heat flows out of D at a rate of $5P$.

(a) Find relation between temperatures of points A, B and C.

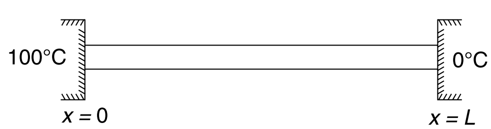
(b) Find temperature of D if temperature of A and M and T_A and T_M respectively.



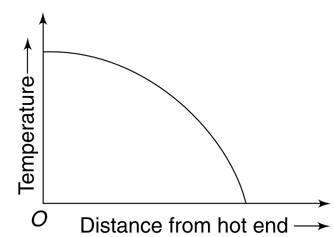
Q. 2: The two ends of a uniform metallic rod are maintained at 100°C and 0°C as shown in the figure. Assume that end of the rod at 100°C is at $x = 0$ and the other end at 0°C is at $x = L$. Plot the variation of temperature as x changes from 0 to L in steady state. Consider two cases.

(a) The rod is perfectly lagged.

(b) The rod is not lagged and surrounding is at 0°C.



Q.3: The ends of a metallic bar are maintained at different temperature and there is no loss/gain of heat from the sides of the bar due to conduction or radiation. In the steady state the temperature variation along the length of the bar is a shown in the figure what do you think about the cross sectional area of the bar?



Q.4: A thick spherical shell of inner and outer radii r and R respectively has thermal conductivity $k = \frac{\rho}{x^n}$, where ρ is a constant and x is distance from the centre of the shell. The inner and outer walls are maintained at temperature T_1 and T_2 ($< T_1$)

(a) Find the value of number n (call it n_0) for which the temperature gradient remains constant throughout the thickness of the shell.

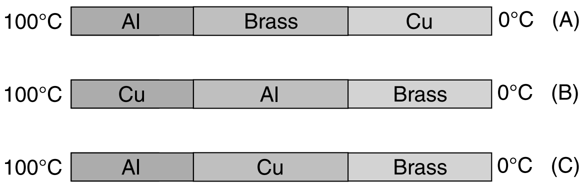
(b) For $n = n_0$, find the value of x at which the temperature is $\frac{T_1 + T_2}{2}$

(c) For $n = n_0$, calculate the rate of flow of heat through the shell.

Q. 5: Three bars of aluminium, brass and copper are of equal length and cross section. The three pieces are joined together as shown in A, B and C and the ends are maintained at 100°C and 0°C. The thermal conductivities of aluminium, brass and copper are in ratio 2 : 1 : 4. Assume no heat loss through curved surface of the bar and that the system is in steady state.

(a) In which of the three cases (A, B or C) the temperature difference across aluminium bar will be maximum?

(b) Draw a graph showing variation of temperature from one end of the bar to another in case B.



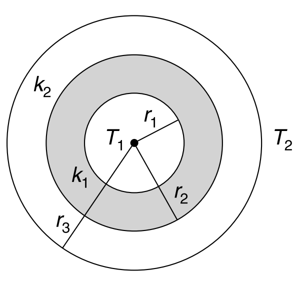
Q. 6: A lake is covered with ice 5 cm thick and the atmospheric temperature above the ice is –10°C. At what rate (in cm/hour) will the ice layer thicken?

Thermal conductivity of ice = 0.005 cgs unit, density of ice = 0.9 g/cc and latent heat of fusion of ice = 80 cal/g.

Q. 7. A liquid having mass m = 250 g is kept warm in a vessel by use of an electric heater. The liquid is maintained at 50°C when the power supplied by heater is 30 watt and surrounding temperature is 20°C. As the heater is switched off, the liquid starts cooling and it was observed that it took 10 second for temperature to fall down from 40°C to 39.9°C. Calculate the specific heat capacity of the liquid. Assume Newton's law of cooling to be applicable.

Q.8: (i) A cylindrical pipe of length L has inner and outer radii as a and b respectively. The inner surface of the pipe is at a temperature T_1 and the outer surface is at a lower temperature of T_2. Calculate the radial heat current if conductivity of the material is K.

(ii) A cylindrical pipe of length L has two layers of material of conductivity K_1 and K_2. (see figure). If the inner wall of the cylinder is maintained at T_1 and outer surface is at T_2 (< T_1), calculate the radial rate of heat flow.



Q. 9: A 3 mm diameter and 5 m long copper wire is insulated using a 2 mm thick plastic cover whose thermal conductivity is K = 0.15 $Wm^{-1}K^{-1}$. The wire has a potential difference of 10 V between its ends and the current through it is 8A. The outer surface of the wire is at 30°C. Neglect convection.

(i) Calculate the temperature at the interface of the wire and the plastic cover.

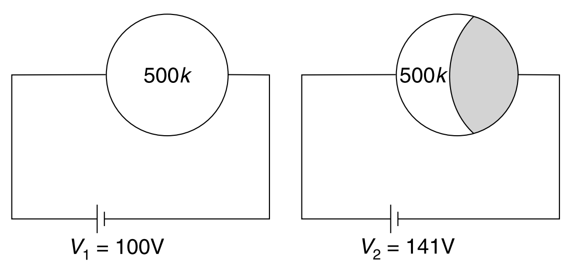
(ii) Determine whether doubling the thickness of the plastic cover will increase or decrease the interface temperature. [Given $\ell n(2.33) = 0.85$]

Q. 10: A potato at initial temperature T_0 is placed inside a hot convection oven maintained at a constant temperature $T_1 (> T_0)$. Assume that the potato receives heat only because of convection phenomenon and the rate at which it receives heat is given as $hA(T_1 - T)$ where h is a constant, A is surface area of the potato and T is instantaneous temperature of the potato. Mass and specific heat capacity of the potato are m and s respectively. In how much time the potato will be at a temperature $T_2 = \dfrac{T_0 + T_1}{2}$? Assume no change in volume of the potato.

Q. 11: What is emissivity of a perfectly reflecting surface?

Q. 12: A body is in thermal equilibrium with surrounding. Absorptive power of the surface of the body is a = 0.5. E is the radiant energy incident in unit time on the surface of the body. How much energy propagates from its surface in unit time?

Q. 13: A copper sphere is maintained at 500 K temperature by connecting it to a battery of emf V_1 = 100 V (see figure). The surrounding temperature is 300 K. When half the surface of the copper sphere is completely blackened (so that the surface behaves almost like a black body), a cell of emf V_2 = 141 V is needed to maintain its temperature at 500 K. Calculate the emissivity of the copper surface.

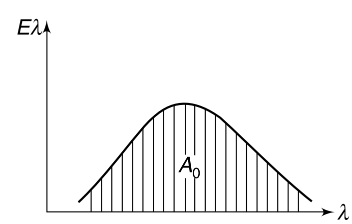


Q. 14: Stefan's contain (σ) derives from other known constant of nature, viz. Boltzmann constant, (k) planck's constant (h) and speed of light in vacuum (c). Value of the constant is

$$\sigma = 5.67 \times 10^{-8}\ Js^{-1}m^{-2}K^{-4}$$

If speed of light were 2% more than its present value, how much different (in percentage) the value of σ would have been?

Q. 15: An iron ball is heated to 727°C and it appears bright red. The plot of energy density distribution versus wavelength is as shown. The graph encloses an area A_0 under it. Now the ball is heated further and it appears bright yellow. Find the area (A) of the energy density graph now.



If the given that wavelengths for red and yellow light are 8000 Å and 6000 Å respectively.

Q. 16: A solid cylinder and a sphere of same material are suspended in a room turn by turn, after heating them to the same temperature. The cylinder and the sphere have same radius and same surface area.

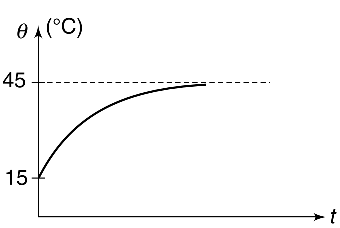
(a) Find the ratio of initial rate of cooling of the sphere to that of the cylinder.

(b) Will the ratio change if both the sphere and the cylinder are painted with a thin layer of lamp black?

Q. 17: A block body at temperature T radiates same amount of energy in the wavelength range λ_1 to $\lambda_1 + \Delta\lambda$ and λ_2 to $\lambda_2 + \Delta\lambda$. It is given that $\Delta\lambda << \lambda_1$ or λ_2 and $\lambda_2 > \lambda_1$. Prove that $\frac{b}{\lambda_1} > T > \frac{b}{\lambda_2}$ where b is Wien's constant.

Q. 18: Majority of radiation from the Sun is in visible and near infra-red (0.7 to 4 μm) region. What can you say about the composition of the radiation from the Earth?

Q. 19: A metal ball of mass 1.0 kg is kept in a room at 15°C. It is heated using a heater. The heater supplies heat to the ball at a constant rate of 24 W. The temperature of the ball rises as shown in the graph. Assume that the rate of heat loss from the surface of the ball to the surrounding is proportional to the temperature difference between the ball and the surrounding. Calculate the rate of heat loss from the ball when it was at temperature of 20°C.



Q. 20: A hot body is suspended inside a room that is maintained at a constant temperature. The temperature difference between the body and the surrounding becomes half in a time interval t_0. In how much time the temperature difference between the body and the surrounding will becomes $\frac{1}{4}$ the original value?

Q. 21: Newton's law of cooling says that the rate of cooling of a body is proportional to the temperature difference between the body and its surrounding when the difference in temperature is small.

(a) Will it be reasonable to assume that the rate of heating of a body is proportional to temperature difference between the surrounding and the body (for small difference in temperature) when the body is placed in a surrounding having higher temperature than the body?

(b) Assuming that our assumption made in (a) is correct estimate the time required for a cup of cold coffee to gain temperature from 10°C to 15°C when it is kept in a room having temperature 25°C. It was observed that the temperature of the cup increases from 5°C to 10°C in 4 min.

Q. 22: "Blue hot is hotter than red hot". Explain.

Q. 23: A planet of radius r_0 is at a distance r from the sun ($r >> r_0$). The sun has radius R. Temperature of the planet is T_0, and that of the surface of the sun is T_s. Calculate the temperature of another planet whose radius is $2r_0$ and which is at a distance $2r$ from the sun. Assume that the sun and the planets are black bodies.

Q. 24: A star having radius R has a small planet revolving around it at a distance d ($>> R$). The star and the planet both behave like black bodies and radiate maximum amount of energy at wavelength λ_s and λ_p respectively.

(i) Find d in terms of other given parameters.

(ii) Show that $\lambda_p >> \lambda_s$

Q. 25: A 20 mm diameter copper pipe is used to carry heated water. The external surface of the pipe is at $T = 80$°C and its surrounding is at $T_0 = 20$°C. The outer surface of the pipe radiates like a black body and also loses heat due to convection. The convective heat loss per unit area per unit time is given by $h(T - T_0)$ where $h = 6\,\text{W (m}^2\text{K)}^{-1}$. Calculate the total heat lost by the pipe in unit time for one meter of its length.

Q. 26: Solar constant, I_s is defined as intensity of solar radiation incident on the Earth. Its value is close to 1.4 kW/m^2. Nearly 68% of this energy is absorbed by the Earth. The average temperature of Earth is about 290 K. Radius of the Earth is $R_e = 6000$ km and that of the Sun is $R_s = 700{,}000$ km. Earth - Sun distance is $r = 1.5 \times 10^8$ km. Assume Sun to be a black body.

(a) Estimate the effective emissivity of earth.

(b) Find the power of the sun.

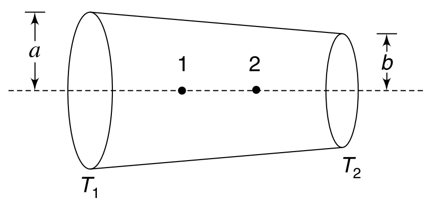
(c) Estimate the surface temperature of the Sun.

LEVEL 2

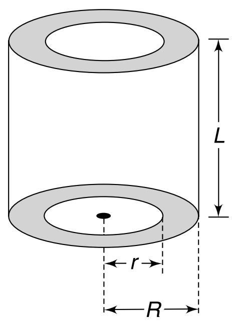
Q. 27: A tapering rod of length L has cross sectional radii of a and $b(< a)$ at its two ends. Its thermal conductivity is k. The end with radius a is maintained at a higher temperature T_1 and the other end is maintained at a lower temperature T_2. The curved surface is insulated.

(i) At which of the two points – 1 and 2 – shown in the figure will the temperature gradient be higher?

(ii) Calculate the thermal resistance of the rod.



Q. 28: A thick cylindrical shell made of material of thermal conductivity k has inner and outer radii r and R respectively and its length is L. When the curved surface of the cylinder are lagged (i.e., given insulation cover) and one end is maintained at temperature T_1 and the other end is maintained at $T_2(< T_1)$; the heat current along the length of the cylinder is H. In another experiment the two ends are lagged and the inner wall and outer wall are maintained at T_1 and T_2 respectively. Find the radial heat flow in this case.



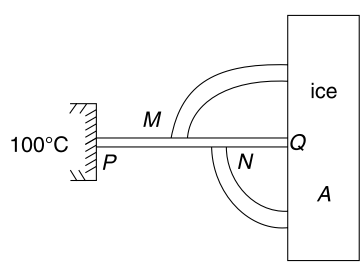
Q. 29: A double pan window used for insulating a room thermally from outside consists of two glass sheets each of area 1 m^2 and thickness 0.01 m separated by 0.05 m thick stagnant air space. In the steady state, the room-glass interface and the glass-outdoor interface are at constant temperatures of 27°C and 0°C respectively. The thermal conductivity of glass is 0.8 Wm^{-1}K^{-1} and of air 0.08 Wm^{-1}K^{-1}. Answer the following questions.

(a) Calculate the temperature of the inner glass-air interface.

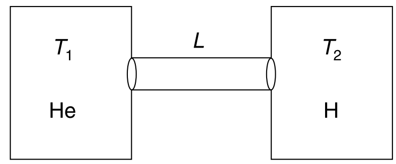
(b) Calculate the temperature of the outer glass-air interface.

(c) Calculate the rate of flow of heat through the window pane.

Q. 30: The container A contains ice at O°C. A conducting uniform rod PQ of length $4R$ is used to transfer heat to the ice in the container. The end P of the rod is maintained at 100°C and the other end Q is kept inside container A. The complete ice melts in 23 minutes. In another experiment, two conductors in shape of quarter circle of radii $2R$ and R are welded to the conductor PQ at M and N respectively and their other ends are inserted inside the container A. All conductors are made of same material and have same cross sectional area. Once again the end P is maintained at 100°C and this time the complete ice melts in t minute. Find t. Assume no heat loss from the curved surface of the rods. [Take $\pi \simeq 3.0$]



Q. 31: Two identical adiabatic containers of negligible heat capacity are connected by conducting rod of length L and cross sectional area A. Thermal conductivity of the rod is k and its curved cylindrical surface is well insulated from the surrounding. Heat capacity of the rod is also negligible. One container is filled with n moles of helium at temperature T_1 and the other one is filled with equal number of moles of hydrogen at temperature $T_2(<T_1)$. Calculate the time after which the temperature difference between two gases will becomes half the initial difference.



Q. 32: A copper slab is 2 mm thick. It is protected by a 2 mm layer of stainless steel on both sides. The temperature on one side of this composite slab is 400°C and 200°C on the other side. Value of thermal conductivities are-

$$k_{cu} = 400 \text{ Wm}^{-1}\text{K}^{-1} \text{ and } k_s = 16 \text{ Wm}^{-1}\text{K}^{-1}$$

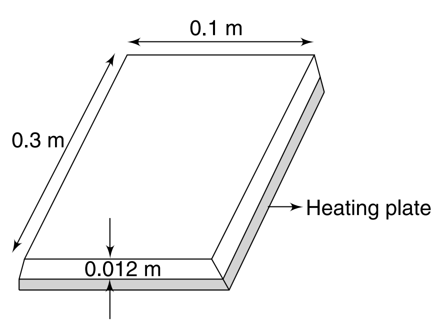
(a) Just by knowing that thermal conductivity of steel is much less than copper, find (approximately) the temperature of the copper slab.

(b) Plot the variation of temperature across the thickness of the composite wall.

Q. 33: A steel plate is 0.3 m long, 0.1 m wide and 0.012 m thick. The plate is placed on a heating plate of identical size maintained at 100°C. The heating plate is receiving energy through a 50 W heater. The heating plate losses heat only to the steel plate which is well insulated from all sides except at the top. The top surface of the steel plate is exposed to an airstream of temperature 20°C. The top surface of the steel plate radiates like a black body. Calculate the rate at which the top surface looses heat due to convection. The surface of steel plate in contact with heating plate is at 100°C.

Thermal conductivity of steel k = 16 Wm^{-1}K^{-1}

Stefan's constant = 5.67 × 10^{-8} Wm^{-2}K^{-4}.



Q. 34: A straight cylindrical wire is connected to a battery. The wire is maintained at constant temperature of T_1 when the room temperature is $T_0(< T_1)$. The wire is disconnected from the battery and half of it is cut-off. The remaining half of the wire is connected to the same battery in the same room. Find the constant temperature (T_2) attained by the wire. Assume that wire loses heat to the atmosphere through radiation from curved surface only.

Q. 35: A spherical shell is kept in an atmosphere at temperature T_0. The wavelength corresponding to maximum intensity of radiation for the shell is λ_0. A point source of constant power is switched on inside the shell. The power radiated by the source is $P = 0.4\sigma SeT_0^4$ where S, e and σ are outer surface area of the shell, emissivity of the outer surface of the shell and Stefan's constant respectively. Calculate the new wavelength (λ) corresponding to the maximum intensity of radiation from the shell. Assume that change in temperature (ΔT) of the shell is small compared to the ambient temperature T_0.

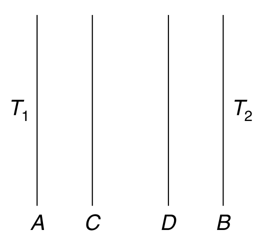
Q. 36: A body at a temperature of 50°C cools to 49°C in time Δt when it is placed in a room maintained at –3°C. The same body cools from 50°C to 49°C in time $\Delta t'$ when placed in a room that is maintained at 24°C. Find $\Delta t'$ in terms of Δt. Assume heat loss through radiation only and the specific heat capacity of the body remains constant with change in temperature.

Q. 37: A plane surface A is at a constant temperature $T_1 = 1000$ K. Another surface B parallel to A, is at a constant lower temperature $T_2 = 300$ K. There is no medium in the space between two surfaces. The rate of energy transfer from A to B is equal to r_1(J/s). In order to reduce rate of heat flow due to radiation, a heat shield consisting of two thin plates C and D, thermally insulated from each other, is placed between A and B in parallel. Now the rate of heat transfer (in steady state) reduces to r_2. Neglect any effect due to finite size of the surfaces, assume all surfaces to be black bodies and take Stefan's constant $\sigma = 6 \times 10^{-8}$ Wm^{-2}K^{-4}. Area of all surfaces $A = 1$m^2.

(i) Find r_1

(ii) Find the ratio $\frac{r_2}{r_1}$

(iii) Find the ratio $\frac{r_2}{r_1}$ if temperature of A and B were 2000 K and 600 K respectively.



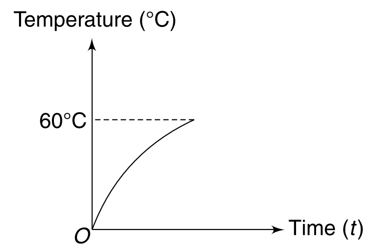
Q. 38: A block is kept in a room which is at 20°C. To raise the temperature of the block, heat is given to it at a constant rate of 600 watt (using an electric heater). The temperature of the block rises with time as shown in the graph. The slope of the graph at time $t = 0$ is 3°C s^{-1}. Once the temperature rises to 60°C, the heater is switched off and another heater is switched on to maintain the temperature of the block at 60°C. This new heater supplies heat at a constant rate of 100 watt. Assume that heat capacity of the block remains constant for the range of temperature involved.

(a) Explain why the slope of the given graph is decreasing with time.

(b) Calculate the heat capacity of the block.

(c) If the 100 W heater is also switched off, what will be initial rate of cooling of the block?

(d) Assuming that rate of heat loss by the block to the surrounding is proportional to difference in its temperature with surrounding, calculate the heat radiated per second by the block when it was at 30°C.



Q. 39: Mass m of a liquid A is kept in a cup and it is at a temperature of 90°C. When placed in a room having temperature of 20°C, it takes 5 min for the temperature of the liquid to drop to 30°C. Another liquid B has nearly same density as that of A and its sample of mass m kept in another identical cup at 50°C takes 5 min for its temperature to fall to 30°C when placed in room having temperature 20°C. If the two liquids at 90°C and 50°C are mixed in a calorimeter where no heat is allowed to leak, find the final temperature of the mixture. Assume that Newton's law of cooling is applicable for given temperature ranges.

Q. 40: Consider the Sun to be a black body at temperature $T_S = 5780$ K. Radius of the Sun is $r_S = 6.96 \times 10^8$ m. The Earth - Sun distance is $R = 1.49 \times 10^{11}$m. Assume that 30% of the solar radiation that hits the earth is scattered back into space without absorption.

(a) Calculate the steady state average temperature of the earth assuming it to be a black body. Take $(0.7)^{\frac{1}{4}} = 0.91$

(b) We know that average temperature of earth is ≃ 288 K. How does this value compare with that obtained in (a)?

The difference is due to greenhouse effect. Comment on the following statement - "Emissivity of earth is reduced more than absorptivity due to green house effect."

Q. 41: A and B are two sphere made of same material. Radius of A is double that of B and initially they are at same temperature (T). Both of them are kept far apart in a room at temperature $T_0(< T)$. Calculate the ratio of initial rate of cooling (i.e. rate of fall of temperature) of sphere A and B if

(a) the spheres are solid,

(b) the spheres are hollow made of thin sheets of same thickness

Q. 42: Heat received by the Earth due to solar radiations is 1.35 KWm^{-2}. It is also known that the temperature of the Earth's crust increases 1°C for every 30 m of depth. The average thermal conductivity of the Earth's crust is $K = 0.75$ $J(msK)^{-1}$ and radius of the Earth is $R = 6400$ km.

(i) Calculate rate of heat loss by the Earth's core due to conduction.

(ii) Assuming that the Earth is a perfect block body estimate the temperature of its surface.

Q. 43: A truck of mass M has 4 wheels. The surface area of the metal disc in each wheel is A. When brakes are applied the brake shoe in each wheel rubs against the metal disc. This produces heat. We will assume that this heat is used to heat the disc in the wheel only. Each disc has mass m and is made of material of specific heat capacity s. One day the truck was going downhill on a road inclined at angle θ to the horizontal. To maintain a constant speed v the driver had to apply brakes. The only other dissipative force, apart from the brake force, on the truck is the air resistance force that is equal to F. Assume no heating of discs due to air friction.

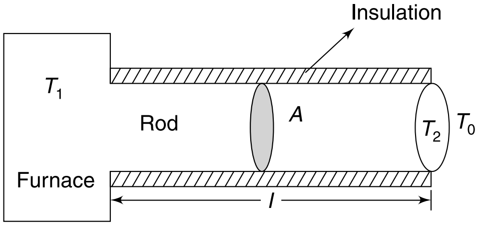
(a) Find the initial rate of rise of temperature of each metal disc after the brakes are applied.

(b) Find the final temperature (T) of the disc assuming that it is a black body and it loses heat only through radiation. Take the atmospheric temperature to be T_0

LEVEL 3

Q. 44: A solid body A of mass m and specific heat capacity 's' has temperature $T_1 = 400$ K. It is placed, at time $t = 0$, in atmosphere having temperature $T_0 = 300$ K. It cools, following Newton's law of cooling and its temperature was found to be $T_2 = 350$ K at time t_0. At time t_0, the body A is connected to a large water bath maintained at atmospheric temperature T_0, using a conducting rod of length L, cross section A and thermal conductivity k. The cross sectional area A of the connecting rod is small compared to the overall surface area of body A. Find the temperature of A at time $t = 2t_0$.

Q. 45: A cylindrical rod of length l, thermal conductivity k and area of cross section A has one end in a furnace maintained at constant temperature. The other end of the rod is exposed to surrounding. The curved surface of the rod is well insulated from the surrounding. The surrounding temperature is T_0 and the furnace is maintained at $T_1 = T_0 + \Delta T_1$. The exposed end of the rod is found to be slightly warmer then the surrounding with its temperature maintained $T_2 = T_0 + \Delta T_2$ [$\Delta T_2 << T_0$]. The exposed surface of the rod has emissivity e. Prove that ΔT_1 is proportional to ΔT_2 and find the proportionality constant.



ANSWERS

1. (a) $T_A = T_B = T_C$ (b) $T_D = 6T_M - 5T_A$

3. Cross sectional area of the bar decreases from hot end to the cold end

4. (a) $n_0 = 2$ (b) $\frac{r+R}{2}$

(c) $\frac{4\pi\rho(T_1 - T_2)}{(R - r)}$

6. 0.5 cm hr^{-1}

7. $8000 \frac{J}{kg°C}$

8. (i) $\frac{(T_1 - T_2)2\pi KL}{\ell n\left(\frac{b}{a}\right)}$

(ii) $\frac{T_1 - T_2}{\frac{1}{2\pi K_1 L}\ell n\left(\frac{r_2}{r_1}\right) + \frac{1}{2\pi K_2 L}\ell n\left(\frac{r_3}{r_2}\right)}$

9. (i) 44.4°C (ii) Increase

10. $t = \frac{ms}{hA}\ell n2$

11. Zero

12. E

13. $\frac{1}{3}$

14. 4% less than its present value

15. $A = \left(\frac{4}{3}\right)^4 A_0$

16. (a) $\frac{3}{4}$ (b) No

18. Earth radiates almost 100% in far infra-red region

19. 4 Watt

20. $t = 2t_0$

21. (a) Yes (b) 5.6 min.

23. $\frac{T_0}{\sqrt{2}}$

24. (i) $d = \frac{R}{2}\left(\frac{\lambda_p}{\lambda_s}\right)^2$

25. 51.7 Wm^{-1}

26. (a) 0.6 (b) 3.96×10^{26}
(c) 5800 K

27. (i) At point 2 (ii) $\frac{L}{k\pi ab}$

28. $\frac{2L^2H}{(R^2 - r^2)\ell n\left(\frac{R}{r}\right)}$

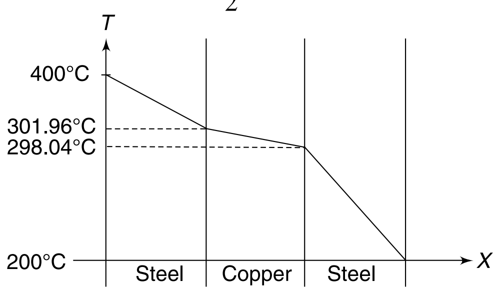
29. (a) 26.5°C (b) 0.5°C
(c) 40.5 J/s

30. 17.5 min

31. $\frac{15nRL}{16kA}\ln 2$

32. (a) Temperature drop will take palace almost entirely in steel. The copper plate will have almost uniform temperature of $\frac{400 + 200}{2} = 300°C$

(b)



33. 30.05 watt.

34. $T_2 = (4T_1^4 - 3T_0^4)^{1/4}$

35. $\frac{10}{11}\lambda_0$

36. $1.55\Delta t$

37. (i) $r_1 = 59514$ W (ii) $\frac{r_2}{r_1} = \frac{1}{3}$
(iii) $\frac{r_2}{r_1} = \frac{1}{3}$

38. (b) 200 J°C^{-1} (c) 0.5°C s^{-1} (d) 25 W

39. 66°C

40. (a) 254 K

41. (a) $\frac{1}{2}$ (b) 1

42. (i) 1.29×10^{13} W (ii) 276 K

43. (a) $\frac{dT}{dt} = \left(\frac{Mg\sin\theta - F}{4m\cdot s}\right)v$

(b) $T = \left[\frac{(Mg\sin\theta - F)v}{\sigma A} + T_0^4\right]^{1/4}$

44. $T = 300 + 50e^{-\left[\frac{\ln 2}{t_0} + \frac{kA}{msL}\right]t_0}$

45. $\frac{4e\sigma LT_0^3}{k} + 1$

SOLUTIONS

1. (a) $K_A = K_B = K$ and $K_C = 3K$

Heat flow rate from A to M is P.

Since rate of heat flow through $CM = 3 \times$ heat flow through AM

This is possible if $T_C = T_A$

$$\therefore \quad T_A = T_B = T_C$$

(b)
$$\frac{5\cdot KA(T_A - T_M)}{L} = \frac{KA(T_M - T_D)}{L}$$

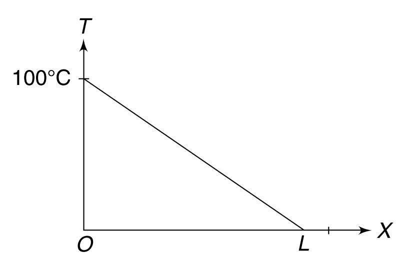
$$\Rightarrow \quad 5T_A - 5T_M = T_M - T_D$$

$$\Rightarrow \quad T_D = 6T_M - 5T_A$$

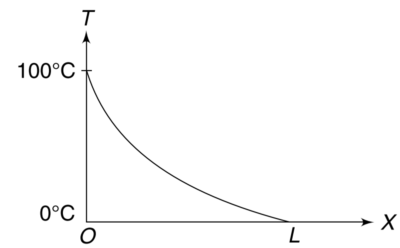
2. (a) For lagged bar, the heat current is constant at all sections.

$$\frac{dQ}{dt} = KA\frac{dT}{dx} = \text{a constant}$$

$$\therefore \quad \frac{dT}{dx} = \text{a constant}$$



(b) For not - lagged bar, as one moves from $x = 0$ to $x = L$, more and more heat is lost to the surrounding [as surface area goes on increasing]. It means the heat current goes on decreasing.



$\therefore \quad KA\left(\dfrac{dT}{dx}\right) = \dfrac{dQ}{dt}$ goes on decreasing from $x = 0$ to $x = L$

$\therefore \quad \dfrac{dT}{dx}$ goes on decreasing, i.e., slope of the graph goes on decreasing

4. (a) Thermal resistance of a shell of radius x and thickness dx will be

$$dR_{th} = \frac{dx}{k4\pi x^2} = \frac{1}{\rho \cdot 4\pi}\frac{x^n}{x^2}dx = \frac{1}{\rho \cdot 4\pi}x^{n-2}\,dx \qquad \text{...(i)}$$

Radial heat current (which is constant) will be

$$\frac{dT}{dR_{th}} = H \text{ (constant)}$$

$$\frac{4\pi\rho dT}{x^{n-2}dx} = H$$

$$\frac{dT}{dx} = \text{a constant if } n = 2$$

$$\therefore \quad n_0 = 2$$

(b) Since temperature gradient is uniform, hence temperature will be $\dfrac{T_1 + T_2}{2}$ at $x = \dfrac{r+R}{2}$

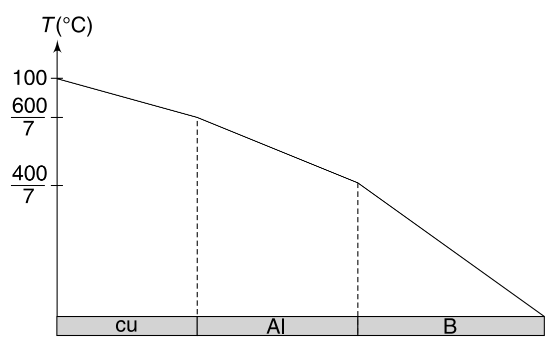
(c) From (i) $dR_{th} = \dfrac{1}{4\pi\rho}dx$

$$\therefore \quad R_{th} = \frac{1}{4\pi\rho}\int_r^R dx = \frac{R-r}{4\pi\rho}$$

$$\therefore \quad H = \frac{T_1 - T_2}{R_{th}} = \frac{4\pi\rho(T_1 - T_2)}{(R - r)}$$

5. (a) In all three cases it will be same.

(b)



6. Consider layer of ice that is x cm thick. Just below the ice temperature is 0°C and above it the temperature is $-\theta$°C. Let A be the area of the ice layer.

Heat conducted through ice in next dt time will be

$$dQ = \frac{kA\theta}{x}dt$$

If ice layer thickens by dx, the mass of ice frozen in time dt will be $= \rho A dx$

$$\therefore \quad dQ = (\rho A dx)L$$

$$\Rightarrow \quad \frac{kA\theta}{x}dt = \rho AL\,dx$$

$$\Rightarrow \quad \frac{dx}{dt} = \frac{k\theta}{x\rho L} = \frac{(0.005 \text{ cal cm}^{-1}\text{s}^{-1}\,°\text{C}^{-1}) \times (10°\text{C})}{(5 \text{ cm})(0.9\text{g cm}^{-3})(80 \text{ cal g}^{-1})} = 0.00014 \text{ cm s}^{-1}$$

$$= 0.00014 \times 3600 \text{ cm hr}^{-1} = 0.5 \text{ cm hr}^{-1}$$

7. $$-ms\frac{d\theta}{dt} = k(\theta - \theta_0)$$

Hence, $$30 \text{ watt} = k(50 - 20) \quad \text{... (i)}$$

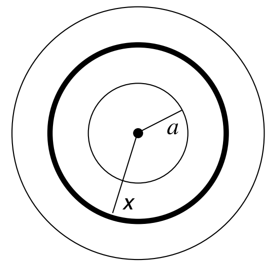
And $$m.s.\frac{0.1}{10} = k(40 - 20) \quad \text{....(ii)}$$

(ii) ÷ (i)

$$\frac{\frac{1}{4}s \times 0.01}{30} = \frac{20}{30}$$

$$s = 8000 \text{ J/Kg°C}$$

8. (i) Consider a layer (shown in dark) of radius x and thickness dx.



Resistance of this layer is $dR = \dfrac{dx}{K \cdot 2\pi Lx}$

∴ Resistance of entire pipe for radial heat flow is

$$R = \frac{1}{2\pi KL}\int_a^b \frac{dx}{x} = \frac{1}{2\pi KL}\,\ell n\left(\frac{b}{a}\right)$$

∴ $$\text{Heat current} = \frac{\Delta T}{R} = \frac{(T_1 - T_2)2\pi KL}{\ell n(b/a)}$$

(ii) In this case two resistances are in series

$$R = R_1 + R_2 = \frac{1}{2\pi K_1 L}\,\ell n\left(\frac{r_2}{r_1}\right) + \frac{1}{2\pi K_2 L}\,\ell n\left(\frac{r_3}{r_2}\right)$$

∴ $$\text{Heat current} = \frac{T_1 - T_2}{\frac{1}{2\pi K_1 L}\,\ell n\left(\frac{r_2}{r_1}\right) + \frac{1}{2\pi K_2 L}\,\ell n\left(\frac{r_3}{r_2}\right)}$$

9. (i) Rate of heat generation

$$H = VI = 10 \times 8 = 80 \text{ W}$$

Thermal resistance of the plastic cover is

$$R = \frac{\ell n(r_2/r_1)}{2\pi KL} = \frac{\ell n(3.5/1.5)}{2\pi \times 0.15 \times 5} = 0.18°\text{C/W}$$

In steady state, the heat produced in the copper wire is conducted by the plastic cover to the atmosphere.

∴ $$H = \frac{\Delta T}{R}$$

⇒ $$\Delta T = H.R = 80 \times 0.18 = 14.4°\text{C}$$

∴ $$\text{Temperature at interface} = 30 + 14.4 = 44.4°\text{C}$$

(ii) If thickness is increased, the resistance R increases.

∴ $$\Delta T = HR \text{ will increase}$$

⇒ Temperature at the interface will increase

10. $$\frac{dQ}{dt} = hA(T_1 - T)$$

$$ms\frac{dT}{dt} = hA(T_1 - T)$$

⇒ $$\int_{T_0}^{T_2} \frac{dT}{T_1 - T} = \frac{hA}{ms}\int_0^t dt$$

$$[\ell n(T_1 - T)]_{T_0}^{T_2} = -\frac{hAt}{ms}$$

$$\Rightarrow \quad \ell n\left(\frac{T_1 - T_2}{T_1 - T_0}\right) = -\frac{hAt}{ms}$$

Put $$T_2 = \frac{T_1 + T_0}{2}$$

$$\ell n\left(\frac{1}{2}\right) = -\frac{hAt}{ms} \Rightarrow \ell n 2 = \frac{hAt}{ms}$$

$$\Rightarrow \quad t = \frac{ms}{hA}\ell n 2$$

12. 0.5 E = energy reflected back

0.5 E = energy radiated

Total energy propagating from the surface of the body = E

13. Power supplied by battery in first case

$$P_1 = \frac{100^2}{R} \qquad [R = \text{ resistance of sphere}]$$

Power supplied after blackening of the half sphere $P_2 = \dfrac{141^2}{R} = \dfrac{(\sqrt{2} \times 100)^2}{R} = 2P_1$

In first case $\quad \sigma eA\,[500^4 - 300^4] = P_1 \qquad$ ...(i)

In second case $\quad \dfrac{\sigma eA}{2}[500^4 - 300^4] + \dfrac{\sigma A}{2}[500^4 - 300^4] = 2P_1 \qquad$ ...(ii)

Dividing (ii) by (i)

$$\frac{\frac{e}{2} + \frac{1}{2}}{e} = 2 \Rightarrow e = \frac{1}{3}$$

14. From dimensional analysis it can be shown that

$$\sigma = a\frac{k^4}{c^2h^3} \qquad [a = \text{constant}]$$

$$\therefore \quad \frac{\Delta\sigma}{\sigma} \simeq -2\frac{\Delta c}{c}$$

$$\therefore \quad \frac{\Delta\sigma}{\sigma} \times 100 = -2\frac{\Delta c}{c} \times 100 = -2 \times 2 = -4\%$$

15. $\lambda_m T = b$ = a constant

$$\therefore \quad \lambda_R\,(727 + 273) = \lambda_y T_y$$

[T_y = temperature when ball is yellow]

$$\therefore \quad T_y = \frac{8000}{6000} \times 1000 = \frac{4000}{3}\,K$$

$$\frac{A}{A_0} = \left(\frac{4000/3}{1000}\right)^4 \Rightarrow A = \left(\frac{4}{3}\right)^4 A_0$$

16. Let length of the cylinder be ℓ

$$2\pi r\ell + 2\cdot\pi r^2 = 4\pi r^2$$

$$\Rightarrow \quad \ell = r$$

$$\frac{m_{sp}}{m_{cy}} = \frac{V_{sp}}{V_{cy}} = \frac{\frac{4}{3}\pi r^3}{\pi r^2\cdot\ell} = \frac{4}{3}$$

T = temperature of sphere/cylinder

$$T_0 = \text{temperature of room}$$

$$-m_{sp} \cdot s \cdot \left(\frac{dT}{dt}\right)_{sp} = e\sigma A(T^4 - T_0^4)$$

$$-m_{cy} \cdot s \cdot \left(\frac{dT}{dt}\right)_{cy} = e\sigma A(T^4 - T_0^4)$$

$\therefore$
$$\frac{-(dT/dt)_{sp}}{-(dT/dt)_{cy}} = \frac{m_{cy}}{m_{sp}} = \frac{3}{4}$$

17. The maximum radiation is given out at wavelength λ_0 such that

$$\lambda_1 < \lambda_0 < \lambda_2$$

But
$$\lambda_0 = \frac{b}{T}$$

$\therefore$
$$\lambda_1 < \frac{b}{T} < \lambda_2$$

$\Rightarrow$
$$\frac{1}{\lambda_1} > \frac{T}{b} > \frac{1}{\lambda_2}$$

$\Rightarrow$
$$\frac{b}{\lambda_1} > T > \frac{b}{\lambda_2}$$

18. Temperature of the Earth << Temperature of the Sun.

$\therefore$ Earth will radiate predominantly in longer wavelength region

Almost all radiation from the earth has $\lambda > 4\,\mu m$.

19. Final constant temperature = 45°C

At 45°C, rate of heat loss = 24W

$$24\text{W} = k(45 - 15) \qquad \text{...(i)}$$

It rate of heat loss at 20°C is P then

$$P = k(20 - 15) \qquad \text{...(ii)}$$

(ii) ÷ (i)

$$\frac{P}{24} = \frac{5}{30} \Rightarrow P = 4 \text{ Watt.}$$

20. Let θ_0 = room temperature

θ_1 = initial temperature of the body

If θ = temperature of the body at time t then

$$-\frac{d\theta}{dt} = k(\theta - \theta_0)$$

$$\int_{\theta_1}^{\theta} \frac{d\theta}{\theta - \theta_0} = -k\int_0^t dt$$

$$\ell n(\theta - \theta_0) - \ell n(\theta_1 - \theta_0) = -kt$$

$\therefore$
$$\frac{\theta - \theta_0}{\theta_1 - \theta_0} = e^{-kt}$$

$$\theta - \theta_0 = (\theta_1 - \theta_0)e^{-kt}$$

$\Rightarrow$
$$\Delta\theta = \Delta\theta_0 e^{-kt}$$

Where $\Delta\theta$ = temperature difference between the body and the surrounding

$$\Delta\theta_0 = \text{initial value of } \Delta\theta$$

As per question

$$\frac{1}{2}\Delta\theta_0 = \Delta\theta_0 e^{-kt_0} \quad \Rightarrow \quad e^{-kt_0} = \frac{1}{2}$$

When $$\Delta\theta = \frac{1}{4}\Delta\theta_0 \Rightarrow \frac{1}{4} = e^{-kt}$$

Obviously, $$t = 2t_0$$

21. (b) The average temperature of the cup between 5°C to 10°C is 7.5°C

The temperature difference with surrounding can be taken as 25°C – 7.5°C = 17.5°C

$$\frac{5°\text{C}}{4\text{min}} = k(17.5) \qquad \text{...(i) } k = a \text{ constant}$$

The average temperature of coffee cup between 10°C and 15°C is 12.5°C

$$\therefore \quad \frac{5°\text{C}}{t} = k(12.5) \qquad \text{...(ii)}$$

(i) ÷ (ii)

$$\frac{t}{4} = \frac{17.5}{12.5} \Rightarrow t = 5.6 \text{ min}$$

23. Power radiated by the sun = $\sigma 4\pi R^2 T_s^4$

Intensity of the sun light at a distance r from the sun is

$$I = \frac{\sigma 4\pi R^2 T_s^4}{4\pi r^2} = \sigma T_s^4 \frac{R^2}{r^2}$$

Power received by the planet = $\pi r_0^2 \cdot \sigma T_s^4 \dfrac{R^2}{r^2}$

If T_0 = temperature of the planet then power radiated by the planet at T_0 must be equal to power received.

$$\therefore \quad \sigma 4\pi r_0^2 T_0^4 = \pi r_0^2 \sigma T_s^4 \frac{R^2}{r^2}$$

$$\therefore \quad T_0 = \left[\frac{R^2}{4r^2} T_s^4\right]^{\frac{1}{4}}$$

$\therefore$ Temperature of the planet does not depend on its radius

If distance r is doubled, the temperature will become $\dfrac{1}{\sqrt{2}}$ times.

24. $$P_{\text{star}} = \sigma A T_s^4 = \sigma 4\pi R^2 T_s^4$$

Intensity of radiation at the surface of the planet is

$$I = \frac{P_{\text{star}}}{4\pi d^2} = \frac{\sigma R^2 T_s^4}{d^2}$$

Energy incident (= energy absorbed) on the planet per unit time

$$P_{\text{planet}} = I\pi r^2 \qquad [r = \text{radius of planet}]$$

$$= \frac{\pi\sigma R^2 T_s^4}{d^2} r^2$$

In thermal equilibrium the planet will radiate same amount of energy per unit time

$$\therefore \quad \sigma 4\pi r^2 T_p^4 = \frac{\pi\sigma R^2 T_s^4 r^2}{d^2}$$

$$\therefore \quad \left(\frac{T_p}{T_s}\right)^4 = \frac{R^2}{4d^2} \qquad \therefore \quad d = \frac{R}{2}\left(\frac{T_s}{T_p}\right)^2$$

$$d = \frac{R}{2}\left(\frac{\lambda_p}{\lambda_s}\right)^2 \qquad [\because\ T\lambda = b = \text{a constant}]$$

(ii) Since $d >> R$

$$\therefore \qquad \lambda_p >> \lambda_s$$

25. $$q_{\text{conv}} = h(T - T_0) = 6(80 - 20) = 360 \text{ Wm}^{-2}$$

For 1 m length of the pipe

$$Q_{\text{conv}} = q_{\text{conv}} A = q_{\text{conv}} \times 2\pi r$$
$$= 360 \times 2 \times 3.14 \times 0.01 = 22.6 \text{ Wm}^{-1}$$
$$q_{\text{rad}} = \sigma(T^4 - T_0^4) = 5.67 \times 10^{-8}(353^4 - 293^4) = 462 \text{ Wm}^{-2}$$

For 1m length of the pipe

$$Q_{\text{rad}} = q_{\text{rad}} A = 462 \times 2 \times 3.14 \times 0.01 = 29.1 \text{ Wm}^{-1}$$

$$\therefore \qquad Q_{\text{conv}} + Q_{\text{rad}} = 22.6 + 29.1 = 51.7 \text{ Wm}^{-1}$$

26. (a) Amount of heat radiated by the Earth = solar flux from the sun

$$e\sigma 4\pi R_e^2 T_e^4 = 0.68 I_s \cdot \pi R_e^2$$

$$e = \frac{0.68 I_s}{4\sigma T_e^4} = \frac{0.68 \times 1.4 \times 10^3 \text{ W/m}^2}{4\left(5.67 \times 10^{-8} \dfrac{W}{m^2K^4}\right)(290K)^4} \simeq 0.6$$

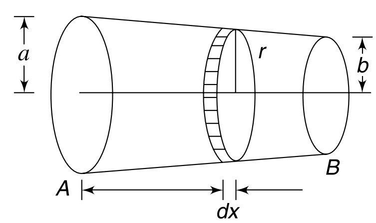
(b) $$P_{\text{sun}} = 4\pi r^2 I_s$$
$$= 4\pi(1.5 \times 10^{11}\text{m})^2 \cdot (1.4 \times 10^3 \text{ W/m}^2) = 3.96 \times 10^{26} \text{ W}$$

(c) $$P_{\text{sun}} = \sigma 4\pi R_s^2 T_s^4$$

$$\Rightarrow \qquad T_s^4 = \frac{3.96 \times 10^{26} \text{ W}}{\left(5.67 \times 10^{-8} \dfrac{W}{m^2K^4}\right)4\pi(7 \times 10^8\text{m})^2} = 1.13 \times 10^{15}$$

$$T_s \simeq 0.58 \times 10^4 = 5800 \text{ K}$$

27. (i) Heat current $\qquad H = kA\left(\dfrac{dT}{dx}\right)$



Heat current is same at every cross section.

Since $A_1 > A_2$

Hence $\qquad \left(\dfrac{dT}{dx}\right)_1 < \left(\dfrac{dT}{dx}\right)_2$

(ii) Radius at a distance x from the end A is

$$r = a - \left(\frac{a-b}{L}\right)x$$

$\therefore$ Area of cross section $\qquad A = \pi r^2 = \pi\left[a - \left(\dfrac{a-b}{L}\right)x\right]^2$

Resistance of a disc shaped element will be

$$dR = \frac{dx}{kA} = \frac{dx}{k\pi\left[a - \left(\dfrac{a-b}{L}\right)x\right]^2}$$

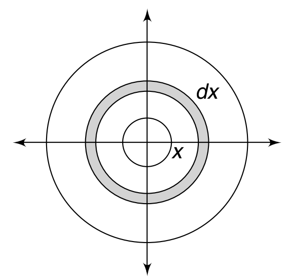
$\therefore$ Resistance of the rod is

$$R = \int dR = \frac{1}{k\pi}\int_0^L \frac{dx}{\left[a - \left(\frac{a-b}{L}\right)x\right]^2}$$

$$= \frac{L}{k\pi(a-b)}\left[\frac{1}{a - \left(\frac{a-b}{L}\right)x}\right]_0^L = \frac{L}{k\pi ab}$$

28. In first case $H = \dfrac{k\pi(R^2 - r^2)\Delta T}{L}$ …(i)

In case of radial heat how the thermal resistance can be calculate as



$$dR_{th} = \frac{dx}{k \cdot 2\pi x \cdot L}$$

$$R_{th} = \frac{1}{2\pi Lk}\int_r^R \frac{dx}{x} = \frac{1}{2\pi Lk}\,\ell n\left(\frac{R}{r}\right)$$

$$\therefore \quad H' = \frac{\Delta T}{R_{th}} = \frac{2\pi Lk\Delta T}{\ell n\left(\frac{R}{r}\right)} \quad \text{...(i)}$$

$$\frac{H'}{H} = \frac{2\pi Lk\Delta T}{\ell n\,\frac{R}{r}} \cdot \frac{L}{k \cdot \pi(R^2 - r^2)\Delta T}$$

$$H' = \frac{2L^2 H}{(R^2 - r^2)\ell n\left(\frac{R}{r}\right)}$$

29. Let T_2 and T_3 be the temperatures of the glass 1-air interface and air-glass 2 interface respectively

The equations of heat flow are $\dfrac{dQ_1}{dt} = \dfrac{dQ_2}{dt} = \dfrac{dQ_3}{dt}$

$$\Rightarrow \quad \frac{dQ}{dt} = \frac{K_g A(T_1 - T_2)}{d_g} = \frac{K_a A(T_2 - T_3)}{d_a} = \frac{K_g A(T_3 - T_4)}{d_g} \quad \text{...(A)}$$

$$\Rightarrow \quad \frac{dQ}{dt} = \frac{K_g A(300 - T_2)}{d_g} = \frac{K_a A(T_2 - T_3)}{d_a} = \frac{K_g A(T_3 - 273)}{d_g}$$

$$\Rightarrow \quad 50(300 - T_2) = T_2 - T_3 \quad \text{...(i)}$$

And $\quad 300 - T_2 = T_3 - 273$ ...(ii)

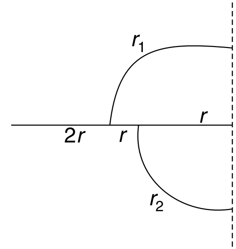
Solve (i) and (ii) to get T_2 and T_3. Put this in (A) to get $\dfrac{dQ}{dt}$

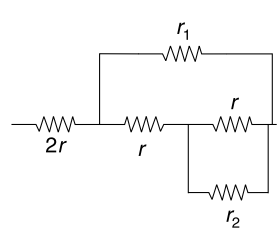
30. Thermal resistance of length R of the rod is $r = \dfrac{R}{kA}$ [k = thermal conductivity]

With only rod PQ, the thermal resistance is $4r$. Heat current $H_1 = \dfrac{\Delta T}{4r}$

$$t_1 \propto \frac{1}{H_1} \quad \Rightarrow \quad t_1 \propto (4r) \quad \text{…(i)}$$

With circular parts added the equivalent thermal resistance will be obtained by series–parallel combination shown in figure.





Where $$r_1 = \frac{\pi R}{kA} = \pi r \simeq 3r$$

$$r_2 = \frac{\pi R/2}{kA} = \frac{\pi}{2}r \simeq \frac{3}{2}r$$

r in parallel to r_2 gives $$\frac{rr_2}{r+r_2} = \frac{\frac{3}{2}r}{1+\frac{3}{2}} = \frac{3r}{5}$$

$\frac{3r}{5}$ and r are in series which gives $\frac{8r}{5}$

$\frac{8r}{5}$ is in parallel to r_1, which gives $\frac{\frac{8r}{5}\times 3r}{\frac{8r}{5}+3r} = \frac{24r}{23}$

$\frac{24r}{23}$ and $2r$ are in series; which gives $\frac{70r}{23}$

$$\therefore \quad H_2 = \frac{\Delta T}{\frac{70r}{23}} \Rightarrow H_2 \propto \frac{23}{70r}$$

$$\therefore \quad t_2 \propto \frac{70r}{23} \qquad \text{...(ii)}$$

From (i) and (ii)

$$\frac{t_2}{t_1} = \frac{70}{23\times 4} \Rightarrow t_2 = \frac{70}{4}\ \text{min}$$

31. Heat is transferred from He to H.

Let temperature of the two containers be θ_1 and θ_2 at time t.

Temperature difference is $\theta = \theta_1 - \theta_2$

In time dt change in temperature difference is

$$d\theta = d\theta_1 - d\theta_2 \qquad \text{...(i)}$$

Heat transferred in time dt is

$$dQ = \frac{kA(\theta_1-\theta_2)}{L}dt = \frac{kA\theta\cdot dt}{L}$$

For helium dQ is negative and for hydrogen it is positive

$$\therefore \quad n(C_v)_{\text{He}}\, d\theta_1 = -dQ$$

$$d\theta_1 = -\frac{2}{3Rn}dQ$$

Similarly for hydrogen $$d\theta_2 = \frac{2dQ}{5R\cdot n}$$

From (i) $$d\theta = \left(-\frac{2}{3Rn} - \frac{2}{5Rn}\right)dQ$$

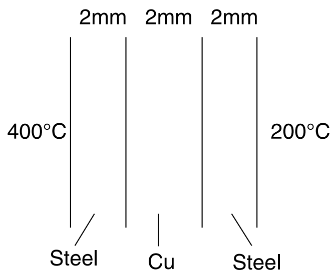
$$d\theta = -\frac{16}{15nR}\cdot\frac{kA\theta}{L}dt$$

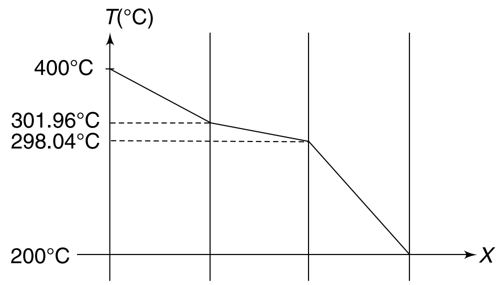
$$\therefore \quad \int_0^t dt = -\frac{15nRL}{16kA}\int_{T_1-T_2}^{\frac{T_1-T_2}{2}}\frac{d\theta}{\theta}$$

$$t = -\frac{15nRL}{16kA}\ln\frac{1}{2} = \frac{15nRL}{16kA}\ln 2$$

32. In steady state heat current is same at any cross section.

$$\therefore \quad \frac{dQ}{dt} = \frac{Ak_s\Delta T_s}{2\text{ mm}} = \frac{Ak_{\text{cu}}\Delta T_{\text{cu}}}{2\text{ mm}}$$





$$\therefore \qquad \Delta T_s = \frac{k_{cu}}{k_s}\Delta T_{cu} \Rightarrow \Delta T_s = \frac{400}{16}\Delta T_{cu}$$

$$\Rightarrow \qquad \Delta T_s = 25\Delta T_{cu}$$

Given, $2\Delta T_s + \Delta T_{cu} = 200°C \quad \therefore \quad 51\Delta T_{cu} = 200°C$

$$\Rightarrow \qquad \Delta T_{cu} = 3.92°C$$

$$\Delta T_s = \frac{200 - 3.92}{2} = 98.04°C$$

33. Let the upper surface of the steel plate be at temperature t°C

$$\frac{dQ}{dt} = \frac{kA(100-t)}{L}$$

$$50 = \frac{16 \times 0.03 \times (100 - t)}{0.012}$$

$$\therefore \qquad t = 98.75°C$$

If heat loss due to convection is H J/s,

$$H + \sigma A(T^4 - T_0^4) = 50$$

$$H + 5.67 \times 10^{-8} \times 0.03[(371.75)^4 - (293)^4] = 50$$

$$H = 30.05 \text{ watt}$$

34. In equilibrium

Electrical power dissipated in the wire = Heat radiated to the surrounding

$$\therefore \qquad \frac{V^2}{R} = e\sigma A(T^4 - T_0^4)$$

V = emf of the battery

R = resistance of the wire

A = Area of curved surface of the wire

For two situations the above equation can be written as

$$\frac{V^2}{R} = e\sigma A(T_1^4 - T_0^4)$$

And $$\frac{V^2}{R/2} = e\sigma \frac{A}{2}(T_2^4 - T_0^4)$$

Taking ratio $$2 = \frac{1}{2}\left(\frac{T_2^4 - T_0^4}{T_1^4 - T_0^4}\right)$$

$$\Rightarrow \qquad T_2 = \left[4T_1^4 - 3T_0^4\right]^{\frac{1}{4}}$$

35. Let the temperature of the shell after the source is switched on be T.

$$T = T_0 + \Delta T$$

Power of the source = net power radiated by the outer surface of the shell

$$0.4\sigma Se\, T_0^4 = e\sigma S(T^4 - T_0^4)$$

$$\Rightarrow \quad 0.4T_0^4 = (T_0 + \Delta T)^4 - T_0^4$$

$$\Rightarrow \quad 0.4T_0^4 = T_0^4\left[\left(1 + \left(\frac{\Delta T}{T_0}\right)\right)^4 - 1\right]$$

$$\Rightarrow \quad 0.4 \simeq 1 + 4\frac{\Delta T}{T_0} - 1 \quad \Rightarrow \quad \frac{\Delta T}{T_0} = \frac{1}{10}$$

$$\Rightarrow \quad T - T_0 = \frac{T_0}{10} \quad \Rightarrow \quad T = \frac{11}{10}T_0$$

Wien's law gives- $\lambda T = \lambda_0 T_0$

$$\lambda = \lambda_0\frac{T_0}{T} = \lambda_0\left(\frac{10}{11}\right)$$

36 Rate of heat loss by a body maintained at temperature T when placed in a room at T_0 is

$$\frac{dQ}{dt} = e\sigma A(T^4 - T_0^4)$$

If temperature difference is small we can write $T = T_0 + \Delta T$

Where $\Delta T << T_0$

$$\therefore \quad T^4 = (T_0 + \Delta T)^4 = T_0^4\left(1 + \frac{\Delta T}{T_0}\right)^4 \simeq T_0^4\left(1 + \frac{4\Delta T}{T_0}\right)$$

$$\therefore \quad \frac{dQ}{dt} \simeq e\sigma AT_0^4\left[1 - \left(1 + \frac{4\Delta T}{T_0}\right)\right] = 4e\sigma AT_0^3\,\Delta T$$

$$\Rightarrow \quad -ms\frac{dT}{dt} = 4e\sigma AT_0^3\,\Delta T$$

$$\Rightarrow \quad -\frac{dT}{dt} = \left(\frac{4e\sigma AT_0^3}{ms}\right)\Delta T$$

$$\Rightarrow \quad -\frac{dT}{dt} = k\Delta T$$

This is Newton's law of cooling. We need to take care that the constant k depends on T_0^3. When room temperature is $-3°C$ and $24°C$ let the value of constant be k_1 and k_2. Then $\frac{k_2}{k_1} = \left(\frac{297}{270}\right)^3 = (1.1)^3 = 1.33$

$\therefore$ When ball is in first room

$$\frac{1°C}{\Delta t} = k_1[49.5 - (-3)] \qquad \text{...(i)}$$

[49.5 = average temperature of the body during cooling]

In second room

$$\frac{1°C}{\Delta t'} = k_2[49.5 - 24] \qquad \text{...(ii)}$$

$\therefore$ (i) ÷ (ii)

$$\frac{\Delta t'}{\Delta t} = \frac{k_1}{k_2} \times \frac{52.5}{25.5}$$

$$\therefore \quad \Delta t' = \frac{1}{1.33} \times \frac{52.5}{25.5}\Delta t = 1.55\Delta t$$

37 (i) When place C and D are not present

$$r_1 = \sigma A(T_1^4 - T_2^4) = 6 \times 10^{-8} \times 1[10^{12} - 3^4 \times 10^8]$$
$$= 6 \times [10000 - 81] = 59514 \text{ Watt}$$

(ii) Let temperature of C and D be T_3 and T_4 respectively in the steady state.

Rate of heat gain by C = rate of heat loss by C

$$\therefore \quad \sigma A(T_1^4 - T_3^4) = \sigma A(T_3^4 - T_4^4)$$
$$\Rightarrow \quad T_1^4 + T_4^4 = 2T_3^4 \quad \text{...(i)}$$

Similarly for plate D

$$\sigma A(T_3^4 - T_4^4) = \sigma A(T_4^4 - T_2^4)$$
$$\therefore \quad T_3^4 + T_2^2 = 2T_2^4 \quad \text{....(ii)}$$

Solving (i) and (ii)

$$T_3^4 = \frac{1}{3}(2T_1^4 + T_2^4)$$
$$\therefore \quad r_2 = \sigma A(T_1^4 - T_3^4) = -\sigma A\left[T_1^4 - \frac{2}{3}T_1^4 - \frac{1}{3}T_2^4\right]$$
$$= \frac{1}{3}\sigma A\left[T_1^4 - T_2^4\right] = \frac{r_1}{3}$$

(iii) $\frac{r_2}{r_1}$ does not depend on values of T_1 and T_2

$\therefore$ Ratio will remain unchanged

38. (b) Initially, block is at room temperature

$$\therefore \quad P_{\text{loss to surrounding}} = 0$$
$$P_{\text{Heater}} = P_{\text{used to increase int. energy}} + P_{\text{loss to surrounding}}$$

At $t = 0$,

$$P_{\text{Heater}} = C\frac{dT}{dt} \qquad [C = \text{heat capacity}]$$
$$600 \text{ W} = C \times 3\frac{°\text{C}}{s}$$
$$\therefore \quad C = 200 \text{ J°C}^{-1}$$

(b) At 60°C

$$P_{\text{loss to surrounding}} = 100 \text{ W}$$

After heater is switched off

$$C\frac{dT}{dt} = -100 \text{ W}$$
$$\frac{dT}{dt} = -0.5\text{°Cs}^{-1}$$

(c) At 60°C, the block is losing energy at a rate of 100 W

$$\therefore \quad k(60 - 20) = 100 \quad \text{...(i)}$$
$$\text{At} \quad 30\text{°C}; \; k(30 - 20) = P_{30\text{°C}} \quad \text{...(ii)}$$

(ii) ÷ (i)

$$P_{30\text{°C}} = 25 \text{ Watt}$$

39. For cooling of liquid A:

$$\frac{\Delta\theta}{\Delta t} = \frac{k}{s_A}(\theta_{av} - \theta_0)$$

$$\frac{(90-30)°\text{C}}{5 \text{ min}} = \frac{k}{s_A}\left(\frac{90+30}{2} - 20\right)$$

$$12 = \frac{k}{s_A}(40)$$

$$\Rightarrow \qquad s_A = \frac{10k}{3} \qquad \text{...(i)}$$

For cooling of liquid B we have

$$\frac{50-30}{5} = \frac{k}{s_B}\left(\frac{50+30}{2} - 20\right)$$

$$4 = \frac{k}{s_B}(20)$$

$$\therefore \qquad s_B = 5k \qquad \text{...(ii)}$$

When the two liquids are mixed

Heat lost by A = Heat gained by B

$$m \cdot s_A(90 - \theta) = m \cdot s_B(\theta - 50)$$

$$\frac{10k}{3}(90 - \theta) = 5k(\theta - 50)$$

$$180 - 2\theta = 3\theta - 150$$

$$\Rightarrow \qquad 5\theta = 330 \quad \Rightarrow \quad \theta = 66°\text{C}$$

Note: Same density of two liquids means that they will occupy same volume in the identical cups. Therefore, the surface area through which the heat leaks is same for both.

40. (a) Power radiated by the Sun is

$$P_S = 4\pi r_S^2 \cdot T_S^4$$

At the Earth, this energy is distributed over sphere of radius R. Energy passing through each square meter of the sphere is given by

$$\frac{P_S}{4\pi R^2} = \frac{r_S^2}{R^2} T_S^4$$

Cross section of the earth is πr_e^2. Amount of solar power absorbed by the earth is

$$P_{\text{ab}} = \pi r_E^2 \cdot \frac{r_S^2}{R^2} \cdot T_S^4 \times 0.7$$

Amount of energy radiated by earth per unit time is $= 4\pi r_E^2 \cdot \sigma T_E^4$

For steady state

$$4\pi r_E^2 \sigma T_E^4 = \pi r_E^2 \frac{r_S^2}{R^2} \cdot T_S^4 \times 0.7$$

$$T_E^4 = 0.7 \frac{r_S^2 T_S^4}{R^2}$$

$$T_E = (0.7)^{\frac{1}{4}} \times (5780 \text{ K}) \sqrt{\frac{(6.96 \times 10^8)^2}{2 \times 1.49 \times 10^{11}}} \simeq 254 \text{ K}$$

(b) The statement is true.

41. $$ms\frac{dT}{dt} = -e\sigma A(T^4 - T_0^4)$$

(a) $$m = \rho\cdot\frac{4}{3}\pi r^3 \text{ and } A = 4\pi r^2$$

$\therefore$ $$\frac{dT}{dt} = -\frac{3e\sigma}{\rho s r}(T^4 - T_0^4)$$

$\therefore$ $$\frac{\left(\frac{dT}{dt}\right)_A}{\left(\frac{dT}{dt}\right)_B} = \frac{r_B}{r_A} = \frac{1}{2}$$

(b) $m = \rho 4\pi r^2 \Delta t$ and $A = 4\pi r^2$

[Δt = thickness of the wall]

$$\frac{dT}{dt} = \frac{-e\sigma(T^4 - T_0^4)}{\rho s \Delta t}$$

This is independent of r.

$\therefore$ Rate of cooling is same for both the sphere.

42. (i) Heat loss due to conduction is

$$\frac{dQ}{dt} = k\cdot(4\pi R^2)\frac{dT}{dR}$$

$$= 0.75 \times 4 \times 3.14 \times (6.4 \times 10^6)^2 \times \frac{1}{30} \qquad \left[\because \frac{dT}{dR} = 1°\text{C}/30 \text{ m}\right]$$

$$= 1.29 \times 10^{13} \text{ Watt}$$

(ii) Power received from the Sun is

$$P_{\text{sun}} = (1.35 \times 10^3) \times \pi R^2$$

$$= 1.35 \times 10^3 \times 3.14 \times (6.4 \times 10^6)^2 \simeq 1.7 \times 10^{17} \text{ W}$$

Heat received from the sun is much larger than heat lost due to conduction.

If equilibrium temperature of the earth is T_0, then it must radiate at a rate = P_{sun} at this temperature.

$\therefore$ $$\sigma A T^4 = 1.7 \times 10^{17}$$

$$5.67 \times 10^{-8} \times 4 \times 3.14 \times (6.4 \times 10^6)^2\, T^4 = 1.7 \times 10^{17}$$

$\therefore$ $$T^4 = 58 \times 10^8$$

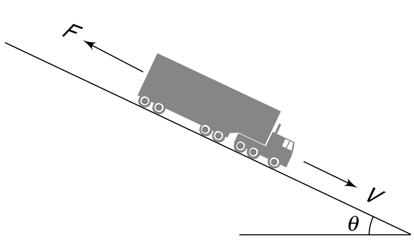
$\therefore$ $$T = (58)^{1/4} \times 10^2 \simeq (7.6)^{1/2} \times 100 \simeq 2.76 \times 100 \simeq 276 \text{ K}$$

43. (a) As per the question the ground friction force acting on wheels are non dissipative. You may assume that wheels are rolling without sliding.

Since, truck is moving with constant speed, power produced by gravitational force is equal to rate of energy dissipation due to air resistance and braking.

$(Mg\sin\theta)v = F\cdot v$ + Rate of heat dissipation in brakes

$\therefore$ Rate of heat dissipation in brakes = $[(Mg\sin\theta) - F]v$



$\therefore$ $$4m\cdot s\frac{dT}{dt} = (Mg\sin\theta - F)v$$

$\therefore$ $$\frac{dT}{dt} = \frac{(Mg\sin\theta - F)v}{4\, m\cdot s}$$

(b) Let final temperature be T

Rate of heat loss from each disc due to radiation $= \sigma A(T^4 - T_0^4)$

This will be equal to rate of heat production in each brake

$$\therefore \qquad \sigma A(T^4 - T_0^4) = [Mg\sin\theta - F]v$$

$$\Rightarrow \qquad T^4 - T_0^4 = \frac{(Mg\sin\theta - F)v}{\sigma A}$$

$$\Rightarrow \qquad T = \left[\frac{(Mg\sin\theta - F)v}{\sigma A} + T_0^4\right]^{1/4}$$

44. For interval 0 to t_0, rate of heat loss by body is

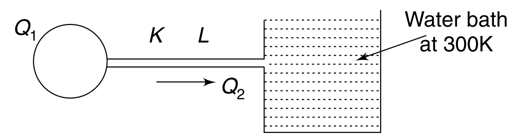
$$-ms\frac{dT}{dt} = b(T - T_0) \qquad \{b = \text{a constant}\}$$

$$\Rightarrow \qquad \int_{T_1=400\,\text{k}}^{T_2=350\,\text{k}} \frac{dT}{T - T_0} = -\frac{b}{ms}\int_{t=0}^{t_0} dt$$

$$\Rightarrow \qquad \ln = (350 - 300) - \ln(400 - 300) = -\frac{bt_0}{ms}$$

$$\Rightarrow \qquad \ln 2 = \frac{bt_0}{ms} \qquad \text{...(i)}$$

During interval t_0 to $2t_0$, body loses heat by radiation as well as conduction



$$-ms\frac{dT}{dt} = \frac{dQ_1}{dt} + \frac{dQ_2}{dt}$$

$$-ms\frac{dT}{dt} = b(T - 300) + \frac{kA(T - 300)}{L}$$

$$\int \frac{dT}{T - 300} = -\int \frac{bdt}{ms} - \int \frac{kAdt}{msL}$$

From (i) $\dfrac{b}{ms} = \dfrac{\ln 2}{t_0}$

$$\int_{350}^{T} \frac{dT}{T - 300} = -\left(\frac{\ln 2}{t_0} + \frac{kA}{msL}\right)\int_{t_0}^{2t_0} dt$$

$$\ln[T - 300] - \ln 50 = -\left(\frac{\ln 2}{t_0} + \frac{kA}{msL}\right)t_0$$

$$\frac{T - 300}{50} = e^{-\left[\frac{\ln 2}{t_0} + \frac{kA}{msL}\right]t_0}$$

$$\Rightarrow \qquad T = 300 + 50\, e^{-\left[\frac{\ln 2}{t_0} + \frac{kA}{msL}\right]t_0}$$

45. In steady state heat conducted from hot end to the exposed end of the rod is being lost as radiation to the surrounding

$$\frac{\Delta Q}{\Delta t} = \frac{kA(T_1 - T_2)}{L} = eA\sigma\left[T_2^4 - T_0^4\right]$$

$$\Rightarrow \qquad T_1 - T_2 = \frac{e\sigma L}{k}\left[(T_0 + \Delta T_2)^4 - T_0^4\right]$$

$$\Rightarrow \qquad (T_0 + \Delta T_1) - (T_0 + \Delta T_2) = \frac{e\sigma L}{k}T_0^4\left[\left(1 + \frac{\Delta T_2}{T_0}\right)^4 - 1\right]$$

$$\Rightarrow \qquad \Delta T_1 - \Delta T_2 \simeq \frac{e\sigma L}{k} T_0^4 \left[1 + \frac{4\Delta T_2}{T_0} - 1\right]$$

$\because \quad \left(1 + \frac{\Delta T_2}{T_0}\right)^4 \simeq 1 + 4\frac{\Delta T_2}{T_0}$ as all other terms will be very small and can be neglected.

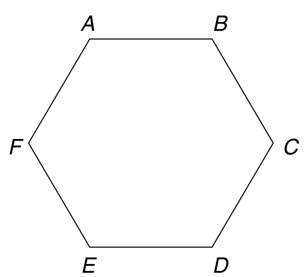
$$\Rightarrow \qquad \Delta T_1 = \left[\frac{4e\sigma L}{k} T_0^3 + 1\right]\Delta T_2$$

CHAPTER 6

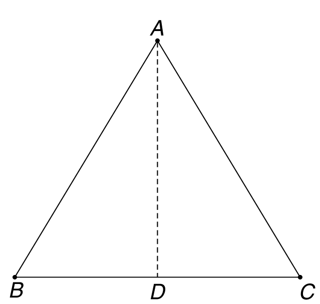
Electrostatics

LEVEL 1

Q. 1: **(a)** Six equal charges have been placed at the vertices of a regular hexagon. Charge at vertex A is moved to the centre of the hexagon and there it experiences a net electrostatic force of magnitude F. Charge at E is also moved to the centre so as to double the magnitude of the charge at the centre. Calculate the magnitude of the electrostatic force that this central charge experiences now.



(b) Three charges of equal magnitude lie on the vertices of an equilateral triangle ABC. All of them are released simultaneously. The charge at A experiences initial acceleration along AD where D is the midpoint of the side BC. Find the direction of initial acceleration of the charge at B.



Q. 2: Two identical small conducting balls have positive charges q_1 and q_2 respectively. The force between the balls when they are placed at a separation is F. The balls are brought together so that they touch and then put back in their original positions. Prove that the force between the balls now, cannot be less than F.

Q. 3: Point charges $-q$, $2q$, $-3q$, q, $-q$, $2q$, $-3q$, q, $-q$, $2q$, $-3q$, and q have been placed at marks 1, 2, 3, 4, 5 .....12 respectively on the circular dial of a clock. Find the electric field intensity at the centre of the dial if distance of each charge from the centre is r.

Q. 4: Let q_1 be a positive charge equal to the magnitude of the total charge on all electrons present in 0.9 mg of pure water and q_2 be the charge on a 6.35 mg copper sphere from which 0.1% of its total electrons have been removed.

Avogadro's number, $N_A = 6 \times 10^{23}$

Molar mass of H_2O = 18 g

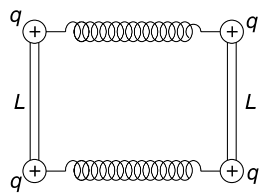
Molar mass of Cu = 63.5 g

(a) Find force between q_1 and q_2 if they are placed at a separation of 1 km.

(b) How does this force compare with the weight of a car having mass of 1200 kg? What conclusion can you draw from the result?

Q. 5: Three point charges $q_1 = q$, $q_2 = q$ and $q_3 = Q$ are placed at points having position vectors $\vec{r}_1$, $\vec{r}_2$ and $(\vec{r}_1 + \vec{r}_2)$ respectively. It is known that $|\vec{r}_1| = |\vec{r}_2| = \vec{r}$. The net electrostatic force on q_3 is $\sqrt{3}$ times the force applied by either of q_1 or q_2 on q_3. Find $\vec{r}_1 \cdot \vec{r}_2$.

Q. 6: Two stiff non conducting rods have length L each and have small balls connected to their ends. The rods are placed parallel to each other and the balls are connected by two identical springs as shown. When each ball is given a charge q, the system stays in equilibrium when it is in the shape of a square. If natural relaxed length of each spring is $L/2$ find the force constant (k) for them.



Q. 7: Five identical charges, q_1 each, are placed at the vertices of a regular pentagon having side length l_1. The net electrostatic force on any of the charges due to other four is F_1. Find the electrostatic force F_2 on any one of the five identical charges, q_2 each, placed at the vertices of a regular pentagon having side length l_2.

Q. 8: A simple pendulum has a bob of mass m which carries a charge q on it. Length of the pendulum is L. There is a uniform electric field E in the region. Calculate the time period of small oscillations for the pendulum about its equilibrium position in following cases:

(i) E is vertically down having magnitude $E = \dfrac{mg}{q}$

(ii) E is vertically up having magnitude $E = \dfrac{2mg}{q}$

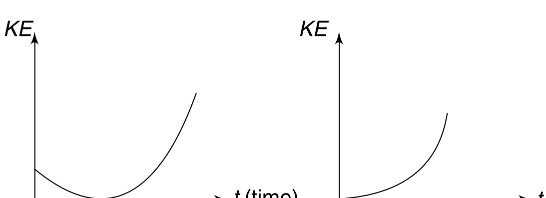
(iii) E is horizontal having magnitude $E = \dfrac{mg}{q}$

(iv) E has magnitude of $E = q$ and is directed upward making an angle of 45° with the horizontal.

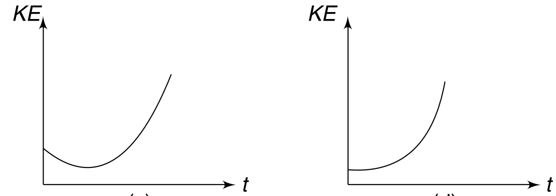
Q. 9: Two identical negative charges are fixed on X axis at equal distances from the origin (O). A particle having positive charge starts at a large distance from O, moves along the Y axis, passes through the origin and moves far away from O in the positive Y direction. Describe qualitatively how its acceleration charges as it moves. Draw a rough graph showing the variation of acceleration (a) vs position of the particle (y). Take the acceleration of the particle to be positive in $+Y$ direction.

Q. 10: An electron is either released from rest or projected with some initial velocity in a uniform electric field. Neglect any other force on the electron apart from electrostatic force. Which of the graphs shown in fig could possibly represent the change in kinetic energy of the electron during its course of motion?

Explain the situation in each case.



(a) (b)

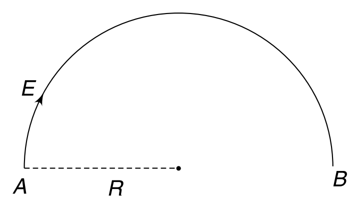


(c) (d)

Q. 11: In a region of space an electric field line is in the shape of a semicircle of radius R. Magnitude of the field at all point is E. A particle of mass m having charge q is constrained to move along this field line. The particle is released from rest at A.

(a) Find its kinetic energy when it reaches point B.

(b) Find the acceleration of the particle when it is at midpoint of the path from A to B.



Q. 12: Electric field $E = -bx + a$ exists in a region parallel to the X direction (a and b are positive constants). A charge particle having charge q and mass m is released from the origin $X = 0$. Find the acceleration of the particle at the instant its speed becomes zero for the first time after release.

Q. 13: A Sphere of radius R_0 carries a volume charge density proportional to the distance (x) from the origin, $\rho = \alpha x$ where α is a positive constant.

(a) Calculate the total charge in the sphere of radius R_0.

(b) The sphere is shaved off so as to reduce its radius. What should be the radius (R) of the remaining sphere so that it contains half the charge of the original sphere?

(c) Find the electric field at a point inside the sphere at a distance $r < R$. Give your answer for original sphere of radius R_0 as well as for the smaller sphere of radius R that was left after shaving off the original sphere.

Q. 14: Draw electric field lines for charge distributions given below–

(a) Two equal point charges placed at a separation.

(b) Two point charges $2q$ and $-q$ placed at a separation. Describe qualitatively how the lines will appear at a very large distance from the two charges?

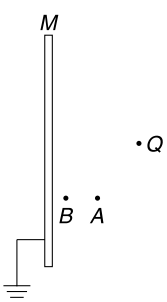
(c) Three point charges, each equal to $+q$ placed at the vertices of an equilateral triangle.

Q. 15: Three conducting concentric spherical shells of radii R, $2R$ and $3R$ carry some charge on them. The potential at the centre is 50 V and that of middle and outer shell is 20 V and 10 V respectively. Find the potential of the inner Shell.

Q. 16: The potential at a point in an electric field is given by $V = Rr$ volt where r is distance of the point from the origin of the co-ordinate system. Find the electric field at a point $r = \left(\hat{i} + 2\hat{j} + 2\sqrt{5}\hat{k}\right)\, m$

Q. 17: Electric pot(e)ntial in a 3 dimensional space is given by $V = \left(\frac{1}{x} + \frac{1}{y} + \frac{2}{z}\right)$ volt where x, y and z are in meter. A particle has charge $q = 10^{-12}$ C and mass $m = 10^{-9}$ g and is constrained to move in xy plane. Find the initial acceleration of the particle if it is released at (1, 1, 1) m.

Q. 18: A metal plate M is grounded. A point charge $+Q$ is placed in front of it. Consider two points A and B as shown in fig. At which point (A or B) is the electric field stronger? At which point is the potential higher?

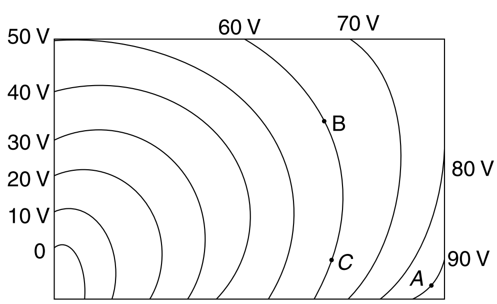


Q. 19: Equipotential lines arising from a static charge distribution has been shown in a certain region of space.

(a) Draw the electric field lines starting from point A

(b) If a charge is released at A, will it move along the field line passing through A?

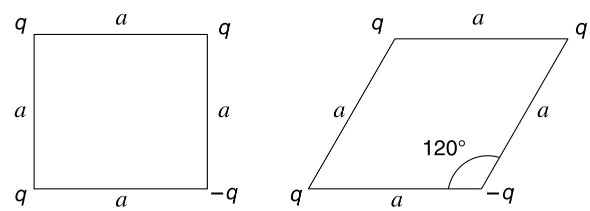
(c) How much work is done by the electric force if a charge moves from B to C? Is the knowledge of path important in this calculation?



Q. 20: (a) The equipotential curves in x, y plane are given by $x^2 - y^2 = V$ when V is potential. Draw the rough sketch of electric field lines in x–y plane.

(b) Repeat the above question if the potential field is given by $V = x^2 + y^2 - 4x + 4$

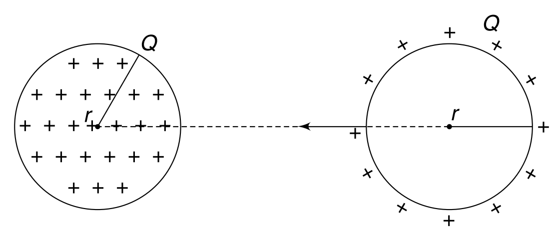
Q. 21: Four point charges q, q, q and $-q$ are placed at the vertices of a square of side length **a**. The configuration is changed and the charge are positioned at the vertices of a rhombus of side length **a** with $-q$ charge at the vertex where angle is 120°. Find the work done by the external agent in changing the configuration.



Q. 22: There is a ball of radius r having uniformly distributed volume charge Q on it and there is a spherical shell of radius r having uniformly distributed surface charge Q on it. The two spheres are far apart.

(a) A point charge q is moved slowly from the centre of the shell (through a small hole in it.) to the centre of the ball. Find work done by the external agent in the process.

(b) The two spheres are brought closer so that their centers are separated by $4r$. Now calculate the amount of work needed in slowly moving a point charge q from the centre of the shell to the centre of the ball. Assume that charge on one ball does not alter the charge distribution of the other. Does your answers in (a) and (b) differ? Why?



Q. 23: A uniformly charged sphere has charge Q. An electron (charge $-e$, mass m) revolves around it in a circular orbit of radius r.

(a) Write the total energy (i.e., sum of its kinetic energy and electrostatic potential energy in the field of the sphere) of the electron.

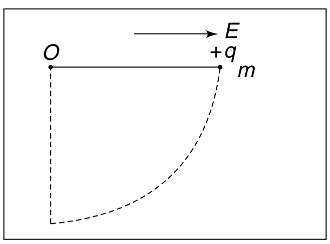
(b) If the time period of revolution of the electron in circular orbit of radius r is T, then find the time period if the orbital radius is made $4r$.

Q. 24: A uniformly charged sphere has radius R and charge Q on it. A negatively charged particle having mass m and charge $-q$ shoots out of the surface of the sphere with speed V. The minimum speed (V_0) for which the particle escapes the attraction of the sphere may be called as escape speed. Will the value of V_0 depend on magnitude of q? [Recall that escape speed of a body from the surface of the earth does not depend on its mass].

Q. 25: There is a uniform horizontal electric field of strength E in a region. A pendulum bob is pulled to make the string horizontal and released. The bob has mass m and charge q.

(a) Find the maximum angle (θ_0) that the bob swings before coming to rest momentarily.

(b) Find E if the bob comes to rest when the string is vertical.



Q. 26: A conducting sphere of radius R carries a charge Q. It is enclosed by another concentric spherical shell of radius $2R$. Charge from the inner sphere is transferred in infinitesimally small installments to the outer sphere. Calculate the work done in transferring the entire charge from the inner sphere to the outer one.

Q. 27: The ratio of energy density in electric field (u_E) to square of potential (V^2) at a point A at a distance x from a static point charge Q is η.

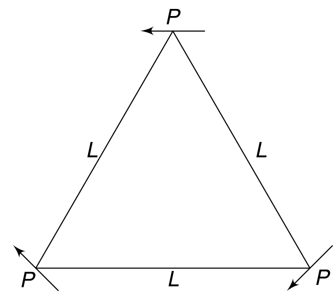
(a) Write the value of the ratio at a distance $2x$ from the point charge Q.

(b) Write the dimensional formula for the ratio η.

Q. 28: (a) Calculate the largest possible electrostatic energy in 1 cm^3 volume of air. The dielectric breakdown of air happens when field exceeds 3×10^6 V/m.

(b) A conducting ball of radius 10 cm is placed in distilled water ($\in_r = 80$) and charged with a charge $Q = 2 \times 10^{-9}$C. Calculate the energy used up in charging the ball.

Q. 29: Three short electric dipoles, each of dipole moment P, are placed at the vertices of an equilateral triangle of side length L. Each dipole has its moment oriented parallel to the opposite side of the triangle as shown in the fig. Find the electric field and potential at the centroid of the triangle.

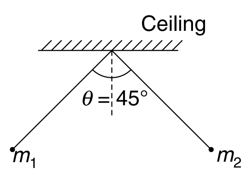


Q. 30: Short electric dipole of dipole moment P is placed at the centre of a ring of radius R having charge Q uniformly distributed on its circumference. The dipole moment vector is along the axis of the ring. Find force on the dipole due to the ring.

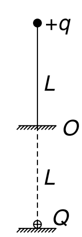
Q. 31: Calculate the electric dipole moment of a system comprising of a charge $+q$ distributed uniformly on a semicircular arc or radius R and a point charge $-q$ kept at its centre.

LEVEL 2

Q. 32: Two unequal masses, $m_1 = 2m$ and $m_2 = m$ have unequal positive charge on them. They are suspended by two mass-less threads of unequal lengths from a common point such that, in equilibrium, both the masses are on same horizontal level. The angle between the two strings is $\theta = 45°$ in this position. Find the Electrostatic force applied by m_1 on m_2 in this position.



Q. 33: A particle of mass m and charge q is attached to a light insulating thread of length L. The other end of the thread is secured at point O. Exactly below point O, there is a small ball having charge Q fixed on an insulating horizontal surface. The particle remains in equilibrium vertically above the ball with the string taut. Distance of the ball from point O is L. Find the minimum value of Q for which the particle will be in a stable equilibrium for any gentle horizontal push given to it.



Q. 34: Four identical charges, Q each, are fixed at the vertices of a square. A free charge q is placed at the centre of the square. Investigate the nature of equilibrium of charge q if it is to be displaced slightly along any of the two diagonals of the square.

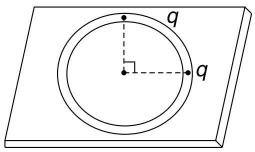
Q. 35: A horizontal circular groove is made in a wooden board. Two positive charges (q each) are placed in the groove at a separation of 90° (see figure). Where shall we place (in the groove) a third charge and what shall be its magnitude such that all three of them remain at rest after they are released. Answer for two cases:

(a) When the third charge is positive.

(b) When the third charge is negative.

Neglect friction and assume that the groove is very thin just wide enough to accommodate the particles.

[Take: $\sin 22.5° = 0.38$; $\cos 22.5° = 0.92$]



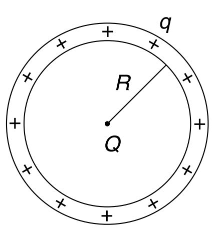
Q. 36: Two identical positive charges Q each, are placed on the x axis at points $(-a, 0)$ and $(a, 0)$. A point charge of magnitude q is placed at the origin. For small displacement along x axis, the charge q executes simple harmonic motion if it is positive and its time period is T_1. If the charge q is negative, it performs oscillations when displaced along y axis. In this case the time period of small oscillations is T_2. Find $\frac{T_1}{T_2}$.

Q. 37: A ring of radius R has uniformly distributed charge q. A point charge Q is placed at the centre of the ring.

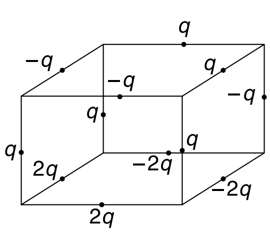
(a) Find the increase in tension in the ring after the point charge is placed at its centre.

(b) Find the increase in force between the two semicircular parts of the ring after the point charge is placed at the centre.

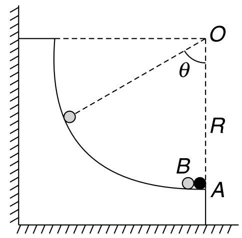
(c) Using the result found in part (b) find the force that the point charge exerts on one half of the ring.



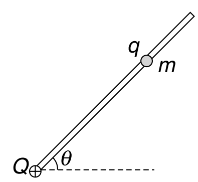
Q. 38: Twelve charges have been placed at the centre of each side of a cube as shown in the figure. Find the magnitude of Electric force acting on a charge Q placed at the centre of the cube. Take the side length of the cube to be r.



Q. 39: A fixed non conducting smooth track is in the shape of a quarter circle of radius R in vertical plane. A small metal ball A is fixed at the bottom of the track. Another identical ball B, which is free to move, is placed in contact with ball A. A charge Q is given to ball A which gets equally shared by the two balls. Ball B gets repelled and ultimately comes to rest in its equilibrium position where its radius vector makes an angle θ ($\theta < 90°$) with vertical. Mass of ball is m. Find charge Q that was given to the balls.

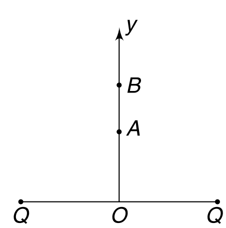


Q. 40: A smooth fixed rod is inclined at an angle θ to the horizontal. At the bottom end of the rod there is a fixed charge $+Q$. There is a bead of mass m having charge q that can slide freely on the rod. The equilibrium separation of the bead from fixed charge Q is x_0. Find the frequency of oscillation of the bead if it is displaced a little from its equilibrium position.



Q. 41: Two charges, Q each, are fixed on a horizontal surface at separation $2a$. Line OY is vertical and is perpendicular bisector of the line joining the two charges. Another particle of mass m and charge q has two equilibrium positions on the line OY, at A and B. The distances OA and OB are in the ratio $1:3\sqrt{3}$

(a) Find the distance of the point on the line OY where the particle will be in stable equilibrium.

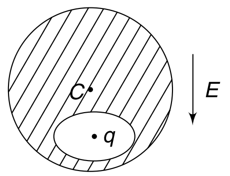


(b) Where will the particle experience maximum electric force – at a point above B or at a point between A and B or somewhere between O and A? Where is the acceleration of particle maximum on y axis from O to B?

Q. 42: A neutral spherical conductor has a cavity. A point charge q is located inside it. It is in equilibrium. An external electric field (E) is switched on that is directed parallel to the line joining the centre of the sphere to the point charge.

(a) What is the direction of acceleration of the charge particle inside the cavity after E is switched on.

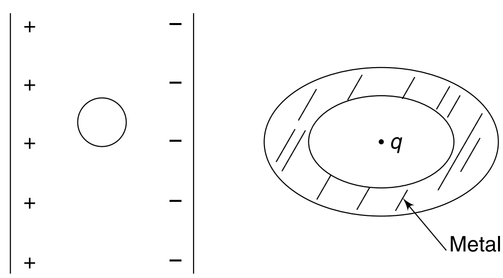
(b) How is the induced charge on the wall of the cavity affected due to the external field.



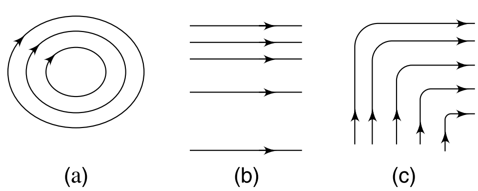
Q. 43: Draw electric field lines in following situations:

(a) A small neutral metal sphere is placed between the plates of an ideal parallel plate capacitor. [see figure]

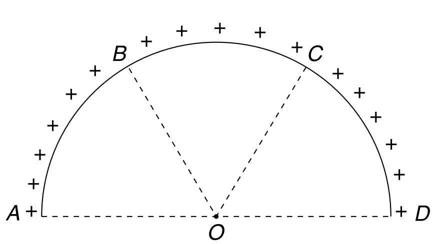
(b) A point charge is trapped inside a cavity in a neutral metal block (see figure)



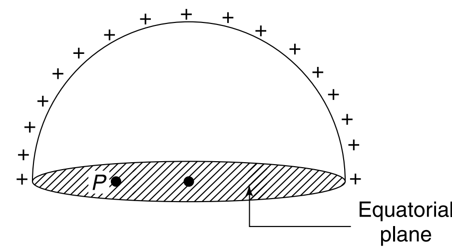
Q. 44: Three configurations of electrostatic field lines have been shown in the figure. Are these configurations possible? Explain your answer.



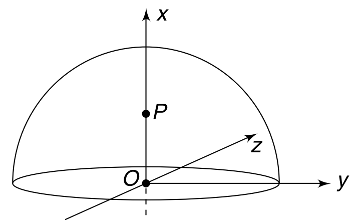
Q. 45: A uniformly charged semicircular ring ($ABCD$) produces an electric field E_0 at the centre O. AB, BC and CD are three equal arcs on the ring. Portion AB and CD are cut from either side and removed. Find the field at O due to remaining part BC.



Q. 46: Consider a uniformly charged hemispherical shell. What can you say about the direction of electric field at points on the equatorial plane. (e.g. point P)

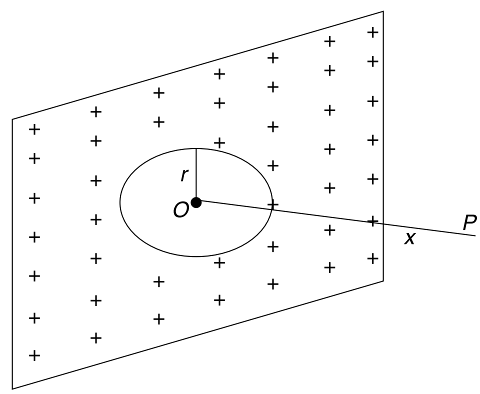


Q. 47: Consider a uniformly charged thin spherical shell as shown in figure. Radius of the shell is R.



The electric field at point $P(x, 0, 0)$ is $\vec{E}$. What is the electric field at a point $Q(-x, 0, 0)$. Given $x < R$.

Q. 48: There is an infinite non conducting sheet of charge having uniform charge density σ. The electric field at a point P at a distance x from the sheet is E_0. Point O is the foot of the perpendicular drawn from point P on the sheet. A circular portion of radius $r << x$ centered at O is removed from the sheet. Now the field at point P becomes $E_0 - \Delta E$. Find ΔE.



Q. 49: A thread having linear charge density λ is in the shape of a circular arc of radius R subtending an angle θ at the centre.

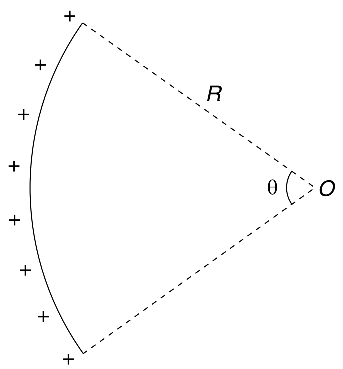
(a) Find the electric field at the centre.

(b) Using the expression obtained in part (a) find the field at the centre if the thread were semicircular

(c) Find the field at centre using the expression obtained in part (a) for the case $\theta \to 0$. Is the result justified?

(d) A thread having total charge Q (uniformly distributed) is in the shape of a circular arc of radius R subtending an angle θ at centre. Write the expression for the field at the center. Obtain the field when $\theta \to 0$.

Make sure you understand the difference in case (c) and (d)

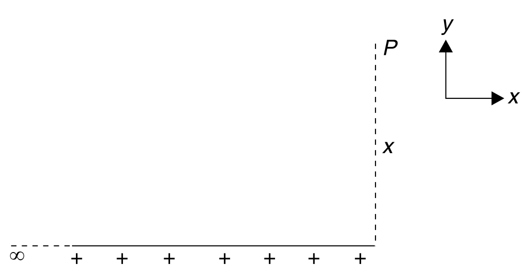


Q. 50: (a) There is an infinitely long thread uniformly charged with linear charge density λ C/m. Using Gauss' law, calculate the electric field (E_0) at a distance x from the thread.

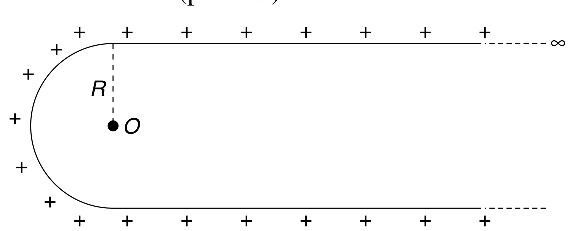
(b) Now consider a semi-infinite uniformly charged thread (linear charge density = λ) as shown in figure. Find the y component of electric field at point P in terms of E_0. Use simple qualitative argument.

(c) For the situation described in (b) calculate the x component of electric field at point P using the method of integration.

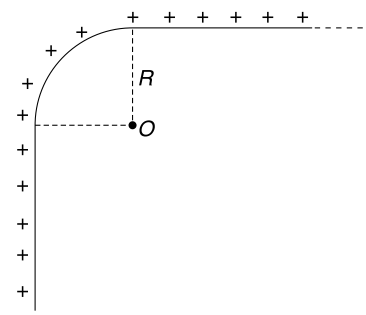
(d) Find the angle that the electric field at P makes with x direction.



Q. 51: An infinitely long line charge is bent in U shape as shown in figure. The semicircular part has radius R and linear charge density is λ C/m. Using the results obtained in last two problems, calculate the electric field intensity at centre of the circle (point O)



Q. 52: Repeat the above problem if the semicircular part is replaced with a quarter circle (see figure).

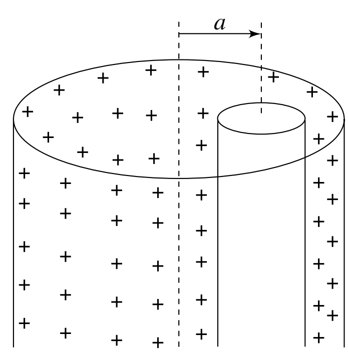


Q. 53: (a) There is a long uniformly charged cylinder having a volume charge density of ρ C/m^3. Radius of the cylinder is R. Find the electric field at a point at a distance x from the axis of the cylinder for following cases

(i) $x < R$ (ii) $x > R$

What is the maximum field produced by the charge distribution at any point?

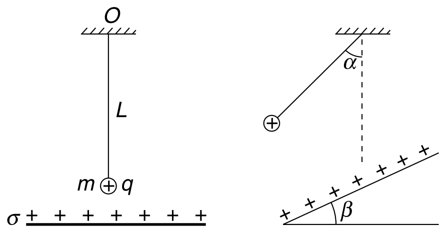
(b) The cylinder described in (a) has a long cylindrical cavity. The axis of cylindrical cavity is at a distance a from the axis of the charged cylinder (see figure). Find electric field inside the cavity.



Q. 54: A pendulum has a bob of mass m carrying a positive charge q. Length of the pendulum string is L. Beneath the pendulum there is a large horizontal dielectric sheet of charge having uniform surface charge density of σ C/m^2. [figure (i)]

(a) Find the time period of small oscillations for the pendulum

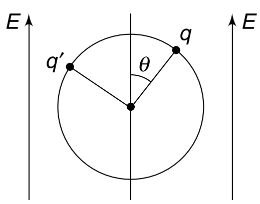
(b) Now the dielectric sheet of charge is tilted so as to make an angle β with horizontal. Find the angle (α) that the thread makes with vertical in equilibrium position. Find time period of small oscillations in this case. [figure (ii)]



Q. 55: A uniform non conducting ring has mass m and radius R. Two point charges q and q' are fixed on its circumference at a separation of $\sqrt{2}R$. The ring remains in equilibrium in air with its plane vertical in a region where exists a uniform vertically upward electric field E. Given $E = \dfrac{4mg}{7q}$.

(a) Find angle θ in equilibrium position (see figure).

(b) The ring is given a small rotation in the plane of the figure and released. Will it perform oscillations?

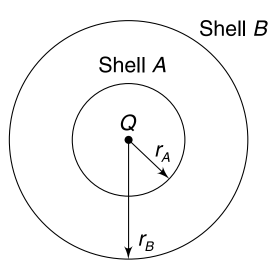


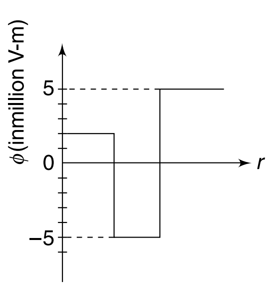
Q. 56: An infinitely long line charge has linear charge density λ C/m. The line charge is along the line $x = 0$, $z = 2$ m. Find the electric field at point (1, 1, 1) m.

Q. 57: A charged particle is placed at the centre of two thin concentric spherical charged shells, made of non-conducting material. Figure A shows cross-section of the arrangement. Figure B gives the net flux ϕ through a Gaussian sphere centered on the particle, as a function of the radius r of the sphere.

(a) Find charge on the central particle and shell A.

(b) In which range of the finite values of r, is the electric field zero?



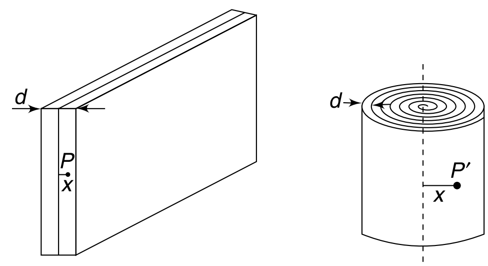


Q. 58: There are two infinite slabs of charge, both of thickness d with the junction lying on the plane $x = 0$. The slab lying in the range $0 < x < d$ has a uniform charge density $+\rho$ and the slab lying in the region $-d < x < 0$ has uniform charge density $-\rho$. Find the Electric field everywhere and plot its variation along the x axis.

Note: This can be used to model the variation of electric field in the depletion layer of a $p-n$ junction.

Q. 59: In an insulating medium (dielectric constant = 1) the charge density varies with y Co-ordinate as $\rho = by$, Where b is a positive constant. The electric field is zero at $y = 0$ and everywhere else it is along y direction. Calculate the electric field as a function of y.

Q. 60: A non conducting sheet of thickness d and large surface area contains a uniformly distributed charge of density ρ throughout its volume. The electric field at a point P inside the sheet at a distance x from the central plane is E_1. Now the sheet is rolled to form a large solid cylinder. Field at a point P' inside the cylinder at a distance x from its axis is E_2. Find the ratio $\dfrac{E_1}{E_2}$



Q. 61: A charge distribution generates a radial electric field $\vec{E} = \dfrac{a}{r^2} e^{-\frac{r}{k}} \hat{r}$, where r is distance from the origin, $\hat{r}$ is a unit vector in radial direction away from the origin and a and k are positive constants. The electric field extends around the origin up to a large distance.

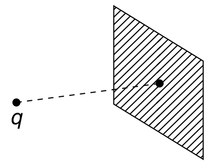
(a) Find the charge (q_0) that must be located at the origin to create such a field

(b) Find the quantity of charge (q) that must be spread around the charge q_0 at origin to create such a field.

Q. 62: A pyramid has four faces, all of them being equilateral triangle of side a. A charge Q is placed at the centre of one of the faces. What is the flux of electric field emerging from any one of the other three faces of the pyramid?

Q. 63: A point charge is placed very close to an infinite plane. What is flux through the plane?

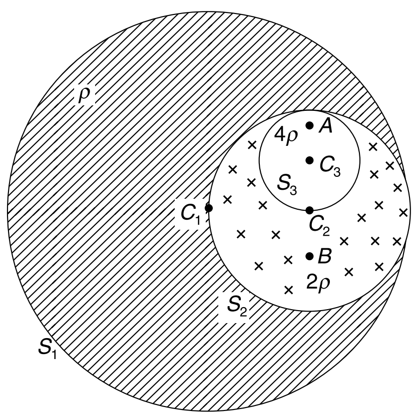
Q. 64: Point charge q is placed at a point on the axis of a square non-conducting surface. The axis is perpendicular to the square surface and is passing through its centre. Flux of Electric field through the square caused due to charge q is ϕ. If the square is given a surface charge of uniform density σ, find the force on the square surface due to point charge q.



Q. 65: In the figure shown, spheres S_1, S_2 and S_3 have radii R, $\frac{R}{2}$ and $\frac{R}{4}$ respectively. C_1, C_2 and C_3 are centers of the three spheres lying in a plane. Angle $\angle C_1C_2C_3$ is right angle. Sphere S_3 has a uniformly spread volume charge density 4ρ. The remaining part of S_2 has uniform charge density of 2ρ and the left over part of S_1 has a uniform charge density of ρ.

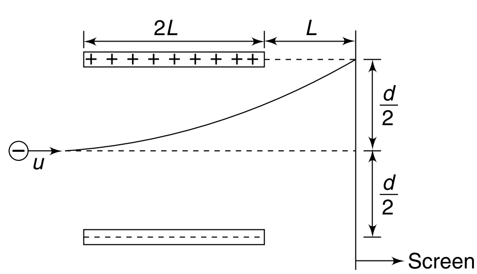
(a) Find electric field at a point A at a distance $\frac{R}{8}$ from C_3 on the line C_2C_3 (see figure)

(b) Find electric field at point B at a distance $\frac{R}{4}$ from C_2 on the line C_3C_2 (see figure)



Q. 66: An electron (charge = e, mass = m) is projected horizontally into a uniform electric field produced between two oppositely charged parallel plates, as shown in figure. The charge density on both plates is $\pm\sigma$ C/m^2 and separation between them is d. You have to assume that only electric force acts on the electron and there is no field outside the plates. Initial velocity of the electron is u, parallel to the plates along the line bisecting the gap between the plates. Length of plates is $2L$ and there is a screen perpendicular to them at a distance L.

(i) Find σ if the electron hits the screen at a point that is at same height as the upper plate.



(ii) Final the angle θ that the velocity of the electron makes with the screen while it strikes it.

Q. 67: A particle is projected at a speed of $u = 40$ m/s in vertically upward direction in a place where exists a horizontal uniform electric field E_0. The specific charge of the particle is $\frac{4g}{3E_0}$.

(a) Find the time after projection, when speed of the particle will be least.

(b) Find the time (after projection) when displacement of the particle becomes perpendicular to its acceleration.

(c) Assuming that the particle has been projected from a great height and the electric field is present in large region, what angle the velocity of the particle will make with horizontal after a long time?

Q. 68: A charged particle having mass m is projected in a uniform electric field with a kinetic energy K_0. After time t_0 it was observed that the kinetic energy of the particle was $\frac{K_0}{4}$ and its velocity was perpendicular to the field.

(a) How much more time is required for the particle to regain its lost kinetic energy?

(b) Write the impulse of the electric force acting on the particle between the two points where its kinetic energy is K_0.

Neglect all other forces on the particle apart from the electrostatic force.

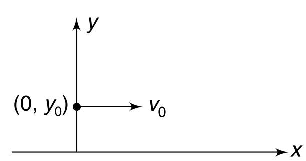
Q. 69: Electric field in xy plane is directed along positive y direction and its magnitude changes with y co-ordinate as

$$E = ay^2$$

A particle having charge q and mass m is projected at point $(0, y_0)$ with velocity

$$\vec{v} = v_0\hat{i}$$

Neglect all other forces on the particle apart from the electric force. Calculate the slope of the trajectory of the particle as a function of y.



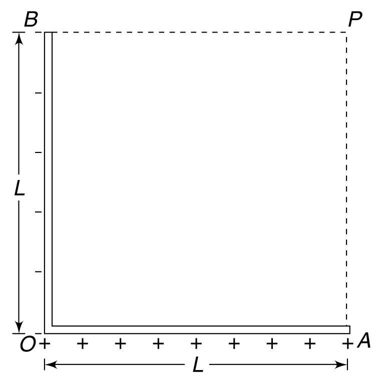
Q. 70: Two point charges $-q$ and $18q$ are placed on x axis at $-x_0$ and x_0 respectively. Draw the variation of potential along x axis assuming $18q$ has been removed. Draw another graph showing variation of potential when only

18q is present. Superimpose the two graphs to obtain the variation of potential along x axis when both charges are present.

Q. 71: L shaped rod has equal and opposite charge ($\pm Q$) spread along its both arms (see figure).

(a) What is direction of electric field at point P?

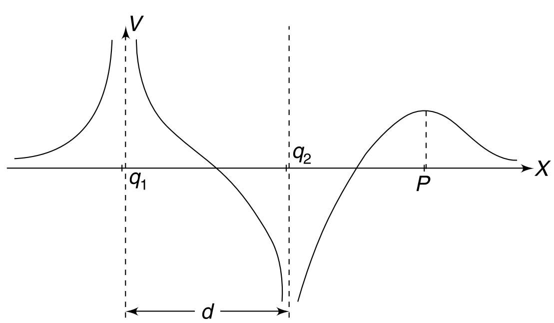
(b) Write electric potential at P.



Q. 72: Two point charge q_1 and q_2 lie on X axis at some separation d. The given figure shows the graphical variation of electric potential due to these two charges along the X axis.

(a) What are signs of $q_1$and q_2? Which charge has larger magnitude?

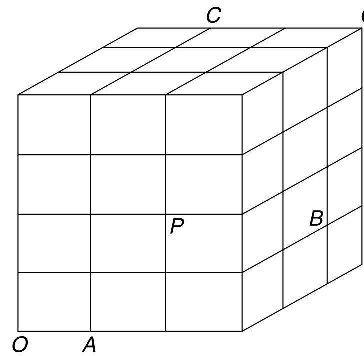
(b) Origin is the point between the charges where potential is zero. Distance of q_2 from origin is $\frac{d}{4}$. Find the distance of point P (marked in figure) from charge q_2.



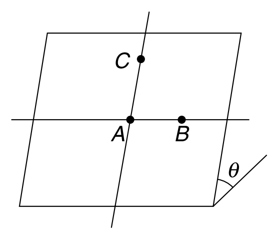
Q. 73: A uniform electric field exists in a region of space. Potential at points O, A, B and C are $V_0 = 0$, and $V_A = -1V$, $V_B = -6V$ and $V_C = -3V$ respectively. All the cubes shown in fig have side length of 1 m.

(a) Find $V_P - V_Q$

(b) Find the smallest distance of a point from O where the potential is $-2V$



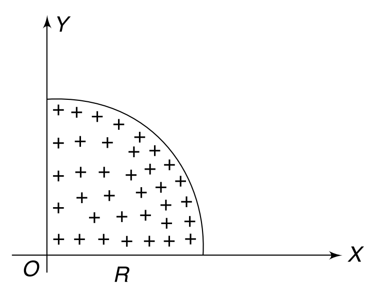
Q. 74: It is known that there exists a uniform electric field in a certain region. Imagine an incline plane (figure) in the region which is inclined at $\theta = 37°$ to the horizontal. When one moves horizontally along the incline from A to B (AB = 1 cm), the electric potential decreases by 10V. Similarly, potential at C (AC = 1 cm) is less than potential at A by 10 V where line AC lies on the incline and is perpendicular to AB. When one moves vertically up from point A to a point D (AD = 1 cm), the potential drops by 10 V again. Find the magnitude of the electric field in the region and the angle that it makes with vertical. $\left[\text{given } \sin 37° = \frac{3}{5}\right]$



Q. 75: The quarter disc of radius R (see figure) has a uniform surface charge density σ.

(a) Find electric potential at a point (O, O, Z)

(b) Find the Z component of electric field at (O, O, Z)



Q. 76: Electric field in a three dimensional space is directed radially towards a fixed point and its magnitude varies with distance (r) from the fixed point as $E = 4r$ V/m

(a) Draw electric field lines to approximately represent such a field.

(b) Calculate the quantity of charge present inside a spherical volume of radius a centered at the fixed point.

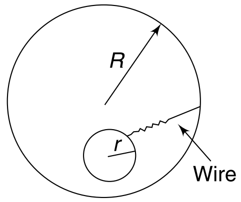
(c) Find potential difference $(V_A - V_B)$ between two points $A(1, 1, \sqrt{2})\,m$ and $B(0, 3, 4)\,m$. Take the fixed point to be the origin.

Q. 77: When a conducting disc is made to rotate about its axis, the centrifugal force causes the free electrons to be pushed toward the edge. This causes a sort of polarization and an electric field is induced. The radial movement of free electrons stops when electric force on an electron balance the centrifugal force. Calculate the potential difference developed across the centre and the edge of a disc of radius R rotating with angular speed ω. [This potential difference is sometimes called as sedimentation potential]. Take mass and charge of an electron to be m and e respectively.

Q. 78: Consider a cube having a uniform volume charge density. Find the ratio of electrostatic potential at the centre to the potential at a corner of the cube.

Q. 79: A metallic sphere of radius R has been charged to a potential of $V = 100$ volt. A thin hemispherical conducting shell has dimensions so that it can just fit on the half of the metallic sphere. The shell is originally grounded. Now, using an insulating handle, it is placed on top of the charged sphere so as to perfectly cover its top half. The shell is removed from the sphere and again grounded. After this the shell is again placed on the sphere, removed and then grounded. The process is continued till the potential of the sphere becomes $V' = 6.25$ volt. How many times the shell was placed on the sphere?

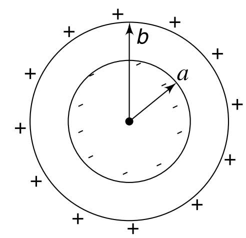
Q. 80: A conducting ball of radius r is charged to a potential V_0. It is enveloped by a thin walled conducting sphere of radius $R (> r)$ and the two spheres are connected by a conducting wire. Find the potential of the outer sphere.



Q. 81: A thick conducting spherical shell of inner radius a and outer radius b is shown in figure. It is observed that the inner face of the shell carries a uniform charge density $-\sigma$. The outer surface also carries a uniform surface charge density $+\sigma$.

(a) Can you confidently say that there must be a charge inside the shell? Find the net charge present on the shell.

(b) Find the potential of the shell.



Q. 82: A conducting bubble of radius a, thickness $t (t << a)$ has potential V. Now the bubble collapses into a droplet. Find the potential of the droplet assuming that there is no leakage of charge.

Q. 83: A point charge q has been placed at a distance x from the centre of a neutral solid conducting sphere of radius $R (x > R)$. Find the potential of the sphere. How will your answer change if the sphere is not solid, rather it is a thin shell of conductor.

Q. 84: There is a hemispherical shell having charge Q uniformly distributed on its surface. Radius of the shell is R. Find electric potential and field at the centre (of the sphere).

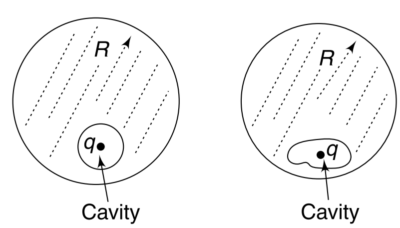
Q. 85: There is a hemisphere of radius R having a uniform volume charge density ρ. Find the electric potential and field at the centre.

Q. 86: Find the potential at a point on the edge of a uniformly charged disc. The surface charge density is σ and radius of the disc is R.

Q. 87: A solid spherical conductor of radius R has a spherical cavity inside it (see figure). A point charge q is placed at the centre of the cavity.

(a) What is the potential of the conductor?

(b) If the charge q is shifted inside the cavity by a distance Δx, how does the potential of the conductor change?

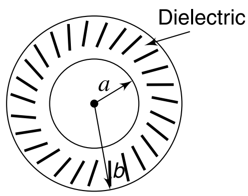


(c) How does your answer to the question (a) and (b) change if the cavity is not spherical and the charge q is placed at any point inside it (see figure)

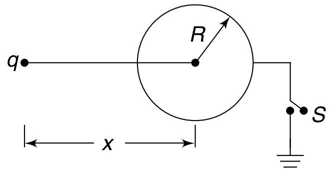
(d) Draw electric field lines in entire space in each case. In which case all field lines are straight lines.

Q. 88: Conducting ball of radius a is surrounded by a layer of dielectric having inner radius a and outer radius b.

The dielectric constant is K. The conducting ball is given a charge Q. Write the magnitude of electric field and electric potential at the outer surface of the dielectric.



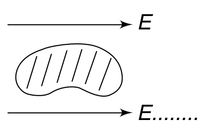
Q. 89: A point charge q is placed at a distance x from the centre of a conducting sphere of radius $R(< x)$.



(a) How much charge will flow through the switch S when it is closed to ground the sphere?

(b) Find the current through the switch S when charge 'q' is moved towards the sphere with velocity V.

Q. 90: A conductor is placed in a uniform external electric field (see figure). Sketch the equipotential surfaces.

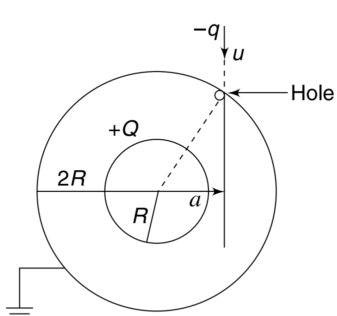


Q. 91: Two concentric spherical shells have radii R and $2R$. The outer shell is grounded and the inner one is given a charge $+Q$. A small particle having mass m and charge $-q$ enters the outer shell through a small hole in it. The speed of the charge entering the shell was u and its initial line of motion was at a distance $a = \sqrt{2}R$ from the centre.

(a) Find the radius of curvature of the path of the particle immediately after it enters the shell.

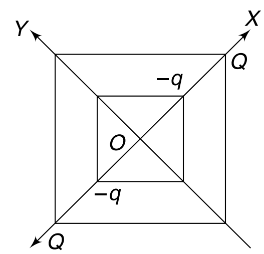
(b) Find the speed with which the particle will hit the inner sphere.

Assume that distribution of charge on the spheres do not change due to presence of the charge particle.



Q. 92: There charges q, $3q$ and $12q$ are to be placed on a straight line AB having 12 cm length. Two of the charges must be placed at end points A and B and the third charge can be placed anywhere between A and B. Find the position of each charge if the potential energy of the system is to be minimum. In the position of minimum potential energy what is the force on the smallest charge?

Q. 93: Two square of sides a and $2a$ are placed in xy plane with their centers at the origin. Two charges, $-q$ each, are fixed at the vertices of smaller square (lying on X axis). Two charges, Q each, are fixed at the vertices of bigger square on the X axis (see figure).



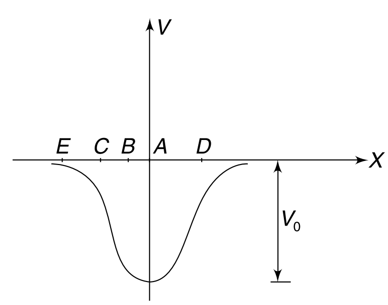
(a) Find work required to slowly move the larger square to infinity from the position shown.

(b) Find work done by the external agent in slowly rotating the inner square by 90° about the Y axis followed by a rotation of 90° about the Z axis.

Q. 94: A certain charge distribution produces electric potential that varies along the X axis as shown in figure. [There is no field in y or z direction]

(a) At which point (amongst A, B, C, D and E) does a negative charge feel the greatest force in positive X direction?

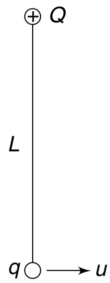
(b) Find the upper limit of the speed that a proton can have, as it passes through the origin, and still remain bound near the origin. Mass and charge of a proton are m and e. How will your answer change for an electron?



Q. 95: A simple pendulum has length L. Bob of the pendulum has mass m and carries a charge q. A point charge Q is fixed at the point of suspension. Find the minimum speed (u) of projection at the lowest point so that the pendulum bob completes the vertical circle. Give your answer for following two cases:

(a) $Q = \dfrac{8\pi\epsilon_0 L^2 mg}{q}$

(b) $Q = \dfrac{2\pi\epsilon_0 L^2 mg}{q}$

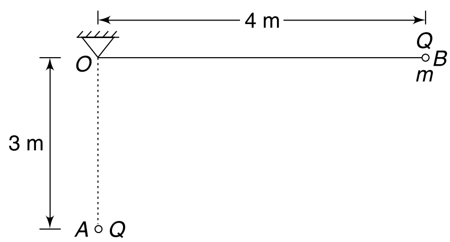


Q. 96: Below the fixed end O of the insulating horizontal thread OB, there is a fixed charge A of $Q = 20\mu c$. At the end B of the thread there is a small mass m carrying charge $Q = 20\mu c$. The mass is released from the position shown and it is found to come to rest when the thread becomes vertical. Assume that the thread does not hit the fixed charge at A. [$g = 10$ m/s^2]

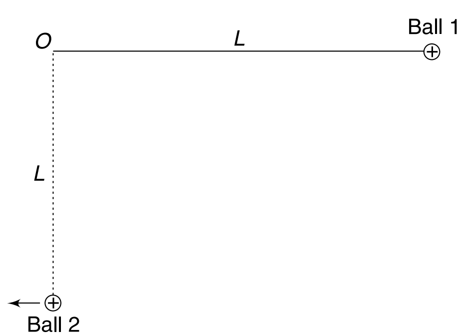
(a) Find mass m.

(b) Find tension in the thread in the equilibrium position when the thread is vertical.

(c) Is the equilibrium mentioned in (b) stable or unstable?



Q. 97: A small positively charged ball of mass m is suspended by an insulating thread of length L. This ball remains in equilibrium with string horizontal when another small charged ball is placed exactly at a distance L below the point of suspension of the first ball. The second ball is slowly moved away from the first ball to a far away point. (The second ball is moved horizontally so that the first ball does not accelerate). As a result the first ball lowers down to the original position of the second ball and the string become vertical. Find the work done by the external agent in removing the second ball.



Q. 98: A particle (A) having charge Q and mass m is at rest and is free to move. Another particle (B) having charge q and mass m is projected from a large distance towards the first particle with speed u.

(a) Calculate the least kinetic energy of the system during the subsequent motion.

(b) Find the final velocity of both the particles.

Consider coulomb force only.

Q. 99: In the last question, the two particles A and B are initially held at a distance $r = \dfrac{qQ}{2\pi\epsilon_0 \text{ mu}^2}$ apart. Particle B is projected directly towards A with velocity u and particle A is released simultaneously. Find the velocity of particle A after a long time. Consider coulomb force only.

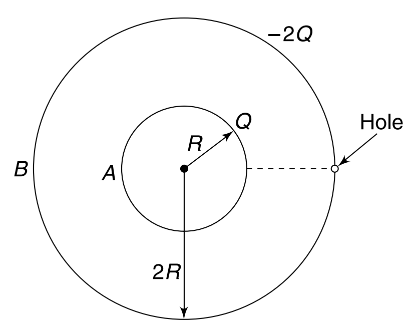
Q. 100: Two positively charged balls having mass m and 2m are released simultaneously from a height h with horizontal separation between them equal to x_0. The ball with mass 2m strikes the ground making an angle of 45° with the horizontal.

(a) At what angle, with horizontal, the other ball hits the ground?

(b) Find the work done by the electrostatic force during the course of fall of the two balls.

Q. 101: A and B are two concentric spherical shells made of conductor. Their radii are R and $2R$ respectively. The two shells have charge Q and $-2Q$ on them. An electron escapes from the surface of the inner shell A and moves towards a small hole in the outer shell B.

(a) What shall be the minimum kinetic energy of the emitted electron so that it can escape to infinity through the small hole in outer shell?



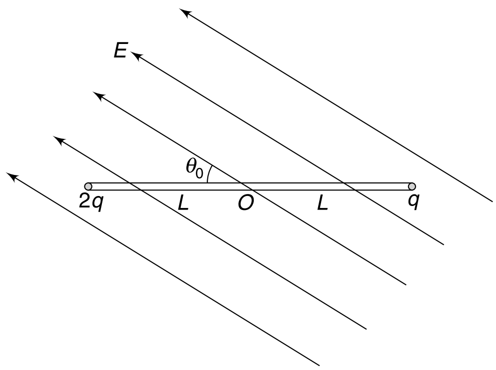
(b) What will be your answer if charge on both the shells were $+Q$? Charge on electron $= e$.

Q. 102: A thin uniform rod of mass M and length $2L$ is hinged at its centre O so that it can rotate freely in horizontal plane about the vertical axis through O. At its ends the insulating rod has two point charges $2q$ and q (see figure). An electric field E is switched on making an angle $\theta_0 = 60°$

with the initial position of the rod. The field is uniform and horizontal.

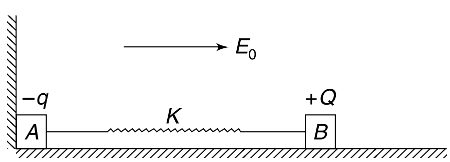
(a) Calculate the maximum angular velocity of the rod during subsequent motion.

(b) Find the maximum angular acceleration of the rod.

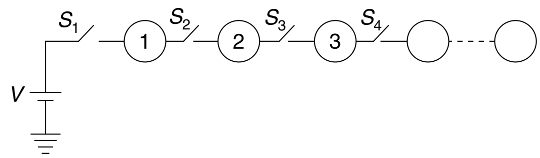


Q. 103: Two charge particles are moving such the distance between them remains constant. The ratio of their masses is 1 : 2 and they always have equal and opposite momentum. The particles interact only though electrostatic force and no other external force is acting on them. The electrostatic interaction energy for the pair of the particles is $-U_0$. Find the kinetic energy of the lighter particle. How does the kinetic energy change with time?

Q. 104: Two blocks A and B are connected by a spring made of a non conducting material. The blocks are placed on a non conducting smooth horizontal surface (see figure). The wall touching A is also non conducting. Block A carries a charge $-q$. There exists a uniform electric field of intensity E_0 in horizontal direction, in the entire region. Find the value of minimum positive charge Q that we must place on block B and release the system so that block A subsequently leaves contact with the wall. Force constant of the spring is k. Neglect interaction between charges on the blocks.



Q. 105: Large number of identical conducting spheres have been laid as shown in figure. Radius of each sphere is R and all of them are uncharged. Switch S_1 is closed to connect sphere 1 to the positive terminal of a V volt cell whose other terminal is grounded. After some time switch S_1 is opened and S_2 is closed. Thereafter, S_2 is opened and S_3 is closed, next S_3 is opened and S_4 is closed. The process is continued till the last switch is closed. Consider the cell and spheres to be your system and calculate the loss in energy of the system in the entire process.



Q. 106: A short electric dipole has dipole moment p. Find the distance of farthest point from the dipole where-

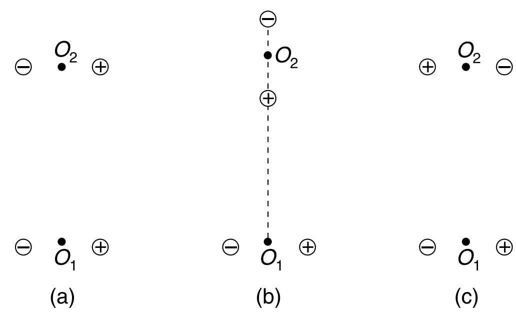
(a) potential due to the dipole is V_0

(b) Electric field due to the dipole is E_0

Q. 107: Two identical electric dipoles are arranged parallel to each other with separation between them large compared to the length of individual dipole. The electrostatic energy of interaction of the two dipoles in this position is U.

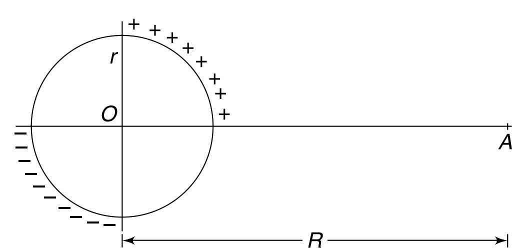
(a) Find work done in slowly rotating one of the dipoles by 90° so as to bring it to position shown in Fig. (b).

(b) Find work done in rotating one of the dipoles by 180° so as to bring it to the position shown in Fig. (c).



O_1 and O_2 are centers of the dipoles.

Q. 108: A ring of radius r has a uniformly spread charge $+q$ on quarter of its circumference. The opposite quarter of the ring carries a charge $-q$ uniformly spread over it. Find the electric potential at a point A shown in the figure. Point A is at a distance $R(>> r)$ from the centre of the ring.



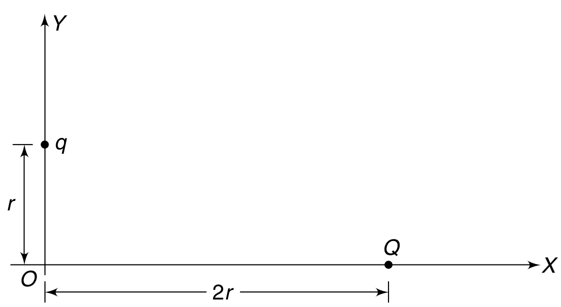
LEVEL 3

Q. 109: Three small equally charged identical conducting balls are suspended from identical insulating threads secured at one point. Length (L) of the threads is large compared to the equilibrium separation (a) between any two balls.

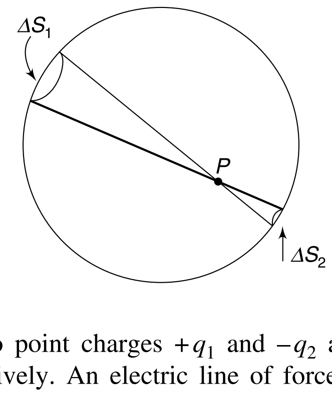
(a) One of the balls is suddenly discharged. Find the separation between the charged balls when equilibrium is restored. Assume that the threads do not interfere and balls do not collide.

(b) If two of the balls are suddenly discharged, how will the balls behave after this? Find the separation between the balls when equilibrium is restored. The threads do not interfere.

Q. 110: Two charged particle of equal mass are constrained to move along X and Y direction. The $X-Y$ plane is horizontal and the tracks are smooth. The particles are released from rest when they were at positions shown in the figure. At the instant distance of q becomes $2r$ from the origin, find the location of charge Q.



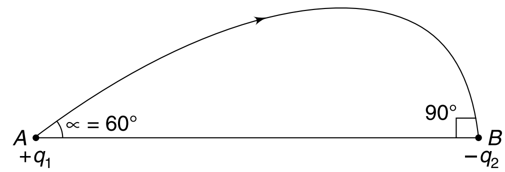
Q. 111: Consider a uniformly charged spherical shell. Two cones having same semi vertical angle, and their common apex at P, intercept the shell. The intercepts have area $\Delta S1$ and $\Delta S2$. For a cone of very small angle, $\Delta S1$ and $\Delta S2$ will be very small and charge on them can be regarded as point charge for the purpose of writing electric field at point P. Prove that the charge on $\Delta S1$ and $\Delta S2$ produce equal and opposite field at P. Hence, argue that field at all points inside the uniformly charged spherical shell is zero.



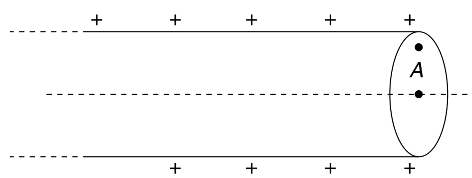
Q. 112: Two point charges $+q_1$ and $-q_2$ are placed at A and B respectively. An electric line of force emerges from q_1 making an angle $\alpha = 60°$ with line AB and terminates at $-q_2$ making an angle of 90° with the line AB.

(a) Find $\left|\frac{q_1}{q_2}\right|$

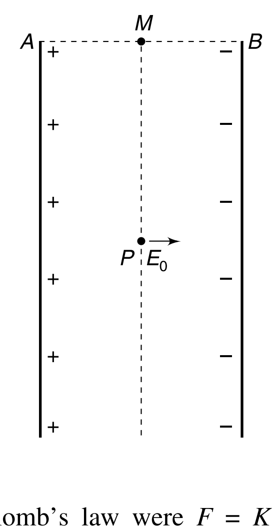
(b) Find the maximum value of angle α at which a line emitted from q_1 terminates on charge q_2.



Q. 113: There is a semi-infinite hollow cylindrical pipe (i.e. one end extends to infinity) with uniform surface charge density. What is the direction of electric field at a point A on the circular end face?

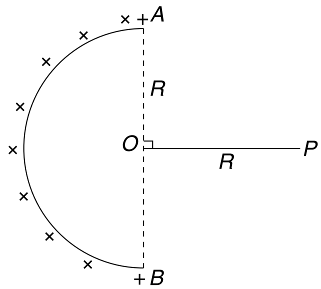


Q. 114: Two equal insulating threads are placed parallel to each other. Separation between the threads (= d) is much smaller than their length. Both the threads have equal and opposite linear charge density on them. The electric field at a point P, equidistant from the threads (in the plane of the threads) and located well within (see figure) is E_0. Calculate the field at mid point (M) of line AB.



Q. 115: If coulomb's law were $F = K\frac{qQ}{r^3}$, calculate the electric field due to a uniformly charged line charge at a distance d from it. The linear charge density on the line charge is λ C/m, and it is of infinite length.

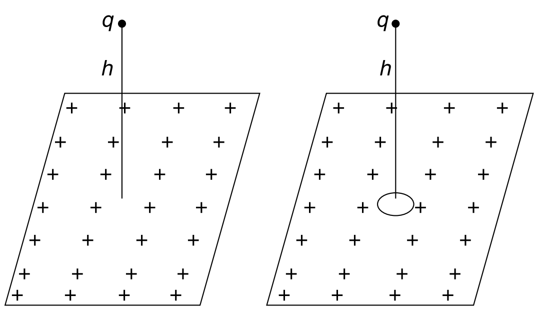
Q. 116: A semicircular ring of radius R carries a uniform linear charge density of λ. P is a point in the plane of the ring at a distance R from centre O. OP is perpendicular to AB. Find electric field intensity at point P.



Q. 117: A small charged ball is in state of equilibrium at a height h above a large horizontal uniformly charged dielectric plate having surface charge density of σ C/m^2.

(a) Find the acceleration of the ball if a disc of radius $r(<< h)$ is removed from the plate directly underneath the ball.

(b) Find the terminal speed (V_0) acquired by the falling ball. Assume that mass of the ball is m, its radius is x and coefficient of viscosity of air is η. Neglect buoyancy and assume that the ball acquires terminal speed within a short distance of its fall.



Q. 118: In a certain region of space the electrostatic field depends only on the coordinates x and y as follows.

$$E = 0 \qquad \text{for } \sqrt{x^2 + y^2} < r_0$$

$$E = a(x\hat{i} + y\hat{j})/(x^2 + y^2), \text{ for } \sqrt{x^2 + y^2} > r_0$$

where a is a positive constant, and $\hat{i}$ and $\hat{j}$ are the unit vectors along the X- and Y-axes. Find the charge within a sphere of radius $2r_0$ with the centre at the origin.

Q. 119: A solid sphere of radius R has total charge Q. The charge is distributed in spherically symmetric manner in the sphere. The charge density is zero at the centre and increases linearly with distance from the centre.

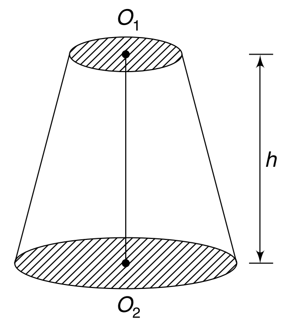
(a) Find the charge density at distance r from the centre of the sphere.

(b) Find the magnitude of electric field at a point 'P' inside the sphere at distance r_1 from the centre.

Q. 120: A ball of radius R carries a positive charge whose volume charge density depends only on the distance r from the ball's centre as: $\rho = \rho_0\left(1 - \frac{r}{R}\right)$

Where r_0 is a constant. Take ε to be permittivity of the ball. Calculate the maximum electric field intensity at a point (inside or outside the ball) due to such a charge distribution.

Q. 121: A frustum is cut from a right circular cone. The two circular faces have radii R and $2R$ and their centers are at O_1 and O_2 respectively. Height of the frustum is $h = 3R$. When a point charge Q is placed at O_1, the flux of electric field through the circular face of radius $2R$ is ϕ_1 and when the charge Q is placed at O_2, the flux through the other circular face is ϕ_2. Find the ratio $\frac{\phi_1}{\phi_2}$.



Q. 122: The electric field in a region of space varies as $E = (x\hat{i} + 2y\hat{j} + 3z\hat{k})$ V/m

(a) Consider an elemental cuboid whose one vertex is at (x, y, z) and the three sides are dx, dy and dz; sides being parallel to the three co-ordinate axes. Calculate the flux of Electric field through the cube.

(b) Using the expression obtained in (a) find the charge enclosed by a spherical surface of radius r, centred at the origin.

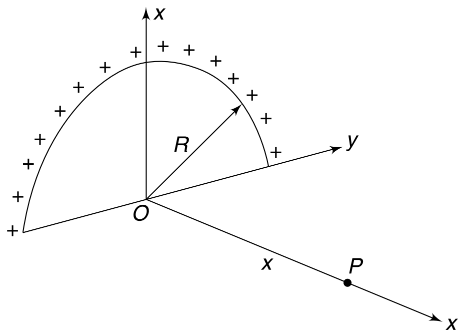
Q. 123: A ball of radius R has a uniformly distributed charge Q. The surrounding space of the ball is filled with a volume charge density $\rho = \frac{b}{r}$, where b is a constant and r is the distance from the centre of the ball.

It was found that the magnitude of electric field outside the ball is independent of distance r.

(a) Find the value of Q.

(b) Find the magnitude of electric field outside the ball.

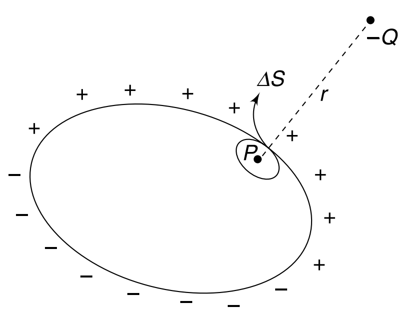
Q. 124: A uniform semicircular ring of radius R is in yz plane with its centre at the origin. The half ring carries a uniform linear charge density of λ.



(a) Find the x, y and z component of Electric field at a point $P(x, 0, 0)$ on the axis of the ring.

(b) Prove that the field at P is directed along a line joining the centre of mass of the half ring to the point P.

Q. 125: A charge $-Q$ is placed at some distance from a neutral conductor. Charge is induced on its surface. In the neighbourhood of a point P on its surface, the charge density is σ C/m^2. Consider a small area ΔS on the surface of the conductor encircling point P. Find the resultant force experienced by the area ΔS due to charge present on the surface elsewhere and the charge $-Q$.



Q. 126: A soap bubble of radius $R = 1$ cm is charged with the maximum charge for which breakdown of air on its surface does not occur. Calculate the electrostatic pressure on the surface of the bubble. It is know that dielectric breakdown of air takes place when electric field becomes larger then $E_0 = 3 \times 10^6$ V/m.

Q. 127: A conducting sphere of radius R is cut into two equal halves which are held pressed together by a stiff spring inside the sphere.

(a) Find the change in tension in the spring if the sphere is given a charge Q.

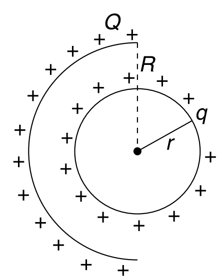
(b) Find the change in tension in the spring corresponding to the maximum charge that can be placed on the sphere if dielectric breakdown strength of air surrounding the sphere is E_0.

Q. 128: Surface tension of a soap solution is T. There is a soap bubble of radius r. Calculate the amount of charge that must be spread uniformly on its surface so that its radius becomes $2r$. Atmospheric pressure is P_0. Assume that air temperature inside the bubble remains constant.

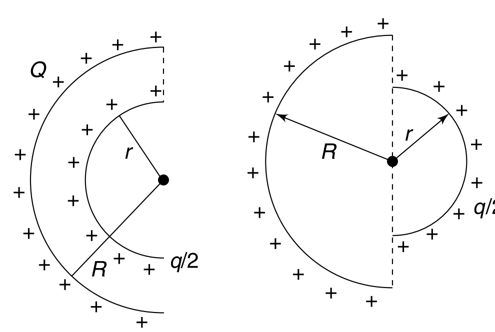
Q. 129: A point charge Q has been placed at a point outside a neutral spherical conductor. The induced charge density at point P on the surface of the conductor is $-\sigma$. The distance of point P from the point charge Q is $2R$ (where R is radius of the conductor).

Find the magnitude and direction of electric field at a point outside the conductor that is very close to its surface near P.

Q. 130: A spherical shell of radius r carries a uniformly distributed surface charge q on it. A hemispherical shell of radius $R(> r)$ is placed covering it with its centre coinciding with that of the sphere of radius r. The hemisphere has a uniform surface charge Q on it. The charge distribution on the sphere and the hemisphere is not affected due to each other. Calculate the force that the sphere will exert on the hemisphere.



Q. 131: In the last question half of the inner sphere is removed along with its charge (i.e., the remaining half has charge $\frac{q}{2}$). Find the force between the bigger and smaller hemispheres in the two cases shown in figure (a) and figure (b).

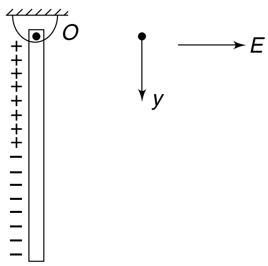


(a) (b)

Q. 132: A charged ball with mass m and charge q is dropped from a height h over a non-conducting smooth horizontal plane. There exists a uniform electric field E_0 in vertically downward direction and the coefficient of restitution between the ball and the plane is e. Find the maximum height attained by the ball after n^{th} collision.

Q. 133: In the last question, the electric field in vertical direction is switched off and a field of same strength (E_0) is switched on in horizontal direction. Find the horizontal velocity of the ball during the n^{th} collision. Also calculate the time interval between n^{th} and $(n + 1)^{\text{th}}$ collision.

Q. 134: A thin insulating rod of mass m and length L is hinged at its upper end (O) so that it can freely rotate in vertical plane. The linear charge density on the rod varies with distance (y) measured from upper end as



$$\left.\begin{aligned} \lambda &= ay^2 \quad ; \quad 0 \le y \le \frac{L}{2} \\ &= -by^n \quad ; \quad \frac{L}{2} < y \le L \end{aligned}\right\}$$

Where a and b both are positive constants. When a horizontal electric field E is switched on the rod is found to remain stationary.

(a) Find the value of constant b in terms of a. Also find n.

(b) Find the force applied by the hinge on the rod, if $EaL^3 = 45$ mg.

Q. 135: Two fixed charges $-2Q$ and Q are located at the points with coordinates $(-3a, 0)$ and $(+3a, 0)$ respectively in the $x - y$ plane.

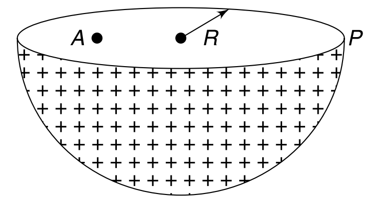
(a) Show that all points in the $x-y$ plane where the electric potential due to the two charges is zero, lie on a circle. Find its radius and the location of its centre.

(b) Give the expression $V(x)$ at a general point on the $x-$axis and sketch the function $V(x)$ on the whole $x-$axis.

(c) If a particle of charge $+q$ starts from rest at the centre of the circle, show by a short quantitative argument that the particle eventually crosses the circle. Find its speed when it does so.

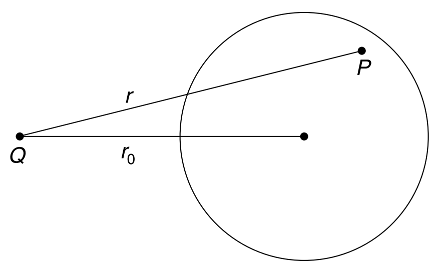
Q. 136: A metallic sphere of radius R has a small bulge of hemispherical shape on its surface. The radius of the bulge is r. If a charge Q is given to the sphere, calculate the quantity of charge on the surface of the bulge. Assume that charge is uniformly distributed on the surface of the bulge (though it is wrong !) and also on the remaining surface of the sphere.

Q. 137: A hemispherical bowl of radius R carries a uniform surface charge density of σ. Find potential at a point P located just outside the rim of the bowl (see figure). Also calculate the potential at a point A located at a distance $R/2$ from the centre on the equatorial plane.

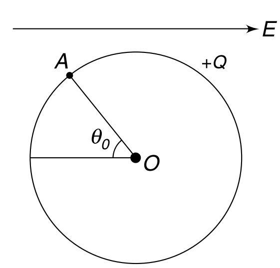


Q. 138: Two infinite line have linear charge densities $-\lambda$ and $+\lambda$. They are parallel to z axis passing through x axis at points $x = -a$ and $x = a$ respectively. Show that the equipotential surface having potential $\dfrac{\lambda \ln(2)}{4\pi\varepsilon_0}$ is a cylinder having radius $2\sqrt{2}a$.

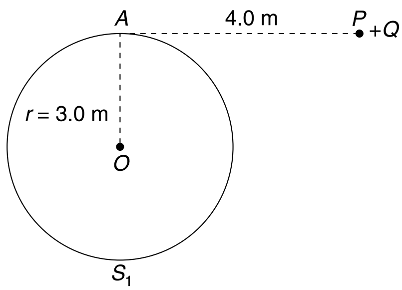
Q. 139: A conducting shell having no charge has radius R. A point charge Q is placed in front of it at a distance r_0 from its centre. Find potential due to charge induced on the surface of the shell at a point P inside the shell. Distance of point P from point charge Q is r.



Q. 140: A conducting sphere of radius R having charge Q is placed in a uniform external field E. O is the centre of the sphere and A is a point on the surface of the sphere such that AO makes an angle of $\theta_0 = 60°$ with the opposite direction of external field. Calculate the potential at point A due to charge on the sphere only.



Q. 141: Consider a solid neutral conducting sphere S_1 of radius $r = 3.0$ m. A point charge $Q = +2\mu C$ is placed at point P such that $AP = 4.0$ m (AP is tangent to the sphere). Charge Q' is induced on the surface of the conducting sphere S_1. Consider another non conducting sphere (S_2) of same radius r. Charge Q' is spread on the surface of S_2 in exactly the same way as it is present on the surface of conducting sphere S_1 [i.e., the distribution of charge on surface of S_2 is exact replica of the induced surface charge on S_1]. There is no other charge in vicinity of S_2. Find the smallest potential (at a point) on the surface of sphere S_2.

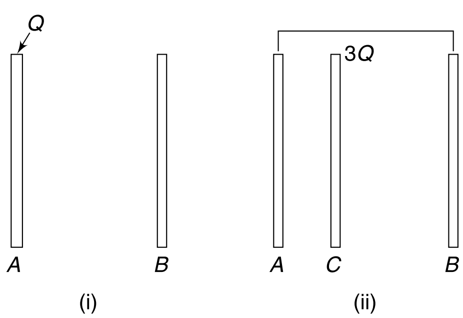


Q. 142: A and B are two large identical thin metal plates placed parallel to each other at a small separation. Plate A is given a charge Q.

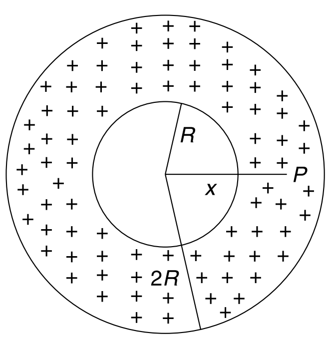
(i) Find the amount of charge on each of the two faces of A and B.

(ii) Another identical plate C having charge $3Q$ is inserted between plate A and B such that distance of C from B is twice its distance from A. Plate A and B is shorted using a conducting wire. Find charge on all six faces of plates A, B and C.

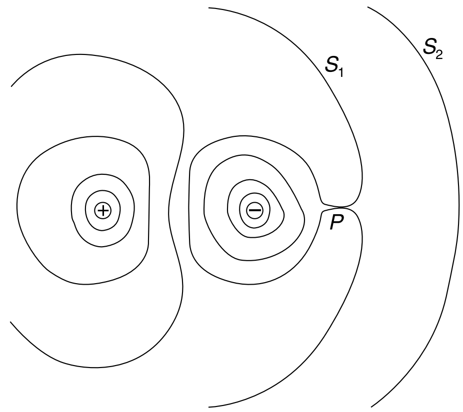
(iii) In the situation described in (ii) the plate A is grounded. Now write the charge on all six faces.



Q. 143: A non conducting shell of inner and outer radii R and $2R$ respectively has a charge Q uniformly distributed in its volume. Find the electric potential at a point P at a distance r_0 from the centre such that $R < r_0 < 2R$.



Q. 144: The figure shows equipotential surfaces due to two point charges $+3Q$ and $-Q$ placed at a separation d.



(a) Will the shape of epuipotential surface be spherical very close to both the point charges? What shape of equipotentials will be seen at very far away points from the pair of charges.

(b) Find the distance of point P from the negative charge.

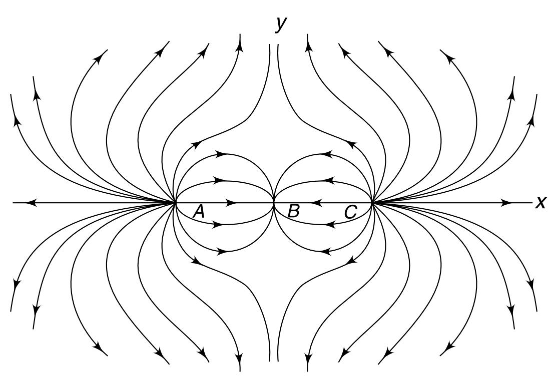
(c) Find the potential of the surface marked as S_1 is the figure.

(d) Consider the surface having zero potential. Write the flux of electric field through this surface.

Q. 145: Three point charge have been placed along the x axis at points A, B and C. Distance $AB = BC$. The field lines generated by the system is shown in figure.

(a) Plot the variation of electric potential along x axis. Show potential on y axis of your graph.

(b) Plot the variation of electric potential along y axis with B as origin. Shows potential on x axis in your graph.



Q. 146: A conducting shell has inner radius R and outer radius $2R$. A charge $+q$ is given to the spherical shell.

(a) Find the electric field at a point which is at a distance x from the centre of the shell. Give your answer for three cases

(i) $x < R$ (ii) $R < x < 2R$ (iii) $x > 2R$

(b) Find the electric potential in all the three cases mentioned in (a)

(c) Find field and potential in all the three cases mentioned in (a) after a point charge $-q$ is introduced at the centre of the shell.

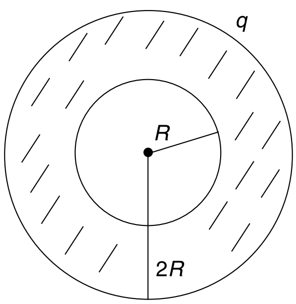
(d) Write the electrical potential energy of the system consisting of the shell and the point charge at its centre.

(e) Find the electrostatic force that the shell exerts on the point charge.

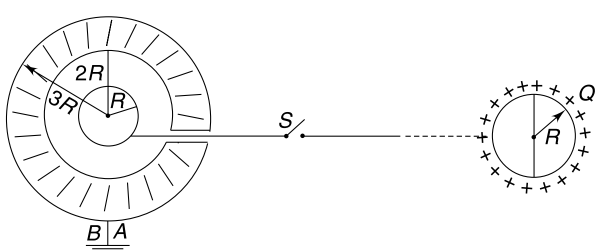
(f) Now, another point charge $+q$ is placed at a distance $4R$ from the centre of the shell. Find electric field and potential in following cases.

(i) $x < R$

(ii) $R < x < 2R$



Q. 147: A solid conducting sphere of radius R is surrounded by a concentric metallic shell of inner radius $2R$ and outer radius $3R$. The shell is earthed. The inner sphere is connected to a switch S by a thin conducting wire through a small hole in the shell. By closing the switch S, the inner sphere is connected to a distant conducting sphere of radius R having charge Q. Find the charge that flows to earth through wire AB.

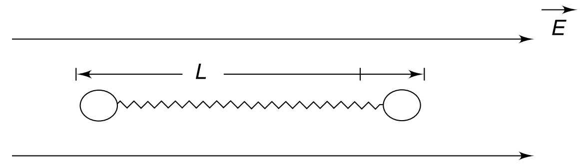


Q. 148: Two small identical conducting balls each of radius r and mass m are placed on a frictionless horizontal table, connected by a light conducting spring of force constant K and un-deformed length $L(L >> r)$. A uniform electric field of strength E is switched on in horizontal direction parallel to the spring.

(a) How much charge will appear on the two balls when they are at separation L.

(b) The system fails to oscillate if $K < K_0$. Find K_0.

(c) Assuming $K = 2K_0$, find the time period of oscillation after the electric field is switched on.



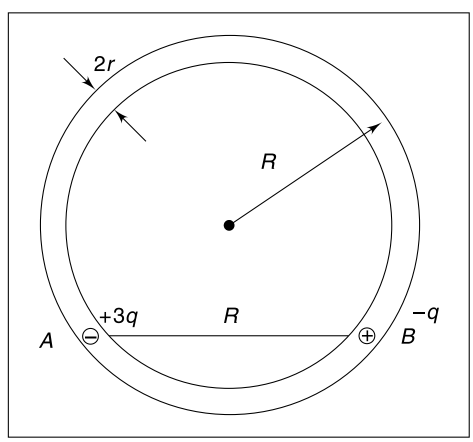
Q. 149: Four charges have equal magnitude. Two of them are positive and remaining two are negative. Charges have been placed on the vertices of a rectangle and the electrostatic potential energy of the system happens to be zero.

(a) Show the arrangement of the charges on the vertices of the rectangle.

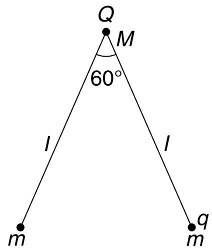
(b) If the smaller side of the rectangle has a length of 1.0 m, show that length of larger side must be less than 2.0 m.

[Actually you need to solve on algebraic polynomial to get the exact length of the larger side. Here I am not exactly interested in making you solve that equation]

Q. 150: On a horizontal table, there is a smooth circular groove of mean radius R. The walls of the groove are non conducting. Two metal balls (each having mass m and radius r) are placed inside the groove with their centers R apart. The balls just fit inside the groove. The two balls are given charge $+3q$ and $-q$ and released from state of rest. Ignore the non-uniformity in charge distribution as the balls come close together and collide. The collision is elastic. Find maximum speed acquired by each ball after they collide for the first time.



Q. 151: A particle having charge Q and mass $M = 2m$ is tied to two identical particles, each having mass m and charge q. The strings are of equal length, l each, and they are inextensible. The system is held at rest on a smooth horizontal surface (with string taut) in a position where the strings make an angle of 60° between them. From this position the system is released.



(a) Find the amplitude of oscillation of M

(b) Find maximum speed acquired by M

(c) Find tension in the string when all the three particles get in one straight line.

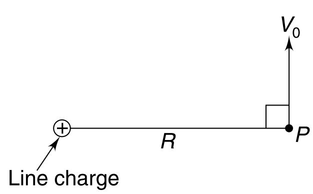
Q. 152: Four point charges $+8\,\mu C$, $-1\,\mu C$, $-1\,\mu C$ and, $+8\,\mu C$ are fixed at the points $-\sqrt{27/2}\,m$, $-\sqrt{3/2}\,m$, $+\sqrt{3/2}\,m$ and $+\sqrt{27/2}\,m$ respectively on the y-axis. A particle of mass 6×10^{-4} kg and $+0.1\,\mu C$ moves along the $x-$direction. Its speed at $x = +\infty$ is v_0. Find the least value of v_0 for which the particle will cross the origin. Find also the kinetic energy of the particle at the origin in this case. Assume that there is no force apart from electrostatic force.

Q. 153: A dielectric disc of radius R and uniform positive surface charge density σ is placed on the ground with its axis vertical. A particle of mass m and positive charge q is dropped, along the axis of the disc from a height H with zero initial velocity. The charge – mass ratio of the particle is $\frac{q}{m} = \frac{4\varepsilon_0 g}{\sigma}$.

(i) Find the value of H if the particle just reaches the disc.

(ii) Sketch the potential energy of the particle as a function of its height and find its height in equilibrium position

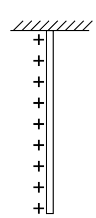
Q. 154: An infinite line charge is perpendicular to the plane of the figure having linear charge density λ. A particle having charge Q and mass m is projected in the field of the line charge from point P. The point P is at a distance R from the line charge and velocity given to the particle is perpendicular to the radial line at P (see figure).



(a) Find the speed of the particle when its distance from the line charge grows to ηR $(\eta > 1)$.

(b) Find the velocity component of the particle along the radial line (joining the line charge to the particle) at the instant its distance becomes ηR.

Q. 155: A uniformly charged non conducting rod is suspended vertically at its end. The rod can swing freely in the vertical plane without any friction. The linear charge density on the rod is λ C/m and it has a uniformly distributed mass of M. Length of the rod is L. A uniform horizontal Electric field (E) is switched on in the region.



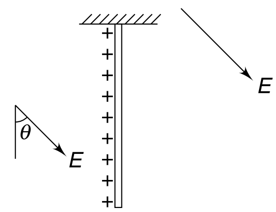
(a) For what value of electric field (call it E_0) the rod just manages to make itself horizontal?

(b) If Electric field E_0 is switched on, what is the maximum angular speed acquired by the rod during its motion?

Q. 156: In the last question let us assume that the uniform electric field makes an angle θ with the vertical in downward direction. With the uniformly charged rod in vertical position the field is switched on. Mass of the rod, its length and charge per unit length is M, L and λ respectively.

(a) Find the strength of field (E) such that the rod can reach the horizontal position if $\theta = 30°$

(b) Find the minimum strength of field (E) such that the rod can reach the horizontal position if $\theta = 60°$



(c) However high the field might be, the rod cannot become horizontal if $\theta < \theta_0$. Find θ_0

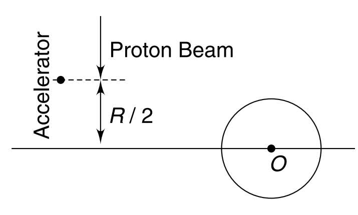
Q. 157: A massless rod of length L has two equal charges (q) tied to its ends. The rod is free to rotate in horizontal plane about a vertical axis passing through a point at a distance $\frac{L}{4}$ from one of its ends. A uniform horizontal electric field (E) exists in the region.

(a) Draw diagrams showing the stable and unstable equilibrium positions of the rod in the field.

(b) Calculate the change in electric potential energy of the rod when it is rotated by an angle $\Delta\theta$ from its stable equilibrium position.

(c) Calculate the time period of small oscillations of the rod about its stable equilibrium position. Take the mass of each charge to be m.

Q. 158: A proton accelerator produces a narrow beam of protons, each having an initial speed of v_0. The beam is directed towards an initially uncharged distant metal sphere of radius R. The sphere is fixed and centered at point O. The initial path of the beam is at a distance of (R/2) from the centre, as indicated in the diagram.



The protons in the beam that collide with the sphere get absorbed and cause it to become uniformly charged. The subsequent potential field at the accelerator due to the sphere can be neglected. Assume the mass of the proton as m_P and the charge on it as e.

(a) After a long time, when the potential of the sphere reaches a constant value, sketch the trajectory of proton in the beam.

(b) Once the potential of the sphere has reached its final, constant value, find the minimum speed v of a proton along its trajectory path.

(c) Find the limiting electric potential of the sphere.

Q. 159: Consider a uniformly charged spherical shell of radius R having charge Q. Charge Q can be thought to be made up of number of point charges q_1, q_2, q_3 ... etc. The electrostatic energy of the charged shell is sum of interaction energies of all possible pairs of charges.

$$U = \sum_{\text{all pairs}} \frac{q_i q_j}{4\pi\varepsilon_0 r_{ij}}$$

Where r_{ij} is distance between q_i and q_j. For continuous charge on the shell, the summation has to be carried through integration.

(a) Calculate the electrostatic energy of the shell. We can term this energy as self energy of the shell.

(b) Calculate the work done in assembling a spherical shell of uniform charge Q and radius R by bringing charges in small installments from infinity and putting them on the shell.

Do you find the answers in (a) and (b) to be same?

Q. 160:

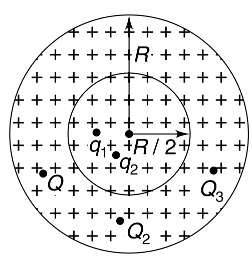
(a) Use the method in part (b) of the previous problem to calculate the electrostatic self energy of a uniformly charged sphere of radius R having charge Q.

(b) Divide the above sphere (mentally) into two regions-spherical concentric part having radius $\frac{R}{2}$ and the remaining annular part $\left(\text{between } \frac{R}{2} \text{ and } R\right)$.

Denote the point charges in sphere of radius $R/2$ by q_1, q_2, q_3 ... etc.

The charges in annular part be denoted by Q_1, Q_2, Q_3 ... etc.

Calculate the electrostatic interaction energy for all pairs like $[(Q_i, Q_j) + (q_i + q_j)]$.



Q. 161: A conducting sphere S_1 of radius r is attached to an insulating handle. Another conducting sphere S_2 of radius R is mounted on an insulating stand. S_2 is initially uncharged.

S_1 is given a charge Q brought into contact with S_2 and removed. S_1 is recharged such that the charge on it is again Q and it is again brought into contact with S_2 and removed. This procedure is repeated n times.

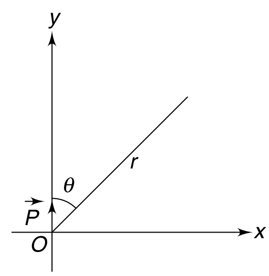
(i) Find the electrostatic energy of S_2 after n such contact with S_1.

(ii) What is the limiting value of this energy as $n \to \infty$?

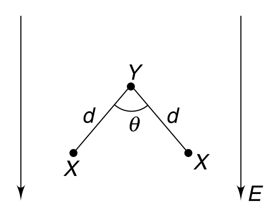
Q. 162: A short electric dipole is placed at the origin of the Co-ordinate system with its dipole moment P along y direction. Give answer to following questions for points which are at large distance r from the origin in $x - y$ plane. [r is large compared to length of the dipole].

(a) Find maximum value of x component of electric field at a point that is at r_0 distance from the origin.

(b) Prove that (for $0 < \theta < 90°$) all the points, where electric field due to the dipole is parallel to x–axis, fall on a straight line. Find the slope of the line.



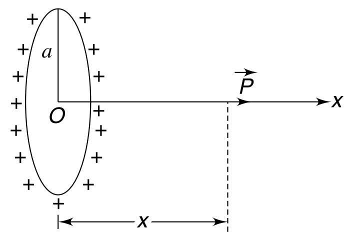
Q. 163: A tri atomic molecule X_2Y has plane structure as shown in figure. Due to difference in electronegativity, charge acquired by each X atom is q and charge on Y atom is $-2q$. The bond length between Y and X is d, and angle between the two bonds is $\theta = 120°$ mass of one atom of X and Y are m and 8m respectively. The molecule is placed in a uniform Electric field E and is making small oscillations about an axis perpendicular to the plane of figure and passing through the centre of mass of the molecule. Find the time period of oscillation.



Q. 164: A short electric dipole (dipole moment P) is placed on the axis of a uniformly charged ring at a distance x from the centre as shown in figure. Radius of the ring is a and charge on it is Q.

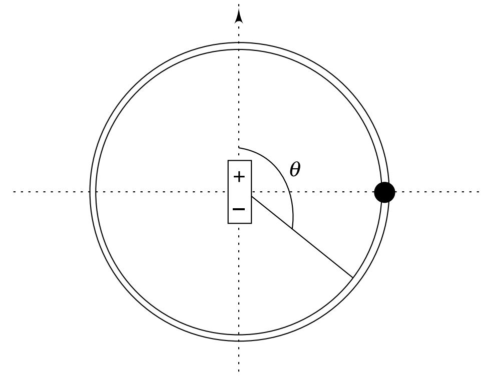
(i) Write the force on the dipole when $x = \frac{a}{2}$ and when $x = a$. Why the direction of force at two points is different?

(ii) Is the force on the dipole zero if $x = 0$? If not, where will you place the dipole so that force on it is zero.



Q. 165: Consider two spheres of the same radius R having uniformly distributed volume charge density of same magnitude but opposite sign ($+\rho$ and $-\rho$). The spheres overlap such that the vector joining the centre of the negative sphere to that of the positive sphere is $\vec{d}$. ($d << R$). Find magnitude of electric field at a point outside the spheres at a distance r in a direction making an angle θ with $\vec{d}$. Distance r is measured with respect to the mid point of the line joining the centers of the two spheres.

Q. 166: A small, electrically charged bead can slide on a circular, frictionless, thin, insulating ring. Charge on the bead is Q and its mass is m . A small electric dipole, having dipole moment P is fixed at the centre of the circle with the dipole's axis lying in the plane of the circle. Initially, the bead is held on the perpendicular bisector of the dipole (see fig.) Ignore gravity and answer the following questions.



(a) Write the speed of the bead when it reaches the position θ shown in the figure.
(b) Find the normal force exerted by the ring on the bead at position θ.
(c) How does the bead move after it is released? Where will the bead first stop after being released?
(d) How would the bead move in the absence of the ring?

Q. 167: A short electric dipole ($\vec{P}$) has been placed in a uniform electric field ($\vec{E}$) with the dipole moment vector ($\vec{P}$) parallel to $\vec{E}$. Show the field lines in the region. Mark the null points (i.e. the points where the field is zero)

ANSWERS

1. (a) $2F$ towards point F
(b) In a direction making an angle of 60° with BA (NOT along BC).

3. Zero.

4. (a) 120000 N
(b) Force is nearly 10 times the weight of the car.
Two conclusions can be drawn from the result-
(i) Electrostatic force is a strong force
(ii) One coulomb is a very large charge.

5. $\frac{r^2}{2}$

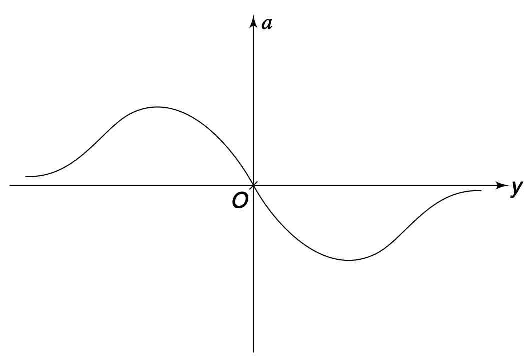
6. $\therefore\ k = \frac{q^2}{4\pi\epsilon_0 L^3}\left(\frac{2\sqrt{2}+1}{\sqrt{2}}\right)$

7. $F_2 = F_1\left(\frac{q_2^2}{q_1^2}\frac{l_1^2}{l_2^2}\right)$

8. (i) $T = 2\pi\sqrt{\frac{L}{2g}}$ (ii) $T = 2\pi\sqrt{\frac{L}{g}}$

(iii) $T = 2\pi\sqrt{\frac{L}{\sqrt{2}g}}$ (iv) $T = 2\pi\sqrt{\frac{L}{g}}$.

9.



10. All graphs are possible.
(a) electron has been projected in the direction of electric field
(b) electron has been released from rest
(c) electron has been projected making an acute angle with field
(d) electron has been projected in a direction making an obtuse angle or right angle to the direction of field

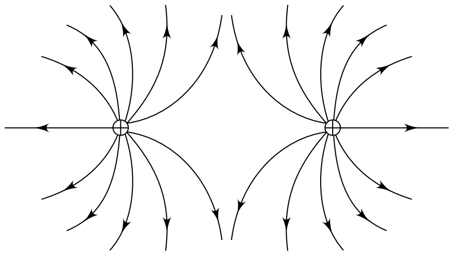
11. (a) $K_B = \pi qER$ (b) $a_t = \frac{qE}{m}\sqrt{1+\pi^2}$

12. $-\frac{qa}{m}$

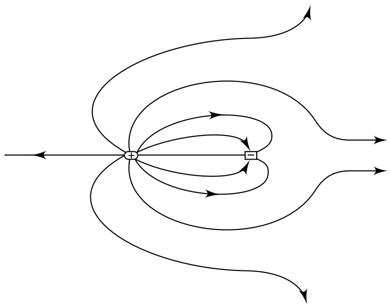
13. (a) $\pi\alpha R_0^4$ (b) $\dfrac{R_0}{(2)^{1/4}}$

(c) $\dfrac{\alpha r^2}{4\in_0}$

14. (a)

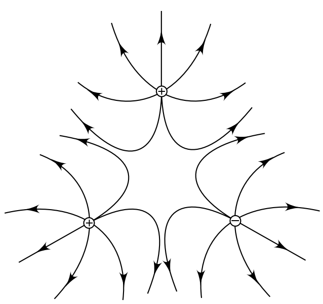


(b)



Field lines will appear to be radial at a large distance.

(c)



15. 50 V

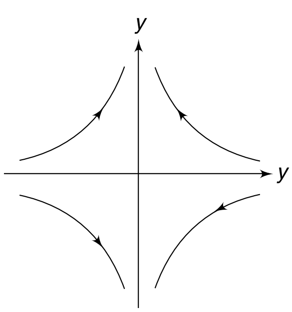
16. $\vec{E} = \left(\hat{i} + 2\hat{j} + 2\sqrt{5}\hat{k}\right)$ V/m

17. $\vec{a} = (\hat{i} + \hat{j})$ m/s^2

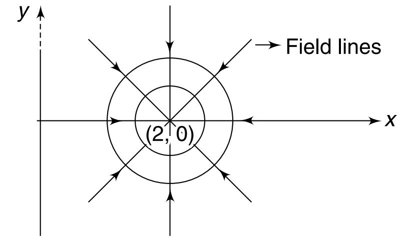
18. $E_A > E_B$ $V_A > V_B$

19. (b) No, (c) Zero, No

20. (a)



(b)



21. $W = -\dfrac{Kq^2}{a}\left(1 - \dfrac{1}{\sqrt{3}}\right)$

22. For both (a) and (b) $W = \dfrac{Qq}{8\pi\in_0 r}$

23. (a) $-\dfrac{Qe}{8\pi\in_0 r}$. (b) $8T$

24. Yes, V_0 depends on q.

$V_0 = \sqrt{\dfrac{2KQq}{mR}}$

25. (a) $\theta_0 = 2\tan^{-1}\left(\dfrac{qE}{mg}\right)$ (b) $E = \dfrac{mg}{q}$

26. $-\dfrac{Q^2}{16\pi\in_0 R}$

27. (a) $\dfrac{\eta}{4}$ (b) $[M^{-1}L^{-5}T^4A^2]$

28. (a) 39.6μJ (b) 2.25×10^{-9}J

29. $E = \dfrac{6\sqrt{3}KP}{L^3}$; $V = 0$

30. $\dfrac{PQ}{4\pi\in_0 R^3}$

31. $P = \dfrac{2Rq}{\pi}$

32. $\left(\dfrac{\sqrt{17} - 3}{2}\right) mg$

33. $\dfrac{32\pi\varepsilon_0 L^2 mg}{q}$

34. Stable equilibrium.

35. (a) $Q = +3.14q$ (b) $Q = -0.22q$

36. $\dfrac{1}{\sqrt{2}}$

37. (a) $\Delta T = \dfrac{KQq}{2\pi R^2}$ (b) $\dfrac{KQq}{\pi R^2}$ (c) $\dfrac{KQq}{\pi R^2}$

38. $\dfrac{8KQq}{r^2}$

39. $Q = 8R\sin\left(\dfrac{\theta}{2}\right)\sqrt{2\pi\varepsilon_0 mg\sin\left(\dfrac{\theta}{2}\right)}$

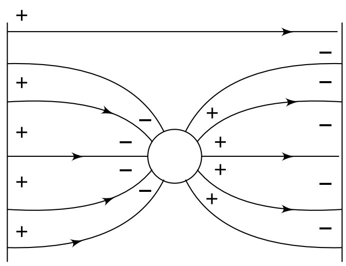
40. $f = \dfrac{1}{2\pi}\sqrt{\dfrac{2g\sin\theta}{x_0}}$

41. (a) $\dfrac{3}{2}a$

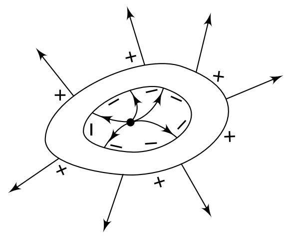
(b) Maximum electric force is between A and B. Maximum acceleration is at O

42. (a) No acceleration (b) Not affected

43. (a)



(b)



44. None of the configurations are possible.

45. $\dfrac{E_0}{2}$ perpendicular to AD

46. Perpendicular to the equatorial plane.

47. $\vec{E}$

48. $\dfrac{\sigma}{4\varepsilon_0}\dfrac{r^2}{x^2}$

49. (a) $E = \dfrac{2K\lambda}{R}\sin\left(\dfrac{\theta}{2}\right)$ (b) $E = \dfrac{2K\lambda}{R}$

(c) $E = 0$ (d) $E = \dfrac{KQ}{R^2}$

50. (a) $E_0 = \dfrac{\lambda}{2\pi\varepsilon_0 x}$ (b) $E_y = \dfrac{E_0}{2}$

(c) $E_x = \dfrac{E_0}{2} = \dfrac{\lambda}{4\pi\varepsilon_0 x}$ (d) 45°

51. Zero

52. $\dfrac{\lambda}{2\sqrt{2}\pi\varepsilon_0 R}$

53. (a) (i) $\dfrac{\rho x}{2\varepsilon_0}$

(ii) $\dfrac{\rho R^2}{2\varepsilon_0 x}$ Field at the surface is maximum

$E_{max} = \dfrac{\rho R}{2\varepsilon_0}$

(b) $\vec{E} = \dfrac{\rho\vec{a}}{2\varepsilon_0}$

54. (a) $T = 2\pi\sqrt{\dfrac{L}{g - \dfrac{q\sigma}{2m\varepsilon_0}}}$

(b) $T = 2\pi\sqrt{\dfrac{L}{\sqrt{g^2 + \left(\dfrac{qE}{m}\right)^2 - 2g\dfrac{qE}{m}\cos\beta}}}$

55. (a) $\theta = \tan^{-1}\left(\dfrac{3}{4}\right)$ (b) yes

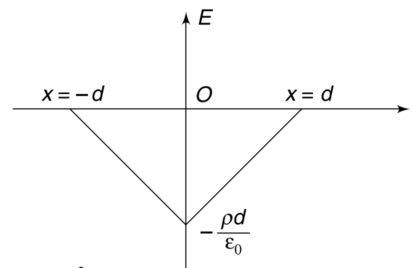
56. $\vec{E} = \dfrac{\lambda}{4\pi\varepsilon_0}(\hat{i} - \hat{k})$

57. (a) 17.7 μC, −53.1μC (b) No where

58. $E = -\dfrac{\rho}{\varepsilon_0}(d + x)$ $\quad -d < x \le 0$

$= -\dfrac{\rho}{\varepsilon_0}(d - x)$ $\quad 0 \le x < d$

$= 0$ $\quad |x| > d$



59. $E = \dfrac{by^2}{2\varepsilon_0}$

60. 2 : 1

61. (a) $q_0 = 4\pi\varepsilon_0 a$ (b) $q = -4\pi\varepsilon_0 a$

62. $\dfrac{Q}{6\varepsilon_0}$

63. $\dfrac{q}{2\varepsilon_0}$

64. $\sigma\phi$

65. (a) $\dfrac{\sqrt{5}}{6}\dfrac{\rho R}{\varepsilon_0}$ (b) $\dfrac{\sqrt{41}}{24}\dfrac{\rho R}{\varepsilon_0}$

66. (i) $\sigma = \dfrac{\varepsilon_0 mdu^2}{8eL^2}$ (ii) $\theta = \tan^{-1}\left(\dfrac{\varepsilon_0 mu^2}{2Le\cdot\sigma}\right)$

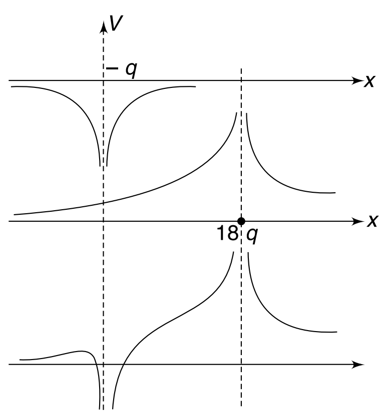
67. (a) $\dfrac{36}{25}$ s (b) $\dfrac{72}{25}$ s

(c) $\tan^{-1}\left(\dfrac{3}{4}\right)$

68. (a) t_0 (b) $\sqrt{6mK_0}$

69. $\sqrt{\dfrac{2qa}{3mV_0^2}(y^3 - y_0^2)}$

70.



71. (a) Parallel to $\overrightarrow{AB}$ (b) Zero.

72. (a) $q_1 \to +\text{ve}$; $q_2 \to -\text{ve}$
q_1 has larger magnitude
(b) $\dfrac{d}{\sqrt{3}-1}$

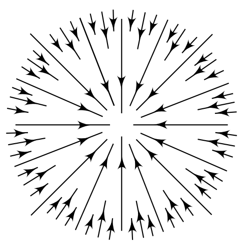
73. (a) 5 V (b) $\sqrt{\dfrac{2}{3}}$ m

74. $E = 15$ V/cm; $\cos^{-1}\left(\dfrac{2}{3}\right)$

75. (a) $V = \dfrac{\sigma}{8\epsilon_0}\left[\sqrt{R^2+Z^2} - Z\right]$

(b) $E_Z = \dfrac{\sigma}{8\epsilon_0}\left[1 - \dfrac{Z}{\sqrt{R^2+Z^2}}\right]$

76. (a)



(b) $-16\pi\epsilon_0 a^3$ (c) -42 Volt

77. $\dfrac{mR^2\omega^2}{2e}$

78. 2 : 1

79. 4

80. $\dfrac{Vr}{R}$

81. (a) Yes. $4\pi\sigma(b^2 - a^2)$ (b) $\dfrac{\sigma b}{\epsilon_0}$

82. $V' = V\left(\dfrac{a}{3t}\right)^{1/3}$

83. $\dfrac{1}{4\pi\epsilon_0}\dfrac{q}{x}$ for both cases.

84. $E = \dfrac{q}{8\pi\epsilon_0 R^2}$; $\dfrac{q}{4\pi\epsilon_0 R}$

85. $E = \dfrac{\rho R}{4\epsilon_0}$; $V = \dfrac{\rho R^2}{4\epsilon_0}$

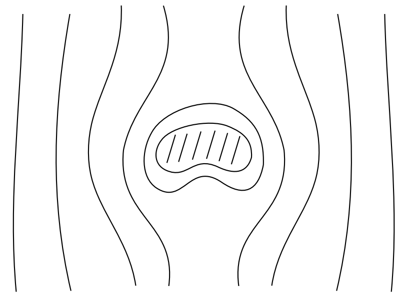
86. $\dfrac{\sigma R}{\pi\epsilon_0}$

87. (a) $\dfrac{1}{4\pi\epsilon_0}\dfrac{q}{R}$ (b) No change
(c) No change.

88. $E = \dfrac{1}{4\pi\epsilon_0}\dfrac{Q}{b^2}$; $V = \dfrac{1}{4\pi\epsilon_0}\dfrac{Q}{b}$

89. (a) $-\dfrac{R}{x}q$ (b) $\dfrac{Rq}{x^2}v$

90.



91. (a) $r = \dfrac{16\sqrt{2}\pi\epsilon_0 R^2 mu^2}{Qq}$ (b) $V = \sqrt{u^2 + \dfrac{Qq}{4\pi\epsilon_0 mR}}$

92. $12q$ and $3q$ at the two ends and q at a distance of 8 cm from $2q$. Force on q is zero.

93. (a) $\dfrac{8\sqrt{2}}{3}\dfrac{kqQ}{a}$ (b) $\left(\dfrac{8\sqrt{2}}{3} - \dfrac{4\sqrt{2}}{\sqrt{5}}\right)\dfrac{kqQ}{a}$

94. (a) D (b) $\sqrt{\dfrac{2eV_0}{m}}$
Electron cannot remain bound for any speed.

95. (a) $\sqrt{4gL}$ (b) $\sqrt{4.5gL}$

96. (a) 72 g (b) 4.32 N
(c) unstable

97. $W_{ext} = -3$ mgL

98. (a) $\dfrac{1}{4}$ mu^2
(b) A moves to right with velocity u. B is at rest.

99. $\left(\dfrac{1+\sqrt{3}}{2}\right)u$

100. (a) $\theta = \tan^{-1}\left(\dfrac{1}{2}\right)$ (b) 6 mgh

101. (a) $\dfrac{KQe}{2R}$ (b) $\dfrac{3}{2}\dfrac{KQe}{R}$

102. (a) $\sqrt{\dfrac{3qE}{ML}}$ (b) $\dfrac{3\sqrt{3}}{2}\dfrac{qE}{ML^2}$

103. $\dfrac{U_0}{3}$

104. $Q = \dfrac{q}{2}$

105. $\dfrac{10}{3}\pi\epsilon_0 RV^2$

106. (a) $\sqrt{\dfrac{KP}{V_0}}$ (b) $\left(\dfrac{2KP}{E_0}\right)^{1/3}$

where $K = \dfrac{1}{4\pi\epsilon_0}$

107. (a) $-\dfrac{KP^2}{r^3}$ (b) $-\dfrac{2KP^2}{r^3}$

108. $V = \dfrac{qr}{\pi^2 \epsilon_0 R^2}$

109. (a) $\left(\dfrac{2}{3}\right)^{\frac{1}{3}} a$ (b) $\left(\dfrac{1}{9}\right)^{\frac{1}{3}} a$

110. Q is at a distance of $4r$ from O.

112. (a) $\left|\dfrac{q_1}{q_2}\right| = \dfrac{2}{1}$ (b) $\alpha_{\max} = 90°$

113. Parallel to the axis of the cylinder.

114. $\dfrac{E_0}{2}$

115. $E = \dfrac{K\lambda\pi}{2d^2}$

116. $\dfrac{1}{4\pi\epsilon_0}\dfrac{\lambda}{R}\ln(\sqrt{2}+1)$

117. (a) $\dfrac{gr^2}{2h^2}$ (b) $V_0 = \dfrac{mgr^2}{12\pi\eta x h^2}$

118. $q = 16\pi\epsilon_0 a r_0$

119. (a) $\rho(r) = \dfrac{Q}{\pi R^4} r$ (b) $\dfrac{Qr_1^2}{4\pi\epsilon_0 R^4}$

120. $E_{\max} = \dfrac{\rho_0 E}{9\varepsilon}$

121. $\left(\dfrac{\sqrt{13}-3}{\sqrt{10}-3}\right)\sqrt{\dfrac{10}{13}}$.

122. (a) $6\,dx\,dy\,dz$ (b) $8\varepsilon_0\pi r^3$

123. (a) $Q = 2\pi b R^2$ (b) $E = \dfrac{b}{2\varepsilon_0}$

124. (a) $E_x = \dfrac{K\lambda R\pi\cdot x}{(R^2+x^2)^{3/2}}$ (b) $E_y = 0$

(c) $E_z = -\dfrac{2K\lambda R^2}{(R^2+x^2)^{3/2}}$

125. $\dfrac{\sigma^2}{2\varepsilon_0}\Delta s$

126. $P = \dfrac{1}{2}\varepsilon_0 E_0^2 = 39.6\ \text{N/m}^2$

127. (a) $F = \dfrac{Q^2}{32\pi\varepsilon_0 R^2}$ (b) $F = \dfrac{\pi\varepsilon_0 E_0^2 R^2}{2}$

128. $8\pi r^2\left[\varepsilon_0\left(7P_0 + \dfrac{12T}{r}\right)\right]^{1/2}$

129. $\dfrac{\sigma}{\varepsilon_0}$ in radially inward direction.

130. $\dfrac{Qq}{4\pi\varepsilon_0 R^2}\left(1 - \dfrac{1}{\sqrt{2}}\right)$

131. $\dfrac{Qq}{8\pi\varepsilon_0 R^2}\left(1 - \dfrac{1}{\sqrt{2}}\right)$ in both cases

132. he^{2n}

133. $V_{xn} = \dfrac{qE}{m}\sqrt{\dfrac{2h}{g}}\left[1 + 2e\left(\dfrac{1-e^{n-1}}{1-e}\right)\right]$ and $T = 2e^n\sqrt{\dfrac{2h}{g}}$

134. (a) $b = \dfrac{a}{15}$; $n = 2$ (b) $\sqrt{2}$ mg

135. (a) radius $4a$, centre at $(5a, 0)$

(b) $V = \dfrac{Q}{4\pi\varepsilon_0}\left[\dfrac{1}{|(x-3a)|} - \dfrac{2}{|(x+3a)|}\right]$

(c) At $x = 9a$ where $V = 0$, the charged particle eventually crosses the circle.

$v = \sqrt{\dfrac{Qq}{8\pi\varepsilon_0 ma}}$

136. $\dfrac{Qr}{2R}$

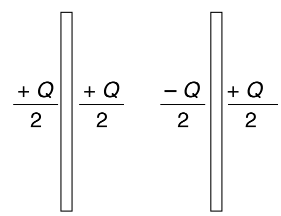
137. At both P and A the potential is $\dfrac{\sigma R}{2\varepsilon_0}$

139. $\dfrac{KQ}{r_0} - \dfrac{KQ}{r}$

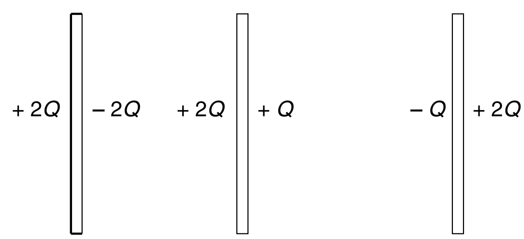
140. $\dfrac{1}{4\pi\varepsilon_0}\dfrac{Q}{R} - \dfrac{ER}{2}$

141. 5400 V.

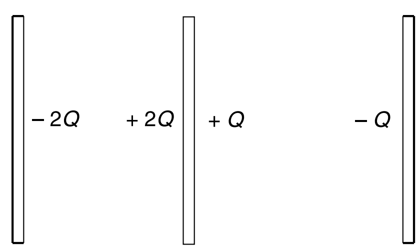
142. (i)



(ii)



(iii)

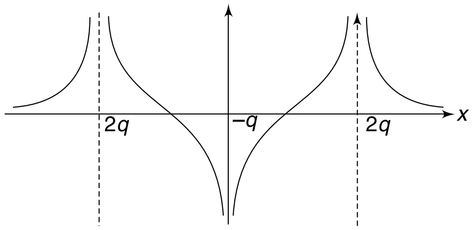


143. $\dfrac{6}{7}\dfrac{KQ}{R} - \dfrac{KQ}{7R^3}\left(\dfrac{r_0^2}{2} - \dfrac{R^3}{r_0}\right)$

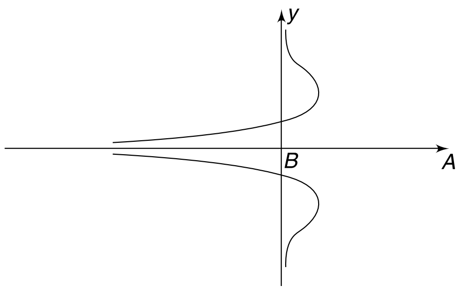
144. (a) Yes, spherical (b) $\dfrac{d}{\sqrt{3}-1}$

(c) $\frac{2KQ}{d}(2-\sqrt{3})$ (d) $\frac{-Q}{\epsilon_0}$

145 (a)



(b)



146. (a) (i) $E = 0$ (ii) $E = 0$ (iii) $E = \frac{Kq}{x^2}$

(b) (i) $V = \frac{Kq}{2R}$; (ii) $V = \frac{Kq}{2R}$ (iii) $V = \frac{Kq}{x}$

(c) (i) $V = \frac{Kq}{x^2}$; $V = Kq\left(\frac{1}{R} - \frac{1}{x}\right)$

(ii) $E = 0;\ V = 0$

(iii) $E = 0;\ V = 0$

(d) $-\frac{Kq^2}{2R}$

(e) zero

(f) (i) $E = \frac{Kq}{x^2};\ V = \frac{Kq}{4R} - \frac{Kq}{x} + \frac{Kq}{R}$

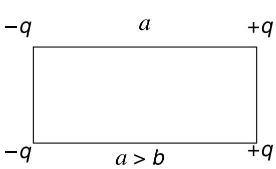
(ii) $E = 0;\ V = \frac{Kq}{4R}$

147. $\frac{2Q}{3}$

148. (a) $\pm 2\pi\epsilon_0 ErL$ (b) $k_0 = 2\pi\epsilon_0 rE^2$

(c) $\frac{1}{E}\sqrt{\frac{\pi m}{\epsilon_0 r}}$

149.



150. $V_0 = \sqrt{\frac{Kq^2(4R-7r)}{2mrR}}$

151. (a) $\frac{\sqrt{3}}{4}l$ (b) $\frac{q}{4}\sqrt{\frac{1}{\pi\varepsilon_0 ml}}$

(c) $\frac{q}{16\pi\varepsilon_0 l^2}(5q+4Q)$

152. $(v_0)_{\min} = 3$ m/s, $K = 3 \times 10^{-4}$ J

153. (i) $H = \frac{4}{3}R$ (ii) $H = \frac{R}{\sqrt{3}}$

154. (a) $V = \sqrt{V_0^2 + \frac{\lambda Q}{\pi\varepsilon_0 m}\ln\eta}$

(b) $V_r = \sqrt{V_0^2\left(1 - \frac{1}{\eta^2}\right) + \frac{\lambda Q}{\pi\varepsilon_0 m}\ln\eta}$

155. (a) $E_0 = \frac{Mg}{\lambda L}$ (b) $\omega = \sqrt{3(\sqrt{2}-1)\frac{g}{L}}$

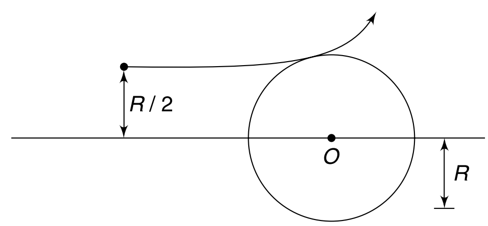
156. (a) Not possible (b) $E = \left(\frac{2}{\sqrt{3}-1}\right)\frac{Mg}{\lambda L}$

(c) $\theta_0 = 45°$

157. (b) $\Delta U = \frac{EqL}{2}(1 - \cos\Delta\theta)$

(c) $T = \pi\sqrt{\frac{5mL}{Eq}}$

158. (a)



(b) $v_0/2$ (c) $\frac{3m_P v_0^2}{8e}$

159. (a) $\frac{Q^2}{8\pi\epsilon_0 R}$ (b) same as in (a)

160. (a) $\frac{3}{5}\frac{Q^2}{4\pi\varepsilon_0 R}$ (b) $\frac{147}{320}\frac{Q^2}{4\pi\varepsilon_0 R}$

161. (i) $U_n = \frac{q_n^2}{8\pi\varepsilon_0 R}$ Here $q_n = \frac{QR}{r}\left[1 - \left(\frac{R}{R+r}\right)^n\right]$

(ii) $U_\infty = \frac{Q^2R}{8\pi\varepsilon_0 r^2}$

162. (a) $\frac{3}{2}\frac{KP}{r_0^3}$ (b) $\frac{1}{\sqrt{2}}$

163. $T = \frac{8\pi}{3}\sqrt{\frac{md}{qE}}$

164. (i) at $x = \frac{a}{2}$; $F = \frac{16}{5^{3/2}}\frac{KQP}{a^3}$

at $x = a$; $F = -\dfrac{KQP}{2^{5/2}a^3}$

(ii) No; Force is zero at $x = \pm\dfrac{a}{\sqrt{2}}$

165. $\dfrac{\rho R^3 d}{3\varepsilon_0 r^3}\sqrt{3\cos^2\theta + 1}$

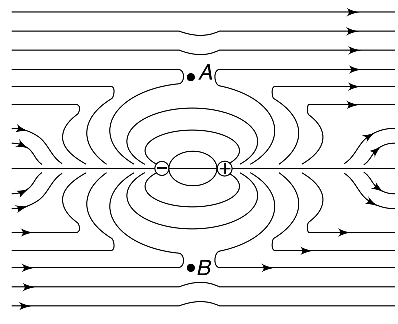
166. (a) $v = \sqrt{-2\dfrac{QKP\cos\theta}{mr^2}}$,

(b) Zero

(c) The bead oscillates. It stops at a point diametrically opposite to its starting point.

(d) Exactly same as it would move in the presence of the ring

167.



SOLUTIONS

1. (a) When there is q charge at the centre and vertices B, C, D, E, F, the net force on the central charge will be towards A (due to charge at D). The force due to charge at B and E cancel out. Similarly, force due to charge at F and C cancel out

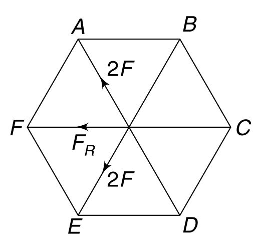
$$\frac{Kq \cdot q}{y^2} = F$$

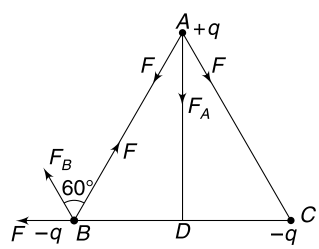
Where y = distance of centre from any vertex.

When charge at centre is $2q$ with no charge at A and E, the force due to charge at C and F cancel out. Force due to charge at B and D each will be

$$\frac{Kq(2q)}{y^2} = 2F$$

$\therefore$ Resultant is $F_R = \sqrt{(2F)^2 + (2F)^2 + 2\cdot(2F)(2F)\cdot\cos 120^\circ} = 2F$ towards point F





(b) Since all charges have same magnitude, the force between any two of them will have equal magnitude. Charge at B and C must attract the charge at A for its acceleration to be along AD.

Hence, charge at B and C must have same sign. They will repeal each other.

Thus charge at B will experience two equal forces at 120° as shown. Resultant is force F_B.

Hence charge at B gets accelerated in a direction making an angle of 60° with BA.

2. The force between two charges is proportional to the product of the charges. $F \propto q_1 q_2$

After touching, the charge on each particle will be $q = \dfrac{q_1 + q_2}{2}$

$$F_{\text{new}} \propto \left(\frac{q_1 + q_2}{2}\right)^2$$

Since arithmetic mean is greater than or equal to geometric mean for two numbers hence

$$\frac{q_1 + q_2}{2} \geq \sqrt{q_1 q_2}$$

$$\left(\frac{q_1 + q_2}{2}\right)^2 \geq q_1 q_2 \quad \Rightarrow \quad F_{\text{new}} \geq F$$

3. There are three $-q$ charges which are kept at angular separation of 120° from each other. The resultant of three co planar vectors of equal magnitude at separation of 120° from each other is zero. Hence, these 3 charges produce zero field. Similarly, three number of $+q$ charges, three number of $2q$ and three number of $-3q$ charges produce zero field at centre.

4. q_1 = positive charge equal to the magnitude of the total charge on all electrons present in 0.9 mg of pure water

= charge on electrons present in electrons of $\dfrac{0.9 \times 10^{-3}}{18} = 5 \times 10^{-5}$ mol water

= charge on $5 \times 10^{-5} \times N_A \times 10$ electrons (since each molecule has 10 electrons)

= charge on 3×10^{20} electrons

$= 3 \times 10^{20} \times 1.6 \times 10^{-19}$ C = 48 C

q_2 = charge on a 6.35 mg copper sphere from which 0.1% of its total electrons have been removed.

= charge on 0.1% of electrons on 10^{-4} mol Cu

= charge on $0.001 \times 10^{-4} \times N_A \times 29$ electrons (since each Cu atom has 29 electrons)

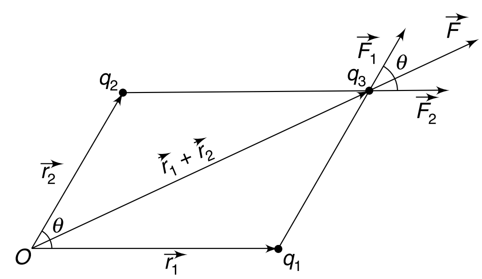
= charge on 1.74×10^{18} electrons

$= 1.74 \times 10^{18} \times 1.6 \times 10^{-19}$ C = 0.27 C

Force between q_1 and q_2 if they are placed at a separation of 1 km is

$$F = 9 \times 10^9 \frac{q_1 q_2}{r^2} = 120000 \text{ N}$$

5. The situation is shown in Fig

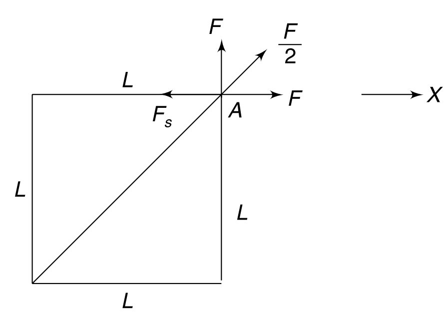


$$|\vec{F}_1| = |\vec{F}_2| = F_0 \text{ (say). Given } F = \sqrt{3}\, F_0$$

$$\sqrt{F_0^2 + F_0^2 + 2F_0^2 \cdot \cos\theta} = \sqrt{3}F_0 \quad \Rightarrow \quad \cos\theta = \frac{1}{2}; \quad \text{i.e, } \theta = 60°$$

$$\therefore \qquad \vec{r}_1 \cdot \vec{r}_2 = r \cdot r \cdot \cos 60° = \frac{r^2}{2}$$

6. Consider the equilibrium of charge at A



F = force due to other charge at distance $L = \dfrac{kq^2}{L^2}$

Force due to diagonally opposite charge $\dfrac{kq^2}{(\sqrt{2}L)^2} = \dfrac{F}{2}$

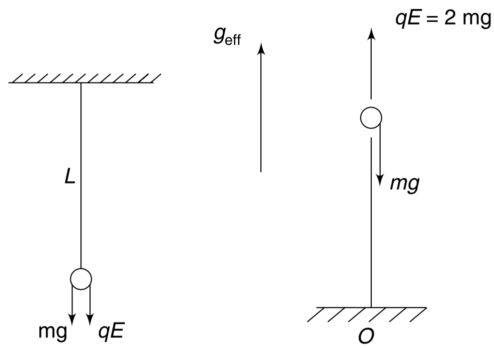
The resultant force in X direction $F_x = F + \dfrac{F}{2}\cos 45° = F\left(1 + \dfrac{1}{2\sqrt{2}}\right)$

This gets balanced by the spring force. $k \cdot \dfrac{L}{2} = F\left(1 + \dfrac{1}{2\sqrt{2}}\right)$

$$kL = \frac{q^2}{4\pi\epsilon_0 L^2}\left(\frac{2\sqrt{2}+1}{\sqrt{2}}\right) \qquad \therefore \quad k = \frac{q^2}{4\pi\epsilon_0 L^3}\left(\frac{2\sqrt{2}+1}{\sqrt{2}}\right)$$

7. From dimensional analysis $\qquad F_1 \; \alpha \; \dfrac{q_1^2}{l_1^2}$ and $F_2 \; \alpha \; \dfrac{q_2^2}{l_2^2}$

$$\Rightarrow \quad \frac{F_2}{F_1} = \left(\frac{q_2}{q_1}\frac{l_1}{l_2}\right)^2 \quad \Rightarrow \quad F_2 = F_1\left(\frac{q_2^2}{q_1^2}\frac{l_1^2}{l_2^2}\right)$$

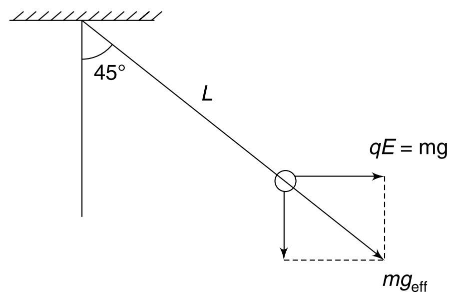


8. (i) In this case effective value of g can be taken as

$$g_{eff} = \frac{mg + gE}{m} = 2g$$

$$\therefore \quad T = 2\pi\sqrt{\frac{L}{2g}}$$

(ii) In this case equilibrium position is as shown in second figure. Effective acceleration is vertically upward given by–



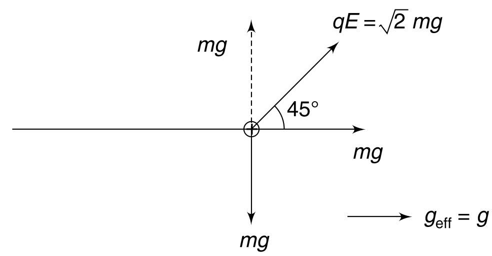
$$g_{eff} = \frac{qE - mg}{m} = g$$

$$T = 2\pi\sqrt{\frac{L}{g}}$$

(iii) Equilibrium position is shown

$$g_{eff} = \sqrt{2}g$$

$$T = 2\pi\sqrt{\frac{L}{\sqrt{2}\,g}}$$



(iv) Equilibrium is shown (thread is horizontal)

$$\therefore \quad T = 2\pi\sqrt{\frac{L}{g}}$$

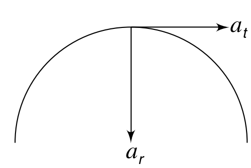
11. (a) Work done by the electric force on the particle

$$W = \int_A^B q\vec{E}\cdot\vec{dl} = qE\cdot\pi R$$

$$\therefore \quad K_B - K_A = qE\pi R$$

$$\therefore \quad K_B = \pi qER \qquad [\because \; K_A = 0]$$

(b) At mid point let the speed of the particle be V.



$$\frac{1}{2}\text{mV}^2 = qE\frac{\pi R}{2} \quad \Rightarrow \quad \frac{V^2}{R} = \frac{\pi qE}{m} = a_r$$

This is radial acceleration of the particle. It arises due to constraining forces.

Tangential acceleration $\quad a_t = \dfrac{qE}{m}$

$\therefore$ Resultant acceleration

$$a = \sqrt{a_r^2 + a_t^2} = \frac{qE}{m}\sqrt{1 + \pi^2}$$

12.

$$F = qE$$

$$ma_0 = q(-bx + a) \qquad [a_0 = \text{accleration}]$$

$$V\frac{dv}{dx} = -\frac{bq}{m}x + \frac{aq}{m}$$

$$\int_0^V V dv = -\frac{bq}{m}\int_0^x x dx + \frac{aq}{m}\int_0^x dx$$

$$\frac{V^2}{2} = -\frac{bqx^2}{2m} + \frac{aqx}{m}$$

$$V = 0 \quad \text{when} \quad \frac{bqx^2}{2m} = \frac{aqx}{m} \quad \Rightarrow x = \frac{2a}{b}$$

Now $$a_0 = \frac{q}{m}(-bx + a)$$

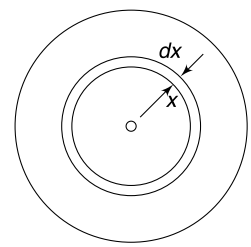
at $$X = \frac{2a}{b}$$

$$\text{acceleration} = \frac{q}{m}(-2a + a) = -\frac{qa}{m}$$

13. (a) Consider an elemental shell of thickness dx as shown. Charge in the shell is $dq = \rho 4\pi x^2 dx = 4\pi\alpha x^3 dx$

$\therefore$ charge in sphere of radius R_0 will be

$$Q_0 = \int dq = 4\pi\alpha \int_0^{R_0} x^3 dx = \pi\alpha R_0^4$$



(b) Charge in sphere of radius R will be $Q = \pi\alpha R^4$

As per question $$Q = \frac{Q_0}{2} \quad \therefore R^4 = \frac{R_0^4}{2} \quad \Rightarrow \quad R = \frac{R_0}{(2)^{1/4}}$$

(c) Field at a distance $r < R$ will depend on the inner charge only. Charge lying outside r will not contribute to the field. Hence, the answer for both cases will be same

$$E = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2} = \frac{1}{4\pi\epsilon_0}\frac{\pi\cdot\alpha r^4}{r^2} = \frac{\alpha r^2}{4\epsilon_0}$$

15. Electric field inside the inner shell is zero. Potential of the inner shell and all points inside it will be constant.

16. $$E = -\frac{dV}{dr} = \frac{25}{r^2}$$

The direction of field is along $\vec{r}$ at any point (why?)

At $$\vec{r} = \hat{i} + 2\hat{j} + 2\sqrt{5}\hat{k}$$

$$r = \sqrt{1^2 + 2^2 + (2\sqrt{5})^2} = 5 \text{ m.}$$

$\therefore$ $$E = \frac{125}{25} = 5 \text{ V/m} \qquad \therefore \quad \vec{E} = 5\left(\frac{\vec{r}}{r}\right) = \hat{i} + 2\hat{j} + 2\sqrt{5}\,\hat{k}$$

17. $$V = \frac{1}{x} + \frac{1}{y} + \frac{2}{z}$$

$$\vec{E} = -\frac{\partial V}{\partial X}\hat{i} - \frac{\partial V}{\partial y}\hat{j} - \frac{\partial V}{\partial z}\hat{k} = \frac{1}{x^2}\hat{i} + \frac{1}{y^2}\hat{j} + \frac{2}{z^2}\hat{k}$$

At (1, 1, 1) m

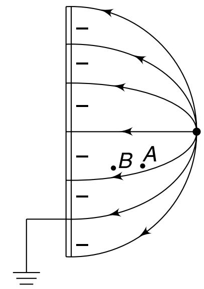
$$\vec{E} = \hat{i} + \hat{j} + 2\hat{k}$$

Resultant force on the particle in XY plane is

$$\vec{F} = q(\hat{i} + \hat{j})$$

$\therefore$ $$\vec{a} = \frac{q}{m}(\hat{i} + \hat{j}) = \frac{10^{-12}\text{C}}{10^{-12}\text{kg}}(\hat{i} + \hat{j}) = (\hat{i} + \hat{j}) \text{ m/s}^2.$$

18. Negative charge gets induced on the metal surface, so as to make its potential zero. Field lines are as shown.



Density of lines is higher at A

$$\therefore \qquad E_A > E_B$$

While moving is direction of the field, the potential drops

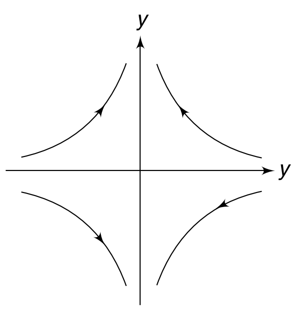
$$\therefore \qquad V_A > V_B$$

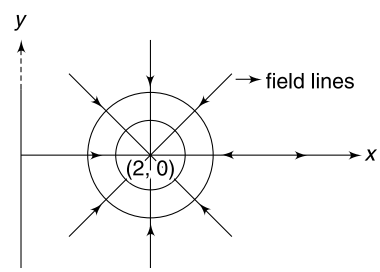
20. (a)
$$E_x = -\frac{\partial V}{\partial x} = -2x$$
$$E_y = -\frac{\partial V}{\partial y} = 2y$$

(b)
$$V = (x-2)^2 + y^2$$

$\therefore$ Equipotentials are circles with centre at (2, 0)

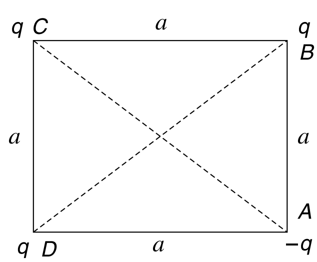
$\therefore$ Field lines are radial directed towards the centre of the circle.

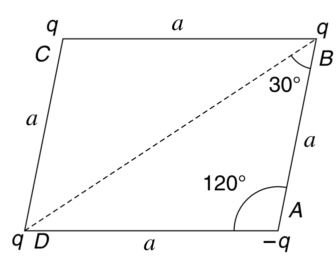




21. On square: There will be six terms in the expression of potential energy for the system, as there are six pairs of interactions.

U_{AB} and U_{AD} are negative whereas U_{BC} and U_{CD} are positive terms of same magnitude. Similarly U_{BD} and U_{AC} cancel out. Hence PE is zero.





On Rhombus: U_{AB} and U_{AD} cancel out with U_{BC} and U_{CD}

Length $BD = 2a\cos 30° = \sqrt{3}a$

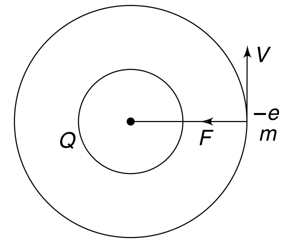
$AC = 2a\cos 60° = a$

$$\therefore \qquad U = U_{BD} + U_{AC} = K\frac{q^2}{\sqrt{3}a} - K\frac{q^2}{a} = -\frac{Kq^2}{a}\left(1 - \frac{1}{\sqrt{3}}\right)$$

$$\therefore \qquad W_{\text{extAgt}} = U_{\text{Rhombus}} - U_{\text{Square}} = -\frac{Kq^2}{a}\left(1 - \frac{1}{\sqrt{3}}\right)$$

22. (a) $W_{ext} = qV_{\text{ball centre}} - qV_{\text{shell centre}}$

$$= q\left[\frac{3}{2}\frac{1}{4\pi\epsilon_0}\frac{Q}{r} - \frac{1}{4\pi\epsilon_0}\frac{Q}{r}\right] = \frac{Qq}{8\pi\epsilon_0 r}$$



(b) $$V_{\text{ball centre}} = \frac{3}{2}\cdot\frac{1}{4\pi\epsilon_0}\frac{Q}{r} + \frac{1}{4\pi\epsilon_0}\frac{Q}{4r} = \frac{7Q}{16\pi\epsilon_0 r}$$

$$V_{\text{shell centre}} = \frac{1}{4\pi\epsilon_0 r}\frac{Q}{r} + \frac{1}{4\pi\epsilon_0}\frac{Q}{4r} = \frac{5Q}{16\pi\epsilon_0 r}$$

$$\therefore \quad W_{ext} = q\left[\frac{7Q}{16\pi\epsilon_0 r} - \frac{5Q}{16\pi\epsilon_0 r}\right] = \frac{Qq}{8\pi\epsilon_0 r}$$

23. (a) Let orbital speed be V. Electrostatic attraction provides the centripetal force.

$$\frac{mV^2}{r} = \frac{1}{4\pi\epsilon_0}\frac{Q(e)}{r^2}$$

$$\therefore \quad V = \sqrt{\frac{Qe}{4\pi\epsilon_0 mr}}$$

$$\therefore \quad E = K + U = \frac{1}{2}\text{mV}^2 + \frac{1}{4\pi\epsilon_0}\frac{Q(-e)}{r}$$

$$= \frac{1}{2}\cdot\frac{1}{4\pi\epsilon_0}\frac{Qe}{r} + \frac{1}{4\pi\epsilon_0}\frac{Q(-e)}{r}$$

$$= -\frac{1}{2}\frac{1}{4\pi\epsilon_0}\frac{Qe}{r} = -\frac{Qe}{8\pi\epsilon_0}$$

(b) $$T = \frac{2\pi r}{V} \Rightarrow T^2 = \frac{4\pi^2 r^2}{V^2} = \frac{4\pi^2 r^2}{Qe}\cdot 4\pi\epsilon_0\, mr = \frac{16\pi^3\epsilon_0 m}{Qe}\cdot r^3$$

$$\therefore \quad T^2 \propto r^3 \quad \therefore \quad \left(\frac{T_2}{T_1}\right)^2 = \left(\frac{r_2}{r_1}\right)^3$$

$$\left(\frac{T_2}{T}\right)^2 = 4^3 \quad \therefore \quad T_2 = 8T$$

24. The particle will escape if

$$\frac{1}{2}mV_0^2 + K\frac{Q(-q)}{R} = 0$$

$$V_0 = \sqrt{\frac{2KQq}{mR}}$$

$$V_0 \propto \sqrt{q}$$

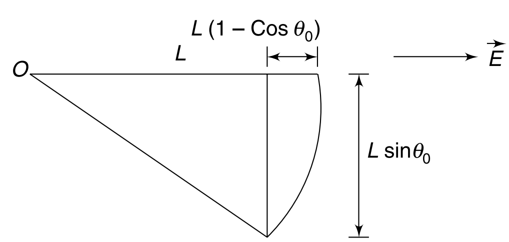
25. (a) let the bob come to rest in the position shown. Using conservation of energy we have

Loss in gravitational PE = Gain in electrostatic PE [$\because$ change in kE = 0]

$$\therefore \quad mg\, L\sin\theta_0 = qEL(1 - \cos\theta_0)$$

$$mg\left(2\sin\frac{\theta_0}{2}\cos\frac{\theta_0}{2}\right) = qE\, 2\cdot\sin^2\frac{\theta_0}{2}$$

$$\Rightarrow \quad \tan\frac{\theta_0}{2} = \frac{qE}{mg} \Rightarrow \theta_0 = 2\tan^{-1}\left(\frac{qE}{mg}\right)$$

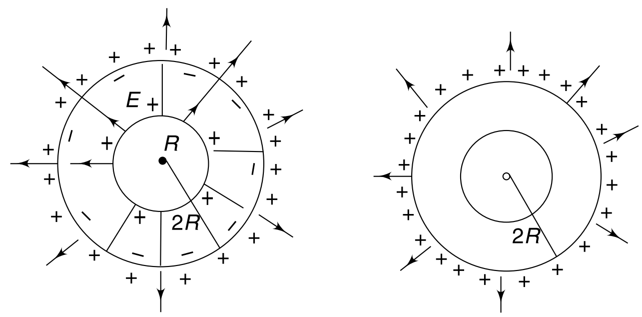


(b) For $\theta_0 = \pi/2$

$$\tan\frac{\pi}{4} = \frac{qE}{mg}$$

$$E = \frac{mg}{q}$$

26. The electric field in initial and final configuration is as shown below:



In initial configuration there is electric field in all region $r > R$. The field varies as $E = \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2}$. In final configuration there is no field in the region $R < r < 2R$. Elsewhere the field is similar to that in initial configuration. Thus initial configuration has more energy stored in electric field given by

$$\Delta E = \int_R^{2R} \frac{1}{2}\epsilon_0 E^2 dV = \frac{1}{2}\epsilon_0 \int_R^{2R}\left(\frac{1}{4\pi\epsilon_0}\frac{Q}{r^2}\right)^2 4\pi r^2 dr$$

$$= \frac{Q^2}{8\pi\epsilon_0}\int_R^{2R}\frac{dr}{r^2} = -\frac{Q^2}{8\pi\epsilon_0}\left[\frac{1}{2R} - \frac{1}{R}\right] = \frac{Q^2}{16\pi\epsilon_0 R}$$

$$W_{ext} = U_f - U_i = -\frac{Q^2}{16\pi\epsilon_0 R}$$

27. (a)

$$u_E = \frac{1}{2}\epsilon_0 E^2 = \frac{1}{2}\epsilon_0\frac{K^2Q^2}{x^4}$$

$$V = K\frac{Q}{x}$$

$$\therefore \quad \frac{u_E}{V^2} = \frac{\frac{1}{2}\epsilon_0\frac{K^2Q^2}{x^4}}{\frac{K^2Q^2}{x^2}} = \frac{\epsilon_0}{2x^2}$$

Given $\quad \frac{\epsilon_0}{2x^2} = \eta$

At a distance $2x$, obviously the value of this ratio will be $\frac{\eta}{4}$

(b)

$$[\eta] = \frac{[\epsilon_0]}{[L^2]} = \frac{[M^{-1}L^{-3}T^4A^2]}{[L^2]} = [M^{-1}L^{-5}T^4A^2]$$

28. (a)

$$\frac{1}{2}\epsilon_0 E^2 = \frac{1}{2}\times 8.8\times 10^{-12}\times(3\times 10^6)^2 \text{ J/m}^3$$

$$= 39.6 \text{ J/m}^3 = 39.6\ \mu\text{J/cm}^3$$

(b) Energy consumed to charge the ball = Electrostatic self energy of the ball

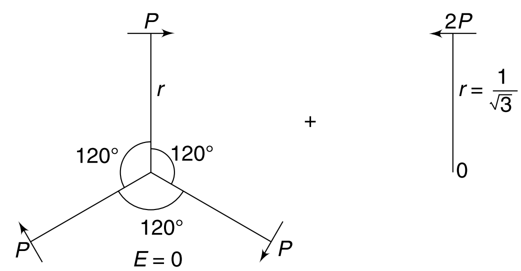
$$= \frac{Q^2}{8\pi\epsilon_0 \epsilon_r R} = \frac{4 \times 10^{-18}}{2 \times 80 \times 0.1} \times 9 \times 10^9 = 2.25 \times 10^{-9}\text{J}$$

29. System is equivalent to the superposition of two systems shown in figure:

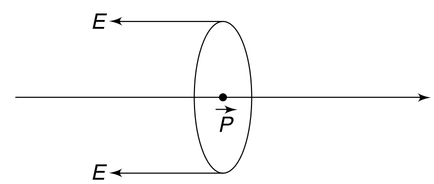
$$E = K\frac{2P}{r^3} \text{ Opposite to } 2\vec{P}$$

$$\therefore \qquad E = 2\frac{KP}{r^3} = \frac{2KP}{\left(\frac{L}{\sqrt{3}}\right)^3} = 6\sqrt{3}\,\frac{KP}{L^3}$$

Potential = 0



30. Each point on the circumference of the ring lies on the perpendicular bisector of the dipole. Field due to dipole at these points is $E = \frac{1}{4\pi\epsilon_0}\frac{P}{R^3}$ directed opposite to $\vec{P}$.



$\therefore$ Force on ring $\qquad F = QE$

$$= \frac{1}{4\pi\epsilon_0}\frac{PQ}{R^3}$$

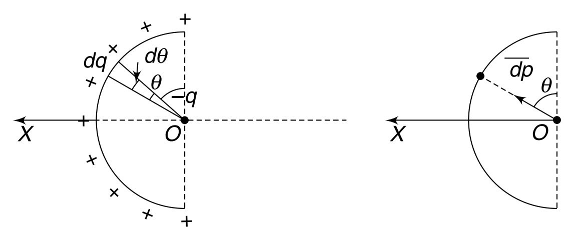
(directed opposite to $\vec{P}$)

From Newton's third law, the ring will exert equal and opposite force on the dipole.

31. A charge $+dq$ on the ring and a charge $-dq$ at O forms a dipole having dipole moment

$$dp = Rdq \qquad \text{(direction as shown in figure)}$$

Summing up all such dipole moment will give the net dipole moment of the system. From symmetry the resultant of all such $\overrightarrow{dp}$ vectors will be along X direction.



$$\therefore \qquad \vec{P} = \int_{0=0}^{\pi} dp \cdot \sin\theta = R\int_{0}^{\pi} dq \sin\theta$$

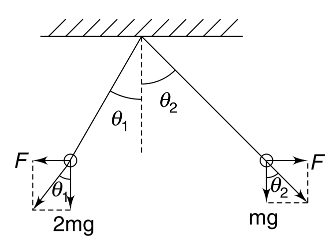
But $\qquad dq = \frac{q}{\pi R}\cdot Rd\theta = \frac{qd\theta}{\pi}$

$$\therefore \qquad \vec{P} = \frac{Rq}{\pi}\int_0^{\pi} \sin\theta\, d\theta = -\frac{Rq}{\pi}\,[\cos\theta]_0^{\pi}$$

$$= -\frac{Rq}{\pi}\,[-1 - 1] = 2\,\frac{Rq}{\pi} \text{ along } OX$$

32. F = Electrostatic force.

For equilibrium



$$\tan\theta_1 = \frac{F}{2mg} \quad \text{and} \quad \tan\theta_2 = \frac{F}{mg}$$

$$\tan(\theta_1 + \theta_2) = \frac{\tan\theta_1 + \tan\theta_2}{1 - \tan\theta_1 \tan\theta_2}$$

$$\tan 45° = \frac{\frac{F}{2\,mg} + \frac{F}{mg}}{1 - \frac{F}{2\,mg}\cdot\frac{F}{mg}}$$

$\Rightarrow$ $$2m^2g^2 - F^2 = 3mgF \Rightarrow F^2 + 3mgF - 2m^2g^2 = 0$$

$\therefore$ $$F = \frac{-3\,mg \pm \sqrt{9m^2g^2 + 8m^2g^2}}{2}$$

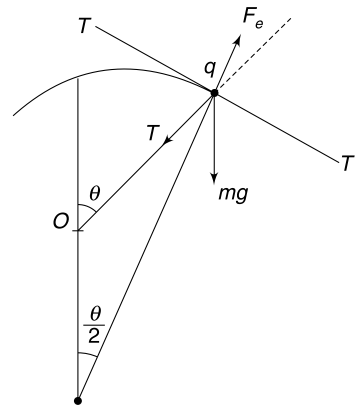
Negative sign of F is not acceptable

$\therefore$ $$F = \left(\frac{\sqrt{17} - 3}{2}\right)\text{mg}$$

33. In equilibrium $\frac{KqQ}{(2L)^2} > mg \Rightarrow Q > \frac{4L^2mg}{Kq}$

Now consider the particle in a slightly displaced position

Equilibrium will be stable if tangential component (along TT) of electrostatic repulsion is greater than the tangential component of mg.



$\Rightarrow$ $$F_e \sin\left(\frac{\theta}{2}\right) > mg\,\sin\theta$$

For small displacement, $$\sin\frac{\theta}{2} \simeq \frac{\theta}{2}$$

$\therefore$ $$K\frac{qQ}{(2L)^2}\cdot\frac{\theta}{2} > mg\theta$$

$\therefore$ $$Q > \frac{8L^2mg}{Kq} = \frac{32\pi\varepsilon_0 L^2 mg}{q}$$

35. For charges to remain at rest, tangential force on each of them must be zero.

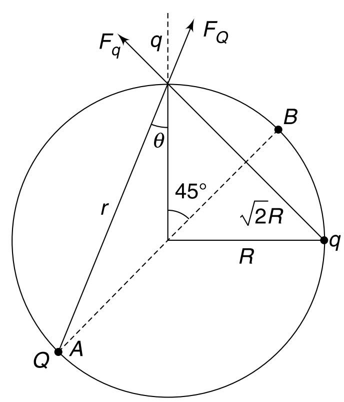
(a) Due to symmetry, the third charge has to be located at A or B.

When it is positive its location is A. For tangential force to be zero on one of the equal charges-

$$F_q\cos 45° = F_Q\cdot\sin\theta$$

$$\theta = \frac{180 - 135}{2} = 22.5°$$

And $$r = 2R\cos\theta$$



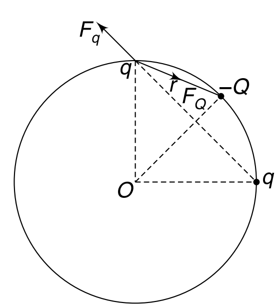
$\therefore$ $$\frac{qq}{(\sqrt{2}R)^2}\cdot\frac{1}{\sqrt{2}} = \frac{qQ}{(2R\,\cos\theta)^2}\cdot\sin\theta$$

$\therefore$ $$\frac{Q}{q} = \sqrt{2}\,\frac{\cos^2\theta}{\sin\theta}$$

$$Q = \left[1.41 \times \frac{(0.92)^2}{0.38}\right]q = (3.14)\,q$$

(b) $$r = 2R\sin(22.5°)$$

$$\frac{F_q}{(\sqrt{2}R)^2}\cdot\frac{1}{\sqrt{2}} = F_Q\cdot\sin(90 - 22.5°)$$



$$\frac{qq}{2\sqrt{2}R^2} = \frac{qQ}{(2R\sin 22.5°)^2}\cos(22.5°)$$

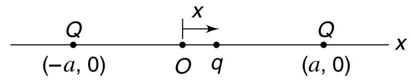
$$Q = \sqrt{2}\,\frac{\sin^2(22.5°)}{\cos\,(22.5°)}q$$

$$= \frac{1.41 \times (0.38)^2}{(0.92)}q$$

$$= 0.22q$$

36. SHM along x

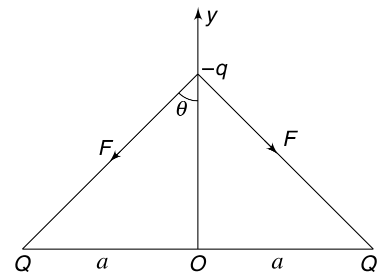
Net force acting on q when it is displaced by x is



$$F = \frac{KQq}{(a+x)^2} - \frac{KQq}{(a-x)^2}$$

$$F = -\frac{4KQqax}{(a^2-x^2)^2} \simeq -\frac{4KQqax}{a^4} \qquad (x << a)$$

$$= -\frac{4KQq}{a^3}x$$



$$\therefore \quad \text{acceleration} = -\frac{4KQq}{ma^3}x$$

$$\therefore \quad T_1 = 2\pi\sqrt{\frac{ma^3}{4KQq}}$$

SHM along y

Resultant force when $-q$ is displaced by y is

$$F_R = -2F\cos\theta = -2\frac{KQq}{(a^2+y^2)}\cdot\frac{y}{\sqrt{a^2+y^2}} = -\frac{2KQqy}{(a^2+y^2)^{3/2}} \simeq -\frac{2KQqy}{a^3}$$

$$\therefore \quad \text{acceleration} = -\frac{2KQq}{ma^3}y$$

$$\therefore \quad T_2 = 2\pi\sqrt{\frac{ma^3}{2KQq}} \qquad \therefore \quad \frac{T_1}{T_2} = \frac{1}{\sqrt{2}}$$

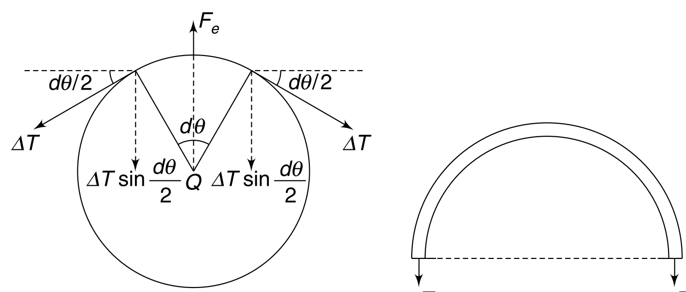
37. With no charge at the centre, there is some tension in the ring due to repulsion of charge present on the ring.

(a) When Q is placed at the centre the tension increases by ΔT. Consider an infinitesimally small element on the ring having angular width $d\theta$ as shown.

$$2\Delta T\sin\left(\frac{d\theta}{2}\right) = F_e$$

$$2\Delta T\cdot\frac{d\theta}{2} = \frac{KQ\lambda R d\theta}{R^2}$$

$$\Delta T = \frac{KQ\lambda}{R} = \frac{KQq}{2\pi R^2}$$



(b) Clearly the answer is $2\Delta T = \dfrac{KQq}{\pi R^2}$

(c) Answer to (b) must be the answer to (c) also.

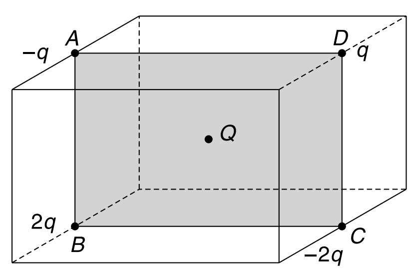
38. Consider 4 charges on the square $ABCD$. Force due to these 4 charges

$$F_1 = 2\sqrt{2}\frac{KQq}{r^2} \text{ parallel to } BC$$

Similarly, force due to 4 charges on square $EFGH$ is

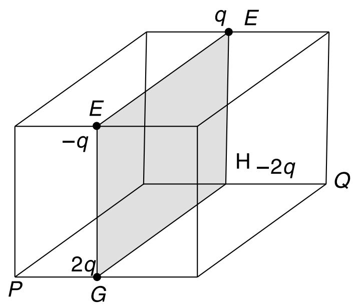
$$F_2 = 2\sqrt{2}\frac{KQq}{r^2} \text{ parallel to } GH$$

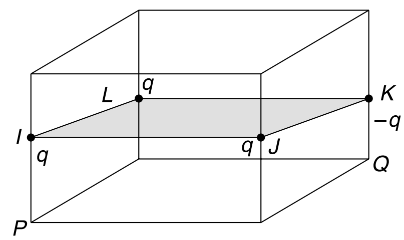
Resultant of $\vec{F}_1$ and $\vec{F}_2$ will be $\dfrac{4KQq}{r^2}$ parallel to PQ.



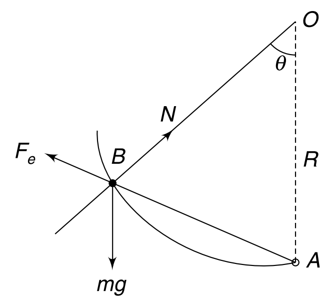
Force due to 4 charges at I, J, K, L is also parallel to PQ and has magnitude $F = \dfrac{4KQq}{r^2}$

Hence, resultant is $\dfrac{8KQq}{r^2}$ along PQ





39 Let charge on both A and B be q



$$AB = 2R\sin\left(\frac{\theta}{2}\right)$$

$$F_e = \frac{Kq^2}{4R^2\sin^2\left(\frac{\theta}{2}\right)}$$

For net tangential force to be zero

$$mg\,\sin\theta = F_e\sin\left(\frac{\pi}{2} - \frac{\theta}{2}\right)$$

$$mg\,\sin\theta = \frac{Kq^2}{4R^2\sin^2\left(\frac{\theta}{2}\right)}\cdot\cos\left(\frac{\theta}{2}\right)$$

$$32\pi\varepsilon_0\,mgR^2\,\sin^3\left(\frac{\theta}{2}\right) = q^2 \qquad \left[\because\ K = \frac{1}{4\pi\varepsilon_0}\right]$$

$$\therefore \qquad q = 4R\sin\left(\frac{\theta}{2}\right)\sqrt{2\pi\varepsilon_0\,mg\,\sin\left(\frac{\theta}{2}\right)}$$

$$\text{Total charge given} = 2q$$

$$Q = 8R\sin\left(\frac{\theta}{2}\right)\sqrt{2\pi\varepsilon_0\,mg\,\sin\left(\frac{\theta}{2}\right)}$$

40. In equilibrium

$$\frac{KQq}{x_0^2} = mg\,\sin\theta \qquad \text{...(1)}$$

If the charge is displaced by $x(<< x_0)$

$$ma = \frac{KQq}{(x_0 + x)^2} - mg\sin\theta$$

$$= \frac{KQq}{x_0^2\left(1 + \frac{x}{x_0}\right)^2} - mg\,\sin\theta$$

$$= \frac{KQq}{x_0^2}\left(1 + \frac{x}{x_0}\right)^{-2} - mg\,\sin\theta$$

Using binomial expansion and neglecting higher order terms-

$$ma = \frac{KQq}{x_0^2}\left[1 - \frac{2x}{x_0}\right] - mg\sin\theta$$

$$ma = -\frac{2KQq}{x_0^2}\cdot x \qquad \text{[using 1]}$$

$$ma = -\left(2\frac{mg\sin\theta}{x_0}\right)\cdot x \qquad \text{[again using 1]}$$

$$\therefore \quad a = -\left(\frac{2g\sin\theta}{x_0}\right)x \qquad \therefore \quad \omega = \sqrt{\frac{2g\sin\theta}{x_0}}$$

$$f = \frac{1}{2\pi}\sqrt{\frac{2g\sin\theta}{x_0}}$$

41. (a) Field due to two $+Q$ charges has a maxima at a distance $y_0 = \frac{a}{\sqrt{2}}$ on the line OY. Equilibrium will be stable only if charged particle is placed above $y_0 = \frac{a}{\sqrt{2}}$ (i.e., at B). Field due to two $+Q$ charges at a distance y on the line OY is-

$$E = \frac{2KQy}{(a^2 + y^2)^{3/2}}$$

At y and $3\sqrt{3}y$ the electric force balances mg.

$$\therefore \quad qE_y = qE_{3\sqrt{3}y} = mg$$

$$\therefore \quad \frac{q\cdot 2KQy}{(a^2 + y^2)^{3/2}} = \frac{q\cdot 2KQ\, 3\sqrt{3}y}{[a^2 + (3\sqrt{3}y)^2]^{3/2}} \Rightarrow a^2 + 27y^2 = 3(a^2 + y^2) \Rightarrow 24y^2 = 2a^2$$

$$y = \frac{a}{\sqrt{12}} \qquad \therefore \quad y_B = 3\sqrt{3}y = \frac{3}{2}a$$

(b) Maximum electric force on the particle is

$$F_0 = \frac{2KQq\,\frac{a}{\sqrt{2}}}{\left(a^2 + \frac{a^2}{2}\right)^{3/2}} = \frac{4KQq}{3\sqrt{3}a^2} = 0.76\frac{KQq}{a^2} \qquad \text{...(i)}$$

Weight of the particle is equal to the electric force at $y = \frac{a}{\sqrt{12}}$

$$mg = \frac{2KQq\,\frac{a}{\sqrt{12}}}{\left(a^2 + \frac{a^2}{12}\right)^{3/2}} = \frac{24KQq}{13\sqrt{13}a^2} = 0.51\frac{KQq}{a^2} \qquad \text{...(ii)}$$

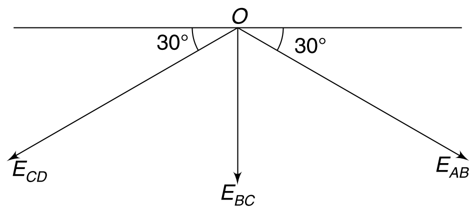
Net force will be maximum at O where electric force is zero and mg remains unbalanced. Particle will have maximum acceleration at O.

44. Electrostatic field is conservative. Field lines cannot form closed loops otherwise the field will perform a net work if a charge is moved in a closed path along the field line. But a conservative force must perform zero work in a closed path. For an electrostatic field $\oint \vec{E}\cdot\vec{dl} = 0$

In Fig. (b) and (c) you can imagine a rectangular path along which $\oint \vec{E}\cdot\vec{dl} \neq 0$

Hence none of the field configuration is possible

45. Field at O due to three segments will have same magnitude (say E) and will have directions as shown.



$$E = |E_{AB}| = |E_{BC}| = |E_{CD}|$$

$\therefore$ $$E_0 = E_{BC} + E_{AB}\sin 30° + E_{CD}\sin 30°$$

$\Rightarrow$ $$E_0 = E + \frac{E}{2} + \frac{E}{2} \qquad \therefore \quad E = \frac{E_0}{2}$$

46. Assume that two perpendicular components of field at P due to the hemisphere are-

E_1 towards centre (say towards right in given Fig.) and E_2 perpendicular to the circular equatorial plane (downward in given Fig.).

Now imagine another hemisphere so as to complete a sphere. This sphere will have its own field at P whose components will be-

E_1 towards centre (towards right in given Fig.) and E_2 perpendicular to the circular equatorial plane(up in given Fig.). Since field inside uniformly charged sphere is zero hence E_1 must be zero.

47. *Hint:* If you consider the complete sphere, field at P as well as Q will be zero.

$\Rightarrow$ $$\vec{E}^{P}_{\text{Hemisphere 1}} + \vec{E}^{P}_{\text{Hemisphere 2}} = 0$$

48. If $r << x$, charge on the removed disc is like a point charge equal to $q = \pi r^2 \cdot \sigma$

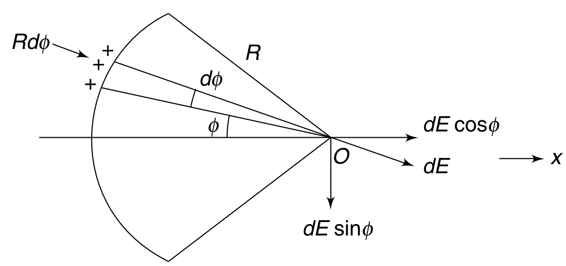
$\therefore$ $$\Delta E = \frac{1}{4\pi\varepsilon_0}\frac{q}{x^2} = \frac{1}{4\pi\varepsilon_0}\frac{\pi r^2\sigma}{x^2} = \frac{\sigma}{4\varepsilon_0}\frac{r^2}{x^2}$$

49. (a) Consider an element of angular width $d\phi$.

Width of element = $Rd\phi$

Charge on element = $\lambda R d\phi$

Field at O due to this element



$$dE = \frac{K\lambda R d\phi}{R^2}$$

Due to symmetry field is along x only. We need to add all $dE\cos\phi$ components.

$$E = \int_{-\theta/2}^{\theta/2} \frac{K\lambda}{R}\cos\phi \, d\phi = \frac{2K\lambda}{R}\sin\left(\frac{\theta}{2}\right)$$

(b) For semicircular thread $\theta = 180°$

$\therefore$ $$E = \frac{2K\lambda}{R}$$

(c) As $\theta \to 0$

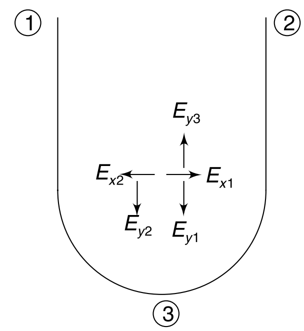
$\sin\left(\frac{\theta}{2}\right) \to 0 \qquad \therefore \quad E = 0$

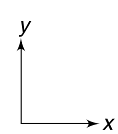
(d) $\lambda = \dfrac{Q}{R\theta}$

$\therefore$ $$E = \frac{2KQ}{R^2}\frac{\sin\left(\frac{\theta}{2}\right)}{\theta} = \frac{KQ}{R^2}\left(\frac{\sin\frac{\theta}{2}}{\frac{\theta}{2}}\right)$$

When $\theta \to 0, \dfrac{\sin\frac{\theta}{2}}{\frac{\theta}{2}} \to 1 \qquad \therefore \quad E = \dfrac{KQ}{R^2}$

51. The x and y components of field due to the three components 1, 2 and 3 have been shown.





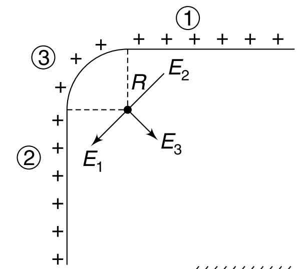
E_{x1} and E_{x2} cancel out.

$$E_{y1} = E_{y2} = \frac{K\lambda}{R}$$

$$E_{y3} = \frac{2K\lambda}{R}$$

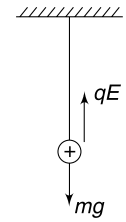
Hence, resultant in y direction in also zero.

52. Resultant is E_3.



$$E_3 = \frac{2K\left(\lambda \frac{\pi R}{2}\right)\sin\left(\frac{\pi}{4}\right)}{\frac{\pi}{2}R^2} = \frac{\lambda}{2\sqrt{2}\pi\varepsilon_0 R}$$

54. (a)

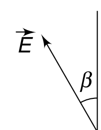


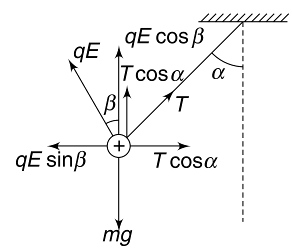
$$g_{\text{eff}} = \frac{mg - qE}{m}$$

$$= g - \frac{q\sigma}{2m\varepsilon_0}$$

$$\therefore \quad T = 2\pi\sqrt{\frac{L}{g - \frac{q\sigma}{2m\varepsilon_0}}}$$

(b) In this case the electric field makes an angle β with vertical. For equilibrium





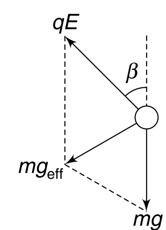
$$T\cos\alpha + qE\cos\beta = mg$$

$$\Rightarrow \quad T\cos\alpha = mg - qE\cos\beta \qquad \ldots(1)$$

$$\text{And} \quad T\sin\alpha = qE\sin\beta \qquad \ldots(2)$$

From (1) and (2)

$$\tan\alpha = \frac{qE\sin\beta}{mg - qE\cos\beta} \quad \text{where} \quad E = \frac{\sigma}{2\varepsilon_0}$$

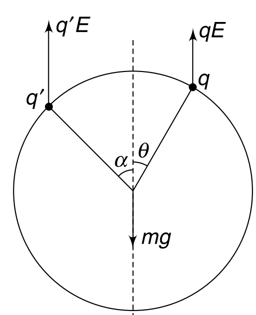


$$g_{\text{eff}}^2 = g^2 + \left(\frac{qE}{m}\right)^2 + 2g\,\frac{qE}{m}\cdot\cos(\pi - \beta)$$

$$\therefore \quad T = 2\pi\sqrt{\frac{L}{\sqrt{g^2 + \left(\frac{qE}{m}\right)^2 - 2g\frac{qE}{m}\cos\beta}}}$$

55. (a) From geometry of the figure $\alpha + \theta = 90°$

Translational equilibrium



$$q'E + qE = mg$$

$$q'E + \frac{4mg}{7} = mg \quad \left[\because \; E = \frac{4mg}{7q}\right]$$

$$q' = \frac{3mg}{7E} \qquad \ldots(1)$$

Rotational equilibrium

$$qE\,R\sin\theta = q'ER\sin\alpha$$

$$\frac{4}{7}\sin\theta = \frac{3}{7}\cos\theta \quad [\because \alpha = 90 - \theta]$$

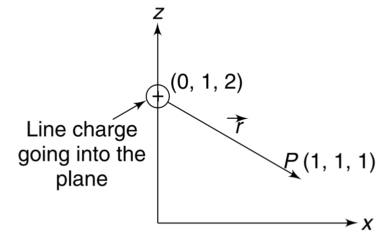
$$\therefore \quad \tan\theta = \frac{3}{4}$$

(b) Suppose the ring is given a small clockwise rotation.

This will cause the anticlockwise torque due to qE to increase whereas the clockwise torque due to $q'E$ will decrease. There will be a net restoring torque.

Hence, there will be oscillations.

56. Foot of the perpendicular from P on the line charge has co-ordinates (0, 1, 2).



$$\therefore \quad \vec{r} = 1\hat{i} + 0\hat{j} - 1\hat{k}$$

$$\hat{r} = \frac{1}{\sqrt{2}}(\hat{i} - \hat{k})$$

$\therefore$ Field at P

$$\vec{E} = \frac{\lambda}{2\pi\varepsilon_0 r}\hat{r} = \frac{1}{2\pi\varepsilon_0\sqrt{2}}\frac{1}{\sqrt{2}}(\hat{i} - \hat{k})$$

$$\therefore \quad \vec{E} = \frac{1}{4\pi\varepsilon_0}(\hat{i} - \hat{k})$$

57. (a) The flux through a spherical surface of radius less than that of the inner sphere is equal to $\frac{q}{\varepsilon_0}$. Where q is the charge on the central particle.

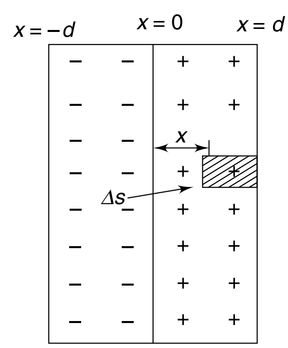
$$\frac{q}{\epsilon_0} = 2 \times 10^6 \text{ V-m}$$

$$\Rightarrow \quad q = 8.85 \times 10^{-12} \times 2 \times 10^6 = 17.7\ \mu\text{C}$$

(b) The flux becomes negative after radius becomes greater than the radius of the inner sphere. It means there is a negative charge on the inner sphere which is larger inmagnitude than the charge on the central particle. The outer shell has still larger positive charge. One can figure out that the electric field exists everywhere.

58. The field on the surface of the composite slab will be zero.

Consider a cylindrical Gaussian surface of cross sectional area ΔS as shown in the figure. One of the circular face of the cylinder is at a distance x from the central plane and the other face is on the surface of the slab.



Let the field at distance x from central plane be E.

From Gauss' Law

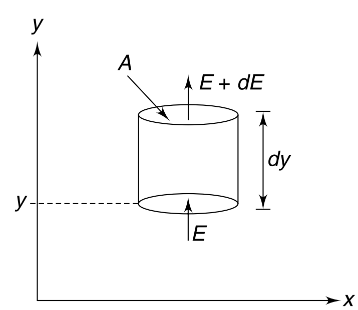
$$E\Delta S = \rho\Delta S(d - x)/\varepsilon_0$$

$$\Rightarrow \quad E = \rho(d - x)/\varepsilon_0$$

The direction will be towards left.

Similarly, one can write field in the region to the left of $x = 0$

59. Consider a cylindrical Gaussian surface as shown.



$$(E + dE)A - EA = \frac{\rho A dy}{\varepsilon_0}$$

$$dE = \frac{by dy}{\varepsilon_0}$$

$$\therefore \quad \int_0^E dE = \frac{b}{\varepsilon_0}\int_0^y y\,dy$$

$$E = \frac{by^2}{2\varepsilon_0}$$

60. Field on the central plane of the sheet = 0

Take a cylindrical Gaussian surface as shown.

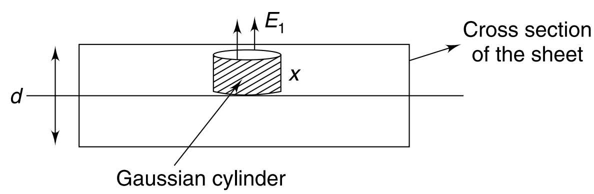
$$E_1\Delta S = \frac{\rho\cdot\Delta S\cdot x}{\varepsilon_0}$$

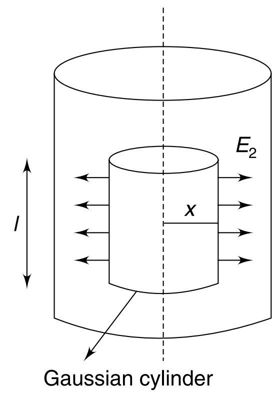
$$E_1 = \frac{\rho x}{\varepsilon_0}$$

By rolling the sheet we have just created a cylinder having uniform volume charge density ρ. Field inside such cylinder can be calculated by considering a co-axial cylindrical Gaussian surface.

$$E_2 \cdot 2\pi x \cdot L = \frac{\rho \pi x^2 L}{\varepsilon_0}$$

$$E_2 = \frac{\rho x}{2\varepsilon_0} \qquad \therefore \quad \frac{E_1}{E_2} = \frac{2}{1}$$





61. Consider a Gaussian surface in the shape of a sphere centered at the origin.
Radius of the sphere is r.
Flux through the sphere

$$\phi = E \cdot 4\pi r^2 = 4\pi a e^{-\frac{r}{k}}$$

Charge inside the Gaussian sphere

$$q = \varepsilon_0 \phi = 4\pi\varepsilon_0 a e^{-\frac{r}{k}}$$

(a) When $r \to 0$

$q \to 4\pi\varepsilon_0 a$

Hence, charge at the origin is $4\pi\varepsilon_0 a$

(b) When $r \to \infty$

$q \to 0$

$\therefore$ Total charge in entire space is zero.

$\therefore$ Charge spread around q_0 is $q = -q_0$.

64. $\phi = \int \vec{E} \cdot \vec{d}s$

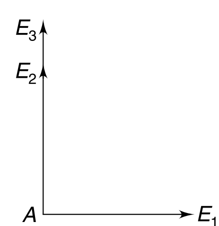
Force $F = \int \sigma \vec{E} \cdot \vec{d}s = \sigma\phi$

65. (a) Field at A:

Due to 'ρ' is $\quad E_1 = \frac{\rho}{3\varepsilon_0}\left(\frac{R}{2}\right)$ parallel to C_1C_2

Due to '2ρ' is $\quad E_2 = \frac{2\rho}{3\varepsilon_0}\left(\frac{R}{4}\right)$ parallel to $C_2C_3 = \frac{\rho}{3\varepsilon_0}\frac{R}{2}$

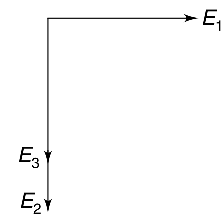
Due to '3ρ' is $\quad E_3 = \frac{4\rho}{3\varepsilon_0}\cdot\frac{R}{8}$ parallel to $C_3A = \frac{\rho}{3\varepsilon_0}\frac{R}{2}$



$\therefore \quad E_A = \sqrt{E_1^2 + (E_2 + E_3)^2} = \frac{\sqrt{5}}{6}\frac{\rho R}{\varepsilon_0}$

(b) Field at B

Due to 'ρ' is $\quad E_1 = \frac{\rho}{3\varepsilon_0}\frac{R}{2}$ parallel to C_1C_2



Due to '2ρ' in entire sphere S_2 is

$$E_2 = \frac{2\rho}{3\varepsilon_0}\cdot\frac{R}{4} \text{ in direction } C_2B$$

Due to '2ρ' in S_3 is

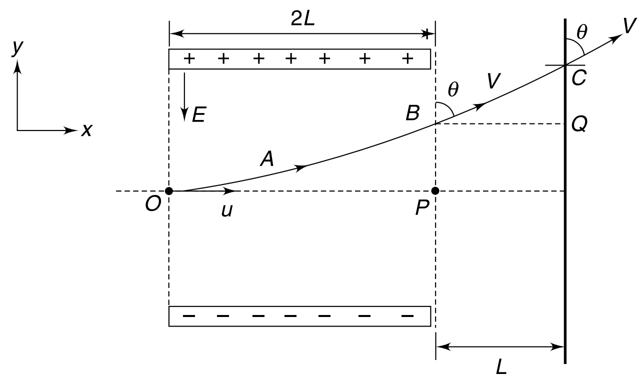
$$E_3 = \frac{1}{4\pi\varepsilon_0}\frac{\frac{4}{3}\pi\left(\frac{R}{4}\right)^3\cdot 2\rho}{\left(\frac{R}{2}\right)^2} \text{ in direction } C_3B$$

$$= \frac{\rho}{3\varepsilon_0}\frac{R}{8}$$

$\therefore$
$$E_B = \sqrt{E_1^2 + (E_2 + E_3)^2}$$

$$= \frac{\sqrt{41}}{24}\frac{\rho R}{\varepsilon_0}$$

66. (a) Acceleration of the electron



$$a_y = \frac{eE}{m} = \frac{e\sigma}{m\varepsilon_0}\,(\uparrow)$$

Path of the electron from O to A to B (i.e. between the plates) is parabolic.

The electron reaches O to B in time $t_1 = \frac{2L}{u}$

At B $\quad V_x = u$

$$V_y = 0 + a_y t_1 = \frac{e\sigma}{m\varepsilon_0}\frac{2L}{u}$$

Also, $\quad PB = \frac{1}{2}a_y t_1^2 = \frac{1}{2}\frac{e\sigma}{m\varepsilon_0}\left(\frac{2L}{u}\right)^2$

$$= 2\frac{e\sigma}{m\varepsilon_0}\frac{L^2}{u^2}$$

Time to travel from B to C

$$t_2 = \frac{L}{u}$$

V_y does not change after the electron crosses the plates

$\therefore$
$$QC = V_y t_2 = \frac{e\sigma}{m\varepsilon_0}\frac{2L}{u}\frac{L}{u} = \frac{2e\sigma}{m\varepsilon_0}\frac{L^2}{u^2}$$

According to the problem

$$PB + QC = \frac{d}{2} \quad \therefore \quad \frac{4e\sigma}{m\varepsilon_0}\frac{L^2}{u^2} = \frac{d}{2}$$

$\therefore$
$$\sigma = \frac{md\varepsilon_0 u^2}{8eL^2}$$

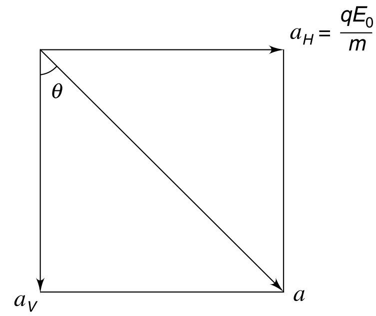
(ii) $\tan\theta = \frac{V_x}{V_y}$

$$= \frac{u(m\varepsilon_0 u)}{2Le\sigma} = \frac{\varepsilon_0 m u^2}{2Le\sigma}$$

67. Acceleration of the particle

Vertical acceleration $a_v = g$

Horizontal acceleration $\qquad a_H = \dfrac{qE_0}{m} = \dfrac{4gE_0}{3E_0} = \dfrac{4g}{3}$

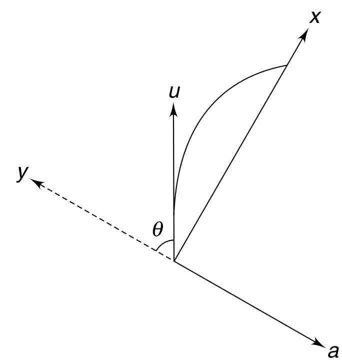


$$a = \sqrt{g^2 + \left(\frac{4g}{3}\right)^2} = \frac{5}{3}\text{ g}$$

$$\tan\theta = \frac{4}{3}$$

Consider, x as direction perpendicular to acceleration and y as direction opposite to a.

Now, the situation is like projectile motion.



(a) $t = \dfrac{u_y}{a} = \dfrac{40\cos\theta}{\frac{5g}{3}} = \dfrac{40 \times 3 \times 3}{5 \times 10 \times 5} = \dfrac{36}{25}$ s

(b) $2t = \dfrac{72}{25}$ s

(c) Velocity will get almost parallel to acceleration.

$\therefore \qquad \alpha = \tan^{-1}\left(\dfrac{3}{4}\right)$

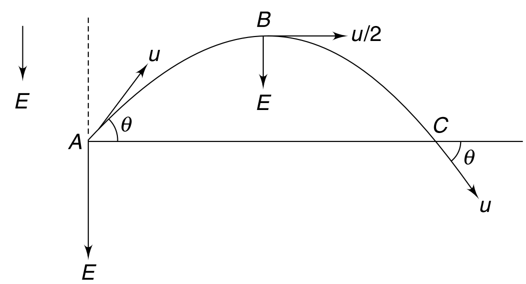
68. (a) Situation is shown in figure

Particle is projected at A. At time t_0 it is at B moving $\perp_r$ to the direction of the field.

At B speed $= \dfrac{u}{2} \Rightarrow \theta = 60°$

Particle will acquire its initial speed when at C.

$\therefore$ Answer to (a) is t_0



(b) Impulse $= \Delta P = 2mu\sin\theta$

$$= 2\sqrt{2mK_0}\cdot\sin 60° = \sqrt{6mK_0}$$

69. Acceleration $a_y = \dfrac{qE}{m} = \dfrac{qay^2}{m}$

$$V_y\frac{dV_y}{dy} = \frac{qay^2}{m}$$

$$\int_0^{V_y} V_y\,dV_y = \frac{qa}{m}\int_{y_0}^{y} y^2 dy$$

$$\frac{V_y^2}{2} = \frac{qa}{3m}(y^3 - y_0^3) \qquad \therefore \quad V_y = \sqrt{\frac{2qa}{3m}(y^3 - y_0^3)}$$

$$\text{Required slope} = \frac{V_y}{V_x} = \frac{V_y}{V_0} = \sqrt{\frac{2qa}{3mV_0^2}(y^3 - y_0^2)}$$

71. One easy way of thinking can be to select one identical element on part OA and OB each. Point P is on perpendicular bisector of the line joining this pair of charges. It is easy to see that field due to such a pair is parallel to AB. All such pairs contribute in same direction. Hence, resultant field is parallel to AB. The pair of charges selected above will produce zero potential at P.

72. (a) Very close to a positive charge the potential will be positive irrespective of other charge in the surrounding. Similarly, potential will be negative at a point very close to a negative charge. Hence, q_1 is positive and q_2 is negative.

Since $V = 0$ at a point that is close to q_2 hence q_2 has smaller magnitude than q_1.

(b) According to the question

$V_{\text{origin}} = 0$

$$K\frac{q_2}{d/4} + K\frac{q_1}{3d/4} = 0 \quad \Rightarrow \quad q_1 = -3q_2$$

If $\qquad q_2 = -q; \quad \text{then} \quad q_1 = +3q$

At P $\qquad \dfrac{dV}{dx} = 0 \quad \Rightarrow \quad E_p = 0$

Let distance of P from q_2 be x.

$$K\frac{|q_2|}{x^2} = K\frac{|q_1|}{(d+x)^2} \quad \Rightarrow \quad (d+x)^2 = 3x^2$$

$$d + x = \pm\sqrt{3}x$$

$$\Rightarrow \qquad \frac{d}{\sqrt{3}-1} = x$$

[–ve sign is not acceptable]

73. (a) Let x, y and z direction be as shown.

$$V_A - V_0 = -E_x\,(1\,\text{m})$$

$$\therefore \qquad E_x = 1\,\text{V/m} \qquad \text{...(1)}$$

$$V_B - V_0 = -E_x\,(3\,\text{m}) - E_y\,(2\,\text{m}) - E_z\,(1\,\text{m})$$

$$-6 = -3E_x - 2E_y - E_z$$

$$2E_y + E_z = 3 \qquad \text{...(2)}$$

And $\qquad V_c - V_o = -E_x(1\,\text{m}) - E_y(3\,\text{m}) - E_z(4\,\text{m})$

$$-3 = -1 - 3E_y - 4E_z \quad \Rightarrow \quad 3E_y + 4E_z = 2 \qquad \text{...(3)}$$

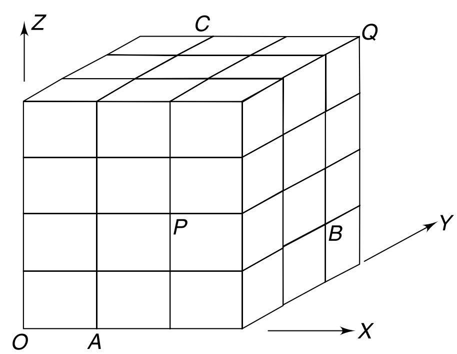
Solving (2) and (3)

$$E_y = 2\,\text{V/m}$$

$$E_z = -1\,\text{V/m}$$

$$V_Q - V_p = -E_x(1\,\text{m}) - E_y(3\,\text{m}) - E_z(2\,\text{m})$$

$$V_Q - V_p = -1 - 6 + 2 \quad \Rightarrow \quad V_P - V_Q = 5\,\text{V}$$



(b) $E = \sqrt{E_x^2 + E_y^2 + E_z^2} = \sqrt{1^2 + 2^2 + 1^2} = \sqrt{6}$ V/m

The potential will change at fastest rate when we move in the direction of field.

$$d = \frac{2\ \text{Volt}}{\sqrt{6}\ \text{Volt/m}} = \sqrt{\frac{2}{3}}\ \text{m}$$

74. Let direction along AB be X and that along AC be Y. Perpendicular to the incline is Z direction.

$$E_x = \frac{\Delta V}{\Delta X} = \frac{10\ \text{V}}{1\ \text{cm}} = 10\ \frac{\text{V}}{\text{cm}}$$

Field is directed along the decreasing potential.

Similarly, $\qquad E_y = \dfrac{\Delta V}{\Delta y} = 10\ \dfrac{\text{V}}{\text{cm}}$

Similarly, $\qquad E_V = 10\ \dfrac{\text{V}}{\text{cm}}$

But $$E_V = E_Z \cos 37° + E_y \sin 37°$$

$$10 = \frac{4}{5}E_z + 10 \times \frac{3}{5} \quad \Rightarrow E_z = 5 \text{ V/cm}$$

$\therefore$ $$E = \sqrt{E_x^2 + E_y^2 + E_z^2} = \sqrt{10^2 + 10^2 + 5^2} = 15 \text{ V/cm}$$

$$\vec{E} = (10\hat{i} + 10\hat{j} + 5\hat{k}) \text{ V/cm}$$

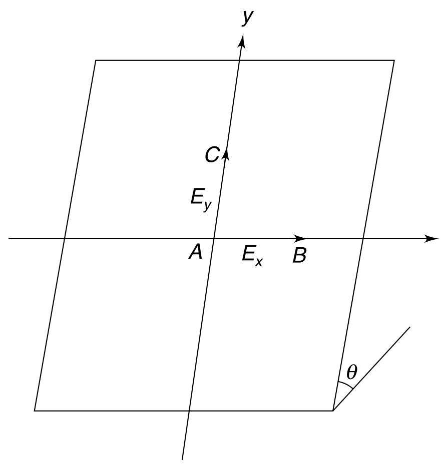
Unit vector along vertical $\hat{n} = \frac{3}{5}\hat{j} + \frac{4}{5}\hat{k}$

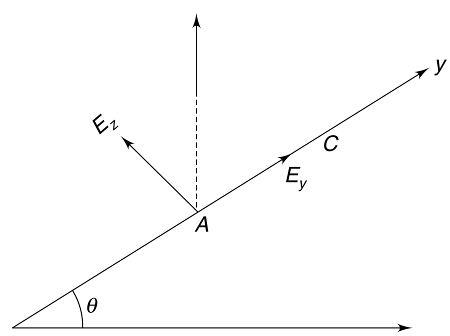
Angle between $\vec{E}$ and $\hat{n}$ is given by

$$\cos\alpha = \frac{\vec{E}\cdot\hat{n}}{E}$$

$$\cos\alpha = \frac{10}{15}$$

$$\alpha = \cos^{-1}\left(\frac{2}{3}\right)$$





75. (a) Potential will be given by

$V = \left(\frac{1}{4}\right)^{\text{th}}$ the potential due to complete disc having charge density σ

$$V = \frac{\sigma}{8\epsilon_0}[\sqrt{R^2 + Z^2} - Z]$$

(b) $E_z = -\frac{\partial V}{\partial Z} = \frac{\sigma}{8\epsilon_0}\left[1 - \frac{Z}{\sqrt{R^2 + Z^2}}\right]$

76. (a) The space has negative charge distribution. Radial field directed towards the fixed point terminate on these negative charges so that density of lines go on decreasing as one moves towards the fixed point

(b) Flux through sphere of radius a is

$$\phi = -(4a)(4\pi a^2) = -16\pi a^3$$

$\therefore$ $$\frac{q}{\epsilon_0} = -16\pi a^3$$

$$q = -16\pi\epsilon_0 a^3$$

(c) $r_A = \sqrt{1^2 + 1^2 + (\sqrt{2})^2} = 2\,\text{m}$

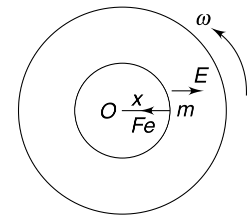
$r_B = \sqrt{0 + 3^2 + 4^2} = 5\,\text{m}$

[All points at a distance of 2 m from origin are equipotential. Similarly, all points at a distance 5 m are at same potential]

$$\therefore \quad V_B - V_A = -\int_{r_A}^{r_B} \vec{E}\cdot\vec{dr} = -\int_2^5 (-4r)dr$$

$$= \frac{4}{2}[r^2]_2^5 = 2 \times 21 = 42 \text{ Volt}$$

$$\therefore \quad V_A - V_B = 42 \text{ Volt}$$



77. For an electron at a distance x from the centre (when it is in equilibrium in the reference frame of the disc)

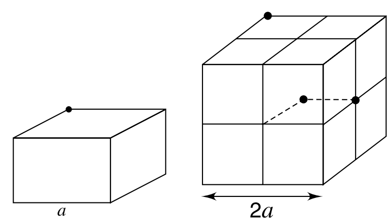
Electric force $(F_e) = m\omega^2 x$

$$eE = m\omega^2 x \qquad \therefore E = \frac{m\omega^2}{e}x$$

$\therefore$ Potential difference between centre (O) and circumference (C) is

$$V_C - V_o = -\int_{x=o}^{R} E\,dx \qquad \therefore \quad V_o - V_c = \frac{m\omega^2}{e}\int_o^R x\,dx = \frac{m\omega^2 R^2}{2e}$$

78. Consider a uniformly charged cube of side length a, and charge density ρ.



From dimensional analysis

$$V_{\text{corner}} \propto \frac{\rho a^3}{a}$$

$$V_{\text{corner}} \propto \rho a^2$$

Now consider a big cube of side length $2a$

$$\frac{V_{\text{corner}}^{\text{Big}}}{V_{\text{corner}}^{\text{Small}}} = \left(\frac{2a}{a}\right)^2 = \frac{4}{1} \qquad ...(1)$$

But it is clear from the figure that

$$V_{\text{centre}}^{\text{Big}} = 8V_{\text{Corner}}^{\text{Small}}$$

$$\therefore \quad \text{from (1)} \quad \frac{V_{\text{Corner}}^{\text{Big}}}{\frac{1}{8}V_{\text{Centre}}^{\text{Big}}} = \frac{4}{1} \Rightarrow \frac{V_{\text{Conrner}}}{V_{\text{Centre}}} = \frac{1}{2}$$

79. Charge on a conductor will always be on its outer surface. Shell is thin and fits perfectly over the sphere. Half the charge on sphere will reside on the shell. As the shell is removed, the charge on it also goes away with it. When grounded, the shell will give away all its charge.

Every time shell is mounted and removed, the charge on sphere become half (or its potential become half)

$$\therefore \quad \frac{100}{2^n} = 6.25 \quad \Rightarrow n = 4$$

80. *Hint:* the entire charge of the inner sphere will get transferred to the outer ball. On the outer sphere the charge will be distributed uniformly on its surface.

81. (a) If there is no charge inside the shell, its entire charge will reside on its outer surface. But the question says that there is charge on the inner wall as well. Hence, there must be a charge inside it.

There is positive charge inside the shell that is equal in magnitude to the charge on the inner wall.

Net charge on the shell = charge on outer surface + charge on inner surface

$$q_{\text{shell}} = 4\pi b^2\sigma - 4\pi a^2\sigma$$

(b) Potential of the shell $V = \frac{1}{4\pi\varepsilon_0}\frac{4\pi b^2\sigma}{b} = \frac{b\sigma}{\varepsilon_0}$

82. Given $V = \frac{1}{4\pi\varepsilon_0}\frac{q}{a}$

Volume of liquid does not change when the bubble collapses to form a drop.

$$4\pi a^2 t = \frac{4}{3}\pi r^3 \text{ where '}r\text{' is the radius of the drop}$$

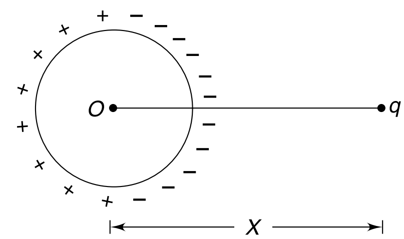
$$r = (3a^2 t)^{1/3}$$

Charge remains conserved. Hence new potential is

$$V' = \frac{1}{4\pi\varepsilon_0}\frac{q}{r} = V\left(\frac{a}{3t}\right)^{1/3}$$

83. Charge is induced on the surface of the sphere but net charge on the surface is zero. Due to induced charge, the potential at the centre of the sphere is zero.

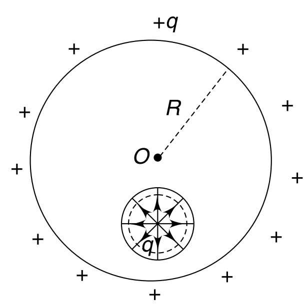
$\therefore$ Potential at O



$$V = \frac{1}{4\pi\epsilon_0}\frac{q}{x}$$

The whole conductor is at a single potential. Even if the sphere is hollow, the charge distribution on the surface will remain identical (there is no reason for it to be different !). Hence answer remains same.

87. In all cases, a charge $-q$ is induced on the inner wall of the cavity and a charge $+q$ is induced on the outer surface. [why exactly $-q$ charge is induced on the cavity wall? why not $-2q$? Explain]. The charge on the outer surface is uniformly distributed in all cases.



The electric field produced by the point charge and the charge on the cavity wall is limited to the space inside the cavity only.

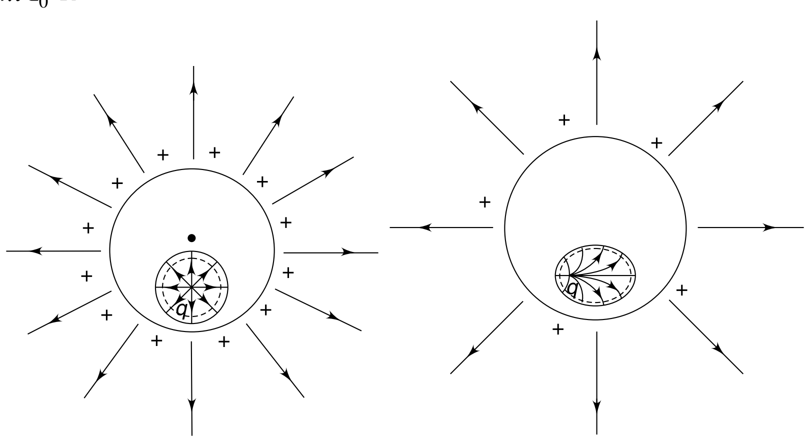
Resultant field due to these two charges is zero at all points outside the cavity!! Field lines starting at the point charge $+q$ and terminating on the cavity walls ($-q$ charge) tells us that no field is created due to this pair of charges outside the cavity.

(a) Field outside the sphere is due to $+q$ charge on its outer surface only

$\therefore$ Outside field resembles a field due to a point charge placed at the centre O of the sphere.

$\therefore$ Potential = work done in slowly moving a unit positive test charge from infinity on to the conductor

$$= \frac{1}{4\pi\epsilon_0}\frac{q}{R}$$



(b) No change

(c) No change

(d) Field line are normal to the conductor surface. These lines are all straight when charge is at the centre of the spherical cavity

88. *Hint:* The net charge inside the ball is Q though charge will get induced on the dielectric surface.

89. Point charge q induces a charge on the surface of the conducting sphere.

Potential of the sphere = potential at the centre

[whole sphere is at one potential]

The induced charge will cause no potential at the centre.

$$\therefore \qquad V_{\text{sphere}} = V_{\text{Centre}} = \frac{Kq}{x}$$

Now, when the sphere is earthed, its potential (i.e, potential at its centre) must become zero. For this the earth supplies a charge q_0.

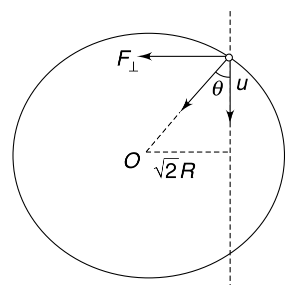
$$V_{\text{sphere}} = V_{\text{centre}} + V_{\substack{\text{induced}\\ \text{charge}}} + V_{q_0}$$

$$O = \frac{Kq}{x} + 0 + \frac{Kq_0}{\text{R}} \quad \Rightarrow q_0 = -\frac{qR}{x}$$

(b) $i = \left|\frac{dq_0}{dt}\right| = \frac{qR}{x^2}\frac{dx}{dt} = \frac{qR}{x^2}\upsilon$

91. Charge $-Q$ is induced on the inner surface of the outer shell. There is no charge on the outer surface of the outer shell as it is grounded.

An electric field exists in the space between the two shells.



$$E = K\frac{Q}{x^2} \quad \text{for } R < x < 2R$$

Just after the charge enters, it experience a force due to this electric field directed towards the centre.

$$F = K\frac{Qq}{(2R)^2}$$

Component of this force perpendicular to the direction of instantaneous velocity u is

$$F_\perp = F\sin\theta = F\frac{\sqrt{2}R}{2R} = \frac{F}{\sqrt{2}}$$

$$= K\frac{Qq}{4\sqrt{2}R^2}$$

If radius of curvature of the path is r

$$\frac{mu^2}{r} = F_\perp$$

$$\therefore \qquad \frac{mu^2}{r} = \frac{KqQ}{4\sqrt{2}R^2}$$

$$\therefore \qquad r = \frac{16\sqrt{2}\pi\epsilon_0 R^2mu^2}{Qq}$$

(b) The potential difference between the two spheres is

$$V_{\text{inner}} - V_{\text{outer}} = \left(K\frac{Q}{R} - \frac{KQ}{2R}\right) - \left(\frac{KQ}{2R} - \frac{KQ}{2R}\right)$$

$$= \frac{KQ}{2R}$$

Energy conservation for the charge entering the shell gives:

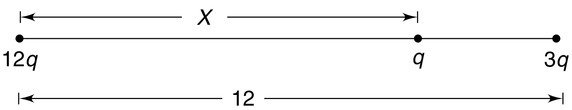
$$\frac{1}{2}mv^2 + (-q)V_{\text{inner}} = \frac{1}{2}mu^2 + (-q)V_{\text{outer}}$$

$$\therefore \quad \frac{1}{2}mv^2 = \frac{1}{2}mu^2 + q(V_{\text{inner}} - V_{\text{outer}})$$

$$\frac{1}{2}mv^2 = \frac{1}{2}mu^2 + \frac{Qq}{8\pi\epsilon_0 R}$$

$$\therefore \quad v = \sqrt{u^2 + \frac{Qq}{4\pi\epsilon_0 mR}}$$

92. The largest charge $12q$ must be placed at one of the corners, and the smallest charge between the two charges.



Let the position be as shown in figure [The trick is to understand that PE of a pair of charges $\propto \frac{qQ}{r^2}$. Therefore, the pair having largest value of qQ must be farthest]

$$U = Kq^2\left[\frac{36}{12} + \frac{12}{X} + \frac{3}{12-X}\right]$$

U will be minimum when $\frac{dU}{dX} = 0$

$$\Rightarrow \quad \frac{12}{X^2} = \frac{3}{(12-X)^2} \quad \Rightarrow \left(\frac{12-X}{X}\right)^2 = \frac{1}{4} \quad \Rightarrow 24 - 2X = X \quad \Rightarrow X = 8\,\text{cm}$$

In this position force on q is $= K\frac{12q^2}{8^2} - K\frac{3q^2}{4^2} = 0$

93. Sum of interaction energy of each $-q$ charge with $+Q$ charge in original position is

$$U_0 = \left[-\frac{KqQ}{\frac{a}{\sqrt{2}}} - \frac{KqQ}{\frac{3a}{\sqrt{2}}}\right] \times 2 = -\frac{8\sqrt{2}}{3}\frac{KqQ}{a}$$

(a) When larger square is shifted to infinity, this interaction energy will become zero.

$$\therefore \quad W_{\text{ext}} = 0 - U_0 = \frac{8\sqrt{2}}{3}\frac{KqQ}{a}$$

(b) First rotation brings the $-q$ charges on z axis at co-ordinates $z = \frac{a}{\sqrt{2}}$ and $z = -\frac{a}{\sqrt{2}}$

The second rotation does not charge the position of two charges.

$\therefore$ distance of each $-q$ charge from every Q charge is

$$r = \sqrt{\left(\frac{a}{\sqrt{2}}\right)^2 + (\sqrt{2}a)^2} = \sqrt{\frac{5}{2}}\cdot a$$

$\therefore$ Interaction energy for $-q$, Q pairs is

$$U = \frac{-KqQ}{\sqrt{\frac{5}{2}}\cdot a} \times 4 = \frac{4\sqrt{2}}{\sqrt{5}}\frac{KqQ}{a}$$

$$\therefore \quad W_{\text{ext}} = U - U_0 = \left[\frac{8\sqrt{2}}{3} - \frac{4\sqrt{2}}{\sqrt{5}}\right]\frac{KqQ}{a}$$

94. (a) The electric field has maximum strength where the magnitude of slope of the V–x graph is maximum. Among marked points, we have large slopes at C and D. Positive slope at D means that field is in negative X direction at that point. At this point a negative charge will feel largest force in positive X direction.

(b) Total energy of the proton shall be less than zero for it to remain bound.

At origin total energy is

$$\frac{1}{2}mv^2 + (-V_0)e \qquad \therefore \quad \frac{1}{2}mv^2 + (-V_0)e \le 0$$

$$v \le \sqrt{\frac{2eV_0}{m}}$$

An electron cannot remain bound as it will always experience a force away from the origin.

95. The tension decreases as the pendulum rises. In case (a) the electrostatic force between the charges is

$$F_0 = \frac{1}{4\pi\epsilon_0}\frac{qQ}{L^2} = 2mg$$

In this case the tension cannot become zero anywhere. Even if the bob rises to the top and has zero speed there, tension is the string will be mg.

Therefore, to complete the circle we need to ensure that the bob has enough speed to rise to the top point.

i.e. $$u = \sqrt{4gL}$$

In case (b)

$$F_e = \frac{mg}{2}$$

If v is speed at the top

$$T + mg - F_e = \frac{mv^2}{L}$$

$$T + \frac{mg}{2} = \frac{mv^2}{L}$$

$\therefore$ for $T \ge 0$

$$v \ge \sqrt{\frac{gL}{2}}$$

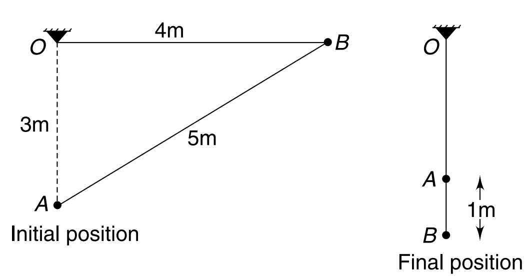
Energy conservation gives

$$u \ge \sqrt{4.5gL}$$

96. (a) **Energy conservation**

$$K\frac{Q \cdot Q}{5} + mg(4) = K\frac{Q \cdot Q}{1}$$

$$\Rightarrow \qquad m = \frac{K}{g}\frac{QQ}{5} = \frac{9 \times 10^9 \times (20 \times 10^{-6})^2}{10 \times 5} = 7.2 \times 10^{-2}\text{ kg} = 72\text{ g}$$



(b) $T = mg + K\dfrac{Q \cdot Q}{1^2} = 7.2 \times 10^{-2} \times 10 + 9 \times 10^9 \times (20 \times 10^{-6})^2 = 4.32\text{ N}$

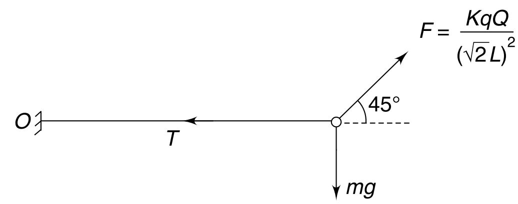
(c) Unstable.

When released, the mass accelerates and then retards to stop when just below A. It means, if disturbed from equilibrium, it will experience a force away from equilibrium.

97. Equilibrium of ball 1.

$$\frac{F}{\sqrt{2}} = mg$$

$$\frac{KqQ}{2\sqrt{2}L^2} = mg \quad ...(1)$$



$$W_{ext} = U_f - U_i$$

$$= 0 - \left[mgL + \frac{KqQ}{\sqrt{2}L}\right] = -[mgL + 2mgL]$$

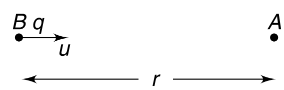
$$= -3mgL$$

98. (a) KE will be least when PE of the system is maximum, i.e. when charges are closest to each other. In this situation both charges will have same velocity (v).

$$mv + mv = mu$$

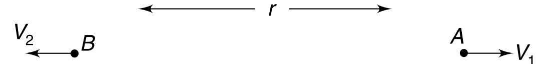
$$\Rightarrow \quad v = \frac{u}{2}$$

$$\therefore \quad KE_{min} = \frac{1}{2}m\left(\frac{u}{2}\right)^2 + \frac{1}{2}m\left(\frac{u}{2}\right)^2 = \frac{1}{4}mu^2$$



(b) A will move to right with velocity u and B will be at rest.

99. Let the final velocity (at ∞ separation) be V_1 and V_2 as shown.



Momentum Conservation

$$mV_1 - mV_2 = mu$$

$$V_1 - V_2 = u \quad ...(1)$$

Energy Conservation

$$\frac{1}{2}mV_1^2 + \frac{1}{2}mV_2^2 = \frac{qQ}{4\pi\epsilon_0 \cdot r} + \frac{1}{2}mu^2$$

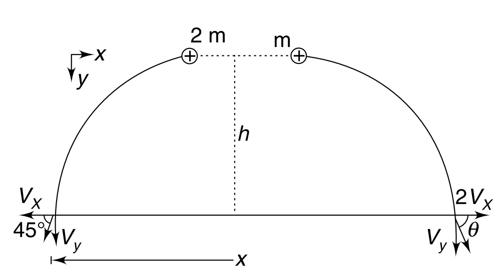
$$\Rightarrow \quad mV_1^2 + m(V_1 - u)^2 = 2mu^2 \quad \left[\because \ r = \frac{qQ}{2\pi\epsilon_0 mu^2}\right]$$

$$\Rightarrow \quad 2V_1^2 - 2uV_1 - u^2 = 0 \quad \Rightarrow V_1 = \left(\frac{1+\sqrt{3}}{2}\right)u$$

100. (a) The y component of velocity of two balls will be same.

$$V_y = \sqrt{2gh}$$

For momentum to remain conserved along horizontal direction, the X component of velocity of the lighter ball will be twice that of the heavier ball (in opposite direction).



Given $\quad \tan 45° = \frac{V_y}{V_x} \Rightarrow V_y = V_x = \sqrt{2gh}$

$$\therefore \quad \tan\theta = \frac{V_y}{2V_x} = \frac{1}{2}$$

(b) **Energy Conservation** $\frac{1}{2}\,2m(V_x^2 + V_y^2) + \frac{1}{2}\,m(V_y^2 + (2V_x)^2) + U_f$

$$= U_i + 3\,\text{mgh}$$

Where U_i and U_f are initial and final electrostatic potential energy of the charge pair

$$V_x = V_y = \sqrt{2gh}$$

$$\therefore \quad 4{\cdot}5m\,(2gh) + U_f = U_i + 3\,\text{mgh}$$

$$\therefore \quad 6\,\text{mgh} = U_i - U_f$$

$$W_{\text{elect}} = -(U_f - U_i) = U_i - U_f$$

$$= 6\,\text{mgh}$$

101. (a) The electric field in the region between the two shells is radially outward and outside the larger shell it is radially inwards. Once the electron crosses the hole it will be automatically pushed to infinity.

The KE of the electron shall be sufficient enough to carry it to the outer shell.

$$K_{\min} + (-e)V_A = (-e)V_B$$

$$K_{\min} = e(V_A - V_B)$$

$K_{\min} = eKQ\left(\frac{1}{R} - \frac{1}{2R}\right)$ (the potential difference depends only on the charge on the inner shell)

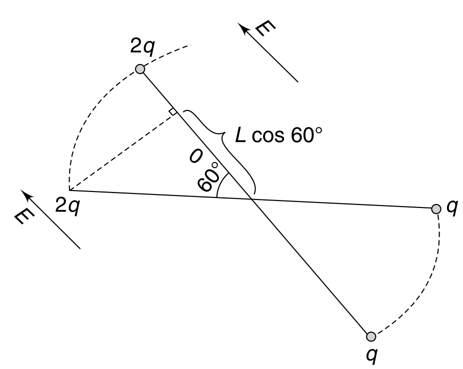
$K_{\min} = KeQ\frac{1}{2R}$

(b) In this case the field is radially outwards throughout. The KE must be large enough to carry the electron to infinity.

$$K_{\min} + (-e)V_A = 0$$

$$K_{\min} = eV_A = eK\left(\frac{Q}{2R} + \frac{2Q}{2R}\right) = \frac{3KQe}{2R}$$

102. (a) ω is maximum when angular acceleration α is zero. This happens when rod becomes parallel to the electric field.



Loss in electrostatic PE

$$\Delta U = 2q\cdot E[L - L\cos 60°] - qE[L - L\cos 60°]$$

$$= \frac{qEL}{2}$$

$$\therefore \qquad \frac{1}{2}I\omega^2 = \frac{qEL}{2}$$

$$\frac{1}{2}\frac{M(2L)^2}{12}\cdot\omega^2 = \frac{qEL}{2}$$

$$\omega = \sqrt{\frac{3qE}{ML}}$$

(b) The rod will perform oscillation with an amplitude of 60° on either side of the direction of $\vec{E}$.

Maximum torque

$$\tau_{\max} = 2qE\sin 60° - qE\sin 60°$$

$$= \frac{\sqrt{3}}{2}qE$$

$$\therefore \qquad \frac{M(2L)^2}{12}\cdot\alpha = \frac{\sqrt{3}}{2}qE$$

$$\alpha = \frac{3\sqrt{3}}{2}\frac{qE}{ML^2}$$

103. Hint: Particles are going in circles about their common centre of mass.

104. A will leave contact with the wall when

$$kx_0 = qE_0$$

$$x_0 = \frac{qE_0}{K} = \text{minimum extension in the spring.}$$

After being released, block B must at least move through a distance x_0 before it comes to rest.

Energy conservation for motion of *B*

Gain in spring potential energy = loss in electrostatic potential energy of B

$$\Rightarrow \qquad \frac{1}{2}kx_0^2 = Q\cdot E_0x_0 \quad \Rightarrow \quad x_0 = \frac{2QE_0}{k}$$

But x_0 must be at least $\frac{qE_0}{k}$

$$\therefore \qquad \frac{2QE_0}{k} = \frac{qE_0}{k} \quad \Rightarrow \quad Q = \frac{q}{2}$$

105. Charge on sphere 1 after S_1 is closed is

$$Q = 4\pi\in_0 RV$$

Now S_1 is opened and S_2 is closed. Charge is shared between sphere 1 & 2.

Charge on 1 & 2 both becomes $\frac{Q}{2}$

Thereafter, S_2 is opened & S_3 closed. Charge on 2 and 3 both will become $\frac{Q}{4}$

This way we can see that final charges on the spheres will be $\frac{Q}{2}, \frac{Q}{4}, \frac{Q}{8}, \frac{Q}{16}$ ...

The final potential of spheres will be $\frac{V}{2}, \frac{V}{4}, \frac{V}{8}, \frac{V}{16} \ldots$

Final energy

$$U = \frac{1}{2} 4\pi \epsilon_0 R \left[\left(\frac{V}{2}\right)^2 + \left(\frac{V}{4}\right)^2 + \left(\frac{V}{8}\right)^2 + \left(\frac{V}{16}\right)^2 + \ldots\right]$$

$$= \frac{1}{2} 4\pi \epsilon_0 R V^2 \left(\frac{1/4}{1 - 1/4}\right) = \frac{1}{6} 4\pi \epsilon_0 R V^2$$

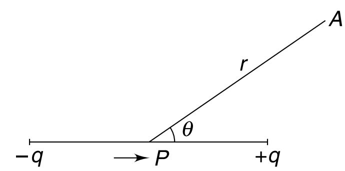
Energy lost by the cell = Work done by the cell

$$= QV = 4\pi \epsilon_0 R V^2$$

Hence, loss in energy of the system

$$\Delta U = 4\pi \epsilon_0 RV^2 - \frac{1}{6} \cdot 4\pi \epsilon_0 RV^2 = \frac{5}{6} 4\pi \epsilon_0 RV^2 = \frac{10}{3} \pi \epsilon_0 RV^2$$

106. (a) Potential at position (r, θ) is



$$V = \frac{KP\cos\theta}{r^2}$$

At a point where potential is V_0

$$r^2 = \frac{KP}{V_0} \cos\theta \quad [\Rightarrow \quad r^2 \propto \cos\theta]$$

Maximum value of $\cos\theta = 1$ (for $\theta = 0°$)

$$\therefore \qquad r_{\max} = \sqrt{\frac{KP}{V_0}}$$

(b) Field at (r, θ) is

$$E = \frac{KP}{r^3} \sqrt{1 + 3\cos^2\theta}$$

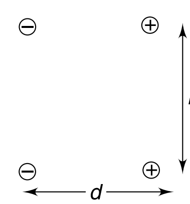
When $E = E_0$

$$r^3 = \frac{KP}{E_0} (1 + 3\cos^2\theta)^{1/2}$$

r is maximum when $\cos\theta = 1$

$$\therefore \qquad r_{\max} = \left(2\frac{KP}{E_0}\right)^{1/3}$$

107.



$$U = K \cdot \frac{q \cdot q}{r} + K\frac{(-q)(-q)}{r} + K\frac{q(-q)}{\sqrt{r^2 + d^2}} + K\frac{(-q)(q)}{\sqrt{r^2 + d^2}}$$

$$= 2Kq^2 \left[\frac{1}{r} - \frac{1}{(r^2 + d^2)^{1/2}}\right]$$

$$= \frac{2Kq^2}{r} \left[1 - \left(1 + \frac{d^2}{r^2}\right)^{-1/2}\right]$$

$$= \frac{2Kq^2}{r} \left[1 - 1 + \frac{1}{2}\frac{d^2}{r^2}\right]$$

$$= \frac{Kq^2 d^2}{r^3} = \frac{KP^2}{r^3} \quad [P = \text{dipole moment of each dipole}]$$

When one of the dipoles is rotated by 90°, interaction energy become zero. It can be easily seen as the potential due to a dipole on its perpendicular bisector is zero.

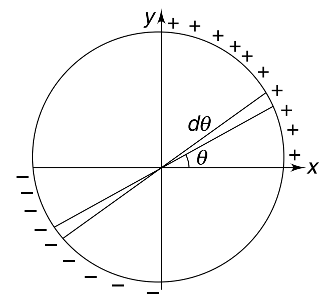
(a) $W = U_{\text{final}} - U_{\text{initial}} = 0 - \dfrac{KP^2}{r^3} = -\dfrac{KP^2}{r^3}$

(b) It can be shown, as above, that potential energy in position shown in figure(c) is

$$U = -\frac{KP^2}{r^3}$$

$$\therefore \quad W = U_{\text{final}} - U_{\text{initial}} = -\frac{KP^2}{r^3} - \frac{KP^2}{r^3} = -\frac{2KP^2}{r^3}$$

108. Consider a segment of ring of angular width $d\theta$. Consider another segment at diametrically opposite end. The two small elements make a dipole. Its dipole moment is



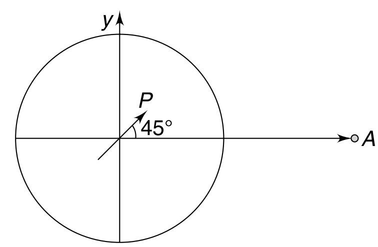
$$dP = (\lambda r d\theta)(2r) = 2\lambda r^2 \, d\theta$$

$$\left[\lambda = \text{linear charge density} = \frac{2q}{\pi r}\right]$$

X component of dipole moment is $dP_x = 2\lambda r^2 \cos\theta d\theta$

Y component is $dP_y = 2\lambda r^2 \sin\theta d\theta$

Adding contributions from all such dipoles



$$P_x = 2\lambda r^2 \int_0^{\pi/2} \cos\theta d\theta = 2\lambda r^2$$

$$P_y = 2\lambda r^2 \int_0^{\pi/2} \sin\theta d\theta = 2\lambda r^2$$

The charge system can be treated like a single dipole of dipole moment

$$P = 2\sqrt{2}\lambda r^2 = \frac{4\sqrt{2}\, q r}{\pi}$$

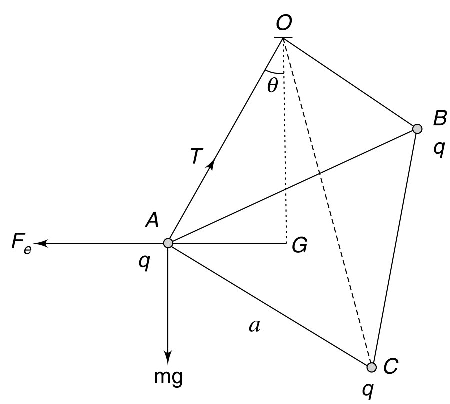
The resultant dipole moment vector make an angle 45° with the x-axis

$\therefore$ Potential at A is

$$V = \frac{1}{4\pi \in_0} \frac{P\cos 45°}{R^2}$$

$$V = \frac{qr}{\pi^2 \in_0 R^2}$$

109. In equilibrium, the three balls are on the vertices of an equilateral triangle in horizontal plane. Resultant electrostatic force on charge A is



$$F_e = K\frac{q^2}{a^2}\cos 30° \times 2 \text{ (Horizontal)}$$

$$= \sqrt{3}\frac{Kq^2}{a^2}$$

$$\therefore \quad \tan\theta = \frac{F_e}{mg} = \frac{\sqrt{3}\,Kq^2}{mga^2}$$

But $$\tan\theta \approx \sin\theta = \frac{AG}{L} = \frac{a}{\sqrt{3}\,L}$$

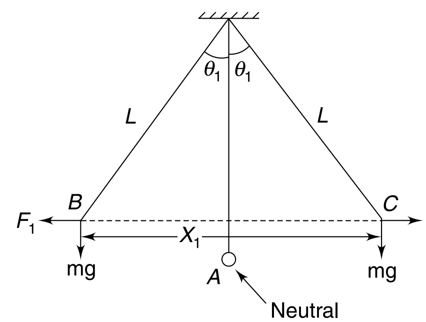
Using above two equations we get

$$a^3 = \frac{3Kq^2L}{mg} \quad ...(1)$$

(a) When one ball gets discharged, it will experience no electric force. It will be in equilibrium with string vertical. The other two balls will be in equilibrium in position shown.

$$\tan\theta_1 = \frac{F_1}{mg}$$

$$\frac{X_1}{2L} = \frac{Kq^2}{mg \cdot X_1^2} \quad [X_1 = \text{separation between } B \text{ \& } C] \qquad ...(2)$$



$$(2) \div (1)$$

$$\left(\frac{X_1}{a}\right)^3 = \frac{2}{3}$$

$$\therefore \quad X_1 = \left(\frac{2}{3}\right)^{\frac{1}{3}} a$$

(b) If B and C are discharged, all the balls will collide at G.

Each of them will acquire charge $\frac{q}{3}$ and then repel each other.

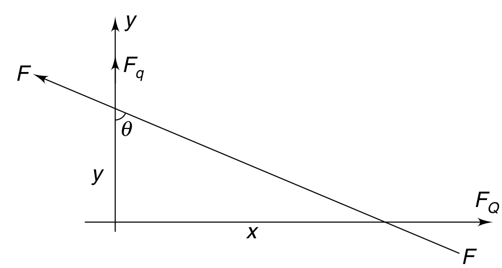
Equilibrium separation (X_2) can be obtained from (1)

$$X_2^3 = \frac{3K\left(\frac{q}{3}\right)^2 L}{mg}$$

$$\therefore \quad \frac{X_2^{\,3}}{a^3} = \frac{1}{9}$$

$$\therefore \quad X_2 = \left(\frac{1}{9}\right)^{\frac{1}{3}} \cdot a$$

110. Force on q along y is $\quad F_q = \frac{KqQ}{(x^2 + y^2)} \cdot \cos\theta = \frac{KqQy}{(x^2 + y^2)^{\frac{3}{2}}}$

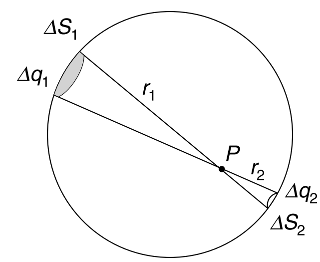


Force on Q along x axis is $\quad F_Q = \frac{KqQ}{(x^2 + y^2)} \sin\theta = \frac{KqQx}{(x^2 + y^2)^{\frac{3}{2}}}$

$$\therefore \quad \frac{F_q}{F_Q} = \frac{y}{x}$$

$\therefore$ acceleration of the particles is in the ratio of their respective distance from origin.
In the interval q moves a distance r, Q will move through $2r$.

111. The solid angle at the apex of the cone is



$$\Delta\Omega = \frac{\Delta S_1}{r_1^2} = \frac{\Delta S_2}{r_2^2}$$

If σ is surface charge density $\quad \frac{\sigma \Delta S_1}{r_1^2} = \frac{\sigma \Delta S_2}{r_2^2}$

$$\frac{\Delta q_1}{r_1^2} = \frac{\Delta q_2}{r_2^2}$$

$$K\frac{\Delta q_1}{r_1^2} = K\frac{\Delta q_2}{r_2^2}$$

$$\therefore \quad E_1 = E_2$$

The entire spherical surface can be divided into many such pair of small area which produce zero resultant at p.

112. (a) Solid angle formed by a cone having semi vertical angle α is

$$\Omega = 2\pi(1 - \cos\alpha) = 2\pi(1 - \cos 60°) = \pi\ Sr$$

Therefore, one fourth of the total number of lines emitted from q_1 = Half the number of lines terminating on q_2

$$\therefore \quad N_1 = 2N_2$$

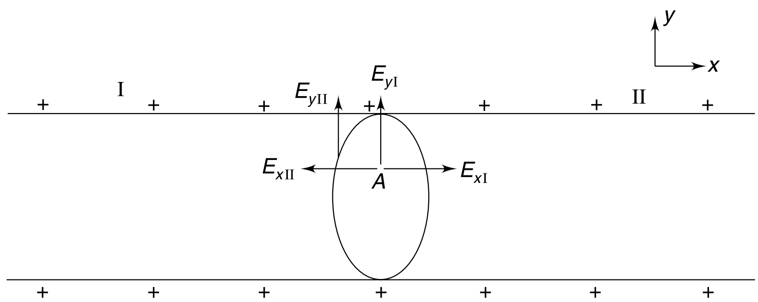
$$\therefore \quad |q_1| = 2|q_2|$$

(b) Half the lines emitted from q_1 do terminate on q_2. Remaining half will go to ∞.

$$\therefore \quad \alpha_{max} = 90°$$

113. Consider the cylindrical shell to be of infinite extent on both sides.

We can imagine it to be made of two identical part I and II.



X & Y components of electric field at A due to part I may have directions as shown. Due to symmetry, the field at A due to part II will have components as shown (E_{XII} & E_{YII}).

Using Gauss law we can shown that field at A is zero.

$$\Rightarrow \quad E_{XI} = E_{XII}$$

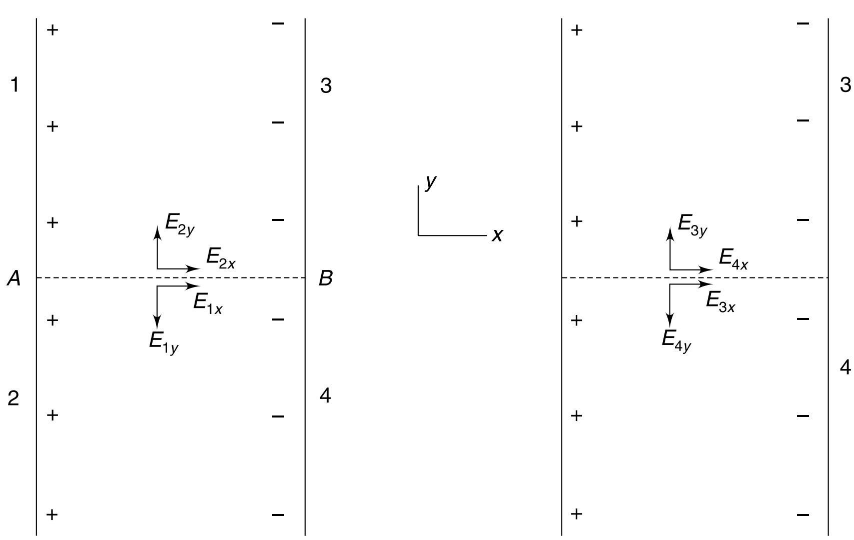
and $$E_{YI} + E_{YII} = 0 \quad \Rightarrow \quad E_{YI} = E_{YII} = 0$$

Hence, field at A due to part I has X component only.

114. At p

$$E_0 = \frac{\lambda}{2\pi\epsilon_0 r} + \frac{\lambda}{2\pi\epsilon_0 r} = \frac{\lambda}{\pi\epsilon_0 \frac{d}{2}} \quad \left[\because \ r = \frac{d}{2}\right]$$

$$E_0 = \frac{2\lambda}{\pi\epsilon_0 d} \qquad ...(1)$$



Had the threads been very long on both sides of AB, the field at M would have been E_0

$$|E_{1Y}| = |E_{2Y}| = |E_{3Y}| = |E_{4Y}|$$

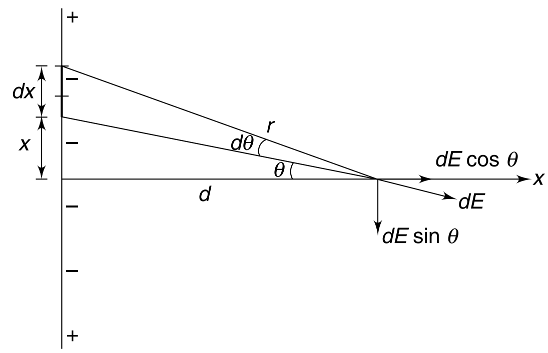
$$|E_{1X}| = |E_{2X}| = |E_{3X}| = |E_{4X}| = \frac{E_0}{4}$$

In the given situation, E_{2Y} and E_{4Y} cancel out and resultant is along AB given by

$$E = E_{2X} + E_{4X} = \frac{E_0}{2}$$

115. ***Note:*** gauss law is not applicable as F is not proportional to $\frac{1}{r^2}$.

Consider an elemental charge as shown in figure.



$$x = d\tan\theta \quad \Rightarrow dx = d\sec^2\theta d\theta$$

$\therefore$ $$dq = \lambda dx = \lambda d\sec^2\theta d\theta$$

Field $$dE = \frac{Kdq}{r^3}$$

Due to symmetry field is along X. Hence resultant field is

$$E = \int dE\cos\theta = \int \frac{Kdq}{r^3}\cdot\frac{d}{r} = K\lambda d^2\int\frac{\sec^2\theta d\theta}{r^4} = K\lambda d^2\int\frac{\sec^2\theta d\theta}{(d\sec\theta)^4}$$

$$= \frac{K\lambda}{d^2}\int\cos^2\theta d\theta$$

For infinite line charge, limit charges from $\theta = -\frac{\pi}{2}$ to $\theta = \frac{\pi}{2}$

$\therefore$ $$E = \frac{K\lambda}{d^2}\int_{-\pi/2}^{\pi/2}\cos^2\theta d\theta = \frac{K\lambda}{2d^2}\int_{-\pi/2}^{\pi/2}(1+\cos 2\theta)d\theta = \frac{K\lambda\pi}{2d^2}$$

116. Consider an element of angular width $d\phi$ as shown in figure.

Charge on the element $dq = \lambda R d\phi$

Distance of the element from point $P = 2R\cos\theta$

Field at P due to the element is $dE = K\frac{dq}{r^2} = K\frac{\lambda R d\phi}{(2R\cos\theta)^2}$

Due to symmetry, field will be along X and we need to add $dE\cos\theta$ for all elements on the ring.

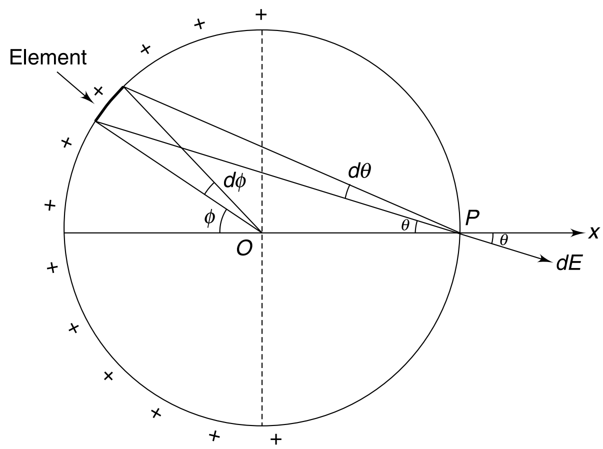
$\therefore$ $$\text{Field} = \int dE\cos\theta = \frac{K\lambda}{4R}\int\sec\theta d\phi$$

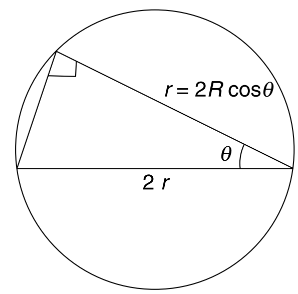
$$= \frac{K\lambda}{2R}\int\sec\theta d\theta \quad [\because \phi = 2\theta,\ d\phi = 2d\theta]$$

θ Changes from $-\frac{\pi}{4}$ to $+\frac{\pi}{4}$ or else, we can integrate from $\theta = 0$ to $\theta = \frac{\pi}{4}$ and double the result.

$$\text{Field} = 2\frac{K\lambda}{2R}\int_0^{\pi/4}\sec\theta d\theta = \frac{K\lambda}{R}[\ln\sec\theta + \tan\theta]_0^{\pi/4}$$

$$= \frac{K\lambda}{R}\left[\ln(\sqrt{2}+1) - \ln 1\right] = \frac{K\lambda}{R}\ln\left(\sqrt{2}+1\right) \text{ along } X$$





117. (a) In equilibrium,

$$mg = qE \quad \left[E = \frac{\sigma}{2\epsilon_0}\right] \qquad ...(1)$$

The removed disc is like a point charge = $\sigma\pi r^2$

$\therefore$ Unbalanced force created by removal of disc

$$= \frac{1}{4\pi\epsilon_0}\frac{\sigma\pi r^2}{h^2}\cdot q = \frac{\sigma r^2}{4\epsilon_0 h^2}\cdot q = \frac{mgr^2}{2h^2} \text{ [where we have used (1)\}]}$$

$\therefore$ acceleration $a = g\dfrac{r^2}{2h^2}$

(b) When terminal speed is acquired

$$6\pi\eta x V_0 = \frac{mgr^2}{2h^2}$$

$$V_0 = \frac{mgr^2}{12\pi\eta x h^2}$$

118. Notice that the field is radial.

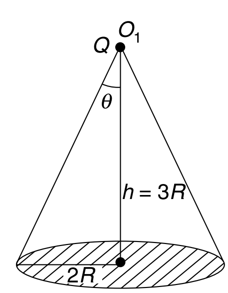
Apply Guass law on a sphere of radius $2r_0$

121. When charge is at O_1 the flux through the face having radius $2R$ is

$$\phi_1 = \frac{q}{\varepsilon_0}\cdot\frac{2\pi(1-\cos\theta)}{4\pi}$$

$$= \frac{q}{2\varepsilon_0}\cdot(1-\cos\theta)$$

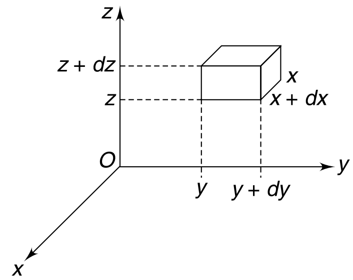
$$= \frac{q}{2\varepsilon_0}\left[1 - \frac{3R}{\sqrt{(2R)^2+(3R)^2}}\right] = \frac{q}{2\varepsilon_0}\left[1-\frac{3}{\sqrt{13}}\right]$$



Similarly, $$\phi_2 = \frac{q}{2\varepsilon_0}\left[1 - \frac{3R}{\sqrt{R^2+(3R)^2}}\right] = \frac{q}{2\varepsilon_0}\left[1-\frac{3}{\sqrt{10}}\right]$$

$$\therefore \qquad \frac{\phi_1}{\phi_2} = \left(\frac{\sqrt{13}-3}{\sqrt{10}-3}\right)\cdot\sqrt{\frac{10}{13}}$$

122. (a) Flux through two faces parallel to $y - z$ plane is



$$= \left[(E_x)_{\text{at}\,(x+dx)} - (E_x)_{\text{at}\,x}\right]\cdot dy\cdot dz$$

$$= [x + dx - x]\,dy\cdot dz = dx\,dy\,dz$$

$$= dV = Vol^m \text{ of cuboid.}$$

Similarly, for faces parallel to xz plane, flux is $= 2\,dV$

And for faces parallel to xy plane, flux is $= 3\,dV$

$$\therefore \qquad d\phi_{\text{cuboid}} = (1 + 2 + 3)\,dV = 6\,dV = 6\,dx\,dy\,dz$$

(b) Charge inside the elemental cuboid

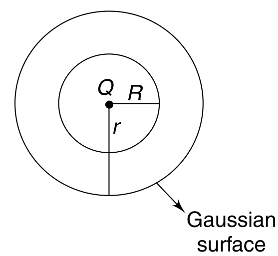
$$dq = 6\,\varepsilon_0 dV$$

Charge per unit volume $\dfrac{dq}{dV} = 6\,\varepsilon_0$

$\therefore$ charge inside the sphere

$$q = \frac{4}{3}\pi r^3\cdot 6\varepsilon_0 = 8\,\varepsilon_0\pi r^3$$

123. The quantity of charge in the space $R < x < r$ is given by



$$q = \int_R^r \rho 4\pi x^2\,dx = 4\pi b\int_R^r \frac{x^2}{x}\,dx$$

$$= 4\pi b\left[\frac{x^2}{2}\right]_R^r = 2\pi b[r^2 - R^2]$$

Let strength of field at a distance r from the centre be E.
Applying Gauss law over spherical surface of radius r we get

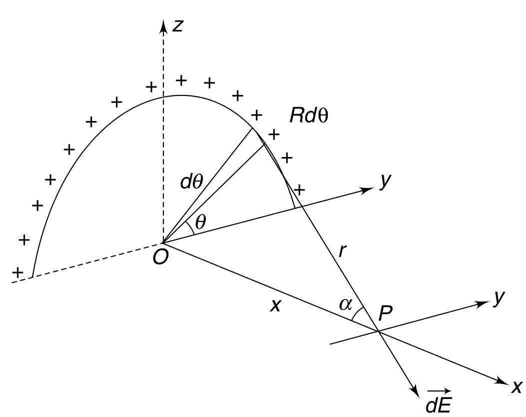
$$E\cdot 4\pi r^2 = \left[Q + 2\pi b[r^2 - R^2]\right]/\varepsilon_0$$

$$E = \left(\frac{Q - 2\pi bR^2}{4\pi\varepsilon_0}\right)\frac{1}{r^2} + \frac{2\pi b}{4\pi\varepsilon_0}$$

For E to not depend on r it is necessary that $Q = 2\pi bR^2$

In this case $\qquad E = \dfrac{2\pi b}{4\pi\varepsilon_0} = \dfrac{b}{2\varepsilon_0}$

124. Consider an element as shown.
Field at P due to this element is



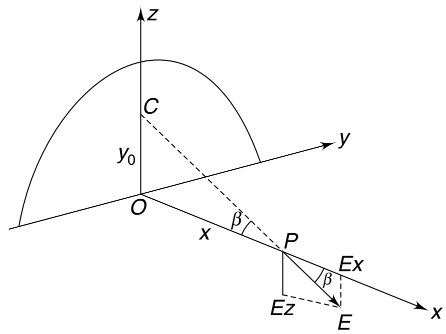
$$dE = K\frac{\lambda R\,d\theta}{r^2} \qquad \left[r = \sqrt{R^2 + x^2}\right]$$

$$dE_x = dE\cos\alpha = \frac{K\lambda R\,d\theta}{r^2}\cdot\frac{x}{r}$$

$$E_x = \int dE_x = \frac{K\lambda R\pi\cdot x}{r^3}$$

$$dE_z = -dE\sin\alpha\cdot\sin\theta = -\frac{K\lambda R}{r^2}\,\frac{R}{r}\,\sin\theta\cdot d\theta$$

$$E_z = -\frac{K\lambda R^2}{r^3}\int_0^{\pi}\sin\theta d\theta = -\frac{2K\lambda R^2}{r^3}$$



From symmetry $E_y = 0$

The resultant field is in xz plane making an angle β with the x axis.

$$\tan\beta = \frac{E_z}{E_x} = \left(\frac{2R}{\pi}\right)\frac{1}{x}$$

Clearly the direction of E is along $\overrightarrow{CP}$

When $OC = y_0 = \frac{2R}{\pi}$

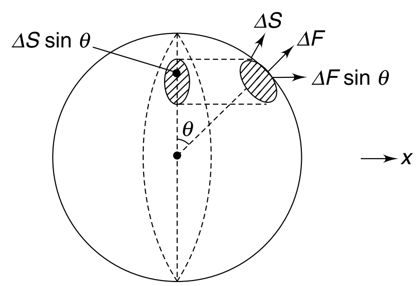
$\therefore$ C is COM of the ring.

126. $P = \frac{\sigma_{max}^{\ 2}}{2\varepsilon_0} = \frac{1}{2}\varepsilon_0 E_0^2 \quad \left[\because \frac{\sigma_{max}}{\varepsilon_0} = E_0\right]$

$$= \frac{1}{2}\times 8.8\times 10^{-12}\times(3\times 10^6)^2 = 39.6\,\text{N/m}^2$$

127. (a) Electrostatic pressure on the surface $P = \frac{\sigma^2}{2\varepsilon_0}$

[charge on a small area $\Delta s = \sigma\Delta s$]



Radial force on the area Δs is $\Delta F = P\Delta s = \frac{\sigma^2\Delta s}{2\varepsilon_0}$.

Due to symmetry, the resultant force on one half is along x direction and is obtained by summing up $\Delta F\sin\theta$

$$\therefore \quad F = \sum\Delta F\sin\theta = \frac{\sigma^2}{2\varepsilon_0}\sum\Delta s\sin\theta$$

$$= \frac{\sigma^2}{2\varepsilon_0}\cdot\pi R^2 = \frac{Q^2}{32\pi\varepsilon_0 R^2} \quad \left[\because \sigma = \frac{Q}{4\pi R^2}\right]$$

Note: Electrostatic force ΔF on a charge $\sigma\Delta s$ is actually force due to repulsion from all the charges on the sphere. It includes interaction with charges present on the half of the sphere on which we are calculating the force. But such internal interaction cancels out in pair and the force F obtained above is the force due to one half of the sphere on the other half.

(b) $$Q_{max} = \sigma_{max}4\pi R^2 = (\varepsilon_0 E_0)4\pi R^2 = 4\pi\varepsilon_0 E_0 R^2$$

$$\therefore \quad F = \frac{\pi\varepsilon_0 E_0^2 R^2}{2}$$

128. Pressure inside a soap bubble of radius r is

$$P_1 = P_0 + \frac{4T}{r}$$

After expansion, the pressure becomes

$$P_2 = \frac{P_1}{8} \quad [\because \text{ volume becomes 8 times and temperature is constant}]$$

$$= \frac{P_0}{8} + \frac{T}{2r}$$

Now, Excess pressure is $\Delta P = P_2 - P_0 = \frac{T}{2r} - \frac{7P_0}{8}$

The electrostatic pressure is $\dfrac{\sigma^2}{2\varepsilon_0}$

$$\therefore \qquad \frac{\sigma^2}{2\varepsilon_0} + \Delta P = \frac{4T}{2r}$$

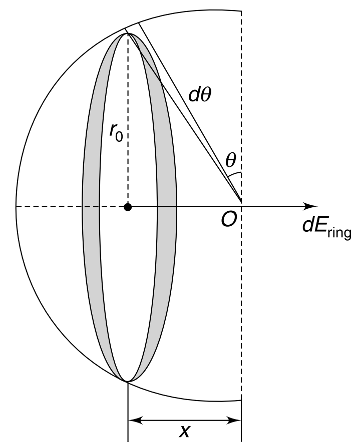
$$\frac{\sigma^2}{2\varepsilon_0} + \frac{T}{2r} - \frac{7P_0}{8} = \frac{4T}{2r}$$

$$\frac{\sigma^2}{\varepsilon_0} = \frac{3T}{r} + \frac{7P_0}{4}$$

$$\sigma = \left[\varepsilon_0\left(\frac{3T}{r} + \frac{7P_0}{4}\right)\right]^{1/2}$$

$$\therefore \qquad Q = \sigma_0 4\pi(2r)^2 = 16\pi r^2\cdot\sigma = 8\pi r^2\left[\varepsilon_0\left(\frac{12T}{r} + 7P_0\right)\right]^{1/2}$$

130. The sphere can be replaced with a point charge (q) at its centre. We can either calculate the force applied by q on each ring element of the hemisphere and add it to get the resultant force or we can calculate electric field produced due to the hemisphere at its centre (by considering similar ring elements). Let's proceed by the second method.



Recall that field due to a charged ring on its axis is given by

$$E = \frac{1}{4\pi\varepsilon_0}\frac{Qx}{(R^2 + x^2)^{3/2}}$$

Consider a ring element on the hemisphere as shown.

$$r_0 = R\cos\theta$$
$$x = R\sin\theta$$

Charge on the ring $= \left(\dfrac{Q}{2\pi R^2}\right)(2\pi r_0)(Rd\theta)$

$= Q\cos\theta d\theta$

$$\therefore \qquad dE_{\text{ring}} = \frac{1}{4\pi\varepsilon_0}\frac{(Q\cdot\cos\theta d\theta)(R\sin\theta)}{[R^2 + R^2\sin^2\theta]^{3/2}} = \frac{Q}{4\pi\varepsilon_0 R^2}\frac{\cos\theta\sin\theta d\theta}{(1 + \sin^2\theta)^{3/2}}$$

$$\therefore \qquad E = \frac{Q}{4\pi\varepsilon_0 R^2}\int_{\theta=0}^{\pi/2}\frac{\cos\theta\sin\theta d\theta}{(1 + \sin^2\theta)^{3/2}}$$

Put $t = 1 + \sin^2\theta$

$$\frac{dt}{d\theta} = 2\sin\theta\cos\theta \Rightarrow \frac{1}{2}dt = \sin\theta\cos\theta d\theta$$

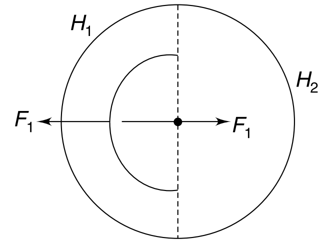
$$\therefore \qquad E = \frac{Q}{8\pi\varepsilon_0 R^2}\left[\frac{dt}{t^{3/2}}\right]_{t=1}^{2}$$

$$E = -\frac{Q}{8\pi\varepsilon_0 R^2}\left[\frac{1}{\sqrt{t}}\right]_1^2 = \frac{Q}{4\pi\varepsilon_0 R^2}\left[1 - \frac{1}{\sqrt{2}}\right]$$

$\therefore$ Required force

$$F = E\cdot q = \frac{Qq}{4\pi\varepsilon_0 R^2}\left(1 - \frac{1}{\sqrt{2}}\right)$$

131. Complete the outer sphere.



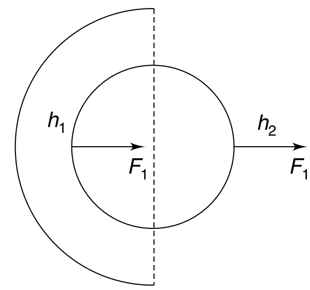
Now this sphere will exert no force on the inner hemisphere.

This means the two hemispheres in the outer sphere exert equal and opposite force on the inner hemisphere.

Call the forces on inner sphere due to two halves H_1 and H_2 of outer sphere as F_1 and F_2.

$$F_1 = F_2 = F \text{ (say)}$$

Now complete the inner sphere with outer sphere as hemispherical



Now, force between one half of inner sphere (h1) and the outer sphere is very much like that between H1 and inner hemisphere in previous drawing.

Force between h_2 and outer hemisphere is like F_2 in previous drawing.

$\therefore$ Resultant force on inner ball due to outer hemisphere is $F_1 + F_2 = 2F\ (\rightarrow)$

This force is already known to us from the previous question.

$\therefore$ Force between the outer and the inner hemispheres = half the answer obtained in last question

$$= \frac{Q}{8\pi\varepsilon_0 R^2}\left(1 - \frac{1}{\sqrt{2}}\right)$$

132. $g_{\text{eff}} = g + \dfrac{qE}{m}$

Velocity just before 1st collision $V_0 = \sqrt{2\,g_{\text{eff}} \cdot h}$

Velocity just after 1st collision $V_1 = eV_0$

Velocity just before 2nd collision $= V_1$

Velocity just after 2nd collision $= eV_1 = e^2 V_0$

Hence, Velocity just after n^{th} collision $V_n = e^n V_0 = e^n \sqrt{2\,g_{\text{eff}} \cdot h}$

$\therefore$ Height attained after n^{th} collision $h_n = \dfrac{V_n^2}{2\,g_{\text{eff}}} = \dfrac{e^{2n} \cdot 2\,g_{\text{eff}} \cdot h}{2\,g_{\text{eff}}} = e^{2n} \cdot h$

133. As done in last Q., the vertical velocity after n^{th} collision

$$V_{yn} = e^n V_0 = e^n \sqrt{2gh}$$

The time of flight between n^{th} and $(n+1)^{\text{th}}$ collision is

$$T = \frac{2V_{yn}}{g} = \frac{2e^n\sqrt{2gh}}{g} = 2e^n\sqrt{\frac{2h}{g}}$$

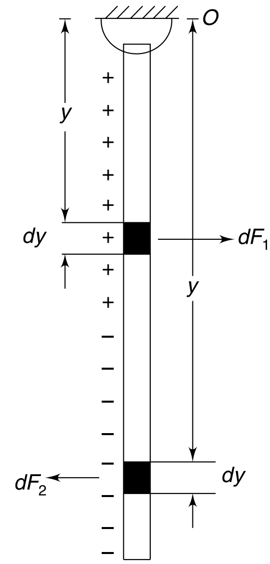
The total time that has passed from start upto n^{th} collision is

$$T_n = \sqrt{\frac{2h}{g}} + 2\cdot\frac{e^1\sqrt{2gh}}{g} + 2\cdot\frac{e^2\sqrt{2gh}}{g} + 2\cdot\frac{e^3\sqrt{2gh}}{g} + \ldots\ldots + 2\cdot\frac{e^{n-1}\sqrt{2gh}}{g}$$

$$= \sqrt{\frac{2h}{g}} + 2\sqrt{\frac{2h}{g}}\left[e + e^2 + e^3 + \ldots\ldots + e^{n-1}\right] = \sqrt{\frac{2h}{g}} + 2\sqrt{\frac{2h}{g}}\left[e\cdot\left(\frac{1-e^{n-1}}{1-e}\right)\right]$$

$\therefore$ Horizontal velocity at the moment of n^{th} collision is

$$V_{xn} = a_x T_n = \frac{qE}{m}\cdot\sqrt{\frac{2h}{g}}\left[1 + 2e\left(\frac{1-e^{n-1}}{1-e}\right)\right]$$



134. (a) From $y = 0$ to $y = \frac{L}{2}$ the rod has positive charge. Force on it (F_1) is towards right due to Electric field.

The lower half has negative charge and electric force (F_2) on it is towards left.

The torque due to F_1 and F_2 must balance.

$$\tau_{F1} = \tau_{F2}$$

$$\int_0^{\frac{L}{2}} E(ay^2 dy) y = \int_{\frac{L}{2}}^{L} E(by^n dy)\, y$$

$$\frac{aL^4}{2^6} = \frac{b}{n+2}\left(\frac{2^{n+2}-1}{2^{n+2}}\right) L^{n+2}$$

Comparing the powers of L we get $n = 2$.

$$\therefore \qquad \frac{a}{2^6} = \frac{b}{4}\cdot\left(\frac{2^4-1}{2^4}\right) \Rightarrow b = \frac{a}{15}$$

(b)
$$F_1 = Ea\cdot\int_0^{L/2} y^2 dy = \frac{EaL^3}{24}$$

and
$$F_2 = Eb\cdot\int_{L/2}^{L} y^2 dy = \frac{Ea}{15}\,\frac{7L^3}{24} = \frac{7}{15}\,\frac{EaL^3}{24}$$

$\therefore$ Net horizontal force

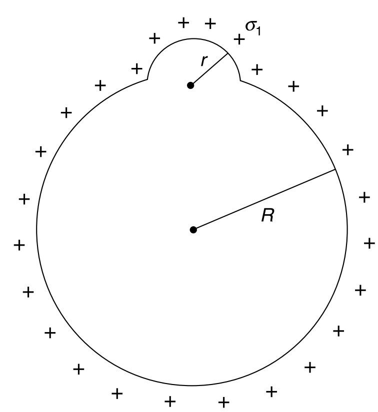
$$F_x = F_1 - F_2 = \frac{EaL^3}{45}(\rightarrow) = \frac{45mg}{45} = mg(\rightarrow)$$

Weight of the rod $= mg(\downarrow)$

$\therefore$ Force by hinge
$$F_x = mg(\leftarrow)$$
$$F_y = mg\,(\uparrow)$$
$$F_{\text{hinge}} = \sqrt{F_x^2 + F_y^2} = \sqrt{2}\,mg$$

136. The bulge will acquire a small charge. Therefore, charge on remaining sphere can be assumed to be $\simeq Q$.

Charge density will be inversely proportional to the radius of curvature of the surface.



$$\therefore \qquad \sigma_1 r = \sigma_2 R$$

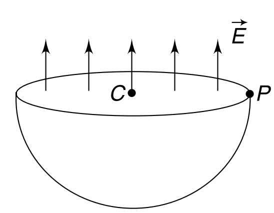
$$\sigma_1\cdot r = \frac{Q}{4\pi R^2}\cdot R$$

$$\sigma_1 = \frac{Q}{4\pi R r}$$

$$\therefore \qquad q = \sigma_1\cdot 2\pi r^2 = \frac{Q}{4\pi R^2}\cdot 2\pi r^2 = \frac{Qr}{2R}$$

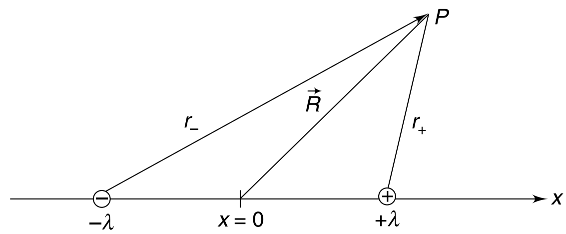
137. Potential at centre

$$V_C = \frac{KQ}{R} = \frac{1}{4\pi\varepsilon_0}\,\frac{2\pi R^2\cdot\sigma}{R} = \frac{\sigma R}{2\varepsilon_0}$$



Potential at point P and A will also be same because the electric field at all points of the circular base is perpendicular to the circular surface. When one moves from C to P, he is travelling on an equipotential surface.

138. Electric field at a distance r from an infinite line charge of charge density λ is



$$E = \frac{\lambda}{2\pi\varepsilon_0 r}$$

Taking potential to be zero at $r = r_0$, we have

$$\int_0^V dV = -\int_{r_0}^{r} E\,dr$$

$$V = -\frac{\lambda}{2\pi\varepsilon_0}\int_{r_0}^{r}\frac{dr}{r}$$

$$V = -\frac{\lambda}{2\pi\varepsilon_0}\ln r + \frac{\lambda}{2\pi\varepsilon_0}\ln r_0$$

$$V = -\frac{\lambda}{2\pi\varepsilon_0}\ln r + C$$

If the line charge has negative charge density the constant will be $-C$ [$\because$ it depends on λ]

Now consider a point P as shown. The position vector of point P is $\vec{R} = x\hat{i} + y\hat{j} + z\hat{k}$

$$\therefore \quad V_P = V_+ + V_-$$

$$= \left[-\frac{\lambda}{2\pi\varepsilon_0}\ln(r_+) + C\right] + \left[\frac{\lambda}{2\pi\varepsilon_0}\ln(r_-) - C\right]$$

$$= \frac{\lambda}{2\pi\varepsilon_0}\ln\left(\frac{r_-}{r_+}\right) = \frac{\lambda}{2\pi\varepsilon_0}\ln\left[\sqrt{\frac{(x+a)^2+y^2}{(x-a)^2+y^2}}\right] = \frac{\lambda}{4\pi\varepsilon_0}\ln\left[\sqrt{\frac{(x+a)^2+y^2}{(x-a)^2+y^2}}\right]$$

For $\quad V_P = \dfrac{\lambda\ln(2)}{4\pi\varepsilon_0}$

$$(x+a)^2 + y^2 = 2(x-a)^2 + 2y^2 \quad \Rightarrow x^2 + y^2 - 6ax + a^2 = 0$$

$$\Rightarrow \quad (x-3a)^2 + y^2 - 8a^2 = 0$$

$$(x-3a)^2 + y^2 = 8a^2$$

Radius of this circle is $2\sqrt{2}\,a$

139. Potential at centre $\quad V_C = V_Q + V_{\text{in}} = K\dfrac{Q}{r_0} + 0 = \dfrac{KQ}{r_0}$

Potential at centre = potential at all the points of the shell

$$\therefore \quad V_P = \frac{KQ}{r_0}$$

$$V_{\text{in}}^P + V_Q^P = \frac{KQ}{r_0} \quad \therefore \quad V_{\text{in}}^P = \frac{KQ}{r_0} - \frac{KQ}{r}$$

140. Potential at A = potential at O

$$V_A = V_0 \qquad [\because \text{ sphere is conducting}]$$

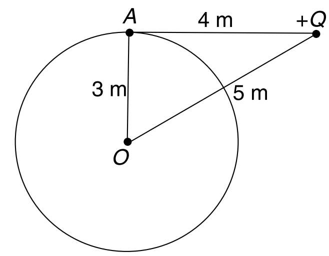
$$V_{AE} + V_{AQ} = V_{OE} + V_{OQ}$$

$$(V_{AE} - V_{OE}) + V_{AQ} = V_{OQ}$$

$$ER\cos\theta_0 + V_{AQ} = \frac{1}{4\pi\varepsilon_0}\frac{Q}{R}$$

$$V_{AQ} = \frac{1}{4\pi\varepsilon_0}\frac{Q}{R} - \frac{ER}{2}$$

141. For sphere S1



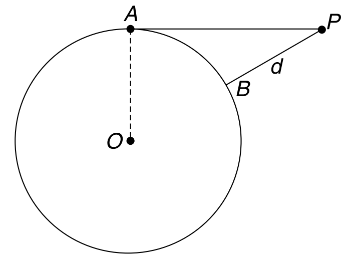
Induced charge will produce zero potential at the centre O.

$\therefore$ potential at $O = \dfrac{KQ}{5} = 9 \times 10^9 \times \dfrac{2 \times 10^{-6}}{5} = 3600$ V

Because it is a conducting sphere, potential at all points in the sphere will be same. Potential at a point B on the surface of the sphere will be

$$V_B = V_Q + V_{\text{inducded}}$$

$$\therefore \quad V_{\text{induced}} = 3600 - \frac{9 \times 10^9 \times 2 \times 10^{-6}}{d}$$



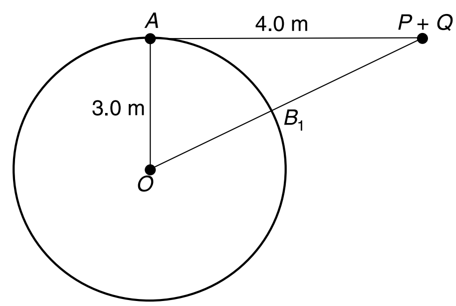
Where d is distance of point B from P.

Potential due to induced charge will be minimum at a point where d is minimum. Such point will lie on straight line OP

Potential due to induced charge will be least at point B_1

$$d = B_1P = 2.0 \text{ m}$$

$$\therefore \quad (V_{\text{induced}}) \text{ at } B_1 = 3600 - \frac{18 \times 10^3}{2} = 5400 \text{ V}$$



Because charge distribution of surface of S_2 is identical to that on S_1, hence 5400 V is least potential on the surface of S_2.

142. The inner faces must have equal and opposite charge. Let it be $\pm x$

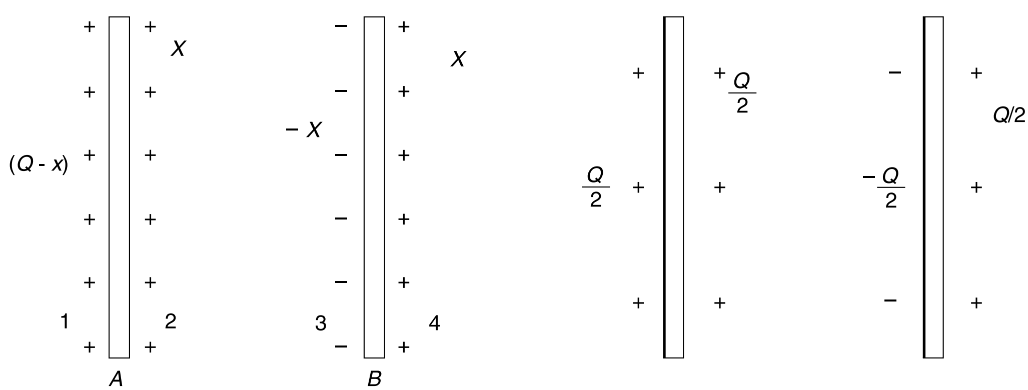
Field inside any of the plate (say B) must be zero.

$$\frac{\sigma_1}{2\epsilon_0} + \frac{\sigma_2}{2\epsilon_0} - \frac{\sigma_3}{2\epsilon_0} - \frac{\sigma_4}{2\epsilon_0} = 0$$

$$Q - x + x - x - x = 0$$

$$\therefore \quad x = \frac{Q}{2}$$

Note: In general if charge Q_1 and Q_2 is given to two plates, charge on both outer surface is same equal to $\dfrac{Q_1 + Q_2}{2}$ and charge on inner faces is $\pm\left(\dfrac{Q_1 - Q_2}{2}\right)$



(ii) The inner faces facing each other must have equal and opposite charge. And the outer walls of A and B must have equal charge to ensure zero field inside the conductor.

Hence, the charge distribution can be as shown in figure below.

$$2y - (3Q - x) - x = Q$$

$$2y = 4Q \quad \Rightarrow y = 2Q \qquad \text{...(i)}$$

$\because \qquad V_A = V_B \qquad \therefore \quad V_C - V_A = V_C - V_B$

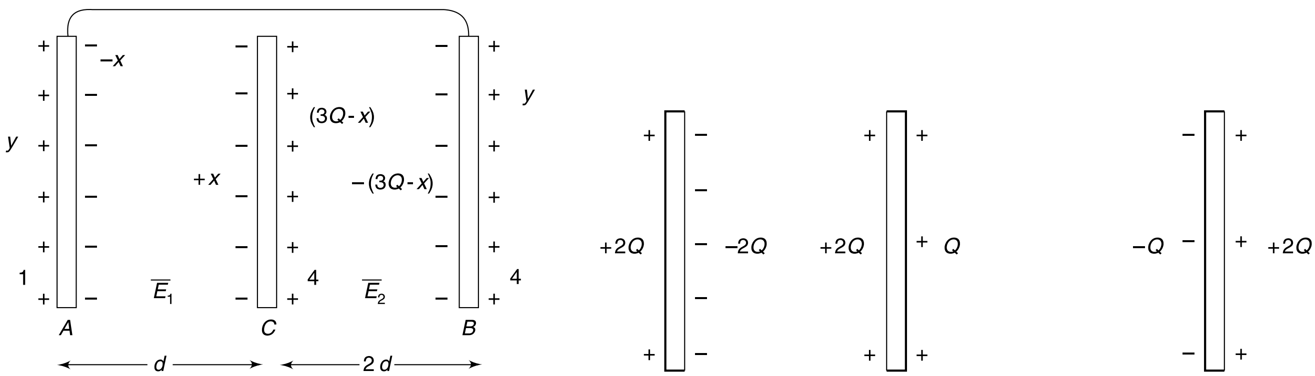
$$E_1 d = E_2(2d)$$

$$\frac{x}{A \cdot \in_0} d = \frac{(3Q - x)2d}{A \cdot \in_0}$$

$$3x = 6Q \Rightarrow x = 2Q$$

From (1) $\qquad y = 2Q$

$\therefore$



(i) When A is grounded, the charge on outer walls of A and B will become zero (so as to make their potentials zero)

For $\qquad V_A = V_B$

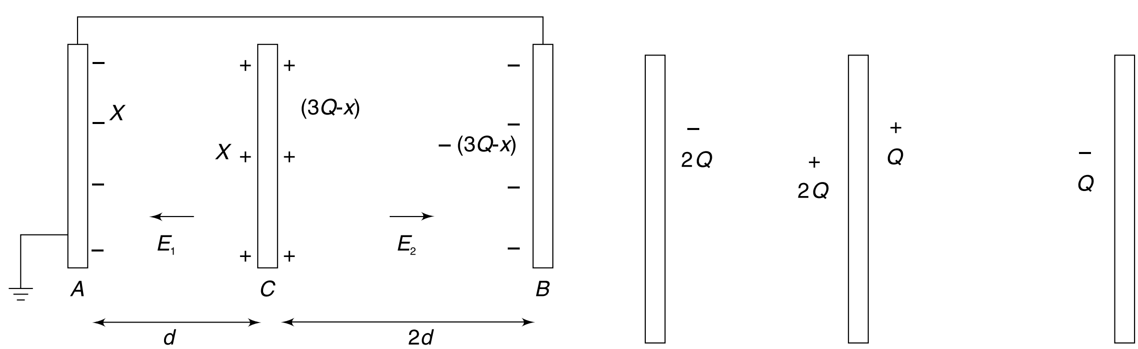
$$E_1 d = E_2 2d$$

$$E_1 = 2E_2$$

$\therefore \qquad x = 2(3Q - x)$

$\therefore \qquad 3x = 6Q$

$$x = 2Q$$



143. Charge density $\rho = \dfrac{Q}{\dfrac{4}{3}\pi(8R^3 - R^3)} = \dfrac{3Q}{28\pi R^3}$

Charge in the region $R < r < x$ $(x < 2R)$ is

$$q = \rho \frac{4}{3}\pi (x^3 - R^3) = \frac{Q}{7R^3}(x^3 - R^3)$$

$\therefore$ Electric field at distance x from the centre is

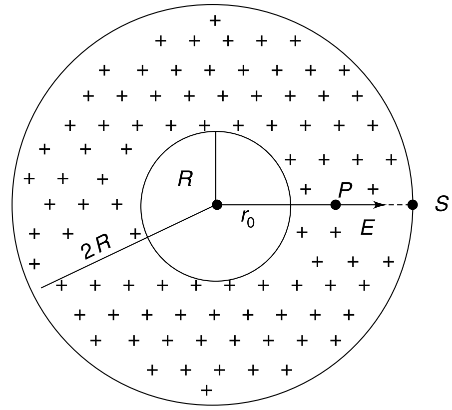
$$E = \frac{Kq}{x^2} = \frac{KQ}{7R^3}\left(x - \frac{R^3}{x^2}\right)$$

$$\int_{V_P}^{V_S} dV = -\int_{r_0}^{2R} E\,dx$$

$$V_S - V_P = -\frac{KQ}{7R^3}\left[\int_{r_0}^{2R} x\,dx - R^3\int_{r_0}^{2R}\frac{dx}{x^2}\right]$$

$$\frac{KQ}{2R} - V_P = -\frac{KQ}{7R^3}\left[2R^2 - \frac{r_0^2}{2} + R^3\left(\frac{1}{2R} - \frac{1}{r_0}\right)\right]$$

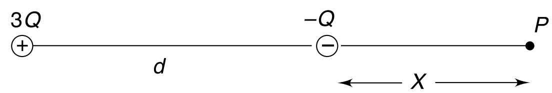
$$V_P = \frac{6KQ}{7R} - \frac{KQ}{7R^3}\left[\frac{r_0^2}{2} - \frac{R^3}{r_0}\right]$$



144. (a) yes. Spherical

(b) At p electric field is zero.

$$\therefore \quad K\frac{Q}{x^2} = K\frac{3Q}{(d+x)^2} \quad \Rightarrow \quad \frac{d+x}{x} = \sqrt{3} \quad \Rightarrow x = \frac{d}{\sqrt{3}-1}$$



(c) Potential of surface S_1 = potential at P

$$V = -\frac{KQ}{x} + K\frac{3Q}{d+x} = \frac{2KQ}{d}(2-\sqrt{3})$$

(d) One can easily show that potential is zero at a point at a distance $\frac{d}{2}$ to the right of $-Q$ charge. Now, $\frac{d}{2} < \frac{d}{\sqrt{3}-1}$.

Hence, this point is to the left of point P. Looking at the given diagram, we can say that zero potential surface will enclose $-Q$ charge only.

$$\therefore \quad \text{Required flux} = -\frac{Q}{\epsilon_0}$$

145. *Hint:* Number of field lines originating at A = no. of lines originating at C = 2 times number of lines terminating at B

$\therefore$ If $\quad q_B = -q$

Then $\quad q_A = q_C = 2q$

For graphs see the answer

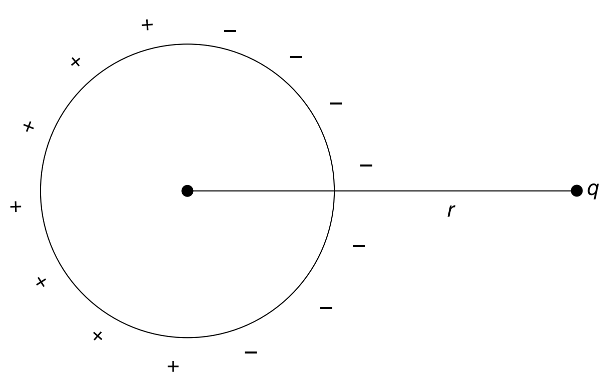
146. Solution for (f)

If a charge q is placed in front of a conducting sphere, the induced charges plus the charge q creates a potential

$V = \frac{Kq}{r}$ at all points of the sphere.

In our case, the point charge $-q$ at the centre of the shell induces $+q$ charge on the inner surface and the charge induced on the outer surface will be just as shown in fig (with $r = 4R$). The system of charge shown in fig above will cause a potential

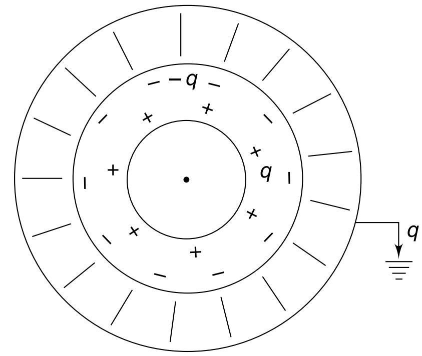
$V = \frac{Kq}{4R}$ at all points $0 \le x \le 2R$.



147. Two sphere of equal size will share charge till both of them acquire same potential. Let charge on inner sphere be q.

A charge $-q$ is induced on the inner surface of the shell.

Potential of the inner sphere = potential of distant sphere



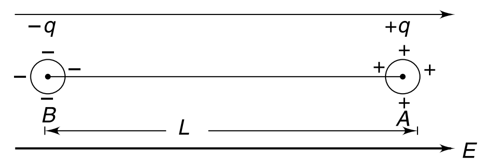
$$= K\frac{q}{R} + K\frac{-q}{2R} = K\frac{Q-q}{R} \quad \Rightarrow \quad q = 2Q - 2q$$

$$\Rightarrow \quad q = \frac{2Q}{3}$$

$\therefore$ charge induced on the inner surface of the shell $= -q = \dfrac{-2Q}{3}$

$\therefore$ charge that flows to earth $= +q = \dfrac{2Q}{3}$

148. (a) Let ball A acquire charge $+q$ and B acquire charge $-q$ after the field is switched on



$$V_B = V_0 - \frac{Kq}{r} + \frac{Kq}{L}$$

$$V_A = V_0 - EL + \frac{Kq}{r} - \frac{Kq}{L}$$

Since balls are connected by conducting spring.

(V_0 = potential of point B in electric field E if our system of balls were not there)

$$V_A = V_B$$

$$\Rightarrow \quad -EL + \frac{Kq}{r} - \frac{Kq}{L} = -\frac{Kq}{r} + \frac{Kq}{L}$$

$$\Rightarrow \quad 2Kq\left[\frac{1}{r} - \frac{1}{L}\right] = EL \quad \Rightarrow \quad \frac{q}{2\pi\epsilon_0}\frac{1}{r} = EL \quad \left[\because \frac{1}{r} >> \frac{1}{L}\right]$$

$$\therefore \quad q = 2\pi\epsilon_0 ErL$$

(b) The electric force on the two balls stretches the spring. As the balls move out, induced charge (q) on them increases (as it is directly proportional to separation L between them)

This increases the electric force stretching the balls. The system can oscillate only if spring force increases at a much faster pace. When separation between the balls increase by x, charge on them change by

$$\Delta q = 2\pi\epsilon_0 rE\cdot x$$

Electric force increases by

$$\Delta qE = 2\pi\epsilon_0 rE^2\cdot x$$

For oscillation

$$kx > 2\pi\epsilon_0 rE^2\cdot x$$

$$k > 2\pi\epsilon_0 rE^2$$

For no oscillation $k < 2\pi\epsilon_0 rE^2$

$$\therefore \quad k_0 = 2\pi\epsilon_0 rE^2$$

(c) Consider the motion of ball A. Let it be in equilibrium when length of spring is l_0

$$k(l_0 - L) = 2\pi\epsilon_0 l_0 rE^2 \qquad ...(1)$$

Consider A to be further displace by x (in the same time B also move by x and spring length increases by $2x$)

$$ma = 2\pi\epsilon_0 rE^2(l_0 + 2x) - k(l_0 + 2x - L)$$

Using (1)

$$ma = -x[2k - 4\pi\epsilon_0 rE^2]$$

$$a = -\left(\frac{2k - 4\pi\epsilon_0 rE^2}{m}\right)x$$

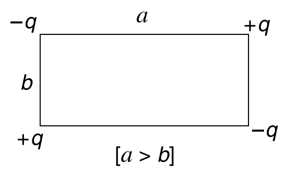
$$= -\left(\frac{8\pi\epsilon_0 rE^2 - 4\pi\epsilon_0 rE^2}{m}\right)x \qquad [\because\ k = 2k_0]$$

$$= -\left(\frac{4\pi\epsilon_0 rE^2}{m}\right)x \qquad [SHM]$$

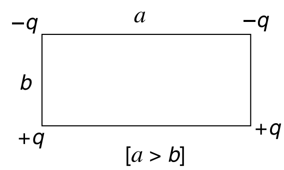
$$\therefore \qquad \omega = \sqrt{\frac{4\pi\epsilon_0 rE^2}{m}}$$

$$T = 2\pi\sqrt{\frac{m}{4\pi\epsilon_0 rE^2}} = \left(\sqrt{\frac{\pi m}{\epsilon_0 \cdot r}}\right)\cdot\frac{1}{E}$$

149. (a) There are $^4C_2 = 6$ terms in expression of potential energy and two of them are positive; remaining 4 being negative. The arrangement shown in Fig. (a) and (b) cannot result in zero potential energy because the negative term is less then positive term.

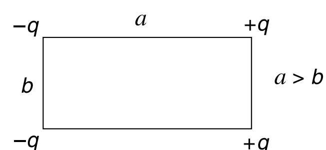


(a)



(b)

An arrangement of kind shown in Fig. (c) can result in zero potential energy



(c)

(b) $$U = K\left[\frac{(-q)(-q)}{b} + \frac{q\cdot q}{b} + \frac{(-q)(q)}{a} + \frac{(-q)(q)}{a} + \frac{(-q)(q)}{\sqrt{a^2+b^2}} + \frac{(-q)(q)}{\sqrt{a^2+b^2}}\right]$$

$$\therefore \quad \frac{2}{b} - \frac{2}{a} - \frac{2}{\sqrt{a^2+b^2}} = 0$$

With $b = 1.0$ m

$$\frac{a-1}{a} = \frac{1}{\sqrt{1+a^2}}$$

$$[a^2 + 1 - 2a][a^2 + 1] = a^2$$

$$a^4 + a^2 - 2a^3 + a^2 + 1 - 2a = a^2$$

$$a^3(a-2) + (a-1)^2 = 0$$

The second term is positive.

First term can be negative only if $a < 2$

$$\therefore \qquad 1 < a < 2$$

150. Let speed of both be V just before collision

Energy Conservation

$$\frac{1}{2}mV^2 \times 2 + \frac{K(3q)(-q)}{2r} = \frac{K(3q)(-q)}{R}$$

$$\therefore \qquad mV^2 = 3Kq^2\left[\frac{1}{2r}-\frac{1}{R}\right] \qquad \text{...(i)}$$

In elastic collision, exchange of velocity will take place and charge on both balls will now be $\frac{3q-q}{2} = q$. Now, the two balls repel each other. Speed will be maximum when separation between the centers of the balls become $2R$.

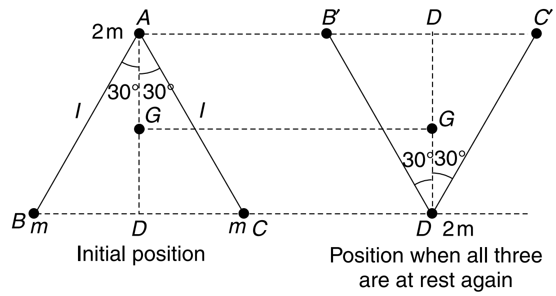
$$\frac{1}{2}mV_0^2 \times 2 + \frac{Kq^2}{2R} = \frac{1}{2}mV^2 \times 2 + \frac{Kq^2}{2r}$$

$$mV_0^2 = 3Kq^2\left[\frac{1}{2r}-\frac{1}{R}\right] + \frac{Kq^2}{2}\left[\frac{1}{r}-\frac{1}{R}\right]$$

$$= \frac{Kq^2}{2rR}[3R - 6r + R - r] = \frac{Kq^2}{2rR}(4R - 7r) \quad \therefore\ V_0 = \sqrt{\frac{Kq^2}{2mrR}(4R-7r)}$$

Note: Energy loss takes place when charge redistribution takes place during collision.

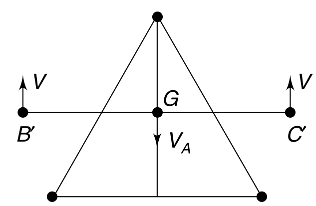
151. (a) In original position, the COM of the system is located at centre of line AD (i.e. at a distance of $AG = \frac{\sqrt{3}}{4}l$ from A). The particles move such that the COM does not get displaced. When all three come to rest again, particle of mass 2 m will be at D [This particle experiences net force along AD and moves on this line itself] and the other two will be symmetrically placed at B' and C'.



$\therefore$ A and D are extreme positions of oscillation of particle of mass 2 m.

$$\therefore \quad \text{Amplitude} = \frac{AD}{2} = \frac{\sqrt{3}}{4}l$$

(b) Speed of the particle of mass $2m$ will be maximum when all three fall in a straight line passing through COM(G)



Momentum conservation $MV_A = mV + mV$

$$2mV_A = 2mV \quad \Rightarrow \quad V_A = V$$

Energy Conservation

$$2.\ K\frac{Qq}{l} + K\frac{q\cdot q}{2l} + \frac{1}{2}mV^2 + \frac{1}{2}mV^2 + \frac{1}{2}(2m)V^2 = K\frac{qq}{l} + 2\cdot\frac{KQq}{l}$$

$$\Rightarrow \qquad 2mV^2 = \frac{Kq^2}{2l} \quad \Rightarrow V = \sqrt{\frac{Kq^2}{4ml}} = \frac{q}{4}\sqrt{\frac{1}{\pi\in_0 ml}} \qquad \text{...(i)}$$

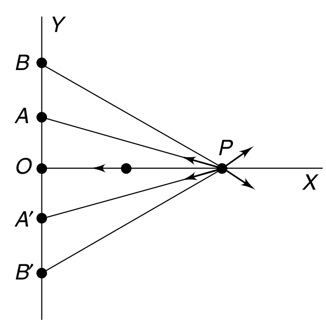
(c) Particle of mass m may be viewed to be rotating about particle of mass M with a velocity of $2V$ (for the moment, when all three are in straight line, the central particle is un accelerated. Frame attached to it is inertial)

$$\therefore \qquad \frac{m(2V)^2}{l} = T - \frac{Kq^2}{(2l)^2} - \frac{KqQ}{l^2}$$

$$\therefore \qquad T = \frac{Kq^2}{l^2} + \frac{Kq^2}{4l^2} + \frac{KqQ}{l^2} \quad \text{[using (i)]}$$

$$= \frac{q}{16\pi\varepsilon_0 l^2}(5q + 4Q)$$

152. $\vec{E}_{P(AA')}$ = Electric field at P due to charges at A and A' $= 2\left(\frac{1}{4\pi\varepsilon_0}\right)\frac{q_A x_P}{(y_A^2 + x_P^2)^{3/2}}(-\hat{i})$



Similarly, $$\vec{E}_{P(BB')} = 2\left(\frac{1}{4\pi\varepsilon_0}\right)\frac{q_B x_P}{(y_B^2 + x_P^2)^{3/2}}(\hat{i})$$

For $$\vec{E}_P = 0$$

$$2\left(\frac{1}{4\pi\varepsilon_0}\right)\frac{q_A x_P}{(y_A^2 + x_P^2)^{3/2}} = 2\left(\frac{1}{4\pi\varepsilon_0}\right)\frac{q_B x_P}{(y_B^2 + x_P^2)^{3/2}} \quad \Rightarrow \quad x = \sqrt{2.5}\ \text{m}$$

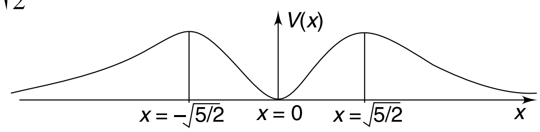
To the left of point P the field is towards left and to the right of point P it is towards right. Hence, v_0 should be just enough to enable the particle to reach P.

[Alternately, $$V(x) = \frac{2}{4\pi\varepsilon_0}\left(\frac{q_B}{(y_B^2 + x^2)^{1/2}} + \frac{q_A}{(y_A^2 + x^2)^{1/2}}\right)$$

Field will be zero at point where

$$\frac{dV(x)}{dx} = 0 \quad \Rightarrow \quad x = 0 \text{ and } x = \pm\sqrt{\frac{5}{2}}$$

V at x equal to infinity is zero.



And at $x = \sqrt{\frac{5}{2}}$, $V(x)$ is positive $\Rightarrow V(x)$ is maximum at $x = \sqrt{\frac{5}{2}}$

So, if the particle is able to cross $x = \sqrt{\frac{5}{2}}$ then it would automatically reach the origin.]

$$\Rightarrow \quad \frac{1}{2}mv_0^2 = \frac{2q}{4\pi\varepsilon_0}\left[\frac{q_A}{\sqrt{y_A^2 + x_P^2}} + \frac{q_B}{\sqrt{y_B^2 + x_P^2}}\right]$$

Solving we get, $v_0 = 3$ m/s

Kinetic energy at origin = loss of $PE = (PE)_P - (PE)_{\text{origin}} = 3 \times 10^{-4}$ J.

153. *Hint:* Apply conservation of energy. Remember there is gravitational PE apart from the electrostatic PE.

154. (a) Electric field due to the line charge at a distance r is

$$E = \frac{\lambda}{2\pi\varepsilon_0 r}$$

Potential difference between two points at distance ηR and R is

$$\Delta V = -\int_R^{\eta R} E\,dr = -\frac{\lambda}{2\pi\varepsilon_0}\int_R^{\eta R}\frac{dr}{r} = -\frac{\lambda}{2\pi\varepsilon_0}\ln\eta$$

Gain in KE of charge = loss in its electrostatic PE

$$\Rightarrow \quad \frac{1}{2}mV^2 - \frac{1}{2}mV_0^2 = \frac{\lambda Q}{2\pi\varepsilon_0}\ln\eta \quad \Rightarrow \quad V = \sqrt{V_0^2 + \frac{\lambda Q}{\pi\varepsilon_0 m}\ln\eta}$$

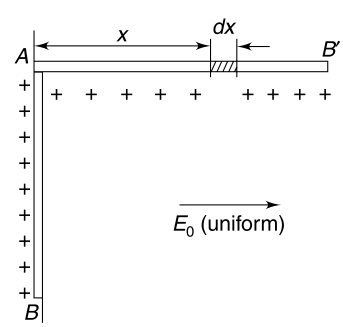
(b) If $V_\perp$ is the velocity component $\perp_r$ to radius, angular momentum conservation about the line charge says

$$mV_\perp \cdot \eta R = mV_0 R \quad \therefore \quad V_\perp = \frac{V_0}{\eta}$$

$$V_\perp^2 + V_r^2 = V^2$$

$$V_r^2 = V^2 - V_\perp^2 = V_0^2 + \frac{\lambda Q}{\pi\varepsilon_0 m}\ln\eta - \frac{V_0^2}{\eta^2}$$

$$\therefore \quad V_r = \sqrt{V_0^2\left(1 - \frac{1}{\eta^2}\right) + \frac{\lambda Q}{\pi\varepsilon_0 m}\ln\eta}$$



155. The torque of electric force on the rod is initially larger than the gravitational force torque. This provides an angular acceleration to the rod and it speeds up. Beyond a certain point, the gravitational force's torque becomes dominant and slows it down to eventually bring the rod at rest when it becomes horizontal.

(a) E_0 can be found by applying law of conservation of energy.

Let potential at line AB be V_0

Electrostatic potential energy of the charge on the rod when it is vertical is $U_i = \lambda L \cdot V_0$

When rod becomes horizontal, we can write its Electrostatic potential energy as

$$U_f = \int_0^L V_x \lambda dx \ [V_x = \text{potential at distance } x \text{ from end } A = V_0 - E_0 x]$$

$$= \int_0^L \lambda(V_0 - E_0 x) dx = \lambda V_0 L - \lambda E_0 \frac{L^2}{2}$$

Loss in Electrostatic potential energy

$$U_i - U_f = \lambda E_0 \frac{L^2}{2}$$

This is equal to gain in gravitational potential energy

$$\lambda E_0 \frac{L^2}{2} = \text{Mg}\frac{L}{2} \quad \Rightarrow E_0 = \frac{\text{Mg}}{\lambda L}$$

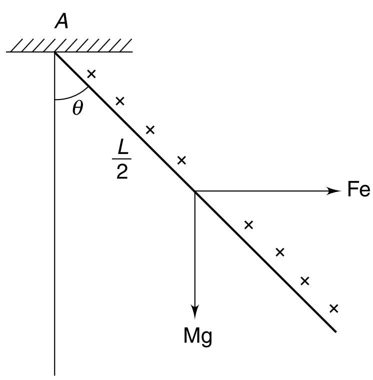
(b) The electric force on the rod is $F_e = \lambda L E_0$. Force can be considered at centre for writing the torque. In equilibrium,

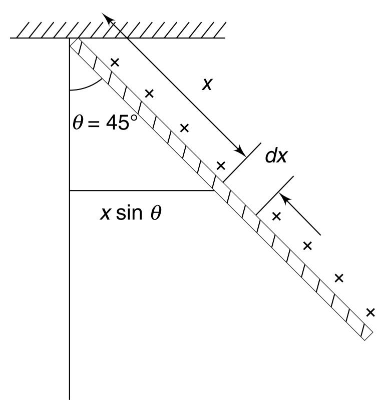
$$\tau_{\text{Fe}} = \tau_{\text{mg}} \text{ [torque about } A]$$

$$\lambda L E_0 \frac{L}{2} \cos\theta = \text{Mg}\frac{L}{2} \sin\theta$$

Putting $E_0 = \dfrac{\text{Mg}}{\lambda L}$ we get $\theta = 45°$

The rod has maximum KE at this position.





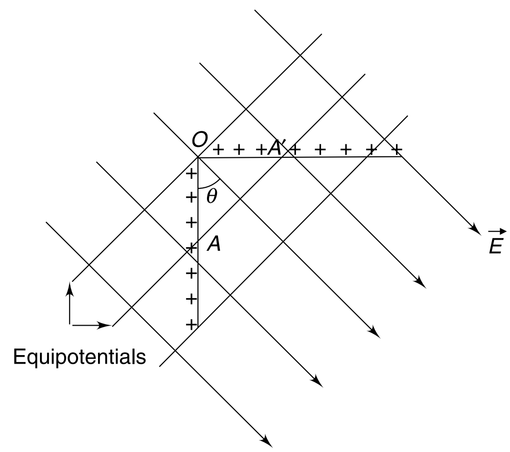
KE = (loss in Electrostatic PE) – (gain in gravitational PE)

$$= \int_0^L E_0 x \sin\theta \lambda dx - \text{Mg}\frac{L}{2}(1 - \cos\theta)$$

$$= \frac{E_0 \lambda}{\sqrt{2}} \frac{L^2}{2} - \text{Mg}\frac{L}{2}\left(1 - \frac{1}{\sqrt{2}}\right) = \frac{\text{Mg}L}{2\sqrt{2}} - \text{Mg}\ \frac{L}{2}\left(1 - \frac{1}{\sqrt{2}}\right)$$

$$\therefore \qquad \frac{1}{2}\left(\frac{ML^2}{3}\right)\omega^2 = MgL\left(\frac{\sqrt{2} - 1}{2}\right) \quad \Rightarrow \omega = \sqrt{3(\sqrt{2} - 1)\frac{g}{L}}$$

156. A small charge on the rod located at A will reach A' when the rod gets horizontal.



If $\theta < 45°$ then A' will lie on an equipotential having higher potential as compared to point A.

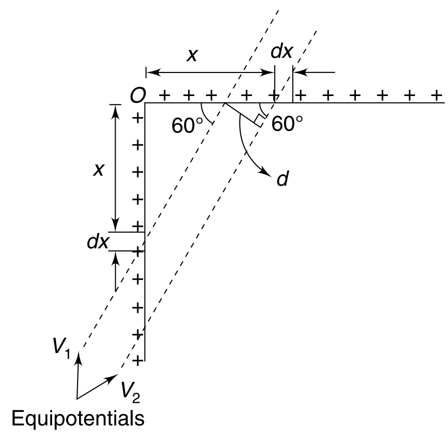
Therefore, the electrostatic potential energy of rod in horizontal position will be more than that in its vertical position. At the same time the gravitational potential energy also increases. This is not possible.

It means rod cannot become horizontal if $\theta = 45°$, however high the field might be.

$\therefore$ Ans. to (a) is that it is not possible for rod to become horizontal.

Ans. to (c) is $\theta_0 = 45°$

(b) With $\theta = 60°$, let's calculate the loss in electrostatic PE when the rod becomes horizontal.



A charge element (λdx) at distance x from O will be on two different equipotentials when the rod is vertical and when it is horizontal. Distance between the equipotentials will be

$$d = \left(x - \frac{x}{\sqrt{3}}\right)\cdot \sin 60° = \left(\frac{\sqrt{3}-1}{2}\right)x$$

$\therefore$ loss in PE of the elemental charge will be

$$dU = \lambda dx Ed = \left(\frac{\sqrt{3}-1}{2}\right)\lambda E x\, dx$$

$\therefore$ Total loss in electrostatic PE of the rod

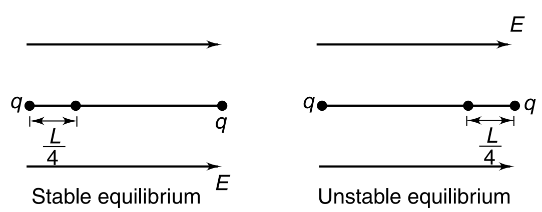
$$\Delta U = \left(\frac{\sqrt{3}-1}{2}\right)\lambda E \int_0^L x\,dx = \left(\frac{\sqrt{3}-1}{4}\right)\lambda E L^2$$

This must be equal to gain in gravitational PE of the rod.

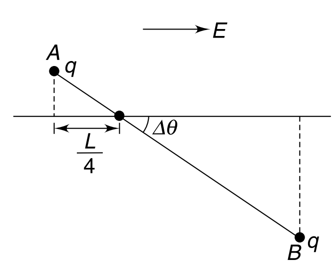
$$\therefore \qquad Mg\frac{L}{2} = \left(\frac{\sqrt{3}-1}{4}\right)\lambda E L^2$$

$$E = \left(\frac{2}{\sqrt{3}-1}\right)\frac{Mg}{\lambda L}$$

157. (a)



(b) When the rod is rotated by $\Delta\theta$, charge at the end A moves $\frac{L}{4}(1 - \cos\Delta\theta)$ in the direction of field and the charge at end B moves $\frac{3L}{4}(1 - \cos\Delta\theta)$ opposite to the field.



Charge at B gains PE (as it moves to a location of higher potential) and the charge at A loses PE.

$$\therefore \qquad \Delta U = q\cdot E\,\frac{3L}{4}(1-\cos\Delta\theta) - q\cdot E\,\frac{L}{4}(1-\cos\Delta\theta)$$

$$= E\,\frac{qL}{4}[3 - 3\cos\Delta\theta - 1 + \cos\Delta\theta]$$

$$\Delta U = \frac{EqL}{4}[2 - 2\cos\Delta\theta] = \frac{EqL}{2}(1 - \cos\Delta\theta) \qquad ...(1)$$

(c) Let the angular speed of the rod be ω when it is at small angular displacement θ with respect to its stable position.

$$KE + PE = a \text{ constant}$$

Taking PE to be zero in position of stable equilibrium

We can write PE at position θ using (1)

$$U = \frac{EqL}{2}(1 - \cos\theta)$$

$$KE = \frac{1}{2}I\omega^2 = \frac{1}{2}\left(\frac{mL^2}{16} + \frac{9mL^2}{16}\right)\omega^2 = \frac{5mL^2}{16}\omega^2$$

$$\therefore \quad \frac{5mL^2}{16}\omega^2 + \frac{EqL}{2}(1 - \cos\theta) = a \text{ constant.}$$

$$\frac{5mL^2}{16}2\omega\frac{d\omega}{dt} + \frac{EqL}{2}\sin\theta\frac{d\theta}{dt} = 0 \qquad \left[\frac{d\theta}{dt} = \omega \text{ and } \frac{d\omega}{dt} = \alpha\right]$$

$$\therefore \quad \alpha = -\left(\frac{4}{5}\frac{Eq}{mL}\right)\theta \qquad [\sin\theta \simeq \theta]$$

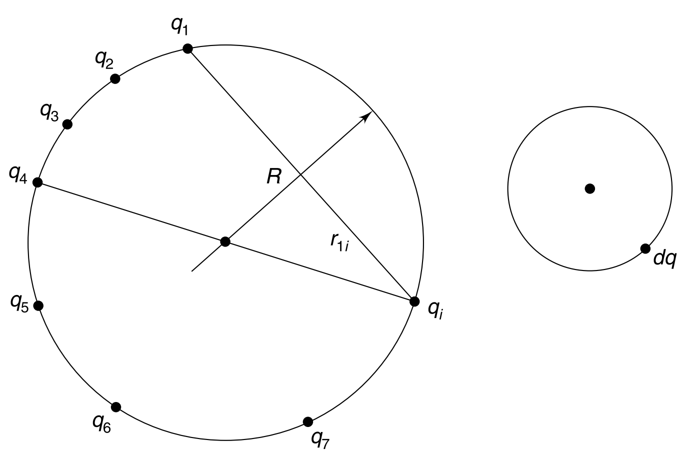
$$\therefore \quad T = 2\pi\sqrt{\frac{5mL}{4Eq}} = \pi\sqrt{\frac{5mL}{Eq}}$$

159. (a) Just remove a small point charge q_i from the shell. This will hardly change the potential of the shell which is equal to

$$V = \frac{1}{4\pi\varepsilon_0}\frac{Q}{R}$$

$$= \frac{1}{4\pi\varepsilon_0}\left[\frac{q_1}{r_{1i}} + \frac{q_2}{r_{2i}} + \frac{q_3}{r_{3i}} + \frac{q_4}{r_{4i}} + ...\right] \; [q_i \text{ is not there}]$$

= potential at the location of q_i due to all other charges



Interaction energy of q_i with all other charges

$$= \frac{1}{4\pi\epsilon_0}\left[\frac{q_1}{r_{1i}} + \frac{q_2}{r_{2i}} + \frac{q_3}{\gamma_{3i}} + ...\right]q_i$$

If we continue like this interaction of each pair will be counted twice. Hence, the interaction energy shall be written as

$$U = \left[\frac{1}{4\pi\varepsilon_0} \sum_{\substack{i=1,2,3...\\ j=1,2,3...}} \frac{q_i q_j}{r_{ij}}\right] \times \frac{1}{2}$$

The summation will be performed using integration.

Interaction energy of dq with all other charges.

$$= \left[\frac{1}{4\pi\varepsilon_0}\frac{Q}{R}\right] dq$$

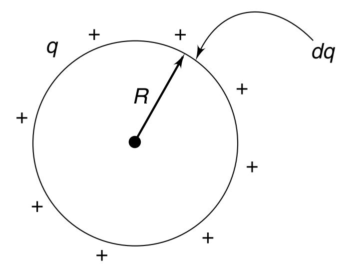
Term inside the bracket is the potential at location of dq due to all other charges

$$\therefore \qquad U = \int_0^Q \frac{1}{4\pi\varepsilon_0}\frac{Q}{R}\, dq \times \frac{1}{2}$$

$$= \frac{Q}{8\pi\varepsilon_0 R}\int_0^Q dq = \frac{Q^2}{8\pi\varepsilon_0 R}$$

(b) Consider that charge 'q' has already been gathered. Potential of the shell is

$$V = \frac{1}{4\pi\varepsilon_0}\frac{q}{R}$$



Work done in bringing next installment of a small charge dq from infinity

$$dW = dq \cdot V = \frac{1}{4\pi\varepsilon_0}\frac{q}{R}\, dq$$

$$\therefore \qquad W = \int dW = \frac{1}{4\pi\varepsilon_0 R}\int_0^Q q\, dq = \frac{Q^2}{8\pi\varepsilon_0 R}$$

The two methods calculate the same thing – self energy of a uniformly charged shell.

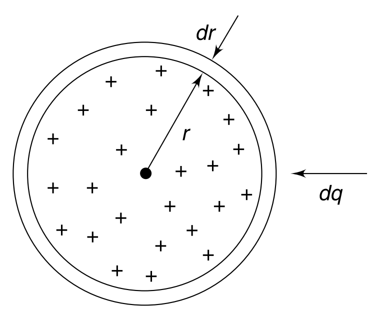
160. (a) Imagine that we assemble the sphere by piling up a succession of thin spherical layers of charge of infinitesimal thickness. At each stage, we gather a small amount of charge and put it in a thin layer from r to $r + dr$. We continue till we arrive at a final radius R.

Let q = charge on sphere of radius r

$$= \rho \cdot \frac{4}{3}\pi r^3$$

dq = charge on layer having thickness dr

$$= \rho \cdot 4\pi r^2 dr$$



Work done in bringing charge dq from ∞ to the sphere of radius r is

$$dU = \frac{1}{4\pi\varepsilon_0}\frac{q}{r}\cdot dq = \frac{4\pi\rho^2 r^4 dr}{3\varepsilon_0}$$

Total energy required to construct sphere of radius R is

$$U = \frac{4\pi\rho^2}{3\varepsilon_0}\int_0^R r^4\, dr = \frac{4\pi\rho^2}{15\varepsilon_0}R^5$$

Put $\qquad \rho = \dfrac{Q}{\frac{4}{3}\pi R^3}$

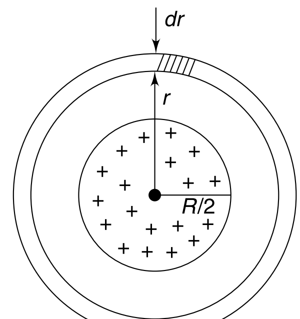
$$U = \frac{3}{5}\frac{Q^2}{4\pi\varepsilon_0 R} \quad ...(A)$$

(b) The desired interaction energy is

$$U^1 = U - U_{qQ}$$

When U = interaction energy of all pairs possible inside the sphere $= \frac{3}{5}\frac{Q^2}{4\pi\varepsilon_0 R}$

U_{qQ} = interaction energy of the sphere of radius $\frac{R}{2}$ (having charge 'q') with the charge in the annular part (i.e., Q_1, Q_2)....



$$q = \frac{4}{3}\pi\left(\frac{R}{2}\right)^3 \cdot \rho$$

Potential due to this charge at radius $r(\geq R/2)$

$$V_r = \frac{1}{4\pi\varepsilon_0}\frac{q}{r} = \frac{R^3\rho}{24\varepsilon_0}\cdot\frac{1}{r}$$

Energy of charge in the layer of thickness dr, in the electrostatic field of q is

$$dU_{qQ} = (\rho 4\pi r^2 dr)\cdot V_r = \frac{\pi R^3 \rho^2}{6\varepsilon_0} r\,dr$$

$$\therefore \quad U_{qQ} = \frac{\pi R^3\rho^2}{6\varepsilon_0}\int_{R/2}^{R} r\,dr = \frac{\pi R^5 r^2}{16\varepsilon_0}$$

$$= \frac{9}{256\pi\varepsilon_0}\frac{Q^2}{R} \quad \left[\because \quad \rho = \frac{Q}{\frac{4}{3}\pi R^3}\right]$$

$$\therefore \quad U^1 = U - U_{qQ}$$

$$= \frac{Q^2}{4\pi\varepsilon_0 R}\left[\frac{3}{5} - \frac{9}{64}\right] = \frac{147}{320}\frac{Q^2}{4\pi\varepsilon_0 R}$$

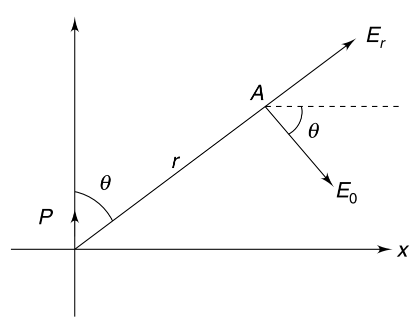
162. We know that field at point A has components (radial and tangential) given by

$$E_r = \frac{K2P\cos\theta}{r^3}; \quad E_\theta = \frac{KP\sin\theta}{r^3}$$

$$\therefore \quad E_x = E_r\sin\theta + E_\theta\cos\theta = K\frac{3P\sin\theta\,\cos\theta}{r^3}$$

$$E_y = E_r\sin\theta - E_\theta\cos\theta = \frac{KP}{r^3}(2\cos^2\theta - \sin^2\theta)$$

$$= \frac{KP}{r^3}(3\cos^2\theta - 1)$$



(a) For a given $r = r_0$, E_x will be maximum when $\theta = 45°$

$$E_{x\max} = \frac{K3P}{r_0^3}\frac{1}{\sqrt{2}}\cdot\frac{1}{\sqrt{2}} = \frac{3}{2}\frac{KP}{r_0^3}$$

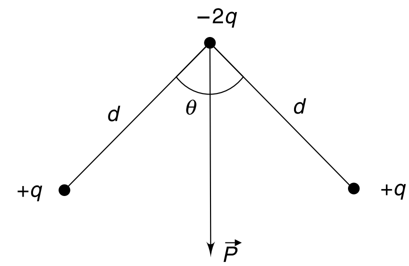
(c) Field is parallel to x axis if $E_y = 0 \quad \Rightarrow 3\cos^2\theta - 1 = 0$

$$\cos\theta = \frac{1}{\sqrt{3}}$$

E_y will be zero at all points on the line defined by $\cos\theta = \frac{1}{\sqrt{3}}$

Slope of line $= \tan(90 - \theta) = \cot\theta = \frac{1}{\sqrt{2}}$.

163. The resultant dipole moment of the molecule is

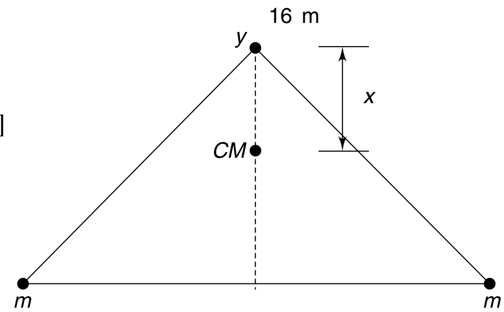


$$P = 2q \cdot d \cdot \cos\left(\frac{\theta}{2}\right) = dq \quad [\because \quad \theta = 120°]$$

Position of COM is as shown.

$$x = \frac{2}{9}d \quad \cos\left(\frac{\theta}{2}\right) = \frac{d}{9}$$

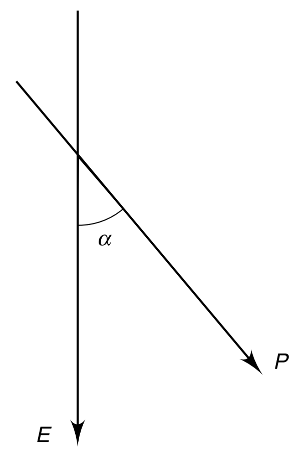
Moment of inertia



$$I_{cm} = I_y - 18m \cdot x^2$$

$$= 2md^2 - 18m \cdot \frac{d^2}{81} = \frac{16}{9}md^2$$

Torque when dipole is at an angle α to the field is



$$\tau = PE\sin\alpha \simeq PE\alpha \text{ [for small } \alpha]$$

$$I_{cm} \cdot \frac{d^2\alpha}{dt^2} = -PE \cdot \alpha$$

$$\frac{d^2\alpha}{dt^2} = -\frac{1}{16/9}\frac{qdE}{md^2}\alpha = -\frac{9}{16}\frac{qE}{md} \cdot \alpha$$

$$\omega = \sqrt{\frac{9qE}{16md}} = \frac{3}{4}\sqrt{\frac{qE}{md}}$$

$$\frac{2\pi}{T} = \frac{3}{4}\sqrt{\frac{qE}{md}}$$

$$T = \frac{8\pi}{3}\sqrt{\frac{md}{qE}}$$

164. Electric field at distance x from the centre is

$$E = KQ\frac{x}{(a^2 + x^3)^{3/2}}$$

Potential energy of dipole placed in this electric field is

$$U = -PE\cos 0^0 = -KQP\frac{x}{(a^2 + x^2)^{3/2}}$$

Force on the dipole is

$$F = -\frac{dU}{dx} = KQP\left[\frac{(a^2 + x^2)^{3/2} - \frac{3}{2}x(a^2 + x^2)^{1/2}(2x)}{(a^2 + x^2)^3}\right]$$

$$= KQP\left[\frac{a^2 + x^2 - 3x^2}{(a^2 + x^2)^{5/2}}\right] = KQP\frac{a^2 - 2x^2}{(a^2 + x^2)^{5/2}}$$

For $x = \frac{a}{2}$

$$F = \frac{16}{5^{3/2}}\frac{KQP}{a^3}$$

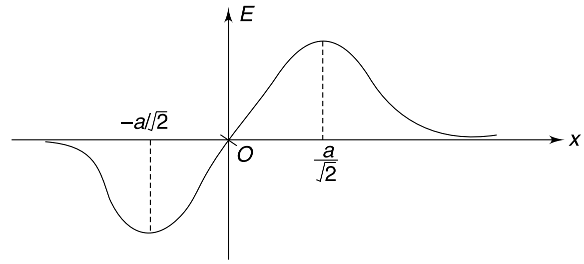
For $x = a$

$$F = -\frac{KQP}{2^{5/2}a^3}$$

Force on the dipole basically depends on $\frac{dE}{dx}$.

(ii) No force is not zero at $x = 0$, because $\dfrac{dE}{dx} \neq 0$

$F = 0$ at $x = \dfrac{a}{\sqrt{2}}$ where $\dfrac{dE}{dx} = 0$



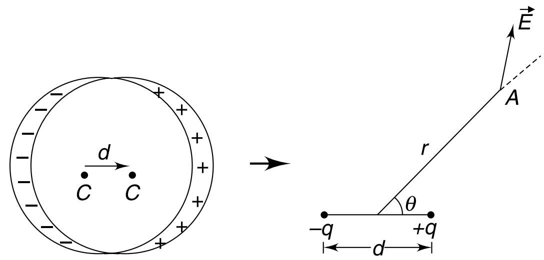
165. For writing field at an outside point, we can consider the charge on each sphere to be located at respective centre.Thus we have a dipole of dipole moment

$$\vec{p} = q\vec{d}$$

Where $$q = \rho \cdot \frac{4}{3}\pi R^3$$

Field at A

$$E = \frac{\rho}{4\pi\varepsilon_0 r^3}\sqrt{3\cos^2\theta + 1} = \frac{\rho}{3\varepsilon_0}\frac{R^3 d}{r^3}\sqrt{3\cos^2\theta + 1}$$

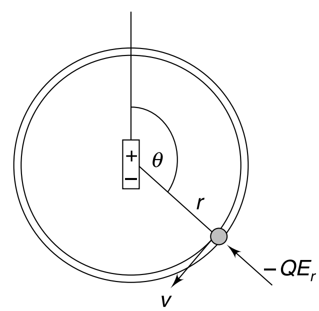


166. (a) Let us apply the law of conservation of energy for the bead having mass m and charge Q:

$$\frac{1}{2}mv^2 + QK\frac{P\cos\theta}{r^2} = \frac{1}{2}mv_0^2 + \frac{QKP\cos(\pi/2)}{r^2} = 0.$$

We can then express the velocity of the bead at angle θ as

$$v = \sqrt{-2\frac{QKP\cos\theta}{mr^2}},\ \left(\frac{\pi}{2} \le \theta \le \pi\right). \qquad ...(1)$$



(b) The circular motion needs a radial force equal to mv^2/r.

The radial component of electric field due to the dipole $E_r = 2\,\dfrac{KP\cos\theta}{r^3}$.

Using the expression of the velocity (equation 1), we notice that the radial force on the bead $(= QE_r)$ is just equal to $-\,mv^2/r$, the required centripetal force. Thus the ring need not exert any force on the bead to sustain circular motion.

(a) The bead would move along a circular path until it reached the point opposite its starting position. The bead would stop there, and then go back to its starting point. It will repeatedly retrace its path executing a periodic motion.

(b) Since the ring does not apply a force on the bead, its absence will make no difference to the motion.

CHAPTER 7

Capacitor

LEVEL 1

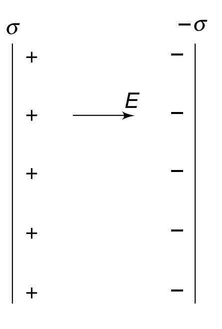
Q. 1: Two students A and B were taught that electric field near a uniformly charged large surface is normal to the surface and is equal to $\frac{\sigma}{2\epsilon_0}$. They were also told that field near the surface of a conductor is $\frac{\sigma}{\epsilon_0}$ normal to the conductor where σ is charge density on the conductor surface. Now both of them were asked to write field between the plates of an ideal parallel plate capacitor having charge density σ and $-\sigma$ on its plates. Student A said that field can be seen as superposition of field due to two large charged surfaces. He wrote the answer as

$$E = \frac{\sigma}{2\epsilon_0} + \frac{\sigma}{2\epsilon_0} = \frac{\sigma}{\epsilon_0}$$

Student B thought that a capacitor has conducting plates and therefore field due to each plate

Must be $\frac{\sigma}{\epsilon_0}$. He wrote his answer as $E = \frac{\sigma}{\epsilon_0} + \frac{\sigma}{\epsilon_0} = \frac{2\sigma}{\epsilon_0}$

Who is wrong and where is the flaw in thinking?



Q. 2: A large parallel plate capacitor has vertical plates with a potential difference of 2000 V between them. Oil drops are sprayed between the plates. Few drops are observed to move with uniform velocities in directions inclined at 45°, 33.7° and 18.4° to the vertical. The space between the plates has air and mass of each drop is $m = 3.3 \times 10^{-15}$ kg. Separation between the plates is 3 cm.

(a) Explain the observations.

(b) From the above observations estimate the magnitude of smallest charge in nature.

tan (33.7°) = 0.67 and tan (18.4°) = 0.333

Q. 3: Electrical susceptibility (χ) of a dielectric material is defined as $\chi = \epsilon_r - 1$ where ϵ_r is its relative permittivity. An isolated parallel plate capacitor carries some charge and the field in the dielectric present between its plates is E. Express the electric field due to induced charge on dielectric surface in terms of χ and E.

Q. 4: Imagin a parallel plate capacitor with a charge $+Q$ on one plate and $-Q$ on the other. Initially, the plates are almost, but not quite, touching. The plates are gradually pulled apart to make the separation d. The separation d is small compared to dimensions of the plates and we can maintain that field between the plates is uniform. Area of each plate is A.

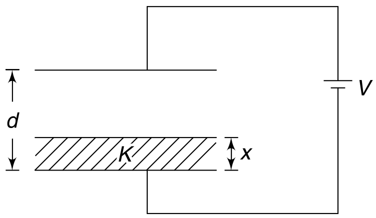
(a) Write the energy stored in the electric field between the plates when separation between then is d.

(b) Assuming that the energy calculated in part (a) can be attributed to the work done by the external agent in pulling the plates apart, calculate the electrostatic attraction force between the plates.

Q. 5: Two conducting plates each having area A are kept at a separation d parallel to each other. The two plates are connected to a battery of emf V. The space between the plates is filled with a liquid of dielectric constant K. The height (x) of the liquid between the plates increases at a uniform rate from zero to d in a time interval t_0.

(a) Write the capacitance of the system as a function of x as the liquid beings to fill the space between the plates.

(b) Write the current through the cell at time t.



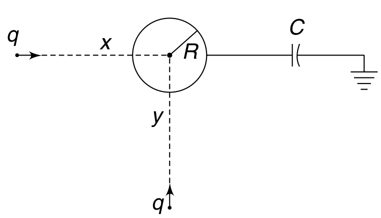
Q. 6: You have been given a parallel plate air capacitor having capacitance C, a battery of emf ε and three dielectric blocks having dielectric constants K_1, K_2 and K_3 such that $K_1 > K_2 > K_3$. Describe a sequence of steps such as connecting or disconnecting the capacitor to the battery, inserting or taking out of one of the dielectrics etc – so that the capacitor ends up having maximum possible energy stored in it. [Each dielectric block fills completely the space between the plates]. Write this maximum energy.

Q. 7: A parallel plate capacitor has two plates of area A separated by a small distance d. The capacitor is charged to a potential difference of V and the battery is disconnected. A metal plate with area A and thickness $\frac{d}{2}$ is fully inserted between the plates, so that it always remains parallel to the plates.

(a) Calculate the work done on the metal slab while it was inserted.

(b) Does the two plates of the capacitor attract of repel the metal plate that is being inserted. Does the answer obtained in part (a) help you in answering this?

Q. 8: A neutral conducting ball of radius R is connected to one plate of a capacitor (Capacitance = C), the other plate of which is grounded. The capacitor is at a large distance from the ball. Two point charges, q each, begin to approach the ball from infinite distance. The two point charges move in mutually perpendicular directions. Calculate the charge on the capacitor when the two point charges are at distance x and y form the centre of the sphere.



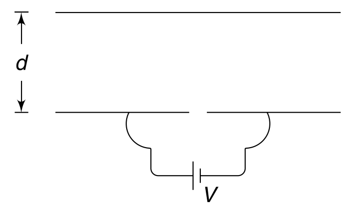
Q. 9: A parallel plate capacitor of capacitance C_0 is charged using a cell of emf V_0. Calculate the work done in reducing the separation between the plates to half its original value if

(a) The cell is disconnected before you start decreasing the plate separation.

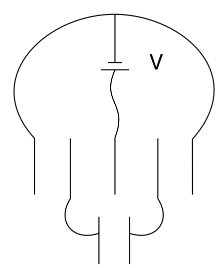
(b) The cell remains connected while you are reducing the separation.

Assume that the plates are moved very slowly.

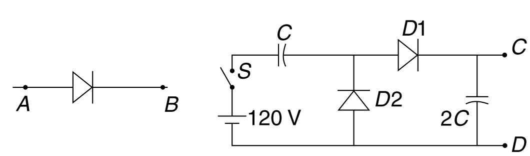
Q. 10: Two large conducting plates, identical in size, are placed parallel to each other at a separation d. Each plate has area A. One of the plates is cut into two equal parts and then a battery of emf V is connected across these two pieces. Find the work done by the battery in supplying charge to the plates.



Q. 11: Seven identical plates, each of area A, are placed as shown. Any two adjacent plates are at separation d. Conducting wires have been used to connect the plates and a cell of emf V volt as shown in the figure. How much charge does the cell supply?

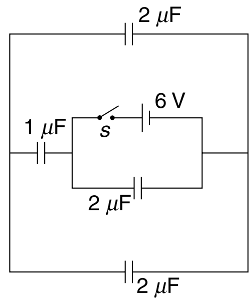


Q. 12: A diode is a device that conducts in one direction only. Figure (a) shows the symbol for a diode. When terminal A is at higher potential than B, the diode conducts; it means current flows from A to B. No current flows if B is kept at higher potential. Find the potential difference between terminals C and D after the switch (S) is closed in the circuit shown in Figure (b).

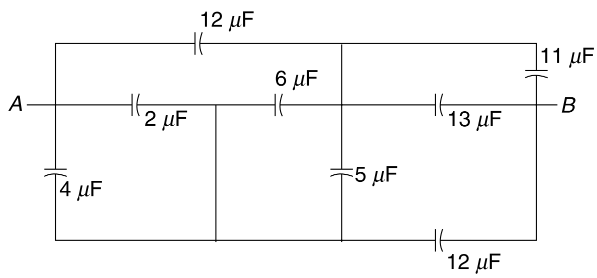


(a) (b)

Q. 13: Find charge supplied by the cell after the switch is closed.



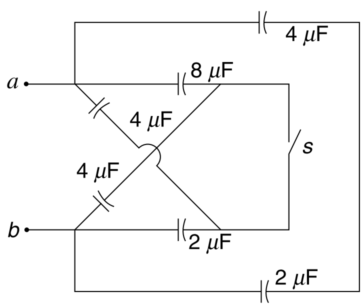
Q. 14: Find the equivalent capacitance across points A and B.



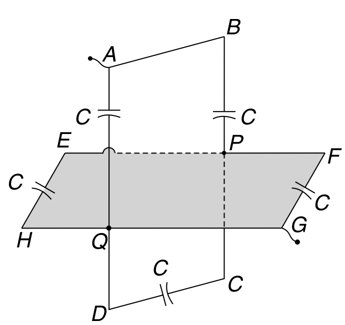
Q. 15: In the circuit shown in the Figure find the equivalent capacitance between points a and b when–

(a) Switch S is open.

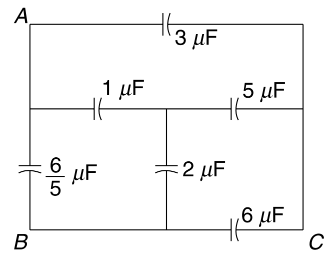
(b) Switch S is closed.



Q. 16: In the circuit shown in the Figure $ABCD$ is a rectangular and vertical frame of conducting wires having three capacitors. $EFGH$ is in horizontal plane having two capacitors. The two rectangular frames are connected at P and Q only.Find equivalent capacitance between A and G if each capacitor has capacitance C.

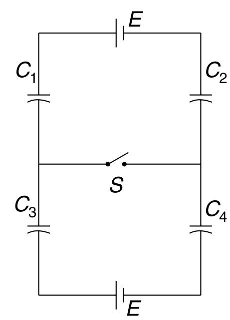


Q. 17: In the circuit shown in the Figure find the ratio of equivalent capacitance between A and B to that between A and C.

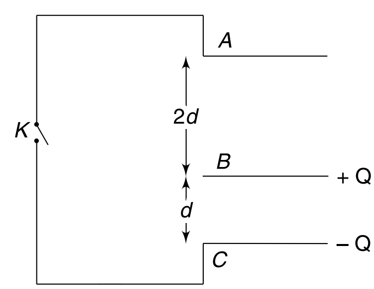


Q. 18: In the circuit shown in Figure $E = 12$ V, $C_1 = 4\ \mu F$, $C_2 = 2\ \mu$F, $C_3 = 6\ \mu$F and $C_4 = 3\ \mu F$.

Find the heat produced in the circuit after switch S is shorted.



Q. 19: Three identical large metal plates each of area S are at distance d and $2d$ from each other as shown. Metal plate A is uncharged, while metal plates B and C have charges $+Q$ and $-Q$ respectively. Metal plates A and C are connected by a conducting wire through a switch K. How much electrostatic energy is lost when the switch is closed?

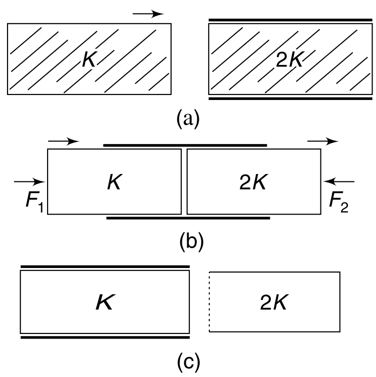


Q. 20: An air core parallel plate capacitor has capacitance C. It is completely filled with a dielectric slab having dielectric constant 2 K. The capacitor is now connected to a battery of emf V. It was planned to replace the dielectric slab of the capacitor while it remains connected to the battery. Another dielectric slab (which fits exactly between the plates) is inserted slowly so as to push out the earlier slab. The new slab has a dielectric constant K [see Figure (a) to (c)]

(a) By energy considerations calculate the mechanical work that must be done against the electric forces in order to complete the process.

(b) Looking at the expression of mechanical work obtained in (a), tell what was the direction of force applied by the external agent - from left to right (as indicated by F_1 in Figure) or from right to left (as indicated by F_2).

(c) Which dielectric slab experienced higher force of attraction from the capacitor plates during the process?

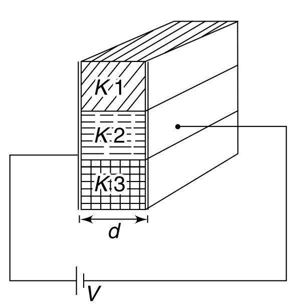


Q. 21: A parallel plate capacitor of plate area A and spacing between the plates d is filled with three dielectrics as shown in the Figure. The dielectric constants of the three dielectrics are $K_1 = K$, $K_2 = 2K$, $K_3 = 3K$. The capacitor is connected to a cell of emf V.

(a) Write the ratio of maximum to minimum charge density on the surface of the capacitor plate.

(b) Calculate the surface charge density of bound (induced) charge on the middle dielectric.

(c) If the three dielectrics occupy equal volume between the plates, calculate the capacitance of the capacitor.



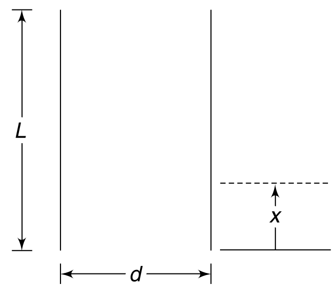
LEVEL 2

Q. 22: A parallel plate capacitor has square plates of side length L. Plates are kept vertical at separation d between them. The space between the plates is filled with a dielectric whose dielectric constant (K) changes with height (x) from the lower edge of the plates as $K = e^{\beta x}$ where β is a positive constant. A potential difference of V is applied across the capacitor plates.

(i) Plot the variation of surface charge density (σ) on the positive plate of the capacitor versus x.

(ii) Plot the variation of electric field between the plates as a function of x.

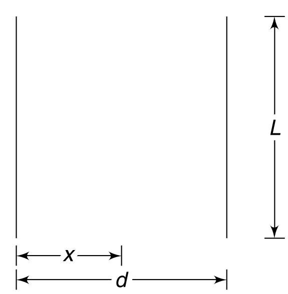
(iii) Calculate the capacitance of the capacitor.



Q. 23: A parallel plate capacitor has square plates of side length L kept at a separation d. The space between them is filled with a dielectric whose dielectric constant changes as $K = e^{\beta x}$ where x is distance measured from the left plate towards the right plate, and β is a positive constant. A potential difference of V volt is applied with left plate positive.

(i) What happens to capacitance of the capacitor if d is increased? What is the smallest possible capacitance that can be obtained by changing d?

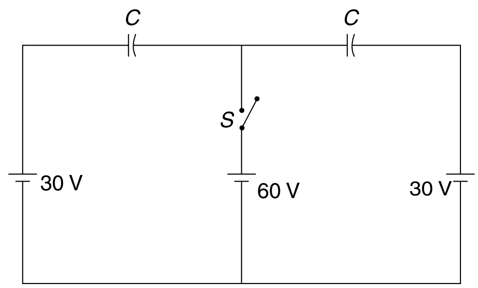
(ii) Write the expression of electric field between the plates as a function of x.



Q. 24: A parallel plate capacitor is to be constructed which can store $q = 10\ \mu$C charge at $V = 1000$ volt. The minimum plate area of the capacitor is required to be A_1 when space between the plates has air. If a dielectric of constant $K = 3$ is used between the plates, the minimum plate area required to make such a capacitor is A_2. The breakdown field for the dielectric is 8 times that of air. Find $\frac{A_1}{A_2}$.

Q. 25: The electric field between the plates of a parallel plate capacitor is E_0. The space between the plates is filled completely with a dielectric. There are n molecules in unit volume of the dielectric and each molecules is like a dumb – bell of length L with its ends carrying charge $+q$ and $-q$. Assume that all molecular dipoles get aligned along the field between the plates. Find the electric field between the plates after insertion of the dielectric.

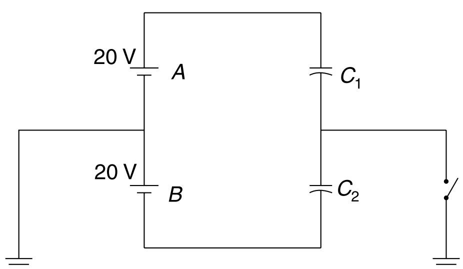
Q. 26: Find heat dissipated in the circuit after switch S in closed. $C = 2\ \mu$F.



LEVEL 2

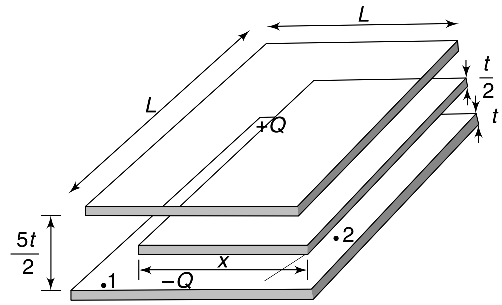
Q. 27: In the circuit shown in the Figure calculate the quantity of charge that flows through the switch after it is closed. Give your answer for following two cases-

(a) $C_1 = C_2 = 2\ \mu$F (b) $C_1 = 2\ \mu$F; $C_2 = 1\ \mu$F



Q. 28: In a parallel plate capacitor the separation between the two plates is maintained by a dielectric of dielectric constant K and thickness d. The dielectric material is not rigid and has a young's modulus of Y. Capacitance of the capacitor is C_0 if applied potential difference is nearly zero. At higher potentials the attractive force between the plates compresses the dielectric (by a small amount) and reduces the gap between the plates. Change in K can be neglected due to compression in the dielectric. Find the change in capacitance when a battery of V volt is connected across it

Q. 29: Two square metal plates have sides of length L and thickness t ($<< L$). They are arranged parallel to each other with their inner faces at a separation of $\frac{5}{2}t$. One of the plates is given a charge $-Q$ and the other one is given a charge $+Q$. A third rectangular metal plate of sides L and x, having thickness $\frac{t}{2}$ is inserted between the plates as shown. The third plate is equidistant from the two plates and parallel to them. Neglect edge effects.



(a) Find the charge density on lower plate at points 1 and 2 shown in Figure.

(b) Find potential difference between the upper plate and the middle plate.

(c) Find electric field between the two outer plates in space where the third plate is not present (i.e., at a point above point 1.)

(d) Find the capacitance of the system across two outer plates.

Q. 30: A hollow spherical conductor of radius R has a charge Q on it. A small dent on the surface decreases the volume of the spherical conductor by 2%. Assume that the charge density on the surface does not change due to the dent and the electric field in the dent region remains same as other points on the surface.

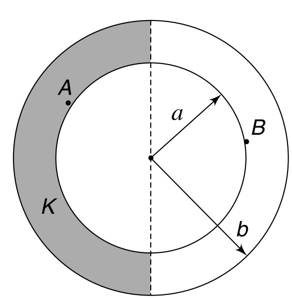
(a) ΔE is the electrostatic energy stored in the electric field in the shallow dent region and E is the total electrostatic energy of the spherical shell. Find the ratio $\frac{\Delta E}{E}$

(b) Using the ratio obtained in part (a) calculate the percentage change in capacitance of the sphere due to the dent.

Q. 31: The space between the conductors of a spherical capacitor is half filled with a dielectric as shown is Figure. The dielectric constant is K.

(a) If a charge is given to the capacitor write the ratio of free charge density on the inner sphere at point A and B.

(b) Write the ratio of capacitance with dielectric and without dielectric.



Q. 32: Two concentric spherical shells have radii a and b ($>a$). Write the capacitance of the system in following cases.

(a) Positive terminal of the battery is connected to the outer shell and its other terminal and the inner shell are grounded.

(b) Positive terminal of the battery is connected to the inner shell and its negative terminal is grounded.

(c) A terminal of the battery is connected to the inner shell and the other terminal along with the outer shell is grounded.

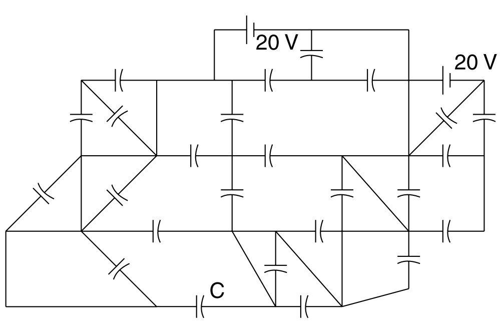
(d) A terminal of the battery is connected to the outer shell and the other terminal is grounded.

Q. 33: Two conducting sphere of radii a and b are placed at separation d. It is given that $d >> a$ and $d >> b$ so that charge distribution on both the sphere remains spherically symmetric. Assume that a charge $+q$ is given to the sphere of radius a and $-q$ is given to the sphere of radius b.

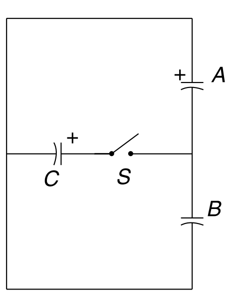
(i) Write the electrostatic energy (U) of the system and calculate the capacitance of the system using the expression of U.

(ii) Prove that the capacitance of the system is given by $\frac{1}{C} = \left(\frac{1}{4\pi\epsilon_0 a} + \frac{1}{4\pi\epsilon_0 b}\right)$ if $d \to \infty$.

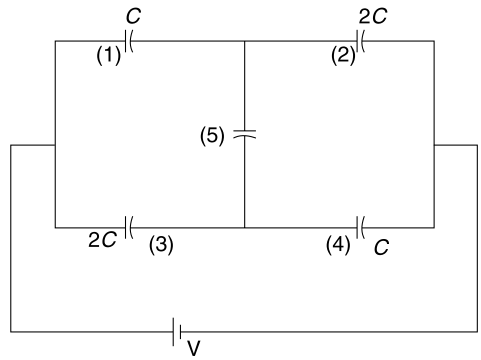
Q. 34: All capacitors in the network given below are identical with capacitance of each being 1 μF. Find the charge on the capacitor marked as C.



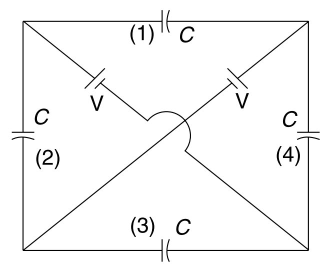
Q. 35: In the given circuit it is known that the capacitor A has a capacitance of 2 μF and carries a charge of 40 μC. Capacitor C has a capacitance of 6 μF and carries a charge of 180 μC. The positive plate of both capacitors has been indicated in the Figure. Capacitance of capacitor B is 3 μF. Calculate charge on B after the switch S is closed.



Q. 36: In the circuit shown in the Figure, find the ratio of potential difference across capacitor 1 and 2. The capacitance values are as indicated in the Figure.



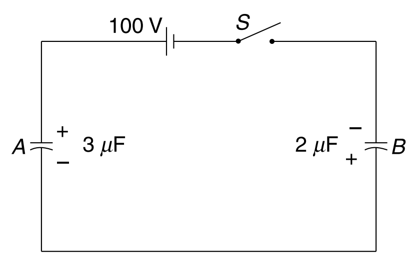
Q. 37: In the circuit shown in the figure, find charge on each capacitor.



Q. 38: Two capacitors A and B with capacitors 3 μF and 2 μF are charged to a potential difference of 100 V and 180 V respectively. One plate of two capacitors are connected as shown. Now switch S is closed so as to connect a cell of 100 V to the free plates of two capacitors.

(a) Find charge on the two capacitors after the switch is closed.

(b) Calculate heat generated in the circuit.



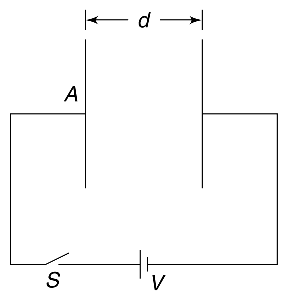
Q. 39: A parallel plate air capacitor has plate area A and separation between the plates d. Switch S is closed to connect the capacitors to a cell of emf V.

(a) Calculate the amount of heat generated in the circuit as the capacitor gets charged.

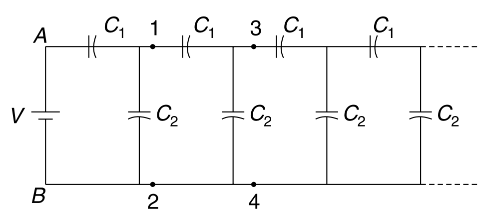
(b) Calculate the force (F) that one capacitor plate exerts on the other.

(c) The distance between the plates is slowly reduced to $\frac{d}{2}$. Calculate the work done by the external agent in the process. For your calculation use the basic definition of work as work = force × displacement.

(d) How much energy is dissipated in the circuit as the distance between the plates is reduced from d to $\frac{d}{2}$? Try to give answer without any calculations. Give reasons. Now use work energy theorem to show that your answer is right.



Q. 40: The circuit shown in the figure continues to infinity. The potential difference between points 1 and 2 is $\frac{V}{2}$, that between points 3 & 4 is $\frac{V}{4}$ and so on ; i.e., the potential difference becomes $\frac{1}{2}$ after every step of the ladder. Find the ratio $\frac{C_1}{C_2}$

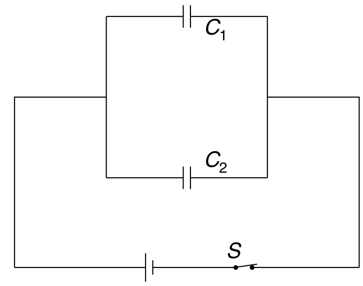


Q. 41: The parallel plate capacitors shown in the Figure have capacitance $C_1 = C$ and $C_2 = 2C$. The switch S is opened. Energy stored in the capacitor system is U_1. Now the separation between the plates of C_1 is slowly reduced to half its original value. Energy stored in the capacitor system now changes to U_2.

(a) Which will be larger - U_1 or U_2? Why?

(b) Calculate work done by the external agent in slowly reducing the distance between the plates of C_1.

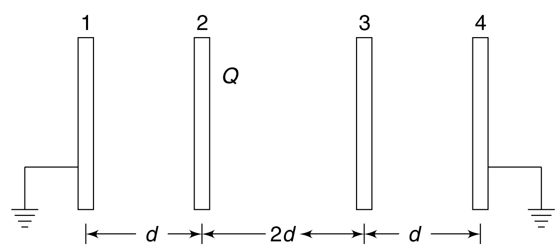
(c) If the plate separation of C_1 is reduced to half and simultaneously the separation between plates of C_2 is doubled, will the energy stored in the capacitor system increase or decrease? Quantify the change in energy.



Q. 42: Four large identical metallic plates are placed as shown in the Figure. Plate 2 is given a charge Q. All other plates are neutral. Now plates 1 and 4 are earthed. Area of each plate is A.

(a) Find charge appearing on right side of plate 3.

(b) Find potential difference between plates 1 and 2.

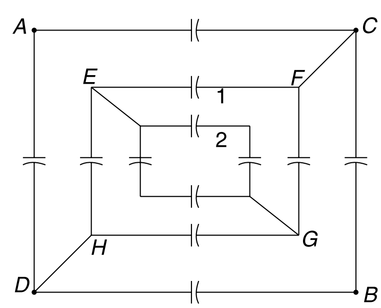


Q. 43: In the circuit shown in the figure all the capacitors have capacitance C.

(a) Find the charge on capacitors marked as 1 and 2 when a battery of emf V is connected across points A and B.

(b) Find the equivalent capacitance across points C and D marked in the Figure.

(c) Find the equivalent capacitance across points E and G.

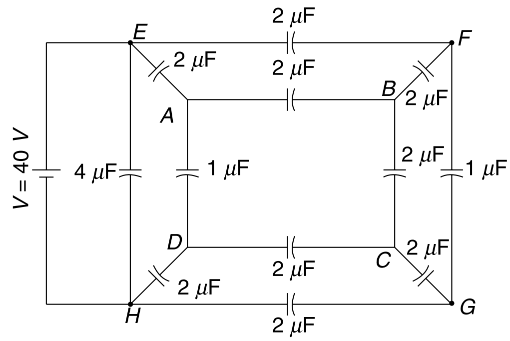


Q. 44: There are nine 2 μF capacitors, two 1 μF capacitors and one 4 μF capacitor in the circuit shown in the Figure.

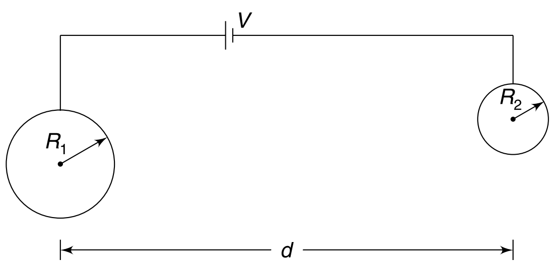
(a) Identify a point in the circuit where potential is same as that of point A.

(b) Identify another pair of points which are having equal potential.

(c) Calculate the charge supplied by the cell to the network of capacitors.



Q. 45: Two solid conducting spheres of radii R_1 and R_2 are kept at a distance d (>> R_1 and R_2) apart. The two spheres are connected by thin conducting wires to the positive and negative terminals of a battery of emf V. Find the electrostatic force between the two spheres



LEVEL 3

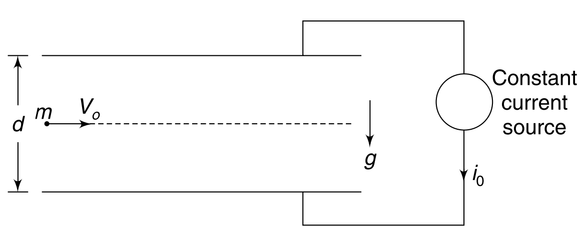
Q. 46: Two identical long metal wires having radius a are held parallel to each other at a separation d (>> a). Calculate the capacitance of the system per unit length.

Q. 47: A particle of mass m and charge $+q$ enters horizontally with speed V_0 midway between the horizontal plates of a parallel plate capacitor at time $t = 0$. Separation between the capacitor plates is 'd' and it starts getting charged, by a constant current source, at time $t = 0$. Plate area of the capacitor is A. It was found that the particle just misses (to hit) the lower plate. Assume that the plates are quite long and acceleration due to gravity is g.

(a) Give a rough sketch of the path of the particle.

(b) Find the constant current (i_0) supplied by the source to the capacitor.

Consider no magnetic force on the charge.



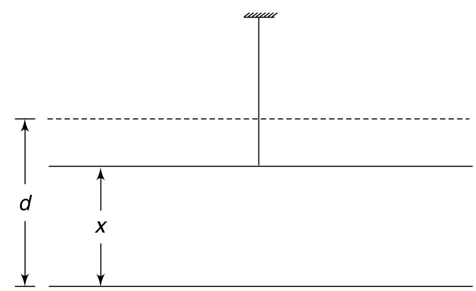
Q. 48: Lower plate of a parallel plate capacitor is fixed on a horizontal insulating surface. The upper plate is suspended above it using on elastic cord of force constant K. The upper plate has negligible mass and area of each plate is A. When there is no charge on the plates the equilibrium separation between them is d. When a potential difference $= V$ is applied between the plates the equilibrium separation changes to x.

(a) Calculate V as a function of x.

(b) Find the value of x for which V is maximum. Calculate the maximum value of V $(= V_{max})$

(c) What will happen if $V > V_{max}$?

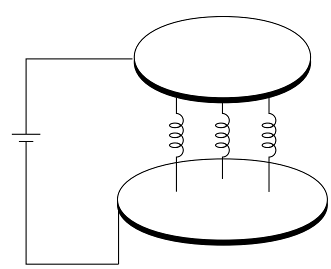
(d) Plot a rough graph showing variation of equilibrium separation (x) with V.



Q. 49: Two identical metal plates with area A and mass m are kept separated by help of three insulating springs as shown in the Figure. The equilibrium separation between the plates is d_0 $(<< \sqrt{A})$ and force constant of each spring is K. When a constant voltage source having emf V is connected to the plates, the equilibrium separation changes to d. Assume that the lower plate is fixed and the upper plate is free to move.

(a) Find V in terms of given parameters.

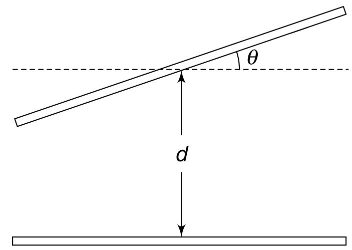
(b) If the upper disc is slightly displaced from its equilibrium position and released, calculate the time period of its oscillation.



Q. 50: One plate of a parallel plate capacitor is tilted by a small angle about its central line as shown in the Figure. The tilt angle θ is small. Both the plates are square in shape with side length a and separation between their centers is d. Find the capacitance of the capacitor.

Given: $\ell n(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} + \ldots$

$\ell n(1 - x) = -\left(x + \frac{x^2}{2} + \frac{x^3}{3} + \ldots\right)$



ANSWERS

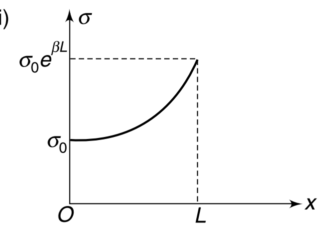
1. B is wrong
2. (b) 1.62×10^{-19} C
3. χE
4. (a) $\frac{Q^2 d}{2\epsilon_0 A}$ (b) $\frac{Q^2}{2\epsilon_0 A}$
5. (a) $C = \frac{K\epsilon_0 A}{Kd - (K-1)x}$

   (b) $V \frac{K\epsilon_0 A d t_0 (K-1)}{[Kdt_0 - (K-1)\, dt]^2}$
6. Insert dielectric with constant k_1, connect the battery, disconnect the battery and remove the dielectric.

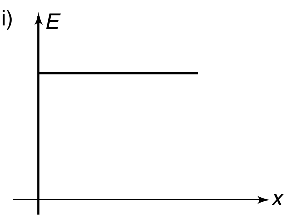
   $U_{\max} = \frac{1}{2} K_1^2 V^2 C$
7. (a) $-\frac{\epsilon_0 A}{4d} V^2$ (b) Attract. Yes
8. $Q = \frac{qRC}{(4\pi\epsilon_0 R + C)}\left(\frac{1}{x} + \frac{1}{y}\right)$
9. (a) $-\frac{1}{4} C_0 V_0^2$ (b) $-\frac{1}{2} C_0 V_0^2$
10. $\frac{\epsilon_0 A}{4d} V^2$
11. $Q = \frac{\epsilon_0 A}{d} V$
12. 40 V
13. 16.8 μc
14. 12 μF
15. (a) $\frac{16}{3}$ μF (b) $\frac{16}{3}$ μF
16. C
17. $\frac{4}{5}$
18. 240 μJ
19. $\frac{Q^2 d}{6\epsilon_0 S}$
20. (a) $\frac{1}{2} KCV^2$

    (b) In the direction indicated by $\vec{F}_1$

    (c) Dielectric having constant $2K$ experienced higher force.
21. (a) $\frac{\sigma_{\max}}{\sigma_{\min}} = \frac{3}{1}$ (b) $\sigma_b = \frac{\epsilon_0 V}{d}(2K - 1)$ (c) $C = \frac{2K\epsilon_0 A}{d}$
22. (i)



(ii)



(iii) $C = \frac{\epsilon_0 L}{\beta d}[e^{\beta L} - 1]$

23. (i) Capacitance decreases, $C_{\min} = \epsilon_0 L^2 \beta$

    (ii) $E = \frac{\beta V e^{-\beta x}}{1 - e^{-\beta d}}$
24. 24
25. $E = E_0 - \frac{qLn}{\epsilon_0}$
26. 1.8 mJ
27. (a) Zero (b) 20 μC
28. $\Delta C \simeq \frac{C_0 K^2 V^2 \epsilon_0}{2d^2 Y}$
29. (a) $\sigma_1 = \frac{4Q}{L(4L + x)}$; $\sigma_2 = \frac{5Q}{L(4L + x)}$

    (b) $\frac{5Qt}{\epsilon_0 L(4L + x)}$

(c) $\dfrac{4Q}{\in_0 L(4L+x)}$

(d) $\dfrac{\in_0 L(4L+x)}{10t}$

30. (a) $\dfrac{1}{150}$ (b) 0.67%

31. (a) $\dfrac{\sigma_A}{\sigma_B} = K$ (b) $\dfrac{1+K}{2}$

32. (a) $4\pi\in_0\left(\dfrac{b^2}{b-a}\right)$ (b) $4\pi\in_0 a$

(c) $4\pi\in_0\left(\dfrac{ab}{b-a}\right)$ (d) $4\pi\in_0 b$

33. (i) $U = \dfrac{q^2}{8\pi\in_0}\left(\dfrac{1}{a}+\dfrac{1}{b}-\dfrac{2}{d}\right)$; $C = \dfrac{4\pi\in_0}{\dfrac{1}{a}+\dfrac{1}{b}-\dfrac{2}{d}}$

34. 20 μF

35. $\dfrac{240}{11}$ μC

36. $\dfrac{V_1}{V_2} = \dfrac{3}{2}$

37. $q_1 = q_3 = 0$; $q_2 = q_4 = CV$

38. (a) $q_A = 84\ \mu$C; $q_B = 144\ \mu$C

(b) 19.44 mJ

39. (a) $\dfrac{1}{2}CV^2$ (b) $\dfrac{\in_0 AV^2}{2d^2}$

(c) $-\dfrac{1}{2}CV^2$ (d) zero

40. $\dfrac{C_1}{C_2} = \dfrac{2}{1}$

41. (a) $U_1 > U_2$ (b) $W_{ext} = -\dfrac{3}{8}CV^2$

(c) No change.

42. (a) $\dfrac{Q}{4}$ (b) $\dfrac{3}{4}\dfrac{Qd}{\in_0 A}$

43. (a) Zero on both (b) 2 C

(c) 2 C

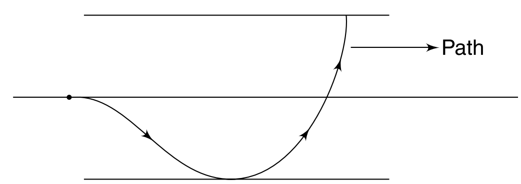
44. (a) F (b) D and G

(c) 208 μC

45. $\dfrac{1}{K}\left(\dfrac{R_1R_2V}{d(R_1+R_2)}\right)^2$

46. $\dfrac{\pi\in_0}{\ln\left(\dfrac{d}{a}\right)}$

47. (a)



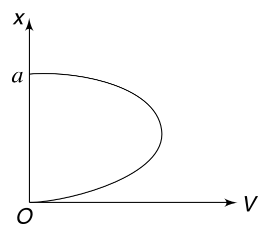
(b) $i_0 = \dfrac{2mgA\in_0}{q}\sqrt{\dfrac{g}{3d}}$

48. (a) $V = \sqrt{\dfrac{2K(d-x)x^2}{\in_0 A}}$

(b) $V_{max} = \sqrt{\dfrac{8Kd^3}{27\in_0 A}}$ at $x = \dfrac{2d}{3}$

(c) System cannot remain in equilibrium.

(d)

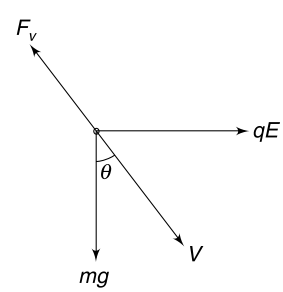


49. (a) $V = \sqrt{\dfrac{6Kd^2(d_0-d)}{\in_0 A}}$ (b) $T = 2\pi\sqrt{\dfrac{m\cdot d}{3K(3d-2d_0)}}$

50. $\dfrac{\in_0 a^2}{d}\left[1+\dfrac{a^2\theta^2}{12d^2}\right]$

SOLUTIONS

2. Electric field between the plates is $E = \dfrac{V}{d} = \dfrac{2000}{3}\dfrac{\text{V}}{\text{cm}}$.



If a drop has charge q, it experiences three force–

(i) Electric force, qE

(ii) Weight, mg

(iii) Viscous force, F_v

F_v is opposite to velocity (V). The resultant force perpendicular to the velocity (V) must be zero.

$$\therefore \qquad mg\sin\theta = qE\cos\theta \qquad \therefore\ \tan\theta = \frac{qE}{mg}$$

$\Rightarrow \qquad q = \dfrac{mg}{E}\tan\theta$

for $\qquad \theta = 45°$

$$q_1 = \frac{mg}{E} \times 1 = \frac{3.3\times 10^{-15}\times 9.8}{\frac{2000}{300}\ \frac{V}{m}} = 4.85 \times 0^{-19}\ \text{C}$$

For $\qquad \theta = 33.7°$

$$q_2 = \frac{mg}{E} \times 0.67 = 4.85 \times 0.67 \times 10^{-19} = 3.25 \times 10^{-19}\ \text{C}$$

For $\qquad \theta = 18.4°$

$$q_3 = 4.85 \times 10^{-19} \times 0.333 = 1.62 \times 10^{-19}\ \text{C}$$

Therefore, the oil drops have acquired different charges during spraying and move in different directions. Smallest charge in nature is LCM of q_1, q_2 and q_3.

$\therefore$ Answer is $\simeq 1.62 \times 10^{-19}$ C

3. If field between the plates is E_0 without any dielectric between them then

$$E = \frac{E_0}{\epsilon_r}$$

But $\qquad E = E_0 - E_{in}$

$\therefore \qquad E_{in} = E_0 - E = \epsilon_r E - E = (\epsilon_r - 1)\ E = \chi E$

4. (a) Energy stored $\qquad U = \dfrac{1}{2}\dfrac{Q^2}{C} = \dfrac{Q^2 d}{2\epsilon_0 A}$

(b) If the required force is F, then $Fd = U$

$$Fd = \frac{Q^2 d}{2\epsilon_0 A} \qquad \therefore \quad F = \frac{Q^2}{2\epsilon_0 A}$$

5. (a) Height of liquid at time t is $x = \dfrac{d}{t_0}t$

We can take the system as two capacitors in series

$$C_1 = \frac{K\epsilon_0 A}{x} \quad \text{and} \quad C_2 = \frac{\epsilon_0 A}{d-x}$$

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{x}{K\epsilon_0 A} + \frac{d-x}{\epsilon_0 A} = \frac{Kd - (K-1)\,x}{K\epsilon_0 A}$$

$\Rightarrow$
$$C = \frac{K\epsilon_0 A}{Kd - (K-1)\,x}$$

(b)
$$i = \frac{dq}{dt} = V\frac{dC}{dt} = V\frac{K\epsilon_0 A\,(K-1)}{[Kd - (K-1)\,x]^2}\,\frac{dx}{dt}$$

$$i = V\frac{K\epsilon_0 A\,(K-1)}{\left[Kd - (K-1)\,\dfrac{dt}{t_0}\right]^2}\,\frac{d}{t_0} = V\frac{K\epsilon_0 A d t_0 (K-1)}{[Kdt_0 - (K-1)\ dt]^2}$$

6.
$$U = \frac{Q^2}{2C}$$

For maximum U, we must have maximum Q and minimum C.

Q will be maximum when battery is connected with capacitance at maximum possible value.

$$C_{\max} = K_1 C \qquad \text{[when dielectric with constant } K_1 \text{ is inserted]}$$

If battery is connected to this capacitor, charge on it

$$Q_{\max} = VC_{\max} = K_1 VC$$

Now battery is disconnected and dielectric is removed.

$$\therefore \qquad U_{\max} = \frac{Q_{\max}^2}{2C} = \frac{K_1^2 V^2 C^2}{2C} = \frac{1}{2} K_1^2 V^2 C$$

7. (a) Capacitance $\quad C_0 = \dfrac{\epsilon_0 A}{d}$

Charge $\quad Q = C_0 V$

Energy in capacitor $\quad U_1 = \dfrac{1}{2} C_0 V^2$

After insertion of the metal plate, capacitance becomes

$$C = \frac{\epsilon_0 A}{d - \dfrac{d}{2}} = \frac{2\epsilon_0 A}{d} = 2C_0$$

$\therefore$ Energy stored in the capacitor becomes

$$U_2 = \frac{Q^2}{2C} = \frac{C_0^2 V^2}{2.2C_0} = \frac{1}{4} C_0 V^2$$

$$\therefore \qquad W = U_2 - U_1$$

$$= -\frac{1}{4} C_0 V^2 = -\frac{\epsilon_0 A}{4d} V^2$$

(b) Since the energy stored in the system has decreased, the electrostatic force must have performed positive work. It means the metal plate being inserted was sucked into the gap between the capacitor plates.

8. Let a charge Q be induced on capacitor plate connected to the ball. Induced charge on the ball is $-Q$.

$\therefore$ Potential at the centre of the ball will be

$$V = K\frac{q}{x} + K\frac{q}{y} + K\frac{-Q}{R} = \frac{q}{4\pi\epsilon_0}\left(\frac{1}{x} + \frac{1}{y}\right) - \frac{1}{4\pi\epsilon_0}\frac{Q}{R}$$

This is the potential of the entire ball and this is the potential of the capacitor plate connected to the sphere. The other plate of the capacitor is at zero potential.

$\therefore$ Potential difference across capacitor plates = V

$$V = \frac{Q}{C} = \frac{q}{4\pi\epsilon_0}\left(\frac{1}{x} + \frac{1}{y}\right) - \frac{1}{4\pi\epsilon_0}\frac{Q}{R}$$

$$\Rightarrow \qquad Q\left(\frac{1}{C} + \frac{1}{4\pi\epsilon_0 R}\right) = \frac{q}{4\pi\epsilon_0}\left(\frac{1}{x} + \frac{1}{y}\right)$$

$$\Rightarrow \qquad Q = \frac{qRC}{(4\pi\epsilon_0 R + C)}\left(\frac{1}{x} + \frac{1}{y}\right)$$

9. (a) $\quad Q_0 = C_0 V_0$

$$U_0 = \frac{1}{2} C_0 V_0^2$$

After the separation is made half new capacitance is

$$C = 2C_0$$

But charge remains fixed.

$$\therefore \qquad U = \frac{Q_0^2}{2(C)} = \frac{C_0^2V_0^2}{2(2C_0)} = \frac{1}{4}C_0V_0^2$$

Work done by the external agent

$$W = \Delta U = U - U_0 = -\frac{1}{4}C_0V_0^2$$

Note: The two plate plates attract each other. The external agent applies force that is opposite to the displacement. Hence, work done is negative.

(b) This time the p.d across the capacitor remains constant.

$$U_0 = \frac{1}{2}C_0V_0^2$$

$$U = \frac{1}{2}(2C_0)V_0^2$$

$$\therefore \qquad \Delta U = \frac{1}{2}C_0V_0^2$$

Battery supplies an additional charge $\Delta Q = 2C_0V_0 - C_0V_0 = C_0V_0$

$$\therefore \qquad W_{batt} = \Delta QV_0 = C_0V_0^2$$

$$\therefore \qquad W + W_{batt} = \Delta U$$

$$\therefore \qquad W = \frac{1}{2}C_0V_0^2 - C_0V_0^2 = -\frac{1}{2}C_0V_0^2$$

10. There are two capacitors in series.
Equivalent capacitance is

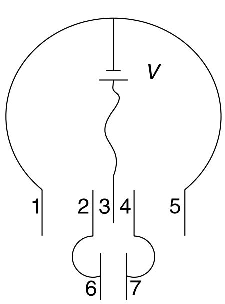
$$C_0 = \frac{1}{2}\left(\frac{\epsilon_0A/2}{d}\right) = \frac{\epsilon_0A}{4d}$$

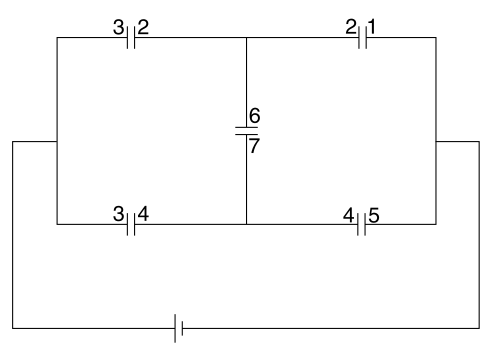
$\therefore$ Charge supplied by the cell

$$Q = C_0V = \frac{\epsilon_0A}{4d}V$$

$$\therefore \qquad W_{batt} = QV = C_0V^2 = \frac{\epsilon_0A}{4d}V^2$$

11. There are five capacitors each of capacitance $C = \dfrac{\epsilon_0A}{d}$





The arrangement is a Wheatstone bridge as shown in figure.
It is a balanced bridge and equivalent capacitance is

$$C_0 = C = \frac{\epsilon_0A}{d}$$

$$\therefore \qquad Q = C_0V = \frac{\epsilon_0A}{d}V$$

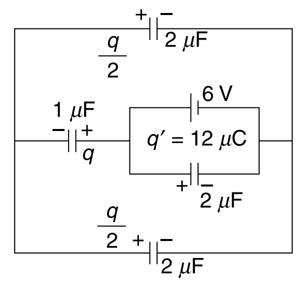
12. The diode $D2$ will never conduct as its end B is at higher potential. It is like an infinite resistance. $D1$ will conduct and the two capacitors will get charged. Both will have same charge.

$$\frac{q}{C} + \frac{q}{2C} = 120$$

$$\therefore \qquad \frac{3q}{2C} = 120$$

$$\Rightarrow \qquad \frac{q}{2C} = 40 \text{ volt}$$

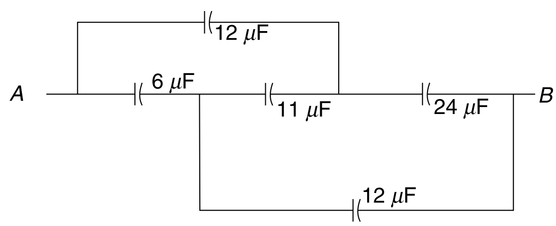
13. The potential difference across 1 μF and 2 μF (uppermost) capacitor must sum up to 6 V



$$\frac{q}{1} + \frac{q/2}{2} = 6 \quad \Rightarrow \quad q = \frac{24}{5}\,\mu\text{C}$$

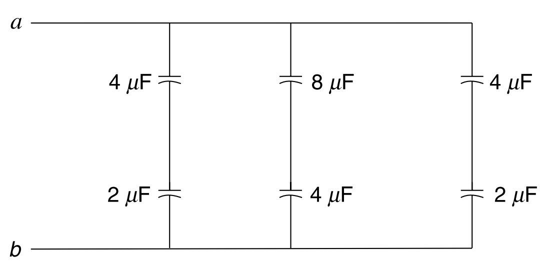
$\therefore$ Answer is $12 + \dfrac{24}{5} = 16.8\ \mu$C

14. 11 μF and 13 μF are in parallel.
6 μF and 5 μF are in parallel.
2 μF and 4 μF are in parallel.

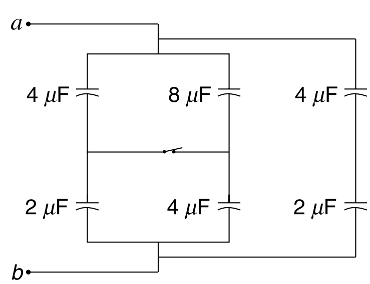


This is a balanced whetstone bridge. 11 μF capacitor can be removed.
12 μF and 24 μF in series = 8 μF.
6 μF and 12 μF in series = 4 μF.
Equivalent = 8 + 4 = 12 μF

15. (a) With switch open the circuit is as shown in the figure. Equivalent capacitance is $C = \dfrac{16}{3}\ \mu$F



(b) With switch closed the circuit is as shown below.



Equivalent is $C = \dfrac{16}{3}\ \mu$F

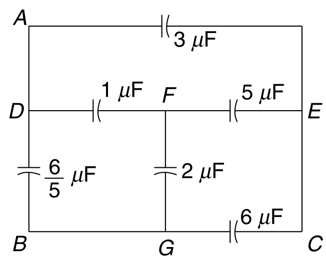
16. Hint: Balanced Wheatstone Bridge.

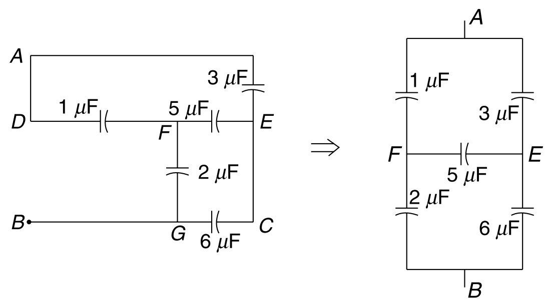
17. Equivalent across *A* & *B*

Remove the $\frac{6}{5}$ μF capacitor between *B* and *D* and the remaining circuit becomes a balanced Wheatstone bridge. [Note *A* & *D* are same points]

The equivalent of Wheatstone bridge is $\frac{8}{3}$ μF. Add to this the removed capacitor of $\frac{6}{5}$ μF in parallel and the equivalent becomes.

$$C_{AB} = \frac{8}{3} + \frac{6}{5} = \frac{58}{15}\ \mu\text{F}$$





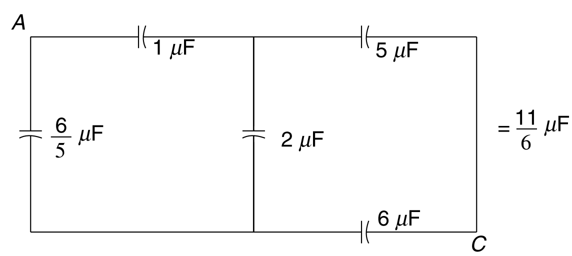
Equivalent across *A* & *C*

Remove the 3 μF capacitor between *A* and *E* (*E* and *C* are same points)

Now remaining circuit is a balanced Wheatstone bridge between *A* and *C* [with the 2 μF capacitor between *F* & *G* being charge less]

The equivalent of Wheatstone bridge is $= \frac{11}{6}$ μF

Add the removed capacitor and the equivalent becomes



$$C_{AC} = \frac{11}{6} + 3 = \frac{29}{6}\ \mu\text{F}$$

$\therefore$ Required ratio is $\quad \frac{58}{15} \times \frac{6}{29} = \frac{4}{5}$

18. With *S* open

Net emf = 0

$\therefore$ Charge on all capacitors = 0

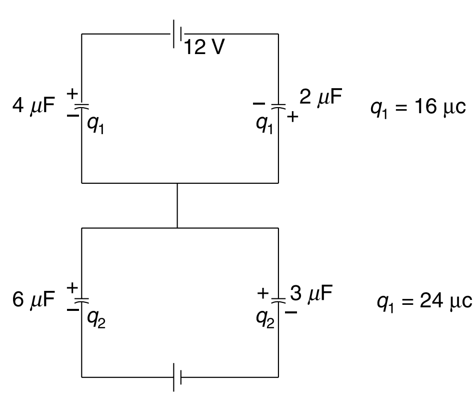
With *S* closed

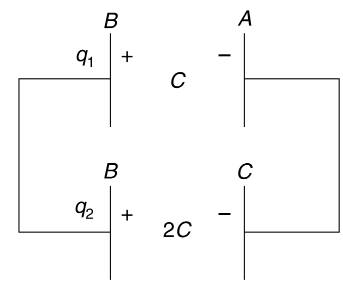
Energy stored in capacitors

$$U = 240\ \mu\text{J}$$

Work done by cells = 480 μJ

Heat produced = 240 μJ





19. Capacitance between A and B is $\quad C = \dfrac{\epsilon_0 S}{2d}$.

Capacitance between B and C is $\dfrac{\epsilon_0 S}{d} = 2C$.

Initial energy stored $\quad U_i = \dfrac{Q^2}{2(2C)} = \dfrac{Q^2}{4C}$.

After the switch is closed, we have two capacitors in parallel, as shown in Figure

$$q_1 + q_2 = Q \qquad \text{...(i)}$$

and $$\frac{q_1}{C} = \frac{q_2}{2C}$$

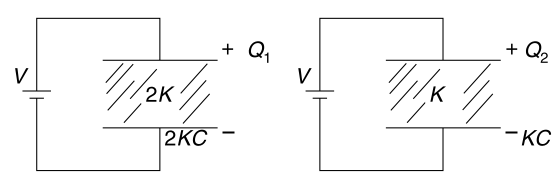
$\Rightarrow$ $$q_1 = \frac{q_2}{2} \qquad \text{...(ii)}$$

Solving (i) and (ii) we get $\quad q_1 = \dfrac{Q}{3};\ q_2 = \dfrac{2Q}{3}$

Final energy stored in the system

$$U_f = \frac{1}{2}\frac{q_1^2}{C} + \frac{1}{2}\frac{q_2^2}{2C} = \frac{Q^2}{6C}$$

$\therefore$ Loss in energy
$$\Delta U = U_i - U_f$$
$$= \frac{Q^2}{4C} - \frac{Q^2}{6C} = \frac{Q^2}{12C}$$
$$= \frac{Q^2}{12\dfrac{\epsilon_0 S}{2d}} = \frac{Q^2 \cdot d}{6\epsilon_0 \cdot S}$$



20. (a) Initially, capacitance = $2KC$

Charge on capacitor $Q_1 = 2KC \cdot V$

Finally, capacitance = KC

Charge on capacitor = $Q_2 = KCV$

$\therefore$ Charge $\Delta q = 2\,KCV - KCV = KCV$ was pushed back from the capacitor plates into the battery. Battery gained energy.

Work done by battery $\quad W_{\text{batt}} = -\Delta qV = -KCV^2$.

Change in electrostatic energy of the capacitor is

$$\Delta U = U_f - U_i = \frac{1}{2}(KC)V^2 - \frac{1}{2}(2\,KC)V^2 = -\frac{1}{2}KCV^2$$

Negative sign shows that the capacitor lost energy. Overall, the capacitor lost $\frac{1}{2}KCV^2$ energy and the battery gained KCV^2 energy ! The gap $\frac{1}{2}KCV^2$ was fulfilled by the work done by the external agent.

Work done by the external agent = $\frac{1}{2}KCV^2$.

(b) Because work done by agent is positive, the force applied must be in the direction of displacement.

$\therefore$ Force was in the direction F_1 indicated in the Figure.

(c) The dielectric slab with constant 2 K experienced higher attractive force. To overcome it the external agent had to apply a rightward force.

21. (a) Electric field between plates is same inside all the dielectrics.

$$E = \frac{V}{d}$$

If σ = free charge density on capacitor plate then $E = \dfrac{\sigma}{K\epsilon_0}$

$$\therefore \quad \frac{\sigma}{K\epsilon_0} = \frac{V}{d} \Rightarrow \sigma = \frac{\epsilon_0 V}{d}K$$

$\therefore$ σ is maximum when $K_3 = 3K$ and it is minimum when $K_1 = K$

$$\therefore \quad \sigma_{max} = \frac{3K\epsilon_0 V}{d}$$

$$\therefore \quad \sigma_{min} = \frac{K\epsilon_0 V}{d}$$

$$\therefore \quad \frac{\sigma_{max}}{\sigma_{min}} = \frac{3}{1}$$

(b) $$\sigma_b = \sigma\left[1 - \frac{1}{2K}\right] = \frac{2\epsilon_0 VK}{d}\left[1 - \frac{1}{2K}\right] = \frac{\epsilon_0 V}{d}(2K - 1)$$

(c) We can consider the capacitor as three parallel plate capacitors placed in parallel.

$$C = C_1 + C_2 + C_3 = \frac{K\epsilon_0 A/3}{d} + \frac{2K\epsilon_0 A/3}{d} + \frac{3K\epsilon_0 A/3}{d}$$

$$\Rightarrow \quad C = \frac{2K\epsilon_0 A}{d}$$

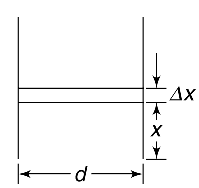
Level 2

22. (i) Consider a plate width of dx at a height x (see Figure)

Capacitance of this segment is

$$\Delta C = \frac{\epsilon_0 K\Delta A}{d} = \frac{\epsilon_0 e^{\beta x}L\Delta x}{d} \quad \text{...(i)}$$

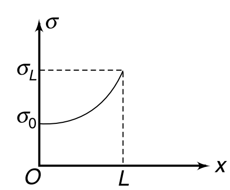
Potential difference across this small capacitor is V.



$$\therefore \quad \Delta q = V\Delta C$$

$$\therefore \quad \sigma = \frac{\Delta q}{\Delta A} = \frac{V\Delta C}{L\Delta x} = \frac{V\epsilon_0 e^{\beta x}}{d}$$

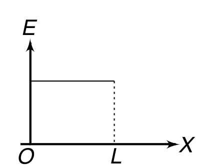
Graph is as shown



$$\sigma_0 = \frac{V\epsilon_0}{d}$$

And $$\sigma_L = \frac{V\epsilon_0}{d}e^{\beta L} = \sigma_0 e^{\beta L}$$

(ii) Electric field remains constant at $E = \dfrac{V}{d}$



(iii) Writing equation (i) for infinitesimal width dx

$$dC = \frac{\epsilon_0 L}{d}e^{\beta x}dx$$

There will be infinite number of such capacitance all in parallel.

$$\therefore \quad C = \int dC = \frac{\epsilon_0 L}{d}\int_0^L e^{\beta x}\cdot dx = \frac{\epsilon_0 L}{\beta d}[e^{\beta L} - 1]$$

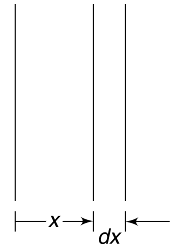
23. (i) We can calculate the capacitance by considering large number of capacitors assembled in series. One such capacitor having plate separation dx is shown in the Figure

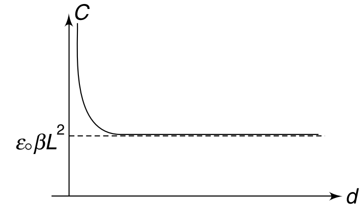
$$dC = \frac{\epsilon_0 KL^2}{dx} = \frac{\epsilon_0 e^{\beta x}L^2}{dx}$$

Capacitance will be given by

$$\frac{1}{C} = \int \frac{1}{dC} = \frac{1}{\epsilon_0 L^2} \int_0^d \frac{dx}{e^{\beta x}}$$

$$= -\frac{1}{\epsilon_0 \beta L^2} [e^{-\beta x}]_0^d = \frac{1}{\epsilon_0 \beta L^2} [1 - e^{-\beta d}]$$





$$\therefore \qquad C = \frac{\epsilon_0 \beta L^2}{1 - e^{-\beta d}}$$

Graph is as shown.

When $d \to \infty$; $C \to \epsilon_0 \beta L^2$

(ii) $$Q = CV = \frac{\epsilon_0 \beta L^2 V}{1 - e^{-\beta d}}$$

$$\sigma = \frac{Q}{L^2} = \frac{\epsilon_0 \beta V}{1 - e^{-\beta d}}$$

$$\therefore \qquad E = \frac{\sigma}{\epsilon_0 K} = \frac{\beta V}{e^{\beta x}[1 - e^{-\beta d}]} = \frac{\beta V e^{-\beta x}}{[1 - e^{-\beta d}]}$$

24. $$C = \frac{K \epsilon_0 A}{d} = \text{a constant}$$

For A to be minimum, d must be minimum. The separation between the plates is limited by the breakdown strength of the dielectric.

For air capacitor $$\frac{V}{d_{\text{min}}} = E_{\text{air}}$$ [E_{air} = Breakdown field for air]

$$\therefore \qquad d_{\text{min}} = \frac{V}{E_{\text{air}}}$$

Now $$\frac{\epsilon_0 A_{\text{min}}}{d_{\text{min}}} = C$$

$$\Rightarrow \qquad A_{\text{min}} = \frac{C}{\epsilon_0} \frac{V}{E_{\text{air}}}$$

$$\therefore \qquad A_1 = \frac{CV}{\epsilon_0 E_{\text{air}}}$$

With dielectric, similar calculation gives

$$A_2 = \frac{CV}{K \epsilon_0 E_{\text{dielec}}}$$

$$\therefore \qquad \frac{A_1}{A_2} = \frac{K E_{\text{dielec}}}{E_{\text{air}}} = 3 \times 8 = 24$$

25. Dipole moment per unit volume (after perfect alignment) in the dielectric = qLn. Net dipole moment induced in dielectric slab is

$$= (qLn)(A \cdot d)$$

$\therefore$ Charge induced on walls of the dielectric

$$Q_{\text{in}} = \pm \frac{qLnAd}{d} = \pm qLnA$$

$\therefore$ Field due to induced charge

$$E_{in} = \frac{Q_{in}}{\epsilon_0 A} = \frac{qLn}{\epsilon_0}$$

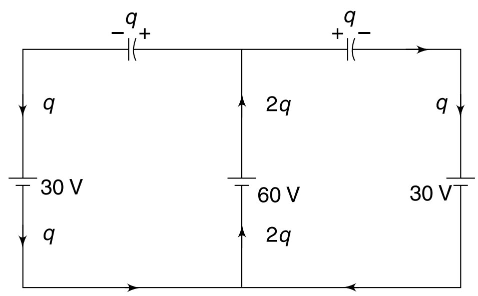
$$\therefore \qquad E = E_0 - E_{in} = E_0 - \frac{qLn}{\epsilon_0}$$

26. With switch S open the potential difference across the group of capacitors is zero. There is no charge on both of them.

With S closed, potential difference across both of them is 30 V .Charge on each of them is

$$q = (2\ \mu F)\ (30\ V) = 60\ \mu C$$

Polarity is as shown. The charge flow in various path has also been shown.



Energy supplied by 60 V cell = $60 \times 2q = 60 \times 120\ \mu J$

Energy absorbed by each of 30 V cell = $30 \times q = 30 \times 60\ \mu J$

Energy stored in each capacitor = $\frac{1}{2}\ CV^2 = \frac{1}{2} \times 2 \times 30^2\ \mu J$

$\therefore$ Heat dissipated $= 60 \times 120 - 2 \times 30 \times 60 - 2 \times 30^2$

$= 60\ [120 - 60 - 30]$

$= 60 \times 30 = 180\ \mu J = 1.8\ mJ$

27. (a) When C_1 and C_2 both are equal, potential difference across each of them is 20 V. The point between the two capacitors is at zero potential and connecting it to earth by closing the switch will make no difference to the circuit. There will be no charge flow.

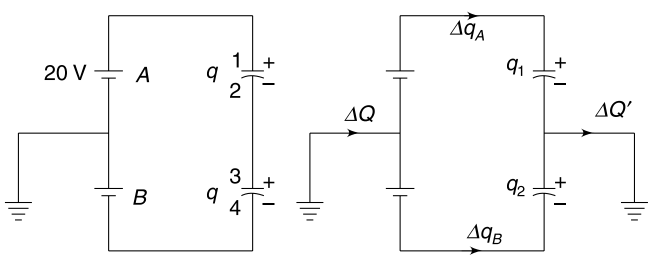
(b) Potential difference across two capacitors is

$$V_1 = 40 \times \frac{C_2}{C_1 + C_2} = \frac{40}{3}\ V$$

And $$V_2 = 40 \times \frac{2}{2+1} = \frac{80}{3}\ V$$

Charge on them is $$q = \frac{80}{3}\ \mu C$$

After the switch is closed, potential difference across both capacitors becomes 20 V. Charges on them are $q_1 = 40\ \mu C$ and $q_2 = 20\ \mu C$



Looking at change in charge on plate 1, it is easy to conclude that a charge

$$\Delta q_A = 40 - \frac{80}{3} = \frac{40}{3}\ \mu C \text{ flows through cell } A.$$

Similarly looking at charge on plate 4, we can conclude that charge flow Δq_B (see Figure) is

$$\Delta q_B = \frac{80}{3} - 20 = \frac{20}{3}\ \mu C$$

$$\therefore \qquad \Delta Q = \Delta q_A + \Delta q_B = \frac{60}{3} = 20\ \mu C$$

Sum of charge on plate 2 and 3 before closing the switch was zero. After closing the switch the charge on these two plates is $= -40 + 20 = -20\ \mu C$

$\therefore$ $\Delta Q'$ (in direction shown) = $20\ \mu C$

28. (a) $$C_0 = \frac{K\epsilon_0 A}{d}$$

When connected to a V volt cell, the plate separation reduces by x.

$$C = \frac{K\epsilon_0 A}{d - x} \quad ...(1)$$

$$\therefore \quad \Delta C = C - C_0 = \frac{K\epsilon_0 A}{d}\left[\frac{1}{1 - \frac{x}{d}} - 1\right] = C_0\left[\left(1 - \frac{x}{d}\right)^{-1} - 1\right]$$

$$= C_0\left[1 + \frac{x}{d} - 1\right] \quad [\because \; x << d]$$

$$\therefore \quad \Delta C = C_0 \frac{x}{d} \quad ...(2)$$

Charge on capacitor plates $Q \simeq C_0 V$

Force between plates $F = \frac{Q^2}{2A\epsilon_0} = \frac{C_0^2 V^2}{2A\epsilon_0}$

Stress on dielectric $\frac{F}{A} = \frac{C_0^2 V^2}{2A^2\epsilon_0}$

$\therefore$ Strain is $\frac{x}{d} = \frac{\text{stress}}{Y} = \frac{C_0^2 V^2}{2A^2\epsilon_0 Y}$

Putting $C_0 = \frac{K\epsilon_0 A}{d}$ we get

$$\frac{x}{d} = \frac{K^2\epsilon_0 V^2}{2d^2 Y}$$

$$\therefore \quad \Delta C \simeq \frac{C_0 K^2 V^2 \epsilon_0}{2d^2 Y}$$

29. The given system is equivalent to the one shown in the Figure. The distance between the plates of C_2 is $\frac{5}{2}t$

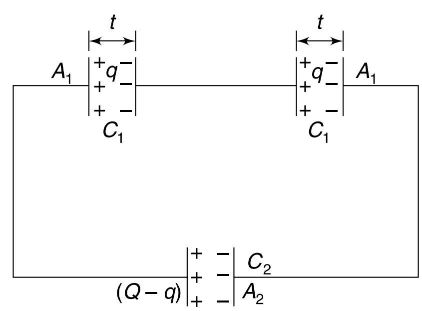


Plate area of capacitors represented by C_1 and C_2 are $A_1 = xL$ and $A_2 = L(L - x)$

$$C_1 = \frac{\epsilon_0 A_1}{t} = \frac{\epsilon_0 Lx}{t}$$

$$C_2 = \frac{\epsilon_0 A_2}{\frac{5t}{2}} = \frac{2\epsilon_0 \cdot L(L - x)}{5t}$$

Let charge on capacitors having capacitance C_1 be q and that on capacitor of capacitance C_2 be $Q - q$.

The equivalent of two C_1 in series is $\frac{C_1}{2}$.

$\therefore$ Potential difference across $\frac{C_1}{2}$ = p.d across C_2.

$$\therefore \quad \frac{q}{C_{1/2}} = \frac{Q - q}{C_2}$$

$$\frac{2q}{\frac{\epsilon_0 Lx}{t}} = \frac{(Q - q)}{\frac{2\epsilon_0 L(L - x)}{5t}}$$

$$\Rightarrow \quad \frac{2q}{x} = \frac{5(Q-q)}{2(L-x)}$$

$$\Rightarrow \quad 4q(L-x) = 5(Q-q)\,x$$

$$\therefore \quad q[4L-4x+5x] = 5Qx$$

$$\therefore \quad q = \frac{5Qx}{4L+x}$$

And
$$Q-q = \frac{4Q(L-x)}{4L+x}$$

(a)
$$\sigma_1 = \frac{Q-q}{L(L-x)} = \frac{4Q}{L(4L+x)}$$

$$\sigma_2 = \frac{q}{L\cdot x} = \frac{5Q}{L(4L+x)}$$

(b) Required p.d = p.d. across $C_1 = \dfrac{q}{C_1}$

$$= \frac{5Qx}{4L+x}\cdot\frac{t}{\epsilon_0 Lx} = \frac{5Qt}{\epsilon_0 L(4L+x)}$$

(c)
$$E = \frac{\sigma_1}{\epsilon_0} = \frac{4Q}{\epsilon_0 L(4L+x)}$$

(d)
$$C = C_2 + \frac{C_1}{2} = \frac{2\epsilon_0 L(L-x)}{5t} + \frac{\epsilon_0 Lx}{2t}$$

$$= \frac{\epsilon_0 L(4L+x)}{10t}$$

30. (a) Energy stored in the electrostatic field produced by the spherical conductor.

$$E = \frac{1}{2}\frac{Q^2}{C} = \frac{Q^2}{8\pi\epsilon_0 R}$$

The electric field on the surface $= \dfrac{1}{4\pi\epsilon_0}\dfrac{Q}{R^2}$

$\therefore$ Energy stored in the electric field in the dent region is

$$\Delta E = \frac{1}{2}\epsilon_0\left(\frac{1}{4\pi\epsilon_0}\frac{Q}{R^2}\right)^2 \Delta V = \frac{Q^2\,\Delta V}{32\pi^2\epsilon_0 R^4} \qquad [\Delta V = \text{Volume of dent}]$$

$$\therefore \quad \frac{\Delta E}{E} = \frac{\Delta V}{4\pi R^3} = \frac{\Delta V}{3\left(\frac{4}{3}\pi R^3\right)} = \frac{\Delta V}{3V} \qquad [V = \text{Volume of the sphere}]$$

$$\therefore \quad \frac{\Delta E}{E} = \frac{1}{3}\times 0{\cdot}02 = \frac{1}{150}$$

(b)
$$E = \frac{Q^2}{2C}$$

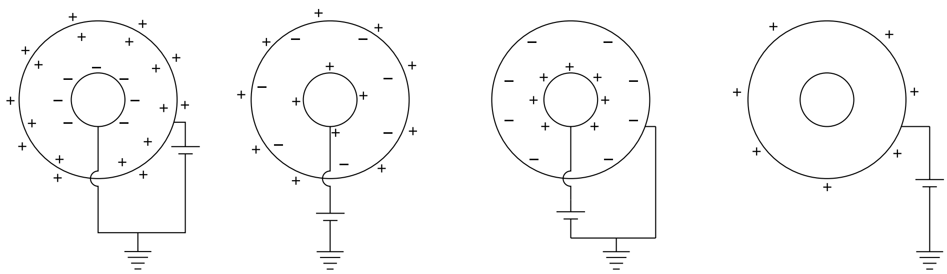
$$\therefore \quad \Delta E = -\frac{Q^2}{2C^2}\,\Delta C \qquad [\text{–sign tells that if } \Delta E \text{ is + ve them } \Delta C \text{ is –ve}]$$

Fractional Decrease in capacitance

$$\frac{\Delta C}{C} = \Delta E \cdot \frac{2C}{Q^2} = \frac{\Delta E}{E} = \frac{1}{150}$$

$$\therefore \quad \frac{\Delta C}{C} \times 100 = \frac{1}{150} \times 100 = \frac{2}{3} = 0.67\%$$

32. The charge distribution on the surfaces of shells in four cases is as shown in the Figures.



(a)
$$C = C_1 + C_2$$
$$= 4\pi\epsilon_0\left(\frac{ab}{b-a}\right) + 4\pi\epsilon_0 b = 4\pi\epsilon_0\left(\frac{b^2}{b-a}\right)$$

(b) In this case C_1 and C_2 are in series

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}$$

$$\Rightarrow \quad \frac{1}{C} = \frac{b-a}{ab}\,\frac{1}{4\pi\epsilon_0} + \frac{1}{4\pi\epsilon_0 b}$$

$$= \frac{1}{4\pi\epsilon_0 b}\left(\frac{b-a}{a} + 1\right) = \frac{1}{4\pi\epsilon_0 b}\left(\frac{b}{a}\right) = \frac{1}{4\pi\epsilon_0 a}$$

$$\therefore \quad C = 4\pi\epsilon_0 a$$

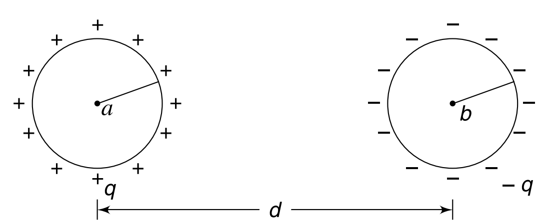
(c)
$$C = C_1 = 4\pi\epsilon_0\left(\frac{ab}{b-a}\right)$$

(d)
$$C = C_2 = 4\pi\epsilon_0 b$$

33. (i) The electrostatic self energy of two conductors are

$$U_1 = \frac{q^2}{8\pi\epsilon_0 a} \quad \text{and} \quad U_2 = \frac{q^2}{8\pi\epsilon_0 b}$$

The interaction energy of two conductors can be written assuming them to be point charges.



$$U_{12} = \frac{q(-q)}{4\pi\epsilon_0 d} = \frac{-q^2}{4\pi\epsilon_0 d}$$

$\therefore$ Total energy of the system is

$$U = U_1 + U_2 + U_{12}$$

$$= \frac{q^2}{8\pi\epsilon_0}\left(\frac{1}{a} + \frac{1}{b} - \frac{2}{d}\right)$$

Now
$$\frac{q^2}{2C} = U \quad \Rightarrow \quad C = \frac{4\pi\epsilon_0}{\frac{1}{a} + \frac{1}{b} - \frac{2}{d}}$$

(ii) If $d \to \infty$

$$C = \frac{4\pi\in_0}{\frac{1}{a}+\frac{1}{b}}$$

$$\therefore \quad \frac{1}{C} = \frac{1}{4\pi\in_0 a} + \frac{1}{4\pi\in_0 b}$$

34. Look at the given figure carefully to find that two plates of the capacitor marked as C are directly connected to the terminals of the battery.

Potential difference across the marked capacitor = 20 V.

$\therefore$ Charge on it = 20 μC

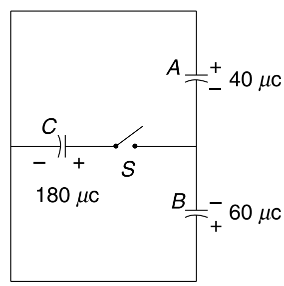
35. Initially B must also have some charge. Potential difference across plates of A is

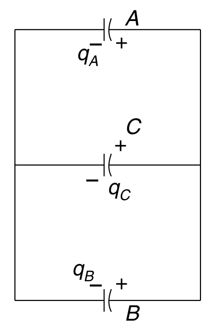
$$V_A = \frac{40\ \mu\text{C}}{2\ \mu\text{F}} = 20\text{ V}$$

$$\therefore \quad V_B = 20\text{ V}$$

Charge on B is (3 μF) (20 V) = 60 μC

Charge on C is 180 μF. Polarity of charges is as shown in the Figure.





After the switch is closed, the three capacitors will be in parallel and they have a total charge of 180 – 40 – 60 = 80 μC to be shared among them.

The charge will be shared in the ratio of capacitance

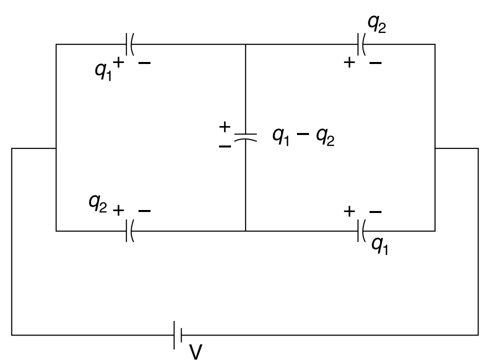
$$\therefore \quad q_A : q_B : q_C = 2:3:6$$

$$\therefore \quad q_A = \frac{2 \times 80}{2+3+6}\ \mu\text{C} = \frac{160}{11}\ \mu\text{C}$$

$$q_B = \frac{3 \times 80}{2+3+6} = \frac{240}{11}\ \mu\text{C}$$

$$q_C = \frac{6 \times 80}{2+3+6} = \frac{480}{11}\ \mu\text{F}$$

36. Due to symmetry charge on capacitor 1 and 4 will be same. Similarly, charge on 2 and 3 will be same. Charge distribution is shown in the Figure.



Note: that charge on capacitor 5 has been written using the fact that sum of charges on three plates connected together must be zero.

Using Kirchhoff's voltage law in loop having capacitors 1, 5 and 3 –

$$\frac{q_1}{C} + \frac{q_1 - q_2}{C} = \frac{q_2}{2C}$$

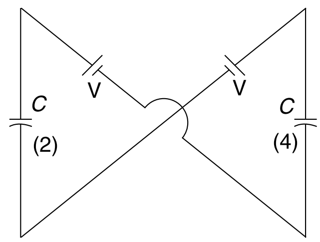
$$\Rightarrow \quad 2q_1 = \frac{3q_2}{2} \Rightarrow \frac{q_1}{q_2} = \frac{3}{4}$$

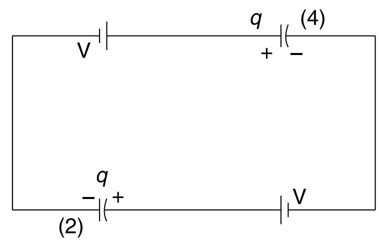
Potential difference across capacitor 1 and 2 will be in ratio

$$\frac{V_1}{V_2} = \frac{q_1/C_1}{q_2/C_2} = \frac{q_1}{q_2}\frac{C_2}{C_1} = \frac{3}{4}\cdot\frac{2C}{C} = \frac{3}{2}$$

37. The terminals of capacitor 1 are in identical position and cannot have different potential. Similarly, there cannot be a potential difference across the terminals of capacitor 3. We can remove capacitors 1 and 3 from the circuit. The effective circuit is as shown in Figure.

Charge on (2) and (4) is $q = CV$

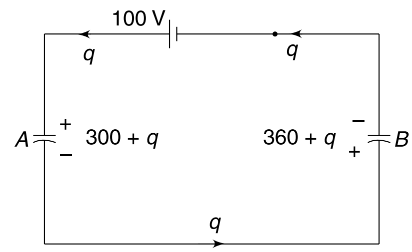




38. Charges on two capacitors are

$$q_A = 100 \times 3 = 300\ \mu\text{C}$$

And $$q_B = 180 \times 2 = 360\ \mu\text{C}.$$



After the switch S is closed let charge q flow in the circuit as shown in the Figure.

Using Kirchhoff's voltage law we can write–

$$\frac{360 + q}{2} + \frac{300 + q}{3} = 100$$

$$\Rightarrow \quad 360 \times 3 + 3q + 600 + 2q = 600$$

$$\Rightarrow \quad q = -216\ \mu\text{C}$$

$\therefore$ Final charge on the two capacitors is

$$q'_A = 300 + q = 84\ \mu\text{C}$$

$$q'_B = 360 + q = 144\ \mu\text{C}$$

Change in electrostatic energy of the capacitor system is

$$\Delta U = \left(\frac{(q'_A)^2}{2 \times 3} + \frac{(q'_B)^2}{2 \times 2}\right) - \left(\frac{q_A^2}{2 \times 3} + \frac{q_B^2}{2 \times 2}\right)$$

$$= \left(\frac{84^2}{6} + \frac{144^2}{4}\right) - \left(\frac{300^2}{6} + \frac{360^2}{4}\right)$$

$$= 1176 + 5184 - 15000 - 32400$$

$$= -41040\ \mu\text{J} = -41.04\ \text{mJ}$$

$\therefore$ Capacitor system has lost 41.04 mJ energy.

Energy gained by the cell $\quad U_{\text{cell}} = 100 \times 216\ \mu\text{J} = 21.6\ \text{mJ}$

$\therefore$ $\quad$ Heat dissipated $= 41.04 - 21.60 = 19.44$ mJ

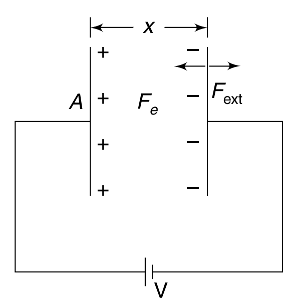
39. (a) Charge on capacitor is $\quad Q = CV \quad \left[C = \frac{\epsilon_0 A}{d}\right]$

$$W_{\text{battery}} = Q \cdot V = CV^2$$

Energy stored in the capacitor $\quad U = \frac{1}{2} CV^2$

$\therefore$ $\quad$ Heat generated $= W_{\text{batt}} - U = \frac{1}{2} CV^2$

(b) Let the distance between the plates be x at some intermediate stage.



$$C = \frac{\epsilon_0 A}{x}$$

$$q = CV = \frac{\epsilon_0 AV}{x}$$

Field due to charge on one plate is $E = \dfrac{q}{2A\epsilon_0}$

$\therefore$ Force between plates $\quad F = Eq = \dfrac{q^2}{2A\in_0} = \dfrac{\in_0 AV^2}{2x^2}$

(c) Assuming that the positive plate is fixed and the negative plate moves slowly towards it, work done by external agent in displacement dx is

$$dW_{\text{ext}} = -|F_{\text{ext}}||dx| = -|F_e||dx| = \frac{\in_0 AV^2}{2}\frac{dx}{x^2}$$

$\therefore$ Work done in reducing the distance from d to $\dfrac{d}{2}$ is

$$W_{\text{ext}} = +\frac{\in_0 AV^2}{2}\int_d^{d/2}\frac{dx}{x^2} = -\frac{\in_0 AV^2}{2d} = -\frac{1}{2}CV^2$$

(e) Energy dissipated must be zero as the plates are moved slowly. Charge flow from battery to the capacitor plates is very slow.

$$W_{\text{ext}} = -\frac{1}{2}CV^2$$

Charge in PE stored in the capacitor

$$\Delta U = \frac{1}{2}2CV^2 - \frac{1}{2}CV^2 = \frac{1}{2}CV^2$$

[Capacitance doubled when d was halved]

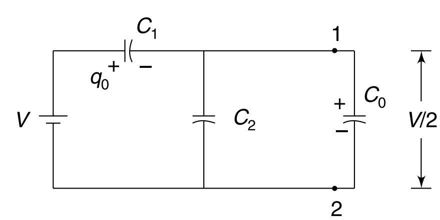
$$W_{\text{batt}} = [(2CV) - CV]\,V = CV^2$$

$\therefore \qquad W_{\text{batt}} + W_{\text{ext}} = \Delta U + \text{Heat generated}$

$\therefore$ Heat generated = 0

40. Let the equivalent capacitance be C_0. Even if we remove one ladder, the equivalent will remain C_0.

The potential drop across C_1 in figure shown is $\dfrac{V}{2}$.



$$\therefore \qquad \frac{q_0}{C_1} = \frac{V}{2} \Rightarrow q_0 = \frac{C_1 V}{2} \qquad ...(1)$$

Also
$$q_0 = C_0 V \qquad ...(2)$$

$$\therefore \qquad \frac{C_1}{2} = C_0 \Rightarrow C_1 = 2C_0 \qquad ...(3)$$

From the given circuit

$$C_0 = \frac{(C_0 + C_2)\,C_1}{C_0 + C_2 + C_1}$$

$$\Rightarrow \qquad C_0^2 + C_0C_2 + C_0C_1 = C_0C_1 + C_1C_2$$

$$\Rightarrow \qquad C_0^2 + C_2C_0 - C_1C_2 = 0$$

$$C_0 = \frac{-C_2 \pm \sqrt{C_2^2 + 4C_1C_2}}{2}$$

Only positive sign is acceptable.

$$\therefore \qquad C_0 = \frac{-C_2 + \sqrt{C_2^2 + 4C_1C_2}}{2}$$

$$\therefore \qquad \frac{C_1}{2} + \frac{C_2}{2} = \frac{\sqrt{C_2^2 + 4C_1C_2}}{2} \qquad \left[\because \frac{C_1}{2} = C_0\right]$$

$$(C_1 + C_2)^2 = C_2^2 + 4C_1C_2$$

$$C_1^2 = 2C_1C_2 \Rightarrow C_1 = 2C_2$$

41. (a) U_1 is larger. The external agent does a negative work as the capacitor plates attract each other. Therefore the energy of the system must decrease.

(b) $$U_1 = \frac{1}{2} C_1V^2 + \frac{1}{2} C_2V^2 = \frac{3C}{2} V^2$$

Charge on capacitors are $\quad Q_1 = C_1V = CV$ and $Q_2 = C_2V = 2CV$

After S is opened the total charge on connected plates of the two capacitors will remain $\pm(Q_1 + Q_2)$.

After separation between plates is halved, the capacitance C_1 becomes $2C$.

$\therefore$ Final charge on both capacitors will be

$$q = \frac{Q_1 + Q_2}{2} = \frac{3CV}{2}$$

Potential difference across both will be

$$V' = \frac{q}{2C} = \frac{3V}{4}$$

$$\therefore \quad U_2 = \frac{1}{2}(2C)\left(\frac{3\text{ V}}{4}\right)^2 + \frac{1}{2}(2C)\left(\frac{3\text{ V}}{4}\right)^2 = \frac{9}{8} CV^2.$$

$$\therefore \text{ Loss in energy} \quad U_1 - U_2 = \left(\frac{3}{2} - \frac{9}{8}\right) CV^2 = \frac{3}{8} CV^2$$

$$\therefore \quad W_{\text{ext}} = -\frac{3}{8} CV^2$$

(c) In this case C_1 becomes $2C$ and C_2 becomes C.

Final common potential difference will be

$$V' = \frac{Q_1 + Q_2}{2C + C} = \frac{CV + 2CV}{3C} = V.$$

This is expected since the situation remains practically same as original, with capacitance getting exchanged.

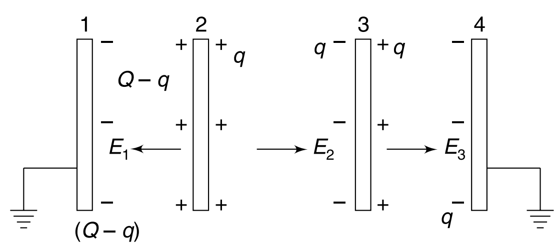
$\therefore$ Final energy $\quad U_3 = U_1$

$\therefore$ No work is done by the external agent.

Note: In a slow process no heat will be dissipated.

42. The opposite faces must have equal and opposite charge. The outer surface of plate 1 & 4 will have no charge. Let charge on right face of plate 2 be q. Charges on other faces are as shown.

$$E_1 = \frac{Q-q}{\epsilon_0 A}; \; E_2 = \frac{q}{\epsilon_0 A}; \; E_3 = \frac{q}{\epsilon_0 A}$$



$$V_1 + E_1d - E_2(2d) - E_3d = V_4$$

But $\quad V_1 = V_4 = 0$

$$\Rightarrow \quad E_1 - 2E_2 - E_3 = 0$$

$$\Rightarrow \quad Q - q - 2q - q = 0$$

$$\Rightarrow \quad q = \frac{Q}{4}$$

Potential difference between plate 1 and 2 is

$$V = E_1 d = \frac{3Qd}{4\epsilon_0 A}$$

43. (a) When a battery is connected across points A and B the circuit is a balanced Wheatstone bridge, with the inner network connected to C and D receiving no charge. Hence charge on 1 and 2 is zero.

(b) When a battery is connected across C and D, the innermost loop will have no charge because the 4 capacitors in loop $EFGH$ and the innermost loop (acting as fifth capacitor of bridge) forms a balanced Wheatstone bridge.

The innermost loop can be removed. It leaves us with 4 parallel branches between C and D each having two capacitors in series.

$\therefore$ Equivalent capacitance $= 4 \cdot \frac{C}{2} = 2C$

(d) In this case the innermost loop (having capacitance $C_1 = C$) is a parallel path across EG. The loop $EFGH$ forms a balanced Wheatstone bridge with the outer loop acting as fifth capacitance between F and H.

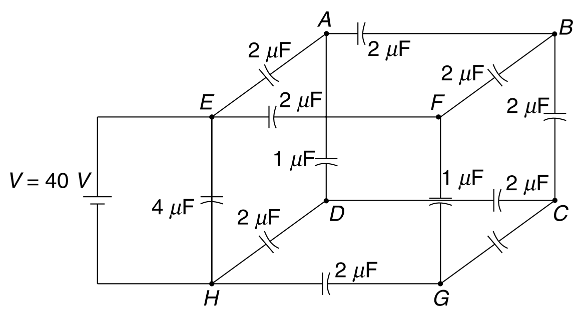
Equivalent of the bridge is $C_2 = C$. This is connected in parallel to C_1

$$\therefore \quad C_{eq} = C_1 + C_2 = 2C$$

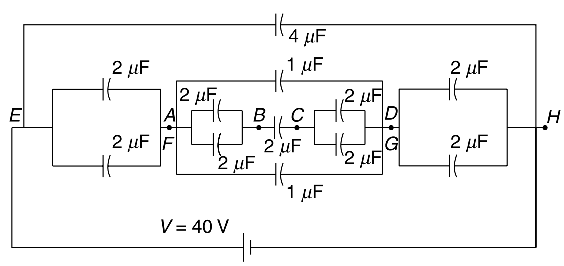
44. Convince yourself that the network obtained by placing 12 capacitors on the edges of a cube (see Figure) is same as the given circuit. Yes, I am asking you to convert a two dimensional circuit into a three dimensional one!

The symmetry becomes obvious.

Point A and F cannot have different potentials and similarly, D and G are equipotential.



Points A and F can be joined together and points D and G can be put together. We can redraw the circuit as shown below.



Equivalent capacitance can be shown to be

$$C_{eq} = \frac{26}{5}\ \mu\text{F}$$

$$\therefore \quad Q = \frac{26}{5} \times 40 = 208\ \mu\text{C}$$

45. let's first calculate the capacitance of the system.

Assuming that the two spheres have charge Q and $-Q$, we can write their potentials as

$$V_1 = \frac{KQ}{R_1} \quad \text{and} \quad V_2 = \frac{K(-Q)}{R_2}$$

Note that charge on one sphere does not have any significant effect on the potential of other since $d >> R_1$ and R_2.

$\therefore$ Potential difference between the two spheres

$$V = V_1 - V_2 = \frac{KQ}{R_1} + \frac{KQ}{R_2}$$

$\therefore$ Capacitance $$C = \frac{Q}{V} = \frac{R_1 R_2}{K(R_1 + R_2)}$$

Charge on the spheres $$Q = CV = \frac{R_1 R_2 V}{K(R_1 + R_2)}$$

$\therefore$ Force between the spheres

$$F = K\frac{Q^2}{d^2} = \frac{K}{d^2}\left[\frac{R_1 R_2 V}{K(R_1 + R_2)}\right]^2$$

$$= \frac{R_1^2 R_2^2 V^2}{Kd^2 (R_1 + R_2)^2}$$

46. The electric field outside a long wire having linear charge density λ is

$$E = \frac{\lambda}{2\pi\epsilon_0 r} \text{ where } r = \text{distance from axis}$$

Electric potential at a point can be written as

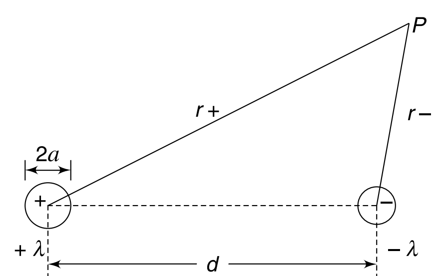
$$V = -\int E dr + C \qquad [C = \text{a constant}]$$

$$= -\frac{\lambda}{2\pi\epsilon_0}\ell nr + \frac{\lambda}{2\pi\epsilon_0}\ell nr_0 \qquad \left[\text{put } C = \frac{\lambda}{2\pi\epsilon_0}\ell nr_0\right]$$

$\therefore$ $$V = \frac{\lambda}{2\pi\epsilon_0}\ell n\frac{r_0}{r}$$

Potential at any point P is presence of both the wires – one having charge $+\lambda\ Cm^{-1}$ and the other having charge $-\lambda\ Cm^{-1}$ is

$$V = \frac{\lambda}{2\pi\epsilon_0}\ell n\left(\frac{r_0}{r_+}\right) - \frac{\lambda}{2\pi\epsilon_0}\ell n\left(\frac{r_0}{r_-}\right)$$



$\therefore$ $$V = \frac{\lambda}{2\pi\epsilon_0}\ell n\left(\frac{r_-}{r_+}\right)$$

Potential on the surface of the left wire is

$$V_1 = \frac{\lambda}{2\pi\epsilon_0}\ell n\left(\frac{d}{a}\right)$$

Potential on the surface of the right wire is

$$V_2 = \frac{\lambda}{2\pi\epsilon_0}\ell n\left(\frac{a}{d}\right)$$

$\therefore$ Potential difference $$V = V_1 - V_2 = \frac{\lambda}{\pi\epsilon_0}\ell n\left(\frac{d}{a}\right)$$

$\therefore$ Capacitance per unit length is

$$C = \frac{\lambda}{V} = \frac{\pi\epsilon_0}{\ell n\left(\frac{d}{a}\right)}$$

47. In time 't' the capacitor acquires a charge Q given by $Q = i_0 t$

Electric field between the plates is $E = \dfrac{Q}{A\epsilon_0} = \dfrac{i_0 t}{A\epsilon_0}$

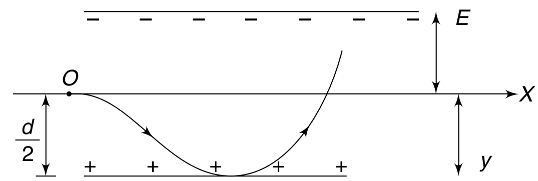
Electric force on the charge $\quad F_e = qE = \dfrac{qi_0 t}{A\epsilon_0}$

Force on the particle in y direction is $F_y = mg - \dfrac{qi_0 t}{A\epsilon_0}$

Acceleration in y direction

$$a_y = g - \frac{qi_0 t}{mA\epsilon_0} = g - kt \qquad \left[\text{where } k = \frac{qi_0}{mA\epsilon_0}\right]$$

The particle initially experiences a downward acceleration. After some time when $kt > g$, its acceleration changes direction. Hence, path of the particle is as shown below.



For y component of motion of the particle

$$\frac{dv_y}{dt} = g - kt$$

$$\therefore \quad \int_0^{V_y} dv_y = g\int_0^t dt - k\int_0^t t\,dt$$

$$v_y = gt - \frac{1}{2}kt^2 \qquad ...(1)$$

$$v_y = 0 \quad \text{when} \quad t_0 = \frac{2g}{k} \qquad ...(2)$$

At time t_0, y – co-ordinate of the particle is $\dfrac{d}{2}$,

$$v_y = gt - \frac{1}{2}kt^2$$

$$\Rightarrow \quad \frac{dy}{dt} = gt - \frac{1}{2}kt^2$$

$$\Rightarrow \quad \int_0^{\frac{d}{2}} dy = g\int_0^{t_0} t\,dt - \frac{1}{2}k\int_0^{t_0} t^2\,dt$$

$$\therefore \quad \frac{d}{2} = g\frac{t_0^2}{2} - \frac{kt_0^3}{6}$$

$$\Rightarrow \quad \frac{d}{2} = g\frac{2g^2}{k^2} - \frac{k}{6}\frac{8g^3}{k^3} \qquad \left[\because \; t_0 = \frac{2g}{k}\right]$$

$$\Rightarrow \quad \frac{d}{2} = \frac{2}{3}\frac{g^3}{k^2}$$

$$\therefore \quad k = \sqrt{\frac{4}{3}\frac{g^3}{d}}$$

$$\Rightarrow \quad \frac{qi_o}{mA\epsilon_0} = \sqrt{\frac{4}{3}\frac{g^3}{d}} \quad \Rightarrow \quad i_o = \frac{mA\epsilon_0}{q}\sqrt{\frac{4}{3}\frac{g^3}{d}}$$

48. (a) Capacitance when separation between the plates is x is given as

$$C = \frac{\epsilon_0 A}{x}$$

Change on plates when applied potential difference is V is

$$Q = CV = \frac{\epsilon_0 AV}{x}$$

Force between the plates

$$F = \frac{1}{2} EQ = \frac{Q}{2\epsilon_0 A} \cdot Q = \frac{Q^2}{2\epsilon_0 A}$$

$\Rightarrow$
$$F = \frac{1}{2} \frac{\epsilon_0 A}{x^2} V^2$$

For equilibrium

$$K(d - x) = \frac{1}{2} \frac{\epsilon_0 A}{x^2} V^2$$

$\Rightarrow$
$$V^2 = \frac{2K(d - x)\, x^2}{\epsilon_0 A}$$

$$V = \sqrt{\frac{2K(d - x)\, x^2}{\epsilon_0 A}}$$

(b) V is maximum when $(d - x)\, x^2$ is maximum. It means when $\frac{d}{dx}[(d - x)\, x^2] = 0$

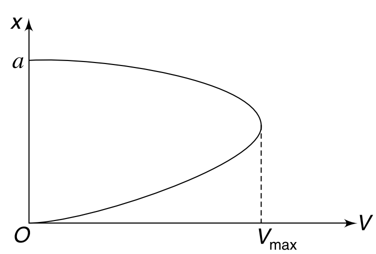
$\Rightarrow$
$$2xd - 3x^2 = 0$$

$\Rightarrow$
$$x = \frac{2d}{3}$$

$\therefore$
$$V_{max} = \sqrt{\frac{2k\left(d - \frac{2d}{3}\right)\left(\frac{2d}{3}\right)^2}{\epsilon_0 A}} = \sqrt{\frac{8Kd^3}{27\epsilon_0 A}}$$

(c) If $V > V_{max}$, the system cannot remain in equilibrium.

(d)



49. (a) With no voltage source

$$mg = F_{spring} \qquad ...(1)$$

With voltage source connected, let the electrostatic force between the two plates in equilibrium be F_0. The rise in spring force balances this F_0.

$$F_0 = 3K(d_0 - d) \qquad ...(2)$$

Electric force between plates, $F_0 = Q \cdot \left(\frac{\sigma}{2\epsilon_0}\right)$

$$= \frac{Q^2}{2\epsilon_0 A} = \frac{C^2 V^2}{2\epsilon_0 A} = \frac{\epsilon_0 A V^2}{2d^2}$$

$\therefore$
$$\frac{\epsilon_0 A V^2}{2d^2} = 3K(d_0 - d) \qquad ...(1)$$

$\therefore$
$$V = \sqrt{\frac{6Kd^2 (d_0 - d)}{\epsilon_0 A}}$$

(c) If plate moves by a small distance Δx the electrostatic force changes by

$$\Delta F_e = \frac{\in_0 AV^2}{2(d+\Delta x)^2} - \frac{\in_0 AV^2}{2d^2} = \frac{\in_0 AV^2}{2d^2}\cdot\left[\left(1+\frac{\Delta x}{d}\right)^{-2} - 1\right]$$

$$= \frac{\in_0 AV^2}{2d^2}\left(-2\frac{\Delta x}{d}\right) = -\frac{\in_0 AV^2}{d^3}\cdot\Delta x$$

Substituting for V we get

$$|\Delta F_e| = \frac{6K(d_0 - d)}{d}\Delta x$$

The spring force changes by

$$|\Delta F_s| = 3K\,\Delta x$$

The two forces ΔF_e & ΔF_s have opposite directions. While approaching the spring pushes the plates away and the electric force causes them to attract more strongly.

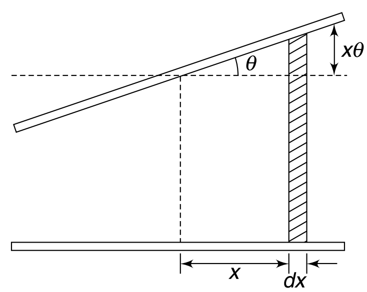
$$\therefore \quad m\frac{d^2x}{dt^2} = -3K\Delta x + \frac{6K(d_0 - d)}{d}\Delta x$$

$$\frac{d^2x}{dt^2} = -\frac{3K}{m}\left[\frac{3d - 2d_0}{d}\right]\Delta x$$

$$\therefore \quad \omega = \sqrt{\frac{3K}{m}\frac{(3d - 2d_0)}{d}}$$

$$T = 2\pi\sqrt{\frac{m\cdot d}{3K(3d - 2d_0)}}$$

50. Consider a strip of width dx in the two plates as shown in the Figure Capacitance of two parallel strips (nearly parallel !) is



$$dC = \frac{\in_0 a\,dx}{d + x\theta}$$

Capacitance of the capacitor can be obtained by summing up the capacitance of all such small capacitor.

$$\therefore \quad C = \int dC = \in_0 a \int_{-\frac{a}{2}}^{+\frac{a}{2}} \frac{dx}{d + \theta x}$$

$$\therefore \quad C = \frac{\in_0 a}{\theta}\left[\ell n(d + \theta x)\right]_{-a/2}^{a/2}$$

$$= \frac{\in_0 a}{\theta}\left[\ell n\left(d + \frac{a\theta}{2}\right) - \ell n\left(d - \frac{a\theta}{2}\right)\right]$$

$$= \frac{\in_0 a}{\theta}\left[\ell n d + \ell n\left(1 + \frac{a\theta}{2d}\right) - \ell n d - \ell n\left(1 - \frac{a\theta}{2d}\right)\right]$$

$$= \frac{\in_0 a}{\theta}\left[\ell n\left(1 + \frac{a\theta}{2d}\right) - \ell n\left(1 - \frac{a\theta}{2d}\right)\right]$$

$$= \frac{\in_0 a}{\theta}\left[\frac{a\theta}{2d} - \frac{1}{2}\left(\frac{a\theta}{2d}\right)^2 + \frac{1}{3}\left(\frac{a\theta}{2d}\right)^3 + \ldots\right]$$

$$+ \frac{\in_0 a}{\theta}\left[\frac{a\theta}{2d} + \frac{1}{2}\left(\frac{a\theta}{2d}\right)^2 + \frac{1}{3}\left(\frac{a\theta}{2d}\right)^3 + \ldots\right]$$

$$\simeq \frac{\in_0 a}{\theta}\left[\frac{a\theta}{d} + \frac{2}{3}\left(\frac{a\theta}{2d}\right)^3\right]$$

$$= \frac{\in_0 a}{\theta}\left[\frac{a\theta}{d} + \frac{1}{12}\,\frac{a^3\theta^3}{d^3}\right]$$

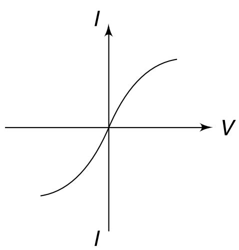
$$= \frac{\in_0 a^2}{d}\left[1 + \frac{a^2\theta^2}{12d^2}\right]$$

CHAPTER 8

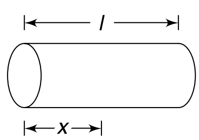
Current Electricity

LEVEL 1

Q. 1: The current – voltage graph for a resistor is as shown in the figure. Is it right to say that resistance decreases as the current through it increases?



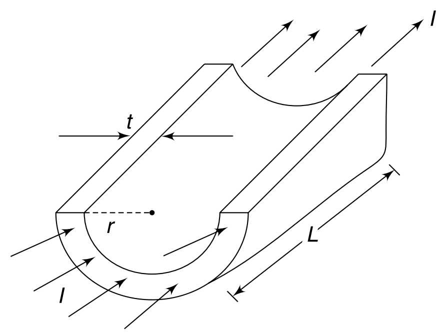
Q. 2: A cylindrical conductor has length l and area of cross section A. Its conductivity changes with distance (x) from one of its ends as $\sigma = \sigma_0 \frac{l}{x}$. [$\sigma_0$ is a constant]. Calculate electric field inside the conductor as a function of x, when a cell of emf V is connected across the ends.



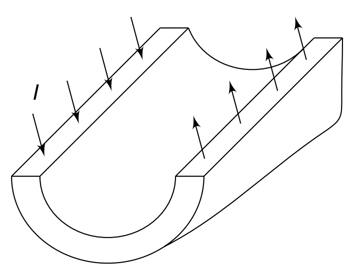
Q. 3: A conductor has density d and molar mass M. A wire made of this conductor has cross sectional radius r. Calculate the drift speed of electrons in this conductor when current through it is I. Assume that each atom contributes one free electron.

Q. 4: A conducting wire of length l and cross sectional area A is used to short the terminals of a cell having emf of ε and internal resistance r. The resistively, density and Molar mass of the material of the wire are r, d and M respectively. Calculate the average time needed for a free electron to travel from positive terminal of the cell to its negative terminal.

Q. 5: A conducting open pipe has shape of a half cylinder of length L. Its semicircular cross section has radius r and thickness of the conducting wall is $t(<<r)$. The resistance of the conductor when the current enters and leaves as shown in Figure (a) is R_1 and its resistance is R_2 when the current is as shown in Figure (b). Find $\frac{R_1}{R_2}$.



(a)



(b)

Q. 6: It was found that resistance of a cylindrical specimen of a wire does not change with small change in temperature. If its temperature coefficient of resistivity is α_R then find its thermal expansion coefficient (α).

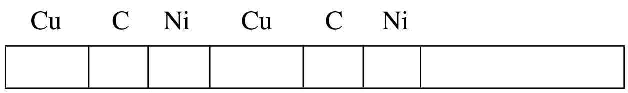
Q. 7: A cylindrical conductor is made so that its resistance is independent of temperature. It is done by stacking layers of copper, carbon and nichrome as shown in Figure. The length of each copper layer is 1 cm and sum of lengths of consecutive carbon and Nichrome Leyers is also 1 cm. Find the length of each Nichrome segment.

Given [ρ = resistivity, α = temperature coefficient of resistivity]

$\rho_{Cu} = 1.7 \times 10^{-8}\ \Omega m^{-1}$; $\rho_C = 5 \times 10^{-5}\ \Omega m^{-1}$

$\rho_{Ni} = 1 \times 10^{-6}\ \Omega m^{-1}$; $\alpha_{Cu} = 3.9 \times 10^{-3}\,°C^{-1}$

$\alpha_C = -5 \times 10^{-4}\ °C^{-1}$; $\alpha_{Ni} = 4 \times 10^{-4}\,°C^{-1}$

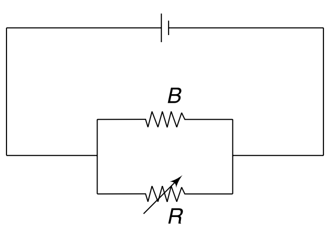


Assume no change in dimensions of the segment due to temperature change.

Q. 8: A bulb B is connected to a source having constant emf and some internal resistance. A variable resistance R is connected in parallel to the bulb. If the resistance R is increased, how will it affect the–

(a) Brightness of the bulb?

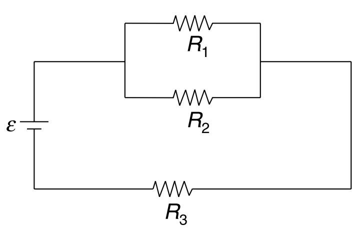
(b) Power spent by the source?



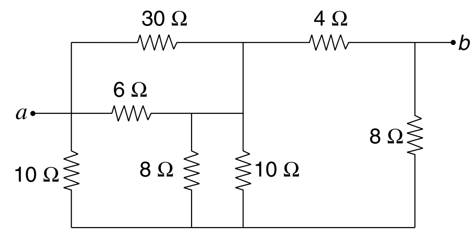
Q. 9: In the circuit shown in the Figure $R_1 = 3\ \Omega$, $R_2 = 2\ \Omega$ and $R_3 = 5\ \Omega$. Emf of the cell is $\varepsilon = 6$ Volt.

(a) Infinitely many 1 Ω resistors are added in parallel to R_1 and R_2. Find current through R_2 and potential drop across it.

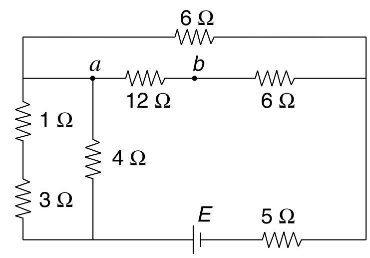
(b) Infinitely many 1 Ω resistors are added in series to R_3. Find the potential drop across R_3 and current through it.



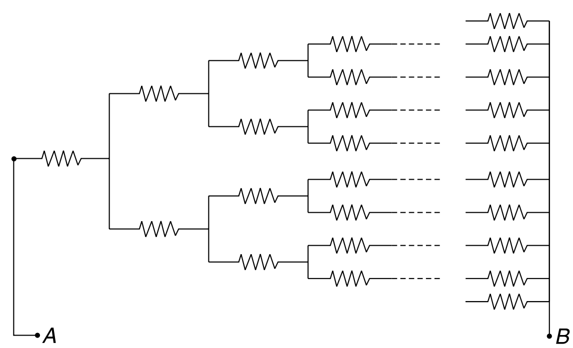
Q. 10: Find the equivalent resistance between point a and b in the network shown in figure.



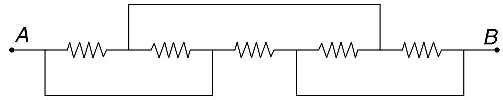
Q. 11: In the circuit shown in the figure, the potential difference between points a and b is $V_a - V_b = 4$ V. Find the emf, E of the battery.



Q. 12: An infinite network of resistances has been made as shown in the Figure. Each resistance is R. Find the equivalent resistance between A and B.



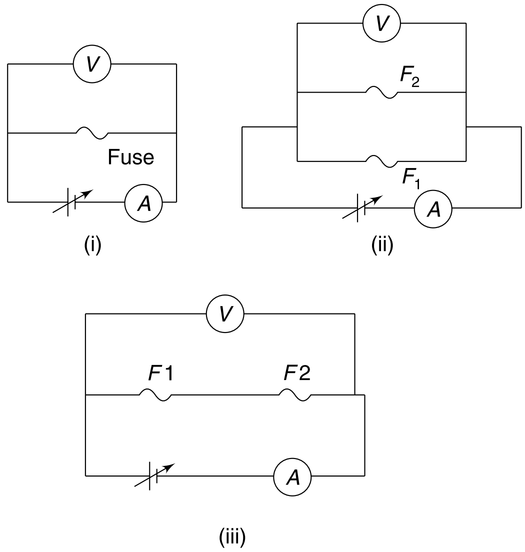
Q. 13: Find equivalent resistance between points A and B in the figure. Each resistance is R.



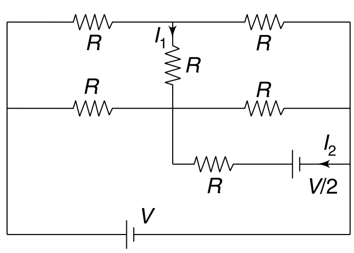
Q. 14: A fuse $F1$ is connected across a source of variable voltage and the voltage is increased gradually. The fuse blows out just when the reading of the voltmeter and ammeter reaches 1.0 V and 1.0 A respectively (see Figure (i)). The experiment is repeated with another fuse $F2$ and the reading of the voltmeter and ammeter when it blows out is 2.4 V and 1.2 A respectively.

(a) The two fuses are connected in parallel as shown in Figure (ii). Voltage is increased gradually. Find the reading of the ammeter when any one of the fuses blows out.

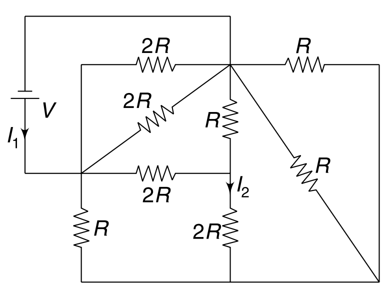
(b) The two fuses are connected in series as shown in Figure (iii). Find the reading of the voltmeter at the point one of the fuses blows out.



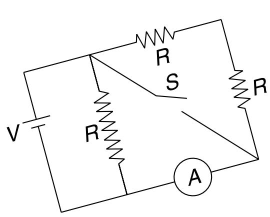
Q. 15: In the circuit shown in the Figure. Find I_1 and I_2.



Q. 16: In the network shown calculate current through the cell (I_1) and the current I_2 through the $2R$ resistance.

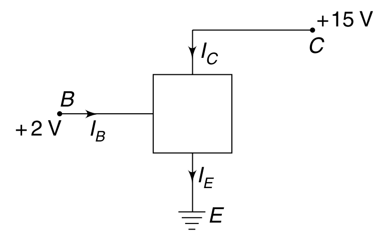


Q. 17: In the circuit shown, $R = 2\ \Omega$ and $V = 20$ volt. With switch S open the reading of ammeter is half its reading when 'S' is closed. Calculate the resistance of the ammeter.

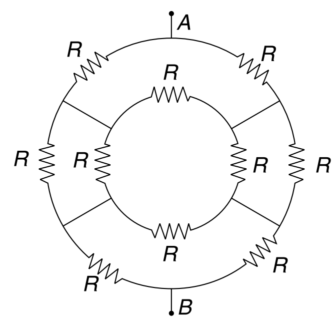


Q. 18: The box shown in the figure has a device which ensures that $I_C = 0.9\ I_E$.

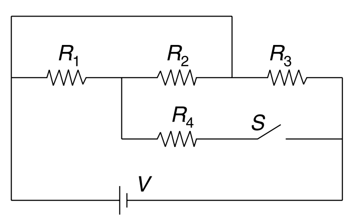
If a small change (ΔI_B) is made in I_B, calculate the corresponding change in I_C.



Q. 19: In the circuit shown in figure find the equivalent resistance across points A and B.



Q. 20: Find the percentage change in power supplied by the cell after the switch 'S' is closed in the circuit shown below. All resistances are identical

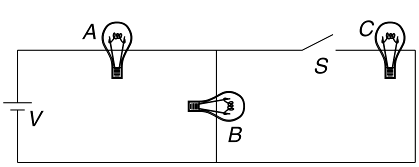


Q. 21: A and B are two identical bulbs of 40 W connected to a $V = 12$ volt cell. Switch S is closed to connect a third bulb C in the circuit. What happens to brightness of bulb A? Answer for two cases:

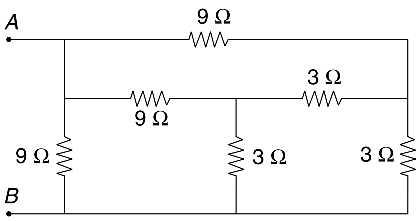
(i) Bulb C is a very high wattage bulb.

(ii) Bulb C is a very low wattage bulb.

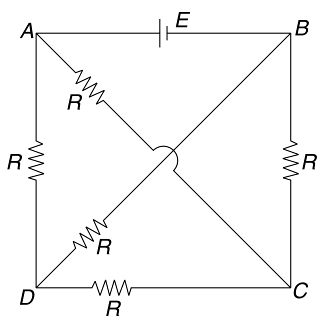
All the three bulbs have rated voltage of 12 volt.



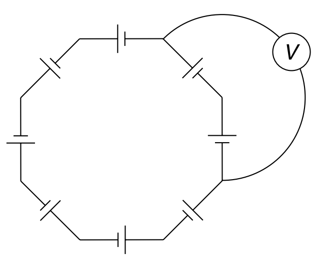
Q. 22: Find equivalent resistance between points A and B in the circuit shown.



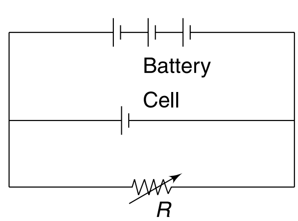
Q. 23: Find current through the cell and potential difference between A and D in the circuit shown in the Figure.



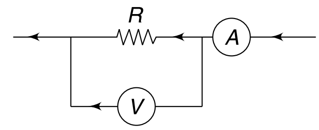
Q. 24: Eight identical 1 volt cells are connected to make a ring as shown in the Figure. An ideal voltmeter is connected as shown. What will be its reading?



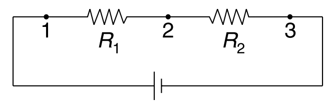
Q. 25: A battery of 120 V and internal resistance $r = 0.5\ \Omega$ is used to charge a 110 V cell in the circuit shown in the figure. Find the range of values of R for which the cell will never get charged.



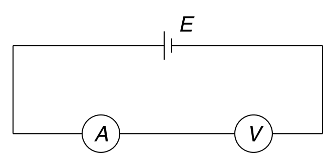
Q. 26: A voltmeter of resistance R_V and an ammeter of resistance R_A are connected as shown in an attempt to measure the resistance R. The measured value of the resistance is $R_M = \dfrac{V}{I_0}$ where V is reading of voltmeter and I_0 is reading of the ammeter. Find the true value of the resistance in terms of R_M, R_V and R_A.



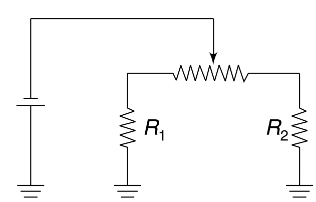
Q. 27: In the circuit shown in the Figure, cell is ideal and $R_2 = 100\ \Omega$. A voltmeter of internal resistance 200 Ω reads $V_{12} = 4$ V and $V_{23} = 6$ V between the pair of points 1-2 and 2-3 respectively. What will be the reading of the voltmeter between the points 1-3.



Q. 28: In the circuit shown, an ideal cell of emf E is connected in series to a non-ideal ammeter and voltmeter. Reading of the voltmeter is V_0. When a resistance is added in parallel to the voltmeter its reading becomes $\dfrac{V_0}{10}$ and the reading of the ammeter becomes 10 times the earlier value. Find V_0 in terms of E.



Q. 29: In the circuit shown, which way would you move the sliding contact, to the left or to the right, in order to increase current through resistance R_1? What will happen to current through R_2 as you move the contact?

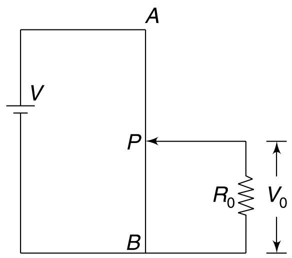


Q. 30: In the circuit shown in Figure. AB is a uniform wire of length L and resistance R. P is a sliding contact. Take the ratio of lengths PB to AB as α.

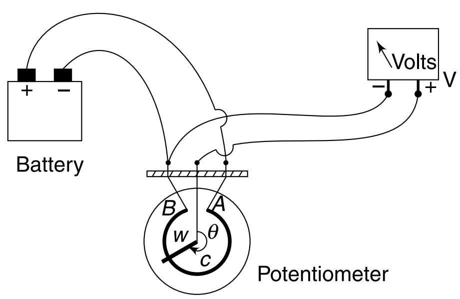
(a) Find the ratio $\dfrac{V_0}{V}$ in terms of R_0, R and α.

(b) Predict the value of $\dfrac{V_0}{V}$ when $R_0 \to \infty$. Use the result obtained in (a) to show that your prediction is correct.

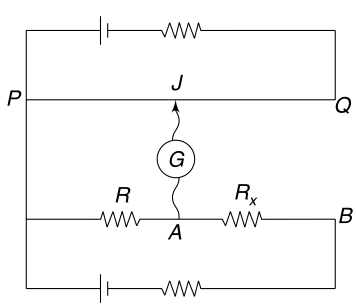
(c) Find the ratio $\frac{V_0}{V}$ when $R_0 = 2R$ and $\alpha = \frac{1}{2}$.



Q. 31: A rotary potentiometer has a circular resistance (C) and a conducting wiper (W) can rotate over it. An ideal battery and an ideal voltmeter are connected to it as shown in the figure. As the wiper is rotated on the circular resistance the reading of the voltmeter changes from zero to 20 V. What is reading of the voltmeter when wiper is at angular position θ = 120°? Assume that resistance C is almost a complete circle and resistance of all connecting wires and knobs in the circuit is zero.



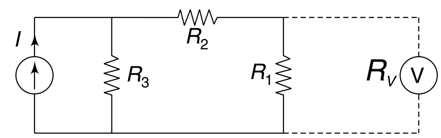
Q. 32: In the figure shown PQ is a potentiometer wire. When galvanometer is connected at A, it shows zero deflection when $PJ = x$. Now the galvanometer is connected to B and it shows zero deflection when $PJ = 3x$. Find the value of unknown resistance R_X in term of R.



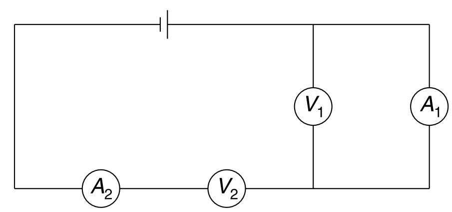
Q. 33: Three resistors R_1 = 3 kΩ, R_2 = 2 kΩ and R_3 = 5 kΩ have been connected to a constant current source as shown in figure. The current source supplies current I = 2 mA to the circuit. A voltmeter with R_V = 6 kΩ internal resistance is connected, as shown, to measure the potential difference across R_1.

(a) Find the percentage error in the measurement of potential difference (V_1) across. R_1 caused due to finite resistance of the voltmeter.

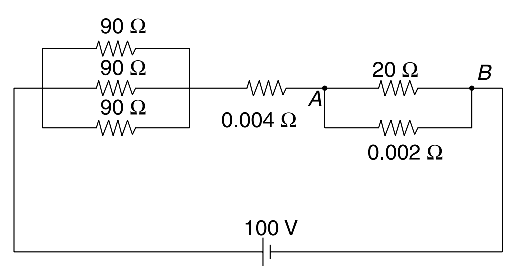
(b) If positions of R_2 and R_3 are interchanged will the percentage error in measurement of V_1 increase or decrease?

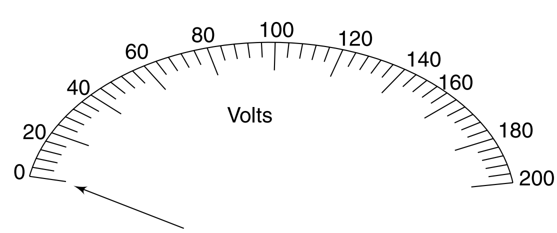


Q. 34: In the circuit shown in Figure the voltmeters V_1 & V_2 are identical. If readings of ammeter A_2, Voltmeter V_2 and V_1 are 400 μA, 100 V and 2 V respectively, find the resistance of ammeter A_1. Is the value realistic?



Q. 35: A lab voltmeter has scale as shown in Figure. A student uses this nearly ideal voltmeter to record potential difference between points A and B in the circuit shown. He could not see any deflection in the pointer and thinks that the voltmeter must be faulty. He replaces the voltmeter with another similar one. What reading will he record this time? Can you give some suggestion to record the potential difference across A and B?





Q. 36: A 10 wire potentiometer has first five wires of cross sectional radius r and next five wires of radius $2r$. The wires are uniform and made of same material. An ideal cell of emf 2V is connected across the potentiometer wire. What length of the potentiometer wire will balance the emf of–

(a) a Daniell cell (emf = 1.0 V)

(b) a Lechlanche cell (emf = 1.5 V)

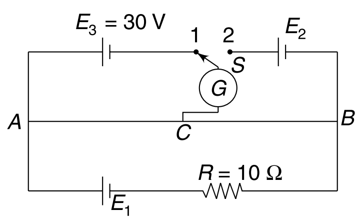
(c) a cell of emf 1.8 V?

Length of each wire is 100 cm.

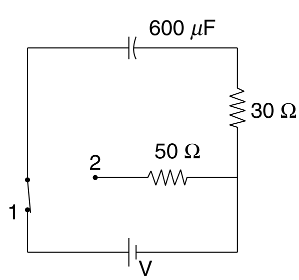
Q. 37: A water heater has a well insulated vertical cylindrical container of radius a in which water is filled to a height h. A resistor made of an allow is used to heat the water in the tank from a temperature θ_1 to θ_2 ($>\theta_1$) in a time interval Δt. The resistor wire has cross sectional radius b and its alloy material has resistivity ρ. Calculate the length of the resistor wire.

Density and specific heat capacity of water are d and s respectively. The power source connected to the resistor has emf ε.

Q. 38: In the circuit shown in Figure AB is a uniform wire of length L = 5m. It has a resistance of 2 Ω/m. When AC = 2.0 m, it was found that the galvanometer shows zero reading when switch s is placed in either of the two positions 1 or 2. Find the emf E_1.



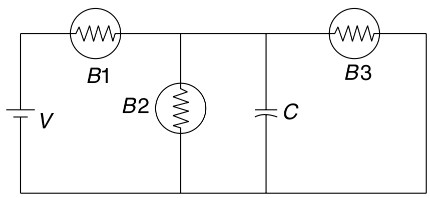
Q. 39: In the circuit shown, after the switch is shifted to position 2 the heat generated in 50 Ω resistance is 6 J. Find the emf (V) of the cell.



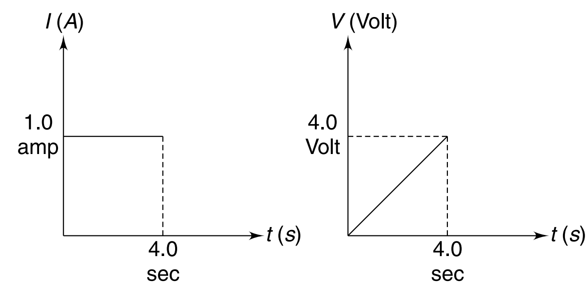
Q. 40: In the circuit shown in Figure $B1$, $B2$ and $B3$ are identical bulbs and C is a parallel plate capacitor.

(a) Which bulb is brightest?

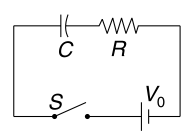
(b) If separation between the plates of the capacitor is increased very slowly to double its value, how does the brightness of each of the bulbs change?



Q. 41: When the applied potential difference across a circuit element is increased linearly with time as shown in the second figure, the current in the circuit remains constant with time as shown in the first Figure. At t = 4 s the lead wire connecting the element to the source breaks. What will be the voltage across the element after t = 4 s? Can you identify the element?



Q. 42: In the circuit shown the switch is closed at time t = 0. Plot the following graphs:



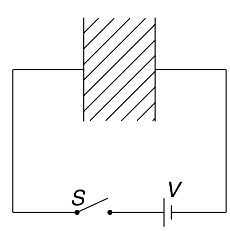
(i) current (I) in the circuit Vs time (t)

(ii) voltage across the capacitor (V_C) Vs 't'

(iii) power absorbed by the capacitor Vs 't'.

Q. 43: A capacitor having initial change q_0 and capacitance C is connected to a variable resistance R. The resistance R can have values ranging from O to R_0. How should we vary R with time(t) such that the discharging current remains constant with time?

Q. 44: A parallel plate capacitor has plate area A and separation between the plates equal to d. A material of dielectric constant K and resistivity ρ is filled between the plates. The switch is closed to connect the capacitor to a cell of emf V.

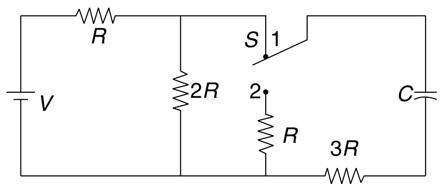


(a) Write the steady state current in the circuit and charge on the capacitor.

(b) When the circuit is in steady state, switch s is opened (at t = 0). Write charge on the capacitor as function of time (t) after this.

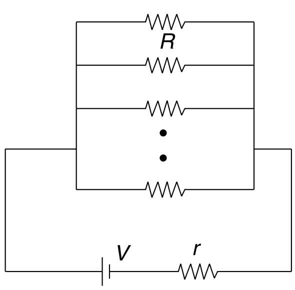
Q. 45: Consider the circuit shown in the figure. The switch has been in position 1 for a long time. Answer following questions:

(a) Find current through $2R$.
(b) Find potential difference across the capacitor.
(c) At $t = 0$ the switch is moved to position 2. What current flows through the capacitor immediately after the switch is placed to position 2?
(d) Draw a graph of current *Vs* time for the current through the capacitor after the switch is moved to position 2. Indicate the time on the graph when the current becomes 37% of its value immediately after the switch is put to position 2.



Q. 46: An electric bulb has a solid cylindrical filament of length l and radius r and it consumes power P when connected to a power source. Another bulb having cylindrical filament of same material, operating at same voltage and emitting the same spectrum of light consumes 8 P power. Find length and radius of the new filament. Assume that the filaments do not radiate from the flat ends and radiation is the only source of heat loss from the filaments.

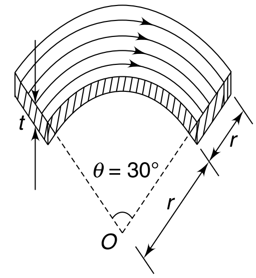
Q. 47: A battery of emf V and internal resistance r is connected to N identical bulbs, all in parallel. Resistance of each bulb is R. It is observed that maximum cumulative power is dissipated in the bulbs if $N = 10$. Can we say that $r = \frac{R}{10}$.



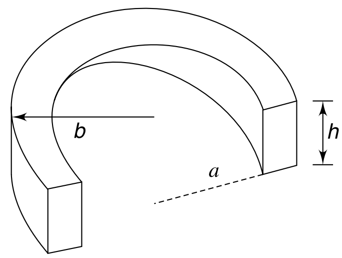
LEVEL 2

Q. 48: A copper wire of length 5 cm carries a current of density $j = 1\,\text{A}\,(\text{mm})^{-2}$. The density and molar mass of copper are 9000 Kg m^{-3} and 63 gmol^{-1}. Each copper atom contributes one free electron. The temperature of the wire is 27°C. Estimate the (average) distance travelled by a free electron during the time it moves from one end of the copper wire to the other end. Assume that thermal motion of electrons are similar to that of molecules of an ideal gas.

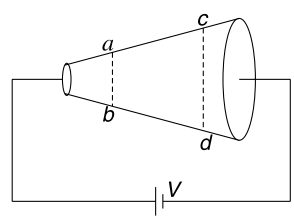
Q. 49: A conducting plate of thickness t is in the shape of an arc. Its inner and outer rim form a 30° arc of radius r and $2r$ respectively at point O (see Figure). Electric current flows through the strip along circular arcs as indicated in the diagram. Find resistance of the plate if its material has conductivity σ.



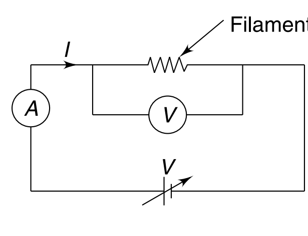
Q. 50: A conductor having resistivity ρ is bent in the shape of a half cylinder as shown in the figure. The inner and outer radii of the cylinder are a and b respectively and the height of the cylinder is h. A potential difference is applied across the two rectangular faces of the conductor. Calculate the resistance offered by the conductor.



Q. 51: A metallic wire has variable cross sectional area. Cross sectional area at cd is twice the area at ab. The wire is connected to a cell as shown. Find the ratio of heat dissipated per unit volume at section cd to that at section ab.

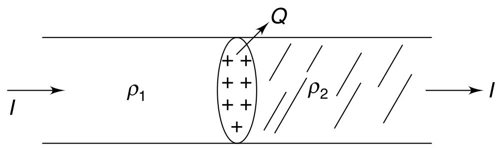


Q. 52: A tungsten filament bulb is connected to a variable voltage supply. The potential difference, V is varied and the current, I and steady temperature T of the filament is recorded. A graph is plotted for $\ln(VI)$ *Vs* $\ln(T)$. Find the slope of the graph. Assume that temperature of filament $T >> T_0$ = atmospheric temperature.



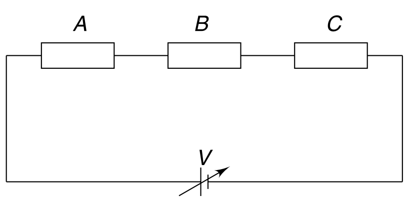
Q. 53: A tungsten filament, carrying current, has a constant diameter except a part of it which has half the diameter of the rest of the wire. If the temperature of thick part is 2000 K, then calculate the temperature of the thin part of the filament. Assume that the temperature is constant within each part and changes suddenly between the parts. Also assume that the temperature of the surrounding is very small compared to temperature of any part of the filament.

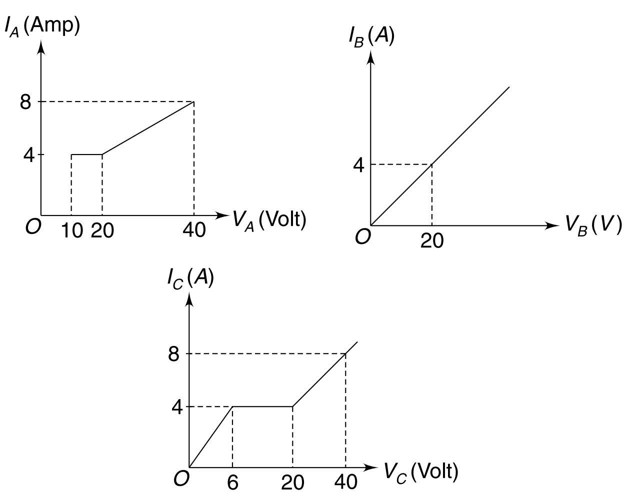
Q. 54: Two cylindrical rods, of different material, are joined as shown. The rods have same cross section (A) and their electrical resistivities are ρ_1 and ρ_2. When a current I is passed through the rods, a charge (Q) gets piled up at the junction boundary. Assuming the current density to be uniform throughout the cross section, calculate Q.



Under what condition the charge Q is negative?

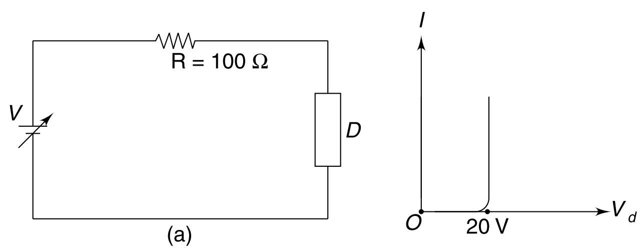
Q. 55: The current (I) – voltage (V) characteristic of three devices A, B and C connected in the circuit is as shown below.





Plot the variation of current through the cell when its emf is changed from 0 to 90 V.

Q. 56: The current – voltage characteristic of an electric device is as shown in figure (b). The device gets damaged if power dissipated in it exceeds 1 Watt. This device is connected to a dc source of variable emf (V) and a resistance (R = 100 Ω) in series. What is possible range of V for which the device remains operational (*i.e.* consumes some power) and safe.

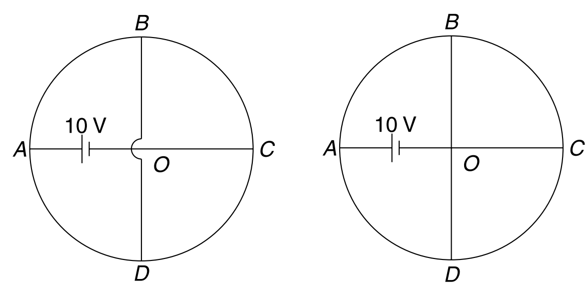


(b)

Q. 57: $ABCD$ is a uniform circular wire of resistance 16 Ω and AOC and BD are two uniform wires forming diameters at right angles, each of resistance 2 Ω. The two straight wires do not touch each other at O. A battery of emf 10 V is placed in AO as shown.

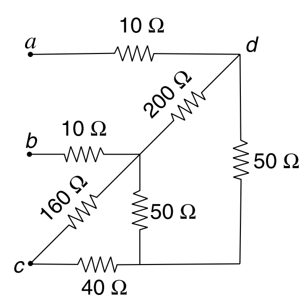
(a) Find the current through the battery (Figure a)

(b) If the straight wires are tied at O so as to form a junction, find the current through the battery (Figure b)

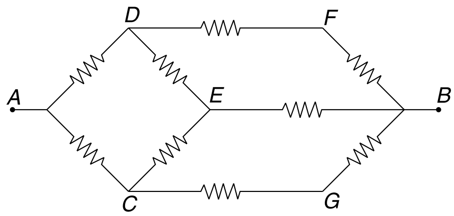


(a) (b)

Q. 58: In the circuit shown in the fig the equivalent resistance between a and b is R_{ab} and the equivalent between a and c is $R_{ac.}$ Find the ratio $R_{ab}:R_{ac}$.



Q. 59: (a) In the circuit shown in figure, all resistances are identical when a 5V supply is connected across AB the current in branch CG is 3 mA. Find the effective resistance of the circuit between A and B.



(b) Twelve equal resistances, each equal to R, are placed along the sides of a cube. Find equivalent resistance across.

(i) Diagonally opposite points of the cube.

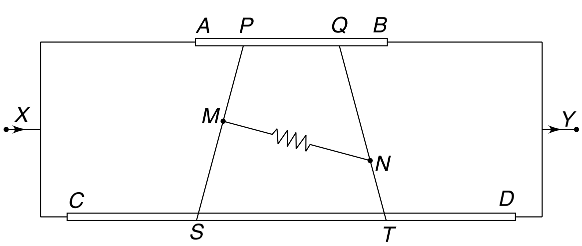
(ii) Diagonally opposite points on one face of the cube.

(iii) End points of a side of the cube.

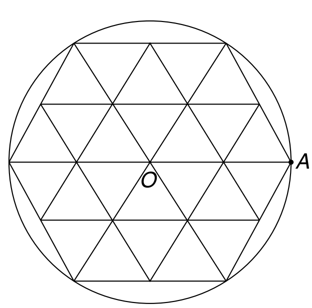
Q. 60: AB and CD are two resistance wires cut from a uniform long wire. Lengths of AB and CD are L and $2L$ respectively and resistance of AB is R. The two resistances are connected in parallel to a supply. P and Q are two points on the resistance AB such that $AP = QB = \frac{L}{3}$. Two conductors PS and QT are connected between two resistors such that no current flows through both the conductors. A resistance R is connected between points M and N as shown. Neglect resistance of PS and QT.

(a) Find the equivalent resistance of the circuit between X and Y.

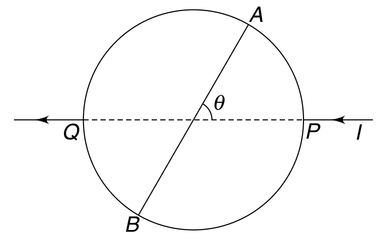
(b) Will there be any current in resistance connected across MN?



Q. 61: In the figure, each segment (side of small triangle) has resistance R and the wire used in the circumference of the circle has negligible resistance. Find equivalent resistance between point O and A.



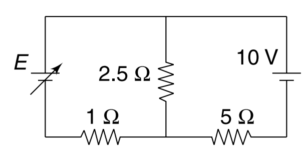
Q. 62: A uniform conducting wire is in the shape of a circle. The same wire has been used to make its diagonal AB. A current I enters at point P and leaves at the diagonally opposite point Q. AB makes an angle θ with the line PQ. Find current (i), through AB as a function of θ. Plot a graph showing variation of i with $\theta\,(0° \le \theta \le 90°)$



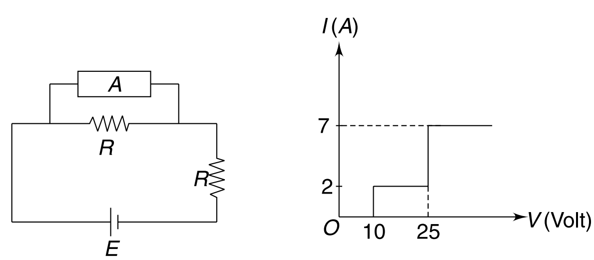
Q. 63: In the circuit shown in Figure the emf E of the battery is increased linearly from zero to 28 V in the interval $0 \le t \le 14$ s.

(a) Find the energy gained by the 10 V cell in the interval $0 \le t \le 14$ s.

(b) At what time the 10 V cell begin to charge?



Q. 64: In the circuit shown in figure (a), the emf of the ideal cell is $E = 100$ V and resistance R is 10 Ω. The current (I) – Voltage (V) characteristic of the circuit contained in box A is as shown in figure (b). Find the potential difference across the box A.

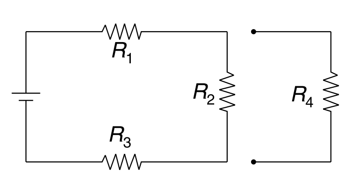


(a) (b)

Q. 65: In the circuit shown in the Fig., $R_1 : R_2 : R_3 = 4 : 1 : 2$.

(a) Will the current through R_1 increase or decrease when a new resistance R_4 is added in parallel to R_2?

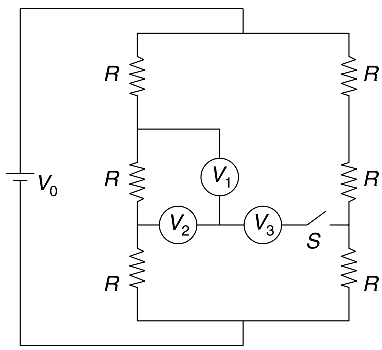
(b) Change in current through R_1 when R_4 is added is found to be $0.2A$. Calculate the current through R_4.



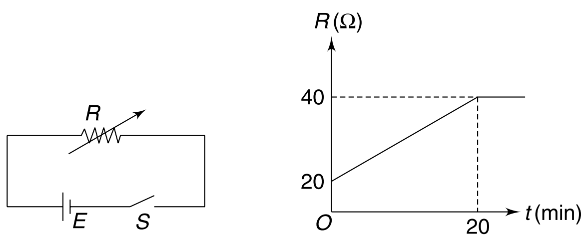
Q. 66: In the circuit shown in figure all resistances are identical (each equal to R) and the cell has an emf of V_0. The three voltmeters V_1, V_2 and V_3 are identical and are nearly ideal.

(a) Find the reading of the voltmeter V_1 when switch 'S' is open.

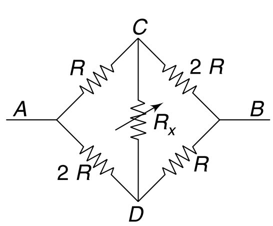
(b) Find the reading of the voltmeter V_1 after the switch is closed.



Q. 67: A chemical cell of emf E has negligible internal resistance. It is connected to a variable resistance (R) which changes linearly from 20 Ω to 40 Ω in 20 minute and thereafter becomes constant. It was found that the cell lost 10% of its total chemical energy in first 20 minute after the switch was closed. How long will the energy in the cell last?



Q. 68: In the circuit shown in the Figure R_X is a variable resistance. Find the equivalent resistance (R_{AB}) between A and B in terms of R and R_X. What are the possible range of values of R_{AB}?

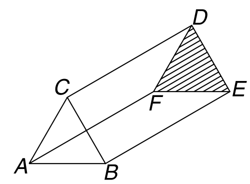


Q. 69: Six identical wires each of resistance R ohm are connected to form the edges of a tetrahedron. Find the equivalent resistance between any two vertices.

Q. 70: A prism shaped network of resistors has been shown in the figure. Each arm (like AB, AC, CD, DF...) has resistance R. Find the equivalent resistance of the network between

(a) A and B

(b) C and D.



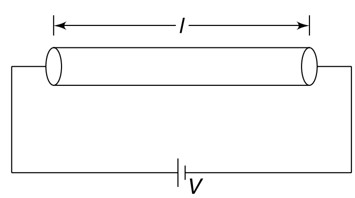
Q. 71: A cylindrical conductor has a resistance R. When the conductor is at a temperature (T) above its surrounding temperature (T_0), the ratio of thermal power dissipated by the conductor to its excess temperature ($\Delta T = T - T_0$) above surrounding is a known constant k. The conductor is connected to a cell of emf V. Initially, the conductor was at room temperature T_0. Mass and specific heat capacity of the conductor are m and s respectively.

(i) Find the time (t) dependence of the temperature (T) of the conductor after it is connected to the cell. Assume no change in resistance due to temperature.

(ii) Find the temperature of the conductor after a long time.

Q. 72: A conductor in the shape of a cylinder of length ℓ and cross sectional radius r is connected to a cell of emf V. The resistivity of the material of the conductor is ρ and does not change much with temperature. The emissivity of the curved surface of the conductor is e. [Take emissivity of the flat circular surfaces to be zero]. In steady state the temperature of the conductor is T when the environmental temperature is T_0. The difference between T and T_0 is much smaller than the environmental temperature. Stefan's Constant is σ.

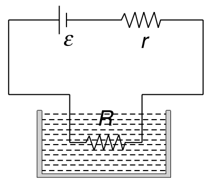
Find the steady state temperature T for the conductor.



Q. 73: In order to heat a liquid an electric heating coil is connected is to a cell of emf $E = 12$ V and internal resistance $r = 1$ Ω. There are three options for selecting the resistance (R) of the heating coil. R can be chosen as 1 Ω, 2 Ω or 4 Ω. The cell has a rating of 2000 mAh (milli Ampere hour) and it is to be used to heat the liquid till it expires. [The cell maintains constant emf till it lasts]

(a) Which value of R will you chose so that maximum heat (H_0) is transferred to the liquid before the cell expires? Calculate H_0.

(b) Which value of R will chose so that heat is transferred to the liquid at fastest possible rate? What percentage of H_0 (as obtained in (a)) is transferred to the liquid in this case by the time the cell expires?

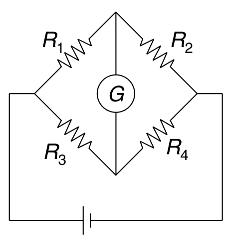


Q. 74: In a wheat stone bridge experiment to determine the unknown resistance R_4, the values of R_1 and R_2 were taken to be 10 Ω and 1 Ω respectively. It was found that the galvanometer will show exact zero deflection when value of R_3 is taken as 643 Ω < R_3 < 644 Ω.

Now R_1 is changed to 100 Ω (R_2 remains unchanged).

(a) If you have been asked to obtain a balanced bridge, which values of R_3 will you try with?

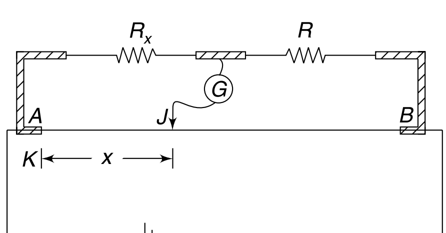
(b) If balance is obtained for 6432 Ω < R_3 < 6433 Ω write the measured value of unknown resistance R_4.



Q. 75: Figure shows an experimental set up to find the value of an unknown resistance (R_x) using a meter bridge. AB is the uniform meter bridge wire of length L = 100 cm. When the sliding jockey is placed at J ($AJ = x$), the galvanometer shows zero deflection. $AJ = x$ is known as balancing length and is measured using a scale having 1 mm as least count.

(a) In one experiment known resistance R was taken to be 20 Ω and balancing length was measured as x = (20.0 ± 0.1) cm. Find the value of R_x.

(b) Show that fractional error in calculated value of R_x is least when $x = \frac{L}{2}$. What shall we do to ensure that x is close to $L/2$?



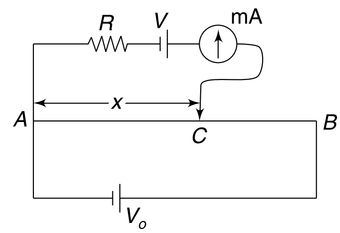
Q. 76: AB is a uniform wire of length L = 100 cm. A cell of emf V_0 = 12 volt is connected across AB. A resistance R, cell of emf V and a milliammeter (which can show deflection in both directions] is connected to the circuit as shown. Contact C can be slid on the wire AB. Distance $AC = x$. The current (I) through the milliammeter is taken positive when the cell of emf V is discharging. A graph of I Vs x has been shown.

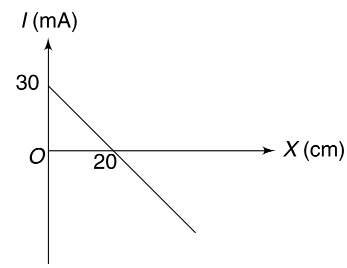
Neglect internal resistance of the cells.

(a) Find V

(b) Find R

(c) Find I when x = 100 cm

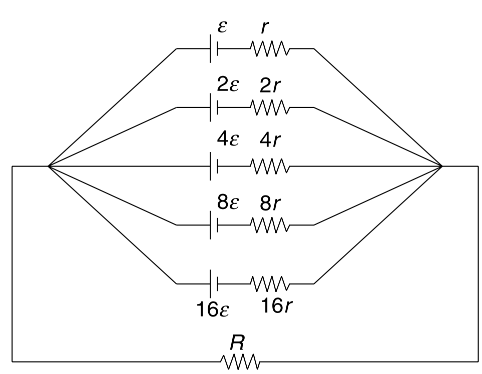




Q. 77: Five cells have been connected in parallel to form a battery. The emf and internal resistances of the cells have been shown in figure. A load resistance R is connected to the battery.

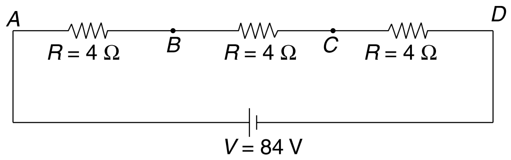
(a) Which of the 5 cells will have maximum current flowing through it?

(b) Find the current through load resistance R.

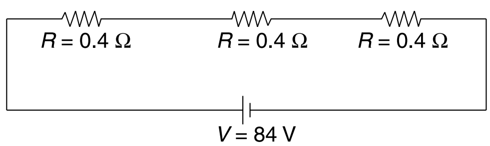


Q. 78: In the circuit shown, when a voltmeter is connected across any one of the three resistances, it shows a reading of 24 V.

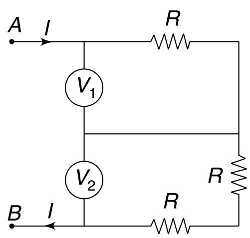
(a) Find the reading of the voltmeter when it is connected between points A and C.



(b) The same voltmeter is used to measure potential difference across resistances shown in figure below. Will the voltmeter be more accurate this time?



Q. 79: In the circuit shown in figure a current $I = 600\ \mu A$ enters through A and leaves through B. Reading of the identical voltmeters V_1 and V_2 are 20 V and 30 V respectively. Find R.

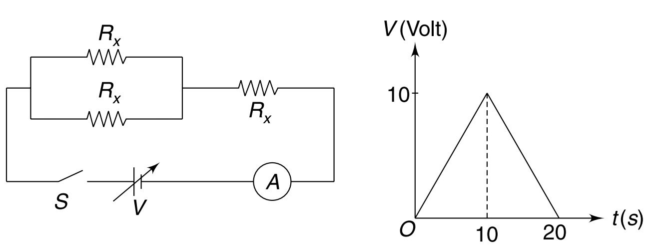


Q. 80: In the circuit shown, each resistor has a resistance R_X which depends on the voltage V_X across it.

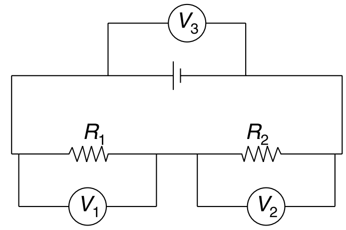
For $\quad V_X \leq 1$ V, $R_X = 1\ \Omega$

and for $\quad V_X > 1$ V, $R_X = 2\ \Omega$.

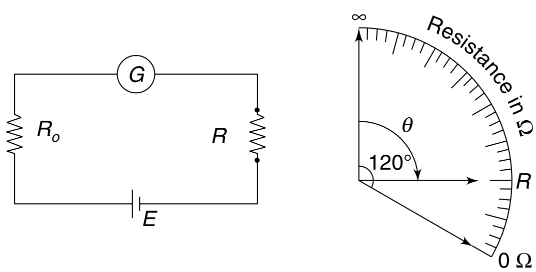
The emf (V) of the source, changes with time (t) after the switch is closed at $t = 0$. The variation of V with time is depicted in the graph. Plot the variation of ammeter reading with time.



Q. 81: In the circuit shown in the figure, two resistors R_1 and R_2 have been connected in series to an ideal cell. When a voltmeter is connected across R_1 its reading is $V_1 = 4.0$ volt and when the same voltmeter is connected across R_2 its reading is $V_2 = 6.0$ volt. The reading of the voltmeter when it is connected across the cell is $V_3 = 12.0$ volt. Find the actual voltage across R_1 in the circuit.



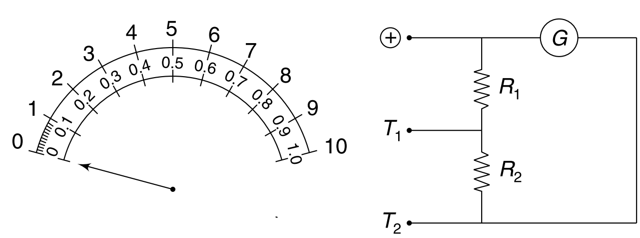
Q. 82: An ohm-meter is a device that measures an unknown resistance. A simple ohm-meter can be constructed using a galvanometer as shown in the figure. The cell used in the circuit has emf $E = 20$ volt. The full scale deflection current and resistance of the galvanometer are 2 mA and 20 Ω respectively. R_0 is a fixed resistance and R is the unknown resistance whose value is directly given by the galvanometer scale. The galvanometer scale is shown in figure. When an unknown resistance R is placed in the circuit, the pointer deflects by $\theta = 90°$. Find R.



Q. 83: To enhance the sensitivity, an Ammeter is to be designed with two kinds of graduation on its scale — 0 to 10 A and 0 to 1 A. For that a galvanometer of resistance 50 Ω and full scale deflection current 1 mA was used along with two resistances R_1 and R_2 as shown. Either of T_1 or T_2 is to be used as negative terminal of the Ammeter.

(a) When measuring a current of the order of 0.1 A, which shall be used as negative terminal — T_1 or T_2?

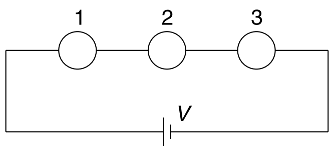
(b) Find the values of R_1 and R_2.

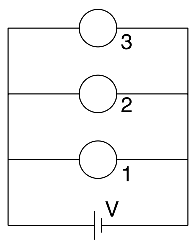


Q. 84: Three ammeters — 1, 2 and 3 have different internal resistances r_1, r_2 and r_3 respectively. Internal resistance r_1 is known but r_2 and r_3 are unknown. The angular deflection of pointer in each ammeter is proportional to the current. Initially, the three ammeters were connected in series to a voltage source (Fig. a) and deflections for the three ammeters were θ_1, θ_2 and θ_3 respectively. The three ammeters were then connected in parallel to the same voltage source (Fig. b). This time the deflections were observed to be θ_1', θ_2' and θ_3' respectively.

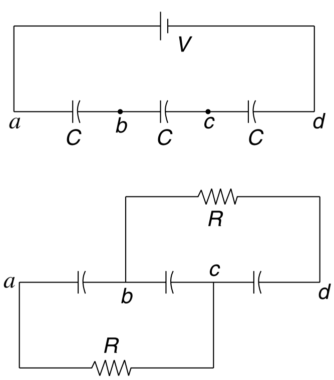
(a) Find r_2 and r_3.

(b) If $\theta_2 = \theta_3$ but $\theta_3' > \theta_2'$ then which one is larger r_2 or r_3?





Q. 85: Three identical capacitors, each of capacitance C are connected in series. The capacitors are charged by connecting a battery of emf V to the terminals (a and d) of the circuit. Now the battery is removed and two resistors of resistance R each are connected as shown. Find the heat dissipated in one of the resistors.



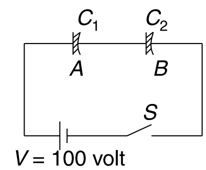
Q. 86: Assume that clouds are distributed around the entire earth at a height of 3000 m above the ground. The atmosphere can be modeled as a spherical capacitor with the earth as one plate and the cloud as other. When the electric field between the earth and the cloud becomes large, the air begins to conduct and the phenomena is called lightning. On a typical day 4×10^5 C of positive charge is spread over the surface of the earth and equal amount of negative charge is there on the cloud. Resistivity of the air is $\rho = 3 \times 10^{13}$ Ωm and radius of the earth = 6000 km.

(a) Find the resistance of the air gap between the earth's surface and cloud.

(b) Estimate the potential difference between the surface of the earth and the cloud.

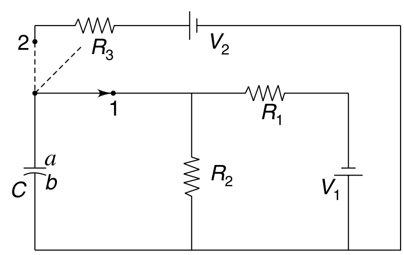
(c) In how much time the capacitor formed between the earth and the cloud will lose 63% of the charge?

Q. 87: The capacitor A shown in Fig. has a capacitance $C_1 = 3\ \mu$F. The dielectric filled in it has a breakdown voltage of 40 V and it has a resistance of 3 MΩ. The capacitor B has a capacitance of $C_2 = 2\ \mu$F and dielectric in it has a resistance of 2 MΩ. Breakdown voltage for B is 50 V. The switch is closed at $t = 0$. Will there be breakdown of any capacitor after the switch is closed? If yes, which will breakdown first and at what time?

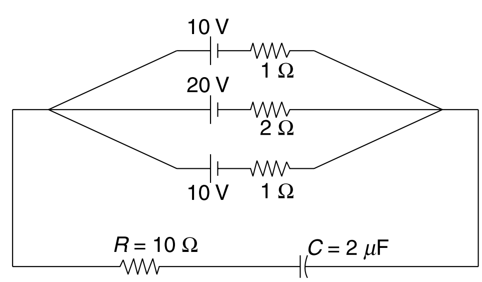


Q. 88: In the circuit shown in fig. the switch is kept closed in position 1 for a long time. At time $t = 0$ the switch is moved to position 2. Write the dependence of voltage (V_C) across the capacitor as a function of time (t). Take V_C to be positive when plate a is positive.

Given: $R_1 = 20\ \Omega$, $R_2 = 60\ \Omega$, $R_3 = 400$ kΩ, $V_1 = 40$ V, $V_2 = 90$ V and $C = 0.5\ \mu$F.

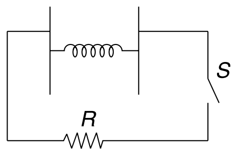


Q. 89: Find the charge on the capacitor in the circuit shown in Fig.



Q. 90: A parallel plate capacitor has its two plates connected to an ideal spring of force constant K. Relaxed length of spring is L and it is made of non conducting material. The area of each plate is A. The capacitor has a charge q_0 on it. To discharge the capacitor through the resistance R, switch S is closed.

If the time constant of the circuit is very large and discharge process is very slow, how much heat will be dissipated in the resistance? Assume that there is no friction and the plates always remains parallel to each other.

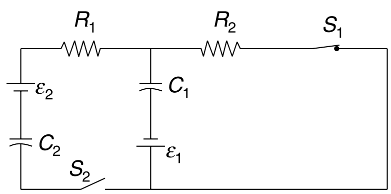


Q. 91: In the last problem calculate the amount of heat dissipated in the resistance assuming that the time constant of the circuit is very small and the discharge process is almost instantaneous.

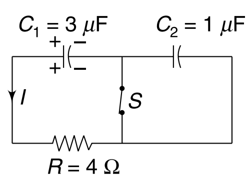
Q. 92: In the circuit shown in the figure $R_1 = R_2 = 5\ \Omega$, $C_1 = C_2 = 2\ \mu F$ and $\varepsilon_1 = \varepsilon_2 = 5$ V. Switch S_1 is kept closed for a long time. Now switch S_2 is also closed. Immediately after S_1 is closed, find

(a) current through R_1

(b) current through R_2



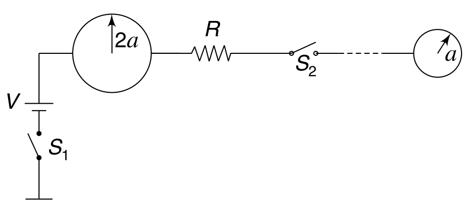
Q. 93: A charged capacitor ($C_1 = 3\ \mu F$) is getting discharged in the circuit shown. When the current I was observed to be $2.5A$, switch 'S' was opened. Determine the amount of heat that will be liberated in the circuit after 'S' is opened.



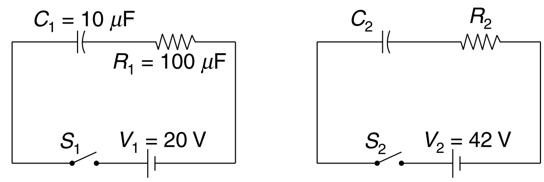
Q. 94: In the Fig. two neutral spherical conductors of radii $2a$ and a are separated by a large distance. Initially, switch S_1 is kept closed and S_2 is open. Now S_1 is opened and S_2 is closed at $t = 0$.

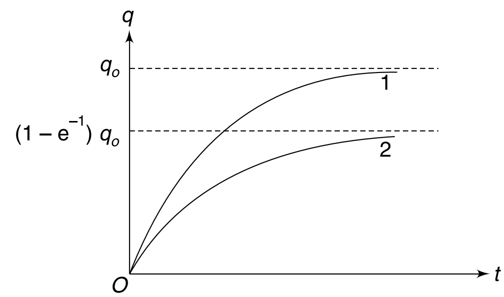
(a) Find the rate of fall in potential of the conductor of radius $2a$ as a function of time.

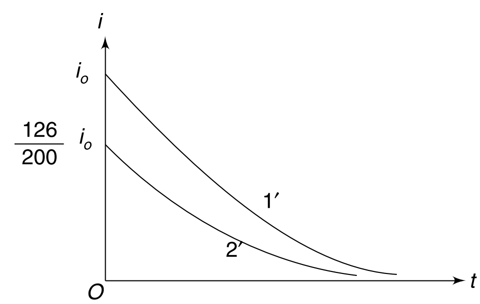
(b) Find the heat dissipated after S_2 is closed.



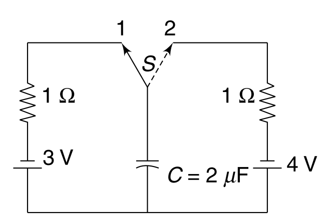
Q. 95: Consider two circuits given below. When switches S_1 and S_2 are closed at $t = 0$, the charge on capacitors C_1 and C_2 change with time as shown in graph 1 and 2 respectively. The current i_1 and i_2, in the two circuits change as shown in graph 1′ & 2′ respectively. Write the variation of current in the second circuit as function of time after the switch is closed at $t = 0$. $e^{-1} = 0.37$







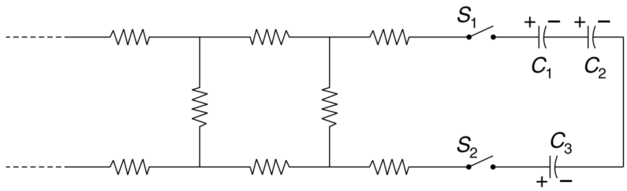
Q. 96: In the circuit shown in Fig. the capacitor is initially uncharged. Two way switch (s) is placed in position 1 for a very short interval of time (Δt) and then is moved to position 2. The switch is held in position 2 for equally short interval Δt and is then moved back to position 1. The process is repeated a large number of times until the charge on the capacitor stops changing. Find this final a value of charge on the capacitor. [Assume that in each contact the charge on the capacitor changes by a very small amount]



Q. 97: An infinite ladder network consisting of all equal resistances, $r = \dfrac{10}{2.732}\ \Omega$ is placed side by side to a capacitor system as shown in fig. Initially, all the switches are kept open and all the three capacitors are given equal charges of 30 μC each. The capacitances are $C_1 = 3\ \mu$F, $C_2 = 6\ \mu$F and $C_3 = 6\ \mu$F. Polarity of charges on the capacitor plates is shown in the Fig. Now all the three switches are closed simultaneously.

(a) Find the magnitude of rate of change of charge on the plates of the capacitors immediately after the switches are closed.

(b) Calculate heat generated in the circuit by the time steady sate condition is established.

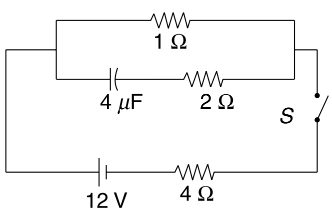


Q. 98: In the circuit shown in the figure, switch S is closed at time $t = 0$. Charge on positive plate of capacitor is q at time t.

(a) Derive a differential equation for q at time t.

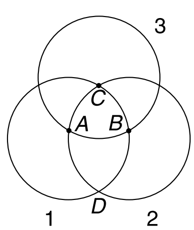
(b) Solve the equation to write q as a function of time.

(c) Put $t = 0$ and $t = \infty$ in your equation to get charge on the capacitor at these times.

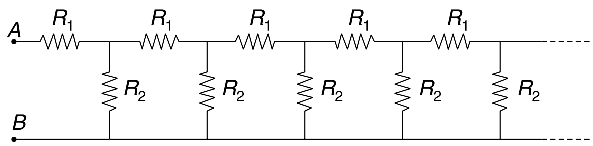


LEVEL 3

Q. 99: Three identical wire rings have been placed symmetrically as shown in the figure A, B and C are centres of the three rings. Resistance of each ring is $3R$. Find the equivalent resistance of this wire mesh across points C and D.

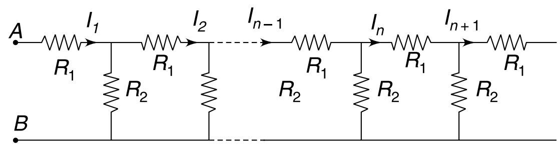


Q. 100: (a) The Fig shows a network consisting of an infinite number of pairs of resistors $R_1 = 2\ \Omega$ and $R_2 = 1\ \Omega$. Since the network is infinite, removing a pair of R_1 and R_2 from either end of the network will not make any difference. Using this calculate the equivalent (R) across points A and B.



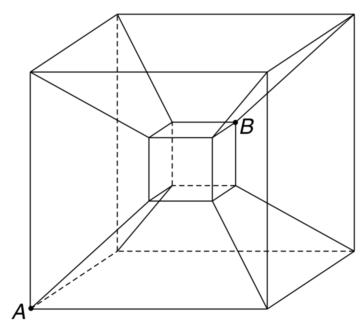
(b) Prove that $I_n = \dfrac{I_{n-1}R_2}{R_2 + R} = \dfrac{I_{n-1}}{\sqrt{3} + 2}$

Where I_n and I_{n-1} represent the current through R_1 in n^{th} and $(n - 1)^{\text{th}}$ segment respectively [see Fig]



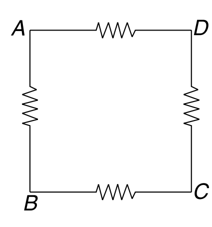
(c) If a 20 V battery is connected across A and B find I_{10}.

Q. 101: Consider the double cube resistor network shown in Fig. Each side of both cubes has resistance R and each of the wires joining the vertices of the two cubes also have same resistance R. Find the equivalent resistance between points A & B.



Q. 102: In the circuit shown in the figure, the power dissipated in the circuit is P_0 if an ideal cell is connected across A and B. Same power is dissipated in the circuit if the same cell is connected across C and D. When the cell was connected across A and D or across B and C, the power dissipated in the circuit is found to be $3P_0$.

Calculate the power dissipated in the circuit if the cell is connected across A and C.

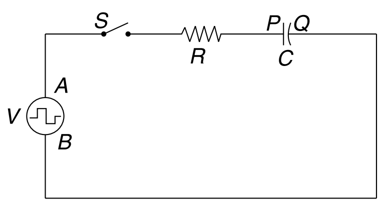


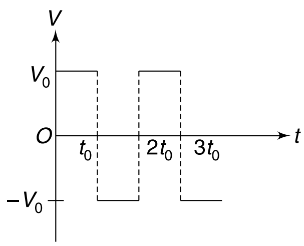
Q. 103: The voltage source shown in the Fig. is a square wave source. Its polarity changes after every $t_0 = 50\ \tau$ second [$\tau = RC$ is time constant of the $R - C$ circuit]. The magnitude of voltage across the source remains constant at V_0. When A is at higher potential compared to B the graph depicts the voltage as positive. Negative voltage means that terminal B is positive. Switch S is closed at $t = 0$.

(a) Taking charge on the capacitor to be positive when plate P is positive, plot the variation of charge on the capacitor as function of time (t).

(b) Write the magnitude of maximum current in the circuit.

(c) Plot the variation of current as function of time (t). Take clockwise current as positive.





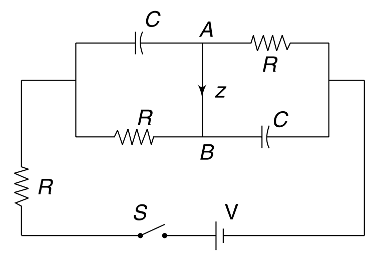
Q. 104: In the circuit shown in the fig, the switch 'S' is closed at time $t = 0$. The current in branch AB is represented by z and is taken to be positive when it is from A to B.

(a) Write the value of z immediately after the switch is closed.

(b) Write the value of z infinite time after the switch is closed.

(c) Write z as a function of time (t) and plot the variation of z with time.

(d) At what time t_0 the current z becomes zero?

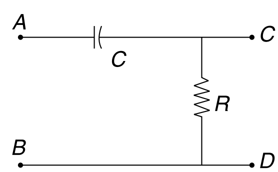


Q. 105: In the circuit shown in the figure, a voltage is applied between points A and B which changes with time as

$$V_0 = kt \text{ for } 0 \le t \le t_0$$

$$= kt_0 \text{ for } t > t_0$$

Plot the variation of potential difference (V) between C and D as a function of time.



ANSWERS

1. No.
2. $\dfrac{2Vx}{l^2}$
3. $\dfrac{IM}{\pi r^2 e \cdot d \cdot N_A}$
4. $t = \dfrac{ledN_A(\rho l + rA)}{\varepsilon M}$
5. $\dfrac{L^2}{\pi^2 r^2}$
6. $\alpha = \alpha_R$
7. 0.98 cm
8. (a) Increases (b) decreases
9. (a) zero, zero (b) zero, zero
10. 6 Ω
11. $E = \dfrac{46}{3}$ V
12. $2R$
13. $R/2$
14. (a) 1.5 A (b) 3.0 V
15. $I_1 = I_2 = 0$
16. $I_1 = \dfrac{2\text{ V}}{R}$; $I_2 = 0$
17. 4 Ω
18. $\Delta I_C = 9\Delta I_B$.
19. $\dfrac{5R}{4}$
20. Power supply increases by 67%

21. (i) A glows near its full brightness
(ii) A becomes slightly more brighter

22. 3.6 Ω

23. $\frac{E}{R}; \frac{E}{2}$

24. Zero

25. $R < 5.5\ \Omega$

26. $R = \frac{R_M R_V}{R_V - R_M}$

27. 12 V

28. $\frac{10E}{11}$

29. To left

30. (a) $\frac{\alpha}{(1-\alpha)\left(\frac{\alpha R}{R_0}+1\right)+\alpha}$

(b) α

(c) $\frac{4}{9}$

31. 13.33 volt

32. $R_X = 2R$

33. (a) 25.9% (b) No change

34. 5.1 kΩ

35. Zero. He shall use a milli voltmeter.

36. (a) 3.125 m (b) 4.688 m
(c) 7.5 m

37. $\frac{\varepsilon^2 b^2}{\rho h d s a^2} \frac{\Delta t}{(\theta_2 - \theta_1)}$

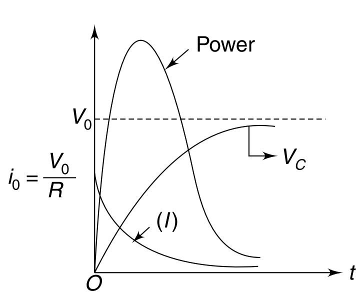
38. 150 V

39. 179 V

40. (a) $B1$ (b) No change

41. 4 V, Capacitor

42.



43. $R = R_0\left[1 - \frac{t}{R_0 C}\right]$

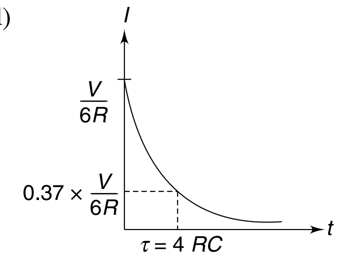
44. (a) $I = \frac{VA}{\rho d}$; $Q = \frac{K\epsilon_0 AV}{d}$

(b) $q = \frac{K\epsilon_0 AV}{d} e^{-\frac{t}{K\rho\epsilon_0}}$

45. (a) $\frac{V}{3R}$ (b) $\frac{2V}{3}$

(c) $\frac{V}{6R}$

(d)



46. $2l$, $4r$

47. No

48. 8×10^4 km

49. $\frac{\pi}{6\ ln 2\ \sigma t}$

50. $R = \frac{\rho\pi}{h \ln(b/a)}$

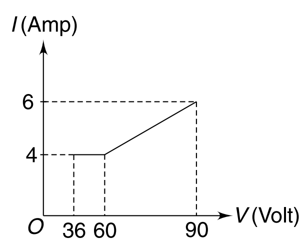
51. $\frac{1}{4}$

52. 4

53. $8^{1/4} \cdot 2000$ K

54. $Q = (\rho_2 - \rho_1)\ \epsilon_0 I$
Q is negative if $\rho_1 > \rho_2$]

55.



56. 20 V < V < 25 V

57. (a) $\frac{5}{3} A$ (b) $\frac{35}{12} A$

58. $\frac{1190}{1189}$

59. (a) 500 Ω

(b) (i) $\frac{5R}{6}$ (ii) $\frac{3R}{4}$ (iii) $\frac{7R}{12}$

60. (a) $\frac{62R}{99}$ (b) Yes

61. $\frac{R}{4}$

62. $i = \dfrac{\pi - 2\theta}{\pi + 4}$

63. (a) zero (b) 7 s

64. 25 V

65. (a) Increase (b) $I_4 = 1.4A$

66. (a) $\dfrac{V_0}{6}$ (b) $\dfrac{2V_0}{9}$

67. $20 + 360\ ln2$

68. (a) $R_{AB} = \left(\dfrac{4R + 3R_x}{3R + 2R_x}\right)R$

(b) $\dfrac{4}{3}R < R_{AB} < \dfrac{3}{2}R$

69. $\dfrac{R}{2}$

70. (a) $\dfrac{8R}{15}$ (b) $\dfrac{3R}{5}$

71. (i) $T = T_0 + \dfrac{V^2}{kR}\left(1 - e^{-\frac{kt}{ms}}\right)$

(ii) $T = T_0 + \dfrac{V^2}{kR}$

72. $T = T_0 + \dfrac{V^2 r}{8e\sigma\rho\ell^2 T_0^3}$

73. (a) 4 Ω, 69.12 kJ (b) 1 Ω, 62.5%

74. (a) We should try with $6430\ \Omega < R_3 < 6440\ \Omega$

(b) $R_4 = 64.325\ \Omega$.

75. (a) $(5.00 \pm 0.03)\ \Omega$

76. (a) 2.4 V (b) 80 Ω

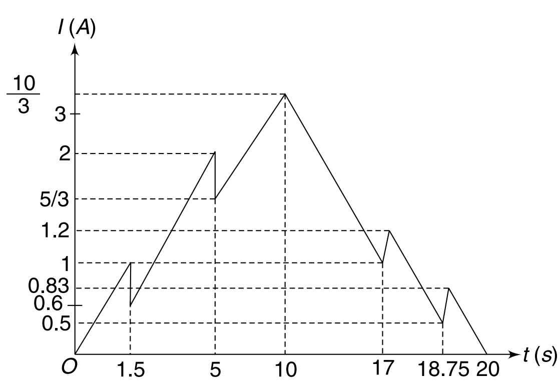
(c) – 120 mA

77. (a) cell with emf $16e$ (b) $I = \dfrac{80\varepsilon}{31R + 16r}$

78. (a) 48 V (b) Yes

79. 50 kΩ

80.



81. 4.8 V

82. 3333 Ω

83. (a) T_2 (b) $R_1 = \dfrac{5}{999}\ \Omega$

$R_2 = \dfrac{5}{111}\ \Omega$

84. (a) $r_2 = r_1\left(\dfrac{\theta_2\,\theta_1'}{\theta_1\,\theta_2'}\right)$; $r_3 = r_1\left(\dfrac{\theta_3}{\theta_1}\cdot\dfrac{\theta_1'}{\theta_3'}\right)$

(b) $r_2 > r_3$

85. $\dfrac{2}{27}CV^2$

86. (a) 199 Ω (b) 3×10^5 V

(c) 265.3 s

87. Capacitor B, $6\ell n6$ sec

88. $V_C = (90 - 120\ e^{-5t})$ volt

89. 24 μC

90. $\dfrac{q_0^2 L}{2\epsilon_0 A} - \dfrac{q_0^4}{8\epsilon_0^2 A^2 k}$

91. $\dfrac{q_0^2 L}{2\epsilon_0 A} - \dfrac{q_0^4}{4\epsilon_0^2 A^2 k}$

92. (a) $1A$ (b) zero

93. 121.87 μJ

94. (a) $\dfrac{V}{(8\pi\epsilon_0 a)R}\ e^{\frac{-3t}{8\pi\epsilon_0 aR}}$

(b) $\dfrac{4\pi\epsilon_0 aV^2}{3}$

95. $i = 0.126\ e^{-1000t}$ [t = in second]

96. 7 μF

97. (a) 1.0 C/s (b) 75 μJ

98. (a) $14\dfrac{dq}{dt} + \dfrac{5}{4}q = 12$ (b) $q = \dfrac{48}{5}\left[1 - e^{-\frac{5t}{56}}\right]$

(c) 0; $\dfrac{48}{5}\ \mu$C.

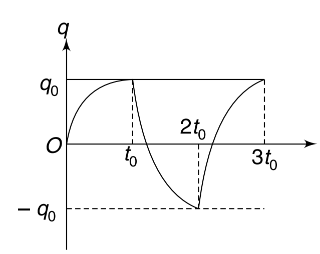
99. $\dfrac{63R}{164}$

100. (a) $R = (1 + \sqrt{3})\Omega$ (c) $\dfrac{20}{(\sqrt{3} + 1)(\sqrt{3} + 2)^9}$

101. $\dfrac{2R}{3}$

102. $\left(\dfrac{15 + 6\sqrt{7}}{16 + 6\sqrt{7}}\right)P_0$

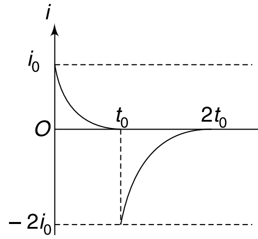
103. (a)



$q_0 = CV_0$

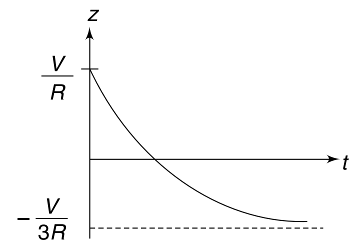
(b) $i_{\max} = \dfrac{2V_0}{R}$

(c)



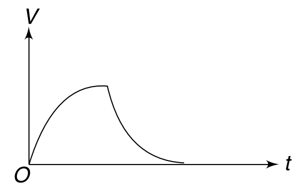
104. (a) $\dfrac{V}{R}$ (b) $-\dfrac{V}{3R}$

(c) $Z = \dfrac{4V}{3R}\left[e^{-\frac{3t}{RC}} - \dfrac{1}{4}\right]$



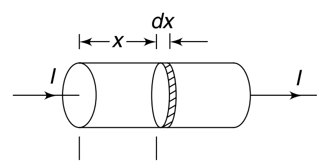
(d) $t_0 = \dfrac{2RC}{3}\ \ell n 2$

105.



SOLUTIONS

2. We will first calculate the resistance of the cylindrical conductor. For this let us consider the cylinder to be made of numerous thin discs. Resistance of one such disc (shown in Figure) will be



$$dR = \frac{dx}{\sigma A} = \frac{xdx}{A\sigma_0 l}$$

Resistances of all such discs are in series for current flowing along the axis of the cylinder.

$\therefore$ $$R = \int dR = \frac{1}{Al\sigma_0}\int_0^l xdx = \frac{l}{2A\sigma_0}$$

Current through the Conductor is

$$I = \frac{V}{R} = \frac{2\ VA\sigma_0}{l}$$

Current density $$j = \frac{I}{A} = \frac{2V\sigma_0}{l}$$

Using ohm's law in microscopic form

$$E = \frac{j}{\sigma} = \frac{2V\sigma_0}{l}\ \frac{x}{\sigma_0 l}$$

$$E = \frac{2\ \text{V}}{l^2}\ x$$

3. Number of atoms in unit volume of the conductor is

$$= \frac{d}{M} N_A \qquad [N_A = \text{Avogadro's number}]$$

$\therefore$ No. of free electrons in unit volume

$$n = \frac{dN_A}{M}$$

Current density $$j = \frac{I}{\pi r^2}$$

$\therefore$ $$nev_d = \frac{I}{\pi r^2}$$

$$v_d = \frac{I}{\pi r^2 e \cdot n} = \frac{IM}{\pi r^2 e \cdot d \cdot N_A}$$

4. Resistence of the wire $R = \frac{\rho \ell}{A}$

Current in the wire $I = \frac{\varepsilon}{R + r} = \frac{\varepsilon}{\frac{\rho \ell}{A} + r}$

Current density $j = \frac{I}{A} = \frac{\varepsilon}{\rho \ell + r \cdot A}$

$$neV_d = \frac{\varepsilon}{\rho \ell + r \cdot A}$$

$$V_d = \frac{\varepsilon}{ne(\rho \ell + r \cdot A)}$$

But n = no. of atoms in unit volume of the Conductor

$$= \frac{d}{M} N_A$$

$\therefore$ $V_d = \frac{\varepsilon M}{edN_A (\rho \ell + rA)}$

$\therefore$ Time to travel a distance ℓ for a free electron is

$$t = \frac{\ell}{V_d} = \frac{ledN_A(\rho \ell + r \cdot A)}{\varepsilon_M}$$

5. $R_1 = \frac{\rho L}{\pi r \cdot t}$

$$R_2 = \frac{\rho \pi r}{tL}$$

$\therefore$ $\frac{R_1}{R_2} = \frac{L^2}{\pi^2 r^2}$

6. A = Area of cross section of wire
L = Length of the wire
ρ = resistivity.

$$R = \frac{\rho L}{A}$$

If temperature changes by $\Delta\theta$, the resistance becomes.

$$R' = \frac{\rho(1 + \alpha_R \Delta\theta)\, L(1 + \alpha \Delta\theta)}{A(1 + 2\alpha \Delta\theta)}$$

$$R' = R\frac{(1 + \alpha_R \Delta\theta)(1 + \alpha \Delta\theta)}{(1 + 2\alpha \Delta\theta)}$$

$$R' = R\frac{1 + (\alpha_R + \alpha)\Delta\theta + \alpha_R \cdot \alpha \cdot \Delta\theta^2}{(1 + 2\alpha \Delta\theta)}$$

$$= R\frac{1 + (\alpha_R + \alpha)\Delta\theta}{1 + 2\alpha \Delta\theta}$$

If $$R' = R \text{ then}$$
$$\alpha = \alpha_R.$$

7. Let R_0 represent the resistance at some initial temperature and R_T be resistance when temperature is changed by T.

$$R_T = R_{0\text{Cu}}(1 + \alpha_{\text{Cu}}T) + R_{0\text{C}}(1 + \alpha_{\text{C}}T) + R_{0\text{Ni}}(1 + \alpha_{\text{Ni}}T)$$

$$\Rightarrow \quad R_{0\text{Cu}} + R_{0\text{C}} + R_{0\text{Ni}} = R_{0\text{Cu}}(1 + \alpha_{\text{Cu}}T) + R_{0\text{C}}(1 + \alpha_{\text{C}}T) + R_{0\text{Ni}}(1 + \alpha_{\text{Ni}}T)$$

$$\Rightarrow \quad R_{0\text{Cu}}\alpha_{\text{Cu}} + R_{0\text{C}}\alpha_{\text{C}} + R_{0\text{Ni}}\alpha_{\text{Ni}} = 0$$

$$\rho_{\text{Cu}} l_{\text{Cu}} \alpha_{\text{Cu}} + \rho_{\text{C}} l_{\text{C}}\ \alpha_{\text{C}} + \rho_{\text{Ni}}\ l_{\text{Ni}}\ \alpha_{\text{Ni}} = 0$$

If $\quad l_{\text{Ni}} = x$ cm $\quad$ then $\quad l_{\text{C}} = (1 - x)$ cm

$$l_{\text{Cu}} = 1 \text{ cm}$$

$$\therefore \quad 1.7 \times 10^{-8} \times 1 \times 3.9 \times 10^{-3} + 5 \times 10^{-5} \times (1 - x)\,(-5 \times 10^{-4}) + 1 \times 10^{-6} \times x \times 4 \times 10^{-4} = 0$$

$$6.63 \times 10^{-11} - 2.5 \times 10^{-8} = \left(-4 \times 10^{-10} - 2.5 \times 10^{-8}\right)x$$

$$x = \frac{249.33}{254} = 0.98 \text{ cm}$$

8. Let resistance of the bulb be R_0.
The equivalent of two parallel resistances is given by

$$\frac{1}{R_{eq}} = \frac{1}{R_0} + \frac{1}{R}$$

If R is increased, R_{eq} must also increase. This increases the overall resistance of the circuit and the current through the cell drops. It means power spent by the cell decrease. Potential drop across the internal resistance of the cell drops and the potential difference across the bulb increases. Hence, the bulb becomes brighter.

9. (a) With addition of each 1 Ω resistor, the equivalent resistance of the resistors in parallel goes on decreasing. If infinite resistors are added the equivalent will become Zero.

The equivalent resistance of entire circuit is = R_3 = 5 Ω.

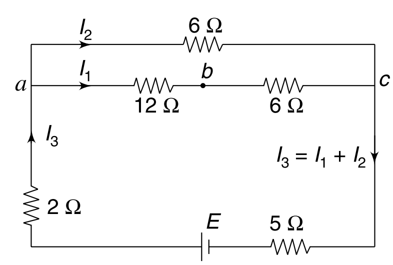
$$\therefore \quad I = \frac{6}{5} = 1.2A$$

Current through all the resistors in parallel is nearly zero since the current of $1.2A$ gets divided into infinitely many parts.

(b) In this case the resistance R_3 along with infinite many 1 Ω resistors has an equivalent → ∞.

∴ Current in the circuit → 0.

∴ Potential drop across each individual resistance is nearly zero.



11.
$$I_1 = \frac{4 \text{ volt}}{12\,\Omega} = \frac{1}{3}\,A$$

$$\therefore \quad V_b - V_c = 6 \times \frac{1}{3} = 2 \text{ V}$$

$$\therefore \quad V_a - V_c = 4 + 2 = 6 \text{ V}$$

$$I_2 = \frac{6\,\text{V}}{6\,\Omega} = 1A.$$

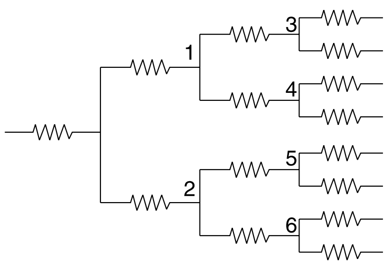
$$I_3 = I_1 + I_2 = \frac{1}{3} + 1 = \frac{4}{3}\,A$$

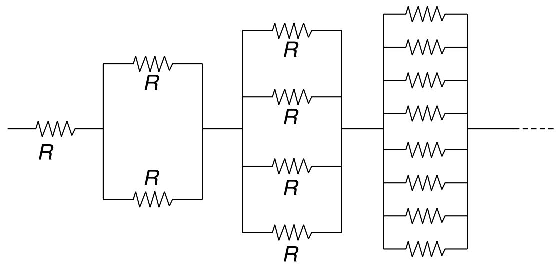
$$V_c - 5 \times \frac{4}{3} + E - 2 \times \frac{4}{3} = V_a$$

$$\Rightarrow \qquad -\frac{20}{3} + E - \frac{8}{3} = V_a - V_c$$

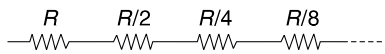
$$\Rightarrow \qquad E = 6 + \frac{20}{3} + \frac{8}{3} = \frac{46}{3} \text{ volt}$$

12. Points 1 and 2 are equipotential. They can be connected together. Points 3, 4, 5 and 6 are equipotential. Tie them together. And so on.



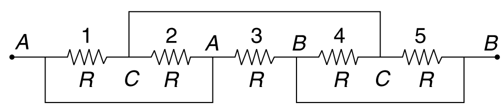


$\Rightarrow$



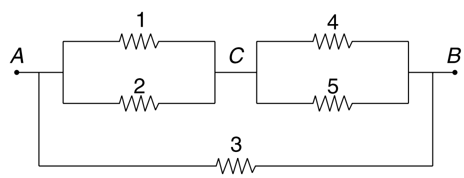
$\therefore$ Equivalent is
$$R_0 = R\left[1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} \ldots \infty\right]$$

$$= R\left[\frac{1}{1 - \frac{1}{2}}\right] = 2R.$$



13. The circuit can be redrawn as shown in the figure.

1 and 2 are in parallel; 4 and 5 are in parallel. Equivalent of each pair is $\frac{R}{2}$. They add to become R which is in parallel to 3.



$$\therefore \qquad R_{eq} = \frac{R}{2}$$

14. Maximum current that the two fuses can tolerate are $I_{1\max} = 1.0\,\text{A}$ and $I_{2\max} = 1.2\,\text{A}$ respectively. Resistances of the two fuses are

$$R_1 = \frac{1.0\ \text{V}}{1.0\,\text{A}} = 1\ \Omega$$

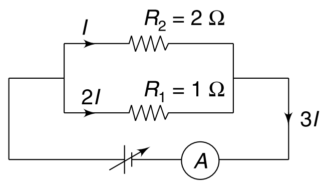
And
$$R_2 = \frac{2.4\ \text{V}}{1.2\,\text{A}} = 2\ \Omega.$$

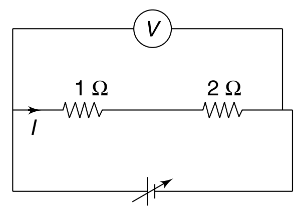
(a) Clearly, F1 blows out when $2I = 1.0\text{A}$

$\therefore$ reading of (A) is 1.5A

(b) F1 blows out when $I = 1.0\text{A}$

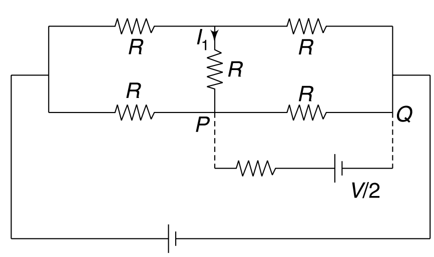
$$\therefore \qquad V = 3 \times 1 = 3.0\text{V}$$





15. The circuit shown is a balanced wheastone bridge with $V_P - V_Q = \frac{V}{2}$.

When a cell having potential difference $\frac{V}{2}$ is added is parallel across P and Q, it will cause so difference to the circuit.



$$\therefore \qquad I_1 = 0$$

$$I_2 = 0.$$

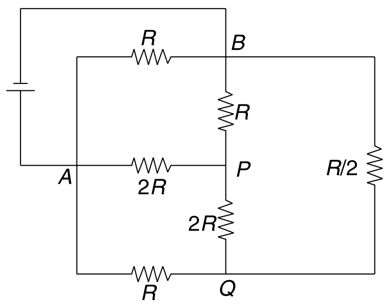
16. The circuit can be redrawn as shown.

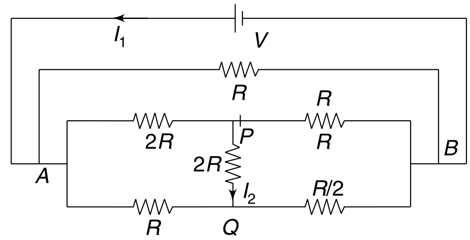
Now you can see a wheatstone bridge (a balanced one) in parallel to R.

$$\therefore \quad I_2 = 0$$

Equivalent resistance is $\frac{R}{2}$

$$\therefore \quad I_1 = \frac{2\text{ V}}{R}$$





17. When 'S' is open, Current through ammeter is

$$i_1 = \frac{20}{2+2+r} = \frac{20}{4+r}$$

When 'S' is closed,

$$i_2 = \frac{20}{r}$$

Given $\quad i_2 = 2i_1$

$$\frac{20}{r} = 2 \times \frac{20}{4+r}$$

$$4 + r = 2r$$

$$\Rightarrow \quad r = 4\Omega$$

18. From Kirchhoff's first law

$$I_E = I_C + I_B$$

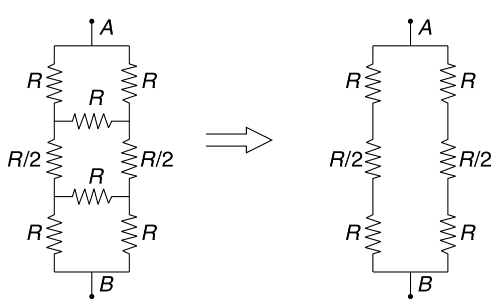
Put $\quad I_E = \frac{I_C}{0.9}$

$$\frac{I_C}{0.9} - I_C = I_B$$

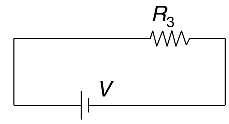
$$I_C = 9I_B$$

$$\therefore \quad \Delta I_C = 9\Delta I_B$$

19. The equivalent circuit is as shown in the figure. Equivalent resistance is $\frac{5R}{4}$.



20. **Switch open**

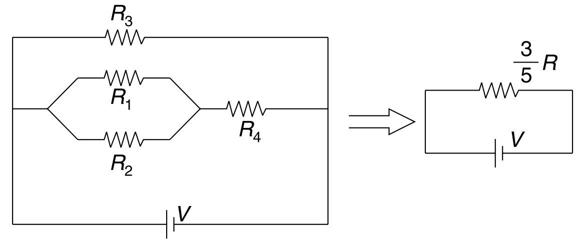


$$P_{\text{cell}} = \frac{V^2}{R}$$

Switch closed

$$P'_{\text{cell}} = \frac{5V^2}{3R} = 1.67\ P_{\text{cell}}$$

21. B and C are in parallel.



Therefore equivalent resistance of B and C will be smaller than that of B.

Hence, the overall resistance decreases.

Current through A will increase and brightness of A will increase.

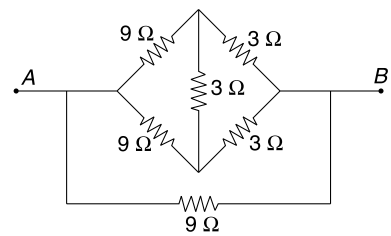
When C has high power rating, its resistance is very small. The equivalent of B and C is even smaller. Bulb A will nearly glow at its full brightness.

If C has low power rating, its resistance is too high.

Thus equivalent resistance of B and C will be only slightly less than that of B.

The brightness of A will increase slightly.

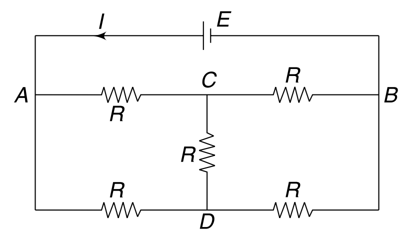
22. The given circuit is a balanced wheatstone bridge with a 9 Ω resistance in parallel.



$$\frac{1}{R_{eq}} = \frac{1}{12} + \frac{1}{12} + \frac{1}{9} = \frac{3+3+4}{36}$$

$$\therefore \qquad R_{eq} = \frac{36}{10} = 3.6\ \Omega$$

23. Circuit is a balanced wheatstone bridge



Resistance between C and D can be removed. Equivalent resistance between A and B is R.

$$\therefore \qquad I = \frac{E}{R}$$

$$V_{AD} = \frac{E}{2}$$

24. Voltmeter has infinite resistance. Current in the loop is $I = \dfrac{12 \text{ volt}}{12r}$ where r is internal resistance of each cell.

Potential difference across the terminals of each cell is $= 1\text{ V} - Ir = 0$

Hence voltmeter will show zero, irrespective of whether it is connected across one or more cells.

25. No current will flow through the cell if potential difference across the terminals of the battery is exactly 110 V.

$$\Rightarrow \qquad 120 - Ir = 110$$

$$\Rightarrow \qquad I = \frac{10}{0.5} = 20\text{ A}$$

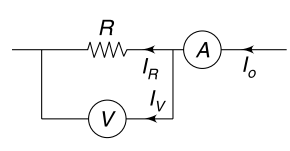
$$\therefore \qquad R = \frac{110\text{ V}}{20\text{ A}} = 5.5\ \Omega$$

If $R > 5.5\ \Omega$ then the potential difference across the cell will be larger than 110 V and it will get charged.

If $R < 5.5\ \Omega$ than the potential difference across the cell will be smaller than 110 V and it will get discharged.

26.

$$R_M = \frac{V}{I_0}$$



$$\Rightarrow \qquad \frac{1}{R_M} = \frac{I_0}{V} = \frac{I_R + I_V}{V} = \frac{I_R}{V} + \frac{I_V}{V}$$

$$\Rightarrow \qquad \frac{1}{R_M} = \frac{1}{R} + \frac{1}{R_V}$$

$$\Rightarrow \qquad \frac{1}{R} = \frac{1}{R_M} - \frac{1}{R_V}$$

$$\Rightarrow \qquad R = \frac{R_M \cdot R_V}{R_V - R_M}$$

27. Let emf of the cell be E.

Current through the voltmeter (when connected between 1-2) is $\frac{V_{12}}{R_V} = \frac{4}{200} = \frac{1}{50}$ A

Current through $R_1 = \frac{4}{R_1}$

$\therefore$ Current through $R_2 = \frac{4}{R_1} + \frac{1}{50}$

$\therefore$ Potential difference across $R_2 = \left(\frac{4}{R_1} + \frac{1}{50}\right) \times 100 = \frac{400}{R_1} + 2$

$$\therefore \qquad E = 4 + \left(\frac{400}{R_1} + 2\right) = 6 + \frac{400}{R_1} \qquad \text{...(1)}$$

When the voltmeter is connected between 2-3

Current through voltmeter $= \frac{6}{200}$ A

Current through $R_2 = \frac{6}{100}$ A

$\therefore$ Current through $R_1 = \frac{6}{200} + \frac{6}{100} = \frac{18}{200} = \frac{9}{100}$ A

$$\therefore \qquad E = \frac{9R_1}{100} + 6 \qquad \text{...(2)}$$

From (1) and (2)

$$\frac{9R_1}{100} + 6 = 6 + \frac{400}{R_1}$$

$$\Rightarrow \qquad R_1 = \sqrt{\frac{40000}{9}} = \frac{200}{3}\ \Omega$$

Put this in (2)

$$E = \frac{9}{100} \times \frac{200}{3} + 6 = 12 \text{ V}$$

When connected across 1-3, the voltmeter will read $E = 12$ V.

28. Let current be I_0 in the original circuit. Potential difference across ammeter is $= E - V_0$

$$\therefore \text{ Resistance of ammeter} \qquad R_A = \frac{E - V_0}{I_0} \qquad \text{...(1)}$$

After the resistance is added to the voltmeter, the potential difference across voltmeter becomes $\frac{V_0}{10}$.

$\therefore$ Potential difference across ammeter $= E - \frac{V_0}{10}$

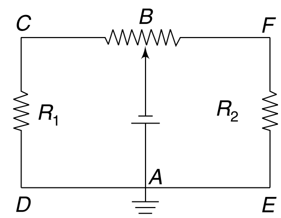
$$\therefore \qquad R_A = \frac{E - V_0/10}{10 I_0} \qquad \text{...(2) [the new current is } 10I_0]$$

From (1) and (2)

$$\frac{10E - V_0}{100I_0} = \frac{E - V_0}{I_0}$$

$$\Rightarrow \qquad V_0 = \frac{90E}{99} = \frac{10E}{11}$$

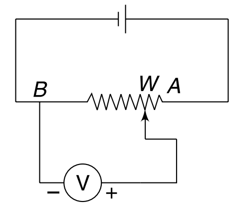
29. The effective circuit is as shown



If contact at B is moved to left, the resistance of loop $ABCD$ decreases and the current in the loop increases.

Resistance of the other loop increases and hence current through R_2 increases.

31. The equivalent circuit is as shown in figure.



When the contact is at A, reading of voltmeter = emf of cell = 20 V.

When the contact is at B, reading of voltmeter = zero.

When $\theta = 120°$; $\qquad R_{AW} = \frac{R}{3}$

Where R = resistance of C

$$R_{BW} = \frac{2R}{3}$$

$$\therefore \qquad \text{Reading} = \frac{2}{3} \times 20 = \frac{40}{3} \text{ volt}$$

32. Let the potential gradient along the potentiometer wire PQ be K.

When galvanometer shows zero deflection the current in two loops are independent of each other. If current in loop having R and R_X is i, then

$$iR = Kx \qquad ...(1)$$

and $\qquad i(R + R_x) = K(3x) \qquad ...(2)$

(2) − (1)

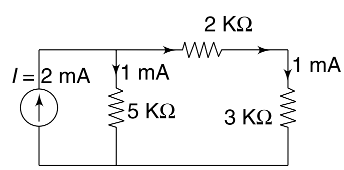
$$iR_X = 2Kx \qquad ...(3)$$

(3) ÷ (1)

$$\frac{R_X}{R} = 2$$

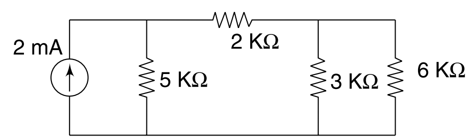
$$\Rightarrow \qquad R_X = 2R.$$

33. **Without Voltmeter**



$$V = (3\ \text{k}\Omega)(1\ \text{mA}) = 3 \text{ Volt.}$$

With Voltmeter

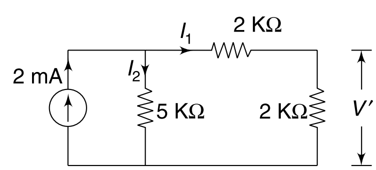


$$4i_1 = 5i_2$$

And $\qquad i_1 + i_2 = 2 \text{ mA}$

$$\therefore \qquad i_1 = \frac{10}{9} \text{ mA}$$

$$\therefore \qquad V' = 2 \times \frac{10}{9} = \frac{20}{9} \text{ Volt.}$$



$$\% \text{ error} = \left|\frac{V - V'}{V}\right| \times 100$$

$$= \frac{3 - \frac{20}{9}}{3} \times 100$$

$$= \frac{7}{27} \times 100 = 25.9\%.$$

34. Reading of ammeter A_2 = current through V_2

$$\therefore \qquad R_{V2} = \frac{V_2}{I_2} = \frac{100}{400 \times 10^{-6}} = 0.25\ \text{M}\Omega$$

$p \cdot d$ across A_1 = reading of V_1 = 2 V

Current through V_1 is

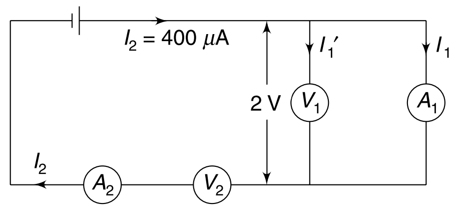
$$I_1' = \frac{V_1}{R_{V1}} \qquad [R_{V1} = R_{V2} = 0.25\ \text{M}\Omega]$$

$$= \frac{2}{0.25 \times 10^{+6}} = 8 \times 10^{-6}\ A = 8\ \mu\text{A}.$$

Current through A_1 is

$$I_1 = 400\ \mu\text{A} - 8\ \mu\text{A} = 392\ \mu\text{A}$$

$$\therefore \qquad R_{A1} = \frac{2\ \text{volt}}{392\ \mu\text{A}} = \frac{2 \times 10^6}{392} = 5102\ \Omega \simeq 5.1\ \text{k}\Omega$$



36. If resistance of first 5 wires is R, then the resistance of next 5 wires must be $\frac{R}{4}$ (since area of cross section is 4 times).

Potential drop across first 5m length

$$= \frac{R}{R + \frac{R}{4}} \times 2 = \frac{8}{5}\ \text{V} = 1.6\ \text{V}$$

Potential drop across next 5 m length = 0.4 Volt

Potential gradient along first 5 wires = $\frac{1.6\ \text{V}}{5\,\text{m}}$

Potential gradient along next 5 wires = $\frac{0.4\ \text{V}}{5\,\text{m}}$

(a) length needed to balance 1.0 V

$$l_1 = \frac{1.0\ \text{V}}{\frac{1.6}{5}\ \frac{V}{m}} = 3.125\ \text{m}$$

(b) $l_2 = \dfrac{1.5\ \text{V}}{\frac{1.6}{5}\ \frac{V}{m}} = 4.688\ \text{m}$

(c) $l_3 = 5\,\text{m} + \dfrac{0.2\,\text{V}}{\frac{0.4\,\text{V}}{5\,\text{m}}}$

$= 5 + 2.5 = 7.5\,\text{m}$

37. Let length of the wire be L.

Resistance of the wire $$R = \frac{\rho L}{\pi b^2}$$

Power dissipated $$P = \frac{\varepsilon^2}{R} = \frac{\varepsilon^2 \pi b^2}{\rho L}$$

Heat dissipated in time Δt

$$Q = P\,\Delta t = \frac{\varepsilon^2 \pi b^2}{\rho L}\,\Delta t \qquad ...(1)$$

Heat required to raise the temperature of water

$$Q = ms\,\Delta\theta = d\cdot(\pi a^2 h)\, s(\theta_2 - \theta_1) \qquad ...(2)$$

From (1) and (2)

$$d(\pi a^2 h)\, s(\theta_2 - \theta_1) = \frac{\varepsilon^2 \pi b^2}{\rho L}\,\Delta t$$

$$\therefore \quad L = \frac{\varepsilon^2 b^2}{\rho h d s a^2}\,\frac{\Delta t}{(\theta_1 - \theta_2)}$$

38. $$V_{AC} = E_3 = 30 \text{ V}$$

$$V_{CB} = E_2$$

$$\therefore \quad \frac{E_2}{30} = \frac{V_{CB}}{V_{AC}}$$

$$\frac{E_2}{30} = \frac{3}{2} \qquad \left[\because \frac{V_{CB}}{V_{AC}} = \frac{l_{CB}}{l_{AC}} = \frac{3}{2}\right]$$

$$\therefore \quad E_2 = 45 \text{ V}$$

$$V_{AB} = 30 + 45 = 75 \text{ V}$$

$$R_{AB} = 2 \times 5 = 10\ \Omega$$

$\therefore$ Current through AB is $\frac{75}{10} = 7.5$ A

$$\therefore \quad E_1 = 7.5 \times (R_{AB} + R)$$

$$= 150 \text{ V.}$$

39. After the switch is moved to position 2, the energy stored in the capacitor gets dissipated in the two resistors. Current through two resistors will be same at any instant; therefore, heat dissipated in them will be in the ratio of their resistances.

$$\therefore \quad \frac{H_{30}}{H_{50}} = \frac{30}{50}$$

$$\Rightarrow \quad H_{30} = \frac{30}{50} \times 6 = 3.6 \text{ J}$$

Total energy stored in the capacitor was

$$U = 6 + 3.6 = 9.6 \text{ J}$$

$$\therefore \quad \frac{1}{2} CV^2 = 9.6$$

$$\frac{1}{2} \times (600 \times 10^{-6})\, V^2 = 9.6$$

$$\therefore \quad V^2 = 3.2 \times 10^4$$

$$\therefore \quad V = 179 \text{ V.}$$

40. (a) There is no current through the capacitor. $B2$ and $B3$ are in parallel. The current through $B1$ gets divided into two parts through $B2$ and $B3$. Hence $B1$ will be brightest.

(b) Still there is no current through the capacitor and there is no change in brightness of bulbs.

41. The element is a capacitor.

In case of capacitor

$$q = CV$$

$$\therefore \quad i = \frac{dq}{dt} = C\left(\frac{dV}{dt}\right)$$

$$\frac{dV}{dt} = \frac{4.0}{4.0} \text{ V/s} = 1.0 \text{ V/s}$$

Therefore, if $C = 1F$ then $I = 1 \times 1 = 1A$ (constant)

42.
$$I = i_0 e^{-t/\tau} \quad \text{...(1)}$$

Where $i_0 = \dfrac{V_0}{R}$ and $\tau = RC$

Charge on capacitor as function of time is

$$q = q_0[1 - e^{-t/\tau}]$$

Where $q_0 = CV$

Energy stored in capacitor

$$U = \frac{q^2}{2C}$$

$$\frac{dU}{dt} = \frac{1}{C} q \frac{dq}{dt} = \frac{qi}{C}$$

$\therefore$ Power absorbed

$$P = \frac{dU}{dt} = \frac{q_0 i_0}{C} e^{-t/\tau}[1 - e^{-t/\tau}] \quad \text{...(3)}$$

For drawing the graph of P versus t you need to acknowledge that P is zero both at $t = 0$ and $t = \infty$.

The graphs have been shown in answer.

43. As the charge on the capacitor decreases, the potential difference across it drops. To keep the current constant the resistance must be decreased progressively.

Initial value of resistance = R_0

Initial current $i_0 = \dfrac{q_0/C}{R_0}$ = a constant. ...(1)

Charge on the capacitor after time t is

$$q = q_0 - i_0 t$$

$\therefore$ Potential difference across the capacitor

$$= \frac{q}{C} = \frac{q_0 - i_0 t}{C}$$

$$\therefore \quad \frac{q_0 - i_0 t}{C} = R i_0$$

$$\therefore \quad R = \frac{q_0}{C i_0} - \frac{t}{C}$$

$$\Rightarrow \quad R = R_0 - \frac{t}{C} \quad \text{[Using (1)]}$$

$$= R_0\left[1 - \frac{t}{R_0 C}\right]$$

44. (a)
$$C = \frac{K\epsilon_0 A}{d}$$

Resistance of dielectric material $R = \frac{\rho d}{A}$

$$I = \frac{V}{R} = \frac{VA}{\rho d}$$

$$Q = CV = \frac{K\epsilon_0 AV}{d}$$

(b) The charge will leak through the capacitor. Let the charge be q after time 't'.

$$\frac{q}{C} = Ri$$

$$\Rightarrow \quad -R\frac{dq}{dt} = \frac{q}{C}$$

$$\Rightarrow \quad \frac{dq}{q} = -\frac{1}{RC}dt$$

$$\Rightarrow \quad \int_Q^q \frac{dq}{q} = -\frac{1}{\tau}\int_0^t dt$$

$$\Rightarrow \quad ln\frac{q}{Q} = -\frac{t}{\tau}$$

$$\Rightarrow \quad q = Qe^{-t/\tau}$$

Where $Q = \frac{K\epsilon_0 AV}{d}$ and $t = RC = K\rho\epsilon_0$

46. Since the two filaments emit same spectrum, their temperature must be same. It means the power radiated is simply proportional to the surface area of the filament.

$P \ \alpha \ 2\pi rl$

$$\Rightarrow \quad P \ \alpha \ rl \qquad ...(1)$$

Power is also given by $P = \frac{V^2}{R} = \frac{V^2\pi r^2}{\rho l}$

$$\therefore \quad P \ \alpha \ \frac{r^2}{l} \qquad ...(2)$$

From (1) and (2) $rl \ \alpha \ \frac{r^2}{l} \Rightarrow l^2 \ \alpha \ r$

Putting this in (1) gives

$$l \ \alpha \ P^{1/3}$$

and $r \ \alpha \ P^{2/3}$

$\therefore$ length of new filament $8^{1/3}l = 2l$

Radius of new filament $8^{2/3}r = 4r$.

47. With 10 bulbs in parallel, their equivalent resistance is $\frac{R}{10}$.

Maximum power transfer theorem tells us that power dissipated in a resistance is maximum when it is equal to internal resistance of the battery. So we are tempted to assume that $r = \frac{R}{10}$. But we need to understand that in this situation, number of bulbs can be integer only and the resistance connected to the battery can be

$R, \frac{R}{2}, \frac{R}{3}, \frac{R}{4}$ ... Only.

We cannot have a resistance equal to $\frac{R}{10.5}$. So it may be that r is actually equal to $\frac{R}{10.5}$ but in no way we can make our resistance equal to that.

48. Number of free electrons per unit volume = Number of atoms per unit volume

$$\therefore \quad n = \frac{\text{Mass of unit volume}}{\text{Molar mass}} \times N_A$$

$$= \frac{9000}{63 \times 10^{-3}} \times 6.02 \times 10^{23} = 8.6 \times 10^{28} \text{ m}^{-3}$$

Drift speed
$$v_d = \frac{j}{ne}$$

$$= \frac{1 \times 10^6 \cdot \text{A/m}^2}{8.6 \times 10^{28} \times 1.6 \times 10^{-19}} = 7.2 \times 10^{-5} \text{ ms}^{-1}$$

Time needed to drift through 5 cm is

$$t = \frac{5 \times 10^{-2}}{7.2 \times 10^{-5}} = 695 \text{ s}$$

The thermal kinetic energy of electrons is of the order of $\frac{3}{2}$ kT where $K = 1.38 \times 10^{-23}$ JK^{-1} is Boltzmann constant.

$$\therefore \quad \frac{1}{2} \text{mv}^2 = \frac{3}{2} \text{kT}$$

$$v = \sqrt{\frac{3\text{kT}}{\text{m}}} = \sqrt{\frac{3 \times 1.38 \times 10^{-23} \times 300}{9.1 \times 10^{-31}}}$$

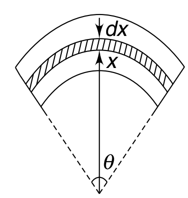
$$= 1.17 \times 10^5 \text{ ms}^{-1}$$

$\therefore$ Distance travelled by electron in $t = 695$ s is

$$S = vt = 1.17 \times 10^5 \times 695 \text{ m}$$

$$= 8.1 \times 10^4 \text{ km}$$

49. Consider a strip of width dx as shown in Figure. Resistance of the strip



$$dR = \frac{x\theta}{\sigma t dx} = \frac{\pi}{6\sigma t} \frac{x}{dx}$$

All such small resistances will be in parallel.

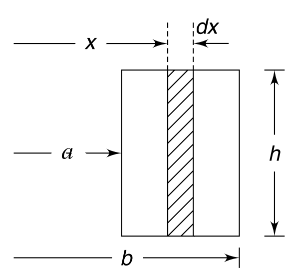
$$\therefore \quad \frac{1}{R} = \frac{1}{dR_1} + \frac{1}{dR_2} + \frac{1}{dR_3} + \ldots$$

$$\Rightarrow \quad \frac{1}{R} = \int \frac{1}{dR} = \frac{6\sigma t}{\pi} \int_r^{2r} \frac{dx}{x}$$

$$\Rightarrow \quad \frac{1}{R} = \frac{6\sigma t}{\pi} [\ln x]_r^{2r} = \frac{6 \ln 2}{\pi} \sigma t$$

$$\therefore \quad R = \frac{\pi}{6 \cdot \ln 2 \sigma t}$$

50. Consider a strip of width dx shown on the rectangular face in the figure



Think of a half cylindrical conductor of radius x and infinitesimally small thickness dx.

Length of this conductor = πx

Cross sectional area of this conductor = $h\,dx$

$\therefore$ Resistance of this thin cylindrical conductor

$$dR = \frac{\rho \pi x}{h\,dx}$$

The given conductor is made of countless number of such thin conductors, all connected in parallel.

$$\therefore \qquad \frac{1}{R} = \int \frac{1}{dR} = \frac{h}{\rho\pi}\int_a^b \frac{dx}{x}$$

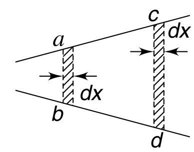
$$= \frac{h}{\rho\pi}\ln\left(\frac{b}{a}\right)$$

$$\therefore \qquad R = \frac{\rho\pi}{h\ln(b/a)}$$

51. let area at $ab = A$

And area at $cd = 2A$

Current in the wire $= I$. Current is same at all cross sections.



Current density at ab $\qquad J_1 = \frac{I}{A}$

Current density at cd $\qquad J_2 = \frac{I}{2A} = \frac{J_1}{2}$

Consider two discs of equal thickness dx at section ab and cd.

Resistance of disc at ab $dR_1 = \sigma\frac{dx}{A}$

Rate of Heat dissipation in disc at $ab = I^2(dR_1)$

Rate of heat dissipation per unit volume $= \frac{I^2(dR_1)}{A\cdot dx}$

$$= \frac{I^2\sigma}{A^2}$$

Similarly rate of heat dissipation per unit volume at section $cd = \frac{I^2\sigma}{(2A)^2}$.

$\therefore$ Required ratio $= \frac{1}{4}$.

52. Power radiated by the filament

$$P = \sigma e(T^4 - T_0^4)$$

$$\simeq \sigma e T^4 \qquad [\because T^4 >> T_0^4]$$

But $\qquad P = VI$

$$\therefore \qquad VI = \sigma e T^4$$

$$\therefore \qquad \ln(VI) = \ln(\sigma e) + 4\ln T$$

$\therefore$ graph has a slope = 4.

53. The power dissipated in a length L of a wire carrying current I is $P = I^2R$.

When the temperature of the wire is constant

$$I^2R = e\sigma\pi d\cdot L\left(T^4 - T_0^4\right)$$

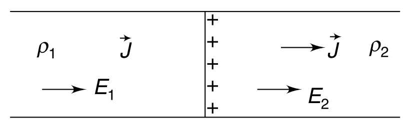
[σ is Stefan's constant and e is emissivity]

$$I^2R \simeq e\sigma\pi d\cdot LT^4$$

$$\Rightarrow \qquad \frac{I^2\rho L}{\pi\left(\frac{d^2}{4}\right)} = e\sigma\pi d\cdot LT^4 \quad\Rightarrow\quad T^4 \propto \frac{1}{d^3}$$

$$\therefore \qquad \left(\frac{T_{\text{thin}}}{2000\text{ K}}\right)^4 = \frac{2^3}{1} \qquad \therefore\ T_{\text{thin}} = 8^{1/4}\ 2000\text{ K}$$

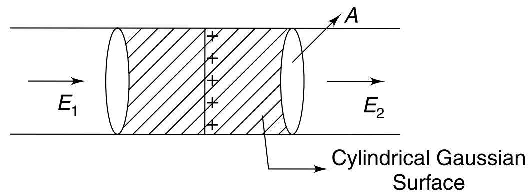
54.
$$J = \frac{E_1}{\rho_1} = \frac{E_2}{\rho_2}$$



$\therefore$
$$E_2 = \frac{\rho_2}{\rho_1} E_1 \qquad ...(1)$$

Consider a Gaussian surface as shown

$$\phi = E_2A - E_1A = \left(\frac{\rho_2}{\rho_1} - 1\right) E_1A$$



$\therefore$
$$\frac{Q}{\epsilon_0} = \left(\frac{\rho_2}{\rho_1} - 1\right) E_1A$$

$$Q = \left(\frac{\rho_2}{\rho_1} - 1\right) \epsilon_0 AE_1$$

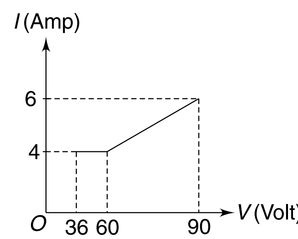
$\because$
$$I = \frac{E_1}{\rho_1} A$$

$\therefore$
$$Q = (\rho_2 - \rho_1)\, \epsilon_0 I$$

Q is positive if $\rho_2 > \rho_1$

Q is negative if $\rho_2 < \rho_1$

55. Minimum current that is possible through A is $4A$. Since three elements are in series, they must have same current.



Hence minimum current in the circuit is $4A$.

For $4A$, current

$$V_{\min} = V_A + V_B + V_C = 10 + 20 + 6 = 36 \text{ V}$$

And
$$V_{\max} = V_A + V_B + V_C = 20 + 20 + 20 = 60 \text{ V}$$

$\therefore$ From 36 V to 60 V supply the current remains $4A$ and below 36 V there is no current.

For $V_A > 20$ V, $V_B > 20$ V, $V_C > 20$ V

All elements have same resistance (the slope of graph is same for all) $R = \frac{20}{4} = 5\ \Omega$.

$\therefore$ The effective resistance is 15 Ω for applied emf > 60 V.

At 90 V, the current is $\frac{90}{15} = 6A$.

56. The device conducts only when voltage across it is 20 V.

$\therefore$
$$V > 20 \text{ V}$$

$$I_{\max} = \frac{P_{\max}}{20 \text{ volt}} = \frac{1 \text{ Watt}}{20 \text{ V}} = \frac{1}{20} A.$$

From the circuit

$$V = 100\, I + 20$$

$\therefore$
$$I = \frac{V - 20}{100}$$

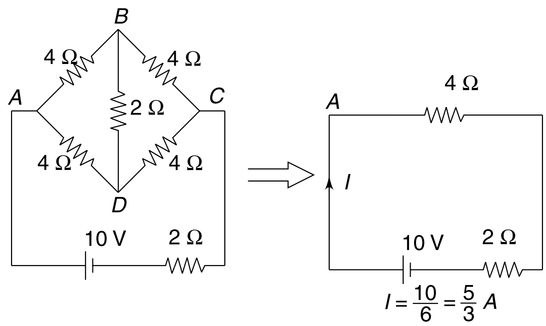
But $$I \le \frac{1}{20}\,A$$

$\therefore$ $$\frac{V-20}{100} \le \frac{1}{20}$$

$$V - 20 \le 5$$

$$V \le 25 \text{ volt.}$$

57. (a) The circuit is a wheatstone bridge connected to a battery with a resistance in series. The bridge is balanced.



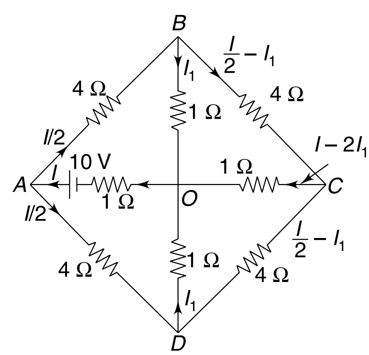
(b) The current distribution is as shown. Current in branch AB and AD are equal due to symmetry. Similarly current in BO and OD are same. Current in BC and DC are also same.

Applying Kirchhoff's second law to loop $OBCO$

$$1 \times I_1 - 4\left(\frac{I}{2} - I_1\right) - 1(I - 2I_1) = 0$$

$$I_1 = \frac{3I}{7} \qquad \text{...(1)}$$

Applying the same law to loop $ABOA$



$$4 \times \frac{I}{2} + 1 \times I_1 + 1 \times I = 10$$

$$3I + I_1 = 10$$

$\Rightarrow$ $$3I + \frac{3I}{7} = 10$$

$\Rightarrow$ $$I = \frac{70}{24} = \frac{35}{12}\,A.$$

58. **Between *a* and *b*.**

40 Ω and 160 Ω in series makes 200 Ω.

200 Ω in parallel with 50 Ω makes $\dfrac{50 \times 200}{50 + 200} = 40\ \Omega$

40 Ω and 50 Ω is series makes 90 Ω.

90 Ω and 200 Ω in parallel make $\dfrac{90 \times 200}{90 + 200} = \dfrac{1800}{29}$.

Now 10 Ω, $\dfrac{1800}{29}$ and 10 Ω are in series

$\therefore$ $$R_{ab} = \frac{2380}{29}\ \Omega$$

Between *a* and *c*

The 10 Ω resistance connected to b is open circuit (not included in the circuit)

Between d and c we have a balanced wheatstone bridge $\left(\frac{50}{40} = \frac{200}{160}\right)$

$\therefore$ 50 Ω resistance can be removed

50 Ω and 40 Ω in series = 90 Ω

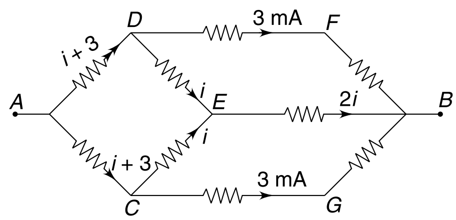
160 Ω and 200 Ω in series = 360 Ω

90 Ω and 360 Ω in parallel = 72 Ω

72 Ω and 10 Ω in series = 82 Ω.

$$\therefore \qquad \frac{R_{ab}}{R_{ac}} = \frac{2380}{19} \times \frac{1}{82} = \frac{1190}{1189}$$

59. (a) let each resistance be R. Due to symmetry current through DF is also 3 mA. If current in CE is i, then current through DE must also be i and current in EB is $2i$. Current in AC and AD is $(i + 3)$ mA



$$V_{CB} = (2R)(3) = iR + 2iR$$

$$\Rightarrow \qquad 3i = 6 \quad \Rightarrow \quad i = 2 \text{ mA.}$$

$\therefore$ Current drawn from the 5V supply is 10 mA

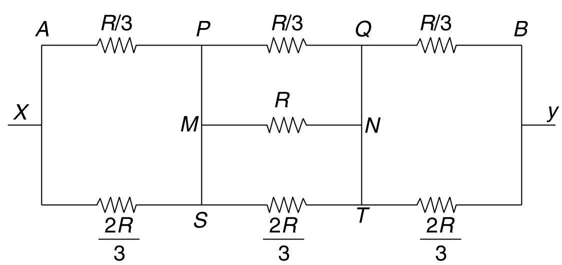
$$\therefore \qquad R_{\text{eff}} = \frac{5 \text{ V}}{10 \text{ mA}} = 500 \ \Omega$$

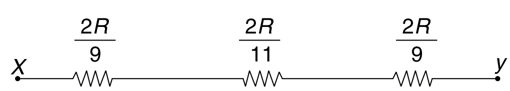
60. $\frac{R}{3}$ is in parallel to $\frac{2R}{3}$.

$\frac{R}{3}$, R and $\frac{2R}{3}$ are in parallel.

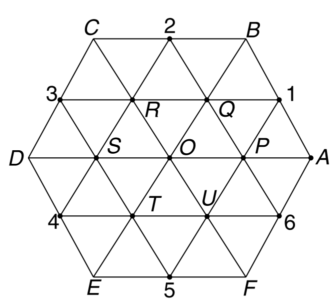
$\frac{R}{3}$ and $\frac{2R}{3}$ are in parallel.

$\therefore \quad R_{eq} = \frac{62R}{99}$

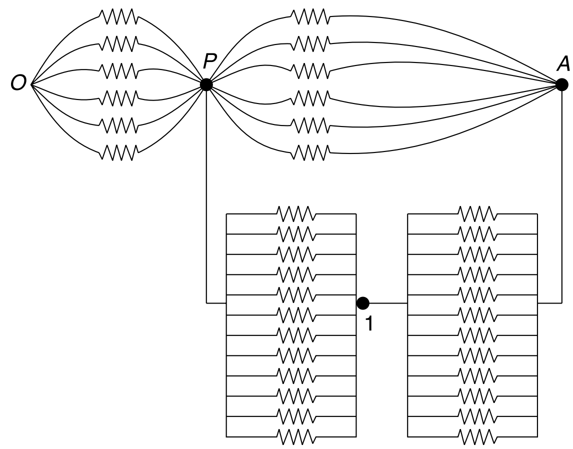




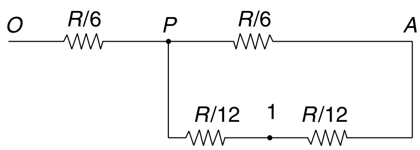
61. Points A, B, C, D, E & F are equipotentials.



From symmetry, points P, Q, R, S, T and U are equipotential and 1, 2, 3, 4, 5 and 6 are also equipotential. The circuit can be redrawn as



$\Rightarrow \therefore \quad R_{eq} = \dfrac{R}{4}$



62. Resistance of PA = Resistance of QB is

$R_1 = K(a\theta)$ $\quad$ K = a constant

$R_1 = C\theta$ $\quad$ a = radius

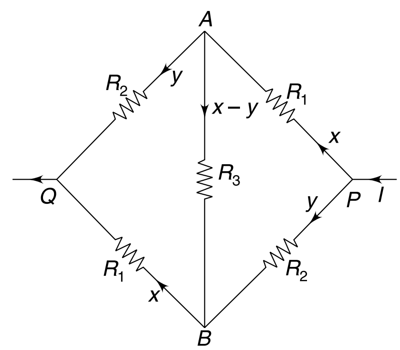
[$C = k \cdot a$ = another constant]

Resistance of PB = Resistance of QA is

$$R_2 = C(\pi - \theta)$$

Resistance of AB

$$R_3 = K(2a) = 2C.$$



The circuit can be redrawn as shown.

Taking into account the symmetry, the current distribution can be assumed as shown in the Figure.

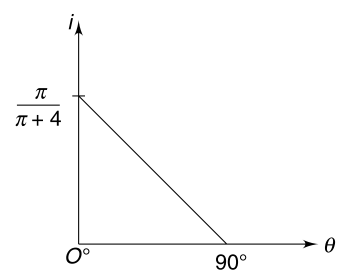
Using Kirchhoff's law in loop ABP we get

$$R_1x + R_3(x - y) = R_2y$$

$$C\theta x + 2C(x - y) = C(\pi - \theta)y$$

$$\Rightarrow \quad (2 + \theta)x = (\pi + 2 - \theta)y \quad \text{...(1)}$$

And $\quad x + y = I \quad$ ...(2)



Solving (1) and (2)

$$x = \frac{(\pi + 2 - \theta)I}{\pi + 4} \quad \text{and} \quad y = \frac{(2 + \theta)I}{\pi + 4}$$

$$\therefore \quad x - y = \frac{\pi - 2\theta}{\pi + 4} \quad \Rightarrow \quad i = \frac{\pi - 2\theta}{\pi + 4}$$

63. (a) Using Kirchhoff's voltage law in the bigger loop we get

$$E - 10 = 5i_1 + I \quad \text{...(1)}$$

In the left loop we get

$$E = (I - i_1)\,2.5 + I$$

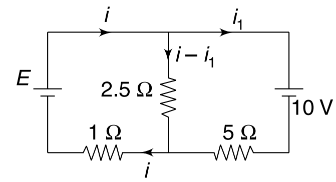
$$\Rightarrow \quad E = 3.5\,I - 2.5i_1 \quad \text{...(2)}$$

Multiplying (1) by 3.5 and subtracting (2) from it

$$2.5\,E - 3.5 = 20\,i_1$$

$$\Rightarrow \quad i_1 = \frac{2.5\,E - 35}{20} = \frac{E - 14}{8} \qquad ...(3)$$

Since $\quad E = 2t$



$$\therefore \quad i_1 = \frac{2t - 14}{8} = \frac{t}{4} - \frac{7}{4}$$

Energy gained by 10 V cell is

$$U = \int_0^{14} i_1 \times 10\, dt = 2.5 \int_0^{14} (t - 7)\, dt$$

$$= 2.5\left[\frac{14^2}{2} - 7 \times 14\right] = 0$$

(b) **From (3)** i_1 is positive when $E > 14$ volt

$\therefore$ Cell starts charging after $t = 7$ s.

Before this it was discharging.

64. Let current in the circuit be I.

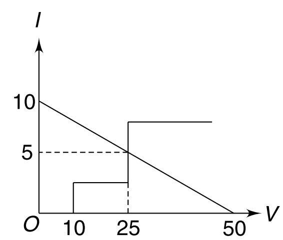
Potential difference across A is $V = E - IR$

or, $\quad V = 100 - 10\,I \qquad ...(1)$

Current through R connected in parallel to A is

$$I' = \frac{V}{R} = \frac{100}{10} - \frac{10I}{10} = 10 - I$$

$\therefore$ Current through the box is



$$i = I - I'$$

$$i = I - (10 - I) = 2I - 10$$

Using (1) $\quad I = 10 - \dfrac{V}{10}$

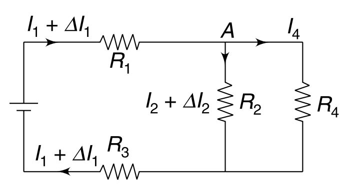
$$\therefore \quad I = 2\left(10 - \frac{V}{10}\right) - 10$$

$$I = 10 - \frac{V}{5} \qquad ...(2)$$

Plot the equation (2) on the given graph

The intersection point gives the correct value of V. Hence answer is 25 V.

65. (a) When R_4 is added, the equivalent resistance of the circuit decreases. Hence, current through R_1 increases.



(b) let $R_1 = 4\,R$, $R_2 = R$, $R_3 = 2\,R$.

Note: Kirchoff's laws are not valid only for currents and voltages, but also for voltage increments ΔV_i and current increments ΔI_i

$$\Delta I_1 R_1 + \Delta I_2 R_2 + \Delta I_3 R_3 = 0$$

$$[\Delta I_1 = \Delta I_3 = 0.2A]$$

$$\therefore \quad 0.2 \times 4R + \Delta I_2 \times R + 0.2 \times 2R = 0$$

$$\Delta I_2 = -1.2A$$

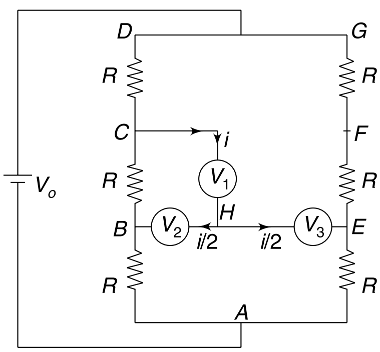
Negative sign means I_2 decreases.

At junction A

$$\Delta I_1 = \Delta I_2 + I_4$$

$$0.2 = -1.2 + I_4$$

$$\therefore \quad I_4 = 1.4\,A.$$



66. (a) When the switch is open, potential drop across each R is $= \dfrac{V_0}{3}$.

Voltmeters are nearly ideal. A very small current will flow through V_1 and V_2 and both of them will show equal reading of $\dfrac{V_0}{6}$.

(b) **After 'S' is closed.**

p.d. across each 'R' is still $\dfrac{V_0}{3}$.

Assuming
$$V_A = 0$$
$$V_B = V_E = \frac{V_0}{3}$$
$$V_C = V_F = \frac{2V_0}{3}$$
$$V_D = V_G = V_0$$

If current through V_1 is i, then current through V_2 and V_3 will be $\dfrac{i}{2}$ as shown. (Remember i is very small] If resistance of each voltmeter is R_V

$$iR_V = V_C - V_H$$
$$iR_V = \frac{2V_0}{3} - V_H \qquad ...(1)$$

And
$$\frac{i}{2}R_V = V_H - V_B$$
$$\frac{i}{2}R_V = V_H - \frac{V_0}{3} \qquad ...(2)$$

(1) + (2)
$$\frac{3}{2}iR_V = \frac{V_0}{3}$$

$\therefore$
$$iR_V = \frac{2V_0}{9}$$

67. Total charge flow in first 20 min can be calculated as–

$$Q_0 = \int_0^t idt = \int_0^{t=20} \frac{E}{R}dt = \int_0^{t=20} \frac{E}{20+t}dt$$

$\Rightarrow$
$$Q_0 = E\,\ell n\left(\frac{20+20}{20}\right) = E\,\ell n(2)$$

After 20 min the resistance remains constant at 40 Ω and current remains constant at $I = \dfrac{E}{40}$

$\therefore$ Charge flow in next Δt time is

$$\Delta Q = I\Delta t = \frac{E\Delta t}{40}$$

Energy lost by a cell is proportional to the charge flow through it. Hence

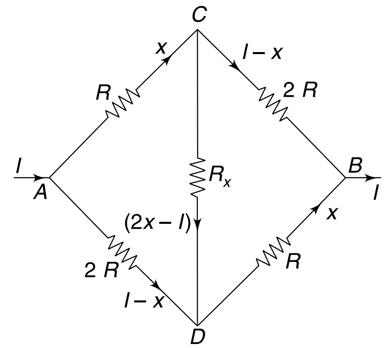
$$\frac{\Delta Q}{Q_0} = \frac{9}{1}$$

$\Rightarrow$
$$\frac{\frac{E\Delta t}{40}}{E\ell n2} = 9 \Rightarrow \Delta t = 330\ \ell n2$$

$\therefore$ Total time for which the cell lasts is

$$T = (20 + 360\ \ell n 2) \text{ minute.}$$

68. Taking into account the symmetry, current distribution is as shown in the Figure.



Using Kirchhoff's voltage law in loop *ACDA*–

$$Rx + R_x(2x - I) = 2R(I - x)$$

$$\Rightarrow \qquad x = \left(\frac{2R + R_x}{3R + 2R_x}\right) I$$

Now
$$V_A - V_B = Rx + 2R(I - x)$$
$$= 2RI - R \cdot x$$
$$= R\left[2I - \frac{2R + R_x}{3R + 2R_x} \cdot I\right]$$
$$= R\left[\frac{4R + 3\,R_x}{3R + 2R_x}\right] I$$

Equivalent resistance
$$R_{AB} = \frac{V_A - V_B}{I}$$

$$\therefore \qquad R_{AB} = \left[\frac{4R + 3R_x}{3R + 2R_x}\right] R$$

$$= \left[\frac{4 + 3\dfrac{R_x}{R}}{3 + 2\dfrac{R_x}{R}}\right] R$$

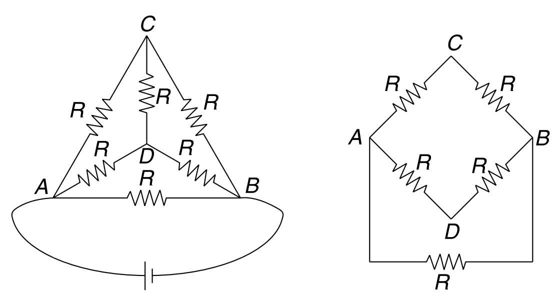
When $\dfrac{R_x}{R} \to 0$

$$R_{AB} = \frac{4}{3} R$$

When $\dfrac{R_x}{R} \to \infty$

$$R_{AB} = \frac{3}{2} R.$$

69. Equivalent circuit is as shown. If the resistance connected directly between *A* and *B* is removed, the remaining circuit is a balanced wheatstone bridge with $V_C = V_D$. Resistance between *C* and *D* can be removed.

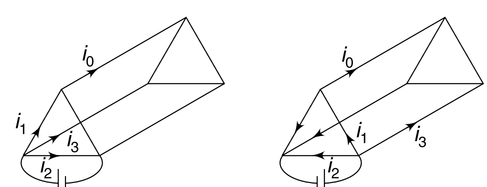


$$R_{eq} = \frac{R}{2}$$

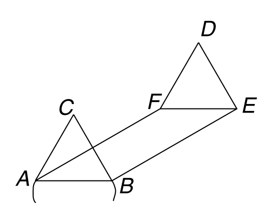
70. **(a)** If a cell is connected across *A* and *B* with *A* positive, then let's assume that current through *CD* is i_0 – from *C* to *D*. If polarity of cell is reversed (*i.e.*, *B* is made positive) the direction of current in each branch must reverse without change in magnitude. But symmetry tells us that direction of current in *CD* will remain same even after the polarity is reversed. (see Fig. (a) & (b))

This can be true only if $i_0 = 0$.

Remove *CD*. Now the circuit is simple series parallel combination of resistors (Fig. (c))



(a) (b)



(c)

$$R_{FE} = \frac{R \cdot 2R}{R + 2R} = \frac{2R}{3}$$

$$R_{AFEB} = R + \frac{2R}{3} + R = \frac{8R}{3}$$

$$R_{AB} = \frac{2R}{3}$$

$$\therefore \quad R_{eq} = \frac{\frac{8R}{3} \cdot \frac{2R}{3}}{\frac{8R}{3} + \frac{2R}{3}} = \frac{16R}{30} = \frac{8R}{15}$$

(b) In this case A and B are equipotential points. Similarly, E and F are equipotential. Remove the arm AB and EF and join A and B together. Also join E and F together. Now, it is a series parallel circuit. Prove yourself that $R_{eq} = \frac{3R}{5}$.

71. (i) Power dissipated in resistance = Rate of heat dissipation to surrounding + Rate of heat absorption by conductor

$$\frac{V^2}{R} = k(T - T_0) + ms\frac{dT}{dt}$$

$$ms\frac{dT}{dt} = \frac{V^2}{R} - k(T - T_0)$$

$$\int_{T_0}^{T} \frac{dT}{\frac{V^2}{R} - k(T - T_0)} = \frac{1}{ms}\int_0^t dt$$

$$\left\{\ell n\left[\frac{V^2}{R} - k(T - T_0)\right]\right\}_{T_0}^{T} = -\frac{kt}{ms}$$

$$\ell n\left[\frac{V^2}{R} - k(T - T_0)\right] - \ell n\left[\frac{V^2}{R}\right] = -\frac{kt}{ms}$$

$$1 - \frac{Rk(T - T_0)}{V^2} = e^{-\frac{kt}{ms}}$$

$$\therefore \quad T - T_0 = \frac{V^2}{kR}\left(1 - e^{-\frac{kt}{ms}}\right)$$

$$T = T_0 + \frac{V^2}{kR}\left(1 - e^{-\frac{kt}{ms}}\right)$$

(ii) when $t \to \infty$; $e^{-\frac{kt}{ms}} \to 0$

$$\therefore \quad T = T_0 + \frac{V^2}{kR}$$

72. Electric power dissipated

$$= \frac{V^2}{R} = \frac{V^2}{\rho\frac{\ell}{\pi r^2}} = \frac{\pi r^2 V^2}{\rho \ell}$$

Radiant power loss

$$= e \cdot \sigma\, 2\pi r\ell\left[T^4 - T_0^4\right]$$

$$= 2\pi e\,\sigma r\ell\left[(T_0 + \Delta T)^4 - T_0^4\right]$$

$$= 2\pi e\sigma r\ell\, T_0^4\left[\left(1 + \frac{\Delta T}{T_0}\right)^4 - 1\right]$$

$$= 2\pi e\sigma r\ell T_0^4\left[1 + \frac{4\Delta T}{T_0} \ldots - 1\right]$$

$$= 2\pi e\sigma r\ell T_0^4\left[\frac{4\Delta T}{T_0}\right]$$

[Because $\Delta T << T_0$]

$$= 8\pi e\sigma r\ell\, T_0^3\, \Delta T$$

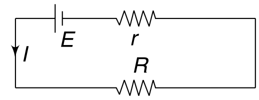
For steady state

$$8\pi e\sigma r\ell T_0^3\, \Delta T = \frac{\pi r^2 V^2}{\rho\ell}$$

$$\Rightarrow \quad \Delta T = \frac{V^2 r}{8e\sigma\rho\ell^2\, T_0^3}$$

$$\Rightarrow \quad T = T_0 + \frac{V^2 r}{8e\sigma\rho\ell^2\, T_0^3}$$

73. (a) Power dissipated in R; $P_R = I^2R$.



Power spent by the cell; $P_{cell} = I^2(R + r)$

Fraction of energy spent by the cell that is dissipated in R is

$$\eta = \frac{P_R}{P_{cell}} = \frac{R}{R + r}$$

$$\eta = \frac{1}{1 + \frac{r}{R}}$$

η will be maximum if R is maximum (*i.e.* = 4Ω)

Total energy dissipated in R will be

$$H_0 = \eta\cdot(EQ) \text{ [}Q\text{ = total charge flow through the cell]}$$

$$= \frac{1}{1+\frac{1}{4}}.\ 12 \times [2000 \times 10^{-3} \times 3600]$$

$$= \frac{4}{5} \times 12 \times 7200 = 69.12 \text{ KJ.}$$

(b) Power transfer to R is maximum when $R = r = 1\Omega$.

In this case, half the energy spent by the cell will get dissipated in R

$$\therefore \quad H = \frac{EQ}{2}$$

$\therefore$ Required answer is

$$\frac{H}{H_0} \times 100$$

$$= \frac{\frac{EQ}{2}}{\eta EQ} \times 100 = \frac{1}{2 \times \frac{4}{5}} \times 100 = 62.5\%$$

74. (a) In first experiment

$$\frac{R_1}{R_2} = 10$$

For balance

$$\frac{R_3}{R_4} = 10$$

$$\therefore \quad R_4 = \frac{R_3}{10}$$

$$\therefore \quad 64.3 < R_4 < 64.4\,\Omega.$$

In second experiment

$$\frac{R_1}{R_2} = 100$$

$\therefore$ For balance $\dfrac{R_3}{R_4} = 100$

$\therefore$ $R_4 = \dfrac{R_3}{100}$

$\therefore$ We should try with $6430 < R_3 < 6440\ \Omega$

$\therefore$ $64.32 < R_4 < 64.33\ \Omega$

$R_4 \simeq 64.325\ \Omega$

75. (a) $\dfrac{R_x}{R} = \dfrac{x}{L - x}$

$R_x = R\left(\dfrac{x}{L - x}\right)$ ...(i)

For $R = 20\,\Omega,\ x = 20$ cm

$R_x = 20\left[\dfrac{20}{100 - 20}\right] = 5\,\Omega$

From (i) $\ell n R_x = \ell n R + \ell n x - \ell n (L - x)$

$\dfrac{dR_x}{R_x} = \dfrac{dx}{x} + \dfrac{dx}{L - x}$

$\dfrac{\Delta R_x}{R_x} = \dfrac{L\Delta x}{x(L - x)}$ [Δx = maximum error in x] ...(2)

For $R_x = 5\,\Omega$; $|\Delta x| = 0.1$ cm; $L = 100$ cm $x = 20$ cm

$\therefore$ $|\Delta R_x| = \dfrac{5 \times 100 \times 0.1}{20 \times 80} = 0.03\ \Omega$

$\therefore$ $R_x = (5.00 \pm 0.03)\ \Omega$

(b) From equation (2), the fractional error $\dfrac{\Delta R_x}{R_x}$ is minimum for a given error in x (*i.e.* for given Δx)

When $x(L - x)$ is maximum.

This happens when $x = \dfrac{L}{2}$

To minimize the error, we need to change the known resistance R so as to bring it close to R_x. In that case $x = \dfrac{L}{2}$.

You can show that if R were equal to 5 Ω in the first part of the question, $R_x = (5.00 \pm 0.02)\ \Omega$.

76. (a) when $I = 0$

$x = 20$ cm

It means $V_{AC} = \dfrac{V_0 x}{L} = \dfrac{12 \times 20}{100} = 2.4$ volt

Since there is no current through the milliammeter

Hence $V_{AC} = V = 2.4$ volt.

(b) When $x = 0$; $I = 30$ mA

$\therefore$ $2.4 = I\,R$

$2.4 = 30 \times 10^{-3} \cdot R$

$R = 80\ \Omega$

(c) When $x = 100$

$$V_{AB} = V_0 = 12 \text{ volt}$$

$\therefore$ $80I = 12 - 2.4$

$$I = 0.12\ A = 120 \text{ mA}$$

This current is in negative direction

77. The internal resistance of equivalent cell is given by

$$\frac{1}{r_0} = \frac{1}{r} + \frac{1}{2r} + \frac{1}{4r} + \ldots + \frac{1}{16r}.$$

$$\frac{1}{r_0} = \frac{1}{r}\left[\frac{1}{1} + \frac{1}{2} + \frac{1}{2^2} + \ldots + \frac{1}{2^4}\right]$$

$$= \frac{1}{r}.\frac{1\left[1 - \left(\frac{1}{2}\right)^5\right]}{\left[1 - \frac{1}{2}\right]}$$

$$= \frac{1}{r}.\frac{31}{16}$$

$\therefore$ $$r_0 = \frac{16r}{31}$$

Equivalent emf $$\varepsilon_0 = \frac{\frac{\varepsilon}{r} + \frac{2\varepsilon}{2r} + \frac{4\varepsilon}{4r} + \frac{8\varepsilon}{8r} + \frac{16\varepsilon}{16r}}{\frac{1}{r_0}}$$

$$= \frac{5\varepsilon}{r}.\frac{16r}{31} = \frac{80\varepsilon}{31}$$

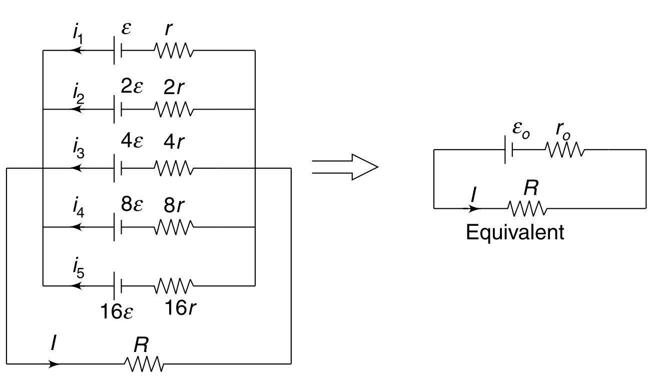
$\therefore$ $$I = \frac{\varepsilon_0}{R + r_0}$$

$$I = \frac{\frac{80\varepsilon}{31}}{R + \frac{16r}{31}} = \frac{80\varepsilon}{31R + 16r}$$

If we write the voltage equation for a loop containing any cell (n^{th} cell) and load resistance R, we get

$$i_n r_n = \varepsilon_n - IR$$

$$i_n = \frac{\varepsilon_n}{r_n} - \frac{IR}{r_n}$$



value of $\frac{\varepsilon_n}{r_n} = \frac{\varepsilon}{r}$ for all cells

$$\therefore \qquad i_n = \frac{\varepsilon}{r} - \frac{IR}{r_n}$$

$\therefore$ i_n is maximum for the cell which has largest value of r_n (*i.e.*, cell having internal resistance $16r$)

78. (a) Let the resistance of voltmeter be r

$$V_{AB} = 24\text{V}; \quad V_{BC} = V_{CD} = 30\text{V}$$

$$\therefore \qquad \frac{R_{AB}}{R} = \frac{24}{30} \Rightarrow R_{AB} = \frac{16}{5}$$

$$\Rightarrow \qquad \frac{4r}{4+r} = \frac{16}{5}$$

$$\Rightarrow \qquad r = 16\ \Omega.$$

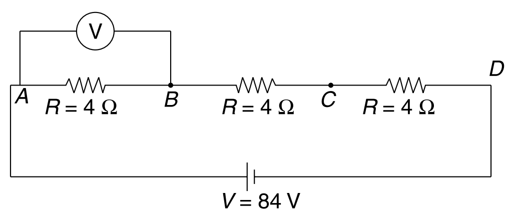
When connected across A and C

$$R_{AC} = \frac{(2R)(r)}{2R+r} = \frac{2 \times 4 \times 16}{2 \times 4 + 16} = \frac{16}{3}\ \Omega$$

$$\therefore \qquad \frac{V_{AC}}{V_{CD}} = \frac{R_{AC}}{R}$$

$$\therefore \qquad V_{AC} = \left(\frac{R_{AC}}{R_{AC}+R}\right) \times 84$$

$$= \frac{16/3}{16/3 + 4} \times 84 = 48\ \text{V}$$



(b) Yes, the voltmeter will now be more close to an ideal voltmeter as its resistance will far exceed the resistance in the circuit.

79. Let resistance of each voltmeter be R_0.

$$Ri = 20 = R_0(I - i) \qquad \text{...(1)}$$

And
$$2Ri' = 30 \qquad \text{...(2)}$$

$$(2) \div (1) \quad \frac{2i'}{i} = \frac{3}{2}$$

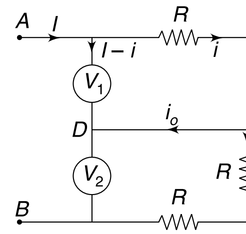
$$i' = \frac{3}{4} i$$

$$\therefore \qquad i_0 = i - \frac{3i}{4} = \frac{i}{4}$$

$\therefore$ current through V_2 is $I - i + \frac{i}{4} = I - \frac{3i}{4}$

$$R_0\left(I - \frac{3i}{4}\right) = 2Ri' = 30 \qquad \text{...(3)}$$

From (1) and (3)



$$\frac{I-i}{I-\frac{3i}{4}} = \frac{2}{3}$$

$$\Rightarrow \quad 3I - 3i = 2I - \frac{3i}{2}$$

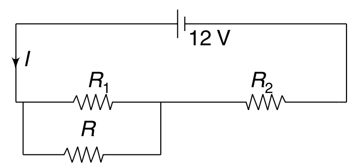
$$\Rightarrow \quad I = \frac{3i}{2} \quad \Rightarrow \quad i = \frac{2}{3} \times 600 = 400\ \mu A$$

$$\because \quad Ri = 20$$

$$R \times 400 \times 10^{-6} = 20$$

$$R = 50\ k\Omega.$$

81. The reading of voltmeter is absolutely correct when it is connected across the battery.



It means the emf of the cell is = 120 volt.

Let the resistance of the voltmeter be R.

The current I is

$$I = \frac{12}{\frac{R_1 R}{R_1 + R} + R_2}$$

$\therefore$ Voltmeter reading = voltage drop across R_1(or R)

$$\Rightarrow \quad 4 = \frac{12}{\frac{R_1 R}{R_1 + R} + R_2} \cdot \frac{R_1 R}{R_1 + R}$$

$$\therefore \quad 1 + \frac{R_2(R_1 + R)}{R_1 R} = 3$$

$$\Rightarrow \quad \frac{R_2(R_1 + R)}{R_1 R} = 2$$

$$\Rightarrow \quad \frac{R_2}{R} + \frac{R_2}{R_1} = 2 \qquad ...(1)$$

Similarly

$$V_2 = \frac{12}{\frac{R_2 R}{R_2 + R} + R_1} \cdot \frac{R_2 R}{R_2 + R}$$

$$\Rightarrow \quad 6 = \frac{12}{1 + \frac{R_1(R_2 + R)}{R_2 R}}$$

$$\Rightarrow \quad \frac{R_1 R_2 + R_1 R}{R_2 R} = 1$$

$$\Rightarrow \quad \frac{R_1}{R} + \frac{R_1}{R_2} = 1 \qquad ...(2)$$

Eliminating R between (1) and (2)

$$\frac{R_1}{R_2} = \frac{2}{3}$$

The actual voltage across R_1 is

$$V_1 = \frac{12 \cdot R_1}{R_1 + R_2} = \frac{12}{1 + \dfrac{R_2}{R_1}} = \frac{12}{1 + \dfrac{3}{2}}$$

$$= \frac{24}{5} = 4.8 \text{ Volt.}$$

82. The ohm meter reads $R = 0\Omega$ when full scale deflection current passes through it.

$$\therefore \quad 2 \text{ mA} = \frac{20 \text{ volt}}{R_0 + 20\ \Omega}$$

$$\therefore \quad R_0 = 9980\ \Omega$$

When $R = \infty$, current in the galvanometer is zero and there is no deflection.

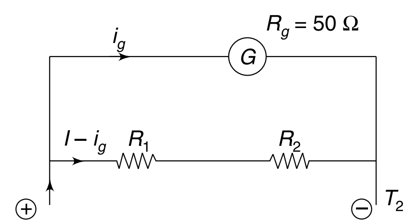
For $\theta = 90°$, current through the galvanometer is

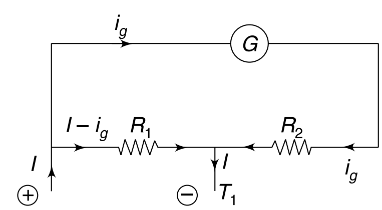
$$i = \frac{2 \text{ mA}}{120°} \times 90° = 1.5 \text{ mA}$$

$$\therefore \quad \frac{20 \text{ volt}}{9980 + 20 + R} = 1.5 \times 10^{-13}$$

$$\Rightarrow \quad 13333 = 10{,}000 + R$$

$$\Rightarrow \quad R = 3333\ \Omega$$





83. (a) A smaller shunt is required to measure higher current. Hence, T_1 shall be (–) terminal if current is higher.

For current of the order of 0.1 A, T_2 shall be the (–) terminal

(b) For the range 0 – 1 A, R_1 and R_2 are in series.

$$i_g R_g = (I - i_g)(R_1 + R_2)$$

$$\Rightarrow \quad R_1 + R_2 = \frac{1 \times 10^{-3} \times 50}{1 - 10^{-3}} = \frac{50}{999} \quad ...(1)$$

For range 0 – 10 A, T_1 is negative terminal. Hence R_g and R_2 are in series and R_1 is in parallel to them.

$$i_g(R_g + R_2) = (I - i_g)R_1$$

$$10^{-3}(50 + R_2) = (10 - 10^{-3})R_1$$

$$\Rightarrow \quad 10^{-3}(R_1 + R_2) + 0.050 = 10R_1$$

Using (1)

$$\frac{0.050}{999} + 0.050 = 10R_1$$

$$\Rightarrow \quad R_1 = \frac{5}{999}\Omega$$

From (1) $\quad R_2 = \dfrac{45}{999} = \dfrac{5}{111}\Omega.$

84. (a) **In series**

current $\quad i = k_1\theta_1 = k_2\theta_2 = k_2\theta_3 \quad ...(1)$

In parallel

$$V_1 = V_2 = V_3$$

$$k_1\ \theta_1' \cdot r_1 = k_2\theta_2' r_2 = k_3\ \theta_3' r_3 \quad ...(2)$$

From (2) $r_2 = r_1\left(\dfrac{k_1\ \theta_1'}{k_2\ \theta_2'}\right)$

$\Rightarrow$ $r_2 = r_1\left(\dfrac{\theta_2\ \theta_1'}{\theta_1\ \theta_2'}\right)$ [using (1)]

Similarly, $r_3 = r_1\left(\dfrac{\theta_3}{\theta_1}\cdot\dfrac{\theta_1'}{\theta_3'}\right)$

(b) From (1), if $\theta_2 = \theta_3$

Then $k_2 = k_3$.

From (2) $k_2\theta_2' r_2 = k_3\theta_3' r_3$

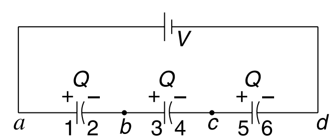
$\Rightarrow$ $\theta_2'\ r_2 = \theta_3'\ r_3$

If $\theta_3' > \theta_2'$

It means $r_2 > r_3$

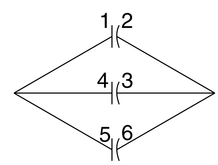
85. Charge on each plate (polarity shown) when capacitors are charged

$$Q = \frac{CV}{3}.$$



The cell is removed and the capacitors are connected in parallel using resistors. Plate 1, 4 and 5 are kept together and 2, 3 and 6 are held together.
Total charge available for distribution amongst the capacitors = $Q + Q - Q = Q$.

When charge flow stops, charge on each capacitor = $\dfrac{Q}{3} = \dfrac{CV}{9}$.



Heat dissipated = loss in energy stored in capacitor system

$$= \frac{1}{2}\cdot\frac{C}{3}V^2 - \frac{1}{2C}\left(\frac{CV}{9}\right)^2 \times 3$$

$$= \left(\frac{1}{6} - \frac{1}{54}\right)CV^2 = \frac{4}{27}CV^2$$

Heat dissipated in each resistor

$$= \frac{2}{27}CV^2$$

86. (a) Resistance of the air gap

$$R = \rho\frac{L}{A} = 3 \times 10^{13}\cdot\frac{3000}{4\pi(6 \times 10^6)^2} = 199\,\Omega$$

(b) The potential difference between two concentric spheres is

$$V = kq\left(\frac{1}{R} - \frac{1}{R+L}\right)$$

$$= \frac{kq}{R}\left[1 - \left(1 + \frac{L}{R}\right)^{-1}\right]$$

$$= \frac{kq}{R}\left[1 - \left(1 - \frac{L}{R}\right)\right] = \frac{kqL}{R^2}$$

$$= \frac{9 \times 10^9 \times 4 \times 10^5 \times 3000}{(6 \times 10^6)^2} = 3 \times 10^5 \text{ volt.}$$

(c) Capacitance; $C = \dfrac{Q}{V} = \dfrac{4 \times 10^5}{3 \times 10^5} = \dfrac{4}{3}$ F.

Time constant $\tau = RC = 199 \times \frac{4}{3} = 265.3$ s.

Time needed for 63% discharging $= \tau = 265.3$ s.

87. The final voltage across A will be

$$V_1 = \frac{C_2}{C_1 + C_2} V = \frac{2}{3+2} \times 100 = 40 \text{ V}$$

and across B it will be

$$V_2 = \frac{C_1}{C_1 + C_2} \cdot V = \frac{3}{3+2} \times 100 = 60 \text{ V}$$

Obviously, capacitor B will breakdown first as soon as voltage across it reaches 50 V (at that time voltage across A will be less than 40 V]

The effective capacitance of the circuit is

$$C = \frac{C_1 C_2}{C_1 + C_2} = \frac{3 \times 2}{3+2} = \frac{6}{5} \mu\text{F}$$

The effective resistance is $R = 3 + 2 = 5$ MΩ.

Time constant of the circuit $\tau = RC = 6$ sec

Charge on capacitors as a function of time

$$q = q_0[1 - e^{-t/\tau}]$$

$$q_0 = CV = \frac{6}{5} \mu\text{F} \times 100 \text{ V} = 120 \ \mu\text{C}$$

$$q = (120 \ \mu C)(1 - e^{-t/\tau})$$

Breakdown of B will take place when

$$Q = 2\mu\text{F} \times 50 \text{ V} = 100 \ \mu\text{C}$$

$$\therefore \quad 100 \ \mu\text{C} = (120 \ \mu\text{C}) \ (1 - e^{-t/\tau})$$

$$\Rightarrow \quad \frac{5}{6} = 1 - e^{-t/\tau}$$

$$\Rightarrow \quad e^{-t/\tau} = \frac{1}{6}$$

$$\frac{t}{\tau} = \ell n 6$$

$$t = \tau \ell n 6 = 6 \ell n 6 \text{ sec.}$$

88. When switch is in position 1, potential difference across R_2 is

$$V = \frac{R_2}{R_1 + R_2} V_1 = \frac{60}{20 + 60} \times 40 = 30 \text{ volt.}$$

This is also the potential difference across the capacitor with plate b positive.

Charge on capacitor is

$$q_0 = CV = (0.5 \ \mu\text{F}) \ (30 \text{ V}) = 15 \ \mu\text{C.}$$

As per sign convention given in the problem, initial charge (q_0) on the capacitor is negative

Let charge on it be q at time t. Then

$$R_3 \frac{dq}{dt} + \frac{q}{C} = V_2$$

$$R_3 \frac{dq}{dt} = V_2 - \frac{q}{C}$$

$$\Rightarrow \quad \int_{q_0}^{q} \frac{dq}{V_2 - \frac{q}{C}} = \frac{1}{R_3} \int_0^t dt$$

$$\Rightarrow \qquad \left[\ell n\left(V_2 - \frac{q}{C}\right)\right]_{q_0}^{q} = -\frac{t}{R_3C}$$

$$\Rightarrow \qquad \ell n\left(V_2 - \frac{q}{C}\right) - \ell n\left(V_2 - \frac{q_0}{C}\right) = -\frac{t}{R_3C}$$

$$\Rightarrow \qquad \frac{V_2 - q/C}{V_2 - q_0/C} = e^{-t/R_3C}$$

$$\Rightarrow \qquad V_2 - \frac{q}{C} = \left(V_2 - \frac{q_0}{C}\right)e^{-\frac{t}{R_3C}}$$

$$\Rightarrow \qquad q = C\left[V_2 - \left(V_2 - \frac{q_0}{C}\right)e^{-t/R_3C}\right]$$

$$\therefore \qquad q = CV_2 - (CV_2 - q_0)\, e^{-t/R_3C}$$

Now $\qquad CV_2 = 0.5\ \mu\text{F} \times 90\ \text{V} = 45\ \mu\text{C}$

$$q_0 = -15\ \mu\text{C}$$

$$R_3C = (400\,\text{k}\Omega)\ (0.5\ \mu\text{F}) = 0.2\ \text{s}$$

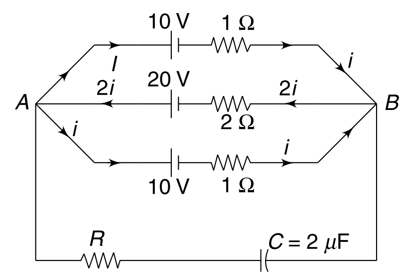
$$\therefore \qquad \frac{1}{R_3C} = \frac{1}{0.2} = 5$$

$$\therefore \qquad q = [45 - 60\,e^{-5t}]\ \mu\text{C}.$$

$\therefore$ Potential difference across C is

$$V = \frac{q}{C} = (90 - 120\ e^{-5t})\ \text{volt}.$$

89. There will be no current through the branch having the capacitor.



Using Kirchhoff's voltage law in a loop containing 20 V cell and one 10 V cell, we get

$$2i(2) + i(1) = 20 - 10$$

$$i = 2A$$

$$\therefore \qquad V_A - V_B = 12\ \text{volt}.$$

[OR, you can find the equivalent emf of the three cells in parallel]

$$\therefore \qquad q = (V_A - V_B)\ C = 24\ \mu\text{C}.$$

90. Force between capacitor plates is

$$F = \frac{q_0^2}{2\epsilon_0 A}$$

The spring must be compressed by x such that $kx = F$

$$\Rightarrow \qquad x = \frac{q_0^2}{2A\epsilon_0 k}$$

$\therefore$ Separation between plates $d = L - x = L - \dfrac{q_0^2}{2A\,\epsilon_0 k}$

Capacitance has initial value of $C_0 = \dfrac{\epsilon_0 A}{d}$

Energy stored in the capacitor is

$$U_0 = \frac{q_0^2}{2C_0} = \frac{q_0^2 d}{2\epsilon_0 A}$$

$$= \frac{q_0^2}{2\epsilon_0 A}\left(L - \frac{q_0^2}{2A\epsilon_0 k}\right) = \frac{q_0^2 L}{2\epsilon_0 A} - \frac{q_0^4}{4\epsilon_0^2 A^2 k}$$

Energy stored in the spring is

$$U_s = \frac{1}{2} kx^2 = \frac{q_0^4}{8\epsilon_0^2 A^2 k}$$

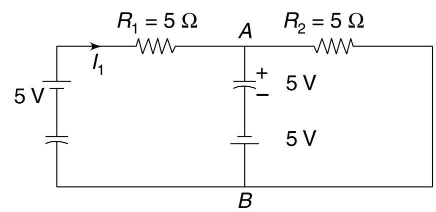
The entire energy stored in the capacitor and the spring gets dissipated as heat in the resistor.

$$\therefore \qquad \text{Heat} = U_0 + U_s = \frac{q_0^2 L}{2\epsilon_0 A} - \frac{q_0^4}{8\epsilon_0^2 A^2 k}$$

91. The spring will get no time to expand during the discharge and the heat dissipated will be = U_0

$$= \frac{q_0^2 L}{2\epsilon_0 A} - \frac{q_0^4}{4\epsilon_0^2 A^2 k}$$

92. With $S1$ closed (and $S2$ open) for long time the charge on capacitance C_1 will become C_1E_1 and potential difference across it will be E_1. Potential difference between A and B is zero. After $S2$ is closed, the potential difference across A and B cannot change instantaneously (since charge on the capacitor cannot change instantaneously)



Hence current through R_2 is zero immediately after S_2 is closed.

For $\qquad V_A - V_B = 0$

$$I_1R_1 = 5, I = 1 \ A$$

93. Just before the switch is opened

$$V_R = IR = 10 \text{ Volt.}$$

$\therefore$ p.d. across C_1 at this instant

$$V_0 = V_R = 10 \text{ Volt.}$$

Energy stored in C_1 at this instant is

$$U_1 = \frac{1}{2} C_1V_0^2 = \frac{1}{2} \times 3 \times 10^{-6} \times 10^2 = 150 \ \mu\text{J.}$$

Charge on C_1 at this instant $\qquad Q_0 = 3 \times 10^{-6} \times 10$

$$= 30 \ \mu\text{C}$$

Now this charge gets shared between C_1 and C_2 so that p.d across both of them becomes equal.
Final common p.d

$$V = \frac{Q_0}{C_1 + C_2} = \frac{30 \ \mu\text{C}}{4 \ \mu\text{F}} = 7.5 \text{ Volt.}$$

$\therefore$ Final energy stored in the capacitor system

$$U_2 = \frac{1}{2} (C_1 + C_2) V^2 = \frac{1}{2} \times 4 \times 10^{-6} \times (7.5)^2$$

$$= 28.13 \ \mu\text{J}$$

$\therefore$ Heat liberated = Energy lost

$$= 150 - 28.13 = 121.87 \ \mu\text{J}$$

94. Conductor of radius $2a$ gets charged to a potential V when S_1 is closed.

Capicitance $\qquad C_1 = 4\pi\epsilon_0(2a) = 8\pi\epsilon_0 a.$

Charge on the sphere $\qquad Q_0 = C_1V = 8\pi\epsilon_0 aV$

After S_2 is closed, both the spheres will acquire same potential. Let final charge on the two spheres be q_1 and q_2.

$$q_1 + q_2 = Q_0 \qquad \text{...(1)}$$

$$\frac{q_1}{8\pi\epsilon_0 a} = \frac{q}{4\pi\epsilon_0 a} \qquad \left[\because C_2 = \text{capacitance of small sphere} = 4\pi\varepsilon_0 a = \frac{C_1}{2}\right]$$

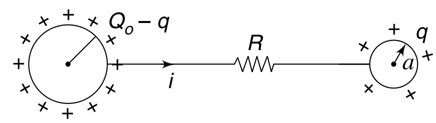
$$q_1 = 2q_2 \qquad \text{...(2)}$$

Solving (1) and (2)

$$q_1 = \frac{2Q_0}{3} = \frac{16\pi\epsilon_0 aV}{3}$$

$$q_2 = \frac{Q_0}{3} = \frac{8\pi\epsilon_0 aV}{3}$$

(a) let the charge on bigger sphere be $(Q_0 - q)$ and that on smaller sphere be q, at time t' after S_2 is closed.



$$\frac{dq}{dt} = i = \text{current through } R$$

Potential difference between two spheres = iR

$$\frac{Q_0 - q}{C_1} - \frac{q}{C_2} = iR$$

$$\frac{Q_0}{C_1} - q\left(\frac{C_1 + C_2}{C_1 C_2}\right) = iR$$

differentiating, w.r.t. time

$$-\frac{C_1 + C_2}{C_1 C_2}\frac{dq}{dt} = R\frac{di}{dt}$$

$$\Rightarrow \qquad -\frac{3}{8\pi\epsilon_0 a}\, i = R\frac{di}{dt}$$

$$\Rightarrow \qquad \int_{i_0}^{i}\frac{di}{i} = -\frac{3}{8\pi\epsilon_0 aR}\int_0^t dt$$

$$\Rightarrow \qquad \ell n\,\frac{i}{i_0} = -\frac{3t}{8\pi\epsilon_0 aR}$$

$$\Rightarrow \qquad i = i_0\, e^{-\frac{3t}{(8\pi\epsilon_0 a)R}} = \frac{V}{R}\, e^{-\frac{3t}{(8\pi\epsilon_0 a)R}}$$

$$i_0 = \text{current at } t = 0^+ = \frac{V}{R}$$

Rate of change of potential

$$V = \frac{Q - q}{C}$$

$$\frac{dV}{dt} = -\frac{1}{C}\frac{dq}{dt} = -\frac{i}{C}$$

$$\left|\frac{dV}{dt}\right| = \frac{i}{C} = \frac{V}{(8\pi\epsilon_0 a)R}\, e^{-\frac{3t}{8\pi\epsilon_0 aR}}$$

(b) Heat dissipated = loss in electrostatic potential energy

$$= \frac{Q_0^2}{2C_1} - \left(\frac{q_1^2}{2C_1} + \frac{q_2^2}{\frac{2\cdot C_1}{2}}\right)$$

$$= \frac{Q_0^2}{2C_1} - \left[\frac{2Q_0^2}{9C_1} + \frac{Q_0^2}{9C_1}\right]$$

$$= \frac{Q_0^2}{2C_1} - \frac{Q_0^2}{3C_1} = \frac{Q_0^2}{6C_1} = \frac{(C_1V)^2}{6C_1}$$

$$= \frac{C_1V^2}{6} = \frac{4}{3}\pi\epsilon_0 aV^2.$$

95. $$q_0 = C_1V_1 = 200\ \mu C$$

Maximum charge on C_2 is $(1 - e^{-1})\, q_0 = 0.63 \times 200 = 126\ \mu C$

$\therefore$ $$C_2V_2 = 126\ \mu C \quad \Rightarrow \quad C_2 = 3\ \mu F$$

$$i_0 = \frac{V_1}{R_1} = 0.2A$$

Maximum current in the second circuit is

$$\frac{126}{200} \times 0.2 = \frac{V_2}{R_2}$$

$\Rightarrow$ $$0.126A = \frac{42}{R_2}$$

$$R_2 = \frac{1000}{3}\ \Omega$$

Time constant of the second circuit is $\tau_2 = R_1C_2 = \frac{1000}{3} \times 3\ \mu s$

$$= 100\ \mu s$$

$$= 10^{-3}\,s$$

$\therefore$ $$i = 0.126\ e^{-t/\tau_2}$$

$$i = 126\ e^{-1000t}$$ [t is in second]

96. Initially when 'S' is in position 1 or 2 the capacitor gets charged. This continues till p.d across capacitor becomes 3.5 volt.

Now, when in position 1, the current is

$$i = \frac{0.5\,V}{1\,\Omega} = 0.5\,A.$$

This current discharges the capacitor. Its charge gets reduced by $\Delta q = i\Delta t = 0.5\,\Delta t$

When in position 2, the current is once again $i = \frac{0.5\ V}{1\Omega} = 0.5$ A but this time it charges the capacitor.

Charge gained in time Δt is

$$\Delta q = 0.5\ \Delta t$$

Therefore, once the capacitor gets charged to 3.5 V, charge on it will become almost constant.

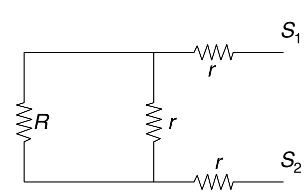
97. (a) Let's first find the equivalent resistance of the infinite ladder network between S_1 and S_2.

Since, there are infinite number of sets of resistance, removing one set will not affect the overall resistance.

Let the equivalent resistance be R.

Then the equivalent of resistance network can be drawn as in figure.

Equivalent resistance between S_1 and S_2 is also $\simeq R$



$\therefore$ $$R = \frac{Rr}{R + r} + 2r$$

or, $$R^2 - 2rR - 2r^2 = 0$$

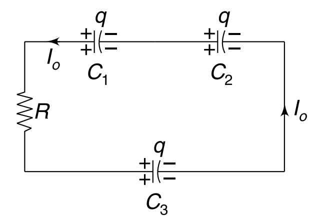
or, $$R = \frac{2r \pm \sqrt{4r^2 + 8r^2}}{2}$$

$$= r(1 + \sqrt{3})$$

[Negative sign is impossible as R cannot be negative]

$$= \frac{10}{2.732} \times 2.732$$

$$= 10\ \Omega.$$

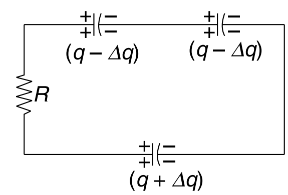


Now at initial moment the circuit arrangement looks like the one shown in second figure.
where I_0 = current just after closing of switches
From Kirchhoff's voltage law, we get

$$\frac{q}{C_1} + \frac{q}{C_2} - \frac{q}{C_3} - I_0R = 0$$

or, $$\frac{30}{3} + \frac{30}{6} - \frac{30}{6} - I_0 \times 10 = 0$$

or, $$I_0 = 1.0 \text{ Amp.}$$ Answers



(b) Initially, the energy stored in the capacitors is

$$U_i = \frac{q^2}{2C_1} + \frac{q^2}{2C_2} + \frac{q^2}{2C_3}$$

$$= \frac{900}{2 \times 3} + \frac{900}{2 \times 6} + \frac{900}{2 \times 6}$$

$$= 300\ \mu\text{J}$$

Let Δq charge flow through the circuit till a steady state is reached. Then charges on C_1, C_2 and C_3 will be as shown in third figure.
For steady state

$$\frac{q - \Delta q}{C_1} + \frac{q - \Delta q}{C_2} - \frac{q + \Delta q}{C_3} = 0$$

or, $$\Delta q = 15\ \mu\text{C}$$

Thus, final energy stored in capacitors is

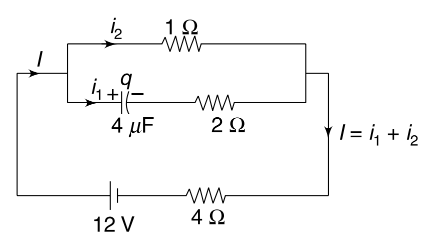
$$U_f = \frac{(q - \Delta q)^2}{2C_1} + \frac{(q - \Delta q)^2}{2C_2} + \frac{(q + \Delta q)^2}{2C_3}$$

$$= \frac{15^2}{2 \times 3} + \frac{15^2}{2 \times 6} + \frac{45^2}{2 \times 6} = 225\ \mu\text{J}$$

$\therefore$ Heat generated = loss in stored energy

$$= U_i - U_f = 75\ \mu\text{J}$$

98. (a) Situation at time t is as shown in the figure.
Applying Kirchhoff's loop law in outer loop we get



$$1(i_2) + 4(i_1 + i_2) = 12$$

$$\Rightarrow \quad 4i_1 + 5i_2 = 12 \quad \Rightarrow \quad i_2 = \frac{12 - 4i_1}{5} \qquad ...(1)$$

Applying Kirchhoff's law in the loop having the capacitor and 1Ω resistor.

$$\frac{q}{C} + 2i_1 = 1(i_2)$$

$$\Rightarrow \quad \frac{q}{4} + 2i_1 = \frac{12 - 4i_1}{5} \qquad [q \text{ is in } \mu\text{C}]$$

$$\Rightarrow \quad 14 i_1 = 12 - \frac{5}{4} q$$

$$\Rightarrow \quad 14 \frac{dq}{dt} = 12 - \frac{5}{4} q$$

(b)
$$\int_0^q \frac{dq}{12 - \frac{5}{4} q} = \frac{1}{14} \int_0^t dt$$

$$\Rightarrow \quad \left[\ell n \left(12 - \frac{5}{4} q \right)_0^q \right] = -\frac{5t}{56}$$

$$\Rightarrow \quad \ell n \left(\frac{12 - \frac{5}{4} q}{12} \right) = -\frac{5t}{56}$$

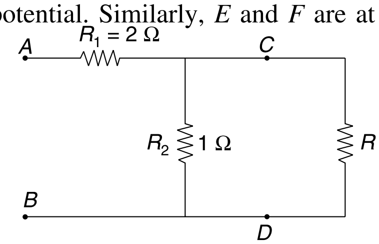
$$\Rightarrow \quad 1 - \frac{5}{48} q = e^{-\frac{5t}{56}}$$

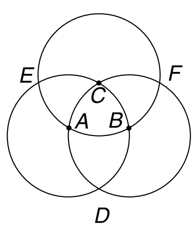
$$\Rightarrow \quad q = \frac{48}{5} \left[1 - e^{-\frac{5t}{56}} \right]$$

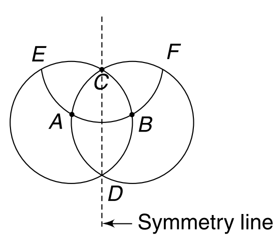
(c) At $t = 0$, $q = 0$

At $t = \infty$, $q = \dfrac{48}{5}$ μC.

99. If a battery is connected across points C and D, points A and B will be at same potential. Similarly, E and F are at same potential. There will be no current in the resitance directly connecting A to B and E to F. The circuit can be redrawn as

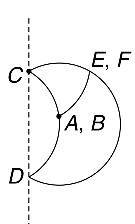


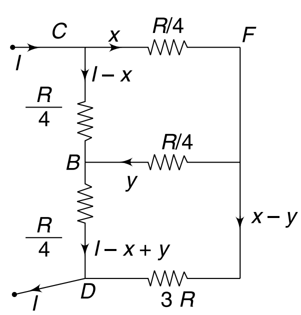




$$R_{AC} = R_{CB} = R_{EC} = R_{FC} = R_{AD} = R_{BD} = \frac{3R}{6} = \frac{R}{2}$$

The circuit can be folded about the line of symmetry shown in the figure so that E overlaps F and A falls on B.





The circuit has been redrawn as shown in figure. We assume a current I flowing into the circuit at C. The current has been distributed using x and y as unknowns.

Using Kirchhoff's laws we can write following equations–

$$2x + y = I \quad ...(1)$$

$$13x - 14y = I \quad ...(2)$$

Solving $$x = \frac{15I}{41} \text{ and } y = \frac{11I}{41}$$

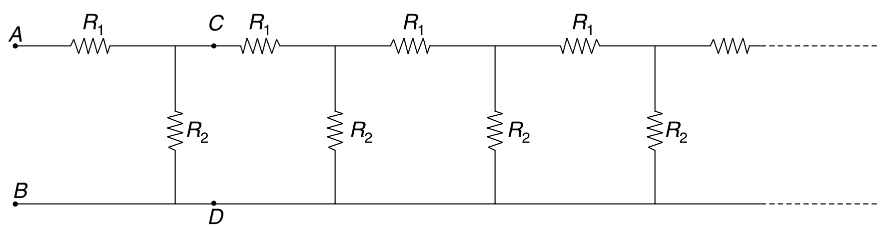
Now $$V_C - V_D = \frac{R}{4}(I - x) + \frac{R}{4}(I - x + y)$$

$$= \frac{R}{4}[2I - 2x + y] = \frac{R}{4}\cdot\left(\frac{63}{41}I\right)$$

$\therefore$ $$\frac{V_C - V_D}{I} = \frac{63}{164}R$$

$\therefore$ Equivalent resistance is $\frac{63R}{164}$.

100. (a) Equivalent across A and $B = R$



If first pair of R_1 and R_2 is removed than equivalent across C and D will remain R. Hence, the circuit can be drawn as shown. Equivalent of this network across A and B is R.

$\therefore$ $$\frac{RR_2}{R + R_2} + R_1 = R$$

Solving $$R = \frac{R_1 \pm \sqrt{R_1^2 + 4R_1R_2}}{2}$$

Since R cannot be negative

$$R = \frac{R_1 + \sqrt{R_1^2 + 4R_1R_2}}{2} = \frac{2 + \sqrt{4 + 4 \times 2 \times 1}}{2}$$

$$= (1 + \sqrt{3})\ \Omega$$

(b) The equivalent across M and N after removing all the previous sections is still R (the network is infinite!)

$$\left(I_{n-1} - I_n\right)R_2 = I_nR$$

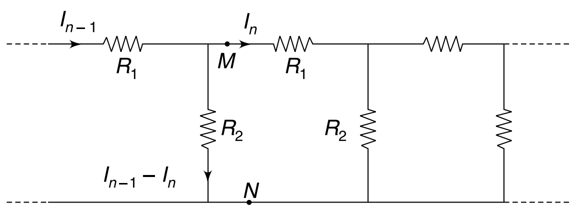
$\Rightarrow$ $$I_n = \frac{I_{n-1}R_2}{R_2 + R} = \frac{I_{n-1}}{\sqrt{3} + 2} \quad ...(1)$$

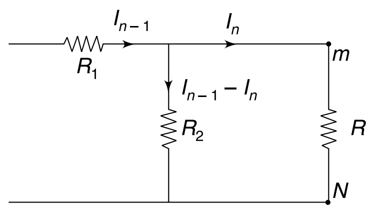
(c) $$I_1 = \frac{20}{R} = \frac{20}{\sqrt{3} + 1}$$

Using (1) $$I_2 = \frac{I_1}{\sqrt{3} + 2} = \frac{20}{(\sqrt{3} + 1)(\sqrt{3} + 2)}$$

$$I_3 = \frac{I_2}{\sqrt{3}+2} = \frac{20}{(\sqrt{3}+1)(\sqrt{3}+2)^2}$$

Similarly, we get $$I_{10} = \frac{20}{(\sqrt{3}+1)(\sqrt{3}+2)^9}$$

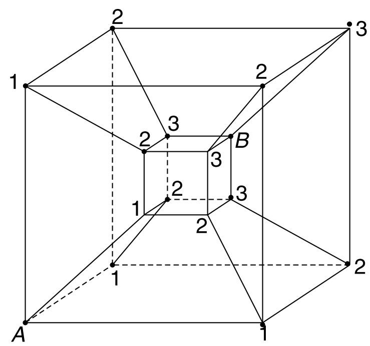


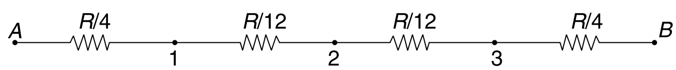


101. Due to Symmetry

Points marked 1 are equipotential points.

Similarly all 2s are at same potential & all 3s are also at same potential.





102. Equivalent resistance across $A - B$ is given by

$$\frac{1}{R_{AB}} = \frac{1}{R_1} + \frac{1}{R_2 + R_3 + R_4} \quad ...(1)$$

Resistance across $C - D$ is

$$\frac{1}{R_{CD}} = \frac{1}{R_3} + \frac{1}{R_2 + R_1 + R_4} \quad ...(2)$$

Same power will be dissipated in two cases if

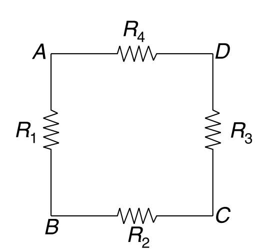
$$R_{AB} = R_{CD} \Rightarrow R_1 = R_3$$

[Have a careful look at (1) and (2). Don't try any mathematics]

Similarly, one can prove that $R_2 = R_4$.

When cell is connected across $B - C$ (or $A - D$) the equivalent resistance is

$$\frac{1}{R_{BC}} = \frac{1}{R_2} + \frac{1}{R_1 + R_2 + R_4}$$



Given $$\frac{\varepsilon^2}{R_{AB}} = P_0 \quad \text{and} \quad \frac{\varepsilon^2}{R_{BC}} = 3P_0$$

$\therefore$ $$\frac{1}{R_{BC}} = \frac{3}{R_{AB}}$$

$$\Rightarrow \quad \frac{1}{R_2} + \frac{1}{2R_1 + R_2} = 3\left[\frac{1}{R_1} + \frac{1}{R_1 + 2R_2}\right]$$

$$\Rightarrow \quad \frac{2R_1 + 2R_2}{2R_1R_2 + R_2^2} = 3\left(\frac{2R_1 + 2R_2}{R_1^2 + 2R_1R_2}\right)$$

$$\Rightarrow \quad R_1^2 + 2R_1R_2 = 3R_2^2 + 6R_1R_2$$

$$\Rightarrow \quad R_1^2 - 4R_2 \cdot R_1 - 3R_2^2 = 0$$

$$\Rightarrow \quad R_1 = \frac{4R_2 \pm \sqrt{16R_2^2 + 12R_2^2}}{2}$$

$$R_1 = 2R_2 + \sqrt{7}R_2 = (2 + \sqrt{7})R_2$$

$$\therefore \quad P_0 = \frac{\varepsilon^2}{R_{AB}} = \varepsilon^2\left[\frac{1}{(2 + \sqrt{7})R_2} + \frac{1}{(4 + \sqrt{7})R_2}\right]$$

$$= \frac{\varepsilon^2}{R_2}\left[\frac{6 + 2\sqrt{7}}{15 + 6\sqrt{7}}\right] = \frac{2\varepsilon^2}{R_2}\frac{(3 + \sqrt{7})}{(15 + 6\sqrt{7})}$$

Resistance across $A - C$ is

$$\frac{1}{R_{AC}} = \frac{1}{R_1 + R_2} + \frac{1}{R_3 + R_4} = \frac{1}{R_1 + R_2} + \frac{1}{R_1 + R_2}$$

$$= \frac{2}{R_1 + R_2} = \frac{2}{(3 + \sqrt{7})R_2}$$

$$\therefore \quad P_{AC} = \frac{\varepsilon^2}{R_{AC}} = \frac{2\varepsilon^2}{(3 + \sqrt{7})R_2}$$

$$= P_0\frac{(15 + 6\sqrt{7})}{(3 + \sqrt{7})(3 + \sqrt{7})}$$

$$= \left(\frac{15 + 6\sqrt{7}}{16 + 6\sqrt{7}}\right)P_0$$

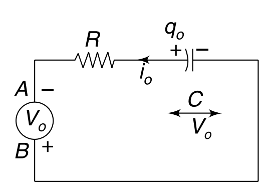
103. After the switch is closed, the capacitor gets (almost) fully charged in an interval of $t_0 = 50\ \tau$. In fact, few time constants are good enough to charge it completely.

During charging the charge on the capacitor varies as

$$q = q_0[1 - e^{-t/\tau}] \qquad [q_0 = CV_0]$$

After the polarity of source gets reversed, the capacitor will get discharged and then it will get recharged with polarity of plates changed. There is enough time to assume that the capacitor is fully charged with polarity changed. The situation at time t_0^+ is shown below. Clearly, the potential difference across R is $2V_0$. Current is $i_0 = \frac{2V_0}{R}$. This is maximum current in the circuit.

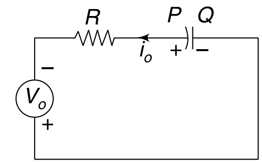
For simplicity in calculation, let's begin counting time at the instant end B becomes positive for the first time. At time 't' the situation is as shown.



Charge on plate Q is taken positive.

$$\frac{q}{C} + V_0 = Ri$$

But $\quad i = -\frac{dq}{dt} \qquad [\because \quad q \text{ is decreasing}]$



$$\therefore \qquad -R\frac{dq}{dt} = \frac{q}{C} + V_0$$

$$\Rightarrow \qquad \int_{q_0}^{q} \frac{dq}{\frac{q}{RC} + \frac{V_0}{R}} = -\int_{0}^{t} dt$$

$$\Rightarrow \qquad \left[\ell n\left[\frac{q}{RC} + \frac{V_0}{R}\right]\right]_{q_0 = CV_0}^{q} = \frac{-t}{RC}$$

$$\Rightarrow \qquad \ell n\left(\frac{q}{RC} + \frac{V_0}{R}\right) - \ell n\left(\frac{CV_0}{RC} + \frac{V_0}{R}\right) = \frac{-t}{RC}$$

$$\Rightarrow \qquad \ell n\left(\frac{\frac{q}{RC} + \frac{V_0}{R}}{2\frac{V_0}{R}}\right) = \frac{-t}{\tau} \qquad [RC = \tau]$$

$$\Rightarrow \qquad \frac{q}{2CV_0} + \frac{1}{2} = e^{-t/\tau}$$

$$\Rightarrow \qquad q = 2CV_0\left[e^{-t/\tau} - \frac{1}{2}\right]$$

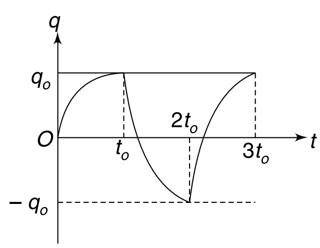
q will became zero when

$$e^{-t/\tau} = \frac{1}{2}$$

$$\therefore \qquad \frac{t}{\tau} = \ell n 2 \quad \Rightarrow \quad t = (\ell n 2)\,\tau$$

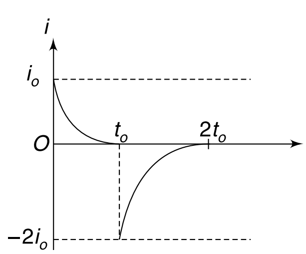
Here we have taken q to be positive when plate Q is positive. But question asks us to take P as positive. This needs to be taken care while plotting the graph.

(a)

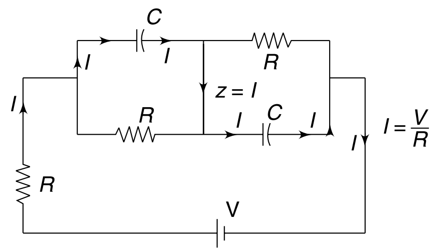


(b) $i_{max} = \dfrac{2V_0}{R}$

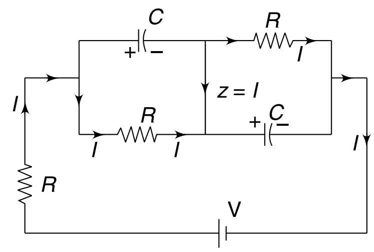
(c) current can be obtained as $\dfrac{dq}{dt}$.



104. (a) at $t = 0^+$ the current flow will be as shown below

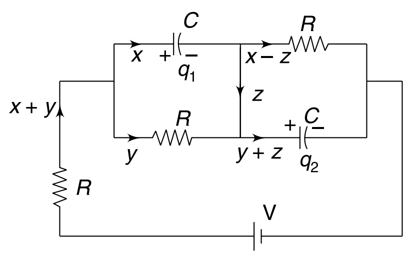


(b) At $t = \infty$, the current flow is as shown below



$$I = \frac{V}{3R} \qquad \text{(in direction shown)}$$

(c) At time t let the charge on capacitors and current in different branches be as shown



The relevant equations are

$$R(x + y) + Ry + R(x - z) = V$$

$$\Rightarrow \qquad 2x + 2y - z = \frac{V}{R} \qquad ...(1)$$

$$\frac{q_1}{C} = Ry \qquad ...(2)$$

$$\frac{q_2}{C} = R(x - z) \qquad ...(3)$$

$$\frac{dq_1}{dt} = x \qquad ...(4)$$

$$\frac{dq_2}{dt} = y + z \qquad ...(5)$$

From (2) $$\frac{dq_1}{dt} = RC\frac{dy}{dt}$$

$$\Rightarrow \qquad x = RC\frac{dy}{dt} \qquad ...(6)$$

From (3) $$\frac{dq_2}{dt} = RC\left(\frac{dx}{dt} - \frac{dz}{dt}\right)$$

$\Rightarrow$ $$y + z = RC\left(\frac{dx}{dt} - \frac{dz}{dt}\right) \quad ...(7)$$

(6) + (7) $$\frac{x + y + z}{RC} = \frac{dx}{dt} + \frac{dy}{dt} - \frac{dz}{dt} \quad ...(8)$$

From equation (1) $$x + y = \frac{V}{2R} + \frac{z}{2}$$

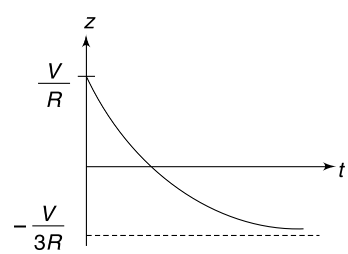
Also $$\frac{dx}{dt} + \frac{dy}{dt} = \frac{1}{2}\frac{dz}{dt}$$

Put both the above results into equation (8)

$$\frac{V}{2R} + \frac{z}{2} + z = \left[\frac{1}{2}\frac{dz}{dt} - \frac{dz}{dt}\right] RC$$

$$\frac{V}{R} + 3z = -RC\frac{dz}{dt}$$

$$\int_{z=\frac{V}{R}}^{z} \frac{dz}{\frac{V}{R} + 3z} = -\frac{1}{RC}\int_{t=0}^{t} dt$$



$$\left[\ell n\left(\frac{V}{R} + 3z\right)\right]_{\frac{V}{R}}^{z} = -\frac{3t}{RC}$$

$$\ell n\left(\frac{V}{R} + 3z\right) - \ell n\left(\frac{4V}{R}\right) = -\frac{3t}{RC}$$

$$\ell n\left(\frac{1}{4} + \frac{3Rz}{4V}\right) = -\frac{3t}{RC}$$

$$z = \frac{4V}{3R}\left[e^{-\frac{3t}{RC}} - \frac{1}{4}\right]$$

(d) $$z = 0 \quad \text{when} \quad e^{-\frac{3t}{RC}} = \frac{1}{4}$$

$$\frac{3t}{RC} = \ell n 4$$

$$t = \frac{2RC}{2}\ \ell n 2$$

105. For $0 \le t \le t_0$

$$V_0 = \frac{q}{C} + Ri$$

Where q and i are instantaneous values of charge on the capacitor and the current.

$$kt = \frac{q}{C} + Ri$$

Differentiate wrt time (t)

$$k = \frac{1}{C}\frac{dq}{dt} + R\frac{di}{dt}$$

$\Rightarrow$ $$k - \frac{i}{C} = R\frac{di}{dt}$$

$$\Rightarrow \quad \int_0^i \frac{di}{k - \frac{i}{C}} = \frac{1}{R}\int_0^t dt$$

$$\Rightarrow \quad \left[\ell n\left(k - \frac{i}{C}\right)\right]_0^i = -\frac{t}{RC}$$

$$\Rightarrow \quad \ell n\left(k - \frac{i}{C}\right) - \ell nk = -\frac{t}{RC}$$

$$\Rightarrow \quad \ell n\left(1 - \frac{i}{kC}\right) = -\frac{t}{RC}$$

$$\Rightarrow \quad i = kC\left(1 - e^{-t/RC}\right)$$

$$\therefore \quad V = Ri = kCR\left(1 - e^{-t/RC}\right)$$

Hence voltage across C and D increases as per above equation till $t = t_0$.
Let the current (i) and voltage V at $t = t_0$ be

$$i_1 = kC\left(1 - e^{-t_0/RC}\right)$$

$$V_1 = kCR\left(1 - e^{-t_0/RC}\right)$$

For $t > t_0$, the applied voltage remains constant at kt_0.
For simplicity, let's begin our time count from this instant itself.

$$kt_0 = \frac{q}{C} + iR$$

Differentiate wrt 't'

$$\frac{1}{C}\frac{dq}{dt} + R\frac{di}{dt} = 0$$

$$\Rightarrow \quad \frac{1}{C}i + R\frac{di}{dt} = 0$$

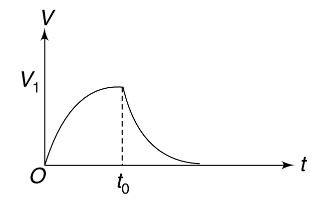
$$\Rightarrow \quad \int_{i_1}^{i} \frac{di}{i} = -\frac{1}{RC}\int_0^t dt$$

$$\Rightarrow \quad \ell n\left(\frac{i}{i_1}\right) = -\frac{t}{RC}$$

$$\therefore \quad i = i_1 e^{-t/RC}$$

$$\therefore \quad V = iR = i_1 R\, e^{-t/RC}$$

Hence V decresses exponentially after $t = t_0$.
The graph will be as shown in the figure

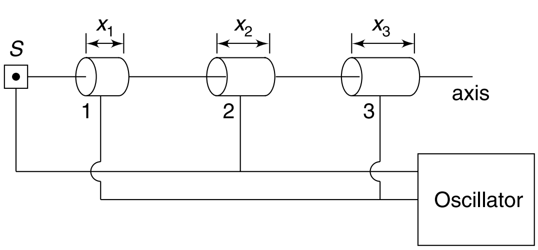


CHAPTER 9

Motion of Charge in Electromagnetic Field

LEVEL 1

Q. 1: A linear oscillator (Linac) is a device which accelerates charged particles in a straight line by means of oscillating electric field. S is a source of ion; emitting ions along a straight line (marked as axis) with negligible kinetic energy. The particles travel along a series of co-axial metallic cylindrical electrodes 1, 2, 3, etc. These electrodes are called drift tubes. These tubes are connected to a high frequency oscillator so that alternate tubes have opposite polarities. Thus in one half cycle if tubes 1, 3, 5… are negative then the source (S), tube 2, 4, 6… are positive. After every half cycle the polarities reverse. Assume that there is no electric field inside the metallic tube and the charged particles maintain a straight line trajectory. Find the ratio of lengths of the tubes $x_1 : x_2 : x_3...$ for the device to work.



Q. 2: A particle of mass m and charge q is projected into a uniform magnetic field $\vec{B} = -B_0\hat{k}$ with velocity $\vec{v} = v_0\hat{i}$ from origin. The position vector of the particle at time t is $\vec{r}$. Find the impulse of magnetic force on the particle by the time $\vec{r}\cdot\vec{v}$ becomes zero for the first time.

Q. 3: In a cyclotron the radius of dees is R and the applied magnetic field has a magnitude of B. Particles having charge q and mass m are accelerated using this cyclotron and the maximum kinetic energy that can be imparted to a particle is k.

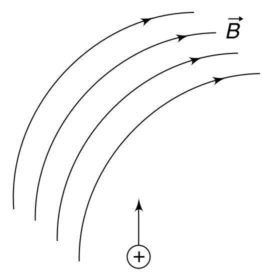
(a) Find k in terms of R, B, q and m.

(b) If protons and alpha particles are accelerated using this cyclotron which particle (proton or alpha) will gain more kinetic energy?

(c) Can we use a cyclotron to impart very high kinetic energy to an electron?

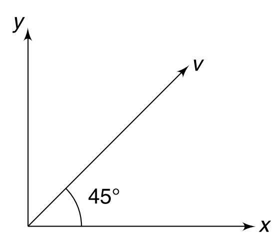
Q. 4: A particle having mass m and charge q is projected at an angle of 75° to the horizontal with a speed u. A uniform electric field $E = \dfrac{mg}{q}$ exists in horizontal direction. Find the time after projection when the velocity of particle makes an angle of 45° with horizontal.

Q. 5: A wide region of space has curved magnetic field lines as shown in the figure. A proton enters into the region as shown. Draw the path of the proton.

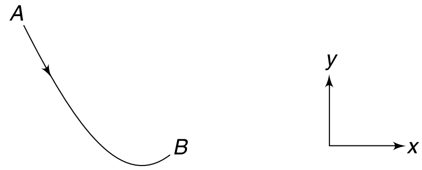


Q. 6: A uniform electric field E exists in a region of space. A charged particle is projected in the plane of electric field making an angle θ with the direction of field. Is it possible to make the particle move in a straight line by applying a uniform magnetic field?

Q. 7: A particle having charge $q = 10\,\mu C$ is moving with a velocity of $v = 10^6$ m/s in a direction making an angle of 45° with positive x axis in the xy plane (see figure). It experiences a magnetic force along negative z direction. When the same particle is projected with a velocity $v' = 10^6$ m/s in positive z direction, it experiences a magnetic force of 10 mN along X direction. Find the direction and magnitude of the magnetic field in the region.



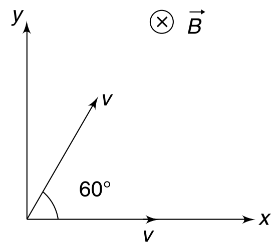
Q. 8: A uniform magnetic field exists perpendicular to the plane of the figure and a uniform electric field exists in the plane of the figure in positive y direction. A charged particle, when projected along xy plane moves along the path shown in the figure. Is the particle positively charged or negatively charged?



Q. 9: Two identical charged particles are projected simultaneously from origin in xy plane. Each particle has charge q and mass m and has been projected with velocity v as shown in the figure. There exists a uniform magnetic field B in negative z direction.

(i) Find $\vec{v}_1 \cdot \vec{v}_2$ at time t where $\vec{v}_1$ and $\vec{v}_2$ are velocities of the particles at time t.

(ii) Find $\vec{r}_1 \cdot \vec{r}_2$ at time $t = \dfrac{\pi m}{qB}$ where $\vec{r}_1$ and $\vec{r}_2$ are the position vectors of the two particles.

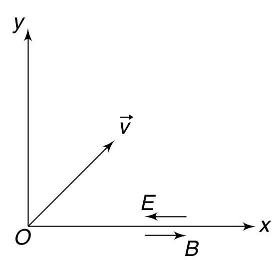


Q. 10: A particle having specific charge σ is projected in xy plane with a speed V. There exists a uniform magnetic field in z direction having a fixed magnitude B_0. The field is made to reverse its direction after every interval of $\dfrac{2\pi}{\sigma B}$. Calculate the maximum separation between two positions of the particle during its course of motion.

Q. 11: A particle having mass m and charge q is projected with a velocity v making an angle θ with the direction of a uniform magnetic field B. Calculate the magnitude of change in velocity of the particle after time $t = \dfrac{\pi m}{2qB}$.

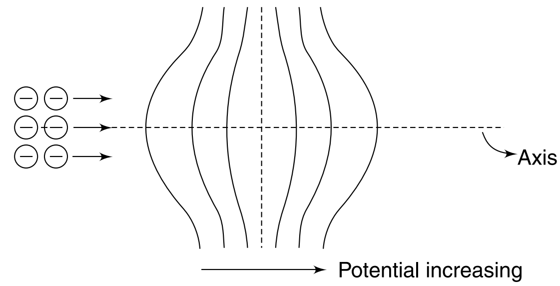
Q. 12: A region has a uniform magnetic field B along positive x direction and a uniform electric field E in negative x direction. A positively charged particle is projected from origin with a velocity $\vec{v} = v_0\hat{i} + v_0\hat{j}$. After some time the velocity of the particle was observed to be $v_0\hat{j}$ while its x co-ordinate was positive.

Write all possible values of $\dfrac{E}{B}$.

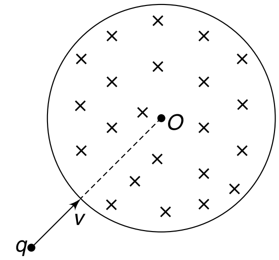


Q. 13: A positively charged particle with specific charge σ leaves the origin with a velocity u directed along positive x direction. The entire space has a uniform electric field (E) and magnetic field (B) directed along the positive y direction. Find the angle that the velocity of the particle makes with y direction at the instant it crosses the y axis for n^{th} time.

Q.14: Euipotential surfaces in a region of electric field are shaped as shown in the figure. The potential of field lines is increasing as one moves to right. A thin beam of electrons enters the region from left along the axis. Give qualitative arguments to show that the region will act as an electrostatic lens trying to focus the beam of electrons.

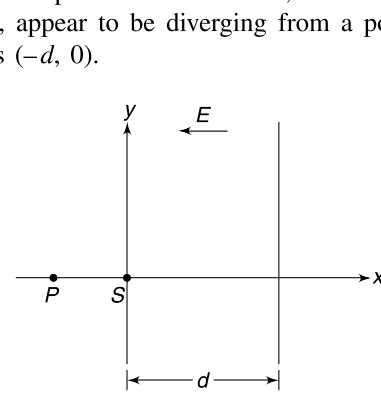


Q. 15: Figure shows a circular region of radius $R = \sqrt{3}\,m$ which has a uniform magnetic field $B = 0.2\ T$ directed into the plane of the figure. A particle having mass $m = 2\,\text{g}$, speed $v = 0.3\,\text{m/s}$ and charge $q = 1\,\text{mC}$ is projected along the radius of the circular region as shown in figure. Calculate the angular deviation produced in the path of the particle as it comes out of the magnetic field. Neglect any other force apart from the magnetic force.



LEVEL 2

Q. 16: A source(s) of electrons is kept at origin of the co-ordinate system and it shoots out electrons in xy plane with very small range of velocities. In the region $x = 0$ to $x = d$ there is a uniform electric field $\vec{E} = -E\hat{i}$ which accelerates the electrons to speeds much larger than their original speed when they are ejected by the source. The electrons emerge from the field region, beyond $x = d$, travelling in straight lines. Prove that paths of the electrons, after they emerge out of the field, appear to be diverging from a point P having co-ordinates $(-d, 0)$.



Q. 17: A uniform electric field, $\vec{E} = -E_0\hat{i}$ and a uniform magnetic field, $\vec{B} = B_0\hat{i}$ exist in region $y > 0$. A particle having positive charge q and mass m is projected from the origin with a velocity $\vec{v} = v_0\hat{i} + v_0\hat{j}$. The velocity of the particle when it leaves the region of fields was found to be $-\vec{v}$.

(i) Find the ratio $\dfrac{E_0}{B_0}$ in terms of v_0.

(ii) Find the co-ordinates of the point where the particle leaves the fields.

(iii) Find the minimum speed of the particle during the course of the motion.

Q. 18: A positively charged ion is released at the origin of a co-ordinate system. The entire space is filled with a uniform electric field directed along positive x axis and a uniform magnetic field directed along positive z direction.

(i) Which (x, y or z) co-ordinate of the particle will always remain zero during its course of motion?

(ii) Which (x, y or z) co-ordinate of the particle will always be negative?

(iii) It is true to say that radius of curvature of the path of the particle is decreasing while its x co-ordinate decreases?

(iv) If the ion is negatively charged will its y co-ordinate ever become negative?

Q. 19: A particle of mass m and charge q is moving in a region where uniform electric field $\vec{E}$ and and uniform magnetic field $\vec{B}$ are present. It is given that $\dfrac{\vec{E}}{|\vec{E}|} = \dfrac{\vec{B}}{|\vec{B}|}$. At time $t = 0$, velocity of the particle is $\vec{v}_0$ and $\vec{v}_0 \cdot \vec{E} = 0$. Write the velocity of the particle at time t.

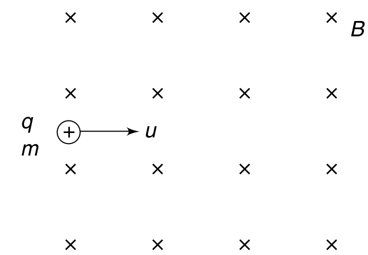
Q. 20: A particle of mass m and charge q is projected into a region having a uniform magnetic field B_0. Initial velocity (v_0) of the particle is perpendicular to the magnetic field. Apart from the magnetic force the particle faces a frictional force which has a magnitude of $f = kv$ where v is instantaneous speed and k is a positive constant.

(a) Find the radius of curvature of the path of the particle after it has travelled through a distance of

$$x_0 = \frac{mv_0}{2k}.$$

(b) Plot the variation of radius of curvature of the path of the particle with time (t).

Q. 21: A uniform magnetic field B_0 exists perpendicular to the plane of the fig. A positively charged particle having charge q and mass m is projected with velocity u into the field. The particle moves in the plane of the fig. During its course of motion the particle is subjected to a friction force which varies with velocity as $\vec{F} = -k\vec{V}$ where k is a positive constant and $\vec{V}$ is instantaneous velocity.



(a) What kind of path the particle will trace? Give qualitative argument to support your answer.

(b) Write the speed of the particle as function of time. Plot the speed–time graph.

(c) Calculate the distance travelled by the particle before it stops.

Assume no other force apart from magnetic force and the friction force.

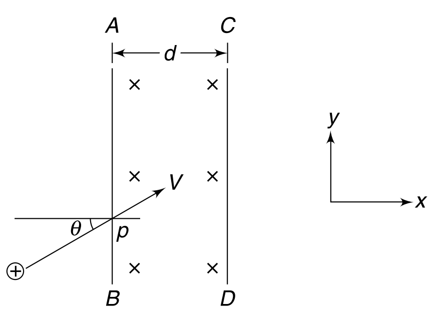
Q. 22: AB and CD are two parallel planes perpendicular to the X axis. There is a uniform magnetic field (B) in the space between them directed in negative Z direction. Width of the region having field is d and rest of the space is having no field. A particle having mass m and charge $+q$ enters the region with a velocity V making an angle θ with the X direction as shown.

(a) Find the values of d for which the particle will come out of the magnetic field crossing CD.

(b) For $d = \left(\dfrac{\sqrt{2}-1}{2}\right)\dfrac{mv}{qB}$ and $\theta = \dfrac{\pi}{6}$ find the angular deviation in the path of the particle.

(c) Find the deviation in path of the particle if

$$d = \frac{5mv}{4qB}(1 - \sin\theta)$$

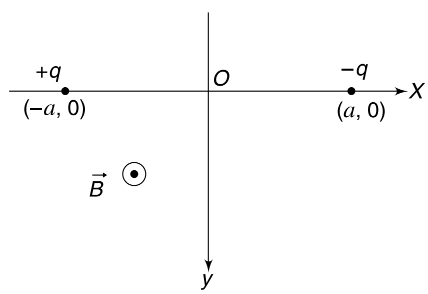


Q. 23: In the last problem, region to the right of CD is filled with a magnetic field of strength twice that in the region between AB and CD. This field exists up to a large distance and both fields are parallel in direction. For $d = \left(\dfrac{\sqrt{2}-1}{2}\right)\dfrac{mv}{qB}$

and $\theta = \frac{\pi}{6}$, find the displacement of the particle (measured from its entry point P) by the time it comes out of the magnetic fields. Find the total time spend by the particle in the two fields.

Q. 24: A particle of mass m is thrown along a horizontal surface with speed u and comes to rest after travelling a distance x_0. A uniform magnetic field B_0 is switched on in vertically downward direction and the particle is projected as before after putting a charge q on it. This time it travelled through a distance X_1 before stopping. Now the experiment is repeated with a uniform electric field E_0 in addition to the magnetic field; electric field being parallel to the magnetic field. Particle having mass m and charge q is projected as earlier and found to travel a distance X_2 before stopping. Find X_1 and X_2.

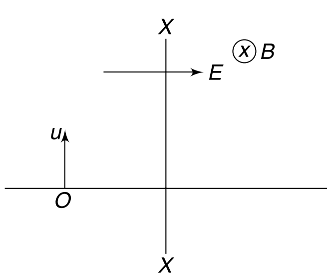
Q. 25: Two particles have equal mass m and electric charge of equal magnitude (q) and opposite sign. The particles are held at rest at co-ordinates $(-a, 0)$ and $(a, 0)$ as shown in the figure. The particles are released simultaneously. Consider only the electrostatic force between the particles and the force applied by the external magnetic field on them.



(a) Find the speed of negatively charged particle as function of its x co-ordinate.

(b) Find the y component of velocity of the negative particle as a function of its x co-ordinate.

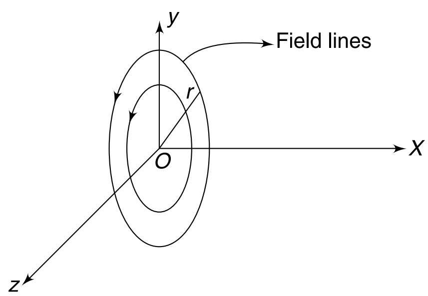
Q. 26: In the fig shown XX represents a vertical plane perpendicular to the plane of the fig. To the right of this plane there is a uniform horizontal magnetic field B directed into the plane of the fig. A uniform electric field E exists horizontally perpendicular to the magnetic field in entire space. A charge particle having charge q and mass m is projected vertically upward from point O. It crosses the plane XX after time T. Find the speed of projection of the particle if it was observed to move uniformly after time T. It is given that $qE = mg$.



Q. 27: A particle of mass m and charge $+q$ is projected from origin with velocity $\vec{V} = V_0\,\hat{i}$ in a magnetic field $\vec{B} = -(B_0 x)\hat{k}$. Here V_0 and B_0 are positive constants of proper dimensions. Find the radius of curvature of the path of the particle when it reaches maximum positive x co-ordinate.

Q. 28: A long uniform cylindrical beam of radius R consists of positively charged particles each of charge q, mass m and velocity V along positive x direction. The axis of the beam is the x-axis. The beam is incident on a region having magnetic field in y-z plane. The magnetic field in the region is confined to $0 \le x \le \Delta x$ (Δx is small). The field lines are circular in yz plane as shown. The magnitude of the field is given by

$B = B_0 r$ where B_0 is a constant and r is distance from the origin.



Show that this magnetic field acts as a converging lens for the ion beam and obtain the expression for focal length. Neglect the divergence in the beam caused due to electromagnetic interaction of the charge particles.

LEVEL 3

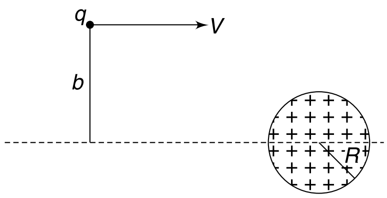
Q. 29: There is a fixed sphere of radius R having positive charge Q uniformly spread in its volume. A small particle having mass m and negative charge $(-q)$ moves with speed V when it is far away from the sphere. The impact parameter (i.e., distance between the centre of the sphere and line of initial velocity of the particle) is b. As the particle passes by the sphere, its path gets deflected due to electrostatics interaction with the sphere.

(a) Assuming that the charge on the particle does not cause any effect on distribution of charge on the sphere, calculate the minimum impact parameter b_0 that allows the particle to miss the sphere. Write the value of b_0 in term of R for the case

$$\frac{1}{2}mV^2 = 100 \cdot \left(\frac{kQq}{R}\right)$$

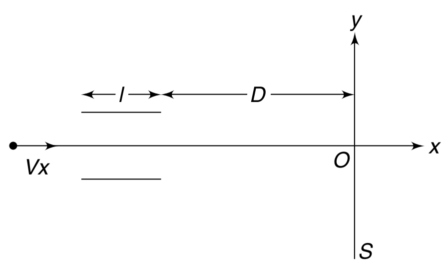
(b) Now assume that the positively charged sphere moves with speed V through a space which is filled with small particles of mass m and charge $-q$. The small particles are at rest and their number density is n [i.e., number of particles per unit volume of space is n].

The particles hit the sphere and stick to it. Calculate the rate at which the sphere starts losing its positive charge $\left(\frac{dQ}{dt}\right)$. Express your answer in terms of b_0.



Q. 30: (a) A charge particle travelling along positive x direction with speed V_x enters a region of width ℓ having a uniform electric field E in positive y direction. A screen is kept, at a distance $D (>> \ell)$ from the region of the field, in yz plane. Find the y co-ordinate of the point where the particle strikes the screen. Charge and mass of the particle is $+q$ and m respectively.

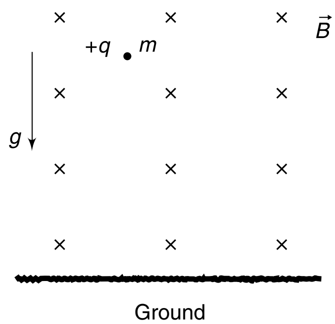
(b) The electric field in the region is replaced with a uniform magnetic field B in negative z direction. Now calculate the y co-ordinate of the point on the screen where the particle hits it. Assume deflection due to field to be small and $D >> \ell$.



(c) Now the field region is filled with a uniform electric (E) and magnetic field (B) both directed in positive y direction. A beam of protons and some other positive ion enters the region travelling along x direction. The particles hit the screen along two curved paths. Explain.

Draw the two curves in yz plane and point out which one represents the protons.

Q. 31: A particle having charge q and mass m is dropped from a large height from the ground. There exists a uniform horizontal magnetic field B in the entire space as shown in the fig. Assume that the acceleration due to gravity remains constant over the entire height involved.



(a) Argue qualitatively that the particle will touch a maximum depth and then start climbing up.

(b) Find the speed of the particle at the moment it starts climbing up.

(c) At what depth from the starting point does the particle starts climbing up?

Q. 32: In a region of space a uniform magnetic field exist in positive z direction and there also exists a uniform electric field along positive y direction. A particle having charge $+q$ and mass m is released from rest at the origin. The particle moves on a curve known as cycloid. If a wheel of radius R were to roll on the X axis, a fixed point on the circumference of the wheel would generate this cycloid. Find the radius R. It is given that strength of magnetic and electric fields are B_0 and E_0 respectively.

ANSWERS

1. $1 : \sqrt{2} : \sqrt{3} : ...$

2. $2mv_0\hat{i}$

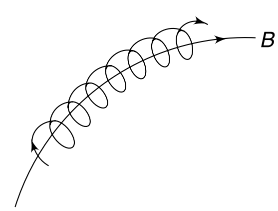
3. (a) $k = \dfrac{R^2q^2B^2}{2m}$

(b) Both will get same energy

(c) No

4. $\dfrac{u}{2\sqrt{2}g}$

5.



6. No

7. 1 mT in negative y direction.

8. Positive.

9. (i) $\dfrac{v^2}{2}$ (ii) $2\left(\dfrac{mv}{qB}\right)^2$

10. $\dfrac{4v}{\sigma B_0}$

11. $\sqrt{2}\, v \sin\theta$

12. $\dfrac{E}{B} = \dfrac{v_0}{2\pi n}$ where $n = 1, 2, 3...$

13. $\alpha = \tan^{-1}\left(\dfrac{uB}{2\pi n E}\right)$

15. $60°$

17. (i) $\dfrac{E_0}{B_0} = \dfrac{2v_0}{\pi}$ (ii) $\left(0,\ 0,\ \dfrac{2mv_0}{qB_0}\right)$

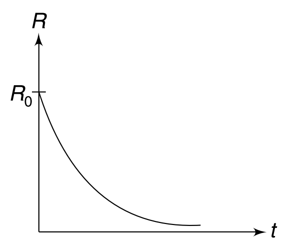
(iii) $v_{\min} = v_0$

18. (i) z (ii) y

(iii) yes (iv) It is always negative

19. $\vec{v}\cos\left(\dfrac{qBt}{m}\right)\vec{v}_0 + \left(\dfrac{qt}{m}\right)(\vec{E}) + \sin\left(\dfrac{qBt}{m}\right)\left(\dfrac{\vec{v}_0 \times \vec{B}}{B}\right)$

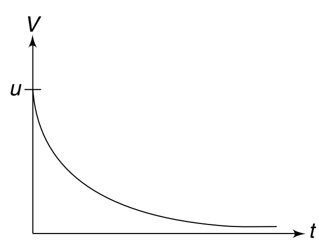
20. (a) $\dfrac{mv_0}{2qB_0}$ (b)



21. (a) Spiral path

(b) $V = u e^{-\frac{kt}{m}}$

(c) $s = \dfrac{mu}{k}$



22. (a) $d < \dfrac{mv}{qB}(1 - \sin\theta)$ (b) $15°$

(c) $\pi - 2\theta$

23. $y = \left(\sqrt{3} - \dfrac{1}{\sqrt{2}}\right)\dfrac{mv}{qB}$; $t = \dfrac{5\pi m}{12 qB}$

24. $X_1 = X_0$; $X_2 = \dfrac{mgX_0}{mg + qE_0}$

25. (a) $V = \sqrt{\dfrac{kq^2}{2m}\left(\dfrac{1}{x} - \dfrac{1}{a}\right)}$ (b) $V_y = \dfrac{qB}{m}(a - x)$

26. $u = 2gT$

27. $R = \sqrt{\dfrac{mV_0}{2qB_0}}$

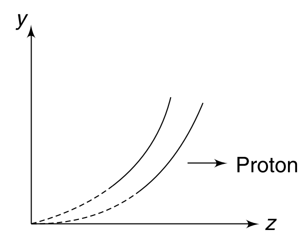
28. $f = \dfrac{mV}{qB_0 \Delta x}$

29. (a) $b_0 = R\sqrt{1 + \dfrac{2KQq}{mRV^2}}$ $b_0 \simeq 1.02\ R$

(b) $qn\pi b_0^2 V$

30. (a) $y = \dfrac{qE\ell D}{mV_x^2}$ (b) $y = \dfrac{D\ell qB}{mV_x}$

(c)



31. $\dfrac{2mg}{qB}$ (c) $h = \dfrac{2m^2 g}{q^2 B^2}$

32. $R = \dfrac{mE_0}{qB_0^2}$

SOLUTIONS

1. Let a positive ion, having charge q and mass m be accelerated during the half cycle when the drift tube 1 is negative. If the applied potential difference is V, the speed (v_1) of the ion on reaching tube 1 is given by

$$\frac{1}{2} m v_1^2 = qV$$

$$\Rightarrow \quad v_1 = \sqrt{\frac{2qV}{m}}$$

The ions travel with constant speed in the field free space inside the tubes. The length of the tube 1 is so chosen such that when it exits from the tube the tube 1 becomes positive and tube 2 becomes negative. The ion is once again accelerated between the tubes and before entering tube 2 it acquires a speed (v_2) given by

$$\frac{1}{2}mv_2^2 = 2qV$$

$$\Rightarrow \qquad v_2 = \sqrt{2}\sqrt{\frac{2qV}{m}} = \sqrt{2}\,v_1$$

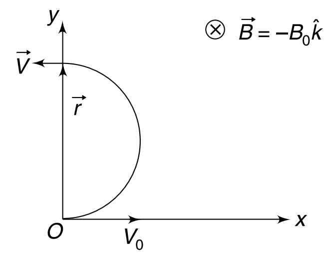
$$\Rightarrow \qquad v_2 = \sqrt{2}\,v_1;\ \text{similarly } v_3 = \sqrt{3}\,v_1$$

The ion must spend equal time inside each tube.

$$\therefore \qquad x_1 : x_2 : x_3 = v_1 : v_2 : v_3 : \ldots$$

$$= 1 : \sqrt{2} : \sqrt{3} : \ldots$$

2. The particle goes in a circular path in xy plane.



$$\vec{r}\cdot\vec{v} = \text{ when } \vec{r} \perp_r \vec{V}.$$

This happens when the particle has completed half circle (see figure)

Change in momentum of particle $\overrightarrow{\Delta p} = -2mv_0\hat{i}$

$\therefore$ Impulse $= -2mv_0\hat{i}$

3. (a) The maximum radius of circular path of the charge is R.

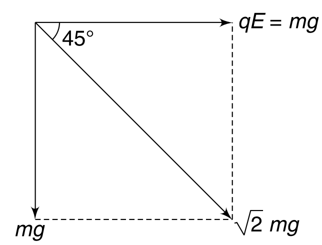
$$\therefore \qquad \frac{mV_{\max}}{qB} = R$$

$$k = \frac{1}{2}mV_{\max}^2 = \frac{R^2q^2B^2}{2m} \qquad \ldots(1)$$

(b) For proton and alpha particle the ratio $\dfrac{q^2}{m}$ is same. Hence both of them will gain same KE.

(c) Equation (i) suggests that an electron can be imparted a large energy as its mass is too small. But this is not true. As the speed electron becomes high, its mass increases due to relativistic effect. The time period of revolution of the electron $\left(T = \dfrac{2\pi m}{qB}\right)$ increases because of this. The synchronization between the source frequency and frequency of revolution of charge gets disturbed and the cyclotron fails to accelerate a very light particle like electron to a very high kinetic energy.

4. Electrostatic force is $F_e = qE = mg$.

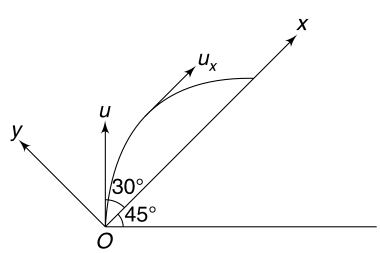


Effective acceleration experienced by the particle is

$$g_{\text{eff}} = \sqrt{2}\,g \qquad \text{(In a direction making 45° to the vertical)}$$

Consider the co-ordinate system as shown in the figure. Velocity will be making 45° to the horizontal when it becomes parallel to OX

$$\therefore \qquad t = \frac{u\sin 30°}{g_{\text{eff}}} = \frac{u}{2\sqrt{2}\,g}$$



8. When particle moves from A to B, radius of curvature of the path is decreasing. It means the particle is slowing down, as it moves from A to B.

9. (i) The two particles will rotate through same angle in time t. Hence angle between their velocities will remain 60°.

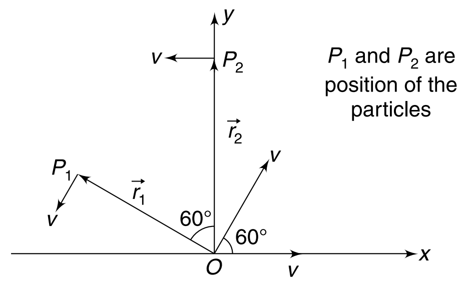
$\therefore \qquad \vec{v}_1 \cdot \vec{v}_2 = v \cdot v \cdot \cos 60^\circ = \dfrac{v^2}{2}$

Note: Speed of charge will remain unchanged in magnetic field.

(ii) Time t given is half the time period of rotation for each charge

$$t = \frac{T}{2}$$

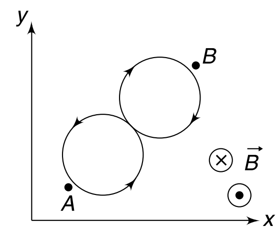
Each charge will complete half circle and will be at a distance $2R = 2\left(\dfrac{mv}{qB}\right)$ from the origin. Their positions are as shown in the figure.



$$\vec{r}_1 \cdot \vec{r}_2 = (2R)(2R)\cos 60^\circ = 2R^2 = 2\left(\frac{mv}{qB}\right)^2$$

10. Path is as shown. Distance AB is the required maximum separation.

$$\therefore \qquad AB = 4R = 4\frac{mv}{qB_0} = \frac{4v}{\sigma B_0}.$$



11. Path is helical, with time period: $T = \dfrac{2\pi m}{qB}$

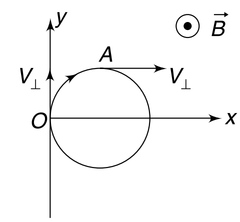
Given time is: $t = \dfrac{T}{4}$

Let the $\vec{B}$ be along + z axis. Velocity component parallel to the field

$$v_{\parallel} = v_z = v\cos\theta$$

This remains unchanged.

Let the component of v perpendicular to $\vec{B}$ be along y direction.



$$v_{\perp} = v_y = v\sin\theta$$

$$\therefore \qquad \text{Initial velocity } \vec{v}_1 = (v\sin\theta)\hat{j} + (v\cos\theta)\hat{k}$$

In time $t = \dfrac{T}{4}$, the particle will rotate through 90° and will be at A. The velocity component in xy plane is always $v_{\perp}$.

$\therefore$ Final velocity at A is

$$\vec{v}_f = (v\sin\theta)\hat{i} + (v\cos\theta)\hat{k}$$

$$\therefore \qquad \overrightarrow{\Delta v} = \vec{v}_f - \vec{v}_i = (v\sin\theta)\hat{i} - (v\sin\theta)\hat{j}$$

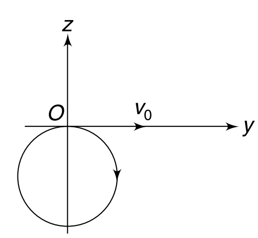
$$\therefore \qquad \left|\overrightarrow{\Delta v}\right| = \sqrt{2}v\sin\theta$$

12. If an observer looks at the particle from a point on positive x axis, he will find it rotating in yz plane with speed v_0 as shown in figure.

While rotating in yz plane the particle also advances in x direction. Due to electric force, x component of velocity decreases.

$$a_x = \frac{qE}{m}$$

$$v_x = v_0 - a_x t$$



v_x becomes zero at time: $t_0 = \dfrac{v_0}{a_x} = \dfrac{mv_0}{qE}$

The particle will be exactly at x axis travelling in y direction if time period of circular motion multiplied by an integer gives t_0.

$$\Rightarrow \qquad n\cdot\frac{2\pi m}{qB} = t_0$$

$$\Rightarrow \qquad n\cdot\frac{2\pi m}{qB} = \frac{mv_0}{qE}$$

$$\Rightarrow \qquad E = \frac{v_0 B}{2\pi n}, \qquad n = 1, 2, 3...$$

$$\Rightarrow \qquad \frac{E}{B} = \frac{v_0}{2\pi n}$$

13. Path is helix with increasing pitch. Axis of the helix is parallel to y axis and plane of circle is parallel to xz plane. The particle will cross the y axis for n^{th} time at time given by

$$t = nT = n\cdot\frac{2\pi m}{qB} = \frac{2\pi n}{\sigma B}$$

At this instant, the y component of velocity is

$$v_y = a_y t = (\sigma E)\left(\frac{2\pi n}{\sigma B}\right) = 2\pi n\left(\frac{E}{B}\right)$$

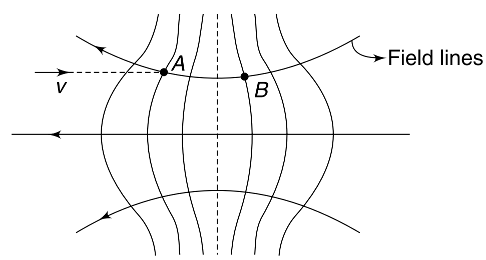
The x component of velocity at this instant will be u and $v_z = 0$

$$\therefore \qquad \tan\alpha = \frac{u}{v_y}$$

$$\therefore \qquad \alpha = \tan^{-1} = \left(\frac{uB}{2\pi n E}\right)$$

14. Electric field lines are perpendicular to the equipotentials as shown in the figure.

An electron at A experiences a force that has a component parallel ($F_{\parallel}$) to the axis and a component perpendicular ($F_{\perp}$) to the axis directed towards the axis.



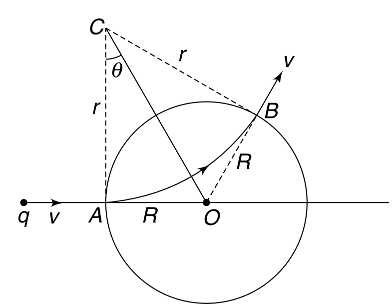
$F_{\parallel}$ speeds up the electron.

$F_{\perp}$ deflects it towards the axis.

However, at a point like B, $F_{\perp}$ is away from the axis. But the electrons have higher speed at B and their upward deflection will be less.

The overall effect will be that the beam of electrons will converge towards the axis.

15. Radius of circular path of the particle is



$$r = \frac{mv}{qB} = \frac{2 \times 10^{-3} \times 0.3}{10^{-3} \times 0.2} = 3\text{m}$$

C is centre of the circular path.

Particle enters the field at A and leaves at B.

$$\tan\theta = \frac{R}{r} = \frac{\sqrt{3}}{3} = \frac{1}{\sqrt{3}}$$

$$\theta = 30°.$$

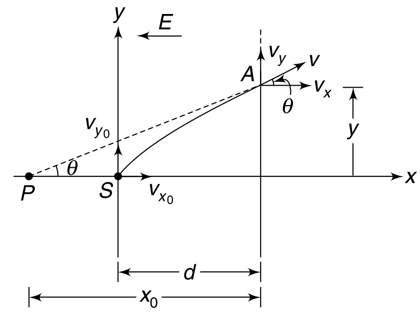
Deviation $= 2\theta = 60°$.

[A careful observation tells that direction of velocity at B is along the radius of the given circle].

16. Velocity of an electron at the source has x and y components: v_{x_0} and v_{y_0}.

When the electron comes out of the field region–

$$v_y = v_{y_0}$$



and $v_x = v_{x0} + \frac{eE}{m}t$; where t is time of travel.

$$\simeq \frac{eE}{m}t$$

$\therefore$ $$\tan\theta = \frac{v_y}{v_x} = \frac{v_{y0}\cdot m}{eEt} \quad \text{...(i)}$$

If tangent to the path at A is extended, it intersects the x axis at P and

$$\tan\theta = \frac{y}{x_0} = \frac{v_{y0}\cdot t}{x_0} \quad \text{...(ii)}$$

From (i) and (ii) $\frac{m}{eE\cdot t} = \frac{t}{x_0}$

$\Rightarrow$ $$x_0 = \frac{eE}{m}t^2$$

But $$d \simeq \frac{1}{2}\left(\frac{eE}{m}\right)t^2$$

$\therefore$ $$x_0 = 2d$$

Hence proved.

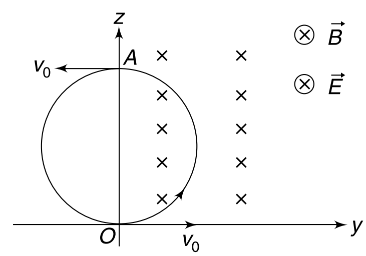
17. When the particle leaves the field, its velocity becomes $-v_0\hat{i} - v_0\hat{j}$.
In $y-z$ plane particle follows a circular path and it moves out of the plane in x direction due to $v_0\,\hat{i}$. The electric force retards it, brings it to rest and then forces it to move in negative x direction.

Since $a_x = -\frac{qE_0}{m}$ = a constant.

$\therefore$ Particle will have $v_x = -v_0$ when it is back to yz plane.

At that moment, its velocity in yz plane must be $-v_0\hat{j}$

$\therefore$ Particle leaves the field at A.



(a) Time of motion in field is $t = \frac{\pi m}{qB_0}$

Also, $$v_x = u_x + a_x t$$

$$-v_0 = v_0 - \frac{qE_0}{m}t$$

$\therefore$ $$t = \frac{2v_0 m}{qE_0}$$

$\therefore$ $$\frac{\pi m}{qB_0} = \frac{2v_0 m}{qE_0}$$

$\therefore$ $$\frac{E_0}{B_0} = \frac{2v_0}{\pi}$$

(ii) Co-ordinates are

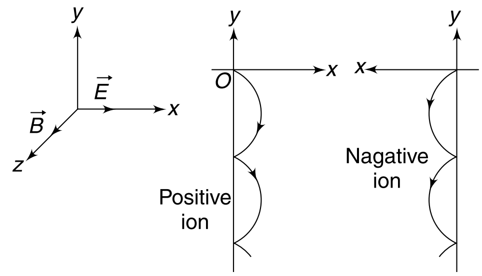
$$x = 0; \quad y = 0$$

$$z = 2R = \frac{2mv_0}{qB_0}$$

(iii) Speed is minimum when $v_x = 0$

$\therefore$ $$v_{min} = v_0$$

18. Path of the particle is a cycloid in xy plane as shown.



z co-ordinate is always zero, and y co-ordinate is always negative.

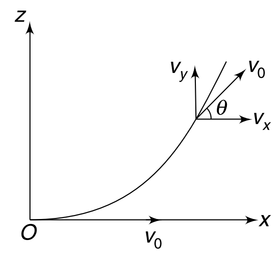
As the x co-ordinate increases the particle speeds up due to electric force. Radius of curvature of the path increases. When the ion moves against the electric field, its speed decreases; and hence the radius of curvature of the path decreases.

For negative ion, path is as shown in third figure.

19. For simplicity let's assume that initial velocity is along x direction. Both $\vec{E}$ and $\vec{B}$ (are parallel) are perpendicular to $\vec{v}_0$. Let $\vec{E}$ and $\vec{B}$ be in positive y direction.

The particle will move in circle in xz plane under influence of magnetic field. Under influence of $\vec{E}$ it acquires an increasing velocity in y direction.

Consider the motion in xz plane.



$$\omega = \frac{qB}{m}$$

At time t $$\theta = \omega t = \frac{qB}{m}t$$

$$v_x = v_0 \cos(\omega t) = v_0 \cos\left(\frac{qB}{m}t\right)$$

$$v_z = v_0 \sin\left(\frac{qB}{m}t\right)$$

And $$v_y = a_y t = \frac{qE}{m}t$$

$\therefore$ $$\vec{v} = v_0 \cos\left(\frac{qB}{m}t\right)\hat{i} + \left(\frac{qEt}{m}\right)\hat{j} + v_0 \sin\left(\frac{qBt}{m}\right)\hat{k}$$

As per our assumption

$$\hat{i} = \frac{\vec{v}_0}{v_0},\ \hat{j} = \frac{\vec{E}}{E}$$

And $$\hat{k} = \frac{v_0 \times \vec{B}}{|\vec{v}_0 \times \vec{B}|} = \frac{\vec{v}_0 \times \vec{B}}{v_0 B}$$

$\therefore$ $$\vec{v} = \cos\left(\frac{qBt}{m}\right)\vec{v}_0 + \left(\frac{qt}{m}\right)\vec{E} + \sin\left(\frac{qBt}{m}\right)\left(\frac{\vec{v}_0 \times \vec{B}}{B}\right)$$

20. (a) The magnetic force does not cause any change in speed and the frictional force will always be against the velocity.

$\therefore$ $$ma = -kv$$

$$v\frac{dv}{dx} = -\frac{k}{m}v$$

$\therefore$ $$\int_{v_0}^{v} dv = -\frac{k}{m}\int_0^{x_0} dx$$

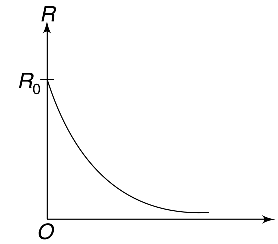
$\Rightarrow$ $$v - v_0 = -\frac{kx_0}{m}$$

$\therefore$ $$v = v_0 - \frac{kx_0}{m}$$

At
$$x_0 = \frac{mv_0}{2k}$$
$$v = \frac{v_0}{2}$$
$$\therefore \quad R = \frac{mv_0/2}{qB_0} = \frac{mv_0}{2qB_0}$$

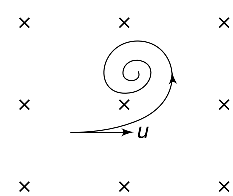
(b) $m\dfrac{dv}{dt} = -kv$

$$\int_{v_0}^{v} \frac{dv}{dv} = -\frac{k}{m}\int_0^t dt$$
$$\ell n\frac{v}{v_0} = -\frac{kt}{m}$$
$$\therefore \quad v = v_0 e^{-\frac{kt}{m}}$$
$$\therefore \quad R = R_0 e^{-\frac{kt}{m}} \text{ where } R_0 = \frac{mv_0}{qB}$$
$$= \text{initial radius}$$

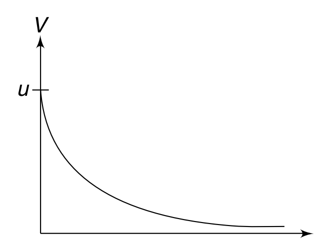


Graph is as given.

21. (a) The particle will follow a spiral path. Magnetic force will always remain perpendicular to the velocity and its only action will be to change the direction of motion.
The frictional force reduces the speed. Due to this the radius of curvature of the path $\left(R = \dfrac{mV}{qB}\right)$ keeps decreasing.



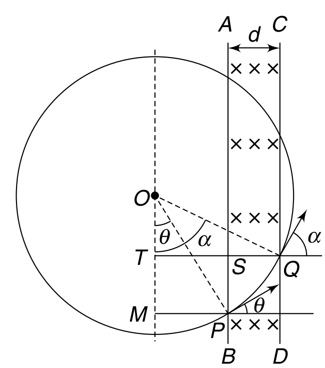
(b)
$$m = \frac{dV}{dt} = -kV$$
$$\int_u^v \frac{dV}{V} = -\frac{k}{m}\int_0^t dt$$
$$\ell n\left(\frac{V}{u}\right) = -\frac{kt}{m}$$
$$V = ue^{-\frac{kt}{m}}$$



(c)
$$V = ue^{-\frac{kt}{m}}$$
$$\frac{ds}{dt} = ue^{-\frac{kt}{m}}$$
$$\int_0^s ds = u\int_0^\infty e^{-\frac{kt}{m}}$$

[particle comes to rest as $t \to \infty$]

$$S = \frac{mu}{k}$$

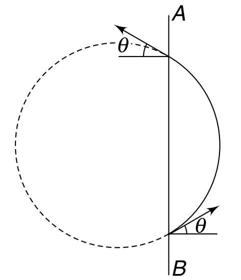


22. The circle shown is the indicative path had there been a magnetic field everywhere. Particle enters the field at P and leaves it at Q. Radius of circle $R = \dfrac{mv}{qB}$

$$MP = TS = R\sin\theta$$
$$TQ = R\sin\theta + d$$

$$\therefore \qquad \sin \alpha = \frac{TQ}{OQ} = \frac{R\sin\theta + d}{R} = \sin\theta + \frac{d}{R}$$

$$\alpha = \sin^{-1}\left[\sin\theta + \frac{d}{R}\right]$$



The particle will emerge to the right of CD as long as $\alpha < \frac{\pi}{2}$

$$\Rightarrow \qquad \sin\theta + \frac{d}{R} < 1$$

$$d < R(1 - \sin\theta)$$

For $$\theta = 30°$$

$$d < \frac{R}{2} \Rightarrow d < \frac{mv}{2qB}$$

For $$d = \left(\frac{\sqrt{2}-1}{2}\right) R \text{ and } \theta = 30°$$

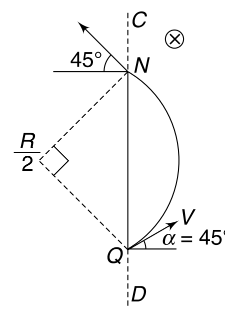
$$\sin\alpha = \sin 30° + \frac{\sqrt{2}-1}{2} = \frac{1}{2} + \frac{\sqrt{2}-1}{2} = \frac{1}{\sqrt{2}}$$

$$\alpha = 45°$$

Deviation = 15°

For $d > R(1 - \sin\theta)$, the particle will emerge out of the field crossing AB.
Deviation in this case will be $= \pi - 2\theta$

23. From the solution of last problem, the displacement in y direction during crossing of first field is



$$y_1 = R\cos\theta - R\cos\alpha$$

$$= R(\cos 30° - \cos 45°) = R\left(\frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}}\right)$$

In the second field the particle enters at Q and leaves at N, while moving on a circle of radius $\frac{R}{2}$ $\left[\because R = \frac{mv}{qB} \text{ and } B \text{ has been doubled}\right]$

Particle completes a quarter circle in the second field.

$$y_2 = QN = \frac{R}{2}\sqrt{2} = \frac{R}{\sqrt{2}}$$

At N, the particle enters the first field at 45° to negative X. It will get deflected (anticlockwise) by 15° as seen in the previous problem. And

$$y_3 = y_1$$

$$\therefore \qquad y = y_1 + y_2 + y_3$$

$$= 2y_1 + y_2$$

$$= 2R\left[\frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}}\right] + \frac{R}{\sqrt{2}}$$

$$= \left(\sqrt{3} - \frac{1}{\sqrt{2}}\right)R$$

Time spent can be calculated as:

$$t = \frac{2\pi m}{qB} \times \left(\frac{15°}{360°}\right) \times 2 + \frac{2\pi m}{q(2B)}\left(\frac{90°}{360°}\right)$$

$$= \frac{\pi m}{6qB} + \frac{\pi m}{4qB} = \frac{5\pi m}{12qB}$$

24. Let coefficient of friction be μ.

$$u^2 = 2(\mu g)X_0 \qquad \ldots(1)$$

$$\mu = \frac{u^2}{2gX_0}$$

With magnetic field switched on, the friction force does not change. Magnetic field does not perform any work on the moving charge. Only friction (which is always tangential to the path) eats away the entire KE

$$\Rightarrow \qquad X_1 = X_0$$

With both the fields switched on, the normal reaction on the particle becomes.

$$N = mg + qE_0$$

$$\therefore \qquad f = \mu(mg + qE_0)$$

$$\therefore \qquad \mu(mg + qE_0)\cdot X_2 = \frac{1}{2}mu^2$$

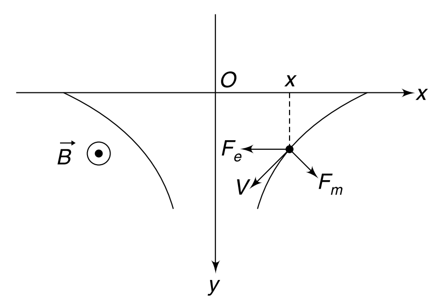
$$\frac{u^2}{2gX_0}(mg + qE_0)X_2 = \frac{1}{2}mu^2$$

$$X_2 = \frac{mgX_0}{mg + qE_0}$$

25. (a) The two particles follow symmetrical curved paths. As the electrostatic attraction speeds them, the magnetic force curves the path.

Magnetic force does not perform any work, therefore conservation of energy gives-

$$\frac{1}{2}mV^2 \times 2 - k\frac{q^2}{(2x)} = -k\frac{q^2}{(2a)}$$



$$\Rightarrow \qquad 2mV^2 = kq^2\left(-\frac{1}{a} + \frac{1}{x}\right)$$

$$\Rightarrow \qquad V = \sqrt{\frac{kq^2}{2m}\left(\frac{1}{x} - \frac{1}{a}\right)}$$

(b) Force on $-q$ is

$$\vec{F} = -F_e\hat{i} + (-q)\,(V_x\hat{i} + V_y\hat{j}) \times (-B\hat{k})$$

$$= (-F_e + qV_yB)\hat{i} - qBV_x\hat{j}$$

$$\therefore \qquad m\frac{dV_y}{dt} = -qBV_x \;\Rightarrow\; m\frac{dV_y}{dt} = -qB\frac{dx}{dt}$$

$$\Rightarrow \qquad \int_0^{V_y} dV_y = -\frac{qB}{m}\int_a^x dx$$

$$\Rightarrow \qquad V_y = \frac{qB}{m}(a - x)$$

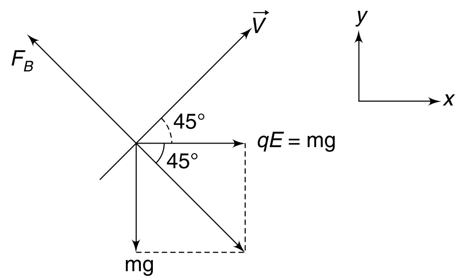
26. After crossing the XX plane, the forces on the particle are

(1) mg $(\downarrow)$

(2) $qE(\rightarrow)$

(3) Magnetic force perpendicular to velocity ($F_B = qVB$)

For particle to move uniformly, the magnetic force must balance the resultant of mg and qE. Hence, direction of velocity must be as shown in figure.



Component of velocity in X and Y direction must be same.

$$V_x = V_y$$

Since acceleration in y direction before entering magnetic field is $a_y = \frac{qE}{m} = \frac{\text{mg}}{m} = g$; we have

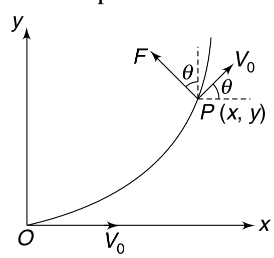
$$u - gT = gT$$

$$u = 2gT$$

27. Particle moves in xy plane. Its path is not a circle because B is not constant. However, speed of the particle remains constant at V_0.

Let velocity at point $P(x, y)$ be inclined to x axis at an angle θ.

Force will be perpendicular to velocity, inclined at θ with y direction.



$$\therefore \quad ma_y = F\cos\theta$$

$$\Rightarrow \quad m\frac{dV_y}{at} = (B_0x)qV_0\cos\theta$$

$$\Rightarrow \quad m\frac{dV_y}{dx}\cdot\frac{dx}{dt} = B_0qx(V_x) \qquad [\because V_0\cos\theta = V_x]$$

But $$\frac{dx}{dt} = V_x$$

$$\therefore \quad \frac{dV_y}{dx} = \frac{B_0q}{m}x$$

$$\Rightarrow \quad \int_0^{v_0} dV_y = \frac{B_0q}{m}\int_0^{x_0} x\,dx$$

Where x_0 = maximum x co-ordinate of the particle where velocity becomes parallel to y axis.

$$\therefore \quad V_0 = \frac{B_0q}{2m}x_0^2$$

$$\Rightarrow \quad x_0 = \sqrt{\frac{2mV_0}{B_0q}}$$

Magnetic force at $x = x_0$ is $F = qV_0(B_0x_0)$

$$\therefore \quad F = qV_0B_0\sqrt{\frac{2mV_0}{B_0q}} = V_0\sqrt{2mqB_0V_0}$$

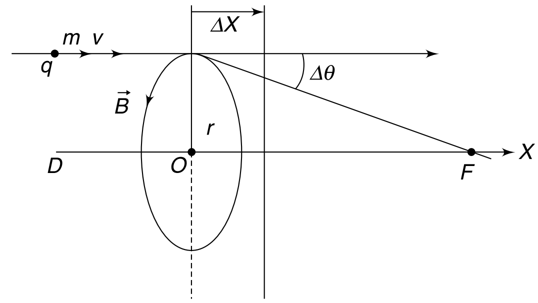
If radius of curvature of path if R then $\frac{mV_0^2}{R} = F$

$$\Rightarrow \quad \frac{mV_0^2}{R} = V_0\sqrt{2mqB_0V_0}$$

$$\Rightarrow \quad R = \sqrt{\frac{mV_0}{2qB_0}}$$

$\Rightarrow$

28. Consider a charge particle at a distance r from the x-axis.



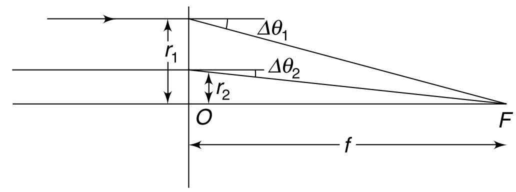
Magnetic force on it is towards O (perpendicular) to its original direction of motion. This will cause the path of the charge to deviate by a small angle $\Delta\theta$.

Time required to cross the field is: $\Delta t \simeq \dfrac{\Delta x}{V}$

Impulse of magnetic force $= qV(B_0 r)\dfrac{\Delta x}{V} = qB_0 r\Delta x$

Charge in momentum of the charge in crossing the field

$$\Delta p = qB_0 r\Delta x$$



This change is perpendicular to original direction of momentum.

$$\therefore \qquad \Delta p \simeq p\Delta\theta$$

$$qB_0 r\Delta x \simeq mV\Delta\theta$$

$$\Rightarrow \qquad \Delta\theta = \frac{qB_0 r\Delta x}{mV}$$

Because $\Delta\theta \;\alpha\; r$, all ions will get focused at one point F or the axis. (see figure.)

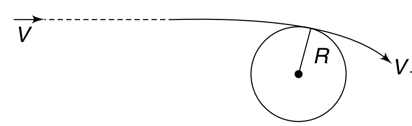
$$f = \frac{r_1}{\Delta\theta_1} = \frac{r_2}{\Delta\theta_2}$$

$$f = \frac{r}{\Delta\theta} = \frac{mV}{qB_0\Delta x}$$

29. (a) Angular momentum of the particle remains conserved about the centre of the sphere.

The particle just misses the sphere means that it grazes the surface of the sphere. We will use angular momentum and energy conservation.

Angular momentum conservation:



$$mbV - mRV_1 \quad \Rightarrow \quad V_1 = \frac{bV}{R}$$

Energy conservation:

$$\frac{1}{2}mV^2 - \frac{kQq}{\infty} = \frac{1}{2}mV_1^2 - K\frac{Qq}{R} \quad \left[K = \frac{1}{4\pi\epsilon_0}\right]$$

$$V_1^2 = V^2 + \frac{2KQq}{mR}$$

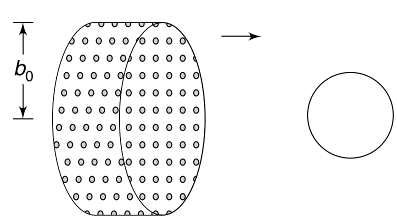
$$\frac{b^2}{R^2}V^2 = V^2 + 2\frac{KQq}{mR}$$

$$b_0 = R\sqrt{1 + \frac{2KQq}{mRV^2}}$$

for $\dfrac{1}{2}mV^2 = 100\dfrac{KQq}{R}$

$$\frac{1}{100} = \frac{2KQq}{mRV^2}$$

$$\therefore \qquad b_0 = R[1 + 0.01]^{1/2} \simeq 1.02\,R$$



(b) In the reference frame of the sphere, the situation is identical to that in part (a). As the sphere moves, in a period Δt it will sweep all the particles in a volume:

$$V = \pi b_0^2\, V\Delta t$$

$\therefore$ Total negative charge deposited on the sphere in interval Δt is

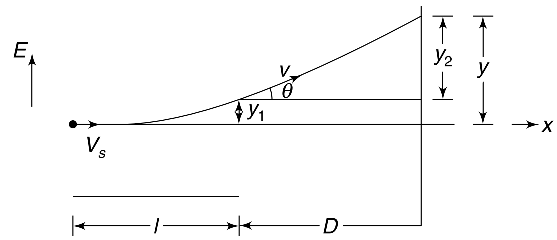
$$\Delta q = q\cdot n\cdot V = qn\pi b_0^2 V\Delta t$$

$$\frac{\Delta q}{\Delta t} = q\ n\ b_0^2\ V$$

$\therefore$ Charge on sphere decreases at rate

$$\left|\frac{\Delta Q}{\Delta t}\right| = qn\pi b_0^2 V.$$

30. (a) Time to cross the region of electric field is



$$t = \frac{\ell}{V_x}$$

$\therefore$
$$y_1 = \frac{1}{2}a_y t^2 = \frac{1}{2}\left(\frac{qE}{m}\right)\left(\frac{\ell}{V_x}\right)^2$$

At the point of leaving the field region velocity makes angle θ with x direction given by

$$\tan\theta = \frac{V_y}{V_x} = \frac{a_y t}{V_x} = \frac{\left(\frac{qE}{m}\right)\left(\frac{\ell}{V_x}\right)}{V_x} = \frac{qE\ell}{mV_x^2}$$

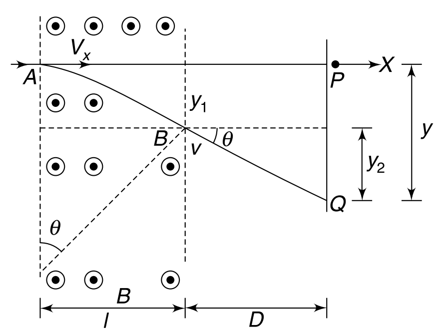
$$y_2 = D\tan\theta = \frac{qE\ell D}{mV_x^2}$$

$\therefore$
$$y = y_1 + y_2 = \frac{qE\ell}{mV_x^2}\left[\frac{\ell}{2} + D\right]$$

If $D >> \ell$

$$y = \frac{qE\ell D}{mV_x^2}$$

(b) Radius of circular path in magnetic field is



$$R = \frac{mV_x}{qB}$$

The deflection angle θ is given by

$$\sin\theta = \frac{\ell}{R} = \frac{\ell qB}{mV_x}$$

For small ℓ, θ can be assumed to be small and y_1 is also small.

$$y \simeq y_2 = D\tan\theta = D\sin\theta = \frac{D\ell qB}{mV_x}$$

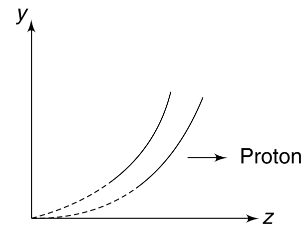
(d) Let the initial direction of travel of charge be along x and the two fields be in y direction. The deflection caused due to two fields will be -

$$y = \frac{qE\ell D}{mV_x^2} \text{ and } z = \frac{qB\ell D}{mV_x}$$

For different values of V_x the ions hit the screen at different points.
Eliminating V_x between the above two expressions gives

$$\frac{z^2}{y} = \frac{D\ell B^2}{E}\left(\frac{q}{m}\right)$$

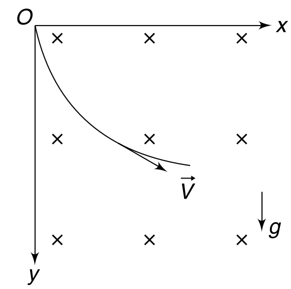
$$z^2 = k\left(\frac{q}{m}\right)y \text{ where } k = \frac{D\,\ell\,B^2}{E} = \text{a constant}$$



This represents an equation of a parabola. For different values of $\frac{q}{m}$ we get different curves.

For hydrogen $\frac{q}{m}$ is highest hence curve 1represents hydrogen and curve 2 represents the ion.

31. (b) Consider the origin at the starting point and the x and y axes as shown. Let the velocity of the particle at any time be $\vec{V} = V_x\hat{i} + V_y\hat{j}$



Force on the particle is:

$$F = \text{mg}\hat{j} + q(V_x\hat{i} + V_x\hat{j}) \times B\hat{k}$$

$$= (\text{mg} - qBV_x)\,\hat{j} + (qBV_y)\hat{i}$$

Hence, $$m\frac{dV_y}{dt} = \text{mg} - qBV_x \quad ...(1)$$

and $$m\,\frac{dV_x}{dt} = qBV_y \quad ...(2)$$

Differentiating equation (1) with respect to time

$$m\frac{d^2V_y}{dt^2} = -qB\,\frac{dV_x}{dt}$$

using (2)

$$\frac{d^2V_y}{dt^2} = -\frac{q^2B^2}{m^2}V_y$$

Solution to this differential equation is (as learnt in chapter of SHM)

$$V_y = V_{y0}\sin\left(\frac{qB}{m}t + \delta\right) \quad ...(3)$$

V_{y0} and δ are constants.

It is known that $V_y = 0$ at $t = 0$

$\therefore$ $$\delta = 0$$

$\therefore$ $$V_y = V_{y0}\sin\left(\frac{qB}{m}t\right)$$

Differentiating wrt time

$$a_y = \frac{dV_y}{at} = V_{y0}\,\frac{qB}{m}\cos\left(\frac{qB}{m}t\right)$$

at time $t = 0$; $a_y = g$

$\therefore$ $$g = \frac{V_{y0}qB}{m}$$

$$V_{y0} = \frac{\text{mg}}{qB}$$

$\therefore$ $$V_y = \frac{\text{mg}}{qB}\sin\left(\frac{qB}{m}t\right) \quad ...(4)$$

$$\frac{dV_y}{dt} = g\cos\left(\frac{qB}{m}t\right)$$

Put in (1)

$$V_x = \frac{mg}{qB}\left[1 - \cos\left(\frac{qB}{m}t\right)\right] \quad ...(5)$$

Time after which the particle starts climbing up is after V_y becomes zero (for the first time after release).

$$V_y = 0$$

$$\Rightarrow \quad \frac{qB}{m}t = \pi \quad \Rightarrow \quad t = \frac{\pi m}{qB}$$

At this time V_x is

$$V_x = \frac{2mg}{qB}$$

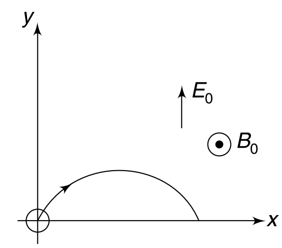
This is the required speed.

(c) Magnetic force does not perform any work during the motion of the charge particle.

$$\therefore \quad mgh = \frac{1}{2}mV_x^2$$

$$h = \frac{2m^2g}{q^2B^2}$$

32. The particle moves in xy plane. Let its velocity at any instant be



$$\vec{V} = V_x\hat{i} + V_y\hat{j}$$

$$\vec{F} = q\vec{E} + q(\overline{V} \times \overline{B})$$

$$= qE_0\hat{j} + q\left(V_x\hat{i} + V_y\hat{j}\right) \times (B_0\,\hat{k})$$

$$= q(E_0 - V_xB_0)\hat{j} + qB_0V_y\hat{i}$$

$$\frac{m\,dV_x}{dt} = qB_0V_y \quad ...(1)$$

And

$$m\frac{dV_y}{dt} = (E_0 - V_xB_0) \quad ...(2)$$

Differentiating (2) wrt time and substituting for $\frac{dV_x}{dt}$ from (1), we get

$$\frac{d^2V_y}{dt^2} = \left(\frac{qB_0}{m}\right)^2 V_y$$

This equation is of the form $\frac{d^2x}{dt^2} = -\omega^2 x$

$$\therefore \quad V_y = V_{y0}\sin(\omega t + \delta) \quad \left[\omega = \frac{qB_0}{m}\right]$$

at $\quad t = 0;\ V_y = 0 \quad \Rightarrow \quad \delta = 0$

$$\therefore \quad V_y = V_{y0}\sin\left(\frac{qB_0}{m}t\right)$$

$$a_y = \frac{dV_y}{dt} = V_{y0}\frac{qB_0}{m}\cos\left(\frac{qB_0}{m}t\right)$$

at $\quad t = 0;\ a_y = \frac{qE_0}{m}$

$$\therefore \quad \frac{qE_0}{m} = \frac{qB_0}{m} V_{y0} \quad \Rightarrow \quad V_{y0} = \frac{E_0}{B_0}$$

$$V_y = \frac{E_0}{B_0} \sin\left(\frac{qB_0}{m} t\right)$$

$$\frac{dy}{dt} = \frac{E_0}{B_0} \sin\left(\frac{qB_0}{m} t\right)$$

$$\int_0^y dy = \frac{E_0}{B_0} \int_0^t \sin\left(\frac{qB_0}{m} t\right) dt$$

$$y = \frac{mE_0}{qB_0^2}\left[1 - \cos\left(\frac{qB_0}{m} t\right)\right]$$

$$\therefore \quad y_{\max} = \frac{2mE_0}{qB_0^2}$$

$$\therefore \quad 2R = \frac{2mE_0}{qB_0^2}$$

$$R = \frac{mE_0}{qB_0^2}$$

Chapter 10

Magnetic Effect of Current

LEVEL 1

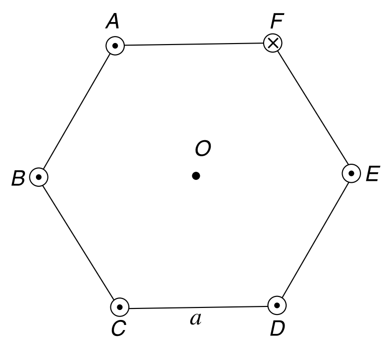
Q. 1: Sketch the magnetic field lines in xy plane for a pair of long parallel wires laid along z direction if-

(a) Both wires carry current in same direction.

(b) Both wire carry current in opposite directions.

Q. 2: A long wire is along $x = 0$, $z = d$ and carries current in positive y direction. Another wire is along $x = y$, $z = 0$ and carries current in direction making acute angle with positive x direction. Both the wires have current I. Find the magnitude of magnetic induction at $(0, 0, 2d)$.

Q. 3: Six long parallel current carrying wires are perpendicular to the plane of the fig. They pass through the vertices of a regular hexagon of side length a. All wires have same current I. Direction of current is out of the plane of the figure in all the wires except the one passing through vertex F; which has current directed into the plane of the figure. Calculate the magnetic induction field at the centre of the hexagon. Also tell the direction of the field.



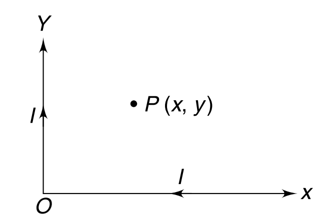
Q. 4: Two infinitely long parallel wires carry current $I_1 = 8$ A and $I_2 = 10$ A in opposite directions. The separation between the wires is $d = 0.12$ m. Find the magnitude of magnetic field at a point P that is at a perpendicular distance $r_1 = 0.16$ m and $r_2 = 0.20$ m respectively from the wires.

Q. 5: A wire frame is in the shape of a regular polygon of 2016 sides. Each side is of length $L = 1$ cm. If a current $I = 5.0$ A is given to the wire frame estimate the magnetic induction field (B) at the centre of the polygon.

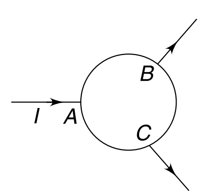
[Take $\pi^2 = 10.08$]

Q. 6: Two coplanar concentric circular wires made of same material have radius R_1 and $R_2 (= 2R_1)$. The wires carry current due to identical source of emf having no internal resistance. Find the ratio of radii of cross section of the two wires if the magnetic induction field at the centre of the circle is zero.

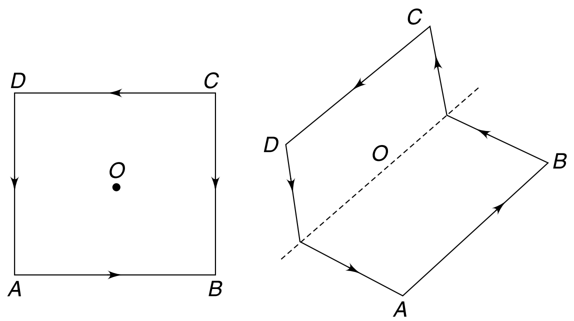
Q. 7: An infinitely long wire carrying current I is bent to from a L shaped wire. Let the bend be the origin and the two arms be along x and y direction (see figure). Calculate the magnitude of magnetic field at point P (in first quadrant) whose co-ordinates are (x, y).



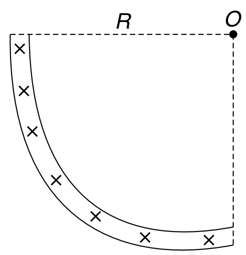
Q. 8: In the figure shown ABC is a circle of radius a. Arc AB and AC each have resistance R. Arc BC has resistance $2R$. A current I enters at point A and leaves the circle at B and C. All straight wires are radial. Calculate the magnetic field at the centre of the circle. Each arc AB, BC and AC subtends 120° at the centre of the circle.



Q. 9: A square loop of side length L carries a current which produces a magnetic field B_0 at the centre (O) of the loop. Now the square loop is folded into two parts with one half being perpendicular to the other (see fig). Calculate the magnitude of magnetic field at the centre O.

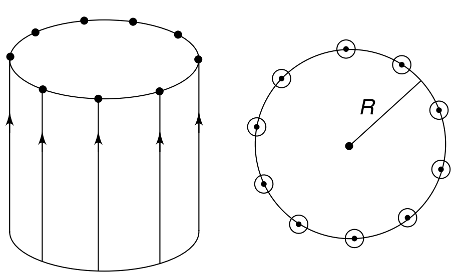


Q. 10: A current I flows in a long straight wire whose cross section is in the shape of a thin quarter ring of radius R. Find the induction of the magnetic field (B) at point O on the axis.

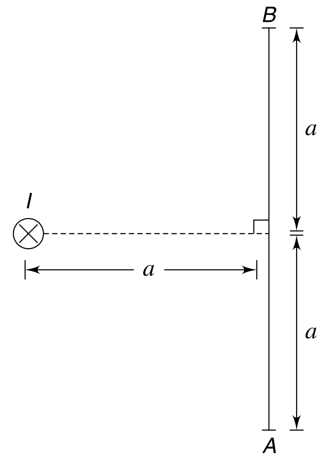


Q. 11: The figure shows a long cylinder and its cross section. There are N (N is a large number) wire on the curved surface of the cylinder at uniform spacing and parallel to its axis. Each wire has current I and cross sectional radius of wires are small compared to radius R of the cylinder. Find magnetic field at a distance x from the axis of the cylinder for

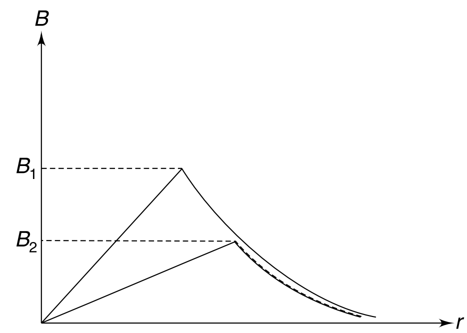
(a) $x < R$ (b) $x > R$



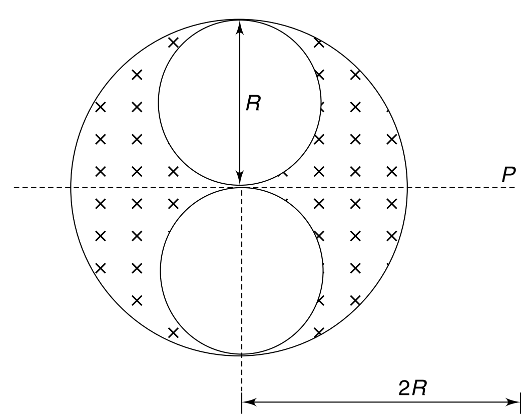
Q. 12: A straight current carrying wire has current I directed into the plane of the fig. There is a line AB of length $2a$ at a distance a from the wire (see fig.). Find the value of line integral $\int_A^B \vec{B}\cdot\vec{dl}$ where $\vec{B}$ represents magnetic field at a point due to current I. Will the value of integral change if a is changed? Length of line AB is always double that of a.



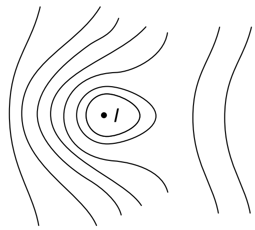
Q. 13: There are two separate long cylindrical wires having uniform current density. The radius of one of the wires is twice that of the other. The fig. shows the plot of magnitude of magnetic field intensity versus radial distance (r) from their axis. The curved parts of the two graphs are overlapping. Find the ratio $B_1 : B_2$.



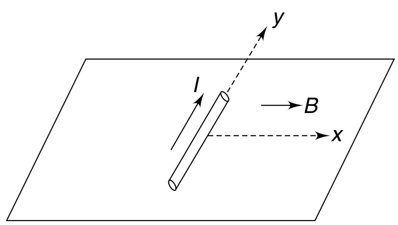
Q. 14: A long cylindrical conductor of radius R has two cylindrical cavities of diameter R through its entire length, as shown in the figure. There is a current I through the conductor distributed uniformly in its entire cross section (apart from the cavity region). Find magnetic field at point P at a distance $r = 2R$ from the axis of the conductor (see figure).



Q. 15: A uniform magnetic field exists in vertical direction in a region of space. A long current carrying wire (having current I) is placed horizontally in the region perpendicular to the figure. The resultant field due to superposition of the uniform field and that due to the current is represented by the field lines shown in the figure. In which direction does the current carrying wire experience the magnetic force?



Q. 16: A conducting wire of length ℓ and mass m is placed on a horizontal surface with its length along y direction. There exists a uniform magnetic field B along positive x direction. With wire carrying a current I in positive y direction, the least value of force required to move it in x and y directions are F_1 and F_2. Now the direction of current in the wire is reversed and the value of two forces becomes F_1' and F_2'. Find the ratio of forces $F_1 : F_2 : F_1' : F_2'$

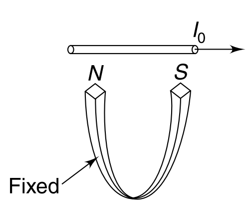


Q. 17: How will the conductor, carrying current I_0 rotate immediately after it is released in following three cases [consider magnetic force only]

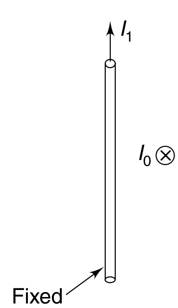
(a) Conductor carrying current I_0 is placed symmetrically above poles of a fixed U shaped magnet (figure a).

(b) Conductor carrying current I_0 is placed symmetrically at a distance from a fixed current (I_1) carrying wire (fig. (b))

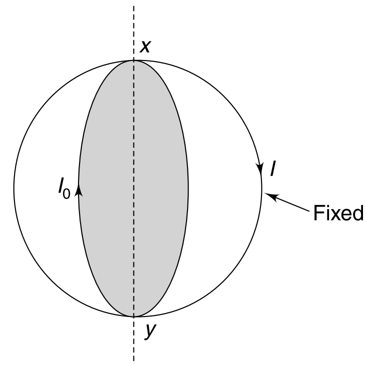
(c) An insulated circular current carrying wire is held fixed in vertical plane. Conductor carrying current I_0 is in the shape of a circle of diameter nearly equal to that of the fixed insulated circle. The planes of the two circles are perpendicular to each other (fig. (c)) with xy as common diameter.



(a)

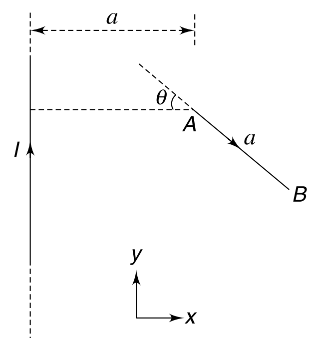


(b)



(c)

Q. 18: A straight wire AB of length a is placed at a distance a from an infinitely long straight wire as shown in the figure. Angle θ is 30°. Find the magnetic force on wire AB if it is also given a current I. Both the wires are in xy plane.

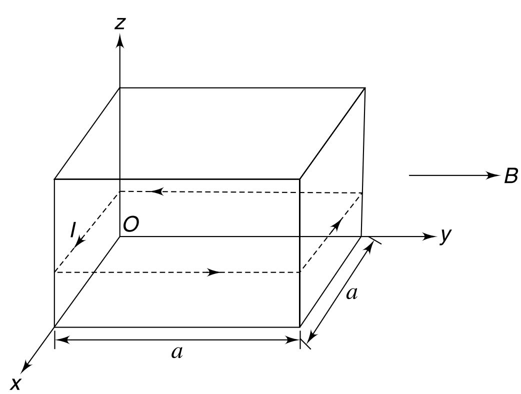


Q.19: A dielectric spherical shell of radius R, having charge Q is rotating with angular speed ω about its diameter. Calculate the magnetic dipole moment (M) of the shell. Write the ratio of M and angular momentum (L) of the rotating shell. This ratio is called gyro-magnetic ratio. Mass of shell is m.

Q. 20: A wooden cubical block of mass m and side a is resting on a horizontal surface. A wire carrying current I, is wrapped around it in from of a square of side a. A uniform magnetic field $\vec{B} = B_0 \hat{j}$ is switched on in the region. Neglect the mass of the wire.

(a) At what distance from the x axis does normal force applied by the horizontal surface on the wooden cube act?

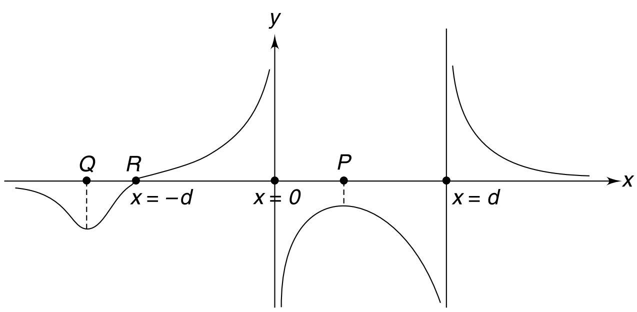
(b) What is the maximum value of current for which the block will not topple?



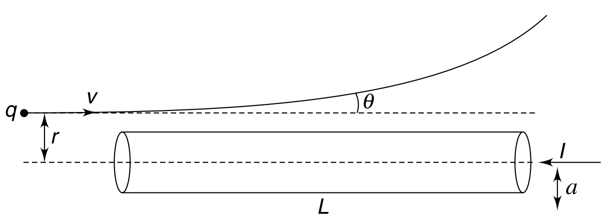
Q.21: A square loop of mass m and side length a lies in xy plane with its centre at origin. It carries a current I. The loop is free to rotate about x axis. A magnetic field $\vec{B} = B_0\hat{j}$ is switched on in the region. Calculate the angular speed acquired by the loop when it has rotated through 90°. Assume no other force on the loop apart from the magnetic force.

LEVEL 2

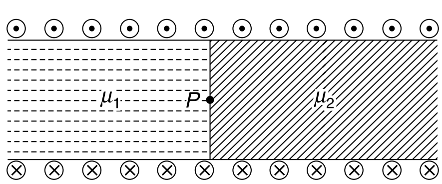
Q. 22: Two long parallel wires are along z direction at $x = 0$ and $x = d$. The magnetic field along x axis has been plotted in the given figure with field (B) positive when it is in positive y direction. The co-ordinate of point R is $x = -d$. Find co-ordinate of points P and Q shown in figure.



Q. 23: A straight wire of length L and radius a has a current I. A particle of mass m and charge q approaches the wire moving at a velocity v in a direction anti parallel to the current. The line of motion of the particle is at a distance r from the axis of the wire. Assume that r is slightly larger than a so that the magnetic field seen by the particle is similar to that caused by a long wire. Neglect end effects and assume that speed of the particle is high so that it crosses the wire quickly and suffers a small deflection θ in its path. Calculate θ.



Q. 24: A long narrow solenoid is half filled with material of relative permeability μ_1 and half filled with another material of relative permeability μ_2. The number of turns per meter length of the solenoid is n. Calculate the magnetic field (B) on the axis of the solenoid at boundary of the two material (i.e. at point P). The current in solenoid coil is I.

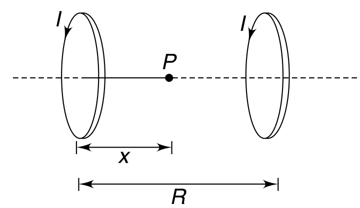


Q. 25: Two identical coils having radius R and number of turns N are placed co-axially with their centres separated by a distance equal to their radius R. The two coils are given same current I in same direction. The configuration is often known as a pair of Helmholtz coil.

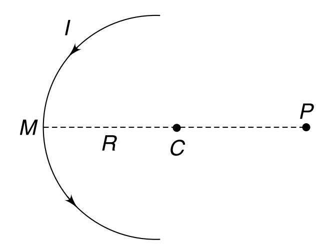
(i) Calculate the magnetic field (B) at a point (P) on the axis between the coils at a distance x from the centres of one of the coils.

(ii) Prove that $\frac{dB}{dx} = 0$ and $\frac{d^2B}{dx^2} = 0$

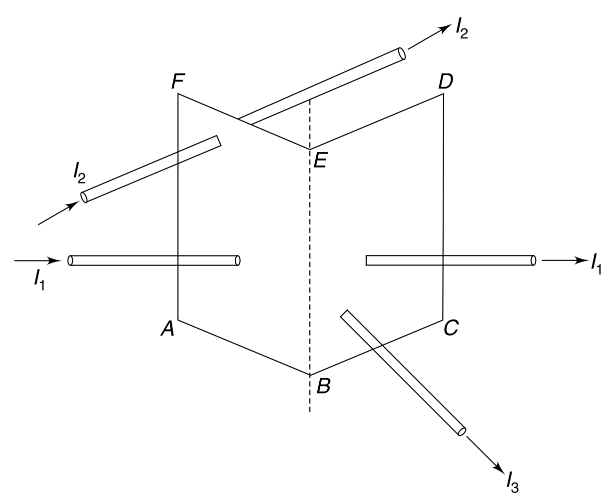
$\left[\text{In fact } \frac{d^3B}{dx^3} \text{ is also equal to zero}\right]$ at the point lying midway between the two coils. What conclusion can you draw from these results?



Q. 26: A current carrying wire is in the shape of a semicircle of radius R and has current I. M is midpoint of the arc and point P lies on extension of MC at a distance $2R$ from M. Find the magnetic field due to circular arc at point P.

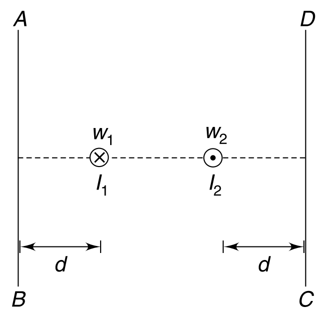


Q. 27: The figure shows three straight current carrying conductors having current I_1, I_2 and I_3 respectively. Calculate line integral of magnetic induction field ($\vec{B}$) along the closed path $ABCDEFA$.

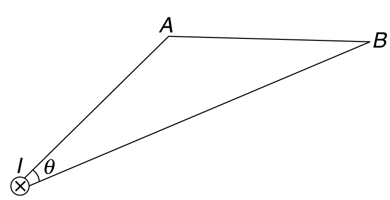


Q.28: A circular coil of N turns carries a current I. Field at a distance x from centre of the loop on its axis is B. Write the value of integral $\int_{-\infty}^{\infty} B\cdot dx$.

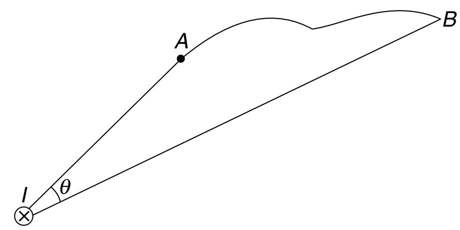
Q. 29: In the figure shown $W1$ represent the cross section of an infinitely long wire carrying current I_1 into the plane of the fig. AB is a line of length L and the wire $W1$ is symmetrically located with respect to the line. The line integral $\int_A^B \vec{B}\cdot\vec{dl}$ along the line from A to B is equal to $-a_0$ where a_0 is a positive number. Another long wire $W2$ is placed symmetrically with respect to AB (see fig) and the value of $\int_A^B \vec{B}\cdot\vec{dl}$ becomes zero. Consider a line DC to the right of $W2$. The line is parallel to AB and has same length. The two wires fall on perpendicular bisector of both lines. If $\int_C^D \vec{B}\cdot\vec{d} = 2a_0$ with both wires $W1$ and $W2$ present, calculate the ratio of current $\frac{I_2}{I_1}$ in the two wires.



Q. 30: (a) A long straight wire carries a current I into the plane of the figure. AB is a straight line in the plane of the figure subtending an angle θ at the point of intersection of the wire with the plane. Find (by integration) the line integral of magnetic field along the line AB.



(b) In the last problem the straight line AB is replaced with a curved line AB as shown in figure. Can you calculate the line integral of magnetic field B along this curved line? If yes, what is its value?



Q. 31: A long straight cylindrical region of radius a curries a current along its length. The current density (J) varies from the axis to the edge of the cylindrical region according to $J = J_0\left(1 - \frac{r}{a}\right)$

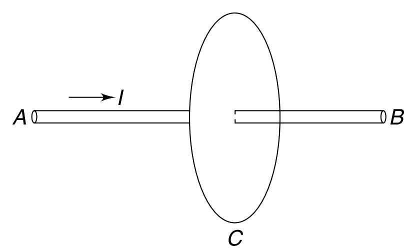
Where r is distance from the axis ($0 \le r \le a$)

(a) Find the mean current density.

(b) Plot the variation of magnetic field (B) with distance r from the axis of the cylinder for $0 \le r \le a$.

Q. 32: A student has studied the use of Ampere's law in calculation of magnetic field (**B**) due to a straight current carrying conductor of infinite length. Now she used similar arguments for calculation of B due to a current carrying conductor (AB) of finite length. She assumes a closed circular path (C) of radius r with the conductor along the axis (see fig.). She argues that because of symmetry the field (B) shall be tangential to C and must have same magnitude at all points on C. Therefore she writes $B = \frac{\mu_0 I}{2\pi r}$

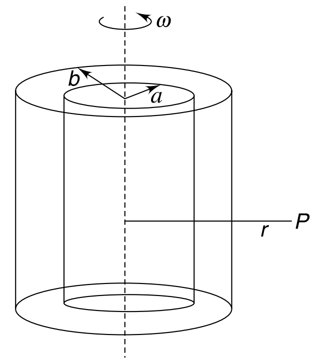
Do you support the answer? Give reasons.



Q. 33: There are two co-axial non conducting cylinders of radii a and $b(>a)$. Length of each cylinder is $L(>>b)$ and their curved surfaces have uniform surface charge densities of $-\sigma$ (on cylinder of radius a) and $+\sigma$ (on cylinder of radius b). The two cylinders are made to rotate with same angular velocity ω as shown in the figure. The charge distribution does not change due to rotation. Find the electric field (E) and magnetic field (B) at a point (P) which is at a distance r from the axis such that

(a) $0 < r < a$ (b) $a < r < b$ (c) $r > b$.

Assume that point P is close to perpendicular bisector of the length of the cylinders



Q. 34: An infinite sheet in xy plane has a uniform surface charge density σ. The thickness of the sheet is infinitesimally small. The sheet begins to move with a velocity $\vec{v} = v\hat{i}$

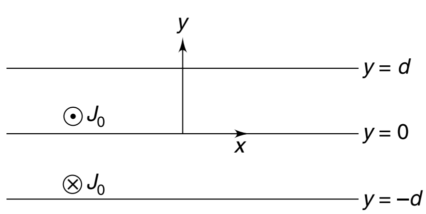
(i) Find the electric field $(\vec{E})$ and magnetic field $(\vec{B})$ above and below the sheet.

(ii) If the velocity of the sheet is changed to $\vec{v} = v\hat{k}$, find the electric and magnetic field above and below the sheet.

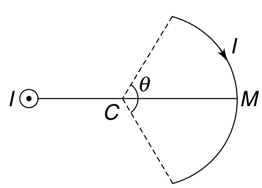
Q. 35: Consider two slabs of current shown in the figure. Both slabs have thickness b in y direction and extend up to infinity in x and z directions. The common face of the two slabs is $y = 0$ plane. The slab in the region $0 < y < d$ has a constant current density $= J_0\hat{k}$ and the other slab in the region $-d < y < 0$ has a constant current density $= J_0(-\hat{k})$.

(a) Find magnetic field at $y = 0$

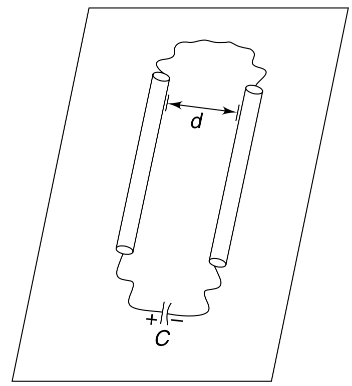
(b) Plot the variation of magnetic field (B) along the y axis.



Q. 36: A current carrying conductor is in the shape of an arc of a circle of radius R subtending an angle θ at the centre (C). A long current carrying wire is perpendicular to the plane of the arc and is at a distance $2R$ from the midpoint (M) of the arc on the line joining the points M and C. Current in the arc as well as straight wire is I. Find the magnetic force on the arc.



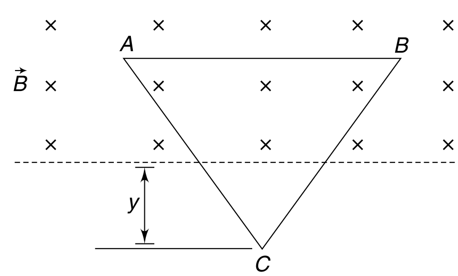
Q. 37: Two long straight conducting wires with linear mass density λ are kept parallel to each other on a smooth horizontal surface. Distance between them is d and one end of each wire is connected to each other using a loose wire as shown in the figure. A capacitor is charged so as to have energy U_0 stored in it. The capacitor is connected to the ends of two wires as shown. The resistance (R) of the entire arrangement is negligible and the capacitor discharges quickly. Assume that the distance between the wires do not change during the discharging process. Calculate the speed acquired by two wires as the capacitor discharges.



Q. 38: A current carrying loop is in the shape of an equilateral triangle of side length a. Its mass is M and it is in vertical plane. There exists a uniform horizontal magnetic field B in the region shown.

(a) The loop is in equilibrium for $y_0 = \frac{\sqrt{3}}{4}a$. Find the current in the loop.

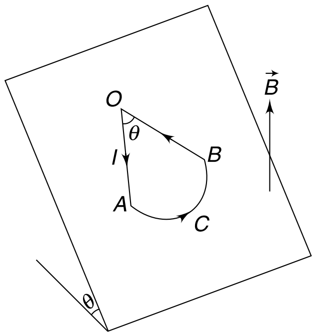
(b) The loop is displaced slightly in its plane perpendicular to its side AB and released. Find time period of its oscillations. Neglect emf induced in the loop. Express your answer in terms of a and g.



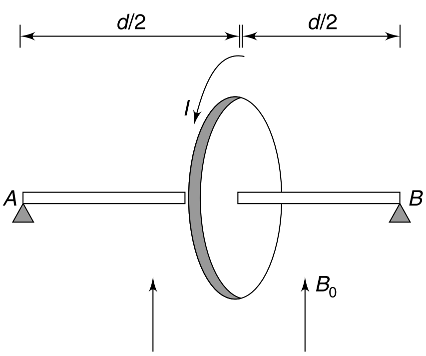
Q. 39: A current loop consist of two straight segments (OA and OB), each of length ℓ, having an angle θ between them and a semicircle (ACB). The loop is placed on an incline plane making an angle θ with horizontal (see figure). The loop carries a current I. A uniform vertical magnetic field B is switched on.

(a) Write the value of magnetic torque on the loop.

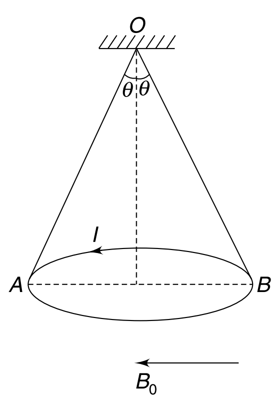
(b) Tell whether the normal contact force between the incline and the loop increases or decreases when magnetic field is switched on. Assume that the loop remains stationary on the incline.



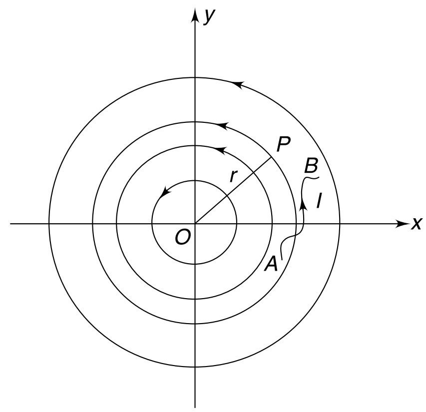
Q. 40: A wooden disc of mass M and radius R has a single loop of wire wound on its circumference. It is mounted on a massless rod of length d. The ends of the rod are supported at its ends so that the rod is horizontal and disc is vertical. A uniform magnetic field B_0 exists in vertically upward direction. When a current I is given to the wire one end of the rod leaves the support. Find least value of I.



Q. 41: A uniform ring of mass M and radius R carries a current I (see figure). The ring is suspended using two identical strings OA and OB. There exists a uniform horizontal magnetic field B_0 parallel to the diameter AB of the ring. Calculate tension in the two strings. [Given $\theta = 60°$]

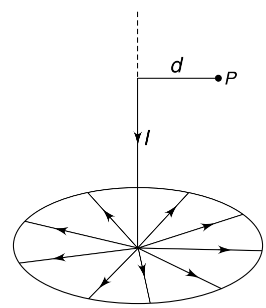


Q. 42: In a two dimensional $x - y$ plane, the magnetic field lines are circular, centred at the origin. The magnitude of the field is inversely proportional to distance from the origin and field at any point P has magnitude given by $B = \frac{k}{r}$; where k is a positive constant. A wire carrying current I is laid in xy plane with its ends at point $A(x_1, y_1)$ and point $B(x_2, y_2)$. Find force on the wire.



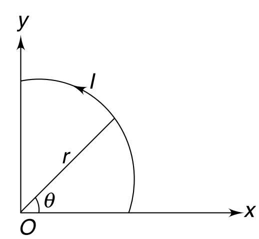
LEVEL 3

Q. 43: A straight current carrying wire has its one end attached to an infinity conducting sheet (shown as a circle in the figure). The other end of the wire goes to infinity and the wire is perpendicular to the sheet. The current spreads uniformly on the surface of the sheet. Calculate the magnitude of magnetic induction field at a point P at a distance d from the straight wire. Current in the wire is I.



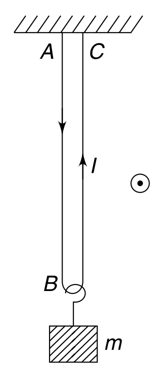
Q. 44: A wire carrying current I is laid in shape of a curve which is represented in plane polar co-ordinate system as

$$r = b + \frac{c}{\pi}\theta \quad \text{for} \quad 0 \le \theta \le \pi/2$$



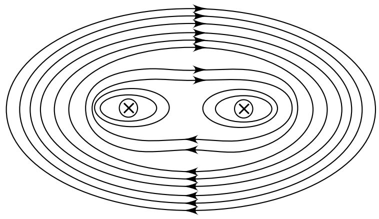
Here b and c are positive constants. θ is the angle measured with respect to positive x direction in anticlockwise sense and r is distance from origin (see figure). Calculate the magnetic field at the origin due to the wire.

Q. 45: A light freely deformable conducting wire with insulation has its two ends (A and C) fixed to the ceiling. The two vertical parts of the wire are close to each other. A load of mass m is attached to the middle of the wire. The entire region has a uniform horizontal magnetic field B directed out of the plane of the figure. Prove that the two parts of the wire take the shape of circular arcs when a current I is passed through the wire. Neglect the magnetic interaction between the two parts of the wire.

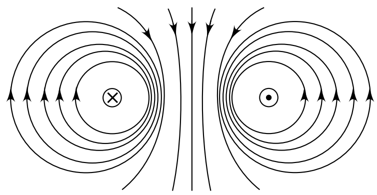


ANSWERS

1. (a)



(b)



2. $\dfrac{\mu_0 I}{2\pi d}\sqrt{\dfrac{5}{4} + \dfrac{1}{\sqrt{2}}}$

3. $\dfrac{\mu_0 I}{\pi a}$ towards midpoint of DE

4. $\sqrt{40}\ \mu$T

5. $1\ \mu$T

6. $\dfrac{r_1}{r_2} = \dfrac{1}{2}$

7. $B = \dfrac{\mu_0 I}{4\pi}\left[\dfrac{1}{x} + \dfrac{1}{y} + \dfrac{x}{y\sqrt{x^2 + y^2}} + \dfrac{y}{x\sqrt{x^2 + y^2}}\right]$

8. $B = 0$

9. $\dfrac{B_0}{\sqrt{2}}$

10. $\dfrac{\sqrt{2}\,\mu_0 I}{\pi^2 R}$

11. (a) 0 (b) $\dfrac{\mu_0 NI}{2\pi x}$

12. $-\dfrac{\mu_0 I}{4}$; No

13. 2 : 1

14. $B = \dfrac{9}{34}\dfrac{\mu_0 I}{\pi R}$

15. Right

16. $1 : 1 : \eta : \eta$ where $\eta = \dfrac{mg - BI\ell}{mg + BI\ell}$

17. (a) Conductor will rotate in Horizontal plane
(b) Conductor will rotate so as to get parallel to fixed wire
(c) Circular conductor will rotate so as to set the current (I_0) parallel to (I_1)

18. $\dfrac{\mu_0 I^2}{2\pi}\ell n\left(\dfrac{2 + \sqrt{3}}{2}\right)\left[\dfrac{\hat{i}}{\sqrt{3}} + \hat{j}\right]$

19. $M = \dfrac{Q\omega R^2}{3}$; $\dfrac{M}{L} = \dfrac{Q}{2m}$

20. (a) $\dfrac{Ia^2 B_0}{mg} + \dfrac{a}{2}$ (b) $I_{max} = \dfrac{mg}{2aB_0}$

21. $\omega = \sqrt{\dfrac{12 I B_0}{m}}$

22. $x_P = \dfrac{d}{\sqrt{2} + 1}$; $x_Q = -\left(\dfrac{d}{\sqrt{2} - 1}\right)$

23. $\theta = \dfrac{\mu_0 I L q}{2\pi m v r}$

24. $\dfrac{1}{2}\mu_0(\mu_1 + \mu_2)\,nI$

25. (i) $\dfrac{\mu_0 NIR^2}{2}\left[\dfrac{1}{(R^2 + x^2)^{3/2}} + \dfrac{1}{[R^2 + (R - x^2]^{3/2}}\right]$
(ii) The magnetic field is constant at points close to mid way between the coils.

26. $\dfrac{\mu_0 I}{4\pi R}\,\ell n(\sqrt{2} + 1)$

27. $\mu_0(I_3 - I_2)$

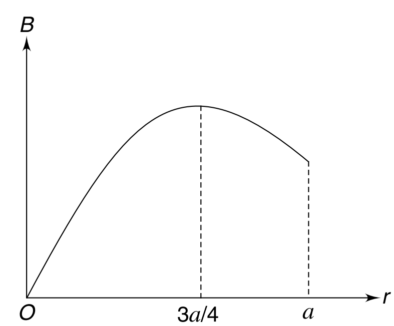
28. $\mu_0 NI$

29. $\dfrac{I_2}{I_1} = 1 + \sqrt{2}$

30. (a) $\dfrac{\theta}{2\pi}\mu_0 I$
(b) Yes, line integral is $\dfrac{\theta}{2\pi}\mu_0 I$

31. (a) $\dfrac{J_0}{3}$

(b)



32. No

33. $E = 0$ for $r < a$

$= \dfrac{a\sigma}{\in_0 r}$ radially inward for $a < r < b$

$= \dfrac{(b - a)\sigma}{\in_0 r}$ radially outward for $r > b$

$B = \sigma\omega(b - a)$ up along the axis for $r < a$

$= b\sigma\omega$ up along the axis for $a < r < b$

$= 0$ for $r > b$

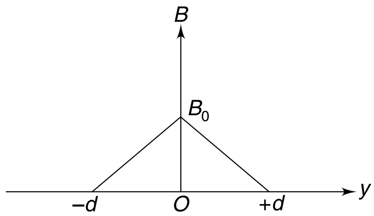
34. (i) $E = \dfrac{\sigma}{2\epsilon_0}$ perpendicularly away from the sheet

$B = \dfrac{\mu_0 \sigma v}{2}$ parallel to the sheet in -ve and +ve y direction

(ii) $E =$ same as in (i); $B = 0$

35. (a) $B_0 = \mu_0 J_0 d$

(b)



36. Zero

37. $\dfrac{\mu_0 U_0}{2\pi\lambda R d}$

38. (a) $I = \dfrac{2Mg}{aB}$ (b) $T = \pi\sqrt{\dfrac{\sqrt{3}a}{g}}$

39. (a) $\tau = IB\ell^2 \sin\theta \cdot \sin\dfrac{\theta}{2}\left[\cos\dfrac{\theta}{2} + \pi\sin\dfrac{\theta}{2}\right]$

(d) No change

40. $I = \dfrac{Mgd}{2\pi R^2 B_0}$

41. $T_{AO} = Mg + \pi IRB_0$

$T_{BO} = Mg - \pi IRB_0$

42. $\vec{F} = \dfrac{kI}{2}\ell n\left(\dfrac{x_2^2 + y_2^2}{x_1^2 + y_1^2}\right)\hat{k}$

43. $\dfrac{\mu_0 I}{2\pi d}$

44. $\dfrac{\mu_0 I}{4c}\ell n\left(1 + \dfrac{c}{2b}\right) \odot$

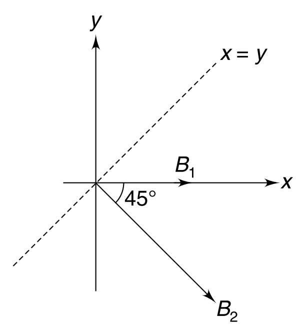
SOLUTIONS

2. Field due to first wire is

$$B_1 = \frac{\mu_0 I}{2\pi d} \text{ along positive } x$$

Field due to second wire is

$$B_2 = \frac{\mu_0 I}{2\pi(2d)} \text{ in the direction shown.}$$



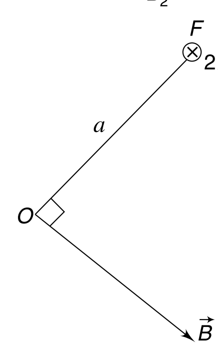
$$\therefore \quad B = \sqrt{B_1^2 + B_2^2 + 2B_1B_2\cos 45^\circ}$$

$$= \frac{\mu_0 I}{2\pi d}\sqrt{1 + \frac{1}{4} + 2\cdot\frac{1}{2}\cdot\frac{1}{\sqrt{2}}} = \frac{\mu_0 I}{2\pi d}\sqrt{\frac{5}{4} + \frac{1}{\sqrt{2}}}$$

3. Imagine a current I at F in the same direction as that of other 5 currents. Such 6 currents will produce zero field at the centre.

Hence field at the centre will be due to a current $2I$ at F directed into the plane of the figure.

$$B = \frac{\mu_0(2I)}{2\pi a} = \frac{\mu_0 I}{\pi a}.$$



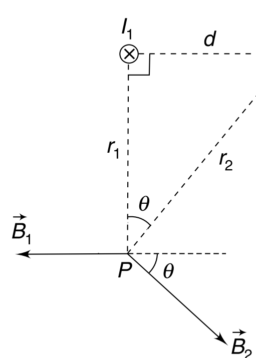
The direction is perpendicular to the line FO. This is a direction towards midpoint of side DE.

4. The situation is as shown in the figure.

Note that $d^2 + r_1^2 = r_2^2$

$$B_1 = \frac{\mu_0 I_1}{2\pi r_1} = \frac{4\pi \times 10^{-7} \times 8}{2\pi \times 0.16} = 10\ \mu\text{T}$$

$$B_2 = \frac{\mu_0 I_2}{2\pi r_2} = \frac{4\pi \times 10^{-7} \times 10}{2\pi \times 0.20} = 10\ \mu\text{T}$$



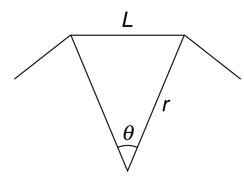
Angle between $\vec{B}_1$ and $\vec{B}_2$ is $180 - \theta$ where $\cos\theta = \dfrac{r_1}{r_2} = \dfrac{0.16}{0.20} = \dfrac{4}{5}$

$\therefore$ Resultant field at P is

$$B = \sqrt{B_1^2 + B_2^2 + 2B_1B_2\cos(180 - \theta)}$$

$$= \sqrt{10^2 + 10^2 - 2 \times 10 \times 10 \times \frac{4}{5}} = \sqrt{40}\ \mu\text{T}$$

5. Since number of sides in the polygon is large we can approximately treat it as a circular loop of radius r given by



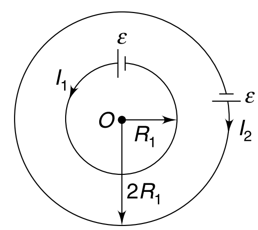
$$r = \frac{L}{\theta} = \frac{1\text{cm}}{\frac{2\pi}{2016}}$$

$$= \frac{1008}{\pi}\text{cm} = \frac{10.08}{\pi}\text{m}$$

$$\therefore \quad B = \frac{\mu_0 I}{2r} = \frac{4\pi \times 10^{-7} \times 5 \times \pi}{2 \times 10.08}$$

$$= 10^{-6}T$$

6. The wires must have current in opposite sense and



$$\frac{\mu_0 I_1}{2R_1} = \frac{\mu_0 I_2}{2\cdot(2R_1)}$$

$$\therefore \quad 2I_1 = I_2 \qquad \text{...(i)}$$

And $$\varepsilon = I_1\frac{\rho(2\pi R_1)}{\pi r_1^2} = I_2\frac{\rho(2\pi 2R_1)}{\pi r_2^2}$$

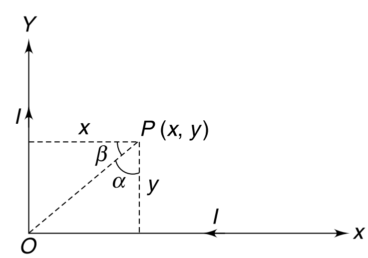
[r_1 and r_2 are radii of cross section]

$$\Rightarrow \quad \frac{r_1^2}{r_2^2} = \frac{1}{2}\frac{I_1}{I_2}$$

$$\Rightarrow \quad \frac{r_1^2}{r_2^2} = \frac{1}{4} \qquad \text{[using (i)]}$$

$$\Rightarrow \quad \frac{r_1}{r_2} = \frac{1}{2}$$

7. Field at P due to wire along x axis is



$$B_1 = \frac{\mu_0 I}{4\pi y}[\sin\alpha + 1] \otimes$$

$$= \frac{\mu_0 I}{4\pi y}\left[\frac{x}{\sqrt{x^2 + y^2}} + 1\right] \otimes$$

Field due to wire along y axis is

$$B_2 = \frac{\mu_0 I}{4\pi x}[\sin\beta + 1]$$

$$= \frac{\mu_0 I}{4\pi x}\left[\frac{y}{\sqrt{x^2 + y^2}} + 1\right] \otimes$$

Resultant field at P

$$B = B_1 + B_2$$

$$= \frac{\mu_0 I}{4\pi}\left[\frac{1}{x} + \frac{1}{y} + \frac{x}{y\sqrt{x^2 + y^2}} + \frac{y}{x\sqrt{x^2 + y^2}}\right] \otimes$$

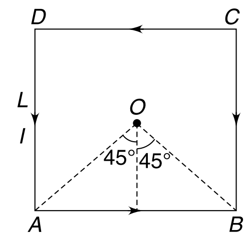
8. Point B and C will have equal potential and there will be no current in arc BC. Current I gets divided into two equal parts at A.

$$\therefore \quad B_{AB} = \frac{\mu_0 (I/2)}{2a} \otimes$$

$$B_{AC} = \frac{\mu_0 (I/2)}{2a} \odot$$

$$\therefore \quad B_{\text{centre}} = 0$$

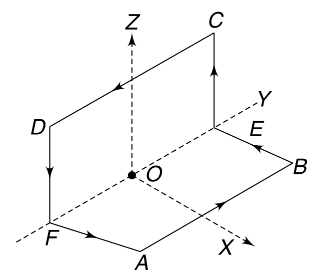
9. $$B_0 = \frac{4\mu_0 I}{4\pi \frac{L}{2}} (\sin 45° + \sin 45°) = \frac{2\sqrt{2}\mu_0 I}{\pi L}$$



After folding

$$\vec{B}_{AB} = \frac{\mu_0 I}{4\pi \frac{L}{2}} (\sin 45° + \sin 45°)\, \hat{k} = \frac{2\sqrt{2}}{4\pi} \frac{\mu_0 I}{L} \hat{k}$$

$$\vec{B}_{BE} = \frac{\mu_0 I}{4\pi \frac{L}{2}} (\sin 0° + \sin 45°)\, \hat{k} = \frac{\sqrt{2}\,\mu_0 I}{4\pi L} \hat{k}$$



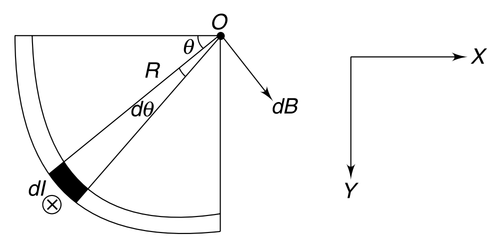
Similarly we have: $$\vec{B}_{EC} = \frac{\sqrt{2}\,\mu_0 I}{4\pi L} \hat{i}$$

$$\vec{B}_{CD} = \frac{2\sqrt{2}}{4\pi} \frac{\mu_0 I}{L} \hat{i}; \; \vec{B}_{DF} = \frac{\sqrt{2}\,\mu_0 I}{4\pi L} \hat{i}; \; \vec{B}_{FA} = \frac{\sqrt{2}\,\mu_0 I}{4\pi L} \hat{k}$$

$$\therefore \quad \vec{B} = \frac{\sqrt{2}\,\mu_0 I}{\pi L} \hat{i} + \frac{\sqrt{2}\,\mu_0 I}{\pi L} \hat{k}$$

$$\therefore \quad |\vec{B}| = \frac{2\mu_0 I}{\pi L} = \frac{B_0}{\sqrt{2}}$$

10. Consider a long wire element as shown. The figure shows the cross section of the wire.



Current through the wire element is $dI = \frac{I}{\frac{\pi}{2}} d\theta = \frac{2I}{\pi} d\theta$

Magnetic field due to the current dI at O is

$$dB = \frac{\mu_0 dI}{2\pi R}$$

$$= \frac{\mu_0 \frac{2I}{\pi} d\theta}{2\pi R} = \frac{\mu_0 I}{\pi^2 R} d\theta$$

x component of the field is

$$dB_x = \frac{\mu_0 I}{\pi^2 R} \sin\theta d\theta$$

$$B_x = \int dB_x = \frac{\mu_0 I}{\pi^2 R} \int_0^{\pi/2} \sin\theta d\theta = \frac{\mu_0 I}{\pi^2 R}$$

From symmetry $B_y = B_x$

$\therefore$ Resultant B at O is

$$B = \sqrt{B_x^2 + B_y^2} = \frac{\sqrt{2}\,\mu_0 I}{\pi^2 R}$$

11. (a) Consider a circle of radius $x < R$ with its axis along the axis of the cylinder. Assuming this circle to be Amperian loop we can write–

$$\oint \vec{B}\cdot\vec{dl} = \mu_0 I_{\text{enclosed}} = 0$$

$$\Rightarrow \quad B = 0$$

(b)

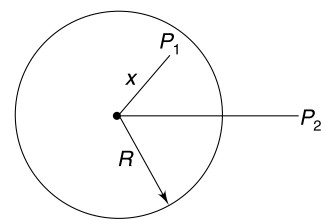
$$\oint \vec{B}\cdot\vec{dl} = \mu_0 NI$$

$$\Rightarrow \quad B2\pi x = \mu_0 NI$$

$$\Rightarrow \quad B = \frac{\mu_0 NI}{2\pi x}$$

13. Field at a point P_1 ($x < R$) is

$$B = \frac{\mu_0 j}{2}x$$



Slope of 1st graph is higher

$$\therefore \quad j_1 > j_2.$$

Also $\quad R_2 = 2R_1$

Field at a point P_2 ($x > R$) is

$$B = \frac{\mu_0 I}{2\pi x} \qquad [I = \text{current in the wire}]$$

Since the two graphs are overlapping

$$\therefore \quad I_1 = I_2$$

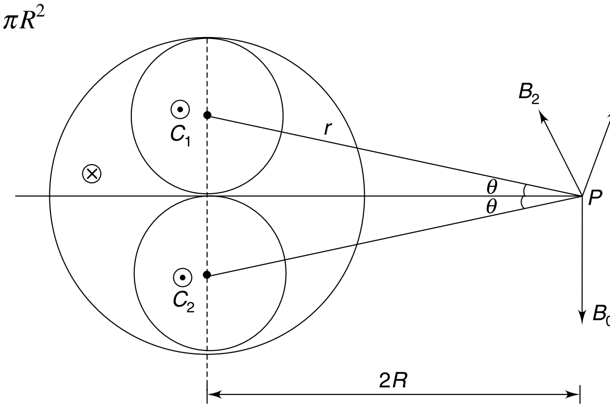
Field on the surface of two wires are–

$$B_1 = \frac{\mu_0 I_1}{2\pi R_1} \quad \text{and} \quad B_2 = \frac{\mu_0 I_2}{2\pi R_2}$$

$$\therefore \quad \frac{B_1}{B_2} = \frac{R_2}{R_1} = \frac{2}{1}$$

14. Current density

$$J = \frac{I}{A} = \frac{I}{\pi\left[R^2 - \frac{R^2}{4} - \frac{R^2}{4}\right]} = \frac{2I}{\pi R^2}$$



Let B_0 = field at P, had there been no cavity and B_1 and B_2 be field at P due to conductors of radius $\frac{R}{2}$ occupying the cavity space and carrying current in direction opposite to the given current.

Field at P will be vector sum of B_0, B_1 and B_2

$$B_0 = \frac{\mu_0(\pi R^2 J)}{2\pi(2R)} = \frac{\mu_0}{4} JR \text{ In the direction shown.}$$

$$B_1 = \frac{\mu_0\left(\pi\frac{R^2}{4}J\right)}{2\pi r} = \frac{\mu_0 R^2 J}{8r} \text{ perpendicular to } C_1P$$

$$B_2 = \frac{\mu_0 R^2 J}{8r} \text{ perpendicular to } C_2P$$

Resultant field at P is

$$B = B_0 - (B_1 + B_2)\cos\theta$$

$$= \frac{\mu_0 JR}{4} - \frac{\mu_0 R^2 J}{4r}\cdot\frac{2R}{r}$$

Put $$r = \sqrt{4R^2 + \frac{R^2}{4}} = \frac{\sqrt{17}}{2} R$$

$\therefore$ $$B = \frac{\mu_0 JR}{4} - \frac{\mu_0 JR^3}{2 \cdot \frac{17}{4} R^2}$$

$\Rightarrow$ $$B = \frac{\mu_0 JR}{4} - \frac{2}{17} \mu_0 JR = \frac{9}{68} \mu_0 JR$$

$\therefore$ $$B = \frac{9}{68} \mu_0 R \frac{2I}{\pi R^2} = \frac{9}{34} \frac{\mu_0 I}{\pi R}$$

15. Let the direction of uniform B be vertically down. The current in the horizontal wire must be towards you. This current will create a field in upward direction to the right of the wire and resultant field will get reduced to the right of the wire. In fact it will be zero at same point. This is what is depicted in the figure.

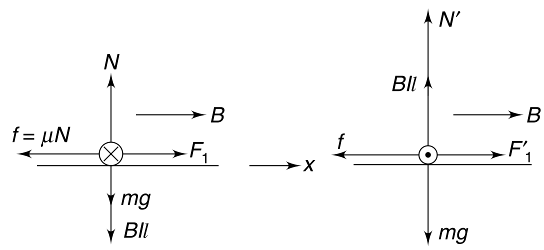
Force on the wire is along $I\vec{L} \times \vec{B}$ which is to right.

16. When current is along positive y, the direction of magnetic force is vertically down.

Hence normal reaction on the wire is

$$N = mg + BI\ell$$
$$F_1 = \mu N = \mu(mg + BI\ell)$$

when we pull the wire in y direction, the friction force will be in negative y direction but its maximum value will remain same as there is no change in N.



$\therefore$ $$F_1 = F_2$$

With direction of current reversed the magnetic force becomes vertically up

$\therefore$ $$N' = mg - BI\ell$$

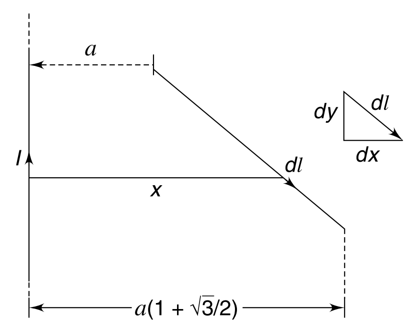
$\therefore$ $$F_1' = F_2' = \mu N' = \mu(mg - BI\ell)$$

$\therefore$ $$\frac{F_1'}{F_1} = \frac{mg - BI\ell}{mg + BI\ell} = \eta \text{ (say)}$$

$\therefore$ $$F_1 : F_2 : F_1' : F_2' = 1 : 1 : \eta : \eta$$

18. An infinitesimal length $d\ell$ on wire AB can be expressed as $\overrightarrow{d\ell} = \hat{i}\,dx - \hat{j}\,dy$

Where $$dy = (\tan 30°)\ dx = \frac{dx}{\sqrt{3}}$$



Field due to long wire at a distance x is

$$\vec{B} = \frac{\mu_0 I}{2\pi x} (-\hat{k})$$

Force on length $d\ell$ of wire AB is

$$\overrightarrow{dF} = I\overrightarrow{d\ell} \times \vec{B} = I\left(\hat{i}\,dx - \hat{j}\frac{dx}{\sqrt{3}}\right) \times \frac{\mu_0 \mathrm{I}}{2\pi x} (-\hat{k})$$
$$= \frac{\mu_0 I^2}{2\pi}\left[\hat{i}\frac{dx}{\sqrt{3} \cdot x} + \hat{j}\frac{dx}{x}\right]$$

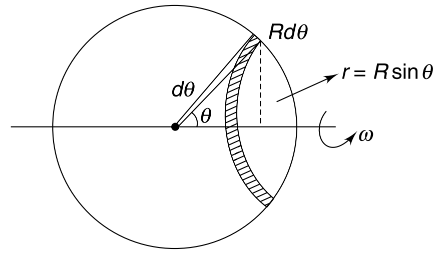
$\therefore$ Force on AB is

$$\vec{F} = \frac{\mu_0 I^2}{2\pi}\left[\frac{\hat{i}}{\sqrt{3}} \int_a^{a\left(1 + \frac{\sqrt{3}}{2}\right)} \frac{dx}{x} + \hat{j} \int_a^{a\left(1 + \frac{\sqrt{3}}{2}\right)} \frac{dx}{x}\right]$$

$$= \frac{\mu_0 I^2}{2\pi} \ell n\left(\frac{2 + \sqrt{3}}{2}\right)\left(\frac{1}{\sqrt{3}}\hat{i} + \hat{j}\right)$$

19. The shell can be throught to be made up of a number of co-axial rings. We will consider one such ring as shown in figure.

Surface charge density $\sigma = \dfrac{Q}{4\pi R^2}$



Charge on ring element $\quad dq = \sigma(2\pi r)(Rd\theta)$

$$= \frac{Q}{2}\sin\theta d\theta$$

This charge rotates with frequency $f = \dfrac{\omega}{2\pi}$

$\therefore$ Current associated with this charge is

$$dI = (dq)\,f = \frac{Q\omega}{4\pi}\sin\theta d\theta$$

Magnetic dipole moment of the elemental ring is

$$dM = (dI)(\pi r^2) = \left(\frac{Q\omega}{4\pi}\sin\theta d\theta\right)(\pi R^2\sin^2\theta)$$

$$\therefore \qquad M = \frac{Q\omega R^2}{4}\int_0^{\pi}\sin^3\theta d\theta$$

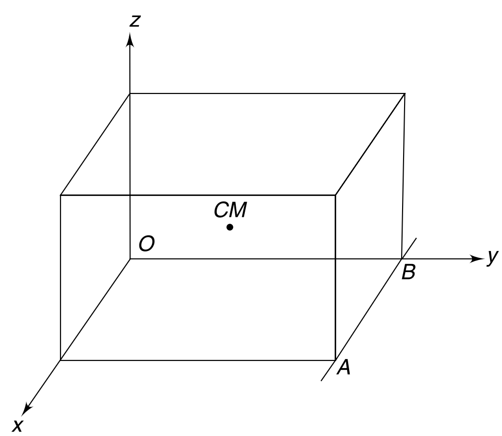
$$= \frac{Q\omega R^2}{4\times 4}\left[\int_0^{\pi} 3\sin\theta d\theta - \int_0^{\pi}\sin 3\theta d\theta\right]$$

$$= \frac{Q\omega R^2}{16}\left[3(2) + \frac{1}{3}(-2)\right]$$

$$M = \frac{Q\omega R^2}{3} \quad \text{and} \quad L = \frac{2}{3}mR^2\omega$$

$$\therefore \qquad \frac{M}{L} = \frac{Q}{2m}$$

20. Magnetic dipole moment of the wire loop is



$$\vec{\mu} = a^2 I\hat{k}$$

Magnetic torque on the loop

$$\vec{\tau} = \vec{\mu}\times\vec{B} = (a^2 I\hat{k})\times(B_0\hat{j})$$

$$= -a^2 I B_0\hat{i}$$

For rotational equilibrium, taking torque about COM we get–

$a^2 IB_0 = Nx$ [$x =$ distance of normal force from centre]

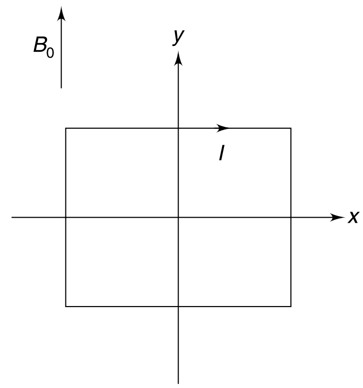
$$\Rightarrow \qquad a^2 IB_0 = mgx \quad\Rightarrow\quad x = \frac{a^2 IB_0}{mg}$$

(a) $\therefore$ Distance of N from x axis $= \dfrac{a^2 IB_0}{mg} + \dfrac{a}{2}$

(b) Block will not topple if $x < \dfrac{a}{2}$

$$\Rightarrow \qquad \frac{a^2 IB_0}{mg} < \frac{a}{2} \quad\Rightarrow\quad I < \frac{mg}{2aB_0}$$

21. $$\vec{\mu}_1 = a^2 I(-\hat{k})$$



Potential energy of the loop

$$U_1 = -\vec{\mu}_1 \cdot \vec{B} = -a^2 I(-\hat{k}) \cdot B_0(\hat{j}) = 0$$

After the loop rotates through 90°

$$\vec{\mu}_2 = a^2 I(\hat{j})$$

$$U_2 = -\vec{\mu}_2 \cdot \vec{B} = -a^2 I B_0$$

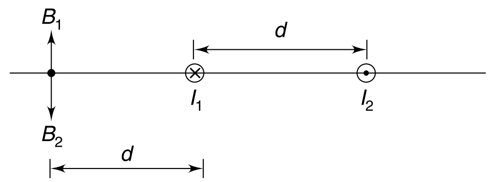
$\therefore$ Loss in $PE = a^2 I B_0$

$\therefore$ Gain in $KE = a^2 I B_0$

$$\frac{1}{2} I_x \omega^2 = a^2 I B_0$$

$$\Rightarrow \quad \frac{1}{2} \frac{ma^2}{6} \omega^2 = a^2 I B_0 \quad \Rightarrow \quad \omega = \sqrt{\frac{12 I B_0}{m}}$$

22. The two wires have current in opposite directions as shown in the figure. Let the currents be I_1 and I_2 Field at R is zero.



$$B_1 = B_2 \text{ at } R$$

$$\Rightarrow \quad \frac{\mu_0 I_1}{2\pi d} = \frac{\mu_0 I_2}{2\pi(2d)} \quad \Rightarrow \quad I_2 = 2I_1$$

At point P field is minimum. Field at a point having coordinate $0 < x < d$ is

$$B = \frac{\mu_0 I_1}{2\pi x} + \frac{\mu_0 I_2}{2\pi(d - x)} \text{ in negative } y \text{ direction}$$

$$= \frac{\mu_0 I_1}{2\pi}\left[\frac{1}{x} + \frac{2}{d - x}\right]$$

B is minimum if $\frac{dB}{dx} = 0$

$$\Rightarrow \quad -\frac{1}{x^2} + \frac{2}{(d - x)^2} = 0$$

$$\Rightarrow \quad \frac{x}{d - x} = \pm \frac{1}{\sqrt{2}}$$

Taking positive sign (as ratio of two positive numbers cannot be negative)

$$\sqrt{2}x = d - x \quad \Rightarrow \quad x = \frac{d}{\sqrt{2} + 1}$$

At point Q, the field is locally maximum. Let distance of point Q from origin be x. Field at such a point is

$$B = \frac{\mu_0 I_2}{2\pi(d + x)} - \frac{\mu_0 I_1}{2\pi x} \text{ in negative } y \text{ direction}$$

$$= \frac{\mu_0 I_1}{2\pi}\left[\frac{2}{d + x} - \frac{1}{x}\right]$$

This is maximum if

$$\frac{dB}{dx} = 0$$

$$\Rightarrow \quad -\frac{2}{(d + x)^2} + \frac{1}{x^2} = 0$$

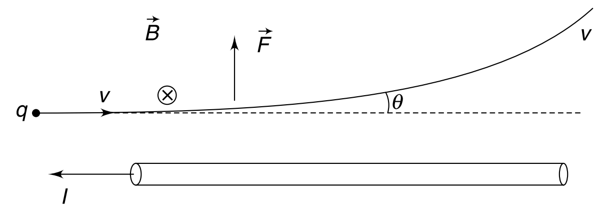
$$\Rightarrow \quad \frac{d + x}{x} = \sqrt{2}$$

$$\Rightarrow \quad d + x = \sqrt{2}x$$

$\Rightarrow$ $$x = \frac{d}{\sqrt{2}-1}$$

$\therefore$ Co-ordinate of point Q is $-\left(\frac{d}{\sqrt{2}-1}\right)$

23. $$B = \frac{\mu_0 I}{2\pi r} \otimes$$

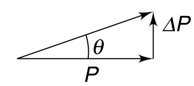


Force on the particle is $$F = qvB = \frac{\mu_0 Iqv}{2\pi r} (\uparrow)$$

This force is always perpendicular to the velocity. Since deflection is small, the force is nearly in ($\uparrow$) direction always.

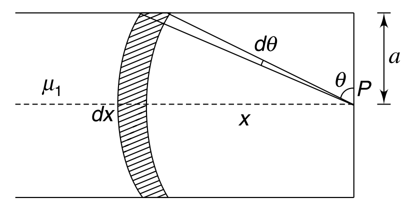
Impulse is $$F\Delta t = \frac{\mu_0 Iqv\Delta t}{2\pi r} = \frac{\mu_0 IqL}{2\pi r} \qquad [\because \; v\Delta t = L]$$

$\therefore$ $$\Delta P = \frac{\mu_0 IqL}{2\pi r}(\uparrow)$$



$$\theta = \frac{\Delta P}{P} = \frac{\mu_0 IqL}{2\pi rmV}$$

24. Let's calculate the field due to one half by way of integration. (This is just to demonstrate ; otherwise the result is obviously half the field due to a solenoid extending to large length on both sides.)



Consider a rig element of angular width $d\theta$ as shown in the figure.

$$x = a\tan\theta$$

$\therefore$ $$dx = a\sec^2\theta d\theta$$

Number of turns in width dx is ndx

Field due to ring at P is along the axis given by

$$dB = \frac{\mu_0\mu_1 ndx Ia^2}{2(a^2+x^2)^{3/2}} = \frac{\mu_0\mu_1 nIa^2}{2}\frac{(a\sec^2\theta d\theta)}{(a\sec\theta)^3} \qquad [\because \; \sqrt{a^2+x^2} = a\sec\theta]$$

$\therefore$ $$dB = \frac{\mu_0\mu_1 nI}{2}\cos\theta d\theta$$

$\therefore$ $$B_1 = \frac{\mu_0\mu_1 nI}{2}\int_0^{\pi/2}\cos\theta d\theta = \frac{1}{2}\mu_0\mu_1 nI$$

Similarly, field due to second half will be

$$B_2 = \frac{1}{2}\mu_0\mu_2 nI$$

Resultant field at P is

$$B = B_1 + B_2 = \frac{1}{2}(\mu_1+\mu_2)\mu_0 nI$$

25. (i) $B = \frac{\mu_0 NIR^2}{2}\left[\frac{1}{(R^2+x^2)^{3/2}} + \frac{1}{(R^2+(R-x^2)^{3/2}}\right]$

[Note that fields due to both the coils are in same direction]

(ii) $\frac{dB}{dx} = \frac{\mu_0 NIR^2}{2}\left[-\frac{3}{2}\frac{2x}{(R^2+x^2)^{5/2}} - \frac{3}{2}\frac{2(R-x)(-1)}{[R^2+\;(R-x^2)^{5/2}}\right]$

$$= \frac{\mu_0 NIR^2}{2}\left[-\frac{3x}{(R^2+x^2)^{5/2}} + \frac{3(R-x)}{[R^2+(R-x)^2]^{5/2}}\right]$$

At $\quad x = \frac{R}{2};\ \frac{dB}{dx} = 0$

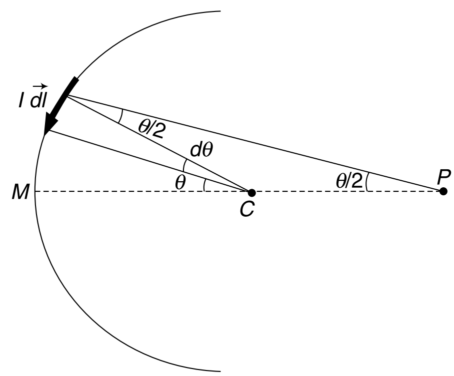
$$\frac{d^2B}{dx^2} = \frac{3\mu_0 NIR^2}{2}$$

$$\left[-\frac{(R^2+x^2)^{5/2} - \frac{5x}{2}(R^2+x^2)^{3/2}(2x)}{(R^2+x^2)^5} + \frac{[R^2+(R-x)^2]^{5/2}(-3)}{[R^2+(R-x)^2]^5} - \frac{3(R-x)\frac{5}{2}[R^2+(R-x)^2]^{3/2}\,2(R-x)(-1)}{[R^2+(R-x)^2]^5}\right]$$

By putting $x = \frac{R}{2}$, one can show that $\frac{d^2B}{dx^2} = 0$

$\frac{dB}{dx} = \frac{d^2B}{dx^2} = 0$ at $x = \frac{R}{2}$ implies that B is constant for small variations in x

26. Point P lies on the circumference of the circle. Consider an element subtending angle $d\theta$ at the centre of the circle. Field at P due to this element is given by Biot – Savart law.



$$dB = \frac{\mu_0 Id\ell}{4\pi}\frac{\sin\left(90+\frac{\theta}{2}\right)}{r^2}$$

Where $\quad d\ell = Rd\theta \quad$ and $\quad r = 2R\cos\left(\frac{\theta}{2}\right)$

$\therefore$ $$dB = \frac{\mu_0 IRd\theta}{4\pi}\frac{\cos\left(\frac{\theta}{2}\right)}{\left[2R\cos\left(\frac{\theta}{2}\right)\right]^2}$$

$$= \frac{\mu_0 I}{16\pi R}\sec\left(\frac{\theta}{2}\right)d\theta$$

Direction of dB is $\odot$

Note: If you consider an identical element in the other quadrant, field at P due to the element is -

$$dB = \frac{\mu_0 IRd\theta}{4\pi}\frac{\sin\left(90-\frac{\theta}{2}\right)}{\left[2R\cos\left(\frac{\theta}{2}\right)\right]^2} = \frac{\mu_0 I}{16\pi R}\sec\left(\frac{\theta}{2}\right)d\theta \odot$$

All elements contribute in same direction and resultant field is

$$B = 2\times\frac{\mu_0 I}{16\pi R}\int_0^{\pi/2}\sec\left(\frac{\theta}{2}\right)d\theta$$

$$= \frac{\mu_0 I}{4\pi R}\left[\ell n\left(\sec\frac{\theta}{2}+\tan\frac{\theta}{2}\right)\right]_0^{\pi/2}$$

$$= \frac{\mu_0 I}{4\pi R}\left[\ell n(\sqrt{2}+1) - \ell n(1+0))\right]$$

$$= \frac{\mu_0 I}{4\pi R}\ell n(\sqrt{2}+1)$$

27.
$$\oint_{ABEFA} \vec{B}\cdot\vec{dl} = -\mu_0(I_1 + I_2)$$

And
$$\oint_{BCDEB} \vec{B}\cdot\vec{dl} = \mu_0(I_1 + I_3)$$

$\therefore$
$$\oint_{ABCDEFA} \vec{B}\cdot\vec{dl} = \oint_{ABEFA} \vec{B}\cdot\vec{dl} + \oint_{BCDEB} \vec{B}\cdot\vec{dl}$$
$$= \mu_0(I_3 - I_2)$$

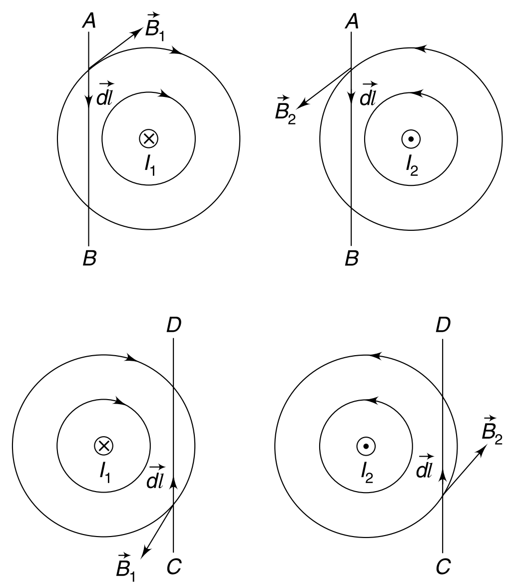
28. Think of a closed rectangular path with its one side of infinite length along the axis of the loop and other parallel side at ∞ distance from the axis. Apart from the side along the axis, the integral $\int \vec{B}\cdot\vec{dl}$ along all three sides will be zero since $B = 0$

Using Ampere's law
$$\oint \vec{B}\cdot\vec{dl} = \mu_0 NI$$
$$\Rightarrow \quad \int_{-\infty}^{\infty} B\cdot dx = \mu_0 NI$$

29. Current in $W2$ must be outward.

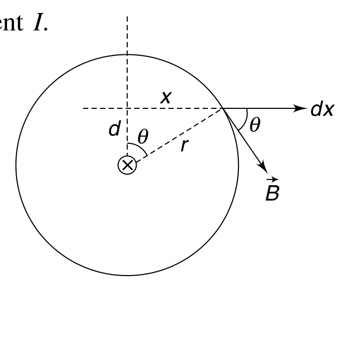
In the field of I_1 we have $\int_A^B \vec{B}_1 \vec{dl} = -a_0$

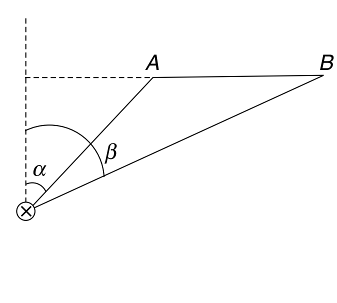
In the field of I_2 we have $\int_A^B \vec{B}_2 \vec{dl} = +a_0$



For CD
$$\int_C^D \vec{B}_1 \vec{dl} = -a\left(\frac{I_1}{I_2}\right)$$
$$\int_C^D \vec{B}_2 \vec{dl} = a_0\left(\frac{I_2}{I_1}\right)$$
$$\therefore \quad \int_C^D \vec{B}\cdot\vec{dl} = a_0\left(\frac{I_2}{I_1} - \frac{I_1}{I_2}\right)$$
$$\Rightarrow \quad 2a_0 = a_0\left(\eta - \frac{1}{\eta}\right) \quad \left[\text{Let } \left(\frac{I_2}{I_1} = \eta\right)\right]$$
$$\Rightarrow \quad \eta^2 - 2\eta - 1 = 0$$
$$\therefore \quad \eta = \frac{2 \pm \sqrt{4 + 4}}{2}$$
$$\therefore \quad \eta = 1 + \sqrt{2} \quad [\eta \text{ cannot be negative}]$$
$$\therefore \quad \frac{I_2}{I_1} = 1 + \sqrt{2}$$

30. (a) The figure shows a circular field line due to current I.





$$B = \frac{\mu_0 I}{2\pi r} = \frac{\mu_0 I}{2\pi d \sec\theta}$$

And
$$x = d\tan\theta$$

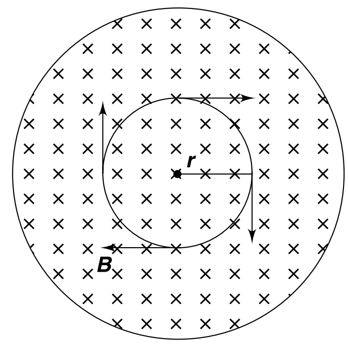
$\therefore$
$$dx = \sec^2\theta d\theta$$
$$\int_A^B \vec{B}\cdot\vec{dx} = \int_A^B B\cdot dx \cos\theta$$

$$= \int_A^B \left(\frac{\mu_0 I}{2\pi r d \sec\theta}\right)(d\sec^2\theta d\theta)\cos\theta$$

$$= \frac{\mu_0 I}{2\pi}\int_\alpha^\beta d\theta = \frac{\mu_0 I}{2\pi}(\beta - \alpha)$$

$$= \frac{\mu_0 I}{2\pi}\theta$$

31. (a) Current $\quad I = \int_0^a (2\pi r dr)\, J = 2\pi J_0 \int_0^a \left(r - \frac{r^2}{a}\right) dr$



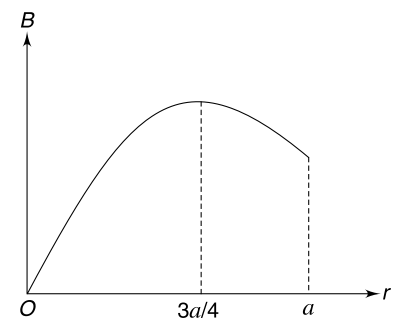
$\Rightarrow \quad I = \frac{1}{3}\pi a^2 J_0$

Mean current density is $\quad \bar{J} = \frac{I}{\pi a^2} = \frac{J_0}{3}$

(b) Current enclosed within distance r from the axis is

$$I_r = \int_0^r (2\pi r dr) J = 2\pi J_0 \int_0^r \left(r - \frac{r^2}{a}\right) dr$$

$$= \pi J_0 r^2 \left(1 - \frac{2r}{3a}\right)$$



Using ampere's law on a circle of radius r gives

$$\oint \vec{B}\cdot\vec{dl} = \mu_0 I$$

$$B\cdot 2\pi r = \mu_0 I_r$$

$$B = \frac{\mu_0 J_0}{2} r\left(1 - \frac{2r}{3a}\right)$$

At $r = 0$; $B = 0$

At $r = a$; $B = \frac{\mu_0 J_0 a}{6}$

The graph is parabolic with maxima at distance given by

$$\frac{dB}{dr} = 0 \quad \Rightarrow \quad 1 - \frac{4r}{3a} = 0$$

$\Rightarrow \quad r = \frac{3a}{4}$

32. If the wire is of finite length there must be a return path for the current which will cause the value and direction of $\vec{B}$ to change from point to point on C. In the expression $\oint \vec{B}\cdot\vec{dl} = \mu_0 I$; $\vec{B}$ is net field at a point produced by all sources.

33. Electric field will be calculated using Gauss' law. We need to consider a co – axial cylindrical Gaussian surface.

For $r < a$; $E = 0$

For $a < r < b$; $\quad E = \frac{-\lambda}{2\pi\epsilon_0 r} = \frac{-2\pi a\sigma}{2\pi\epsilon_0 r}$

$$= \frac{a\sigma}{\epsilon_0 r} \text{ (radially inward)}$$

For $r > b$; $\quad E = \frac{-2\pi a\sigma + 2\pi b\sigma}{2\pi\epsilon_0 r}$

$$E = \frac{(b-a)\sigma}{\epsilon_0 r} \text{ radially outward}$$

For magnetic field, we can consider the two cylinders just like ideal solenoids.

Field inside an ideal solenoid = $\mu_0 ni$ (parallel to the axis)

For inner cylinder $ni = (2\pi a)(1)(\sigma)\left(\frac{\omega}{2\pi}\right) = a\sigma\omega$

$\therefore$ For $r < a$

$$B = b\sigma\omega - a\sigma\omega = \sigma\omega(b-a)\ (\uparrow)$$ [sense of current is opposite in two solenoids]

For $a < r < b$

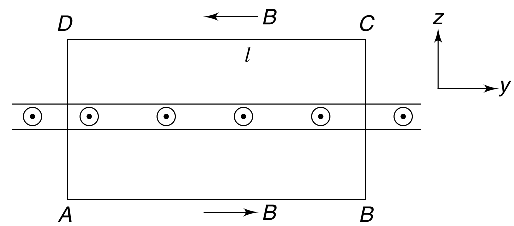
$$B = b\sigma\omega\ (\uparrow)$$ [There is no field due to a solenoid at an outside point]

For $r > b$

$$B = 0$$

34. (i) The $\overline{E}$ has got nothing to do with the motion of the sheet.

$\therefore$ $E = \dfrac{\sigma}{2\epsilon_0}$ perpendicular to the sheet away from it.



The motion of the sheet in x direction creates a current in x direction.

The symmetry (with right hand rule) shows that $\vec{B}$ will be in negative y direction above the sheet and in positive y direction below the sheet. Consider rectangular Amperian loop $ABCD$ as shown in the figure. AB and CD are equidistant from the surface.

$$\oint_{ABCD} \vec{B}\cdot\vec{dl} = 2B\ell$$

Current through the loop $I = \sigma\ell v$

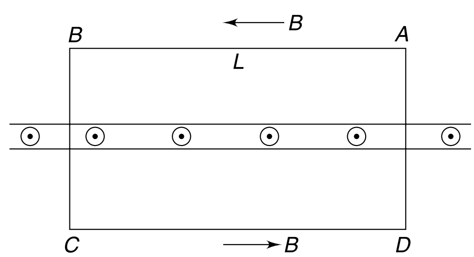
$\therefore$ Ampere's law gives

$$2B\ell = \mu_0 I$$

$$2B\ell = \mu_0\sigma\ell v \qquad \therefore \quad B = \frac{\mu_0\sigma v}{2}$$

(ii) E will not change. There will be no B.

35. (a) Due to symmetry the field at all point will be along $\pm x$ direction only.



Let's first find field due to one slab only. Consider an Amperian loop in shape of a rectangle as shown in the figure. Line AB and CD are located symmetrically wrt the slab. Field at AB and CD are of equal strength but in opposite directions.

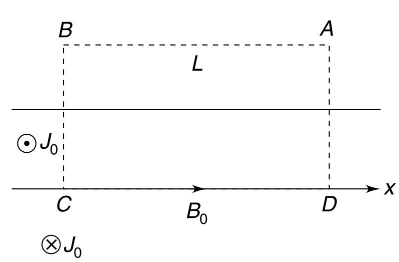
$$\oint \vec{B}\cdot\vec{dl} = \int_A^B \vec{B}\cdot\vec{dl} + \int_B^C \vec{B}\cdot\vec{dl} + \int_C^D \vec{B}\cdot\vec{dl} + \int_D^A \vec{B}\cdot\vec{dl}$$

$$= BL + 0 + BL + + 0 = 2BL$$

Ampere's law gives

$$2BL = \mu_0 J_0 dL$$

$$\Rightarrow \qquad B = \frac{\mu_0 J_0 d}{2}$$



This is independent of distance of line AB from the slab.

It is obvious that $B = 0$ due to both the slabs at all points outside the slabs. Now consider a rectangular loop $ABCD$ with its side CD along x axis as shown.

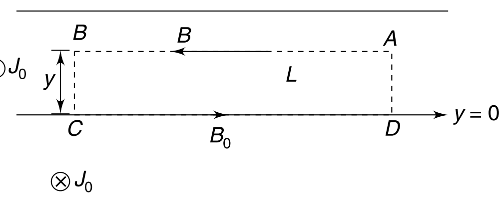
$$\oint_{ABCD} \vec{B}\cdot\vec{dl} = \int_A^B \vec{B}\cdot\vec{dl} + \int_B^C \vec{B}\cdot\vec{dl} + \int_C^D \vec{B}\cdot\vec{dl} + \int_D^A \vec{B}\cdot\vec{dl}$$

$$= 0 + 0 + B_0L + 0 = B_0L$$

Where B_0 is field at $y = 0$

Using Ampere's law gives

$$B_0L = \mu_0(J_0 dL) \Rightarrow B_0 = \mu_0 J_0 d$$



(b) Once again consider an Amperian loop as shown in figure. Let the field be B in negative x direction at line AB.

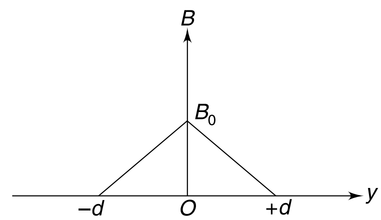
$$\oint_{ABCD} \vec{B}\cdot\vec{dl} = BL + 0 + B_0L + 0 = BL + B_0L$$

Using Ampere's law

$$BL + B_0L = \mu_0(J_0 yL)$$

$$\Rightarrow \quad B + \mu_0 J_0 d = \mu_0 J_0 y$$

$$\Rightarrow \quad B = -\mu_0 J_0 (d - y)$$

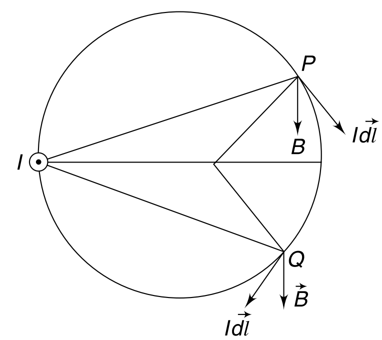


Negative sign indicates that field is in positive x direction. Similarly, field at points

$$-d < y < 0 \text{ will be}$$

$$B = \mu_0 J_0 (d + y) \text{ in positive } x \text{ direction}$$

36. Consider two symmetrically located elements ($I\vec{dl}$) on the arc at points P and Q. The direction of $I\vec{dl} \times \vec{B}$ at these two locations is opposite Hence, the force on the arc will be zero.



37. We will assume that the capacitor discharges quickly and there is no appreciable displacement of the wires in that interval.

Let initial charge on the capacitor be Q_0. Current at time t is

$$I = \frac{Q_0}{RC} e^{-t/RC}$$

Force between two parallel wires per unit length is

$$F = \frac{\mu_0 I^2}{2\pi d} = \frac{\mu_0 Q_0^2}{2\pi d R^2 C^2} e^{-2t/RC}$$

$$\Rightarrow \quad \lambda dv = \frac{\mu_0 Q_0^2}{2\pi d R^2 C^2} e^{-2t/RC}\, dt$$

$$\Rightarrow \quad \int_0^v dv = \frac{\mu_0 Q_0^2}{2\pi\lambda d R^2 C^2} \int_0^\infty e^{-2t/RC}\, dt$$

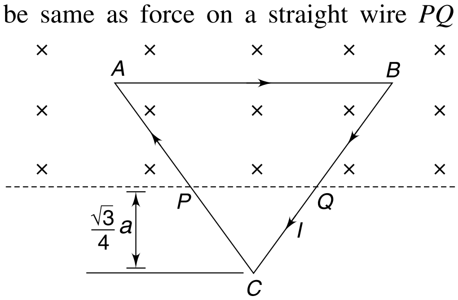
$$\Rightarrow \quad v = \frac{\mu_0 Q_0^2}{4\pi\lambda RCd} = \frac{\mu_0 U_0}{2\pi\lambda Rd} \qquad \left[\because \; U_0 = \frac{Q_0^2}{2C}\right]$$

38. (a) The current in the loop is clockwise. Magnetic force on the loop will be same as force on a straight wire PQ (current from P to Q)

Length $\quad PQ = 2\cdot\frac{\sqrt{3}}{4} a \tan 30° = \frac{a}{2}$

For equilibrium $\quad I\left(\frac{a}{2}\right) B = Mg$

$$\Rightarrow \quad I = \frac{2Mg}{aB} \qquad \text{...(i)}$$



(b) Assume that the loop is displaced downward by a small distance y.

$$M\frac{d^2y}{dt^2} = Mg - I\left[2\left(\frac{\sqrt{3}}{4}a + y\right)\tan 30°\right]B$$

$$\Rightarrow \quad M\frac{d^2y}{dt^2} = Mg - I\frac{a}{2}B - \frac{2IB}{\sqrt{3}}y$$

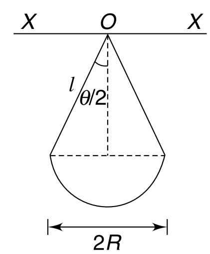
$$\Rightarrow \quad \frac{d^2y}{dt^2} = -\frac{2IB}{\sqrt{3}\,M}y \qquad \text{[using (i)]}$$

This is equation of SHM

$$\omega = \sqrt{\frac{2IB}{\sqrt{3}\,M}}$$

$$\Rightarrow \quad T = 2\pi\sqrt{\frac{\sqrt{3}\,M}{2IB}} = \pi\sqrt{\frac{\sqrt{3}\,a}{g}} \qquad \text{[using (i)]}$$

39. Area of the loop



$$A = \frac{1}{2}(2R)\left(\ell\cos\frac{\theta}{2}\right) + \pi R^2$$

Where $\quad R = \ell\sin\frac{\theta}{2}$

$$\therefore \quad A = \ell^2\sin\frac{\theta}{2}\cos\frac{\theta}{2} + \pi\ell^2\sin^2\frac{\theta}{2}$$

$$= \ell^2\sin\frac{\theta}{2}\left[\cos\frac{\theta}{2} + \pi\sin\frac{\theta}{2}\right]$$

Area vector is normal to the incline plane. Angle between $\vec{A}$ and $\vec{B}$ is θ.

$$\mu = IA$$

$$\tau = IAB\sin\theta = IB\ell^2\sin\theta\cdot\sin\frac{\theta}{2}\left[\cos\frac{\theta}{2} + \pi\sin\frac{\theta}{2}\right]$$

This torque is along XX shown in the figure. It will try to rotate the loop (about XX) so as to lift it up from the plane with support at O. But net force on the loop is zero.

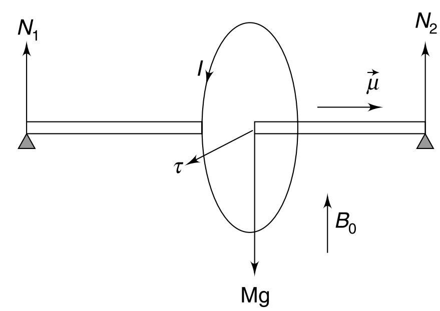
$\therefore$ Normal contact force will not change

40. Magnetic dipole moment of loop is $\mu = I\pi R^2$ (direction as shown)

Magnetic torque on the loop is

$$\tau = \mu B_0\sin 90° = I\pi R^2 B_0 \text{ (in the direction shown)}$$

Balancing torque on the disc gives



$$\tau + N_2\frac{d}{2} = N_1\frac{d}{2}$$

$$\Rightarrow \quad N_1 - N_2 = \frac{2I\pi R^2B_0}{d} \qquad \text{...(i)}$$

Also $\quad N_1 + N_2 = Mg \qquad$ ...(ii)

Solving (i) and (ii) gives

$$N_1 = \frac{Mg}{2} + \frac{I\pi R^2B_0}{d}$$

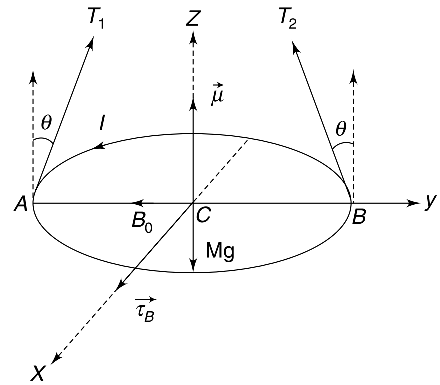
$$N_2 = \frac{Mg}{2} - \frac{I\pi R^2B_0}{d}$$

$$N_2 < N_1$$

N_2 will become zero if

$$\frac{I\pi R^2 B_0}{d} = \frac{Mg}{2}$$

$$\Rightarrow \qquad I = \frac{Mgd}{2\pi R^2 B_0}$$



41. Magnetic dipole moment $\vec{\mu} = I \cdot \pi\ R^2 \hat{k}$

Magnetic torque $\vec{\tau}_B = \vec{\mu} \times \vec{B}$

$$= I\pi R^2 B_0\, (\hat{k} \times (-\hat{j})) = I\pi R^2 B_0\ (\hat{i})$$

To counterbalance this torque we must have $T_1 > T_2$

Torque about centre C

$$-(T_1 \cos\theta \cdot R - T_2 \cos\theta R)\,\hat{i} + I\pi R^2 B_0 \hat{i} = 0$$

$$\therefore \qquad T_1 - T_2 = \frac{I\pi R B_0}{\cos\theta} \qquad \text{...(i)}$$

And $(T_1 + T_2)\cos\theta = Mg$

$$\Rightarrow \qquad T_1 + T_2 = \frac{Mg}{\cos\theta} \qquad \text{...(ii)}$$

Solving (i) and (ii)

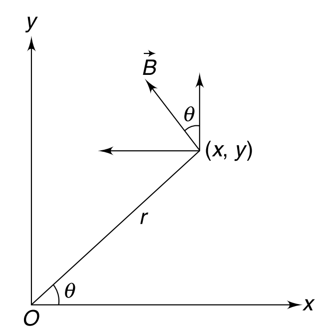
$$T_1 = \frac{1}{2\cos\theta}(Mg + \pi IRB_0) = Mg + \pi IRB_0 \ \text{ and } T_2 = Mg - \pi IRB_0$$

42. Field at any point (x, y) can be written as

$$B = -B\sin\theta\hat{i} + B\cos\theta\hat{j}$$

$$= -\frac{k}{\sqrt{x^2+y^2}} \cdot \frac{y}{\sqrt{x^2+y^2}}\hat{i} + \frac{k}{\sqrt{x^2+y^2}} \cdot \frac{x}{\sqrt{x^2+y^2}}\hat{j}$$

$$= -\frac{ky}{(x^2+y^2)}\hat{i} + \frac{kx}{(x^2+y^2)}\hat{j}$$



A small segment of the wire at point (x, y) can be taken as

$$\vec{dl} = dx\hat{i} + dy\hat{j}$$

Force on segment $\vec{dl}$ is

$$d\vec{F} = I\vec{dl} \times \vec{B}$$

$$= I\frac{kx\,dx}{x^2+y^2}\hat{k} + I\frac{ky\,dy}{x^2+y^2}\hat{k}$$

$\therefore$ Force on wire AB is

$$\vec{F} = \int d\vec{F} = kI \int_{A(x_1, y_1)}^{B(x_2, y_2)} \frac{xdx + ydy}{x^2+y^2}\hat{k}$$

If $x^2 + y^2 = t$

then $2x\,dx + 2y\,dy = dt$

$$x\,dx + y\,dy = \frac{1}{2}dt$$

$$\therefore \qquad \vec{F} = \frac{kI}{2}\int_A^B \frac{dt}{t} \cdot \hat{k}$$

$$= \frac{kI}{2}[\ell nt]_{x_1,\, y_1}^{x_2,\, y_2}\hat{k}$$

$$= \frac{kI}{2}\ell n\left(\frac{x_2^2 + y_2^2}{x_1^2 + y_1^2}\right)\hat{k}$$

43. Consider an Amperian loop in shape of a circle passing through P. Due to symmetry field at all points on the circle will have same magnitude and tangential direction.

$$\therefore \qquad \oint \vec{B}\cdot\vec{dl} = \mu_0 I$$

$$\therefore \qquad B = \frac{\mu_0 I}{2\pi d}$$

44. Unit vectors along $\vec{r}$ and perpendicular to $\vec{r}$ (known as azimuthal direction) are $\hat{r}$ and $\hat{\theta}$ respectively. A small element of length $d\ell$ along the curve can be represented as

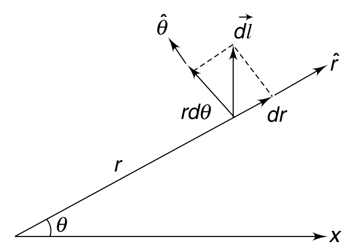
$$\vec{dl} = (dr)\hat{r} + (rd\theta)\ \hat{\theta}$$

Field due to such an element at O is

$$\vec{dB} = \frac{\mu_0}{4\pi} I \frac{\vec{dl} \times (-\hat{r})}{r^2}$$

$$= \frac{\mu_0 I}{4\pi}\frac{1}{r^2}[(dr)\hat{r} + (rd\,\theta)\,\hat{\theta}) \times (-\hat{r})]$$

$$= \frac{\mu_0 I}{4\pi}\frac{I d\theta}{r}\hat{k}$$



$$\therefore \qquad B = \frac{\mu_0 I}{4\pi}\int_0^{\pi/2}\frac{d\theta}{b + \frac{c}{\pi}\theta} \odot$$

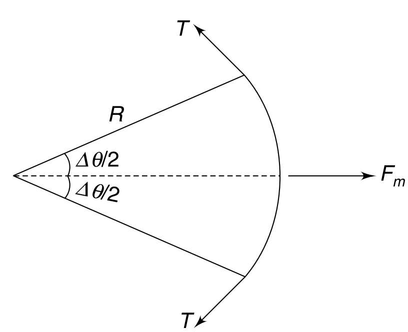
$$= \frac{\mu_0 I}{4\pi}\frac{\pi}{c}\left[\ell n\left(b + \frac{c}{\pi}\theta\right)\right]_0^{\pi/2}$$

$$= \frac{\mu_0 I}{4c}\left[\ell n\left(b + \frac{c}{2}\right) - \ell nb\right]$$

$$B = \frac{\mu_0 I}{4c}\left[\ell n\left(1 + \frac{c}{2b}\right)\right] \odot$$

45. Magnetic force on any small element of the wire is perpendicular to its length. It means that the tension will be constant along the wire.

Consider a small segment of the wire subtending a small angle $\Delta\theta$ at the centre of curvature. Magnetic force is



$$F_m = I(R\Delta\,\theta)B$$

This is balanced by components of tension

$$= 2\cdot T\sin\left(\frac{\Delta\theta}{2}\right) = T\Delta\theta$$

$$\therefore \qquad T\Delta\theta = I(R\Delta\,\theta)B \quad \Rightarrow \quad T = IRB$$

$$\text{Or,} \qquad R = \frac{T}{IB} = \text{constant}$$

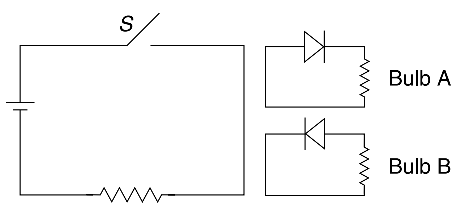
Radius of curvature remains constant means that the two segments will take the shape of circular arcs.

Chapter 11

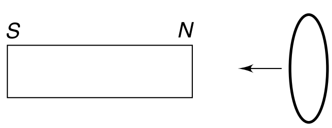
Electromagnetic Induction

Level 1

Q. 1: In the Figure shown, when the switch S is closed, one of the two bulbs glows momentarily. Which one?



Q. 2: A short circuited coil is moved towards a fixed magnet at a constant velocity. When the coil is at a distance x from the magnet it experiences a magnetic force F. Now the number of turns in the coil is doubled and the experiment is repeated with the coil moving with same constant velocity towards the magnet. What will be the magnetic force on the coil when it is at a distance x from the magnet?



Q. 3: A horizontal conducting loop of radius R is fixed in air. A uniformly charged rod having charge Q on it is held vertically above the conducting loop at a height h above it. The rod is released and it begins to fall along the axis of the loop. Calculate the emf induced in the conducting loop.

Q. 4: (a) When a small magnet is moved towards a solenoid, an emf is induced in it. However, if a magnet is moved inside the hole of a toroid, no emf is induced. Explain.

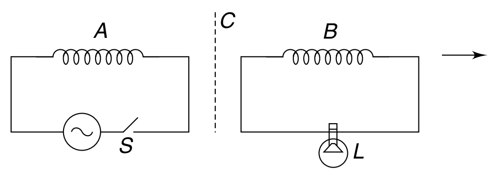
(b) A wire is kept horizontal along North-South direction. It is allowed to fall freely. How much emf will be induced across the ends of the wire due to presence of Earth's magnetic field.

Q. 5: Coil A is connected to an ac source through a switch S. Coil B, kept close to A, is connected to a low voltage bulb (L). Explain the following observations.

(i) When S is closed, the bulb lights up.

(ii) With S closed if B is moved away from A, the bulb gets dimmer.

(iii) With S closed, if a copper plate C is inserted between the coils, the bulb gets dimmer.



Q. 6: A coil is rotated with constant angular speed in a uniform magnetic field about an axis that is perpendicular to the field. Tell, with reason, if the following statement is true–

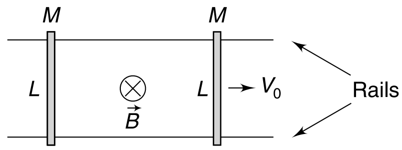
"The emf induced in the coil is maximum at the instant the magnetic flux through the coil is zero".

Q. 7: ABC is a triangular frame made of a conducting wire and is right angled at B. Its side BC is vertical and AB is horizontal. The frame is placed in a uniform magnetic field B in vertically upward direction. The frame is rotated about its side BC with constant angular speed ω. Resistance of the entire frame is R. Neglect gravity. Length $AB = L$; $BC = 2L$.

(a) Find the torque needed to keep the frame rotating with constant angular speed.

(b) Find the potential difference between points A and C.

Q. 8: Two parallel conducting rails are separated by a distance L. Two identical conducting wires are placed on the rails perpendicular to them. Each wire has mass M. One of the wires is given a velocity v_0 parallel to the rails away from the other wire. There is a uniform magnetic field directed into the plane of the figure everywhere. Answer following questions for the system comprising of two wires. [Ignore friction and self inductance]

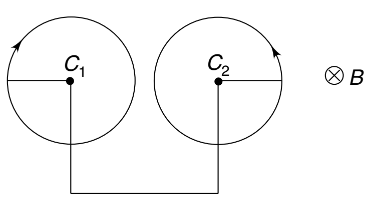


(i) Will the momentum of the system decrease with time?

(ii) What is kinetic energy of the system in steady state?

Q. 9: A stretchable conducting ring is in the shape of a circle. It is kept in a uniform magnetic field (B) that is perpendicular to the plane of the ring. The ring is pulled out uniformly from all sides so as to increase its radius at a constant rate $\frac{dr}{dt} = V$ while maintaining its circular shape. Calculate the rate of work done by the external agent against the magnetic force when the radius of the ring is r_0. Resistance of the ring remains constant at R.

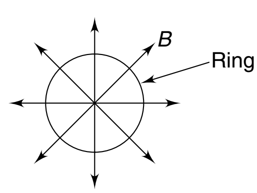
Q. 10: Two rods, each of length $L = 0.6$ m, are rotating in same plane about their ends C_1 and C_2. The distance between C_1 and C_2 is 1.201 m. The rods rotate in opposite directions with same angular speed ω and they are found to be in position shown in the Figure at a given instant of time. The ends C_1 and C_2 are connected using a conducting wire. There exists a uniform magnetic field perpendicular to the figure having magnitude $B = 5.0$ T. At what angular speed ω do we expect to see sparks in air? The dielectric breakdown of air happens if electric field in it exceeds 3×10^6 Vm^{-1}.



Q. 11: A vertical cylindrical region has a horizontal radial magnetic field inside it. A wire ring is made of a wire of cross section S, density d and resistivity ρ. Radius of the ring is r. The ring is kept horizontal with its centre on the axis of the cylindrical region and released. The field strength at all points on the circumference of the ring is B. At a certain instant velocity of the ring is v downward. Find

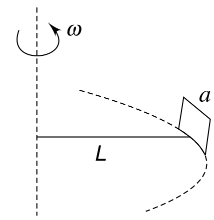
(a) Current in the ring and

(b) Acceleration of the ring at the instant.

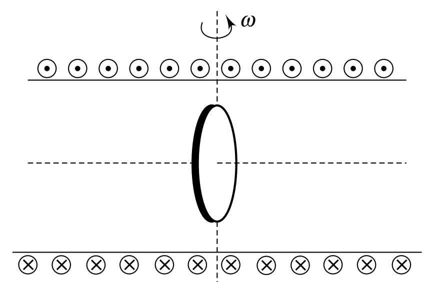


Top view

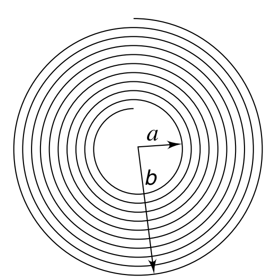
Q. 12: A conducting square loop of side length a is held vertical with the help of a non conducting rod of length L. The rod is made to rotate in horizontal circle about a vertical axis through its end. The loop rotates with the rod while its plane always remains perpendicular to the rod. The resistance of the loop is R and angular speed of the rod is ω. There is a uniform horizontal magnetic field B in the entire space. Find the average rate of heat dissipation in the loop during one rotation.



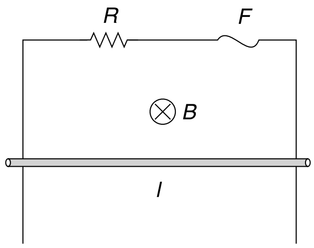
Q. 13: A long solenoid has n turns per unit length and carries a current $i = i_0 \sin \omega t$. A coil of N turns and area A is mounted inside the solenoid and is free to rotate about its diameter that is perpendicular to the axis of the solenoid. The coil rotates with angular speed ω and at time $t = 0$ the axis of the coil coincides with the axis of the solenoid as shown in figure. Write emf induced in the coil as function of time.



Q. 14: A flat coil, in the shape of a spiral, has a large number of turns N. The turns are wound tightly and the inner and outer radii of the coil are a and b respectively. A uniform external magnetic field (B) is applied perpendicular to the plane of the coil. Find the emf induced in the coil when the field is made to change at a rate $\frac{dB}{dt}$.



Q. 15: A pair of long conducting rails are held vertical at a separation $l = 50$ cm. The top ends are connected by a resistance of $R = 0.5\,\Omega$ and a fuse (F) of negligible resistance. A conducting rod is free to slide along the rails under gravity. The whole system is in a uniform horizontal magnetic field $B = 1.5T$ as shown. Resistance of the rod and rails are negligible and the rod remains horizontal as it moves. The rod is released from rest. Find the minimum mass of the rod that will ensure that the fuse blows out. It is known that rating of the fuse is $4A$.

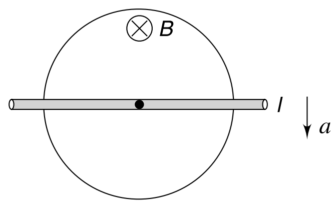


Q. 16: A thin uniform conducting rod of mass M and length L oscillates in a vertical plane about a fixed horizontal axis passing through its top end. The rod is oscillating with angular amplitude θ_0. A uniform horizontal magnetic field B, perpendicular to the plane of the oscillation is switched on.

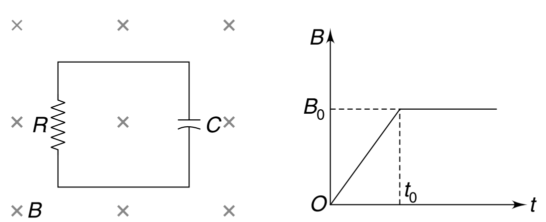
(a) Calculate the maximum emf induced between the ends of the rod.

(b) If the rod has a finite thickness (any real life rod will definitely have a thickness), what difference in oscillation is expected in absence of magnetic field and in presence of magnetic field. Describe qualitatively.

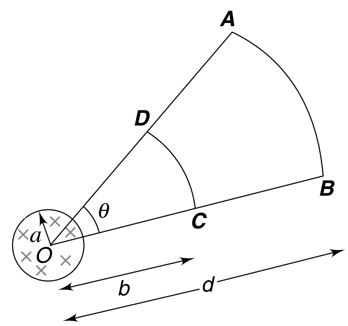
Q. 17: A wire ring of radius R is fixed in a horizontal plane. The wire of the ring has a resistance of $\lambda \Omega m^{-1}$. There is a uniform vertical magnetic field B in entire space. A perfectly conducting rod (l) is kept along the diameter of the ring. The rod is made to move with a constant acceleration a in a direction perpendicular to its own length. Find the current through the rod at the instant it has travelled through a distance $x = \frac{R}{2}$.



Q. 18: A conducting loop having resistance R contains a capacitor of capacitance C. A uniform magnetic field B is applied perpendicular to the plane of the loop. The magnetic field is made to change with time as depicted in the graph [$t_0 = 2RC$]. Plot the variation of charge on the capacitor as a function of time (t).



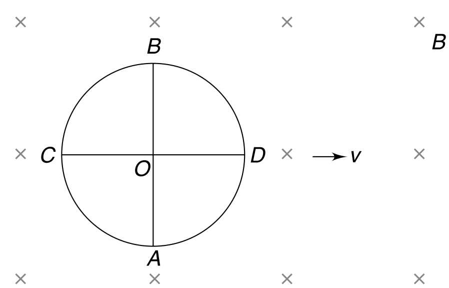
Q. 19: There is a uniform magnetic field (B) perpendicular to the plane of the figure in a circular region of radius a centred at O. $ABCD$ is a conducing loop in the plane of the figure with its arms BC and DA along two radial lines from O having an angle θ between them. AB is circular arc of radius $OB = d$ centred at O. CD is also a circular arc of radius $OC = b$ centred at O. The magnetic field is changed at a rate $\frac{dB}{dt}$.



(a) Find emf induced in the loop $ABCD$.

(b) Find emf induced in arc AB.

Q. 20: A uniform conducting wire is used to make a ring and its two diameters AB and CD. The ring is placed in a uniform magnetic field B perpendicular to its plane. The resistance of the wire per unit length is λ Ω/m.



(a) Find the current in BO and BC when the ring is moved with constant velocity v in its plane.

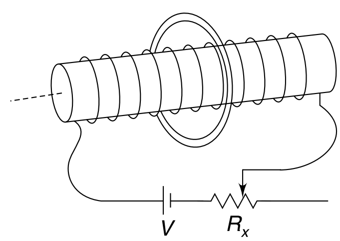
(b) Find current in BO and BC when the ring is kept stationary but the magnetic field is increased at a constant rate of $\frac{dB}{dt} = \alpha$ T/s. Assume that magnetic field is confined inside the circular ring only. Take radius of the ring to be a.

Q. 21: A bar magnet is kept along the axis of a conducting loop. When the magnet is moved along the axis, a current is seen in the loop. Which force is responsible for driving electrons in the loop if the observer is in

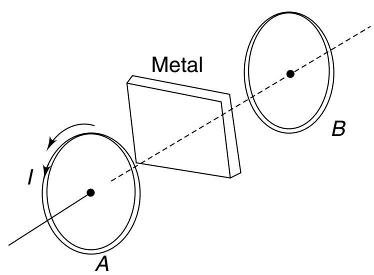
(a) Reference frame attached to the loop.

(b) Reference frame attached to the magnet.

Q. 22: A long narrow solenoid of radius a has n turns per unit length and resistance of the wire wrapped on it is R. The solenoid is connected to a battery of emf V through a variable resistance R_x. There is a conducting ring of radius $2a$ held fixed around the solenoid with its axis coinciding with that of the solenoid. The relaxation time of free electrons inside the material of the conducting ring at given temperature is τ and specific charge of an electron is α. The variable resistance R_x is changed linearly with time from zero to R in an interval Δt. Calculate the drift speed of the free electrons in the ring during this interval. [Neglect inductance]



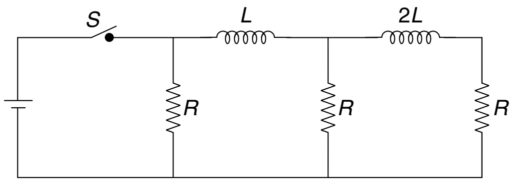
Q. 23: Two coils A and B are mounted co-axially some distance apart. Coil A is given a current that changes sinusoidally with time. A current gets induced in coil B. How does the magnitude of current in coil B change if a metal plate is placed between the two coils.



Q. 24: A thin conducting ring of radius R has a thin layer of insulation on its inner face. A superconducting ring of radius R is pressed into it. The insulation layer separates the two rings. Find the self inductance of the conducting ring.

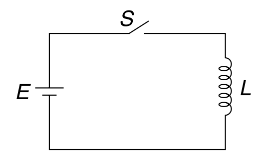
Q. 25: A certain volume of copper is drawn into a wire of radius a and is wrapped in shape of a helix having radius r (>> a). The windings are as close as possible without overlapping. Self inductance of the inductor so obtained is L_1. Another wire of radius $2a$ is drawn using same volume of copper and wound in the fashion as described above. This time the inductance is L_2. Find $\dfrac{L_1}{L_2}$.

Q. 26: In the circuit shown in figure, the current through each resistor is I. Find the currents through the resistors immediately after the switch 'S' is opened. How much heat will get dissipated in the circuit after the switch is opened?

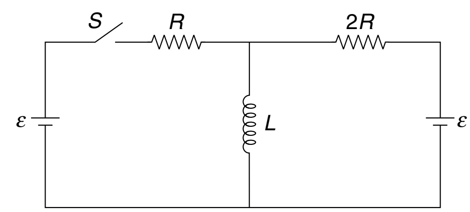


Q. 27: "The current through an inductor cannot change instantaneously. However, the potential difference across an inductor can change abruptly". Do you agree with this statement. Give reason.

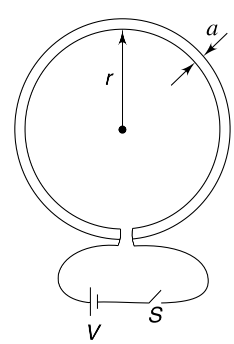
Q. 28: Consider an ideal inductor (having no resistance) of inductance L which is connected to an ideal cell (no resistance) of emf E by closing the switch S at time $t = 0$. Plot the variation of current through the inductor versus time.



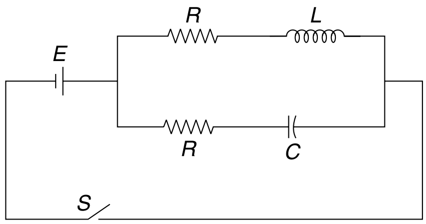
Q. 29: The circuit shown in the Figure is in steady state. Find the rate of change of current through L immediately after the switch S is closed.



Q. 30: A circular loop of radius r is made of a wire of circular cross section of diameter 'a'. When current I flows through the loop the magnetic flux linked with the loop due to self induced magnetic field is given by $\phi = \mu_0 r\left[\ln\left(\frac{16r}{a}\right) - \frac{7}{4}\right] I$. The resistivity of the material of the wire is ρ and $r >> a$. Switch S is closed at time $t = 0$ so as to connect the loop to a cell of emf V. Find the current in the loop at time t.

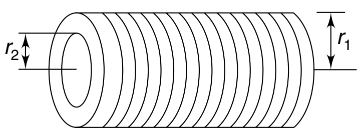


Q. 31: In the circuit shown, find the value of resistance R in terms of inductance L and capacitance C such that the current through the cell remains constant forever after the switch is closed.



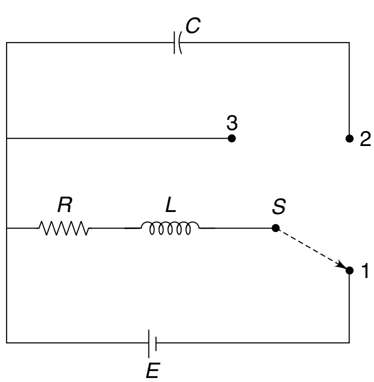
Q. 32: A proton beam has a circular cross sectional area A and it carries a current I distributed uniformly over its cross section. Calculate the magnetic energy per unit length of the beam within the beam.

Q. 33: Two long co-axial solenoids have radii, and number of turns per unit length equal to r_1, r_2 and n_1, n_2 respectively where suffix 1 refers to the outer solenoid and 2 refers to the inner solenoid. Length of both is l. The current in the outer solenoid is made to grow as $I_1 = kt$ where t is time. Resistance of the wire used in inner solenoid is R. Write the current induced in the inner solenoid assuming that it is shorted.



Q. 34: A capacitor of capacitance C is having a charge Q_0. It is connected to a pure inductor of inductance L. The inductor is a solenoid having N turns. Find the magnitude of magnetic flux through each of the N turns in the coil at the instant charge on the capacitor becomes $\frac{Q_0}{2}$.

Q. 35: In the circuit arrangement shown in Figure, the three way switch S is kept in position 1 for a long time.



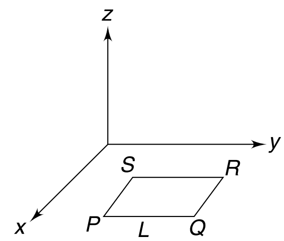
(a) Find the potential difference across the inductor immediately after the switch is thrown from position 1 to position 2.

(b) After being left in position 2 for a long time, the switch is moved to position 3. Find the potential difference across R immediately after the switch is moved to position 3.

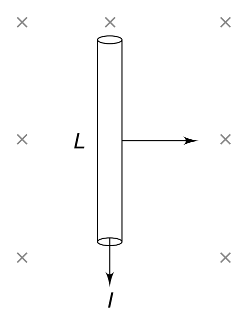
Assume no time lag in moving the switch from one position to another.

LEVEL 2

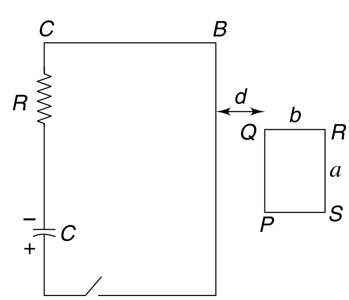
Q 36: A rectangular conducting loop $PQRS$ is kept in xy plane with its adjacent sides parallel to x and y axes. A magnetic field $\vec{B}$ is switched on in the region which varies in position and time as $\vec{B} = [B_0 \sin(ky - \omega t)]\,\hat{k}$. It was found that the emf induced in the loop is always zero. Express the length (L) of the loop in terms of constant k and ω.



Q. 37: A conducting rod of length L is carrying a constant current I. There exists a magnetic field B perpendicular to the rod. Due to the magnetic force the rod moved through a distance x in a direction perpendicular to the field (B) as well as its own length. The rod acquires a kinetic energy. A student says that a magnetic force ILB acted on the rod and it performed a work $W = ILBx$ on the rod. But we know that magnetic force is always perpendicular to the velocity of the charge and it cannot perform work on a moving charge. Which agency has actually spent energy to impart kinetic energy to the rod?



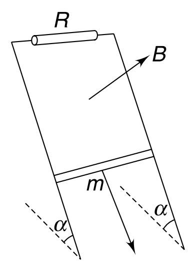
Q. 38: In a rectangular circuit $ABCD$ a capacitor having capacitance $C = 20\ \mu$F is charged to a potential difference of 100 V. Resistance of circuit is $R = 10\ \Omega$. Another rectangular conducting loop $PQRS$ is kept side by side to the first circuit with sides AB and PQ parallel and close to each other. The length and width of rectangle $PQRS$ is $a = 10$ cm and $b = 5$ cm respectively and AB as well as BC is large compared to a. PQ is located near the centre of side AB with $d = 5$ cm. The loop $PQRS$ has 25 turns and the wire used has resistance of $1\ \Omega\,\text{m}^{-1}$. The switch S is closed at time $t = 0$. Neglect self inductance of loops.



(a) Find the current in *ABCD* at $t = 200\ \mu s$

(b) Find the current in *PQRS* at $t = 200\ \mu s$.

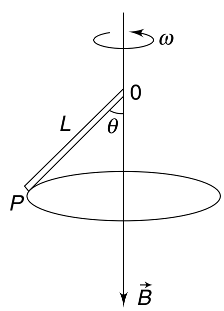
Q. 39: A copper connector of mass m, starting from rest, slides down two conducting bars set at angle α to the horizontal, due to gravity (see Figure). At the top the bars are interconnected through a resistance R. The separation between the bars is equal to l. The system is located in uniform magnetic field of induction B, perpendicular to the plane in which the connector slides. The resistance of the bars the connector and the sliding contacts, as well as the self inductance of the loop, are assumed to be negligible. The coefficient of friction between the connector and the bars is equal to $\frac{1}{2}\tan\alpha$



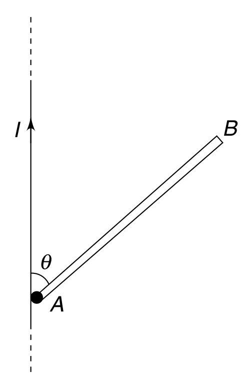
(a) Find the steady-state velocity of the connector.

(b) How will your answer differ if the magnetic field was in opposite direction.

Q. 40: A conducting rod (OP) of length L rotates in form of a conical pendulum with an angular velocity ω about a fixed vertical axis passing through its end O. There is a uniform magnetic field B in vertically downward direction. The rod makes an angle θ with the direction of the magnetic field. Calculate the emf induced across the ends of the rod. Which end is positive?



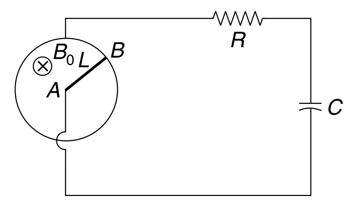
Q. 41: A conducting rod AB of mass M and length L is hinged at its end A. It can rotate freely in the vertical plane (in the plane of the Figure). A long straight wire is vertical and carrying a current I. The wire passes very close to A. The rod is released from its vertical position of unstable equilibrium. Calculate the emf between the ends of the rod when it has rotated through an angle θ (see Figure).



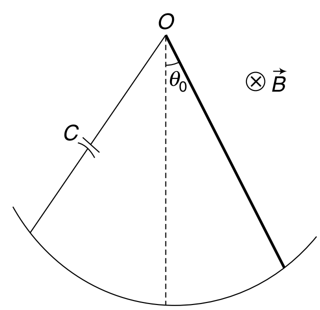
Q. 42: A perfectly conducting ring of radius L is kept fixed on a horizontal surface. A vertical uniform magnetic field B_0 exists in the region. A conducting rod (AB) of length L is hinged at the centre of the ring at A and its other end (B) touches the ring. The ring and the end A of the rod are connected to an external circuit having resistance R and capacitance C. The rod is made to rotate at a constant angular speed ω_0. Neglect friction and self inductance of the circuit.

(i) Find work done by the external agent in rotating the rod by the time the capacitor acquires a charge q_0.

(ii) Find heat generated in resistance R by the time the capacitor acquires a charge q_0.



Q. 43: A conducting rod of mass M and length L can oscillate like a pendulum in a vertical plane about point O. The lower end of the rod glides smoothly on a circular conducting arc of radius L. The circular arc is connected to point O of the rod through a capacitor of capacitance C. The entire device is kept in a uniform horizontal magnetic field B directed into the plane of the Figure. Disregard resistance of any component. The rod is deflected through a small angle θ_0 from vertical position and released at time $t = 0$.

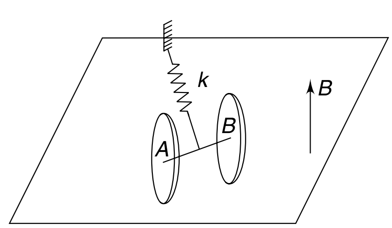


(a) Write the deflection angle (θ) of the rod as a function of time t.

(b) If the capacitor is replaced with a resistor what kind of motion do you expect?

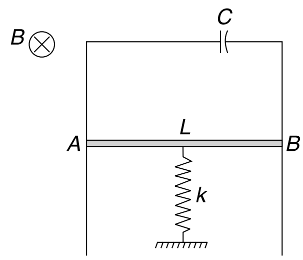
Give qualitative description only.

Q. 44: Two identical thin circular, metal plates are at a small separation d. They are connected by a thin conducting rod (AB) of length d. Each plate has area A. An ideal spring of stiffness k is connected to a rigid support and midpoint of rod AB as shown in the figure. Spring is made of insulating material. The system is on a smooth horizontal surface. The entire region has a uniform vertical upward magnetic field B.



The discs are pulled away from the support and released. Find time period of oscillations. Mass of the two disc plus rod system is M. Neglect any eddy current.

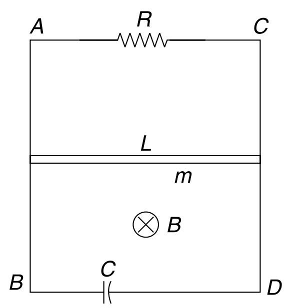
Q. 45: Two metal bars are fixed vertically and are connected on top by a capacitor of capacitance C. A sliding conductor AB can slide freely on the two bars. Length of conductor AB is L and its mass is m. It is connected to a vertical spring of force constant k. The conductor AB is released at time $t = 0$, from a position where the spring is relaxed. Taking initial position of the conductor as origin and downward direction as positive x axis, write the x co-ordinate of the conductor as a function of time. The entire space has a uniform horizontal magnetic field B. Neglect resistance and inductance of the circuit and assume that the bar AB always remains horizontal.



Q. 46: Two long fixed parallel vertical conducting rails AB and CD are separated by a distance L. They are connected by resistance R and a capacitance C at two ends as shown in the figure. There is a uniform magnetic field B directed horizontally into the plane of the figure. A horizontal metallic bar of length L and mass m can slide without friction along the rails. The bar is released from rest at $t = 0$. Neglect resistance of bar and rails and also neglect the self inductance of the loop.

(a) Find the maximum speed acquired by the bar after it is released.

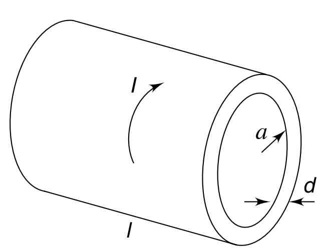
(b) Find the speed of the bar as a function of time t.



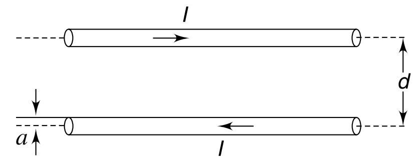
Q. 47: A hollow cylinder made of material of resistivity ρ has length ℓ, radius a and wall thickness d ($\ell >> a >> d$). A current I flows through the cylinder in tangential direction and is uniformly distributed along its length.

(a) Find the emf developed along the circumference of the cylinder if the current changes at a rate $\frac{dI}{dt} = \beta$

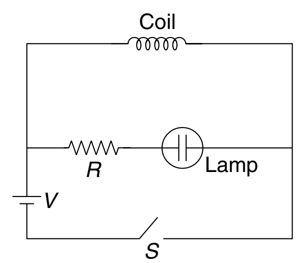
(b) Assume that no external source is present and the current at time $t = 0$ is I_0. The current decays with time. Write I as a function of time.



Q. 48: A long straight wire of cross sectional radius a carries a current I. The return current is carried by an identical wire which is parallel to the first wire. The centre to centre distance between the two wires is d. Find the inductance (L) of a length x of this arrangement. Neglect magnetic flux inside the wires.

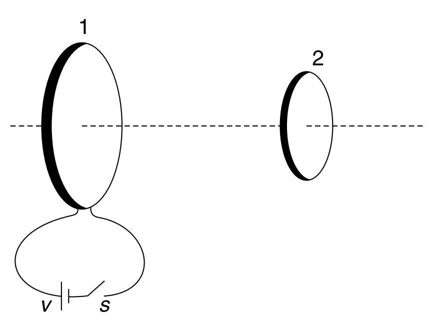


Q. 49: In the circuit shown in the figure the neon lamp lights up if potential difference across it becomes 60 V and goes out if the potential difference falls below 30 V. The inductor coil has a very small resistance and emf of the cell is $V = 4$ volt. The lamp does not light when the switch is closed. The neon light flashes once when the switch is opened. Explain why?

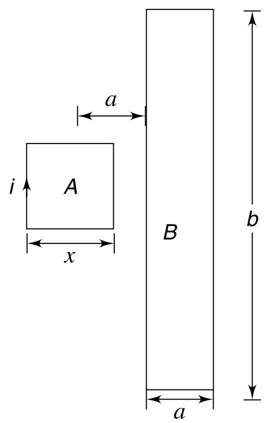


Q. 50: Determine the mutual inductance of a toroid and an infinite straight wire located along the central axis of the toroid. The toroid has a rectangular cross section with inner and outer radii a and b respectively. The width of the rectangular cross section parallel to the straight wire is h. Total turns in the toroid is N.

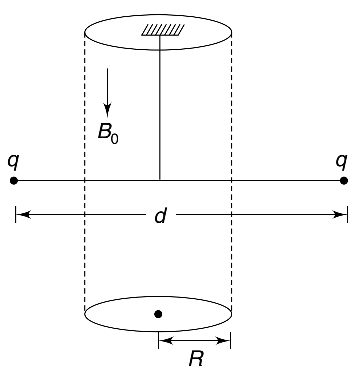
Q. 51: Two coils – 1 and 2 – are mounted co axially as shown in the figure. The resistance of the two coils are R_1 and R_2 and their self inductances are L_1 and L_2 respectively. Switch S is closed at time $t = 0$ to connect the coil 1 to an ideal cell of emf V. It is observed that by the time current reaches its steady value in coil 1, the quantity of charge that flows in coil 2 is Q_0. Calculate the mutual inductance (M) between the two coils.



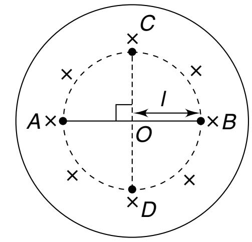
Q. 52: A rectangular conducting loop B has its side lengths equal to b and a ($<< b$). In the plane of the loop there is another very small loop A in shape of a square of side length x. Loop A is placed symmetrically with respect to B with its centre at a distance a from one of its longer side (see Figure). There is a current (i) in loop A which is made to increase at a constant rate of $\frac{di}{dt} = \alpha\ As^{-1}$. Calculate the emf induced in the bigger loop.



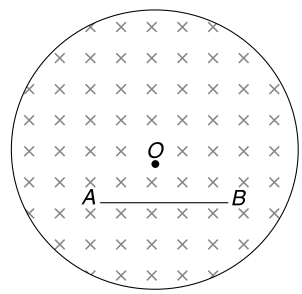
Q. 53: Two identical small balls, made of insulting material are attached to the ends of a light insulating rod of length d. The rod is suspended from the ceiling by a thin torsion free fibre as shown in figure. Each ball is given a charge q. There is a uniform magnetic field B_0, pointing vertically down, in a cylindrical region of radius R. The fibre is along the axis of the cylindrical region. The system is initially at rest. Now the magnetic field is suddenly switched off. Calculate the angular velocity acquired by the system. Each ball has mass m.



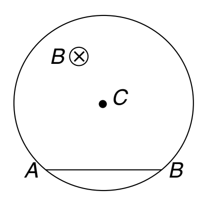
Q. 54: A massless non conducting rod AB of length $2l$ is placed in uniform time varying magnetic field confined in a cylindrical region of radius ($R > l$) as shown in the figure. The center of the rod coincides with the centre of the cylindrical region. The rod can freely rotate in the plane of the Figure about an axis coinciding with the axis of the cylinder. Two particles, each of mass m and charge q are attached to the ends A and B of the rod. The time varying magnetic field in this cylindrical region is given by $B = B_0\left[1 - \frac{t}{2}\right]$ where B_0 is a constant. The field is switched on at time $t = 0$. Consider: $B_0 = 100T$, $l = 4$ cm, $\frac{q}{m} = \frac{4\pi}{100}$ C/kg. Calculate the time in which the rod will reach position CD shown in the figure for the first time. Will end A be at C or D at this instant?



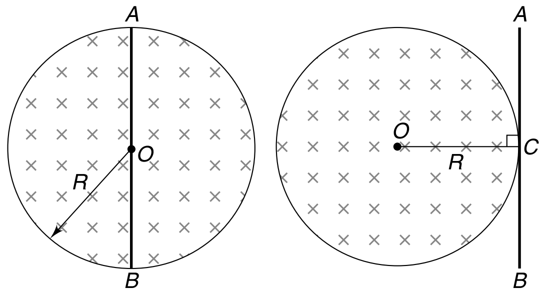
Q. 55: A cylindrical region of radius R is filled with a uniform magnetic field B as shown in the figure. A metal wire (AB) of length L is placed inside the field such that its ends are symmetrically located with respect to the centre (O) of the circular cross section of the region. If the magnetic field is changed at a rate $\frac{dB}{dt}$ the emf induced in the metal wire is ε. Find change in value of ε if the wire is displaced by a small distance ΔL parallel to its own length. Assume that the wire remains inside the field region.



Q. 56: A cylindrical volume of radius R has a uniform axial magnetic field B, which is increasing at a rate of $\frac{dB}{dt} = \alpha\ \text{Ts}^{-1}$. A chord ($AB$) of the circular cross section of the cylindrical region has length L. Calculate the line integral of induced electric field $\left(\int_A^B \vec{E}\,\vec{dl}\right)$ as one moves along the chord from A to B. Try to find the answer without actually performing the integration. Is the value of integral same if one moves along the arc from A to B?



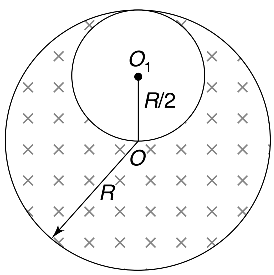
Q. 57: A uniform magnetic field B exists in a circular region of radius R. The field is perpendicular to the plane and is increasing at a constant rate of $\frac{dB}{dt} = \alpha$. There is a straight conducting rod AB of length $2R$. Find the emf induced in the rod when it is placed as shown in figure (a) & (b). Point C is midpoint of the rod in figure (b)



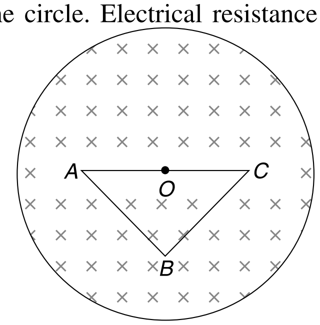
(a) (b)

Q. 58: A thin beam of n identical positively charged particle are constrained to move in a circular orbit of radius R in a particle accelerator. Each particle has charge q and mass m and the current in the circular orbit is I_0. The magnetic flux through the circular path is made to increases at a constant rate of β Wbs^{-1}. Calculate the current after the particles complete one turn.

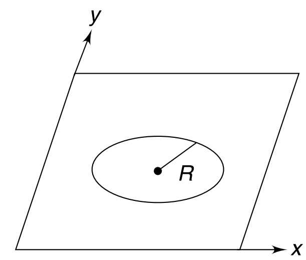
Q. 59: There is a long cylinder of radius R having a cylindrical cavity of radius $R/2$ as shown in the figure. Apart from the cavity, the entire space inside the cylinder has a uniform magnetic field parallel to the axis of the cylinder. The magnetic field starts changing at a uniform rate of $\frac{dB}{dt} = k$ T/s. Find the induced electric field at a point inside the cavity.



Q. 60: A uniform magnetic field B exists in a region of circular cross section of radius R. The field is directed perpendicularly into the plane of the figure. There is a triangular circuit ABC made of a uniform wire placed in the circular region. The triangle is a right angled isosceles triangle with equal sides $AB = BC = \ell$. The hypotenuse AC has its midpoint at the centre of the circle. Electrical resistance of the triangular circuit per unit length is $\frac{r_0}{\ell}$. If the magnetic field is changed at a constant rate of $\frac{dB}{dt} = \alpha$, find the potential difference between points C and A and that between B and A.



Q. 61: A conducting ring of mass $m = \pi$ kg and radius $R = \frac{1}{2}$ m is kept on a flat horizontal surface (xy plane). A uniform magnetic field is switched on in the region which changes with time (t) as $\vec{B} = (2\hat{j} + t^2\hat{k})$ T. Resistance of the ring is $r = \pi$ Ω and $g = 10$ ms^{-2}.



(a) Calculate the induced electric field at the circumference of the ring at the instant it begins to topple.

(b) Calculate the heat generated in the ring till the instant it starts to topple.

Q. 62: A tightly wound solenoid of length ℓ and cross sectional area A_1 is partly inserted co-axially into another tightly wound solenoid of length ℓ and cross section A_2 ($> A_1$). The centres of the two solenoids are separated by x ($< \ell$). Assume that $A_2 << x^2$, $A_2 << l^2$. Calculate the total magnetic field energy inside the solenoid when both of them carry same current (I) in the same sense. Number of turns in each solenoid is N.

Q. 63: In the last problem calculate the emf induced in the outer coil if the inner coil is pulled out at a speed v. How much emf will be induced in the inner coil?

Q. 64: In the last problem, calculate the force needed to pull out the inner coil at constant speed. The outer coil remains fixed.

Q. 65: Two ends of an inductor of inductance L is connected to two parallel conducting rails. A conducting wire of length ℓ (that is equal to separation between the rails) can slide on the rails without friction. The wire has mass m. It is projected with a velocity v_0 parallel to the rails (see Figure). Neglect self inductance and resistance of the loop.

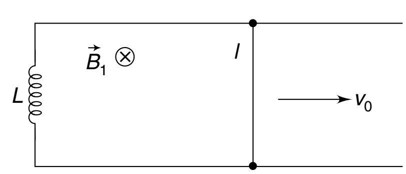
(a) Find velocity of the wire as a function of time.

(b) Write current through the wire at time t.

(c) Find speed of the wire as a function of its displacement.

(d) Is the current in the conductor zero when it stops? If no, find this current.

(e) Will the conductor move after it stops?

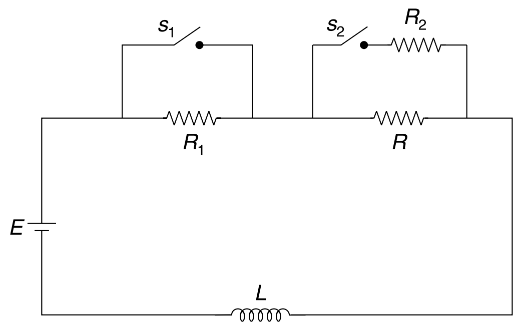


Q. 66: In the circuit shown in Figure, switch S_1 is kept closed and S_2 open for a long time. Resistance $R_1 = 1000$R and $R_2 = 10^{-3}$R.

(a) Switch S_1 is opened at time $t = 0$. Write the current and potential drop across R_1 immediately after S_1 is opened. What will be value of current after a long time?

(b) Switch S_2 is closed (S_1 is already closed since long) at time $t = 0$. Write current and potential drop across

R_2 immediately after this operation. Write current through the inductor as a function of time.



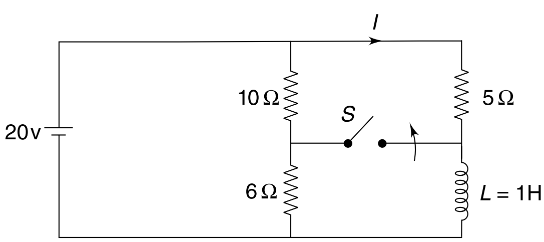
Q 67: A ring is made of a nearly superconducting material. Inductance of the ring is $L = 0.5$ H. A current allowed to decay in the ring was observed to remain constant for a month. The instrument used to measure current could detect any change if it is greater than 1%. Estimate the resistance of the ring considering it as a LR circuit with decaying current.

Q. 68: In the circuit shown, the switch 'S' has been closed for a long time and then opens at $t = 0$.

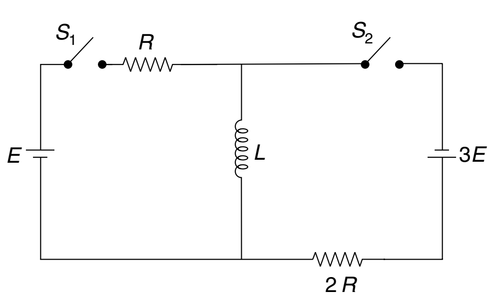
(a) Find the current through the inductor just before the switch is opened.

(b) Find the current I a long time after the switch is opened.

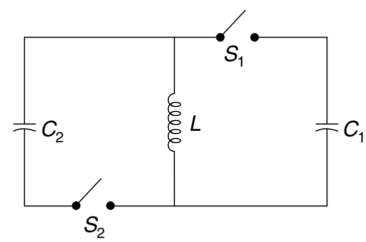
(c) Find current I as a function of time after the switch is opened. Also write the current through the cell as a function of time.



Q. 69: In the circuit shown, switch S_2 is open and S_1 is closed since long. Take $E = 20$ V, $L = 0.5$ H and $R = 10\ \Omega$. Find the rate of change of energy stored in the magnetic field inside the inductor, immediately after S_2 is closed.



Q. 70: The circuit shown in the figure is used to transfer energy from one capacitor to another. Initially capacitor of capacitance $C_1 = C_0$ is charged to a potential difference of V_0. Switch S_1 is closed at time $t = 0$. After some time S_1 is opened and S_2 is closed simultaneously. At time $t = T$, S_2 was opened and it was found that the potential difference across capacitor of capacitance $C_2 = \frac{C_0}{9}$ was $3V_0$. Find the smallest possible value of time T. The coil has inductance L. Assume no resistance.

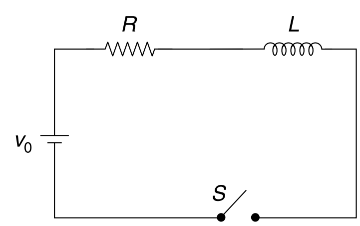


Q. 71: A pure inductor coil having inductance L is connected to a resistance R and a cell of emf V_0 as shown in the figure. Switch 'S' is closed at $t = 0$.

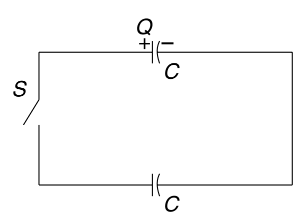
(a) Plot the variation of voltage across the resistance and the inductance as a function of time

(b) Find the time (t_1) when the two curves, obtained in (a) intersect.

(c) A student decides to start counting time from the instant the current becomes half its maximum value. Show the graphical plot of current vs time as obtained by this student.



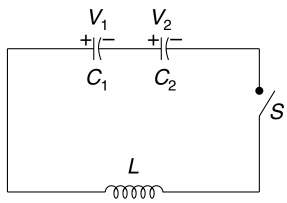
Q. 72: The circuit shown in figure has two identical capacitors one of which carries a charge Q and the other is having no charge. Switch S is closed. Find the maximum value of current in the circuit if self inductance of the loop is L and each capacitor has capacitance C. Neglect resistance of the loop.



Q. 73: The capacitors shown in the circuit have capacitance $C_1 = C$ and $C_2 = 3C$ and they have been charged to potentials $V_1 = 2V_0$ and $V_2 = 3V_0$ respectively. Switch S is closed to connect them to the inductor L.

(a) Find the maximum current through the inductor

(b) Find potential difference across C_1 and C_2 when the current in the inductor is maximum.

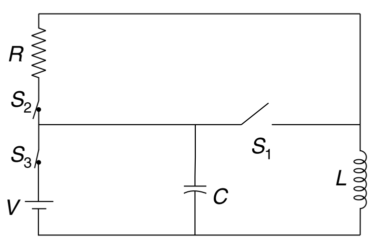


Q. 74. In the circuit shown in figure S_1 is open and, S_2 and S_3 are closed. The circuit is in steady state. At time $t = 0$, switch S_1 is closed and S_2 and S_3 are opened simultaneously.

$V = 100$ volt, $R = 10\ \Omega$, $C = 100\ \mu$F, $L = 0.03$ H

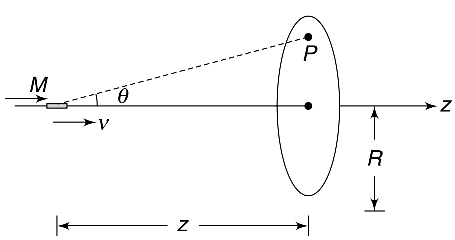
(a) Find the maximum charge that will appear on the capacitor at any time.

(b) Find the time at which the charge on the capacitor will become zero for the first time.



LEVEL 3

Q. 75: A short bar magnet having magnetic dipole moment M is moving along the axis of a fixed conducting (non magnetic) ring of radius R. The axis of the ring is along z direction.

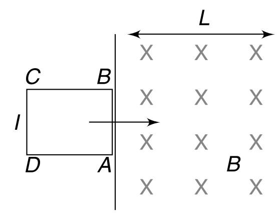


(a) Write the z component of magnetic field due to the magnetic dipole at a point P in the plane of the ring, at the instant the magnet is at a distance z from the centre of the ring. Position of point P can be defined in terms of angle θ as shown.

(b) Write the magnetic flux through the ring due to the magnetic field produced by the magnet as a function of z.

(c) Write the magnitude of emf induced in the ring at the instant shown if speed of the magnet at the moment is v.

Q. 76: A region of width L contains a uniform magnetic field B directed into the plane of the figure. A square conducting loop of side length ℓ ($< L$) is kept with its side AB at the boundary of the field region (see Figure). The loop is pushed into the field region with a speed such that it just manages to exit the field region. Calculate the time needed for the entire loop to enter the field region after it is pushed. Mass and resistance of the loop is M and R respectively.

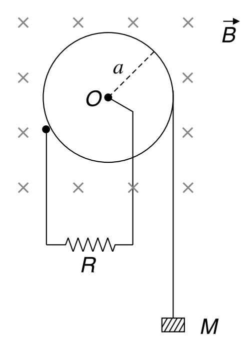


Q. 77: A metallic pulley is in the shape of a disc of radius a. It can rotate freely about a horizontal axis passing through its centre. The moment of inertia of the pulley about this axis is I. A light string is tightly wrapped around the pulley with its one end connected to a block of mass M. The centre of the pulley and its circumference are connected to a resistance R as shown. The contact of resistance at the circumference does not cause any friction. A uniform magnetic field B is switched on which is parallel to the axis of rotation of the pulley (see Figure). The mass M is allowed to fall. Assume that resistivity of the material of the pulley is negligible.

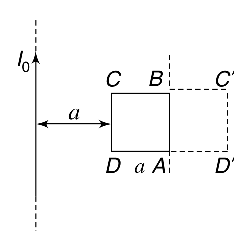
(a) Find the acceleration of the block of mass M at the instant its velocity becomes v_0.

(b) Assuming that the block can fall through a large distance, calculate the terminal speed (v_T) that it will acquire.

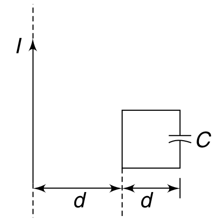
(c) Find the rate of change of kinetic energy of the pulley at the instant speed of the falling block is $\frac{v_T}{2}$.



Q.78: Figure shows a square conducting frame and a long wire-both lying in the same plane. The side length of the square loop is 'a' and it is at a distance 'a' from the long wire which is having a steady current I_0. The inductance and resistance of the square loop are L and R respectively. The loop is turned by 180° about its side AB so as to bring it to final position $ABC'D'$ at rest. Calculate the net charge that flow past a side of the loop.



Q.79: A square loop of side length d has a capacitor of capacitance C and the resistance of the loop is R. In the plane of the loop there is a long straight current carrying wire having current I. The distance of the straight wire from the loop is d as shown in figure. The current in the straight wire is made to grow with time as $I = \alpha t$ where $\alpha = 2\ \mathrm{As}^{-1}$. Neglect self inductance of the loop.

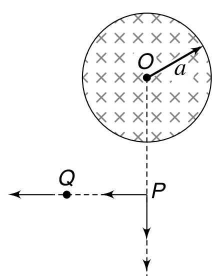


(a) Find the charge on the capacitor as a function of time (t).

(b) Find the heat generated in the loop as a function of time.

(c) Who supplies energy for heat dissipation?

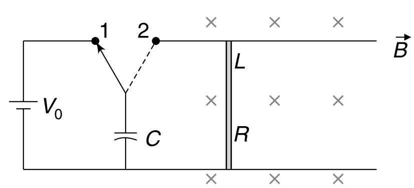
Q.80: There exists a uniform magnetic field perpendicular to the plane of the figure in a cylindrical region of radius a. The magnetic field is increasing at a constant rate of αTs^{-1}. A particle having charge q is at a point P outside the field region. The particle is slowly moved to infinity. Calculate work done by the external agent on the particle if-



(a) the particle is moved in radial direction OP.

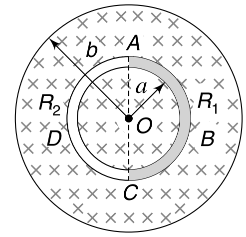
(b) the particle is moved in a direction perpendicular to OP along PQ.

Q.81: The Figure shows an electromagnetic gun. A bar of mass m, resistance R and length L is free to slide on two smooth rails separated by a distance L. A uniform magnetic field B is present perpendicular to the plane of the figure. A capacitor of capacitance C is charged using a battery of emf V_0 by placing switch (S) at 1. To fire the gun (i.e., to impart a kinetic energy to the rod) the switch is shifted to position 2 after the capacitor is fully charged. The rails end abruptly at the point where the speed of the rod becomes maximum. The efficiency of the gun can be defined as the kinetic energy imparted to the bar divided by the energy spent by the battery while charging the capacitor. Calculate the efficiency of the gun. Neglect self inductance of the circuit.



Q.82: A thin solenoid is made of a large number of turns of very thin wire tightly wound in several layers. The radii of innermost and outermost layers are a and b respectively and the length of the solenoid is L ($L >> a, b$). The total number of turns is N. Calculate the self inductance of the solenoid. Neglect edge effects.

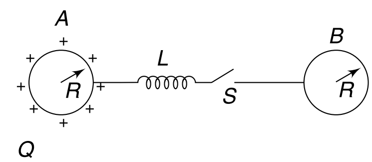
Q.83: A uniform magnetic field B exists perpendicular to the plane of the Figure in a circular region of radius b with its centre at O. A circular conductor of radius a ($< b$) and centre at O is made by joining two semicircular wires ABC and ADC. The two segments have same cross section but different resistances R_1 and R_2 respectively. The magnetic field in increased with time and there is an induced current in the conductor.



(a) Find the ratio of electric fields inside the conductors ABC and ADC.

(b) Explain why the electric field in two conductors is different despite the fact that the magnetic field is symmetrical.

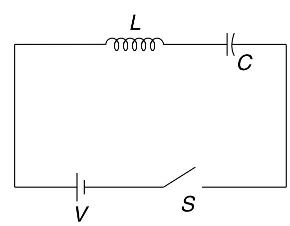
Q.84: Two conducting sphere of radius R are placed at a large distance from each other. They are connected by a coil of inductance L, as shown in the figure. Neglect the resistance of the coil. The sphere A is given a charge Q and the switch 'S' is closed at time $t = 0$. Find charge on sphere B as a function of time. At what time charge on B is $\frac{Q}{2}$?



Q.85: In the circuit shown in the figure, switch S is closed at time $t = 0$.

(a) Write current in the circuit and charge on capacitor as a function of time. Draw the graphical plot for the same.

(b) Find maximum charge on the capacitor. What is potential difference across the inductor when charge on the capacitor is maximum?

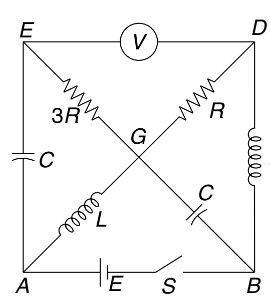


Q.86: In the circuit shown, switch S is kept closed and the circuit is in steady state.

(a) Find reading of the ideal voltmeter

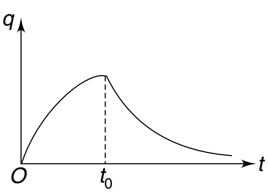
(b) Now the switch is opened. Find the reading of the voltmeter immediately after the switch is opened.

(c) Fid the heat dissipated in resistance R after the switch is opened.



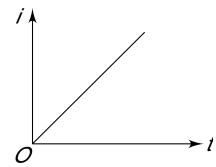
ANSWERS

1. Bulb A
2. 2F
3. Zero
4. (b) Zero
6. True
7. (a) Zero (b) $\frac{1}{2} B\omega L^2$
8. (i) No (ii) $\frac{1}{4} mV_0^2$
9. $\frac{4\pi^2 B^2 r^2 v^2}{R}$
10. $\omega = 1667 \text{ rads}^{-1}$
11. (a) $\frac{BvS}{\rho}$ (b) $g - \frac{B^2 v}{\rho d}$
12. $\frac{a^4 B^2 \omega^2}{2R}$
13. $\mu_0 i_0 NnA \cos(2\omega t)$
14. $\frac{\pi N}{3} (a^2 + b^2 + ab) \frac{dB}{dt}$
15. 0.3 kg
16. (a) $\frac{B}{2} \sqrt{3gL^3(1 - \cos\theta_0)}$

    (b) In presence of field the oscillations will die
17. $\frac{18B}{5\pi} \frac{\sqrt{3aR}}{\lambda}$
18. 

19. (a) Zero (b) $\frac{a^2\theta}{2} \frac{dB}{dt}$
20. (a) Zero in both BO and BC

    (b) Zero in BO and $\frac{a\alpha}{2\lambda}$ in BC
21. (a) Induced electric field applies the force

    (b) Magnetic force.
22. $\frac{\alpha \mu_0 anV\tau}{8R\Delta t}$
23. Current decreases.
24. Zero
25. 8 : 1
26. $2I, I, I, 3LI^2$
27. Statement is true.
28. 

29. $\frac{2\varepsilon}{3L}$
30. $i = \frac{Va^2}{8\rho r} [1 - e^{-t/\tau}]$

    Where $\tau = \frac{\mu_0 a^2}{8\rho} \left[\ln\left(\frac{16r}{a}\right) - \frac{7}{4}\right]$
31. $R = \sqrt{\frac{L}{C}}$
32. $\frac{\mu_0 I^2}{16\pi}$
33. $i = \frac{k\pi\mu_0 n_1 n_2 r_2^2 l}{R} (1 - e^{-t/\tau})$ where $\tau = \frac{\pi\mu_0 n_2^2 r_2^2 l}{R}$
34. $\frac{1}{2}\sqrt{\frac{3L}{C}} \frac{Q_0}{N}$
35. (a) E (b) zero
36. $L = n\frac{2\pi}{k}$; $n = 1, 2, 3, \ldots$
37. The source that is driving current through the rod.
38. (a) 3.7A (b) 35 μA
39. (a) $v = \frac{mgR\sin\alpha}{2B^2\ell^2}$ (b) No difference
40. $\frac{1}{2} B\omega L^2 \sin^2\theta$; O is positive
41. $\frac{\mu_0 I}{2\pi \sin\theta} \sqrt{3gL(1 - \cos\theta)}$
42. (i) $\frac{1}{2} q_0 B_0 \omega_0 L^2$ (ii) $\frac{1}{2} q_0 \left[B_0\omega_0 L^2 - \frac{q_0}{C}\right]$
43. (a) $\theta = \theta_0 \cos\omega t$ where $\omega = \left[\frac{6Mg}{4ML + 3B^2L^3C}\right]^{1/2}$

    (b) Oscillations die after some time.
44. $T = 2\pi\sqrt{\frac{M + \epsilon_0 AB^2 \cdot d}{k}}$
45. $x = \frac{mg}{k}(1 - \cos\omega t)$ where $\omega = \sqrt{\frac{k}{m + B^2L^2C}}$

46. (a) $\dfrac{mgR}{B^2L^2}$

(b) $v = \dfrac{mgR}{B^2L^2}\left[1 - e^{-\frac{B^2L^2t}{R(m + B^2L^2C)}}\right]$

47. (a) $\dfrac{\mu_0\pi a^2\beta}{\ell}$ (b) $I = I_0 e^{-\frac{2\rho t}{\mu_0 ad}}$

48. $L = \dfrac{\mu_0 x}{\pi}\,\ell n\left(\dfrac{d - a}{a}\right)$

50. $M = \dfrac{\mu_0 Nh\,\ell n\,(b/a)}{2\pi}$

51. $\dfrac{R_1R_2Q_0}{V}$

52. $\dfrac{\mu_0 a\alpha}{4\pi}$

53. $\omega = \dfrac{2qR^2B_0}{md^2}$

54. 1 s, end A

55. No change

56. $\dfrac{L\alpha}{4}\sqrt{4R^2 - L^2}$

57. (a) zero (b) $\dfrac{\pi R^2}{4}\,\alpha$

58. $I = \sqrt{I_0^2 + \dfrac{n^2q^3\beta}{2\pi^2R^2m}}$

59. $\dfrac{kR}{4}$ perpendicular to OO_1

60. $V_{CA} = \dfrac{\ell^2\alpha}{2(\sqrt{2} + 1)}$; $V_{BA} = \dfrac{\ell^2\alpha}{4(\sqrt{2} + 1)}$

61. (a) 10 V/m (b) $\dfrac{2\pi}{3}$ kJ

62. $\dfrac{\mu_0 I^2N^2}{2\ell^2}\,[3A_1 - 2A_1x + A_2\ell]$

63. $\dfrac{\mu_0 N^2A_1vI}{\ell^2}$; Same as in the outer coil

64. $F = \dfrac{\mu_0 N^2I^2A_1}{\ell^2}$

65. (a) $v = v_0\cos\omega t$

(b) $i = v_0\sqrt{\dfrac{m}{L}}\,\sin\omega t$ where $\omega = \dfrac{B\ell}{\sqrt{mL}}$

(c) $v = \sqrt{v_0^2 - \dfrac{B^2\ell^2}{mL}\,x^2}$

(d) No, $v_0\sqrt{\dfrac{m}{L}}$

(e) Yes

66. (a) Current $= \dfrac{E}{R}$, $V_{R1} = 1000\,E$, $I_\infty = \dfrac{E}{1001R} \simeq 10^{-3}\,\dfrac{E}{R}$

(b) At $t = 0^+ : i_0 = \dfrac{E}{R}$, $V_{R_2} = 10^{-3}E$,

$i = \dfrac{1000\,E}{R}\left[1 - e^{-\frac{Rt}{1000\,L}}\right]$

67. $1.9 \times 10^{-9}\ \Omega$

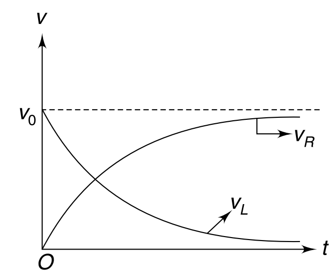
68. (a) $6A$ (b) $4A$

(c) $I = 4 + 2e^{-5t}$, current through cell $= \dfrac{21}{4} + 2e^{-5t}$

69. -40 J/s

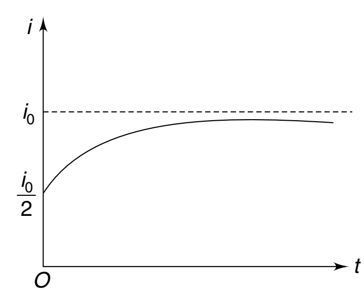
70. $T_{\min} = \dfrac{2\pi}{3}\sqrt{LC_0}$

71. (a)



(b) $t_1 = \dfrac{L}{R}\,\ell n 2$

(c)



72. $i_{\max} = \dfrac{Q}{\sqrt{2LC}}$

73. (a) $\dfrac{5V_0}{2}\sqrt{\dfrac{3C}{L}}$ (b) $\dfrac{7V_0}{4}, \dfrac{7V_0}{4}$

74. (a) 0.02 C (b) 0.9 ms

75 (a) $B_z = \dfrac{\mu_0}{4\pi r^3}\,[2\cos^2\theta - \sin^2\theta]$

(b) $\phi = \dfrac{\mu_0 M R^2}{2(R^2 + z^2)^{3/2}}$

(c) $E_{in} = \dfrac{3\mu_0 M R^2 vz}{2(R^2 + z^2)^{5/2}}$

76. $\dfrac{MR\ell n2}{B^2\ell^2}$

77. (a) $\dfrac{Mga^2 - \dfrac{B^2a^4v_0}{4R^2}}{I + Ma^2}$ (b) $\dfrac{4MgR^2}{B^2a^2}$

(c) $\dfrac{IM^2g^2R^2}{B^2a^2(I + Ma^2)}$

78. $\dfrac{\mu_0 I_0 a}{2\pi R}\,\ell n3$

79. (a) $q = \varepsilon C(1 - e^{-t/RC})$

(b) $H = \dfrac{\varepsilon^2 C}{2}\left(1 - e^{-\frac{2t}{RC}}\right)$ where $\varepsilon = \dfrac{\mu_0 \alpha \cdot d\,\ell n2}{2\pi}$

80. (i) zero (ii) $\dfrac{\pi}{4} qa^2\alpha$

81. $\dfrac{mB^2L^2C}{2(m + B^2L^2C)^2}$

82. $\dfrac{\mu_0 \pi N^2(3a^2 + 2ab + b^2)}{6L}$

83. (a) $\dfrac{E_1}{E_2} = \dfrac{R_1}{R_2}$

84. $q = \dfrac{Q}{2}\left[1 - \cos\left(\sqrt{\dfrac{2K}{LR}}\,t\right)\right]$; $t = \dfrac{\pi}{2}\sqrt{\dfrac{LR}{2K}}$ where $K = \dfrac{1}{4\pi\epsilon_0}$

85. $q = 2CV\sin^2\left(\dfrac{\omega t}{2}\right)$, $i = i_0 \sin\omega t$ Where $\omega = \dfrac{1}{\sqrt{LC}}$

86. (a) E

(b) $2E$ with polarity reversed

(c) $\dfrac{1}{2}E^2\left(\dfrac{L}{R^2} + C\right)$

SOLUTIONS

2. If number of turns is doubled the total flux doubles and hence emf induced in the coil doubles. But this doubles the resistance also so the current wouldn't change. The force doubles since number of turns doubled.

3. The lines of magnetic field produced by the falling rod lie in horizontal plane. The flux through the area bound by the loop is zero at any instant.

4. (a) Moving magnet inside a toroid does not cause a change in flux through any turn.

(b) Neither horizontal nor vertical component of earth's magnetic field is intercepted by the wire.

5. (i) Coil creates a magnetic field that is time changing (since A is connected to an ac source). This causes the flux through B to change. Emf is induced in B.

(ii) With B moving away, the flux linked with B decreases. Thus emf induce in it also decreases.

(iii) The magnetic field strength at B will get reduced substantially due to eddy current in the Cu plate. The bulb will get dimmer.

6. When angle between the normal to the coil and $\vec{B}$ is θ.

$$\phi = BA\cos\theta$$

and
$$-\frac{d\phi}{dt} = BA\sin\theta\cdot\frac{d\theta}{dt}$$

$\therefore$
$$\varepsilon_{in} = BA\omega\sin\theta \quad \left[\frac{d\theta}{dt} = \omega\right]$$

$\therefore$ ε_{in} is maximum when $\theta = 90°$

Flux $\phi = BA\cos 90° = 0$ at this instant.

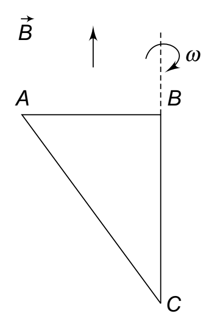
7. (a) There is no flux change in loop ABC.

$\therefore$ Induced emf = 0

Hence there is no current and no torque.

(b) $\varepsilon_{AB} = \varepsilon_{AC} = \dfrac{1}{2}B\omega L^2$

[End B is positive]



8. (i) The current induced in the loop will cause a magnetic force to act on both the wires. This force is equal and opposite on the two wires. Hence, momentum of the system will remain conserved.

(ii) Due to magnetic force the wire on the left speeds up and that on the right slows down.

When speed of both becomes same, emf induced in both of them will be equal. This will make the net emf in the loop equal to zero. There will be no current after this.

$\therefore$ Final speed of both wires is $V = \dfrac{V_0}{2}$

$$\therefore \qquad KE = \frac{1}{2} M\left(\frac{V_0}{2}\right)^2 \times 2 = \frac{1}{4} MV_0^2$$

9. Flux: $\phi = B \cdot \pi r^2$

Induced emf: $\varepsilon = \left|\dfrac{d\phi}{dt}\right| = 2\pi Br \dfrac{dr}{dt} = 2\pi Brv.$

Induced current: $I = \dfrac{\varepsilon}{R} = \dfrac{2\pi Brv}{R}$

Rate of work done = Rate of heat dissipated in Joule heating = I^2R

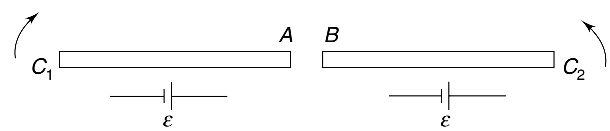
$$= \frac{4\pi^2 B^2 r^2 v^2}{R}$$

10. The emf induced in both rods is $\varepsilon = \dfrac{1}{2} B\omega L^2$ with polarity as shown in the figure.

Since C_1 and C_2 are at same potential, the difference in potential between the tips A and B will be

$$V_{AB} = 2\varepsilon = B\omega L^2$$

In the position shown in the Figure, the gap between the tips A and B is smallest (= 0.001 m). A spark will jump across the gap if



$$\frac{V_{AB}}{0.001} \geq 3 \times 10^6$$

$$\Rightarrow \qquad B\omega L^2 \geq 3 \times 10^3$$

$$\Rightarrow \qquad \omega \geq \frac{3 \times 10^3}{5 \times (0.6)^2} = 1667 \text{ rads}^{-1}$$

11. (a) Induced emf in a small element of length dl is $d\varepsilon = Bvdl$

$$d\varepsilon = Bvdl$$

Total emf in the ring is $\varepsilon = Bv \int dl = BvL$

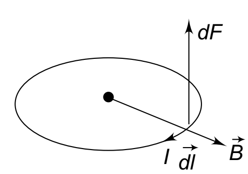
Where L is circumference of the ring.

Resistance of the ring $R = \dfrac{\rho L}{S}$

$\therefore$ Induced current $I = \dfrac{\varepsilon}{R} = \dfrac{BvS}{\rho}$

The current will be clockwise (seen from top).

(b) Magnetic force on each small element of the ring is vertically upward.



$$\therefore \qquad F_B = IB\int dl = IBL$$

$$\Rightarrow \qquad F_B = \frac{B^2 vSL}{\rho}$$

Mass of the ring $m = dLS$

$$\therefore \qquad ma = mg - F_B$$

$$\Rightarrow \qquad dLS\,a = dLS\,g - \frac{B^2 vSL}{\rho}$$

$$\Rightarrow \qquad a = g - \frac{B^2 v}{\rho d}$$

12. Let the area vector of the loop be parallel to $\vec{B}$ at time $t = 0$.
Angle between the two vectors at time t will be $\theta = \omega t$.
Flux through the loop at time t will be

$$\phi = a^2 B \cos(\omega t)$$

$\therefore$ Induced emf in the loop is

$$\varepsilon = -\frac{d\phi}{dt} = a^2 B\omega \sin(\omega t)$$

Current in the loop $\qquad I = \dfrac{\varepsilon}{R} = \dfrac{a^2 B\omega}{R}\sin(\omega t)$

Rate of heat dissipation

$$P = I^2 R = \frac{a^4 B^2 \omega^2}{R}\sin^2(\omega t)$$

Average rate of heat dissipation in one rotation

$$P_{av} = \frac{a^4 B^2 \omega^2}{R} < \sin^2(\omega t) >$$

$$= \frac{1}{2}\frac{a^4 B^2 \omega^2}{R}$$

Note: Average of $\sin^2\theta$ in one cycle (i.e., $0 \le \theta \le 2\pi$) is $\frac{1}{2}$.

13. Field inside the solenoid at time t is

$$B = \mu_0 ni = \mu_0 ni \sin\omega t$$

Let at time $t = 0^+$, the area vector of the coil be in the direction of $\vec{B}$.
At time t, the area vector has rotated by $\theta = \omega t$.
$\therefore$ Flux linked with one turn of coil is

$$\phi = BA\cos(\omega t) = \mu_0 n i_0 A \sin\omega t \cdot \cos\omega t$$

$$= \frac{1}{2}\mu_0 i_0 nA \sin(2\omega t)$$

$\therefore$ Emf induced in the coil is

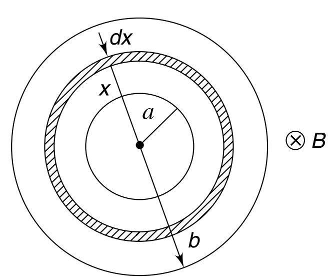
$$\varepsilon = N\frac{d\phi}{dt} = \mu_0 i_0 nNA\cos(2\omega t)$$

14. Consider a circular strip of radius x and width dx.

Number of turns in the strip $\qquad dN = \dfrac{N\,dx}{(b-a)}$

Flux linked with a circular loop of radius x is $\phi = B\cdot\pi x^2$

$$\Rightarrow \qquad \frac{d\phi}{dt} = \pi x^2 \frac{dB}{dt}$$



Emf induced in coil of width dx will be

$$d\varepsilon = (dN)\left(\frac{d\phi}{dt}\right) = \frac{N\,dx}{b-a}\cdot\pi x^2\frac{dB}{dt}$$

$\therefore$ Emf induced in the complete coil is

$$\varepsilon = \int d\varepsilon = \frac{\pi N}{(b-a)}\frac{dB}{dt}\int_a^b x^2 dx$$

$$= \frac{\pi N}{(b-a)}\frac{dB}{dt}\cdot\frac{(b^3-a^3)}{3}$$

$$= \frac{\pi N}{3}(a^2+b^2+ab)\frac{dB}{dt}$$

15. When current in the circuit is $I = 4A$, magnetic force on the rod is

$$F = IlB = 4 \times 0.5 \times 1.5 = 3 \text{ N}$$

The weight of the rod must exceed F to blow out the fuse.

$$\therefore \quad mg \geq 3$$

$$\therefore \quad m \geq \frac{3}{10} = 0.3 \text{ kg}$$

16. (a) Angular speed of the rod will be maximum when it is vertical.

Apply energy conservation between the extreme position of the rod and its vertical position.

$$\frac{1}{2}I\omega^2 = Mg\frac{L}{2}(1-\cos\theta_0)$$

$$\Rightarrow \quad \frac{1}{3}ML^2\omega^2 = MgL(1-\cos\theta_0)$$

$$\Rightarrow \quad \omega = \sqrt{\frac{3g}{L}(1-\cos\theta_0)}$$

$\therefore$ From the concept of motional emf

$$\varepsilon = \frac{1}{2}B\omega L^2$$

$$\varepsilon = \frac{B}{2}\sqrt{3gL^3(1-\cos\theta_0)}$$

(b) If the rod has finite thickness, eddy currents will be induced in it. Energy will get dissipated as heat. The amplitude of oscillation will decrease and the rod will stop oscillating after some time.

17. Speed of the rod when it has travelled a distance $x = \frac{R}{2}$ is

$$v = \sqrt{2ax} = \sqrt{aR}$$

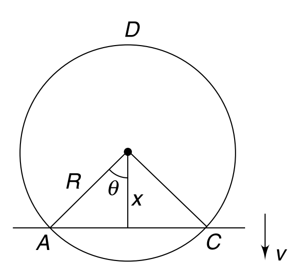
Length of rod inside the ring $\quad L = AC = 2\sqrt{R^2 - \frac{R^2}{4}} = \sqrt{3}R$

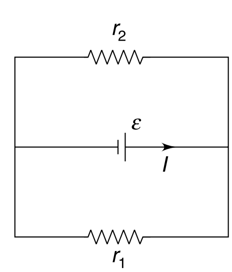
Emf induced in the rod at this instant is

$$E = BvL = B\cdot\sqrt{aR}\sqrt{3}\cdot R = \sqrt{3a}\,B\cdot R^{3/2}$$

$$\sin\theta = \frac{x}{R} = \frac{1}{2} \quad \Rightarrow \quad \theta = 30°$$

$\therefore$ Length of arc $\quad ABC = \frac{2\pi R}{6} = \frac{\pi R}{3}$





Length of arc $\quad ADC = \frac{5}{6}\cdot 2\pi R = \frac{5\pi R}{3}$

Resistance of arc ABC; $\quad r_1 = \frac{\pi R}{3}\lambda$

Resistance of arc ADC; $\quad r_2 = \frac{5\pi R}{2}\lambda$

Equivalent resistance $\quad r = \frac{r_1 r_2}{r_1 + r_2} = \frac{5\pi}{18}\lambda R$

$\therefore$ Current through the rod $\quad I = \dfrac{E}{r} = \dfrac{18B}{5\pi}\dfrac{\sqrt{3aR}}{\lambda}$

18. Flux through the loop $\quad \phi = BA$

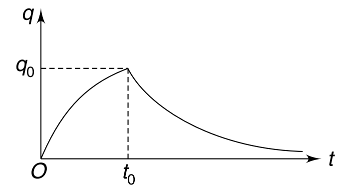
Emf induced $\quad |\varepsilon| = \dfrac{d\phi}{dt} = A\dfrac{dB}{dt}$

For $0 < t < t_0$

$$\varepsilon = A\frac{B_0}{t_0} = \text{a constant}$$

For $t > t_0$

$$\varepsilon = 0$$



The charge acquired by the capacitor in time t_0 is

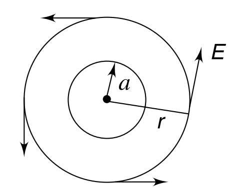
$$q_0 = C\varepsilon(1 - e^{-2}) \qquad [\because \; t_0 = 2\tau]$$

After this there is no emf in the loop and the capacitor begins to discharge. Therefore, the graph will be as shown in the figure.

19. (a) There is no flux linked to the loop. Hence no emf is induced.

(b) Electric field at a distance r from centre O is given by

$$E2\pi r = \pi a^2\frac{dB}{dt}$$



$$\Rightarrow \qquad E = \frac{a^2}{2r}\frac{dB}{dt}$$

Emf induced in arc AB

$$\varepsilon_{AB} = \left(\frac{a^2}{2d}\frac{dB}{dt}\right)(d\theta) = \frac{a^2\theta}{2}\frac{dB}{dt}$$

This does not depend on d. This implies that same emf is induced in CD. Hence there is no emf in loop $ABCD$.

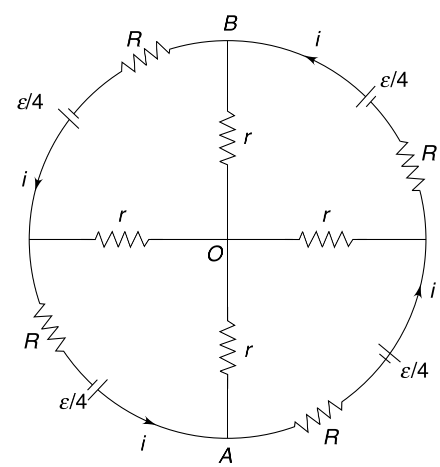
Note: Electric field lines are normal to arms BC and AD. There is no emf in these arms.

20. (a) No emf is induced in the loop. There will be no current anywhere. [But there is an emf between B and A. There is an emf between B and D.]

(b) Emf is induced in the ring. However, there is no emf induced in BA and CD [because induced electric field has circular field lines which are normal to BC and CD].

$$\varepsilon_{in} = \pi a^2 \cdot \frac{dB}{dt} = \pi a^2 \cdot \alpha$$

The effective circuit is as shown in figure.



In the figure- $R = \lambda\dfrac{\pi a}{2}$ and $r = \lambda a$

$$iR = \frac{\varepsilon}{4}$$

$$\Rightarrow \qquad i\frac{\lambda\pi a}{2} = \frac{\pi a^2\alpha}{4}$$

$$i = \frac{a\alpha}{2\lambda}$$

22. Rate of change of current $\quad \dfrac{\Delta I}{\Delta t} = \dfrac{\dfrac{V}{R} - \dfrac{V}{2R}}{\Delta t} = \dfrac{V}{2R\Delta t}$

Flux through ring

$$\phi = \pi a^2(\mu_0 nI)$$

$$\frac{d\phi}{dt} = \mu_0\pi a^2 n\frac{dI}{dt} = \mu_0\pi a^2 n\frac{V}{2R\Delta t}$$

$\therefore$ Induced electric field in the ring can be calculated as

$$E_{\text{in}} \cdot 2\pi(2a) = \frac{\pi\mu_0 a^2 nV}{2R\Delta t}$$

$$\therefore \quad E_{\text{in}} = \frac{\mu_0 anV}{8R\Delta t}$$

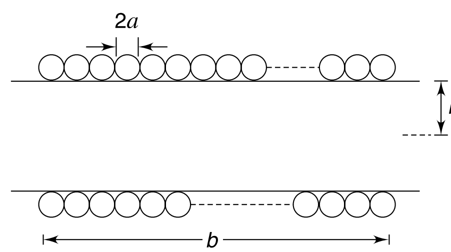
$$\therefore \quad v_d = \frac{eE_{\text{in}}}{m}\tau = \frac{\alpha\mu_0 anVt}{8R\Delta t}$$

23. Eddy currents are induced in the metal plate due to time changing magnetic field created by the current in A. As per Lenz's law the field produced by the eddy current will oppose the cause of induction. The coil B now faces the resultant field due to coil A and that due to the eddy current. Therefore, the rate of change of flux through B gets reduced. Hence current in B reduces.

24. When a current (I) is given to the conducting ring, a current is induced in the superconducting ring such that it cancels the flux due to current I. Thus flux linked with the ring is always zero. Hence self inductance is always zero.

25. Length of wire $l = \dfrac{\text{volume}\,(V)}{\pi a^2}$

Winding is as shown in the Figure.



Number of turns $\quad N = \dfrac{l}{2\pi r}$

Length of the helix $\quad b = 2a \cdot N = \dfrac{al}{\pi r}$

Number of turns per meter length $\quad n = \dfrac{1}{2a}$

$\therefore$ Self inductance $\quad L = \pi\mu_0 n^2 r^2 b$

$$= \pi\mu_0 \cdot \left(\frac{1}{2a}\right)^2 \cdot r^2 \cdot \frac{al}{\pi r}$$

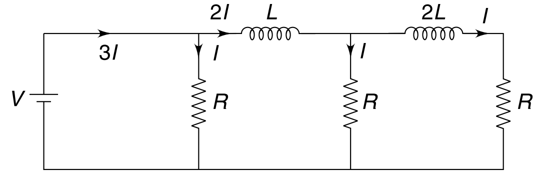
$$= \frac{1}{4}\mu_0 r \frac{l}{a} = \frac{1}{4}\mu_0 r \frac{V}{a(\pi a^2)}$$

$$= \frac{\mu_0 r}{4\pi}\frac{V}{a^3}$$

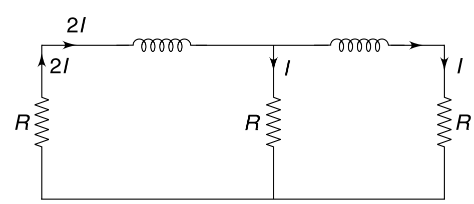
$$\therefore \quad L \propto \frac{1}{a^3}$$

$$\therefore \quad \frac{L_1}{L_2} = \left(\frac{2a}{a}\right)^3 = \frac{8}{1}$$

26. In steady state the current in different branches are as shown.



Immediately after opening the switch, the current through inductors will not change. Current will be as shown below.



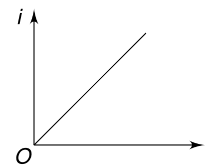
Note that current through a resistor can change abruptly.

Heat loss = Energy stored in magnetic field

$$= \frac{1}{2}L(2I)^2 + \frac{1}{2}2L(I)^2 = 3LI^2$$

28. At time t after closing the switch

$$V - L\frac{di}{dt} = 0$$



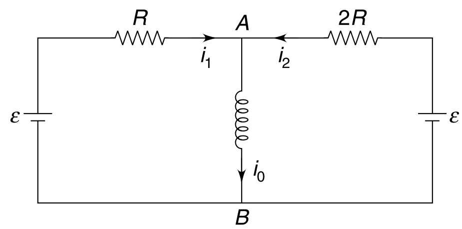
$$\Rightarrow \quad \int_0^i di = \frac{V}{L}\int_0^i dt$$

$$\Rightarrow \quad i = \frac{V}{L}t$$

The current will approach to infinity as $t \to \infty$. In practice, this never happens as there indeed is some resistance in the circuit.

29. In steady state the current through the inductor is $i_0 = \frac{\varepsilon}{2R}$.

The current through the inductor will not change immediately after the switch is closed.



Let $\quad V_{AB} = V$

In left loop $\quad Ri_1 + V = \varepsilon \quad$ ...(i)

In right loop $\quad 2Ri_2 + V = \varepsilon \quad$ ...(ii)

From (i) and (ii)

$$i_1 + i_2 = \frac{\varepsilon - V}{R} + \frac{\varepsilon - V}{2R}$$

Since $\quad i_1 + i_2 = i_0 = \frac{\varepsilon}{2R} \quad \left[\text{Because } i_0 = \frac{\varepsilon}{2R}\right]$

$$\therefore \quad \frac{\varepsilon - V}{R} + \frac{\varepsilon - V}{2R} = \frac{\varepsilon}{2R}$$

$$\Rightarrow \quad V = \frac{2\varepsilon}{3}$$

$$\Rightarrow \quad L\frac{di}{dt} = \frac{2\varepsilon}{3} \quad \Rightarrow \quad \frac{di}{dt} = \frac{2\varepsilon}{3L}$$

At this instant the potential difference across R is $\varepsilon - \frac{2\varepsilon}{3} = \frac{\varepsilon}{3}$. This potential difference will eventually become ε. It means current is increasing.

$\frac{di}{dt}$ is positive.

30. Self inductance of the loop is

$$L = \mu_0 r\left[\ln\left(\frac{16r}{a}\right) - \frac{7}{4}\right]$$

Resistance of the loop is $\quad R = \dfrac{\rho \cdot 2\pi r}{\pi\left(\frac{a}{2}\right)^2} = \dfrac{8\rho r}{a^2}$

The circuit is a simple $L - R$ circuit, hence

$$i = \frac{V}{R}[1 - e^{-t/\tau}] = \frac{Va^2}{8\rho r}[1 - e^{-t/\tau}]$$

Where $$\tau = \frac{L}{R} = \frac{\mu_0 a^2}{8\cdot\rho}\left[\ln\left(\frac{16r}{a}\right) - \frac{7}{4}\right]$$

31. Cell is connected in parallel to $R - L$ path and $R - C$ path.

Current through inductor and capacitor at time t after the switch is closed is

$$i_L = \frac{E}{R}\left[1 - e^{-\frac{Rt}{L}}\right]$$

$$i_C = \frac{E}{R}e^{-\frac{t}{RC}}$$

Current through the cell is

$$i = i_L + i_C = \frac{E}{R}\left[1 - e^{-\frac{Rt}{L}} + e^{-\frac{t}{RC}}\right]$$

This will be independent of time if $e^{-\frac{Rt}{L}} = e^{-\frac{t}{RC}}$

$$\Rightarrow \qquad \frac{R}{L} = \frac{1}{RC} \quad \Rightarrow \quad R = \sqrt{\frac{L}{C}}$$

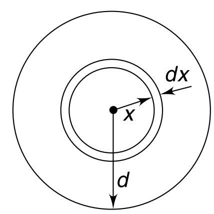
32. Let a = radius of circular area A

Current density $$j = \frac{I}{A} = \frac{I}{\pi a^2}$$

Magnetic field at a distance x from the axis $(0 < x < a)$ is given by Ampere's law as $B\cdot 2\pi x = \mu_0 j\cdot\pi x^2$

$$\Rightarrow \qquad B = \left(\frac{\mu_0 j}{2}\right)x = \frac{\mu_0 I}{2\pi a^2}x$$

Energy density at radial distance x is $u_B = \frac{1}{2\mu_0}B^2 = \frac{\mu_0 I^2}{8\pi^2 a^4}x^2$



Magnetic energy in annulus region of thickness dx and length L is

$$dU = u_B\cdot 2\pi x\cdot dx\cdot L$$

$$= \frac{\mu_0 I^2 x^2}{8\pi^2 a^4}\cdot 2\pi x\cdot dx\cdot L = \frac{\mu_0 I^2 L}{4\pi a^4}x^3\cdot dx$$

$\therefore$ Energy stored in cylindrical region of radius a and length L is

$$U = \int dU = \frac{\mu_0 I^2 L}{4\pi a^4}\int_0^a x^3\cdot dx = \frac{\mu_0 I^2 L}{16\pi}$$

Energy per unit length is $= \frac{\mu_0 I^2}{16\pi}$

33. Mutual inductance $$M = \pi\mu_0 n_1 n_2 r_2^2 l$$

Emf induced in inner coil is

$$|\varepsilon_2| = M\frac{dI_1}{dt} = k\pi\mu_0 n_1 n_2 r_2^2 l$$

Thus a constant emf is induced in the inner coil which has a resistance R and self inductance

$$L_2 = \pi\mu_0 n_2^2 r_2^2 l$$

Therefore, current in it at time t will be $i = i_0(1 - e^{-t/\tau})$

Where $$i_0 = \frac{\varepsilon_2}{R} = \frac{k\pi\mu_0 n_1 n_2 r_2^2 l}{R}$$

And $$\tau = \frac{L_2}{R} = \frac{\pi\mu_0 n_2^2 r_2^2 l}{R}$$

34. $$Q = Q_0 \cos(\omega t) \quad \text{where} \quad \omega = \frac{1}{\sqrt{LC}}$$

When $$Q = \frac{Q_0}{2};\ \cos(\omega t) = \frac{1}{2} \qquad \text{...(i)}$$

Flux linked to the coil is $$\phi = LI = L\frac{dQ}{dt}$$

$\therefore$ $$|\phi| = LQ_0\omega|\sin(\omega t)|$$

When $$\cos(\omega t) = \frac{1}{2};\ \sin(\omega t) = \frac{\sqrt{3}}{2}$$

$\therefore$ $$|\phi| = \frac{\sqrt{3}}{2}LQ_0\omega$$

$$\text{Flux through each turn} = \frac{|\phi|}{N} = \frac{\sqrt{3}LQ_0\omega}{2N} = \frac{1}{2}\sqrt{\frac{3L}{C}}\frac{Q_0}{N}$$

35. (a) Long time after switch is placed in position 1, current through the circuit is $I = \dfrac{E}{R}$

Potential difference across R is $V_R = E$

When switch is moved to position 2, current (through L) cannot change instantly. Drop across R remains E. Capacitor is uncharged, therefore, emf induced in the inductor must be E.

(b) When left in position 2 the LCR circuit will oscillate with decreasing energy. After a long time, the circuit will lose all its energy.

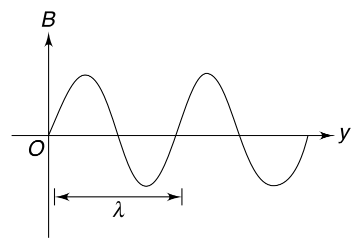
$\therefore$ There is no current after the switch is moved to position 3.

36. At any time (say at $t = 0$) the magnetic field varies sinusoidally with y (see Figure).

The pattern progresses to right with time.

The wavelength of this sinusoidal B is

$$\lambda = \frac{2\pi}{k}$$



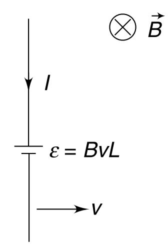
If the length (L) of the loop is such that there are exactly integral number of wave inside it, the total flux will remain constant (= 0) with time

$\Rightarrow$ $$L = n\lambda$$

$\Rightarrow$ $$L = n\frac{2\pi}{k};\ n = 1, 2, 3, \ldots$$

37. From the theory of motional emf we know that an emf will be induced in the rod with its upper end positive. The source driving the current is doing work against this emf. Rate at which work is being performed is

$$\varepsilon I = BvLI$$



$\therefore$ Work done in time 't' is

$$W = BLI\int v\,dt = BLIx$$

38. (a) $PQRS$ is a discharging RC circuit

$\therefore$ $$q = q_0 e^{-t/RC}$$

And $$i = \frac{dq}{dt} = \frac{q_0}{RC}e^{-t/RC} = \frac{CV}{RC}e^{-t/RC}$$

$$i = \frac{V}{R}e^{-t/RC} = \frac{100}{10}e^{-t/RC} = 10e^{-t/RC}$$

Time constant $\tau = RC = 10 \times 20 \times 10^{-6} = 200\ \mu s$

$\therefore$ $$i = 10e^{-\frac{t}{200}} \text{ where } t \text{ is in } \mu s$$

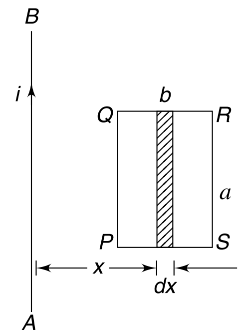
At $\qquad t = 200\ \mu s$

$$i = 10e^{-1} = 0.37 \times 10 = 3.7A$$

(b) Current in wire AB will have meaningful field inside the loop $PQRS$. All other arms of large loop are far away from $PQRS$.

Field at a distance x from AB is

$$B \simeq \frac{\mu_0 i}{2\pi x}$$



Flux through an element of width dx is

$$d\phi = Ba\,dx = \frac{\mu_0 ia}{2\pi}\frac{dx}{x}$$

$\therefore$ Flux through $PQRS$ is

$$\phi = \frac{\mu_0 ia}{2\pi}\int_d^{d+b}\frac{dx}{x} = \frac{\mu_0 ia}{2\pi}\,\ell n\left(\frac{d+b}{d}\right)$$

$$\Rightarrow \quad \frac{d\phi}{dt} = \frac{\mu_0 a}{2\pi}\,\ell n\left(\frac{d+b}{d}\right)\frac{di}{dt}$$

$$\Rightarrow \quad \frac{d\phi}{dt} = \frac{\mu_0 a}{2\pi}\,\ell n\left(\frac{d+b}{d}\right)\left(-\frac{10}{200\times 10^{-6}}\,e^{-\frac{t}{200\mu s}}\right)$$

At $\qquad t = 200\ \mu s$

$$\frac{d\phi}{dt} = -2\times 10^{-7}\times 0.1\ \ell n\left(\frac{5+5}{5}\right)\frac{3.7}{200\times 10^{-6}}$$

$$\therefore \quad \varepsilon_{in} = 3.7\ell n2 \times 10^{-4} = 0.26\text{ mV}$$

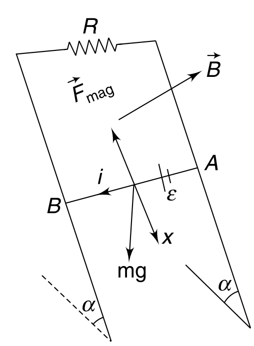
Resistance of loop $\qquad PQRS = 1 \times 25 \times 0.3 = 7.5\ \Omega$

$$\therefore \text{ Current} \quad I = \frac{\varepsilon_{in}}{7.5} = \frac{0.26\text{ mV}}{7.5} = 35\,\mu\text{A}$$

39. From lenz's law the current through the connector is directed from A to B. Here $\varepsilon = vB\ell$ between A and B. Where v is the velocity of the connector at any moment.

For the connector, $\qquad F_x = ma_x$

Or, $\qquad mg\sin\alpha - i\ell B - \mu mg\cos\alpha = ma_x$



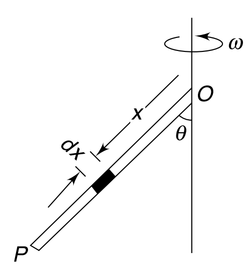
For steady state, acceleration of the rod must be equal to zero

Hence, $\qquad \frac{1}{2}mg\sin\alpha = i\ell B \qquad$ ...(i)

But $\qquad i = \frac{\varepsilon_{in}}{R} = \frac{vB\ell}{R} \qquad$ ...(ii)

From (i) and (ii) $\qquad v = \frac{mgR\sin\alpha}{2B^2\ell^2}$

40. Consider an elemental length dx. Velocity of the element is perpendicular to the plane of the figure and its magnitude is



$$v = \omega(x\sin\theta)$$

Emf induced in the element is

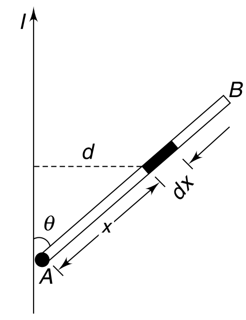
$$d\varepsilon = Bv(dx)\sin\theta \qquad [\theta \text{ is angle between } dx \text{ and } B]$$

$$= B\omega\sin^2\theta\, x dx$$

$$\therefore \qquad \varepsilon = B\omega\sin^2\theta\int_0^L x\,dx = \frac{1}{2}B\omega\sin^2\theta(L^2)$$

The free electrons experience force towards P. Hence O is positive.

41. We will apply energy conservation to find the angular speed (ω) of the rod.



$$\frac{1}{2}I_A\omega^2 = \text{loss in gravitational } PE$$

$$\frac{1}{2}\left(\frac{ML^2}{3}\right)\omega^2 = Mg\,\frac{L}{2}(1-\cos\theta)$$

$$\Rightarrow \qquad \omega = \sqrt{\frac{3g(1-\cos\theta)}{L}} \qquad \text{...(i)}$$

Now consider an element of length dx on the rod.

Speed of the element is $\quad v = \omega x$

Magnetic field at the location of the element is

$$B = \frac{\mu_0 I}{2\pi d} = \frac{\mu_0 I}{2\pi x\sin\theta}$$

$\therefore$ *Emf* induced in the element is

$$d\varepsilon = Bv\,dx = \frac{\mu_0 I\omega}{2\pi\sin\theta}\,dx$$

$\therefore$ *Emf* in the rod is

$$\varepsilon = \int d\varepsilon = \frac{\mu_0 I\omega}{2\pi\sin\theta}\int_0^L dx = \frac{\mu_0 I\omega L}{2\pi\sin\theta}$$

$$= \frac{\mu_0 I}{2\pi\sin\theta}\sqrt{3gL(1-\cos\theta)}$$

42. (i) *Emf* induced in the rod $\quad \varepsilon = \frac{1}{2}B_0\omega_0 L^2$

This is constant since ω_0 is constant. If charge q_0 is forced through this emf, work done (by imaginary cell created in the rod is)

$$W = q_0\varepsilon = \frac{1}{2}q_0B_0\omega_0L^2$$

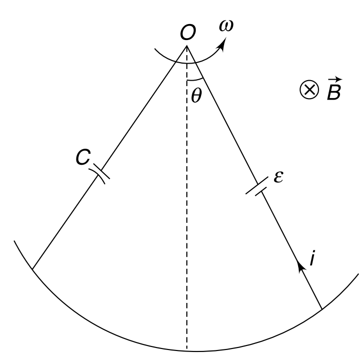
This is the work done by the external agent in keeping the rod rotating with a constant speed. Mechanical work by the agent gets converted into the electrical energy.

(ii) W = energy stored in capacitor + Heat dissipated in R

$$\therefore \qquad \frac{1}{2}q_0B_0\omega_0L^2 = \frac{q_0^2}{2C} + H$$

$$\therefore \qquad H = \frac{1}{2}q_0\left[B_0\omega_0L^2 - \frac{q_0}{C}\right]$$

43. (a) For any position θ of the rod, let its angular speed be $\omega\left(=\frac{d\theta}{dt}\right)$



emf induced $\quad \varepsilon = \frac{1}{2}B\omega L^2$

Charge on the capacitor is

$$q = C\varepsilon = \frac{1}{2}BC\omega L^2$$

Current in the circuit

$$i = \frac{dq}{dt} = \frac{1}{2} BCL^2 \frac{d\omega}{dt} = \frac{1}{2} BCL^2 \cdot \alpha \qquad [\alpha = \text{angular acceleration}]$$

Using $\tau = I\alpha$ for rod we get

$$\frac{1}{3} ML^2 \cdot \alpha = -Mg \frac{L}{2} \sin\theta - (BLi) \frac{L}{2}$$

$$\frac{1}{3} ML^2 \cdot \alpha = -Mg \frac{L}{2} \sin\theta - \frac{1}{4} B^2L^4C \cdot a$$

$\therefore$
$$\left[\frac{1}{3} ML^2 + \frac{1}{4} B^2L^4C\right] \alpha \simeq -Mg \frac{L}{2} \theta$$

$$\alpha = -\left(\frac{6Mg}{4ML + 3B^2L^3C}\right) \theta$$

Thus motion of the rod is simple harmonic. Initially, it is at positive extreme.

$\therefore$
$$\theta = \theta_0 \cos\omega t \text{ where } \omega = \left(\frac{6Mg}{4ML + 3B^2L^3C}\right)^{1/2}$$

(b) In presence of a resistor the mechanical energy of the system will get dissipated as heat and the oscillations will die.

44. Capacitance of disc system $\qquad C = \dfrac{\epsilon_0 A}{d}$

When speed of the system is v, the emf induced between two plates is

$$\varepsilon = Bvd$$

$\therefore$ Charge on the capacitor $\qquad q = C\varepsilon = \epsilon_0 ABv$

Energy stored in the capacitor is $\qquad U = \dfrac{1}{2} C\varepsilon^2$

$\Rightarrow \qquad U = \dfrac{\epsilon_0 A}{2} B^2v^2 \cdot d$

Force associated with this energy is

$$F = -\frac{dU}{dx} = -\frac{\epsilon_0 A}{2} B^2d\, 2v \frac{dv}{dx}$$

$$= -\epsilon_0 AB^2 \cdot d \cdot v \frac{dv}{dt} \cdot \frac{dt}{dx}$$

But $\qquad \dfrac{dt}{dx} = \dfrac{1}{v}$

$\therefore \qquad F = -\epsilon_0 AB^2 d \dfrac{dv}{dt}$

Alternatively, this force can be worked out as magnetic force on the rod ($= Bid$) when the current through it is $i = \dfrac{dq}{dt}$

When the system is displaced by x from its equilibrium and has a speed v; we have

$$M \frac{dv}{dt} = -kx - \epsilon_0 AB^2 \cdot d \frac{dv}{dt}$$

$\Rightarrow \qquad (M + \epsilon_0 AB^2 \cdot d) \dfrac{dv}{dt} = -kx$

$\Rightarrow \qquad \dfrac{dv}{dt} = -\left(\dfrac{k}{M + \epsilon_0 AB^2 \cdot d}\right) \cdot x \quad [SHM]$

$$\therefore \qquad T = 2\pi\sqrt{\frac{M + \in_0 AB^2 \cdot d}{k}}$$

45. At any time t let the velocity of the conductor be v

Induced emf, $\varepsilon = BLv$

If charge on the capacitor at time t is q then

$$q = C\varepsilon = BLCv$$

Current through the conductor is $i = \dfrac{dq}{dt} = BLC\dfrac{dv}{dt}$

Force equation for the conductor is

$$m\frac{dv}{dt} = mg - kx - BiL \qquad \text{[where } x = \text{displacement]}$$

$$m\frac{dv}{dt} = mg - kx - B^2L^2C\frac{dv}{dt}$$

$$\Rightarrow \qquad \frac{dv}{dt}[m + B^2L^2C] = mg - kx$$

In equilibrium $\dfrac{dv}{dx} = 0 \quad \Rightarrow \quad kx_0 = mg$ ...(i)

If x' is displacement of the bar measured from the equilibrium position then $x' = x - x_0$

$$[m + B^2L^2C]\frac{dv}{dt} = mg - k(x_0 + x')$$

$$\Rightarrow \qquad \frac{d^2x'}{dt^2} = -\left(\frac{k}{m + B^2L^2C}\right)x' \qquad [\because \ mg = kx_0]$$

Solution to this equation is of the form

$$x' = A\sin(\omega t + \delta) \text{ where } \omega = \sqrt{\frac{k}{m + B^2L^2C}}$$

$$\therefore \qquad x = x_0 + x' = \frac{mg}{k} + A\sin(\omega t + \delta)$$

$$\therefore \qquad v = \frac{dx}{dt} = A\omega\cos(\omega t + \delta)$$

At $t = 0,\ v = 0$ $\qquad \therefore \ \delta = \dfrac{\pi}{2}$

$$\therefore \qquad x = \frac{mg}{k} + A\cos\omega t$$

At $t = 0;\ x = 0 \quad \Rightarrow \quad A = -\dfrac{mg}{k}$

$$\therefore \qquad x = \frac{mg}{k}(1 - \cos\omega t)$$

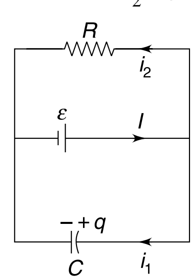
46. When speed of the bar is v emf induced in it is $\varepsilon = BLv$

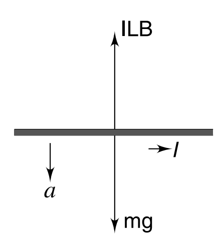
Let charge on the capacitor and current through it be q and i_1. Current in R be i_2. Current through the bar is $I = i_1 + i_2$

$$i_2 = \frac{BLv}{R}$$

and $q = BLvC \quad \Rightarrow \quad \dfrac{dq}{dt} = BLC\dfrac{dv}{dt}$

$$\Rightarrow \qquad i_1 = BLC\frac{dv}{dt}$$





Force equation for the bar is

$$m\frac{dv}{dt} = mg - ILB$$

$$\Rightarrow \quad m\frac{dv}{dt} = mg - \left(\frac{BLv}{R} + BLC\frac{dv}{dt}\right)LB \qquad [\because \; I = i_2 + i_1]$$

$$\Rightarrow \quad (m + B^2L^2C)\frac{dv}{dt} = mg - \frac{B^2L^2}{R}v$$

$$\Rightarrow \quad \int_0^v \frac{dv}{mg - \frac{B^2L^2}{R}v} = \left(\frac{1}{m + B^2L^2C}\right)\int_0^t dt$$

$$\Rightarrow \quad \left[\ell n\left(mg - \frac{B^2L^2}{R}v\right)\right]_0^v = -\frac{B^2L^2t}{R(m + B^2L^2C)}$$

$$\Rightarrow \quad \ell n\left[\frac{mg - \frac{B^2L^2v}{R}}{mg}\right] = -\frac{B^2L^2t}{R(m + B^2L^2C)}$$

$$\therefore \quad 1 - \frac{B^2L^2v}{mgR} = e^{-\frac{B^2L^2t}{R(m + B^2L^2C)}}$$

$$\Rightarrow \quad v = \frac{mgR}{B^2L^2}\left[1 - e^{-\frac{B^2L^2t}{R(m + B^2L^2C)}}\right]$$

when $t \to \infty$; $v \to \dfrac{mgR}{B^2L^2}$

47. (a) Drawing analogy from the case of an ideal solenoid we can write the magnetic field inside the cylinder as:

$$B = \mu_0\frac{I}{\ell} \qquad \left[\because \begin{array}{l} B = \mu_0 ni \\ ni = I/\ell \end{array}\right]$$

Flux linked with the cylinder is

$$\phi = BA = \pi a^2\frac{\mu_0 I}{\ell}$$

Self inductance is $\quad L = \mu_0\dfrac{\pi a^2}{\ell}$

Induced emf $\quad \varepsilon = \left|\dfrac{d\phi}{dt}\right| = \dfrac{\mu_0\pi a^2}{\ell}\dfrac{dI}{dt} = \dfrac{\mu_0\pi a^2\beta}{\ell}$

(b) Resistance of cylinder $R = \rho\dfrac{2\pi a}{\ell d}$

$$\therefore \quad \varepsilon = IR$$

$$-\frac{\mu_0\pi a^2}{\ell}\frac{dI}{dt} = I\frac{\rho 2\pi a}{\ell d}$$

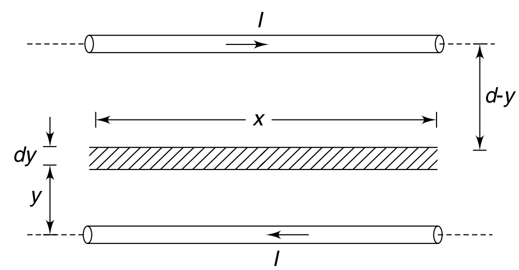
$$\Rightarrow \quad \frac{dI}{dt} = -\frac{2\rho}{\mu_0 ad}I$$

$$\therefore \quad \int_{I_0}^{I}\frac{dI}{I} = -\frac{2\rho}{\mu_0 ad}\int_0^t dt$$

$$\Rightarrow \quad \ell n\left(\frac{I}{I_0}\right) = \frac{-2\rho t}{\mu_0 ad}$$

$$\Rightarrow \quad I = I_0 e^{-\frac{2\rho t}{\mu_0 ad}}$$

48. Consider a strip of width dy as shown in figure.



Magnetic field at the strip due to current in two wires is

$$B = \frac{\mu_0 I}{2\pi}\left(\frac{1}{y} + \frac{1}{d-y}\right) \otimes$$

$\therefore$ Flux through the strip of area xdy is

$$d\phi = \frac{\mu_0 I}{2\pi}\left(\frac{1}{y} + \frac{1}{d-y}\right) x\,dy$$

Flux linked with the area between two wires in a length x will be

$$\phi = \frac{\mu_0 I x}{2\pi}\left[\int_a^{d-a} \frac{dy}{y} + \int_a^{d-a} \frac{dy}{d-y}\right]$$

$$= \frac{\mu_0 I x}{2\pi}\left[\ell n\left(\frac{d-a}{a}\right) - \ell n\left(\frac{d-d+a}{d-a}\right)\right]$$

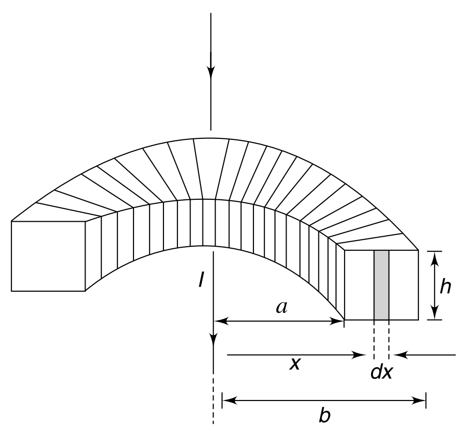
$$= \frac{\mu_0 I x}{2\pi}[2\ell n(d-a) - 2\ell n a]$$

$$= \frac{\mu_0 I x}{\pi}\,\ell n\left(\frac{d-a}{a}\right)$$

$$\therefore \qquad L = \frac{\phi}{I} = \frac{\mu_0 x}{\pi}\,\ell n\left(\frac{d-a}{a}\right)$$

49. When switch is closed the induced emf in the inductor is 4 volt and begins to fall thereafter. In steady state there is some large current (i_0) flowing through it. The bulb will not light as the potential difference across it never reaches 60 V. When the switch is opened the cell gets disconnected and the current in the inductor must go to zero. The decay path of the current has a huge resistance in form of the neon lamp. The time constant is small which means that $\frac{di}{dt}$ is large. A big emf is induced in the inductor which causes the lamp to flash.

50. We have drawn half of the toroid and the straight wire. This diagram will give you a fair idea of the situation. Consider a current I in the straight wire. Consider a strip of width dx on the cross section of the toroid at radial distance x.



Magnetic field at this location $\qquad B = \frac{\mu_0 I}{2\pi x}$

Flux through the strip $= B(hdx) = \frac{\mu_0 I h dx}{2\pi x}$

Flux through the entire cross section is

$$\phi = \frac{\mu_0 h I}{2\pi}\int_a^b \frac{dx}{x} = \frac{\mu_0 h}{2\pi}\,\ell n\left(\frac{b}{a}\right)\cdot I$$

Total flux linked with all turns of the coil

$$\phi_0 = N\phi = \frac{\mu_0 N h\,\ell n(b/a)}{2\pi}\cdot I$$

Comparing with $\qquad \phi = MI$

$$M = \frac{\mu_0 N h\,\ell n(b/a)}{2\pi}$$

51. Current in coil 1 will attain its steady value $I_{01} = \frac{V}{R_1}$ after infinite time.

Let current in the two coils be i_1 and i_2 at time t. Using Kirchhoff's loop rule to the second coil we can write-

$$M\frac{di_1}{dt} + L_2\frac{di_2}{dt} - R_2 i_2 = 0$$

$$\Rightarrow \quad M di_1 + L_2 di_2 = R_2 i_2 dt$$

$$\Rightarrow \quad M\int_0^{I_{01}} di + L_2\int_0^{I_{02}} di_2 = R_2\int_{t=0}^{\infty} i_2 dt$$

$$\Rightarrow \quad M I_{01} + L_2 I_{02} = R_2 Q_0$$

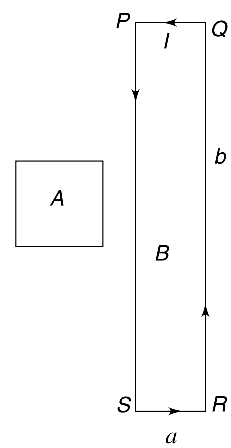
But $\quad I_{02} = 0$ and $I_{01} = \dfrac{V}{R_1}$

$$\therefore \quad M\frac{V}{R_1} = R_2 Q_0$$

$$\Rightarrow \quad M = \frac{R_1 R_2 Q_0}{V}$$

52. To calculate the mutual inductance of the pair of loops, let us assume that there is a current I in the bigger loop. Field due to QP and SR will be very small at the location of loop A due to large distance. Loop A is very close to arm PS and RQ. We can consider PS and RQ to be of infinite length for writing magnetic field at location of A.

$$B_{PS} = \frac{\mu_0 I}{2\pi a} \otimes \quad \text{and} \quad B_{RQ} = \frac{\mu_0 I}{2\pi(2a)} \odot$$



$$\therefore \quad B = \frac{\mu_0 I}{4\pi a} \otimes$$

Since loop A is small, we can consider the field to be uniform throughout its area.

$$\therefore \quad \phi_A = x^2 \cdot B = \frac{\mu_0 I}{4\pi a} x^2$$

Comparing with $\phi = MI$

Mutual inductance of coils $M = \dfrac{\mu_0 x^2}{4\pi a}$

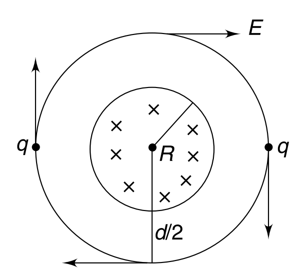
M does not change if loop A is made primary.

$\therefore$ emf induced in B when current is made to change in A is

$$\varepsilon_B = M\frac{di_A}{dt} = \frac{\mu_0 x^2 \alpha}{4\pi a}$$

53. When the magnetic field is switched off, an electric field is induced. The electric field (E) on a circle of radius $\dfrac{d}{2}$ is given by $E \cdot 2\pi\left(\dfrac{d}{2}\right) = \pi R^2 \dfrac{\Delta B}{\Delta t}$

$$\therefore \quad E = \frac{R^2}{d}\frac{\Delta B}{\Delta t}$$



Electric force on each ball; $\quad F_e = Eq$

Impulse on each ball $= Eq\Delta t = q \cdot \dfrac{R^2}{d}(\Delta B)$

Angular impulse on each ball $= q \cdot \dfrac{R^2}{d}(\Delta B) \cdot \dfrac{d}{2}$

Total angular impulse on both balls $= qR^2 \Delta B$

Since $\quad \Delta B = B_0$

$\therefore \quad$ Angular Impulse $= qR^2 B_0$

Change in angular momentum = qR^2B_0

$$2 \cdot m\left(\frac{d}{2}\right)^2 \omega = qR^2B_0$$

$$\Rightarrow \qquad \omega = \frac{2qR^2B_0}{md^2}$$

54. The magnetic field is decreasing with time. Using Faraday's law(or Lenz's law) one can see that the emf induced in a closed path will be in clockwise sense. It means that the induced electric field is clockwise. Therefore, end A will reach point C.

Magnitude of induced electric field is-

$$E = \frac{l}{2}\left|\frac{dB}{dt}\right| = \frac{l}{2}\frac{B_0}{2} = \frac{B_0 l}{4} = \frac{B_0 \times 0.04}{4} = 1 \text{ V/m.}$$

$$\tau = 2Eq \times l$$

$$\therefore \qquad \alpha = \frac{2Eql}{I} = \frac{2Eql}{2 \times ml^2} = \frac{Eq}{ml}$$

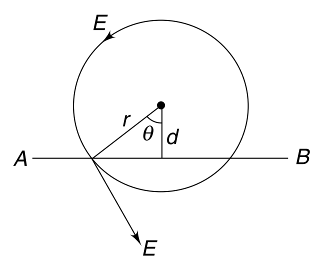
$$\therefore \qquad \theta = \frac{1}{2}\alpha t^2$$

$$t^2 = \frac{2\theta}{\alpha} = \frac{2 \times \frac{\pi}{2}}{\alpha} = \frac{\pi ml}{Eq}$$

$$= \frac{\pi l}{E} \times \frac{1}{q/m} = \frac{\pi \times 0.04}{1} \times \frac{1}{\frac{4\pi}{100}}$$

$$\therefore \qquad t^2 = 1 \qquad \therefore \quad t = 1 \text{ s.}$$

55. The electric field induced at a distance r from the centre is given by



$$\oint \vec{E}\,\vec{dl} = -\frac{d\phi}{dt}$$

$$\Rightarrow \qquad E2\pi r = \pi r^2\frac{dB}{dt} \quad \Rightarrow \quad E = \frac{r}{2}\frac{dB}{dt}$$

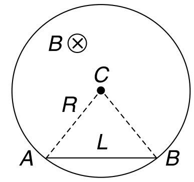
Component of E along AB is

$$E_{AB} = E\cos\theta = \frac{1}{2}(r\cos\theta)\frac{dB}{dt} = \frac{d}{2}\left(\frac{dB}{dt}\right) = a \text{ constant.}$$

The emf induced in wire AB is given by $E_{AB}\,L$ and it will not change even if the wire is displaced by a small amount parallel to itself.

56. Consider a triangular closed loop ABC

Area of the loop



$$A = \frac{1}{2} \times L \times \sqrt{R^2 - \left(\frac{L}{2}\right)^2}$$

Flux through loop ABC is

$$\phi = \frac{L}{4}\sqrt{4R^2 - L^2} \cdot B$$

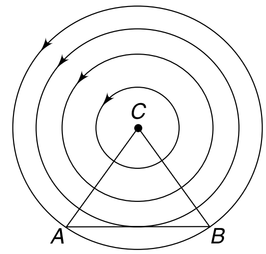
Emf induced in the loop is $\qquad \varepsilon = \dfrac{d\phi}{dt} = \dfrac{L}{4}\sqrt{4R^2 - L^2} \cdot \alpha$

Because of symmetry the induced field lines will be circular. It means field lines of E are perpendicular to CA and CB. The emf in the loop ABC can also be written as

$$\int_C^A \vec{E}\cdot\vec{dl} + \int_A^B \vec{E}\cdot\vec{dl} + \int_B^C \vec{E}\cdot\vec{dl} = \frac{L\alpha}{4}\sqrt{4R^2 - L^2}$$

$$\Rightarrow \quad 0 + \int_A^B \vec{E}\cdot\vec{dl} + 0 = \frac{L\alpha}{4}\sqrt{4R^2 - L^2}$$

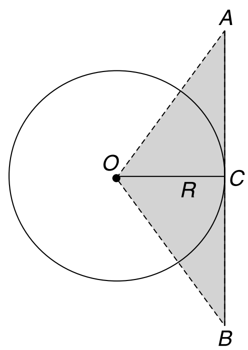
$$\therefore \quad \int_A^B \vec{E}\cdot\vec{dl} = \frac{L\alpha}{4}\sqrt{4R^2 - L^2}$$



Note: Alternative way of finding the answer can be found in the result obtained in last problem.

57. (a) Electric field lines are perpendicular to the rod at all points in Figure (a)

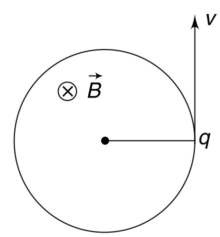
(b) Emf induced in AB = rate of change of flux through the shaded region.



$$\varepsilon = \left|\frac{d\phi}{dt}\right| = \frac{\pi R^2}{4}\cdot\alpha$$

58. $I_0 = nqf_0$. Where f_0 = frequency of rotation

$$I_0 = nq\frac{\omega_0}{2\pi} \quad \text{...(i)}$$



Due to change in magnetic flux an electric field is induced.

$$2\pi RE = \frac{d\phi}{dt} \quad \Rightarrow \quad E = \frac{\beta}{2\pi R}$$

It will be wise to assume that the magnetic field is assisting in providing the necessary centripetal force. The situation can be as shown in the figure.

If B is made to increase, it will create an electric field in anticlockwise sense. This field will speed up the particle.

Tangential acceleration $a_t = \frac{qE}{m} = \frac{qB}{2\pi Rm}$

Angular acceleration $$a = \frac{a_t}{R} = \frac{qB}{2\pi R^2 m}$$

Angular speed after one rotation is given by

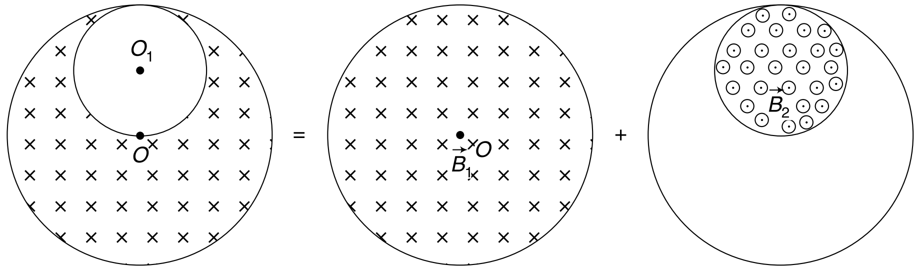
$$\omega^2 = \omega_0^2 + 2\alpha(2\pi)$$

Using (i)

$$\left(\frac{2\pi I}{nq}\right)^2 = \left(\frac{2\pi I_0}{nq}\right)^2 + \frac{2q\beta}{R^2 m}$$

$$\therefore \quad I = \sqrt{I_0^2 + \frac{n^2q^3\beta}{2\pi^2R^2m}}$$

59. We can consider the entire system of magnetic field to be superposition of fields as shown below.



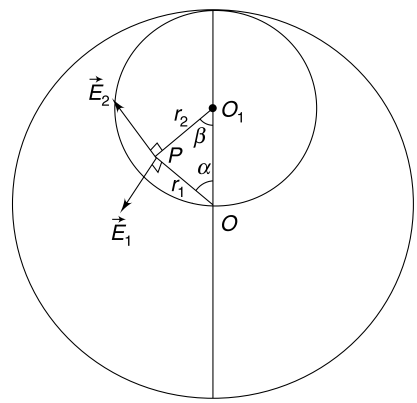
$$\frac{dB_1}{dt} = \frac{dB_2}{dt} = \alpha$$

Say, both fields are increasing.

Induced electric field inside a cylindrical region having time changing magnetic field is given by

$$E = \frac{r}{2}\left(\frac{dB}{dt}\right) \text{ where } r \text{ is distance from the axis of the cylinder.}$$

Induced Field at point P (inside cavity)



(i) Due to changing $\vec{B}_1$ is $\vec{E}_1$ (perpendicular to OP)

$$E_1 = \frac{r_1}{2}\frac{dB_1}{dt} = \frac{r_1}{2}k$$

(ii) Due to changing $\vec{B}_2$ is $\vec{E}_2$ (perpendicular to O_1P)

$$E_2 = \frac{r_2}{2}\frac{dB_2}{dt} = \frac{r_2}{2}k$$

Vector sum of E_1 and E_2 is the resultant field at P

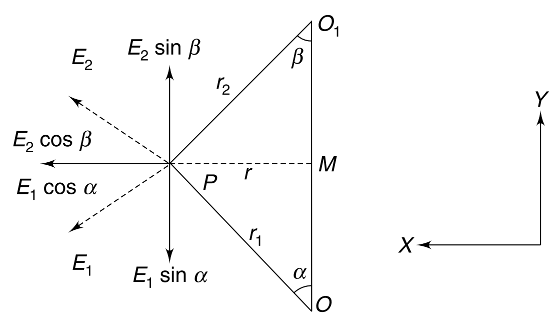
$$E_y = E_2 \sin\beta - E_1 \sin\alpha$$

$$= \frac{r_2}{2}k\frac{r}{r_2} - \frac{r_1}{2}k\frac{r}{r_1} = 0$$

$$E_x = E_1\cos\alpha + E_2\cos\beta$$

$$= \frac{r_1}{2}k\frac{OM}{r_1} + \frac{r_2}{2}k\frac{O_1M}{r_2}$$

$$= \frac{k}{2}(OM + O_1M) = \frac{kR}{4}$$



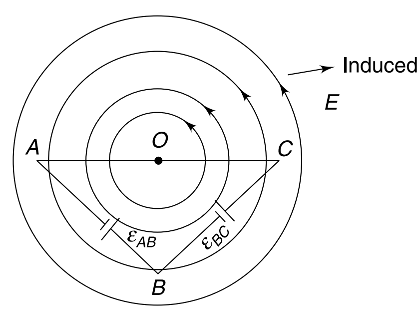
Hence, resultant field inside the cavity is uniform perpendicular to line OO_1

60. Faraday's law gives-

Induced $\quad$ $\text{emf} = \left|-\frac{d\phi}{dt}\right|$

$$\varepsilon = \frac{d}{dt}(B \times \text{Area})$$

$$= (\text{Area})\left(\frac{dB}{dt}\right) = \frac{1}{2}\ell^2 \cdot \alpha$$



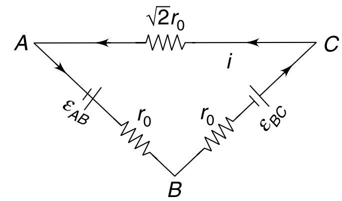
The time changing magnetic field induces an electric field having circular lines. Because there is no radial component of induced field E, there is no emf induced in the segment AC.

Segment AB and BC are symmetrical

$\therefore$ emf induced in AB and BC

$$\varepsilon_{AB} = \varepsilon_{BC} = \frac{\varepsilon}{2} = \frac{\ell^2\alpha}{4}$$

From point of view of current electricity, the circuit is as shown in figure.



Current in the triangle

$$i = \frac{\varepsilon}{2r_0 + \sqrt{2}r_0} = \frac{\ell^2\alpha}{2\sqrt{2}r_0(\sqrt{2}+1)}$$

$$V_{CA} = i\sqrt{2}r_0 = \frac{\ell^2\alpha}{2(\sqrt{2}+1)}$$

$$V_{BA} = \varepsilon_{AB} - ir_0$$

$$= \frac{\ell^2\alpha}{4} - \frac{\ell^2\alpha}{2\sqrt{2}(\sqrt{2}+1)} = \frac{\ell^2\alpha}{4(\sqrt{2}+1)}$$

61. Flux through the ring is $\phi = \vec{B}\cdot\vec{A} = (2\hat{j} + t^2\hat{k})\cdot(\pi R^2\hat{k}) = \pi R^2 t^2$

$\therefore$ Induced emf $|\varepsilon| = \dfrac{d\phi}{dt} = 2\pi R^2 t$

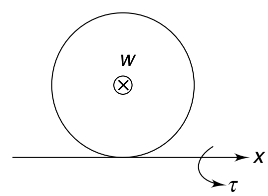
If induced electric field is E_{in} then

$$E_{\text{in}}2\pi R = \varepsilon$$

$$E_{\text{in}}2\pi R = 2\pi R^2 t$$

$$\therefore \quad E_{\text{in}} = Rt \qquad \text{...(i)}$$

Current in the ring is $i = \dfrac{\varepsilon}{r} = \dfrac{2\pi R^2 t}{r}$ [Clockwise]



Magnetic dipole moment $\vec{\mu} = (\pi R^2 i)(-\hat{k}) = \dfrac{2\pi^2R^4t}{r}(-\hat{k})$

Torque on the ring $\vec{\tau} = \vec{\mu}\times\vec{B} = \dfrac{4\pi^2R^4t}{r}(\hat{i})$

The ring will begin to topple when magnetic torque exceeds the torque due to weight about an axis tangential to the ring and parallel to x axis. In the Figure the weight (W) of the ring is acting into the plane of the Figure.

$$\frac{4\pi^2R^4t}{r} = mgR$$

$$\Rightarrow \quad t_0 = \frac{mg\cdot r}{4\pi^2R^3} = \frac{\pi\times 10\times\pi}{4\pi^2\times\left(\frac{1}{2}\right)^3} = 20 \text{ sec.}$$

Induced field at t_0 from equation (i) is

$$E_{\text{in}} = \frac{mgr}{4\pi^2R^2} = \frac{\pi\times 10\times\pi}{4\pi^2\left(\frac{1}{2}\right)^2} = 10 \text{ V/m}$$

Heat generated till t_0 is

$$H = \int_0^{t_0} i^2 r\,dt = \frac{4\pi^2R^4}{r}\int_0^{t_0} t^2\,dt = \frac{4}{3}\frac{\pi^2R^4}{r}t_0^3$$

$$= \frac{4}{3}\frac{\pi^2}{\pi}\times\left(\frac{1}{2}\right)^4\times(20)^3 = \frac{2\pi}{3}\text{ kJ}$$

62. Field due to one coil is $B = \mu_0\left(\dfrac{N}{\ell}\right)I$

Field in overlapping region of length $(\ell - x)$ is $= 2B$

$\therefore$ Energy = energy in overlapping region + energy in non overlapping region of outer solenoid + energy in non overlapping region of inner solenoid

$$E = \frac{(2B)^2}{2\mu_0}\cdot A_1(\ell - x) + \frac{B^2}{2\mu_0}[A_2x + (A_2 - A_1)(\ell - x)] + \frac{B^2}{2\mu_0}A_1x$$

$$\therefore \quad E = \frac{B^2}{2\mu_0}[3A_1\ell - 2A_1x + A_2\ell] = \frac{\mu_0 N^2 I^2}{2\ell^2}[3A_1\ell - 2A_1x + A_2\ell]$$

63. The outer coil is intersected by the inner coil's flux = BA_1

In time dt this flux is intercepted by fewer turns of the outer coil. Number of turns of outer coil which lose the flux due to inner coil in interval dt is = $\frac{N}{\ell}v\,dt$

$\therefore$ emf induced in outer coil

$$\varepsilon_{in} = BA_1\frac{N}{\ell}v = \frac{\mu_0 NIA_1Nv}{\ell^2} = \frac{\mu_0 N^2 A_1 vI}{\ell^2}$$

The other coil will also have same emf.

Put your own arguments for this.

64. Mechanical power supplied by external agent = Rate of change of magnetic field energy in the solenoids + Rate of work done by the power sources driving current in the two coils.

$$F\cdot v = \frac{dE}{dt} + \varepsilon_{in}\cdot I + \varepsilon_{in}\cdot I$$

$$= \frac{dE}{dx}\cdot\frac{dx}{dt} + 2\varepsilon_{in}\cdot I \qquad \left(\text{from Q. 62 } E = \frac{\mu_0 N^2 I^2}{2\ell^2}[3A_1\ell - 2A_1x + A_2\ell]\right)$$

$$= -\frac{\mu_0 N^2 I^2 A_1}{\ell^2}\cdot v + 2\frac{\mu_0 N^2 I^2 A_1 v}{\ell^2} = \frac{\mu_0 N^2 I^2 A_1 v}{\ell^2} \quad (\text{from Q. 63 we get } \varepsilon_{in})$$

$$\therefore \quad F = \frac{\mu_0 N^2 I^2 A_1}{\ell^2}$$

65. (a) Let the velocity at time t be v. Induced emf $\varepsilon = B\ell v$

$$\therefore \quad L\frac{di}{dt} = B\ell v \qquad \text{...(i)}$$

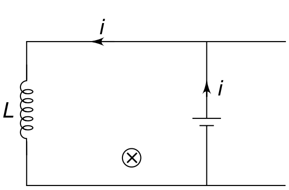
Magnetic force on the wire is = $i\ell B$

$$\therefore \quad m\frac{dv}{dt} = -i\ell B \qquad \text{...(ii)}$$

Differentiating equation (ii) with respect to time t

$$B\ell\frac{di}{dt} = -m\frac{d^2v}{dt^2}$$

Putting in (i)



$$L\frac{m}{B\ell}\frac{d^2v}{dt^2} = -B\ell v$$

$$\therefore \quad \frac{d^2v}{dt^2} = -\frac{B^2\ell^2}{mL}v$$

Solution to this equation is

$$v = v_0\sin(\omega t + \delta) \text{ where } \omega = \frac{B\ell}{\sqrt{mL}}$$

At $t = 0$; $v = v_0$

This means $\quad \delta = \frac{\pi}{2}$ and $v = v_0\cos(\omega t)$

(b) From (ii) $$i = -\frac{m}{\ell B}\frac{dv}{dt}$$

$$= \frac{mv_0\omega}{\ell B}\sin\omega t = \frac{mv_0}{\ell B}\frac{B\ell}{\sqrt{mL}}\sin\omega t$$

$$= v_0\sqrt{\frac{m}{L}}\sin\omega t$$

(c) $$L\frac{di}{dt} = B\ell v \Rightarrow L\frac{di}{dt} = B\ell\frac{dx}{dt}$$

$$\Rightarrow \quad L\,di = B\ell\,dx \Rightarrow L\int_0^i di = B\ell\int_0^x dx$$

$$\Rightarrow \quad Li = B\ell x \text{ where } x = \text{displacement}$$

Magnetic force on the conductor will be

$$F = ilB = \frac{B^2\ell^2}{L}x$$

$$\therefore \quad mv\frac{dv}{dx} = -\frac{B^2\ell^2}{L}\cdot x$$

$$\Rightarrow \quad \int_{v_0}^{v} v\,dv = -\frac{B^2\ell^2}{mL}\int_0^x x\,dx$$

$$\Rightarrow \quad v^2 = v_0^2 - \frac{B^2\ell^2}{mL}x^2 \Rightarrow v = \sqrt{v_0^2 - \frac{B^2\ell^2}{mL}x^2}$$

(d) $v = 0$ when $$x = \frac{v_0}{B\ell}\sqrt{mL}$$

At this instant $$\frac{di}{dt} = 0$$

But $$i = \frac{B\ell}{L}x = v_0\sqrt{\frac{m}{L}}$$

(e) Since $i \neq 0$, the conductor will experience magnetic force and will move towards left.

66. (a) Initial current in the circuit is $i_0 = \frac{E}{R}$

After S_1 is opened, the resistance in the circuit suddenly rises to $R_1 + R = R_1'$ (say) $= 1001R$
But the current will not change suddenly.

$\therefore$ Drop across R_1' is $= i_0R_1' = \frac{E}{R}1001R = 1001E$

$\therefore$ emf induced in the inductor $= 1000E$

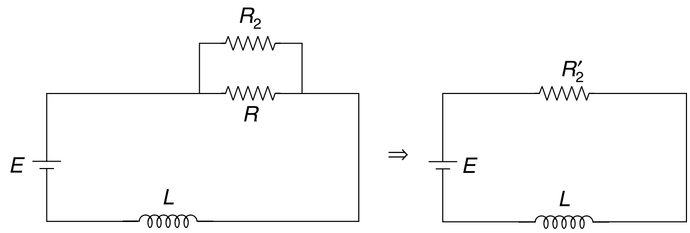
Potential drop across $R_1 = 1000E$

[$1000E$ plus a drop of E across R makes it $1001E$]

At $t\to\infty$: $$I = \frac{E}{1001R}$$

(b) Initial current $$i_0 = \frac{E}{R}$$

With S_1 closed and S_2 closed, the circuit changes to as shown in figure.



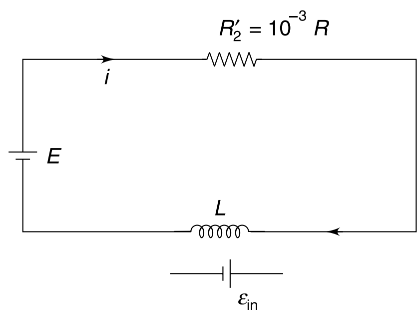
$$R_2' = \frac{RR_2}{R + R_2} = \frac{R_2}{1 + \dfrac{R_2}{R}} \simeq R_2 = 10^{-3} R$$

At time $t = 0^+$, the current is $\quad i_0 = \dfrac{E}{R}$

$\therefore$ Potential drop across R_2'

$$V_{R_2'} = \frac{E}{R}(10^{-3} R) = 10^{-3} E$$

At time '*t*'



i will increase to acquire final value of $I_\infty = \dfrac{E}{10^{-3} R} = 1000\,\dfrac{E}{R}$

$$E - R_2' i - L\frac{di}{dt} = 0$$

$$\Rightarrow \qquad L\frac{di}{dt} = E - R_2' i$$

$$\Rightarrow \qquad \int_{\frac{E}{R}}^{i} \frac{di}{E - R_2' i} = \frac{1}{L}\int_0^t dt$$

$$\Rightarrow \qquad \ell n(E - R_2' i) - \ell n\left(E - \frac{E R_2'}{R}\right) = -\frac{R_2'}{L} t$$

$$\Rightarrow \ell n(E - 10^{-3} Ri) - \ell n(E - 10^{-3} E) = 10^{-3}\,\frac{Rt}{L}$$

$$\Rightarrow \qquad \ell n(E - 10^{-3} Ri) - \ell n(E) = -\frac{Rt}{1000L}$$

$$\Rightarrow \qquad 1 - \frac{10^{-3} Ri}{E} = e^{-\frac{Rt}{1000L}}$$

$$\Rightarrow \qquad i = \frac{E}{10^{-3} R}\left[1 - e^{-\frac{Rt}{1000L}}\right]$$

Hence
$$i = \frac{1000E}{R}\left[1 - e^{-\frac{Rt}{1000L}}\right]$$

67. Decaying current in a LR circuit is given by

$$I = I_0 e^{-t/\tau} \simeq I_0\left[1 - \frac{\tau}{t}\right] \qquad \text{...(i)}$$

[$\because \; e^{-x} \simeq 1 - x$ for small x]

Notice that $\tau = \frac{L}{R}$ is expected to be a very large number owing to a small resistance of the superconducting ring.

From (i) $\frac{\Delta I}{\Delta t} = -\frac{I_0}{\tau}$

Negative sign indicates that ΔI is negative. Let's talk only about magnitude of change in current.

$$\frac{\Delta I}{I_0} = \frac{\Delta t}{\tau}$$

As per the question, no change in current was observed for a month.

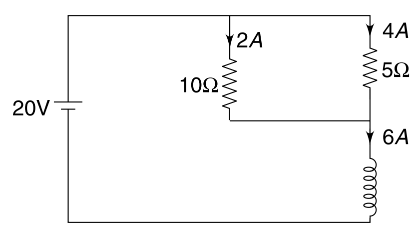
It means maximum value of $\frac{\Delta I}{I_0}$ is 0.01 when Δt = 1 month

$$\therefore \qquad 0.01 = \frac{30 \times 24 \times 60 \times 60}{\frac{L}{R}}$$

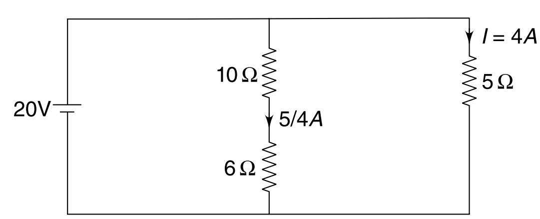
$$\therefore \qquad R = \frac{0.01 \times 0.5}{30 \times 24 \times 60 \times 60} = 1.9 \times 10^{-9}\ \Omega$$

68. (a) When the switch 'S' is in closed position for a long time the circuit is in steady condition and the inductor acts as a short circuit.

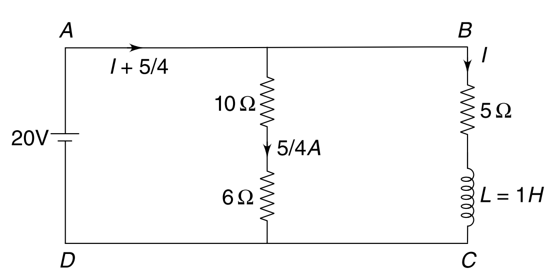
There is no potential difference across the inductor. It means no potential difference across 6Ω resistance. There is no current through 6Ω resistance. Effective circuit at $t = 0^-$ is as shown in Figure. Required answer is $6A$.



(b) At $t \to \infty$ once again the circuit will be in steady condition. $I = 4A$.



(c) At time 't' the circuit parameters are as shown in figure.



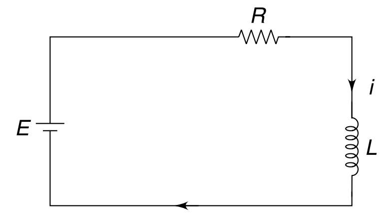
Applying Kirchhoff's Voltage law (KVL) in loop $ABCDA$:

$$5I + L\frac{dI}{dt} = 20$$

$$\int_6^I \frac{dI}{5I-20} = -\int_0^t dt \qquad \text{[At } t = 0 \text{ current through inductor is } 6A]$$

Solving $\quad I = 4 + 2e^{-5t}$

Current through cell $= \frac{5}{4} + I = \frac{21}{4} + 2e^{-5t}$



69. Just before closing S_2, the effective circuit is as shown.

$$i = \frac{E}{R} = \frac{20}{10} = 2A$$

Immediately, after closing S_2, the current through inductor will remain $i = 2A$. The effective circuit has been shown in figure.

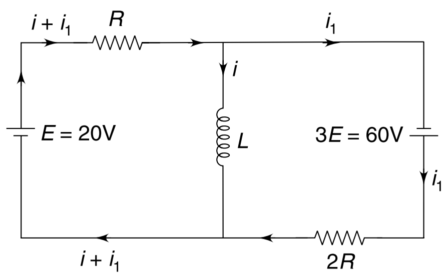
Applying KVL to the bigger loop

$$(i + i_1)R + i_1(2R) = 4E$$

$$3iR = 4E - iR$$

$$\Rightarrow \quad 30i_1 = 80 - 20 \qquad [\because \; iR = 2 \times 10 = 20]$$

$$\Rightarrow \quad i_1 = 2A$$



Applying KVL to loop containing L and cell of emf $3E$

$$3E + L\frac{di}{dt} = 2Ri_1$$

$$60 + 0.5\frac{di}{dt} = 40$$

$$\Rightarrow \quad \frac{di}{dt} = -40 \text{ A/s}$$

Negative sign says that current through the inductor is decreasing

Energy stored in magnetic field

$$U_B = \frac{1}{2}Li^2$$

$$\frac{dU_B}{dt} = Li\frac{di}{dt} = 0.5 \times 2 \times (-40) = -40 \text{ J/s}$$

Negative sign says that energy stored in the inductor is decreasing.

70. Initial energy in C_1 is $\quad U_1 = \frac{1}{2}C_0V_0^2$

Final energy in C_2 is $\quad U_2 = \frac{1}{2}\frac{C_0}{9}(3V_0)^2 = \frac{1}{2}C_0V_0^2$

Therefore, complete energy of C_1 gets transferred to C_2

The time period of LC circuit consisting of L and C_1 is

$$T_1 = 2\pi\sqrt{LC_0}$$

The entire energy of the capacitor can get transferred to the inductor in smallest interval of $t_1 = \frac{T_1}{4} = \frac{\pi}{2}\sqrt{LC_0}$

At time t_1 the current through L is maximum and switch S_1 is opened and S_2 is closed.

Now the time period of LC circuit having L and C_2 is

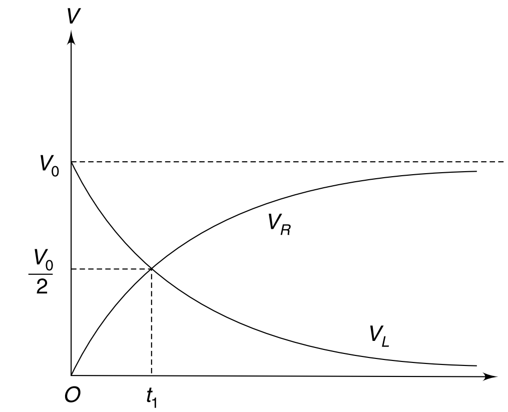
$$T_2 = 2\pi\sqrt{\frac{LC_0}{9}} = \frac{2\pi}{3}\sqrt{LC_0}$$

The inductor will transfer its entire energy to C_2 in smallest time given by

$$t_2 = \frac{T_2}{4} = \frac{\pi}{6}\sqrt{LC_0}$$

$\therefore$ $$T_{\min} = t_1 + t_2 = \frac{2\pi}{3}\sqrt{LC_0}$$

71. (a) Current $$i = i_0[1 - e^{-t/\tau}]$$



Where $$t_0 = \frac{V_0}{R} \quad \text{and} \quad \tau = \frac{L}{R}$$

$$V_R = iR = i_0R\,[1 - e^{-t/\tau}] = V_0[1 - e^{-t/\tau}]$$

$$V_L = V_0 - V_R = V_0e^{-t/\tau}$$

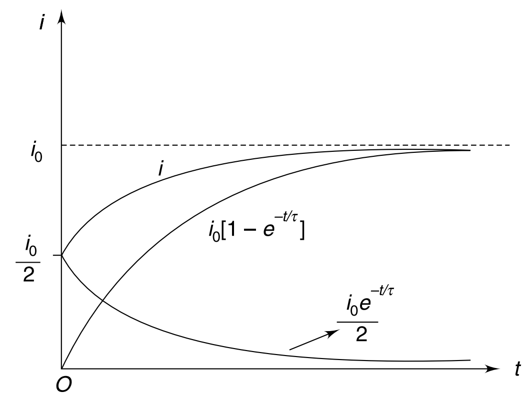
Graph is as shown.

(b) The two curves intersect when

$$V_R = V_L \;\Rightarrow\; V_0[1 - e^{-t/\tau}] = V_0e^{-t/\tau}$$

$$\Rightarrow \quad e^{-t/\tau} = \frac{1}{2} \;\Rightarrow\; \frac{t}{\tau} = \ell n\,2$$

$$\Rightarrow \quad t_1 = \tau\ell n2 = \frac{L}{R}\,\ell n2$$



(c) $$L\frac{di}{dt} + Ri = V_0$$

$$\int \frac{di}{V_0 - R_i} = \frac{1}{L}\int dt$$

$$[\ell n\,[V_0 - Ri]]_{\frac{i_0}{2}}^{i} = -\frac{R}{L}\,[t]_0^t \qquad \left[\because \text{ at } \begin{array}{l} t = 0 \\ i = \frac{i_0}{2} \end{array}\right]$$

$$\Rightarrow \quad \ell n\left(\frac{V_0 - Ri}{V_0 - R\,\frac{i_0}{2}}\right) = -\frac{t}{\tau} \qquad \left[\tau = \frac{L}{R}\right]$$

$$\Rightarrow \quad V_0 - Ri = \left(V_0 - R\,\frac{i_0}{2}\right)e^{-t/\tau}$$

$$\Rightarrow \quad i = i_0[1 - e^{-t/\tau}] + \frac{i_0}{2}\,e^{-t/\tau} \qquad \left[\text{where } i_0 = \frac{V_0}{R}\right]$$

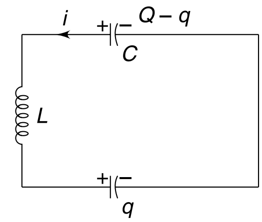
72. Let charge on two capacitor at some time be $(Q - q)$ and q as shown in Figure.

Energy is conserved in the circuit and magnetic energy will be maximum (i.e. the current will be maximum) when energy stored in the two capacitor is minimum.

Energy stored in capacitors is

$$E = \frac{(Q-q)^2}{2C} + \frac{q^2}{2C} = \frac{1}{2C}\,[2q^2 - 2Qq + Q^2]$$

E is minimum when $$\frac{dE}{dq} = 0$$



$$\Rightarrow \quad 4q - 2Q = 0 \quad \Rightarrow \quad q = \frac{Q}{2}$$

$$E_{\min} = \frac{Q^2}{4C}$$

Magnetic energy is $\frac{1}{2} Li_{\max}^2 = \frac{Q^2}{2C} - \frac{Q^2}{4C}$

$$\Rightarrow \quad i_{\max} = \frac{Q}{\sqrt{2LC}}$$

73. When current through L is maximum $\frac{dI}{dt} = 0$

$\Rightarrow$ emf induced in L is zero at that instant. At this time a total charge $Q = 9CV_0 - 2CV_0 = 7CV_0$ is distributed on the two capacitors so that potential difference across them is same (V)

$$CV + 3CV = 7CV_0$$

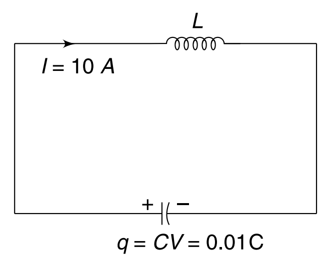
$$\Rightarrow \quad V = \frac{7V_0}{4}$$

Energy conservation

$$\frac{1}{2} LI_{\max}^2 = \left[\frac{1}{2}C\,(2V_0)^2 + \frac{1}{2}3C\,(3V_0)^2\right] - \left[\frac{1}{2}CV^2 + \frac{1}{2}\,(3C)\,V^2\right]$$

Solving $I_{\max} = \frac{5V_0}{2}\sqrt{\frac{3C}{L}}$

74. (a) At time $t = 0^+$, situation is as shown below



$$q = CV = 100 \times 10^{-6} \times 100 = 0.01 \text{ C}$$

Energy stored in the capacitor

$$U_1 = \frac{1}{2} CV^2 = \frac{1}{2} \times 100 \times 10^{-6} \times 100^2 = 0.5\,\text{J}$$

Energy stored in the Inductor

$$U_2 = \frac{1}{2} LI^2 = \frac{1}{2} \times 0.03 \times 10^2 = 1.5\,\text{J}$$

When charge on capacitor become maximum energy stored in is $= U_1 + U_2 = 2\,\text{J}$

$$\frac{q_0^2}{2C} = 2$$

$$\Rightarrow \quad q_0^2 = 4 \times 100 \times 10^{-6}$$

$$\Rightarrow \quad q_0 = 2 \times 10^{-2} = 0.02\,C$$

(b) Angular frequency of oscillation $\omega = \frac{1}{\sqrt{LC}} = \frac{10^3}{\sqrt{3}}$ rad/s

Charge on capacitor changes with time according to $q = q_0 \sin(\omega t + \delta)$

At $t = 0$, $q = 0.01$

$$\therefore \quad 0.01 = 0.02 \sin \delta$$

$$\Rightarrow \quad \delta = \pi/6 \quad \text{or} \quad 5\pi/6$$

We take $\delta = \frac{5\pi}{6}$ as charge on the capacitor is initially decreasing (look at the direction of current initially)

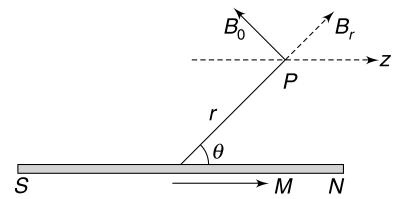
$$\therefore \quad q = 0.02 \sin\left(\omega t + \frac{5\pi}{6}\right)$$

q is zero when $\omega t + \frac{5\pi}{6} = \pi$

$$\Rightarrow \quad \omega t = \pi/6$$

$$\frac{10^3}{\sqrt{3}}\, t = \frac{\pi}{6} \quad \Rightarrow \quad t = \frac{3.14 \times 1.73}{6 \times 10^3} = 0.9 \text{ ms}$$

75. (a) We know the two components of field due to a dipole as



$$B_r = \frac{\mu_0}{4\pi}\frac{2M\cos\theta}{r^3} \quad \text{and} \quad B_\theta = \frac{\mu_0}{4\pi}\frac{M\sin\theta}{r^3}$$

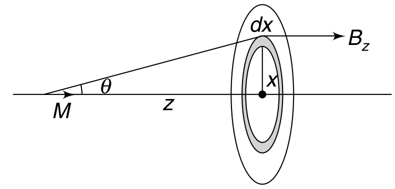
$\therefore$
$$B_z = B_r\cos\theta - B_\theta\sin\theta$$

$$= \frac{\mu_0}{4\pi r^3}[2\cos^2\theta - \sin^2\theta] \qquad \text{...(i)}$$

(b) If we consider the point P to be at a distance x from the centre of the ring, then

$$\sin\theta = \frac{x}{\sqrt{z^2 + x^2}}; \ \cos\theta = \frac{z}{\sqrt{z^2 + x^2}}$$

$\therefore$ equation (i) becomes



$$B_z = \frac{\mu_0 M}{4\pi(z^2 + x^2)^{3/2}}\left[\frac{3z^2}{x^2 + z^2} - 1\right]$$

Flux through a region lying between radius $x + dx$ and x is

$$d\phi = 2\pi x dx B_z$$

$\therefore$ Flux through ring is

$$\phi = \int_{x=0}^{x=R} 2\pi x\,dx \cdot B_z$$

$$= \frac{\mu_0 M}{2}\left[3z^2\int_0^R \frac{x\,dx}{(z^2 + x^2)^{5/2}} - \int_0^R \frac{x\,dx}{(z^2 + x^2)^{3/2}}\right]$$

$$= \frac{\mu_0 M}{2}\frac{R^2}{(R^2 + z^2)^{3/2}}$$

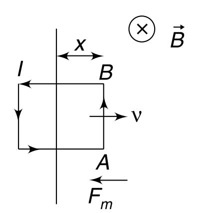
[Integration can be done by method of substitution. Put $z^2 + x^2 = t$]

(c)
$$E_{\text{in}} = \left|\frac{d\phi}{dt}\right| = \left|\frac{d\phi}{dz}\frac{dz}{dt}\right| = v\left|\frac{d\phi}{dz}\right|$$

$$= \frac{3\mu_0 MR^2 vz}{2[R^2 + z^2]^{5/2}}$$

76. Let the speed of the loop be v when it penetrates x inside the field region. Motional emf in arm AB is $\varepsilon = B\ell v$

$\therefore$ Current
$$I = \frac{B\ell v}{R}$$



Magnetic force (opposing the motion) on arm AB is $F_m = I\ell B = \dfrac{B^2\ell^2 v}{R}$

$\therefore$
$$M\frac{dv}{dt} = -\frac{B^2\ell^2 v}{R}$$

$\therefore$
$$\frac{dv}{dt} = -\frac{B^2\ell^2 v}{MR} \qquad \text{...(i)}$$

If we write acceleration as $\dfrac{dv}{dt} = v\dfrac{dv}{dx}$ then

$$v\frac{dv}{dx} = -\frac{B^2\ell^2}{MR}v$$

$\Rightarrow$
$$dv = -\frac{B^2\ell^2}{MR}dx$$

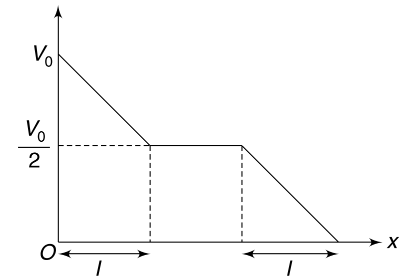
$\Rightarrow \qquad \int_{v_0}^{v} dv = -\left(\frac{B^2\ell^2}{MR}\right)\int_0^x dx \qquad [v_0 = \text{initial speed}]$

$$v = v_0 - \left(\frac{B^2\ell^2}{MR}\right)x \qquad \text{...(ii)}$$

Speed decreases linearly with x.

When the loop in completely inside the field region; there is no current, no force and hence no change in speed. When arm AB begins to exit the field region, emf is induced again and the loop beings to retard once again.

If we wish that the loop just manages to come out of the field region then its $v - x$ graph will look as shown.



v vs x graph will have the same slope as given by equation (ii), even when the loop is exiting. Hence, the speed when it is completely inside the field region is $\frac{v_0}{2}$

From (i) $\qquad \frac{dv}{dt} = -\frac{B^2\ell^2}{MR}v$

$\therefore \qquad \frac{dv}{v} = -\frac{B^2\ell^2}{MR}dt$

$$\int_{v_0}^{v_0/2}\frac{dv}{v} = -\frac{B^2\ell^2}{MR}\int_0^t dt$$

$$\ell n\left(\frac{1}{2}\right) = -\frac{B^2\ell^2}{MR}t$$

$\therefore \qquad t = \frac{MR\ell n2}{B^2\ell^2}$

77. (a) velocity of block $= v_0$

Angular velocity of the pulley $\qquad \omega = \frac{v_0}{a}$

Emf developed across the circumference and centre $\varepsilon = \frac{1}{2}B\omega a^2$

Current in the resistance, $\qquad i = \frac{B\omega a^2}{2R}$

Power dissipated in R is

$$P_R = i^2R = \frac{B^2\omega^2a^4}{4R^2}$$

Mechanical power delivered to pulley = rate of change of its kinetic energy

$$P_{\text{mech}} = \frac{d}{dt}\left(\frac{1}{2}I\omega^2\right) = I\omega\frac{d\omega}{dt} \qquad \text{...(i)}$$

$P_R + P_{\text{mech}}$ = power delivered by string tension to the pulley

$$= T\cdot a\cdot\omega$$

$$= \left[Mg - M\frac{dv}{dt}\right]a\omega \qquad \left[\because\ T = Mg - M\frac{dv}{dt}\right]$$

$\therefore \qquad \frac{B^2\omega^2a^4}{4R^2} + I\omega\frac{d\omega}{dt} = Mga\omega - Ma\omega\frac{d(\omega a)}{dt} \qquad [\because\ v = \omega a]$

$\Rightarrow \qquad [I + M\cdot a^2]\frac{d\omega}{dt} = Mga - \frac{B^2a^4\omega}{4R^2}$

Acceleration of mass M is $\left(a\frac{d\omega}{dt}\right)$

$$\therefore \qquad a_M = \frac{Mga^2 - \dfrac{B^2a^5\omega}{4R^2}}{I + Ma^2}$$

When $$\omega = \frac{v_0}{a}$$

$$a_M = \frac{Mga^2 - \dfrac{B^2a^4v_0}{4R^2}}{I + Ma^2} \qquad \text{...(i)}$$

(b) When the block acquires terminal speed $a_M = 0$

$$\Rightarrow \qquad Mga^2 = \frac{B^2a^4v_T}{4R^2}$$

$$\Rightarrow \qquad v_T = \frac{4MgR^2}{B^2a^2}$$

(c) When $\omega \cdot a = \dfrac{v_T}{2}$

$$\omega = \frac{2MgR^2}{B^2a^3}$$

$$\frac{d\omega}{dt} = \frac{a_M}{a} = \frac{Mga - \dfrac{B^2a^3}{4R^2}\cdot\dfrac{2MgR^2}{B^2a^2}}{I + Ma^2}$$

$$= \frac{Mga}{2(I + Ma^2)}$$

From (i) $P_{\text{mech}} = I\omega \dfrac{d\omega}{dt}$

$$= \frac{2IMgR^2}{B^2a^3}\cdot\frac{Mga}{2(I + Ma^2)}$$

$$= \frac{I\cdot M^2g^2R^2}{B^2a^2(I + Ma^2)}$$

78. Let ϕ = flux through the loop due to current I_0 and I = instantaneous current in the loop. The emf induced in the loop will be

$$\varepsilon = -\frac{d\phi}{dt} - L\frac{dI}{dt}$$

$$\Rightarrow \qquad IR = -\frac{d\phi}{dt} - L\frac{dI}{dt} \quad \Rightarrow \quad I\,dt = -\frac{1}{R}\,d\phi - \frac{L}{R}\,dI$$

$$\Rightarrow \qquad \int I\,dt = -\frac{1}{R}\int d\phi - \frac{L}{R}\int dI$$

$$\Rightarrow \qquad \Delta q = -\frac{\Delta\phi}{R} - \frac{L\Delta I}{R}$$

$$|\Delta q| = \frac{\Delta\phi + L\Delta I}{R}$$

The loop is at rest initially as well as finally; hence

$$\Delta I = I_f - I_i = 0 - 0 = 0$$

$$\therefore \qquad |\Delta q| = \left|\frac{\Delta\phi}{R}\right| = \frac{|\phi_f - \phi_i|}{R} \qquad \text{...(i)}$$

Flux in initial as well as final position can be obtained by integrating the flux over a strip of width dx.

$$d\phi = \frac{\mu_0 I_0}{2\pi x}\, a\, dx$$

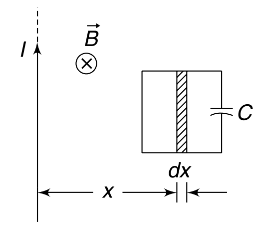
$$\phi_i = \frac{\mu_0 I_0 a}{2\pi}\int_a^{2a}\frac{dx}{x} = \frac{\mu_0 I_0 a}{2\pi}\,\ell n2$$

$$\phi_f = -\frac{\mu_0 I_0 a}{2\pi}\int_{2a}^{3a}\frac{dx}{x} = -\frac{\mu_0 I_0 a}{2\pi}\,\ell n\left(\frac{3}{2}\right) = \frac{\mu_0 I_0 a}{2\pi}\,\ell n\left(\frac{2}{3}\right)$$

$$|\phi_f - \phi_i| = \frac{\mu_0 I_0 a}{2\pi}\,\ell n3$$

$$\therefore \qquad |\Delta q| = \frac{\mu_0 I_0 a\,\ell n3}{2\pi\cdot R}$$

79. (a) Magnetic field due to straight wire at a distance x from it is $B = \dfrac{\mu_0 I}{2\pi x}$



Flux linked with a strip of area $d\cdot(dx)$ [see figure] will be

$$d\phi = \frac{\mu_0 I\cdot d}{2\pi}\,\frac{dx}{x}$$

Flux through the loop

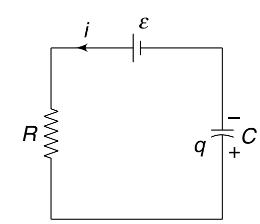
$$\phi = \int d\phi = \frac{\mu_0 I\cdot d}{2\pi}\int_a^{2a}\frac{dx}{x} = \frac{\mu_0 I\cdot d}{2\pi}\,\ell n2$$

$$\therefore \qquad \phi = \frac{\mu_0 \alpha d\,\ell n2}{2\pi}\cdot t \qquad [\because \; I = \alpha t]$$

Flux is time dependent. An emf is induced in the loop. Magnitude of emf is

$$\varepsilon = \frac{d\phi}{dt} = \frac{\mu_0 \alpha\cdot d\cdot \ell n2}{2\pi}$$

From Lenz's law the sense of emf is so as to drive an anticlockwise current in the loop. Let charge on the capacitor and current in the loop be q and i at any time t



$$iR + \frac{q}{C} = \varepsilon$$

$$\therefore \qquad R\frac{dq}{dt} + \frac{q}{C} = \varepsilon$$

$$\therefore \qquad \int_0^q \frac{dq}{\varepsilon - \dfrac{q}{C}} = \frac{1}{R}\int_0^t dt$$

$$\Rightarrow \qquad \left[\ell n\left(\varepsilon - \frac{q}{C}\right)\right]_0^q = -\frac{1}{RC}\,t$$

$$\Rightarrow \qquad \ell n\left(\varepsilon - \frac{q}{C}\right) - \ell n\varepsilon = -\frac{t}{RC} \;\Rightarrow\; \ell n\left(1 - \frac{q}{C\varepsilon}\right) = -\frac{t}{RC}$$

$$\therefore \qquad q = C\varepsilon[1 - e^{-t/RC}] = \frac{\mu_0 \alpha\cdot d\cdot \ell n2C}{2\pi}(1 - e^{-t/RC})$$

(b) $$i = \frac{dq}{dt} = \frac{\mu_0 \alpha \cdot d \cdot \ell n2}{2\pi R} e^{-t/RC}$$

$\therefore$ Heat generated

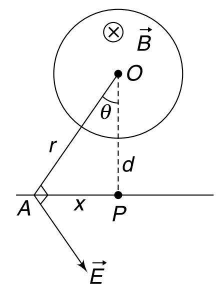
$$H = \int_0^t i^2 R\, dt = \left(\frac{\mu_0 \alpha \cdot d \cdot \ell n2}{2\pi R}\right)^2 \cdot R \int_0^t e^{-\frac{2t}{RC}} dt$$

$$= \left(\frac{\mu_0 \alpha \cdot d \cdot \ell n2}{\pi}\right)^2 \cdot \frac{C}{8}\left[1 - e^{-\frac{2t}{RC}}\right] \Rightarrow H = \frac{\varepsilon^2 C}{2}\left(1 - e^{-\frac{2t}{RC}}\right)$$

(d) The time changing current in the loop opposes the current in straight wire. The source driving the current in the straight wire has to do more work maintaining or increasing the current. This work done provides energy to the capacitor and produces heat as well.

80. (i) The induced electric field will be perpendicular to OP at all points in this path. Hence the particle experiences no force and work done in moving it to infinity is zero.

(ii) Induced electric field at a point A shown in the figure can be calculated as



$$E \cdot 2\pi r = \pi a^2 \frac{dB}{dt}$$

$$\Rightarrow \quad E = \frac{a^2 \alpha}{2r}$$

Electric force on the charge along AP is

$$F_e = qE\cos\theta = \frac{qa^2 \alpha \cdot d}{2r^2}$$

The external agent must apply equal and opposite force to keep the charge moving without gaining any kinetic energy. Work done by the external agent in small displacement dx will be

$$dW = \frac{qa^2 \alpha d}{2r^2} dx$$

But $$x = d\tan\theta \Rightarrow dx = d\sec^2\theta\, d\theta \text{ and } r = d\sec\theta$$

$$\therefore \quad dW = \frac{qa^2\alpha d}{2(d\sec\theta)^2} \cdot d\sec^2\theta\, d\theta = \frac{qa^2\alpha}{2} d\theta$$

$$\therefore \quad W = \frac{qa^2\alpha}{2}\int_0^{\pi/2} d\theta = \frac{\pi q a^2 \alpha}{4}$$

81. Charge on capacitor $Q_0 = CV_0$

After the switch is put to position 2, there is a current through the bar and thereby a magnetic force acts on it, pushing it away from the capacitor. The bar accelerates, its speed increases. Now, there is an induced emf in the bar which is increasing with increasing speed. At the instant induced emf becomes equal to the potential difference across the capacitor, the current will become zero. At this instant speed of the bar is maximum. As per the question, the rails come to an end and the bar is thrown out.

Let q_0 = charge on the capacitor when the bar acquires maximum speed ($u_{\max}$).

$$BLu_{\max} = \frac{q_0}{C} \qquad \text{...(i)}$$

Force equation for moving bar is

$$m\frac{du}{dt} = BiL$$

$$\Rightarrow \quad m\frac{du}{dt} = BL\left(-\frac{dq}{dt}\right) \Rightarrow mdu = -BLdq$$

$$\Rightarrow \qquad m\int_0^{u_{max}} du = -BL\int_{Q_0}^{q_0} dq \quad \Rightarrow \quad mu_{max} = BL(Q_0 - q_0)$$

$$\Rightarrow \qquad mu_{max} = BLQ_0 - B^2L^2Cu_{max}$$

$$\Rightarrow \qquad u_{max} = \frac{BLQ_0}{m + B^2L^2C} = \frac{BLCV_0}{m + B^2L^2C}$$

$\therefore$ Kinetic energy
$$K = \frac{1}{2}mu_{max}^2 = \frac{m}{2}\left(\frac{BLCV_0}{m + B^2L^2C}\right)^2$$

Energy spent by the battery in charging the capacitor is $Q_0V_0 = CV_0^2$

$\therefore$ Efficiency is

$$\eta = \frac{K}{CV_0^2} = \frac{mB^2L^2C}{2(m + B^2L^2C)^2}$$

82. Number of turns in a layer of radius r and thickness dr is $dN = \frac{N\,dr}{(b-a)}$

Field produced due to such a layer is

$$dB = \frac{\mu_0 dNI}{L} = \frac{\mu_0 NI}{(b-a)L}dr$$

For $r < a$, the resultant field due to all layers is

$$B = \int_{r=a}^{r=b} dB = \frac{\mu_0 NI}{(b-a)L}(b-a) = \frac{\mu_0 NI}{L}$$

For $a < r < b$

$$B = \int_r^b dB = \frac{\mu_0 NI}{(b-a)L}(b-r)$$

For $r > b$ $\qquad B = 0$

$\therefore$ Flux linked with a single turn of radius r is

$$\phi_0 = \frac{\mu_0 NI}{L}\pi a^2 + \frac{\mu_0 NI}{(b-a)L}\int_a^r (b-r)\,2\pi r\,dr$$

$$= \frac{\mu_0 \pi NI(3br^2 - 2r^3 - a^3)}{3(b-a)L}$$

$\therefore$ Total flux linked with the coil is

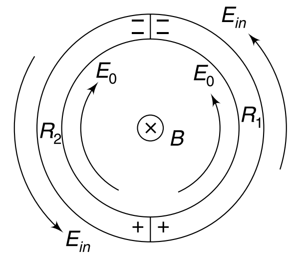
$$\phi = \int_{r=a}^{b} \phi_0\, dN = \frac{\mu_0 \pi N^2 I}{3(b-a)^2 L}\int_a^b (3br^2 - 2r^3 - a^3)\,dr$$

$$= \frac{\mu_0 \pi N^2(3a^2 + 2ab + b^2)}{6L} I$$

$$\Rightarrow \qquad L_{self} = \frac{\mu_0 \pi N^2(3a^2 + 2ab + b^2)}{6L}$$

83. (a) Current (and hence current density) in the entire loop must be same.

From microscopic form of Ohm's law we can write



$$\sigma_1 E_1 = \sigma_2 E_2$$

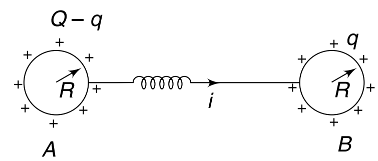
$$\Rightarrow \quad R_2 E_1 = R_1 E_2 \quad \Rightarrow \quad \frac{E_1}{E_2} = \frac{R_1}{R_2}$$

(b) The induced electric field must be uniform everywhere in the circular conductor. It is given by

$$2\pi a E_{\text{in}} = \pi a^2 \frac{dB}{dt} \quad \Rightarrow \quad E_{\text{in}} = \frac{a}{2}\left(\frac{dB}{dt}\right)$$

There is accumulation of charge at the junctions which produce additional electric field in the conductor. If $R_1 > R_2$ then $E_1 > E_2$. In this case the charge at the upper junction is negative and charge at the lower junction will be positive. In figure E_0 is electric field due to charge.

84. Let q = charge on B at time 't'



$$V_A - V_B = L\frac{di}{dt}$$

$$\Rightarrow \quad K\frac{Q-q}{R} - K\frac{q}{R} = L\frac{d^2q}{dt^2}$$

[Spheres are at large distance. Hence charge on one does not affect the potential of other]

$$\Rightarrow \quad \frac{d^2q}{dt^2} = -\frac{2K}{LR}\left[q - \frac{Q}{2}\right] \qquad \text{...(i)}$$

Let $q - \frac{Q}{2} = x$ which means $\frac{d^2q}{dt^2} = \frac{d^2x}{dt^2}$

In terms of x, the differential equation (i) becomes

$$\frac{d^2x}{dt^2} = -\frac{2K}{LR}x$$

Solution to this equation is

$$x = x_0 \sin(\omega t + \delta) \qquad \left[\omega = \sqrt{\frac{2K}{LR}}\right]$$

$$\Rightarrow \quad q - \frac{Q}{2} = x_0 \sin(\omega t + \delta)$$

$$\Rightarrow \quad q = \frac{Q}{2} + x_0 \sin(\omega t + \delta)$$

Current $\quad i = \frac{dq}{dt} = x_0 \omega \cos(\omega t + \delta)$

Just after closing the switch (at $t = 0^+$) the current is zero

$$\therefore \quad \delta = \frac{\pi}{2}$$

$$\therefore \quad q = \frac{Q}{2} + x_0 \cos \omega t$$

Also, at $t = 0$, $q = 0$

$$\therefore \quad x_0 = -Q/2$$

$$\therefore \quad q = \frac{Q}{2}[1 - \cos \omega t]$$

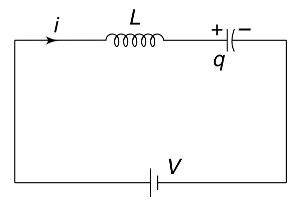
$$q = \frac{Q}{2} \text{ means } \cos \omega t = 0$$

$$\Rightarrow \qquad \omega t = \frac{\pi}{2}$$

$$\Rightarrow \qquad \sqrt{\frac{2K}{LR}}\, t = \frac{\pi}{2}$$

$$\Rightarrow \qquad t = \frac{\pi}{2}\sqrt{\frac{LR}{2K}}$$

85. Let i be current and q charge on the capacitor at time 't'



$$L\frac{di}{dt} + \frac{q}{C} = V$$

Differentiating wrt time we get $L\dfrac{d^2i}{dt^2} = -\dfrac{i}{C}$

Solution to this equation is $i = i_0 \sin(\omega t + \delta)$ $\qquad \left[\omega = \dfrac{1}{\sqrt{LC}}\right]$

At $t = 0$, $\quad i = 0 \qquad \therefore \quad \delta = 0$

$$\therefore \qquad i = i_0 \sin \omega t \qquad \text{...(i)}$$

$$\Rightarrow \qquad \frac{dq}{dt} = i_0 \sin \omega t$$

$$\Rightarrow \qquad \int dq = i_0 \int \sin \omega t\, dt$$

$$\Rightarrow \qquad q = -\frac{i_0}{\omega}\cos \omega t + c$$

At $t = 0$, $q = 0 \qquad \therefore \quad c = \dfrac{i_0}{\omega}$

$$\therefore \qquad q = \frac{i_0}{\omega}[1 - \cos \omega t] \qquad \text{...(ii)}$$

Maximum charge on capacitor $\qquad q_0 = \dfrac{2i_0}{\omega}$

Charge is maximum when entire work done by the battery is stored as energy in the electric field of the capacitor.

$$Vq_0 = \frac{q_0^2}{2C}$$

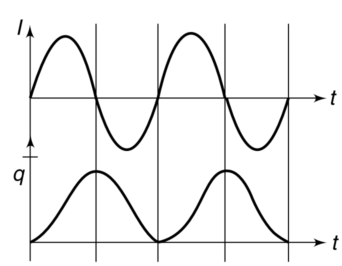
$$\therefore \qquad q_0 = 2CV$$

When charge on capacitor is maximum, potential difference across it is 2 V. Induced emf in the inductor at this instant is V

$$\therefore \qquad q = CV[1 - \cos \omega t] = 2CV \sin^2\left(\frac{\omega t}{2}\right)$$

$$i = i_0 \sin \omega t$$

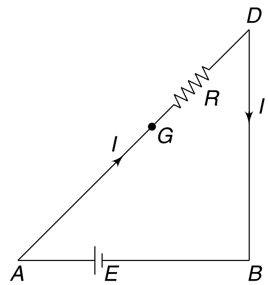
Graphs are as shown.



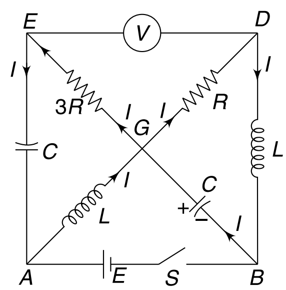
86. (a) There is no current through capacitors, voltmeters and inductors are zero resistance, when the circuit is in steady state. The effective circuit is as shown.

$$V_E = V_G \quad \text{and} \quad V_G - V_D = E$$

$\therefore$ Reading of voltmeter $= E$



(b) The current through inductors and the voltages across the capacitors cannot change immediately. Current, before the switch is opened, is $I = \dfrac{E}{R}$ through both inductors.



After opening the switch, the current in R and the inductor between B and D must be same (since voltmeter does not conduct). It implies that current through R is still I and $V_G - V_D = RI = E$

The current I in GD will loop through $GDBG$. The current I in the inductor between A and G must loop through $AGEA$.

$\therefore$ $$V_{GE} = 3R \cdot I = 3E$$

$\therefore$ $$V_D - V_E = 2E$$

(c) When switch was opened the capacitor between A and E was uncharged. The circuit is effectively two disjoint loops – BGD and AGE

Energy stored in L and C gets dissipated in R.

$\therefore$ $$U_R = \frac{1}{2}LI^2 + \frac{1}{2}CE^2$$

$$= \frac{1}{2}L\left(\frac{E}{R}\right)^2 + \frac{1}{2}CE^2 = \frac{1}{2}E^2\left(\frac{L}{R^2} + C\right)$$

CHAPTER 12

Alternating Current

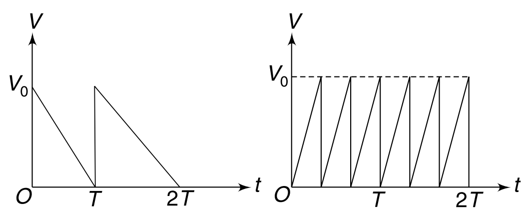
LEVEL 1

Q. 1: An electric appliance draws 3A current from a 200 V, 50 Hz power supply.

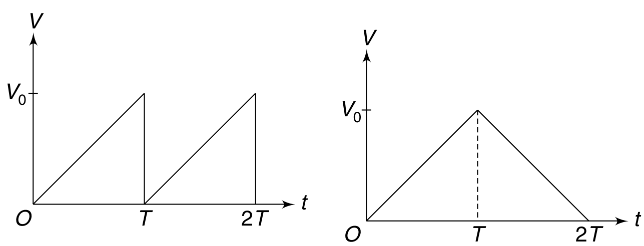
(a) Find the average of square of the current.

(b) Find the amplitude of the supply voltage.

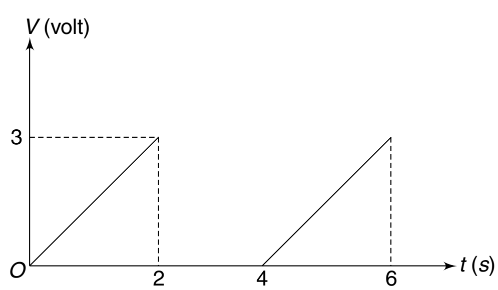
Q. 2: Which of the two waveforms shown in Figure has a higher average value?



Q. 3: Which of the two waveforms shown in Figure has a higher rms value?



Q. 4: A voltage waveform is as shown in the Figure calculate the ratio of rms value and average value of the voltage.

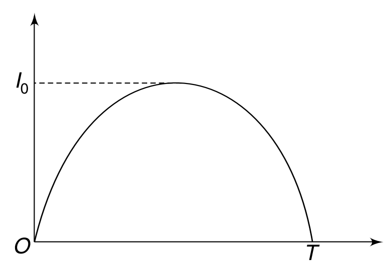


Q. 5: The graph given here represents an idealized sinusoidal current flow between a cloud at a height of 1 km and the earth, during a lightning discharge. Value of I_0 is 157 kA and T = 0.2 ms. Assume that discharge happens when the electric field in the air between the cloud and the earth becomes equal to the breakdown field of air i.e., $E_0 = 3 \times 10^6$ V/m

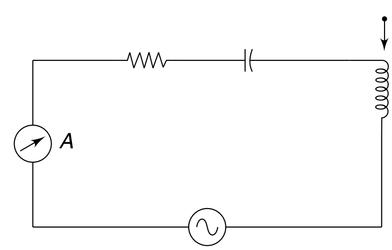
(a) Calculate the total charge flow due to lightning.

(b) Calculate the average current between the cloud and the earth during the lightning.

(c) Assume that the entire charge on the cloud is released during the lightning and estimate the capacitance of the cloud – Earth system.



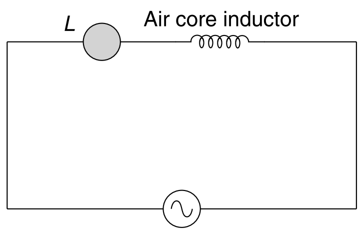
Q. 6: In the circuit shown, the frequency of the source is adjusted so that the reading of the *ac* ammeter is maximum. The inductor shown is a short coil in vertical orientation. A steel ball is dropped through the coil. How is the reading of ammeter affected when the ball enters the coil?



Q. 7: A lamp L is connected to an ac source along with an air core inductor as shown in the Figure. How is the brilliance of the lamp affected if core made of following material is inserted inside the inductor?

(a) Iron (b) Copper (c) Iron sheets pasted together with insulation in between [laminated Iron core]

In which case the lamp will be least bright?

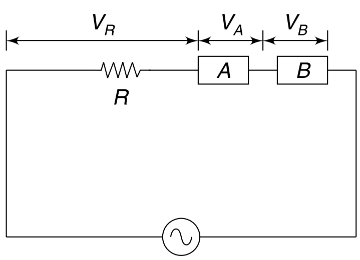


Q. 8: In the circuit shown in the Figure, the voltage across resistance R, box A and box B are represented as

$v_R = V\sin(\omega t)$, $v_A = \sqrt{2}\,V\sin\left(\omega t + \dfrac{\pi}{4}\right)$ and

$v_B = V\sin\left(\omega t + \dfrac{\pi}{2}\right)$

Find the phase difference between current and the applied voltage.



Level 2

Q. 9: A series LCR circuit has 120 Ω resistance. When the angular frequency of the source is 4×10^5 rad s^{-1} the voltage across resistance, inductance and capacitance are 60 V, 40 V and 40 V respectively. At what angular frequency of the source the current in the circuit will lag behind the source voltage by $\dfrac{\pi}{4}$.

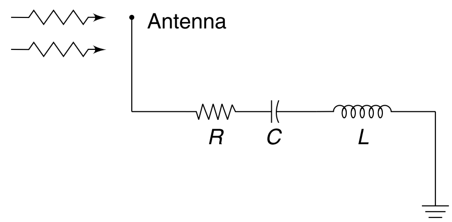
Q. 10: A resistance (R), inductance (L) and capacitance (C) are connected in series to an *ac* source of voltage V having variable frequency. Calculate the energy delivered by the source to the circuit during one period if the operating frequency is twice the resonance frequency.

Q. 11: A FM radio receiver has a series LCR circuit with $L = 1\ \mu$H, and $R = 100\ \Omega$. The antenna receives radio waves and induces a sinusoidally alternating emf of amplitude 10 μV. The induced voltage is fed to the series LCR circuit. The capacitance in the circuit is adjusted to a value of $C = 2$ pF.

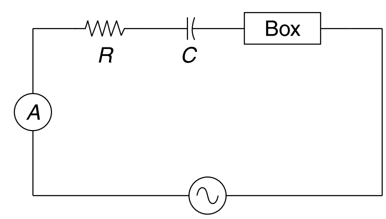
(a) Find the frequency of radio wave to which the radio will tune to.

(b) Find the rms current in the circuit.

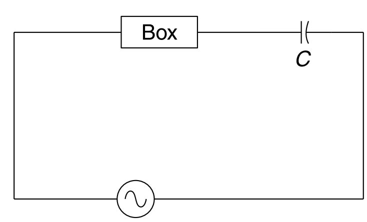
(c) Find quality factor of the resonance.



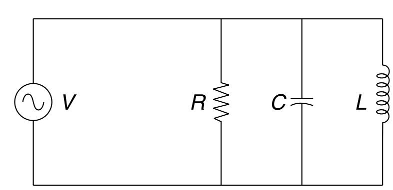
Q. 12: In the circuit shown in Figure, the source has a rating of 15 V, 100 Hz. The resistance R is 3 Ω and the reactance of the capacitor is 4 Ω. It is known that the box certainly contains one or more element (resistance, capacitance or inductance). Which element/s are present inside the box?



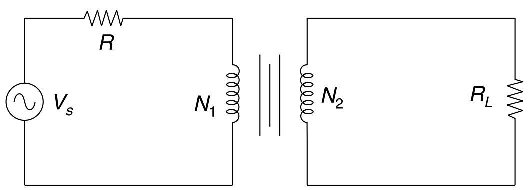
Q. 13: A box has a large electric circuit inside it. When it was connected to an ac generator it was found that it was putting a lot of load on the generator and the power factor of the box was $\dfrac{1}{\sqrt{2}}$. A capacitor of capacitance C was connected in series with the box and the power factor of the circuit became equal to the ideal value. Find the impedance of the box. The generator has an angular frequency of ω.



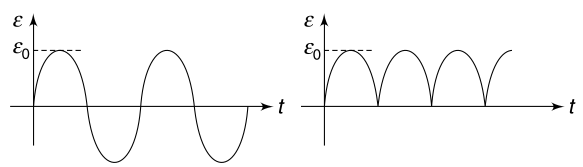
Q. 14: In the circuit shown the source voltage is given as $v = V_0 \sin \omega t$. Find the current through the source as a function of time.



Q. 15: In the circuit shown the transformer is ideal with turn ratio $\frac{N_1}{N_2} = \frac{5}{1}$. The voltage of the source is V_s = 300 Volt. The voltage measured across the load resistance R_L = 100 Ω is 50 Volt. Find the value of resistance R in the primary circuit.



Q. 16: A transformer with 20 turns in its secondary coil is used to step down the input 220 V ac emf. The output of the transformer is fed to a rectifier circuit which convents the ac input into dc output. The input and output of rectifier are as shown in Figure (there is no change in peak voltage). The rectifier output has an average emf of 8.98 volt. Calculate the number of turns in the primary coil of the transformer.



Rectifier input Rectifier output

Q. 17: Two boys are holding a wire, standing 4 m apart. The wire sags in the shape of a circular arc with a sag of 1.0 m. The students rotate the wire about the horizontal line connecting their hands, as if they were playing jump rope. The boys rotate the wire at a speed of 4 revolutions per second. The earth's magnetic field at the location is 4×10^{-5} T. Calculate the rms value of emf developed between the ends of the wire. Assume that the shape of the wire is maintained as it is rotated.

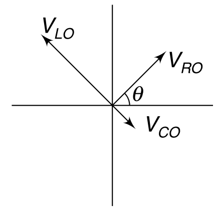
Q. 18: In a series LCR circuit the phasors corresponding to voltage across resistance, capacitor and inductor at an instant are as shown in the Figure and have amplitudes of V_{RO} = 4 volt, V_{CO} = 3 volt, and V_{LO} = 6 volt

(a) Is the source frequency larger than or lesser than the resonance frequency? Does the current (I) lead or lag the source voltage?

(b) Find the voltage amplitude of the source.

(c) If V_R phasor makes $\theta = 53°$ at time $t = 0$, write the source voltage as function of time. Take angular frequency of the source to be ω

$$\left[\sin 53° = \frac{4}{5}\right]$$

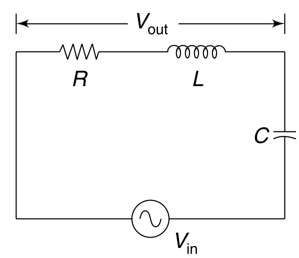


Q. 19: A series LCR circuit having resistance R, capacitance C and inductance L has a voltage source of angular frequency ω and voltage V_{in}. Output voltage (V_{out}) is taken as voltage across the resistor and inductor combined.

(a) Find $\eta = \frac{V_{out}}{V_{in}}$

(b) Find η in the limit of large $\omega\left(\omega >> \frac{1}{RC}, \frac{1}{\sqrt{LC}}, \frac{R}{L}\right)$

(c) Find η in the limit of small $\omega\left(\omega << \frac{1}{RC}, \frac{1}{\sqrt{LC}}, \frac{R}{L}\right)$.



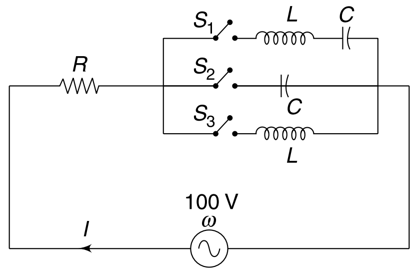
Q. 20: In a series LCR circuit, the frequencies at which the current amplitude is $\frac{1}{\sqrt{2}}$ times the current amplitude at resonance are f_1 and $f_2 (> f_1)$. Find the frequency bandwidth of resonance which is defined as $\Delta f = f_2 - f_1$. Express your answer in terms of R and L. Assume that resonance frequency $f_0 >> \Delta f$.

Q. 21: A series RLC circuit is in resonance with a source of frequency ω_0 = 10 MHz. The current amplitude in the circuit is I_0. It was found that when a different source of frequency $\omega = \omega_0 + \Delta\omega$ [$\Delta\omega$ = 10 KHz] was used the current amplitude in the circuit was only 1% of I_0. Find the inductance in the circuit if it is known that resistance in the circuit is R = 0.314 Ω.

Q. 22: In the circuit shown in the figure, one of the three switches is kept closed and other two are open. The value of resistance is R = 20 Ω. When the angular frequency(ω) of the 100 V source is adjusted to 500 rad/s, 1000 rad/s and 2000 rad/s it was found that the current I was 4A, 5A and 4A respectively.

(a) Which switch is closed? (S_1, S_2 or S_3)

(b) Find the value of L and C.



Q. 23: A village with a demand of 800 kW electric power at 220 V is located 30 km from an electric plant generating power at 440 V. The resistance of the two wire line carrying power is 0.25 Ω/km. The village gets power from the line through a 4000 V – 220 V step down transformer at a sub-station in the village. Assume negligible power loss in the transformers.

(a) Estimate the power loss in form of heat in the transmission line.

(b) How much power must the plant supply?

(c) What is the input and output voltage of step-up transformer at the plant ?

(d) What difference will it make if the village receives power through a 40,000 V – 220 V step down transformer

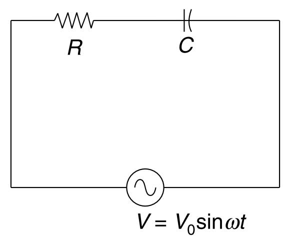
Level 3

Q. 24: A resistance R and a capacitor having capacitance C are connected to an alternating source having emf $v = V_0 \sin(\omega t)$. It is given that $\omega = \dfrac{1}{\sqrt{3}RC}$

(a) Plot the variation of power supplied by the source as a function of time. Mark the maximum and minimum values of power in the graph.

(b) How does the plot change if capacitor is removed and only R remains connected to the source?

(c) Plot the graph when only C remains connected to source and R is removed.



ANSWERS

1. (a) $9A^2$ (b) 282 V

2. Both have same average

3. Both have same value

4. 1.63

5. (a) $Q = 20$ C (b) 100 kA (c) 6.67 nF

6. Reading of ammeter decreases.

7. (i) Brilliance of lamp decreases
(ii) Brilliance increase
(iii) Brilliance decrease
The lamp is least bright with laminated iron core.

8. 45°

9. $\omega = 8 \times 10^5$ rad s^{-1}

10. $\dfrac{\pi R\sqrt{LC}V^2}{R^2 + 2.25\dfrac{L}{C}}$

11. (a) 112.9 MHz (b) 70 nA (c) 7.07

12. The box has L and C in series

13. $\dfrac{\sqrt{2}}{\omega C}$

14. $i = V_0\left[\dfrac{1}{R}\sin\omega t + \left(\omega C - \dfrac{1}{\omega L}\right)\cos\omega t\right]$

15. 500 Ω

16. 440

17. 0.97 mV

18. (a) Source frequency is greater than resonant frequency. Current lags.
(b) 5 volt
(c) $V_0 \cos\omega t$

19. (a) $\sqrt{\dfrac{1 + \left(\dfrac{\omega L}{R}\right)^2}{1 + \left(\dfrac{\omega L}{R} - \dfrac{1}{\omega RC}\right)^2}}$
(b) 1 (c) ωRC

20. $\Delta f = \dfrac{R}{2\pi L}$

21. 0.25 mH

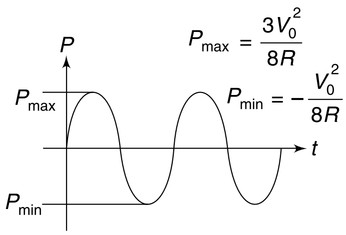
22. (a) S_1 (b) $L = 100$ mH, $C = 100\ \mu$F

23. (a) 600 kW (b) 1400 kW

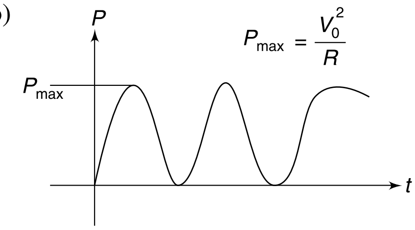
(c) 440 V – 7000 V

(d) Line power loss will drop to 6 kW. Power supply needed from plant = 806 kW

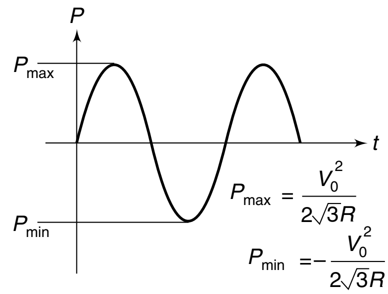
24. (a)



(b)



(c)



SOLUTIONS

1. (a) RMS current = 3*A*

$$\Rightarrow \qquad \sqrt{\langle i^2\rangle} = 3A$$

$$\Rightarrow \qquad \langle i^2\rangle = 9A^2$$

(b)

$$V_0 = \sqrt{2}V_{\text{rms}} = \sqrt{2} \times 200 = 282 \text{ V}$$

2. Area under the two graphs is same if time interval T is considered. Hence average will be same for both.

4. Time period $T = 4$ s

$$V = 1.5t \quad \text{for} \quad 0 \le t \le 2$$
$$= 0 \quad \text{for} \quad 2 < t \le 4$$

$$\therefore \qquad V_{av} = \frac{1}{T}\int_0^T V dt = \frac{1}{4}\int_0^2 1.5t\ dt = \frac{1}{4} \times \left[\frac{1.5t^2}{2}\right]_0^2 = 0.75 \text{ volt}$$

$$V_{\text{rms}} = \left[\frac{1}{T}\int_0^T V^2 dt\right]^{1/2} = \left[\frac{1}{4}\int_0^2 (1.5t)^2 dt\right]^{1/2} = 1.22 \text{ V}$$

$$\therefore \qquad \frac{V_{\text{rms}}}{V_{av}} = \frac{1.22}{0.75} = 1.63$$

5. (a) Charge released

$$Q = \int_0^T I_0 \sin\left(\frac{\pi}{T}t\right) dt$$

$$= \frac{2I_0T}{\pi} = \frac{2 \times 157 \times 10^3 \times 0.2 \times 10^{-3}}{3.14}$$

$$= 20 \text{ C}$$

(b) Average current $\qquad I_{av} = \dfrac{Q}{T} = \dfrac{2I_0}{\pi} = 100$ kA

(c) Electric field at the time of discharge $E = 3 \times 10^6$ V/m

Potential difference between the cloud & the earth

$$V = Ed = 3 \times 10^6 \times 10^3 = 3 \times 10^9 \text{ volt}$$

$\therefore$ Capacitance $$C = \frac{Q}{V} = \frac{20}{3 \times 10^9} = 6.67 \times 10^{-9} \text{ F}$$

6. Steel is ferromagnetic. Its presence inside the coil increases the inductance. The circuit is initially in resonance as the current is maximum. Reactance of the circuit is zero. The reactance of the circuit becomes non zero as the ball enters the coil. Thus impedance increases and current decrease.

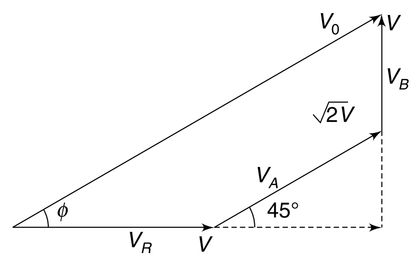
7. With Fe core, inductance of the inductor increases. This is due to strong magnetization of the ferromagnetic material. Due to this inductive reactance (X_L) increases. This increases the impedance (Z) of the circuit. The current drops and the lamp becomes dull.

When Cu core is inserted, periodically changing $\vec{B}$ produces an eddy current in it. The field produced by eddy current will weaken the field inside the coil. This decreases the inductance and hence X_L. Bulb becomes brighter.

Eddy current effect also happens in case of iron core but the property of ferromagnesian proves to be much stronger.

In case of laminated Fe core, the effect of eddy current gets reduced further. This laminated core results in sharp increases in impedance. The brilliance of the bulb decreases a lot.

8. In the Figure shown, V_0 represents the phasor associated with the source voltage.



$$\tan\phi = \frac{2V}{2V} = 1$$

$\Rightarrow$ $$\phi = 45°$$

The current phasor will be parallel to V_R hence the angle between V_0 and I is 45°

9. With $\omega_0 = 4 \times 10^5$ rad s^{-1}, the circuit is in resonance ($\because$ $V_L = V_C$)

$\therefore$ $$Z = R = 120\ \Omega$$

$\therefore$ RMS current $$I = \frac{V}{Z} = \frac{60}{120} = 0.5 \text{ A}$$

[Note that source voltage = V_R in resonance]

Now $$IX_L = V_L \Rightarrow 0.5\,(\omega L) = 40$$

$\Rightarrow$ $$L = \frac{40}{0.5 \times 4 \times 10^5} = 0.2 \text{ mH}$$

And $$IX_C = V_C$$

$\Rightarrow$ $$0.5\,\frac{1}{\omega C} = 40 \Rightarrow C = \frac{0.5}{4 \times 10^5 \times 40} = 31.25 \text{ nF}$$

If current lags behind the voltage by $\pi/4$ we must have

$$\tan\left(\frac{\pi}{4}\right) = \frac{X_L - X_L}{R}$$

$$\omega L - \frac{1}{\omega C} = 1 \times 120$$

$$0.2 \times 10^{-3}\,\omega - \frac{1}{\omega \times 31.25 \times 10^{-9}} = 120$$

$$0.2 \times 31.25 \times 10^{-12}\,\omega^2 - 120 \times 31.25 \times 10^{-9}\,\omega - 1 = 0$$

$\Rightarrow$ $$6.25\omega^2 - 3.75 \times 10^6\omega - 10^{12} = 0$$

Solving this quadratic equation gives

$$\omega = 8 \times 10^5 \text{ rad s}^{-1}$$

10. Resonance angular frequency $\omega_0 = \dfrac{1}{\sqrt{LC}}$

When $\omega = 2\omega_0 = \dfrac{2}{\sqrt{LC}}$ then–

$$X_L = L\omega = \frac{2L}{\sqrt{LC}} = 2\sqrt{\frac{L}{C}} \quad \text{and} \quad X_C = \frac{1}{\omega C} = \frac{\sqrt{LC}}{2C} = \frac{1}{2}\sqrt{\frac{L}{C}}$$

Impedance at frequency 2ω will be

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$\Rightarrow \quad Z^2 = R^2 + \left(2 - \frac{1}{2}\right)^2 \frac{L}{C} = R^2 + 2.25\,\frac{L}{C}$$

Power supplied is

$$P = VI\cos\phi = \frac{V^2}{Z}\cos\phi$$

$$= \frac{V^2}{Z}\cdot\frac{R}{Z} = \frac{V^2R}{Z^2}$$

$$\therefore \quad P = \frac{V^2R}{R + 2.25\left(\dfrac{L}{C}\right)}$$

This is actually average power (averaged over one cycle).

$\therefore$ Energy supplied is

$$E = P.T = P\,\frac{2\pi}{\omega} = \pi\sqrt{LC}\,P$$

$$\therefore \quad E = \frac{\pi R\sqrt{LC}\,V^2}{R^2 + 2.25\left(\dfrac{L}{C}\right)}$$

11. (a) Resonance frequency

$$f_0 = \frac{1}{2\pi}\sqrt{\frac{1}{LC}}$$

$$= \frac{1}{2\pi}\sqrt{\frac{1}{10^{-6} \times 2 \times 10^{-12}}}$$

$$= \frac{10^9}{2\pi \times 1.41} = 1.129 \times 10^8 \text{ Hz}$$

$$\simeq 112.9 \text{ MHz}$$

(b) Since circuit is in resonance $Z = R = 100\ \Omega$

$$\therefore \quad I = \frac{V}{Z} = \frac{\dfrac{10}{\sqrt{2}} \times 10^{-6}}{100}$$

$$\therefore \quad I = 70 \text{ nA}$$

(c) $$Q = \frac{\omega_0 L}{R} = \frac{1}{R}\sqrt{\frac{L}{C}} = \frac{1}{100}\sqrt{\frac{10^{-6}}{2 \times 10^{-12}}}$$

$$= 7.07$$

12. Impedance of the circuit without the box is

$$Z = \sqrt{R^2 + X_c^2} = \sqrt{3^2 + 4^2} = 5\ \Omega$$

$\therefore$ Voltage across R and C combination is $= 5\ \Omega \times 3\ \text{A} = 15$ V

Hence, voltage drop across the box = 0

The box must have a series combination of L and C with $X_L = X_C$

13. The box has inductive load apart from resistance.

$$\tan\phi = \frac{X_L}{R}$$

$$1 = \frac{X_L}{R} \qquad \left[\because \cos\phi = \frac{1}{\sqrt{2}}\right]$$

$$\therefore \quad X_L = R \qquad \text{...(i)}$$

After addition of the capacitor the power factor becomes 1 (the ideal value). This is possible if

$$X_L = X_C = \frac{1}{\omega C} \qquad \text{...(i)}$$

$$\therefore \quad Z_{\text{Box}} = \sqrt{R^2 + X_L^2} = \sqrt{X_L^2 + X_L^2}$$

$$= \sqrt{2}\,X_L = \sqrt{2}\,X_C = \frac{\sqrt{2}}{\omega C}$$

14. The instantaneous current in three branches are–

$$i_R = \frac{V_0}{R}\sin\omega t$$

$$i_C = \frac{V_0}{X_C}\cos\omega t$$

And $$i_L = -\frac{V_0}{X_L}\cos\omega t$$

$$\therefore \quad i = i_R + i_C + i_L$$

$$= V_0\left[\frac{1}{R}\sin\omega t + \left\{\frac{1}{X_C} - \frac{1}{X_L}\right\}\cos\omega t\right]$$

$$= V_0\left[\frac{1}{R}\sin\omega t + \left(\omega C - \frac{1}{\omega_L}\right)\cos\omega t\right]$$

15. $$V_1 = \left(\frac{N_1}{N_2}\right)V_2 \quad [V_2 = V_L = 50\ \text{V}]$$

And $$V_s = I_1 R + \left(\frac{N_1}{N_2}\right)V_2$$

Current in secondary circuit is

$$I_2 = \frac{V_2}{R_L}$$

And $$I_1 = \left(\frac{N_2}{N_1}\right)I_2 = \frac{N_2}{N_1}\frac{V_2}{R_L}$$

$$\therefore \quad V_s \left(\frac{N_2}{N_1}\right)\left(\frac{V_2}{R_L}\right)R + \left(\frac{N_2}{N_1}\right)V_2$$

$$\Rightarrow \qquad R = \frac{N_1 R_L}{N_2 V_2}\left[V_s - V_2\left(\frac{N_1}{N_2}\right)\right]$$

$$= 5 \times \frac{100}{50}[300 - 50 \times 5]$$

$$= 100 \times 50 = 500\ \Omega$$

16. Average of rectifier output is

$$\frac{2\varepsilon_0}{\pi} = 8.98$$

$$\therefore \qquad \varepsilon_0 = \frac{8.98 \times 3.14}{2} = 14.1 \text{ volt}$$

$\therefore$ rms value of transformer output (i.e., same as rectifier input) $= \frac{\varepsilon_0}{\sqrt{2}} = \frac{14.1}{1.41} \simeq 10$ volt

For transformer

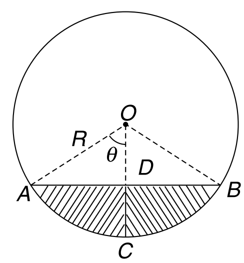
$$\frac{\varepsilon_{\text{input}}}{\varepsilon_{\text{output}}} = \frac{N_P}{N_S}$$

$$\frac{220}{10} \times 20 = N_P$$

$$\therefore \qquad N_P = 440$$

17. Let's Calculate the area occupied by the rotating wire

A and B are boys and ACB is the wire



$$AD = \frac{AB}{2} = \frac{4}{2} = 2.0 \text{ m}$$

$$CD = 1.0 \text{ m}$$

In ΔOAD: $\qquad R^2 = (R-1)^2 + 2^2 \quad \Rightarrow \quad R = \frac{5}{2}$ m

$$\tan\theta = \frac{2}{2.5} = \frac{4}{5} \quad \Rightarrow \quad \theta \simeq 40°$$

Area of sector $OACB = \frac{80}{360} \times \pi R^2 = \frac{2}{9} \times 3.14 \times \frac{25}{4} = 4.36 \text{ m}^2$

Area of $\Delta OAB = \frac{1}{2} \times 4 \times \frac{3}{2} = 3 \text{ m}^2$

$\therefore$ Area of $ACBDA = 4.36 - 3 = 1.36 \text{ m}^2$ (= A say)

Angular speed of rotation $\qquad \omega = 4 \times 2\pi = 8\pi$ rad s^{-1}

Flux through area A at any instant can be written as

$$\phi = BA\sin(\omega t + \delta)$$

$\therefore$ Induced emf $\qquad \varepsilon = \frac{d\phi}{dt} = BA\omega\cos(\omega t + \delta)$

$$\overline{\varepsilon^2} = B^2A^2\omega^2\overline{\cos^2(\omega t + \delta)} = \frac{1}{2}B^2A^2\omega^2$$

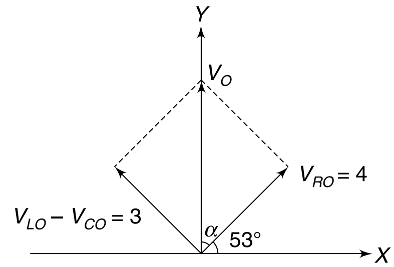
$$\therefore \qquad \varepsilon_{\text{rms}} = \frac{1}{\sqrt{2}}BA\omega = \frac{1}{\sqrt{2}} \times 4 \times 10^{-5} \times 1.36 \times 8\pi$$

$$= 96.6 \times 10^{-5} \text{ V} = 0.97 \text{ mV}$$

18. (a) $V_L > V_C \quad \Rightarrow \quad X_L > X_C \quad \Rightarrow \quad \omega L > \frac{1}{\omega L} \quad \Rightarrow \quad \omega > \frac{1}{\sqrt{LC}}$

The (source) voltage phasor will be represented by vector sum of the three given phasors. It will be somewhere between V_{RO} and V_{LO}. It means the voltage across R (which is in phase with current) is behind the source voltage. It means current lags in the circuit.

(b) Voltage amplitude of the source is given by



$$V_0 = \sqrt{V_{RO}^2 + (V_{LO} - V_{CO})^2} = \sqrt{4^2 + (6-3)^2} = 5 \text{ volt}$$

(c) If V_0 makes angle α with V_{RO} then $\tan \alpha = \frac{3}{4} \Rightarrow \alpha = 37°$

This implies that at $t = 0$, phasor V_0 is along Y. The phase angle of this phasor at $t = 0$ is $\pi/2$

$$\therefore \quad V = V_0 \sin\left(\omega t + \frac{\pi}{2}\right) = V_0 \cos \omega t$$

19. (a)

$$\frac{V_{\text{out}}}{V_{\text{in}}} = \frac{I\sqrt{R^2 + X_L^2}}{I\sqrt{R^2 + (X_L - X_C)^2}}$$

$$= \sqrt{\frac{R^2 + (\omega L)^2}{R^2 + \left(\omega L - \frac{1}{\omega L}\right)^2}}$$

$$= \sqrt{\frac{1 + \left(\frac{\omega L}{R}\right)^2}{1 + \left(\frac{\omega L}{R} - \frac{1}{\omega RC}\right)^2}}$$

(b) when ω is large $\frac{\omega L}{R} >> \frac{1}{\omega RC}$

$$\therefore \quad \frac{\omega L}{R} - \frac{1}{\omega RC} \simeq \frac{\omega L}{R}$$

$$\therefore \quad \frac{V_{\text{out}}}{V_{\text{in}}} \simeq 1$$

(c) When ω is small $\quad \frac{\omega L}{R} - \frac{1}{\omega RC} \simeq -\frac{1}{\omega RC}$

Also $\frac{\omega L}{R} << 1$ and $\frac{1}{\omega RC} >> 1$

$$\therefore \quad \frac{V_{\text{out}}}{V_{\text{in}}} \simeq \sqrt{\frac{1}{1 + \left(\frac{1}{\omega RC}\right)^2}} \simeq \omega RC$$

20. In resonance $\quad f_0 = \frac{1}{2\pi\sqrt{LC}}$ and $I_0 = \frac{V_0}{R}$

At a different frequency ($\omega = 2\pi f$), the current amplitude is

$$\frac{V_0}{Z} = \frac{V_0}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$$

Given

$$\frac{V_0}{\left[\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}\right]} = \frac{1}{\sqrt{2}} \frac{V_0}{R}$$

$$\Rightarrow \quad R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2 = 2R^2$$

$$\Rightarrow \qquad \omega L - \frac{1}{\omega L} = \pm R$$

$$2\pi f_1 L - \frac{1}{2\pi f_1 C} = -R \qquad \text{...(i)}$$

And
$$2\pi f_2 L - \frac{1}{2\pi f_2 C} = R \qquad \text{...(ii)}$$

Diving (i) by f_2 and (ii) by f_1 we get

$$2\pi L \frac{f_1}{f_2} - \frac{1}{2\pi C f_1 f_2} = -R \qquad \text{...(iii)}$$

$$2\pi L \frac{f_2}{f_1} - \frac{1}{2\pi C f_1 f_2} = R \qquad \text{...(iv)}$$

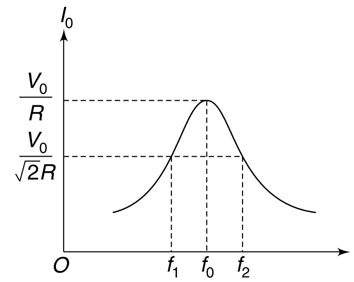
(iv) – (iii) gives

$$2\pi L\left(\frac{f_2^2 - f_1^2}{f_1 f_2}\right) = 2R$$

$$\Rightarrow \qquad \frac{(f_2 - f_1)\,(f_2 + f_1)}{f_1 f_2} = \frac{R}{\pi L}$$

$$f_1 + f_2 \simeq 2f_0 \quad \text{and} \quad f_1 f_2 \simeq f_0^2$$

$$\therefore \qquad \Delta f = \frac{R}{2\pi L}$$



21. Current amplitude at resonance frequency (ω_0) is $I_0 = \dfrac{V_0}{R}$

Current amplitude at frequency $\omega = \omega_0 + \Delta\omega$ is

$$I = \frac{V_0}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$$

It is required that
$$I = \frac{I_0}{100}$$

$$\Rightarrow \qquad I^2 = \frac{I_0^2}{10^4}$$

$$\Rightarrow \qquad \frac{1}{R^2 + \left(\omega L - \frac{1}{\omega L}\right)^2} = \frac{1}{10^4 R^2}$$

$$\Rightarrow \qquad R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2 = 10^4 R^2 \quad \Rightarrow \quad \left(\omega L - \frac{1}{\omega C}\right)^2 \simeq 10^4\, R^2$$

$$\Rightarrow \qquad \left(\omega L - \frac{1}{\omega C}\right) = 100\; R$$

We have chosen positive sign because $\omega L > \dfrac{1}{\omega C}$ at frequency higher than resonance frequency

$$\therefore \qquad (\omega_0 + \Delta\omega)\, L - \frac{1}{(\omega_0 + \Delta\omega)\, C} = 100R$$

$$\Rightarrow \qquad \omega_0 L + \Delta\omega L - \frac{1}{\omega_0 C}\left[1 + \frac{\Delta\omega}{\omega_0}\right]^{-1} = 100R$$

$$\Rightarrow \qquad \omega_0 L + \Delta\omega L - \frac{1}{\omega_0 C}\left[1 - \frac{\Delta\omega}{\omega_0}\right] = 100R$$

$$\Rightarrow \quad \Delta\omega L + \frac{\Delta\omega}{\omega_0^2 C} = 100R \qquad \left[\because \quad \omega_0 L = \frac{1}{\omega_0 C}\right]$$

$$\Rightarrow \quad 2\Delta\omega L = 100R$$

$$\Rightarrow \quad L = \frac{50R}{\Delta\omega} = \frac{50 \times 0.314}{2 \times 3.14 \times 10^4}$$

$$L = 2.5 \times 10^{-4} \text{ H} = 0.25 \text{ mH}$$

22. When current is $5A$, the impedance is

$$Z = \frac{V}{I} = \frac{100}{5} = 20\ \Omega$$

This is equal to R.
Hence for $\omega_0 = 1000$ rad/s

$$Z = R = 20\ \Omega$$

This is possible only if S_1 is closed and

$$\omega_0 L = \frac{1}{\omega_0 C}$$

$$\Rightarrow \quad LC = \frac{1}{10^6} \qquad \text{...(i)}$$

When $\omega = 500$ rad/s

$$I = 4A \quad \Rightarrow \quad \frac{100}{Z} = 4A$$

$$\Rightarrow \quad Z = 25\ \Omega$$

$$\sqrt{R^2 + X^2} = 25 \quad \Rightarrow \quad X = 15\ \Omega$$

$$\therefore \quad \frac{1}{\omega C} - \omega L = 15$$

[Note that X is not $\omega L - \frac{1}{\omega C}$, since on increasing the frequency from 500 rad/s to 1000 rad/s the impedance is decreasing]

$$\Rightarrow \quad \frac{1}{500C} - 500L = 15 \qquad \text{...(ii)}$$

Solving (i) and (ii) $L = 10$ mH, $C = 100\ \mu$F

23. rms current in line $\quad I = \dfrac{800 \times 10^3 \text{ W}}{4000 \text{ V}} = 200$ A

(a) Line loss $= 1^2R \qquad [R = 0.25 \times 60 = 15\ \Omega]$

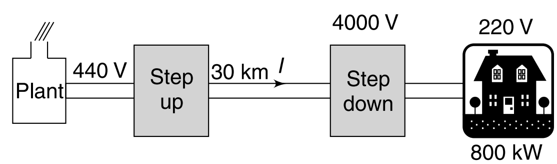
$= (200)^2 \times 15 = 600$ kW

(b) $P_{\text{plant}} = 800 + 600 = 1400$ kW

(c) Voltage drop on the line $= 200 \times 15 = 3000$ V

$\therefore$ Transformer needed at plant is 440 V – 7000 V

(d) High voltage transmission will cause low power loss



24. (a) Current leads voltage by phase angle

$$\tan\phi = \frac{X_C}{R} = \frac{1}{\omega CR} = \sqrt{3} \quad \Rightarrow \quad \phi = \pi/3$$

$$\therefore \quad v = V_0 \sin\omega t \quad \text{and} \quad i = i_0 \sin(\omega t + \pi/3) \qquad \left[\text{where} \quad i_0 = \frac{V_0}{\sqrt{R^2 + X_C^2}} = \frac{V_0}{\sqrt{R^2 + 3R^2}} = \frac{V_0}{2R}\right]$$

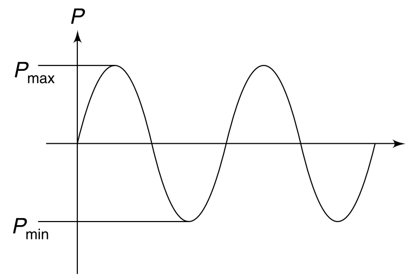
Power supplied by the source

$$P = vi = V_0 i_0 \sin \omega t \sin (\omega t + \pi/3)$$

$$= V_0 i_0 \left[\cos \frac{\pi}{3} \cdot \sin^2 \omega t + \sin \frac{\pi}{3} \cdot \sin \omega t \cdot \cos \omega t\right]$$

$$= \frac{V_0 i_0}{2}\left[\cos \frac{\pi}{3} (1 - \cos 2\omega t) + \sin \frac{\pi}{3} \cdot \sin 2\omega t\right]$$

$$= \frac{V_0 i_0}{2}\left[\cos \frac{\pi}{3} - \cos\left(2\omega t + \frac{\pi}{3}\right)\right]$$



$\therefore$ $$P_{max} = \frac{V_0 i_0}{2}\left(\cos \frac{\pi}{3} + 1\right) = \frac{3}{4} V_0 i_0 = \frac{3V_0^2}{8R}$$

And $$P_{min} = \frac{V_0 i_0}{2}\left(\cos \frac{\pi}{3} - 1\right) = -\frac{V_0 i_0}{4} = -\frac{V_0^2}{8R}$$

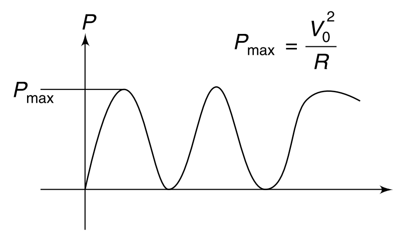
Power oscillates with frequency twice that of current and voltage

(b) with only 'R' the phase difference between current and voltage will be $\phi = 0$

$\therefore$ $$P = v_i = \frac{v^2}{R} = \frac{V_0^2}{R} \sin^2 \omega t$$

$$= \frac{V_0^2}{2R} [1 - \cos 2\omega t]$$

$$P_{max} = \frac{V_0^2}{R}; P_{min} = 0$$



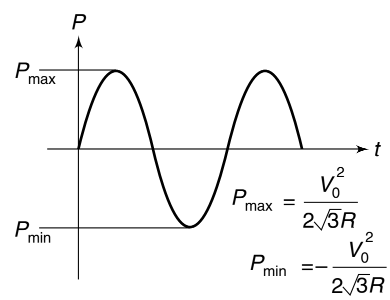
(c) with only 'C'

$$P = \frac{V_0 i_0}{2} \sin (2\omega t)$$

$$P_{max} = \frac{V_0 i_0}{2}; P_{min} = -\frac{V_0 i_0}{2} = -\frac{V_0^2}{2X_C}$$

$$= -\frac{V_0^2 \omega C}{2} = -\frac{V_0^2}{2\sqrt{3} R}$$

$$P_{max} = \frac{V_0^2}{2\sqrt{3} R}$$



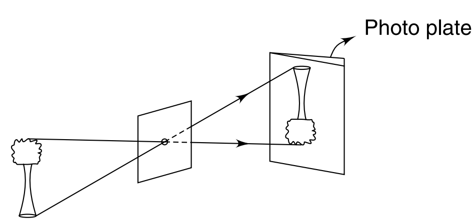
CHAPTER 13

Geometrical Optics

Level 1

Q. 1: A man of height 1.8 m is standing in front of a wall. The sun is exactly behind him. His shadow has a length 1.5 m on the ground and 0.75 m on the wall. Find the length of his shadow on the ground if the wall is removed.

Q. 2: Light travels in a straight line. This principle is illustrated in a pinhole camera. In this simple device the image of an object is formed on a photographic plate by light passing through a small hole.



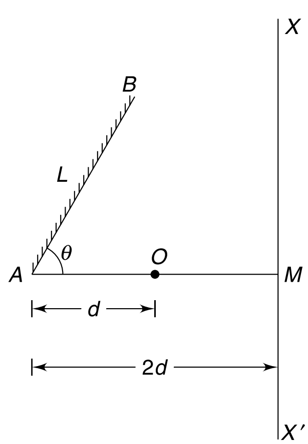
In one experiment, 5 cm high image of a tree was obtained on a photo plate placed at a distance of 15 cm from the pin hole. Actual height of the tree is 20 m.

(a) Find the distance of the pinhole from the tree.

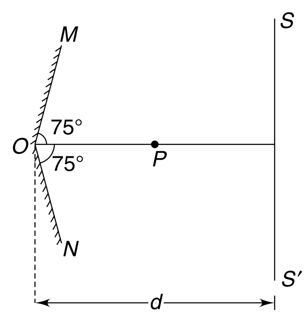
(b) How is the size of image affected if the photo plate is moved away from the pinhole?

(c) What will happen if a large hole is made in place of a pin hole?

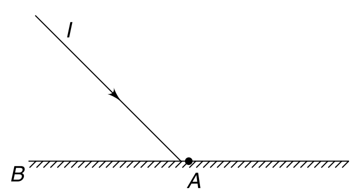
Q. 3: A point object O is kept in front of a plane mirror AB having length $L = 2$ m. The line AOM makes an angle $\theta = 60°$ with the mirror. An observer is walking along the line XMX' (perpendicular to AOM). Find the length of his path along XMX' in which he can see the image of the object. Given $AO = d = 1$ m and $AM = 2d$.



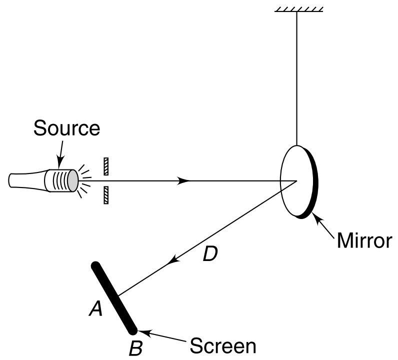
Q. 4: Two large plane mirrors OM and ON are arranged at 150° as shown in the figure. P is a point object and SS' is a long line perpendicular to the line OP. Find the length of the part of the line SS' on which two images of the point object P can be seen.



Q. 5: I is a ray incident on a plane mirror. Keeping the incident ray fixed, the mirror is rotated by an angle θ about an axis passing through A perpendicular to the plane of the Fig. show that the refracted ray rotates through an angle 2θ. Does your answer differ if the mirror is rotated about an axis passing through B?



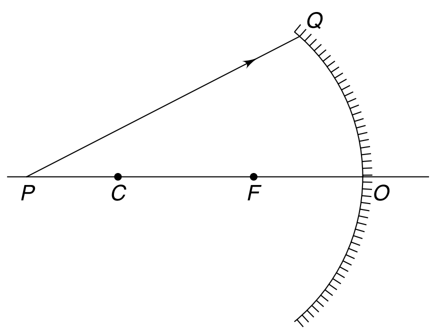
Q. 6: The Fig. shows a device used to measure small twist in a thread. A plane mirror is suspended from a twist free thread. A light ray striking the mirror is reflected on to a screen placed at a distance $D = 1$ m from the mirror.



As the thread is twisted by an angle θ (so that mirror rotates by θ), the light spot on the screen moves from A to B such that $AB = 0.5$ cm. Find θ.

Q. 7: While looking at her face in a mirror, Hema notes that her face is highly magnified when she is close to the mirror. As she backs away from the mirror, her image first gets blurry, then disappears when she is at a distance of 45 cm from the mirror. Explain the happenings? What will happen if she moves beyond 45 cm distance from the mirror?

Q. 8: We know that parallel light rays which are inclined to the principal axis of a spherical mirror, after reflection converge at a point in the focal plane of the mirror. With this knowledge explain how you will trace the reflected ray for incident ray PQ shown in the Fig. F is focus and C is centre of curvature of the mirror.



Q. 9: The inner surface of the wall of a sphere is perfectly reflecting. Radius of the sphere is R. A point source S is placed at a distance $R/2$ from the centre of the sphere. Consider the reflection of light from the farthest wall followed by reflection from the nearest wall. Where is the image of the source? Consider paraxial rays only.

Q. 10: A concave mirror forms a real image, on a screen of thrice the linear dimension of a real object placed on its principal axis. The mirror is moved by 10 cm along its principal axis and once again a sharp image of the object is obtained on the screen. This time the image is twice as large as the object. Find the focal length of the mirror.

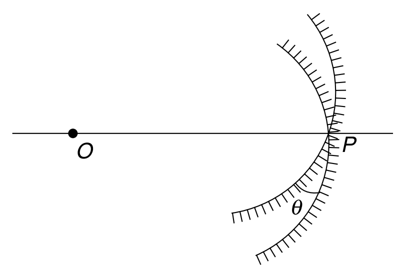
Q. 11: Plot the graphs of $\left|\frac{1}{v}\right|$ vs $\left|\frac{1}{u}\right|$ where v is image distance and u is object distance for the conditions given below:

(a) for concave mirror when image is real
(b) for concave mirror when image is virtual
(c) for convex mirror when image is virtual
(d) for convex mirror when image is real.

Q. 12: A point object O is placed at a distance of 60 cm from a concave mirror of radius of curvature 80 cm.

(a) At what distance from the concave mirror should a plane mirror be kept so that rays converge at O itself after getting reflected from the concave mirror and then from the plane mirror?
(b) Will the position of the point where the rays meet change if they are first reflected from the plane mirror?

Q. 13: A point object (O) is placed at the centre of curvature of a concave mirror. The mirror is rotated by a small angle θ about its plole (P). Find the approximate distance between the object and its image. Focal length of the mirror is f.



Q. 14: A one eyed demon has a circular face of radius $a_0 = 10$ cm. The eye is located at the centre of the face. At what distance from his face he must hold a convex mirror of 5 cm aperture diameter so as to see his complete face? Focal length of the mirror is 10 cm.

Q. 15: A small insect starts walking away from a concave mirror along its principal axis. At a point (P) 20 cm from the mirror the image flips upside down.

(a) What can you say about the size of the image at the instant it flips upside down – it is very large, very small or of the size similar to the insect?
(b) Find the distance of the insect from point (P) where its image is thrice as large as the insect.

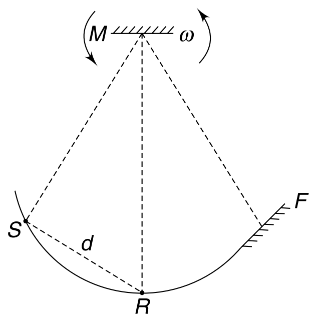
Q. 16: A piece of metal is cut from a hollow metal sphere of radius R and is polished on both sides. A boy looks at the metal piece and finds his image 13 cm behind the metal piece. His friend flips the mirror, keeping its position unchanged and now the boy finds his image to be 52 cm behind the mirror. Find R.

Q. 17: The aperture diameter of a spherical mirror is $D = \eta R$ where η is a positive number less than 2 and R is radius of curvature of the mirror. Consider a wide parallel beam of light incident on the mirror parallel to its principal axis.

(a) Find minimum value of η for which marginal rays start getting reflected twice.

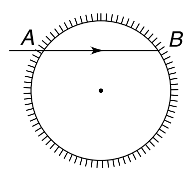
(b) Find minimum value of η for which marginal rays undergo three reflections.

Q. 18: A source of laser (S), a receiver (R) and a fixed mirror (F) – all lie on an arc of a circle of radius $R = 0.5$ km. The distance between the source and the receiver is $d = 0.5$ m. At the centre of the circle there is a small mirror M which is rotating with angular speed ω (see figure). Find smallest value of ω is if it is seen that the source shoots a laser pulse which gets reflected at M, then gets reflected at F and finally gets reflected at M to be received by the receiver.



Q. 19: A real object is kept at a distance a from the focus of a concave mirror on the principal axis. A real image is formed at a distance b from the focus. Plot the variation of b with a. If b versus a graph is given to you, how will you find the focal length of the mirror? Explain.

Q. 20: A light ray enters horizontally into a vertical cylinder through a small hole at A. The ray is initially travelling along a chord (AB) whose length is $\left(\frac{\sqrt{5}+1}{2}\right)R$ where R is the radius of the cylinder. After how many reflections on the inner wall of the cylinder the light ray will be incident at point A?

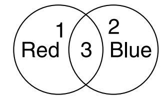


$$\left[\text{Given } \cos 36° = \frac{\sqrt{5}+1}{4}\right]$$

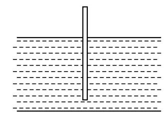
Q. 21: A ship has a green light ($\lambda = 510$ nm) on its mast. What colour would be observed for this light by a diver deep inside water. Refractive index for water is $\mu = \frac{4}{3}$.

Q. 22: A vertical rod is partially submerged in an aquarium. You look at the aquarium from some distance. Does the underwater part of the rod appear to be closer than, farther than or the same distance as the top of the rod.

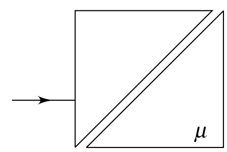
Q. 23: Two transparent plastic sheets of red and blue colour overlap as shown in the Fig. An observer looks at a clear sky through the sheets. What can you say about the colour and brightness of light coming through sections 1, 2 and 3 (see Fig.)



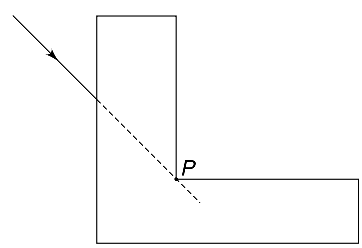
Q. 24: A wooden stick of length 100 cm is floating in water while remaining vertical. The relative density of the wood is 0.7. Calculate the apparent length of the stick when viewed from top (close to the vertical line along the stick) Refractive index of water $= \frac{4}{3}$.



Q. 25: A glass cube is cut symmetrically into two halves and the two parts are kept at a small separation between them. Calculate the angular deviation suffered by a light ray incident normally on one of the faces of the cube.



Q. 26: A glass block of refractive index $\mu = 1.5$ has an L cross section with both arms identical. A light ray enters the block from left at an angle of incidence of 45°, as shown in the figure. If the block was absent the ray would pass through the point P. Calculate the angle at which the ray will emerge from the bottom face after refraction through the block.



Q. 27: A ray of light passes through a rectangular glass block placed in air. Which diagram shows a possible path of a ray?

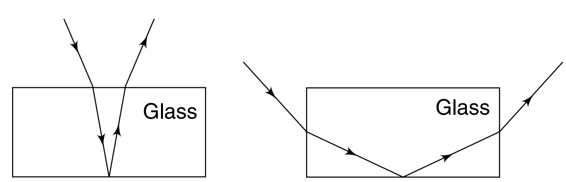
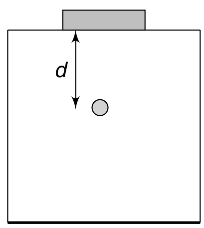


Fig. (a) Fig. (b)

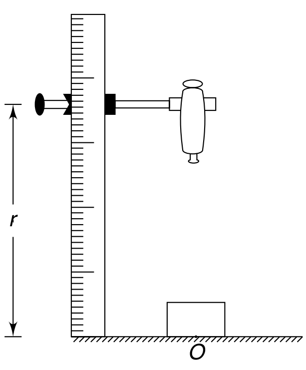
Q. 28: A ray of light travelling in air is incident on a composite transparent slab at an angle of incidence i = 45°. The composite slab consist of 100 parallel faced slices of equal thickness. The refractive index of the n^{th} slice (counting from the one on which light is incident) is given by $\mu_n = 1.0 + 0.01n$. The medium on the other side of the slab has refractive index of $\sqrt{2}$. Calculate the angular deviation suffered by the ray as it comes out of the slab.

Q. 29: Prove that it is impossible to see through adjacent sides of a square block of glass with index of refraction 1.5.

Q. 30: A large transparent cube (refractive index = 1.5) has a small air bubble inside it. When a coin (diameter 2 cm) is placed symmetrically above the bubble on the top surface of the cube, the bubble cannot be seen by looking down into the cube at any angle. However, when a smaller coin (diameter 1.5 cm) is placed directly over it, the bubble can be seen by looking down into the cube. What is the range of the possible depths d of the air bubble beneath the top surface?



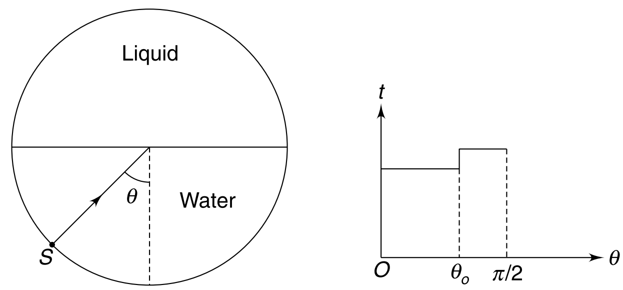
Q. 31: A travelling microscope can move vertically along a scale. It is focused at a mark O on the table and the reading on the vertical scale is r_1. Now a glass slab is placed over mark O and the microscope has to be moved up to bring the mark in focus again. This time the scale reads r_2. Lycopodium powder is spread over the top of the glass slab and the microscope is moved up once again to bring the powder particles in sharp focus. This time the vertical scale reads r_3. Find the refractive index of the material of the glass slab.



Q. 32: A vertical beam of light of cross sectional radius $\frac{R}{2}$ is incident symmetrically on the curved surface of a glass hemisphere of refractive index $\mu = \frac{3}{2}$. Radius of the hemisphere is R and its base is on a horizontal table. Find the radius of luminous spot formed on the table.

$$\sin 20° = \frac{1}{3} \text{ and } \sin 80° = 0.98$$

Q. 33: A horizontal cylindrical tank is half full of water $\left(\text{refractive index} = \frac{4}{3}\right)$. The space above the water is filled with a light liquid of unknown refractive index (μ). A small laser source (s) can move along the curved bottom of the cylinder and aims a light beam towards the centre of the cylinder. The time needed by the laser beam to travel from the source to the rim of the cylinder depends on position (θ) of the source as shown in the graph. Find μ, it is given that $\sin\theta_0 = \frac{5}{6}$.



Q. 34: A man standing on sea-shore sees an elongated image (shown by dashed line) of a floating object AB. In fact he finds the image to be oscillating due to air turbulence. Figure (ii) gives three plots (a, b and c) of height from the water surface vs air temperature. Which one best illustrates the air-temperature condition that can create this image? [Many people have seen sea monsters due to this phenomena!]

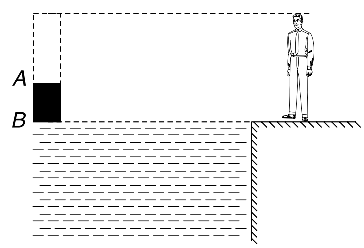


Fig. (a)

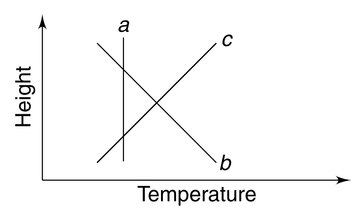
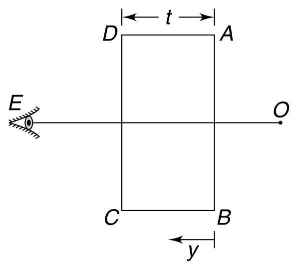


Fig. (b)

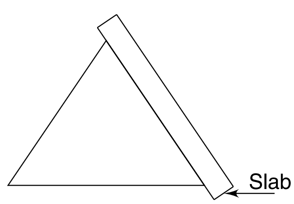
Q. 35: The atmosphere of earth extends upto height H and its refractive index varies with depth y from the top as $\mu = 1 + \frac{y}{H}$. Calculate the apparent thickness of the atmosphere as seen by an observer in space.

Q. 36: A glass slab is placed between an object (O) and an observer (E) with its refracting surfaces AB and CD perpendicular to the line OE. The refractive index of the glass slab changes with distance (y) from the face AB as $\mu = \mu_0(1 + y)$. Thickness of the slab is t. Find how much closer (compared to original distance) the object appears to the observer. Consider near normal incidence only.

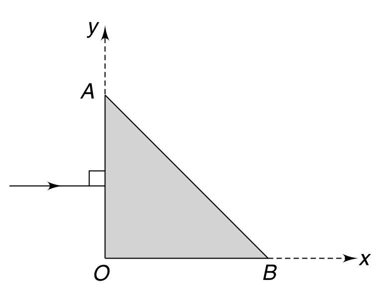


Q. 37: An equilateral prism deviates a ray through 40° for two angles of incidence. The two incidence angels differ by 20°. Find their values.

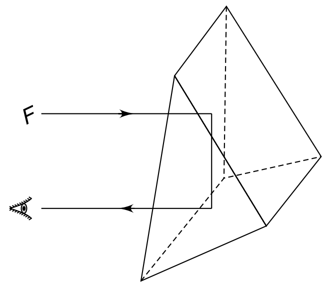
Q. 38: An equilateral glass prism can produce a minimum deviation of 30° to the path of an incident ray. A transparent slab of refractive index 1.5 is placed in contact with one of the refracting faces of the prism. Thickness of the slab is 3 cm. Now calculate the minimum possible deviation which can be produced by this prism-slab combination.



Q. 39: A triangular medium has varying refracting index $\mu = \mu_0 + ax$ where x is the distance (in cm) along x-axis from origin and $\mu_0 = 4/3$. A ray is incident normally on face OA at the mid-point of OA. Find the range of value of a so that light does not escape through face AB when it falls first time on the face AB.



Q. 40: Letter F is kept in front of a right triangular psim. The light rays enter perpendicular to the large rectangular face, is reflected twice by small rectangular faces and exits perpendicularly to the large rectangular face (see Fig.). Draw the image of the letter seen by the eye.



Q. 41: A prism (apex angle $A \leq 90°$) produces minimum deviation that is equal to the apex angle. What can be said about the refractive index of the material of the prism.

Q. 42: Limiting angle of a prism is defined as the largest angle of the prism (A) for which no emergent ray is obtained.

(a) Find the limiting angle (A_0) for a glass prism having refractive index μ.

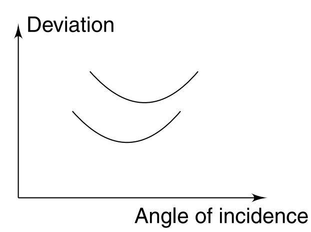
(b) A prism has limiting angle for light of wavelength λ_0. Can there be any emergent ray for light of wavelength $\lambda < \lambda_0$.

Q. 43: An isosceles glass prism has one of its faces silvered. A light ray is incident normally on the other face which is identical in size to the silvered face. The light ray is reflected twice on the same sized faces and emerges through the base of the prism perpendicularly. Find the minimum value of refractive index of the material of the prism.

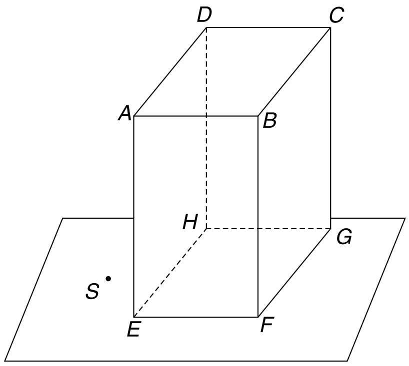
Q. 44: A parallel beam of light falls normally on the first face of a prism of small refracting angle. At the second refracting face it is partly reflected and partly transmitted. The reflected light strikes the first face again and emerges from it making an angle of 4° with the reversed direction of incident beam. The deviation suffered by refracted ray is 1° from original direction of incident ray. Find the refractive index of glass of the prism and the angle of the prism.

Q. 45: A ray of light is incident upon one face of a prism in a direction perpendicular to the other refracting face. The critical angle for glass – air interface is 30°. Find the angle of the prism (assuming it to be less than 90°) if the ray fails to emerge from the other face.

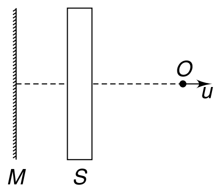
Q. 46: The plot of deviation versus angle of incidence for two prisms made of same material has been shown in the Figure. Which of the two graph corresponds to the prism of higher refracting angle (A)?



Q. 47: A large rectangular glass block of refractive index μ is lying on a horizontal surface as shown in Figure. Find the minimum value of μ so that the spot S on the surface cannot be seen through top plane '$ABCD$' of the block.



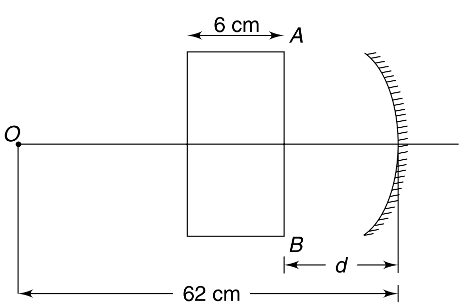
Q. 48: Object O is placed in front of a plane mirror M. A glass slab S having thickness t = 3 cm and refractive index μ = 1.8 is placed between the object and the mirror. The refracting faces of the slab are parallel to the mirror surface. The object is made to move with a velocity of u = 2 m/s perpendicular to the mirror surface. Find the speed of the image formed after



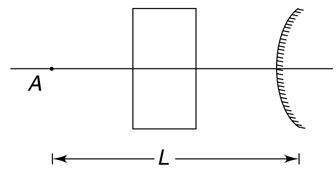
(a) refraction from the slab followed by reflection from the mirror.

(b) refraction from the slab followed by reflection from the mirror followed by the refraction from the slab.

Q. 49: A point object O is placed at a distance of 62 cm in front of a concave mirror of focal length f = 20 cm. A glass slab of refractive index $\mu = \frac{3}{2}$ and thickness 6 cm is inserted between the object and the mirror. Let's define final image as image formed after the light ray originating from O passes through the slab, gets reflected from the mirror and then again passes through the slab. At what distance d from the mirror, the face AB of the slab can be placed so that the final image is formed inside the slab itself?

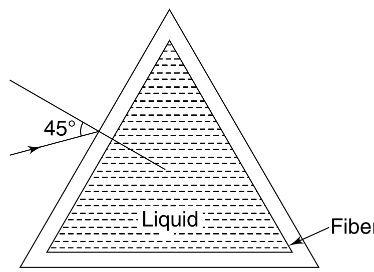


Q. 50: A point object (A) is kept at a distance (L) from a convex mirror on its principal axis. A glass slab is inserted between the object and the mirror with its refracting surfaces perpendicular to the principal axis of the mirror. The thickness of the slab is 6 cm and its refractive index is $\frac{3}{2}$. The image formed after refraction through the slab, reflection from the mirror followed by refraction through the slab is a virtual image at a distance of 10 cm from the pole of the mirror (on its principal axis). Consider paraxial rays only and calculate the distance (L) of the object from the mirror. Focal length of the mirror is f = 20 cm.

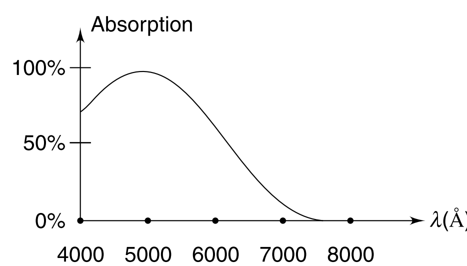


Q. 51: The two perpendicular faces of a right angled isosceles prism are silvered. Prove that a light ray incident on the third face (hypotenuse face) will emerge from the prism parallel to the initial direction.

Q. 52: An equilateral prism has its faces made of a transparent fiber sheet (having refractive index = 1.25) having thickness of 1 mm. The fibre prism is filled with a liquid of refractive index $\sqrt{2}$. Find the deviation of a light ray incident on one face of such a prism at an angle of 45°.



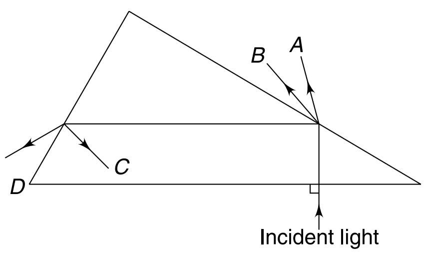
Q. 53: The Figure shows the absorption spectrum for a body. What is the colour of the body?



Q. 54: Two thin prisms are combined such that they neither produce any average deviation nor do they cause any dispersion when white light is incident on the combination. Angle of one prism is A = 2° and refractive index of its glass for red, yellow and violet lights are 1.49, 1.50 and 1.51 respectively. Find the dispersive power of the glass of the other prism.

Q. 55: White light is incident on a glass prism as shown. Four easily identifiable colors – red, green, yellow and blue get separated as *A*, *B*, *C* and *D*.

Which of the rays (*A*, *B*, *C* and *D*) correspond to which color?

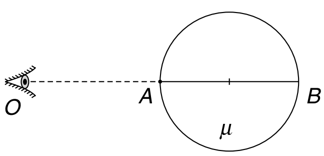


Q. 56: A concave spherical surface of radius of curvature R = 20 cm separates two media *A* and *B* having refractive indices $\mu_A = \frac{4}{3}$ and $\mu_B = \frac{3}{2}$ respectively. A point object is placed on the principal axis. Find the distance of the object from the surface so that its image is virtual when

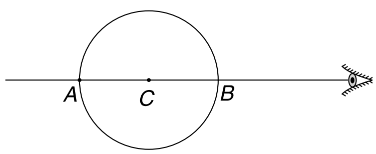
(a) the object is in medium *A*.

(b) the object is in medium *B*.

Q. 57: A transparent ball of radius *R* is viewed by an observe *O* along its diameter *AB*. The observe *O* sees the distance *AB* to be infinitely large. Find the refractive index of the material of the ball.



Q. 58: A glass (μ = 1.5) sphere of radius *R* is viewed from outside along a diameter. Calculate the distance between two points (say *P* and *Q*) lying on the line *AB* whose images are seen at centre *C* and point *A* respectively.



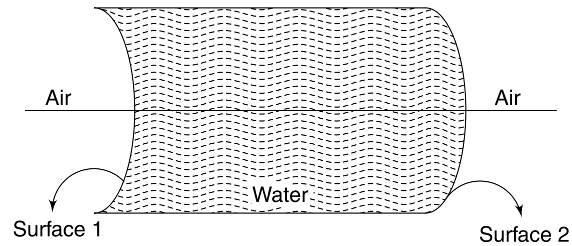
Q. 59: A concave spherical surface of radius of curvature 10 cm separates two mediums *X* and *Y* of refractive indices 4/3 and 3/2 respectively. Centre of curvature of the surface lies in the medium *X*. An object is placed in medium *X*. Will the image be real or virtual?

Q. 60: A region bounding water has air on two sides. Tell the nature (real or virtual) of the image for following cases- (The object is real and lies on the principal axis (see fig) in all cases.)

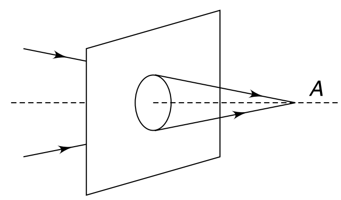
(a) The object is to the left of surface 1 and the image to be considered is formed after the first refraction.

(b) The object is to the left of surface 1 and the image to be considered is formed after two refractions.

(c) The object is to the right of second surface and the image to be considered is formed after one refraction.

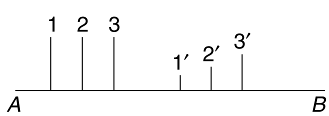


Q. 61: A converging beam of light rays passes through a round opening in a screen. The beam converges at a point *A* which is at a perpendicular distance of 15 cm from the screen and lies slightly above the central axis of the circular opening. A convex lens of focal length 30 cm is inserted in the opening. At what distance from the screen do the rays converge now?

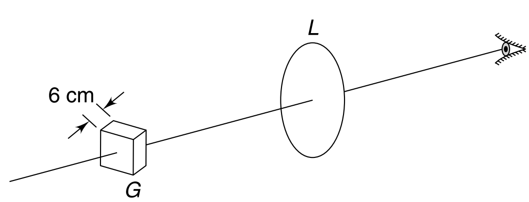


Q. 62: In the Figure *AB* is the principal axis of an optical element (a lens or a mirror). For position 1, 2 and 3 of a real object, the corresponding position of images are 1′, 2′ and 3′ respectively. Size of image at 3′ is largest and that at 1′ is smallest.

Identify the optical element and indicate its position.



Q. 63: A transparent glass slab (*G*) of thickness 6 cm is held perpendicular to the principal axis of a convex lens (*L*) as shown in the Figure. The refractive index of the material of the glass is $\frac{3}{2}$ and its nearer face is at a distance 40 cm from the lens. Focal length of the lens is 20 cm. Find the thickness of the glass slab as observed through the lens.

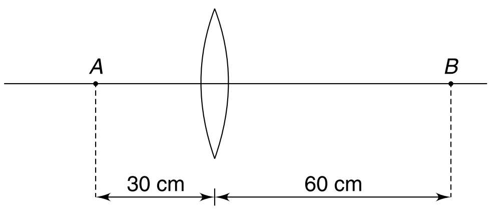


Q. 64: An object 240 cm in front of a lens forms a sharp image on a screen 12 cm behind the lens. A glass slab 1 cm thick, having refractive index 1.50 is placed between the lens and the screen with its refracting faces perpendicular to the principal axis of the lens.

(a) By how much distance the object must be moved so as to again cast a sharp image on the screen?

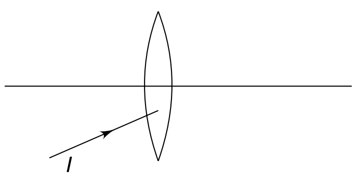
(b) Another identical glass slab is interposed between the object and the lens. How much further the object shall be moved so as to form a sharp image on the screen.

Q. 65: Two point objects A and B are kept on the principal axis of a convex lens as shown. Image of both the objects is formed at same position. Find the focal length of the lens.

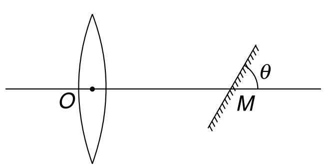


Q. 66: In which case does a light ray pass through the centre of a thin lens without deviation?

Q. 67: The medium on both sides of a convex lens is same. A light ray (I) is incident on it as shown. Draw the path of the ray after it emerges from the lens. Write each step of your construction. The focal length of the lens is known.

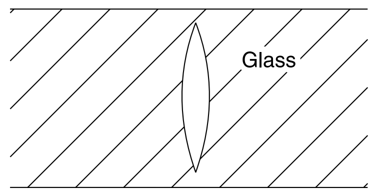


Q. 68: A horizontal parallel beam of light passes though a vertical convex lens of focal length 40 cm. Behind the lens there is a plane mirror making an angle θ with the principal axis of the lens. The mirror intersects the principal axis at M. Distance between the optical centre of the lens and point M is OM = 20 cm. The light beam reflected by the mirror converges at a point P. Distance OP is 20 cm. Find θ.

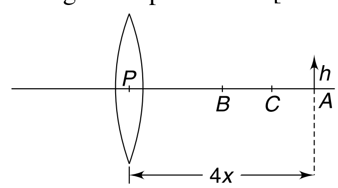


Q. 69: A cylindrical tube has a length of 60 cm. Three identical convex lenses, each of focal length f = 10 cm are fixed inside the tube; one at each of the ends and one at the centre. One end of the tube is placed 10 cm away from a point source. The device casts an image of the object. How much does the image shift when the tube is moved away from the source by 10 cm.

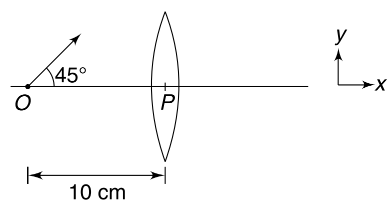
Q. 70: There is an air lens in an extended glass medium. The radius of curvature of both the curved surfaces is R, and refractive index of the glass is $\frac{3}{2}$. Power of this air lens is P. Find the refractive index of the material to be filled inside the lens so that its power becomes $-P$.



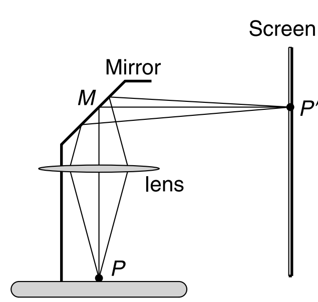
Q. 71: A thin converging lens forms a real image of an object located far away from the lens. The image is formed at A at a distance $4x$ from the lens and height of the image is h. A thin diverging lens of focal length x is placed at $B\,[PB = 2x]$ and a converging lens of focal length $2x$ is placed at $C\,[PC = 3x]$. The principal axes of all lenses coincide. Find the height of the final image formed.



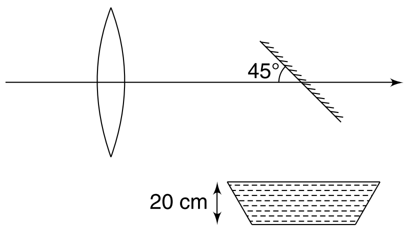
Q. 72. A point object (O) lies at a distance of 20 cm on the principal axis of a convex lens of focal length f = 10 cm. The object begins to move in a direction making an angle of 45° with the principal axis. At what angle with the principal axis does the image beings to move?



Q. 73: The lens in an overhead projector forms an image P' of a point P on a transparency. If the screen is moved away from the projector, how should we move the lens to keep the image on the screen in focus?



Q. 74: A small object is at the bottom of a container which has water filled up to a height of 20 cm. There is a plane mirror inclined at 45° to the horizontal above the container. A convex lens having focal length 15 cm is at a distance of 50 cm from the mirror. The horizontal principal axis of the lens is at a distance of 45 cm from the bottom of the container. Find the distance of the image (from the lens) of the object as seen by an observer to the left of the lens. Light rays from the object hit the lens only after they are reflected from the mirror. (use $\mu_{\text{water}} = 4/3$)

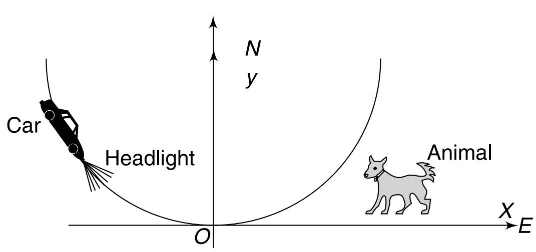


Q. 75: The principal axis of a thin equi-convex lens is the x-axis. The co-ordinate of a point object and its image are (−20 cm, 1 mm) and (25 cm, 2 mm) respectively. Find the focal length of the lens.

Level 2

Q. 76: Sunrays pass through a pinhole in the roof of a hut and produce an elliptical spot on the floor. The minor and major axes of the spot are 6 cm and 12 cm respectively. The angle subtended by the diameter of the sun at our eye is 0.5°. Calculate the height of the roof.

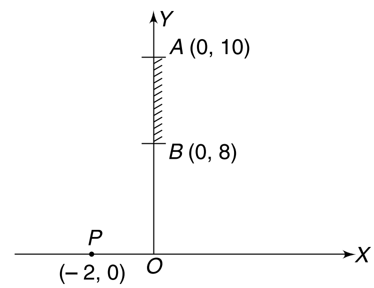
Q. 77: A car is travelling at night along a highway shaped like a parabola with its vertex at the origin of the co-ordinate system. The car starts at a point 200 m West and 200 m North of the origin and travels in easterly direction. There is an animal standing 200 m East and 100 m North of the origin. At what point on the highway will the car's headlight illuminate the animal?



Q. 78: Consider the situation shown in the Figure. The mirror AB forms image of point object P. Co-ordinates of A, B and P are (0, 10) m, (0, 8) m and (−2, 0) m respectively. Two observers O_1 and O_2 are located at (−2, 10) m and (−1, 13) m respectively.

(a) One of the two observers cannot see the image of point P. Identify the observer.

(b) To ensure that both the observers are able to see the image, it was decided to use a longer mirror. Keeping the upper end of the mirror fixed at A (0, 10) m, what is the minimum length of mirror required so that both observers can see the image of point P?



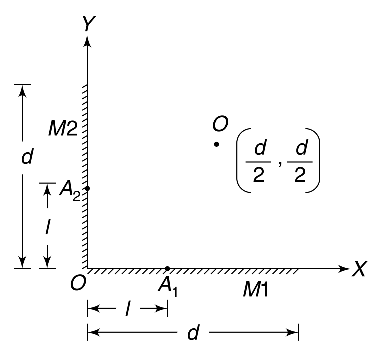
Q. 79: Two plane mirrors $M1$ and $M2$ of length d each are placed at right angle as shown. A point object O is placed symmetrically with respect to the mirrors at co-ordinates $\left(\frac{d}{2}, \frac{d}{2}\right)$

(a) How many images of O will be seen?

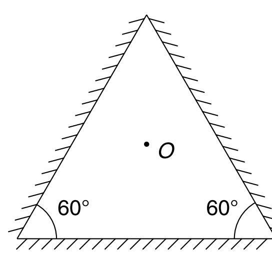
(b) Show that all the images lie on a circle.

(c) Length l $(= OA_1 = OA_2)$ of both the mirrors is cut and removed.

Find least value of l such that only two images of the object are formed.



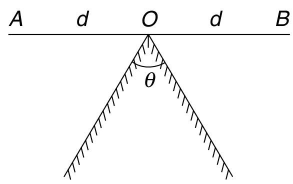
Q. 80: Three plane mirrors are kept as shown in the Figure A point object (O) is kept at the centroid of the triangle seen in the Figure. How many images will be formed?



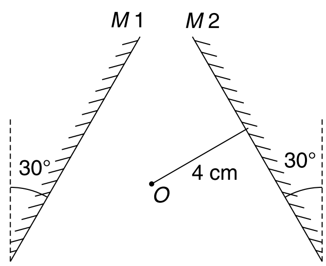
Q. 81: Two plane mirrors are joined together as shown. Two point objects A and B are placed symmetrically such that $OA = OB = d$. [AOB is a straight line]

(a) If the images of A and B coincide find θ (call it θ_0).

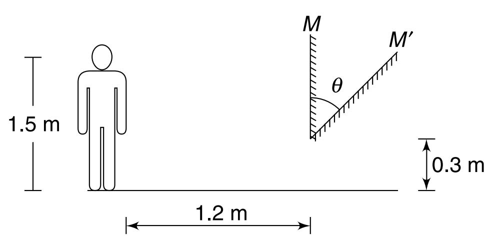
(b) Keeping the position of objects unchanged the angle between the two mirrors is increased to $\theta = \frac{4}{3}\theta_0$. Now find the distance between the images of A and B.



Q. 82: Two plane mirrors $M1$ and $M2$ are inclined at 30° to the vertical. A point object (O) is placed symmetrically between them at a distance of 4 cm from each mirror. Find the distance of the object from the second image formed in mirror $M1$.



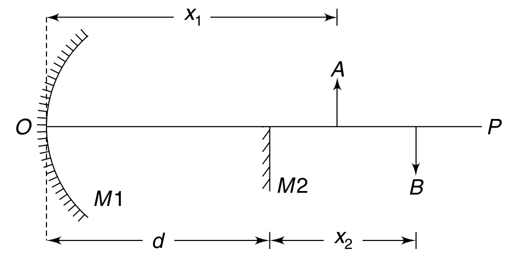
Q. 83: The distance between the eye and the feet of a boy is 1.5 m. He is standing on a flat ground and *a* vertical plane mirror M is placed at a distance of 1.2 m from the boy, with its lower edge at a height of 0.3 m from the ground. Now the mirror is tilted about is lower edge as shown in the Figure. Find maximum value of angle θ for which the image of feet remains visible to the boy. $\left[\text{Take } \tan 15^\circ \simeq \frac{1}{4}\right]$



Q. 84: OP is the principal axis of a concave mirror $M1$. Just below the axis a plane mirror $M2$ is placed at a distance d from the concave mirror. Two small pins A and B are placed on the principal axis as shown. By moving $M2$ and changing d, the virtual image of A formed in mirror $M1$ and the virtual image of B formed in mirror $M2$ were made to coincide.

(a) Calculate the focal length of the concave mirror if it was found that the images coincide when separation between the mirrors was d_0.

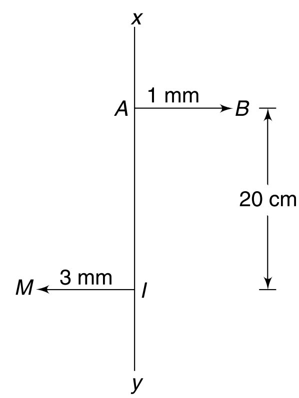
(b) Can the two virtual images be observed by the eye simultaneously?



Q. 85: A real object AB has its image as IM when placed in front of a spherical mirror. XY is the principal axis of the mirror.

(a) Draw a ray diagram to locate the position of the mirror and its focus.

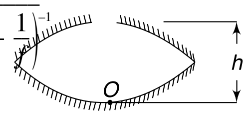
(b) Find the focal length of the mirror.



Q. 86: Two spherical concave mirrors of equal focal length are put against each other with their reflecting surfaces facing each other. The upper mirror has an opening at its centre. A small object (O) is kept at the bottom of the cavity so formed (see Figure). The top mirror produces a virtual image of the object and the lower mirror then creates a real image of the virtual image. The second image is created just outside the cavity mouth. This creates an optical illusion as if the object is raised above its original position.

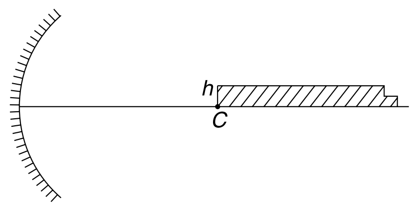
(a) Prove that the height h shown in the Figure is related to focal length of the mirror as

$$\frac{1}{f} = \frac{1}{h} + \frac{1}{h + \left(\frac{1}{h} - \frac{1}{f}\right)^{-1}}$$

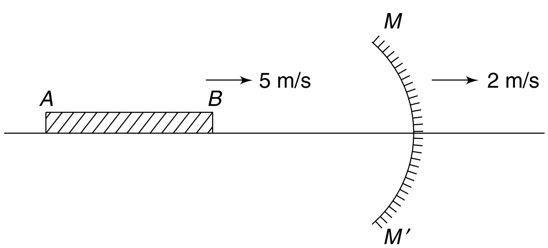


(b) Rewrite the above equation in terms of $x = \frac{h}{f}$ and solve it for x.

Q. 87: A long rectangular strip is placed on the principal axis of a concave mirror with its one end coinciding with the centre of curvature of the mirror (see Figure). The width (h) of the object is very small compared to the focal length (f) of the mirror. Calculate the area of the image formed.



Q. 88: A pencil (AB) of length 20 cm is moving along the principal axis of a concave mirror MM', with a velocity 5 m/s approaching the mirror. The mirror itself is moving away from the pencil at a speed of 2 m/s. Find the rate of change of length of the image of the pencil at the instant end A is at a distance of 60 cm from the mirror.



Q. 89: A small object of height h is placed perpendicular to the principal axis of a convex mirror of focal length f at a distance x from the pole of the mirror. An observer is located on the principal axis at a distance L from the pole of the mirror. (Take $x, f, L >> h$].

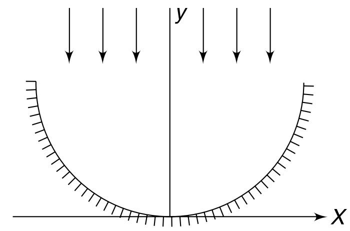
(a) Calculate the angle α formed by the image at the eye of the observer

(b) If the convex mirror is replaced by a plane mirror, with all other things remaining unchanged, calculate the angle β formed by the image at the eye.

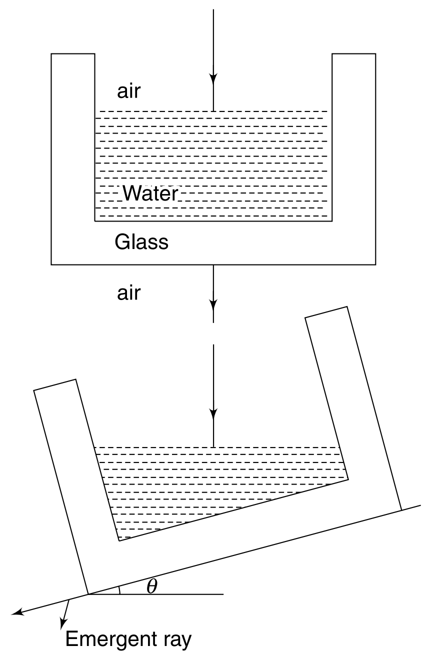
(c) Justify the statement "objects are closer than they appear" written on the rear view mirror of your car.

Q. 90: Consider a large parabolic mirror whose section can be represented by $y = kx^2$, where k is a positive number. Show that a parallel beam of light travelling in negative y direction, after reflection from the mirror, gets focused at a point $\left(0, \frac{1}{4k}\right)$.

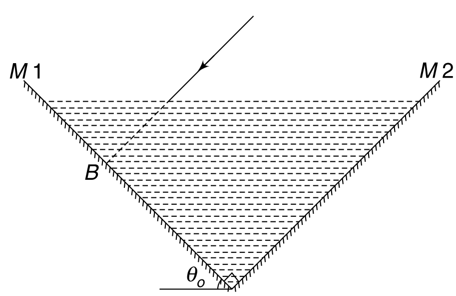
If you need a parallel beam of light, will you prefer a parabolic reflector over a spherical one?



Q. 91: Figure shows a glass ($\mu_g = 1.5$) vessel, partly filled with water $\left(\mu_w = \frac{4}{3}\right)$. A ray of light is incident normally on the water surface and passes straight through. The vessel is tilted slowly till angle θ such that the light ray is emergent grazing the lower surface of the glass. Find θ.



Q. 92: Two plane mirrors $M1$ and $M2$, placed at right angles, form two sides of a container. Mirror $M1$ is inclined at an angle θ_0 to the horizontal. A light ray AB is incident normally on $M1$. Now the container is filled with a liquid of refractive index μ so that the ray AB is first refracted, then reflected at $M1$. The ray is next reflected at $M2$ and then comes out of the liquid surface making an angle θ with the normal to the liquid surface. Find θ.

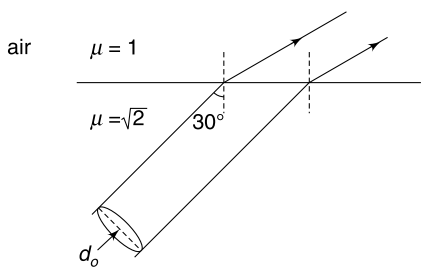


Q. 93: When the sun appears to be just on horizon, it is in fact below the horizon. This is because the light from the sun bends when it enters the earth's atmosphere. Let us assume that atmosphere is uniform and has index of refraction equal to μ. It extends upto a height h ($<< R$ = radius of earth) above the earth's surface. In absence of atmosphere how late would we see the sunrise compared to what we see now? Take time period of rotation of earth to be T.

Calculate this time for following data

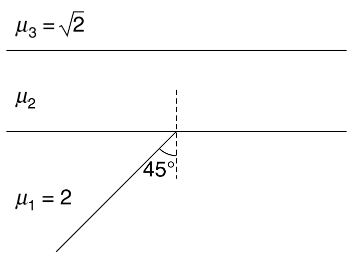
$R = 6400$ km; $\mu = 1.0003$; $h = 20$ km; $T = 24$ hr.

Q. 94: Intensity of a light beam can be defined as amount of light energy incident in unit time on a unit area held normal to the direction of beam. A light beam of intensity I_0 has a circular cross section of diameter d_0. This beam is travelling in a medium of refractive index $\mu = \sqrt{2}$ and gets incident on the medium – air boundary at an angle of incidence $i = 30°$. Assume that entire light energy gets transmitted into air. Find the intensity (I) of transmitted light beam.



Q. 95: In the diagram shown, a light ray is incident on the lower medium boundary at an angle 45° with the normal. Find the deviation suffered by the light ray if–

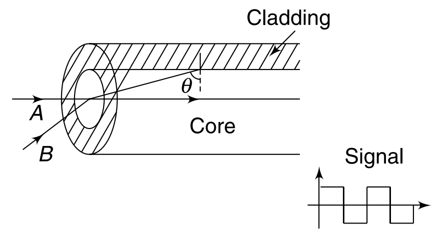
(a) $\mu_2 > \sqrt{2}$ (b) $\mu_2 < \sqrt{2}$



Q. 96: A fibre optic cable has a transparent core of refractive index 1.6 and the cladding has a refractive index of 1.5. An optical signal travels along path A and another signal travels along path B such that it strikes the core – cladding interface at an angle of incidence θ that is just greater than the critical angle. Length of the cable is 1500 m.

(a) Find the time difference between the two signals reaching the other end of the cable.

(b) A digital signal shown in the Figure. is transmitted through the cable. Find maximum frequency so that the crest from path A never arrives with a trough from path B at the receiving end of the cable.



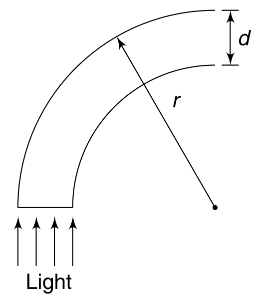
Q. 97: An optical fibre has diameter d and is made of material of refractive indeed μ. It is surrounded by air. Light is made to enter through one end of the fibre as shown. The fiber is in the shape of a circular bend of outer radius r.

(a) Find least value of $r (= r_o)$ for which no light can escape out of the fibre.

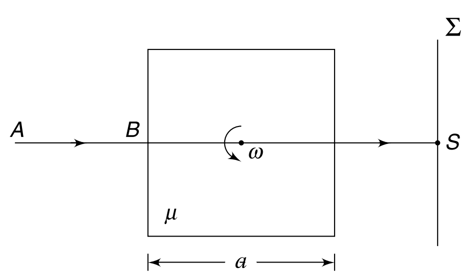
Calculate r_o for $d = 200$ μm and $\mu = 1.4$.

(b) How is value of r_o affected as d is made smaller?

(c) For sharper bends, shall we have higher μ or smaller μ?

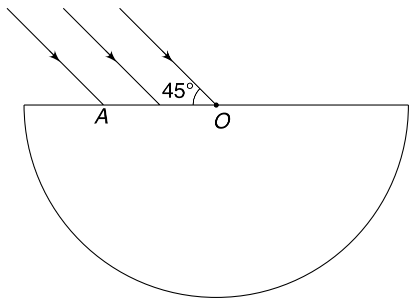


Q. 98. A glass cube has side length a and its refractive index is $\mu = \frac{3}{2}$. A ray of light (AB) is incident normally on one of its face and after passing through the cube it forms a spot S on screen Σ. The cube begins to rotate with angular speed ω about its central axis as shown in the Figure. Immediately after the cube begins to rotate, find the speed of the spot S on the screen.

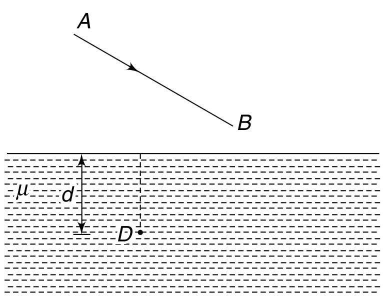


Q.99: A single ray traverses a glass plate (thickness = t) with plane surfaces that are parallel to each other. The emergent ray is parallel to the incident ray but suffers a lateral displacement d. Assuming that glass plate (refractive index μ) is placed in air, find the dependence of d on angle of incidence i. Plot the variation of d with i (changing from 0° to 90°)

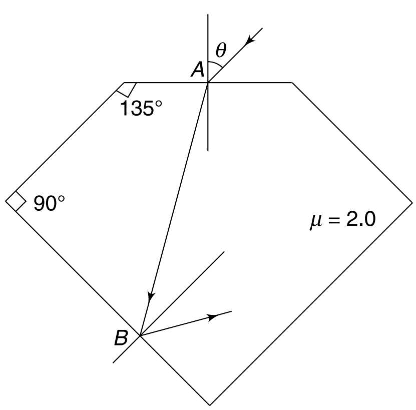
Q. 100: A transparent semicylinder has refractive index $\mu = \sqrt{2}$. A parallel beam of light is incident on its plane surface making an angle of 45° with the surface. The incident beam extends from O to A on the plane surface. Find the maximum width OA (in terms of radius R of the cylinder) so that no ray suffers total internal reflection at the curved surface. [O is the centre of the circular cross section of the cylinder]



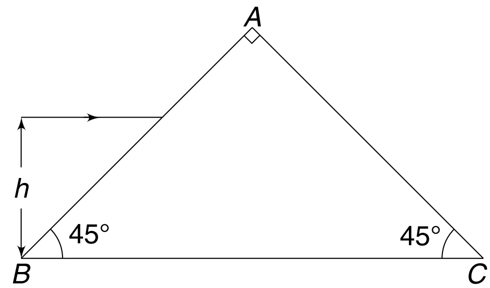
Q. 101: A diver D is still under water $\left(\mu = \frac{4}{3}\right)$ at a depth d = 10 m. A bird is diving along line AB at a constant velocity in air. When the bird is exactly above the diver he sees it at a height of 50 m from himself and velocity of the bird appears to be inclined at 45° to the horizontal. At what distance from the diver the bird actually hits the water surface.



Q. 102: Light is incident at point A on one of the faces of a diamond crystal (μ = 2.0). Find the maximum allowed value of angle of incidence θ so that light suffers total internal reflection at point B.



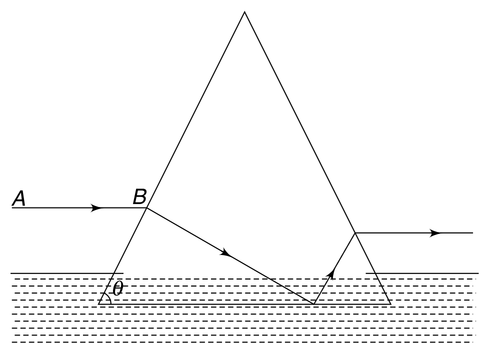
Q. 103: ABC is a glass prism with $\angle A$ = 90° and other two angles 45° each.



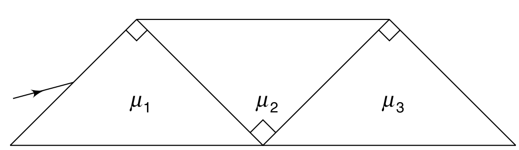
(a) Prove that any light ray that enters the prism through face AB will emerge out through the face AC if refractive index of the glass of the prism is $\mu \geq \sqrt{2}$.

(b) A ray of light is incident parallel to BC at a height h = 3.0 cm from BC. Find the height above BC at which the emergent ray leaves the surface AC. It is given that $\mu = \sqrt{2}$ and length BC = 20 cm.
[Take $\tan 15° \simeq 0.25$]

Q. 104: An isosceles glass prime $\left(\text{refractive index} = \frac{3}{2}\right)$ has its base just submerged in water $\left(\text{refractive index} = \frac{4}{3}\right)$. The base of the prism is horizontal. A horizontal light ray AB is incident on the prism and takes a path shown in figure to emerge out of the prism. Find the maximum value of base angle θ of the prism for which total internal reflection can take place at the base.



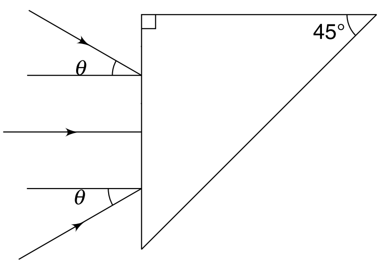
Q. 105: Three right angled prisms are glued as shown in the figure. An incident ray passes undeviated through the system. Express the refractive index (μ_2) of the middle prism in terms of μ_1 and μ_3.



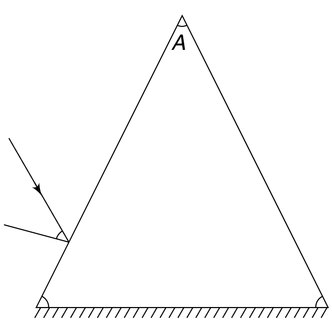
Q. 106: A prism has refracting angle of 60° and its material has refractive index 1.5 and 1.6 for red and violet light respectively. A parallel beam of white light is incident on one face of the prism such that the red light undergoes minimum deviation. Find the angle of incidence (i) and the angular width (θ) of the spectrum obtained.

Given: $\sin(49°) = 0.75°$; $\sin(28°) = 0.47$; $\sin(32°) = 0.53$; $\sin(58°) = 0.85$

Q. 107: A converging beam of light is incident on a right angled isosceles prism as shown in the Figure. The marginal rays in the beam are incident at angle $\pm\theta$. The refractive index for the glass of the prism is $\mu = 1.49 \left(= \dfrac{1}{\sin 42°}\right)$. Find the maximum value of θ for which no light comes out of the hypotenuse surface.



Q. 108: The isosceles prism shown in the Figure has one of its face silvered. A light ray is incident on the prism as shown. Prove that the deviation suffered by the ray is independent of the wavelength of the light.



Q. 109: An equilateral prism is made of glass whose refractive index for red and violet light is 1.510 and 1.550 respectively. White light is incident at an angle of incidence i and the prism is set to give minimum deviation for red light. Find

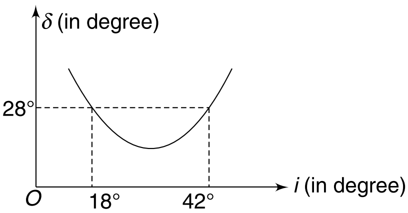
(a) angle of incidence

(b) angular dispersion (i.e., angular width of the spectrum).

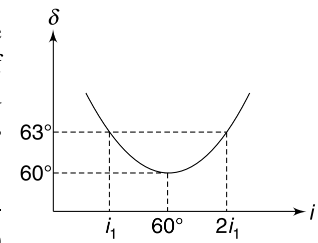
Given

$\theta =$	28°	32°	50°	55°
$\sin\theta =$	0.487	0.529	0.755	0.819

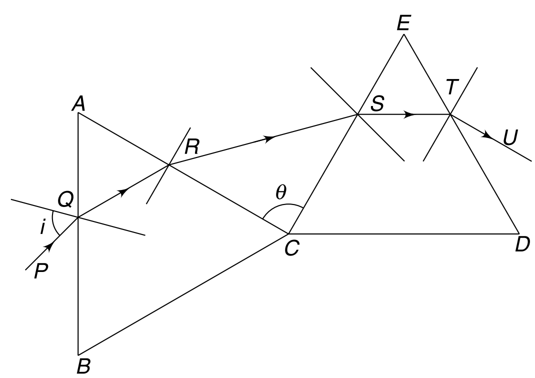
Q. 110: The plot of deviation (δ) *vs* angle of incidence (i) for a prism is as shown in the figure. Find the angle of the prism (A).



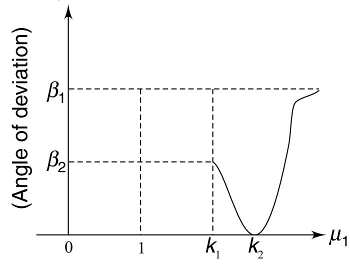
Q. 111: For a glass prism the plot of deviation (δ) vs angle of incidence (i) is as shown. Find the refractive index of the glass and value of angle i_1.



Q. 112: Two identical equilateral glass (refractive index = $\sqrt{2}$) prisms ABC and CDE are kept such that the angle between faces AC and CE is θ. A ray of light is incident at an angle i at the face AB and traverses through the two prisms along the path $PQRSTU$. Find the value of angle i and θ such that angle between the incident ray PQ and emergent ray TU is minimum.



Q. 113: An equilateral prism of refractive index $\mu = \dfrac{4}{\sqrt{3}}$ is kept in a medium of refractive index μ_1. Consider a light ray to be normally incident on one of the refracting faces. The diagram shows variation of magnitude of angle of deviation (β) with respect to μ_1.



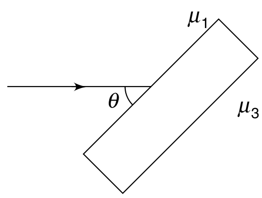
(a) Find value of k_2.

(b) Find value of k_1.

Q. 114: A rectangular glass block (μ = 1.5) is on top of a sheet of paper on which there on which there is a small dot. There is a layer of liquid between the paper and the glass block. The dot is visible through a vertical face up to a point where the angle of emergence of light (starting from the dot) is 30°. Find the refractive index (μ_0) of the liquid. Can we see the dot through a vertical face if the liquid layer is replaced with air?

$$\sin^{-1}\frac{2}{3} = 42°$$

Q. 115: A light ray travelling in a medium of refractive index μ_1 is incident on a parallel faced glass slab making an angle of θ with the glass surface. The refractive index of the medium on the other side of the glass slab is $\mu_3 (> \mu_1)$. Find the angular deviation suffered by the light ray.

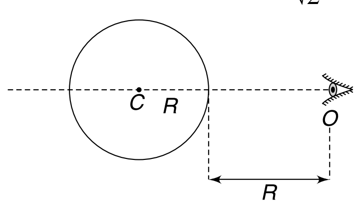


Q. 116: The Figure shows the equatorial circle of a glass sphere of radius R having centre at C. The eye of an observer is located in the plane of the circle at a distance R from the surface. A small insect is crawling along the equatorial circle.

(a) Calculate the length (L) of the are on the circle where the insect lies where its image is visible to the observer.

(b) Calculate the value of L when the eye is brought very close to the sphere.

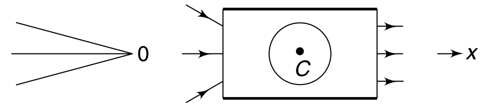
Refractive index of the glass is $\mu = \frac{1}{\sqrt{2}}$



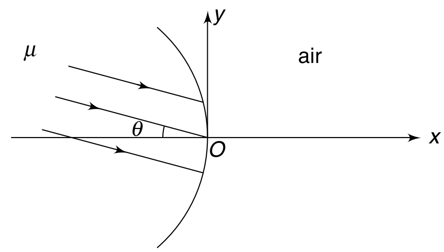
Q. 117: On a hot summer day in a desert the refractive index of the atmosphere changes with height (y) above the surface of the earth as $\mu = \mu_0(1 + by)^{1/2}$ where μ_0 is the refractive index at the surface and $b = 6 \times 10^{-4}$ m^{-1}. A man of height 1.5 m is standing on a straight level road. Calculate the distance beyond which he cannot see a point on the road.

Q. 118: The angle of minimum deviation caused by a prism is equal to the angle of the prism. What are the possible values of refractive index of the material of the prism?

Q. 119: A beam of light rays converges to a point O on x-axis as shown. The angle of convergence is small. A cube of glass of refractive index μ = 1.5 and side length 40 cm containing a concentric spherical air cavity of radius 10 cm is to be placed in the path of the converging beam so that the beam emerging from the cube is parallel to x-axis. At what point C on x-axis should the centre of the cube be placed to achieve this? Give the x coordinate of C taking O as origin.

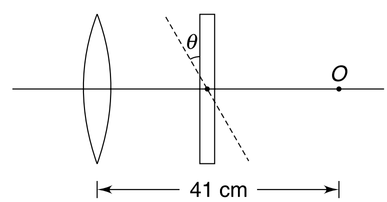


Q. 120: A spherical surface of radius R separates air from a medium of refractive index μ. Parallel beam of light is incident, from medium side, making a small angle θ with the principal axis of the spherical surface. Find the co-ordinates of the point where the rays will focus in air.



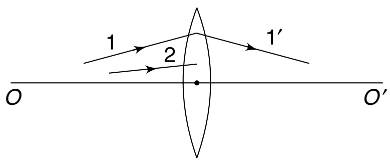
Q. 121: A point object O is placed at a distance of 41 cm from a convex lens of focal length f = 20 cm on its principal axis. A glass slab of thickness 3 cm and refractive index μ = 1.5 is placed between the lens and the object with its faces perpendicular to the principal axis of the lens. Image of the object is formed at point I_1. Now the glass slab is tilted by an angle of θ = 1° (as shown in the Figure) and the final image is formed at I_2. Calculate the distance between points I_1 and I_2.

Consider only paraxial rays for the lens and near normal incidence for the glass slab.



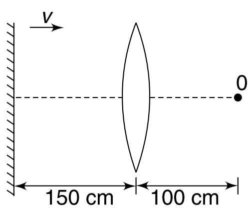
Q. 122: A man is Standing on the peak of a mountain and finds that evening sun rays are nearly horizontal. At a horizontal distance of 6 km from him, its raining and he sees a beautiful rainbow. The sun rays entering water drop get refracted, reflected and refracted to form a rainbow. The red light is emitted from a drop upto a maximum angle of 42° with respect to the incident sunlight. In front of the man there is a flat valley at a depth of 0.5 km from the mountain peak. What fraction of the complete circular arc of the rainbow is visible to the man?

Q. 123: A light ray 1 after passing through a lens kept in air, goes along path 1′. OO' is the principal axis. Draw the refracted path of light ray 2. Write all steps used in construction.



Q. 124: An observer holds in front of himself a thin equi convex lens. R is the radius of curvature of each face. He sees two images of his nose, one erect and the other inverted. Explain the formation of these images and assuming the refractive index of glass to be 1.50 prove that he will see two erect images if the distance of the lens is less than $0.25R$ from his nose.

Q. 125: An observer is standing at a point O, at a distance of 100 cm from a convex lens of focal length 50 cm. A plane mirror is placed behind the lens at a distance of 150 cm from the lens. The mirror now starts moving towards the right with a velocity of 10 cm/s. What will be the magnitude of velocity (in m/s) of her own image as seen by the observer, at the moment when the mirror just starts moving?

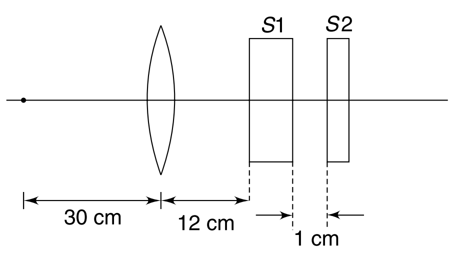


Q. 126: A small object (A) is placed on the principal axis of an equiconvex lens at a distance of 30 cm. The refractive index of the glass of the lens is 1.5 and its surfaces have radius of curvature R = 20 cm. Two glass slabs $S1$ and $S2$ have been placed behind the lens as shown in Figure. Thickness of the two slabs is 6 cm and 4 cm respectively and their refractive indices are $\frac{3}{2}$ and 2 respectively.

(a) Find the distance of the final image measured from the lens. Also find the magnification.

(b) How does the position of the image change if the slab $S2$ is moved to left so as to put it in contact with $S1$.

(c) How does the position of the image change if the two slabs in contact are together moved to right by a distance of 100 cm.



Q. 127: A virtual image is formed by a lens for a real object.

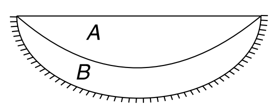
Take $\left|\frac{v}{f}\right| = y$ and $\left|\frac{u}{f}\right| = x$ and draw y vs x graph if

(a) lens is diverging

(b) lens is converging.

Q. 128: An equiconvex lens of refractive index 1.5 has its two surfaces having radius of curvature of 30 cm. A point object has been placed on the principal axis at a distance of 60 cm from the lens. Find the distance of image from the lens formed by the rays which suffer refraction at first surface, reflection at second surface, again a reflection on the first surface and finally a refraction from the second surface.

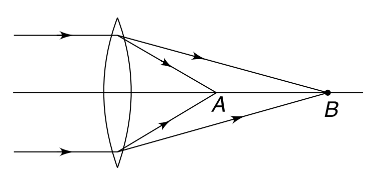
Q. 129: A thin plano convex lens A has material of refractive index μ_A = 1.8 and its curved surface has radius of curvature R. A thin layer of transparent material B is laid over the curved surface of A. The refractive index of B is μ_B = 1.2 and the curved surface of B that is not touching a has radius of curvature $\frac{R}{2}$. This surface of B is silvered. A point object is kept at a distance of 10 cm on the principal axis of the system (above the plane surface) and its image is formed at a distance of 40 cm above the plane surface. Find R.



Q. 130: The refractive index of light in glass varies with its wavelength according to equation

$$\mu(\lambda) = a + \frac{b}{\lambda^2}$$

where a and b are positive constants.

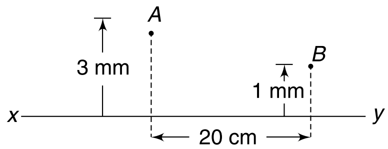


A nearly monochromatic parallel beam of light is incident on a thin convex lens as shown. The wavelength of incident light is $\lambda_0 \pm \Delta\lambda$ where $\Delta\lambda << \lambda_0$. The light gets focused on the principal axis of the lens over a region AB. If the focal length of the lens for a light of wavelength λ_0 is f_0, find the spread AB.

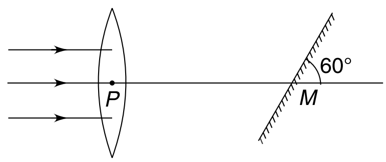
Q. 131: An image B is formed of a real point object A by a lens whose optic axis is XY.

(a) Draw a ray diagram to locate the lens and its focus point.

(b) If A and B are separated by 20 cm along the axis, find the focal length of the lens.



Q. 132: A horizontal parallel beam of light passes through a vertical convex lens of focal length f. The optical centre of the lens is P. A small plane mirror is placed at point M inclined at 60° to the axis of the lens. Distance $PM = f/2$. The mirror reflects the light passing through the lens and forms an image at point I. Find distance PI.



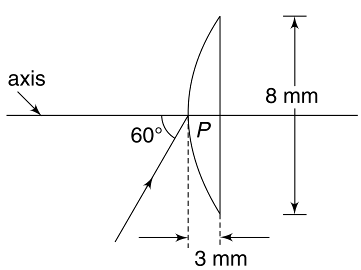
Q. 133: A plane convex lens has aperture diameter of 8 mm and thickness of the lens at the centre is 3 mm. The refractive index of the material of the lens is $\sqrt{3}$. A light ray is incident at mid point P of the curved surface at an angle of incidence of 60°.

(a) Calculate the angular deviation suffered by the ray as it passes through the lens.

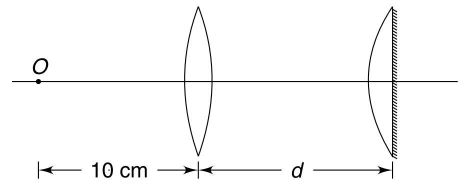
(b) Find the lateral shift in the path of the ray as it passes through the lens.

(c) Find the radius of curvature of the curved surface.

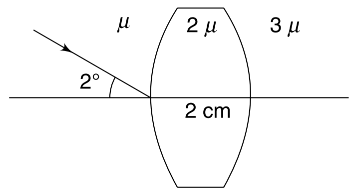
(d) If a narrow beam of light is incident at P parallel to the axis shown, where will it get focused. Take $2\sqrt{3}(\sqrt{3} - 1) \simeq 2.5$



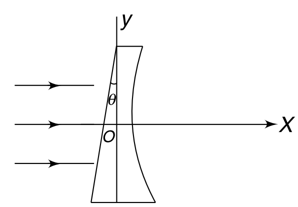
Q. 134: A convex lens of focal length 20 cm and another planocovex lens of foal length 40 cm are placed co-axially. The plane surface of the plano convex lens is silvered. An object O is kept on the principal axis at a distance of 10 cm from the convex lens (see Figure). Find the distance d between the two lenses so that final image is formed on the object itself.



Q. 135: An equiconvex lens has its two surfaces of radius of curvature R = 10 cm. Thickness of the lens at its centre is 2 cm. A light ray is incident making an angle of 2° with the optic axis of the lens. Find the angle that the emergent ray will make with the optic axis. The refractive index of all media is as marked in the Figure.



Q. 136: Due to manufacturing defect, the plane surface of a thin plano-concave lens has been made tilted at a small angle θ outwards from its usual place. The spherical surface has radius of curvature R and refractive index of the material of the lens is μ. A parallel beam of light is incident as shown. In the co-ordinate system shown find the co-ordinates of the point where the rays will focus or appear to be diverging from.



Q. 137: Two lenses L_1 and L_2 are used to make a telescope. The larger lens L_1 is a convex lens with both surfaces having radius of curvature equal to 0.5 m. The smaller lens L_2 has two surfaces with radius of curvature 4 cm. Both the lenses are made of glass having refractive index 1.5. The two lenses are mounted in a tube with separation between them equal to 1 cm less than the sum of their focal length.

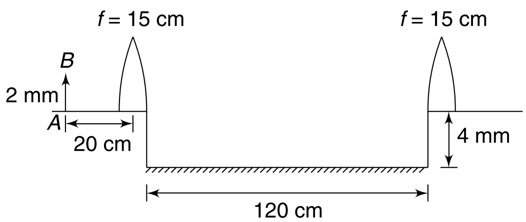
(a) Find the position of the image formed by such a telescope for an object at a distance of 100 m from the objective lens L_1.

(b) What is the size of the image if object is 1 m high? Do you think that lateral magnification is a useful way to characterize a telescope?

(c) Angular magnification is defined as ratio of angle subtended by the image at the eyepiece to the angle subtended by the distant object at an unaided eye. Find the angular magnification of the above mentioned telescope.

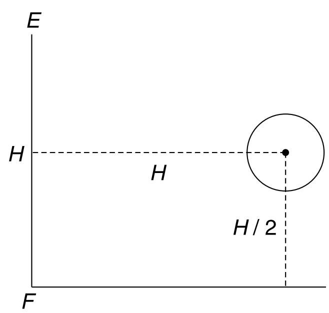
Q. 138: A convex lens of focal length 15 cm is split into two halves and the two halves are placed at a separation of 120 cm. Between the two halves of convex lens a plane mirror is placed horizontally and at a distance of 4 mm below the principal axis of the lens halves. An object AB of length

2 mm is placed at a distance of 20 cm from one half lens as shown in figure. Find the distance of final image of the point A from the principal axis.



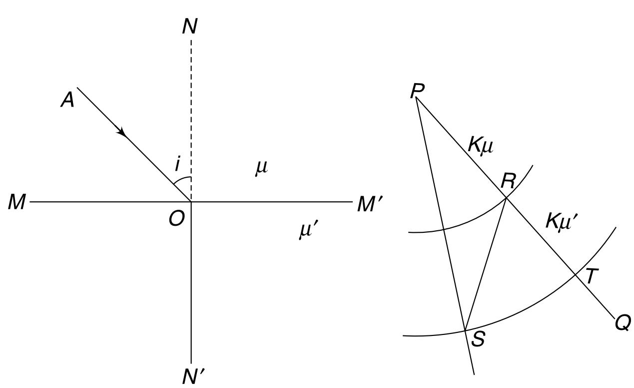
Level 3

Q. 139: In the Figure FE is a man of height H standing on a floor. E is eye of the man and F is his foot. The distance between eye and the head is negligible. A steel ball of radius r is suspended in front of him. The distance of the ball from the man is H and height of the centre of the ball from the floor is $\frac{H}{2}$. It is given that $r << H$. The surface of the ball acts like a mirror and the man sees his image in it. Calculate the angle subtended by the image at the eye of the man.



Q. 140: An observer views his own image in a convex mirror of radius of curvature R. If the least distance of distinct vision for the observer is d, calculate the maximum possible magnification.

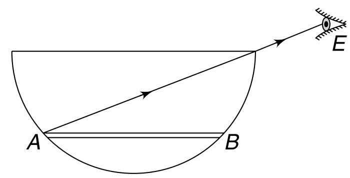
Q. 141: The Figure (a) shows two media having refractive index μ and μ' with MM' as boundary. A ray of light AO is incident at the boundary and gets refracted.



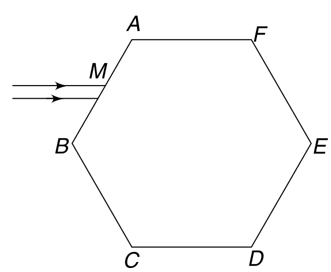
(a) (b)

In order to construct the refracted ray graphically a teacher suggests to draw a line PQ parallel to the incident ray AO. With P as centre, two circular arcs are drawn having radii $k\mu$ and $k\mu'$ where k is a positive number. The arcs intersect the line PQ at R and T respectively. From R, a line is drawn parallel to normal NN' and it intersects the other arc at S. Prove that the refracted ray is parallel to line PS.

Q. 142: A stick is placed inside a hemispherical bowl as shown in Figure. The stick is horizontal and has a length of $2a$. Eye of an observer is located at E such that it can just see the end A of the stick. A liquid is filled upto edge of the bowl and the end B of the stick becomes visible to the observer. Radius of the bowl is R. Find the refractive index (μ) of the liquid.

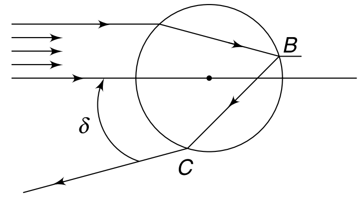


Q. 143: The cross section of a prism is a regular hexagon. A narrow beam of light strikes a face of the prism just below the midpoint (M) of the edge AB. The beam is parallel to the top and bottom faces of the prism. Final the minimum value of refractive index of the material of the prism for which the emergent beam will be parallel to the incident beam.

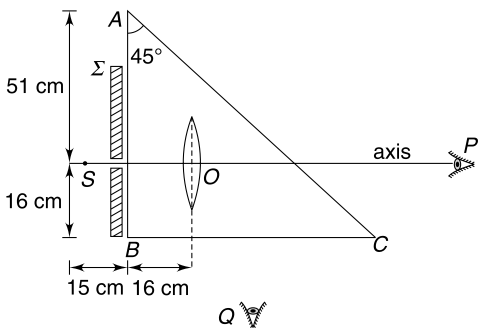


Q. 144: A parallel beam of light is incident on a spherical drop of water (μ = 4/3). Consider refraction of light at air-water interface, then reflection at water-air boundary (of course there will be refracted light energy in air as well), followed by refraction at water-air interface. Path of a typical ray refracted at A, then reflected at B and finally refracted at C has been shown in the Figure. Find the maximum value of angle δ.

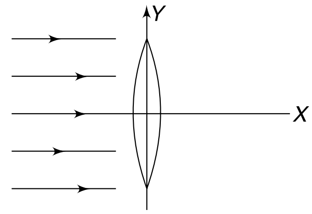
use $\sin^{-1}\sqrt{\frac{20}{27}} = 60°$ and $\sin^{-1}\sqrt{\frac{5}{12}} = 40°$.



Q. 145: An isosceles right angled triangular glass ($\mu = 1.6$) prism has a cavity inside it in the shape of a thin convex lens whose both surfaces have radius of curvature equal to 20 cm. The cavity has been filled with a transparent liquid of refractive index 2.4. S is a point source and Σ is an opaque sheet having a small hole such that the source and the hole both lie on the principal axis of the lens. The small hole in the opaque sheet is just to ensure that only paraxial rays are incident on the optical system. However, the size of hole is large enough to neglect diffraction effects. Will the observers at P and Q be able to see the image of source S? Where is the image located?



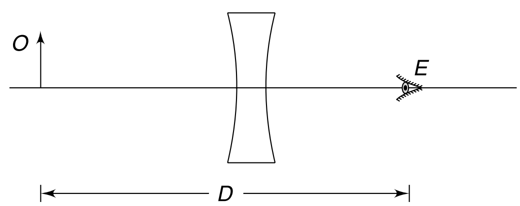
Q. 146: Monochromatic light rays parallel to the principal axis (the x axis) are incident on a convex lens of focal length f. If the lens oscillates such that it tilts up to a small angle θ on either side of the y axis, then find the distance between the extreme positions of the image. Take $\sec\theta \simeq 1 + \frac{\theta^2}{2}$ for small θ.



Q. 147: O is a small object placed at a distance D from the eye E of an observer. A concave lens of focal length $f(\angle D)$ is placed near to the eye and image of the object is viewed. Now the lens is moved towards the object O, away from the eye.

(a) Show that the angle subtended by the image at the eye first decreases and then increases as the lens is moved away from the eye.

(b) Find the distance of the lens from the object when the apparent size of image is smallest.



ANSWERS

1. 2.57 m

2. (a) 60 m (b) Size of image increases (c) Image will get blurred.

3. $\frac{10}{\sqrt{3}}$ m

4. $\frac{2d}{\sqrt{3}}$

5. No

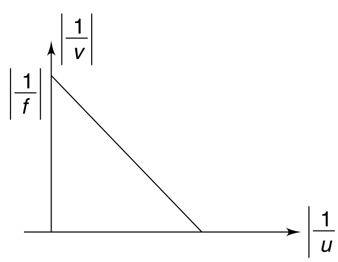
6. 0.14°

7. Image gets inverted beyond 45 cm.

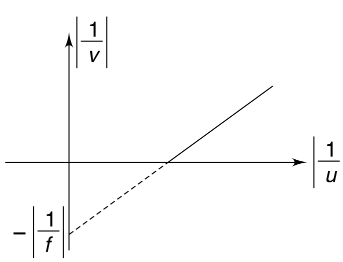
9. At a distance $\frac{5R}{6}$ from the nearest wall

10. 60 cm

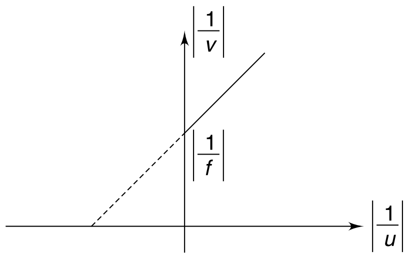
11. (a)



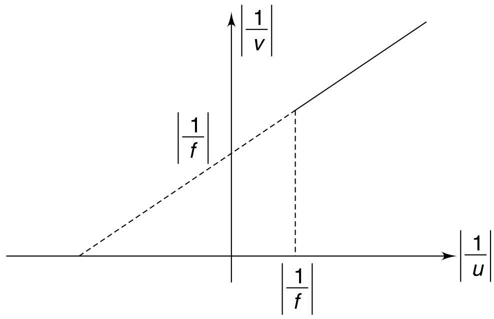
(b)



(c)



(d)



12. (a) 90 cm (b) No.

13. $4f\theta$

14. 20 cm

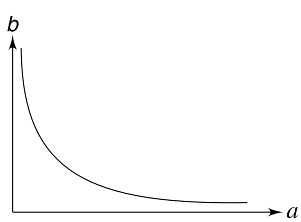
15. (a) very large (b) 6.67 cm on both sides of P.

16. 69.33 cm.

17. (a) $\sqrt{2}$ (b) $\sqrt{3}$

18. 150 rad s^{-1}

19.



20. 9

21. green

22. Underwater part is closer

23. Section 1: Red colour, less bright than usual

Section 2: Blue colour, less bright than usual

Section 3: Dark, almost no light.

24. 82.5 cm

25. Zero.

26. 45°

27. Fig. (b)

28. 15°

30. $\frac{3\sqrt{5}}{8} < d < \frac{\sqrt{5}}{2}$ cm

31. $\mu = \frac{r_3 - r_1}{r_3 - r_2}$

32. 0.34 R

33. $\mu = \frac{10}{9}$

34. C

35. H ln 2

36. $t - \frac{1}{\mu_0} \ln(1 + t)$

37. 60° and 40°

38. 30°

39. $a > \frac{2}{9}$

40. E

41. $\sqrt{2} < \mu < 2$

42. (a) $A_0 = 2\sin^{-1}\left(\frac{1}{\mu}\right)$ (b) No

43. $\mu > \frac{1}{\sin 72°}$

44. $\mu = 2$, $A = 1°$

45. $\cot A < (\sqrt{3} - 1)$ where A is angle of the prism.

46. Upper graph

47. $\mu_{min} = \sqrt{2}$

48. (a) 2 m/s (b) 2 m/s

49. 26 cm < d < 32 cm

50. L = 32 cm

52. 30°

53. Red

54. 0.04

55. $A \rightarrow$ red, $B \rightarrow$ yellow, $C \rightarrow$ Blue, $D \rightarrow$ green.

56. (a) For all distance

(b) For all distance

57. $\mu = 2$

58. $\frac{R}{2}$

59. Image is always virtual

60. (a) Virtual

(b) May be real or virtual

(c) May be real or virtual

61. 10 cm

62. Concave lens. Lens is to the right of 3′

63. 3.3 cm

64. (a) 320 cm (b) $\frac{1}{3}$ cm

65. $f = 40$ cm

66. when media on both sides of the lens is same.

68. 60°

69. The image does not move.

70. 2

71. $2h$

72. 45° downward

73. Moved down

74. 18 cm

75. 90 cm

76. 3.44 m

77. $X = 58.6$ m; $Y = 17.17$ m.

78. (a) O_1 will not see the image (b) 5 m.

79. (a) 3 (c) $l = \frac{d}{3}$

80. 12

81. (a) $\theta_0 = 90°$ (b) d

82. $8\sqrt{3}$ cm

83. 15°

84. (a) $|f| = \frac{(x_2 - d_0)\, x_1}{x_2 - (d_0 + x_1)}$ (b) No.

85. Mirror is at a distance of 10 cm from A. Concave mirror, $f = 7.5$ cm

86. (b) $x = 1$

87. $\frac{1}{2} fh$

88. $\frac{9}{4}$ m/s

89. (a) $\alpha = \dfrac{h}{L + x + \dfrac{Lx}{f}}$ (b) $\beta = \dfrac{h}{L + x}$

91. $\theta = \sin^{-1}\left(\frac{3}{4}\right)$

92. θ_0

93. $\Delta t = \frac{T}{2\pi}\left[\sin^{-1}\left(\frac{\mu R}{R + h}\right) - \sin^{-1}\left(\frac{R}{R + h}\right)\right]$

$= 0.9$ minute.

94. $I = \frac{3}{2} I_0$

95. (a) 45° (b) 90°

96. (a) 0.53 μs (b) 0.94 MHz

97. (a) $r_0 = \frac{\mu d}{\mu - 1}$

0.7 mm

(b) r_0 gets smaller

(c) higher μ.

98. $\frac{a\omega}{3}$

99. $d = t \sin i\left[1 - \frac{\cos i}{\sqrt{\mu^2 - \sin^2 i}}\right]$

100. $\sqrt{\frac{2}{3}}\, R$

101. 24.62 m

102. $\theta_{max} = \sin^{-1}(2 \sin 15°)$

103. (b) 1 cm

104. $\theta_{max} = \cos^{-1}\left(\sqrt{\frac{17}{21}}\right)$

105. $\mu_2 = \sqrt{\mu_1^2 + \mu_3^2 - 1}$

106. $i = 49°$; $\theta = 9°$

107. 4.47°

109. 5°

110. 32°

111. $\mu = \sqrt{3}$ and $i_1 = 41°$

112. $i = 45°$, $\theta = 90°$

113. (a) $\frac{4}{\sqrt{3}}$ (b) $\frac{8}{3}$

114. $\mu_0 = \sqrt{2}$, No

115. $\delta = \frac{\pi}{2} - \theta - \sin^{-1}\left(\frac{\mu_1}{\mu_3}\cos\theta\right)$

116. (a) $\frac{\pi R}{3}$ (b) πR

117. 100 m

118. $\sqrt{2} < \mu < 2$

119. 0

120. $x = \frac{R}{\mu - 1}$; $y = R\left(\frac{\mu}{\mu - 1}\right)$

121. $\frac{\pi}{180}$ cm

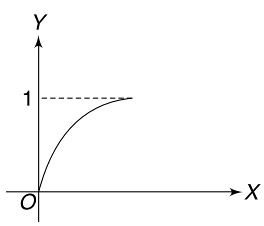
122. 0.53

124. Reflection from the two surfaces results in formation of two images.

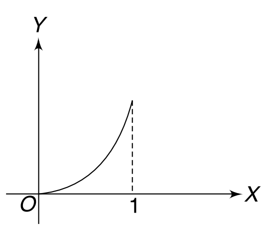
125. 20/9 = 2.22 cm/s

126. (a) 60 cm; $m = -2$ (b) No change (c) No change

127. (a)



(b)



128. $\frac{60}{13}$ cm

129. 48 cm.

130. $f_0\left[\frac{4b\Delta\lambda}{\lambda_0(a\lambda_0^2 + b - \lambda_0^2)}\right]$

131. (b) concave lens, focal length = 15 cm

132. $f/2$

133. (a) 0 (b) $\sqrt{3}$ mm

(c) $\frac{25}{6}$ mm

(d) At a distance $\left(3 + \frac{7}{\sqrt{3}}\right)$ mm from P.

134. $d = 20$ cm

135. $\left(\frac{8}{15}\right)^{\circ}$

136. $\left(\frac{-R}{\mu - 1}, R\theta\right)$

137. (a) 8.8 cm from L_2 in between the two lenses (b) 1.6 cm, No (c) 18.2

138. 8/3 mm

139. $\theta = \frac{2r}{5H}$

140. $\frac{R}{\sqrt{R^2 + d^2} + d}$

142. $\mu = \sqrt{\frac{R + a}{R - a}}$

143. $\mu_{\min} = \frac{\sqrt{13}}{2}$

144. 40°

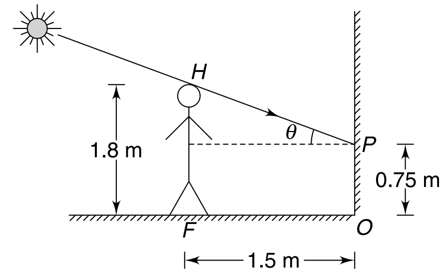
145. Observer at Q sees the image at a distance 6.88 cm from face BC, inside the prism.

146. $\frac{f\theta^2}{2}$

147. (b) $\frac{D}{2}$

SOLUTIONS

1. If sun rays make angle θ with horizontal

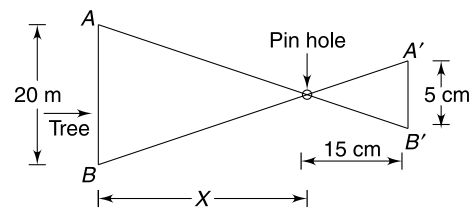


$$\tan\theta = \frac{1.05}{1.5} = \frac{7}{10}$$

When there is no wall, length of shadow (L) is given as

$$\frac{1.8}{L} = \tan\theta$$

$$L = \frac{1.8 \times 10}{7} = \frac{18}{7} = 2.57 \text{ m}$$



2. (a) $$\frac{x}{15 \text{ cm}} = \frac{20 \text{ m}}{5 \text{ cm}}$$

$\Rightarrow$ $$x = 60 \text{ m}$$

3. One extreme ray OA, after reflection reaches P on the line XMX'

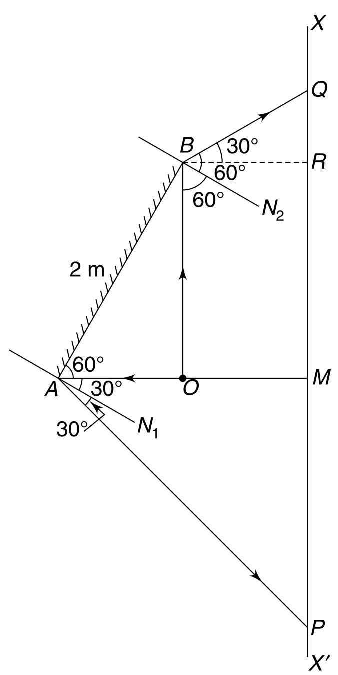
$$MP = AM \cdot \tan 60° = 2\sqrt{3} \text{ m}.$$

The other extreme ray OB, after reflection reaches the observe at Q.

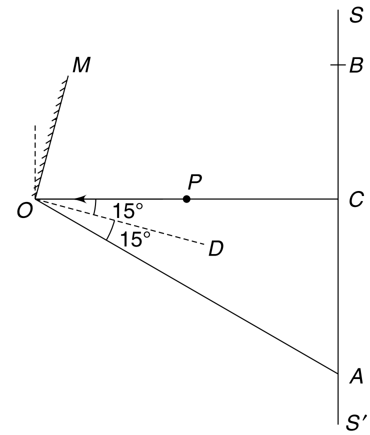
$$QR = BR \cdot \tan 30° = \frac{1}{\sqrt{3}} \text{ m}.$$

$\therefore$ $$PQ = 2\sqrt{3} + 2\sin 60° + \frac{1}{\sqrt{3}}$$

$$= 2\sqrt{3} + \sqrt{3} + \frac{1}{\sqrt{3}} = 3\sqrt{3} + \frac{1}{\sqrt{3}} = \frac{10}{\sqrt{3}} \text{ m}.$$



4. OD is normal to mirror OM at O. The ray PO is reflected along OA.



$$\frac{AC}{d} = \tan 30° \Rightarrow AC = \frac{d}{\sqrt{3}}$$

Various rays starting from P and incident on the mirror OM, after getting reflected from OM, reach the line SS' at all points except below point A.
Similarly rays incident on ON will reach all points on line SS' except above B, where

$$CB = \frac{d}{\sqrt{3}}$$

$\therefore$ An observe located between A and B will be able to sea two images of P.

6. When mirror rotates by θ, the reflected ray rotated by 2θ.

$$\therefore \quad D(2\theta) = AB$$

$$\theta = \frac{0.5 \text{ cm}}{2 \times 100 \text{ cm}} = 0.0025 \text{ radian}$$

$$= 0.14°$$

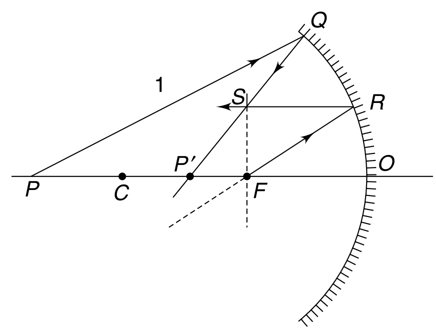
7. Since image is magnified, the mirror must be concave.

Image is magnified and erect when the object is between pole and focus of the mirror. As the object moves away, the magnification increases sharply and becomes infinite near the focus. Highly magnified image will appear to be blurred and when the object is at focus the image disappear. In fact image is at great distance behind the mirror when object is just about to cross the focus and moves to a great distance on the other side as the object crosses the focus.

Focal length of mirror = 45 cm.

Beyond 45 cm, the image gets inverted.

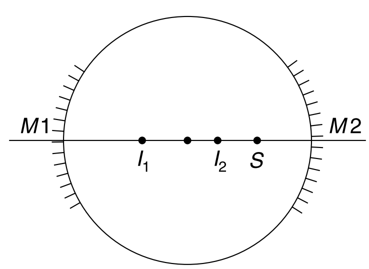
8. Draw a ray FR parallel to PQ. This ray will get parallel to principal axis after reflection. This reflected ray intersects the focal plane at S. Reflect ray at Q will also pass through S.



Hence QSP' is the reflected ray.

P' is image of point P.

9. For mirror M1 (The farthest wall)



$$u = -\frac{3R}{2}; \quad f = \frac{-R}{2}$$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} = -\frac{2}{R} + \frac{2}{3R} = \frac{-4}{3R}$$

$$v = -\frac{3R}{4} \text{ (Image at } I_1)$$

This image acts as an object for mirror $M2$

$$u = -\left(2R - \frac{3R}{4}\right) = \frac{-5R}{4}$$

$$\therefore \quad \frac{1}{v} = \frac{-2}{R} + \frac{4}{5R}$$

$$v = -\frac{5R}{6} \text{ (image at } I_2)$$

$\therefore$ Distance of image from the nearest wall is $\frac{5R}{6}$.

10. Let x = object distance and

y = image distance, initially.

$$y = 3x$$

$$-\frac{1}{f} = -\frac{1}{x} - \frac{1}{3x} \Rightarrow f = \frac{3x}{4} \quad ...(1)$$

The mirror must be moved away from the object so that the object position moves towards centre of curvature and the size of image decreases.

Now $x' = x + 10$ = object distance

$y' = 2x'$ = image distance

$$\therefore \quad \frac{1}{x'} + \frac{1}{2x'} = \frac{1}{f}$$

$$\Rightarrow \quad f = \frac{2x'}{3} \quad ...(2)$$

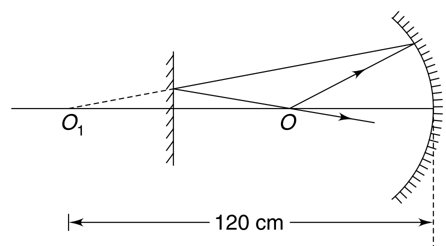
From (1) and (2) $\frac{3}{4}x = \frac{2x'}{3}$

$$\Rightarrow \quad \frac{3}{4}x = \frac{2}{3}(x + 10)$$

$$\Rightarrow \quad 9x = 8x + 80 \Rightarrow x = 80 \text{ cm.}$$

From (1) $f = 60$ cm.

12. (a) For concave mirror



$$u = -60 \text{ cm}; \quad f = -\frac{R}{2} = -40 \text{ cm}$$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} = -\frac{1}{40} + \frac{1}{60} = \frac{-3+2}{120}$$

$$v = -120 \text{ cm}$$

If O_1 is the image of O formed by the concave mirror, distance between O and O_1 is 60 cm.

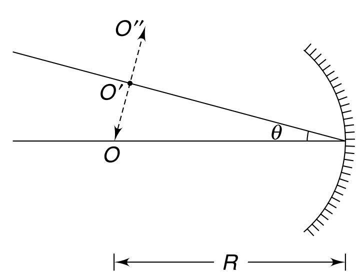
The plane mirror shall be placed exactly midway between O and O_1.

It means the required distance is 90 cm.

(b) Position will not change. It follows from the principle of reversibility of the optical path. (See Figure)

13.
$$OO' = R\theta = 2f\theta$$

If $00'$ is an object then its image is $O'O''$ with



$$O'O'' = OO'$$

$$\therefore \quad OO'' = 4f\theta$$

14. AB is face and E is eye

$$AB = 2a_0 = 20 \text{ cm}$$

$$d = 5 \text{ cm}$$

$A'B'$ is the image.

From similarity of triangles
$$\frac{x+y}{h} = \frac{x}{d} \quad ...(1)$$

From mirror equation

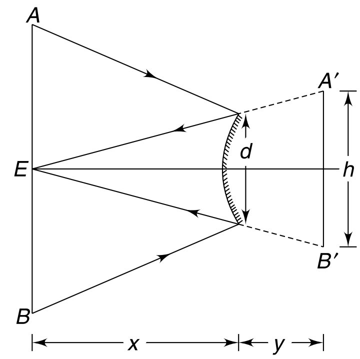
$$\frac{1}{y} + \frac{1}{-x} = \frac{1}{f}$$

$$\Rightarrow \quad y = \frac{xf}{x+f} \quad \text{...(2)}$$

and magnification

$$\frac{h}{2a_0} = \frac{y}{x}$$

$$\Rightarrow \quad h = \frac{y \cdot 2a_0}{x} = 2a_0 \frac{f}{f+x} \quad \text{...(3)}$$



Put (2) and (3) in (1)

$$\frac{x + \dfrac{xf}{f+x}}{2a_0 \dfrac{f}{f+x}} = \frac{x}{d}$$

$$\Rightarrow \quad \frac{2f + x}{2a_0 f} = \frac{1}{d}$$

$$\Rightarrow \quad \frac{2 \times 10 + x}{2 \times 10 \times 10} = \frac{1}{5}$$

$$20 + x = 40$$

$$x = 20 \text{ cm}$$

15. (a) The image flips at focus of the mirror. Between the pole and the focus the image is erect and as soon as the insect crosses the focus its image gets inverted.

For an object near the focus, magnification is large. Hence, image size is very large.

(b) There are two positions of the object for which we get magnified image of same size. Once when object is between focus and pole and other when the object is between focus and the centre of curvature.

<u>For object between focus and the pole</u>

Let distance of insect from pole be x.

$$u = -x, \quad v = +3x, \quad f = -20 \text{ cm}$$

$$\therefore \quad \frac{1}{+3x} + \frac{1}{-x} = \frac{1}{-20}$$

$$\frac{-2}{3x} = -\frac{1}{20}$$

$$\Rightarrow \quad x = \frac{40}{3} \text{ cm} = 13.33 \text{ cm.}$$

Distance of object from focus (P) is $= 20 - 13.33$

$= 6.67$ cm

<u>When the insect is between F and C</u>

$$\frac{1}{-3x} + \frac{1}{-x} = \frac{1}{-20}$$

$$\Rightarrow \quad x = \frac{80}{3} = 26.67 \text{ cm}$$

Distance from focus = 6.67 cm

16. In both cases image is virtual. Virtual image in a concave mirror is magnified. Hence, in the context of the problem it is clear that image distance is 52 cm when the boy is looking at the concave side. And image distance is 13 cm when he is looking into the convex side.

For concave mirror we have

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \quad \frac{1}{52} + \frac{1}{u} = \frac{2}{-R} \quad ...(1)$$

For convex mirror

$$\frac{1}{13} + \frac{1}{u} = \frac{2}{R} \quad ...(2)$$

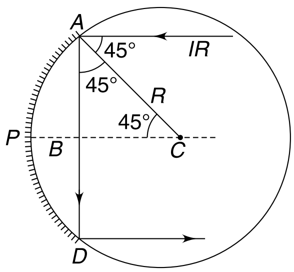
(2) – (1) gives

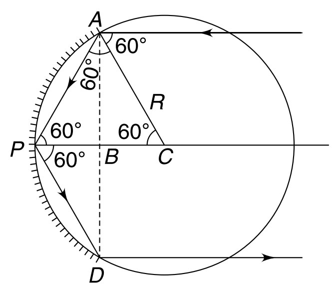
$$\frac{1}{13} - \frac{1}{52} = \frac{4}{R}$$

$$\Rightarrow \quad \frac{3}{52} = \frac{4}{R}$$

$$\Rightarrow \quad R = \frac{52 \times 4}{3} = 69.33 \text{ cm}$$

17.





$AB = \frac{R}{\sqrt{2}} \quad \therefore \quad D = AD = \sqrt{2}R$ $\qquad$ $AB = \frac{\sqrt{3}}{2}R \quad \therefore \quad D = AD = \sqrt{3}R$

18. $\angle SMR = \theta \simeq \frac{d}{R} = \frac{0.5}{0.5 \times 10^3} = 10^{-3}$ rad. The laser light after getting reflected at M, takes $\Delta t = \frac{2R}{c}$ time to return back to M. If the mirror M was at rest, the light would have been reflected back along MS. For the incident light FM to be reflected along MR, the mirror M must rotate through an angle equal to $\frac{\theta}{2}$. Recall that if the mirror rotates by an angle α, the reflected ray rotates by 2α.

$$\Delta t = \frac{2R}{c} = \frac{2 \times 0.5 \times 10^3}{3 \times 10^8} = \frac{1}{3} \times 10^{-5} \text{ s.}$$

In time Δt, mirror rotates by $\quad \Delta\theta = \frac{\theta}{2}$

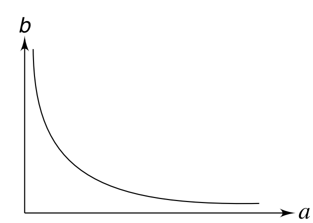
$$= \frac{1}{2} \times 10^{-3} \text{ rad.}$$

$$\therefore \quad \omega = \frac{\Delta\theta}{\Delta t} = \frac{3}{2} \times 10^2 = 150 \text{ rad s}^{-1}.$$

19. $\quad u = f + a \quad$ and $\quad v = f + b$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \quad \frac{1}{f+b} + \frac{1}{f+a} = \frac{1}{f}$$



Simplifying this gives $ab = f^2$

This is known as Newton's formula.

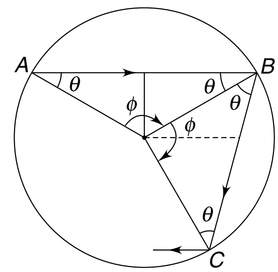
Since ab = a constant the graph is a rectangular hyperbola.

Draw a line through the origin inclined at 45° to the x-axis. This line intersects the graph at P. x or y co-ordinate of P gives the focal length of the mirror.

20.

$$\cos\theta = \frac{\frac{(\sqrt{5}+1)}{4}R}{R}$$

$$= \frac{\sqrt{5}+1}{4}$$



$\therefore$ $\theta = 36°$

Clearly $\phi = 2 \times 54° = 108°$

Ray will be incident at A when

$$n\phi = m \times 360° \text{ [}m, n \text{ are integers]}$$

Obviously, $m = 3, \quad n = 10$

$\therefore$ light ray will suffer 9 reflections before reaching A again.

22. An object in a denser medium appears to be closer when viewed from a rarer medium.

23. A red sheet transmits red component and absorbs all other colours. Similarly, blue sheet absorbs everything apart from blue.

24. 30% length of the stick will lie outside water.

The lower end will be at a distance 70 cm from the surface of the water.

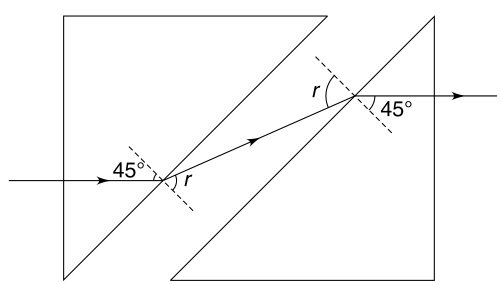
Apparent depth of the lower end is

$$h = \frac{70}{\frac{4}{3}} \text{ cm} = 52.5 \text{ cm}$$

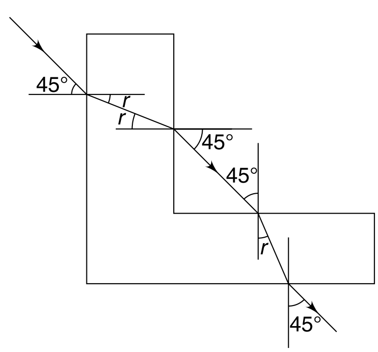
$\therefore$ Apparent length of the stick = 30 cm + 52.5

= 82.5 cm.

25. Path of the ray is as shown in the Figure.



26. The path of the light ray is as shown.



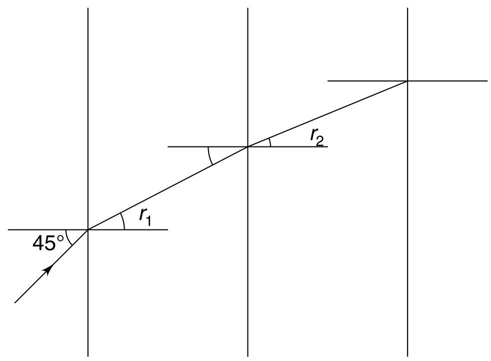
28. Since all refracting surfaces are parallel to each other.

$$1 \cdot \sin 45° = \mu_1 \sin r_1 = \mu_2 \sin r_2 \; ... = \sqrt{2} \, \sin e$$

$\therefore$ $$\sin e = \frac{1}{2}$$

$\Rightarrow$ $e = 30°$ where e = angle of emergence into medium of refractive index $\sqrt{2}$

Deviation $$\delta = i - e = 15°.$$



29. We are basically supposed to prove that no light ray entering the face AB can leave the glass block through face BC.

Consider a light ray incident on face AB at an angle of incidence $i \to 90°$.

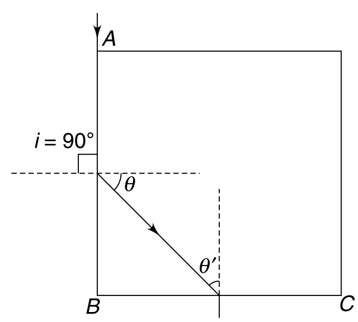
$$\mu \sin \theta = 1 \cdot \sin 90°$$

$$\sin \theta = \frac{2}{3}$$

Now $$\sin \theta' = \sin(90 - \theta)$$

$$= \cos \theta$$

$\therefore$ $$\sin \theta' = \sqrt{1 - \sin^2 \theta} = \sqrt{1 - \frac{4}{9}} = \frac{\sqrt{5}}{3}$$



Critical angle is $$C = \sin^{-1}\left(\frac{2}{3}\right)$$

But $$\frac{\sqrt{5}}{3} > \frac{2}{3}$$

$\therefore$ $$\sin \theta' > \sin C$$

$\therefore$ $$\theta' > C$$

$\therefore$ Light ray will suffer total internal reflection at face BC and no light will pass through BC.

If i is made smaller, θ will become smaller. This will make θ' even larger.

Hence no light ray incident on face AB can come out of face BC.

30. The depth of the bubble shall not be more than that shown in the figure , when the coin of diameter 2 cm is placed above it. A larger depth will mean that that the angle of incidence at points not covered by the coin can be lesser than the critical angle (C).

$$\mu \sin C = 1 \quad \Rightarrow \quad \tan C = \frac{1}{\sqrt{\mu^2 - 1}} \quad \Rightarrow \quad \tan C = \frac{2}{\sqrt{5}}$$

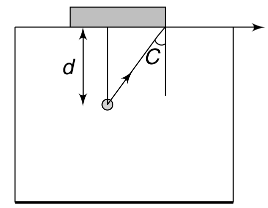
$\Rightarrow$ $$\frac{1}{d} = \frac{2}{\sqrt{5}} \quad \Rightarrow \quad d = \frac{\sqrt{5}}{2}$$

The depth at which the bubble will be just visible if the smaller coin is placed can be similarly calculated as –

$$\tan C = \frac{2}{\sqrt{5}}$$

$$\Rightarrow \quad \frac{3/4}{d} = \frac{2}{\sqrt{5}} \Rightarrow d = \frac{3\sqrt{5}}{8}$$

If the bubble happens to be below this depth it will be certainly visible.



Therefore, $\frac{3\sqrt{5}}{8} < d < \frac{\sqrt{5}}{2}$ cm

31. Actual thickness of the glass slab is

$$t = r_3 - r_1$$

Apparent thickness of the slab is

$$t' = r_3 - r_2$$

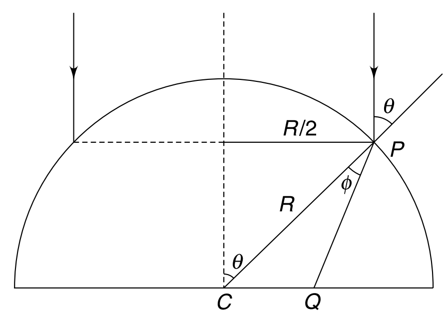
$$\therefore \quad \mu = \frac{t}{t'} = \frac{r_3 - r_1}{r_3 - r_2}$$

32. Apply snell's law to refraction of extreme rays.

$$1 \cdot \sin\theta = \mu \sin\phi$$

$$\Rightarrow \quad \frac{1}{2} = \frac{3}{2}\sin\phi \quad \left[\because \sin\theta = \frac{1}{2}\right]$$

$$\Rightarrow \quad \sin\phi = \frac{1}{3} \Rightarrow \phi = 20°$$



Using sine rule in ΔCPQ

$$\frac{R}{\sin[180° - 60° - 20°]} = \frac{CQ}{\sin 20°}$$

$$CQ = \frac{\sin 20°}{\sin 80°} \cdot R$$

$$= \frac{0.33}{0.98} R$$

$$= 0.34\ R.$$

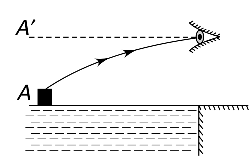
33. At angle of incidence > θ_0, total internal refraction takes place.
When light travels all throughout in water it takes more time in covering a distance = $2R$ (R = radius).
When light gets refracted into the liquid, it takes lesser time to travel through a distance $2R$.

$$\theta_0 = \text{critical angle}$$

$$\mu_\omega \sin\theta_0 = \mu \cdot \sin 90°$$

$$\frac{4}{3} \times \frac{5}{6} = \mu$$

$$\Rightarrow \quad \mu = \frac{10}{9}.$$



34. With rise in temperature density (and hence refractive index) of air decreases. Light ray from A takes a curved path as shown and appears to come from A'.

35. If we have multiple layers of different media with parallel surfaces, the apparent depth is given by (see Figure (*a*))–

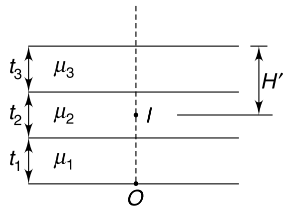


Fig. (a)

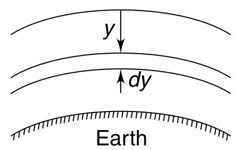


Fig. (b)

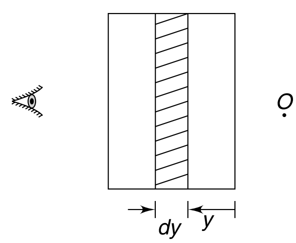
$$H' = \frac{t_1}{\mu_1} + \frac{t_2}{\mu_2} + \frac{t_3}{\mu_3} + \ldots$$

The atmosphere is like infinite number of thin layers. Hence apparent depth is given by

$$H' = \int_0^H \frac{dy}{\mu} = \int_0^H \frac{dy}{1 + \frac{y}{H}}$$

$\Rightarrow$ $$H' = H \ln 2.$$

36. Shift in position of the object caused due to thickness dy of the slab is



$$ds = dy\left(1 - \frac{1}{\mu}\right)$$

$$= dy\left(1 - \frac{1}{\mu_0(1 + y)}\right)$$

Total shift due to all such layers is

$$S = \int ds = \int_o^t dy - \frac{1}{\mu_0}\int_o^t \frac{dy}{1 + y}$$

$$= t - \frac{1}{\mu_o}[\ln(1 + y)]_o^t$$

$$= t - \frac{\ln(1 + t)}{\mu_o}$$

37.

$$\delta = i + e - A$$

$$40° = i + e - 60°$$

$$i + e = 100° \quad \ldots(1)$$

If $$i - e = 20° \quad \ldots(2)$$

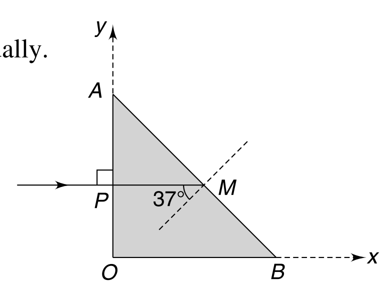
Solving (1) and (2) $$i = 60°$$

$$e = 40°.$$

38. Glass slab causes no deviation.

The net deviation is the sum of deviations produced by the two elements individually.

39.



$$PM = \frac{3}{2} \text{ cm}$$

$$37° > C$$

$\Rightarrow$ $$37° > \sin^{-1}\left(\frac{1}{\mu_0 + a\left(\frac{3}{2}\right)}\right)$$

$$\Rightarrow \qquad \frac{3}{5} > \frac{1}{\mu_0 + \frac{3}{2}a}$$

$$\Rightarrow \qquad 3\mu_o + \frac{9a}{2} > 5$$

$$\frac{9a}{2} > 5 - 3 \times \frac{4}{3}$$

$$a > \frac{2}{9}$$

41. The deviation is minimum when angle of incidence (i) = angle of emergence (e)

$$\delta = i + e - A$$

For minimum deviation ($\delta_{\min} = A$)

$$A = i + i - A$$

$$\Rightarrow \qquad i = A.$$

For angle of incidence $i = A$, Snell's law at first refracting surface gives.

$$1 \cdot \sin A = \mu \sin\left(\frac{A}{2}\right) \quad \left[\because \quad r = \frac{A}{2} \text{ for minimum deviation}\right]$$

$$\therefore \qquad \mu = 2\cos\left(\frac{A}{2}\right)$$

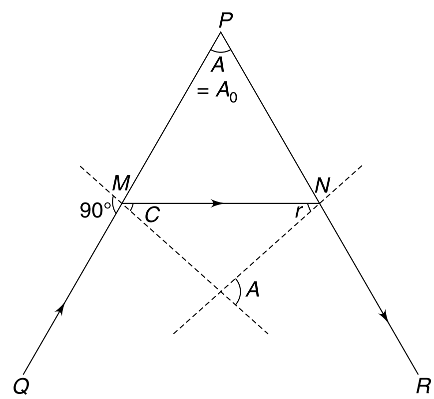
When $A \to 0$

$\mu \to 2.$

When $i = A \to 90°$

$\mu \to \sqrt{2}$

42. Ray QM is at grazing incidence. If angle of prism (A) is increased keeping ray QM at grazing incidence, the angle of incidence on the other face (r) will go on increasing. At a certain value of $A (= A_0)$ $\angle r$ will become equal to critical angle (C) and the emergent ray will be NR. If A is increased beyond this, total internal reflection will occur at N and no emergent ray will be there.



From simple geometry $A_o = 2c = 2\sin^{-1}\left(\frac{1}{\mu}\right)$

43. Angle of incidence on silvered face is $r_2 = A = 180 - 2\theta$

$\angle PRQ = 180° - 90° - 2r_2 = 90 - (360° - 4\theta) = 4\theta - 270°.$

Now $\qquad r_3 = 90° - \angle PRQ = 360° - 4\theta \qquad$ ...(1)

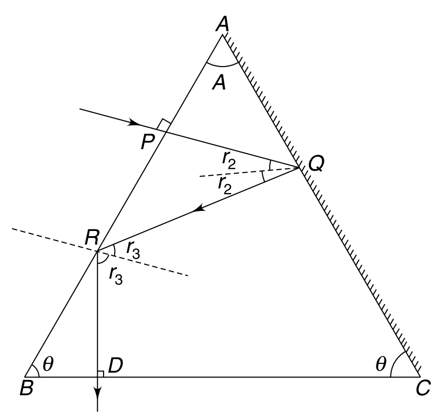
Also $\qquad \angle BRD = 90° - r_3$

$\therefore \qquad 90 - \theta = 90 - r_3$

$\Rightarrow \qquad r_3 = \theta$

From (1) $\qquad 5\theta = 360° \quad \Rightarrow \quad \theta = 72°$

For total internal reflection at R



$$r_3 > C$$

$$\Rightarrow \qquad \sin 72° > \sin C$$

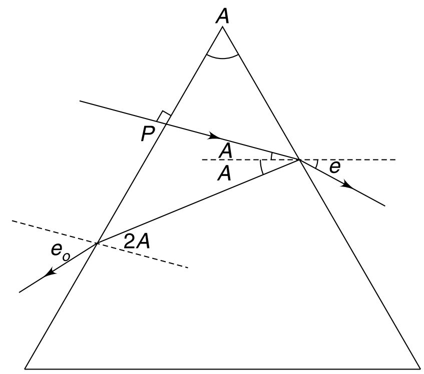
$$\Rightarrow \qquad \sin 72° > \frac{1}{\mu}$$

$$\Rightarrow \qquad \frac{1}{\sin 72°} < \mu$$

44. $e = A + 1°$ and $e_0 = 4°$.

Using snell's law for two refractions.

$$\mu = \frac{\sin(A + 1°)}{\sin A} = \frac{\sin 4°}{\sin(2A)}$$



All angles are small therefore [$\sin\theta \simeq \theta$]

$$\frac{A + 1°}{A} = \frac{4°}{2A}$$

$\Rightarrow$ $$A = 1°$$

and $$\mu = \frac{4°}{2°} = 2$$

45. From geometry angle of incidence $i = A$.

Light will not emerge out of the second face if

$$r_2 > C$$

$\Rightarrow$ $$A - r_1 > C$$

$\Rightarrow$ $$r_1 < A - C$$

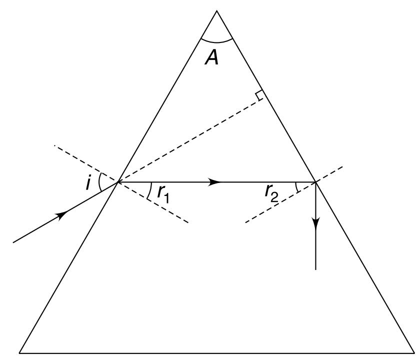
$\Rightarrow$ $$\sin r_1 < \sin(A - C) \quad ...(1)$$

Using Snell's law

$$\mu \sin r_1 = 1 \cdot \sin A$$

$\Rightarrow$ $$\sin r_1 = \frac{1}{\mu}\sin A$$

$\Rightarrow$ $$\sin r_1 = \sin C \cdot \sin A \quad ...(2)$$



From (1) and (2)

$$\sin A \sin C < \sin A \cos C - \sin C \cos A$$

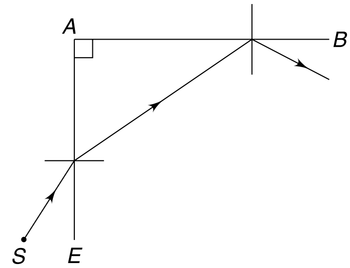
$\Rightarrow$ $$1 < \cot C - \cot A$$

$\Rightarrow$ $$\cot A < \cot C - 1$$

$\Rightarrow$ $$\cot A < \sqrt{3} - 1$$

47. The spot S will not be visible if any ray of light entering through face $AEHD$ of the slab fails to emerge out of face $ABCD$.

We can apply the condition of no emergence for a prism of angle $A = 90°$.



$\Rightarrow$ $$\frac{A}{2} > C$$

$$45° > C$$

$$\sin 45° > \sin C$$

$$\frac{1}{\sqrt{2}} > \frac{1}{\mu}$$

$$\boxed{\sqrt{2} < \mu}$$

48. The glass slab causes a shift but the shift does not depend on the distance between the slab and the object. It means, even if the distance of the object changes from the slab the shift caused will remain same.

Presence of glass slab makes no difference to the velocity of the image.

If we are looking at traffic through a thick glass door, we see the cars moving at their true speed.

49. The action of slab is to cause a shift in the direction of incident ray.

$$\text{shift } S = t\left(1 - \frac{1}{\mu}\right) = 6\left(1 - \frac{2}{3}\right)$$

$$= 2 \text{ cm}$$

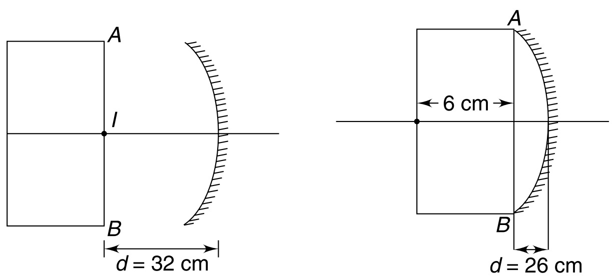
After refraction from the slab, image is formed at distance of 60 cm from the mirror. This acts as object for mirror.

$$\frac{1}{v} + \frac{1}{-60} = \frac{1}{-20}$$

$$\Rightarrow \quad v = -30 \text{ cm}$$

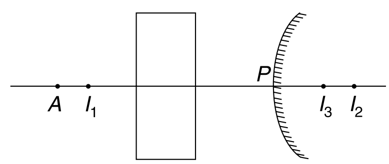
After reflection from the mirror, image is formed at a distance of 30 cm to the left of the mirror. This act as an object for the slab and its image is shifted by 2 cm. Final image is at a distance of 32 cm from the mirror. For this image to be inside the slab.

$$26 < d < 32$$



50. Shift caused by the slab; $S = t\left(1 - \frac{1}{\mu}\right) = 6\left(1 - \frac{2}{3}\right)$

$$= 2 \text{ cm}$$



I_1, I_2 and I_3 are images formed after the light rays get refracted from the slab, reflected from the mirror, and finally after refraction through the slab respectively.

$$I_2I_3 = AI_1 = 2 \text{ cm}$$

Given $\quad PI_3 = 10$ cm

$\therefore \quad PI_2 = 12$ cm = image distance for the mirror.

PI_1 = object distance from the mirror.

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{12} + \frac{1}{u} = \frac{1}{20}$$

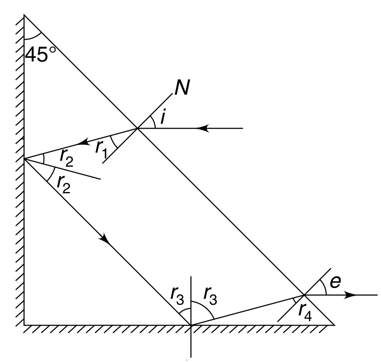
$$\therefore \quad \frac{1}{u} = \frac{1}{20} - \frac{1}{12} = \frac{3-5}{60}$$

$$u = -30 \text{ cm}$$

$$\therefore \quad PI_1 = 30 \text{ cm}$$

$$\therefore \quad PA = 32 \text{ cm}$$

51. The path of the ray is shown is the Figure.



$$r_2 + r_1 = 45°$$

$$\therefore \quad r_2 = 45° - r_1$$

$$r_3 = 90° - r_2 = 45° + r_1$$

$$(90 - r_3) + 45° + (90° + r_4) = 180°$$

$$\therefore \quad r_4 = r_3 - 45° = 45° + r_1 - 45° = r_1$$

From Snell's law (since $r_1 = r_4$) we can easily prove that $i = e$.

Hence incident and emergent rays are parallel.

52. The parallel faced sheet will not result in any deviation.

$$\therefore \quad 1 \cdot \sin 45° = \sqrt{2} \cdot \sin r_1$$

$$r_1 = 30°$$

$$r_2 = A - r_1 = 30°$$

$$\therefore \quad e = 45°$$

$$\therefore \quad \delta = i + e - A = 45° + 45° - 60° = 30°.$$

54. The two prisms must be combined with their vertex placed opposite to each other. For no deviation we must have

$$A(\mu_y - 1) = A'(\mu_y' - 1) \qquad ...(1)$$

For no dispersion

$$(\delta_v - \delta_R) = (\delta_v' - \delta_R')$$

$$\Rightarrow \quad A(\mu_v - \mu_R) = A'(\mu_v' - \mu_R')$$

$$\Rightarrow \quad A\omega(\mu_y - 1) = A'\omega'(\mu_y' - 1)$$

using (1) we get $\omega = \omega'$

$$\therefore \quad \omega' = \omega = \frac{\mu_v - \mu_R}{\mu_y - 1} = \frac{1.51 - 1.49}{1.5 - 1} = \frac{0.02}{0.5}$$

$$= 0.04$$

55.

$$\lambda_R > \lambda_Y > \lambda_G > \lambda_B$$

$$\mu_R < \mu_Y < \mu_G < \mu_B$$

$$C_R > C_Y > C_G > C_B \quad [C = \text{Critical Angle}]$$

$\therefore$ Red and yellow are least likely to suffer TIR.

Since $\mu_y > \mu_R$, the yellow color suffers more deviation.

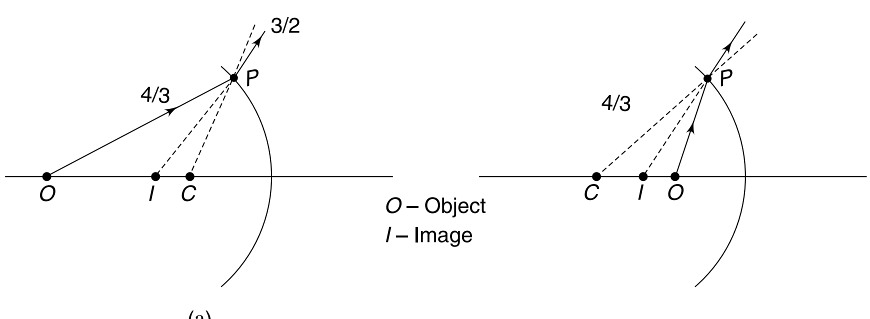
A is red

B is yellow.

Out of Blue and green, critical angel for green is higher. It is less likely to suffer TIR.

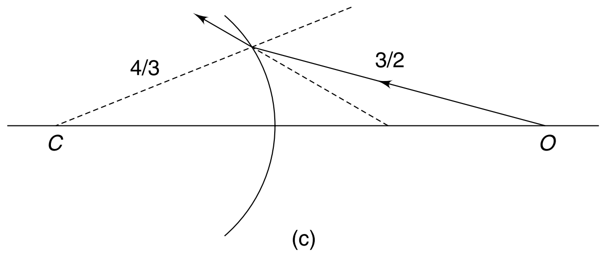
$\therefore$ D is green.

56. (a) When object is in medium A, the image is always virtual as shown in Figure (a) and (b)



(a) (b)

(b) As shown in Figure (c), the image is virtual for all positions of O.



(c)

57. Image of B is formed at infinity

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

$$\frac{1}{\infty} - \frac{\mu}{(-2R)} = \frac{1-\mu}{(-R)}$$

$$\frac{\mu}{2} = -1 + \mu$$

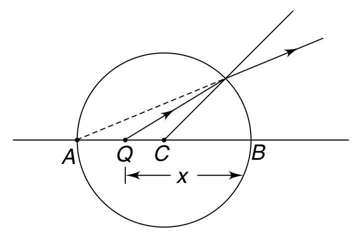
$$\Rightarrow \quad \frac{\mu}{2} = 1$$

$$\Rightarrow \quad \mu = 2$$

58. Image of point C is formed at C itself as light rays diverging from C will not suffer any deviation at the spherical refracting surface.

$\therefore$ Point P is same as point C.

Let point Q be at a distance x from B as shown.



using
$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

$$\frac{1}{-2R} - \frac{1.5}{-x} = \frac{1-1.5}{-R}$$

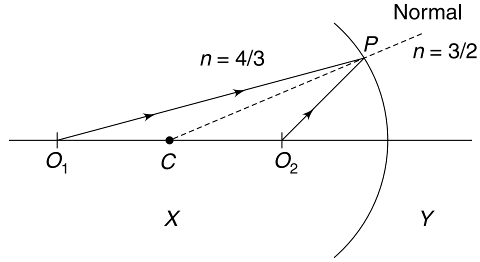
$$\Rightarrow \quad \frac{1}{2R} - \frac{3}{2x} = -\frac{1}{2R}$$

$$\Rightarrow \quad \frac{3}{2x} = \frac{1}{R}$$

$$\Rightarrow \quad x = \frac{3R}{2}$$

Distance between C and Q is $\frac{R}{2}$.

59. Consider two point objects O_1 and O_2. Incident rays from O_1 and O_2 at point P shall both bend towards normal. The corresponding refracted rays will intersect the principal axis, when extended, in the left medium. Therefore image formed under given condition shall always be virtual.



60. **(a)** Consider object on left side of spherical surface 1 separating two media.

If real object is in rarer medium i.e., $\mu_2 > \mu_1$

Then $$\frac{\mu_2}{v_1} = \frac{\mu_2 - \mu_1}{(-R)} + \frac{\mu_1}{(-u)}$$

v_1 is definitely negative. Hence image shall be virtual.

(b) For second refraction

$$\frac{\mu_1}{v_2} = \frac{\mu_1 - \mu_2}{(-R)} + \frac{\mu_2}{(-u_2)} = \frac{\mu_2 - \mu_1}{R} - \frac{\mu_2}{u_2}$$

Thus v_2 can be positive or negative. It means the image can be real or virtual.

(c) It can be concluded as above that the image can be real or virtual.

61. Point A will act like a virtual object for the lens. It is off the principal axis.

$$u = +15 \text{ cm}, f = +30 \text{ cm}$$

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{v} = \frac{1}{15} + \frac{1}{30} = \frac{2+1}{30}$$

$$\therefore \quad v = +10 \text{ cm}$$

Therefore, the rays converge at 10 cm from the screen.

63. Image of the front face is formed at a distance of 40 cm on the other side of the lens.

The rear face of the glass slab appears to be shifted by $s = t\left(1 - \frac{1}{\mu}\right) = 6\left(1 - \frac{1}{3/2}\right) = 2$ cm nearer to the lens.

For lens the distance of rear face of the slab is 40 + 6 – 2 = 44 cm.

Its image is formed at a distance v from lens given by

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \quad \frac{1}{v} - \frac{1}{-44} = \frac{1}{20}$$

$$\Rightarrow \quad \frac{1}{v} = \frac{1}{20} - \frac{1}{44} = \frac{11-5}{220} = \frac{3}{110}$$

$$\therefore \quad v = \frac{110}{3} = 36.7 \text{ cm.}$$

$\therefore$ Thickness of slab in the image formed by the lens is = 40 – 36.7 = 3.3 cm.

64. (a) $$\frac{1}{f} = \frac{1}{12} - \frac{1}{-240} = \frac{21}{240}$$

Shift caused by glass slab $$S = t\left(1 - \frac{1}{\mu}\right)$$

$$= 1\left(1 - \frac{2}{3}\right) = \frac{1}{3} \text{ cm.}$$

$\therefore$ Image must be formed at a distance of $12 - \frac{1}{3} = \frac{35}{3}$ cm from the lens

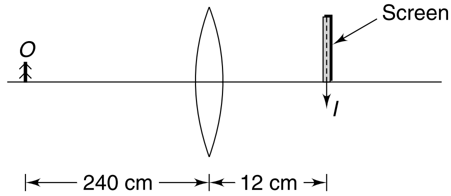
$$\frac{3}{35} - \frac{1}{u} = \frac{21}{240}$$

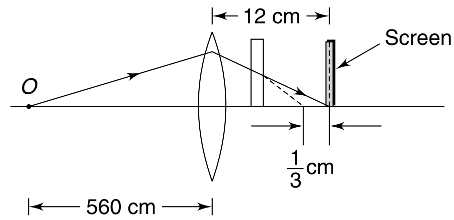
$$\Rightarrow \quad \frac{1}{u} = -\frac{3}{1680}$$

$$\therefore \quad u = -560 \text{ cm}$$

$\therefore$ Object shall be moved further away by

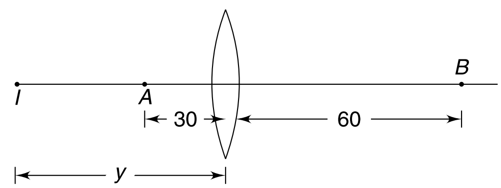
$$560 - 240 = 320 \text{ cm}$$





(b) The object shall be moved further away from the lens by $\frac{1}{3}$ cm.

65. It shall be noted that $30 < f < 60$ cm. This will ensure that image of A is to the left of the lens (virtual image) which can coincide with the real image of B.



Let distance of image from the lens be y.

For A: $$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \quad \frac{1}{-y} - \frac{1}{-30} = \frac{1}{f}; \quad \Rightarrow \quad \frac{1}{y} - \frac{1}{30} = -\frac{1}{f} \qquad ...(1)$$

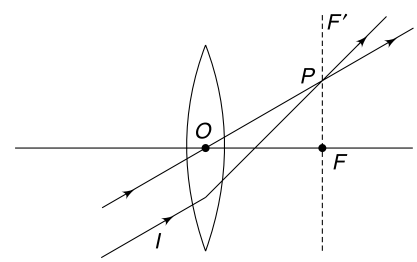
For B: $$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{y} - \frac{1}{-60} = \frac{1}{f} \quad \Rightarrow \quad \frac{1}{y} + \frac{1}{60} = \frac{1}{f} \qquad ...(2)$$

(2) – (1) gives $$\frac{1}{60} + \frac{1}{30} = \frac{2}{f}$$

$$\Rightarrow \quad f = 40 \text{ cm}$$

67. Consider a ray passing through optical centre and parallel to the given ray. This ray will pass undeviated. Draw the focal plane FF'. Let the ray intersect this plane at P.

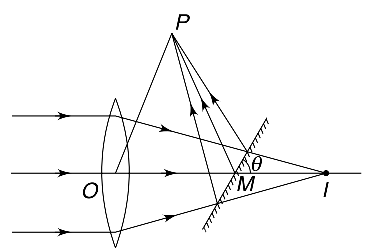


The incident ray, after refraction will pass through point P.

68. $OI = 40$ cm; $MI = MP = 20$ cm.

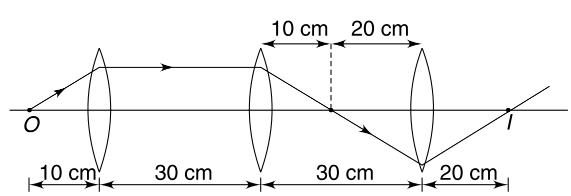
ΔOPM is equilateral

$$\theta = 60°.$$



69. The diagram explains the position of image (I).

If the tube is moved away by 10 cm, the object distance for the first lens becomes 20 cm. From the principle of reversibility of path, we can conclude that final image will be formed at a distance of 10 cm from the third lens. Hence image position does not change.



70. For air lens $$P = \left(\frac{\mu_a}{\mu_g} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$= \left(\frac{1}{3/2} - 1\right)\left(\frac{2}{R}\right)$$

$$= -\frac{2}{3R}$$

If a material of refractive index μ is filled in the cavity

$$P' = \left(\frac{\mu}{3/2} - 1\right)\left(\frac{2}{R}\right)$$

$$\frac{2}{3R} = \left(\frac{2\mu}{3} - 1\right)\frac{2}{R}$$

$\Rightarrow$ $$1 = 2\mu - 3$$

$\Rightarrow$ $$\mu = 2$$

71. For, lens at B, A is object.

$\therefore$ $$\frac{1}{v_B} - \frac{1}{2x} = \frac{1}{-x}$$

$$\frac{1}{v_B} = -\frac{1}{2x} \Rightarrow v_B = -2x$$

$\therefore$ Image is formed at P and magnification is $m_1 = \left(\frac{-2x}{2x}\right) = -1$

The image at P acts as object for the lens at C.

$$\frac{1}{v_C} - \frac{1}{-3x} = \frac{1}{2x}$$

$$\frac{1}{v_C} = \frac{1}{2x} - \frac{1}{3x} = \frac{1}{6x}$$

$$v_C = 6x$$

Find image is at a distance $6x$ from C.

Magnification $$m_2 = \left(\frac{6x}{-3x}\right) = -2$$

$\therefore$ $$M = m_1 m_2 = 2$$

$\therefore$ Find height of image = $2h$.

72. The image of the object is formed on the other side of the lens at a distance of 20 cm.

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$\Rightarrow$ $$-\frac{1}{v^2}\frac{dv}{dt} = -\frac{1}{u^2}\frac{du}{dt} \Rightarrow \frac{dv}{dt} = \left(\frac{v}{u}\right)^2\frac{du}{dt}$$

When $$v = u;\ \frac{dv}{dt} = \frac{du}{dt}$$

$\therefore$ $$V_{IX} = V_{OX}$$

Magnification $$m = \frac{v}{u}$$

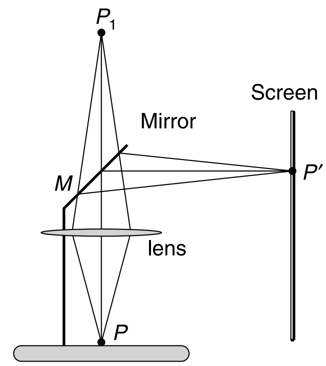
$\therefore$ $$\frac{y_I}{y_o} = \frac{v}{u}$$

$$\frac{dy_I}{dt} = \frac{v}{u}\frac{dy_0}{dt} + y_0\frac{d}{dt}\left(\frac{v}{u}\right).$$

In given position $$y_0 = 0$$

$\therefore$ $$V_{YI} = \left(\frac{v}{u}\right)V_{Y0} = \left(\frac{20}{-20}\right)V_{Y0} = -V_{Y0}$$

$\therefore$ Velocity of image will also make 45° with the principal axis (in downward direction)



73. In absence of mirror the image formed by mirror would have been at P1 with distance MP1 equal to MP'. If screen is moved away the distance MP1 must also increase. For this to happen the object must come closer to the lens. Thus the lens must be moved down.

74. Depth of image from water surface formed due to refraction = $\frac{3}{4} \times 20 = 15$cm. This image acts as an object at a distance of 40 cm from the mirror.

The mirror forms a virtual image at a distance of 40 cm that lies on the principal axis of the lens. This acts as an object for the lens at a distance of 90 cm.

Applying lens formula–

$$\frac{1}{v} - \frac{1}{-90} = \frac{1}{15}$$

$$\therefore \qquad v = 18 \text{ cm}$$

Thus final image is to the left of the lens at a distance of 18 cm.

75. Since magnification is positive, the object is at a distance less than its focal length and the image is virtual. The X co–ordinate of the lens is less than -20 cm. Let it be $-x_0$.

$$u = -(x_0 - 20),\ v = -(x_0 + 25)$$

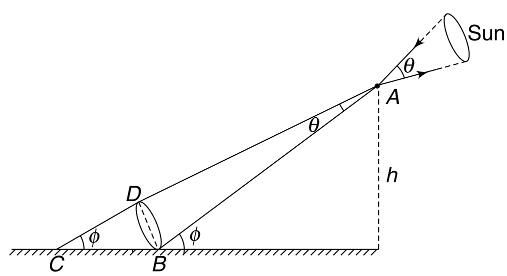
Also $$\left|\frac{v}{u}\right| = 2 \Rightarrow x_0 + 25 = 2(x_0 - 20) \Rightarrow x_0 = 65 \text{ cm}$$

$$\therefore \qquad u = -(65 - 20) = -45\text{cm and } v = -(65 + 25) = -90 \text{ cm}$$

Lens formula gives $$\frac{1}{f} = \frac{1}{-90} - \frac{1}{-45} \Rightarrow f = 90 \text{ cm}$$

76. The cone of rays passing through the hole at A produce an elliptical spot on the floor.

The circular base having diameter BD will get projected on the floor as an ellipse.



CB = major axis = 12 cm

DB = Minor axis = 6 cm

$$DB \simeq AB(\theta) = \frac{h\theta}{\sin\phi} \Rightarrow h = (6 \text{ cm})\frac{\sin\phi}{\theta}$$

But $$\sin\phi = \frac{DB}{CB} = \frac{6}{12} = \frac{1}{2}$$

$$\therefore \qquad h = \frac{(6 \text{ cm})}{2 \times \theta}$$

$$= \frac{3 \text{ cm} \times 180}{0.5 \times \pi} \qquad \left[0.5° = \frac{0.5 \times \pi}{180} \text{ radian}\right]$$

$$= 344 \text{ cm}$$
$$= 3.44 \text{ m}$$

77. For light to fall on the animal, tangent to the parabola must pass through the animal (200, 100)

Equation of parabola $y = ax^2$

because (–200, 200) lies on this parabola

$$\therefore \quad 200 = a(-200)^2 \Rightarrow a = \frac{1}{200}$$

$$\therefore \quad y = \frac{x^2}{200}$$

Let the required point has co-ordinates (x_0, y_0). Slope of line passing through (x_0, y_0) and (200, 100) is

$$m = \frac{y_0 - 100}{x_0 - 200} = \frac{ax_0^2 - 100}{x_0 - 200} = \frac{\frac{x_0^2}{200} - 100}{x_0 - 200}$$

This should be equal to $\left.\frac{dy}{dx}\right|_{x_0, y_0} = 2ax_0$

$$= \frac{x_0}{100}$$

$$\therefore \quad \frac{\frac{x_0^2}{200} - 100}{x_0 - 200} = \frac{x_0}{100}$$

$$\left(\frac{x_0^2}{200} - 100\right) \times 100 = x_0^2 - 200x_0$$

$$\Rightarrow \quad x_0^2 - 400x_0 + 20000 = 0$$

Solving $x_0 = 58.6$ m [other value of x_0 is not acceptable. Why?]

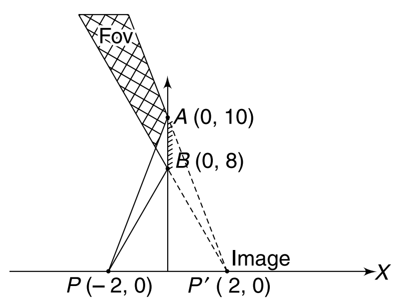
$$\therefore \quad y_0 = \frac{(58.6)^2}{200} = 17.17 \text{ m.}$$

78. (a) Image is visible only if the observer lies in the field of view (FOV) region

Equation of line BP' is $y_1 = -4x + 8$

equation of line AP' is $y_2 = -5x + 10$

Point O_1 lies outside the FOV whereas O_2 lies inside it. [x – co-ordinate for O_1 is –2. Putting in equation the two lines gives $y_1 = 16$ and $y_2 = 20$. The y co-ordinate of O_1 does not lie in this range. Similarly for O_2 – $y_1 = 12$; $y_2 = 15$ and y co-ordinate of O_2 lies in this range $12 < 13 < 15$. Hence, O_2 is in FOV. He will see the image].



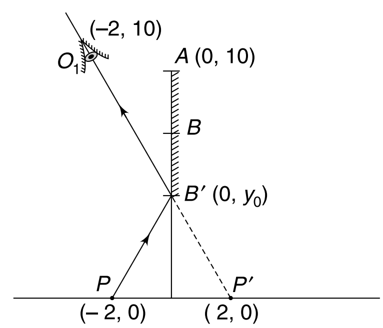
(b) Let the mirror be extended upto B' $(0, y_0)$.

O_1 will be just able to see the image if ray reflected at B' passes through $O_1(-2, 10)$. Equation of line O_1P' is

$$y = -2.5\,x + 5$$

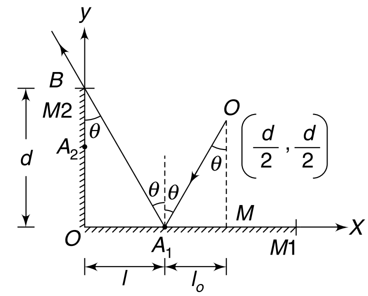
$$\therefore \quad y \text{ intercept} = 5 = y_0$$

Hence length of mirror $AB' = 5$ m



79. (a) Ray OA_1 incident on $M1$ just misses mirror $M2$. In this case third image will not be formed. Similarly, if OA_2 (= OA_1) is removed, no ray incident on $M2$ can hit mirror $M1$.

From geometry



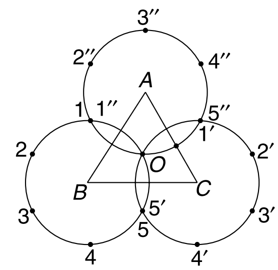
$$\tan\theta = \frac{l}{d} = \frac{l_0}{d/2}$$

$\Rightarrow$ $$l = 2l_0$$

$$l + l_0 = \frac{d}{2}$$

$\therefore$ $$l = \frac{2}{3}\frac{d}{2} = \frac{d}{3}$$

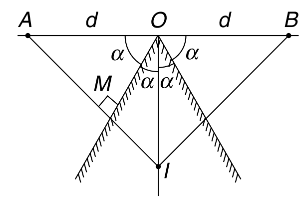
80. Images formed due to multiple reflections in mirrors AB and BC will lie on a circle centred at B. Position of images are 1, 2, 3, 4, 5. Images due to reflections in BC and CA are 1′, 2′, 3′, 4′ and 5′. Images due to reflections in AB and AC are 1″, 2″, 3″ 4″ and 5″.



1 and 1″, 1′ and 5″ and 5 and 5′ coincide.

$\therefore$ Total number of images = 12.

81. (a) Line joining the object and the image is perpendicular to the mirror, and object and the image are at equal distance from the mirror.



$$\Delta AOM \cong \Delta IOM$$

$\therefore$ All angles marked as α in the figure are equal.

$$4\alpha = 180° \quad \Rightarrow \quad \alpha = 45°$$

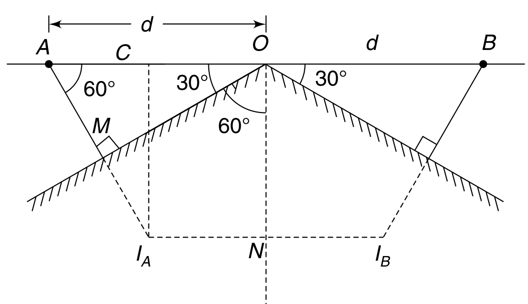
$\therefore$ $$\theta_0 = 90°$$

(b) $$\theta = \frac{4}{3}\,\theta_0 = 120°$$

$$AM = d\sin 30° = \frac{d}{2} \quad \Rightarrow \quad AI_A = 2AM = d$$

$\therefore$ $$AC = (AI_A)\cos 60° = \frac{d}{2} \quad \Rightarrow \quad OC = \frac{d}{2}\ (= NI_A)$$

$\therefore$ $$I_AI_B = d.$$

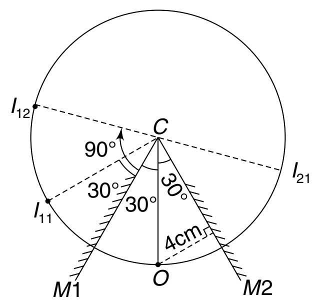


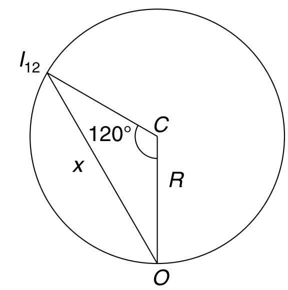
82. All the images lie on a circle of radius

$$R = OC = \frac{4}{\sin 30°} = 8 \text{ cm.}$$

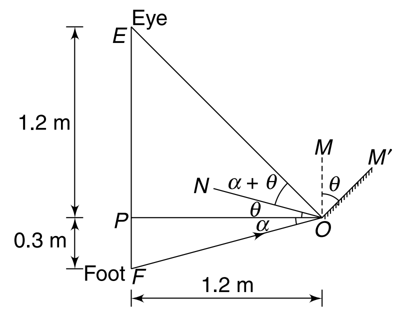
Position of second image (I_{12}) formed in $M1$ has been shown in the Figure. It is actually the image of first image (I_{21}) formed in $M2$. Required distance is

$$x = 2R\sin 60° = 2 \times 8 \times \frac{\sqrt{3}}{2} = 8\sqrt{3} \text{ cm.}$$





83. When mirror is in position OM, normal to it is OP. When mirror is rotated, the normal to it rotates to position ON.



$$FO = \text{incident ray from foot to the lower edge.}$$

$$OE = \text{reflected ray.}$$

$$\angle FON = \text{angle of incidence} = \alpha + \theta$$

$$= \text{angle of reflection} = \angle NOE$$

From Figure $\quad EP = PO = 1.2 \text{ m}$

$\therefore \quad \angle EOP = 45°$

$\therefore \quad 2\theta + \alpha = 45°$

But $\quad \tan\alpha = \dfrac{0.3}{1.2} = \dfrac{1}{4} \Rightarrow \alpha = 15°$

$\therefore \quad \theta = 15°$

If mirror is rotated beyond this angle, reflected rays will pass above the eye.

84. (a) Image formed by $M2$ is x_2 behind it. It means the image is $x_2 - d_0$ behind the mirror $M1$.

For concave mirror (with A as object)

$$u = -x_1; \quad v = x_2 - d_0$$

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

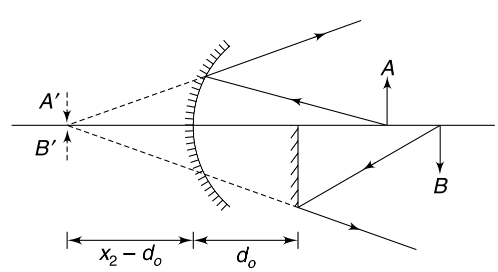
$$\frac{1}{f} = -\frac{1}{x_1} + \frac{1}{x_2 - d_0}$$

$$f = \frac{(x_2 - d_0)\, x_1}{-x_2 + d_0 + x_1}$$

$$= -\frac{(x_2 - d_0)\, x_1}{x_2 - (d_0 + x_1)}$$

$$\therefore \quad |f| = \frac{(x_2 - d_0)\, x_1}{x_2 - (d_0 + x_1)}$$

(b) Rays incident on $M1$ will be reflected above the principal axis and rays reflected from $M2$ will be found only below the axis. There is no region where both set of reflected rays are present. An observer cannot see both the images simultaneously. To compare the position of the images the observer should move his eye in the vertical plane and view the images in turn.



85. (a) **Steps:**

(1) Mirror is concave because image is enlarged and inverted.

Magnification = 3, *i.e.* greater than 1. It means image distance is larger than object distance.

Hence mirror is above AB in our diagram.

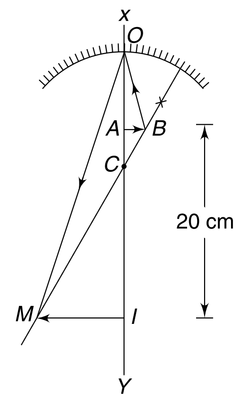
(2) $OI = 3\ OA$ where O is pole of the mirror.

$AI = 20$ cm

$\therefore\quad AO = 10$ cm

It means pole (O) of the mirror is 10 cm from A.

(3) To complete the ray diagram we can draw the incident ray BO which is reflected as OM.



(b) $\dfrac{MI}{AB} = \dfrac{CI}{CA}$

$$3 = \frac{20 - CA}{CA}$$

$\therefore\quad CA = 5$ cm

$\therefore\quad OC = 10 + 5 = 15$ cm

$\therefore\quad$ Focal length $f = \dfrac{OC}{2} = 7.5$ cm.

86. (a) The image formed by upper mirror is virtual. It means it is above the mirror. Say it is at a distance y from the mirror.

$$\frac{1}{y} + \frac{1}{-h} = \frac{1}{-f} \Rightarrow \frac{1}{y} = \frac{1}{h} - \frac{1}{f} \qquad \text{...(1)}$$

This image is at a distance $(y + h)$ from the lower miror and acts as an object. Image of this object is just above the mouth. This implies that image distance $\simeq h$

$$\therefore \qquad \frac{1}{-h} + \frac{1}{-(y+h)} = \frac{1}{-f}$$

$$\therefore \qquad \frac{1}{h} + \frac{1}{h + \left(\frac{1}{h} - \frac{1}{f}\right)^{-1}} = \frac{1}{f} \quad \text{Proved.}$$

(b) Multiply the above equation by f

$$\frac{f}{h} + \frac{1}{\frac{h}{f} + \left(\frac{f}{h} - 1\right)^{-1}} = 1$$

$$\Rightarrow \qquad \frac{1}{x} + \frac{1}{x + \left(\frac{1}{x} - 1\right)^{-1}} = 1$$

$$\Rightarrow \qquad \frac{1}{x} + \frac{1}{x + \frac{x}{1-x}} = 1$$

$$\Rightarrow \qquad \frac{1}{x} + \frac{1-x}{x - x^2 + x} = 1$$

$$\Rightarrow \qquad x - x^2 + x + x - x^2 = x^2 - x^3 + x^2$$

$$\Rightarrow \qquad x^3 - 4x^2 + 3x = 0$$

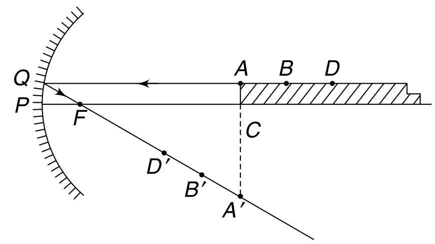
$$\Rightarrow \qquad x[x^2 - 4x + 3] = 0 \quad \text{But} \quad h \neq 0 \quad \Rightarrow \quad x \neq 0$$

$\therefore \qquad x^2 - 4x + 3 = 0 \Rightarrow x = 1; x = 3$

$$x = 3 \Rightarrow h = 3f$$

This means that first image cannot be virtual.

Hence $x = 1$



87. AQ is an incident ray parallel to the principal axis which gets reflected as QF. [F is the focus]. Image of A is at A'. Image of B, D etc. will be nearer to the focus on the line QF.

A point that is at infinite distance along ABD will have its image at F.

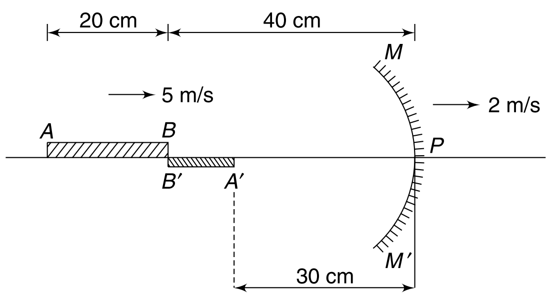
Hence the image of the rectangular strip will be a triangle $A'CF$.

$$\text{Area of image} = \frac{1}{2} \times FC \times CA'$$

$$= \frac{1}{2} fh.$$

88. Point B is at the centre of curvature of the mirror and its image will be at the same position.

Image of point A is formed at distance v_A given by mirror formula.



$$\frac{1}{v_A} + \frac{1}{-60} = \frac{1}{-20}$$

$$\frac{1}{v_A} = \frac{1}{60} - \frac{1}{20}; \Rightarrow v_A = -30 \text{ cm}$$

And $$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$-\frac{1}{v^2}\frac{dv}{dt} - \frac{1}{u^2}\frac{du}{dt} = 0$$

$$\frac{dv}{dt} = -\left(\frac{v}{u}\right)^2 \frac{du}{dt}$$

$$V_{Im} = -\left(\frac{v}{u}\right)^2 \cdot V_{Om} \qquad \begin{bmatrix} V_{Im} = \text{velocity of image wrt mirror} \\ V_{Om} = \text{velocity of object wrt mirror} \end{bmatrix}$$

$$V_{B'} - V_m = -\left(\frac{40}{40}\right)^2 \cdot [V_B - V_m]$$

$$V'_B - 2 = -[5 - 2]$$

$$V'_B = -1 \text{ m/s} = 1 \text{ m/s} (\rightarrow)$$

For A' (Image of A)

$$V_{A'} - V_m = -\left(\frac{30}{60}\right)^2 [V_A - V_m]$$

$$V_{A'} - 2 = -\frac{1}{4}[5 - 2]$$

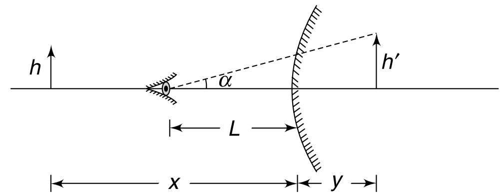
$$V_{A'} = +\frac{5}{4} \text{ m/s} = \frac{5}{4} \text{ m/s } (\rightarrow)$$

$\therefore$ length $A'B'$ is increasing at a rate of

$$1 + \frac{5}{4} = \frac{9}{4} \text{ m/s.}$$

89. (a) Let h and h' be the height of the object and image respectively. y is image distance.

From mirror formula



$$\frac{1}{y} + \frac{1}{-x} = \frac{1}{f}$$

$$\Rightarrow \quad y = \frac{fx}{f+x} \quad \text{...(1)}$$

And $$\frac{h'}{h} = \frac{y}{x}$$

$$\therefore \quad \tan\alpha \simeq \alpha = \frac{h'}{L+y} = \frac{\frac{yh}{x}}{L+y}$$

$$= \frac{h}{x\left[\frac{L}{y} + 1\right]}$$

$$= \frac{h}{\frac{L(f+x)}{f} + x} = \frac{h}{L + x + \frac{Lx}{f}} \quad \text{[using (1)]}$$

(b) For a plane mirror $y = x$ and $h' = h$

$$\therefore \quad \beta = \frac{h'}{y+L} = \frac{h}{x+L}$$

(c) Clearly $\alpha < \beta$.

It means images in convex mirror subtend smaller angles at our eyes and hence appear to be smaller. We perceive them to be at larger distance.

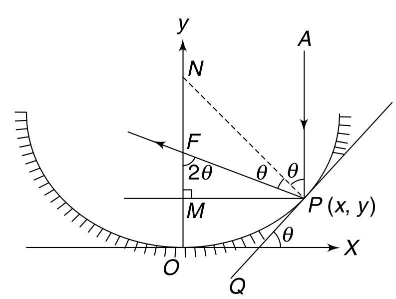
90. Consider a light ray incident on the mirror at point $P(x, y)$ on the mirror.

$AP \rightarrow$ incident ray, $PN \rightarrow$ normal

$PF \rightarrow$ reflected ray, $PQ \rightarrow$ tangent at P

$$y = kx^2$$

$$\therefore \quad \tan\theta = \frac{dy}{dx} = 2kx \quad \text{...(1)}$$



In ΔFMP $$\tan 2\theta = \frac{PM}{FM} = \frac{x}{f-y}$$

$$[f = OF]$$

$$\Rightarrow \quad \frac{x}{f-y} = \tan 2\theta$$

$$\Rightarrow \quad f = y + \frac{x}{\tan 2\theta} = y + \frac{x[1 - \tan^2\theta]}{2\tan\theta}$$

$$= kx^2 + \frac{x[1 - (2kx)^2]}{2\cdot(2\cdot kx)} = \frac{1}{4k} \text{ (independent of } (x, y)!)$$

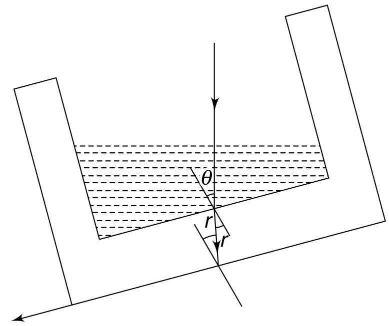
91. $$\mu_w \sin\theta = \mu_g \sin r$$

$$\frac{4}{3}\sin\theta = \frac{3}{2}\sin r \qquad ...(1)$$

And $$\frac{3}{2}\sin r = 1\cdot\sin 90°$$

$$\sin r = \frac{2}{3} \qquad ...(2)$$

From (1) and (2) $$\sin\theta = \frac{3}{4}$$

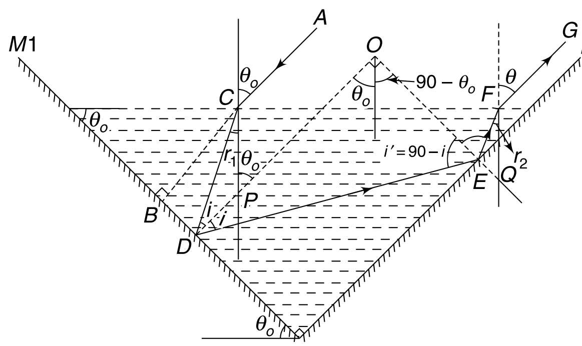


92. Angle of incidence at liquid surface = θ_0

DO is normal to *M*1 and hence parallel to *AC*

EO is normal to *M*2

$$\angle DOE = 90°$$



In ***ΔCDP***

$$i + r_1 = \theta_0 \qquad ...(1)$$

Angle of incidence on ***M*2**

$$i' = 90 - i \qquad [\because \text{ In } \Delta DOE - \angle DOE = 90°]$$

In ***ΔQEF*** $$\angle EQF = 90° - \theta_0$$

$$\angle FEQ = 180° - (90 - i) = 90 + i$$

$\therefore$ $$r_2 = 180° - [90 - \theta_0 + 90 + i] = \theta_0 - i$$

$$= r_1 \qquad [\text{from (1)}]$$

Because $$r_2 = r_1$$

$\therefore$ $$\theta = \theta_0$$

93 In Figure *AB* is incident ray reaching the observe at *C* after undergoing refraction at *B*.

Snell's law

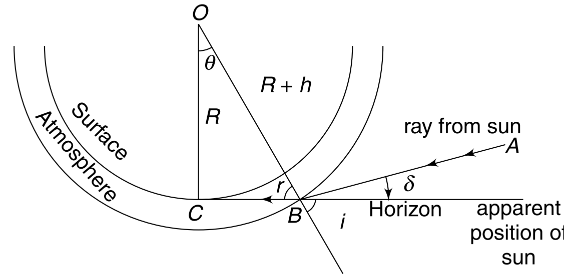
$$\sin i = \mu \sin r$$

$$\sin i = \left(\frac{\mu R}{R + h}\right)$$

$$i = \sin^{-1}\left(\frac{\mu R}{R + h}\right)$$

$$\delta + r = \sin^{-1}\left(\frac{\mu R}{R + h}\right)$$

$$\delta = \sin^{-1}\left(\frac{\mu R}{R + h}\right) - \sin^{-1}\left(\frac{R}{R + h}\right)$$



In presence of atmosphere the sun becomes visible when it is δ below the horizon. In absence of atmosphere, the observer at C will be able to see the sun only after the earth rotates further by an angle δ.

$$\therefore \quad \Delta t = \frac{\delta}{\frac{2\pi}{T}} = \frac{T}{2\pi}\left[\sin^{-1}\left(\frac{\mu R}{R+h}\right) - \sin^{-1}\left(\frac{R}{R+h}\right)\right]$$

$$\simeq 0.9 \text{ min.}$$

94.

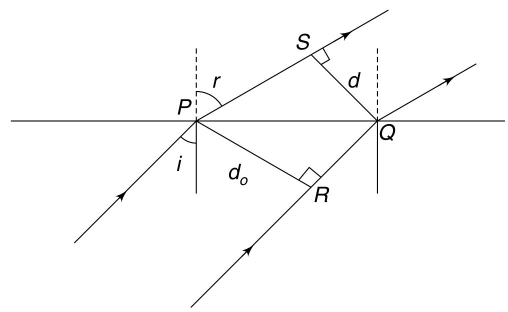
$$\sqrt{2}\sin 30° = 1\cdot \sin r \quad \Rightarrow \quad r = 45°$$

$$\frac{PR}{PQ} = \cos i \quad \Rightarrow \quad PQ = \frac{d_0}{\cos 30°} = \frac{2d_0}{\sqrt{3}}$$

$$QS = d = \text{diameter of refracted light}$$

$$\frac{d}{PQ} = \cos r \quad \Rightarrow \quad PQ = \frac{d}{\cos 45°} = \sqrt{2}d.$$

$$\therefore \quad \sqrt{2}d = \frac{2d_0}{\sqrt{3}} \quad \Rightarrow \quad d = \sqrt{\frac{2}{3}}\, d_0$$



The refracted beam has lesser diameter.

Intensity of refracted beam

$$I = \frac{I_0\pi\left(\frac{d_0}{2}\right)^2}{\pi\left(\frac{d}{2}\right)^2} = I_0\cdot\left(\frac{d_0}{d}\right)^2$$

$$\therefore \quad I = \frac{3}{2} I_0$$

95. Angle of refraction at first refraction is given by

$$\mu_1 \sin 45° = \mu_2 \sin r \quad \Rightarrow \quad 2\frac{1}{\sqrt{2}} = \mu_2 \sin r$$

$$\Rightarrow \quad \mu_2 \sin r = \sqrt{2} \qquad \text{...(i)}$$

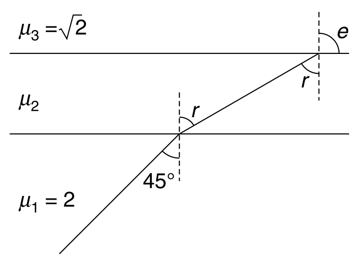
For $\mu_2 < \sqrt{2}$, $\sin r > 1$ which is not possible. It means the ray will suffer total internal reflection.

Deviation in this case will be 90°.

For $\mu_2 > \sqrt{2}$, the ray will pass on to the second interface with angle of refraction equal to r.

Applying snell's law at second interface–

$$\Rightarrow \quad \mu_2 \sin r = \sqrt{2}\sin e \qquad \text{...(ii)}$$



From (i) and (ii)

$$\sqrt{2}\sin e = \sqrt{2} \quad \Rightarrow \quad e = 90°$$

Deviation is clearly 45°

96. (a) $\theta \simeq C$

$$\therefore \quad 1.6\sin\theta = 1.5\sin 90°$$

$$\sin\theta = \frac{15}{16}$$

A distance x along the length of the cable means the ray along B must have travelled a path length of $\frac{x}{\sin\theta} = \frac{16x}{15}$.

$$\therefore \quad \text{Total path length for } B \text{ is} \quad x_B = \frac{16}{15}\times 1500$$

$$= 1600 \text{ m.}$$

$$\therefore \quad \Delta x = 100 \text{ m.}$$

$$\therefore \qquad \Delta t = \frac{100 \text{ m}}{V} = \frac{100 \text{ m}}{\dfrac{3 \times 10^8}{1.6} \text{ ms}^{-1}} = 0.53 \times 10^{-6} \text{ s}$$

$$= 0.53 \ \mu\text{s}.$$

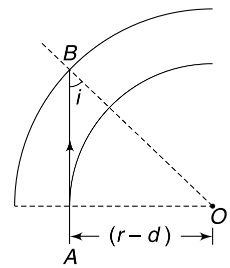
(b) Crest from path A will reach with the trough from path B if

Δt = Half the time period of the signal.

$\therefore$ Time period of signal $= 2\Delta t = 1.06\ \mu$s.

$$\therefore \qquad f = \frac{1}{T} = \frac{1}{1.06 \times 10^{-6}} = 0.94 \text{ MHz}.$$

97. (a) The minimum radius r would be when the angle of incidence of the ray AB is such that it is totally reflected.



$$\Rightarrow \qquad i = C$$

$$\sin C = \frac{r_o - d}{r_o}$$

and

$$\mu \sin C = 1 \cdot \sin 90°$$

$$\Rightarrow \qquad \sin C = \frac{1}{\mu}$$

$$\therefore \qquad \frac{r_o - d}{r_o} = \frac{1}{\mu}$$

$$\Rightarrow \qquad r_o = \frac{\mu d}{(\mu - 1)}$$

for $d = 200\ \mu$m and $\mu = 1.4$

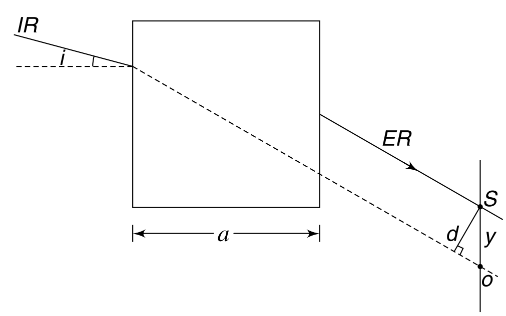
$$r_o = \frac{1.4 \times 200}{0.4} = 700\ \mu\text{m} = 0.7 \text{ mm (very small !!)}$$

(b) as $d \to 0$; $r_o \to 0$

(c)
$$r_o = \frac{d}{1 - \dfrac{1}{\mu}}$$

$\therefore$ for smaller r_o, μ shall be larger

98. For angle of incidence i, the incident ray (IR) and the emergent ray (ER) has been shown in the Figure. The lateral shift when i is small is given by



$$d = \left(1 - \frac{1}{\mu}\right) i$$

The spot at S is at a distance y from O (the point where incident ray will hit the screen if there is no shift).

$$y = \frac{d}{\cos i}$$

When $i \to 0$; $y \to d$.

$$\therefore \qquad y = a\left(1 - \frac{1}{\mu}\right) i$$

$$\frac{dy}{dt} = a\left(1 - \frac{1}{\mu}\right) \frac{di}{dt}$$

$$V = a\left(1 - \frac{1}{\mu}\right) \omega = \frac{a\omega}{3}$$

99.

$$AB = \frac{t}{\cos r}$$

$$AC = AB \sin(i - r)$$

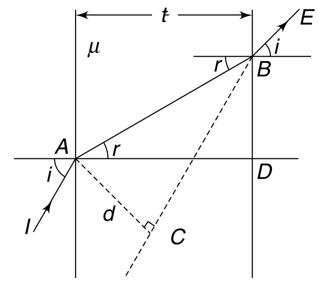
$$d = \frac{t}{\cos r}\cdot \sin(i - r) \quad ...(1)$$

$$= \frac{t}{\cos r}[\sin i \cos r - \cos i \sin r]$$

$$= t\sin i\left[1 - \frac{1}{\mu}\frac{\cos i}{\cos r}\right] \quad \left[\because \frac{\sin i}{\sin r} = \mu\right]$$

$$= t\sin i\left[1 - \frac{\cos i}{\sqrt{\mu^2 - \sin^2 i}}\right] \quad ...(2)$$

$$\left[\because \sin r = \frac{\sin i}{\mu}, \cos r = \sqrt{1 - \frac{\sin^2 i}{\mu^2}}\mu, \cos r = \sqrt{\mu^2 - \sin^2 i}\right]$$



For $i \to 0$ $\quad d = t(i - r)$ [from (1)]

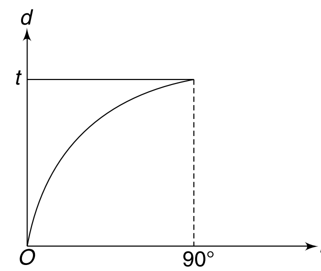
$$d \simeq ti\left(1 - \frac{r}{i}\right) = ti\left(1 - \frac{1}{\mu}\right)$$

[$\because$ for $i \to 0$ $\sin(i - r) = i - r$ and $\cos r \to 1$]

For $i \to \frac{\pi}{2}$, d is maximum

$$d_{\max} = t \quad \text{[from (2)]}$$

$\therefore$ graph is as shown.



100. Snell's law

$$\mu\sin r = 1\cdot\sin 45°$$

$$\therefore \quad \sin r = \frac{1}{2}$$

$$r = 30°.$$

Light ray incident at O will not suffer any deviation at the curved surface.

Consider another ray incident at point A on the plane surface. For this ray to not suffer a TIR at B

$$\alpha \le C$$

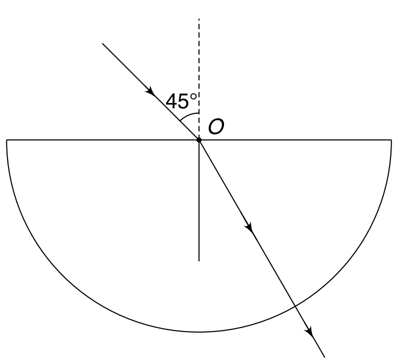
$$\alpha \le \sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$$

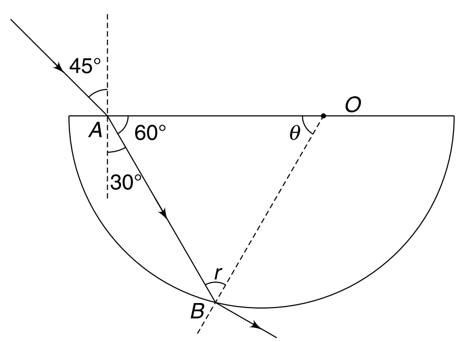
$$\alpha \le 45°$$

$$(180° - 60° - \theta) < 45°$$

$$-\theta \le -75°$$

$$\theta \ge 75°$$





For θ = 75°

$$\frac{OA}{\sin 45°} = \frac{R}{\sin 60°}$$

$$OA = \frac{2R}{\sqrt{3}} \cdot \frac{1}{\sqrt{2}} = \sqrt{\frac{2}{3}} \cdot R$$

101. $$h_{\text{app}} = \mu h$$

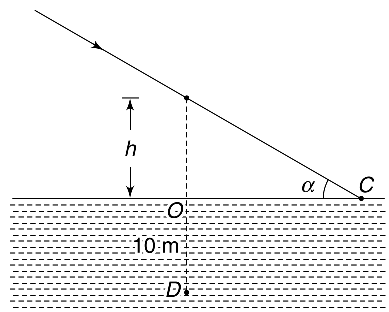
$$h_{\text{app}} = 50 - d = 40 \text{ m} \quad \text{and} \quad \mu = \frac{4}{3}$$

$\therefore$ $$h = 30 \text{ m}$$

Now, $$h_{\text{app}} = \mu h$$

$$V_{y\,\text{app}} = \mu V_y$$

$$V_{x\,\text{app}} = V_x$$



Given $$\frac{V_{y\,\text{app}}}{V_{x\,\text{app}}} = \tan 45° = 1$$

$\therefore$ $$\frac{\mu V_y}{V_x} = 1 \quad \Rightarrow \quad \tan\alpha = \frac{1}{\mu} = \frac{3}{4}$$

$$\alpha = 37°$$

α is the true angle that the line of motion of the bird makes with horizontal.

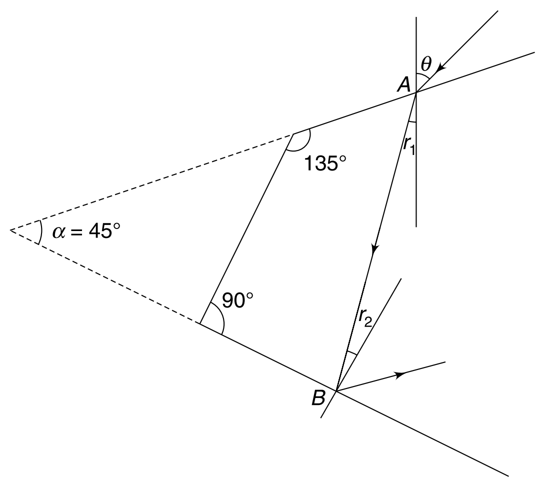
$\therefore$ $$OC = h \tan a = 30 \cdot \frac{3}{4}$$

$$= 22.5 \text{ m}$$

$\therefore$ $$DC = \sqrt{(22.5)^2 + 10^2}$$

$$= 24.62 \text{ m.}$$

102. For TIR at B, $r \geq C$



$$r_2 \geq \sin^{-1}\left(\frac{1}{\mu}\right)$$

$$r_2 \geq \sin^{-1}\left(\frac{1}{2}\right)$$

$$r_2 \geq 30°$$

$$\alpha - r_1 \geq 30°$$

$$45° - r_1 \geq 30°$$

$$15° \geq r_1$$

$$\sin(15°) \geq \sin(r_1) \quad [\mu \sin r_1 = 1 \cdot \sin\theta]$$

$$\sin 15° \geq \frac{1}{2}\sin\theta$$

$$2\sin 15° \geq \sin\theta$$

$$\sin^{-1}(2\sin 15°) \geq \theta$$

103.(a) For any light ray $$i' = 90° - \alpha$$

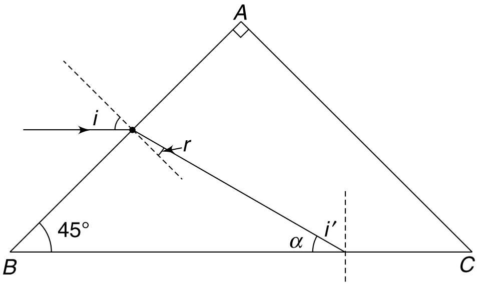
$$= 90° - [180° - 45° - (90° + r)]$$

$$= 45° + r$$

$\therefore$ Minimum value of $$i' = 45°$$

when $r = 0$, i.e., when $i = 0$

Total internal reflection will occur at BC if



$$45° \geq C$$

$$\Rightarrow \quad \sin 45° \geq \sin C$$

$$\Rightarrow \quad \frac{1}{\sqrt{2}} \geq \frac{1}{\mu}$$

$$\Rightarrow \quad \sqrt{2} \leq \mu.$$

(b) $PQ \rightarrow$ incident ray

$ST \rightarrow$ emergent ray

$$1 \cdot \sin 45° = \mu \cdot \sin r$$

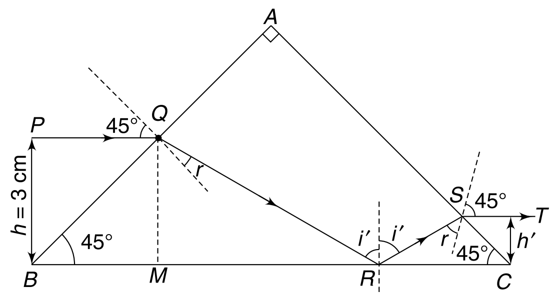
$$\Rightarrow \quad r = 30° \quad [\because \quad \mu = \sqrt{2}]$$

Path of light ray is as shown.

$$\Delta BQR \sim \Delta CSR$$

$$\therefore \quad \frac{BQ}{SC} = \frac{BR}{RC} \Rightarrow \frac{\sqrt{2}h}{\sqrt{2}h'} = \frac{BR}{RC}$$

$$\Rightarrow \quad h' = 3\left(\frac{RC}{BR}\right) \quad [\because \quad h = 3 \text{ cm}] \qquad ...(1)$$



$$i' = 45° + r = 75°$$

$$\angle QRM = 15°$$

$$\therefore \quad \frac{QM}{MR} = \tan 15° \Rightarrow MR = \frac{3}{0.25} = 12 \text{ cm}$$

$$BM = 3 \text{ cm}$$

$$\therefore \quad BR = 3 + 12 = 15 \text{ cm}$$

$$\therefore \quad RC = 20 - 15 = 5 \text{ cm}$$

$$\therefore \text{ from (1)} \quad h' = 3 \times \frac{5}{15} = 1 \text{ cm}$$

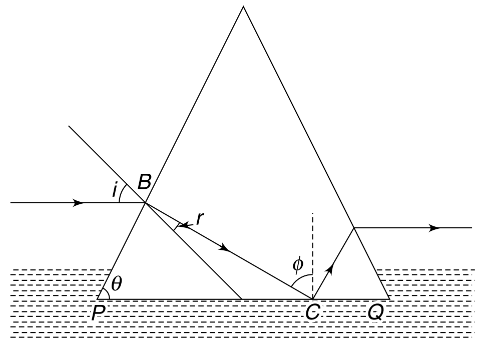
104. Angle of incidence at the first refracting surface is

$$i = \frac{\pi}{2} - \theta$$

Snell's law gives

$$\mu_g \sin r = \sin i$$

$$\frac{3}{2} \sin r = \cos \theta \qquad ...(1)$$



In ΔPBC

$$\angle B + \angle P + \angle C = 180°$$

$$90 + r + \theta + 90 - \phi = 180°$$

$$\Rightarrow \quad \phi = r + \theta.$$

For total internal reflection at the base.

$$\phi > C$$

$$r + \theta > C$$

$$\Rightarrow \quad \sin(r+\theta) > \sin C = \frac{4/3}{3/2} = \frac{8}{9}$$

$$\Rightarrow \quad \sin(r+\theta) > \frac{8}{9}$$

$$\Rightarrow \quad \cos\theta\cdot\sin r + \sin\theta\cos r > \frac{8}{9}$$

$$\frac{2}{3}\cos^2\theta + \sin\theta\cdot\sqrt{1-\frac{4}{9}\cos^2\theta} > \frac{8}{9} \quad \text{[using (1)]}$$

$$\Rightarrow \quad \sin^2\theta\left(1-\frac{4}{9}\cos^2\theta\right) > \left(\frac{8}{9}-\frac{2}{3}\cos^2\theta\right)^2$$

$$\cos^2\theta = x$$

$$\Rightarrow \quad (1-x)\left(1-\frac{4}{9}x\right) > \left(\frac{8}{9}-\frac{2}{3}x\right)^2$$

$$\Rightarrow \quad 9(1-x)(9-4x) > (8-6x)^2$$

$$\Rightarrow \quad 9[9-13x+4x^2] > (64+36x^2-96x)$$

$$\Rightarrow \quad \frac{17}{21} > x \quad \Rightarrow \quad \cos\theta < \sqrt{\frac{17}{21}}$$

$$\therefore \quad \theta_{max} = \cos^{-1}\sqrt{\frac{17}{21}}$$

105. Using Snell's law for four successive refractions we get

$$\sin i = \mu_1 \sin r_1 \qquad ...(1)$$

$$\mu_1\cdot\sin(90-r_1) = \mu_2\sin r_3$$

$$\Rightarrow \quad \mu_2\sin r_3 = \mu_1\cos r_1 \qquad ...(2)$$

$$\mu_2\sin r_4 = \mu_3\sin r_5$$

$$\Rightarrow \quad \mu_2\sin(90-r_3) = \mu_3\sin r_5$$

$$\Rightarrow \quad \mu_2\cos r_3 = \mu_3\sin r_5 \qquad ...(3)$$

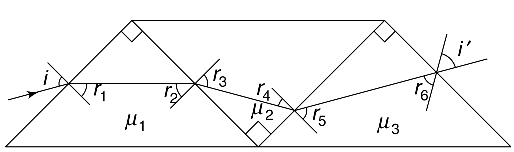
And $\quad \mu_3\sin r_6 = \sin i'$

for emergent ray to be parallel to the incident ray

$$i' = 90 - i$$

$$\therefore \quad \mu_3\sin(90-r_5) = \sin(90-i)$$

$$\Rightarrow \quad \cos i = \mu_3\cos r_5 \qquad ...(4)$$



Square all the equations -1, 2, 3 and 4 and add them

$$\sin^2 i + \mu_2^2 \sin^2 r_3 + \mu_2^2 \cos^2 r_3 + \cos^2 i = \mu_1^2 \sin^2 r_1 + \mu_1^2 \cos^2 r_1 + \mu_3^2 \sin^2 r_5 + \mu_3^2 \cos^2 r_5$$

$$\Rightarrow \quad 1 + \mu_2^2 = \mu_1^2 + \mu_3^2$$

$$\Rightarrow \quad \mu_2 = \sqrt{\mu_1^2 + \mu_3^2 - 1}$$

106. For minimum deviation of red light

$$r_1 = r_2 = \frac{A}{2} = 30^\circ$$

Applying Snell's law at first refracting face

$$(1)\sin i = \mu \sin r_1$$

$$\sin i = 1.5 \cdot \sin 30^\circ = 0.75$$

$$\therefore \quad i = \sin^{-1}(0.75) = 49^\circ$$

Deviation of red light $\delta_R = 2i - A$

$$= 2 \times 49^\circ - 60^\circ = 38^\circ$$

For violet light

$$(1)\sin 49^\circ = (1.6)\sin r_1$$

$$\therefore \quad \sin r_1 = \frac{0.75}{1.6} = 0.47$$

$$\therefore \quad r_1 = 28^\circ$$

$$\therefore \quad r_2 = A - r_1 = 60 - 28 = 32^\circ$$

For refraction at second face

$$1.6 \cdot \sin(32^\circ) = (1)\sin(e)$$

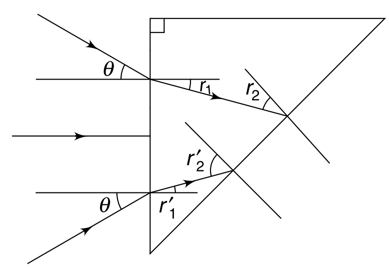
$$\therefore \quad \sin(e) = 1.6 \times 0.53 = 0.85$$

$$\therefore \quad e = 58^\circ$$

$$\therefore \quad \delta_v = i + e - A = 49^\circ + 58^\circ - 60^\circ = 47^\circ$$

$\therefore$ Angular width $(\theta) = \delta_v - \delta_R = 47^\circ - 38^\circ = 9^\circ$

107. Critical angle $C = \sin^{-1}\left(\frac{1}{\mu}\right) = 42^\circ$



The Figure shows the extreme rays.

From geometry one can easily show that

$$r_2' = 45^\circ + r_1' \quad [r_1' = r_1]$$

$$\therefore \quad r_2' > C$$

$\therefore$ Lower extreme ray will suffer TIR.

For upper extreme ray

$$r_2 = 45^\circ - r_1$$

For TIR $r_2 \geq C$

$$\therefore \quad 45^\circ - r_1 \geq C$$

$$\therefore \quad 45^\circ - C \geq r_1$$

$$3^\circ \geq r_1$$

Snell's law

$$1 \cdot \sin\theta = \mu \sin r_1$$

For small angles $1(\theta) \simeq \mu(r_1)$

$$\therefore \quad \theta \leq 1.49 \times 3^\circ = 4.47^\circ$$

108. $$\delta = 180° - \phi \quad ...(1)$$

In polygon *PBFD*

$$A + (90 + i) + \phi + (90 + e) = 360°$$

$$\phi = 180° - i - e - A$$

From (1)

$$\delta = i + e + A$$

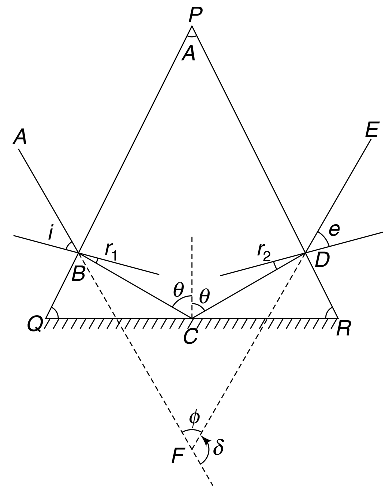
$\because \Delta QBC \sim \Delta CDR$

$\therefore\ r_1 = r_2$

$\therefore\ i = e$

$\therefore\ \delta = 2i + A$

This is independent of μ.



Note: Such prisms are known as reflecting prism which do not produce dispersion

109. (a) Minimum deviation for red

$$\frac{\sin\left(\frac{A+\delta}{2}\right)}{\sin\left(\frac{A}{2}\right)} = \mu_r$$

$$\frac{\sin\left(\frac{60+\delta}{2}\right)}{\sin\left(\frac{60}{2}\right)} = 1.510$$

$$\therefore \quad \sin\left(30 + \frac{\delta}{2}\right) = 0.755$$

$$30 + \frac{\delta}{2} = 50$$

$$\delta = 40°$$

$$i + e - A = 40°$$

$$2i = 40 + 60 \quad \text{[In minimum deviation } i = e\text{]}$$

$$i = 50°$$

Refraction for violet

1st refraction $$\mu_v \sin r_1 = 1 \times \sin i$$

$$\Rightarrow \quad \sin r_1 = \frac{\sin 50°}{1.550} = \frac{0.755}{1.550}$$

$$= 0.487$$

$$\therefore \quad r_1 = 28°$$

$$\therefore \quad r_2 = A - r_1 = 32°$$

2nd refraction $$1.55 \sin 32° = 1 \times \sin e$$

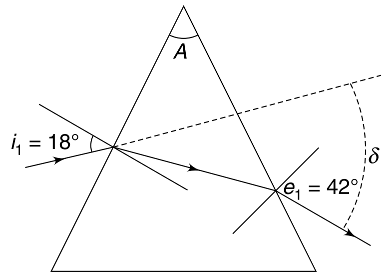
$$\sin e = 1.55 \times 0.529 = 0.819$$

$$e = 55°$$

$$\therefore \quad \text{Dispersion} \quad \theta = 55° - 50°$$

$$= 5°$$

110. Path of the light ray is optically reversible. For two angle of incidence $i_1 = 18°$ and $i_3 = 42°$, the deviation suffered by the light ray is same. It means that for incidence angle, $i_1 = 18°$, the angle of emergence, $e_1 = 42°$ and vice–versa.



$$\delta = i + e - A$$

$\Rightarrow$ $$28° = 18° + 42° - A$$

$\Rightarrow$ $$A = 32°$$

111. $$\delta_m = 60° \quad \text{when} \quad i = 60°$$

For minimum deviation $$i = e = 60°$$

$\therefore$ $$\delta_m = 2i - A$$

$\Rightarrow$ $$60° = 2 \times 60° - A \quad \Rightarrow \quad A = 60°$$

$$\mu = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)} = \frac{\sin\left(\frac{60 + 60}{2}\right)}{\sin\left(\frac{60}{2}\right)}$$

$\therefore$ $$\mu = \frac{\sin 60°}{\sin 30°} = \sqrt{3}$$

When angle of incidence is i_1, the angle of emergence is $e = 2i_1$.

$\therefore$ $$\delta = i_1 + 2i_1 - A$$

$$63° = 3i_1 - 60°$$

$\therefore$ $$i_1 = 41°$$

112. For overall deviation to be minimum, the light ray must suffer minimum deviations at both the prisms.

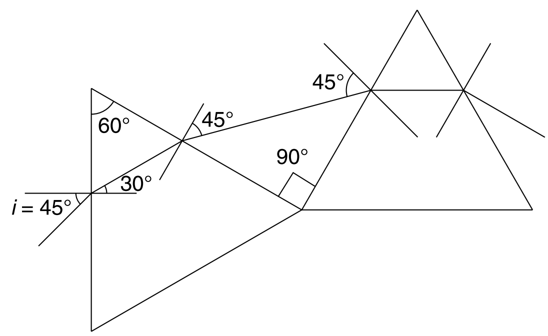
For minimum deviation at first prism

$$r_1 = r_2 = \frac{A}{2} = 30°$$

$\therefore$ Snell's law for first refraction gives (1) $\sin i = \sqrt{2}\,\sin 30°$

$\Rightarrow$ $$\sin i = \frac{1}{\sqrt{2}}$$

$\Rightarrow$ $$i = 45°$$



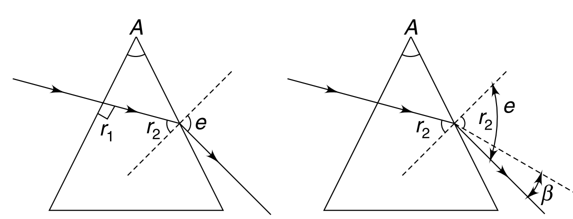
For angle of incidence to be 45° at the second prism, θ shall be 90° as shown in the diagram.

113. (a) There is no deviation when refractive index of surrounding medium is same as that of prism glass.

$\therefore$ $$\mu_1 = \frac{4}{\sqrt{3}} \quad \therefore \quad k_2 = \frac{4}{\sqrt{3}}$$

(b) It can be seen from graph when $\mu_1 < k_1$, light will not emerge.

$\therefore$ $$r_2 = C \quad \text{when} \quad \mu_1 = k_1$$



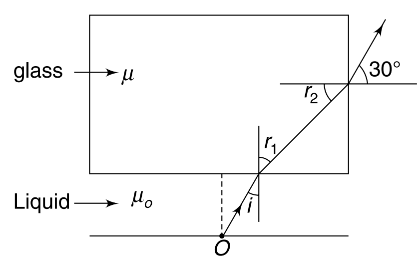
$$\sin r_2 = \sin C = \frac{\mu}{\mu_1}$$

$$\sin 60° = \frac{\mu}{\mu_1}$$

$$\frac{\sqrt{3}}{2} = \frac{4}{\sqrt{3}} = \times \frac{1}{\mu_1}$$

$$\mu_1 = \frac{8}{3}$$

$\therefore$ $$k_1 = \frac{8}{3}$$



114. $$i_{\max} = 90°$$

$\therefore$ $$\mu_0 \sin 90° = \mu \sin (r_{1\max})$$

$\Rightarrow$ $$\sin (r_{1\max}) = \frac{2\mu_0}{3} \qquad ...(1)$$

But $\qquad r_1 + r_2 = 90°$

$\therefore \qquad r_{2\,\min} = 90° - r_{1\,\max} \qquad ...(2)$

Using Snell's law at vertical face

$$\frac{3}{2}\sin r_2 = 1\cdot\sin 30°$$

$$\Rightarrow \quad \sin r_2 = \frac{1}{3}$$

$$\Rightarrow \quad \sin(90 - r_1) = \frac{1}{3} \quad \Rightarrow \quad \cos r_1 = \frac{1}{3}$$

$$\Rightarrow \quad \sqrt{1 - \frac{4\mu_0^2}{9}} = \frac{1}{3} \qquad \text{[using (i)]}$$

$$\Rightarrow \quad \mu_0 = \sqrt{2}.$$

When there is no liquid layer; $\mu = 1$

$$r_{1\,\max} = \sin^{-1}\left(\frac{2}{3}\right) \simeq 42°$$

$$\therefore \quad r_{2\,\min} = 48°$$

But critical angle $\qquad C = \sin^{-1}\left(\frac{2}{3}\right) = 42°$

Hence $r_{2\,\min} > C$ and light cannot come out of the vertical face. The dot cannot be seen.

115. Angle of incidence $i = 90 - \theta$.

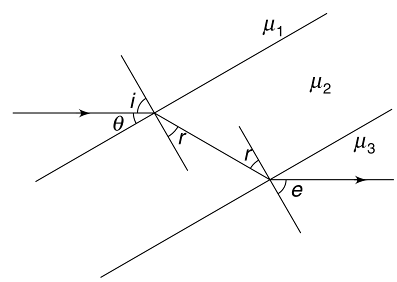
Snell's law for first refraction gives

$$\mu_2 \sin r = \mu_1 \sin i \quad \Rightarrow \quad r = \sin^{-1}\left(\frac{\mu_1}{\mu_2}\sin i\right)$$

At the second face the angle of incidence is r.

$$\therefore \quad \mu_2 \sin r = \mu_3 \sin e \quad \Rightarrow \quad \mu_1 \sin i = \mu_3 \sin e$$

$$\Rightarrow \quad e = \sin^{-1}\left(\frac{\mu_1}{\mu_3}\sin i\right) \qquad ...(1)$$



Since i and e are angles made by incident ray with normal to the glass slab, the deviation must be

$$\delta = i - e$$

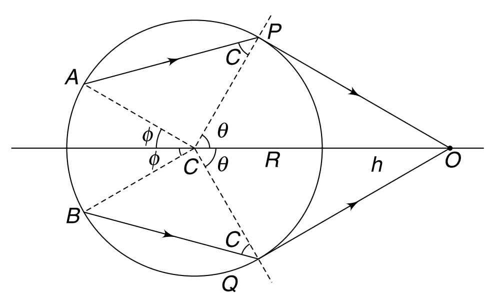
$$= i - \sin^{-1}\left(\frac{\mu_1}{\mu_3}\sin i\right)$$

(Note that equation (1) tells you that $e < i$ since $\mu_3 > \mu_1$]

putting $\qquad i = \frac{\pi}{2} - \theta$ gives

$$\delta = \frac{\pi}{2} - \theta - \sin^{-1}\left(\frac{\mu_1}{\mu_3}\cos\theta\right)$$

116. AB is the arc on which the insect is visible. Ray AP (and BQ) are incident at critical angle $\left(C = \sin^{-1}\frac{1}{\sqrt{2}} = \frac{\pi}{4}\right)$ and come out tangentially to reach the observer at O. This follows from principle of reversibility of the path of light rays. Rays starting from O cannot reach the back surface of the sphere beyond arc AB.



Clearly $\qquad \cos\theta = \frac{CP}{CO} = \frac{R}{R+h} \qquad ...(1)$

(a) when $\qquad h = R; \quad \cos\theta = \frac{1}{2} \quad \Rightarrow \quad \theta = \frac{\pi}{3}$

$$\therefore \quad \phi = \pi - (\pi - 2C) - \theta = 2\cdot\frac{\pi}{4} - \frac{\pi}{3}$$

$$= \frac{\pi}{6}$$

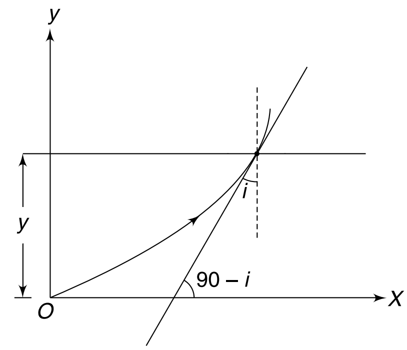
$\therefore \quad \text{arc } AB = 2 \times \frac{\pi R}{6} = \frac{\pi R}{3}$

(b) When $h \to 0$; $\cos\theta \to 1$

$\Rightarrow \quad \theta \to 0$

$\therefore \quad \phi = \pi - (\pi - 2C) = 2C = \frac{\pi}{2}$

$\therefore \quad \text{arc } AB = R(2\phi) = \pi R.$



117. A light ray starting from a point on the road will take a curved path to reach the eye of the man.

Consider a light ray starting from point O at grazing incidence and travelling up.

Let the angle of incidence at height y be i.

$$\mu_0 \sin 90° = \mu \sin i$$

$$\mu_0 = \mu_0 (1 + by)^{1/2} \sin i$$

$$\sin i = \frac{1}{(1 + by)^{1/2}}$$

$$\therefore \quad \cot i = \frac{\sqrt{by}}{1}$$

But $$\frac{dy}{dx} = \tan(90 - i) = \cot i$$

$$\therefore \quad \frac{dy}{dx} = \sqrt{by}$$

$$\int_0^y \frac{dy}{\sqrt{y}} = \sqrt{b} \int_0^x dx$$

$$2[\sqrt{y}]_0^y = \sqrt{b}\, x$$

$$\therefore \quad x = \frac{2\sqrt{y}}{b}$$

Value of x when $y = 1.5$ m is

$$x = 2\sqrt{\frac{1.5}{6 \times 10^{-4}}}$$

$$= 100 \text{ m.}$$

This is maximum value of x as we have considered a ray starting at grazing incidence.

118. $$\mu = \frac{\sin \frac{(A + \delta_m)}{2}}{\sin\left(\frac{A}{2}\right)}$$

Since $$\delta_m = A$$

$$\therefore \quad \mu = \frac{\sin A}{\sin \frac{A}{2}} = 2 \cos \frac{A}{2} \qquad \text{...(1)}$$

Also $A < 2C$, where C is critical angle for glass air interface. If this condition is not met there will be no emergent ray.

$$\therefore \quad A < 2C \quad \Rightarrow \quad \frac{A}{2} < C$$

$$\Rightarrow \quad \sin\left(\frac{A}{2}\right) < \sin C$$

$$\Rightarrow \quad \sin \frac{A}{2} < \frac{1}{\mu} \qquad \text{...(2)}$$

From (1) and (2) $\sin^2 \frac{A}{2} + \cos^2 \frac{A}{2} < \frac{1}{\mu^2} + \left(\frac{\mu}{2}\right)^2$

$$\Rightarrow \quad 1 < \frac{4 + \mu^4}{4\mu^2}$$

$$\Rightarrow \quad \mu^4 - 4\mu^2 + 4 > 0$$

$$\Rightarrow \quad (\mu^2 - 2)^2 > 0$$

$$\Rightarrow \quad \mu^2 > 2$$

$$\Rightarrow \quad \mu > \sqrt{2}$$

119. Four refractions take place. The final emerging ray is perpendicular to the glass slab. Incident ray must also be perpendicular. It means that the third refraction at the back of the bubble must form image at infinity. Using the formula for refraction at curved surface gives the position of object for this refraction.

$$\frac{\mu}{\infty} - \frac{1}{u_3} = \frac{(1.5 - 1)}{(-10)} \Rightarrow u_3 = 20 \text{ cm}$$

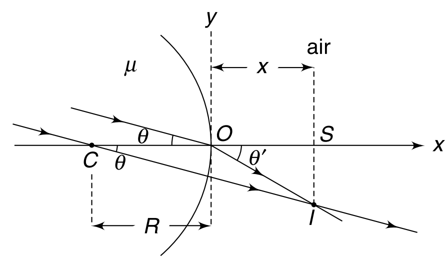
Therefore the image formed after second refraction is at a distance of 40 cm from the refracting surface. Using the formula again

$$\frac{1}{40} - \frac{1.5}{u_2} = \frac{1 - 1.5}{10} \Rightarrow u_2 = 20 \text{ cm}$$

This means that the image was formed at a distance of 30 cm from the first refracting surface after first refraction. The virtual object for this refraction must be at a distance $= \frac{30}{3/2} = 20$ cm from the first face.

Thus C must be at O.

120. Figure shows two rays – one through O and the other through the centre of curvature (C) of the surface. The second ray passes undeviated. The two rays intersect at I.



Snell's law gives

$$\sin\theta' = \mu \sin\theta$$

$$\Rightarrow \quad \theta' \simeq \mu\theta$$

$$SI = x\tan\theta' = (x + R)\tan\theta$$

$$\Rightarrow \quad x\theta' = (x + R)\,\theta$$

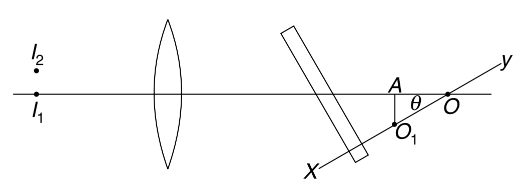
$$\Rightarrow \quad x\mu\theta = (x + R)\,\theta$$

$$\Rightarrow \quad (\mu - 1)\,x = R$$

$$\Rightarrow \quad x = \frac{R}{\mu - 1}$$

And $$SI \simeq (x + R)\,\theta = R\left(\frac{\mu}{\mu - 1}\right)$$

121. The glass slab shifts the position of the object (for the lens) by a distance of $t\left(1 - \frac{1}{\mu}\right) = 3\left(1 - \frac{2}{3}\right) = 1$ cm towards the lens.



since object is now at a distance of 40 cm (= 2f) from the lens, its image is formed on the other side of the lens at a distance of 40 cm. When the glass slab is tilted slightly, the normal from the object on the slab is line XY.

Now the shift is OO_1

$$\therefore \quad OO_1 = 1 \text{ cm}$$

$$OA = OO_1 \cos\theta \simeq OO_1 = 1 \text{ cm} \quad [\cos\theta \simeq 1]$$

$$AO_1 = OO_1 \sin\theta \simeq 1 \cdot \theta = \frac{\pi}{180} \text{ cm} \qquad \left[\because \quad 1° = \frac{\pi}{180} \text{ radian}\right]$$

Distance of A from lens = 40 cm = $2f$.

$\therefore$ Final image is at 40 cm from lens.

Height of image = AO_1 [since magnification = 1]

$$\therefore \quad I_1I_2 = \frac{\pi}{180} \text{ cm.}$$

122. Radius of the ring of the rainbow

$$\frac{R}{6} = \tan 42°$$

$$R = 0.9 \times 6 = 5.4 \text{ km}$$

Arc ABC is visible to the man

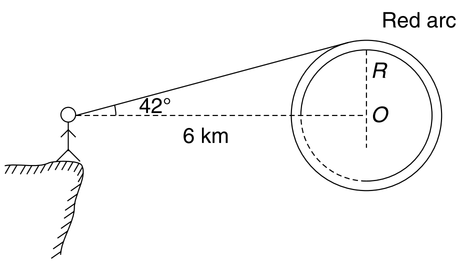
$$d = 0.5 \text{ km}$$

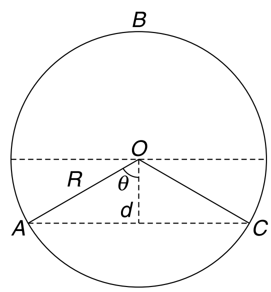
$$\cos\theta = \frac{0.5}{5.4} = 0.09$$

$$\theta = 85°$$

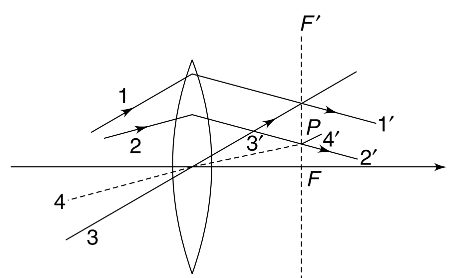
$$\therefore \quad \text{Required fraction} = \frac{190}{360} = 0.53$$

$$= 53\%$$





123. Draw a ray 3 parallel to 1 and passing through the optical centre of the lens. This ray passes undeviated. Intersection of 1′ and 3′ occurs in focal plane of the lens. Mark the focal plane FF'. Draw another ray 4 through the optical centre that is parallel to ray 2. Ray 4 passes undeviated and intersects the focal plane at P. The refracted ray will pass through point P.



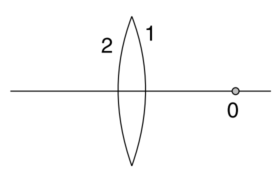
124. The light rays reflected from surface 1 form an erect image as the surface can be treated like a convex mirror.

The other image that he can see is formed by combination of three events – refraction at face 1, reflection at face 2 and refraction at face 2. We can treat the system as a convex lens whose one surface is silvered.

The system can be replaced with an equivalent mirror whose power is given by–

$$P_{\text{eqMirror}} = 2P_L + P_M$$

$$\Rightarrow \quad \frac{1}{-f_{\text{eq}M}} = 2\frac{2(\mu - 1)}{R} + \frac{2}{R} = \frac{4}{R}$$



$$\Rightarrow \quad f_{\text{eq}M} = -\frac{R}{4}$$

If the nose is kept at a distance less that focal length of the equivalent mirror, the image formed is virtual , erect and magnified.

125. Lens formula gives $\frac{1}{v} - \frac{1}{-100} = \frac{1}{50} \Rightarrow v = 100$ cm

The mirror forms an image 50 cm behind itself and the image is moving to right with a velocity of 20 m/s. The image formed by the mirror will act as moving object for the lens.

Again applying the lens formula gives–

$$\frac{1}{v} - \frac{1}{-200} = \frac{1}{50} \Rightarrow v = \frac{200}{3} \text{ cm}$$

For lens $$\frac{dv}{dt} = \left(\frac{v}{u}\right)^2 \frac{du}{dt} = \left(\frac{200/3}{200}\right)^2 \cdot 20 = \frac{20}{9} \text{ cm/s}$$

126. For lens $$\frac{1}{f} = (\mu - 1)\left(\frac{2}{R}\right) = (1.5 - 1)\left(\frac{2}{R}\right)$$

$\therefore$ $$f = R = 20 \text{ cm}$$

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} - \frac{1}{-30} = \frac{1}{20} \Rightarrow \frac{1}{v} = \frac{1}{20} - \frac{1}{30} = \frac{3-2}{60}$$

$\therefore$ $$v = 60 \text{ cm}$$

Shift due to $S1$ is $$t_1\left(1 - \frac{1}{\mu_1}\right) = 6\left(1 - \frac{2}{3}\right) = 2 \text{ cm}$$

Shift due to $S2$ is $$t_2\left(1 - \frac{1}{\mu_2}\right) = 4\left(1 - \frac{1}{2}\right) = 2 \text{ cm.}$$

$\therefore$ Final image will be at a distance of 60 + 2 + 2 = 64 cm from the lens.

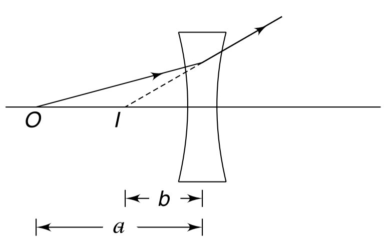
Position of image will remain unchanged in case (b) and (c).

glass slab does not produce any magnification.

$\therefore$ $$m = \frac{v}{u} \text{ (for lens)}$$

$$= \frac{60}{-30} = -2.$$

127.(a)



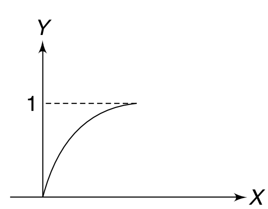
$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{-b} - \frac{1}{-a} = \frac{1}{-f}$$

$\therefore$ $$\frac{1}{b} - \frac{1}{a} = \frac{1}{f}$$

$$\frac{f}{b} - \frac{f}{a} = 1$$

$$\left|\frac{v}{f}\right| = \frac{b}{f} \text{ and } \left|\frac{u}{f}\right| = \frac{a}{f}$$

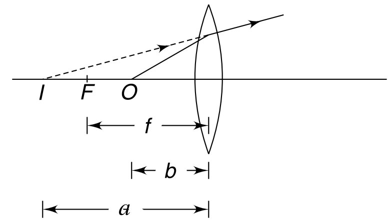


$\therefore$ $$\frac{1}{y} - \frac{1}{x} = 1 \Rightarrow y = \frac{x}{x+1}$$

When $x \to 0; \quad y \to 0$

$x \to \infty; \quad y \to 1$

(b) For converging lens



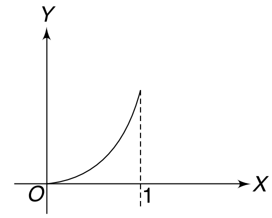
$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{-b} - \frac{1}{-a} = \frac{1}{f}$$

$$\frac{f}{b} - \frac{f}{a} = -1$$

$$\left|\frac{v}{f}\right| = \frac{b}{f}$$

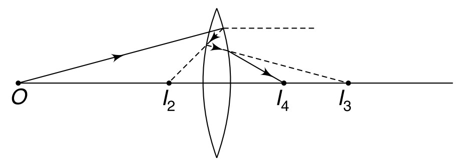
$$\left|\frac{u}{f}\right| = \frac{a}{f}$$



$\therefore$ $$\frac{1}{y} - \frac{1}{x} = -1 \Rightarrow y = \frac{x}{1-x} \quad [x < 1]$$

When $x \to 0; \quad y \to 0$

$x \to 1; \quad y \to \infty$



128. First refraction

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

$$\frac{1.5}{v_1} - \frac{1}{-60} = \frac{1.5 - 1}{30}$$

$\Rightarrow$ $$v_1 = \infty$$

Reflection at second surface

Mirror formula $$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v_2} + \frac{1}{\infty} = \frac{1}{-15} \quad \left(\because \; f = \frac{R}{2}\right)$$

$\therefore$ $$v_2 = -15 \text{ cm} \quad (\text{Image at } I_2)$$

Reflection at first surface

Object is I_2

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v_3} + \frac{1}{15} = \frac{1}{-15} \Rightarrow v_3 = -\frac{15}{2} \text{ cm} \quad (\text{Image at } I_3)$$

Refraction at second surface

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

$$\frac{1}{v_4} - \frac{1.5}{15/2} = \frac{1 - 1.5}{-30}$$

$\Rightarrow$ $$v_4 = \frac{60}{13} \text{ cm. (Image at } I_4)$$

129. The system acts as a mirror of focal length f.

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{-40} + \frac{1}{-10} = \frac{1}{f} \Rightarrow \frac{1}{f} = -\frac{5}{40} \Rightarrow f = -8 \text{ cm}$$

The system is like a concave mirror of focal length 8 cm.

For lens A, $\dfrac{1}{f_A} = (\mu_A - 1)\left(\dfrac{1}{\infty} - \dfrac{1}{-R}\right) = \dfrac{0.8}{R}$

$$\therefore \qquad f_A = \frac{R}{0.8}$$

For lens B, $\dfrac{1}{f_B} = (\mu_B - 1)\left(\dfrac{1}{-R} - \dfrac{1}{-R/2}\right) = \dfrac{0.2}{R}$

$$\therefore \qquad f_B = \frac{R}{0.2}$$

For the given system, power of equivalent mirror is given by.

$$P_{eqM} = 2P_A + 2P_B + P_M$$

$$\text{Power of mirror} = -\frac{1}{f}$$

$$\text{Power of lens} = \frac{1}{f}$$

$$\therefore \qquad -\frac{1}{(-8)} = \frac{2 \times 0.8}{R} + \frac{2 \times 0.2}{R} - \frac{1}{-R/4}$$

$$\frac{1}{8} = \frac{1.6}{R} + \frac{0.4}{R} + \frac{4}{R} = \frac{6}{R}$$

$$\therefore \qquad R = 48 \text{ cm}$$

130. Given $\qquad \mu = a + \dfrac{b}{\lambda^2}$

$$\Rightarrow \qquad \Delta\mu = \frac{-2b}{\lambda^3}\Delta\lambda$$

$\therefore$ Refractive index for incident light is

$$\left(\mu_0 - \frac{2b}{\lambda_0^3}\Delta\lambda\right) \le \mu_0 = a + \frac{b}{\lambda_0^2} \le \left(\mu_0 + \frac{2b}{\lambda_0^3}\Delta\lambda\right)$$

$$\therefore \qquad \text{Spread of refractive index} = \Delta\mu_0 = \frac{4b}{\lambda_0^3}\Delta\lambda$$

$$= \mu_A - \mu_B$$

Focal length of lens is given by

$$\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \qquad ...(1)$$

A change in μ by $\Delta\mu$ will change the focal length by Δf, which is obtained by differentiating (1)

$$-\frac{\Delta f}{f^2} = \Delta\mu\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$= \left(\frac{\Delta\mu}{\mu - 1}\right)\frac{1}{f}$$

$$\therefore \qquad \Delta f_0 = -f_0\left(\frac{\Delta\mu_0}{\mu_0 - 1}\right)$$

Which represents the spread

$$\therefore \qquad f_B - f_A = f_0\left(\frac{\mu_A - \mu_B}{\mu_0 - 1}\right)$$

$$= f_0\left(\frac{4b\,\Delta\lambda}{\lambda_0^3\left[a + \dfrac{b}{\lambda_0^2} - 1\right]}\right)$$

$$= f_0\left[\frac{4b\,\Delta\lambda}{\lambda_0(a\lambda_0^2 + b - \lambda_0^2)}\right]$$

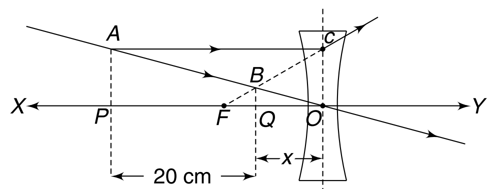
131.(a) **Steps:**

(1) Join A to B. Extend the line so that it intersects XY at O. O is the optical centre of the lens.

(2) Since image is on same side of the principal axis and closer to it, the lens must be a diverging one.

(3) Draw AC parallel to XY.

(4) Joint C to B and extend. It meets XY at F. F is the focus.



(b) $\Delta AOP \sim \Delta BOQ$

$$\frac{AP}{BQ} = \frac{20 + x}{x}$$

$\therefore$ $$3 = \frac{20 + x}{x}$$

$\Rightarrow$ $$x = 10 \text{ cm}$$

$\therefore$ $$u = -30 \text{ cm}; \quad v = -10 \text{ cm}; \quad f = ?$$

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} \Rightarrow \frac{1}{f} = \frac{1}{-10} + \frac{1}{30} = \frac{-3+1}{30}$$

$$f = -15 \text{ cm}$$

Alter: $$\Delta FBQ \sim \Delta FCO$$

$$\frac{BQ}{CO} = \frac{FQ}{FO}$$

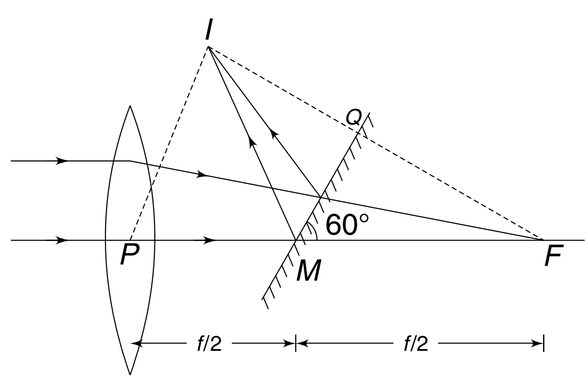
$$\frac{1}{3} = \frac{FQ}{FQ + 10}$$

$\Rightarrow$ $$FQ = 5 \text{ cm}$$

$\Rightarrow$ $$FO = 15 \text{ cm.}$$

132. Focus point F acts as object for the plane mirror. Its image is formed at I.

From geometry, $\angle IMQ = 60°$ and $MF = MI = f/2$



In ΔPIM $$IM = f/2$$

$$PM = f/2$$

$\therefore$ $$IM = PM$$

and $$\angle PMI = 60°$$

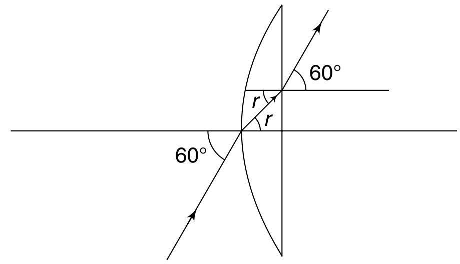
Hence, ΔPMI is equilateral.

$\therefore$ $$PI = f/2$$

133.(a) The ray diagram makes it clear that the emergent ray will be parallel to the incident ray. Hence deviation angle = 0.

(b) Using Snell's law

$$1 \cdot \sin 60° = \sqrt{3}\,\sin r \quad \Rightarrow \quad r = 30°$$



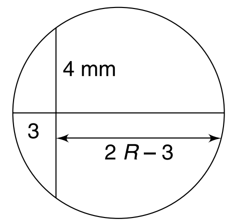
The lateral shift produced will be same as deviation caused due to a parallel faced glass slab of thickness $t = 3$ mm.

$$S = t\frac{\sin(i-r)}{\cos r} = 3\frac{\sin(60^\circ - 30^\circ)}{\cos 30^\circ}$$

$$= \sqrt{3} \text{ mm}$$

(c) From geometry $\quad 3(2R-3) = 4 \times 4$

$$\Rightarrow \quad R = \frac{25}{6} \text{ mm}$$



(d) Refraction at curved surface

$$\frac{\sqrt{3}}{v} - \frac{1}{\infty} = \frac{\sqrt{3}-1}{25/6}$$

$$\Rightarrow \quad \frac{\sqrt{3}}{v} = \frac{6(\sqrt{3}-1)}{25}$$

$$\Rightarrow \quad v = \frac{25}{2\sqrt{3}(\sqrt{3}-1)} \simeq 10 \text{ mm}$$

For refraction at second (plane) surface the object distance will be

$$u = +7 \text{ mm}$$

$$\frac{1}{v} - \frac{\sqrt{3}}{7} = \frac{1-\sqrt{3}}{\infty} \quad [\because \ R = \infty]$$

$$\frac{1}{v} = \frac{\sqrt{3}}{7} \quad \Rightarrow \quad v = \frac{7}{\sqrt{3}} \text{ mm}$$

$\therefore$ Rays will get focused at a distance $\left(3 + \frac{7}{\sqrt{3}}\right)$ mm from P.

134. Focal length of the plano convex lens is given by $\frac{1}{f} = \frac{\mu - 1}{R}$

where R is radius of curvature of its curved surface.

$$\therefore \quad \frac{1}{40} = \frac{\mu-1}{R} \quad \text{...(1)}$$

For the convex lens.

$$\frac{1}{v} - \frac{1}{-10} = \frac{1}{20} \quad \Rightarrow \quad V = -20 \text{ cm.}$$

Image formed by the lens (at a distance 20 cm from the lens) acts as an object for the curved surface of the plano convex lens.

Applying the formula for refraction at the curved surface, we get.

$$\frac{\mu}{v'} - \frac{1}{-(d+20)} = \frac{\mu - 1}{R}$$

If $v' = \infty$, the light rays will be incident normally on the silvered surface and after reflection they will retrace back their paths to O.

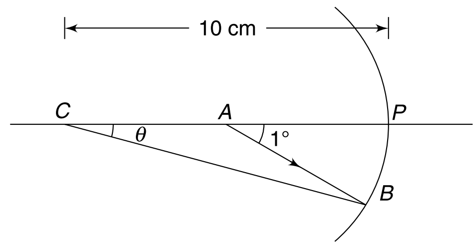
$$\therefore \quad \frac{\mu}{\infty} + \frac{1}{d+20} = \frac{1}{40}$$

$$\Rightarrow \quad d = 20 \text{ cm.}$$

135. Using Snell's law at first surface we get $\mu \sin 2° = 2\mu \sin r$.

For small angles $\sin\theta \simeq \theta$

$$\therefore \quad r \simeq 1°.$$



The incident ray on the second surface and the normal to the surface has been shown in the Figure.

C is centre of curvature of the second surface and CB is normal at B. AB is the incident ray.

$$CB = CP = 10 \text{ cm}$$

and
$$AP = 2 \text{ cm}$$

$$\therefore \quad \theta = \left(\frac{1}{5}\right)^{\circ}$$

$$\text{Angle of incidence} = \angle ABC = 1° - \left(\frac{1}{5}\right)^{\circ} = \left(\frac{4}{5}\right)^{\circ}$$

Using Snell's law

$$2\mu \sin\left(\frac{4°}{5}\right) = 3\mu \sin r'$$

$$\Rightarrow \quad r' = \frac{2}{3} \times \frac{4}{5} = \left(\frac{8}{15}\right)^{\circ}$$

136. The plane surface is AB.

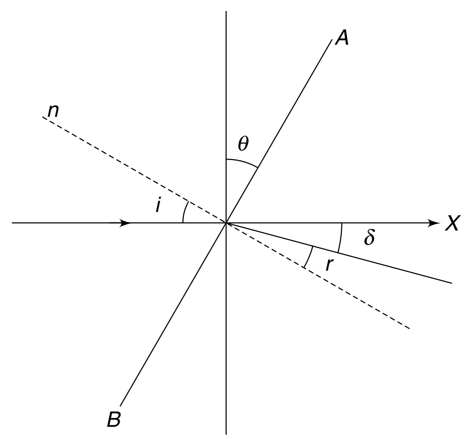
n is normal to the surface.

Angle of incidence $i = \theta$

Snell's law $\sin\theta = \mu \sin r$

$$\Rightarrow \quad \theta \simeq \mu r$$

$$\therefore \quad \text{Deviation} \quad \delta = \theta - r = \theta\left(1 - \frac{1}{\mu}\right)$$



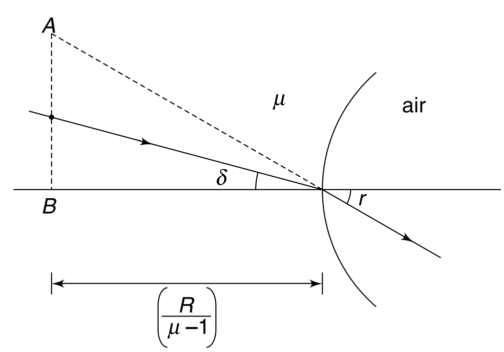
Now consider refraction at the curved surface

$$\frac{\mu_2}{v} - \frac{\mu_1}{\mu} = \frac{\mu_2 - \mu_1}{R}$$

$$\frac{1}{v} - \frac{\mu}{-\infty} = \frac{1-\mu}{R}$$

$$\therefore \quad v = \frac{R}{1-\mu} = -\left(\frac{R}{\mu - 1}\right)$$

$\therefore$ X co-ordinate of the image is $-\left(\dfrac{R}{\mu-1}\right)$



Snell's law gives
$$\mu \sin\delta = 1 \cdot \sin r$$

$$\mu \cdot \delta = r$$

$$\Rightarrow \quad r = (\mu - 1)\,\theta$$

Rays appear to be diverging from A.

$$BA = |v|\, r$$

$$= \left(\frac{R}{\mu-1}\right)(\mu - 1)\,\theta = R\theta.$$

$\therefore$ Co-ordinates of point A are $\left(\dfrac{-R}{\mu-1}, R\theta\right)$

137.(a) For L_1

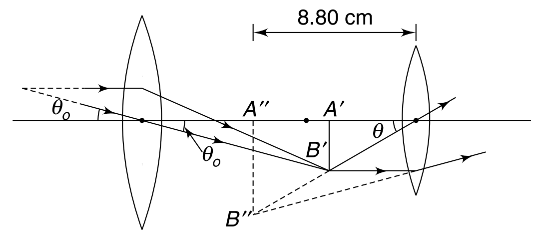
$$\frac{1}{f_1} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) = (1.5 - 1)\left(\frac{1}{50} - \frac{1}{-50}\right)$$

$\therefore \quad f_1 = 50$ cm

Similarly, for L_2; $f_2 = 4$ cm

$\therefore$ Distance between the lenses = 53 cm.

For L_1



$$u_0 = -100,00 \text{ cm}; \; v_0 = \text{image distance}$$

$$f_0 = 50 \text{ cm}$$

$$\therefore \quad \frac{1}{v_0} - \frac{1}{-100,00} = \frac{1}{50}$$

$$\Rightarrow \quad v_0 = 50.25 \text{ cm}$$

For L_2

$$u_e = -2.75 \text{ cm}$$

[$\because$ distance between lenses = 53 cm]

$$\therefore \quad \frac{1}{v_e} - \frac{1}{-2.75} = \frac{1}{4}$$

$$v_e = -8.80 \text{ cm}$$

Final image is located between the two lenses at a distance of 8.80 from L_2.

(b) magnification $\quad m = m_1 \times m_2$

$$= -\frac{50.25}{100,00} \times \frac{8.80}{2.75} = -0.016$$

The image is 1.6 cm high, inverted (–sign) and virtual.

$$\text{Angular magnification} = \frac{\theta}{\theta_0} \simeq \frac{\tan\theta}{\tan\theta_0} = \frac{A''B''/8.80}{100/100,00}$$

$$= \frac{1.6 \times 100}{8.8} = 18.2$$

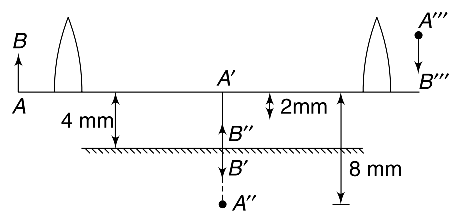
Linear magnification for a telescope is usually very small since objects are at large distances (near infinity). More useful parameter is angular magnification or magnifying power.

138. The first half–lens forms the image $A'B'(= 6 \text{ mm})$.

$$\therefore \quad \frac{1}{v} - \frac{1}{-20} = \frac{1}{15} \Rightarrow v = +60 \text{ cm.}$$

$$\therefore \quad m_1 = -3.$$

The plane mirror forms the image $A''B''$ with A'' located 8 mm below the principal axis. The second half lens forms the image $A'''B'''$. Now $u = -60$ cm



$$\Rightarrow \quad v = +20 \text{ cm}$$

$$\therefore \quad m_2 = -\frac{1}{3}$$

$\therefore$ A is located at $\frac{8}{3}$ mm above the principal axis.

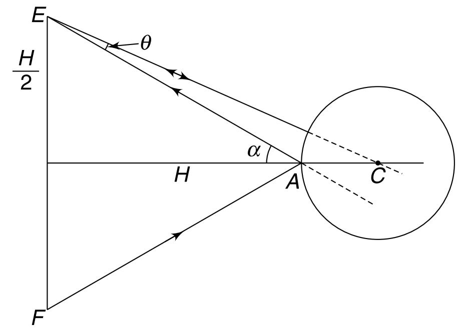
139. A light ray FA will get reflected and reach the eye of the man travelling along AE. It means image of F will be seen on line EA extended. Similarly a ray from E heading towards the centre of the ball will get reflected back along the same line (CE) and therefore image of E will lie somewhere on the line EC.

$\therefore$ Angle subtended by the image at the eye is $\angle AEC = \theta$ (say)

In ΔAEC: $\quad \angle C \simeq \alpha \quad$ [$\because$ $AC = r$ is very small]

$$AE = \sqrt{H^2 + \frac{H^2}{4}} = \frac{\sqrt{5}}{2} H$$

Applying sine rule in ΔAEC



$$\frac{\sin\theta}{AC} = \frac{\sin\alpha}{AE}$$

$\Rightarrow$
$$\sin\theta = \frac{r}{AE}\cdot\frac{H/2}{AE} = \frac{rH}{2(AE)^2}$$

$\Rightarrow$
$$\sin\theta = \frac{rH}{2\,\frac{5}{4}\,H^2} = \frac{2r}{5H}$$

Since θ is small $\sin\theta \simeq \theta$

$\therefore$
$$\theta \simeq \frac{2r}{5H}$$

140. A convex mirror will always form a virtual image on the other side of the mirror.

Let distance of observer from mirror be x and the distance of image from the mirror be y.

[Both x and y are positive numbers]

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$\therefore$
$$\frac{1}{y} - \frac{1}{x} = \frac{2}{R}$$

$$\frac{x}{y} - 1 = \frac{2x}{R} \Rightarrow \frac{x}{y} = \frac{2x+R}{R}$$

magnification
$$|m| = \frac{y}{x} = \frac{R}{2x+R} \qquad \text{...(1)}$$

But
$$y + x \geq d$$

$\Rightarrow$
$$\frac{Rx}{2x+R} + x \geq d$$

$$2x^2 + 2Rx \geq 2dx + Rd$$

$$2x^2 + 2(R-d)x - Rd \geq 0$$

Roots of $2x^2 + 2(R-d)x - Rd = 0$ are

$$x = \frac{-2(R-d) \pm \sqrt{4(R-d)^2 + 8Rd}}{4}$$

$\therefore$
$$x_1 = -\left[\frac{(R-d) + \sqrt{(R-d)^2 + 2Rd}}{2}\right]$$

and
$$x_2 = \frac{\sqrt{(R-d)^2 + 2Rd} - (R-d)}{2}$$

Our x is positive. Hence for (2) to hold $x \geq x_2$

$$m = \frac{R}{2x+R}$$

M is maximum when x is minimum ($\Rightarrow \ x = x_2$)

$\therefore$
$$m_{max} = \frac{R}{\sqrt{(R-d)^2 + 2Rd} - (R-d) + R}$$

$$= \frac{R}{\sqrt{R^2 + d^2} + d}$$

141. RS is parallel to NN'

PQ is along incident ray

$$\therefore \quad \angle QRS = i$$

In ΔPSR
$$\frac{PR}{\sin(\angle PSR)} = \frac{PS}{\sin[\pi - i]}$$

$$\Rightarrow \quad \frac{k\mu}{\sin(\angle PSR)} = \frac{k\mu'}{\sin i}$$

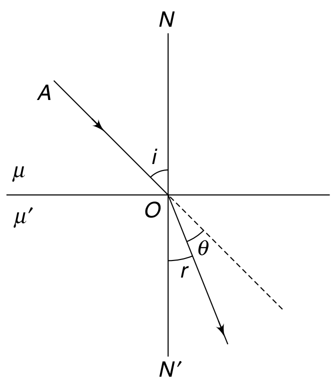
$$\Rightarrow \quad \mu' \sin(\angle PSR) = \mu \sin i$$

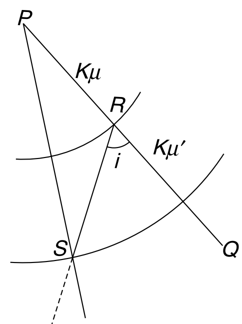
From Snell's law
$$\mu' \sin r = \mu \sin i$$

$$\therefore \quad \angle PSR = r$$

Angle between RS (which is parallel to NN') and PS is r.

$\therefore$ PS is in the direction of refracted ray.





142. $BP \to$ incident ray, $PE \to$ refracted ray.

$$\mu \sin i = 1 \cdot \sin r$$

$$\therefore \quad \mu = \frac{\sin r}{\sin i} = \frac{AQ/AP}{BQ/BP} = \frac{AQ}{AP} \times \frac{BP}{BQ}$$

$$\Rightarrow \quad \mu = \frac{AS + SQ}{\sqrt{(AS + SQ)^2 + (PQ)^2}} \times \frac{\sqrt{(PQ)^2 + (SQ - SB)^2}}{SQ - SB}$$

$$= \frac{a + R}{\sqrt{(a + R)^2 + (R - h)^2}} \times \frac{\sqrt{(R - h)^2 + (R - a)^2}}{R - a}$$

$$\mu = \left(\frac{R + a}{R - a}\right) \frac{\sqrt{(R - h)^2 + (R - a)^2}}{\sqrt{(R - h)^2 + (R + a)^2}}$$

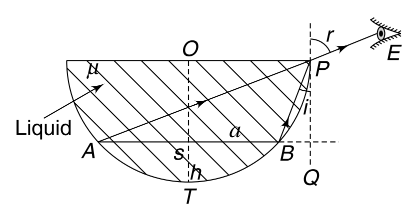
$$= \left(\frac{R + a}{R - a}\right) \sqrt{\frac{2R^2 + h^2 + a^2 - 2Rh - 2Ra}{2R^2 + h^2 + a^2 - 2Rh + 2Ra}}$$

But
$$(R - h)^2 + a^2 = R^2$$

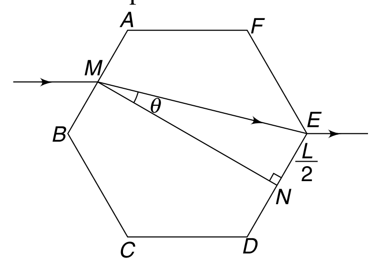
$$\Rightarrow \quad a^2 + h^2 = 2Rh$$

$$\therefore \quad \mu = \left(\frac{R + a}{R - a}\right) \sqrt{\frac{2R(R - a)}{2R(R + a)}}$$

$$\Rightarrow \quad \mu = \sqrt{\frac{R + a}{R - a}}$$



143. For light beam to emerge parallel to the incident direction it must strike the face DE which is parallel to face AB. It means incident ray at M must go to E (or below it).



Let side length of hexagon be b.

$$MN = \sqrt{3}b$$

$$ME =$$

$$\therefore \sqrt{\left(\frac{b}{2}\right)^2 + (\sqrt{3}b)^2} = \frac{\sqrt{13}}{2}\, b$$

Using Snell's law

$$1 \cdot \sin 30° = \mu \cdot \sin\theta$$

$$\Rightarrow \quad \frac{1}{2} = \mu\, \frac{1}{\sqrt{13}}$$

$$\Rightarrow \quad \mu = \frac{\sqrt{13}}{2}$$

144. The deviation suffered by the ray, incident at angle i is

$$\phi = (i - r) + (180° - 2r) + (i - r)$$

$$= 180° - 4r + 2i$$

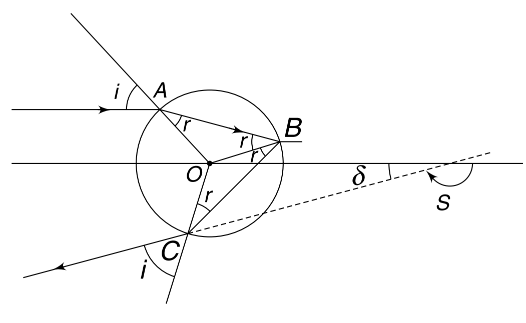
$$\therefore \quad \delta = 180° - \phi = 4r - 2i$$

δ is maximum when $\dfrac{d\delta}{di} = 0$

$$\Rightarrow \quad 4\,\frac{dr}{di} - 2 = 0$$

$$\Rightarrow \quad \frac{dr}{di} = \frac{1}{2} \qquad \text{...(1)}$$

Snell's law



$$1 \cdot \sin i = \mu \sin r$$

$$\sin i = \frac{4}{3}\sin r$$

$$\sin r = \frac{3}{4}\sin i \qquad \text{...(2)}$$

$$\Rightarrow \quad \cos r\,\frac{dr}{di} = \frac{3}{4}\cos i$$

$$\therefore \text{ from (1)} \quad \frac{3\cos i}{4\cos r} = \frac{1}{2}$$

$$3\cos i = 2\cos r$$

$$3\cos i = 2\sqrt{1 - \sin^2 r}$$

$$3\cos i = 2\sqrt{1 - \frac{9}{16}\sin^2 i} \qquad \text{[using (2)]}$$

$$(6\cos i)^2 = 16 - 9\sin^2 i$$

$$36(1 - \sin^2 i) = 16 - 9\sin^2 i$$

$$27\sin^2 i = 20$$

$$\Rightarrow \quad \sin i = \sqrt{\frac{20}{27}}; \quad i = 60°$$

$$\text{from (2)} \quad \sin r = \frac{3}{4}\sqrt{\frac{20}{27}}$$

$$= \sqrt{\frac{5}{12}} \quad \Rightarrow \quad r = 40°$$

$$\therefore \qquad \delta_{max} = 4r - 2i = 4 \times 40 - 2 \times 60$$

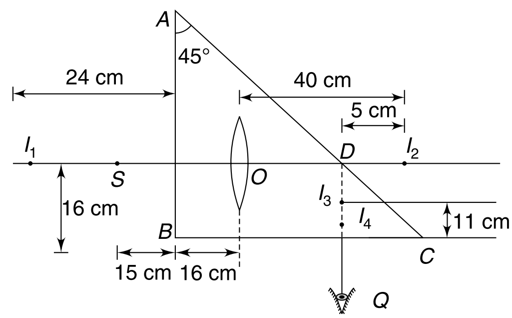
$$= 40°$$

145. Refraction at AB

Image is formed a I_1 at a distance $15 \times 1.6 = 24$ cm from AB

Refraction at lens

Focal length of the lens.



$$\frac{1}{f} = \left(\frac{2.4}{1.6} - 1\right)\left(\frac{2}{20}\right)$$

$$f = +20 \text{ cm}$$

Object distance = 24 + 16 = 40 cm

Lens formula $\qquad \frac{1}{v} - \frac{1}{-40} = \frac{1}{+20}$

$$v = +40 \text{ cm}$$

Image is at I_2.

$$OD = 35 \text{ cm (from geometry of the Figure)}$$

$$\therefore \qquad DI_2 = 5 \text{ cm.}$$

Light ray after passing through the lens will suffer total internal reflection at face AC. This is because angle of incidence is close to 45°, as we are considering paraxial rays.

$$45° > \sin^{-1}\left(\frac{1}{1.6}\right)$$

Surface AC may be treated as a mirror for which object is at I_2.

Image is formed at I_3

$$DI_3 = 5 \text{ cm}$$

Observer at P will not be able to see image as light rays suffer TIR at surface AC. Observer at Q see the image at I_4. Image I_4 is above surface BC by a distance $= \frac{11}{1.6} = 6.88$ cm.

146. Consider the incident ray FA on the lens. F is focus point and angle θ is small. The emergent ray AB is parallel to x direction. Now rotate the entire figure by an angle θ about point O. The incident ray FA is now parallel to x direction but not parallel to the principal axis.

$$FA = f\sec\theta$$

Since $FA\ F'O$ is a parallelogram hence

$$OF' = FC = f\sec\theta = f'$$

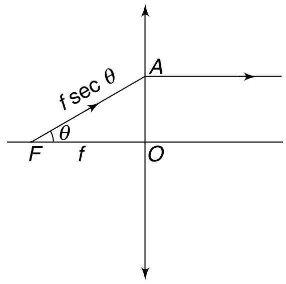
$\therefore$ shift in position of image is

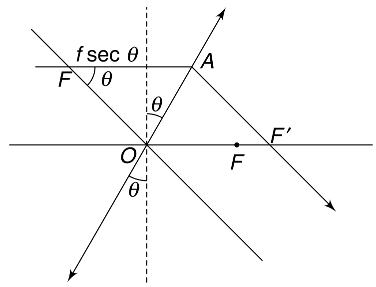
$$\Delta s = f' - f = f(\sec\theta - 1)$$

$$= f\left[1 + \frac{\theta^2}{2} - 1\right] = \frac{f\theta^2}{2}$$

If the whole Figure in (a) is rotated by θ in opposite direction, then also the image will be formed at F'.

Hence the image oscillates between F and F' and the required answer is $\frac{f\theta^2}{2}$





147.(a) When distance of object from lens is x, let the distance of image be y.

Lens formula, $$\frac{1}{-y} - \frac{1}{-x} = \frac{1}{-f}$$

$$\frac{1}{y} = \left(\frac{1}{x} + \frac{1}{f}\right); \quad \Rightarrow \quad y = \frac{xf}{x+f}$$

Magnification $$m = \frac{y}{x} = \frac{f}{x+f}$$

Height of image is

$$h_i = mh_0 \qquad [h_0 = \text{height of object}]$$

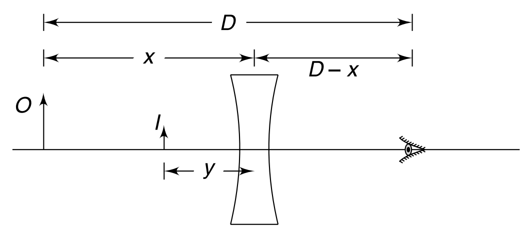
$$h_i = \frac{fh_0}{x+f}$$

Angle subtended by the image at the eye

$$\theta = \frac{h_i}{D-x+y} = \frac{fh_0}{(x+f)(D-x+y)}$$

Let $$(x+f)\,(D-x+y) = Z$$

$$Z = \left(D - x + \frac{xf}{x+f}\right)(x+f)$$

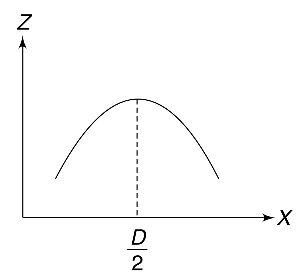


$$Z = -x^2 + Dx + Df$$

'Z' is maximum when

$$\frac{dZ}{dx} = 0$$

$$x = \frac{D}{2}$$



When lens is moved away from the eye, x decreases. This means that Z first increases and then decreases. When Z increases 'θ' decreases and when Z decreases 'θ' increases.

(b) The apparent size of image depends on θ.

$\therefore$ Size of image will first decrease, become minimum and then will become larger and larger.

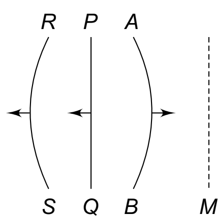
Apparent size is smallest when $\theta = \frac{D}{2}$.

Chapter 14

Wave Optics

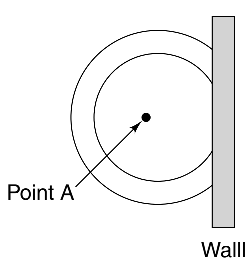
Level 1

Q. 1: In the figure shown M represents a mirror. AB is an incident wavefront. Which type of mirror is M if the reflected wave front is



(a) PQ (b) RS

Q. 2: On the surface of a calm lake a source at A is causing disturbance. The circular ripples formed get reflected at a wall in the lake. The reflected ripples can be thought to be generated from a virtual source. Indicate the position of virtual source in the diagram.



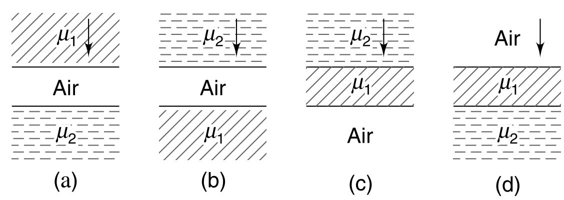
Q. 3: A wavefront expressed by $x + y - z = 4$ is incident on a plane mirror which is lying parallel to xy plane. Write a unit vector in the direction of reflected ray.

Q. 4: A parallel beam of light is travelling along a direction making an angle of 30° with the positive X direction and 60° with positive Y direction. Wavelength is λ. Find the phase difference between points having co-ordinates $(1, \sqrt{3}, 2)$ and $(0, 0, 0)$.

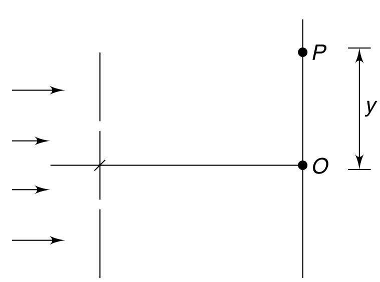
Q. 5: A point source of light is being moved closer to a thin concave lens from a large distance on its principal axis. Is the radius of curvature of the refracted wave front, close to the lens, increasing or decreasing?

Q. 6: Three coherent sources S_1, S_2 and S_3 can throw light on a screen. With S_1 switched on intensity at a point P on the screen was observed to be I. With only S_2 on, intensity at P was $2I$ and when all three are switched on the intensity at P becomes zero. Intensity at P is I when S_1 and S_2 are kept on. Find the phase difference between the waves reaching at P from sources S_1 and S_3.

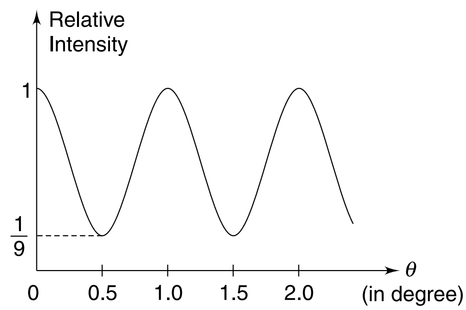
Q. 7: Three media are arranged in various ways as shown in figures a, b, c and d. Light of wavelength λ is incident perpendicularly on the boundary of the middle layer and interference between waves reflected at the boundaries of the middle layer is studied. In which of the four cases the reflected light is eliminated by destructive interference when the thickness of middle layer approaches zero. $\mu_1 = 1.5$ and $\mu_2 = 1.8$



Q. 8: In young's double slit experiment, when the slit plane is illuminated with light of wavelength λ_1, it was observed that point P is closest point from central maximum O, where intensity was 75% the intensity at O. When the light of wavelength λ_2 is used, point P happens to be the nearest point from O where intensity is 50% of that at O. Find the ratio $\frac{\lambda_1}{\lambda_2}$.



Q. 9: In young's double slit experiment relative intensity at a point on the screen may be defined as ratio of intensity at that point to the maximum intensity on the screen. Light of wavelength 7500 Å passing through a double slit, produces interference pattern of relative intensity variation as shown in Fig. θ on horizontal axis represents the angular position of a point on the screen.

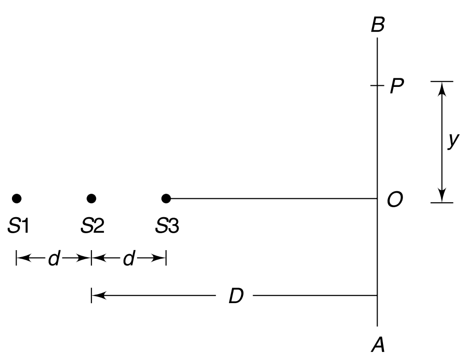


(a) Find separation d between the slits.

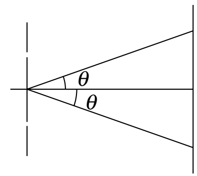
(b) Find the ratio of amplitudes of the two waves producing interference pattern on the screen.

Q. 10(a): A monochromatic point source (S_1) is at a distance d from a screen. Another identical source (S_2) is at a large distance from the screen. The two sources are on a line which is perpendicular to the screen. Sources are coherent. What is the shape of interference fringes on the screen?

(b): Three identical coherent sources are placed on a straight line with two neighbouring ones separated by a distance $d = 0.06$ mm. The sources produce monochromatic light of wavelength $\lambda = 550.4$ nm. A line AB is located at a distance $D = 2.50$ m from the sources and is perpendicular to line joining the three sources. Intensity of light at P with any one of the sources switched on is I_o. Find the intensity when all three sources are switched on. Distance of point P from O is $y = 1.72$ cm (see figure)

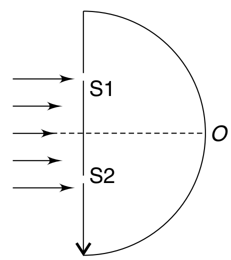


Q. 11: In Young's double-slit experiment, the separation between two slits is $d = 0.32$ mm and the wavelength of light used is $\lambda = 5000$ Å. Find the number of maxima in the angular range $-\sin^{-1}(0.6) \le \theta \le \sin^{-1}(0.6)$.



Q. 12: In young's double slit experiment (with identical slits) the intensity of a maxima is I. P is a point on the screen where 10^{th} maxima is formed with light of wavelength $\lambda = 6000$ Å. Find the intensity at point P if the entire experimental set up is submerged in water of refractive index $\mu = \frac{4}{3}$. Assume that intensity due to individual slits remains unchanged after the system is dipped in water.

Q. 13: Coherent light of wavelength $\lambda = 500$ nm is sent through two narrow parallel slits in a large vertical wall. The two slits are 5 μm apart. In front of the wall there is a semi cylindrical screen with its horizontal axis at the line running on the wall parallel to the slits and midway between them. Radius of the cylindrical screen is $R = 2.0$ m. Find the vertical height of the second order interference maxima from the centre (O) of the screen.



Q. 14: In a Young double slit experiment, the two slits are named as A and B. Two transparent films of thickness t_1 and t_2 having refractive indices μ_1 and μ_2 placed in front of the slits A and B respectively. It is given that $\mu_1 t_1 = \mu_2 t_2$ and $t_1 < t_2$

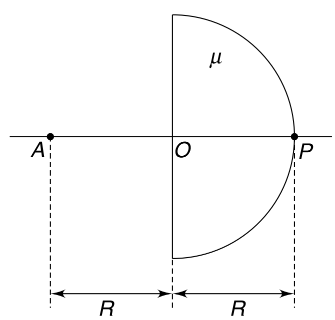
In which direction will the central maximum shift after the two films are placed?

Level 2

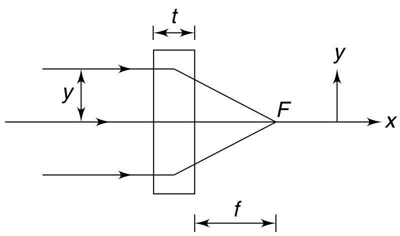
Q. 15: A point source (A) is kept on the axis of a hemispherical paperweight made of glass of refractive index $\mu = \frac{3}{2}$. The distance of the point source from the centre (O) of the sphere is R where R is radius of the hemisphere. Use paraxial approximations for answering following questions

(a) Find the change in radius of curvature of the wavefronts just after they enter the glass at O.

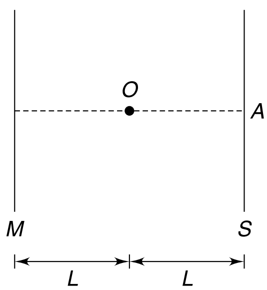
(b) Find the radius of curvature of the wavefronts at point P just outside the glass.



Q. 16: A parallel beam of light travelling in x direction is incident on a glass slab of thickness t. The refractive index of the slab changes with y as $\mu = \mu_0\left(1 - \frac{y^2}{y_0^2}\right)$ where μ_0 is the refractive index along x axis and y_0 is a constant. The light beam gets focused at a point F on the x axis. By using the concept of optical path length calculate the focal length f. Assume $f >> t$ and consider y to be small.



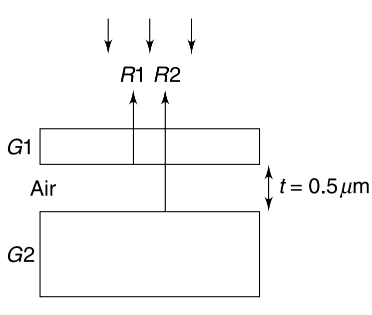
Q. 17: In the figure shown, O is a point source of light and S is a screen placed at a distance L from the source. Intensity of light at point A on the screen due to the source is $81I$ where I is some unit. Now a large mirror (M) is placed behind the source at a distance L from it. The mirror reflects 100% of the light energy incident on it. Calculate the intensity at point A.



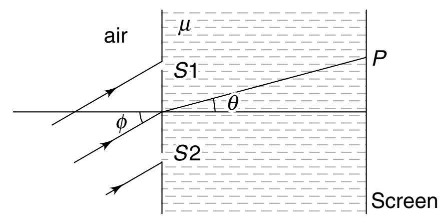
Q. 18: In a routine Young's double slit experiment the parallel beam of light incident on the slit plane is a mixture of wavelength distributed between $\lambda + \Delta\lambda$ and $\lambda - \Delta\lambda$. Because of this the interference fringes are poorly defined compared to the ideal case of monochromatic light. It is understood that the central maxima and the first order maxima will be well resolved if there is no overlapping in the first order minima and the first order maxima. Write the values of $\Delta\lambda$ so that the first order maxima is well resolved from the central fringe.

Q. 19: A thin glass slab $G1$ is held over a large glass slab $G2$, creating an air gap of uniform thickness $t = 0.5\ \mu$m between them. Electromagnetic wave having wavelengths ranging from 0.4 μm to 1.15 μm is incident normally on the slab $G1$. When interference between waves reflected from boundaries of air gap (the two reflected waves are shown in fig as $R1$ and $R2$) was studied, it was found that only two wavelengths interfered constructively. One of these two wavelengths is $\lambda_1 = 0.04\ \mu$m.

Find the other wavelength (λ_2) that interferes constructively.



Q. 20: Light is incident at an angle ϕ with the normal to a vertical plane containing two narrow slits ($S1$ and $S2$) at separation d. The medium to the left of slit plane is air and wavelength of the incident light is λ. The medium to the right of the slit plane has refractive index μ. Find all values of angular position (θ) of a point P where we will observe constructive interference. Wavelength of incident light is λ.

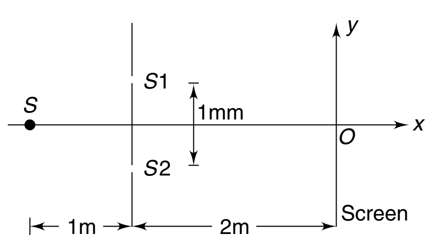


Q. 21: In a Young's double slit experiment set up source S of wavelength $\lambda = 500$ nm illuminates two symmetrically located slits $S1$ and $S2$. The source S oscillates about its shown position parallel to the screen according to the equation $y = (0.5\text{ mm})\sin(\pi t)$.

Where t is time in second. Distances are as marked in the figure.

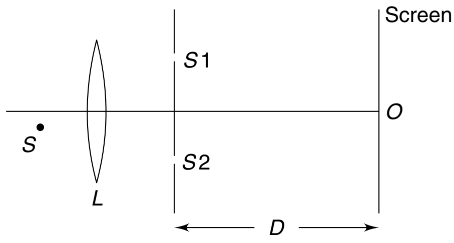
(a) Write the y co-ordinate (y') of central maximum as a function of time

(b) Find least value of time (t) at which the intensity becomes maximum at a point on the screen that is exactly in front of slit $S1$.



Q. 22: In a young's double slit experiment the distance between slits $S1$ and $S2$ is d and distance of slit plane from the screen is D (>> d). The point source of light (S) is placed a distance $\frac{d}{2}$ below the principal axis in the focal plane of the convex lens (L). The slits $S1$ and $S2$ are located symmetrically with respect to the principal axis of the lens. Focal length of the lens is f (>> d).

Find the distance of the central maxima of the fringe pattern from the centre (O) of the screen.



Q. 23: A Soap bubble has a thickness of 90 nm and its refractive index is μ = 1.4. What colour does the bubble appear to be at a point on its surface closest to an observer when it is illuminated by white light?

Q. 24: A parallel beam of white light falls from air on a thin film whose refractive index is $\sqrt{3}$. The medium on both sides of the film is air. The angle of incdidence is 60°. Find the minimum film thickness if reflected light is most intense for λ = 600 nm.

Q. 25: A thin film having refractive index μ = 1.5 has air on both sides. It is illuminated by white light falling normally on it. Analysis of the reflected light shows that the wavelengths 450 nm and 540 nm are the only missing wavelengths in the visible portion of the spectrum. Assume that visible range is 400 nm to 780 nm.

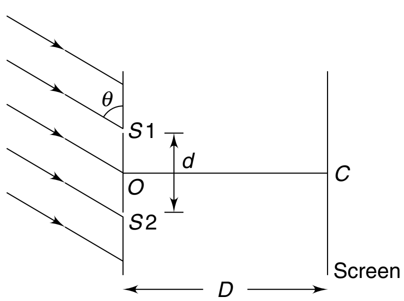
(a) Find thickness of the film.

(b) Which wavelengths are brightest in the interference pattern of the reflected light?

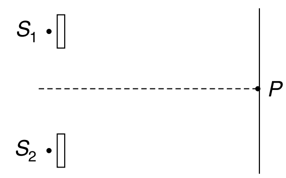
Q. 26: In a double slit experiment a parallel beam of light strikes the slit plane at an angle θ as shown in the figure. The two slits are covered with transparent plastic sheets of equal thickness t but of different refractive indices 1.2 and 1.5. The central maxima is formed at the centre of the screen at C.

(a) Which sheet was used to cover the slit $S1$?

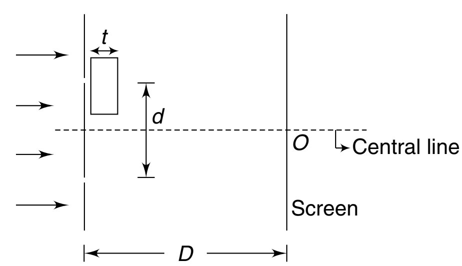
(b) Find θ.



Q. 27: In Young's double slit experiment, the upper slit is covered by a thin glass plate of refractive index 1.4 while the lower slit is covered by another glass plate having same thickness as the first one but having refractive index 1.7. Interference pattern is observed using light of wavelength 5400 Å. It is found that the point P on the screen where the central maximum fell before the glass plates were inserted, now has $\frac{3}{4}$ the original intensity. It is also observed that what used to be 5th maximum earlier, lies below the point P while the 6th minimum lies above P. Calculate the thickness of glass plates. Absorption of light by glass plate may be neglected.



Q. 28: In Young's double slit experiment a transparent sheet of thickness t and refractive index μ is placed in front of one of the slits and the central fringe moves away from the central line. It was found that when temperature was raised by $\Delta\theta$ the central fringe was back on the central line (at C). It is known that temperature coefficient of linear expansion of the material of the transparent sheet is α. A young scientist modeled that the refractive index of the material changes with temperature as $\Delta\mu = -\gamma\Delta\theta$. Find $\Delta\theta$ in terms of other given quantities. D and d are given and have usual meaning.

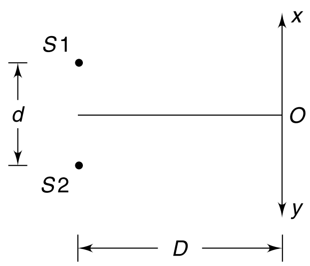


Q. 29: In Young's double slit experiment a monochromatic light of wavelength λ from a distant point source is incident upon the two identical slits.The interference pattern is viewed on a distant screen. Intensity at a point P is equal to the intensity due to individual slits (equal to I_0). A thin piece of glass of thickness t and refractive index μ is placed in front of the slit which is at larger distance from point P; perpendicular to the light path. Assume no absorption of light energy by the glass.

(a) Write intensity at point P as a function of t.

(b) Write all values of t for which the intensity at P is minimum.

Q. 30: In the shown fig. $S1$ and $S2$ are two identical coherent sources of sound, separated by a distance d. A receiver moves along the line XY (which is parallel to line $S1\,S2$) to detect the intensity of sound at various points on the line. Distance of line XY from line $S1S2$ is $D(>> d)$. Point O is the foot of perpendicular bisector of the line $S1S2$ on XY. Distance of first intensity maxima (on line XY) measured from O is y. Find percentage change in value of y if the temperature of air increases by 1%.

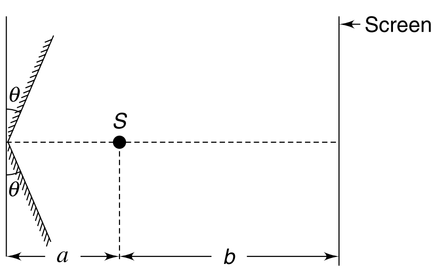


Q. 31: Two plane mirrors, a source S of light, emitting wavelengths of $\lambda_1 = 4000$ Å and $\lambda_2 = 5600$ Å and a screen are arranged as shown in figure. The angle θ shown is 0.05 radian and distance a and b are 1 cm and 38 cm respectively.

(a) Find the fringe width of interference pattern formed on screen by the blue light.

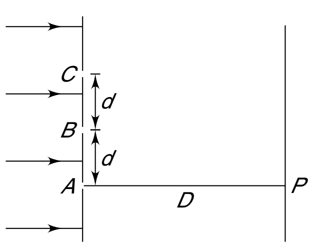
(b) Calculate the distance of first black line from central bright fringe.

(c) Find the distance between two black lines which are nearest to the central bright fringe.



Q. 32: Three narrow slits A, B and C are illuminated by a parallel beam of light of wavelength λ, P is a point on the screen exactly in front of point A. Slit plane is at a distance D from the screen ($D >> \lambda$). It is know that

$$BP - AP = \frac{\lambda}{3}$$

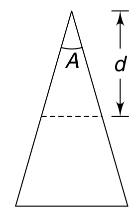


(a) Find d in terms of D and λ

(b) Write the phase difference between waves reaching at P from C and A.

(c) If intensity of P due to any of the three slits individually is I_0, find the resultant intensity at P.

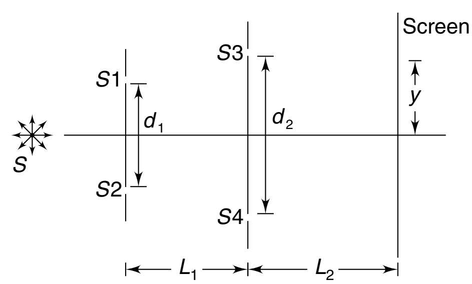
Q. 33: A very thin prism has an apex angle A and its material has refractive index $\mu = 1.48$. Light is made to fall on one of the refracting faces at near normal incidence. Interference results from light reflected from the outer surface and that emerging after reflection at the inner surface. When violet light of wavelength $\lambda = 400$ nm is used, the first constructive interference band is observed at a distance $d = 3.0$ cm from the apex of the prism.



(a) Find the apex angle A.

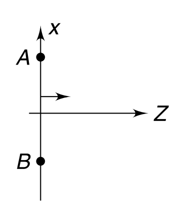
(b) If red light ($\lambda = 800$ nm) is used, at what distance from the apex will we observe the first constructive interference band.

Q. 34: In the arrangement shown, S is a point source of monochromatic light. $S1$ and $S2$ are two slits located symmetrically with respect to the source with separation between them d_1. Parallel to this slit plane there are two more slits ($S3$ and $S4$) at separation d_2. These slits are also symmetrically located with respect to S. A screen is at a distance L_2 from this slit plane. How does the intensity on the screen change with y and d_1?

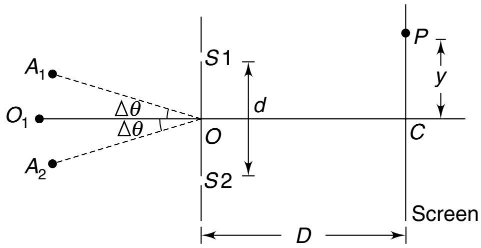


Level 3

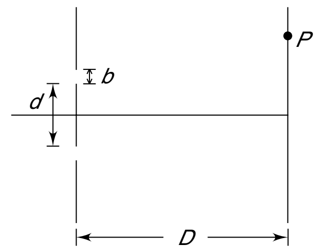
Q. 35: The refractive index of a medium changes as $\mu = \mu_0\left[1 - \frac{x^2 + y^2}{d^2}\right]^{1/2}$ where μ_0 is the refractive index on the z axis. A plane wavefront (AB) is incident along z axis as shown in the figure. Draw the wavefront at a later time Δt. Is the wavefront getting focused?



Q. 36: In the arrangement shown in the figure, $S1$ and $S2$ are two parallel slits at a separation d. There is a screen at a distance D (>> d) from the slits. Two point sources $A1$ and $A2$ have been placed symmetrically with respect to the slits at a large distance with $\langle A_1 OO_1 = \langle A_2 OO_1 = \Delta\theta$. The sources are monochromatic giving out light of wavelength λ but they are incoherent. Write the intensity of light at a point P on the screen as a function of y. Take I_0 to be maximum intensity on the screen when one of the two sources ($A1$ and $A2$) is switched on. Assume y to be small.



Q. 37 The figure represents two identical slits in a Young's double slit experiment. The width of each slit is b and distance between the centres of the two slits is d. Consider a point P on the screen that is close to centre of the screen. Δx_t represents the optical path difference to point P from the top edges of the two slits and Δx_b represents the optical path difference to P from the bottom edges of the two slits. Find $\Delta x_b - \Delta x_t$.

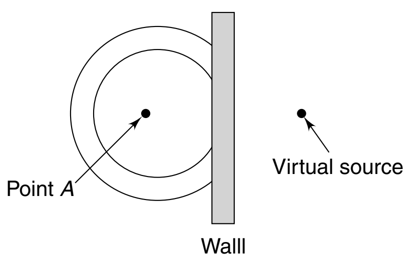


It is given that $D >> d$.

ANSWERS

1. (a) Concave (b) Plane

2.



3. $\frac{1}{\sqrt{3}}(\hat{i} + \hat{j} + \hat{k})$

4. $\frac{2\sqrt{3}\pi}{\lambda}$

5. Decreasing.

6. $\pi/2$

7. a, b.

8. $\frac{3}{2}$.

9. (a) 0.04 mm (b) 2 : 1

10. (a) Circular
(b) I_0

11. 769.

12. $\frac{I}{4}$.

13. 0.4 m

14. shift towards A

15. (a) $\frac{R}{2}$ (b) $10R$.

16. $\frac{y_0^2}{2\mu_o t}$

17. 90 I

18. $\Delta\lambda < \frac{\lambda}{3}$.

19. 0.67 μm

20. $\theta = \sin^{-1}\left[\frac{1}{\mu d}(n\lambda - d\sin\phi)\right]$; $[n = 0, \pm 1, \pm 2 ...]$

21. (a) $y' = -\sin \pi t$ (b) $\frac{1}{6}$ s

22. $\frac{Dd}{2f}$

23. Green

24. 100 nm.

25. (a) 900 nm (b) 415 nm, 491 nm, 600 nm, 771 nm

26. (a) Sheet of $\mu = 1.5$ (b) $\cos^{-1}\left(\frac{3t}{10d}\right)$

27. 9.3 μm

28. $\Delta\theta = \left[\frac{\mu - 1}{(\mu - 1)\alpha - \gamma}\right]$

29. (a) $4I_0\cos^2\left(\frac{\pi}{3} + \frac{\pi}{\lambda}(\mu - 1)t\right)$

(b) $t = \frac{(2n-1)\lambda}{2(\mu - 1)} - \frac{\lambda}{3(\mu - 1)}$ where $n = 1, 2, 3...$

30. 0.5%

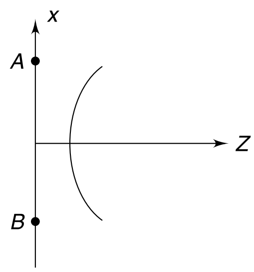
31. (a) 80 μm (b) 280 μm (c) 560 μm

32. (a) $\sqrt{\dfrac{2D\lambda}{3}}$ (b) $\dfrac{8\pi}{3}$ (c) $3I_0$

33. (a) 1.3×10^{-4} degree (b) 6 cm.

34. Intensity $\propto \cos^2\left(\dfrac{\pi d_1 d_2}{2\lambda L_1}\right) \cos^2\left(\dfrac{\pi d_2 y}{\lambda L_2}\right)$

35.



Yes the wavefront is getting focussed.

36. $I_0 \cos^2\left[\dfrac{2\pi}{\lambda} d\left(\dfrac{yd}{D} + \sin\Delta\theta\right)\right]$

$$+ I_0 \cos^2\left[\frac{2\pi}{\lambda} d\left(\frac{yd}{D} - \sin\Delta\theta\right)\right]$$

37. $\dfrac{db}{D}$

SOLUTIONS

3. Light ray is normal to the wavefront.

Incident ray is parallel to $\vec{a} = \hat{i} + \hat{j} - \hat{k}$.

The normal to the surface is parallel to z axis. The component of $\vec{a}$ parallel to xy plane does not change during reflection and the z component gets reversed.

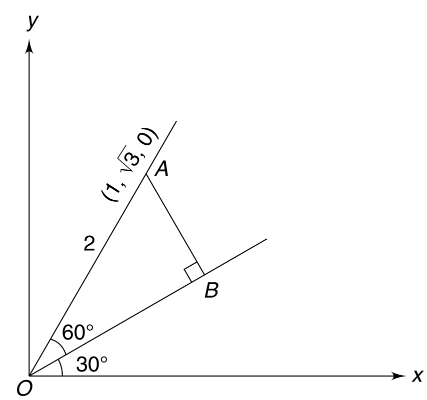
Therefore a vector in the direction of reflected ray is

$$\vec{b} = \hat{i} + \hat{j} + \hat{k}$$

unit vector along $\vec{b}$ is

$$\hat{b} = \frac{1}{\sqrt{3}}(\hat{i} + \hat{j} + \hat{k})$$

One more answer is possible if the light is assumed to be travelling in opposite direction.



4. Wave front is a plane perpendicular to line OB. Hence phase at point $(1, \sqrt{3}, 2)$ is same as that at point A $(1, \sqrt{3}, 0)$.

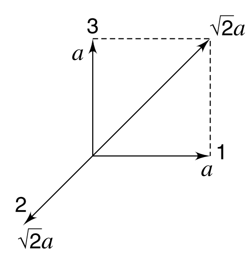
$$OB = OA \cos 30° = \sqrt{3}$$

$$\therefore \quad \Delta\phi = \frac{2\pi}{\lambda}\sqrt{3}.$$

5. When object is at large distance its image is at focus. The diverging wave fronts have their centre of curvature at the focus of the lens. As the object gets closer the image also gets closer to the lens. It means that the radius of diverging wavefronts will be decreasing.

6. Resultant intensity due to $S1$ and $S2$ is I and adding third to this makes the resultant zero. This implies that intensity due to $S3$ alone is I.

Amplitudes of waves reaching P from the three sources $S1$, $S2$ and $S3$ can be written as a, $\sqrt{2}a$ and a respectively. These three waves can produce zero resultant if phase difference between the first and third wave is $\pi/2$. The figure explains the situation.



7. **Hint:** Light suffers a phase change of π when it gets reflected while travelling from a rarer to a denser medium.

8. $\left(\dfrac{I_P}{I_{\max}}\right)_{\lambda_1} = \cos^2\dfrac{\phi}{2} = 0.75$

[ϕ = phase difference between two waves arriving at P]

$$\therefore \qquad \cos\left(\frac{\phi}{2}\right) = \frac{\sqrt{3}}{2}$$

$$\phi = \pi/3$$

$$\Rightarrow \qquad \frac{2\pi}{\lambda_1}(\Delta x)_1 = \pi/3$$

$$\Rightarrow \qquad (\Delta x)_1 = \lambda_1/6 \qquad ...(1)$$

Similarly, $\left(\dfrac{I_P}{I_{max}}\right)_{\lambda 2} = \cos^2\dfrac{\phi'}{2} = 0.5$

$$\therefore \qquad \cos\frac{\phi'}{2} = \frac{1}{\sqrt{2}}$$

$$\phi' = \pi/2$$

$$\therefore \qquad (\Delta x)_2 = \frac{\lambda_2}{4}$$

Because $(\Delta x)_1 = (\Delta x)_2$ ...(2)

$$\frac{\lambda_1}{6} = \frac{\lambda_2}{4} \Rightarrow \frac{\lambda_1}{\lambda_2} = \frac{3}{2}$$

9. (a) At $\theta = 0.5°$, we have first minima

$$\therefore \quad d\sin\theta = \frac{\lambda}{2}$$

$$d\theta \simeq \frac{\lambda}{2}$$

$$d = \frac{7500 \times 10^{-7}\text{mm}}{2 \times \left(0.5 \times \dfrac{3.14}{180}\text{rad}\right)} = 0.04 \text{ mm}$$

(b) $\dfrac{I_{max}}{I_{min}} = \left(\dfrac{a_1 + a_2}{a_1 - a_2}\right)^2$

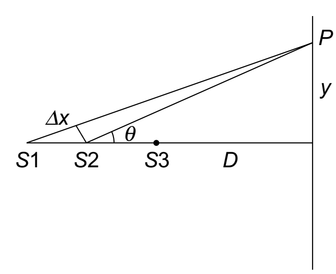
$$9 = \left(\frac{a_1 + a_2}{a_1 - a_2}\right)^2$$

$$\Rightarrow \frac{a_1 + a_2}{a_1 - a_2} = 3$$

$$\Rightarrow \frac{a_1}{a_2} = \frac{2}{1}$$

10. (a) **Hint:** The wave from S_2 can be treated as a plane wave. All points on the screen illuminated due to S_2 will have same phase. Due to S_1 alone the locus of points of constant phase are circles.

(b) Angular position of P is



$$\tan\theta = \frac{y}{D}$$

$$\Rightarrow \qquad \theta \simeq \frac{y}{D} \quad [\because y << D]$$

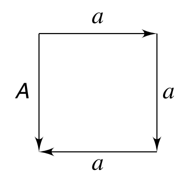
Path difference between waves reaching P from $S1$ and $S2$ is $\Delta x = d\sin\theta = d\cdot\theta$

$$\therefore \text{ Phase difference } \delta = \frac{2\pi}{\lambda}\Delta x = \frac{2\pi}{\lambda}d\cdot\theta$$

$$= \frac{2\pi}{\lambda}\frac{d\cdot y}{D} = \frac{2\pi \times 0.06 \times 17.2}{550.4 \times 10^{-6} \times 250}$$

$$= 1.5\pi = \frac{3\pi}{2}$$

Similarly phase difference between waves from $S2$ and $S3$ will be $\frac{3\pi}{2}$.



Using pharor method of amplitude addition, the resultant amplitude at P is

$$A = a \text{ [see fig]}$$

Where a = amplitude due to individual waves

$\therefore$ Resultant intensity at P is I_0

11. Position of n^{th} order maxima is given by

$$d \sin\theta = n\lambda; \quad n = 0, \pm 1, \pm 2 \ldots.$$

At $\sin\theta = 0.6$; $d = 0.32$ mm; $\lambda = 5000$ Å we have

$$n = \frac{d\sin\theta}{\lambda} = \frac{0.32 \times 0.6}{5000 \times 10^{-7}} = 384$$

It means there are 384 maxima in the range $0 < \theta \leq \sin^{-1}(0.6)$. By symmetry we have same number of maxima on the other side and there is one central maxima (corresponding to $n = 0$)

Therefore, total number of maxima = 384 + 384 + 1

= 769

12. Path difference of two interfering waves at P is $\Delta x = 10\lambda = 60000$ Å.

In water the wavelength of light changes to $\lambda' = \frac{\lambda}{\mu} = \frac{6000}{4/3} = 4500$ Å.

Phase difference between waves at P is

$$\lambda = \frac{2\pi}{\lambda'}\Delta x = \frac{2\pi}{4500} \times 60000 = \frac{80\pi}{3} = 26\pi + \frac{2\pi}{3}$$

Intensity at P will be

$$I' = 4I_0 \cos^2\left(\frac{\delta}{2}\right) = 4I_0 \cos^2\left[13\pi + \frac{\pi}{3}\right]$$

$\Rightarrow$ $$I' = 4I_0 \cdot \frac{1}{4}$$

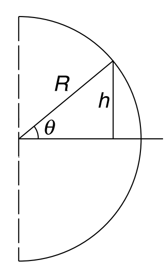
It is given that $4I_o = I$

$\therefore$ $$I' = \frac{I}{4}.$$

13. The second order maxima occurs in direction given by

$$d\sin\theta = 2\lambda$$

$\Rightarrow$ $$\sin\theta = \frac{2\lambda}{d} = \frac{2 \times 500 \times 10^{-9}}{5 \times 10^{-6}} = 0.2$$



The required height is $h = R\sin\theta = 2 \times 0.2 = 0.4\,\text{m}$

14. Extra optical path length for wave from A; $\Delta x_1 = t_1(\mu_1 - 1)$

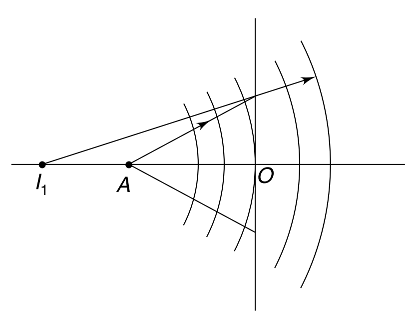
Extra optical path length for wave from B; $\Delta x_2 = t_2(\mu_2 - 1)$

Under given conditions $\Delta x_1 > \Delta x_2$ Hence the central fringe will shift towards A

15. (a) Due to refraction at plane surface the image (I_1) is formed at a distance $\mu R = \frac{3}{2}R$ from O (to left of O).

Refracted rays appear to be diverging from I_1. Hence radius of curvature of refracted wavefronts at $O = OI_1 = \frac{3}{2}R$.

Change in radius of wavefronts $= \frac{3}{2}R - R = \frac{R}{2}$



(b) I_1 acts as object for refraction at curved surface. Image position (I_2) can be calculated as

$$\frac{1}{V} - \frac{\mu}{-\left(\frac{3}{2}R + R\right)} = \frac{1 - \mu}{-R}$$

$$\frac{1}{V} = \frac{1}{2R} - \frac{3}{5R} = -\frac{1}{10R}$$

$\therefore$ $$V = -10R.$$

Image is at a distance of $10R$ from P (to left).

All emergent wave fronts appear to be diverging from I_2. Hence radius of curvature is $10R$.

16. Optical path length of ray along x axis and a ray at a height y will be same.

$\therefore$ $$\mu_0 t + f \simeq \mu t + (y^2 + f^2)^{\frac{1}{2}}$$

$$\left[\mu_0 - \mu_0\left(1 - \frac{y^2}{y_0^2}\right)\right]t = (y^2 + f^2)^{\frac{1}{2}} - f$$

$$\mu_0 \frac{y^2}{y_0^2} t = f\left[\left(1 + \frac{y^2}{f^2}\right)^{1/2} - 1\right]$$

$$\mu_0 \frac{y^2}{y_0^2} t \simeq f \frac{y^2}{2f^2}$$

$\Rightarrow$ $$f = \frac{y_0^2}{2\mu_0 t}$$

17. Imagine the image of O to be another point source. The mirror reflects the light as if it is diverging from the image.

The wavefronts around a point source are spherical in shape and the intensity falls off with distance (x) from the source as

$$I \propto \frac{1}{x^2}$$

Image, as a source, is at a distance 3L from the point A.

$\therefore$ Intensity due to it will be $\frac{1}{9}$th of that caused by the original source.

$\therefore$ Intensity at $A = 81I + \frac{81I}{9}$

$= 90\ I$

18. The given condition implies the angular position (θ_1) of first minima for $\lambda + \Delta\lambda$ should be less than the angular position (θ_2) for first maxima of $\lambda - \Delta\lambda$. This means

$$\theta_1 < \theta_2$$

$$\sin\theta_1 < \sin\theta_2$$

$$d\sin\theta_1 < d\sin\theta_2$$

$$\frac{1}{2}(\lambda + \Delta\lambda) < (\lambda - \Delta\lambda)$$

$$3\Delta\lambda < \lambda$$

$$\Delta\lambda < \frac{\lambda}{3}$$

19. Let thickness of air gap be t.

On reflection $R2$ suffers a phase change of π.

$\therefore$Condition for constructive interference is

$$2t = (2n_1 + 1)\frac{\lambda_1}{2} \quad \text{where } \lambda_1 = 0.4\ \mu\text{m}$$

$\therefore$ $$2t = (2n_1 + 1)\ 0.2 \qquad ...(1)$$

For other wavelength

$$2t = (2n_2 + 1)\frac{\lambda_2}{2} \quad ...(2)$$

From (1), $2 \times 0.5\ \mu m = (2n_1 + 1)\frac{0.4}{2}\ \mu m$

$\Rightarrow \quad n_1 = 2$

According to the question; $\lambda_2 > \lambda_1$

(Since $\lambda_1 = 0.4\ \mu m$ is the smallest incident wavelength)

$\therefore \quad n_2 < n_1$

$\therefore \quad n_2 = 1$

$\therefore \quad \lambda_2 = \frac{4t}{2n_2 + 1} = \frac{4 \times 0.5\ \mu m}{3} = 0.67\ \mu m$

20. Path difference between light incident on two slits is

$$\Delta x_1 = d\sin\phi$$

Geometrical path difference between S_2P and S_1P is $\Delta x_2 = d\sin\theta$

The equivalent length in air will be

$$(\Delta x_2)_{air} = \mu d\sin\theta.$$

Therefore, total path difference between two waves reaching at P will be

$$\Delta x = \Delta x_1 + (\Delta x_2)_{air}$$
$$= d\sin\phi + \mu d\sin\theta$$

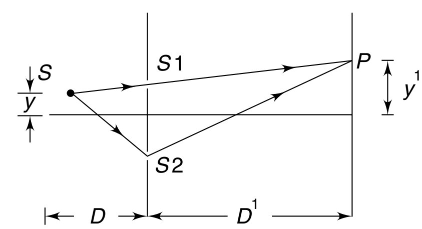
For a maxima at P, we must have

$$\Delta x = n\lambda; \quad n = 0, 1, 2, 3\ldots$$
$$d\sin\phi + \mu d\sin\theta = n\lambda$$

$\Rightarrow \quad \mu d\sin\theta = n\lambda - d\sin\phi$

$\Rightarrow \quad \theta = \sin^{-1}\left[\frac{1}{\mu d}(n\lambda - d\sin\phi)\right]$

21. (a) Displacement of the source at time t is



$$Y = (0.5\text{ mm})\sin(\pi t)$$

Path difference of the two waves reaching at P is

$$\Delta x = \frac{yd}{D} + \frac{y'd}{D'}$$

For central maximum $\Delta x = 0$

$\Rightarrow \quad y' = -\frac{D'}{D}y = -\left(\frac{2}{1}\right)0.5\sin\pi t$

$\therefore \quad y' = -\sin(\pi t)$ in mm

(b) $y' = \frac{d}{2}$ for a point exactly in front of $S1$

$\therefore \quad \Delta x = \frac{yd}{D} + \frac{\frac{d}{2}d}{D'}$

For maximum intensity $\Delta x = n\lambda$

$$\frac{yd}{D} + \frac{\frac{d^2}{2}}{D'} = n\lambda$$

$\Rightarrow \quad 0.5\sin\pi t + \frac{(1)^2}{4} = n \times 500 \times 10^{-6} \times 10^3$

$\Rightarrow \quad 0.5\sin\pi t + 0.25 = 0.5n$

$$\Rightarrow \qquad \sin(\pi t) = \frac{0.5n - 0.25}{0.5}$$

For minimum 't' we have $n = 1$

$$\therefore \qquad \sin(\pi t) = \frac{0.5 - 0.25}{0.5} = 0.5$$

$$\Rightarrow \qquad \pi t = \frac{\pi}{6}$$

$$\Rightarrow \qquad t = \frac{1}{6}\ s$$

22. All rays transmitted through the lens will be parallel since the source (s) is in focal plane.

$$\tan\phi = \frac{d/2}{f}$$

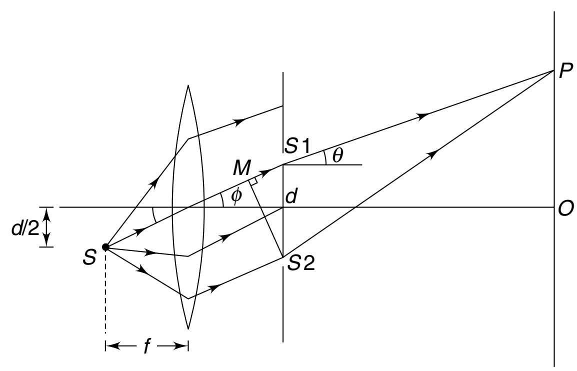
Path difference in waves reaching the slits $S1$ and $S2$ is

$$\Delta x_1 = S_1M = d\sin\phi \simeq d\tan\phi = \frac{d^2}{2f}$$

Central maxima will be formed at a point P where

$$S_2P - S_1P = \Delta X_1$$

If angular position of point P is θ then,



$$S_2P - S_1P = d\sin\theta$$

$$\therefore \qquad d\sin\theta = \frac{d^2}{2f}$$

$$\Rightarrow \qquad \sin\theta = \frac{d}{2f}$$

$$\therefore \qquad \tan\theta \simeq \frac{d}{2f}$$

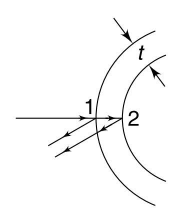
$$\frac{\text{OP}}{\text{D}} = \frac{d}{2f}$$

$$\text{OP} = \frac{Dd}{2f}.$$

23. Colours are seen due to interference between the light waves reflected from the outer and inner surfaces of the liquid film. Light wave, reflected at 1 undergoes a phase change of π and for near normal incidence (as said in the question. Read carefully) the path difference between two interfering waves will be

$$\Delta x = \mu(2t)$$

If constructive interference occurs for wavelength λ then



$$2\mu t = \frac{\lambda}{2}, \frac{3\lambda}{2}, \frac{5\lambda}{2} \ldots$$

$$\Rightarrow \qquad \lambda = 4\mu t, \frac{4\mu t}{3}, \frac{4\mu t}{5} \ldots$$

$$= (4 \times 1.4 \times 90), \frac{4 \times 1.4 \times 90}{3}, \frac{4 \times 1.4 \times 90}{5} \ldots$$

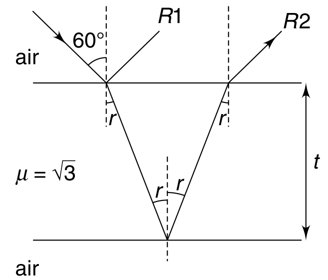
$$= 510 \text{ nm}, 168 \text{ nm}, 101 \text{ nm} \ldots$$

510 nm is visible green light and all other wavelength, for which constructive interference takes place, are invisible.

$\therefore$ The point will appear green.

24. 1. $\sin 60° = \sqrt{3}\,\sin r \quad \Rightarrow \quad r = 30°$.

The optical path difference between $R1$ and $R2$ is given by



$$\Delta x = 2\mu t \sec r - 2t \tan r \sin i$$

$$= 2\sqrt{3}\,(t \sec 30°) - 2t \tan 30° \cdot \sin 60° = 3t$$

∴ For constructive interference, we have

$$3t = \frac{\lambda}{2}; \frac{3\lambda}{2} \ldots$$

For minimum t

$$3t = \frac{\lambda}{2}$$

$$\Rightarrow \qquad t = \frac{\lambda}{6} = 100 \text{ nm}.$$

25. (a) Condition for destructive interference in the thin film will be

$$2\mu t = \lambda, 2\lambda, 3\lambda \ldots$$

$$\Rightarrow \qquad \lambda = \frac{2\mu t}{n} \quad \text{where} \quad n = 1, 2, 3 \ldots$$

For $\lambda_1 = 450$ nm

$$450 \text{ nm} = \frac{2\mu t}{n} \qquad \ldots(1)$$

For $\lambda_2 = 540$ nm

$$540 \text{ nm} = \frac{2\mu t}{m} \qquad \ldots(2)$$

Where n and m are integers. $[n = m + 1]$

$$(1) \div (2) \quad \frac{450}{540} = \frac{m}{n}$$

$$\Rightarrow \qquad \frac{5}{6} = \frac{m}{n}$$

$$\therefore \qquad m = 5, n = 6$$

Putting in (1) $\qquad t = \dfrac{450 \times 6}{2 \times 1.5} = 900$ nm.

(b) Condition for constructive interference is

$$2\mu t = \frac{\lambda}{2}, \frac{3\lambda}{2} \ldots$$

$$\Rightarrow \qquad 2\mu t = \left(m + \frac{1}{2}\right)\lambda \qquad [m = 0, 1, 2, \ldots]$$

$$\Rightarrow \qquad \lambda = \frac{2\mu t}{m + \frac{1}{2}} = \frac{2 \times 1.5 \times 900 \text{ nm}}{m + \frac{1}{2}} = \frac{2700}{m + \frac{1}{2}} \text{ nm}.$$

For $m = 0$; $\lambda = 5400$ nm
For $m = 1$; $\lambda = 1800$ nm
For $m = 2$; $\lambda = 1080$ nm
For $m = 3$; $\lambda = 771$ nm
For $m = 4$; $\lambda = 600$ nm
For $m = 5$; $\lambda = 491$ nm
For $m = 6$; $\lambda = 415$ nm
For $m = 7$; $\lambda = 360$ nm

Out of all these 415 nm, 491 nm, 600 nm, and 771 nm fall in visible range.

26. Path difference between light wave reaching at $S2$ and $S1$ is.

$$\Delta x = d\cos\theta$$

To the left of the slit plane the optical path from $S1$ to C must be large by the same amount.

For this the upper slit must be covered using the sheet of refractive index 1.5.

The geometrical path length from $S1$ to C and from $S2$ to C is same. The difference in optical path arises due to different refractive indices of the two sheets.

$$\therefore \quad \Delta x = (1.5 - 1.2)t$$

$$\therefore \quad d\cos\theta = \frac{3}{10}t$$

$$\therefore \quad \cos\theta = \frac{3t}{10d}$$

$$\Rightarrow \quad \theta = \cos^{-1}\left(\frac{3t}{10d}\right)$$

27. After plates are inserted

$$\Delta x = (\mu_2 - \mu_1)t = (1.7 - 1.4)t = 0.3t$$

Since point P lies between original 5^{th} maximum and 6^{th} minimum

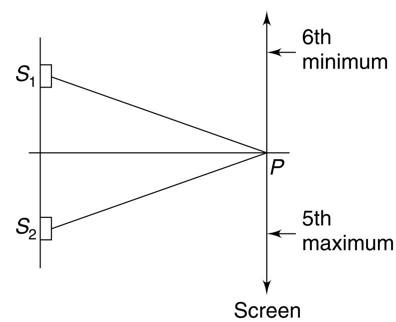
$\therefore$ path difference of waves reaching P can be written as

$$\Delta x = 5\lambda + \Delta$$

$$\therefore \quad 0.3t = 5\lambda + \Delta \qquad ...(a)$$

It is given that

$$\frac{I}{I_0} = \frac{3}{4} = \cos^2\left[\frac{\pi\Delta}{\lambda}\right]$$



$$\Rightarrow \quad \frac{\cos\pi\Delta}{\lambda} = \frac{\sqrt{3}}{2}$$

$$\text{so,} \quad \frac{\pi\Delta}{\lambda} = \frac{\pi}{6}; \qquad \therefore \ \Delta = \frac{\lambda}{6}$$

$$\therefore \text{ from (a)} \quad 0.3t = 5\lambda + \frac{\lambda}{6}$$

$$\Rightarrow \quad t = \frac{31 \times 5400 \times 10^{-10}}{6 \times 0.3} = 9.3 \times 10^{-6} \text{ m}$$

28. Shift in fringe pattern caused by the transparent sheet is

$$S = \frac{D}{d}(\mu - 1)t$$

$$\therefore \quad \Delta S = \frac{D}{d}[\Delta(\mu - 1)t + (\mu - 1)\Delta t]$$

$$= \frac{D}{d}[\Delta\mu t + (\mu - 1)\Delta t]$$

$$= \frac{D}{d}[-t\cdot\gamma\Delta\theta + (\mu - 1)t\alpha\Delta\theta]$$

For the central fringe to be back at its original location.

$$S + \Delta s = 0$$

$$\Rightarrow \quad \frac{D}{d}(\mu-1)t + \frac{D}{d}[-t\gamma\Delta\theta + (\mu-1)t\alpha\Delta\theta] = 0$$

$$[-\gamma + (\mu-1)\alpha]\Delta\theta = -(\mu-1)$$

$$\Rightarrow \quad \Delta\theta = \frac{(\mu-1)}{(\mu-1)\alpha-\gamma}$$

29. (a) If δ is phase difference between the two waves arriving at P then

$$I_0 = 4I_0\cos^2\left(\frac{\delta}{2}\right)$$

$$\Rightarrow \quad \delta = \frac{2\pi}{3}$$

After placing the glass piece the new phase difference will be –

$$\delta' = \frac{2\pi}{3} + \frac{2\pi(\mu-1)t}{\lambda}$$

(b) Intensity at P will be- $4I_0\cos^2\left(\frac{\pi}{3} + \frac{\pi}{\lambda}(\mu-1)t\right)$

Intensity at P will be minimum (= 0) if

$$\frac{\pi}{3} + \frac{\pi}{\lambda}(\mu-1) = (2n-1)\frac{\pi}{2} \text{ where } n = 1, 2, 3...$$

$$t = \frac{(2n-1)\lambda}{2(\mu-1)} - \frac{\lambda}{3(\mu-1)} \text{ where } n = 1, 2, 3...$$

30. From the theory of interference, the position of first maxima (from O) is given as

$$y = \frac{D\lambda}{d}$$

or, $$\Delta y = \frac{D}{d}\Delta\lambda$$

$$\therefore \quad \frac{\Delta y}{y} = \frac{\Delta\lambda}{\lambda} \quad ...(i)$$

Now, velocity of sound is proportional to square root of its temperature

$$\therefore \quad V = K'\sqrt{T}; \quad \text{where } K' \text{ is a constant.}$$

The frequency of source remains unaltered.

$$\therefore \quad \lambda v = K'\sqrt{T}$$

or, $$\lambda = \frac{K'}{v}\sqrt{T} = K\sqrt{T} \quad \left[\text{where } K = \frac{K'}{v} = \text{constant}\right]$$

$$\therefore \quad \Delta\lambda = \frac{K\Delta T}{2\sqrt{T}}$$

or, $$\frac{\Delta\lambda}{\lambda} = \frac{\Delta T}{2T} = \frac{0.01}{2} \quad \left[\because \frac{\Delta T}{T} = 0.01\right]$$

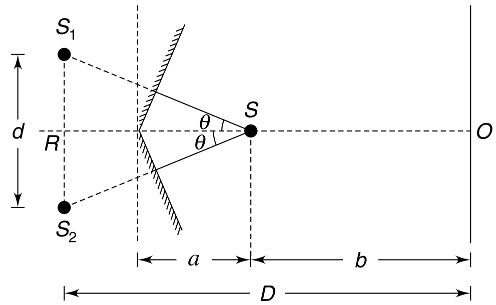
$$= 0.005 \quad ...(ii)$$

Form (i) and (ii)

$$\frac{\Delta y}{y} = 0.005$$

$$\therefore \quad \%\text{ change} = \frac{\Delta y}{y} \times 100 = 0.5\% \text{ (increment).}$$

31. Interference is due to reflected waves, therefore, images S_1 and S_2 of source S behave like two coherent sources.



Distance of S from each mirror $= a\cos\theta$.

$$\therefore \quad SS_1 = SS_2 = 2a\cos\theta$$

$$\therefore \quad S_1S_2 = d = SS_1\sin\theta + SS_2\sin\theta = 4a\cos\theta\sin\theta$$

Since θ is small

$$\cos\theta \simeq 1 \quad \text{and} \quad \sin\theta \simeq \theta$$

$$d = 4a\theta$$

Distance $RS = SS_1\cos\theta = 2a\cos^2\theta \simeq 2a.$

$$\therefore \quad D = RO = RS + SO = (2a + b)$$

$$\therefore \quad \text{Fringe width} \quad \beta = \frac{D\lambda}{d} = \frac{(2a+b)\lambda}{4a\theta}$$

(a) For blue light $\lambda = 4000$ Å, $a = 1$ cm, $b = 38$ cm

$$\therefore \quad \beta_1 = \frac{2\times 1 + 38}{4\times 1\times 0.05} \times 4000 \times 10^{-10}\text{ m}$$

$$= \frac{40}{20} \times 4 \times 10^{-5}\text{ m} = 80\ \mu\text{m}$$

(b) Fringe width for light of wavelength 5600 Å is

$$\beta_2 = \frac{2\times 1 + 38}{4\times 1\times 0.05} \times 5600 \times 10^{-10}\text{ m} = 112\ \mu\text{m}.$$

A black line is formed at the position where dark fringes are formed for both the wavelengths.
Let the first black line be at a distance y from the central fringe. Let this be the position of m^{th} dark fringe of 4000 Å and n^{th} dark fringe of 5600 Å light.

$$\therefore \quad y = \left(m - \frac{1}{2}\right)\beta_1 = \left(n - \frac{1}{2}\right)\beta_2$$

$$\Rightarrow \quad \frac{2m-1}{2n-1} = \frac{\beta_2}{\beta_1}$$

For first black line, y should be minimum possible which corresponds to least possible integral values of m and n.

$$\therefore \quad \frac{2m-1}{2n-1} = \frac{7}{5}$$

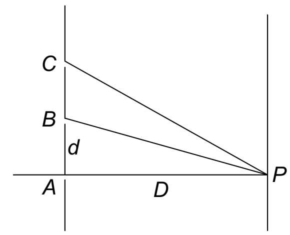
or, $\quad m = 4,\ n = 3$

$\therefore$ position of first black line $y = \left(m - \frac{1}{2}\right)\beta_1 = 280\ \mu$m.

(c) Since the interference pattern is symmetric about the central fringe, there will be one dark line at a distance y above it and the other will be y below it.

$\therefore$ Relevant answer is $2y = 560\ \mu$m.

32. (a)



$$BP - AP = \frac{\lambda}{3}$$

$$\sqrt{D^2 + d^2} - D = \frac{\lambda}{3}$$

$$\Rightarrow \quad D\left[1 + \frac{d^2}{D^2}\right]^{1/2} - D = \frac{\lambda}{3}$$

$$\Rightarrow \quad D\left[1 + \frac{d^2}{2D^2}\right] - D = \frac{\lambda}{3}$$

$$\left[d << D \text{ as } BP - AP = \frac{\lambda}{3} << D\right]$$

$$\therefore \quad d = \sqrt{\frac{2D\lambda}{3}} \qquad ...(a)$$

(b) Path difference for two waves

$$\Delta x_{CA} = CP - AP$$

$$= \sqrt{D^2 + (2d)^2} - D$$

$$= D\left(1 + \frac{4d^2}{D^2}\right)^{1/2} - D$$

$$\simeq D\left(1 + \frac{2d^2}{D^2}\right) - D = \frac{2d^2}{D}$$

$$= \frac{4\lambda}{3}\left[\because \quad d = \sqrt{\frac{2D\lambda}{3}}\right]$$

$\therefore$ phase difference

$$\Delta\phi_{CA} = \frac{2\pi}{\lambda}\cdot\Delta x_{CA} = \frac{2\pi}{\lambda}\cdot\frac{4\lambda}{3} = \frac{8\pi}{3}$$

(c) Path difference at P for waves from B and A

$$\Delta x_{BA} = \frac{\lambda}{3}$$

$\therefore$ phase difference

$$\Delta\phi_{CA} = \frac{2\pi}{\lambda}\cdot\Delta x_{BA} = \frac{2\pi}{\lambda}\cdot\frac{\lambda}{3} = \frac{2\pi}{3}$$

The three waves at P can be represented as

$$y_1 = a\sin\omega t$$

$$y_2 = a\sin\left(\omega t + \frac{2\pi}{3}\right)$$

$$y_3 = a\sin\left(\omega t + \frac{8\pi}{3}\right) = a\sin\left(\omega t + \frac{2\pi}{3}\right)$$

$\therefore$ Resultant wave is

$$y = y_1 + y_2 + y_3$$

$$= a\sin\omega t + 2a\sin\left(\omega t + \frac{2\pi}{3}\right)$$

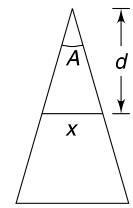
$$= A\sin(\omega t + \delta)$$

Where $$A = \sqrt{a^2 + (2a)^2 + 2\cdot a\cdot 2a\cdot\cos\frac{2\pi}{3}} = \sqrt{3}a.$$

Intensity is proportional to square of amplitude.

$$\therefore \qquad I = 3I_0.$$

33. (a)
$$x \simeq A \cdot d \qquad \ldots(1)$$



Path difference between the two interfering waves is

$$\Delta x = 2\mu x$$

For constructive interference the condition is

$$2\mu x = n\lambda + \frac{\lambda}{2}$$

$$n = 0, 1, 2, \ldots$$

Since reflection at the first surfaces causes an additional phase change of π.

For first maxima–

$$2\mu x = \frac{\lambda}{2} \quad \Rightarrow \quad 2 \times 1.48x = \frac{400 \times 10^{-9}}{2}$$

$$\Rightarrow \qquad x = 6.8 \times 10^{-8} \text{ m} = 68 \text{ nm}$$

Form (1)

$$A = \frac{x}{d}$$

$$= \frac{68 \times 10^{-9}}{3 \times 10^{-2}} = 22.7 \times 10^{-7} \text{ rad}$$

$$= 2.3 \times 10^{-6} \text{ rad} \simeq 1.3 \times 10^{-4} \text{ degree.}$$

(b) For red light

$$2\mu x = \frac{\lambda_R}{2} \quad \Rightarrow \quad x = 136 \text{ nm}$$

$$\therefore \qquad d = \frac{x}{A} = 6 \text{ cm}$$

34. Let I_0 = intensity at S_3 (and S_4) due to either of S_1 or S_2.

Path difference between light waves reaching S_3 (or S_4) = $d_1\left(\frac{d_2}{2L_1}\right)$

Phase difference
$$\Delta\phi = \frac{2\pi}{\lambda} \cdot \frac{d_1 d_2}{2L_1} = \frac{\pi d_1 d_2}{\lambda L_1}$$

Intensity at slits S_3 and S_4 will be

$$I = 4I_0 \cos^2\left(\frac{\Delta\phi}{2}\right) = 4I_0 \cos^2\left(\frac{\pi d_1 d_2}{2\lambda L_1}\right)$$

Now, S_3 and S_4 are two coherent sources producing interference fringes on the screen. At distance y from the centre

$$\Delta x' = d_2 \frac{y}{L_2}$$

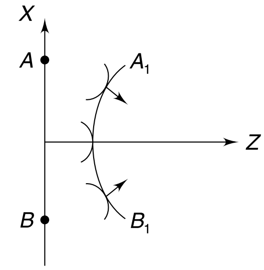
$$\Delta\phi' = \frac{2\pi}{\lambda} \frac{d_2 y}{L_2}$$

$\therefore$ Intensity at distance y will be proportional to

$$I' \propto I \cdot \cos^2 \frac{\Delta\phi'}{2}$$

$$\Rightarrow \qquad I' \propto \cos^2\left(\frac{\pi d_1 d_2}{2\lambda L_1}\right) \cdot \cos^2\left(\frac{\pi d_2 y}{\lambda L_2}\right)$$

35. The refractive index decrease as x and y increase. The speed of wave increases as we move away from the z axis. When we draw spherical wavefronts originating from different points on AB, we get spheres of increasing radius as we move away from the z axis. When we draw common tangent to all these spheres, the resulting wavefront is as shown in the figure. Draw one more wavefront and you will realize that it is converging.



For the above construction to be valid, Δt should be very small so that we can treat the secondary wavelets to be spherical.

36. Assume that only A_1 is on.

Path difference $\quad A_1S_2 - A_1S_1 = d\sin\Delta\theta$

Path difference $\quad S_2P - S_1P = \dfrac{yd}{D}$

$\therefore$ Total path difference

$$\Delta x = A_1S_2P - A_1S_1P$$

$$= d\sin\Delta\theta + \frac{yd}{D}$$

$$\therefore \qquad \delta = \frac{2\pi}{\lambda}d\left[\sin\Delta\theta + \frac{y}{D}\right]$$

$\therefore$ Intensity at P when only A_1 is on is

$$I_1 = I_0\cos^2\delta = I_0\cos^2\left[\frac{2\pi}{\lambda}d\left(\sin\Delta\theta + \frac{y}{D}\right)\right]$$

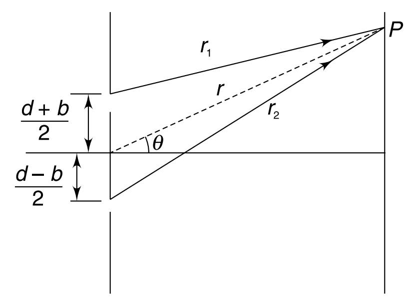
Similarly, intensity at P when only A_2 is on is

$$I_2 = I_0\cos^2\left[\frac{2\pi}{\lambda}d\left(\frac{y}{d} - \sin\Delta\theta\right)\right]$$

Since sources are incoherent; the resultant intensity when both are on is

$$I = I_1 + I_2 = I_0\cos^2\left[\frac{2\pi}{\lambda}d\left(\sin\Delta\theta + \frac{y}{D}\right)\right] + I_0\cos^2\left[\frac{2\pi}{\lambda}d\left(\frac{y}{D} - \sin\Delta\theta\right)\right]$$

37. The path length from top edges of the slits to P is r_1 and r_2. Using law of cosines:



$$r_1^2 = r^2 + \left(\frac{d+b}{2}\right)^2 - 2\cdot\left(\frac{d+b}{2}\right)\cdot r\cdot\cos\left(\frac{\pi}{2} - \theta\right)$$

$$= r^2 + \left(\frac{d+b}{2}\right)^2 - (d+b)r\cdot\sin\theta$$

And
$$r_2^2 = r^2 + \left(\frac{d-b}{2}\right)^2 - 2\cdot\left(\frac{d-b}{2}\right)r\cdot\cos\left(\frac{\pi}{2} + \theta\right)$$

$$= r^2 + \left(\frac{d-b}{2}\right)^2 + (d-b)r\cdot\sin\theta$$

$$r_2^2 - r_1^2 = -db + 2d\cdot r\cdot\sin\theta$$

$$\Rightarrow \qquad (r_2 + r_1)(r_2 - r_1) = 2d\ r\sin\theta - d\cdot b$$

$$\Rightarrow \qquad (r_2 - r_1) = \Delta x_t = d\sin\theta - \frac{d\cdot b}{2r} \qquad [\because\ r_1 + r_2 = 2r]$$

Similarly, consider r_3 and r_4 as path length from the bottom edge of the two slits and prove that

$$r_4 - r_3 = \Delta x_b = d\sin\theta + \frac{d\cdot b}{2r}$$

$$\therefore \qquad \Delta x_b - \Delta x_t = \frac{db}{r} \simeq \frac{db}{D}.$$

Chapter 15

Wave Particle Duality and Atomic Physics

LEVEL 1

Q. 1: A beam of red light (λ_r = 800 nm) is made up of a stream of photons. In the diagram shown the size of dots represent the photon energy and the spacing represents the spatial distribution of photons.

Copy the diagram in your notebook; and just below it draw a similar diagram representing a beam of blue light (λ_b = 400 nm) having same intensity.

Q. 2: The brightness of an incandescent light bulb is controlled by a regulator. The regulator is a variable resistor connected in series with the bulb. What happens to the colour of the light given off by the bulb as the resistance of the regulator is increased? What happens to the average energy of photons given out by the bulb? Does it increase?

Q. 3: Find the change in energy of a photon of blue light (λ = 400 Å) when the light enters into a medium of refractive index $\mu = \frac{4}{3}$.

Q. 4: A 0.6×10^{-3} W laser is aimed at the Moon. The wavelength of emitted light is 600 nm and the laser beam spreads out of the source at a divergence angle of $\theta = 0.5 \times 10^{-3}$ rad. The Earth-Moon distance is nearly 4×10^8 m. Calculate the maximum number of photons arriving per second per square meter on the Moon? Neglect any absorption by the atmosphere.

Q. 5: A laser beam of wavelength $\lambda = 5 \times 10^{-7}$ m strikes normally a blackened plate and produces a force of 10^{-5} N. Mass of the plate is 10 g and its specific heat capacity is 400 J kg^{-1} K^{-1}. At what rate will the temperature of the plate rise? Assume no heat loss to the surrounding.

Q. 6: When photons of wavelength λ_1 = 2920 Å strike the surface of metal A, the ejected photoelectron have maximum kinetic energy of k_1 eV and the smallest de-Broglie wavelength of λ. When photons of wavelength λ_2 = 2640 Å strike the surface of metal B the ejected photoelectrons have kinetic energy ranging from zero to $k_2 = (k_1 - 1.5)$ eV. The smallest de-Broglie wavelength of electrons emitted from metal B is 2λ. Find

(a) Work functions of metal A and B.

(b) k_1

Take hc = 12410 eV Å

Q. 7: Work function of metal X is equal to 3.5 eV and work function of material Y is equal to ionization energy of He^+ ion in its first excited state. Light of same wavelength is incident on both X and Y. The maximum kinetic energy of photoelectrons emitted from X is twice that of photoelectrons emitted from Y. Find the wavelength of incident light. hc = 12400 eV Å.

Q. 8: A particle of mass m is allowed to fall freely under gravity on an elastic horizontal surface. The quantum effect become important if the smallest de-Broglie wavelength of the particle is of the same order as the height from which it was dropped. Write the mechanical energy of the particle if quantum effects become important.

Q. 9: A microscope can make you "see" something as small as the wavelength of wave used to make the observation. Calculate the minimum energy of an electron needed in an electron microscope to see a hydrogen atom.

Q. 10: Electrons originally at rest are accelerated through a potential difference of V volts. The applied potential difference is measured using a voltmeter having a least count of ΔV. The electrons in the beam have a de-Broglie wavelength of $\lambda \pm \Delta\lambda$ [$\Delta\lambda << \lambda$]. Find $\left|\frac{\Delta\lambda}{\lambda}\right|$.

Q. 11: Calculate the ratio of de-Broglie wavelength of molecules of hydrogen and oxygen kept in two separate jars at 27°C and 127°C respectively.

Q. 12: Calculate de-Broglie wavelength of hydrogen molecules (H_2) present in a container at 400 K. Assume that molecules travel at rms speed.

Boltzmann constant $k = 1.38 \times 10^{-23}$ JK^{-1}

Mass of H_2 molecule = 2.0 amu

Q. 13: In a nuclear physics experiment, electrons are to be accelerated so as to have de-Broglie wavelength of the order of the diameter of a heavy nucleus ($\sim 10^{-14}$ m). Determine the momentum of the electron needed to make the wavelength of electrons equal to diameter. Use the classical formula for momentum, $p = m_0 v$ to determine the speed of electrons. Is this speed permissible? Now use the relativistic mass of the electrons in the expression of momentum to find the speed of electrons. Relativistic mass is given by $m = \dfrac{m_0}{\sqrt{1 - v^2/c^2}}$.
Where m_0 is rest mass and v is speed of the particle.

Q. 14: A parallel beam of monochromatic light of frequency v is incident on a surface. Intensity of the beam is I and area of the surface is A. Find the force exerted by the light beam on the surface for following cases–

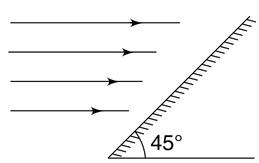
(i) the surface is perfectly absorbing and the light beam is incident normally on it.

(ii) the surface is perfectly reflecting and the light beam is incident normally.

(iii) the surface is perfectly absorbing and the light beam is incident at an angle of incidence θ.

(iv) the surface is perfectly reflecting and the light beam is incident at an angle of incidence θ.

Q. 15: A horizontal beam of light in incident on a plane mirror inclined at 45° to the horizontal. The percentage of light energy reflected from the mirror is 80%. Find the direction in which the mirror will experience force due to the incident light.



Q. 16: Intensity of sunlight on the surface of the earth is $I = 1400$ W/m^2 (neglecting atmospheric absorption).

(a) Find the Wattage of the Sun.

(b) Assuming that light emitted from the sun is monochromatic having wavelength $\lambda = 6000$ Å, estimate the number of photons emitted from the sun in one second.

(c) According to mass energy equivalence principle, estimate the decrease in mass of the sun in one second.

Given: $h = 6.64 \times 10^{-34}$ J s, $c = 3 \times 10^8$ m/s

Q. 17: A perfectly absorbing solid sphere with a known fixed density, hovers stationary above the sun. This is because the gravitational attraction of the sun is balanced by the pressure due to the sun's light. Assume the sun is far enough away so that it closely approximates a point source of light. Find the radius of the sphere and prove that it is independent of the distance of the sphere from the Sun.

Q. 18: Prove that Bohr's condition for quantization of angular momentum in hydrogen atom can be obtained by requiring an integer number of standing waves around an electron orbit. Use de-Broglie wavelength as the wavelength of wave associated with electron.

Q. 19: The circumference of circular orbit of electron in a He^+ ion is five times the de-Broglie wavelength associated with the electron. Find the radius of the orbit.

Q. 20: A gas of hydrogen like atoms can absorb radiations having photons of energy 68 eV. Consequently, the atoms emit radiations of only three different wavelengths. All the wavelengths are equal or smaller then the absorbed photon.

(a) Find the initial state of gas atoms.

(b) Identify the gas.

(c) Find the minimum wavelength of emitted radiations.

(d) Find the ionisation energy of the atom.

Q. 21: A particle of mass m moves in a circular orbit of radius r. It is under the influence of a force field where its potential energy is given as $U = ar^2$ where a is a positive constant. Assume that Bohr's model of quantization of angular momentum is applicable and calculate the radius of n^{th} allowed orbit.

Q. 22: Hydrogen like atoms (atomic number = Z) in a sample are in excited state with principal quantum number n. The emission spectrum of the sample has 15 different lines. The second most energetic photon emitted by the sample has energy of 27.2 eV. Find Z.

Q. 23: A μ – meson (charge = $-e$, mass = 208 m_e) moves in a circular orbit around a heavy nucleus having charge $+3e$. Find the quantum state n for which the radius of the orbit is same as that of first Bohr orbit for hydrogen atom.

Q. 24: A hydrogen like atom (atomic number = Z) is in higher excited state of quantum number n. This excited atom can make a transition to first excited state by successively emitting two photons of energy 22.94 eV and 5.15 eV. The atom from the same state n can make transition to second excited state by successively emitting two photons of energies 2.4 eV and 8.7 eV. Find values of n and Z.

Q. 25: A system of positron ($_{+1}e^0$) and electron ($_{-1}e^0$) is called Positronium atom. The radii of Positronium atom is expanded but energy levels are reduced by a factor of x compared to H – atom.

(a) Find x

(b) Is it true to say that the entire spectrum of Positronium will be shifted towards longer wavelength compared to H – atom.

Q. 26: A negatively charged muon (mass 207 times the mass of an electron) is captured in a Bohr's orbit of high principal quantum number (n). The atom thus formed is called mesic atom. The muon in high energy state cascades down to lower orbits by emitting photons.

(a) The emitted photons are X rays. Why?

(b) Find the atomic number (Z) of a mu-mesic atom in which the orbit with $n = 1$ will just touch the nuclear surface. Assume that the nucleus has equal number of protons and neutrons.

Q. 27: Consider a gas of hydrogen atom filling a cubical box of side length 1 m. Assume that all hydrogen atoms are in their ninth excited state and they fill up the space like footballs filling up a room. Estimate the number of hydrogen atoms in the room.

Q. 28: Hydrogen atoms in ground state are bombarded by electrons accelerated through a potential difference of V volts. For what range of values of V, will we get only one spectral line in the spectrum of hydrogen atoms excited by impacts of electrons? Take ground state energy of hydrogen atom to be – 13.6 eV.

Q. 29: A glass tube has Be^{+++} ions excited to their second excited state. The radiation coming out of this tube is incident on a metal plate having work function $\phi = 5$ eV. Find the minimum de-Broglie wavelength of emitted photoelectrons.

Q. 30: An electron in an excited hydrogen atom makes transition from a state of quantum number n to a state of quantum number (n – 1).

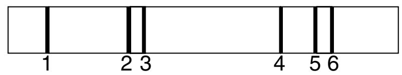
(a) Show that the frequency of emitted radiation is intermediate between the frequencies of orbital revolution in initial and final states.

(b) What is the relationship between frequency of emitted photon and the orbital frequency of the electron when n is large?

Q. 31: Hydrogen atoms in very high quantum state have been observed in radio astronomy. The wavelength of radiation emitted when hydrogen atom makes transition from $n = 110$ state to $n = 109$ state is $\lambda = 5.8$ cm. Imagine a He atom in which one of the electrons has been excited to $n = 110$ state. What will be wavelength of emitted radiation if the electron makes transition to $n = 109$?

Q. 32: A moving hydrogen atom absorbs a photon and comes to rest. What is the maximum possible wavelength of photon? [hc = 12420 eV Å]

Q. 33: A tube contains a sample of hydrogen atoms which are all in their third excited state. The atoms de-excite and a spectrum of the radiation emitted is obtained. The spectrum is shown in the given figure.



(a) Which of the lines (1, 2, 3, 4, 5 or 6) represent a transition from quantum state $n = 3$ to $n = 2$.

(b) Which of the lines represent the one with second smallest wavelength?

Q. 34 The biding energy of an electron in the ground state of He atom is 25 eV. Find the energy required to remove both the electrons.

Q. 35: Which of the followings can excite a hydrogen atom in ground state?

(i) A photon having 11 eV energy

(ii) A neutron having 11 eV kinetic energy

(iii) An electron having 11 eV kinetic energy

Give reasons for your answer.

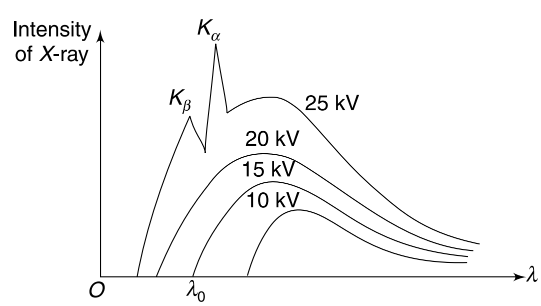
Q. 36: Absorption spectrum is obtained for a sample of gas having atomic hydrogen in ground state. Which lines of the spectrum will be in the range 94 nm to 122 nm.

Take Rydberg's constant $R = 1.1 \times 10^7$ m^{-1}

Q. 37: Radiation coming out of a sample of hydrogen gas excited to first excited state is used for illuminating certain metal plate. When the radiation from some unknown hydrogen like gas excited to the same level is used to expose the same plate, it is found that the de–Broglie wavelength of the fastest photoelectron has decreased 2.3 times. It is given that the energy corresponding to the longest wavelength of the Lyman series of the unknown gas is 3 times the ionization energy of hydrogen (13.6 eV). Find the work function (W) of photoelectric plate in eV. (Take $(2.3)^2 = 5.25$.

Q. 38: An X ray tube with Copper target is found to emit characteristic X rays other than only due to Copper. The k_α lie of Copper has a wavelength of 1.5405 Å. The other K_α line observed is having a wavelength of 1.6578 Å. Identify the impurity.

Q. 39: X-ray spectrum obtained from a metallic target has been shown in the figure. The spectrum has been obtained for four different accelerating voltages in the X-ray tube: 25 kV, 20 kV, 15 kV and 10 kV.



(a) Find the value of λ_0 shown in the graph.

(b) What can you say about the energy of a K-shell electron in the target atom?

Q. 40: When the voltage applied to an X-ray tube increases from 10 to 20 kV the wavelength difference between the k_α line and the short wave cut-off of continuous X-ray spectrum increases by a factor of 3.0. Identify the target element.

Take $\frac{4}{3R} \simeq 1200$ Å where R is Rydberg's contant and $hc = 12.4$ keV Å

Q. 41: An X-ray tube has nickel as target. The wavelength difference between the k_α X-ray and the cut-off wavelength of continuous X-ray spectrum is 84 pm. Find the accelerating voltage applied in the X-ray tube.

[Take $hc = 12400$ eV Å]

Q. 42: How many elements have k_α lines between 241 pm and 180 pm?

Take Rydberg's constant $R = 1.09 \times 10^7$ m^{-1}

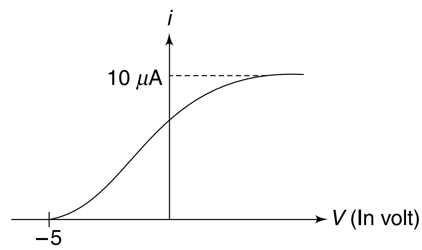
Q. 43: In an electron capture process, a nucleus captures an electron from K shell of the atom. The electron is having a binding energy of B_0. Followed by this capture of electron, several photons are emitted due to electronic transitions. What will be Sum of energy of all such photons emitted?

LEVEL 2

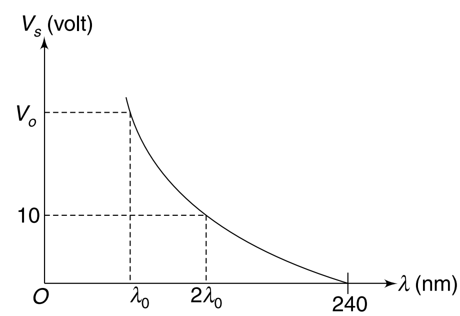
Q. 44: A normal human eye can detect yellow light if more than 10 photons enter into it per second. A star is generating as much power as the Sun and is emitting predominantly yellow light ($\lambda = 6000$ Å). How far is the star if our eye is barely able to see it? It is given that intensity of solar light on surface of the earth is $I = 1400$ Wm^{-2} and the distance of the Sun from the Earth is $r = 1.5 \times 10^{11}$ m. The diameter of pupil of our eye is $d = 6$ mm.

Q. 45: A microwave oven operates at 2.5 GHz. Assume that 5% of microwave photons are absorbed by 200 ml of water kept inside the oven. The time needed to warm the water from 20°C to 70°C is 2 minute. Specific heat capacity of water is 4.2 Jg^{-1} °C^{-1}. Neglect any heat loss by water to the surrounding. Calculate the number of photons emitted per second per kilowatt of power consumed by the microwave. Efficiency of the oven to convent electrical energy into microwave energy is 70%.

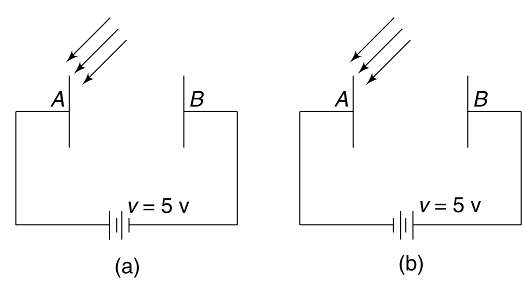
Q. 46: In a photoelectric experiment a metal plate of work function $\phi = 2.0$ eV is irradiated with beam of monochromatic light. The photoelectric current (i) versus the applied potential difference (V) graph is as shown in the figure. It is known that the efficiency of photoemission is 10^{-3}%. Calculate the power of light incident on the metal plate.



Q. 47: In a photoelectric experiment light of different wavelengths are used on a metal surface. For each wavelength the stopping potential difference is recorded. The given graph shows the variation of stopping potential difference (V_s) versus the wavelength (λ) of light used. Find the value of V_0 shown in the graph. Given $h = 4 \times 10^{-15}$ eVs and $c = 3 \times 10^8$ ms^{-1}.



Q. 48: In a photoelectric experiment, a monochromatic light is incident on the metal plate A. It was observed that with $V = 5$ volt, the maximum kinetic energy of photoelectrons striking plate B was 1 eV. The polarity of the applied potential difference was as shown in figure (a). With polarity of the applied potential difference reversed (as shown in figure (b)) and frequency of incident light doubled, it was observed that in saturation state, the kinetic energy of electrons striking plate B ranged between 5 eV to 20 eV. Find the work function of metal used in plate A.



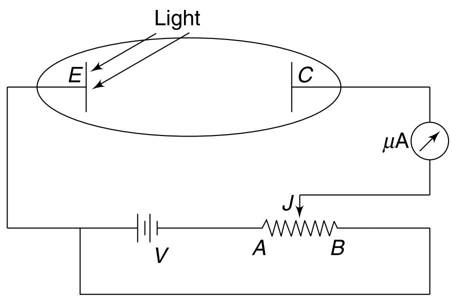
Q. 49: In a photoelectric experimental set-up ultraviolet light of wavelength 350 nm is incident on emitter plate (E). The work function of the emitter plate is $\phi = 2.2$ eV.

AB is a uniform wire resistor having length $L = 100$ cm. Resistance of the wire AB is 10 Ω and the emf of the battery is $V = 10$ volt. The sliding contact J can be moved along the wire AB. When the sliding contact is placed at end B of the wire resistor the micro-ammeter shows a reading of $i = 6\ \mu A$. For answering the following questions assume that photoelectric current is very small compared to the current through the cell.

(a) Find the reading of the ammeter when the slider is moved to end A of the wire.

(b) The slider is moved gradually away from A. Plot the variation of ammeter reading versus distance of the slider measured from A.

(c) Find the maximum speed with which an electron will hit the collector plate (C) if slider J is placed at a distance of 95 cm from A.



Q. 50: A point source is emitting 0.2 W of ultraviolet radiation at a wavelength of $\lambda = 2537$ Å. This source is placed at a distance of 1.0 m from the cathode of a photoelectric cell. The cathode is made of potassium (Work function = 2.22 eV) and has a surface area of 4 cm^2.

(a) According to classical theory, what time of exposure to the radiation shall be required for a potassium atom to accumulate sufficient energy to eject a photoelectron. Assume that radius of each potassium atom is 2 Å and it absorbs all energy incident on it.

(b) Photon flux is defined as number of light photons reaching the cathode in unit time. Calculate the photon flux.

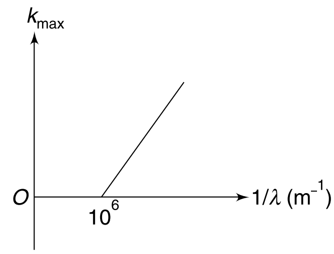
(c) Photo efficiency is defined as probability of a photon being successful in knocking out an electron from the metal surface. Calculate the saturation photocurrent in the cell assuming a photo efficiency of 0.1.

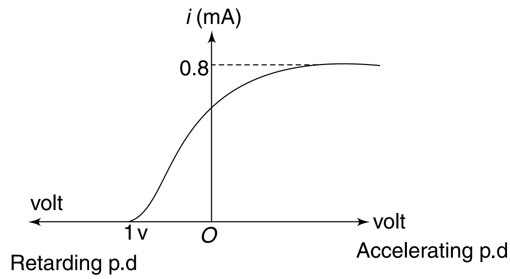
(d) Find the cut – off potential difference for the cell.

Q. 51: In photoelectric experiment set up, the maximum kinetic energy (k_{max}) of emitted photoelectrons was measured for different wavelength (λ) of light used. The graph of k_{max} vs $\frac{1}{\lambda}$ was obtained as shown in first figure. In the same setup, keeping the wavelength of incident light fixed at λ, the applied potential difference was varied and the photoelectric current was recorded. The result has been shown in graph in second figure.

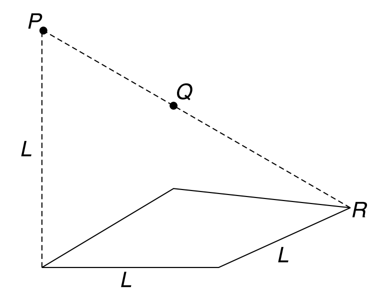
(a) Find λ is Å

(b) Taking the photo efficiency to be 2% (i.e. percentage of incident photons which produce photoelectrons) find the power of light incident on the emitter plate in the experiment. [Take hc = 12400 eV Å]



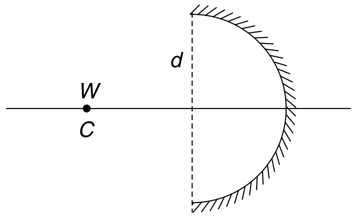


Q. 52: A square plate of side length L absorbs all radiation incident on it. A point source of light is placed at a point P which is directly above a corner of the plate at a height L. The incident light on the plate produces a force of magnitude F_0 on the plate. Calculate the magnitude of force on the plate if the source is moved to a point Q (Q is midpoint of the line PR).

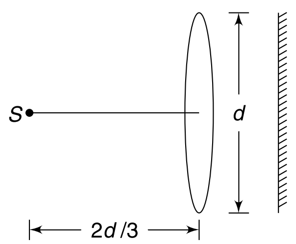


Q. 53: A light photon of wavelength λ_0 is absorbed by an atom of mass m at rest in ground state. The atom is free to move. The atom after absorbing the incident photon, emits another photon in direction opposite to that of the incident photon and returns back to its ground state. In the process the atom acquires a kinetic energy k. Find λ_0 in terms of k and m.

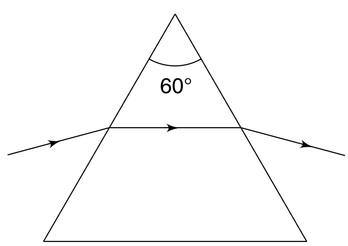
Q. 54: A spherical concave mirror has aperture diameter d and radius of curvature R. A point source of light is placed at its centre of curvature. Source emits power W and the mirror surface is completely reflecting. Find force on the mirror due to light incident on it.



Q. 55: A point source (S) of light having power 500 W is kept at the focus of a lens of aperture diameter d. The focal length of the lens is $\frac{2d}{3}$. Assume that 40% of the incident light energy is transmitted through the lens and the complete transmitted light is incident normally on a perfectly reflecting surface placed behind the lens. Calculate the force on the reflecting surface.



Q. 56: An equilateral glass prism kept on a table has refractive index of $\mu = \sqrt{2}$. It is illuminated by a narrow laser beam having power P_0 and wavelength λ. The path of the laser beam inside the prism is parallel to the base of the prism. Calculate change in weight of the prism due to the incident laser beam.



Q. 57: A proton X is projected in a region of uniform electric field E with speed u. Another proton Y is projected simultaneously with same speed u in a separate region having a uniform magnetic field 'B'. It was found that both X and Y had same de-Broglie wavelength after time T. Specific charge on a proton is σ (C/kg).

(a) Find the angle α at which the proton X was projected with respect to the direction of electric field.

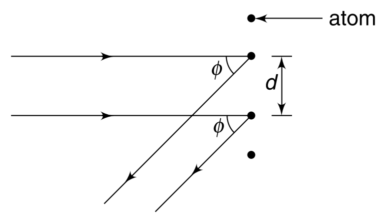
(b) Find the displacement of the proton X in time T.

Q. 58: It is possible for a photon to materialize into an electron and a position. The process is called pair production.

(a) Using conservation of energy and momentum prove that pair production cannot occur in empty space.

(b) Argue qualitatively that such pair production is possible near a nucleus.

Q. 59: The figure shows the reflection of a plane wave by two neighbouring atoms on the surface of a crystal. The depicted waves are actually de-Broglie wave of electrons accelerated from rest by a potential difference of V volt. Neglect relativistic effects and take mass and charge of an electron to be m and e respectively. Calculate ϕ for the case of a constructive interference.



Q. 60: A particle of mass m is trapped between two perfectly rigid parallel walls. The particle bounces back and forth between the walls without losing any energy. From a wave point of view, the particle trapped between the walls is like a standing wave in a stretched string between the walls. Distance between the two walls is L.

(a) Calculate the energy difference between third energy state and the ground state (lowest energy state) of the particle.

(b) Calculate the ground state energy if the mass of the particle is m = 1 mg and separation between the walls is L = 1 cm.

Q. 61: A moving H-atom makes a head on perfectly inelastic collision with a stationary Li^{++} ion. Just before collision, both - H and Li^{++} are in their first excited state. Immediately after collision H was found to be in ground state and Li^{++} was in its second excited state. Find the kinetic energy of H – atom before collision. No photon is emitted in the process. Assume that mass of Li^{++} is nearly 7 times the mass of H.

Q. 62: In a Lithium atom there is one electron in second orbit. The interaction of this electron with the two inner electrons can be accounted for by assuming that this electron (the electron in second orbit) sees a nuclear attraction of Z' protons (where $Z' < Z = 3$). Experimentally, it is known that 5.39 eV of energy is needed to remove the outermost electron from the atom.

(a) Find Z'

(b) Find the longest wavelength of photon that can be absorbed by the electron in $n = 2$ state.

Q. 63: In a hypothetical hydrogen-like atom the wavelength in $A°$ for the spectral lines for transitions from $\boldsymbol{n = P}$ to $\boldsymbol{n = 1}$ are given by $\lambda = \dfrac{\mathbf{1500}p^2}{p^2 - \mathbf{1}}$ where $\boldsymbol{p}$ is an integer larger than 1. Find the energy of three lowest states for this atom. What is the ionization potential of this atom?

Q. 64: Assume that structure of hydrogen atom is governed by classical mechanics. An electron is circulating around a proton and it is radiating energy at a rate given by $\dfrac{dE}{dt} = \dfrac{e^2 a^2}{6\pi c^3 \in_0}$ where a is the acceleration of the electron. Assume that speed of the electron is v ($<< c$).

(a) Estimate the fraction of kinetic energy lost by the electron per revolution in terms of v. Make suitable assumptions.

(b) Is the energy loss per revolution large? Is it safe to assume that orbit is circular during a small time interval?

Q. 65: In a hypothetical hydrogen atom the electrostatic potential energy of interaction of proton and electron is given by $U = U_0 \ell n\left(\dfrac{r}{r_0}\right)$ where U_0 and r_0 are constants and r is radius of circular orbit of electron. For such hydrogen atom the energy difference between n^{th} and m^{th} state is represented by ΔE_{nm}. Calculate the ratio $\Delta E_{12} : \Delta E_{24}$ Assume Bohr's assumption of angular momentum quantization to hold.

Q. 66: When radiations of wavelength λ_1, λ_2 ($= 0.6\ \lambda_1$), λ_3 ($= 4\lambda_2$) and λ_4 are incident on a metal plate, photoelectrons with maximum kinetic energy of -5.2 eV are emitted for λ_1, 12 eV are emitted for λ_2, 0.95 eV are emitted for λ_4 and no electron is emitted for λ_3. It is known that a hydrogen like atom (atomic number Z) in a higher excited state of quantum number n can make a transition to first excited state by successively emitting two photons of wavelength λ_1 and λ_2 respectively. Alternatively, the atom from same excited state can make a transition to the second excited state by successively emitting two photons of wavelength λ_3 and λ_4 respectively.

(a) Find the work function of the metal.

(b) Find the values of n and Z for the atom.

Q. 67: An electron approaches a fixed proton from a large distance with a kinetic energy of 12.2 eV. The electron gets captured by the proton to form a hydrogen atom in an excited state and a photon of wavelength $\lambda_1 = 796$ Å is emitted. Later, the hydrogen atom de-excites by emitting a single photon of wavelength λ_2. Find λ_2. [Take $h = 6.62 \times 10^{-34}$ Js, $c = 3 \times 10^8$ m/s]

Q. 68: A free hydrogen atom in its ground state is at rest. A neutron having kinetic energy k_0 collides head one with the atom. Assume that mass of both neutron and the atom is same.

(a) Find minimum value of k_0 so that this collision can be inelastic.

(b) If $k_0 = 25$ eV, find the kinetic energy of neutron after collision if its excites the hydrogen atom to its second excited state.

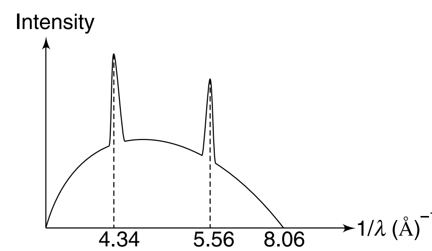
Take ionization energy of hydrogen atom in ground state to be 13.6 eV.

Q. 69: The X-ray spectrum of a metallic target has been shown in figure

(a) What is the accelerating potential difference for bombarding electrons?

(b) Two characteristic X-rays have been shown in the figure one of them is k_α X-ray and the other one is k_β X-ray. What is wavelength of k_α X-ray?

(c) Find the atomic number of the target atom.



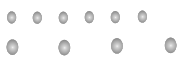
Q. 70: In a X-ray tube, after an electron has been removed from an inner shell of a target atom, an electron from outer shell falls into the vacancy and the excess energy is usually released in form of a photon. Many a times an atom chooses to release this excess energy by ejecting another electron. The ejected electron is called Auger electron. In one such event, the bombarding electron removed an electron from K shell of a target atom. An electron from L shell falls to occupy the vacant K shell position, and the excess energy is used by the atom to eject an Auger electron from L shell. What will be kinetic energy of such an Auger electron if the ionization energies for K and L shell of the atom is E_k and E_L respectively.

LEVEL 3

Q. 71: A hydrogen atom in ground state is moving with a kinetic energy of 30 eV. It collides with a deuterium atom in ground state at rest. The hydrogen atom is scattered at right angle to its original line of motion. Assume that energy of n^{th} state in both the atoms is given by $E_n = -\dfrac{13.6}{n^2}$ eV and the mass of deuterium is twice that of hydrogen. Write the maximum and minimum possible kinetic energy of deuterium after collision.

Q. 72: A hydrogen atom at rest de-excites from $n = 2$ state to $n = 1$ state. Calculate the percentage error in calculation of momentum of photon if recoil of atom is not taken into account. The energy equivalent of rest mass of hydrogen atom is 940 MeV.

ANSWERS

1. 

2. It becomes reddish. Average photon energy decreases.

3. No change.

4. $5.7 \times 10^4\ m^{-2}s^{-1}$

5. $750\ Ks^{-1}$

6. (a) $\phi_A = 2.25$ eV, $\phi_B = 4.2$ eV
(b) 2 eV

7. 523 Å

8. $E = \left(\dfrac{mg^2h^2}{2}\right)^{1/3}$

9. 0.15 keV

10. $\dfrac{\Delta\lambda}{\lambda} = \dfrac{\Delta V}{2V}$

11. $\dfrac{8}{\sqrt{3}}$

12. 1.02 Å

13. No

14. (i) $\dfrac{IA}{c}$ (ii) $\dfrac{2IA}{c}$
(iii) $\dfrac{IA\cos\theta}{c}$ (iv) $\dfrac{2IA\cos^2\theta}{c}$

15. $\theta = \tan^{-1}(0.8)$ with horizontal

16. (a) 3.96×10^{26} W (b) 1.19×10^{45}
(c) 4.4×10^9 kg/s.

17. $R = \dfrac{3P_{sun}}{16\pi cdGM_{sun}}$

19. 6.613 Å

20. (a) 2 (b) $Z = 6$
(c) $\lambda_{min} = 28.5\ Å$ (d) 489.6 eV

21. $r = \left[\dfrac{n^2h^2}{8am\pi^2}\right]^{1/4}$

22. $Z = 3$

23. $n \simeq 25$

24. $n = 7$; $Z = 3$

25. (a) 2 (b) yes

26. (b) $z \simeq 48$

27. 8×10^{23}

28. 10.2 volt < V ≤ 12.09 volt.

29. 0.9 Å

30. (b) The two are same

31. 5.8 cm

32. 1218 Å

33. (a) 2 (b) 5

34. 79.4 eV

35. An electron having 11 eV kinetic energy.

36. 94.7 nm, 96.9 nm, 102.2 nm, 121 nm

37. 3 eV

38. $_{28}Ni$

39. (a) 0.08 nm
(b) Energy of an electron in K-shell lies between −25 keV and −20 keV.

40. Cu ($Z = 29$)

41. 14.8 kV

42. 4 elements ($Z = 24, 25, 26, 27$)

43. B_0

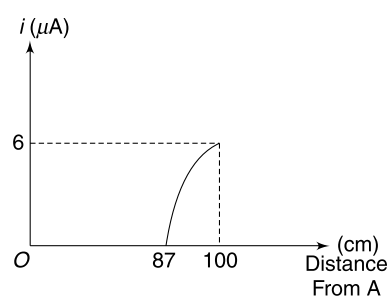
44. 1.64×10^{19} m

45. 4.22×10^{26}

46. 7 W

47. 25 Volt

48. $\phi = 3$ eV

49. (a) Zero
(b) 

(c) 0.8 eV

50. (a) 178 s (b) $8.12 \times 10^{12}\ s^{-1}$
(c) 65 nA (d) 2.68 V

51. (a) 5536 Å (b) 0.089 W

52. $4F_0$

53. $\lambda_0 = \dfrac{2h}{\sqrt{2mk} + \dfrac{k}{c}}$

54. $\dfrac{Wd^2}{8R^2c}$

55. 1.33×10^{-7} N

56. $\dfrac{2P_0}{c}\cos 15°$

57. (a) $\dfrac{\pi}{2} + \sin^{-1}\left(\dfrac{\sigma ET}{2u}\right)$ (b) $\sqrt{u^2 - \dfrac{\sigma^2 E^2 T^2}{4}} \cdot T$

59. $\phi = \sin^{-1}\left(\dfrac{nh}{d\sqrt{2\text{ meV}}}\right)$

60. (a) $\dfrac{\text{h}^2}{\text{mL}^2}$ (b) 5×10^{-54} J

61. 7.77 keV

62. (a) $Z' = 1.26$ (b) $\lambda_{\max} = 415$ nm

63. −8.28 eV, −2.07 eV, −0.92 eV. Ionization Potential = 8.28 eV

64. (a) $\dfrac{8\pi}{3}\left(\dfrac{v}{c}\right)^3$ (b) No, Yes

65. 1

66. (a) 5 eV (b) $n = 6$, $Z = 3$

67. 1217 Å

68. (a) 20.4 eV (b) 8.72 eV or 4.19 eV

69. (a) 10^5 volt (b) $\lambda_{k\alpha} = 0.23$ Å

(c) $Z = 74$

70. $E_k - 2E_L$

71. 20 eV, 13.2 eV

72. $5.4 \times 10^{-7}\%$

SOLUTIONS

2. The heat dissipated in the filament will decrease and the temperature of the bulb will fall. More light of longer wavelength will be emitted (Wien's law). A photon of longer wavelength has lesser energy.

4. Number of photons emitted (per second) from the source is

$$N = \frac{\text{Power}}{\dfrac{hc}{\lambda}} = \frac{0.6 \times 10^{-3} \times 600 \times 10^{-9}}{6.63 \times 10^{-34} \times 3 \times 10^8}$$

$$= 1.8 \times 10^{15} \text{ per second}$$

Beam diameter $d = r\theta = 4 \times 10^8 \times 0.5 \times 10^{-3}$

$$= 2 \times 10^5 \text{ m}$$

Area illuminated on the moon, $A = \pi\left(\dfrac{d^2}{4}\right)$

$$= 3.14 \times \frac{4 \times 10^{10}}{4} = 3.14 \times 10^{10} \text{ m}^2$$

∴ Number of photons incident per unit area per second

$$= \frac{1.8 \times 10^{15}}{3.14 \times 10^{10}} = 5.7 \times 10^4 \text{ m}^{-2}\text{s}^{-1}$$

5. Force = (Change in momentum of one photon) × (number of photons striking per second)

$$\Rightarrow \quad 10^{-5} = \left(\frac{h}{\lambda}\right)(n) = \frac{hc}{\lambda}\frac{n}{c} = \frac{E}{c}$$

Where E = energy incident per second.

$$\therefore \quad E = 10^{-5} \times 3 \times 10^8 = 3 \times 10^3 \text{ Js}^{-1}$$

Now $ms\dfrac{d\theta}{dt} = 3 \times 10^3$

$$\Rightarrow \quad \frac{d\theta}{dt} = \frac{3 \times 10^3}{10 \times 10^{-3} \times 400} = 750 \text{ Ks}^{-1}$$

6. Energy of photons incident on A is

$$E_1 = \frac{hc}{\lambda_1} = \frac{12410}{2920} = 4.25 \text{ eV}$$

$$\therefore \quad k_1 = 4.25 - \phi_1 \qquad \text{...(i)}$$

Energy of photons incident on B is

$$E_2 = \frac{hc}{\lambda_2} = \frac{12410}{2640} = 4.70 \text{ eV}$$

$$\therefore \quad k_2 = 4.70 - \phi_2$$

$$\therefore \quad k_1 - 1.5 = 4.70 - \phi_2 \qquad \text{...(ii)}$$

Combining (i) and (ii) gives

$$\phi_2 - \phi_1 = 1.95 \text{ eV} \qquad \text{... (iii)}$$

The smallest de-Broglie wavelength corresponds to the electron with maximum kinetic energy

$$\therefore \quad \lambda = \frac{h}{\sqrt{2mk_1}}$$

And
$$2\lambda = \frac{h}{\sqrt{2mk_2}}$$

Taking ratio
$$2 = \sqrt{\frac{k_1}{k_2}} = \sqrt{\frac{k_1}{k_1 - 1.5}}$$

$$\Rightarrow \quad 4 = \frac{k_1}{k_1 - 1.5} \quad \Rightarrow \quad k_1 = 2 \text{ eV}$$

Put this in (i) to get $\phi_1 = 2.25$ eV

Substituting in (iii) gives

$$\phi_2 = 4.2 \text{ eV.}$$

7.
$$\phi_x = 3.5 \text{ eV}$$

$$\therefore \quad E - 3.5 = k_x$$

Where E = energy of incident photon

k_x = maximum KE of emitted photoelectrons.

Also
$$\phi_Y = \frac{13.6}{(2)^2}(2)^2 = 13.6 \text{ eV}$$

$$\therefore \quad E - 13.6 = k_y$$

Given
$$k_x = 2k_y$$

$$\Rightarrow \quad E - 3.5 = 2E - 27.2$$

$$\Rightarrow \quad E = 23.7 \text{ eV}$$

$$\Rightarrow \quad \frac{hc}{\lambda} = 23.7 \text{ eV}$$

$$\therefore \quad \lambda = \frac{12400 \text{ eVÅ}}{23.7 \text{ eV}} = 523 \text{ Å}$$

8. Mechanical energy
$$E = mgH = \frac{p^2}{2m}$$

p = momentum at the lowest point (i.e., maximum momentum of the particle)

$$\therefore \quad mgH = \frac{p^2}{2m} = \frac{(h/\lambda)^2}{2m} \qquad \left[\because \; p = \frac{h}{\lambda}\right]$$

When quantum effects are considerable $\lambda \simeq H$

$$\therefore \quad mgH = \frac{h^2}{2mH^2}$$

$$\therefore \quad H = \left(\frac{h^2}{2m^2 g}\right)^{1/3}$$

$$\therefore \quad E = mgH = \left(\frac{mg^2 h^2}{2}\right)^{1/3}$$

9. Diameter of H – atom $\simeq 1.0$ Å

De-Broglie wavelength of electron having kinetic energy k is

$$\lambda = \frac{h}{\sqrt{2mk}}$$

One can "see" hydrogen atom if

$$\lambda = 1\text{Å}$$

$$\Rightarrow \quad \frac{h}{\sqrt{2mk}} = 10^{-10} \quad \Rightarrow \quad k = \frac{(6.63 \times 10^{-34})^2}{10^{-20} \times 2 \times 9.1 \times 10^{-31}}$$

$$k = 2.4 \times 10^{-17} \text{ J} = \frac{2.4 \times 10^{-17}}{1.6 \times 10^{-19}} \text{ eV}$$

$$= 1.5 \times 10^2 \text{ eV} \simeq 0.15 \text{ keV}$$

10. KE of an electron accelerated through a potential difference of V is

$$k = \text{eV}$$

$$\Rightarrow \quad \frac{p^2}{2m} = \text{eV} \quad \Rightarrow \quad p = \sqrt{2\text{ meV}}$$

$$\therefore \quad \lambda = \frac{h}{p} = \frac{h}{\sqrt{2\text{ meV}}}$$

$$\Rightarrow \quad \frac{\Delta\lambda}{\lambda} = -\frac{1}{2}\frac{\Delta V}{V}$$

11.

$$V_{\text{rms}} = \sqrt{\frac{\gamma RT}{M}}$$

$$\therefore \quad \frac{V_H}{V_0} = \sqrt{\frac{T_H}{M_H} \cdot \frac{M_0}{T_0}} = \sqrt{\frac{300}{2} \cdot \frac{32}{400}} = \sqrt{12}$$

$$\frac{\lambda_H}{\lambda_0} = \frac{h}{m_H V_H} \cdot \frac{m_0 V_0}{h} = \frac{m_0}{m_H} \frac{V_0}{V_H}$$

$$= \frac{32}{2} \cdot \frac{1}{\sqrt{12}} = \frac{8}{\sqrt{3}}$$

12. Mean kinetic energy of one atom $= \frac{3}{2} kT$

$$p = \sqrt{2m\left(\frac{3}{2} kT\right)}$$

$$= \sqrt{3 \times 2 \times 1.66 \times 10^{-27} \times 1.38 \times 10^{-23} \times 400}$$

$$= 6.50 \times 10^{-24} \text{ kg m/s}$$

$$\therefore \quad \lambda = \frac{h}{p} = \frac{6.63 \times 10^{-34}}{6.50 \times 10^{-24}}$$

$$= 1.02 \times 10^{-10} \text{ m} = 1.02 \text{ Å}$$

13.
$$p = \frac{h}{\lambda} = \frac{6.63 \times 10^{-34}}{10^{-14}\ \text{m}}\ \text{J-s} = 6.63 \times 10^{-20}\ \text{kg ms}^{-1}$$
$$m_0 v = p$$
$$v = \frac{6.63 \times 10^{-20}\ \text{J-s}}{9.1 \times 10^{-31}\ \text{kg}} = 7.3 \times 10^{10}\ \text{m/s}$$

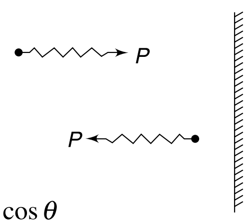
This speed is not possible as it is larger than speed of light.
$$m = \frac{m_0}{\sqrt{1 - v^2/c^2}}$$
$$\therefore \quad \frac{m_0}{\sqrt{1 - v^2/c^2}} \cdot v = p$$
$$\frac{9.1 \times 10^{-31}\, v}{\sqrt{1 - \frac{v^2}{(3 \times 10^8)^2}}} = 6.63 \times 10^{-20}$$
$$\Rightarrow \quad (1.37 \times 10^{-12})^2 v^2 = 1 - 1.11 \times 10^{-17} v^2$$
$$\Rightarrow \quad [1.11 \times 10^{-17} + 1.88 \times 10^{-24}]\, v^2 = 1$$
$$\Rightarrow \quad v^2 \simeq \frac{10^{17}}{1.11} = 0.9 \times 10^{17}$$
$$v = 3 \times 10^8\ \text{m/s}$$

14. (i) Incident power $= IA$

Momentum incident per unit time $= \frac{IA}{c}$ $\quad \left[\because\ p = \frac{E}{c}\right]$

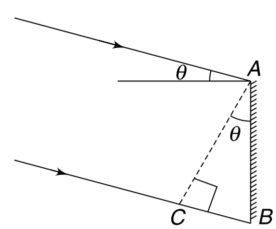
$\therefore$ Force $\quad F = \frac{IA}{c}$

(ii) Momentum transferred to the wall per unit time



$$= \frac{2IA}{c}$$
$$\therefore \quad F = \frac{2IA}{c}$$

(iii) Energy incident on the surface AB in unit time = Energy incident on AC in unit time $= IA\cos\theta$ [where $A\cos\theta$ = area of surface AC]

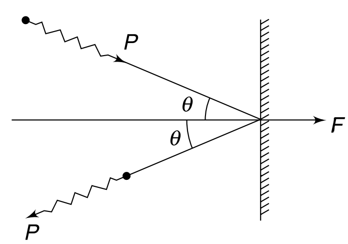


Momentum transferred per unit time $= \frac{IA\cos\theta}{c}$

$\therefore \quad F = \frac{IA\cos\theta}{c}$ [Force is along the direction of the beam]

(iv) Incident power $= IA\cos\theta$

Momentum incident per unit time $= \frac{IA\cos\theta}{c}$

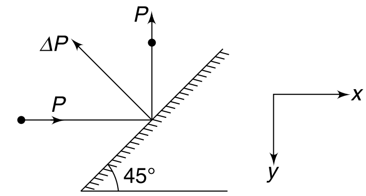


Momentum transferred per unit time to the surface
$$= 2\left(\frac{IA\cos\theta}{c}\right)\cos\theta$$

Since change in momentum of each photon is $2p\cos\theta$ if p is momentum of each photon.
$$\therefore \quad F = \frac{2IA}{c}\cos^2\theta$$

Force is normal to the surface, i.e., along the direction of change in momentum.

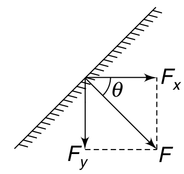
15. If n is number of photons incident per second on the mirror, $0.8n$ is the number that suffers reflection. $0.2n$ is absorbed.



The reflected photons suffer change in momentum in a direction perpendicular to the mirror.

If each incident photon has momentum $= p$, then change in momentum of each incident photon is $\Delta p = \sqrt{2}\,p$ (normal to mirror)

And change in momentum of each absorbed photon is $= p$ (in horizontal direction)



$$\therefore \quad F_x = 0.8n\left(\sqrt{2}\,p\right)\cos 45° + 0.2\cdot n\cdot p = np$$

$$F_y = 0.8n\left(\sqrt{2}\,p\right)\sin 45° = 0.8\,np$$

Alternate:

All photons lose their entire momentum in x direction.

$$\therefore \quad F_x = np$$

$0.8n$ photons acquired a momentum p in upward direction

$$\therefore \quad F_y = 0.8\,np$$

Resultant makes an angle θ with horizontal where

$$\tan\theta = \frac{F_y}{F_x} = 0.8.$$

16. (a) Power of the Sun

$$p = 4\pi r^2\cdot I$$

$$p = 4 \times 3.14 \times (1.5 \times 10^{11})^2 \times 1400$$

$$= 3.96 \times 10^{26}\ \text{W}$$

(b) Energy of one photon

$$E = \frac{hc}{\lambda} = \frac{6.64 \times 10^{-34} \times 3 \times 10^8}{6000 \times 10^{-10}}$$

$$= 3.32 \times 10^{-19}\ \text{J}$$

Number of photons emitted per second

$$n = \frac{3.96 \times 10^{26}}{3.32 \times 10^{-19}} = 1.19 \times 10^{45} \text{ photons per second}$$

(c) $1\ \text{kg} = 1 \times (3 \times 10^8)^2\ \text{J/c}^2$

$= 9 \times 10^{16}\ \text{J/c}^2$

$\therefore$ Decrease in mass of the sun in one second

$$= \frac{3.96 \times 10^{26}}{9 \times 10^{26}} = 4.4 \times 10^9\ \text{kg}$$

17. The radiation pressure is give by $P = \frac{I}{c}$.

If r is distance of the sphere from the centre of the Sun and R is radius of the sphere itself then-

$$\frac{GM_{sun}m}{r^2} = \frac{I}{c}\,\pi R^2$$

But $m = \frac{4\pi R^3}{3}d$ and $I = \frac{P_{sun}}{4\pi R^2}$

$$\therefore \quad \frac{GM_{sun}\dfrac{4\pi R^3}{3}d}{r^2} = \frac{\dfrac{P_{sun}}{4\pi r^2}}{c}\,\pi R^2$$

$$\therefore \quad R = \frac{3P_{sun}}{16\pi c d G M_{sun}}$$

This is independent of distance from the sun.

18. $$2\pi r = n\lambda$$

$\Rightarrow$ $$2\pi r = n\frac{h}{mv}$$

$\Rightarrow$ $$mvr = \frac{nh}{2\pi}$$

19. As per question $n = 5$; $Z = 2$

$\therefore$ $$r = 0.529\frac{n^2}{Z}\text{ Å}$$

$$= 0.529 \times \frac{25}{2} = 6.613\text{ Å}$$

20. (a) Since 3 radiations are emitted, the final state is $n = 3$ and initial state must be is $n = 2$; because the emitted radiations have equal or smaller wavelength than of absorbed photon.

(b) $13.6Z^2\left[\frac{1}{2^2} - \frac{1}{3^2}\right] = 68 \quad \Rightarrow Z = 6$

(c) $\frac{1}{\lambda_{\min}} = RZ^2\left[\frac{1}{1^2} - \frac{1}{3^2}\right] \quad \Rightarrow \lambda_{\min} = 28.5Å$

(d) $E = 13.6Z^2 = 13.6 \times 6^2 = 489.6$ eV

21. $$U = ar^2$$

$\Rightarrow$ $$F = -\frac{dU}{dr} = -2ar$$

Negative sign indicates that force is towards the centre of the circle. It must be centripetal force.

$\therefore$ $$\frac{mV^2}{r} = 2ar$$

$\Rightarrow$ $$v = \sqrt{\frac{2a}{m}}\cdot r \quad \text{...(i)}$$

Given: $$mvr = n\frac{h}{2\pi}$$

$\Rightarrow$ $$m\left(\sqrt{\frac{2a}{m}}\cdot r\right)r = \frac{nh}{2\pi}$$

$\Rightarrow$ $$r = \left[\frac{n^2h^2}{8am\pi^2}\right]^{1/4}$$

22. Number of lines in the spectrum = ${}^nC_2$

$\Rightarrow$ $$\frac{n(n-1)}{2} = 15 \quad \Rightarrow \quad n = 6$$

The second most energetic photon will be emitted due to transition $n = 6 \rightarrow n = 2$ [The most energetic photon will result from $n = 6 \rightarrow n = 1$ transition.

$\therefore$ $$13.6\left[\frac{1}{2^2} - \frac{1}{6^2}\right]Z^2 = 27.2$$

$\Rightarrow$ $$Z^2 = 2 \times \frac{36}{8}$$

$\Rightarrow$ $$Z = 3$$

23. $$\frac{mv^2}{r} = \frac{Ze^2}{4\pi\varepsilon_0 r^2}$$

and $$mvr = \frac{nh}{2\pi}$$

Solving the above two equations we get

$$r = \frac{n^2 h^2 \varepsilon_0}{Z \cdot \pi m e^2}$$

Put $Z = 3$; $m = 208\, m_e$

$$r_\mu = \frac{n^2 h^2 \varepsilon_0}{624\, \pi m_e e^2}$$

According to the question

$$\frac{n^2 h^2 \varepsilon_0}{624\, \pi m_e e^2} = \frac{h^2 \varepsilon_0}{\pi m_e e^2}$$

$\Rightarrow$ $$n^2 = 624$$

$\Rightarrow$ $$n \simeq 25.$$

24. $$13.6 \left[\frac{1}{2^2} - \frac{1}{n^2}\right] Z^2 = 22.94 + 5.15 = 28.09$$

$\Rightarrow$ $$\left(\frac{1}{4} - \frac{1}{n^2}\right) Z^2 = 2.065 \qquad \ldots(i)$$

$\Rightarrow$ $$E_n - E_2 = 28.09 \text{ eV}$$

$$E_n - E_3 = 2.4 + 8.7 = 11.1 \text{ eV}$$

$\Rightarrow$ $$E_3 - E_2 = 16.99 \text{ eV}$$

$\Rightarrow$ $$13.6\, Z^2 \left[\frac{1}{4} - \frac{1}{9}\right] = 16.99$$

$\Rightarrow$ $$Z^2 \simeq 9$$

$$Z = 3$$

Put in (i) to get $n = 7$

25. (a) Reduced mass $\mu = \dfrac{m \cdot m}{m + m} = \dfrac{m}{2}$

$\therefore$ Energy values are $\dfrac{1}{2}$ compared to hydrogen.

(b) Transition between any two energy states in Positronium will result in emission of photon having energy half that of corresponding photon in hydrogen.

$\therefore$ Each line in spectrum will have twice the wavelength of the corresponding lines in H – atom.

26. (a) For hydrogen like atom the energy in n^{th} state is $E_n = -13.6\, \dfrac{\mu Z^2}{n^2}$

For mesic atom $\mu \simeq 200\, m_e$

The transition energies are enhanced by a factor of 200, so that emitted radiation falls in X – ray region instead of UV, IR or visible part of electromagnetic spectrum.

(b) Nuclear radius $$R = R_0 A^{1/3}$$

$$= (1.2 \times 10^{-15})\ (2Z)^{1/3}$$

Atomic radius of first orbit is

$$r_1 = \frac{0.529 \times 10^{-10}}{200 \times Z}$$

Since $$r_1 = R$$

$\therefore$ $$\frac{0.529 \times 10^{-10}}{200 Z} = (1.2 \times 10^{-15})(2Z)^{1/3}$$

$\Rightarrow$ $$Z^{4/3} = \frac{0.529 \times 10^5}{1.2 \times (2)^{1/3} \times 200}$$

$$\Rightarrow \qquad Z^4 = \left(\frac{0.529 \times 10^3}{1.2 \times 2}\right)^3 \times \frac{1}{2} = 5.35 \times 10^6$$

$$\Rightarrow \qquad Z^2 = 2.31 \times 10^3$$

$$\Rightarrow \qquad Z \simeq 48$$

27. Radius of each atom

$$r = r_0 \cdot n^2 = 0.53\,\text{Å} \times (10)^2$$

$$= 53\ \text{Å}$$

No. of atoms in one line

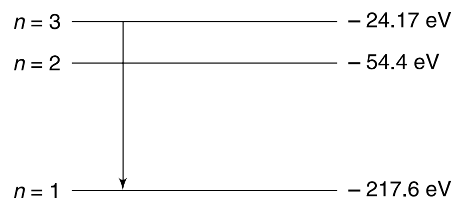
$$N = \frac{1\ \text{m}}{2 \times 53\ \text{Å}} = \frac{10^{10}}{106} = 9.4 \times 10^7$$

$\therefore$ No. of atoms in the box

$$= N^3 = (9.4 \times 10^7)^3 = 8 \times 10^{23}$$

29. Energy levels for Be^{+++} ($Z = 4$) are as shown.

Transition ($3 \rightarrow 1$) releases maximum energy photons



$$E = -24.17 - (-217.6) = 193.4\ \text{eV}$$

When these photons hit the metal plate, electrons are emitted with maximum KE given by

$$K_{max} = E - \phi = 193.4 - 5 = 188.4\ \text{eV}$$

$$= 188.4 \times 1.6 \times 10^{-19}\ \text{J} = 3.01 \times 10^{-17}\ \text{J}$$

De-Broglie wavelength of such electrons will be minimum

$$\lambda_{min} = \frac{h}{\sqrt{2mK_{max}}} = \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 3.01 \times 10^{-17}}}$$

$$= 0.90\ \text{Å}$$

30. (a) Radius of orbit for n^{th} state and speed of electron in the orbit is given by

$$r = \frac{\varepsilon_0 h^2 n^2}{\pi m Z e^2}$$

$$v = \frac{Ze^2}{2\varepsilon_0 hn}$$

$\therefore$ Frequency of orbital revolution of electron in n^{th} state is

$$f_n = \frac{v}{2\pi r} = \frac{me^4 Z^2}{4\varepsilon_0^2 h^3 n^3} \qquad \text{...(i)}$$

Frequency of emitted radiation can be obtained as

$$hf_0 = E_n - E_{n-1}$$

$$hf_0 = \frac{me^4}{8\varepsilon_0^2 h^2} Z^2 \left[\frac{1}{(n-1)^2} - \frac{1}{n^2}\right]$$

$$f_0 = \frac{me^4}{8\varepsilon_0^2 h^3} Z^2 \left[\frac{2n-1}{(n-1)^2 n^2}\right] \qquad \text{...(ii)}$$

You can show (or else verify) that

$$\frac{1}{n^3} < \frac{2n-1}{(n-1)^2 n^2} < \frac{1}{(n-1)^3}$$

Hence, from (i) and (ii)

$$f_n < f_0 < f_{n-1}$$

(b) When $n >> 1$

$$f_0 \simeq \frac{me^4 Z^2}{8\varepsilon_0^2 h^3} \frac{2}{n^3} = \frac{me^4 Z^2}{4\varepsilon_0^2 h^3 n^3}$$

Therefore, for large quantum number the frequency of emitted radiation is equal to the orbital frequency of the electron. It means for large quantum numbers the results predicted by quantum mechanics are nearly same as that given by classical mechanics.

31. The outer electron will effectively see attraction of one positive charge. The electron present in first orbit will screen the attraction of two protons. Since outer electron is in an orbit of large radius it will see one positive charge at nucleus.

32. The hydrogen atom is having momentum equal to that of the photon and the two momenta are in opposite directions.

The kinetic energy of hydrogen atom will be negligible compared to the energy of photon. The entire photon energy is absorbed as excitation energy by the hydrogen atom. Hence minimum energy of photon must be 10.2 eV.

$$\lambda_{max} = \frac{hc}{10.2 \text{ eV}} = \frac{1240 \text{ eVÅ}}{10.2 \text{ eV}} = 1218 \text{ Å}.$$

33. There are 6 lines in spectrum. 4, 5 and 6 represent transitions: $2 \to 1$, $3 \to 1$, $4 \to 1$ respectively. The wavelength difference between 4 and 5 will be higher than the difference between 5 and 6.

Line 2 and 3 are the first two lines of Balmer series and line 1 is the first line of Paschen series. The gap between 3 and 4 is large which means energy difference of photon corresponding to 3 and 4 is large. In fact, among all emitted photons, the energy difference between these two photons is largest.

34. When one e^- is removed the remaining atom is hydrogen like atom with $Z = 2$.

Binding energy of such atom will be

$$= 13.6 \times 2^2 = 54.4 \text{ eV}$$

$\therefore$ Total energy needed $= 54.4 + 25 = 79.4$ eV

36. Absorption lines will correspond to Lyman series in the emission spectrum.

$$\frac{1}{\lambda} = R\left[\frac{1}{1^2} - \frac{1}{n^2}\right] \qquad n = 2, 3, 4, 5...$$

For $n = 2$; $\lambda = \dfrac{4}{3R} = 1.21 \times 10^{-7}\text{m} = 121\,\text{nm}$

Similarly, for $n = 3$; $\lambda = 102.2$ nm

for $n = 4$; $\lambda = 96.9$ nm

for $n = 5$; $\lambda = 94.7$ nm

for $n = 6$; $\lambda = 93.5$ nm

$\therefore$ Required answers are 94.7 nm, 96.9 nm, 102.2 nm and 121 nm.

37. We have,

$$10.2 = W + K_1 \qquad ...(i)$$

and

$$10.2\, Z^2 = W + K_2 \qquad ...(ii)$$

Also

$$\lambda = \frac{h}{P} = \frac{h}{\sqrt{2mK}}$$

$\therefore$

$$\frac{\lambda_1}{\lambda_2} = \sqrt{\frac{K_2}{K_1}} = 2.3 \quad \Rightarrow \quad K_2 = 5.25\, K_1 \qquad ...(iii)$$

Also $10.2\, Z^2$ = energy corresponding to longest wavelength of the Lyman series = 3×13.6

$\Rightarrow$ $Z = 2$.

$\therefore$ From equations (i), (ii) and (iii)

$$W = 3 \text{ eV}.$$

38. $$\frac{1}{\lambda_{k\alpha}} = \frac{3}{4}R(Z-1)^2$$

$\therefore$ $$\frac{\lambda}{\lambda_{\text{cu}}} = \frac{(29-1)^2}{(Z-1)^2}$$

$\therefore$ $$(Z-1)^2 = 28^2 \times \frac{1.5404}{1.6578}$$

$\therefore$ $$Z - 1 = 27$$

$\therefore$ $$Z = 28$$

40. $$\lambda_k = \frac{4}{3R(Z-1)^2} = \frac{1200}{(Z-1)^2}\ \text{Å}$$

$$\lambda_c = \frac{12.4}{V} \quad \text{where } V \text{ is in } kV$$

$\therefore$ $$\frac{1200}{(Z-1)^2} - \frac{12.4}{V} = \Delta\lambda$$

$\Rightarrow$ $$\frac{1200}{(Z-1)^2} - \frac{12.4}{10} = \Delta\lambda \qquad \text{...(i)}$$

and $$\frac{1200}{(Z-1)^2} - \frac{12.4}{20} = 3\Delta\lambda \qquad \text{...(ii)}$$

Eliminating $\Delta\lambda$ between (i) and (ii) we get

$$Z = 28.8 \simeq 29$$

41. For nickel $Z = 28$

$$\lambda_k = \frac{4}{3R(Z-1)^2} = \frac{4}{3 \times 1.09 \times 10^7 \times (28-1)^2}\ \text{m}$$

$$= \frac{4000}{3 \times 1.09 \times 27^2}\ \text{Å}$$

$$= \frac{1223}{27^2}\ \text{Å} = 1.68\ \text{Å}$$

$$\lambda_k - \lambda_c = 0.84$$

$\Rightarrow$ $$\lambda_c = 0.84\ \text{Å}$$

$\Rightarrow$ $$\frac{hc}{eV} = 0.84\ \text{Å}$$

$\Rightarrow$ $$\frac{12400\ \text{eV Å}}{V} = 0.84\ \text{Å}$$

$\Rightarrow$ $$V = 14762\ \text{volt} = 14.8\ \text{kV}.$$

42. $$\lambda_k = \frac{4}{3R(Z-1)^2} = \frac{4}{3 \times 1.09 \times 10^7 (Z-1)^2}\ \text{m.}$$

$$= \frac{4000}{3 \times 1.09} \frac{1}{(Z-1)^2}\ \text{Å}$$

$$= \frac{1223}{(Z-1)^2}\ \text{Å}$$

For $$\lambda_k = 2.41\ \text{Å}$$

$$(Z-1)^2 = \frac{1223}{2.41} = 507$$

$$\Rightarrow \qquad Z - 1 = 22.53 \quad \Rightarrow \quad Z = 23.53$$

Similarly for $\lambda_k = 1.80$ Å

$$(Z-1)^2 = \frac{1223}{1.8} = 679.4$$

$$\Rightarrow \qquad Z = 27.06$$

$\therefore$ The elements have $Z = 24, 25, 26, 27$.

44. Power of the sun $P = I.4\pi r^2$

The star has same power but it is at a distance x from us.

$\therefore$ Intensity of star light at earth is

$$I_S = \frac{P}{4\pi x^2}$$

$\therefore$ Energy from star that enters an eye in one second is

$$E = I_S \cdot \pi\left(\frac{d}{2}\right)^2 = \frac{P}{16}\frac{d^2}{x^2}$$

This must be at least equal to 10 times the energy of a photon of yellow light

$$\frac{Pd^2}{16x^2} = 10\,\frac{hc}{\lambda}$$

$$\Rightarrow \qquad x^2 = \frac{Pd^2\lambda}{160\,hc} = \frac{I.4\pi r^2 d^2\lambda}{160\,hc}$$

$$x^2 = \frac{1.4 \times 10^3 \times 4 \times 3.14 \times (1.5 \times 10^{11})^2 \times (6 \times 10^{-3})^2 \times (6 \times 10^{-7})}{160 \times 6.63 \times 10^{-34} \times 3 \times 10^8}$$

$$= 2.68 \times 10^{38}$$

$$\Rightarrow \qquad x = 1.64 \times 10^{19}\text{ m}$$

45. Energy needed to warm the water

$$E_0 = ms\Delta\theta = (200\text{ g})(4.2\text{ Jg}^{-1}{}^\circ\text{C}^{-1})(50^\circ\text{C})$$

$$= 4.2 \times 10^4\text{ J}$$

Energy of one photon, $E = hv = 6.63 \times 10^{-34} \times 2.5 \times 10^9 = 1.66 \times 10^{-24}$ J

Number of microwave photons absorbed by water $= \dfrac{4.2 \times 10^4}{1.66 \times 10^{-24}} = 2.53 \times 10^{28}$

Number of photons emitted by the oven in 2 min is $= \dfrac{2.53 \times 10^{28}}{0.05} = 5.06 \times 10^{29}$

Number of photons emitted per second $= \dfrac{5.06 \times 10^{29}}{2 \times 60} = 4.22 \times 10^{27}$

Photon energy emitted per second by oven $= \dfrac{1}{0.05} \times \dfrac{4.2 \times 10^4\text{ J}}{120\text{ s}} = 7000$ W

Electrical energy consumed per second $= \dfrac{7000}{0.7} = 10000$ W $= 10$ kW

Number of photons emitted per kW of power consumed $= \dfrac{4.22 \times 10^{27}}{10} = 4.22 \times 10^{26}$

46. Stopping potential difference = 5 volt

$\therefore$ Energy of incident photon $h\upsilon = 5 + \phi = 7$ eV

Saturation current is $i_s = 10\,\mu$A

$\therefore$ Number of electrons emitted per second is

$$n_e = \frac{i_s}{e} = \frac{10 \times 10^{-6}}{1.6 \times 10^{-19}} = \frac{1}{1.6} \times 10^{14}$$

No of photons incident per second

$$n_p = n_e \times \frac{100}{10^{-3}} = 10^5 n_e = \frac{10^{19}}{1.6}$$

∴ Power incident = $(7\,\text{eV})\,n_p$

$$= (7 \times 1.6 \times 10^{-19}\text{ J}) \times \left(\frac{10^{19}}{1.6}\text{ s}^{-1}\right)$$

$$= 7\text{ J s}^{-1} = 7\text{ W}$$

47. Threshold wavelength $\lambda_{th} = 240\,\text{nm}$

∴ Work function, $\phi = \dfrac{hc}{\lambda_{th}}$

$$= \frac{(4 \times 10^{-15}\,\text{eVs})(3 \times 10^8\,\text{m s}^{-1})}{240 \times 10^{-9}} = 5\text{ eV}$$

With light of wavelength $2\lambda_0$, the maximum kinetic energy of emitted electrons is 10 eV.

∴ Energy of photon of wavelength $2\lambda_0$ is = 10 + 5 = 15 eV

∴ Energy of photon of wavelength λ_0 is = 15 × 2 = 30 eV

With light of wavelength λ_0, the maximum kinetic energy of emitted electrons is = 30 – 5 = 25 eV

∴ $V_0 = 25\text{ V}$

48. In the experiment performed in figure (a), the applied potential difference is retarding.

∴ Electrons lose 5 eV kinetic energy in travelling from *A* to *B*.

Hence K_{max} for electrons emitted from plate *A* must be 6 eV. An electron emitted with KE of 6 eV will reach plate *B* with a KE of 1 eV

$$h\upsilon = \phi + 6\text{ eV} \quad \ldots\text{(i)}$$

In experiment in figure (b), the applied potential difference is accelerating. Electrons gain a KE of 5 eV in travelling from plate *A* to *B*.

∴ Maximum KE of an electrons emitted from *A* is 15 eV.

$$2h\upsilon = \phi + 15\text{ eV} \quad \ldots\text{(ii)}$$

Solving (i) & (ii)

$$\phi = 3\text{ eV}$$

49. When slider is at A, the applied potential difference across the photoelectric cell is a retarding potential difference of 10 V.

Energy of incident photon

$$E = \frac{hc}{\lambda} = \frac{1240\text{ eVnm}}{350\text{ nm}} = 3.5\text{ eV}$$

Maximum KE of emitted photoelectrons is

$$\text{KE}_{max} = \frac{hc}{\lambda} - \phi = 3.5 - 2.2 = 1.3\text{ eV}$$

∴ Stopping potential difference $V_S = 1.3\text{ V}$

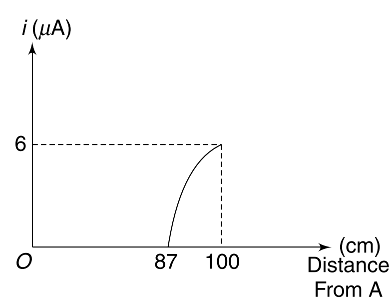
(a) The applied potential difference is

$$V(= 10\ V) > V_S\ (= 1.3\text{ V})$$

Hence there is no photocurrent and reading of Ammeter is zero.

(b) When slider is at a distance of 13 cm from *B*, the applied retarding potential difference is $V_B - V_J = 1.3$ V

When slider moves from A through 87 cm, there is no photocurrent because applied retarding potential difference is higher than V_S.



After 87 cm, the photocurrent increases as applied potential difference becomes less than V_S.

(c) Applied retarding potential difference = 0.5 V

$\therefore$ Maximum KE of electrons hitting plate C

$$= K_{max} - 0.5 \text{ eV}$$
$$= (1.3 - 0.5)\text{eV} = 0.8 \text{ eV}$$

50. (a) Energy required to eject a photoelectron is $\phi = 2.2$ eV

$$= 2.22 \times 1.6 \times 10^{-19} \text{ J} = 3.552 \times 10^{-19} \text{ J}$$

According to classical theory, energy flow is a continuous process and a photo electron will be ejected if a potassium atom receives this amount of energy over a length of time. The potassium atom is at a distance of 1.0 m from the source.

Hence, intensity of ultraviolet radiation on the potassium surface is

$$I = \frac{W}{4\pi r^2} = \frac{0.2}{4\pi(1)^2} = \frac{0.2}{4\pi} \text{ J/m}^2\text{/s}$$

Cross sectional area of one K atom is

$$A = \pi(2 \times 10^{-10}\text{m})^2 = 4\pi \times 10^{-20} \text{ m}^2$$

Hence, the exposure time is given as

$$t = \frac{3.552 \times 10^{-19} \text{ J}}{\frac{0.2}{4\pi} \text{Jm}^{-2}\text{s}^{-1} \times 4\pi \times 10^{-20}\text{m}^2} = 178 \text{ s}$$

(b) Energy of photon

$$E = hv = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{2537 \times 10^{-10}}$$
$$= 7.84 \times 10^{-19} \text{ J} = 4.9 \text{ eV}$$

Let N be the number of photons reaching the cathode per second. Then intensity at cathode is $I' = NE$

But energy falling per second on cathode is $\frac{0.2}{4\pi} \times 4 \times 10^{-4}$ J

$$\therefore \quad N = \frac{0.2 \times 4 \times 10^{-4}}{4\pi \times 7.84 \times 10^{-19}} = 8.12 \times 10^{12} \text{ photons/s}$$

(c) 10 % of photons are able to eject electrons. Hence, current is

$$i = 0.1 \times N \times 1.6 \times 10^{-19} = 65 \text{ nA}$$

(d) Cut off potential

$$V_0 = \frac{hv - \phi}{e} = \left(\frac{4.90 - 2.22}{e}\right) \text{eV} = 2.68 \text{ volts}$$

51. (a) Photoelectric equation is $k_{max} = \frac{hc}{\lambda} - \phi$

Graph between k_{max} and $\frac{1}{\lambda}$ is a straight line. $k_{max} = 0$ for wavelength given by-

$$\frac{1}{\lambda_0} = \frac{\phi}{hc}$$

$$\phi = \frac{hc}{\lambda_0} = (12400 \text{ eVÅ}) (10^6 \times 10^{-10} \text{ Å}^{-1})$$

$$\left[\text{From graph } \frac{1}{\lambda_0} = 10^6\text{m}^{-1} = 10^6 \times 10^{-10} \text{ Å}^{-1}\right]$$

$$\therefore \quad \phi = 1.24 \text{ eV}$$

When light of wavelength λ is used the stopping potential is $V_S = 1$ volt

$\therefore \quad k_{max} = 1 \text{ eV}$

$\therefore$ Energy of incident photon

$$E = \frac{hc}{\lambda} = \phi + k_{max} = 2.24 \text{ eV}$$

$$\therefore \quad \lambda = \frac{hc}{2.24 \text{ eV}} = \frac{12400 \text{ eV Å}}{2.24 \text{ eV}} = 5536 \text{ Å}$$

(b) Saturation current $i_s = 0.8 \times 10^{-3}$ A

Number of photoelectrons emitted is

$$n_e = \frac{i_s}{e} = \frac{0.8 \times 10^{-3}}{1.6 \times 10^{-19}} \text{ s}^{-1} = 5 \times 10^{15} \text{ s}^{-1}$$

Since out of 100 photons, only 2 emit electrons; number of photons incident will be-

$$N_P = 50 \times n_e = 2.5 \times 10^{17} \text{ s}^{-1}$$

Incident power $\quad P = EN_P = (2.24 \times 1.6 \times 10^{-19})(2.5 \times 10^{17})$ J/s

$$= 8.96 \times 10^{-2} = 0.089 \text{ watt.}$$

52. Point Q is the centre of a cube of side length L, one of the faces of which is the plate. One sixth of the total radiation strikes the plate.

Point P is centre of a cube of side length 2 L and the given plate is $\frac{1}{4}$ of one face of cube. Hence radiation striking the plate is $\frac{1}{24}$ of the total radiation emitted.

$\therefore \quad F = 4F_0$

53. Momentum conservation gives

$$\frac{h}{\lambda_1} = -\frac{h}{\lambda_2} + \sqrt{2mk} \qquad \text{...(i)}$$

Energy conservation given

$$\frac{hc}{\lambda_1} = \frac{hc}{\lambda_2} + k$$

$$\Rightarrow \quad \frac{h}{\lambda_1} = \frac{h}{\lambda_2} + \frac{k}{c} \qquad \text{...(ii)}$$

Add (i) and (ii)

$$\frac{2h}{\lambda_1} = \sqrt{2mk} + \frac{k}{c}$$

$$\Rightarrow \quad \lambda_1 = \frac{2h}{\sqrt{2mk} + \frac{k}{c}}$$

You must be able to argue why the wavelength of emitted photon will be different from the incident one?

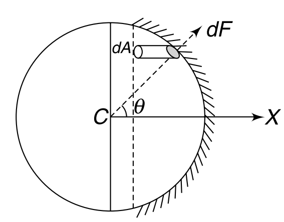
54. Intensity of light on the mirror surface $I = \frac{W}{4\pi R^2}$

Consider a small patch of area dS on the surface of the mirror. Energy incident per unit time on this area is

$$dE = IdS = \frac{WdS}{4\pi R^2}$$

Momentum incident on area $\quad dS = \frac{dE}{c} = \frac{WdS}{4\pi R^2 c}$

Light is reflected back towards the centre of the sphere, hence change in momentum per unit time for area dS is



$$= \frac{2WdS}{4\pi R^2 c}$$

This is equal to force on dS.

$$dF = \frac{2\,WdS}{4\,\pi R^2 c}$$

By symmetry the resultant force is along X direction

$$\therefore \qquad dF_X = dF\cos\theta = \frac{2W(dS\cos\theta)}{4\,\pi R^2 c}$$

Projection of dS on vertical plane (in figure) is $dA = dS\cos\theta$

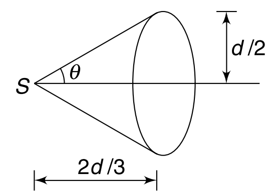
$$\therefore \qquad F_X = \frac{2W}{4\,\pi R^2 c}\int dS\cos\theta = \frac{2W}{4\,\pi R^2 c}\int dA$$

$$= \frac{2W}{4\,\pi R^2 c}\,\pi\left(\frac{d}{2}\right)^2 = \frac{Wd^2}{8R^2 c}$$

55.

$$\tan\theta = \frac{d}{2}\times\frac{3}{2d} = \frac{3}{4}$$

$$\therefore \qquad \cos\theta = \frac{4}{5}$$



$\therefore$ Solid angle subtended by the lens at the source S is given by

$$\Omega = 2\pi(1-\cos\theta) = 2\pi\left(1-\frac{4}{5}\right) = \frac{2\pi}{5}\ sr.$$

$\therefore$ Power incident on lens

$$P_0 = \frac{500}{4\pi}\Omega = 50 \text{ watt}$$

Transmitted power (P) = 0.4 × 50 = 20 W

The transmitted beam strikes the reflecting surface normally. Momentum incident per unit time on the reflecting surface $= \frac{P}{c}$

$$\therefore \qquad \text{Force} = \frac{2P}{c} = \frac{2\times 20}{3\times 10^8} = 1.33\times 10^{-7}\text{ N}$$

56. The prism is set for minimum deviation condition

$$r_1 = r_2 = A \quad\Rightarrow\quad r_1 = r_2 = 30^\circ$$

Angle of incidence (i) = Angle of emergence (e)

Snell's law gives-

$$1\cdot\sin i = \sqrt{2}\sin 30^\circ \quad\Rightarrow\quad i = 45^\circ$$

$\therefore$ Deviation

$$\delta = i + e - A$$

$$= 45^\circ + 45^\circ - 60^\circ = 30^\circ$$

The incident photons have momentum directed at 15° to the horizontal (up with horizontal) and the emergent photons have their momentum making 15° to the horizontal (down).

$\therefore$ Change in momentum of one photon is

$$\Delta P_{\text{one}} = 2P\cos 15^\circ \text{ (in vertical direction)}$$

$$= \frac{2h}{\lambda}\cos 15^\circ$$

Number of photons incident per second is

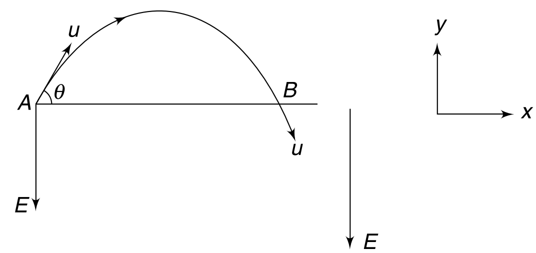
$$n = \frac{P_0}{\frac{hc}{\lambda}} = \frac{P_0\lambda}{hc}$$

$\therefore$ Vertically downward force on the prism applied by the incident beam is

$$F = n\Delta P_{\text{one}} = \frac{P_0 \lambda}{hc} \frac{2h}{\lambda} \cos 15°$$

$$= \frac{2P_0}{c} \cos 15°$$

57. Speed of 'Y' does not change. de – Broglie wavelength of 'X' will becomes equal to that of 'Y' when speed of 'X' again become equal to its initial speed 'u'. This is possible in a situation shown in figure. Proton starts from A with speed 'u' and reaches point B with same speed 'u'. [compare this to a projectile]



(a) T = time of flight from A to B

$$T = \frac{2u_y}{a} \qquad \left[a = \frac{eE}{m}\right]$$

$\therefore$ $$u_y = \frac{eET}{2m}$$

$\Rightarrow$ $$u \sin\theta = \frac{eET}{2m}$$

$$\theta = \sin^{-1}\left(\frac{eET}{2mu}\right)$$

Desired angle $$\alpha = \frac{\pi}{2} + \theta = \frac{\pi}{2} + \sin^{-1}\left(\frac{eET}{2mu}\right)$$

(b) $$u_x = \sqrt{u^2 - u_y^2} = \frac{\pi}{2} + \sin^{-1}\left(\frac{\sigma ET}{2u}\right)$$

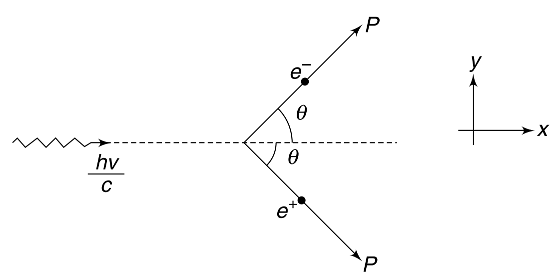
$$AB = u_x \cdot T$$

$$= \sqrt{u^2 - u_y^2} \cdot T$$

$$= \sqrt{u^2 - \frac{e^2E^2T^2}{4m^2}} \cdot T$$

$$= \sqrt{u^2 - \frac{\sigma^2E^2T^2}{4}} \cdot T$$

58. (a) To conserve momentum, the two particles (electron and positron) will move with equal momentum (P) making equal angle with the original line of motion of the photon.



For conservation of momentum along X

$$2P\cos\theta = \frac{h\nu}{c}$$

$\Rightarrow$ $$h\nu = 2Pc\cos\theta$$

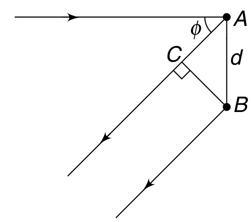
$\Rightarrow$ $$h\nu = 2(mV)\,c\cos\theta = 2mc^2\left(\frac{V}{c}\right)\cos\theta$$

Since $\frac{V}{c} < 1$ and $\cos\theta \le 1$

$\therefore$ $$h\nu < 2mc^2$$

But energy conservation requires that $h\nu$ must be equal to rest mass energy ($2mc^2$) of the particle plus their kinetic energies. Hence it is impossible to conserve both energy and momentum unless some other particle is also involved.

(b) In presence of a nucleus the process is possible. The massive nucleus can carry a lot of momentum despite having negligible kinetic energy. Now the pair particles are required to carry less momentum and it is possible to conserve momentum and energy.



59. Path difference between the wave reflected at A and B is

$$\Delta x \simeq AC = d \sin\phi$$

For constructive interference $\Delta x = n\lambda \qquad n = 1, 2, 3, 4 \ldots$

$$\Rightarrow \qquad d\sin\phi = n\lambda$$

But
$$\lambda = \frac{h}{P} = \frac{h}{\sqrt{2mK}} = \frac{h}{\sqrt{2\text{meV}}}$$

$$\therefore \qquad d\sin\phi = \frac{nh}{\sqrt{2\text{meV}}}$$

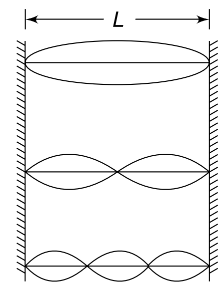
$$\therefore \qquad \phi = \sin^{-1}\left(\frac{nh}{d\cdot\sqrt{2\text{meV}}}\right)$$

60. The permitted wavelengths are

$2L, L, \frac{2L}{3} \ldots$

i.e.,
$$\lambda = \frac{2L}{n} \quad \text{where} \quad n = 1, 2, 3, 4 \ldots$$

de-Broglie wavelength will be given by-



$$\lambda_d = \frac{h}{P} = \frac{h}{\sqrt{2mK_n}} \quad [K_n \text{ is KE for } n^{\text{th}} \text{ mode}]$$

$$\therefore \qquad \frac{h}{\sqrt{2mK_n}} = \frac{2L}{n}$$

$$\therefore \qquad K_n = \frac{n^2h^2}{8\text{ mL}^2}$$

Since the particle has only KE, hence energy of the particle is

$$E_n = \frac{n^2h^2}{8\text{ mL}^2};\ n = 1, 2, 3, 4$$

(a) $E_3 - E_1 = (3^2 - 1^2)\dfrac{h^2}{8\text{ mL}^2} = \dfrac{h^2}{\text{mL}^2}$

(b) $E_1 = \dfrac{h^2}{8\text{ mL}^2} = \dfrac{(6.6 \times 10^{-34})^2}{8 \times 10^{-6} \times (10^{-2})^2} = 5 \times 10^{-54}$ J

Note: Quantization of energy is conspicuous only when m and L both are very small.

61. Energy absorbed by Li^{++} to move from $n = 2$ to $n = 3$ state is

$$\Delta E = E_3 - E_2 = 13.6\,(3)^2\left[\frac{1}{2^2} - \frac{1}{3^2}\right]$$

$$= \frac{13.6 \times 9 \times 5}{36} = 17 \text{ eV}$$

Energy released by H atom in transition $n = 2$ to $n = 1$ is

$$\Delta E' = E_2 - E_1 = 10.2 \text{ eV}$$

Thus $\Delta E - \Delta E' = 6.8$ eV must come from loss in KE in collision.

Momentum conservation warrants that momentum of H and Li^{++} after collision = momentum of hydrogen before collision (say = P).

$$\text{Loss in KE} = \frac{P^2}{2m_{\text{H}}} - \frac{P^2}{2(m_{\text{H}} + m_{\text{Li}})}$$

$$\Rightarrow \quad 6.8 \text{ eV} = \frac{P^2}{2m_H}\left[1 - \frac{m_H}{m_H + m_{Li}}\right]$$

$$\Rightarrow \quad 6.8 \text{ eV} = K_H\left[\frac{m_{Li}}{m_H + m_{Li}}\right]$$

$$\Rightarrow \quad K_H = 6.8 \times \frac{8m_H}{7m_H} = 6.8 \times \frac{8}{7} = 7.77 \text{ keV}$$

62. (a) $E'_\infty - E'_2 = 5.39$ eV

$$\Rightarrow \quad 0 + 13.6\,\frac{(Z')^2}{2^2} = 5.36$$

$$\Rightarrow \quad Z' = 1.26$$

It is logical to expect $1 < Z' < 3$ since we do not expect the screening by two electrons to be perfect.

(c) By absorbing longest wavelength photon the electron will transit from $n = 2$ to $n = 3$

$$\therefore \quad \frac{hc}{\lambda} = E'_3 - E'_2$$

$$= Z'^2 \times 13.6\left(\frac{1}{2^2} - \frac{1}{3^2}\right)$$

$$= (1.26)^2 \times 13.6 \times \frac{5}{36}$$

$$= 2.99 \text{ eV}$$

$$\therefore \quad \lambda = \frac{hc}{2.99 \text{ eV}} = \frac{12420\,\text{eVÅ}}{2.99\,\text{eV}} = 4154 \text{ Å} \simeq 415 \text{ nm}$$

63. As we know energy of a photon is given by $E = \dfrac{hc}{\lambda}$

From the given condition $\lambda = \dfrac{1500p^2}{p^2 - 1}$

Hence, $E = \dfrac{hc}{\lambda} = \dfrac{hc}{1500}\left(1 - \dfrac{1}{p^2}\right) \times 10^{10}$ J; converted into eV it becomes

$$= \frac{hc}{(1500)(1.6 \times 10^{-19})}\left(1 - \frac{1}{p^2}\right) \times 10^{10} \text{ eV}$$

$$= 8.28\left(1 - \frac{1}{p^2}\right) \text{ eV}$$

Hence energy of n^{th} state is given by $E_n = -\dfrac{8.28}{n^2}$

i.e., ionization energy $= 8.28$ eV

Hence, ionization potential $= 8.28$ V

The energy of three lowest states are obtained by putting $n = 1, 2, 3$ in the above expression. Values are -8.28 eV, -2.07 eV, -0.92 eV.

$n = 3$ _____ -0.92 eV

$n = 2$ _____ -2.07 eV

$n = 1$ _____ -8.28 eV

64. (a) We assume that orbit is circular (though the electron will spiral down into the nucleus). If orbital radius at an instant is r then equating the electrostatic force to the centripetal force we get

$$\frac{mv^2}{r} = \frac{1}{4\pi\epsilon_0}\frac{e^2}{r^2} \Rightarrow a = \frac{v^2}{r} = \frac{e^2}{4\pi\epsilon_0 mr} \qquad \text{...(i)}$$

And $$v = \sqrt{\frac{e^2}{4\pi\epsilon_0 r}}$$

Kinetic energy

$$E = \frac{1}{2}mv^2 = \frac{1}{2}\cdot\frac{1}{4\pi\epsilon_0}\frac{e^2}{r} = \frac{e^2}{8\pi\epsilon_0 r} \quad ...(ii)$$

Time period of circular motion is

$$T = \frac{2\pi r}{v}$$

Energy loss in one revolution is $\Delta E \simeq \left(\frac{dE}{dt}\right) T$

$$= \frac{e^2 a^2}{6\pi\epsilon_0 c^3}\cdot\frac{2\pi r}{v} = \frac{e^2 r}{3\epsilon_0 c^3 v}\left(\frac{v^2}{r}\right)^2 \quad \left[\because \ a = \frac{v^2}{r}\right]$$

$$= \frac{e^2 v^3}{3\varepsilon_0 c^3 r}$$

$$\frac{\Delta E}{E} = \frac{\Delta E}{\frac{1}{2}mv^2} = \frac{e^2 v^3}{3\epsilon_0 c^3 r}\times\frac{8\pi\epsilon_0 r}{e^2} \quad \text{[using (ii)]}$$

$$= \frac{8\pi}{3}\left(\frac{v}{c}\right)^3$$

(b) Since $v << c$

$\therefore \quad \frac{\Delta E}{E} << 1$

This means that fraction of energy lost per revolution is small. And because of this reason the radius of the circular path is changing slowly. We can assume the path to be circular for a small interval of time.

65. $$U = U_0 \ell n r - U_0 \ell n r_0$$

$\Rightarrow$ $$\frac{dU}{dr} = \frac{U_0}{r}$$

$\therefore$ Force between the particles is

$$F = -\frac{dU}{dr} = -\frac{U_0}{r}$$

Negative sign indicates attraction

$\therefore$ $$\frac{mv^2}{r} = \frac{U_0}{r} \quad ...(i)$$

And $$mvr = \frac{nh}{2\pi} \quad ...(ii)$$

From (i) and (ii) $$r = \frac{nh}{2\pi\sqrt{mU_0}}$$

Kinetic energy of electron $$K = \frac{1}{2}mv^2 = \frac{U_0}{2} \quad \text{[using (i)]}$$

Energy of electron in nth orbit is

$$E_n = \text{KE} + \text{PE} = \frac{U_0}{2} + U_0 \ell n\left(\frac{r}{r_0}\right)$$

$$= \frac{U_0}{2} + U_0 \ell n\left[\frac{nh}{2\pi\sqrt{mU_0}}\cdot\frac{1}{r_0}\right]$$

$$= \frac{U_0}{2}\left[1 + \ell n\left(\frac{n^2 h^2}{4\pi^2 m U_0 r_0^2}\right)\right]$$

$$\Delta E_{nm} = E_n - E_m$$

$$= \frac{U_0}{2}\ell n\left(\frac{n^2 h^2}{4\pi^2 m U_0 r_0^2}\right) - \frac{U_0}{2}\,\ell n\left(\frac{m^2 h^2}{4\pi^2 m U_0 r_0^2}\right)$$

$$= \ell n\left(\frac{n}{m}\right)^2 = 2\ell n\left(\frac{n}{m}\right)$$

$\therefore$ $$\frac{\Delta E_{12}}{\Delta E_{24}} = \frac{\ell n\left(\frac{1}{2}\right)}{\ell n\left(\frac{2}{4}\right)} = 1$$

66. Let E_1, E_2, E_3 and E_4 be energy of photons of wavelength λ_1, λ_2, λ_3 and λ_4 respectively.

$$E_1 = \phi + 5.2 \text{ eV} \quad \text{...(i)}$$

$$E_2 = \phi + 12 \text{ eV} \quad \text{...(ii)}$$

$$E_4 = \phi + 0.95 \text{ eV} \quad \text{...(iii)}$$

But $$\lambda_2 = 0.6\,\lambda_1$$

$\Rightarrow$ $$E_1 = 0.6\,E_2 \quad \text{...(iv)}$$

From (i) and (ii)

$$0.6 = \frac{\phi + 5.2}{\phi + 12}$$

Solving we get: $\phi = 5$ eV

$\therefore$ $$E_1 = 10.2 \text{ eV};\ E_2 = 17 \text{ eV};\ E_4 = 5.95 \text{ eV}$$

And $\because \lambda_3 = 4\lambda_2$

$\therefore$ $$E_2 = 4E_3 \quad \Rightarrow \quad E_3 = \frac{17}{4} = 4.25 \text{ eV}$$

For atom

$$-13.6Z^2\left[\frac{1}{n^2} - \frac{1}{4}\right] = 10.2 + 17$$

$\Rightarrow$ $$Z^2\left[\frac{1}{4} - \frac{1}{n^2}\right] = 2 \quad \text{...(A)}$$

Also $$-13.6Z^2\left[\frac{1}{n^2} - \frac{1}{9}\right] = 4.25 + 5.95$$

$\Rightarrow$ $$Z^2\left[\frac{1}{9} - \frac{1}{n^2}\right] = 10.2 \quad \text{...(B)}$$

Solving (A) and (B)

$$Z = 3 \text{ and } n = 6$$

67. Energy conservation gives

Energy of H atom in nth state + $\frac{hc}{\lambda_1}$ = 12.2 eV

$\Rightarrow$ $$-\frac{13.6}{n^2}\text{ eV} + \frac{hc}{\lambda_1} = 12.2 \text{ eV}$$

$\Rightarrow$ $$-\frac{13.6}{n^2} + \frac{6.62 \times 10^{-34} \times 3 \times 10^8}{796 \times 10^{-10} \times 1.6 \times 10^{-19}} = 12.2$$

$$\Rightarrow \qquad \frac{13.6}{n^2} = 15.6 - 12.2 = 3.4 \text{ eV}$$

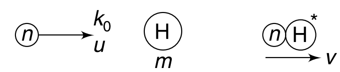
$$\therefore \qquad n = 2$$

Hydrogen atom formed is in 1st excited state.

$$\therefore \qquad \frac{hc}{\lambda_2} = 10.2 \text{ eV}$$

$$\lambda_2 = \frac{6.62 \times 10^{-34} \times 3 \times 10^8}{10.2 \times 1.6 \times 10^{-19}} = 1217 \text{ Å}$$

68. (a) For collision to be inelastic, the atom must get excited. Minimum excitation energy for hydrogen atom is



$$\Delta E = -\frac{13.6}{4} - (-13.6) = 10.2 \text{ eV}$$

In a collision, maximum loss of KE (i.e. maximum energy available for excitation of the atom) will happen when both the particles move together with same speed after collision.

$$2mv = mu$$

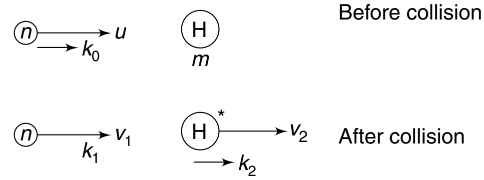
$$\Rightarrow \qquad v = \frac{u}{2}$$

$$\text{Loss in KE} = \frac{1}{2}mu^2 - \frac{1}{2}(2m)\left(\frac{u}{2}\right)^2 = \frac{1}{4}mu^2 = \frac{k_0}{2}$$

$$\Rightarrow \qquad \frac{k_0}{2} = 10.2$$

$$\Rightarrow \qquad k_0 = 20.4 \text{ eV} = \text{minimum } k_0 \text{ for excitation of hydrogen atom.}$$

(b) Excitation energy



$$\Delta E = -\frac{13.6}{9} - (-13.6) = 12.09 \text{ eV}$$

Momentum conservation gives-

$$mv_1 + mv_2 = mu$$

$$\Rightarrow \qquad \sqrt{2mk_1} + \sqrt{2mk_2} = \sqrt{2mk_0}$$

$$\Rightarrow \qquad k_1 + k_2 + 2\sqrt{k_1 k_2} = k_0 = 25 \qquad \text{...(i)}$$

Energy conservation gives-

$$k_1 + k_2 + \Delta E = k_0$$

$$\Rightarrow \qquad k_1 + k_2 = 25 - 12.09 = 12.91 \text{ eV} \qquad \text{...(ii)}$$

From (i) and (ii)

$$2\sqrt{k_1 k_2} = 12.09$$

$$(k_1 - k_2)^2 = (k_1 + k_2)^2 - 4k_1 k_2$$

$$= (12.91)^2 - (12.09)^2 = 20.5$$

$$\therefore \qquad k_1 - k_2 = \pm 4.53 \text{ eV} \qquad \text{...(iii)}$$

Solving (ii) and (iii)

$$k_1 = 8.72 \text{ eV}; \; k_2 = 4.19 \text{ eV}$$

$$\text{Or,} \qquad k_1 = 4.19 \text{ eV}; \; k_2 = 8.72 \text{ eV}$$

69. (a) From figure

$$\frac{1}{\lambda_{\min}} = 8.06 \text{ (Å)}^{-1}$$

If the bombarding electrons have KE of E (in eV units) then

$$\frac{12420}{\lambda_{\min}} = E \quad [\text{Where } \lambda_{\min} \text{ is in Å}]$$

$\therefore \qquad E = 12420 \times 8.06 \simeq 10^5$ eV

$\therefore$ Accelerating potential difference is 10^5 volt.

(b) k_a will have larger wavelength $\left(\text{and hence smaller } \frac{1}{\lambda}\right)$

$$\therefore \qquad \frac{1}{\lambda_{k\alpha}} = 4.34\ (\text{Å})^{-1} \quad \Rightarrow \quad \lambda_{k\alpha} = 0.23\,\text{A}^\circ$$

(c) We know

$$\frac{1}{\lambda_{k\alpha}} = (Z-1)^2 R\left[\frac{1}{1^2} - \frac{1}{2^2}\right]$$

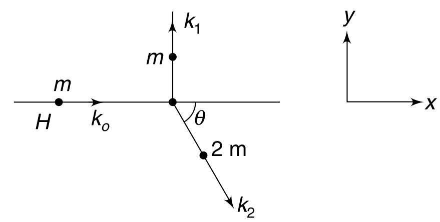
$$\Rightarrow \qquad 4.34 \times 10^{10}\ \text{m} = (Z-1)^2 \times 1.09 \times 10^7 \times \frac{3}{4}$$

$$\therefore \qquad (Z-1) = 73$$

$$Z = 74$$

71. Let the initial line of motion of hydrogen atom be along x direction and its kinetic energy be $k_0 = 30$ eV After collision the kinetic energy of hydrogen and deuterium be k_1 and k_2 respectively.

For conservation of momentum along x and y we must have



$$\sqrt{2mk_0} = \sqrt{2(2m)k_2}\cos\theta \qquad \text{...(i)}$$

$$\sqrt{2mk_1} = \sqrt{2(2m)k_2}\sin\theta \qquad \text{...(ii)}$$

Squaring and adding the above two equations gives

$$k_0 + k_1 = 2k_2$$

$$\therefore \qquad 2k_2 - k_1 = 30 \text{ eV} \qquad \text{...(iii)}$$

The sum $k_1 + k_2$ will be equal to 30 eV minus the excitation energy of atoms. Let's check all possible cases.

Case-1: None of the atoms is excited.

$$k_1 + k_2 = 30 \qquad \text{...(iv)}$$

Solving (iii) and (iv)

Gives $\qquad k_2 = 20$ eV and $k_1 = 10$ eV

Case-2: One of the atoms is excited to $n = 2$ state.

$$k_1 + k_2 = 30 - 10.2 = 19.8 \text{ eV} \qquad \text{...(v)}$$

Solving (iii) and (v) gives

$$k_2 = 16.6 \text{ eV and } k_1 = 3.2 \text{ eV}$$

Case-3: One of the atoms is excited to $n = 3$ state.

$$k_1 + k_2 = 30 - 12.09 = 17.91 \text{ eV} \qquad \text{...(vi)}$$

Solving (iii) and (vi) gives

$$k_2 = 15.97 \text{ eV and } k_1 = 1.94 \text{ eV}$$

Case-4: Now consider the case when one atom is excited to $n = \infty$ state

$$k_1 + k_2 = 30 - 13.6 = 16.4 \text{ eV} \qquad \text{... (vii)}$$

Solving (iii) and (vii) gives

$$k_2 = 15.47 \text{ eV},\ k_1 = 0.93$$

Values of kinetic energies are positive. It means one atom can get excited to all possible states. There can be infinite cases between case 3 and 4.

Case-5: Both atoms are excited to $n = 2$ state.

$$k_1 + k_2 = 30 - 2 \times 10.2 = 9.6 \text{ eV} \qquad \text{... (viii)}$$

Solving (iii) and (viii) gives

$$k_2 = 13.2 \text{ eV and } k_1 = -3.6 \text{ eV}$$

Negative KE is meaningless. Therefore, excitation of both atoms is not possible

$$\therefore \qquad (k_2)_{max} = 20 \text{ eV}$$

$$\text{And} \qquad (k_2)_{min} = 13.2 \text{ eV}$$

72. The initial momentum of the atom is zero. Thus if the emitted photon carries momentum P, the atom must recoil with equal momentum.

For energy to remain conserved we must have

$$\text{KE}_{\text{atom}} + \text{E}_{\text{Photon}} = \Delta E \qquad [\Delta E = 10.2 \text{ eV}]$$

$$\therefore \qquad \frac{P^2}{2m} + P \cdot c = \Delta E \quad \Rightarrow \quad P^2 + (2mc)\, P - 2m\,\Delta E = 0$$

$$\Rightarrow \qquad P = \frac{-2mc \pm (4m^2c^2 + 8m\Delta E)^{1/2}}{2}$$

$$= mc\left[-1 + \left(1 + \frac{2\,\Delta E}{mc^2}\right)^{1/2}\right]$$

We have retained + sign only because magnitude of momentum cannot be negative.

$$\text{Now} \qquad \frac{2\Delta E}{mc^2} = \frac{20.2\,\text{eV}}{940\,\text{MeV}} = 2.2 \times 10^{-8}$$

$$\therefore \qquad \frac{\Delta E}{mc^2} << 1$$

$$\therefore \qquad P \simeq mc\left[-1 + 1 + \frac{\Delta E}{mc^2} - \frac{1}{8}\left(\frac{2\Delta E}{mc^2}\right)^2\right]$$

We have neglected all other higher order terms.

$$\therefore \qquad P = mc\left[\frac{\Delta E}{mc^2} - \frac{1}{8}\left(\frac{2\Delta E}{mc^2}\right)^2\right]$$

$$P = \frac{\Delta E}{c}\left[1 - \frac{\Delta E}{2mc^2}\right]$$

$\frac{\Delta E}{c} = P_0$ = momentum of photon if recoil of atom is not taken into account.

$$\therefore \qquad P = P_0\left[1 - \frac{\Delta E}{2mc^2}\right]$$

$$\therefore \qquad \frac{P - P_0}{P_0} = -\frac{\Delta E}{2mc^2}$$

$$\text{Percentage error} = \frac{\Delta E}{2mc^2} \times 100$$

$$= \frac{10.2 \text{ eV}}{2 \times 940 \times 10^6 \text{ eV}} \times 100$$

$$= 5.4 \times 10^{-7}\%$$

CHAPTER 16

Nuclear Physics

LEVEL 1

Q. 1: Assume that an electron and a position pair is formed from a gamma ray photon having energy 3.0 MeV. Find the total kinetic energy of the positron and the electron that is formed.

Q. 2: (a) Why is an alpha particle, rather than neutrons and protons, emitted from an unstable nucleus?

(b) A nucleus ^{A}X alpha decays to produce Y. The volume of nucleus Y is nearly 56 times the volume of alpha particle. Find the mass number (A) of X.

Q. 3: A radionuclide has a half life of 10 s. Which of the following statements are correct?

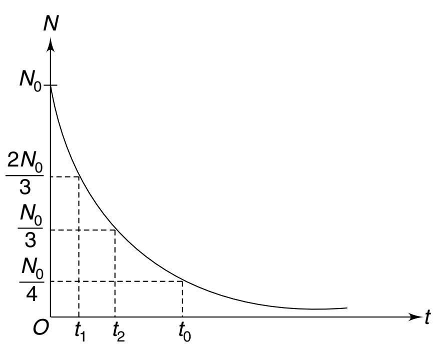
(a) In a sample, probability that a nucleus will decay in next 10 s is 0.5.

(b) In a sample, the probability that a nucleus which has survived first 10 s, will decay in next 10 s is 0.5.

(c) The probability that a nucleus which has survived first 10 s, will decay in next 10 s is 0.25.

(d) If a sample of radionuclide has 4 nuclei, two nuclei will decay in next 10 s.

Q. 4: The population of active nuclei in a sample of radioactive isotope has been plotted with time in figure. Look at the graph carefully and find the value of $t_2 - t_1$. It is given that $t_0 = 50$ minute.



Q. 5: (a) Assume that a particular nucleus, in a radioactive sample, has a probability p of decaying in the interval t to $t + \Delta t$. Taking that the nucleus actually does not decay in the above interval, what is probability that it will decay in the interval $t + \Delta t$ to $t + 2\Delta t$?

(b) Find the mean life of $^{55}C_0$ if its activity is known to decreases 4% per hour. $\ell n\left(\frac{25}{24}\right) = 0.04$.

Q. 6: A radioactive sample is a mixture of two radioactive species – one having population N_1 has a mean life of t_1 and the other one having population N_2 has nuclei having mean life of t_2. Find the effective mean life of the nuclei in the mixture.

Q. 7: An alpha active element has molar mass M and has a half life of τ. A large plate of thickness a is made from the radioactive element. Density of the plate is ρ. Calculate the number of alpha particles emitted in unit time from unit surface area of the plate. Avogadro's number is N_A.

Q. 8: A radioactive isotope X decays into a stable nucleus Y. Draw graphs showing the variation of rate of formation of Y versus time and the variation of population of Y versus time. Initial population of Y is zero.

Q. 9: An excited $^{69}Z\overset{*}{n}$ Nucleus at rest decays to ground state ^{69}Zn by way of emitting a gamma photon. The energy equivalent of difference in mass of $Z\overset{*}{n}$ and Zn is 0.44 MeV. Calculate the kinetic energy of recoil of Zn nucleus. Rest mass of ^{69}Zn is $M_0 = 68.927$ u.

Q. 10: The α-decay of nucleus X^{230} is accompanied by emission of two groups of α-particles - one group having α-particles of kinetic energy 6.7 MeV and other having α-particles of kinetic energy 5.2 MeV. Following the emission of these particles the daughter nuclei are found in ground and excited states. Find the energy of gamma photon that is subsequently emitted by de-excitation of the excited daughter nuclei. Assume that nuclei of X^{230} are at rest.

Q. 11: In a nuclear reactor, the neutrons produced in fission reaction have high kinetic energy of the order of few MeV, whereas the U^{235} nucleus fissions only with a slow neutron. In a reactor heavy water (D_2O) is used to slow down the neutrons. A neutron hitting a deuterium nucleus slows down quickly. Calculate now many head on, elastic collisions must a neutron have with deuterium nuclei to reduce its energy from 1 MeV to 0.025 eV. Assume that neutron does not lose energy by any other way apart from collisions with deuterium.

[Take $\log_{10} 2 = 0.301$ and $\log_{10} 3 = 0.477$]

Q. 12: Consider the nuclear reaction

$$^{7}_{3}\text{Li} + ^{1}_{1}\text{H} \longrightarrow 2\, ^{4}_{2}\text{He}$$

The relevant atomic masses are:

$^{7}\text{Li}: 7.018$ u; $^{1}\text{H}: 1.008$ u; $^{4}\text{He}: 4.004$ u.

One gram of ^{1}H is completely consumed in the reaction along with sufficient amount of ^{7}Li. Find energy released.

Q. 13: Beryllium nucleus ($^{9}_{4}\text{Be}$) when bombarded by alpha particle undergoes a nuclear reaction producing $^{12}_{6}\text{C}$. Binding energy per nucleon for different species are as under.

$BE_{Be}: 6.4763$ MeV; $BE_{He}: 7.0819$ MeV;

$BE_{C}: 7.6885$ MeV

(a) Is there any other product (X) that is formed apart from ^{12}C?

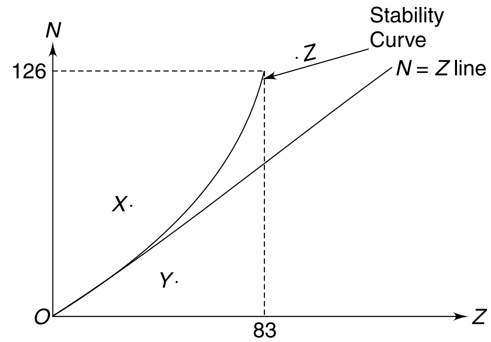
(b) Calculate the energy released in the reaction. Why don't we require Binding energy of X in this calculation?

Q. 14: A sample contains two isotopes – one stable and the other unstable. Number of stable nuclei is equal to N_s and number of unstable nuclei is equal to N_u. After a time the activity of the sample decreased to one third of the initial activity while the total number of nuclei (excluding decayed nuclei) became half. Find the initial value of the ratio N_s/N_u.

Q. 15: In a beam of neutrons the particles are having a kinetic energy of 0.0327 eV. If the half life of neutron is 693 second, what fraction of neutrons will decay before the beam advances by 10 m?

Given mass of neutron = 1.675×10^{27} kg.

Q. 16: The figure shows a plot of number of neutron (N) versus number of protons (Z) in all stable nuclei found in nature. What kind of decay are nuclei *X*, *Y* and *Z* expected to show?



Q. 17: Which of the following isotopes is more stable and why?

(a) $^{7}_{3}\text{Li}$ or $^{8}_{3}\text{Li}$

(b) $^{13}_{6}\text{C}$ or $^{15}_{6}\text{C}$

(c) What kind of decay will be shown by the unstable nuclei in the above sets (a) and (b)?

Q. 18: Consider the position emission reaction

$$^{A}_{Z}\text{X} \longrightarrow {}^{A}_{Z-1}\text{Y} + {}^{0}_{+1}\text{e} + \nu$$

Atomic mass of X and Y are m_x and m_y respectively and mass of a position is m_e.

(a) Write the disintegration energy (Q) of the reaction.

(b) If $^{A}_{Z}\text{X}$ is $^{11}_{6}\text{C}$ and atomic mass of ^{11}C is 11.011434 u, atomic mass of ^{11}B is 11.009305 u, mass of positron is $m_e = 0.000549$ u, then find the maximum kinetic energy of emitted positron. [$(1u)c^2 = 930$ MeV]

Q. 19: Tritium ($^{3}_{1}\text{H}$) is a radioactive isotope of hydrogen. It has a half life of about 12 years. It is produced in the upper atmosphere and is brought to the earth by rain. This mechanism replenishes the Tritium in water on the Earth. Estimate the age of a bottle of wine whose $^{3}_{1}\text{H}$ radiation is about $\left(\frac{1}{5}\right)^{\text{th}}$ of that of a new bottle. [$\log_{10} 2 = 0.303$].

Q. 20: In the thorium decay series that begins with ^{232}Th; ^{228}Th alpha-decays into ^{224}Ra with a half life of 1.9 yr. ^{224}Ra alpha-decays into ^{220}Rn with a half life of 3.7 d, which in turn decays into ^{216}Po with a half life of 55 s. If the nuclides are in radioactive equilibrium in a mineral sample that contains 1.0 g of ^{220}Rn, what are the masses of ^{224}Ra and ^{228}Th in the sample?

Q. 21: Nuclei of ^{64}Cu can decay by electron capture (probability 61%) or by β^+ decay (39%). Half life of ^{64}Cu is 12.7 hour. Find the partial half life for electron capture decay process.

Q. 22: In a sample there are two radioactive nuclides *X* and *Y*. Initially, population of *X* is 2.5 times that of *Y*. It was observed that 3 days later the population of *X* becomes 5.0 times that of population of *Y*. Find the half life of *X* if half life of *Y* is 2.0 days.

Q. 23: It is known that the ratio of ^{14}C to ^{12}C atoms in living beings and atmosphere is a constant. In one experiment it was found that the radiocarbon activity of a living thing is 16 disintegration per minute per gram of carbon content. Find the ratio of ^{14}C to ^{12}C atoms in your body. Half life of beta decay of ^{14}C is 5760 yr = 3.03×10^9 minute.

Q. 24: Taking the data from last problem and assuming that 18% of our body mass is made up of carbon, calculate the number of beta particles being emitted from the body of a man of mass 50 kg in one minute. Assume beta activity due to radiocarbon only.

Q. 25: Consider the nuclear decay reaction.

$$^{A}_{Z}X \rightarrow {}^{A}_{Z-1}Y + {}^{0}_{+1}e + \upsilon$$

For the above β^+ decay reaction to be feasible what should be the minimum difference in atomic masses of X and Y? Rest mass of υ is nearly zero.

Q. 26: The ratio (by weight) for U^{238} and Pb^{226} in a rock sample is 4 : 3. Assume that originally the rock had only U^{238} and the entire Pb^{226} is product of decay of U^{238}. The intermediate radioactive nuclides in the chain are very small in quantity in the rock.

Find the age of the rock if half life of U^{238} is 4.5×10^9 year.

[$\log_{10} 1.79 = 0.25$, $\log_{10} 2 = 0.3$]

LEVEL 2

Q. 27: A radioactive species has decay constant of $\lambda = 10^{-2}\,s^{-1}$. Probability that a particular nucleus decays in time *t* is *p*. Draw a graph showing variation of *p* with *t*. Assume that the selected nucleus survives till last.

Q. 28: A radioactive substance has a half life of t_0. Two particular nuclei – let's name them as nucleus *A* and *B*-have not decayed over a particular observation period of $6t_0$.

(a) What is the probability that A will decay over a further period of $3t_0$?

(b) What is the probability that both *A* and *B* will survive over a further period of $3t_0$?

Q. 29: In a sample of radioactive material the probability that a particular nucleus will decay in next 2 hour is 8×10^{-4}. Find the half life of the radioactive sample

[Take $\ell n2 = 0.693$, $\ell n(0.9992) = -8 \times 10^{-4}$]

Q. 30: Atomic mass of $^{7}_{3}Li$ is 7.01600 u and that of $^{7}_{4}Be$ is 7.01693 u. Mass of one electron in atomic mass unit is 0.00055 u. If it is observed that $^{7}_{4}Be$ decays into $^{7}_{3}Li$, what kind of decay will it be?

Q. 31: $_{15}P^{32}$ is beta active and has a short half life of 14 days. The $_{16}S^{32}$ nucleus produced is in ground state. On an average, the emitted beta particles have 50% of the energy released in the reaction, remaining energy being carried by antineutrino. A liquid has small amount of P in it and it was found that the liquid was receiving heat at a rate of about 10 mW due to decay of P^{32}. It is to be noted that antineutrino do not interact with matter easily and it would be safe to assume that they escape out of the container. Estimate the number of P^{32} atom in the liquid. Relevant masses are

$$P^{32} : 31.9740\,u \quad S^{32} : 31.9720\,u \quad e : 0.0005\,u$$

$$1 \text{ day} = 86400\,s \quad 1u = \frac{931\,MeV}{c^2}$$

Q. 32: A free neutron at rest, decays into three particles: a proton, an electron and an anti neutrino.

$$^{1}_{0}n \rightarrow {}^{1}_{1}p + {}^{0}_{-1}e + \overline{\nu}$$

The rest masses are: $m_n = 939.5656\ MeV/c^2$

$m_p = 938.2723\ MeV/c^2 \quad m_e = 0.5109\ MeV/c^2$

In a particular decay, the antineutrino was found to have a total energy (including rest mass energy) of 0.0004 MeV and the momentum of proton was found to be equal to the momentum of electron. Find the kinetic energy of the electron.

Q. 33: A radioactive nucleus *A* decay to *C* after emitting two α and three β particles. Another nucleus *B* decays to *C* by emitting one α and five β particles. Half life of *A* and *B* are 1hr and 2hr respectively and all intermediates have negligible half lives. At time $t = 0$, population of nuclei of *A* and *B* are $4N_0$ and N_0 respectively and population of *C* is zero.

(a) Find the difference in atomic numbers and mass numbers of *A* and *B*.

(b) Find the population of *C* when both *A* and *B* have equal population.

(c) Find time t_0 when both *A* and *B* have equal rate of decay.

Q. 34: A nucleus at rest decays by emitting an α particle. The de-Broglie wavelength of emitted α particle is $\lambda = 5.76$ femtometer. Mass of the daughter nucleus and α particle is 223.610 u and 4.002 u. Find the mass of the parent nucleus.

Q. 35: A radioactive sample consists of two isotopes one of them decays by α–emission with a half life of $\tau_1 = 405\,s$ and the other one decays by β emission with half life of

$\tau_1 = 1620\,s$. At time $t = 0$, probabilities of getting α and β particles from the sample are equal.

(a) If a particle coming out of the sample is detected at $t = 1620\,s$, what is the probability that it is α particle?

(b) Find the time when total number of surviving nuclei is half the initial quantity.

Given: If $x^4 + 4x - 2.5 = 0$ then $x = 0.59$

$\log_{10} 2 = 0.301$; $\log_{10} 5.9 = 0.774$

Q. 36: A deuterium nucleus at rest absorbs a Gamma ray photon of energy 2.21 MeV and splits into a proton and a neutron. The neutron was found to be at rest after the disintegration. Calculate the mass of proton if it is known that mass of deuterium and hydrogen atom is 2.014740 u and 1.008145 u respectively. [$1\,u = 931$ MeV/c^2]

Q. 37: Two isotopes of radioactive Radon gas - R_n^{222} and R_n^{220} are mixed in atomic ratio 1000 : 1 and kept in a 831 cm^3 container at 10^5 N/m^2 pressure and 27°C temperature. The two isotopes have half life of 4 days and 1 minute respectively. Calculate the activity of Radon gas sample in unit of curie.

Q. 38: The Uranium −238 decay series (also known as $4n + 2$ series. Can you tell why?) ends at a stable isotope $_{82}Pb^{206}$. Lead −206 is found in a certain Uranium ore due to disintegration of Uranium.

(a) How many alpha and beta particles are emitted in the series decay of one $_{92}U^{238}$ nucleus into $_{82}Pb^{206}$?

(b) What is the age of uranium ore if it now contains 8g of Pb^{206} for every 10g of U^{238}? Half life of U^{238} is 4.5 billion year.

[Take $\ell n2 = 0.6930$; $\ell n(1.9243) = 0.6546$]

Q. 39: The half lives of two longest – lived radioactive isotopes of phosphorus P^{32} and P^{33}, are 14 days and 28 days respectively. *A* sample has been prepared by mixing the two isotopes in the ratio of 4 : 1 of their atoms. Initial activity of the mixed sample is 9.0 mCi. Find the activity of the sample after 80 days.

Take $\ell n2 = 0.7$ and $e^{-2} = 0.14$

Q. 40: A diatomic molecule moving at a speed u absorbs a photon of wavelength λ and then dissociates into two identical atoms. One of the atoms is found to be moving with a speed v in a direction perpendicular to the initial direction of motion of the molecule. Take mass of the molecule to be M and calculate the binding energy of the molecule. Assume that momentum of absorbed photon is negligible compared to that of the molecule.

Q. 41: *A* $^{14}_{7}N$ nucleus, when bombarded by $^{4}_{2}He$, converts into $^{17}_{8}O$ according to the following reaction

$$^{14}_{7}N + ^{4}_{2}He \rightarrow ^{17}_{8}O + ^{1}_{1}H$$

Atomic masses are:

^{14}N : 14.003 u; ^{4}He : 4.003 u; ^{17}O : 16.999 u; ^{1}H : 1.008 u

(a) Find the Q value of the reaction

(b) Assume that a ^4He nucleus collides with a free ^{14}N nucleus originally at rest. Calculate the minimum kinetic energy (k_0) that ^4He must have so as to cause this reaction.

(c) If the ^{17}O nucleus has an excitation energy of 1.0 MeV, find the minimum kinetic energy (k_0') that ^4He must have.

[Take $1\,u = 930$ Me V/c^2]

Q. 42: Following reactions are known as inverse beta decay

$$p + \bar{v} \rightarrow n + e^+$$

$$n + v \rightarrow p + e^-$$

These reactions have extremely low probabilities. Because of this, neutrinos and antineutrinos are able to pass through vast amount of matter without any interaction. In an experiment to detect neutrinos, large number of neutrinos coming out from beta decays of a radioactive material were made to pass through a tank of water, containing a cadmium compound in solution, which provided the protons to interact with antineutrinos. Immediately after a proton absorbed a neutrino to yield a positron and a neutron, the positron encountered an electron and both got annihilated. The gamma ray detectors surrounding the tanks responded to the resulting photons. This confirmed that the above reaction has taken place.

(a) How many gamma ray photons are produced when an electron annihilates with a positron? What is energy of each photon? Take the mass of an electron to be 0.00055 u.

(b) The neutron produced in the above reaction was captured by ^{112}Cd to form ^{113}Cd. The atomic masses of these two isotopes of cadmium are 111.9028 u and 112.9044 u respectively. Mass of a neutron is 1.0087 u. Find the Q value of this reaction. Assume half of this energy is excitation energy of ^{113}Cd. If the nucleus de-excites by emitting a gamma ray photon find its wavelength.

Q. 43: Consider the β^- decay of $^{27}_{12}Mg$

$$^{27}_{12}Mg \rightarrow ^{27}_{13}A\ell^* + ^{0}_{-1}e + \bar{v}$$

The atomic masses of Mg^{27} and $A\ell^{27}$ are 26.98434 u and 26.98154 u respectively. Mass of electron is 0.00055 u. The $A\ell^*$ nucleus has excitation energy of 1.015 MeV.

(a) Find the Q value of the β^- decay.

(b) What is maximum possible kinetic energy of emitted β^- particle?

(c) Find the smallest wavelength photon that $A\ell^*$ can emit when it de-excites.

(d) As an alternative to gamma decay, the excited $A\ell^*$ nucleus returns to its ground state by giving up its excitation energy to an atomic electron which has a binding energy of 2300 eV. The process is known as internal conservation. Find the kinetic energy of the emitted electron.

Q. 44: Moderator is used in a nuclear reactor to slow down fast neutrons generated after a fission event in the reactor. Consider a fast neutron (mass m_n) moving at speed v. The neutron hits a moderator nucleus (Mass M), which is originally static, and gets scattered elastically by 180°.

(a) Find the ratio of kinetic energy of the neutron after and before collision.

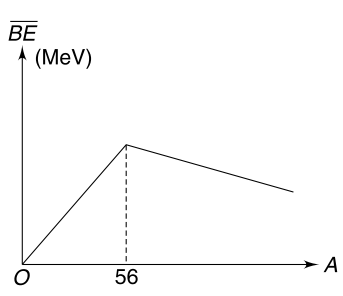
(b) Show that when $M >> m_n$, the neutron does not lose much energy and therefore, a moderator should have a small mass number.

Q. 45: Two radioactive elements A and B are present in a mixture. Both are beta active. Half lives of A and B are 30 minute and 60 minute respectively. The initial activity of the sample was measured to be 40 decay per second and after an hour only 30 beta particles were detected in an interval of 2 second. Assume that law of radioactivity decay is obeyed.

(a) Find the ratio of number of active nuclei of A to that of B in the mixture initially.

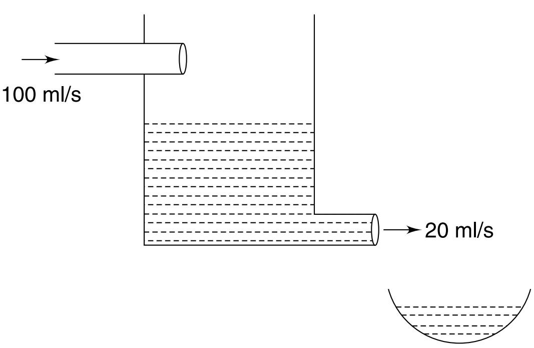
(b) The molar mass of both A and B is closed to 200g. The mixture contains other substance (non radioactive) apart from A and B. What fraction of the entire mixture is radioactive (fraction by weight) if it is given what mass of the mixture is $1\,\mu$g.

Q. 46: The graph of binding energy per nucleon ($\overline{BE}$) versus mass number (A) for various nuclei has been approximated as made up of two linear components as shown in figure. The slope of line for $0 < A < 56$ is 0.16 and slope of line for $A > 56$ is -0.012. The binding energy of ${}^4_2\text{He}$ nucleus is 11 times the value predicted by the shown graph due to its very stable structure. Estimate the minimum mass number (A) of a nucleus for it to exhibit alpha decay.



Q. 47: A rock sample has radioactive isotope A that decays into a stable Product B. Half life of A is 10^3 year. It was found that the rock sample contained n_1 moles of A and $\frac{n_1}{4}$ moles of B. Plot the variation of population of B with time. After how much time the rock will have equal population of A and B? [Given $2^{0.68} \simeq 1.6$]

Q. 48: A 100 ml solution of a radioactive material has activity A_0. The solution is kept in a container and is diluted by adding water at a constant role of 100 ml/sec. From a small hole in the container 20 ml/sec of solution is taken out simultaneously. Find the activity of the 100 ml solution that is taken out in 5 sec. Assume that the half life of radioactive material is very large and that the solution always remains homogeneous.



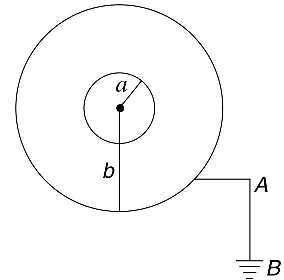
Q. 49: The radio isotope ${}^{64}_{29}\text{Cu}$ can decay through positron emission and electron capture. In both the processes neutrino (v) is emitted. Calculate the difference in maximum possible kinetic energy of emitted neutrinos in the two processes.

Q. 50: A β active radioactive source is in the form of a conducting sphere of radius a. It is surrounded by a concentric conducting shell of radius $b (> a)$. The shell is grounded. β particles are emitted with kinetic energy ranging from E_1 to $E_2 (> E_1)$.

(a) Find the maximum potential that will be acquired by the sphere of radius a.

(b) Find the total charge that will flow through the grounding wire AB.

(c) Find the final maximum charge on the outer sphere.



Q. 51: A radioactive isotope X decays simultaneously by two ways α and β decay. The decay constants of α and β decays are λ_1 and λ_2 and the products formed are Y and Z respectively. Both Y and Z are stable. Initially, there was no presence of Y and Z in the sample. The process of α decay stops after time t_0. Find the ratio of mass of Y to mass of Z in the sample at time $2t_0$. The molar masses of Y and Z are M_y and M_z respectively.

Q. 52: The sun has a power of $P = 3.85 \times 10^{26}$ W and the only source of energy in it is the following reaction

$$4\,{}^1\text{H} \rightarrow {}^4\text{He} + 2e^+ + 2\nu_e$$

The 'electron neutrinos (ν_e)' are nearly mass less and carry negligible energy. However, they are able to escape from the Sun and have been detected on the earth. The Earth-Sun distance is $r = 1.5 \times 10^{11}$m and masses of a hydrogen atom, helium atom and a positron are 1.6740×10^{-27} kg, 6.6450×10^{-27} kg and 0.0009×10^{-27} kg respectively.

(a) Calculate the flux density (i.e. number of neutrinos arriving at the Earth) in units of $m^{-2}\,s^{-1}$.

(b) While travelling from the Sun to the Earth, some of the electron neutrinos (ν_e) are converted into other types of neutrinos - ν_0. The detector on the Earth has $\frac{1}{5}$th efficiency for detecting ν_0 as compared to its efficiency to detect ν_e. Had there been no conversions of ν_e, we expect to detect N_1 neutrinos in a year. However, due to conversion, we detect only N_2 neutrinos (ν_0 and ν_e combined) per year. What fraction (f) of ν_e gets converted into ν_0. Express your answer in terms of N_1 and N_2.

Q. 53: Consider a hypothetical case of a radionuclide A which produces a daughter B which is turn decays to a stable product C. The half life of B is 5 times that of A. Take initial population of both B and C to be zero.

(a) Using qualitative arguments only, plot the variation of population of A, B and C with time.

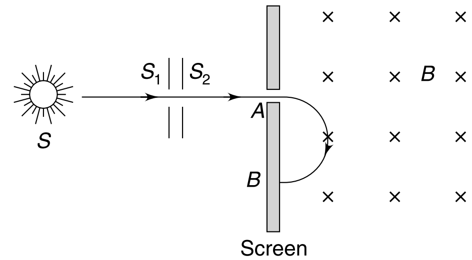
(b) Find the ratio of population of B to that of A at the instant the population of B becomes maximum.

(c) Describe qualitatively how the slope of the graph of population of C changes with time.

Q. 54: A source(s) of beta particles contains a radioactive isotope X. The emitted beta particles are passed through two parallel slits S_1 and S_2 to get a narrow parallel beam. This beam is made to enter a uniform magnetic field (B). The particles follow semicircular trajectories and are found to hit the screen at all points from A to B. The distance AB is equal to $2a$. The reaction for decay of a nucleus of X is

$$\text{X} \rightarrow \text{Y} + {}^{0}_{-1}e + \bar{\nu}$$

Calculate the minimum difference in mass of a nucleus of X and Y. Mass of an electron is m and charge on it is e.



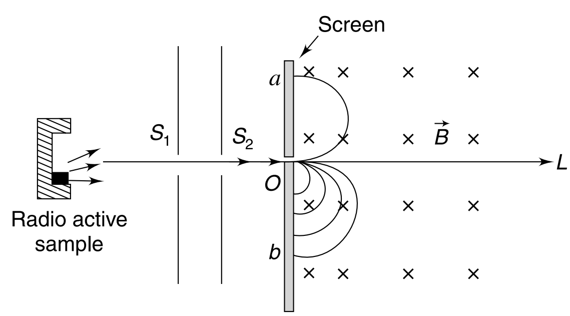
Q. 55: A sample of radioactive material has an alpha emitter (X) and a beta emitter (D). The energy of decay reactions are Q_1 and Q_2 respectively.

$$ {}^{A}_{Z}X \rightarrow {}^{A-4}_{Z-2}Y + {}^{4}_{2}\text{He} + Q_1$$

$$ {}^{A'}_{Z'}D \rightarrow {}^{A'}_{Z'+1}B + {}^{0}_{-1}e + \bar{\nu} + Q_2$$

The radiation coming out from the sample is passed through two parallel slits S_1 and S_2 to get a narrow parallel beam of alpha and beta particles. This beam is allowed to enter perpendicularly into a region of uniform magnetic field. The particles after taking a semicircular path strike the screen at point a above the line L and, below the line L particles are found to strike the screen everywhere from O to b. The distance Oa and Ob and r_1 and r_2 respectively. Find the ratio $\frac{r_1}{r_2}$.

Take $\frac{Q_1}{Q_2} = k$ and assume that mass of alpha particle is 4 time the mass of a proton which is η times the mass of an electron.



Q. 56: U^{238} decays in 14 steps that together constitute the net reaction

$${}_{92}\text{U}^{238} \rightarrow {}^{206}_{82}\text{Pb} + \_\_\_ + 6\bar{\nu} + 8\left({}^{4}_{2}\text{He}\right)$$

(a) Fill in the blank in the above reaction.

(b) The atomic masses of U^{238}, Pb^{206} and He^4 are 238.050783 u, 205.974449 u, and 4.002603 u respectively. The energy equivalent of mass of one electron is 0.51 MeV. With this data calculate the Q value of the reaction.

(c) U^{238} has a long half life of $t = 4.47 \times 10^9$ yr. The other members in the chain reaction have relatively small half lives. As soon as a nucleus of U^{238} decays the remaining steps are completed almost instantaneously. With this assumption, calculate the power output from a 100 kg block of pure U^{238}. The density of U^{238} is 18700 kg/m^3.

Q. 57: Some radioisotopes are used for medical imaging. A radioisotope is injected into the blood of a person. Malignant tissues absorb and retain this isotope much more efficiently than other healthy tissues. A sample of radio isotope having half life of 5 hour and activity of 22 mCi is injected into the blood of a patient. The isotope enters into thyroid glands through the blood streams. The isotope radiates gamma photons each having energy of 2×10^{-8} μJ.

(a) Qualitatively sketch the amount of radio isotope in the thyroid glands as a function of time.

(b) Sometime after the injection a scan of the glands is taken over a period of 20 minute. The total energy emitted by the glands was found to be 6×10^{-4} J. Assume that the thyroid glands are quickly saturated (with isotopes) after the injection and almost 50% of the injected isotopes are absorbed by the glands. Make an estimate of the time after the injection when the scan was performed.

Q. 58: A radioactive element is produced in nuclear reactor at a constant rate $\boldsymbol{R}$ (= numbers of nuclei per second). Its half-life is $\boldsymbol{T}_{1/2}$. How much time, in terms of $\boldsymbol{T}_{1/2}$, is required to produce 50% of the equilibrium quantity of the radioactive element? Is the result dependent on R?

Q. 59: Nuclei of radioactive element A are being produced at rate t^2 at time t. The element A has decay constant equal to λ. If N is the number of active nuclei of element A at any time t then $\frac{dN}{dt}$ was found to be minimum at $t = t_0$. Find N at time $t = t_0$.

LEVEL 3

Q. 60: The power radiated by a star and its mass m are related by $\frac{P}{P_0} = \left(\frac{m}{M_0}\right)^{7/2}$ where P_0 and M_0 are power radiated by the Sun and mass of the Sun respectively. Assume that the fraction of mass lost by the star since its birth is $\alpha (<< 1)$. Calculate the age of the star in terms of α, M_0 P and P_0.

Q. 61: Rate of evaporation (volume evaporated per unit time) of water kept in an earthen pot is proportional to the volume of water present in the pot. It is observed that it takes 48 hrs for 75% of the water kept in the pot to evaporate. The empty pot was placed below a tap from which water leaks in small drops each of volume v_0. The drops fall at a uniform rate of n drops per hour. Calculate the volume of water in the pot 24 hrs after it was kept below the tap.

Q. 62: (a) The rest mass energy of an electron is $m_0c^2 = 0.51$ MeV. Can a photon of this energy create an electron in free space?

(b) A single photon of energy $2m_0c^2$ or greater has enough energy to form an electron (e^-), positron (e^+) pair. But prove that this process cannot occur in free space.

(c) The pair production is possible in presence of a third particle. Assume that a photon having frequency v collides with a nucleus of rest mass M_0 (at rest) and creates an e^-, e^+ pair at rest. Calculate the mass M_0 of the nucleus.

Neglect relativistic effect in all questions.

Q. 63: A radioactive nucleus A decays into B with a decay constant λ_1. B is also radioactive and it decays into C with a decay constant of λ_2. At time $t = 0$, activity of A was recorded as A_1^0 and that of B was $A_2^0 = 0$, in a sample.

(a) If N_1 and N_2 are population of nuclei of A and B respectively at time t, set up a differential equation in N_2.

(b) The solution to the differential equation obtained in part (a) is given by

$$N_2 = \frac{\lambda_1}{\lambda_2 - \lambda_1} N_1^0 (e^{-\lambda_1 t} - e^{-\lambda_2 t})$$

Where N_1^0 is initial population of A. Based on this answer following questions–

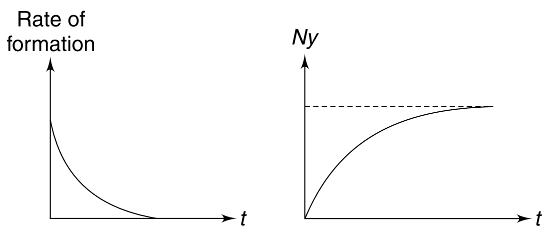
(i) If half life of A is quite large compared to B (for example half life of A is 8 hr and that of B is 1 hr) show that ratio of activity of B to that of A approaches a constant value after a sufficiently long time. Find this ratio, in terms of λ_1 and λ_2.

(ii) If half life of A is infinitely large compared to that of B (for example, A has half life ~10^9 yr and for B half life ~10^5 yr), show that both A and B will have equal activity after a very long time.

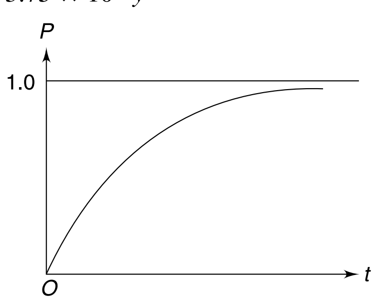
(iii) If $\lambda_1 > \lambda_2$, find the time at which activity of B reaches a maximum.

ANSWERS

1. 1.98 MeV.
2. (b) $A = 228$
3. (a), (b)
4. 25 min.
5. (a) p (b) 25 hr
6. $\dfrac{N_1 t_1 + N_2 t_2}{N_1 + N_2}$
7. $\dfrac{\ell n 2}{2} \dfrac{a\rho N_A}{\tau M}$
8.

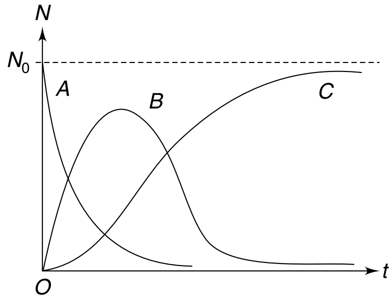


9. 1.5 eV
10. 1.53 MeV
11. 8
12. 1.62×10^{12} J
13. (a) Yes, it is neutron (b) 5.65 MeV
14. $N_s/N_u = 1/3$
15. 4×10^{-6}
16. $X \rightarrow$ negative β decay $Y \rightarrow$ positron emission $Z \rightarrow$ alpha decay
17. (a) ${}^{7}_{3}\text{Li}$ (b) ${}^{13}_{6}\text{C}$
    (c) ${}^{8}_{3}\text{Li} \rightarrow$ negative β decay; ${}^{15}_{6}\text{C} \rightarrow$ negative β decay
18. (a) $Q = [m_x - m_y - 2m_e]c^2$ (b) 0.959 MeV
19. 27.6 yr
20. 5.9 kg, 1149 kg
21. 20.8 hr.
22. 6 days
23. 1.37×10^{-12}
24. 1.41×10^{5} per minute
25. Twice the mass of an electron.
26. 3.75×10^{9} y
27.



28. (a) $\dfrac{7}{8}$ (b) $\dfrac{1}{64}$
29. 1732.5 hr
30. Electron capture
31. 1.56×10^{17}
32. 0.7820 MeV
33. (a) $Z_A - Z_B = 4$; $A_A - A_B = 4$ (b) $\dfrac{9N_0}{2}$ (c) 6 hr
34. 227.62 u
35. (a) $\dfrac{1}{9}$ (b) 1215 s
36. 1.008968 u
37. 7.3×10^{6} Ci
38. (a) 8α and 6β (b) 4.25×10^{9} yr
39. 0.3 mCi
40 $BE = \dfrac{hc}{\lambda} - \dfrac{1}{2}Mu^2 - \dfrac{1}{2}Mv^2$
41. (a) 0.93MeV (b) $k_0 = 1.20$ MeV (c) $k'_0 = 2.20$ MeV
42. (a) 0.51 MeV (b) 6.60 MeV, 0.004 Å
43. (a) 2.0925 MeV
    (b) $KE_\beta < 1.0775$ MeV
    (c) 0.0122 Å
    (d) 1.013 MeV
44. $\left(\dfrac{M - m_n}{M + m_n}\right)^2$
45. (a) $\dfrac{1}{2}$ (b) 5.2×10^{-11}
46. 216
47. 680 y
48. $A_0\left[1 - \left(\dfrac{5}{7}\right)^{1/4}\right]$
49. $(k_{\max})_{Ec} - (k_{\max})_{\beta^+} = 2m_e c^2 = 1.02$ MeV
50. (a) $V_0 = \dfrac{E_2}{e}$ (b) zero (c) $q = -\dfrac{4\pi\epsilon_0 abE_2}{e(b-a)}$
52. (a) 5.9×10^{14} m^{-2} s^{-1} (b) $\dfrac{5}{4}\left(1 - \dfrac{N_2}{N_1}\right)$

53. (a)



(b) $\dfrac{N_B}{N_A} = 5$

(c) The slope first increases. It becomes maximum when N_B is maximum. Thereafter the slope decreases.

54. $\dfrac{a^2e^2B^2}{2mc^2}$

55. $\dfrac{r_1}{r_2} = \sqrt{\eta k\left(\dfrac{A-4}{A}\right)}$

56. (a) $6\,{}_{-1}^{\;0}e$ (b) 51.71 MeV

(c) 0.01 Watt.

57. (b) 20 hours

58. $\boldsymbol{T}_{1/2}$, No

59. $\dfrac{2t_0 - \lambda t_0^2}{\lambda^2}$

60. $\dfrac{\alpha M_0 c^2}{P_0^{2/7}\,P^{5/7}}$

61. $\dfrac{12\,nv_0}{\ell n 2}$

62. (a) No (b) $M_0 = \dfrac{h^2v^2}{2c^2[hv - 2m_0c^2]}$

63. (a) $\dfrac{dN_2}{dt} = -\lambda_2 N_2 + \lambda_1 N_1^0\, e^{-\lambda_1 t}$

(b) $\dfrac{A_2}{A_1} = \dfrac{\lambda_2}{\lambda_2 - \lambda_1}$

(c) $\left(\dfrac{1}{\lambda_2 - \lambda_1}\right)\ell n\left(\dfrac{\lambda_2}{\lambda_1}\right)$

SOLUTIONS

1. Conservation of energy

$$hv = 2m_0c^2 + E_1 + E_2$$

$$m_0 = 9.1\times10^{-31}\text{ kg} = \text{rest mass of an electron/positron}$$

$$m_0c^2 = \frac{(9.1\times10^{-31}\text{ kg})(3\times10^8\text{ m/s})^2}{1.6\times10^{-13}}\text{ MeV} = 0.51\text{ MeV}$$

$\therefore$ $$2m_0c^2 = 1.02\text{ MeV}$$

$\therefore$ $$E_1 + E_2 = 3.0 - 1.02 = 1.98\text{ MeV}$$

2. (b) $$\frac{\text{Volume of } Y}{\text{Volume of alpha}} = 56$$

$\Rightarrow$ $$\frac{\frac{4}{3}\pi\left[R_0A_1^{1/3}\right]^3}{\frac{4}{3}\pi\left[R_0\cdot 4^{1/3}\right]^3} = 56$$

$\Rightarrow$ $$A_1 = 56\times4 = 224$$

Mass number of X will be $$A = A_1 + 4 = 228$$

3. Radioactivity is a statistical phenomenon. The law of radioactivity is applicable for a large sample size. Probability of decay of a nucleus remains same at all time and does not depend on history. Hence (a) and (b) are true.

4. In interval 0 to t_0 the population becomes $\frac{1}{4}$ of original.

$$\therefore \quad t_0 = 2 \times \text{Half life}$$

$$\therefore \quad \text{Half life} = 25 \text{ min}$$

In interval t_1 to t_2 the population becomes half.

$$\therefore \quad t_2 - t_1 = 25 \text{ min}$$

5. (a) A nucleus doses not have a memory and its decay probability per unit time is constant until it actually does decay.

(b) Initial activity $\quad A_0 = \lambda N_0$

Activity after time t is $\quad A = A_0 e^{-\lambda t}$

Fractional decreases in activity $= \dfrac{A_0 - A}{A_0}$

$$\therefore \quad \frac{4}{100} = \frac{A_0 - A_0 e^{-\lambda t}}{A_0} = \frac{1 - e^{-\lambda t}}{1}$$

$$\Rightarrow \quad e^{-\lambda t} = \frac{24}{25}$$

$$\therefore \quad \lambda = \frac{1}{t} \ell n\left(\frac{25}{24}\right)$$

Put $\quad t = 1$ hour

$$\lambda = \ell n\left(\frac{25}{24}\right) = 0.04$$

$$\therefore \quad \text{mean life} = \frac{1}{\lambda} = \frac{1}{0.04} \text{ hr} = 25 \text{ hr}$$

6. S_1 = sum of life of N_1 nuclei of 1st species
S_2 = sum of life of N_2 nuclei of 2nd species

$$t_1 = \frac{S_1}{N_1} \quad \text{and} \quad t_2 = \frac{S_2}{N_2}$$

For mixture, mean life will be

$$t = \frac{S_1 + S_2}{N_1 + N_2} = \frac{N_1 t_1 + N_2 t_2}{N_1 + N_2}$$

7. Mass of area A of the plate $(A \cdot a)\rho$.

Number of moles of nuclei $= \dfrac{A.a\rho}{M}$

Number of active nuclei $= \dfrac{Aa\rho}{M} N_A$

Activity $= \lambda N = \dfrac{\ell n 2}{\tau} \times \dfrac{Aa\rho}{M} N_A$

$\therefore$ Number of α particles emitted per unit area per unit time

$$= \frac{1}{2} \frac{\ell n 2}{\tau} \frac{a\rho N_A}{M}$$

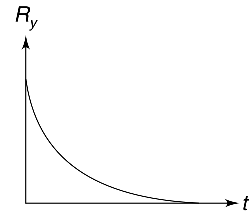
The factor $\frac{1}{2}$ is to account for the fact that the plate has two surfaces and activity must be same on both sides.

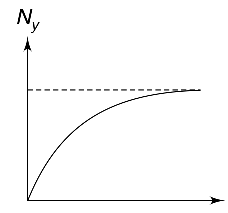
8. Rate of formation of Y = rate of decay of X

$$R_y = \lambda N_0 e^{-\lambda t}$$

Population of $Y = N_0$ – Surviving population of X

$\Rightarrow$
$$N_y = N_0 - N_0 e^{-\lambda t}$$
$$= N_0(1 - e^{-\lambda t})$$





9. $^{69}Zn^* \longrightarrow Zn + \gamma$

To conserve momentum

$$P_{Zn} = P_\gamma = \frac{E}{c}$$

Where
$$E = [M_0^* - M_0]c^2 = 0.44 \text{ MeV}$$

Note that kinetic energy of recoiling Zn nucleus will be very small and entire energy released is with γ photon.

$\therefore$
$$K_{zn} = \frac{(P_{zn})^2}{2M_0} = \frac{E^2}{2c^2 M_0}$$
$$= \frac{(0.44 \text{ MeV})^2}{2 \times 931.5 \times 68.927} = 1.5 \times 10^{-6} \text{ MeV}$$
$$= 1.5 \text{ eV}$$

10. For the decay

$$X^{230} \longrightarrow Y^{226} + He^4 + Q$$

Kinetic energy of α-particle

$$K_\alpha = \left(\frac{A-4}{A}\right)Q$$

$\Rightarrow$
$$Q = \frac{6.7 \times 230}{226} = 6.82 \text{ MeV}$$

For the decay

$$X^{230} \rightarrow {}^*Y^{226} + He^4$$

Momentum conservation requires

$$|P_{Y^*}| = |P_\alpha|$$
$$\sqrt{2m_y \cdot K_y} = \sqrt{2m_\alpha K_\alpha}$$

$\Rightarrow$
$$K_y = \frac{4 \times 5.2}{226} = 0.09 \text{ MeV}$$

$\Rightarrow$ Excitation energy of Y^* is

$$\Delta E = Q - K_\alpha - K_y$$
$$= 6.82 - (5.2 + 0.09) = 1.53 \text{ MeV}$$

Y^* de-excites by emitting a gamma photon of energy $\Delta E = 1.53$ MeV

11. If can be shown using classical mechanics that energy of neutron after a collision

$$= \frac{\text{Energy before collision}}{9}$$

(For proving this take mass of deuterium to be twice the mass of a neutron.)

$\therefore$
$$0.025 \text{ eV} = \frac{1 \times 10^6 \text{ eV}}{9^n} \quad (n = \text{required number of collisions})$$
$$9^n = 4 \times 10^7$$

$$n \log_{10} 9 = \log_{10} 4 + 7$$

$$n \times 2 \times 0.477 = 2 \times 0.301 + 7$$

$$\Rightarrow \quad n \simeq 8.$$

12. $\Delta m = 7.018 + 1.008 \times 2 - 4.004$

$= 0.018$ u

Energy released in one reaction $= \Delta mc^2$

$= (0.018 \times 1.67 \times 10^{-27} \text{ kg})\ (3 \times 10^8 \text{ m/s})^2$

$= 2.7 \times 10^{-12}$ J

1 g H $= 6.023 \times 10^{23}$ nuclei of H

$\therefore$ Energy released $= 2.7 \times 10^{-12} \times 6.023 \times 10^{23}$

$= 1.62 \times 10^{12}$ J

13. ${}^{9}_{4}\text{Be} + {}^{4}_{2}\text{He} \longrightarrow {}^{12}_{6}\text{C} + {}^{1}_{0}\text{n}$

Conservation of charge and mass number can be used to predict that particle X is neutron,

There is no binding energy for a single particle. X (i.e., neutron) is free particle not attached to any other particle. It has no binding energy.

Energy released in reaction = BE of product – BE of reactant

$= 12(7.6885) - 9(6.4763) - 4(7.0819) = 5.6477$ MeV

14. Let N denote the total no of nuclei. At any time T–

$$N = N_s + N_u.$$

After a time T since the activity decreased to one third, number of unstable nuclei will also become one third. So after a time T

$$N/2 = N_s + N_u/3$$

Solving the two equations we get $N_s/N_u = 1/3$

15. Speed of neutrons is given by–

$$\frac{1}{2} mv^2 = 0.0327 \times 1.6 \times 10^{-19}$$

$$\therefore \quad v^2 = \frac{2 \times 0.0327 \times 1.6 \times 10^{-19}}{1.675 \times 10^{-27}} = 625 \times 10^4$$

Or, $v = 2500$ m/s

with this speed, the time taken by the neutrons to travel a distance of 10 m is,

$$\Delta t = \frac{10}{2500} = 4 \times 10^{-3} \text{ s}$$

This time is very small compared to the half life hence, the fraction of neutrons decayed in time Δt can be calculated as

$$\frac{\Delta N}{N} = \lambda \Delta t \quad \text{But,} \quad \lambda = \frac{0.693}{T_{1/2}}$$

$$\therefore \quad \frac{\Delta N}{N} = \frac{0.693}{T_{1/2}} \Delta t = \frac{0.693}{693} \times (4 \times 10^{-3}) = 4 \times 10^{-6}.$$

16. X has excess neutrons where as Y has excess protons. Z is oversized.

Only α decay leads to size reduction. Negative β decay leads to lowering of $\frac{N}{Z}$ ratio. Positron emission leads to increase of $\frac{N}{Z}$ ratio.

18. (a) If m_x^n, m_y^n are nuclear masses of X and Y, then

$$\Delta m = (m_x^n) - (m_y^n + m_e)$$
$$= (m_x - Zm_e) - (m_y - (Z-1)m_e + m_e)$$
$$= m_x - m_y - 2m_e$$

$\therefore$ $$Q = \Delta mc^2 = [m_x - m_y - 2m_e]c^2$$

(b) $$\Delta m = [11.011434 - 11.009305 - 2 \times 0.000549]$$
$$= 0.001031 \text{ u}$$
$$Q = 0.001031 \times 930 \text{ MeV} = 0.959 \text{ MeV}$$

Maximum energy of emitted positron $\simeq Q = 0.959$ MeV

19. Because tritium in water is being replenished, we assume that the amount is constant until the wine is made, and then it decays. After n half-lives the activity becomes–

$$A = \frac{A_0}{2^n}$$

$\therefore$ $$\frac{1}{5} = \left(\frac{1}{2}\right)^n$$

$\therefore$ $$\log_{10} 5 = n \log_{10} 2$$

$\Rightarrow$ $$\log_{10} 10 - \log_{10} 2 = n \log_{10} 2$$

$\Rightarrow$ $$(n+1)\log_{10} 2 = \log_{10} 10$$

$\Rightarrow$ $$n + 1 = \frac{1}{0.303} = 3.3$$

$\Rightarrow$ $$n = 2.3$$

$\therefore$ $$\text{Age of wine} = 2.3 \times 12 = 27.6 \text{ yr.}$$

20. $$\underset{N_1}{^{228}\text{Th}} \xrightarrow{t_1 = 1.9\text{y}} \underset{N_2}{^{224}\text{Ra}} \xrightarrow{t_2 = 3.7\text{d}} \underset{N_3}{^{220}\text{Rn}} \xrightarrow{t_3 = 55\text{s}} \underset{N_4}{^{216}\text{Po}}$$

In radioactive equilibrium

$$N_1 \lambda_1 = N_2 \lambda_2 = N_3 \lambda_3$$

$\Rightarrow$ $$\frac{N_1}{t_1} = \frac{N_2}{t_2} = \frac{N_3}{t_3}$$

No. of nuclei in a 'm' gram sample of an isotope

$$N = \frac{m}{M} N_A$$

M = molar mass; N_A = Avogadro's number.

$\therefore$ $$\text{using } \frac{N_2}{t_2} = \frac{N_3}{t_3}$$

$$\frac{m_{Ra}}{224 \times (3.7\text{d})} = \frac{(1.0\text{g})}{220 \times (55\text{s})}$$

$$m_{Ra} = \frac{224}{220} \times \frac{3.7 \text{ d}}{55 \text{ s}} \text{ g}$$
$$= 5918 \text{ g} = 5.9 \text{ kg}$$

$$\text{Similarly,} \qquad \frac{N_1}{t_1} = \frac{N_3}{t_3}$$

$$\frac{m_{\text{Th}}}{232 \times 1.9y} = \frac{1.0 \text{ g}}{220 \times 55 \text{ s}}$$

$$\Rightarrow \qquad m_{\text{th}} = 1148849 \text{ g} = 1149 \text{ kg}$$

21.
$$\lambda = \frac{\ell n2}{12.7} = 5.46 \times 10^{-2} \text{ h}^{-1}$$

$$\lambda_{EC} + \lambda_\beta = \lambda$$

$$\lambda_{EC} + \left(\frac{39}{61}\right)\lambda_{EC} = \lambda$$

$$\therefore \qquad \lambda_{EC} = \frac{61\lambda}{100} = 3.33 \times 10^{-2} \text{ h}^{-1}$$

$$\therefore \qquad t^{EC}_{\frac{1}{2}} = \frac{\ell n2}{\lambda_{EC}} = 20.8 \text{ h.}$$

22.
$$N_{xo} = 2.5\, N_{yo}$$

After three days
$$N_x = 5\, N_y$$

$$\therefore \qquad N_{xo}e^{-\lambda_x(3d)} = 5\, N_{yo}\, e^{-\lambda_y(3d)}$$

$$\Rightarrow \qquad 2.5\, e^{-3\lambda_x} = 5\, e^{-3\lambda_y}$$

$$\Rightarrow \qquad -3\lambda_x = \ell n(2) - 3\lambda_y$$

$$3\frac{\ell n2}{t_x} = -\,\ell n(2) + 3\frac{\ell n2}{t_y}$$

$$\therefore \qquad \frac{1}{t_x} = -\frac{1}{3} + \frac{1}{t_y}$$

$$t_x = \frac{3\,t_y}{3 - t_y} = \frac{3 \times 2}{3 - 2} = 6 \text{ days}$$

23. No. of ^{14}C atom in 1 g carbon

$$N = \frac{A}{\lambda} = \frac{t_{1/2}\, A}{0.693}$$

$$= \frac{(3.03 \times 10^9 \text{ min}) \times 16}{0.693} = 6.99 \times 10^{10}$$

There are $\frac{1}{12}$ mol of C atoms in 1 g.

$\therefore$ Total no of C atoms in 1 g is

$$\text{No} = \frac{1}{12} \times 6.02 \times 10^{23}$$

$\therefore$ Required ratio

$$= \frac{6.99 \times 10^{10}}{\frac{6.02}{12} \times 10^{23} - 6.99 \times 10^{10}}$$

$$\simeq \frac{6.99 \times 10^{10}}{\frac{6.02}{12} \times 10^{23}} = 1.37 \times 10^{-12}$$

24. Mass of carbon in body $= \frac{18}{100} \times 50 \times 10^3$ gram.

No. of atoms of C in the body

$$= \frac{18 \times 50 \times 10^3}{100} \times \frac{1}{12} \times 6.02 \times 10^{23}$$

No. of atoms of ^{14}C in the body

$$N \simeq \frac{18 \times 50 \times 10^3 \times 6.02 \times 10^{23}}{100 \times 12} \times 1.37 \times 10^{-12}$$

$$= 6.18 \times 10^{14}$$

$\therefore$ Activity = no of β particle emitted per second.

$$= \lambda N$$

$$= \frac{0.693}{3.03 \times 10^9} \times 6.18 \times 10^{14}$$

$$= 1.41 \times 10^5 \text{ per minute}$$

25. ${}^{A}_{Z}X \rightarrow {}^{A}_{Z-1}Y + {}^{0}_{+1}e + \upsilon$

$$\Delta m = (m_x - Zm_e) - [m_y - (Z-1)m_e + m_e]$$

$$= m_x - m_y - 2m_e$$

For reaction to be feasible

$$\Delta Q > 0 \Rightarrow \Delta mc^2 > 0$$

$$\Rightarrow \quad m_x - m_y > 2m_e$$

26. Let N_0 = initial count of U^{238} nuclei

N = number of surviving nuclei of U^{238} after time t

$N_0 - N$ number of Pb nuclei at time t

$$\therefore \quad \frac{m_u}{m_{pb}} = \frac{4}{3} \Rightarrow \frac{238\,N}{226(N_0 - N)} = \frac{4}{3}$$

$$\therefore \quad \frac{N_0 - N}{N} = \frac{238 \times 3}{226 \times 4} = 0.79$$

$$\therefore \quad \frac{N_0}{N} = 1.79$$

$$\Rightarrow \quad \frac{N_0}{N_0\,e^{-\lambda t}} = 1.79 \Rightarrow e^{-\lambda t} = \frac{1}{1.79}$$

$$\Rightarrow \quad -\lambda t = \ell n\,1 - \ell n\,1.79$$

$$\therefore \quad t = \frac{\ell n 1.79}{\lambda} = \frac{\ell n 1.79}{\ell n 2} \times t_{1/2}$$

$$t = \frac{0.25}{0.3} \times 4.5 \times 10^9$$

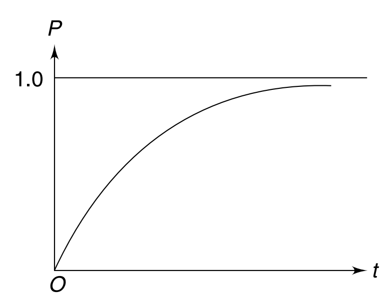
$$= 3.75 \times 10^9 \text{ y}$$

27.

$$N = N_0 e^{-\lambda t}$$

Probability that a nucleus decays in time 't' is

$$P = \frac{N_0 - N}{N_0} = 1 - \frac{N}{N_0} = 1 - e^{-\lambda t}$$



28. (a) A single radioactive nucleus has no memory of its past.

In time $3t_0$ the population will become $\frac{1}{8}$.

It means $\frac{7}{8}$ of the population will decay

$\therefore$ Probability that A will decay $= \frac{7}{8}$.

(b) Probability that A will survive $= \frac{1}{8}$

Probability that B will survive $= \frac{1}{8}$

Both events are independent

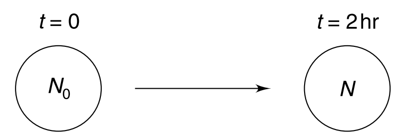
$\therefore$ Probability that both A and B will survive

$$= \frac{1}{8} \times \frac{1}{8} = \frac{1}{64}$$

29. Probability that a particular nucleus will decay in next 2 hr is

$$\frac{N_0 - N}{N_0} = 8 \times 10^{-4} \quad [N = \text{surviving population after } 2\,\text{hr}]$$

$$N_0[1 - 8 \times 10^{-4}] = N$$



$$\Rightarrow \quad 1 - 8 \times 10^{-4} = e^{-\lambda t} \qquad \left[\because \frac{N}{N_0} = e^{-\lambda t}\right]$$

$$\Rightarrow \quad e^{-\lambda t} = 0.9992 \quad \Rightarrow \quad -\lambda t = \ell n(0.9992)$$

$$-\frac{\ell n 2}{t_{1/2}} \times 2 = \ell n(0.9992) \qquad [\because \ t = 2\text{hr}]$$

$$t_{1/2} = \frac{2\ell n 2}{\ell n(0.9992)} \text{ hr}$$

$$= \frac{2 \times 0.693}{8 \times 10^{-4}} = 1732.5 \text{ hr}$$

30. The two possible decay modes are β^+ decay and electron capture.

(i) β^+ decay: $\quad {}^7_4\text{Be} \rightarrow {}^7_3\text{Li} + {}^{0}_{+1}\text{e} + \text{v}$

Q value for this decay is

$$Q = [m_{\text{Be}} - m_{\text{Li}} - 2m_e]\, c^2$$

$$= [7.01693 - 7.01600 - 2 \times 0.00055]\ u \times c^2$$

$$= -\ 0.00017\ u \times c^2$$

Negative Q values means this decay is not allowed.

(ii) Electron capture: $\quad {}^7_4\text{Be} + {}^{-1}_{0}\text{e} \rightarrow {}^7_3\text{Li} + v$

Q value for this decay is

$$Q = [m_{\text{Be}} - m_{\text{Li}}]\, c^2$$

$$Q = 0.00093\ u \times c^2$$

Hence, only electron capture is allowed

31. The reaction is

$${}_{15}\text{P}^{32} \rightarrow {}_{16}\text{S}^{32} + {}_{-1}\text{e}^0 + \overline{v}$$

$$\Delta m = (31.9740) - (31.9720 + 0.0005) = 0.0015\,\text{u}$$

$\therefore$ Energy released in the reaction

$$Q = 0.0015 \times 931 = 1.396 \text{ MeV} \simeq 1.4 \text{ MeV}$$

Average energy carried by ${}_{-1}e^0$ is 0.7 MeV

$$\therefore \quad 0.7 \times 10^6 \times 1.6 \times 10^{-19} \times \frac{dN}{dt} = 10 \times 10^{-3}$$

$$\Rightarrow \quad \frac{dN}{dt} = \frac{10^{12}}{11.2} \quad \Rightarrow \quad \lambda N = \frac{10^{12}}{11.2}$$

$$\Rightarrow \quad N = \frac{10^{12}}{11.2} \times \frac{14 \times 86400}{\ell n 2}$$

$$= 1.56 \times 10^{17}$$

32. From conservation of energy

$$m_n c^2 = m_P c^2 + k_P + m_e c^2 + k_e + m_\nu c^2 + k_\nu$$

$$939.5656 = 938.2723 + 0.5109 + 0.0004 + (k_P + k_e)$$

$$[\because \quad m_\nu c^2 + k_\nu = 0.0004 \text{ MeV}]$$

$$\Rightarrow \quad k_P + k_e = 0.0004 \text{ MeV}$$

$$\Rightarrow \quad \frac{P^2}{2m_P} + \frac{P^2}{2m_e} = 0.7820 \text{ MeV}$$

$$\Rightarrow \quad \frac{P^2}{2m_e}\left[1 + \frac{m_e}{m_p}\right] = 0.7820$$

$$k_e = \left(\frac{m_P}{m_p + m_e}\right) \times 0.7820 \simeq 0.7820 \text{ MeV}$$

33. (a)

$$Z_C = Z_A - 2 \times 2 + 3 \times 1 = Z_A - 1$$

Also $$Z_C = Z_B - 2 \times 1 + 5 \times 1 = Z_B + 3$$

$$\therefore \quad Z_A - 1 = Z_B + 3 \quad \Rightarrow \quad Z_A - Z_B = 4$$

$$A_C = A_A - 4 \times 2 = A_A - 8$$

And $$A_C = A_B - 1 \times 4 = A_B - 4$$

$$\therefore \quad A_A - 8 = A_B - 4 \quad \Rightarrow \quad A_A - A_B = 4$$

(b) For A $4N_0 \xrightarrow{1\,\text{hr}} 2N_0 \xrightarrow{1\,\text{hr}} N_0 \xrightarrow{1\,\text{hr}} \frac{N_0}{2} \xrightarrow{1\,\text{hr}} \frac{N_0}{4}$

For B $N_0 \xrightarrow{2\,\text{hr}} \frac{N_0}{2} \xrightarrow{2\,\text{hr}} \frac{N_0}{4}$

$\therefore$ After 4 hr both have same population. At this instant population of C = sum of population of decayed nuclei of A and B

$$= 4N_0 - \frac{N_0}{4} + N_0 - \frac{N_0}{4} = \frac{9N_0}{2}$$

(c) $$\lambda_A N_A = \lambda_B N_B$$

$$\Rightarrow \qquad \frac{\ell n2}{T_A} N_A = \frac{\ell n2}{T_B} N_B$$

$$\Rightarrow \qquad \frac{1}{1}\frac{4N_0}{(2)^{t_0/1}} = \frac{1}{2}\frac{N_0}{(2)^{t_0/2}} \quad \Rightarrow \quad 8 = 2^{t_0/2}$$

$$\Rightarrow \qquad \frac{t_0}{2} = 3 \quad \Rightarrow t_0 = 6 \text{ hr.}$$

34. Momentum of α particle

$$P = \frac{h}{\lambda} = \frac{6.63 \times 10^{-34}}{5.76 \times 10^{-15}} = 1.15 \times 10^{-19} \text{ kg ms}^{-1}$$

The daughter product will also have same momentum. Kinetic energy of products is

$$k = k_\alpha + k_{\text{daughter}} = \frac{P^2}{2m} + \frac{P^2}{2M}$$

$$\Rightarrow \qquad k = \frac{P^2}{2}\left(\frac{M+m}{Mm}\right)$$

$$= \left(\frac{1.15 \times 10^{-19}}{2}\right)^2 \times \frac{223.61 + 4.002}{223.61 \times 4.002 \times 1.67 \times 10^{-27}}$$

$$= 1.0 \times 10^{-12} \text{ J} = \frac{10^{-12}}{1.6 \times 10^{-13}} = 6.25 \text{ MeV}$$

Since k is equal to energy of the reaction

$$\therefore \qquad \Delta mc^2 = 6.25 \text{ MeV}$$

$$\Rightarrow \qquad \Delta m = \frac{6.25}{931} = 0.0067 \text{ u}$$

$\therefore$ Mass of parent = 4.002 + 223.610 + 0.0067 = 227.62 u

35. (a) Initial activity of both isotopes is same

$$A_0 = N_{01}\lambda_1 = N_{02}\lambda_2$$

$$\frac{N_{01}}{N_{02}} = \frac{\lambda_2}{\lambda_1} = \frac{\tau_1}{\tau_2} = \frac{405}{1620} = \frac{1}{4}$$

At time $t = 1620$ s

$$A_1 = \frac{N_{01}}{2^4}\lambda_1 = \frac{A_0}{16}$$

$$A_2 = \frac{N_{02}}{2}\lambda_2 = \frac{A_0}{2}$$

Total activity $A = A_1 + A_2 = \dfrac{9A_0}{16}$

Probability of getting $\alpha = \dfrac{A_1}{A} = \dfrac{1}{9}$

(b) $$\frac{N_{01}}{N_{02}} = \frac{1}{4}$$

If $N_{01} + N_{02} = N_0$ then

$$N_{01} = \frac{N_0}{5};\; N_{02} = \frac{4N_0}{5}$$

At time t $N_1 + N_2 = \dfrac{N_0}{2}$

$$\Rightarrow \quad \frac{N_0}{5}\left(\frac{1}{2}\right)^{\frac{t}{405}} + \frac{4N_0}{5}\left(\frac{1}{2}\right)^{\frac{t}{1620}} = \frac{N_0}{2}$$

Let $\left(\dfrac{1}{2}\right)^{\frac{t}{1620}} = x$

Then $x^4 + 4x - 2.5 = 0 \Rightarrow x = 0.59$

$$\therefore \quad \left(\frac{1}{2}\right)^{\frac{t}{1620}} = 0.59$$

$$\Rightarrow \quad \frac{t}{1620}[\log_{10} 1 - \log_{10} 2] = \log_{10}(0.59)$$

$$\Rightarrow \quad t = \frac{1620 \times [\log_{10} 10 - \log_{10} (5.9)]}{\log_{10} 2} = 1215\,\text{s}$$

36. ${}_1H^2 + \gamma \rightarrow {}_1^1H + {}_0^1n$ ...(i)

Momentum of γ photon $P = \dfrac{E}{c}$

$$P = \frac{2.22 \times 1.6 \times 10^{-13}}{3 \times 10^8} = 1.18 \times 10^{-21} \text{ kg m/s}$$

The proton formed after reaction must have this momentum. Hence, kinetic energy of proton is

$$KE_P = \frac{P^2}{2m_p} = \frac{(1.18 \times 10^{-21})^2}{2 \times 1.67 \times 10^{-27} \text{ kg}} = 4.17 \times 10^{-16} \text{ J}$$

$$= \frac{4.17 \times 10^{-16}}{1.6 \times 10^{-13}} \text{ MeV} = 2.6 \times 10^{-3} \simeq 0.003 \text{ MeV}$$

The KE carried by proton is small (and we could neglect it !)
From reaction given in (i)

$$2.014740 + \frac{2.21}{931} = 1.008145 + M_n$$

[We can use atomic masses in place of nuclear masses.Why?]

$\therefore \quad M_n = 1.008968\,\text{u}$

37. Number of moles $n = \dfrac{PV}{RT}$

$$= \frac{\left(10^5 \dfrac{N}{m^2}\right)(831 \times 10^{-6} \text{ m}^3)}{\left(8.31 \dfrac{\text{J}}{\text{mol.K}}\right)(300 \text{ K})}$$

$$\frac{1}{30} = \frac{6 \times 10^{23}}{30} = 2 \times 10^{22} \text{ atoms}$$

Let number of atoms of R_n^{222} and R_n^{220} be N_1 and N_2 respectively.

$$N_1 + N_2 = 2 \times 10^{22}$$

$$\frac{N_1}{N_2} = \frac{1000}{1}$$

$$\Rightarrow \qquad N_1 \simeq 2 \times 10^{22} \quad \text{and} \quad N_2 \simeq 2 \times 10^{19}$$

Rate constant
$$\lambda_1 = \frac{0.693}{4 \times 24 \times 60 \times 60} = 2.0 \times 10^{-6}\ \text{s}^{-1}$$

$$\lambda_2 = \frac{0.693}{1 \times 60} = 1.16 \times 10^{-2}\ \text{s}^{-1}$$

$\therefore$ Activity
$$\begin{aligned} A &= \lambda_1 N_1 + \lambda_2 N_2 \\ &= 2.0 \times 10^{-6} \times 2 \times 10^{22} + 1.16 \times 10^{-2} \times 2 \times 10^{19} \\ &= 4 \times 10^{16} + 23 \times 10^{16} \\ &= 27 \times 10^{16}\ \text{dps} \\ &= \frac{27 \times 10^{16}}{3.7 \times 10^{10}} = 7.3 \times 10^{6}\ \text{Ci} \end{aligned}$$

38. (b) 206 gram Pb is product of 238 gram U

$\therefore$ 0.8 g Pb is product of $\frac{238}{206} \times 0.8$ g of U

$$= 0.9243 \text{ g U}$$

For each g of U that is present in the ore now there was 1 + 0.9243 = 1.9243 g U present originally.

Now,
$$m = m_0\, e^{-\lambda t}$$

$$1 = 1.9243\, e^{-\frac{0.693}{4.5 \times 10^9} t}$$

$$t = \frac{\ell n(1.9243) \times 4.5 \times 10^9}{0.693}$$

$$= \frac{0.6546 \times 4.5 \times 10^9}{0.693} = 4.25 \times 10^9 \text{ yr}$$

39. Let A_{10} initial activity of P^{32}

$$A_{20} = \text{initial activity of } P^{33}$$

Let λ_1, λ_2 be respective decay constants

$$\lambda_1 = \frac{\ell n2}{14 \text{ days}};\ \lambda_2 \frac{\ell n2}{28 \text{ days}}$$

$$\therefore \qquad \frac{\lambda_1}{\lambda_2} = 2$$

$$\frac{A_{10}}{A_{20}} = \frac{\lambda_1 N_{10}}{\lambda_2 N_{20}} = 2 \times 4 = 8$$

$$A_{10} + A_{20} = 9\text{mCi}$$

$$\therefore \qquad 9A_{20} = 9\text{mCi}$$

$$\therefore \qquad A_{20} = 1\text{mCi and } A_{10} = 8\text{mCi}$$

Activity after 80 days

$$A = A_{10} e^{-\left(\frac{\ell n2}{14} \times 80\right)} + A_{20}\, e^{-\left(\frac{\ell n2}{28} \times 80\right)}$$

$$= 8\, e^{-\frac{0.7 \times 80}{14}} + 1e^{-\frac{0.7 \times 80}{28}}$$

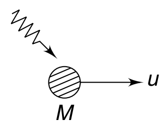
$$= 8e^{-4} + e^{-2}$$

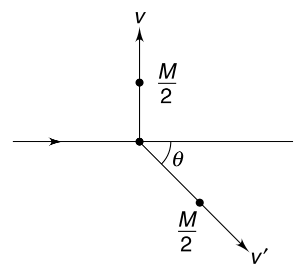
$$= e^{-2}[8e^{-2} + 1] = 0.14\,[8 \times 0.14 + 1]$$

$$= 0.3 \text{ mCi}$$

40. Momentum conservation

v' = velocity of the second atom





$$Mu = \frac{M}{2} v' \cos\theta$$

$\Rightarrow$ $$v' \cos\theta = 2u \quad \text{...(i)}$$

And $$\frac{M}{2} v' \sin\theta = \frac{M}{2} v$$

$\Rightarrow$ $$v' \sin\theta = v \quad \text{...(ii)}$$

From (i) and (ii) $v' = \sqrt{(2u)^2 + v^2}$

Energy conservation

$$\frac{1}{2}Mu^2 + \frac{hc}{\lambda} = BE + \frac{1}{2}\frac{M}{2}v^2 + \frac{1}{2}\frac{M}{2}(v')^2$$

$\therefore$ $$BE = \frac{1}{2}Mu^2 + \frac{hc}{\lambda} - \frac{M}{4}v^2 - \frac{M}{4}(4u^2 + v^2)$$

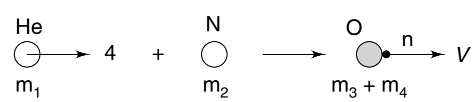
$$= \frac{hc}{\lambda} - \frac{Mu^2}{2} - \frac{Mv^2}{2}$$

41. $$\Delta m = (m_N + m_{He}) - (m_0 + m_H)$$

$$= (14.003 + 4.003) - (16.999 + 1.008)$$

$$= -0.001u$$

$\therefore$ $$Q = -0.001 \times 930 = -0.93 \text{ MeV}$$



The negative sign implies that reaction is endoergic.

(b) Maximum loss in KE will happen when the collision is perfectly inelastic.

$$m_1 u = (m_3 + m_4)v \qquad \text{[Momentum conservation]}$$

$\Rightarrow$ $$v = \frac{m_1 u}{m_3 + m_4}$$

And, energy conservation

$$Q = \frac{1}{2}m_1 u^2 - \frac{1}{2}(m_3 + m_4)\left(\frac{m_1 u}{m_3 + m_4}\right)^2$$

$$= \frac{1}{2}m_1 u^2 \left[\frac{m_3 + m_4 - m_1}{m_3 + m_4}\right]$$

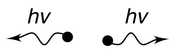
$$m_3 + m_4 \simeq m_1 + m_2$$

$$\therefore \qquad Q = \frac{1}{2} m_1 u^2 \left[\frac{m_2}{m_1 + m_2}\right]$$

$$\therefore \qquad \frac{1}{2} m_1 u^2 = Q\left[\frac{m_1 + m_2}{m_2}\right]$$

$$k_0 \simeq 0.93 \times \left[\frac{4 + 14}{14}\right]$$

$$= 1.20 \text{ MeV}$$

42. (a) ${}^{0}_{-1}e + {}^{0}_{+}e \rightarrow$ 

Two gamma photons of same energy must be produced in order to conserve momentum. Energy of each photon

$$m_e c^2 = 0.00055 \times 930 = 0.51 \text{ MeV}$$

(b)

$$\Delta m = 111.9028 + 1.0087 - 112.9044 = 0.0071\text{u}$$

$$Q = \Delta mc^2 = 0.0071 \times 930 \text{ MeV} = 6.60 \text{ MeV}$$

$$\lambda = \frac{hc}{3.30 \text{ MeV}} = \frac{12400 \text{ eV Å}}{3.3 \times 10^6 \, eV} = 0.004 \text{ Å}$$

43. (a)

$$\Delta m = 26.98434 - (26.98154 + 0.00055) = 0.00225\,\text{u}$$

$$Q = \Delta mc^2 = 0.00225 \times 930\,\text{MeV} = 2.0925 \text{ MeV}$$

(b) $KE_\beta < [Q - \text{excitation energy of } A\ell^*]$

$$KE_\beta < 1.0775 \text{ MeV}$$

(c) Smallest wavelength photon will have energy equal to excitation energy

$$\Rightarrow \qquad \frac{hc}{\lambda} = 1.015 \text{ MeV}$$

$$\Rightarrow \qquad \frac{12400 \text{ eV Å}}{\lambda} = 1.015 \times 10^6 \text{ eV}$$

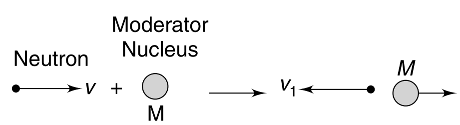
$$\Rightarrow \qquad \lambda = 0.0122 \text{ Å}$$

(d) The emitted electron has a kinetic energy equal to the lost nuclear excitation energy minus the binding energy of electron in the atom.

$$KE_{\text{electron}} = 1.015 \text{ MeV} - 2300 \text{ eV}$$

$$= 1.013 \text{ MeV}$$

44. (a) Momentum conservation



$$m_n v = -m_n v_1 + M v_2$$

$$v_2^2 = \left(\frac{m_n}{M}\right)^2 (v + v_1)^2 \qquad \text{...(i)}$$

Conservation of energy

$$\frac{1}{2} m_n v^2 = \frac{1}{2} m_n v_1^2 + \frac{1}{2} M v_2^2$$

$$v_2^2 = \left(\frac{m_n}{M}\right)\left(v^2 - v_1^2\right) \qquad \text{...(ii)}$$

From (i) and (ii)

$$\frac{m_n}{M}\left(v^2 - v_1^2\right) = \left(\frac{m_n}{M}\right)^2 (v + v_1)^2$$

$$\Rightarrow \qquad (v - v_1) = \frac{m_n}{M}(v + v_1)$$

$$\Rightarrow \qquad \frac{v_1}{v} = \frac{M - m_n}{m_n + M}$$

For the neutron

$$\frac{KE_{\text{after collison}}}{KE_{\text{before collison}}} = \frac{\frac{1}{2}m_n v_1^2}{\frac{1}{2}m_n v^2} = \left(\frac{M - m_n}{M + m_n}\right)^2$$

(b) If $M >> m_n$, the above ratio is close to 1 and energy lost by the neutron is small.

45. Let initial population of A and B be N_A and N_B respectively.

$$A_0 = \frac{dN_A}{dt} + \frac{dN_B}{dt}$$

$$40 = \lambda_A N_\text{A} + \lambda_B N_B$$

But $\qquad \lambda = \frac{\ell n2}{t_{1/2}} \qquad \therefore \quad \lambda_A = 2\lambda_B$

$$\therefore \qquad 40 = \lambda_A N_\text{A} + \frac{\lambda_A}{2}N_B \qquad ...(i)$$

After one hour population of $A = \frac{N_A}{4}$; Population of $B = \frac{N_B}{2}$

$\therefore$ Activity $= \lambda_A \frac{N_\text{A}}{4} + \lambda_B \frac{N_B}{2}$

$$15 = \lambda_A \frac{N_A}{4} + \frac{\lambda_A}{2}\frac{N_B}{2} \qquad ...(ii)$$

Solving (i) & (ii)

$$\lambda_A N_\text{A} = 20 \quad \text{and} \quad \lambda_A N_\text{B} = 40$$

$$\Rightarrow \qquad \frac{N_A}{N_B} = \frac{1}{2}$$

Also $\qquad \lambda_A N_A = 20$

$$\frac{0.693}{30 \times 60} N_A = 20 \quad \Rightarrow \quad N_A = 5.2 \times 10^4$$

And $\qquad N_B = 2N_A = 1.04 \times 10^5$

Total number of active nuclei $= 1.56 \times 10^5$

Mass of active nuclei $= \frac{1.56 \times 10^5}{6.02 \times 10^{23}} \times 200$ g

$= 5.2 \times 10^{-17}$ g

$\therefore$ Required fraction $= \frac{5.2 \times 10^{-17}}{10^{-6}} = 5.2 \times 10^{-11}$

46. $\overline{BE}$ value for mass no 56 is

$$\overline{BE}_{56} = 0.16 \times 56 = 8.96 \text{ MeV}$$

$\overline{BE}$ value for ${}_2^4$He as predicted by graph is

$$\overline{BE}_\alpha = 0.16 \times 4 = 0.64$$

$\therefore$ $$\overline{BE}_\alpha = 11 \times 0.64 = 7.04 \text{ MeV}$$

For $A > 56$ the equation of the line can be written as

$$\overline{BE} = -0.012\,A + C$$

For $$A = 56;\ \overline{BE} = 8.96$$

$\therefore$ $$C = 9.63 \text{ MeV}$$

$\therefore$ $$\overline{BE} = -\,0.012A + 9.63 \text{ for } A > 56$$

α – decay reaction is ${}^{A}_{z}X \rightarrow {}^{A-4}_{z-2}Y + {}^{4}_{2}\text{He}$

For this reaction to take place $BE_y + BE_\alpha > BE_x$

$$(A-4)(-0.012A + 9.63) + 7.04 \times 4 > A(-0.\ 012A + 9.63)$$

$\Rightarrow$ $$0.048A - 38.52 + 28.16 > 0$$

$\Rightarrow$ $$A > 215.8$$

$\therefore$ $$A_{\min} = 216$$

47. When population become equal both A and B have number of moles equal to

$$n = \frac{n_1 + n_1/4}{2} = \frac{5}{8}n_1$$

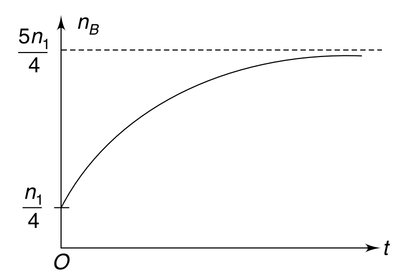
For $$A:\quad N = N_0\, e^{-\lambda t}$$

$$\frac{5}{8}\,n_1 = n_1 e^{-\lambda t} \quad\Rightarrow\quad \ell n\frac{8}{5} = \lambda t$$

$\Rightarrow$ $$(\ell n2)\,\frac{t}{t_{1/2}} = \ell n(1.6)$$

$\Rightarrow$ $$t \cdot (\ell n2) = t_{1/2} \times 0.68\,(\ell n2)$$

$\Rightarrow$ $$t = 1000 \times 0.68 = 680\,y$$



Graphical plot of variation of population of B is as shown.

Population of $B = \dfrac{n_1}{4}$ + population of A that decays

$$n_B = \frac{n_1}{4} + n_1\left[1 - e^{-\lambda t}\right]$$

$$n_B = n_1\left[\frac{5}{4} - e^{-\lambda t}\right]$$

48. Let number of nuclei initially in the sample be N_0. At time 't' let the number of nuclei in the container be N. Volume of sample inside the container at time t is

$$v = 1000 + 100t - 20t = 1000 + 80t$$

$\therefore$ No. of nuclei present in 20 ml solution at time 't' is $\dfrac{N}{1000 + 80t}20$

$\therefore$ $$-\frac{dN}{dt} = \frac{20N}{1000 + 80t}$$

$$\therefore \quad \int_{N_0}^{N} \frac{dN}{N} = -\int_0^5 \frac{2dt}{100 + 8t}$$

$$\ell n \frac{N}{N_0} = -\frac{1}{4}[\ell n(50 + 4t)]_0^5$$

$$\ell n \frac{N}{N_0} = -\frac{1}{4}\ell n\frac{7}{5}$$

$$\Rightarrow \quad \frac{N}{N_0} = \left(\frac{5}{7}\right)^{1/4} \Rightarrow N = N_0\left(\frac{5}{7}\right)^{1/4}$$

$\therefore$ Population in 100 ml solution taken out is $N_0 - N = N_0\left[1 - \left(\frac{5}{7}\right)^{1/4}\right]$

$$\therefore \quad A = A_0\left[1 - \left(\frac{5}{7}\right)^{1/4}\right]$$

49. The β^+ decay equation is

$$^{64}_{29}\text{Cu} \rightarrow {}^{64}_{28}\text{Ni} + {}^{0}_{-1}e + \text{v} + Q_1$$

The energy of the reaction is

$$Q_1 = (m_{\text{Cu}} - m_{\text{Ni}} - 2m_e)c^2$$

Where 'm' represents the atomic mass. Emitted neutrino can have KE

$$0 \le k_1 < Q_1$$

The EC decay equation is

$$^{64}_{29}\text{Cu} + {}^{0}_{-1}e \rightarrow {}^{64}_{28}\text{Ni} + v + Q_2$$

The energy of the reaction is

$$Q_2 = (m_{\text{Cu}} - m_{\text{Ni}})c^2$$

The neutrino can have KE

$$0 \le k_2 < Q_2$$

$$\therefore \quad k_2 - k_1 = Q_2 - Q_1 = 2m_ec^2 = 1.02 \text{ MeV}$$

50. (a) When potential of inner sphere is V_0 such that $eV_0 = E_2$,

then no β particle will be able to touch the outer sphere and will be pushed back to the inner sphere due to the electric field between the spheres. Hence, the final maximum potential of inner sphere is

$$V_0 = \frac{E_2}{e}$$

(b) $$V_0 = \frac{q}{4\pi\epsilon_0}\left(\frac{1}{a} - \frac{1}{b}\right)$$

Since inner surface of outer sphere and the inner sphere both will have equal and opposite charge

$$\therefore \quad q = \frac{4\pi\epsilon_0 V_0 ab}{b - a} = \frac{4\pi\epsilon_0 abE_2}{e(b - a)}$$

Charge $-q$ accumulates on the inner wall of the outer sphere due to accumulation of β particles emitted by the inner sphere. There is no supply of charge from the earth.

$\therefore$ Charge flow in wire $AB = 0$

51. Initial population of $X = N_0$

Effective decay constant $\lambda = \lambda_1 + \lambda_2$

Population of X after time t_0 is

$$N_{t_0} = N_0 e^{-(\lambda_1 + \lambda_2)t_0}$$

$\therefore$ Number of nuclei decayed by time t_0 is

$$= N_0[1 - e^{-(\lambda_1 + \lambda_2)t_0}]$$

Number of nuclei of Y formed is

$$N_y = \frac{\lambda_1}{\lambda_1 + \lambda_2} N_0[1 - e^{-(\lambda_1 + \lambda_2)t_0}]$$

Number of nuclei of Z at time t_0 is

$$N_z = \frac{\lambda_2}{\lambda_1 + \lambda_2} N_0[1 - e^{-(\lambda_1 + \lambda_2)t_0}]$$

In next interval of t_0, number of nuclei of Z produced = number of nuclei of X decayed

$$= N_{t_0} - N_{t_0} e^{-\lambda_2 t_0}$$

$$= N_0 e^{-(\lambda_1 + \lambda_2)t_0} [1 - e^{-\lambda_2 t_0}]$$

$\therefore$ At the end of interval $2t_0$ we have

$$\frac{\text{mass of } Y}{\text{mass of } Z} = \frac{M_y\left[\frac{\lambda_1 N_0}{\lambda_1 + \lambda_2}\left(1 - e^{-(\lambda_1 + \lambda_2)t_0}\right)\right]}{M_z\left[\frac{\lambda_2 N_0}{\lambda_1 + \lambda_2}\left(1 - e^{-(\lambda_1 + \lambda_2)t_0}\right) + N_0\, e^{-(\lambda_1 + \lambda_2)t_0}\left(1 - e^{-\lambda_2 t_0}\right)\right]}$$

$$= \frac{M_y\left[\lambda_1\left(1 - e^{-(\lambda_1 + \lambda_2)t_0}\right)\right]}{M_z\left[\lambda_2\left(1 - e^{-(\lambda_1 + \lambda_2)t_0}\right) + (\lambda_1 + \lambda_2)e^{(-\lambda_1 + \lambda_2)t_0}\left(1 - e^{-\lambda_2 t_0}\right)\right]}$$

52. (a) Mass defect in the given reaction is

$$\Delta m = 4\mathrm{m_H} - \mathrm{m_{He}}$$

$$= 4 \times 1.674 \times 10^{-27} - 6.645 \times 10^{-27}$$

$$= 5.1 \times 10^{-29} \text{ kg}$$

Energy released per reaction $\quad \Delta E = \Delta m c^2$

$$= 5.1 \times 10^{-29} \times (3 \times 10^8)^2$$

$$= 4.6 \times 10^{-12} \text{ J}$$

$\therefore$ Number of reactions occurring in the Sun per second is $= \dfrac{3.85 \times 10^{26}}{4.6 \times 10^{-12}}$

Number of neutrinos emitted per second

$$= \frac{3.85 \times 10^{26}}{4.6 \times 10^{-12}} \times 2$$

$\therefore$ Flux density

$$= \frac{3.85 \times 10^{26} \times 2}{4.6 \times 10^{-12}} \times \frac{1}{4\pi(1.5 \times 10^{11})^2}$$

$$= 5.9 \times 10^{14} \text{ m}^{-2} \text{ s}^{-1}$$

(c) Let N_0 = actual number of neutrinos arriving.

If efficiency for detecting is η then

$$N_1 = \eta N_0 \qquad \text{...(i)}$$

Due to conversion, out of N_0, fN_0 are v_0 and $(1 - f)N_0$ are v_e arriving at the detector.

$$\therefore \qquad N_2 = \frac{\eta}{5} f N_0 + \eta(1-f)N_0$$

$$\Rightarrow \qquad N_2 = f\frac{N_1}{5} + (1-f)N_1$$

$$\Rightarrow \qquad f = \frac{5}{4}\left(1 - \frac{N_2}{N_1}\right)$$

53. $A \xrightarrow{\lambda_A} B \xrightarrow{\lambda_B} C$

$t = 0 : \; N_0 \qquad 0 \qquad 0$

$t \quad : \; N_A \qquad N_B \qquad N_C$

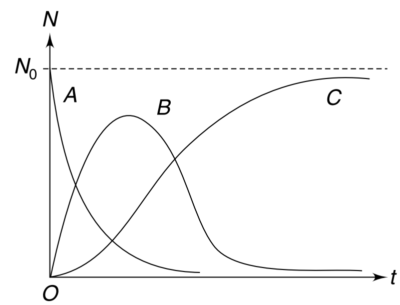
As per question $t_B = 5t_A$

$$\frac{\ell n2}{\lambda_B} = 5\,\frac{\ell n2}{\lambda_A}$$

$$\therefore \qquad \frac{\lambda_B}{\lambda_A} = \frac{1}{5}$$

(a) Initially, population of A is large. It decays fast. Population of B builds up fast. With decreasing population of A the rate of formation of B decreases, while its rate of decay will keep rising till its population gets maximum. Population of B starts decreasing afterward due to small population of A (i.e. small rate of formation of B). Population of C rises and eventually becomes N_0.

(b) When population of B is maximum



$$\frac{dN_B}{dt} = 0$$

$$\Rightarrow \qquad \lambda_A N_A = \lambda_B N_B$$

$$\Rightarrow \qquad \frac{N_B}{N_A} = \frac{\lambda_A}{\lambda_B} = 5$$

(c) Rate of formation of C = rate of decay of B

$$\frac{dN_C}{dt} = \lambda N_B$$

$\therefore$ Slope of N_C vs t graph rises till N_B becomes maximum. Thereafter, the slope of the graph decreases.

54. Radius of circular paths of beta particles ranges from zero to a

If v_0 is maximum speed of a particles then

$$\frac{mv_0}{eB} = a \quad \Rightarrow \quad mv_0 = aeB$$

Kinetic energy of such a particle is

$$k_0 = \frac{P^2}{2\text{m}} = \frac{a^2e^2B^2}{2\text{m}}$$

This is maximum KE of emitted beta particles. But $k_0 < Q$, where Q is energy of reaction.

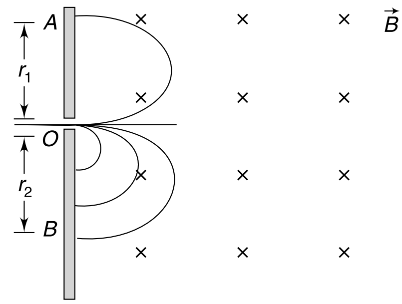
$$\therefore \qquad k_0 < (m_x - m_y)c^2$$

$$\Rightarrow \qquad \frac{a^2e^2B^2}{2mc^2} < (m_x - m_y)$$

55. ${}^{A}_{Z}X \rightarrow {}^{A-4}_{Z-2}Y + {}^{4}_{2}\text{He} + Q_1$

KE of α particles is given by-

$$k_\alpha = \left(\frac{A-4}{A}\right)Q_1$$



The particles striking the screen at A are α particles

$$\therefore \qquad r_1 = \frac{\sqrt{2m_\alpha k_\alpha}}{q_\alpha \cdot B} = \frac{\sqrt{2(4\text{m})\left(\frac{A-4}{A}\right)Q_1}}{2eB}$$

$$= \frac{\sqrt{2m\frac{(A-4)}{A}Q_1}}{eB}$$

Particle striking the screen at B is most energetic β particle

$$^{A}_{Z}D \rightarrow {}^{A}_{Z+1}B + {}^{0}_{-1}e + \bar{v} + Q_2$$

The maximum KE of β particles can be just less then Q_2

$$\therefore \qquad r_2 = \frac{\sqrt{2m_\beta \cdot k_\beta}}{q_\beta \cdot B} = \frac{\sqrt{2m_e Q_2}}{eB}$$

$$\frac{r_1}{r_2} = \sqrt{\frac{m}{m_e}\left(\frac{A-4}{A}\right)\frac{Q_1}{Q_2}} = \sqrt{\eta k\left(\frac{A-4}{A}\right)}$$

56. (a) $6\,{}^{0}_{-1}e$

(b) The reaction is

$$_{92}\text{U}^{238} \rightarrow {}^{206}_{82}\text{Pb} + 6\,{}^{0}_{-1}e + 6\bar{v} + 8({}^{2}_{4}\text{He})$$

$$Q = \left[M_{\text{U}_{238}} - 8M_{\text{He}} - M_{\text{Pb}} - 6m_e\right]c^2$$

All masses indicated in capital M represent nuclear masses.

$$\therefore \qquad Q = [(m_\text{U} - 92m_e) - 8(m_\text{He} - 2m_e) - (m_\text{Pb} - 82m_e) - 6m_e]\,c^2$$

All mass indicated in small case 'm' represent atomic masses.

$$\therefore \qquad Q = [m_\text{U} - 8m_\text{He} - m_\text{Pb}]\,c^2$$

$$= [238.050783 - 8 \times 4.002603 - 205.974449]\,u\left(931.5\ \frac{\text{MeV}}{u}\right)$$

$$= 5.71 \text{ MeV}$$

(d) If activity of the 100 kg sample is A, then power released will be

$$P = A \cdot Q$$

$$\lambda = \frac{\ell n2}{t_{1/2}} = \frac{0.693}{4.47 \times 10^9} = 1.55 \times 10^{-10}\ \text{yr}^{-1}$$

Population $\qquad N = \dfrac{100 \times 10^3}{238} \times 6.02 \times 10^{23} = 2.53 \times 10^{26}$

$$\therefore \qquad A = \lambda N = 1.55 \times 10^{-10} \times 2.53 \times 10^{26} \text{ decays per year.}$$

$$= 3.92 \times 10^{16} \text{ decays per year}$$

$$\therefore \qquad P = AQ = [3.92 \times 10^{16} \times 51.71]\ \frac{\text{MeV}}{\text{yr}}$$

$$= 3.92 \times 51.71 \times 10^{16} \times 1.6 \times 10^{-13}\ \text{J yr}^{-1}$$

$$= 3.24 \times 10^5\ \text{J yr}^{-1}$$

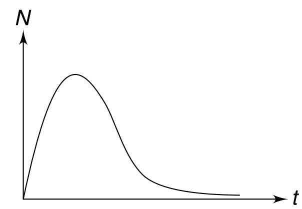
$$= \frac{3.24 \times 10^5}{60 \times 60 \times 24 \times 365} = \text{J s}^{-1}$$

$$= 0.01 \text{ Watt.}$$

57. (a) First the population in glands will increases as the blood flow brings in more and more isotopes and they get absorbed. The gland will get saturated and after that we will see a fall in population.

(b) Activity of the isotope present in the gland is

$$A = \frac{(6 \times 10^{-4} \text{ J})}{(20\text{ min}) \times (2 \times 10^{-8} \times 10^{-6} \text{ J})} = 1.5 \times 10^9 \text{ per minute}$$



Original activity of the gland $= \frac{22\text{ mCi}}{2} = 11\text{ mCi}$

$$= 11 \times 3.7 \times 10^7 = 4.07 \times 10^8 \text{ per second}$$

$$= 24.4 \times 10^9 \text{ per minute}$$

$$\therefore \qquad \frac{A}{A_0} = \frac{1.5}{24.4} \simeq \frac{1}{16}$$

$\therefore$ Four half lives have passed.

Hence Answer is 20 hour.

58. At the instant the population of the element is N the rate of increase of element is $\frac{dN}{dt} = R - \lambda N$

$$\frac{dN}{R - \lambda N} = dt \Rightarrow \int_0^N \frac{dN}{R - \lambda N} = \int_0^t dt$$

so that
$$N = \frac{R}{\lambda}(1 - e^{-\lambda t})$$

The equilibrium value when time becomes infinite ($t \to \infty$) is $= R/\lambda$. Setting N equal to 1/2 of this value gives

$$\frac{1}{2}\left(\frac{R}{\lambda}\right) = \frac{R}{\lambda}(1 - e^{-\lambda t})$$

$$\Rightarrow \qquad e^{-\lambda t} = \frac{1}{2} \quad \Rightarrow \quad t = \frac{\ln 2}{\lambda} = T_{1/2}$$

The result is independent of R.

59.
$$\frac{dN}{dt} = t^2 - \lambda N$$

For $\frac{dN}{dt}$ to be minimum $\frac{d^2N}{dt^2} = 0$

$$\Rightarrow \qquad \frac{d^2N}{dt^2} = 2t - \lambda(t^2 - \lambda N) = 0$$

Or,
$$N = \frac{2t_0 - \lambda t_0^2}{\lambda^2}$$

60.
$$P = P_0\left(\frac{m}{M_0}\right)^{7/2}\left(-\frac{dm}{dt}\right)c^2 = \frac{P_0}{M_0^{7/2}} \cdot m^{7/2}$$

$$\therefore \qquad \frac{dm}{m^{7/2}} = -\frac{P_0}{M_0^{7/2} \cdot c^2}\, dt$$

$$\therefore \qquad \int_{m_0}^{m} \frac{dm}{m^{7/2}} = -\frac{P_0}{M_0^{7/2}\, c^2}\int_0^t dt$$

Where m_0 = mass of the star at its birth.

$$\therefore \quad \left[\frac{1}{m^{5/2}}\right]_{m_0}^{m} = \frac{5P_0 t}{2M_0^{7/2}\, c^2}$$

$$\Rightarrow \quad \frac{1}{m^{5/2}} - \frac{1}{m_0^{5/2}} = \frac{5P_0 t}{2M_0^{7/2}\, c^2}$$

$$\Rightarrow \quad \frac{1}{m^{5/2}}\left[1 - \left(\frac{m}{m_0}\right)^{5/2}\right] = \frac{5P_0 t}{2M_0^{7/2}\, c^2} \qquad \text{...(i)}$$

Fraction of mass lost by the star since birth is

$$\alpha = \frac{m_0 - m}{m_0} = 1 - \frac{m}{m_0}$$

$$\Rightarrow \quad \frac{m}{m_0} = 1 - \alpha$$

$$\therefore \quad 1 - \left(\frac{m}{m_0}\right)^{5/2} = 1 - (1-\alpha)^{5/2} \simeq 1 - \left[1 - \frac{5}{2}\alpha\right] = \frac{5}{2}\alpha$$

$\therefore$ Equation (i) becomes

$$\frac{5\alpha}{2m^{5/2}} = \frac{5P_0 t}{2M_0^{7/2}\, c^2}$$

$$\therefore \quad t = \frac{\alpha M_0^{7/2}\, c^2}{P_0 m^{5/2}}$$

But $\quad \dfrac{P}{P_0} = \left(\dfrac{m}{M_0}\right)^{7/2} \quad$ But $\quad \Rightarrow \quad m = M_0\left(\dfrac{P}{P_0}\right)^{2/7}$

$$\therefore \quad t = \frac{\alpha M_0^{7/2}\, c^2}{P_0 M_0^{5/2}}\left(\frac{P_0}{P}\right)^{5/7} = \frac{\alpha M_0 c^2}{P_0^{7/2} \cdot P^{5/7}}$$

61. Tine required for evaporation to reduce the volume to half is $t_{1/2} = 24$ hr.

$$\therefore \quad \lambda = \frac{\ell n 2}{t_{1/2}} = \frac{\ell n 2}{24}$$

Volume added per hour to the pot = nv_0

$$\therefore \quad \frac{dv}{dt} = nv_0 - \lambda v$$

$$\int_{v=0}^{v} \frac{dv}{nv_0 - \lambda v} = \int_0^t dt \quad \Rightarrow \quad [\ell n(nv_0 - \lambda v)]_{v=0}^{v} = -\lambda t$$

$$\Rightarrow \quad \ell n\left(\frac{nv_0 - \lambda v}{nv_0}\right) = -\lambda t \quad \Rightarrow \quad 1 - \frac{\lambda v}{nv_0} = e^{-\lambda t}$$

$$\Rightarrow \quad v = \frac{nv_0}{\lambda}(1 - e^{-\lambda t})$$

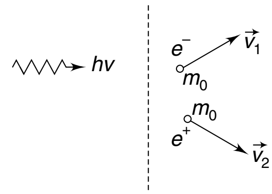
At $\quad t = 24$ hr; $\lambda t = \ell n 2$

$$\therefore \quad v = \frac{24 nv_0}{\ell n 2}[1 - e^{-\lambda n 2}]$$

$$= 24\frac{nv_0}{\ell n 2}\left[1 - \frac{1}{2}\right] = \frac{12}{\ell n 2} nv_0$$

62. (a) No. The process will violate the principle of conservation of charge.

(b) Let us assume that the process takes place as described.



Momentum Conservation

$$\frac{hv}{c} = |m_0(\vec{v}_1 + \vec{v}_2)| \qquad ...(i)$$

Energy Conservation

$$hv = m_0c^2 + m_0c^2 + \frac{1}{2}m_0v_1^2 + \frac{1}{2}m_0v_2^2$$

$$\Rightarrow \quad \frac{hv}{c} = m_0(c + c) + \frac{1}{2}m_0\frac{(v_1^2 + v_2^2)}{c} \qquad ...(ii)$$

Both the equations cannot be true simultaneously because

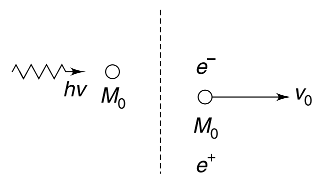
$$m_0(c + c) > |m_0(\vec{v}_1 + \vec{v}_2)|$$

Hence, the process cannot occur.

(c) **Momentum Conservation**

$$\frac{hv}{c} = M_0v_0 \qquad ...(i)$$

Energy Conservation



$$hv = 2m_0c^2 + \frac{1}{2}M_0v_0^2$$

$$\Rightarrow \quad \frac{(M_0v_0)^2}{M_0} = 2hv - 4m_0c^2$$

$$\Rightarrow \quad \frac{h^2v^2}{c^2M_0} = 2hv - 4m_0c^2$$

$$\therefore \quad M_0 = \frac{h^2v^2}{2c^2[hv - 2m_0c^2]}$$

63. (a)

$$A \xrightarrow{\lambda_1} B \xrightarrow{\lambda_2} C$$

$t = 0$	N_1^0	$N_2^0 = 0$
t	N_1	N_2

$$\left.\begin{aligned} N_1 &= N_1^0 e^{-\lambda_1 t} \\ \Rightarrow A_1 &= A_1^0 e^{-\lambda_1 t} \end{aligned}\right\} \qquad ...(i)$$

The required differential equation for population of B is

$$\frac{dN_2}{dt} = \lambda_1N_1 - \lambda_2N_2$$

$$\frac{dN_2}{dt} = \lambda_1N_1^0 e^{-\lambda_1 t} - \lambda_2N_2$$

(b) Solution to above equation (under condition $N_2^0 = 0$) is

$$N_2 = \frac{\lambda_1}{\lambda_2 - \lambda_1} N_1^0(e^{-\lambda_1 t} - e^{-\lambda_2 t})$$

$$\Rightarrow \quad A_2 = \lambda_2N_2 = \frac{\lambda_2}{\lambda_2 - \lambda_1} A_1^0 (e^{-\lambda_1 t} - e^{-\lambda_2 t}) \qquad ...(ii)$$

(ii) Under given condition, (λ_2 is quite large compared to λ_1) after a long time

$$e^{-\lambda_1 t} >> e^{-\lambda_2 t} \therefore A_2 \simeq \left(\frac{\lambda_2}{\lambda_2 - \lambda_1}\right) A_1^0 e^{-\lambda_1 t}$$

$\Rightarrow$ $$A_2 = \left(\frac{\lambda_2}{\lambda_2 - \lambda_1}\right) A_1$$

$$\frac{A_2}{A_1} = \frac{\lambda_2}{\lambda_2 - \lambda_1} = \text{constant}$$

(ii) Here $\lambda_2 >> \lambda_1$

$\therefore$ From equation (ii)

$$A_2 \simeq \frac{\lambda_2}{\lambda_2} A_1^0 (e^{-\lambda_1 t} - 0)$$

$\Rightarrow$ $$A_2 = A_1^2 \, e^{-\lambda_1 t} = A_1$$

(iii) $$A_2 = \frac{\lambda_2 A_1^0}{\lambda_2 - \lambda_1} [e^{-\lambda_1 t} - e^{-\lambda_2 t}]$$

$$\frac{dA_2}{dt} = 0$$

$$-\frac{\lambda_1 \lambda_2 A_1^0}{\lambda_2 - \lambda_1} e^{-\lambda_1 t} + \frac{\lambda_2^2 A_1^0}{\lambda_2 - \lambda_1} e^{-\lambda_2 t} = 0$$

$\Rightarrow$ $$\lambda_1 e^{-\lambda_1 t} = \lambda_2 e^{-\lambda_2 t}$$

$\therefore$ $$t = \frac{1}{\lambda_2 - \lambda_1} \ln\left(\frac{\lambda_2}{\lambda_1}\right)$$

This is the time when A_2 is maximum.